Transseries

The polynomial–logarithmic calculus,
series reversal at infinity,
and the inversion of rapidly growing functions

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Abstract

This volume is the single consolidated treatment of the `series-and-transseries` corpus. It develops one formal scale — the polynomial–logarithmic transseries, whose monomials are $X^{-\mu}(\log X)^k$ — from its ring structure through arithmetic, differentiation, composition and reversal at infinity; it then specializes that calculus to the Lambert W function and the Wright ω function, converts formal identities into Poincaré asymptotics with certificates, and finally applies the whole apparatus to invert a dozen rapidly growing functions and sequences to all orders.

The organizing observation is that inversion is not a collection of tricks. For every subject treated here the same three steps suffice: isolate a *dominant block* that can be inverted in closed form; check that its normalized slope is a unit; and revert the remainder by one triangular recurrence whose only division is by that slope. What changes between subjects is which block, and what the slope is. For the polynomial–logarithmic scale the block is $X +$ (lower) and the slope is 1; for a phase $aX + b\log X$ the block is inverted by W and the slope is $a + b/X$; for $ax^p(\log x - \beta)$, where the unknown multiplies the logarithm, the block is again inverted by W but by a different substitution and the slope is $v + 1$; for a pure exponential the block is a logarithm and the slope is constant.

The first eight parts develop the calculus. The remaining parts apply it: to the reversal of $x + W(x)$; to four combinatorial sequences — the rooted trees of A000081, the double factorial, the partition numbers, the swing factorial; to four special functions — Euler's Γ, the Barnes G-function, the hyperfactorial K-function, the subfactorial; and to a real-argument Fibonacci function whose inverse, alone among them, converges. A synthesis part compares all of them and records what the common frame does not explain.

Every part of this volume is a consolidation of independently written articles; the provenance appendix lists each source and the part that absorbed it, and the repair appendix lists every correction made to a source. Where several sources proved the same thing, one presentation is kept and the others cite it; where a source asserted a named result without proof, a proof is supplied or the statement is demoted to a remark that says what is actually established.

Contents

Part I

Orientation

Chapter 1

What a transseries is, and why one is needed

This part replaces four independently written expository introductions: `Transseries_Tutorial-1`, `-2`, `Transseries_Tutorial-3` (*Transseries for Mere Mortals*) and `-4`. Between them they state 58 named results, of which 21 are already theorems of the later parts of this volume and are cited here rather than restated. Of the remainder, most are definitions — the vocabulary is genuinely common to all four — and those are given once. Five of the named theorems are labelled in their sources "stated without proof" or "schematic"; Section 2.4 says which, and why they stay quoted. Everything else stated here as a theorem is proved here.

1.1 An expansion needs a scale

A limit remembers too little. Knowing that $f(x) \to 0$ says nothing about whether f is $1/x$, e^{-x} or $1/\log x$, and those three behave so differently that no single statement covers them. An asymptotic expansion records more: it compares f against a fixed yardstick.

Definition 1.1 (Asymptotic scale)**.** Let $\mathcal{S}$ be a filter base on a set — for us always $x \to +\infty$ along a real ray, or $x \to \infty$ in a sector. A sequence $(\varphi_n)_{n\geq 0}$ of eventually nonvanishing functions is an *asymptotic scale* if

$$\varphi_{n+1} = o(\varphi_n) \qquad \text{for every } n \geq 0. \tag{1.1}$$

Lean status. Formal as `Fabius.IsAsymptoticScale` in `TransseriesScale.lean`, for an arbitrary filter l on an arbitrary type with values in an arbitrary normed field — the filter base $\mathcal{S}$ of the definition is the filter itself, so nothing is lost and the sector case is included. The structure has exactly the two fields the definition asks for: `eventually_ne`, that each φ_n is eventually nonvanishing, and `isLittleO_succ`, which is (1.1). Concrete instances are supplied in `TransseriesPolyLogScale.lean`: `Fabius.isAsymptoticScale_plMonomial` for any lexicographically decreasing sequence of exponent pairs, and the two ladders used most often, `Fabius.isAsymptoticScale_plMonomial_pow` for X^{-n} and `Fabius.isAsymptoticScale_plMonomial_log` for $X^c(\log X)^{-n}$.

Definition 1.2 (Poincaré asymptotic expansion)**.** Given a scale (φ_n) and constants a_n, write

$$f \sim \sum_{n\geq 0} a_n\varphi_n \quad \text{to mean} \quad f - \sum_{n<N} a_n\varphi_n = \mathcal{O}(\varphi_N) \ \text{ for every fixed } N. \tag{1.2}$$

The number of retained terms is fixed *before* the limit is taken. No claim is made that the infinite series converges, and in the cases that matter in this volume it does not.

Lean status. Formal as `Fabius.IsPoincareExpansion` in `TransseriesScale.lean`, transcribing (1.2) literally: for every N, the function $x \mapsto f(x) - \sum_{n<N} a_n\varphi_n(x)$ is $\mathcal{O}(\varphi_N)$. The quantifier order that the definition emphasizes — N fixed before the limit — is the quantifier order of the formal statement, and no summability or convergence hypothesis appears anywhere in it.

Proposition 1.3 (Uniqueness relative to a scale)**.** *The coefficients in* (1.2) *are uniquely determined by f and the scale:*

$$a_N = \lim \frac{f - \sum_{n<N} a_n\varphi_n}{\varphi_N}, \tag{1.3}$$

the limit being along $\mathcal{S}$. In particular two different sequences of coefficients cannot represent the same f on the same scale.

Proof. Induction on N. For $N = 0$, (1.2) at $N = 1$ gives $f - a_0\varphi_0 = \mathcal{O}(\varphi_1) = o(\varphi_0)$, so $f/\varphi_0 \to a_0$; this both proves the formula and shows a_0 is determined. Suppose $a_0, \ldots, a_{N-1}$ are determined. Applying (1.2) at $N+1$ to $g := f - \sum_{n<N} a_n\varphi_n$ gives $g - a_N\varphi_N = \mathcal{O}(\varphi_{N+1}) = o(\varphi_N)$, whence $g/\varphi_N \to a_N$, which is (1.3). □

Lean status. Formal in `TransseriesScale.lean`. The explicit limit (1.3) is `Fabius.IsPoincareExpansion.tendsto_coeff`, and the uniqueness assertion is `Fabius.IsPoincareExpansion.coeff_unique`, proved from it by strong induction with `tendsto_nhds_unique`; the filter must be nontrivial (`Filter.NeBot`) for the second, as it must be informally, since on the trivial filter every sequence expands every function. The step that does the work is isolated as `Fabius.IsPoincareExpansion.isLittleO_succ_remainder`: the $(N+1)$-st condition turns the N-th remainder into $a_N\varphi_N + o(\varphi_N)$, which is the whole content of the argument and the only place the scale property is used.

Remark 1.4 (Formal crosswalk). The sequence-indexed content of Theorems 1.1 to 1.3 and Equations (1.1) to (1.3) is Exact in `TransseriesScale.lean`. The structure `Fabius.IsAsymptoticScale` records eventual nonvanishing and the successive little-o law; `Fabius.IsPoincareExpansion` records the first-omitted-term remainder convention, with finite sums `Fabius.poincarePartialSum` and their boundary/successor laws `Fabius.poincarePartialSum_zero` and `Fabius.poincarePartialSum_succ`; `Fabius.IsPoincareExpansion.isLittleO_succ_remainder` sharpens the next remainder to little-o; `Fabius.IsPoincareExpansion.tendsto_coeff` is exactly (1.3), with the literal quotient form supplied by `Fabius.IsPoincareExpansion.tendsto_coeff_div`; and `Fabius.IsPoincareExpansion.coeff_unique` proves coefficientwise uniqueness. The last theorem assumes a nontrivial filter `[l.NeBot]`, which makes the limiting direction meaningful. This is the ordered sequence version used throughout the corpus; it does not package the unordered-set, finite-maximal, positive-monomial, or nonzero-coefficient conventions of Theorem 4.6.

Remark 1.5 (Uniqueness is relative, and that is the whole point). Theorem 1.3 says the coefficients are unique *given the scale.* It does not say the expansion determines f. It cannot: the two functions f and $f + \mathrm{e}^{-x}$ have identical expansions on the scale $(x^{-n})_{n\geq 0}$, because $\mathrm{e}^{-x} = \mathcal{O}(x^{-N})$ for every N. This is the single fact that makes transseries necessary, and Section 1.2 develops it.

1.2 Beyond all algebraic orders

Definition 1.6 (Flatness)**.** A function ε is *flat* relative to a scale (φ_n) if $\varepsilon = \mathcal{O}(\varphi_n)$ for every n. Relative to the power scale (x^{-n}) this is often written $\varepsilon = \mathcal{O}(x^{-\infty})$.

Lean status. Formal as `Fabius.IsFlat` in `TransseriesFlat.lean`, stated for the abstract scale above rather than only for the power scale, so it applies verbatim to every scale used in this volume. The power-scale instance $\varepsilon = \mathcal{O}(x^{-\infty})$ is `Fabius.powScale`, defined through `Fabius.plMonomial` so that the dominance results of `TransseriesPolyLogScale.lean` apply to it unchanged, with `Fabius.powScale_eq_rpow` the bridge to the plain real power.

Proposition 1.7 (The invisible-function problem). e^{-x} *is flat relative to the power scale* $(x^{-n})_{n\geq 0}$. *Relative to that scale, the flat functions form a vector space and are closed under multiplication by anything of at most polynomial growth. Consequently a Poincaré expansion on the power scale determines* f *only up to a flat function, and no refinement of the coefficients can recover the difference.*

Proof. For any N, $x^N \mathrm{e}^{-x} \to 0$ as $x \to +\infty$, since e^x exceeds $x^{N+1}/(N+1)!$; hence $\mathrm{e}^{-x} = o(x^{-N})$ for every N. Closure under scalar multiplication and finite sums follows from the corresponding rules for $\mathcal{O}$. For the multiplier assertion, let $h = \mathcal{O}(x^M)$ for some integer $M \geq 0$ and let ε be flat on the power scale. Given N, flatness at index $N + M$ gives $\varepsilon = \mathcal{O}(x^{-(N+M)})$, and therefore $h\varepsilon = \mathcal{O}(x^M x^{-(N+M)}) = \mathcal{O}(x^{-N})$. This holds for every N, so $h\varepsilon$ is flat. Finally, if $f - \sum_{n<N} a_n x^{-n} = \mathcal{O}(x^{-N})$ for every N, adding a flat ε preserves every one of those estimates. Thus f and $f + \varepsilon$ have the same coefficient sequence, and the expansion cannot separate them. □

Lean status. Formal in `TransseriesFlat.lean`, all three clauses, and the third in a sharper form than stated here. That the flat functions form a vector space is `Fabius.flatSubmodule`, a genuine `Submodule` over the coefficient field, with `Fabius.isFlat_zero`, `Fabius.IsFlat.add`, `Fabius.IsFlat.neg`, `Fabius.IsFlat.sub` and `Fabius.IsFlat.const_smul` as its elementary closure laws. Closure under multiplication is stated through an abstract hypothesis, `Fabius.AbsorbsScale`: multiplying by h can be compensated by moving far enough along the scale, $h\varphi_m = \mathcal{O}(\varphi_n)$ for some m depending on n. `Fabius.IsFlat.mul_absorbsScale` is then the closure law, and `Fabius.absorbsScale_of_isBigO_pow` is the concrete statement of the proposition, that anything of at most polynomial growth absorbs the power scale — there the compensation is exact rather than an estimate, since $X^N \cdot X^{-(n+N)} = X^{-n}$ identically for $X > 0$. Flatness of e^{-x} itself is `Fabius.isFlat_exp_neg`.

The third clause is formal as an *equivalence*, `Fabius.isPoincareExpansion_iff_isFlat_sub`: given one function with the coefficients a, the functions with those same coefficients are precisely the ones differing from it by a flat function. So the flat functions are exactly the invisible ones, not merely some of them, which is a little more than the proposition claims; the one-directional half is `Fabius.IsPoincareExpansion.add_isFlat` and the converse `Fabius.isFlat_sub_of_isPoincareExpansion`. A pleasant corollary of the same bookkeeping is `Fabius.isPoincareExpansion_zero_iff`: being flat *is* having the identically zero Poincaré expansion. None of this needs the scale to be an asymptotic scale, nor the filter to be nontrivial, both of which Theorem 1.3 does need — invisibility is a statement about $\mathcal{O}$ alone.

Remark 1.8 (Editorial boundary of the multiplier claim). An earlier version asserted polynomial-growth multiplier closure for the flat functions of an arbitrary asymptotic scale. The scale axioms alone do not give the index shift used in the proof: for a general scale one must separately know that, for every N, some later m satisfies $h\varphi_m = \mathcal{O}(\varphi_N)$. The power scale has this compatibility with $m = N + M$. The proposition is therefore restricted to the power scale, which is also the only scale needed for its consequence here.

Lean status. Exact, by composition, in `TransseriesFlat.lean`. The definition Theorem 1.6 is `Fabius.IsFlat`; its vector-space operations are `Fabius.isFlat_zero`, `Fabius.IsFlat.add`, `Fabius.IsFlat.neg`, `Fabius.IsFlat.sub`, and `Fabius.IsFlat.const_smul`. The concrete exponential estimate is `Fabius.isFlat_exp_neg_rpow_atTop`. The corrected multiplier clause on the power scale is `Fabius.IsFlat.smul_of_isBigO_inv_pow`; the more general and explicitly necessary scale-compatibility premise is isolated in `Fabius.IsFlat.smul_of_scale_absorption`. Finally, `Fabius.IsPoincareExpansion.add_flat`, `Fabius.IsPoincareExpansion.sub_same_coeff_isFlat`, and `Fabius.IsPoincareExpansion.iff_sub_isFlat` prove precisely that adding a flat remainder preserves all coefficients, and characterize equality of a fixed coefficient sequence by a flat difference.

Remark 1.9 (This volume is a catalogue of the phenomenon). Almost every chapter of this volume is, at bottom, an instance of Theorem 1.7. In Part L the entire block structure of the inverse of $x + W(x)$

is flat relative to the inverse-logarithmic scale, so an expansion in powers of $1/\log z$ — correct to every order — sees none of it. In Part Q the constant $\frac{1}{2}$ in Γ_+^{-1} is flat relative to the outer expansion of the Lambert core. In Part U the third family of sectors is flat relative to the first two. In Part S the whole subfactorial correction is flat relative to every fixed exponential. The remedy is always the same: enlarge the scale until the invisible term becomes a monomial of it.

1.3 Divergence is not failure

Remark 1.10 (The Euler example). The classical illustration is the Euler equation $x^2y' + y = x$, whose formal solution is $\sum_{n\geq 0}(-1)^n n!\, x^{n+1}$. The series diverges for every $x \neq 0$, and yet its truncations approximate a genuine solution extremely well: the error after N terms is about $N!\,|x|^{N+1}$, which for fixed x first decreases and then grows. Stopping at the smallest term — optimal truncation — leaves an error of order $\mathrm{e}^{-1/|x|}$, exponentially small. The rigorous version of this, for the Gevrey-1 remainder bounds this volume actually needs, is Theorem J.54; the point to carry forward is that a divergent expansion can be more informative than a convergent one, and that the size of its least term is itself a meaningful quantity.

Remark 1.11 (Where the exponentially small term goes). The $\mathrm{e}^{-1/|x|}$ left over by optimal truncation is exactly a flat function in the sense of Theorem 1.6. It is not noise: it is a term of the answer, on a scale the power series cannot express. A transseries is the bookkeeping that lets one write it down — and then, if one wishes, expand *its* coefficient in powers again, producing a second series, and so on.

Chapter 2

The algebra of monomials

The vocabulary below is common to all four source tutorials and to the calculus parts of this volume; it is fixed once here.

2.1 Ordered monomials and well-based support

Definition 2.1 (Ordered monomial group)**.** A *monomial group* is a totally ordered abelian group $(\mathfrak{M}, \cdot, \prec)$: multiplication of monomials is the group law, and $\mathfrak{m} \prec \mathfrak{n}$ means $\mathfrak{m}$ is asymptotically negligible against $\mathfrak{n}$. Compatibility means $\mathfrak{m} \prec \mathfrak{n}$ implies $\mathfrak{m}\mathfrak{p} \prec \mathfrak{n}\mathfrak{p}$ for every $\mathfrak{p}$.

Lean status: Mathlib. This is Mathlib's hypothesis package for Hahn series, written additively: a `LinearOrder` on Γ together with `IsOrderedCancelAddMonoid`, whose `add_le_add_left` field is precisely the compatibility demanded here. Nothing in the corpus needs to be added; the note is placed because the crosswalk should say which formal object the definition names. One convention must be inverted when reading Mathlib against this volume: Mathlib indexes a Hahn series by *exponents* and orders them so that the leading term sits at the *smallest* index, whereas $\prec$ here orders *monomials* so that the leading one is the greatest. The two are the same structure through $\mathfrak{m} \mapsto -(\text{exponent})$, and every statement transcribed in either direction has to reverse the order.

Remark 2.2 (Why the ordering cannot be Archimedean). In $\mathfrak{M}$ the monomials x^{-1} and e^{-x} satisfy $\mathrm{e}^{-x} \prec x^{-n}$ for every n; so no multiple of e^{-x} ever reaches x^{-1}, and the ordered group is not Archimedean. This is the algebraic shadow of Theorem 1.7, and it is why the value group of a transseries field is large and the field is not a Hardy field of finite rank.

Definition 2.3 (Well-based support and Hahn series)**.** A subset $S \subseteq \mathfrak{M}$ is *well-based* (equivalently *reverse well-ordered*) if every nonempty subset of S has a $\prec$-greatest element; equivalently, S contains no strictly increasing infinite sequence. Given a coefficient field $\mathbb{K}$, the *Hahn series* field $\mathbb{K}[[\mathfrak{M}]]$ consists of the formal sums $\sum_{\mathfrak{m}\in S} c_{\mathfrak{m}}\mathfrak{m}$ with S well-based. The *leading monomial* of a nonzero such series is the greatest element of its support, and its coefficient the *leading coefficient.*

Lean status: Mathlib. `HahnSeries` Γ R is exactly this object, a coefficient function $\Gamma \to R$ carrying a proof that its support is `Set.IsPWO`. Well-basedness in the sense used here — no strictly increasing infinite sequence — is `Set.IsPWO` read through the order reversal noted at Theorem 2.1; on a linear order it coincides with `Set.IsWF`, and Mathlib supplies both readings of the same field as `HahnSeries.isPWO_support` and `HahnSeries.isWF_support`. The leading monomial is `HahnSeries.orderTop`, valued in `WithTop` Γ so that the zero series is handled rather than excluded, and the leading coefficient is `HahnSeries.leadingCoeff`. That $\mathbb{K}[[\mathfrak{M}]]$ is a field when $\mathbb{K}$ is one and $\mathfrak{M}$ is a linearly ordered abelian group — asserted but not proved in this section — is `HahnSeries.instField`, machine-checked in Mathlib.

The next two lemmas are what make that definition usable: they are the reason products and infinite sums of Hahn series exist. Both are stated in the sources; one of them is stated there without proof.

Lemma 2.4 (Neumann's lemma)**.** *Let $A, B \subseteq \mathfrak{M}$ be well-based. Then $A \cdot B$ is well-based, and every $\mathfrak{c} \in A \cdot B$ has only finitely many factorizations $\mathfrak{c} = \mathfrak{a}\mathfrak{b}$ with $\mathfrak{a} \in A$, $\mathfrak{b} \in B$.*

Proof. Write the group additively for the proof, so $A + B$ and $\mathfrak{c} = a + b$.

Finiteness of factorizations. Suppose $c = a_i + b_i$ for infinitely many distinct pairs (a_i, b_i), $i \in \mathbb{N}$. Every sequence in a totally ordered set has a monotone subsequence, so we may assume (a_i) is monotone. It cannot be strictly increasing, since A is well-based; so (a_i) is non-increasing. Then $b_i = c - a_i$ is non-decreasing, and because the pairs are distinct and determined by a_i, infinitely many b_i are distinct, giving a strictly increasing sequence in B — contradicting that B is well-based.

$A + B$ is well-based. Suppose $c_1 \prec c_2 \prec \cdots$ in $A + B$, with $c_i = a_i + b_i$. Passing to a subsequence, assume (a_i) monotone; as above it is non-increasing. Then $b_i = c_i - a_i$ is strictly increasing, since c_i is strictly increasing and $-a_i$ is non-decreasing — again contradicting that B is well-based. □

Lean status. Exact in `TransseriesWellBased.lean` in Mathlib's partially-well-ordered orientation, for an arbitrary commutative monoid that is partially ordered and order-cancellative. The first half is `Fabius.neumann_isPWO`, that a product of well-based sets is well-based, and the second is `Fabius.neumann_finite_factorizations`, that $\{(\mathfrak{m}, \mathfrak{n}) \in A \times B : \mathfrak{m}\mathfrak{n} = \mathfrak{c}\}$ is finite for every $\mathfrak{c}$ — which is exactly what makes the convolution defining a product of Hahn series a finite sum. Both are bridges to Mathlib (`Set.IsPWO.mul` and `Set.MulAntidiagonal.finite_of_isPWO`) rather than new arguments; the point of recording them is that the volume notes one of the two is asserted without proof in its sources, and here both are machine-checked. Mathlib's `Set.IsPWO` rules out decreasing chains and selects least elements; the manuscript's literal greatest-element/no-increasing convention is recorded by the public wrappers `Fabius.neumann_isPWO_orderDual` and `Fabius.neumann_finite_factorizations_orderDual`. The manuscript's totally ordered abelian group is a specialization of their more general partially ordered, order-cancellative commutative-monoid setting. This is an exact order-theoretic crosswalk; it does not construct a Hahn-series type.

Lemma 2.5 (Dickson's lemma)**.** *Every subset of $\mathbb{N}^k$ has finitely many minimal elements for the componentwise order; equivalently, $\mathbb{N}^k$ contains no infinite antichain.*

Proof. Let $(v_i)_{i\geq 1}$ be any sequence in $\mathbb{N}^k$. Its first coordinates form a sequence in $\mathbb{N}$, which has a non-decreasing subsequence; within that subsequence the second coordinates again have a non-decreasing subsequence; after k such extractions we obtain a subsequence non-decreasing in every coordinate, so its first two terms satisfy $v \leq v'$ componentwise. Hence no infinite antichain exists, and a set whose minimal elements were infinite would contain one. □

Lean status. Formal in `TransseriesWellBased.lean` as `Fabius.dickson_isPWO`, that every subset of $\mathbb{N}_0^k$ is partially well-ordered by the componentwise order, with `Fabius.dickson_isPWO_pi` the same for an arbitrary finite index type and `Fabius.dickson_antichain_finite` the antichain form. Again a bridge rather than a proof: the content is Mathlib's `Pi.wellQuasiOrderedLE`.

Remark 2.6 (Formal crosswalk). The finite-grid statement Theorem 2.5 is Exact in `TransseriesWellBased.lean`: `Fabius.dickson_antichain_finite` is its displayed antichain formulation, while `Fabius.dickson_isPWO` and `Fabius.dickson_isPWO_pi` supply the equivalent partially-well-ordered forms, the latter for an arbitrary finite coordinate type.

The same module contains the full multiplicative Neumann mechanism in `Fabius.neumann_isPWO` and `Fabius.neumann_finite_factorizations`, together with the literal reverse-order wrappers `Fabius.neumann_isPWO_orderDual` and `Fabius.neumann_finite_factorizations_orderDual` and the generated additive twins `Fabius.neumann_add_isPWO` and `Fabius.neu` `mann_finite_decompositions`. Thus Theorem 2.4 is Exact: `OrderDual` records precisely this chapter's greatest-element/no-increasing convention, and the printed total order is a specialization of the stronger partial-order hypotheses. This status covers the support and finite-factorization assertions, not a construction of the ambient Hahn-series object.

Remark 2.7 (What these two are for). Theorem 2.4 is what makes the *product* of two Hahn series well defined: each coefficient of the product is a finite sum. It is used in this volume at Theorem Q.13, where it is exactly the hypothesis that lets a triangular recurrence terminate at each exponent, and it is the reason that theorem needs only well-basedness and not the stronger local finiteness one source assumed. Theorem 2.5 is what makes a *grid* — a support contained in a finitely generated monoid — behave like a finite object, and it underlies the closure properties that the calculus parts use without further comment.

Definition 2.8 (Summable family)**.** A family $(f_j)_{j\in J}$ of Hahn series is *summable* if the union of their supports is well-based and each monomial lies in only finitely many supports. Its sum is then defined coefficientwise.

Lean status: Mathlib. `HahnSeries.SummableFamily` is this definition verbatim, with the two conditions as its two fields: `isPWO_iUnion_support'`, that the union of the supports is well-based, and `finite_co_support'`, that each monomial has a nonzero coefficient in only finitely many members of the family. The coefficientwise sum is `HahnSeries.SummableFamily.hsum`. This is one of the two places where the present chapter's foundations are already carried out in full generality in Mathlib, the other being Theorem 2.3; the corpus contribution in this area is Theorems 2.4 and 2.5, which name the order-theoretic facts that make the definitions non-vacuous.

2.2 Height, depth, and how monomials compare

Definition 2.9 (Exponential height and logarithmic depth)**.** The *exponential height* of a transmonomial is the number of nested exponentials it carries; the *logarithmic depth* is the number of nested logarithms. Thus $x = \mathrm{e}^{\log x}$ has height 0 and depth 0 in the usual normalization, e^x has height 1, and $\log\log x$ has depth 2.

Proposition 2.10 (Height wins)**.** *For all $N, M \geq 0$,*

$$x^N(\log x)^M = o\big(\mathrm{e}^{\varepsilon x}\big) \qquad \text{for every } \varepsilon > 0, \tag{2.1}$$

and consequently no product of powers and logarithms can compare with a monomial of greater exponential height. Dually, $(\log x)^M = o(x^\delta)$ for every $\delta > 0$: a greater logarithmic depth is always negligible against any positive power.

Proof. Taking logarithms, $N\log x + M\log\log x - \varepsilon x \to -\infty$, since $\log x = o(x)$ and $\log\log x = o(\log x)$; exponentiating gives (2.1). The dual statement is the same computation with $M\log\log x - \delta\log x \to -\infty$. □

Lean status. Formal in `TransseriesHeight.lean`, both halves and with the exponents where the proposition puts them. (2.1) is `Fabius.isLittleO_pow_mul_log_pow_exp`, that $x^N(\log x)^M = o(\mathrm{e}^{\varepsilon x})$ for every $\varepsilon > 0$ and all N, M; the dual half is `Fabius.isLittleO_log_pow_rpow`, that $(\log x)^M = o(x^\delta)$ for every $\delta > 0$, with the special case $\delta = 1$ recorded separately as `Fabius.isLittleO_log_pow_id`. The consequence drawn in the text — that no product of powers and logarithms compares with a monomial of greater exponential height — is not formalized as a statement about heights, there being no formal notion of transmonomial in the corpus; what is formal is the inequality that would prove it in any given case.

Remark 2.11 (Formal crosswalk). The two displayed asymptotic estimates in Theorem 2.10 are Exact in `TransseriesHeight.lean` for natural N, M: `Fabius.isLittleO_pow_mul_log_pow_exp` proves (2.1), and `Fabius.isLittleO_log_pow_rpow` proves the dual logarithm-versus-power estimate; `Fabius.isLittleO_log_pow_id` is the unit-power specialization. The full proposition remains Partial because the Lean module does not define general transmonomial height or logarithmic depth and therefore does not formalize the prose consequences for arbitrary nested exponentials and logarithms.

Remark 2.12 (The comparison algorithm). Theorem 2.10 is the top of a decision procedure: to compare two transmonomials, compare exponential heights first; on a tie, compare the exponents recursively; on a further tie, compare powers; and only at the bottom compare logarithmic depths. That is exactly the order in which the scale part of this volume organizes the polynomial–logarithmic monomials, and it is why the degree profile there is a lexicographic invariant rather than a single number.

Remark 2.13 (Branch hypotheses, and which way to get them wrong). Every inverse-function statement in this volume carries a hypothesis choosing a branch, and the same question about that hypothesis has arisen independently in three different chapters — for the Lambert equation of Theorem J.23, for the shifted inverse of Theorem L.1, and for the Cayley tree function of Theorem M.8. It is worth settling once.

The branch point of W at $z = -\mathrm{e}^{-1}$ belongs to the principal branch: $W_0(-\mathrm{e}^{-1}) = -1$, and the principal branch is defined and monotone on the *closed* ray $[-\mathrm{e}^{-1}, \infty)$. So a statement *about* W_0 — monotonicity, concavity, a bound, a functional equation — should take the closed hypothesis, and loses nothing by it. But a statement asserting that the two branches are *distinct* must take the open one. At $z = -\mathrm{e}^{-1}$ the two branches coincide, so a phase with two positive roots for z inside the punctured interval has exactly one at the endpoint: the assertion $X_0 < |b|/a < X_{-1}$ is not vacuous there, it is false.

The asymmetry is what makes this worth stating rather than remembering. A hypothesis that is too weak yields a theorem that is *wrong* at one point; a hypothesis that is too strong yields one that is merely *vacuous* there. Only the second failure is safe, so where the two readings compete, take the stronger hypothesis and weaken it at the point of use — passing $z > -\mathrm{e}^{-1}$ to a lemma that asks only $z \geq -\mathrm{e}^{-1}$ costs one token and no generality. The corresponding formal hypotheses are the closed `Set.Ici` for statements about the branch and the open `Set.Ioo` for statements separating two of them; both readings typecheck, which is precisely why the distinction has to be made by the author rather than by the checker.

2.3 How to read this volume

The parts that follow fall into three groups.

The calculus develops one scale in operational depth: the polynomial–logarithmic transseries, whose monomials are $X^{-\mu}(\log X)^k$. It builds the rings, then arithmetic and differentiation, then composition, then reversal at infinity; then it specializes to W and the Wright ω function; then it converts formal identities into Poincaré asymptotics with certificates; then algorithms and diagnostics. A reader who wants the machinery should read these in order.

The inversion apparatus is a single part standing between the calculus and its applications. It contains what the calculus does not: the exponential–power model, the monomial α-reduction, the axiomatized dominant core, perturbed inversion around it, and — the part with no analogue in the calculus — the theory of inverting a *sequence*, where three different inverse objects must be distinguished and a staircase theorem relates them.

The applications are independent of one another. Each takes a rapidly growing function or sequence, identifies its dominant block, inverts that block in closed form, and reverts the rest. A reader interested in one subject can read its part alone, following the citations back into the calculus as needed.

Remark 2.14 (The one thing worth carrying through all of it). Every part of this volume repeats one manoeuvre: *normalize until what remains is an ordinary reversion.* Whether the normalization is the substitution $u = \mathrm{e}^{-\lambda x}$ of Theorem T.6, the Lambert coordinate of Section Q.3, the change $q = L/(L - v) - \mathrm{e}^{-v}$ of Theorem L.7, or the saddle $r = W_0(n)$ of Theorem U.3, the shape is the same: find the block that can be inverted exactly, check that its normalized slope is a unit, and revert the remainder by a triangular recurrence whose only division is by that slope. What differs between chapters is which block and what the slope is.

2.4 What is quoted rather than proved

Five statements appear in the source tutorials as named theorems that they do not prove, and this volume does not prove them either. They are recorded here, demoted from theorems to attributed quotations, so that nothing in the volume carries the authority of a proof it does not have.

1. *Real closedness of the transseries field.* The field of logarithmic-exponential transseries is real closed. This is a theorem of the literature (van der Hoeven; Aschenbrenner–van den Dries–van der Hoeven), and neither the sources nor this volume proves it. Nothing in the volume depends on it.

2. *Closure under antidifferentiation.* Every transseries has a transseries antiderivative. Again a theorem of the literature; the calculus part of this volume proves the corresponding statement only for the polynomial–logarithmic scale, where it is elementary and where a genuine obstruction — the appearance of a new logarithm — is visible and is handled explicitly.

3. *The formal Taylor principle* and *the formal contraction principle*, stated schematically in two of the sources. This volume proves the versions it uses — the composition part for the first, and the reversal part for the second — on the polynomial–logarithmic scale with explicit hypotheses. The schematic general forms are not established here.

4. *Generalized Borel reconstruction.* That a resurgent germ is recovered from its Borel transform by a directional Laplace integral, with Stokes jumps across singular directions, is quoted schematically. This volume uses only the elementary Watson-type statements it proves, and every hyperasymptotic claim in the applications is labelled as quoted or as open — see Theorem Q.45.

Remark 2.15 (Why this list exists). A tutorial may reasonably state a deep theorem without proof, to orient the reader. A reference volume may not do so silently, because a later chapter may then appear to rest on it. Listing the five here, and checking that no result of this volume uses any of them, is cheaper than tracking the dependency later.

Chapter 3

Reader's guide and the structure of this volume

3.1 The eight parts

The volume is arranged so that each part depends only on those before it. One of those dependencies is lighter than the ordering suggests: a reader who wants only the Lambert expansion may pass from Part I directly to Parts IV and V, because Part V uses Parts II and III only through the two statements quoted in reading path 1 below.

Part	Subject	What it establishes
I	The scale	The generators $t = X^{-1}$, $L = \log X$; dominance and the rank-two valuation; the rings $\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$, $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$, $\mathscr{PL}_{\mathrm{rat},\mathbb{K}}$, $\mathscr{PL}_{\mathrm{itLaur},\mathbb{K}}$; coefficient support, leading block, leading monomial and leading scalar; exclusive truncation $\mathrm{Trunc}_{<N} F$; t-adic completeness and formal summability.
II	Arithmetic and differential calculus	Cauchy convolution block by block; the exact unit criterion and the reciprocal and quotient recurrences; the distinguished derivation $D_X = t\partial_L - t^2\partial_t$ and the Euler operator ϑ; repeated derivatives $\Delta_{n,k}$; antidifferentiation and the resonant block; powers, logarithms and exponentials of small series.
III	Composition	Admissible inner arguments; substitution on the generators; the exact formal Taylor formula and its formal summability; finite coefficient formulas; associativity and the identity; the near-identity group $\mathcal{G}_{\mathrm{poly},\mathbb{K}}$; the exact points at which composition leaves the class.
IV	Series reversal	Existence and uniqueness of $F^{\langle -1\rangle\circ}$ on $\mathcal{G}_{\mathrm{poly},\mathbb{K}}$; four algorithms — contractive fixed point, triangular coefficient matching, Newton iteration with precision doubling, and Lagrange–Bürmann at infinity; residual certificates $\mathrm{Err}_{\mathrm{rev}}(F, H)$; reversion with a nontrivial leading monomial.
V	Wright omega, the Lambert polynomials, and Lambert W	The reversion of $X + L$; the canonical zero-based family $\mathsf{L}_n^{\mathrm{Lam}}$ and the scaled integer family $\mathsf{L}_n^{\mathrm{Lam,sc}}$; recurrence, generating equation, operator formula, closed form through unsigned Stirling cycle numbers, degree law, Laplace transform; all complex branches, the lower real branch W_{-1}, and the square-root branch point.
VI	From formal transseries to analytic asymptotics	What a formal identity does and does not give; blockwise Poincaré expansions with the logarithmic degree printed; transport of asymptotics through the algebraic operations; from a formal residual to an analytic inverse error; convergent versus divergent expansions; branch consistency in sectors.
VII	Algorithms, certificates, diagnostics	Sparse data structures for $n \mapsto p_n(L)$; truncated primitives and their cost; reversion strategies compared; verification invariants, residual certificates, and a decision procedure for the recurring failure modes.
VIII	Extensions and the wider landscape	Puiseux and Hahn exponent groups; rational-log and Laurent-log coefficients; iterated logarithms and multiseries; power-log and Dulac classes at zero; the relation to the logarithmic-exponential transseries field and to Hardy fields [20, 13, 33, 1].

3.2 Appendices

- *Bell polynomials and Faà di Bruno bookkeeping.* The partial and complete exponential Bell polynomials $\mathcal{B}^{\mathrm{exp}}_{n,k}$ and $\mathcal{B}^{\mathrm{exp}}_{n}$ in exactly the form used by the composition and reversion formulas of Parts III and IV.

- *Closed-form coefficient extraction.* Explicit formulas for $[t^n]F$ and $[t^n L^j]F$ after each operation, with the conventions of Theorem 3.1 in force.
- *Lambert coefficient tables.* $\mathsf{L}_n^{\mathrm{Lam}}$ and $\mathsf{L}_n^{\mathrm{Lam,sc}}$ through a stated order, with Stirling diagonals $\left[{n \atop k}\right]$ and a reproducible residual check for each row.
- *Operational formula sheet.* One entry per operation: hypotheses, formula, cost, and the certificate that validates a computed truncation.
- *Exercises with solutions.*
- *Notation concordance.* The map from each source draft's local letters to the normative catalogue commands.
- *Provenance.* Which source contributed which result, and the editorial repairs applied to it.

3.3 Reading paths

1. *Only the Lambert W expansion.* Read Part I $\to$ Part IV $\to$ Part V. Part I supplies the scale, the ring and the valuation; Part IV supplies the one theorem this path needs, namely that every element of $\mathcal{G}_{\mathrm{poly},\mathbb{K}}$ has a compositional inverse in $\mathcal{G}_{\mathrm{poly},\mathbb{K}}$ computable to any order by a finite calculation; Part V performs the reversion of $X + L$ and reads off every branch. The only imports from Parts II and III along this path are the block derivative (4.14) and the formal Taylor formula, both quoted with full statements where used.
2. *The full calculus.* Read Parts I–IV in order. This is the self-contained algebraic core.
3. *Analytic use.* After Part I, read Part VI. A reader who intends to convert a formal truncation into a numerical bound should read Part VI before computing anything, because none of the formal machinery by itself licenses an analytic claim.
4. *Implementation.* Read Part I, then the recurrences of Parts II–IV, then Part VII with the formula-sheet appendix.
5. *Context.* Read Part VIII for the embedding of this class into Hahn series, Dulac classes, Hardy fields, and the full logarithmic-exponential transseries field.

Structural checkpoint

After Part I a reader should be able to: write an element of $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ in the normal form (4.13); compute ν_t, lm, lc, $\deg_{\mathrm{L}}$ and the log-degree profile d_F of a given series; decide from the leading block alone whether the series is a unit; say which of $\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$, $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$, $\mathscr{PL}_{\mathrm{rat},\mathbb{K}}$, $\mathscr{PL}_{\mathrm{itLaur},\mathbb{K}}$ a given expression belongs to; and state, for a displayed truncation, whether its tail is a formal valuation tail or an analytic remainder.

3.4 Standing conventions and normalizations

Remark 3.1 (Normalizations adopted, and how they differ from the source drafts). This volume merges six independently written treatments. Where their conventions collided, the notation catalogue of the corpus is the arbiter, and the following choices are in force everywhere.

1. *Regime and field.* Unless a statement says otherwise, $X \to +\infty$ along the positive real ray, $\mathbb{K}$ is a field of characteristic zero, and $t = X^{-1}$, $L = \log X$. Complex work names its sector and its branch of log. Local letters used by individual sources — λ, ℓ, x — are renamings of L or X and are not additional generators.

2. *Truncation is exclusive.* $\operatorname{Trunc}_{<N} F = \sum_{n<N} p_n(L)t^n$ keeps every logarithmic monomial in every retained block; "through outer order t^{N-1}" means the exclusive cutoff $< N$. One source uses the inclusive $[F]_{\le N}$; the conversion is $[F]_{\le N} = \operatorname{Trunc}_{<N+1} F$, a one-block difference that must be applied whenever a formula is transported from that source.

3. *Coefficient extraction.* $[t^n]F = p_n(L)$ and $[t^n L^j]F = a_{n,j}$, with the series written to the right of the bracket; an absent coefficient is 0. The same bracket glyph also denotes unsigned Stirling cycle numbers $\left[{n \atop k}\right]$ with two integer indices; the roles are separated by argument type and never abbreviated to the bare glyph in prose.

4. *Lambert polynomials.* The canonical zero-based family $\mathsf{L}_n^{\mathrm{Lam}}$ of (4.10) is used throughout, with $\mathsf{L}_0^{\mathrm{Lam}}(u) = -u$. The degree law $\deg \mathsf{L}_n^{\mathrm{Lam}} = n$ is asserted only for $n \ge 1$. The scaled family is $\mathsf{L}_n^{\mathrm{Lam,sc}} = n!\, \mathsf{L}_n^{\mathrm{Lam}} \in \mathbb{Z}[u]$ for $n \ge 1$; it is a storage normalization, not a different coefficient sequence. The bare symbol P_n is not reserved for this family and is not used for it.

5. *The three O-roles* are kept apart exactly as in the warning box of Section 4.8.

6. *Two inverses.* F^{-1} is the multiplicative reciprocal and $F^{\langle -1\rangle\circ}$ the compositional inverse. They are unrelated operations, and the ambiguous notation F^{-1} is used for neither.

7. *Filtration, not ideal.* In $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ the generator t is a unit, so the principal ideal generated by t^N is all of $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$, and $t^N \mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ is an additive valuation-filtration tail rather than an ideal. Agreement below outer order N is therefore phrased as $\nu_t(F - G) \ge N$, as $\mathcal{O}_t^{\mathrm{form}}(t^N)$, or as equality of exclusive truncations — never as a congruence modulo an ideal. Genuine quotient-ring language is legitimate only in $\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$, where (t^N) is a proper ideal.

8. *Coefficient blocks are polynomials.* In $\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ and $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ every p_n lies in $\mathbb{K}[L]$, so logarithmic exponents are nonnegative and each block is finite. Membership implies no uniform bound on $\deg_L p_n$; any such bound is a hypothesis or a proved invariant, never part of the definition.

Part II

The polynomial–logarithmic scale

Chapter 4

Why powers and logarithms belong in one expansion

This volume is about a single asymptotic scale and the exact algebra it supports. The scale is generated by two symbols,

$$t := X^{-1}, \qquad L := \log X, \qquad X \to +\infty, \tag{4.1}$$

and its monomials are the products $t^n L^j$. A reader who has met only power series may reasonably ask why two generators are needed at all and, if two, why exactly these two. This chapter answers both questions. It exhibits the smallest problem that forces the scale — inverting $y \mapsto y + \log y$ — and then proves three separate minimality statements: a scale of pure powers stalls after one term, a scale of pure logarithms cannot even begin, and the ring generated by t and L is the smallest subring of germs at $+\infty$ that contains a logarithm and is closed under differentiation.

Throughout this chapter X is a real variable tending to $+\infty$, all logarithms are the real logarithm on $(0, \infty)$, and $f \sim g$ means $f/g \to 1$. Complex branches are introduced only in Part V, where every branch of W and every choice of log is printed with the statement it belongs to.

4.1 The inversion problem that starts everything

The whole subject is visible in one implicit equation with one logarithmic term.

Proposition 4.1 (The basic inverse and its elementary envelope)**.** *The map $\Phi(y) = y + \log y$ is a strictly increasing bijection from $(0, \infty)$ onto $\mathbb{R}$. Denote its inverse by ω, so that*

$$\omega(X) + \log \omega(X) = X \qquad (X \in \mathbb{R}). \tag{4.2}$$

Then ω is real analytic and strictly increasing, $\omega(X) \to +\infty$ as $X \to +\infty$, and for every $X \geq 1$, writing $L = \log X$,

$$X - L \;\leq\; \omega(X) \;\leq\; X. \tag{4.3}$$

Moreover $W_0(z) = \omega(\log z)$ for every $z > 0$, where W_0 is the principal real branch of Lambert's function, defined by $W_0(z)\mathrm{e}^{W_0(z)} = z$.

Proof. On $(0, \infty)$ we have $\Phi'(y) = 1 + 1/y > 0$, so Φ is strictly increasing and real analytic with nowhere vanishing derivative; $\Phi(y) \to -\infty$ as $y \downarrow 0$ and $\Phi(y) \to +\infty$ as $y \to +\infty$, so Φ is a bijection onto $\mathbb{R}$. The inverse function theorem gives that ω is real analytic and strictly increasing, and $\omega(X) \to +\infty$ as $X \to +\infty$.

Fix $X \geq 1$ and put $y = \omega(X)$. If $y < 1$ then $\log y < 0$ and hence $X = y + \log y < 1$, a contradiction; so $y \geq 1$ and $\log y \geq 0$. From $y = X - \log y$ we get $y \leq X$; then $\log y \leq \log X = L$, whence $y = X - \log y \geq X - L$. This is (4.3).

For the last claim let $z > 0$ and put $w = \omega(\log z)$. Then $w > 0$ and $w + \log w = \log z$, so $\log(w\mathrm{e}^w) = \log z$ and therefore $w\mathrm{e}^w = z$. Since $v \mapsto v\mathrm{e}^v$ is strictly increasing on $(-1, \infty)$ and $w > 0 > -1$, the value w is the principal branch $W_0(z)$. □

Lean status. Exact over the reals in `WrightOmega.lean`. Rather than construct the inverse of Φ from scratch, ω is *defined* there by the last relation of the proposition, `Fabius.wrightOmega X` $= W_0(\mathrm{e}^X)$, which makes $W_0(z) = \omega(\log z)$ a rewriting step (`Fabius.principalLambertW_eq_wrightOmega_log`) and leaves every other clause a consequence of the Lambert API already in the corpus. The defining equation (4.2) is `Fabius.wrightOmega_add_log`, valid at every real X with no hypothesis; that Φ is a left inverse, hence that ω really does invert it on $(0, \infty)$, is `Fabius.wrightOmega_leftInverse`, proved from the uniqueness of the principal branch above -1. Strict monotonicity is `Fabius.wrightOmega_strictMono`, positivity `Fabius.wrightOmega_pos`, and the anchor $\omega(1) = 1$ `Fabius.wrightOmega_one`. The envelope (4.3) is `Fabius.wrightOmega_envelope`, and it is obtained in a sharper order than the proof above uses: the upper half `Fabius.wrightOmega_le_self` needs only $\log \omega(X) \geq 0$, and the lower half `Fabius.sub_log_le_wrightOmega` then follows from it by monotonicity of the logarithm, with no appeal to the asymptotics of W. Divergence `Fabius.tendsto_wrightOmega_atTop` is deduced not from the envelope but from a hypothesis-free linear minorant proved along the way, `Fabius.add_one_div_two_le_wrightOmega`: since $\log y \leq y - 1$, the defining equation gives $(X + 1)/2 \leq \omega(X)$ for *every* real X. Finally `Fabius.analyticAt_wrightOmega` proves real analyticity at every real X, by composing the real-analytic principal Lambert branch with the real exponential. This is exactly the real analyticity asserted here; it makes no complex holomorphy or branch-domain claim.

The envelope (4.3) already contains the message of this volume in miniature. It says that $\omega(X)$ equals X up to a correction of size at most $\log X$; so the answer to a question posed purely in powers of X is not expressible purely in powers of X. The next subsection makes the correction explicit to two further orders.

Remark 4.2 (Names). The function ω is Wright's omega function [7, 5]. Working with ω rather than with W removes one layer of logarithms from every display: the two nested logarithms $\log z$ and $\log \log z$ of the classical Lambert expansion become the single large variable X and the single logarithmic coefficient variable L. Every result about W in Part V is obtained from a result about ω by substituting a branch coordinate for X.

4.2 Two orders by hand

The following computation is the motivating one. It is short enough to do on paper, and it already displays every structural feature of the scale: the corrections are inverse powers of X; their coefficients are polynomials in $\log X$; and the logarithmic degree grows with the inverse power.

Proposition 4.3 (Two orders of the basic inverse)**.** *With $L = \log X$ and ω as in Theorem 4.1, as $X \to +\infty$ along the real axis,*

$$\omega(X) = X - L + \frac{L}{X} + \frac{1}{X^2}\left(\frac{L^2}{2} - L\right) + \mathcal{O}_{X\to+\infty}\left(\frac{L^3}{X^3}\right). \tag{4.4}$$

Moreover there is $X_0 \geq 4$ such that for $X \geq X_0$

$$\frac{1}{2} \cdot \frac{L}{X} \;\leq\; \omega(X) - X + L \;\leq\; 2 \cdot \frac{L}{X}. \tag{4.5}$$

In particular $\omega(X) - X \sim -L$ and $\omega(X) - X + L \sim L/X$.

Proof. Choose $X_0 \geq 4$ so large that $L \geq 1$ and $\varepsilon := L/X \leq \frac{1}{2}$ for $X \geq X_0$, and assume $X \geq X_0$ throughout. Write

$$r := \omega(X) - X + L, \qquad u := \frac{L-r}{X},$$

so that $\omega(X) = X(1-u)$. By (4.3) we have $0 \leq r \leq L$, hence

$$0 \leq u \leq \varepsilon \leq \tfrac{1}{2}. \tag{4.6}$$

Since $\omega(X) > 0$, we may take logarithms in $\omega(X) = X(1-u)$: $\log \omega(X) = L + \log(1-u)$. Substituting this into (4.2) and solving for $\omega(X)$ gives $\omega(X) = X - L - \log(1-u)$, that is,

$$r = -\log(1-u) = \sum_{k\geq 1} \frac{u^k}{k}, \tag{4.7}$$

the series converging because $0 \leq u \leq \frac{1}{2}$.

First estimate. For $0 \leq u \leq \frac{1}{2}$ one has $u \leq -\log(1-u) \leq 2u$: the quotient $-\log(1-u)/u = \sum_{k\geq 1} u^{k-1}/k$ is nondecreasing on $[0, \frac{1}{2}]$, equals 1 at $u = 0$ and equals $2\log 2 < 2$ at $u = \frac{1}{2}$. Hence by (4.7) and (4.6), $u \leq r \leq 2u \leq 2\varepsilon$. Feeding $r \leq 2\varepsilon$ back into the definition of u gives $u = (L-r)/X \geq (L - 2L/X)/X \geq \frac{1}{2}\varepsilon$, because $X \geq 4$. Therefore

$$\tfrac{1}{2}\varepsilon \leq u \leq \varepsilon, \qquad \tfrac{1}{2}\varepsilon \leq r \leq 2\varepsilon, \tag{4.8}$$

which is (4.5).

Second estimate. From (4.7) and $0 \leq u \leq \frac{1}{2}$ we get the two tail bounds

$$\left| r - u - \frac{u^2}{2} \right| = \left| \sum_{k\geq 3} \frac{u^k}{k} \right| \leq \frac{u^3}{3} \sum_{k\geq 0} u^k \leq \frac{2}{3} u^3 \leq \frac{2}{3}\varepsilon^3,$$

$$|r - u| \leq \frac{u^2}{2} \cdot \frac{1}{1-u} \leq u^2 \leq \varepsilon^2.$$

Next, $u = \varepsilon - r/X$ with $0 \leq r \leq 2\varepsilon$, so $|u - \varepsilon| \leq 2\varepsilon/X \leq 2\varepsilon^2$, where the last step uses $1/X \leq L/X = \varepsilon$, valid because $L \geq 1$. Combining this with the second tail bound gives $r = \varepsilon + \mathcal{O}_{X\to+\infty}\big(\varepsilon^2\big)$, whence $r/X = \varepsilon/X + \mathcal{O}_{X\to+\infty}\big(\varepsilon^2/X\big)$ and consequently

$$u = \varepsilon - \frac{r}{X} = \varepsilon - \frac{\varepsilon}{X} + \mathcal{O}_{X\to+\infty}\left(\frac{\varepsilon^2}{X}\right). \tag{4.9}$$

Squaring (4.9) and using $r = \mathcal{O}_{X\to+\infty}(\varepsilon)$,

$$u^2 = \left(\varepsilon - \frac{r}{X}\right)^2 = \varepsilon^2 - \frac{2\varepsilon r}{X} + \frac{r^2}{X^2} = \varepsilon^2 + \mathcal{O}_{X\to+\infty}\left(\frac{\varepsilon^2}{X}\right).$$

Adding the three contributions,

$$r = u + \frac{u^2}{2} + \mathcal{O}_{X\to+\infty}\big(\varepsilon^3\big) = \varepsilon - \frac{\varepsilon}{X} + \frac{\varepsilon^2}{2} + \mathcal{O}_{X\to+\infty}\left(\frac{\varepsilon^2}{X}\right) + \mathcal{O}_{X\to+\infty}\big(\varepsilon^3\big).$$

Substituting $\varepsilon = L/X$, the three main terms are L/X, $-L/X^2$ and $L^2/(2X^2)$, while the two error terms are $\mathcal{O}_{X\to+\infty}\big(L^2/X^3\big)$ and $\mathcal{O}_{X\to+\infty}\big(L^3/X^3\big)$; both are $\mathcal{O}_{X\to+\infty}\big(L^3/X^3\big)$ because $L \geq 1$. Since $\omega(X) = X - L + r$, this is exactly (4.4).

The two asymptotic equivalences follow: $\omega(X) - X = -L + r$ with $0 \leq r \leq 2L/X = o_{X\to+\infty}(L)$, and $\omega(X) - X + L = r \sim \varepsilon = L/X$. $\square$

Lean status: partial, with both concluding equivalences Exact. `WrightOmegaTwoOrders.lean` proves `Fabius.self_sub_wrightOmega_isEquivalent_log`, its signed companion, and `Fabius.wrightOmega_residual_isEquivalent`. These are precisely $\omega(X) - X \sim -\log X$ and $\omega(X) - X + \log X \sim (\log X)/X$. The explicit second-order polynomial and $O(L^3/X^3)$ remainder, and the displayed numerical envelope, are not formalized, so the whole proposition remains Partial.

Remark 4.4 (What the two coefficients are). The three coefficient blocks produced by the computation are the members $n = 0, 1, 2$ of the canonical zero-based Lambert family; the next two members, recorded here for orientation but derived only in Part V, continue the pattern:

$$\begin{aligned} \mathsf{L}_0^{\mathrm{Lam}}(u) &= -u, \qquad \mathsf{L}_1^{\mathrm{Lam}}(u) = u, \qquad \mathsf{L}_2^{\mathrm{Lam}}(u) = \frac{u^2}{2} - u, \\ \mathsf{L}_3^{\mathrm{Lam}}(u) &= \frac{u^3}{3} - \frac{3}{2}u^2 + u, \qquad \mathsf{L}_4^{\mathrm{Lam}}(u) = \frac{u^4}{4} - \frac{11}{6}u^3 + 3u^2 - u, \end{aligned} \tag{4.10}$$

defined and studied in Part V. In that normalization (4.4) reads

$$\omega(X) = X + \sum_{n=0}^{2} \mathsf{L}_n^{\mathrm{Lam}}(L)\, t^n + \mathcal{O}_{X\to+\infty}\left(L^3 t^3\right),$$

the block $n = 0$ being the term $-L$. The degree law $\deg \mathsf{L}_n^{\mathrm{Lam}} = n$ holds for $n \geq 1$ only: $\mathsf{L}_0^{\mathrm{Lam}}(u) = -u$ has degree one, so a blanket statement "$\mathsf{L}_n^{\mathrm{Lam}}$ has degree n" over $n \in \mathbb{N}_0$ is false and is not made anywhere in this volume.

Remark 4.5 (The same computation for W). Substituting $X = \log z$ into (4.4) and using $W_0(z) = \omega(\log z)$ from Theorem 4.1 gives, with the iterated logarithms $L_1 = \log z$ and $L_2 = \log L_1$,

$$W_0(z) = L_1 - L_2 + \frac{L_2}{L_1} + \frac{L_2^2/2 - L_2}{L_1^2} + \mathcal{O}_{z\to+\infty}\left(\frac{L_2^3}{L_1^3}\right), \tag{4.11}$$

the classical large-argument expansion [5, 11]. This is the sense in which the algebra of this volume sits one level below Lambert's function: the two nested logarithms of z become the single pair (X, L). Here L_1 and L_2 are iterated logarithms and never Lambert polynomials; the Lambert family is always written $\mathsf{L}_n^{\mathrm{Lam}}$.

4.3 Asymptotic scales, dominance, and blocks

Before showing that no one-generator scale works, we fix what a scale is and prove the one comparison on which everything rests.

Definition 4.6 (Asymptotic scale and expansion; working definition)**.** An *asymptotic scale* at $+\infty$ is a set $\mathfrak{M}$ of functions, each positive on some half-line, such that for all distinct $m, m' \in \mathfrak{M}$ either $m/m' \to 0$ or $m'/m \to 0$ as $X \to +\infty$. A function f defined near $+\infty$ has the *asymptotic expansion* $\sum_{i\geq 1} c_i m_i$ with respect to $\mathfrak{M}$ if $c_i \in \mathbb{R} \setminus \{0\}$, the $m_i \in \mathfrak{M}$ satisfy $m_{i+1}/m_i \to 0$, and for every index N present in the (finite or infinite) list,

$$f - \sum_{i\leq N} c_i m_i = o_{X\to+\infty}(m_N)\,.$$

A finite such expansion is *maximal* when it admits no further term, that is, when the final remainder is not asymptotically equivalent to a constant multiple of any element of $\mathfrak{M}$.

Lean status. Partial. The ordered-sequence core is formalized by `Fabius.IsAsymptoticScale` and `Fabius.IsPoincareExpansion` in `TransseriesScale.lean`; Theorem 1.4 records the exact five-declaration API. The power–logarithmic sequence is supplied by all four declarations in `TransseriesPolyLogScale.lean`: `Fabius.isLittleO_plMonomial`, `Fabius.isAsymptoticScale_plMonomial`, `Fabius.isAsymptoticScale_plMonomial_pow`, and `Fabius.isAsymptoticScale_plMonomial_log`. The definition printed here also asks for an unordered set of eventually positive monomials, nonzero coefficients, finite as well as infinite lists, maximality of finite expansions, and a little-o remainder after the retained last term. Those data are not bundled by the sequence API, whose remainder convention is the sharper first-omitted-term $O(\varphi_N)$ relation.

This is the coarse blockwise notion adequate for a motivating discussion. Part VI replaces it by the sharper statements needed for analytic work, including remainder estimates with the logarithmic degree printed and the exact sense in which a truncated polynomial-logarithmic series approximates a function.

Lemma 4.7 (Lexicographic dominance of power-logarithmic monomials)**.** *Let $(a,b),(a',b')\in\mathbb{R}^2$. As $X\to+\infty$,*

$$\frac{X^a(\log X)^b}{X^{a'}(\log X)^{b'}}\longrightarrow\begin{cases}0, & \text{if } a<a',\ \text{or } a=a' \text{ and } b<b',\\ 1, & \text{if } (a,b)=(a',b'),\\ +\infty, & \text{if } a>a',\ \text{or } a=a' \text{ and } b>b'.\end{cases}$$

Consequently $\{X^a(\log X)^b : (a,b)\in\mathbb{R}^2\}$ is an asymptotic scale in the sense of Theorem 4.6, totally ordered by the lexicographic order on (a,b); and in the generators (4.1)*, for $n,m,j,k\in\mathbb{Z}$,*

$$t^nL^j\succ t^mL^k\quad\Longleftrightarrow\quad n<m,\ \text{ or }\ (n=m\ \text{ and }\ j>k).\tag{4.12}$$

Proof. Put $s=\log X$, so $s\to+\infty$, and write the ratio as

$$X^{a-a'}(\log X)^{b-b'}=\mathrm{e}^{(a-a')s}s^{\,b-b'}=\exp\bigl((a-a')s+(b-b')\log s\bigr).$$

If $a<a'$ then $(a-a')s+(b-b')\log s\to-\infty$, because $\log s=o_{s\to+\infty}(s)$ and $a-a'<0$; hence the ratio tends to 0. The case $a>a'$ is the reciprocal statement. If $a=a'$ the ratio is $s^{\,b-b'}$, which tends to 0, equals 1, or tends to $+\infty$ according as $b<b'$, $b=b'$, or $b>b'$. Since the trichotomy is exhaustive, distinct exponent pairs give ratios tending to 0 or to $+\infty$, which is exactly the scale condition of Theorem 4.6.

For (4.12), apply the trichotomy with $(a,b)=(-n,j)$ and $(a',b')=(-m,k)$: the monomial t^nL^j dominates t^mL^k exactly when $-n>-m$, i.e. $n<m$, or when $n=m$ and $j>k$. ☐

Lean status. The displayed three-way limit is Exact in `TransseriesScaleDominance.lean`, with real exponents in both slots: `Fabius.plMonomial` is $X^a(\log X)^b$, and its three cases are `Fabius.tendsto_plMonomial_div_atTop_zero`, `Fabius.tendsto_plMonomial_div_atTop_one`, and `Fabius.tendsto_plMonomial_div_atTop`. The workhorse is `Fabius.tendsto_plMonomial_atTop_zero`, that $X^c(\log X)^d\to0$ for every $c<0$ and every real d. Positivity past $X=1$ is `Fabius.plMonomial_pos`, and the quotient rewrite is `Fabius.plMonomial_div_eventuallyEq`. The forward direction of (4.12) is `Fabius.plMonomial_generators_dominance`.

The scale consequence is Exact for every chosen lexicographically decreasing sequence: `Fabius.isLittleO_plMonomial` and `Fabius.isAsymptoticScale_plMonomial` in `TransseriesPolyLogScale.lean`, with the power and logarithmic ladders `Fabius.isAsymptoticScale_plMonomial_pow` and `Fabius.isAsymptoticScale_plMonomial_log`. The compound lemma as printed is nevertheless only Partial: Lean does not package the unordered full set as the set-indexed scale of Theorem 4.6, nor does the named generator theorem state the reverse implication in (4.12). Those unformalized packaging clauses must not be inferred from the exact limit and sequence results.

The dominance order is not a convention chosen to make bookkeeping convenient. The grouping of monomials into inverse-power blocks is intrinsic.

Proposition 4.8 (Blocks are the logarithmic-comparability classes)**.** *Let $\mu=t^nL^j$ and $\nu=t^mL^k$ with $n,m,j,k\in\mathbb{Z}$. The following are equivalent.*

1. $n = m$.
2. *There is an integer* $N \geq 0$ *with* $(\log X)^{-N} \leq \mu/\nu \leq (\log X)^N$ *for all sufficiently large* X.

Hence "differ by at most a fixed power of $\log X$*" is an equivalence relation on the group of monomials* $t^n L^j$*, its classes are exactly the sets* $\{t^n L^j : j \in \mathbb{Z}\}$ *for fixed* n*, and* $\succ$ *descends to the total order of* $-n$ *on the classes.*

Proof. (i)⇒(ii): if $n = m$ then $\mu/\nu = L^{j-k}$, so $N = |j-k|$ works, with equality on one side.

(ii)⇒(i): suppose $n \neq m$. If $n > m$ then, with $s = \log X$, we have $\mu/\nu = \mathrm{e}^{(m-n)s} s^{j-k}$ and, for every N,

$$\frac{\mu/\nu}{s^{-N}} = \mathrm{e}^{(m-n)s} s^{j-k+N} \longrightarrow 0$$

by Theorem 4.7, since $m - n < 0$; so the lower bound in (ii) fails for every N. If $n < m$, the same computation applied to ν/μ shows that the upper bound fails.

For the last sentence: the relation is reflexive ($N = 0$) and symmetric (replace N by N), and transitive with N replaced by the sum of the two constants; by the equivalence just proved, its classes are the fibres of $t^n L^j \mapsto n$. That $\succ$ induces the order of $-n$ on classes is (4.12). □

Lean status. Formal in `TransseriesBlockClasses.lean`, and generalized past the statement above. The equivalence as displayed, for integer exponents, is `Fabius.exists_log_bracket_div_iff`; it rests on `Fabius.exists_log_bracket_iff`, the same statement with *real* exponents — $X^a(\log X)^b$ is trapped between two fixed powers of $\log X$ if and only if $a = 0$. The generalization is not decoration: with real exponents the quantifier over N never has to be handled, because $a \neq 0$ destroys the bracket for every N at once through the two dominance limits, whereas the integer form invites an induction on N. `Fabius.plMonomial_zero_left` is the degenerate case $a = 0$, and the last clause — that $\succ$ descends to the order of $-n$ on classes — is `Fabius.plMonomial_generators_dominance` in `TransseriesScaleDominance.lean`.

Theorem 4.8 is the structural justification for the normal form used throughout this volume: a polynomial-logarithmic transseries is written

$$F = \sum_{n \geq n_0} p_n(L)\, t^n, \qquad p_n \in \mathbb{K}[L], \quad n_0 \in \mathbb{Z}, \tag{4.13}$$

one finite polynomial per comparability class, the classes indexed by the integer n and ordered by $-n$. The formal ring housing (4.13) is $\mathscr{PL}_{\mathrm{poly},\mathbb{K}} = \mathbb{K}[L]((t))$, with power-series subring $\mathscr{PL}_{\mathrm{pow},\mathbb{K}} = \mathbb{K}[L][[t]]$; their construction, valuation, leading data and truncation topology occupy the rest of Part I.

Remark 4.9 (Where the class ends). Three failures of closure mark the exact boundary of (4.13), and each names its own minimal enlargement.

- $(L+1)^{-1}$ is not a polynomial in L. Unrestricted division needs the coefficient field $\mathbb{K}(L)$, giving $\mathscr{PL}_{\mathrm{rat},\mathbb{K}} = \mathbb{K}(L)((t))$; expanding those rational coefficients at $L = \infty$ gives instead $\mathscr{PL}_{\mathrm{itLaur},\mathbb{K}} = \mathbb{K}((L^{-1}))((t))$. These are different objects: $1/(L+1)$ is exact in the first and becomes $L^{-1} - L^{-2} + L^{-3} - \cdots$ in the second.
- $\log L$ arises from composing with an inner argument whose leading behaviour is logarithmic, and requires a second logarithmic level.
- $X^{1/2}$ arises from inverting a leading behaviour that is not linear: a leading power X^k with $k \neq 1$ inverts to one with exponents in $k^{-1}\mathbb{Z}$. This requires a Puiseux or Hahn exponent group.

Negative powers of L do not belong to $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$. A display that mixes L^{-1} into a coefficient block has already moved to $\mathscr{PL}_{\mathrm{rat},\mathbb{K}}$ or $\mathscr{PL}_{\mathrm{itLaur},\mathbb{K}}$ and must say so. These enlargements are the subject of Part VIII.

4.4 Why one generator is never enough

We now show that neither obvious one-generator scale can carry the expansion (4.4). The two failures are of different kinds and deserve different names: the power scale *stalls*, the logarithmic scale is *blind*.

Proposition 4.10 (Both one-generator scales fail)**.** *Set* $\psi_1(X) = \omega(X) - X$ *and* $\psi_2(X) = \omega(X) - X + \log X$, *and let*

$$\mathfrak{M}_{\text{pow}} = \{X^a : a \in \mathbb{R}\}, \qquad \mathfrak{M}_{\log} = \left\{(\log X)^b : b \in \mathbb{R}\right\},$$

both asymptotic scales by Theorem 4.7.

1. (The power scale stalls.) *With respect to* $\mathfrak{M}_{\text{pow}}$, *the function* ω *has the one-term expansion* X, *and that expansion is maximal. Indeed* $\psi_1 \sim -\log X$, *so* $X^a = o_{X\to+\infty}(|\psi_1|)$ *for every* $a \le 0$ *while* $\psi_1 = o_{X\to+\infty}(X^a)$ *for every* $a > 0$*; no element of* $\mathfrak{M}_{\text{pow}}$ *is asymptotically proportional to* ψ_1.
2. (The logarithmic scale is blind.) *With respect to* $\mathfrak{M}_{\log}$, *the function* ω *has no expansion at all, because* $\omega(X)/(\log X)^b \to +\infty$ *for every* $b \in \mathbb{R}$. *Moreover the residual* ψ_2 *is strictly positive for large* X *and satisfies* $\psi_2 = o_{X\to+\infty}\big((\log X)^b\big)$ *for* every $b \in \mathbb{R}$*; so* $\mathfrak{M}_{\log}$ *assigns* ψ_2 *the empty expansion and cannot distinguish* ψ_2 *from* 0.
3. (One mixed monomial repairs both failures.) $\psi_2(X) \sim X^{-1} \log X$.

Proof. (i) By Theorem 4.3, $\psi_1 \sim -\log X$. For $a \le 0$ we have $X^a \le 1$ for $X \ge 1$ while $|\psi_1| \to +\infty$, so $X^a/|\psi_1| \to 0$; for $a > 0$, $\psi_1/X^a \to 0$ by Theorem 4.7 applied to $\log X$ and X^a. Hence there are no $a \in \mathbb{R}$ and $c \ne 0$ with $\psi_1 \sim cX^a$: for $a > 0$ the quotient ψ_1/X^a tends to 0, and for $a \le 0$ it is unbounded. Finally $\omega \sim X$ by (4.3), and $\omega - X = \psi_1 = o_{X\to+\infty}(X)$, so X is a legitimate first — and, by the above, last — term.

(ii) By (4.3), $\omega(X) \ge X - \log X$, and $(X - \log X)/(\log X)^b = X(\log X)^{-b} - (\log X)^{1-b} \to +\infty$ by Theorem 4.7, since the first summand dominates the second. So no leading term exists. For the second claim, (4.5) gives $\frac{1}{2}\log X/X \le \psi_2(X) \le 2\log X/X$ for $X \ge X_0$; in particular $\psi_2 > 0$. For any b, $\psi_2(X)(\log X)^{-b} \le 2(\log X)^{1-b}/X \to 0$, again by Theorem 4.7.

(iii) This is the last assertion of Theorem 4.3. □

Lean status: Exact. In `TransseriesWrightOmegaTerms.lean`: part (i) is `Fabius.not_isEquivalent_pure_power_wrightOmega_sub`, no cX^a with $c \ne 0$ is asymptotic to ψ_1, obtained in one line from the rigidity of the second term; part (ii) is `Fabius.tendsto_wrightOmega_div_plMonomial_zero_atTop`, $\omega(X)/(\log X)^b \to +\infty$ for every real b, together with `Fabius.isLittleO_wrightOmega_residual_plMonomial_zero`, $\psi_2 = o_{X\to+\infty}\big((\log X)^b\big)$ for every real b; part (iii) is `Fabius.wrightOmega_residual_isEquivalent` in `WrightOmegaTwoOrders.lean`. The one clause not carried over is the strict positivity of ψ_2 for large X, which is a consequence of the equivalence in (iii) and is not stated separately.

Remark 4.11 (The two failures are not symmetric). Part (i) is a *gap*: the true size of the correction, $\log X$, lies strictly between X^0 and X^a for every $a > 0$, so the power scale simply runs out of monomials at the place where one is needed. Part (ii) is a *loss of resolution*: the logarithmic scale has monomials on both sides of ψ_2, but every one of them is too coarse, so an expansion exists and carries no information. A treatment that reports only "a pure power series is impossible" has stated the weaker of the two obstructions.

4.5 The smallest closed class

The negative results say what fails. The positive statement is an exact minimality theorem, and it is where the pair (t, L) is singled out. Let $\mathcal{G}$ denote the ring of germs at $+\infty$ of real functions smooth on some half-line, with the derivation $D_X = \mathrm{d}/\mathrm{d}X$.

Theorem 4.12 (The two-generator ring is the smallest differentially closed class containing a logarithm)**.** *Inside $\mathcal{G}$, with $t = X^{-1}$ and $L = \log X$:*

1. *The smallest D_X-stable subring of $\mathcal{G}$ containing $\mathbb{Q}$ and X is $\mathbb{Q}[X]$. Every unbounded element of it grows at least as fast as X; in particular it contains no element asymptotic to a nonzero multiple of $\log X$.*
2. *The smallest D_X-stable subring of $\mathcal{G}$ containing $\mathbb{Q}$ and L is $\mathbb{Q}[t, L]$.*
3. *The smallest D_X-stable subring of $\mathcal{G}$ containing $\mathbb{Q}$, X and L is $\mathbb{Q}[X, t, L] = \mathbb{Q}[t, t^{-1}][L]$, the ring of Laurent polynomials in t with coefficients polynomial in L.*
4. *t and L are algebraically independent over $\mathbb{R}$. Hence $\mathbb{R}[t, L]$ is a genuine two-variable polynomial ring, and each of its elements has a* unique *representation $\sum_{n\geq 0} p_n(L)t^n$ with $p_n \in \mathbb{R}[L]$ and only finitely many $p_n \neq 0$.*
5. *For $n \in \mathbb{Z}$ and $p \in \mathbb{Q}[L]$,*

$$D_X\big(t^n p(L)\big) = t^{n+1}\big(p'(L) - np(L)\big). \tag{4.14}$$

The same statements hold with $\mathbb{Q}$ replaced by any subfield of $\mathbb{R}$. Abstractly, for a field $\mathbb{K}$ of characteristic zero, the ring in (iii) is the Laurent polynomial ring $\mathbb{K}[t, t^{-1}][L]$ carrying the unique derivation with $D_X t = -t^2$ and $D_X L = t$.

Proof. We first verify (v). As germs, $D_X t = -X^{-2} = -t^2$ and $D_X L = X^{-1} = t$; by the product and chain rules,

$$D_X\big(t^n p(L)\big) = nt^{n-1}(-t^2)p(L) + t^n p'(L)\, t = t^{n+1}\big(p'(L) - np(L)\big).$$

(i) $\mathbb{Q}[X]$ contains $\mathbb{Q}$ and X and is D_X-stable, since $D_X X = 1$. Conversely any D_X-stable subring containing $\mathbb{Q}$ and X contains all powers of X and all rational combinations of them, hence contains $\mathbb{Q}[X]$. A nonzero element of $\mathbb{Q}[X]$ is a polynomial of some degree d; if $d = 0$ it is a constant, and if $d \geq 1$ it is asymptotic to a nonzero multiple of X^d. Since $\log X/X^d \to 0$ for $d \geq 1$ and $\log X \to \infty$, no element is asymptotic to a nonzero multiple of $\log X$.

(ii) Put $R = \mathbb{Q}[t, L]$. It contains $\mathbb{Q}$ and L. It is D_X-stable: a derivation is determined on a finitely generated ring by its values on generators, and $D_X t = -t^2 \in R$, $D_X L = t \in R$, so $D_X R \subseteq R$. Conversely let S be D_X-stable with $\mathbb{Q} \cup \{L\} \subseteq S$. Then $t = D_X L \in S$, so S contains the subring generated by $\mathbb{Q}$, t and L, namely R.

(iii) Put $R' = \mathbb{Q}[X, t, L]$. Since $Xt = 1$, R' is the ring of Laurent polynomials in t with coefficients in $\mathbb{Q}[L]$. It is D_X-stable by $D_X X = 1$, $D_X t = -t^2$, $D_X L = t$. Any D_X-stable subring containing $\mathbb{Q}$, X and L contains $D_X L = t$, hence contains R'.

(iv) Suppose $P \in \mathbb{R}[T, \Lambda]$ is nonzero with $P(t, L) = 0$ as a germ, i.e. $P(X^{-1}, \log X) = 0$ for all large X. Write $P = \sum_{n\geq 0} T^n q_n(\Lambda)$ with $q_n \in \mathbb{R}[\Lambda]$, and let n_0 be minimal with $q_{n_0} \neq 0$. Multiplying the relation by X^{n_0} gives, for all large X,

$$q_{n_0}(\log X) = -\sum_{n>n_0} X^{-(n-n_0)} q_n(\log X).$$

Each summand on the right is X^{-k} with $k \geq 1$ times a polynomial in $\log X$, hence tends to 0 by Theorem 4.7; the sum is finite, so the right side tends to 0. The left side is a nonzero polynomial evaluated along $\log X \to +\infty$, so it tends to $\pm\infty$ or is a nonzero constant. This contradiction proves algebraic independence; uniqueness of the representation follows, since two representations of the same germ would differ by a nontrivial polynomial relation.

For the abstract version, $\mathbb{K}[t, t^{-1}][L]$ is the localization at t of the polynomial ring $\mathbb{K}[t, L]$. A derivation of a polynomial ring is determined freely by its values on the variables, and it extends uniquely to any localization by the quotient rule; taking $D_X t = -t^2$, $D_X L = t$ gives the derivation in question. $\square$

Lean status: partial; (i)–(iii) *and* (v) *formal.* The block law (4.14) is formal in `TransseriesDifferentialBlock.lean`, and in greater generality than stated here: no ambient $\mathcal{G}$ and no construction of $\mathbb{K}[t, t^{-1}][L]$ is needed, only a commutative ring with a derivation d and two elements satisfying $d\,t = -t^2$ and $d\,L = t$ — the abstract shape of $t = X^{-1}$, $L = \log X$. (This route was chosen because Mathlib has `LaurentPolynomial` but no derivation on it, so the concrete ring would have had to be built before anything could be said about it; on the abstract statement the concrete ring is merely one model.) For $n \geq 0$ the law is `Fabius.derivation_block`, which identifies the right-hand side with the block operator $\partial_L - n$ of Theorem 4.16; the integer form of item (v), with $n \in \mathbb{Z}$ and t a unit, is `Fabius.derivation_block_zpow`, and the auxiliary powers are `Fabius.derivation_pow_t`, `Fabius.derivation_zpow_t`, `Fabius.derivation_val_inv` and `Fabius.derivation_pow_inv`. The resonant case $n = 0$, where the law degenerates to $d(p(L)) = t\,p'(L)$, is `Fabius.derivation_block_zero`, and the two antidifferentiation statements it controls are `Fabius.exists_block_primitive` (for $(n : F) \neq 0$, with the primitive given explicitly by the Neumann sum) and `Fabius.exists_block_primitive_resonant`.

The three minimality statements are formal as well, in `TransseriesDifferentialClosure.lean` and as genuine `IsLeast` assertions rather than inclusions in one direction: `Fabius.isLeast_adjoin_singleton_of_derivation_eq_one` is (i), `Fabius.isLeast_adjoin_pair_of_derivation_log` is (ii), and `Fabius.isLeast_adjoin_triple_of_derivation_log` is (iii).

What is *not* formalized is item (iv) and the growth clause of (i). Both are statements about the concrete ring of germs rather than about an abstract differential ring: the growth clause says every unbounded element of $\mathbb{Q}[X]$ grows at least like X, and (iv) is the algebraic independence of t and L over $\mathbb{R}$, a transcendence statement that is not implied by anything above and that the unique block representation depends on.

Remark 4.13 (Reading the theorem). Part (i) says that the pure power world is closed but expressively poor: a logarithm can never be produced from X by ring operations and differentiation, so if a problem's answer contains a logarithm, that logarithm must enter through an operation — inversion, antidifferentiation, composition — that leaves the class. Part (ii) says the converse asymmetry does not exist: once $\log X$ is admitted, X^{-1} is forced immediately, and then all of $\mathbb{Q}[t, L]$. There is no "logarithms only" differential algebra. This is the precise sense in which powers and logarithms belong in one expansion. The ambient object $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}} = \mathbb{K}[L]((t))$ of Part I is obtained from the Laurent ring of (iii) by completing its *polynomial* subring $\mathbb{K}[L][t]$ in the t-adic topology, which gives $\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}} = \mathbb{K}[L][[t]]$, and then inverting t. Equivalently, $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ is the completion of $\mathbb{K}[t, t^{-1}][L]$ for the valuation ν_t. One may not complete $\mathbb{K}[t, t^{-1}][L]$ t-adically: there t is a unit, so (t^N) is the whole ring for every N and the t-adic completion is trivial.

Corollary 4.14 (Both exponent directions are forced by ω). *Let $S \subseteq \mathbb{R}^2$ and put $\mathfrak{M}_S = \{X^a(\log X)^b : (a, b) \in S\}$, an asymptotic scale by Theorem 4.7. If ω has an asymptotic expansion with respect to $\mathfrak{M}_S$ with at least three terms, then*

$$(1, 0) \in S, \qquad (0, 1) \in S, \qquad (-1, 1) \in S,$$

with coefficients $1, -1, 1$ respectively. Consequently the subgroup of $(\mathbb{R}^2, +)$ generated by S contains $\mathbb{Z}^2$; in particular S is contained neither in $\mathbb{R} \times \{0\}$ (a scale of pure powers) nor in $\{0\} \times \mathbb{R}$ (a scale of pure logarithms).

Proof. By Theorem 4.7, a monomial $X^a(\log X)^b$ can be asymptotically equivalent to $cX^{a'}(\log X)^{b'}$ with $c \in (0, \infty)$ only if $(a, b) = (a', b')$ and $c = 1$, since otherwise the ratio tends to 0 or to $+\infty$.

By Theorem 4.6, the first term satisfies $\omega \sim c_1 m_1$. By (4.3), $\omega \sim X = X^1(\log X)^0$, so $m_1 \sim c_1^{-1} X$; by the previous paragraph $m_1 = X$ and $c_1 = 1$, whence $(1, 0) \in S$. The first partial sum is therefore exactly X, and the second term satisfies $\omega - X \sim c_2 m_2$. By Theorem 4.3, $\omega - X \sim -\log X$; as

$m_2 > 0$, this forces $c_2 < 0$ and $m_2 \sim |c_2|^{-1} \log X$, hence $m_2 = \log X$ and $c_2 = -1$, so $(0, 1) \in S$. The second partial sum is exactly $X - \log X$, and the third term satisfies $\omega - X + \log X \sim c_3 m_3$; by Theorem 4.3 this residual is $\sim X^{-1} \log X$, so $m_3 = X^{-1} \log X$, $c_3 = 1$ and $(-1, 1) \in S$. Finally $(1, 0)$ and $(0, 1)$ generate $\mathbb{Z}^2$. □

Lean status: Exact for monomial scales. The uniqueness engine is `Fabius.isEquivalent_const_mul_plMonomial_iff` in `TransseriesMonomialUniqueness.lean`, stated with a coefficient on *both* sides and no sign hypothesis: for $c, c' \neq 0$, an asymptotic equivalence $cX^a(\log X)^b \sim c'X^{a'}(\log X)^{b'}$ forces $a = a'$, $b = b'$ and $c = c'$. No case split on the sign of c/c' is needed: in the divergent case, multiplying the quotient by the inverse constant would hand a divergent function a finite limit. The three rigidity statements that are the content of the corollary are `Fabius.exponents_of_wrightOmega`, `Fabius.exponents_of_wrightOmega_sub` and `Fabius.exponents_of_wrightOmega_residual` in `TransseriesWrightOmegaTerms.lean`: any scaled monomial with a nonzero coefficient representing ω, $\omega - X$ or $\omega - X + \log X$ has exponent pair $(1, 0)$, $(0, 1)$, $(-1, 1)$ and coefficient $1, -1, 1$ respectively. The "consequently" clause is `Fabius.not_pure_of_wrightOmega_three_terms`. The hypothesis "an expansion with at least three terms" is rendered as the three equivalences of the residuals, which is what the proof uses.

Remark 4.15 (Scope of the corollary). The corollary is stated for monomial scales in the two generators, and the restriction is not cosmetic. Nothing above rules out an exotic scale built from other generators — for instance one containing $X + X^{1/2}$ and $X^{1/2}$ — in which the first terms of ω look different. What *is* ruled out, by Theorem 4.10, is any scale of powers alone or of logarithms alone, whatever the exponents; and what is asserted by Theorem 4.12 is that among differentially closed rings the two-generator ring is minimal. Together these are the honest content of the phrase "smallest closed class". A stronger uniqueness statement — that $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ is the *unique* minimal differential ring in which the reversion of $X + L$ exists — is not proved in this volume and should not be inferred from it.

4.6 Where a logarithm is created

Theorem 4.12(i) says a logarithm cannot be manufactured from powers by differentiation. It can be manufactured by *anti*differentiation, and by the discrete analogue of antidifferentiation, summation. These two mechanisms account for nearly every occurrence of the scale in practice, so we isolate them.

Lemma 4.16 (Block antidifferentiation and the resonant block)**.** *Let $\mathbb{K}$ be a field of characteristic zero and work in the differential ring $\mathbb{K}[t, t^{-1}][L]$ of Theorem 4.12(iii). Let $c \in \mathbb{K}$ and consider the $\mathbb{K}$-linear operator $\partial_L - c$ on $\mathbb{K}[L]$.*

1. *If $c \neq 0$, then $\partial_L - c$ is a degree-preserving bijection of $\mathbb{K}[L]$ onto itself. Consequently, for $n \neq 1$ and $p \in \mathbb{K}[L]$, the element $t^n p(L)$ has a unique antiderivative of the shape $t^{n-1} q(L)$ with $q \in \mathbb{K}[L]$, and $\deg q = \deg p$.*
2. *If $c = 0$, then ∂_L is surjective on $\mathbb{K}[L]$ with kernel $\mathbb{K}$, and $\deg q = \deg(\partial_L q) + 1$ for nonconstant q. Consequently $t\, p(L)$ has antiderivative $q(L)$ with $q' = p$, unique up to an additive constant; when $p \neq 0$, the explicit primitive with zero constant term has $\deg q = \deg p + 1$.*

In particular $t = X^{-1}$ integrates to $L = \log X$. The outer level $n = 1$ is the unique block at which antidifferentiation raises logarithmic degree.

Proof. (i) Write $q = \sum_{i=0}^{d} q_i L^i$ with $q_d \neq 0$. Then $(\partial_L - c)q = -cq_d L^d +$ (terms of degree $< d$), so the operator preserves degree, hence is injective and maps the finite-dimensional space of polynomials of degree at most d into itself injectively, therefore bijectively; taking the union over d gives bijectivity on $\mathbb{K}[L]$. For the antiderivative statement, apply (4.14) with $n - 1$ in place of n: $D_X(t^{n-1} q) = t^n \big(q' - (n-1)q\big) = t^n (\partial_L - c) q$ with $c = n - 1 \neq 0$; solve $(\partial_L - c)q = p$, which has a unique solution of the same degree.

(ii) ∂_L is surjective on $\mathbb{K}[L]$ because $\partial_L\big(L^{i+1}/(i+1)\big) = L^i$, which uses $\operatorname{char}\mathbb{K} = 0$; its kernel is $\mathbb{K}$; and it lowers degree by exactly one on nonconstant polynomials. Taking $n = 1$ in (4.14), with $n - 1 = 0$, gives $D_X q(L) = t\, q'(L)$, so the antiderivatives of $t\, p(L)$ are exactly the q with $q' = p$. If $p \neq 0$, termwise integration with zero constant term raises degree by one. The case $p = 1$, $q = L$ is $\int X^{-1}\, \mathrm{d}X = \log X$. □

Lean status. Partial overall. The polynomial-operator core is formalized in `TransseriesBlockAntiderivative.lean`. Its complete three-definition, twelve-theorem surface is `Fabius.blockOperator`, `Fabius.blockAntiderivative`, `Fabius.resonantAntiderivative`, `Fabius.sum_sub_sum_shift`, `Fabius.blockOperator_zero`, `Fabius.blockOperator_sub`, `Fabius.blockOperator_blockAntiderivative`, `Fabius.blockOperator_surjective`, `Fabius.natDegree_C_mul_of_ne_zero`, `Fabius.natDegree_blockOperator`, `Fabius.blockOperator_injective`, `Fabius.blockOperator_bijective`, `Fabius.derivative_resonantAntiderivative`, `Fabius.derivative_surjective`, and `Fabius.natDegree_resonantAntiderivative`. Thus the nonresonant operator has an explicit finite-Neumann-series inverse and preserves `natDegree`; the chosen resonant primitive exists and raises `natDegree` by one for a nonzero input. In the companion module `TransseriesDifferentialBlock.lean`, the complete twelve-theorem surface is `Fabius.derivation_pow_t`, `Fabius.derivation_block`, `Fabius.derivation_zpow_block`, `Fabius.exists_zpow_block_primitive`, `Fabius.existsUnique_zpow_block_primitive`, `Fabius.exists_block_primitive`, `Fabius.derivation_block_zero`, `Fabius.exists_block_primitive_resonant`, `Fabius.derivation_val_inv`, `Fabius.derivation_pow_inv`, `Fabius.derivation_zpow_t`, and `Fabius.derivation_block_zpow`. These cover the natural- and integer-power block laws, the nonresonant primitive and its conditional uniqueness, the resonant primitive, and inverse-power/unit reformulations. `Fabius.derivation_zpow_block` makes the displayed integer identity (4.14) Exact in an ambient field with nonzero power generator. For $n \neq 1$, `Fabius.exists_zpow_block_primitive` proves source-shaped existence and degree preservation, while `Fabius.existsUnique_zpow_block_primitive` proves uniqueness under an explicit injectivity hypothesis on evaluation at L. The compound lemma nevertheless asserts these consequences inside the concrete ring $\mathbb{K}[t, t^{-1}][L]$, including the needed faithful evaluation and the resonant ambient kernel/uniqueness conclusions; that Laurent model and those remaining links are not packaged. Thus the compound lemma remains Partial even though its displayed integer block equation and nonresonant conditional primitive API are Exact. The nonresonant operator core needs only a field and $c \neq 0$; characteristic zero is needed for resonant surjectivity and the printed resonant degree claims.

Lemma 4.17 (The discrete counterpart: a harmonic increment makes a logarithm)**.** *Let $(w_n)_{n \geq n_1}$ be real with $w_n \to +\infty$ and suppose that for constants $c_0 > 0$ and $c_1 \in \mathbb{R}$,*

$$w_{n+1} = w_n + c_0 + \frac{c_1}{w_n} + \mathcal{O}_{n\to\infty}\big(w_n^{-2}\big)\,. \tag{4.15}$$

Then there is a constant C with

$$w_n = c_0 n + \frac{c_1}{c_0}\log n + C + o_{n\to\infty}(1)\,. \tag{4.16}$$

Proof. Since $w_n \to +\infty$, the last two terms of (4.15) tend to 0, so $w_{n+1} - w_n \to c_0$. By the Stolz–Cesàro theorem applied to the increments, $w_n/n \to c_0$; in particular $w_n \sim c_0 n$ and $1/w_n = \mathcal{O}_{n\to\infty}\big(n^{-1}\big)$.

Put $v_n = w_n - c_0 n$. Then $v_{n+1} - v_n = c_1/w_n + \mathcal{O}_{n\to\infty}\big(w_n^{-2}\big) = \mathcal{O}_{n\to\infty}\big(n^{-1}\big)$; summing, $v_n = \mathcal{O}_{n\to\infty}(\log n)$.

Now bootstrap. Since $v_n = \mathcal{O}_{n\to\infty}(\log n) = o_{n\to\infty}(n)$, we have $|v_n/(c_0 n)| \leq \frac{1}{2}$ for large n, and

$$\frac{1}{w_n} = \frac{1}{c_0 n}\cdot\frac{1}{1 + v_n/(c_0 n)} = \frac{1}{c_0 n} - \frac{v_n}{c_0^2 n^2} + \mathcal{O}_{n\to\infty}\left(\frac{v_n^2}{n^3}\right) = \frac{1}{c_0 n} + \mathcal{O}_{n\to\infty}\left(\frac{\log n}{n^2}\right).$$

Hence $v_{n+1} - v_n = \dfrac{c_1}{c_0 n} + \mathcal{O}_{n\to\infty}\left(\dfrac{\log n}{n^2}\right)$. Summing from n_1 to $n-1$, using $\sum_{k=n_1}^{n-1} k^{-1} = \log n + \gamma - \sum_{k<n_1} k^{-1} + o_{n\to\infty}(1)$ and the absolute convergence of $\sum_k k^{-2}\log k$ (so that its tail is $o_{n\to\infty}(1)$), we obtain (4.16) for a suitable constant C. □

Lean status: partial, leading order only. In `TransseriesHarmonicIncrement.lean`. What is proved is that the hypothesis (4.15) forces $w_n/n \to c_0$: `Fabius.tendsto_div_atTop_of_harmonic_increment`, which needs $w_n \to +\infty$ and the error bound but *not* $c_0 > 0$. Its engine is a Stolz–Cesàro step stated for an arbitrary real sequence, `Fabius.tendsto_div_atTop_of_tendsto_sub`: if the increments $w_{n+1} - w_n$ converge to c then $w_n/n \to c$, with no positivity, monotonicity or divergence hypothesis — the telescoped sum of the increments is $w_n - w_0$, so this is Cesàro convergence of the increments together with $w_0/n \to 0$.

What is *not* formalized is the rest of (4.16): the logarithmic correction $(c_1/c_0)\log n$, the constant C, and the $o(1)$ remainder. Closing that gap needs the Euler–Mascheroni comparison $\sum_{k<n} w_k^{-1} = c_0^{-1}\log n + C' + o(1)$; Mathlib does carry the required machinery (`harmonic` $n - \log n \to \gamma$ in `Mathlib/NumberTheory/Harmonic/EulerMascheroni.lean`), so it is reachable, but it is separate work.

Example 4.18 (Iterating the sine). Let $x_0 \in (0,\pi)$ and $x_{n+1} = \sin x_n$. Since $0 < \sin x \le \min(x,1)$ for $x \in (0,\pi)$, the sequence stays in $(0,1] \subset (0,\pi)$ and decreases; its limit ℓ satisfies $\ell = \sin\ell$, so $x_n \downarrow 0$. Put $u_n = x_n^{-2}$. From $\sin x = x\big(1 - x^2/6 + x^4/120 + \mathcal{O}_{x\to0}(x^6)\big)$ and $(1+a)^{-2} = 1 - 2a + 3a^2 + \mathcal{O}_{a\to0}(a^3)$ with $a = -x^2/6 + x^4/120$,

$$\left(\frac{\sin x}{x}\right)^{-2} = 1 + \frac{x^2}{3} + \left(\frac{3}{36} - \frac{1}{60}\right)x^4 + \mathcal{O}_{x\to0}(x^6) = 1 + \frac{x^2}{3} + \frac{x^4}{15} + \mathcal{O}_{x\to0}(x^6)\,.$$

Multiplying by u_n and using $u_n x_n^2 = 1$,

$$u_{n+1} = u_n + \frac{1}{3} + \frac{1}{15u_n} + \mathcal{O}_{n\to\infty}(u_n^{-2})\,.$$

Theorem 4.17 with $c_0 = 1/3$, $c_1 = 1/15$ gives $u_n = n/3 + \frac{1}{5}\log n + C + o_{n\to\infty}(1)$, and therefore

$$x_n = \sqrt{\frac{3}{n}}\left(1 - \frac{3}{10}\cdot\frac{\log n}{n} + \mathcal{O}_{n\to\infty}\left(\frac{1}{n}\right)\right). \tag{4.17}$$

The coefficient of n^{-1} is a polynomial of degree one in $\log n$, and the whole expression is a Puiseux-logarithmic monomial times a polynomial-logarithmic factor: with $X = n$, $t = X^{-1}$, $L = \log X$, the right side of (4.17) is $\sqrt{3}\,t^{1/2}\big(1 - \frac{3}{10}Lt + \cdots\big)$, an element of the Puiseux enlargement $\mathbb{K}[L]\big(\!\big(t^{1/2}\big)\!\big)$. Continuing the same bootstrap produces further blocks, each again a polynomial in $\log n$. This is de Bruijn's example [10, Ch. 8].

Example 4.19 (Iterating a parabolic germ). Let $f(z) = z - z^2 + az^3 + \mathcal{O}_{z\to0}(z^4)$ be an analytic germ at 0 tangent to the identity, and let $z_n = f^{\circ n}(z_0)$ for $z_0 > 0$ small enough that $z_n \downarrow 0$. With $w_n = 1/z_n$, the expansion $\big(1 - z + az^2 + \mathcal{O}_{z\to0}(z^3)\big)^{-1} = 1 + z + (1-a)z^2 + \mathcal{O}_{z\to0}(z^3)$ gives

$$w_{n+1} = w_n\big(1 + z_n + (1-a)z_n^2 + \mathcal{O}_{n\to\infty}(z_n^3)\big) = w_n + 1 + \frac{1-a}{w_n} + \mathcal{O}_{n\to\infty}(w_n^{-2})\,,$$

so Theorem 4.17 yields $w_n = n + (1-a)\log n + C + o_{n\to\infty}(1)$ and hence

$$z_n = \frac{1}{n} - \frac{(1-a)\log n}{n^2} + \mathcal{O}_{n\to\infty}\left(\frac{1}{n^2}\right).$$

The same logarithm appears in the Fatou coordinate Φ of f, the solution of $\Phi\circ f = \Phi + 1$, whose leading behaviour as $z \downarrow 0$ is $\Phi(z) = 1/z - (1-a)\log(1/z) + \text{const} + o_{z\downarrow0}(1)$; one checks directly that this shape reproduces the increment 1 to the displayed order. Normal-form and conjugacy theory for such germs is one of the classical homes of this scale.

Section summary

Two mechanisms create the logarithm, and they are one mechanism in continuous and in discrete clothing. Antidifferentiation preserves logarithmic degree on every outer block except t^1, where it raises it by one (Theorem 4.16); summation of increments is bounded except for the harmonic one, which produces $\log n$ (Theorem 4.17). Once one logarithm is present, Theorem 4.12(ii) forces the whole ring $\mathbb{K}[t, L]$.

4.7 Provenance: where these expansions arise

4.7.1 Implicit equations carrying a logarithmic term

The generic source is a relation in which the unknown occurs both linearly and inside a logarithm,

$$y + \alpha \log y + \frac{p_1(\log y)}{y} + \frac{p_2(\log y)}{y^2} + \cdots = X, \qquad \alpha \in \mathbb{K}, \quad p_i \in \mathbb{K}[\,\cdot\,], \tag{4.18}$$

whose two-term case $y + \alpha \log y = X$ is equivalent to the exponentiated form $y^\alpha \mathrm{e}^y = \mathrm{e}^X$. Dominant balance gives $y \sim X$; substituting that into the logarithm creates $\log X$; iterating creates powers of X^{-1} carrying polynomials in $\log X$. Part IV turns this heuristic into a theorem: read as a transseries in the variable y, the left side of (4.18) lies in the near-identity class $\mathcal{G}_{\mathrm{poly},\mathbb{K}}$, and $\mathcal{G}_{\mathrm{poly},\mathbb{K}}$ is a group under composition, so the solution is again of the same shape.

A classical instance outside the Lambert family is the size of the n-th prime. Inverting $\pi(p_n) = n$ against the logarithmic-integral approximation gives Cipolla's expansion

$$\frac{p_n}{n} = \log n + \log\log n - 1 + \frac{\log\log n - 2}{\log n} + \mathcal{O}_{n\to\infty}\left(\frac{(\log\log n)^2}{(\log n)^2}\right). \tag{4.19}$$

Read with $X = \log n$ and $L = \log\log n$, the right side of (4.19) is a truncation of an element of $X + \mathbb{K}[L][[t]] = \mathcal{G}_{\mathrm{poly},\mathbb{K}}$: the very class that houses ω. Quantile asymptotics for laws with exponential-type tails have the same shape, for the same reason — one inverts a relation in which the unknown appears both linearly and logarithmically.

4.7.2 Dulac maps and first-return maps of hyperbolic sectors

Let v be a real analytic planar vector field with a hyperbolic saddle at the origin, and let D be the transition (Dulac, or corner) map from a transversal on one separatrix to a transversal on the other, written in coordinates $x \downarrow 0$ on the transversals. Then D has an asymptotic expansion in the power-logarithmic scale at zero,

$$D(x) \sim \sum_{i\geq 0} x^{\lambda_i} P_i\big(\log(1/x)\big), \qquad 0 < \lambda_0 < \lambda_1 < \cdots \to +\infty, \quad P_i \in \mathbb{R}[\,\cdot\,], \tag{4.20}$$

where λ_0 is the hyperbolicity ratio of the saddle. The logarithms are forced precisely by resonances among the exponents: for a nonresonant saddle the P_i are constants. Composing finitely many such maps with regular transition maps produces the first-return map of a hyperbolic polycycle, whose asymptotic expansion is again of the form (4.20). The finiteness of the number of limit cycles accumulating on such a polycycle — Dulac's problem, settled independently by Écalle [12] and by Ilyashenko — turns on how much of this formal expansion can be promoted to an analytic statement. Normal forms for the associated germs are studied in [23, 27], the ε-neighbourhood length of an orbit of a Dulac map is computed in the same scale [24], and closure properties of the ambient logarithmic transseries algebras are in [26].

The change of variable $X = 1/x$, $L = \log X = \log(1/x)$ carries (4.20) to $\sum_i t^{\lambda_i} P_i(L)$, which is the normal form (4.13) with a real exponent set; Part VIII treats that Hahn enlargement. The core of this volume keeps the integer lattice, where the algorithms are exact and finite.

A common trap

Part of the dynamical literature uses $-1/\log x$ as its small logarithmic generator, so that the generator tends to 0 with x, rather than $\log(1/x)$, which tends to $+\infty$. The two conventions differ by an inversion of the coefficient variable, and a formula transported between them without applying that inversion will be wrong. In this volume $L = \log X$ always tends to $+\infty$.

4.7.3 Iteration, normal forms, and recursions

Theorems 4.18 and 4.19 are the model cases: whenever the increment of a slowly diverging quantity carries a term of size $1/w$, the partial sums manufacture a logarithm, and the subsequent expansion is organized in inverse powers of n with polynomial coefficients in $\log n$. In the analytic classification of parabolic germs the same logarithm is the logarithmic term of the Fatou coordinate; in the formal classification of maps tangent to the identity at infinity the conjugating series is an element of $\mathcal{G}_{\mathrm{poly},\mathbb{K}}$. Symbolic algorithms for computing such expansions — and for the broader exp-log asymptotic inversion problem — are due to Salvy and Shackell [29, 30]; Part VII specializes their framework to the present class.

4.7.4 The Fabius–Rvachev endpoint, where this corpus meets the scale

The corpus in which this volume sits studies the Fabius function F_{bd} and the Rvachev function up. Their contact with the polynomial-logarithmic scale is at the endpoints, and it is structural rather than decorative: the endpoint problem is an inversion problem of exactly the type analysed above, and its solution needs two generators.

The corpus proves, and has verified in Lean, the effective flatness family

$$F_{\mathrm{bd}}(x) \le 2^{\binom{n+1}{2}} x^n, \qquad F_{\mathrm{bd}}(x) \le \frac{2^{\binom{n+1}{2}}}{n!} x^n, \qquad \text{valid for all } n \in \mathbb{N}_0,\ x \ge 0 \text{ with } 2^n x \le 1. \tag{4.21}$$

Optimizing the free index n in the sharper of the two turns the family into a quadratic-in-log decay law.

Proposition 4.20 (A two-term endpoint upper bound from the flatness family). *Write $\ell = \log(1/x)$ and $a = \log 2$, and set $C = \frac{1}{2} + 3/a$. For all $0 < x \le \frac{1}{4}$,*

$$\log F_{\mathrm{bd}}(x) \le -\frac{\ell^2}{2a} - \frac{\ell \log \ell}{a} + C\ell. \tag{4.22}$$

Proof. Put $m = \ell/a$; since $x \le \frac{1}{4}$ we have $\ell \ge 2a$, hence $m \ge 2$. Let $n = \lfloor m \rfloor \ge 2$ and $\theta = m - n \in [0, 1)$. The hypothesis $2^n x \le 1$ is equivalent to $na \le \ell$, i.e. to $n \le m$, which holds. Taking logarithms in the second bound of (4.21) and using $\log x = -\ell$,

$$\log F_{\mathrm{bd}}(x) \le A_1 + A_2, \qquad A_1 := \frac{an(n+1)}{2} - n\ell, \qquad A_2 := -\log n!.$$

The quadratic part. Using $\ell = am$ and $n = m - \theta$,

$$A_1 = \frac{an}{2}\big(n + 1 - 2m\big) = -\frac{a}{2}(m-\theta)(m+\theta-1) = -\frac{a}{2}\big(m^2 - m + \theta - \theta^2\big).$$

Since $\theta - \theta^2 \geq 0$ on $[0, 1)$, and since $am^2/2 = \ell^2/(2a)$ and $am/2 = \ell/2$,

$$A_1 \leq -\frac{\ell^2}{2a} + \frac{\ell}{2}.$$

The factorial part. From $\mathrm{e}^n \geq n^n/n!$ we get $n! \geq n^n \mathrm{e}^{-n}$, hence $A_2 \leq -n \log n + n$. As $m \geq 2$ and $0 \leq \theta < 1$, we have $0 \leq \theta/m \leq \frac{1}{2}$, so $\log(1 - \theta/m) \geq -2\theta/m \geq -2/m$ and therefore

$$n \log n = (m - \theta)\big(\log m + \log(1 - \theta/m)\big) \geq (m - \theta) \log m - 2.$$

Using $0 \leq \theta < 1$, $\log m \geq \log 2 > 0$ and $n \leq m$, this gives $A_2 \leq -m \log m + \log m + 2 + m$. Now

$$m \log m = \frac{\ell}{a}\big(\log \ell - \log a\big) = \frac{\ell \log \ell}{a} - \frac{\ell \log a}{a},$$

and $a = \log 2 < 1$ gives $\log a < 0$, so $-m \log m = -\ell \log \ell/a + \ell \log a/a \leq -\ell \log \ell/a$. Finally $\log m \leq m = \ell/a$ and $2 \leq m = \ell/a$, so

$$A_2 \leq -\frac{\ell \log \ell}{a} + \frac{3\ell}{a}.$$

Adding the two estimates gives (4.22) with $C = \frac{1}{2} + 3/a$. ☐

Corollary 4.21 (The inverse endpoint lives on a Puiseux-logarithmic scale)**.** *Let Q_F be the inverse of the strictly increasing restriction $F_{\mathrm{bd}}|_{[0,1]}$, and put $B = \log(1/y)$. Then, as $y \downarrow 0$,*

$$\log \frac{1}{Q_F(y)} \leq \sqrt{2B \log 2} + \mathcal{O}_{y \downarrow 0}(1)\,. \tag{4.23}$$

Proof. Let $x = Q_F(y)$, so that $y = F_{\mathrm{bd}}(x)$ and $x \downarrow 0$ as $y \downarrow 0$; for y small, $x \leq \frac{1}{4}$. Put $\ell = \log(1/x)$ and $a = \log 2$. Since $\ell \geq 2a > 1$ we have $\log \ell > 0$, so Theorem 4.20 gives

$$B = \log \frac{1}{y} = -\log F_{\mathrm{bd}}(x) \geq \frac{\ell^2}{2a} + \frac{\ell \log \ell}{a} - C\ell \geq \frac{\ell^2}{2a} - C\ell.$$

Hence $\ell^2 - 2aC\ell - 2aB \leq 0$, so $\ell \leq aC + \sqrt{a^2C^2 + 2aB} \leq 2aC + \sqrt{2aB}$, using $\sqrt{p+q} \leq \sqrt{p} + \sqrt{q}$. As $\ell = \log(1/Q_F(y))$, this is (4.23). ☐

Lean status: covered, by a sharper statement. The corpus does not carry this corollary as such; what it has is strictly stronger, a two-sided asymptotic law with an explicit error term. `Fabius.log_fabiusInv_sub_fabiusInverseLogAsymptoticMain_isBigO` in `FabiusInverseAsymptotic.lean` gives the main term $-\sqrt{2 \log 2 \cdot B} + \frac{1}{2} \log B - 1 - \frac{1}{2} \log \log 2$ with error $\mathcal{O}(\log B/\sqrt{B})$, from which the corollary above follows by discarding everything past the leading term. No separate module is needed for it, and the crosswalk records the implication rather than a restatement.

Remark 4.22 (What the corpus adds, and what must not be inferred). Theorems 4.20 and 4.21 are one-sided: they rest on a family of upper bounds, and an upper bound for F_{bd} yields only an upper bound for $\log(1/Q_F(y))$. The corpus obtains the matching lower bounds by a lattice computation along the dyadic scale ladder, giving the two-sided laws

$$\log F_{\mathrm{bd}}(x) = -\frac{\ell^2}{2 \log 2} - \frac{\ell \log \ell}{\log 2} + \mathcal{O}_{x \downarrow 0}(\ell)\,,$$

$$\log \frac{1}{Q_F(y)} = \sqrt{2B \log 2} - \tfrac{1}{2} \log B + \mathcal{O}_{y \downarrow 0}(1)\,,$$

with $\ell = \log(1/x)$ and $B = \log(1/y)$. Those results are quoted here, not proved; their proofs belong to the endpoint volumes of the corpus.

Two structural points connect them to the present volume. First, the inverse law is not a polynomial-logarithmic series in the core sense. Read with $X = B$, $t = X^{-1}$ and $L = \log X$, its displayed terms are $\sqrt{2\log 2}\, X^{1/2} - \frac{1}{2}L + \mathcal{O}_{X\to+\infty}(1)$, an element of the Puiseux enlargement $\mathbb{K}[L]\big((t^{1/2})\big)$ of Part VIII, not of $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$. The half-integer exponent is the signature of inverting a leading behaviour that is quadratic rather than linear, a situation treated in Part IV under reversion beyond the near-identity class. Second, the corpus records that the refined dyadic endpoint expansions carry *log-periodic* corrections — bounded one-periodic functions of $\ell/\log 2$ — and that the natural coordinate for the associated saddle equations is the lower real branch W_{-1}. A bounded oscillating function of $\ell/\log 2$ is not a monomial of any power-logarithmic scale: by Theorem 4.7 it is not asymptotic to a constant multiple of any $t^n L^j$, so it cannot appear as a coefficient in (4.13). Its presence is a genuine departure from the class studied here, and the algebra of this volume applies to such an expansion only after the periodic factor has been separated out. The appearance of W_{-1}, by contrast, is squarely inside the volume: that branch is expanded in Part V.

4.8 Formal, asymptotic, convergent

The material above has used three logically independent kinds of statement, and the rest of the volume depends on never confusing them.

A common trap

A displayed infinite expansion can have any of three statuses.

1. A *formal identity*: an equality of coefficient arrays in a series ring. Written with the formal tail $\mathcal{O}_t^{\mathrm{form}}(t^N)$, which means $\nu_t(F-G) \geq N$ and asserts no numerical bound, no convergence, and no uniformity in L.

2. An *asymptotic expansion*: a hierarchy of remainder estimates for an actual function on a stated domain. Written $\mathcal{O}_{X\to+\infty}(A(X))$, or $\mathcal{O}_{X\to\infty,\ X\in S}(A(X))$ in a sector, with the limit and the domain printed.

3. A *convergent expansion*: an actual infinite sum, with its domain of convergence stated.

A fourth use of the same glyph, algorithmic complexity $\mathcal{O}(N^2)$, is none of the three. No one of these four meanings may be inferred from another; in particular a bare $\mathcal{O}$ is never acceptable for an asymptotic claim in this volume.

For the Lambert expansion all three statuses are in fact available on suitable domains, which is exceptional rather than typical: the large-argument Lambert series is not only asymptotic but convergent in appropriate real and complex regions, a fact requiring separate analytic work [5, 11]. What is elementary, and worth isolating, is a statement about the *coefficients*: the formal series underlying (4.11), namely $L_1 + \sum_{n\geq 0} \mathsf{L}_n^{\mathrm{Lam}}(L_2)L_1^{-n}$, collapses to its leading term at $z = \mathrm{e}$, where $L_1 = 1$ and $L_2 = 0$. Indeed $\mathsf{L}_n^{\mathrm{Lam}}(0) = 0$ for every $n \in \mathbb{N}_0$: in the defining formula $\mathsf{L}_n^{\mathrm{Lam}}(u) = \big((-1)^{n+1}/(n+1)!\big)\big[\prod_{j=0}^{n-1}(\partial_u - j)\big]u^{n+1}$ each of the n factors either differentiates or rescales, so every monomial of the result has degree at least $(n+1) - n = 1$ and $\mathsf{L}_n^{\mathrm{Lam}}$ has no constant term. Hence every correction block vanishes at $z = \mathrm{e}$ and the series returns the correct value $W_0(\mathrm{e}) = 1$. This is an identity between coefficients, not a convergence statement: a convergence threshold is a different assertion and is not proved in this volume; Part VI states what is proved and cites what is not.

Section summary

The inverse of $y \mapsto y + \log y$ cannot be expanded in powers of X alone — the correction $-\log X$ falls in a gap of that scale — nor in powers of $\log X$ alone, which cannot even see the correction $X^{-1}\log X$. It can be expanded in the two-generator scale $t^n L^j$, and that scale is forced: the smallest differentially closed ring containing a logarithm is $\mathbb{K}[t, L]$. Dominance is lexicographic, one inverse power of X defeating every fixed power of $\log X$, so a series is organized into inverse-power blocks, each a polynomial in L. The same scale organizes implicit equations with a logarithmic term, Dulac and first-return maps, parabolic normal forms, recursions with a harmonic increment, and the endpoint asymptotics of the Fabius–Rvachev system.

Chapter 5

The formal polynomial–logarithmic rings

5.1 Generators, coefficients, and what "formal" means here

Throughout this volume $\mathbb{K}$ is a field of characteristic zero. The default analytic regime is the positive ray $X \to +\infty$; for complex work a sector and a branch of $\log$ must be named before any analytic statement is made, and the algebra developed in this part is branch-neutral until such a choice is imposed.

Definition 5.1 (Generators)**.** Set

$$t := X^{-1}, \qquad L := \log X. \tag{5.1}$$

In the formal algebra t and L are two *independent* generators: the underlying additive structure of every class below is a space of families of scalars indexed by exponent pairs, and no algebraic relation between t and L is imposed.

Independence is a statement about the ring operations only, and it is exactly as strong as it needs to be.

Remark 5.2 (The two-variable illusion). For addition and multiplication, treating t and L as unrelated indeterminates is harmless and is what makes the coefficient bookkeeping finite. For differentiation and for composition it is false. The analytic relation $L = \log X$, $t = X^{-1}$ re-enters through the distinguished derivation

$$D_X = t\,\partial_L - t^2\,\partial_t, \qquad D_X t = -t^2, \qquad D_X L = t,$$

studied in Part II, and through the substitution rule $t \mapsto G^{-1}$, $L \mapsto \log G$ of Part III. Every "shifted" operator $\partial_L - n$ that appears later is a trace of this coupling. A calculation that silently uses $\partial_L L = 1$ together with $\partial_t L = 0$ inside a differentiation is using the coupling, and must say so.

Formal is not analytic

Every statement in this part is an identity between formal objects: two elements are equal when all their coefficients agree. No such statement asserts that substituting a numerical value of X makes an infinite sum converge, nor that any function has the displayed expansion. The passage from a formal identity to an asymptotic or convergent statement is a separate theorem with separately printed hypotheses.

5.2 The four ambient classes

Definition 5.3 (The four classes)**.** Define

$$\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}} := \mathbb{K}[L][[t]] = \left\{\textstyle\sum_{n\geq 0} p_n(L)t^n : p_n \in \mathbb{K}[L]\right\}, \tag{5.2}$$

$$\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}} := \mathbb{K}[L]((t)) = \left\{\textstyle\sum_{n\geq n_0} p_n(L)t^n : n_0 \in \mathbb{Z},\ p_n \in \mathbb{K}[L]\right\}, \tag{5.3}$$

$$\mathscr{P}\mathscr{L}_{\mathrm{rat},\mathbb{K}} := \mathbb{K}(L)((t)), \tag{5.4}$$

$$\mathscr{P}\mathscr{L}_{\mathrm{itLaur},\mathbb{K}} := \mathbb{K}((L^{-1}))((t)). \tag{5.5}$$

In each case the outer construction is a formal Laurent (or power) series in t; the coefficient object at level n is called the *block* at t^n. Addition is blockwise and multiplication is the Cauchy convolution

$$\Big(\sum_i p_i t^i\Big)\Big(\sum_k q_k t^k\Big) = \sum_n \Big(\sum_{i+k=n} p_i q_k\Big) t^n,$$

in which every inner sum is finite because the outer supports are bounded below.

Two facts are definitional but deserve to be said once, because almost every later argument uses them. First, each block is a *single* finite polynomial (resp. rational function, resp. Laurent series in L^{-1}); no uniform bound on its degree is implied. Second, only finitely many negative powers of t, that is finitely many positive powers of X, may occur.

Proposition 5.4 (Localization)**.** $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}} = \mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}[t^{-1}]$*: the natural map from the localization of* $\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ *at the multiplicative set* $\{t^m : m \geq 0\}$ *to* $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ *is a ring isomorphism. The same holds with* $\mathbb{K}[L]$ *replaced by* $\mathbb{K}(L)$ *or* $\mathbb{K}((L^{-1}))$*.*

Proof. The map sends F/t^m to $t^{-m}F$; it is well defined and a ring homomorphism. It is surjective because an element of $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ with $\nu_t \geq -m$ is t^{-m} times an element of $\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$. It is injective because t^{-m} is invertible in $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$, so $t^{-m}F = 0$ forces $F = 0$, whence $F/t^m = 0$ in the localization. □

Proposition 5.5 (Strict inclusions)**.** *There are inclusions of* $\mathbb{K}$*-algebras*

$$\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}} \subsetneq \mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}} \subsetneq \mathscr{P}\mathscr{L}_{\mathrm{rat},\mathbb{K}} \subsetneq \mathscr{P}\mathscr{L}_{\mathrm{itLaur},\mathbb{K}}, \tag{5.6}$$

each given by the identity on coefficients except the last, which is the coefficientwise Laurent expansion at $L = \infty$*. All three inclusions are strict, and:*

(i) $\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ *and* $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ *are integral domains and are not fields;*

(ii) $\mathscr{P}\mathscr{L}_{\mathrm{rat},\mathbb{K}}$ *and* $\mathscr{P}\mathscr{L}_{\mathrm{itLaur},\mathbb{K}}$ *are fields.*

Proof. The first two inclusions. The first is the inclusion $E[[t]] \subset E((t))$ for $E = \mathbb{K}[L]$, a subring inclusion by Theorem 5.3; the second is the blockwise extension of the coefficient inclusion $\mathbb{K}[L] \subset \mathbb{K}(L)$, hence an injective ring homomorphism. Strictness: $t^{-1} \in \mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}} \setminus \mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$, and $L^{-1} \in \mathscr{P}\mathscr{L}_{\mathrm{rat},\mathbb{K}} \setminus \mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ because $L^{-1} \notin \mathbb{K}[L]$.

The third inclusion. The polynomial ring $\mathbb{K}[L]$ sits inside $\mathbb{K}((L^{-1}))$ (a polynomial has finitely many positive and no negative powers of L). Every nonzero $p = aL^d + \cdots \in \mathbb{K}[L]$, $a \in \mathbb{K}^\times$, is a unit there: writing $u = L^{-1}$ we have $p = au^{-d}(1 + w)$ with $w \in u\mathbb{K}[u]$, and $1 + w$ is invertible in $\mathbb{K}[[u]]$ by the geometric series $\sum_{r\geq 0}(-w)^r$, which is u-adically summable (Theorem 7.23 with coefficient ring $\mathbb{K}$ and u in place of t). By the universal property of the field of fractions, the inclusion $\mathbb{K}[L] \hookrightarrow \mathbb{K}((L^{-1}))$ extends uniquely to a field embedding $\mathbb{K}(L) \hookrightarrow \mathbb{K}((L^{-1}))$. Extending blockwise gives $\mathscr{P}\mathscr{L}_{\mathrm{rat},\mathbb{K}} \hookrightarrow \mathscr{P}\mathscr{L}_{\mathrm{itLaur},\mathbb{K}}$. Strictness is Theorem 5.7.

(i). A Laurent series ring $E((t))$ over an integral domain E is an integral domain: if $F, G \neq 0$, then by Theorem 6.2 (i) the block of FG at level $\nu_t(F) + \nu_t(G)$ is the product of two nonzero elements of E, hence nonzero. Neither $\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ nor $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ is a field: if $LG = 1$ with $G = \sum_n q_n(L)t^n$, comparison of the blocks at t^0 gives $Lq_0(L) = 1$ in $\mathbb{K}[L]$, which is impossible on degree grounds.

(ii). Let E be a field and $F \in E((t))$ nonzero, with $\nu = \nu_t(F)$ and block $c = p_\nu \in E^\times$. Then $F = ct^\nu(1 + U)$ with $U = \sum_{n\geq 1}(p_{\nu+n}/c)t^n \in tE[[t]]$, and $\sum_{r\geq 0}(-U)^r$ is t-adically summable (Theorem 7.23), so F is invertible with $F^{-1} = c^{-1}t^{-\nu}\sum_{r\geq 0}(-U)^r$. Apply this with $E = \mathbb{K}(L)$ and $E = \mathbb{K}((L^{-1}))$. □

Corollary 5.6 (Domains). *All four classes are integral domains, and ν_t and the rank-two valuation of Theorem 6.6 are defined on each of them.*

Proof. Immediate from Theorem 5.5, since a subring of a field is a domain and $\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}} \subset \mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}} \subset \mathscr{P}\mathscr{L}_{\mathrm{itLaur},\mathbb{K}}$. □

Proposition 5.7 (The iterated field is strictly larger than the rational one). *The element*

$$\Phi := \sum_{k\geq 0} L^{-k^2} \in \mathbb{K}((L^{-1})) \subset \mathscr{P}\mathscr{L}_{\mathrm{itLaur},\mathbb{K}} \tag{5.7}$$

does not lie in the image of $\mathbb{K}(L)$*; consequently* $\mathscr{P}\mathscr{L}_{\mathrm{rat},\mathbb{K}} \subsetneq \mathscr{P}\mathscr{L}_{\mathrm{itLaur},\mathbb{K}}$.

Proof. Put $u = L^{-1}$, so that $\mathbb{K}((L^{-1})) = \mathbb{K}((u))$, the image of $\mathbb{K}(L)$ is the subfield $\mathbb{K}(u)$, and $\Phi = \sum_{k\geq 0} c_k u^k \in \mathbb{K}[[u]]$ with $c_k = 1$ if k is a perfect square and $c_k = 0$ otherwise. Suppose $\Phi \in \mathbb{K}(u)$. Write $\Phi = A/B$ in lowest terms with $A, B \in \mathbb{K}[u]$. If $B(0) = 0$, then $u \mid B$; since $\Phi \in \mathbb{K}[[u]]$ we get $A = B\Phi \in u\mathbb{K}[[u]] \cap \mathbb{K}[u]$, so $u \mid A$, contradicting coprimality. Hence $B(0) = b_0 \neq 0$. Let $B = \sum_{i=0}^m b_i u^i$ and $D = \deg A$. Comparing coefficients of u^k in $B\Phi = A$ gives, for every $k > D$,

$$b_0 c_k = -\sum_{i=1}^m b_i c_{k-i}.$$

Therefore, if $k > D + m$ and $c_{k-1} = \cdots = c_{k-m} = 0$, then $c_k = 0$, and by induction $c_l = 0$ for all $l \geq k$. Consecutive squares differ by $(k+1)^2 - k^2 = 2k + 1$, so the sequence (c_k) has runs of zeros of every length; choose one of length m beginning beyond $D + m$. Then $c_l = 0$ for all large l, contradicting $c_{k^2} = 1$ for arbitrarily large k. □

Proposition 5.8 (The coefficient extensions are not localizations). *Let $S = \mathbb{K}[L] \setminus \{0\}$. Then the localization $S^{-1}\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ is a proper subring of $\mathscr{P}\mathscr{L}_{\mathrm{rat},\mathbb{K}}$. Explicitly,*

$$\Phi_0 := \sum_{n\geq 0} \frac{t^n}{L + n} \in \mathscr{P}\mathscr{L}_{\mathrm{rat},\mathbb{K}} \tag{5.8}$$

admits no representation F/q with $F \in \mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ and $q \in S$.

Proof. If $\Phi_0 = F/q$ then $q\Phi_0 = F \in \mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$, so for every $n \geq 0$ the block $q/(L+n)$ lies in $\mathbb{K}[L]$, that is, $L + n$ divides q. Since $\operatorname{char}\mathbb{K} = 0$, the polynomials $L + n$, $n \geq 0$, are pairwise non-associate irreducibles, so q would have infinitely many pairwise non-associate irreducible factors; hence $q = 0$, a contradiction. □

Remark 5.9 (What each step buys, and what it costs). Table 5.1 records the trade-offs. The two extensions on the right are not interchangeable: $\mathscr{P}\mathscr{L}_{\mathrm{rat},\mathbb{K}}$ keeps rational dependence on L *exact* but does not resolve the secondary asymptotic scale, while $\mathscr{P}\mathscr{L}_{\mathrm{itLaur},\mathbb{K}}$ resolves that scale but replaces an exact coefficient such as $(L+1)^{-1}$ by the infinite tail $L^{-1} - L^{-2} + L^{-3} - \cdots$ and therefore needs a second, inner cutoff before anything can be stored or printed.

Table 5.1: The four ambient classes. "Finite record" means the data a computer must hold to determine the object up to a declared precision.

Class	Buys	Costs
$\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}} = \mathbb{K}[L][[t]]$	$t^N\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ is a genuine ideal, so residues modulo t^N form a ring; complete and separated; local finiteness is automatic	does not contain X; not stable under multiplication by t^{-1}
$\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}} = \mathbb{K}[L]((t))$	smallest of the four containing X; still exact polynomial blocks; finite record $n \mapsto p_n$	not a field; t is a unit, so $t^N\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ is only a filtration tail, not an ideal
$\mathscr{P}\mathscr{L}_{\mathrm{rat},\mathbb{K}} = \mathbb{K}(L)((t))$	a field; exact rational dependence on L; unrestricted division of blocks	not a localization of $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ (Theorem 5.8); a block has no intrinsic asymptotic expansion until one is chosen
$\mathscr{P}\mathscr{L}_{\mathrm{itLaur},\mathbb{K}} = \mathbb{K}((L^{-1}))((t))$	a field; the Hahn field of the rank-two value group (Theorem 5.19); the secondary scale in L^{-1} is resolved	blocks are infinite objects, so a finite record needs an inner L^{-1}-cutoff as well as the outer t-cutoff; the embedding of $\mathscr{P}\mathscr{L}_{\mathrm{rat},\mathbb{K}}$ destroys exactness

Remark 5.10 (The classes are not local rings). The subset $t\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ is a prime ideal *of* $\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$, with $\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}/t\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}} \cong \mathbb{K}[L]$; since $\mathbb{K}[L]$ is not a field, $t\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ is not maximal and $\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ is not a local ring. In $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ the set $t\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ is not an ideal at all (Theorem 7.11). By contrast $\mathbb{K}(L)[[t]]$ and $\mathbb{K}((L^{-1}))[[t]]$ are discrete valuation rings with residue fields $\mathbb{K}(L)$ and $\mathbb{K}((L^{-1}))$.

5.3 Coefficient support

Expanding every block into monomials gives the double form

$$F = \sum_{n \ge n_0} p_n(L)t^n = \sum_{\substack{n\in\mathbb{Z}\\ j\in\mathbb{N}_0}} a_{n,j}\, t^n L^j, \qquad p_n(L) = \sum_{j\ge 0} a_{n,j} L^j. \tag{5.9}$$

Definition 5.11 (Coefficient support)**.** For $F \in \mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ written as in (5.9),

$$\mathrm{supp}_{\mathrm{coeff}}(F) := \{(n,j) \in \mathbb{Z}\times\mathbb{N}_0 : a_{n,j} \ne 0\}.$$

The zero series has empty coefficient support. This is support in an exponent-index lattice; it is unrelated to the pointwise support of a function.

Definition 5.12 (Dominance)**.** For $(n,j), (m,k) \in \mathbb{Z}\times\mathbb{Z}$,

$$t^n L^j \succ t^m L^k \quad\Longleftrightarrow\quad n < m, \text{ or } (n = m \text{ and } j > k). \tag{5.10}$$

We write $\prec$, $\preccurlyeq$ and $\asymp$ accordingly.

Lemma 5.13 (Analytic justification of dominance)**.** *Let $\alpha, \alpha', \beta, \beta'$ be fixed reals with either $\alpha > \alpha'$, or $\alpha = \alpha'$ and $\beta > \beta'$. Then, on the part $X > \mathrm{e}$ of the positive ray, where $\log X > 1$ and the quotient below is positive,*

$$\frac{X^{\alpha}(\log X)^{\beta}}{X^{\alpha'}(\log X)^{\beta'}} \longrightarrow +\infty \qquad (X \to +\infty).$$

Consequently (5.10) *orders the monomials $t^n L^j = X^{-n}(\log X)^j$ by decreasing eventual size.*

Proof. For $X > \mathrm{e}$ we have $\log X > 1$, so the quotient is positive and

$$\log \frac{X^{\alpha}(\log X)^{\beta}}{X^{\alpha'}(\log X)^{\beta'}} = (\alpha-\alpha')\log X + (\beta-\beta')\log\log X.$$

In the first case, since $\log\log X = o_{X\to+\infty}(\log X)$ and $\alpha - \alpha' > 0$, the right-hand side tends to $+\infty$. In the second case, $\alpha = \alpha'$ and $\beta > \beta'$, so the expression reduces to $(\beta - \beta')\log\log X \to +\infty$. $\square$

Remark 5.14 (Fixed exponents only). Theorem 5.13 is the two-generator fragment of Hardy's comparison of orders of infinity [20]: it compares two monomials with *fixed* exponents. It says nothing about a family in which the logarithmic exponent grows with the inverse-power level; see Theorem 7.9 for what this costs.

Theorem 5.15 (Characterization of admissible supports). *For a subset $S \subseteq \mathbb{Z} \times \mathbb{N}_0$ the following are equivalent.*

(a) $S = \operatorname{supp}_{\mathrm{coeff}}(F)$ *for some* $F \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$.

(b) *The projection* $\{n : (n,j) \in S$ *for some* $j\}$ *is bounded below, and for each* n *the fiber* $S_n = \{j : (n,j) \in S\}$ *is finite.*

(c) *Every nonempty subset of* S *has a* $\succ$*-greatest element* (*reverse well-ordering*).

(d) S *can be enumerated as a strictly* $\succ$*-decreasing sequence* $s_0 \succ s_1 \succ s_2 \succ \cdots$*, finite or of order type* ω*, exhausting* S.

Proof. (a)⇔(b). By Theorem 5.3 an element of $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ is exactly a family $(a_{n,j})$ whose blocks $p_n = \sum_j a_{n,j} L^j$ are polynomials — that is, have finite support in j — and which vanishes for $n < n_0$. Conversely any S with property (b) is the support of $\sum_{(n,j)\in S} t^n L^j$.

(b)⇒(d). If $S = \emptyset$ there is nothing to prove. Otherwise let $n_0 = \min\{n : S_n \neq \emptyset\}$, which exists because the projection is a nonempty subset of $\mathbb{Z}$ bounded below. List first the finitely many elements of $\{n_0\} \times S_{n_0}$ in order of decreasing j, then those of $\{n_0+1\} \times S_{n_0+1}$, and so on. By (5.10) the resulting list is strictly $\succ$-decreasing, every element of S occupies a finite position, and every element occurs.

(d)⇒(c). Given $\emptyset \neq T \subseteq S$, let $k = \min\{i : s_i \in T\}$; then $s_k \succ s_i$ for every other $s_i \in T$, so s_k is the greatest element of T.

(c)⇒(b). If the projection of S were unbounded below, then S itself would have no greatest element, since any $(n,j) \in S$ is dominated by some $(n',j') \in S$ with $n' < n$. If some fiber S_n were infinite, then $\{n\} \times S_n$ would be a nonempty subset with no greatest element, because an infinite subset of $\mathbb{N}_0$ is unbounded above. $\square$

Remark 5.16 (Order type). Condition (d) is the sharp form of the "well-based support" hypothesis of Hahn's construction [19, 13] in the present rank-two, integer-lattice case: the support of an element of $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ is not merely reverse well-ordered, it has order type at most ω. The same is *not* true in $\mathscr{PL}_{\mathrm{itLaur},\mathbb{K}}$, where fibers may be infinite downwards; see Theorem 5.19.

Example 5.17 (A support lattice). The table below marks with • the pairs (n,j) of a support whose log-degree profile is $d_n = 1, 2, 3, 3, 4, 5, 5$ for $n = 0, \dots, 6$. Each column is finite; the columns start at $n = 0$; the greatest monomial of any nonempty sub-diagram is found by moving as far left as possible and then as far up as possible.

$j=5$	·	·	·	·	·	•	•
$j=4$	·	·	·	·	•	•	•
$j=3$	·	·	•	•	•	•	•
$j=2$	·	•	•	•	•	•	•
$j=1$	•	•	•	•	•	•	•
$j=0$	•	•	•	•	•	•	•
	$n=0$	1	2	3	4	5	6

5.4 The Hahn-series viewpoint

Let $\Gamma = \mathbb{Z} \times \mathbb{Z}$ be the additive group of exponent pairs of the monomials $t^n L^j$, $n, j \in \mathbb{Z}$, ordered by the rank-two valuation

$$v(t^n L^j) = (n, -j), \tag{5.11}$$

with Γ lexicographically ordered. Thus a smaller value means a more dominant monomial, and (5.11) is exactly Theorem 5.12 rewritten. The general framework — an ordered monomial group, a coefficient field, and a support condition making convolution locally finite — is Hahn's [19]; its transseries specialization is developed in [13, 33, 1]. The present rank-two case is the smallest nontrivial instance, and the whole of Section 5.3 is the statement that its support condition can be checked coordinatewise.

Lemma 5.18 (Rank two). *The convex subgroups of the ordered abelian group Γ are precisely $0 \subset \{0\} \times \mathbb{Z} \subset \Gamma$. Hence Γ has rank two, and v is a valuation of rank two.*

Proof. $\{0\} \times \mathbb{Z}$ is convex: if $(0, b) < (a, c) < (0, b')$ then $a = 0$ by the lexicographic rule. Let H be a convex subgroup containing some (a, b) with $a \neq 0$; replacing (a, b) by its negative we may assume $a > 0$. For any $(c, d) \in \Gamma$ choose N with $Na > |c|$; then $-N(a, b) < (c, d) < N(a, b)$ and both bounds lie in H, so $(c, d) \in H$ by convexity. Thus $H = \Gamma$. A convex subgroup contained in $\{0\} \times \mathbb{Z}$ and containing some $(0, b) \neq 0$ is all of $\{0\} \times \mathbb{Z}$ by the same argument applied inside that subgroup. □

Theorem 5.19 (The iterated field is the Hahn field of Γ). *Let $\mathbb{K}[[t^{\mathbb{Z}} L^{\mathbb{Z}}]]$ denote the Hahn field of formal sums $\sum_{(n,j)} a_{n,j} t^n L^j$ with coefficients in $\mathbb{K}$ whose support is reverse well ordered for $\succ$, with the convolution product. Then*

$$\mathbb{K}[[t^{\mathbb{Z}} L^{\mathbb{Z}}]] = \mathscr{PL}_{\mathrm{itLaur},\mathbb{K}}$$

as $\mathbb{K}$-algebras: a family $(a_{n,j})_{(n,j)\in\Gamma}$ has reverse well-ordered support if and only if its n-projection is bounded below and every fiber S_n is bounded above in j; such families are exactly the elements of $\mathbb{K}((L^{-1}))((t))$, and the two products agree.

Proof. Support condition. Suppose the support S is reverse well ordered. If its n-projection were unbounded below, S would have no $\succ$-greatest element, as in the proof of Theorem 5.15. If some fiber S_n were unbounded above in j, then $\{n\} \times S_n$ would have no greatest element. Conversely, assume the two conditions and let $\emptyset \neq T \subseteq S$. The n-projection of T is a nonempty subset of $\mathbb{Z}$ bounded below, so has a least element n_1; the fiber T_{n_1} is a nonempty subset of $\mathbb{Z}$ bounded above, so has a greatest element j_1; and (n_1, j_1) is $\succ$-greatest in T.

Identification of the underlying sets. A family whose n-projection is bounded below by n_0 and whose fibers are bounded above is precisely a family $(p_n)_{n\geq n_0}$ with $p_n = \sum_j a_{n,j} L^j \in \mathbb{K}((L^{-1}))$, that is, an element of $\mathbb{K}((L^{-1}))((t))$; and conversely.

Products. In $\mathscr{PL}_{\mathrm{itLaur},\mathbb{K}}$ the coefficient of t^n in a product is the finite sum $\sum_{i+k=n} p_i q_k$ (finite because both outer supports are bounded below), and the coefficient of L^j in $p_i q_k$ is the finite sum $\sum_{j_1+j_2=j} a_{i,j_1} b_{k,j_2}$ (finite because both inner supports are bounded above). This is exactly the Hahn convolution $\sum_{(n_1,j_1)+(n_2,j_2)=(n,j)} a_{n_1,j_1} b_{n_2,j_2}$, whose local finiteness has just been verified. □

Corollary 5.20 (Where the smaller classes sit). *Under the identification of Theorem 5.19: $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ is the subring of elements with* finite *fibers contained in $\mathbb{N}_0$; $\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ is the further subring with n-projection contained in $\mathbb{N}_0$; and $\mathscr{PL}_{\mathrm{rat},\mathbb{K}}$ is the set of elements of $\mathscr{PL}_{\mathrm{itLaur},\mathbb{K}}$* every *block of which lies in the subfield $\mathbb{K}(L) \subset \mathbb{K}((L^{-1}))$.*

Proof. The first two statements are Theorem 5.15 (b) together with Theorem 5.3. The third is the definition of $\mathbb{K}(L)((t))$ transported along the blockwise embedding of Theorem 5.5. □

Remark 5.21. The third description cannot be improved to "the subring generated by $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ and the inverses of the nonzero elements of $\mathbb{K}[L]$": every element of that subring is a finite sum of finite products and therefore has a *common* denominator across all blocks, whereas Φ_0 of (5.8) has none, by Theorem 5.8.

5.5 Why the iteration order is part of the type

Both $\mathbb{K}((L^{-1}))((t))$ and $\mathbb{K}((t))((L^{-1}))$ are fields of formal sums of the monomials t^nL^j, and both are naturally sets of families $(a_{n,j}) \in \mathbb{K}^\Gamma$. They impose *different* support conditions, and neither is contained in the other.

Proposition 5.22 (Incomparability). *Regard both fields as sets of families $(a_{n,j})_{(n,j)\in\Gamma}$. Then*

$$\mathscr{PL}_{\mathrm{itLaur},\mathbb{K}} = \{(a_{n,j}) : \ n \textit{ bounded below};\ \textit{for each } n,\ j \textit{ bounded above}\}\,, \tag{5.12}$$

$$\mathbb{K}((t))((L^{-1})) = \{(a_{n,j}) : \ j \textit{ bounded above};\ \textit{for each } j,\ n \textit{ bounded below}\}\,. \tag{5.13}$$

Neither set contains the other. Explicit witnesses:

$$A = \sum_{n\geq 0} t^nL^n \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}} \subset \mathscr{PL}_{\mathrm{itLaur},\mathbb{K}}, \qquad B = \sum_{r\geq 0} t^{-r}L^{-r} \in \mathbb{K}((t))((L^{-1})), \tag{5.14}$$

with $A \notin \mathbb{K}((t))((L^{-1}))$ and $B \notin \mathscr{PL}_{\mathrm{itLaur},\mathbb{K}}$.

Proof. The two displayed descriptions are the definitions of an outer Laurent series whose coefficients are inner Laurent series, read in the two possible orders. For A: its support is $\{(n,n) : n \geq 0\}$, whose j-coordinate is unbounded above, so A violates the first condition in (5.13); but its n-coordinate is bounded below and each fiber is a single point, so $A \in \mathscr{PL}_{\mathrm{itLaur},\mathbb{K}}$ — indeed $A \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$, since its block at t^n is the polynomial L^n. For B: its support is $\{(-r,-r) : r \geq 0\}$, whose n-coordinate is unbounded below, so $B \notin \mathscr{PL}_{\mathrm{itLaur},\mathbb{K}}$; but its j-coordinate is bounded above by 0 and each of its fibers over a fixed $j = -r$ is the single point $n = -r$, so B satisfies (5.13). □

Remark 5.23 (The common part, and why we choose t outermost). Intersecting (5.12) and (5.13) gives the families supported in a quadrant $\{n \geq n_0\} \times \{j \leq j_0\}$. These form a subring of both fields, but not a field: $1 + tL$ lies in it and its inverse $\sum_{r\geq 0}(-1)^r t^r L^r$ does not. We take t outermost because the intended hierarchy is $t \prec L^{-m} \prec 1 \prec L^m \prec t^{-1}$ for every fixed $m > 0$; this is the formal counterpart of $X^{-1} = o_{X\to+\infty}((\log X)^{-m})$, and it is exactly the statement that the convex subgroup $\{0\} \times \mathbb{Z}$ of Theorem 5.18 is the *smaller* one. The same phenomenon — that an iterated series field remembers the order in which its scales were adjoined — is standard in the general theory [14, 33]; what Theorem 5.22 adds is that here the two orders are *incomparable*, not merely distinct.

5.6 The Puiseux-log extension

Definition 5.24 (Puiseux-log ring). For $s \in \mathbb{N}_{>0}$ let

$$\mathbb{K}[L]\big((t^{1/s})\big) = \Big\{\textstyle\sum_{r\geq r_0} p_r(L)\, t^{r/s} : r_0 \in \mathbb{Z},\ p_r \in \mathbb{K}[L]\Big\}\,, \tag{5.15}$$

the polynomial-logarithmic Laurent ring in the generator $t^{1/s}$. Its monomials are $t^\alpha L^j$ with $\alpha \in s^{-1}\mathbb{Z}$ and $j \in \mathbb{N}_0$.

Proposition 5.25 (Ramified tower). *For $s \mid s'$ the substitution $t^{1/s} \mapsto (t^{1/s'})^{s'/s}$ is an injective ring homomorphism $\mathbb{K}[L]((t^{1/s})) \hookrightarrow \mathbb{K}[L]((t^{1/s'}))$, and these maps are compatible, so*

$$\mathbb{K}[L]\big((t^{1/\infty})\big) \coloneqq \bigcup_{s\in\mathbb{N}_{>0}} \mathbb{K}[L]\big((t^{1/s})\big)$$

is a ring, containing $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}} = \mathbb{K}[L]((t^{1/1}))$. *Every element of the union has all its exponents in a single* $s^{-1}\mathbb{Z}$*; consequently the union is* strictly *smaller than the Hahn ring of well-ordered supports in* $\mathbb{Q} \times \mathbb{N}_0$.

Proof. The substitution is injective because it is injective on exponents and the identity on coefficients, and it is a ring homomorphism because it is induced by a monoid homomorphism of monomials; compatibility is immediate. A directed union of rings along injective homomorphisms is a ring. For strictness, consider $\sum_{k\geq 1} t^{1-1/k}$, whose support $\{1 - 1/k : k \geq 1\} \subset \mathbb{Q}$ is well ordered in the direction of increasing smallness and has no bounded denominator; it therefore lies in the Hahn ring but in no $\mathbb{K}[L]((t^{1/s}))$. □

Remark 5.26 (What ramification costs). Formally, nothing in Section 5.3 changes: the support is a subset of $s^{-1}\mathbb{Z} \times \mathbb{N}_0$ with finite fibers and lower-bounded projection, and Theorem 5.15 applies verbatim after rescaling. Analytically a branch of $X^{1/s}$ must be chosen, and that choice is data, not a convention. Fractional exponents arise in this volume only where they are forced — at a degenerate leading derivative, and at the square-root branch point of W. The corresponding classes near a finite point, where one writes $\sum_{\lambda\in S} z^{\lambda} p_{\lambda}(\log z)$ as $z \to 0^{+}$, are the power–log and Dulac series of local dynamics [23, 24, 27, 26]; the change of variable relating them to the present classes is carried out in Part V.

Chapter 6

Valuation, leading data, and the degree profile

Throughout this chapter R denotes any one of the four classes of Theorem 5.3, with coefficient ring $E \in \{\mathbb{K}[L], \mathbb{K}(L), \mathbb{K}((L^{-1}))\}$, and $F = \sum_{n \geq n_0} p_n t^n$ denotes the block representation of an element of R.

6.1 The outer valuation

Definition 6.1 (Outer valuation)**.** For nonzero $F \in R$ set

$$\nu_t(F) := \min\{n \in \mathbb{Z} : p_n \neq 0\}, \qquad \nu_t(0) := +\infty. \tag{6.1}$$

The symbol $+\infty$ obeys $+\infty + m = +\infty$ and $+\infty > m$ for every $m \in \mathbb{Z}$.

The minimum exists because the set of occurring n is a nonempty subset of $\mathbb{Z}$ bounded below. The convention $\nu_t(0) = +\infty$ is not cosmetic: it is what makes the laws below hold without case distinctions, and it is what makes $\{F : \nu_t(F) \geq N\}$ an additive subgroup.

Theorem 6.2 (Valuation laws)**.** *Let $F, G \in R$. Then*

(i) $\nu_t(FG) = \nu_t(F) + \nu_t(G)$;

(ii) $\nu_t(F + G) \geq \min\{\nu_t(F), \nu_t(G)\}$;

(iii) $\nu_t(F) = +\infty$ *if and only if* $F = 0$, *and* $\nu_t(-F) = \nu_t(F)$.

Moreover the inequality in (ii) *is* strict *if and only if*

$$\nu_t(F) = \nu_t(G) < +\infty \quad \textit{and} \quad p_{\nu_t(F)} + q_{\nu_t(G)} = 0, \tag{6.2}$$

where q_n denotes the blocks of G. In particular ν_t satisfies the axioms of a valuation with values in $\mathbb{Z} \cup \{+\infty\}$ on each of the four classes, and $\{F \in R : \nu_t(F) \geq 0\} = E[[t]]$. When E is a field — that is, for $R = \mathscr{PL}_{\mathrm{rat},\mathbb{K}}$ and $R = \mathscr{PL}_{\mathrm{itLaur},\mathbb{K}}$ — R is a field (Theorem 5.5 (ii)), ν_t is surjective onto $\mathbb{Z}$, and ν_t is a discrete valuation of R in the usual sense, with valuation ring $E[[t]]$. For $R = \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ and $R = \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ the same laws hold, but R is not a field and the phrase "valuation ring" is not available.

Proof. (iii) is immediate from (6.1).

(i) If $F = 0$ or $G = 0$ both sides are $+\infty$. Otherwise put $m = \nu_t(F)$, $r = \nu_t(G)$. For $n < m + r$ the block of FG at t^n is $\sum_{i+k=n} p_i q_k$; in each summand either $i < m$ or $k < r$ (else $i + k \geq m + r > n$), so every summand vanishes. At $n = m + r$ the only surviving pair is (m, r), giving the block $p_m q_r$, which is nonzero because E is an integral domain. Hence $\nu_t(FG) = m + r$.

(ii) and the strictness criterion. Write $\mu = \min\{\nu_t F, \nu_t G\}$. For $n < \mu$ both p_n and q_n vanish, so the block of $F + G$ at t^n vanishes; this is (ii). For the criterion, suppose first $\nu_t F \neq \nu_t G$, say $\nu_t F < \nu_t G$. Then the block of $F+G$ at $t^{\nu_t F}$ is $p_{\nu_t F} + q_{\nu_t F} = p_{\nu_t F} \neq 0$, so equality holds. If $\nu_t F = \nu_t G = \mu < +\infty$, the block of $F + G$ at t^μ is $p_\mu + q_\mu$, and the inequality is strict exactly when this vanishes. If $\mu = +\infty$ then $F = G = 0$ and both sides are $+\infty$.

Finally, (i)–(iii) are the axioms of a valuation with values in $\mathbb{Z} \cup \{+\infty\}$; the set $\{\nu_t \geq 0\}$ is, by (6.1), the set of series with no negative blocks, namely $E[[t]]$. That $E((t))$ is a field when E is one — so that ν_t is then a discrete valuation of a field with valuation ring $E[[t]]$ — is Theorem 5.5 (ii), whose proof uses only Theorem 7.23 and not the present theorem. The argument used about E only that it is an integral domain, so the same laws hold in $E[[t]]$ and $E((t))$ for every integral domain E; this is the form in which they are applied in Theorem 5.5. □

Example 6.3 (Strictness needs equal valuations and full cancellation). With $F = (L^2 + L)t^3 + t^4$ and $G = (-L^2 - L)t^3 + t^5$ we have $\nu_t F = \nu_t G = 3$ but $\nu_t(F + G) = 4$. Cancelling only part of a block is not enough: $F' = (L^2 + L)t^3$ and $G' = (-L^2 + 2)t^3$ give $F' + G' = (L + 2)t^3$, still of valuation 3. And with $F'' = t^3$, $G'' = -t^4$ the valuations differ, so $\nu_t(F'' + G'') = 3$ whatever the blocks are.

6.2 The rank-two valuation and the three leading objects

Definition 6.4 (Leading data). Let $F \in R$ be nonzero and $\nu = \nu_t(F)$. Write the block p_ν in its L^{-1}-expansion inside $\mathbb{K}((L^{-1}))$ — for $E = \mathbb{K}[L]$ this is the polynomial itself, and for $E = \mathbb{K}(L)$ it is the image under the embedding of Theorem 5.5 — so that $p_\nu = aL^d +$ (strictly lower powers of L) with $a \in \mathbb{K}^\times$ and $d = \deg_{\mathrm{L}} p_\nu$ as in Theorem 6.6. Then:

- the *leading coefficient block* of F is the whole coefficient p_ν;
- the *leading monomial* is $\mathrm{lm}(F) = t^\nu L^d$;
- the *scalar leading coefficient* is $\mathrm{lc}(F) = a$.

For $F = 0$ none of the three is defined.

Remark 6.5 (Leading block, leading monomial, leading scalar). p_ν is a polynomial (or rational function, or Laurent series in L^{-1}); $\mathrm{lm}(F)$ is a monomial; $\mathrm{lc}(F)$ is a scalar. The distinction is not pedantic. For $F = t(L + 1)$ we have $\mathrm{lc}(F) = 1 \in \mathbb{K}^\times$ and $\mathrm{lm}(F) = tL$, yet $t^{-1}F = L + 1$ is not invertible in $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$: what governs invertibility is the leading *block*, not the leading scalar. Every "the leading coefficient is 1, so we may divide" argument must name which of the three objects it means.

Definition 6.6 (Rank-two valuation). For nonzero $F \in R$ set

$$v(F) := \big(\nu_t(F),\, -\deg_{\mathrm{L}} p_{\nu_t(F)}\big) \in \Gamma = \mathbb{Z} \times \mathbb{Z}, \tag{6.3}$$

lexicographically ordered, and $v(0) := \infty_\Gamma$, a symbol larger than every element of Γ and absorbing under addition. Here $\deg_{\mathrm{L}}$ of a nonzero block means the largest j with a nonzero coefficient of L^j; for $E = \mathbb{K}((L^{-1}))$ this exists because inner supports are bounded above (Theorem 5.19), and for $E = \mathbb{K}(L)$ it is deg numerator minus deg denominator, computed in $\mathbb{K}((L^{-1}))$ under the embedding of Theorem 5.5. Thus $v(t^n L^j) = (n, -j)$, matching (5.11), and ν_t is the coarsening of v obtained by projecting to the first coordinate.

Theorem 6.7 (Rank-two valuation laws and leading-term arithmetic). *For $F, G \in R$,*

(i) $v(FG) = v(F) + v(G)$ *— equivalently,* $\mathrm{lm}(FG) = \mathrm{lm}(F)\,\mathrm{lm}(G)$ *for nonzero F, G — and, in addition,*

$$\mathrm{lc}(FG) = \mathrm{lc}(F)\,\mathrm{lc}(G) \qquad (F, G \neq 0);$$

(ii) $v(F+G) \geq \min\{v(F), v(G)\}$*, with equality unless* $v(F) = v(G) < \infty_\Gamma$ *and* $\mathrm{lc}(F) + \mathrm{lc}(G) = 0$*;*

(iii) *if* $\mathrm{lm}(F) \neq \mathrm{lm}(G)$*, then* $\mathrm{lm}(F+G)$ *is the* $\succ$*-greater of* $\mathrm{lm}(F)$ *and* $\mathrm{lm}(G)$*, and* $\mathrm{lc}(F+G)$ *is the corresponding leading scalar.*

Consequently v *is a valuation on* R *with values in the lexicographically ordered group* Γ*. Its image is all of* Γ *exactly when* E *is* $\mathbb{K}(L)$ *or* $\mathbb{K}((L^{-1}))$*; on* $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ *the image is* $\{(n,-d) : n \in \mathbb{Z},\ d \in \mathbb{N}_0\}$ *and on* $\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ *it is* $\{(n,-d) : n, d \in \mathbb{N}_0\}$*. In every case the image generates* Γ*, so the associated value group is* Γ*, which has rank two by Theorem 5.18.*

Proof. (i) The cases involving 0 are handled by the absorbing convention. Let $F, G \neq 0$, $m = \nu_t F$, $r = \nu_t G$. By Theorem 6.2 (i) the leading block of FG is $p_m q_r$. In E — a polynomial ring, a rational function field, or a field of Laurent series in L^{-1} — the top L-degree is additive and the top coefficients multiply, because $\mathbb{K}$ is a domain: $\deg_{\mathrm{L}}(p_m q_r) = \deg_{\mathrm{L}} p_m + \deg_{\mathrm{L}} q_r$ and the coefficient of the top power is the product of the two top coefficients. Hence $v(FG) = (m+r, -\deg_{\mathrm{L}} p_m - \deg_{\mathrm{L}} q_r) = v(F) + v(G)$, which is the displayed statement about lm and lc.

(ii)–(iii) If $\nu_t F < \nu_t G$, then $v(F) < v(G)$ lexicographically and the block of $F+G$ at $t^{\nu_t F}$ equals $p_{\nu_t F}$, so $v(F+G) = v(F) = \min$. If $\nu_t F = \nu_t G = \mu$ but $\deg_{\mathrm{L}} p_\mu > \deg_{\mathrm{L}} q_\mu$, then again $v(F) < v(G)$, and the top L-term of $p_\mu + q_\mu$ is that of p_μ, so $v(F+G) = v(F)$. These two cases cover $\mathrm{lm}(F) \neq \mathrm{lm}(G)$ and give (iii). In the remaining case $\mathrm{lm}(F) = \mathrm{lm}(G) = t^\mu L^d$, the coefficient of L^d in $p_\mu + q_\mu$ is $\mathrm{lc}(F) + \mathrm{lc}(G)$; if this is nonzero then $v(F+G) = v(F) = v(G)$, and if it vanishes then $v(F+G) > v(F)$.

The image of v. $v(t^n L^d) = (n, -d)$, and $t^n L^d$ lies in $\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ for $n, d \geq 0$, in $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ for $n \in \mathbb{Z}$, $d \geq 0$, and in $\mathscr{P}\mathscr{L}_{\mathrm{rat},\mathbb{K}} \subset \mathscr{P}\mathscr{L}_{\mathrm{itLaur},\mathbb{K}}$ for all $n, d \in \mathbb{Z}$; conversely $\deg_{\mathrm{L}}$ of a block of an element of $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ is a nonnegative integer and $\nu_t \geq 0$ on $\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$, which gives the reverse inclusions. Since $(1,0)$ and $(0,-1)$ lie in every one of these images, each generates Γ. Rank two is then Theorem 5.18. □

Remark 6.8 (What ν_t forgets). ν_t is deliberately coarser than v: it ignores the internal L-ordering of a block. That is precisely what makes it usable as a computational filtration — the set $\{\nu_t \geq N\}$ is a *finitely describable* tail, whereas $\{v \geq (N, -d)\}$ cuts a block in half and is not stable under the operations of Part II. All truncation, completeness and summability statements in this volume are for ν_t; the finer v is used only to compare individual monomials.

6.3 Uniqueness of the block representation

The block form (5.9) is unique by construction if one *defines* an element to be its family of coefficients. What actually needs proof is that an element presented as a formally summable sum of monomials — which is how every later formula presents its output — has uniquely determined blocks, independently of the order of summation. The proof uses the summability calculus of Section 7.7; we state it here because it belongs to the leading-data discussion.

Theorem 6.9 (Uniqueness of the block representation)**.** *Let* $(a_{n,j})$ *be a family of scalars indexed by* $\mathbb{Z} \times \mathbb{N}_0$ *whose index set satisfies Theorem 5.15 (b). Then the family of monomial terms* $\left(a_{n,j} t^n L^j\right)$ *is formally summable in* $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$*, its sum* F *is independent of the enumeration, and*

$$[t^n]F = \sum_j a_{n,j} L^j, \qquad [t^n L^j]F = a_{n,j} \qquad \textit{for all } (n,j).$$

In particular, if $\sum a_{n,j} t^n L^j = \sum b_{n,j} t^n L^j$ *with both index sets admissible, then* $a_{n,j} = b_{n,j}$ *for all* (n,j)*: the monomials* $t^n L^j$ *are linearly independent over* $\mathbb{K}$ *in the strong, summable sense.*

Proof. Summability: for $N \in \mathbb{Z}$, a term $a_{n,j} t^n L^j$ has $\nu_t = n$, and the terms with $n < N$ are indexed by $\bigcup_{n_0 \leq n < N} \{n\} \times S_n$, a finite union of finite sets by Theorem 5.15 (b). Hence only finitely many terms have $\nu_t < N$, which is Theorem 7.21. By Theorem 7.22 (ii)–(iii) the sum exists, is independent of the enumeration, and satisfies $[t^n]F = \sum_{(n',j)}[t^n]\big(a_{n',j}t^{n'}L^j\big) = \sum_j a_{n,j}L^j$, a finite sum. Extracting the coefficient of L^j from this polynomial gives $[t^nL^j]F = a_{n,j}$. The last assertion follows because F determines all its coefficients. $\square$

Proposition 6.10 (Leading-block factorization). *Let E be a* field *and $F \in E((t))$ nonzero. Then there are unique $\nu \in \mathbb{Z}$, $c \in E^\times$ and $U \in tE[[t]]$ with*

$$F = c\,t^\nu\,(1+U), \tag{6.4}$$

namely $\nu = \nu_t(F)$, $c = p_\nu$, and $U = \sum_{n\geq 1}(p_{\nu+n}/c)t^n$. The factorization is available in $\mathscr{PL}_{\mathrm{rat},\mathbb{K}}$ and in $\mathscr{PL}_{\mathrm{itLaur},\mathbb{K}}$; over $E = \mathbb{K}[L]$ it is available with $c = p_\nu$ and $U \in t\mathbb{K}(L)[[t]]$, but in general not *with $U \in t\mathbb{K}[L][[t]]$.*

Proof. Existence over a field is the displayed formula, which makes sense because $c = p_\nu \in E^\times$. Uniqueness: applying ν_t to (6.4) and using Theorem 6.2 (i) with $\nu_t(1+U) = 0$ gives $\nu = \nu_t(F)$; comparing blocks at t^ν gives $c = p_\nu$; and then $U = c^{-1}t^{-\nu}F - 1$ is determined. For the last assertion take $F = L + t \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$. A factorization $F = L(1+U)$ with $U = \sum_{n\geq 1} u_n(L)t^n \in t\mathbb{K}[L][[t]]$ would force, at t^1, $1 = Lu_1(L)$ in $\mathbb{K}[L]$, which is impossible; and c must equal L by the uniqueness just proved. $\square$

6.4 The log-degree profile

Definition 6.11 (Log-degree profile). For $F \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ define $d_F : \mathbb{Z} \to \mathbb{N}_0 \cup \{-\infty\}$ by

$$d_F(n) = \begin{cases} \deg_{\mathrm{L}} p_n, & p_n \neq 0,\\ -\infty, & p_n = 0, \end{cases} \tag{6.5}$$

with the conventions $-\infty + m = -\infty$ and $\max(-\infty, m) = m$. Absent blocks are extended by zero, so $d_F(n) = -\infty$ for $n < \nu_t(F)$.

The profile is genuinely extra data: $\nu_t(F)$ is the first n at which $d_F(n) > -\infty$, and nothing more about d_F can be recovered from it.

Proposition 6.12 (Profile under the ring operations). *For $F, G \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ and every $n \in \mathbb{Z}$,*

$$d_{F+G}(n) \leq \max\{d_F(n), d_G(n)\}, \tag{6.6}$$

$$d_{FG}(n) \leq \max_{i+k=n}\big(d_F(i) + d_G(k)\big), \tag{6.7}$$

where in (6.7) *the maximum is over the finitely many pairs with $i \geq \nu_tF$ and $k \geq \nu_tG$, and is $-\infty$ if there are none. Equality holds in* (6.6) *unless the top L-terms cancel, and in* (6.7) *unless the sum of the products of top coefficients over the maximizing pairs vanishes.*

Proof. For a sum of polynomials the degree is at most the maximum of the degrees, with equality unless the coefficients of that maximal power cancel; apply this to $p_n + q_n$ for (6.6) and to $\sum_{i+k=n} p_iq_k$ for (6.7), using $\deg_{\mathrm{L}}(p_iq_k) = \deg_{\mathrm{L}} p_i + \deg_{\mathrm{L}} q_k$ (valid with the $-\infty$ convention). The index set is finite because $i \geq \nu_tF$ and $k = n - i \geq \nu_tG$ force $\nu_tF \leq i \leq n - \nu_tG$. $\square$

Definition 6.13 (Series of bounded slope). For $\sigma \geq 0$ let

$$\mathcal{D}_\sigma := \{F \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}} :\ \exists C \in \mathbb{R}\ \forall n,\ d_F(n) \leq \sigma n + C\}.$$

Proposition 6.14 (Slope transport). *Let $F, G \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ be nonzero with $m = \nu_t F$, $r = \nu_t G$, and suppose*

$$d_F(n) \le A + \alpha n \ (n \ge m), \qquad d_G(n) \le B + \beta n \ (n \ge r),$$

with $\alpha \ge \beta \ge 0$. Then

$$d_{FG}(n) \le \big(A + B - (\alpha - \beta)r\big) + \alpha n \qquad (n \ge m + r). \tag{6.8}$$

In particular each $\mathcal{D}_\sigma$ is a subring of $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ containing $\mathbb{K}[L]$ and $t^{\pm 1}$, and $\mathcal{D}_\sigma \subseteq \mathcal{D}_{\sigma'}$ for $\sigma \le \sigma'$. If moreover $r \ge 0$ — in particular if $F, G \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ — then (6.8) *implies the simpler bound $d_{FG}(n) \le A + B + \alpha n$.*

Proof. By (6.7), for $i + k = n$ with $i \ge m$, $k \ge r$,

$$d_F(i) + d_G(k) \le A + B + \alpha i + \beta k = A + B + \beta n + (\alpha - \beta)i \le A + B + \beta n + (\alpha - \beta)(n - r),$$

using $i = n - k \le n - r$ and $\alpha - \beta \ge 0$; this is (6.8). For the subring claim, closure under subtraction follows from (6.6), and closure under multiplication from (6.8), applied with $\alpha = \beta = \sigma$ (so that the correction term $-(\alpha - \beta)r$ vanishes); the constants A, B are fixed for given F, G. For the monotonicity, let $\sigma \le \sigma'$ and $F \in \mathcal{D}_\sigma$, say $d_F(n) \le \sigma n + C$ for all n, and put $m = \nu_t F$. For $n < m$ the left-hand side is $-\infty$ and nothing is to be proved; for $n \ge m$ one has $\sigma n + C = \sigma' n + C + (\sigma - \sigma')n \le \sigma' n + C + (\sigma - \sigma')m$, because $\sigma - \sigma' \le 0$. Hence $F \in \mathcal{D}_{\sigma'}$ with the constant $C + (\sigma - \sigma')m$. ☐

Example 6.15 (The naive slope bound fails on the Laurent ring). Take $F = \sum_{n \ge 0} L^n t^n$ with $A = 0$, $\alpha = 1$, $m = 0$, and $G = t^{-2}$ with $B = 0$, $\beta = 0$, $r = -2$. Then $FG = \sum_{n \ge -2} L^{n+2} t^n$, so $d_{FG}(n) = n + 2$. The naive bound $A + B + \max(\alpha, \beta)n = n$ is violated for every n, while (6.8) gives $0 + 0 - (1 - 0)(-2) + n = n + 2$, attained.

Remark 6.16 (Degree bounds are hypotheses, never definitions). No class in Theorem 5.3 imposes any bound on d_F. A statement of the form $d_F(n) \le \sigma n + C$ is either an explicit hypothesis, or an invariant proved for the particular construction at hand — as it is for the Lambert polynomials, where $\deg \mathsf{L}_n^{\mathrm{Lam}} = n$ for $n \ge 1$. It is never available merely because the object lies in $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$. Two consequences are worth isolating. First, membership in $\mathcal{D}_\sigma$ must be re-proved after every operation that is not covered by Theorem 6.14; the subring property gives sums and products only. Second, a formal tail $\mathcal{O}_t^{\mathrm{form}}\big(t^N\big)$ becomes an analytic estimate only after a degree bound on the discarded blocks is supplied; see Theorem 7.9.

Example 6.17 (Leading data and profile of a typical element). Let

$$F = X^2 + 3XL + 7 + (L^3 - 2L)t + \sum_{n \ge 2} (L^n + 1)t^n = t^{-2} + 3Lt^{-1} + 7 + (L^3 - 2L)t + \sum_{n \ge 2} (L^n + 1)t^n.$$

Then $\nu_t(F) = -2$; the leading block is the constant polynomial 1; $\mathrm{lm}(F) = t^{-2} = X^2$; $\mathrm{lc}(F) = 1$; and $v(F) = (-2, 0)$. The profile is

$$d_F(-2) = 0, \quad d_F(-1) = 1, \quad d_F(0) = 0, \quad d_F(1) = 3, \quad d_F(n) = n \ (n \ge 2),$$

and $-\infty$ for $n < -2$. It is not monotone, and the smallest affine bound of slope 1 is $d_F(n) \le n + 2$; thus $F \in \mathcal{D}_1$. Here the leading block happens to be a nonzero scalar, so F is invertible in $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ by the unit criterion of Part II. Delete the term X^2: the leading block of $F - t^{-2}$ becomes $3L$, whose scalar leading coefficient is still $3 \in \mathbb{K}^\times$ — and yet $3L$ is not a unit of $\mathbb{K}[L]$, so the reciprocal of $F - t^{-2}$ leaves $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$. This is Theorem 6.5 in a single example.

Chapter 7

Truncation, formal tails, and summability

7.1 The exclusive truncation operator

Definition 7.1 (Exclusive outer truncation)**.** For $N \in \mathbb{Z}$ and $F = \sum_{n \geq n_0} p_n(L)t^n \in \mathscr{PL}_{\text{poly},\mathbb{K}}$ put

$$\operatorname{Trunc}_{<N} F \coloneqq \sum_{n<N} p_n(L)t^n. \tag{7.1}$$

The sum is finite because $n_0 \leq n < N$; it is empty, hence 0, when $N \leq n_0$, and $\operatorname{Trunc}_{<N} 0 = 0$. Every retained block is retained *whole*: all powers of L at a retained level survive. The phrase "through outer order t^{N-1}" means the exclusive cutoff $< N$.

Remark 7.2 (Reconciliation with the inclusive convention)*.* Source S6 writes $[F]_{\leq N} = \sum_{n \leq N} p_n(L)t^n$, which is *inclusive*. The translation is

$$[F]_{\leq N} = \operatorname{Trunc}_{<N+1} F, \qquad \operatorname{Trunc}_{<N} F = [F]_{\leq N-1}. \tag{7.2}$$

Any statement imported from S6 — in particular its residual and stopping criteria — must absorb this one-block shift. We use (7.1) exclusively.

Proposition 7.3 (Elementary properties)**.** *For $F, G \in \mathscr{PL}_{\text{poly},\mathbb{K}}$, $c \in \mathbb{K}$ and $M, N \in \mathbb{Z}$:*

- (i) $\operatorname{Trunc}_{<N} \cdot$ *is $\mathbb{K}$-linear and idempotent, and* $\operatorname{Trunc}_{<N}(\operatorname{Trunc}_{<M} F) = \operatorname{Trunc}_{<\min(M,N)} F$*;*
- (ii) $\nu_t\big(F - \operatorname{Trunc}_{<N} F\big) \geq N$*, and* $\nu_t(\operatorname{Trunc}_{<N} F) \geq \nu_t(F)$*;*
- (iii) $\operatorname{Trunc}_{<N} F = \operatorname{Trunc}_{<N} G$ *if and only if* $\nu_t(F - G) \geq N$*;*
- (iv) $F = G$ *if and only if* $\operatorname{Trunc}_{<N} F = \operatorname{Trunc}_{<N} G$ *for every* $N \in \mathbb{Z}$*;*
- (v) $\operatorname{Trunc}_{<N}(cF + G) = c \operatorname{Trunc}_{<N} F + \operatorname{Trunc}_{<N} G$*.*

Proof. All five are read off blockwise from (7.1). For (iii), $F - G$ has all blocks below N equal to $p_n - q_n$, which vanish for all $n < N$ exactly when the two truncations agree; and $\nu_t(F - G) \geq N$ says precisely that those blocks vanish. For (iv), if the truncations agree for all N then $\nu_t(F - G) \geq N$ for all N, so $F - G = 0$ by Theorem 6.2 (iii). □

Proposition 7.4 (Truncated product)**.** *Let $F, G \in \mathscr{PL}_{\text{poly},\mathbb{K}}$ be nonzero, $m = \nu_t F$, $r = \nu_t G$, and $N \in \mathbb{Z}$. Then*

$$\operatorname{Trunc}_{<N}(FG) = \operatorname{Trunc}_{<N}\big(\operatorname{Trunc}_{<N-r} F \cdot \operatorname{Trunc}_{<N-m} G\big). \tag{7.3}$$

If in addition $m \geq 0$ and $r \geq 0$ — in particular if $F, G \in \mathscr{PL}_{\text{pow},\mathbb{K}}$ — then also

$$\operatorname{Trunc}_{<N}(FG) = \operatorname{Trunc}_{<N}\big(\operatorname{Trunc}_{<N} F \cdot \operatorname{Trunc}_{<N} G\big). \tag{7.4}$$

Proof. Write $F = \mathrm{Trunc}_{<N-r} F + F_1$ and $G = \mathrm{Trunc}_{<N-m} G + G_1$, so that $\nu_t F_1 \geq N - r$ and $\nu_t G_1 \geq N - m$ by Theorem 7.3 (ii). Then

$$FG - \mathrm{Trunc}_{<N-r} F \, \mathrm{Trunc}_{<N-m} G = F_1 G + \mathrm{Trunc}_{<N-r} F \, G_1.$$

By Theorem 6.2, $\nu_t(F_1 G) \geq (N-r)+r = N$ and $\nu_t\big(\mathrm{Trunc}_{<N-r} FG_1\big) \geq m+(N-m) = N$, using $\nu_t(\mathrm{Trunc}_{<N-r} F) \geq m$. Hence the difference has $\nu_t \geq N$, and (7.3) follows from Theorem 7.3 (iii). When $m, r \geq 0$ the same computation with N in place of $N-r$ and $N-m$ gives $\nu_t(F_1 G) \geq N+r \geq N$ and $\nu_t(\mathrm{Trunc}_{<N} FG_1) \geq m + N \geq N$, which is (7.4). □

Example 7.5 (The symmetric law fails on the Laurent ring)**.** Let $F = t^{-1} + 1$, $G = t^{-1}$ and $N = 0$. Then $FG = t^{-2} + t^{-1}$, so $\mathrm{Trunc}_{<0}(FG) = t^{-2} + t^{-1}$. On the other hand $\mathrm{Trunc}_{<0} F = t^{-1}$ and $\mathrm{Trunc}_{<0} G = t^{-1}$, whose product truncates to t^{-2}. The block t^{-1} has been lost, because the term 1 of F, discarded at cutoff 0, meets the pole t^{-1} of G. Formula (7.3) keeps it: $m = -1$, $r = -1$, so one must retain F below $N - r = 1$ and G below $N - m = 1$, giving $(t^{-1} + 1)t^{-1} = t^{-2} + t^{-1}$.

Remark 7.6 (Precision accounting)*.* Theorem 7.4 is the prototype of every precision contract in this volume: an operation must declare how much input precision it consumes. Multiplication consumes $-\nu_t$ of the *other* factor; equivalently, each negative power of t in one factor costs one block of precision in the other. Composition with an inner argument of negative outer order amplifies the requested precision by a factor, and is analysed in Part III.

7.2 The formal tail token

Definition 7.7 (Formal big-O)**.** For $F, G \in \mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ and $N \in \mathbb{Z}$, the statement

$$F = G + \mathcal{O}_t^{\mathrm{form}}\big(t^N\big) \tag{7.5}$$

means any of the following four equivalent things:

$\nu_t(F - G) \geq N$; $\quad F - G \in t^N \mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$; $\quad \mathrm{Trunc}_{<N} F = \mathrm{Trunc}_{<N} G$; $\quad [t^n](F - G) = 0$ for all $n < N$.

The token $\mathcal{O}^{\mathrm{form}}$ is the same operator with its arguments left implicit.

Proposition 7.8 (Equivalence, and the calculus of the token)**.** *The four conditions in Theorem 7.7 are equivalent. Moreover, for $F = G + \mathcal{O}_t^{\mathrm{form}}\big(t^N\big)$ and any $H, H' \in \mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ and $M \in \mathbb{Z}$ — where, in* (ii) *and* (iii)*, a tail exponent that evaluates to $+\infty$ (which happens only through a vanishing factor) is to be read as the assertion that the difference in question is* 0*:*

(i) $F + H = G + H + \mathcal{O}_t^{\mathrm{form}}\big(t^N\big)$;

(ii) $HF = HG + \mathcal{O}_t^{\mathrm{form}}\big(t^{N+\nu_t H}\big)$*; in particular multiplication by H with $\nu_t H < 0$* loses $|\nu_t H|$ *blocks of precision;*

(iii) *if also* $H = H' + \mathcal{O}_t^{\mathrm{form}}\big(t^M\big)$*, then* $FH = GH' + \mathcal{O}_t^{\mathrm{form}}\big(t^K\big)$ *with* $K = \min\{N + \nu_t H,\ M + \nu_t G\}$*;*

(iv) $\mathcal{O}_t^{\mathrm{form}}\big(t^N\big) + \mathcal{O}_t^{\mathrm{form}}\big(t^M\big) = \mathcal{O}_t^{\mathrm{form}}\big(t^{\min(N,M)}\big)$ *as a statement about the set* $t^N \mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}} + t^M \mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}} = t^{\min(N,M)} \mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$.

Proof. Equivalence: $\nu_t(F - G) \geq N$ says all blocks of $F - G$ below N vanish, which is the fourth condition and, by Theorem 7.3 (iii), the third; and a series with $\nu_t \geq N$ is t^N times a series with $\nu_t \geq 0$, that is, an element of $t^N \mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$, and conversely.

(i) is Theorem 7.3 (v). (ii): $H(F - G)$ has $\nu_t = \nu_t H + \nu_t(F - G) \geq \nu_t H + N$ by Theorem 6.2 (i). (iii): write $FH - GH' = H(F - G) + G(H - H')$ and apply (ii) twice together with Theorem 6.2 (ii). (iv): the inclusion $t^N \mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}} + t^M \mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}} \subseteq t^{\min} \mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ is clear, and the reverse holds because one of the two summands already equals $t^{\min} \mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$. □

Remark 7.9 (What the token does not assert). $F = G + \mathcal{O}_t^{\mathrm{form}}(t^N)$ asserts *coefficient agreement below outer order* N and nothing else. It does not assert a numerical bound, a rate, convergence, uniformity in L, or the existence of any function with the displayed expansion. Concretely, the element

$$T = \sum_{n \geq N} L^{n^2} t^n$$

satisfies $T = \mathcal{O}_t^{\mathrm{form}}(t^N)$, yet at $X = \mathrm{e}^{10}$, so that $L = 10$, its very first term is $\mathrm{e}^{-10N} 10^{N^2}$, which for $N = 5$ exceeds 1.9×10^3 and grows without bound in N. The formal tail is small in the filtration, not in size. An estimate of size requires, in addition, a bound on d_T on the discarded range — see Theorem 6.16 — and then a separate analytic theorem.

Remark 7.10 (Contrast with the blockwise analytic statement). The analytic counterpart, proved for actual functions in Part V, reads as follows. Let S be an unbounded subset of the positive ray (or, for complex work, a sector on which a branch of log has been fixed), let $f : S \to \mathbb{C}$, and let $\sum_{n \geq n_0} p_n(L) t^n \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ be a candidate expansion. One says that f has this blockwise asymptotic expansion on S if for every $N > n_0$ there is an exponent $M_N \in \mathbb{N}_0$ with

$$f(X) - \sum_{n_0 \leq n < N} p_n(\log X) X^{-n} = \mathcal{O}_{X \to +\infty,\ X \in S}\big(X^{-N} (\log X)^{M_N}\big) .$$

The exponent M_N is part of the statement; it is exactly the data that Theorem 7.9 shows the formal token cannot supply.

7.3 Why the Laurent tail is not an ideal

Proposition 7.11 (The tail is a filtration, not an ideal). *Fix $N \in \mathbb{Z}$ and set $\mathfrak{F}_N = t^N \mathscr{PL}_{\mathrm{pow},\mathbb{K}} \subseteq \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$.*

(i) *t is a unit of $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$, and the ideal of $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ generated by t^N is all of $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$: $(t^N) = \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$.*

(ii) *$\mathfrak{F}_N$ is an additive subgroup and a $\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$-submodule of $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$, but for no N is it an ideal of $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$; indeed $t^{-1} \cdot t^N = t^{N-1} \notin \mathfrak{F}_N$.*

(iii) *$\mathfrak{F}_N = \{F \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}} : \nu_t F \geq N\}$, the chain $(\mathfrak{F}_N)_{N \in \mathbb{Z}}$ is decreasing, exhaustive $\big(\bigcup_N \mathfrak{F}_N = \mathscr{PL}_{\mathrm{poly},\mathbb{K}}\big)$ and separated $\big(\bigcap_N \mathfrak{F}_N = 0\big)$.*

(iv) *In $\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$, by contrast, $t^N \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ for $N \geq 0$ is a genuine ideal and*

$$\mathscr{PL}_{\mathrm{pow},\mathbb{K}} / t^N \mathscr{PL}_{\mathrm{pow},\mathbb{K}} \;\cong\; \mathbb{K}[L][t]/(t^N)$$

as $\mathbb{K}[L]$-algebras.

Proof. (i) $t \cdot t^{-1} = 1$ in $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$, so $t \in \mathscr{PL}^{\times}_{\mathrm{poly},\mathbb{K}}$ and hence $t^N \in \mathscr{PL}^{\times}_{\mathrm{poly},\mathbb{K}}$; an ideal containing a unit is the whole ring.

(ii) $\mathfrak{F}_N$ is closed under addition and under multiplication by elements of $\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ because $\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ is a ring and $t^N \mathscr{PL}_{\mathrm{pow},\mathbb{K}} \cdot \mathscr{PL}_{\mathrm{pow},\mathbb{K}} = t^N \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$. It is not an ideal of $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$: $t^N \in \mathfrak{F}_N$, $t^{-1} \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$, and $\nu_t(t^{N-1}) = N - 1 < N$, so $t^{N-1} \notin \mathfrak{F}_N$ by (iii).

(iii) $F \in t^N \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ means $F = t^N G$ with $\nu_t G \geq 0$, i.e. $\nu_t F \geq N$ by Theorem 6.2 (i); conversely $t^{-N} F \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ whenever $\nu_t F \geq N$. The chain is decreasing since $\nu_t \geq N + 1$ implies $\nu_t \geq N$; exhaustive because every element has finite ν_t or is 0; separated because $\nu_t F \geq N$ for all N forces $\nu_t F = +\infty$, i.e. $F = 0$.

(iv) For $N \geq 0$, $t^N \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ is the image of $\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ under multiplication by t^N, hence an ideal of $\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$. The inclusion $\mathbb{K}[L][t] \hookrightarrow \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ induces a $\mathbb{K}[L]$-algebra homomorphism $\varphi : \mathbb{K}[L][t]/(t^N) \to \mathscr{PL}_{\mathrm{pow},\mathbb{K}} / t^N \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$. It is surjective because $F \equiv \mathrm{Trunc}_{<N} F$

$\pmod{t^N \mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}}$ and $\mathrm{Trunc}_{<N} F \in \mathbb{K}[L][t]$ for $F \in \mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$. It is injective because a polynomial of t-degree $< N$ lying in $t^N \mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ has all blocks below N zero, hence is 0. □

The three legitimate replacements

On $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$, never write "modulo t^N" or use quotient-ring language. Use instead, in decreasing order of explicitness:

1. the valuation statement $\nu_t(F - G) \geq N$;
2. the formal tail token $F = G + \mathcal{O}_t^{\mathrm{form}}(t^N)$ of Theorem 7.7;
3. equality of exclusive truncations $\mathrm{Trunc}_{<N} F = \mathrm{Trunc}_{<N} G$.

These three are equivalent by Theorem 7.8. Quotient-algebra language becomes legitimate only after the finite Laurent principal part has been removed — that is, after multiplying by $t^{-\nu_t F}$ and working in $\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$, where Theorem 7.11 (iv) applies.

7.4 Coefficient extraction

Definition 7.12 (Extraction brackets). For $F \in \mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ written as in (5.9),

$$[t^n]F := p_n(L) \in \mathbb{K}[L], \qquad [t^n L^j]F := a_{n,j} \in \mathbb{K}. \tag{7.6}$$

The monomial is written to the *left* of the bracket and the series to the right. An absent coefficient is 0.

Proposition 7.13 (Extraction is linear and filtration-continuous). *Each map $[t^n]$ and $[t^n L^j]$ is $\mathbb{K}$-linear, and $[t^n]F = 0$ for every $F \in \mathfrak{F}_N$ with $N > n$. Consequently, if $F = G + \mathcal{O}_t^{\mathrm{form}}(t^N)$ then $[t^n]F = [t^n]G$ for all $n < N$, and F is determined by the family $\big([t^n L^j]F\big)_{(n,j)}$.*

Proof. Linearity is blockwise addition. If $\nu_t F \geq N > n$ then $p_n = 0$. The remaining statements follow from Theorem 7.7 and Theorem 6.9. □

Remark 7.14 (Bracket collisions). The glyph $[\ \cdot\ ]$ carries four unrelated meanings in this corpus: coefficient extraction $[t^n]F$; the inclusive truncation $[F]_{\leq N}$ of S6, banished here by Theorem 7.2; the unsigned Stirling cycle number $\left[{n \atop k}\right]$; and the Gaussian binomial coefficient. They are told apart by argument type, and we always print the semantic form.

7.5 The separate logarithmic cutoff

A log-degree cap is not a truncation in the dominance order

Bounding $\deg_L p_n$ is an additional, independent approximation. It is not part of the t-adic topology, and — this is the essential point — inside a fixed block the discarded high powers of L are the *dominant* terms, not the negligible ones: by Theorem 5.12, $t^n L^j \succ t^n L^k$ whenever $j > k$. Discarding them therefore removes the largest part of the block.

Example 7.15 (A log cap destroys a retained block). Let $F = G = 1 + L^3 t \in \mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$. Then $FG = 1 + 2L^3 t + L^6 t^2$, so $\mathrm{Trunc}_{<2}(FG) = 1 + 2L^3 t$. Capping the logarithmic degree at 2 *before* multiplying replaces both factors by 1 and returns $\mathrm{Trunc}_{<2}(FG) = 1$: an entire retained t-block has been lost, not merely perturbed. Capping *after* multiplying returns 1 as well. No t-adic precision statement detects the error, because both answers agree modulo $\mathcal{O}_t^{\mathrm{form}}(t^1)$ and disagree at the retained level t^1.

Remark 7.16 (When an inner cutoff is legitimate). In $\mathscr{PL}_{\mathrm{itLaur},\mathbb{K}}$ the situation reverses: a block is a Laurent series in L^{-1} whose discarded terms L^{-j} with j large *are* the small ones, so an inner cutoff is a genuine approximation in the rank-two order of Theorem 6.6. A finite record in $\mathscr{PL}_{\mathrm{itLaur},\mathbb{K}}$ therefore needs a two-dimensional precision region: an outer bound $n < N$ and a level-dependent inner bound $j > J(n)$, with the staircase J chosen stable under the intended convolutions. An outer cutoff alone never makes an infinite Laurent-log block finite. In $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$, by contrast, a log-degree cap is legitimate only when a proved bound on d_F says that the capped powers are absent.

7.6 Completeness and separatedness

Definition 7.17 (The t-adic topology). Give $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ the topology defined by the filtration $(\mathfrak{F}_N)_{N\in\mathbb{Z}}$ of Theorem 7.11, equivalently by the ultrametric

$$d(F,G) = 2^{-\nu_t(F-G)}, \qquad 2^{-\infty} := 0. \tag{7.7}$$

A sequence $(F_r)_{r\geq 0}$ is *Cauchy* if for every N there is R with $\nu_t(F_r - F_s) \geq N$ for all $r, s \geq R$; it *converges* to F if $\nu_t(F_r - F) \to +\infty$. We say (F_r) *stabilizes coefficientwise* if for every n the block $[t^n]F_r$ is eventually independent of r.

That (7.7) is an ultrametric — in particular $d(F,H) \leq \max\{d(F,G), d(G,H)\}$ — is Theorem 6.2 (ii) restated, and $d(F,G) = 0$ iff $F = G$ is Theorem 6.2 (iii).

Theorem 7.18 (Completeness and separatedness). (i) Separatedness. $\bigcap_{N\geq 0} t^N \mathscr{PL}_{\mathrm{pow},\mathbb{K}} = 0$ *in* $\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$*, and* $\bigcap_{N\in\mathbb{Z}} \mathfrak{F}_N = 0$ *in* $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$.

(ii) Cauchy = stabilizing truncations. (F_r) *is Cauchy if and only if for every* N *the truncations* $\mathrm{Trunc}_{<N} F_r$ *are eventually independent of* r.

(iii) Completeness. $\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ *and* $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ *are complete: every Cauchy sequence converges, to a unique limit. The same holds for* $E[[t]]$ *and* $E((t))$ *for any coefficient ring* E*; in particular for* $\mathscr{PL}_{\mathrm{rat},\mathbb{K}}$ *and* $\mathscr{PL}_{\mathrm{itLaur},\mathbb{K}}$.

(iv) Coefficientwise stabilization is weaker on the Laurent ring. *Every Cauchy sequence stabilizes coefficientwise. On* $\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ *the converse holds. On* $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ *it fails; it is restored exactly under the additional hypothesis that* $\inf_r \nu_t(F_r) > -\infty$.

Proof. (i) If $F \in \mathfrak{F}_N$ for every N then $\nu_t F \geq N$ for every N by Theorem 7.11 (iii), so $\nu_t F = +\infty$ and $F = 0$.

(ii) Immediate from Theorem 7.3 (iii): the condition $\nu_t(F_r - F_s) \geq N$ is exactly $\mathrm{Trunc}_{<N} F_r = \mathrm{Trunc}_{<N} F_s$.

(iii) Let (F_r) be Cauchy. Apply (ii) with $N = 0$: there is R_0 with $\mathrm{Trunc}_{<0} F_r = \mathrm{Trunc}_{<0} F_{R_0}$ for $r \geq R_0$. Set

$$n_0 = \min\{0,\ \nu_t(\mathrm{Trunc}_{<0} F_{R_0})\} \in \mathbb{Z},$$

a genuine integer (equal to 0 if the truncation vanishes). For $r \geq R_0$ all blocks of F_r below n_0 vanish, since they agree with those of $\mathrm{Trunc}_{<0} F_{R_0}$ below 0 and there are none below n_0; hence $\nu_t(F_r) \geq n_0$. This is the step at which the Cauchy hypothesis, and not merely coefficientwise stabilization, is used: it supplies a *common* lower support bound.

Next, for each $n \geq n_0$, choosing $N = n+1$ in (ii) gives R_n with $[t^n]F_r = [t^n]F_{R_n} =: p_n$ for $r \geq R_n$. Since $\nu_t F_r \geq n_0$ for $r \geq R_0$, we have $p_n = 0$ for $n < n_0$, so $F := \sum_{n\geq n_0} p_n t^n$ is a well-defined element of $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ (and of $\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ if $n_0 \geq 0$, which is the case when all $F_r \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$). Given $N \geq n_0$, take $R = \max\{R_0, R_{n_0}, \ldots, R_{N-1}\}$; then $\mathrm{Trunc}_{<N} F_r = \mathrm{Trunc}_{<N} F$ for $r \geq R$, i.e. $\nu_t(F_r - F) \geq N$. Hence $F_r \to F$. Uniqueness of the limit is (i). The argument used nothing about $\mathbb{K}[L]$ beyond its being a ring, so it applies verbatim to $E[[t]]$ and $E((t))$.

(iv) If $F_r \to F$ then $[t^n]F_r = [t^n]F$ for r large, by Theorem 7.13; so Cauchy sequences stabilize coefficientwise. On $\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ suppose conversely that (F_r) stabilizes coefficientwise. Given $N \ge 0$, the finitely many blocks $n = 0, 1, \dots, N-1$ all stabilize, say beyond R; then $\mathrm{Trunc}_{<N} F_r$ is constant for $r \ge R$, and (ii) gives Cauchy. The same argument works on $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ as soon as $\nu_t(F_r) \ge n_1$ for all r, because then only the finitely many blocks $n_1 \le n < N$ matter. Failure without that hypothesis is Theorem 7.19. That the hypothesis is not merely sufficient but necessary — which is what "exactly" asserts — follows from (iii): a Cauchy sequence satisfies $\nu_t(F_r) \ge n_0$ for all $r \ge R_0$, and the finitely many terms with $r < R_0$ each have $\nu_t(F_r) \in \mathbb{Z} \cup \{+\infty\}$, so $\inf_r \nu_t(F_r) > -\infty$. □

Example 7.19 (Coefficientwise stabilization without convergence)**.** Let

$$F_r = \sum_{n=-r}^{-1} t^n \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}} \qquad (r \ge 0),$$

so $F_0 = 0$. For each fixed n the block $[t^n]F_r$ equals 1 for all $r \ge -n$ when $n < 0$, and 0 for all r when $n \ge 0$: the sequence stabilizes coefficientwise. But $\nu_t(F_{r+1} - F_r) = -(r+1) \to -\infty$, so (F_r) is not Cauchy, and there is no $F \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ with $[t^n]F = 1$ for every $n < 0$, since such an F would have infinitely many negative blocks. The obstruction is exactly the missing common lower bound: $\inf_r \nu_t(F_r) = -\infty$.

Remark 7.20 (Separatedness is what makes residuals certificates). Separatedness says that an element with $\nu_t = +\infty$ is 0. Every verification scheme in this volume rests on it: to prove $F = G$ it suffices to prove $\mathrm{Trunc}_{<N}(F - G) = 0$ for every N, which is a sequence of finite computations. Completeness says the converse construction is available: a consistent prescription of blocks defines an element. Together they are the formal substitute for "the iteration converges".

7.7 Formal summability

Definition 7.21 (Formally summable family)**.** Let I be an index set. A family $(S_i)_{i\in I}$ in $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ is *formally summable* if

$$\text{for every } N \in \mathbb{Z}, \quad \{i \in I : \nu_t(S_i) < N\} \text{ is finite.} \tag{7.8}$$

For a sequence indexed by $\mathbb{N}_0$ this is the condition $\nu_t(S_r) \to +\infty$. Its sum is defined blockwise by

$$[t^n] \sum_{i\in I} S_i := \sum_{i\in I} [t^n] S_i, \tag{7.9}$$

in which all but finitely many terms vanish.

Theorem 7.22 (The summability calculus)**.** *Let $(S_i)_{i\in I}$ be formally summable in $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$. Then:*

(i) (7.9) *defines an element $S = \sum_{i\in I} S_i \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$, with $\nu_t(S) \ge \inf_i \nu_t(S_i) > -\infty$;*

(ii) *for every N, $\mathrm{Trunc}_{<N} S = \sum_{i\in I_N} \mathrm{Trunc}_{<N} S_i$ where $I_N = \{i : \nu_t(S_i) < N\}$ is finite; equivalently S is the limit of the net of finite partial sums, and for $I = \mathbb{N}_0$, $\sum_{r\le R} S_r \to S$ in the t-adic topology;*

(iii) *the sum does not depend on the enumeration of I: for any bijection $\pi : I \to I$, $\sum_i S_{\pi(i)} = S$;*

(iv) *summable families form a $\mathbb{K}$-module and $\sum$ is $\mathbb{K}$-linear on it;*

(v) *for $H \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$, the family (HS_i) is summable and $\sum_i HS_i = H \sum_i S_i$;*

(vi) *if $(T_k)_{k\in J}$ is also summable, then $(S_iT_k)_{(i,k)\in I\times J}$ is summable and $\sum_{(i,k)} S_iT_k = \bigl(\sum_i S_i\bigr)\bigl(\sum_k T_k\bigr)$.*

Proof. (i) Fix n. Terms with $[t^n]S_i \neq 0$ satisfy $\nu_t S_i \leq n$, and by (7.8) with $N = n+1$ there are finitely many such i; so (7.9) is a finite sum in $\mathbb{K}[L]$ and defines a block P_n. Applying (7.8) with $N = 0$ shows that $\{i : \nu_t S_i < 0\}$ is finite, hence

$$\mu := \inf_i \nu_t(S_i) \;\geq\; \min\left\{0,\; \min_{i:\nu_t S_i<0} \nu_t S_i\right\} \;>\; -\infty,$$

with the convention that the inner minimum is 0 when its index set is empty. Thus $\mu \in \mathbb{Z}$ unless every $S_i = 0$, in which case $\mu = +\infty$. For $n < \mu$ every $[t^n]S_i$ vanishes, so $P_n = 0$; thus $S = \sum_{n\geq\mu} P_n t^n \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ and $\nu_t S \geq \mu$.

(ii) For $n < N$ and $i \notin I_N$ we have $\nu_t S_i \geq N > n$, so $[t^n]S_i = 0$; hence $[t^n]S = \sum_{i\in I_N}[t^n]S_i$ for all $n < N$, which is the displayed identity. Consequently, for the partial sums $\Sigma_R = \sum_{r\leq R} S_r$ with $I = \mathbb{N}_0$, choosing R with $I_N \subseteq \{0,\ldots,R\}$ gives $\mathrm{Trunc}_{<N}\,\Sigma_{R'} = \mathrm{Trunc}_{<N}\,S$ for all $R' \geq R$, i.e. $\nu_t(\Sigma_{R'} - S) \geq N$.

(iii) The defining formula (7.9) is a sum over a finite subset of I determined by n alone, and a finite sum in $\mathbb{K}[L]$ is independent of order; and π preserves the summability condition.

(iv) If (S_i) and (S_i') are summable then $\nu_t(cS_i+S_i') \geq \min\{\nu_t S_i, \nu_t S_i'\}$, so $\{i : \nu_t(cS_i + S_i') < N\} \subseteq \{i : \nu_t S_i < N\} \cup \{i : \nu_t S_i' < N\}$ is finite. Linearity of $\sum$ is linearity of (7.9) blockwise.

(v) $\nu_t(HS_i) = \nu_t H + \nu_t S_i$, so $\{i : \nu_t(HS_i) < N\} = \{i : \nu_t S_i < N - \nu_t H\}$ is finite (if $H = 0$ the claim is trivial). For the identity, fix n and compute

$$[t^n]\sum_i HS_i = \sum_i \sum_{a+b=n} [t^a]H\,[t^b]S_i = \sum_{a+b=n} [t^a]H \sum_i [t^b]S_i = [t^n]\Big(H\sum_i S_i\Big),$$

where the interchange is legitimate because every sum involved is finite: the outer index a ranges over $\nu_t H \leq a \leq n - \mu$, and for each b only finitely many i contribute.

(vi) $\nu_t(S_iT_k) = \nu_t S_i + \nu_t T_k$. Let $\mu_S = \inf_i \nu_t S_i$ and $\mu_T = \inf_k \nu_t T_k$, both integers by (i). If $\nu_t S_i + \nu_t T_k < N$, then $\nu_t S_i < N - \mu_T$ and $\nu_t T_k < N - \mu_S$; so

$$\{(i,k) : \nu_t(S_iT_k) < N\} \subseteq \{i : \nu_t S_i < N - \mu_T\} \times \{k : \nu_t T_k < N - \mu_S\},$$

a product of two finite sets. For the value, apply (v) with $H = S_i$ to get $\sum_k S_iT_k = S_i\sum_k T_k$ for each i; then Theorem 7.26, whose proof uses only (i) and (ii) above, and (v) again with $H = \sum_k T_k$, give $\sum_{(i,k)} S_iT_k = \sum_i\big(S_i\sum_k T_k\big) = \big(\sum_i S_i\big)\big(\sum_k T_k\big)$. □

Corollary 7.23 (Geometric summability)**.** *Let E be a commutative ring with* 1 *and let $U \in tE[[t]]$, i.e. $\nu_t U \geq 1$. Then $(U^r)_{r\geq0}$ is formally summable in $E((t))$, $\sum_{r\geq0} U^r \in E[[t]]$, and*

$$(1-U)\sum_{r\geq0} U^r = 1.$$

This applies in particular to $E = \mathbb{K}$, $\mathbb{K}[L]$, $\mathbb{K}(L)$ and $\mathbb{K}((L^{-1}))$, and — reading L^{-1} for t — inside $\mathbb{K}[[L^{-1}]] \subset \mathbb{K}((L^{-1}))$.

Proof. By Theorem 7.27 the summability calculus of Theorem 7.22 is available in $E((t))$ for every commutative ring E. The Cauchy product gives the subadditivity $\nu_t(FG) \geq \nu_t F + \nu_t G$ there — this is the first half of the computation in the proof of Theorem 6.2 (i), which needs no domain hypothesis — so $\nu_t(U^r) \geq r\,\nu_t U \geq r \to +\infty$ and (7.8) holds. Write $V = \sum_{r\geq0} U^r$. By Theorem 7.22 (iv),(v), $V - UV = \sum_{r\geq0} U^r - \sum_{r\geq0} U^{r+1} = \sum_{r\geq0}\big(U^r - U^{r+1}\big)$. Fix $N \geq 1$; for $N \leq 0$ both $\mathrm{Trunc}_{<N}(V-UV)$ and $\mathrm{Trunc}_{<N}\,1$ vanish, since $V - UV$ and 1 lie in $E[[t]]$. Only the indices $r < N$ contribute to $\mathrm{Trunc}_{<N}\,\cdot$, because $\nu_t(U^r - U^{r+1}) \geq r$ (Theorem 7.22 (ii)), and the resulting finite sum telescopes:

$$\mathrm{Trunc}_{<N}(V-UV) = \mathrm{Trunc}_{<N}\Big(\sum_{0\leq r<N}\big(U^r - U^{r+1}\big)\Big) = \mathrm{Trunc}_{<N}(1-U^N) = \mathrm{Trunc}_{<N}\,1,$$

the last step because $\nu_t(U^N) \geq N$. Since this holds for every N, Theorem 7.3 (iv) gives $V - UV = 1$. □

Theorem 7.24 (Three formulations of summability). *Let $(S_i)_{i\in I}$ be a family in $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$. Consider:*

(a) Valuation form. *For every N, $\{i : \nu_t S_i < N\}$ is finite (Theorem 7.21).*

(b) Hahn form. $\bigcup_i \operatorname{supp}_{\mathrm{coeff}}(S_i)$ *satisfies Theorem 5.15 (b), and every monomial $t^n L^j$ lies in* $\operatorname{supp}_{\mathrm{coeff}}(S_i)$ *for only finitely many i.*

(c) Block-local finiteness with a common bound. *For every n the set $\{i : [t^n]S_i \neq 0\}$ is finite, and* $\inf_i \nu_t(S_i) > -\infty$.

Then (a), (b) and (c) are equivalent. If the family lies in $\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$, the lower-bound clause in (c) is automatic and may be dropped. On $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ it may not: block-local finiteness alone does not imply summability.

Proof. (a)⇒(c). $\{i : [t^n]S_i \neq 0\} \subseteq \{i : \nu_t S_i \leq n\}$, finite by (a) with $N = n + 1$; and the lower bound was produced in the proof of Theorem 7.22 (i).

(c)⇒(a). Let $\mu = \inf_i \nu_t S_i \in \mathbb{Z}$. For $N \leq \mu$ the set in (a) is empty. For $N > \mu$,

$$\{i : \nu_t S_i < N\} \subseteq \bigcup_{n=\mu}^{N-1} \{i : [t^n]S_i \neq 0\},$$

a finite union of finite sets.

(a)⇒(b). The union of supports has n-projection bounded below by μ. Its fiber over n is the union of the fibers of the finitely many S_i with $\nu_t S_i \leq n$, each finite, hence finite. The monomial condition follows from $\{i : (n,j) \in \operatorname{supp}_{\mathrm{coeff}}(S_i)\} \subseteq \{i : \nu_t S_i \leq n\}$.

(b)⇒(a). Let n_0 bound the n-projection of the union from below and let U_n be its (finite) fiber over n. Given N,

$$\{i : \nu_t S_i < N\} \subseteq \bigcup_{n=n_0}^{N-1} \bigcup_{j\in U_n} \{i : (n,j) \in \operatorname{supp}_{\mathrm{coeff}}(S_i)\},$$

a finite union — finitely many n, finitely many j for each — of finite sets.

The $\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ case. There $\nu_t S_i \geq 0$ for all i, so $\inf_i \nu_t S_i \geq 0$. □

Example 7.25 (Two families that just fail). *Local finiteness without a common lower bound.* Let $S_r = t^{-r}$, $r \geq 0$, in $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$. Every block index n is hit by at most one member, so block-local finiteness holds; but $\nu_t S_r = -r \to -\infty$, the family is not summable, and the would-be sum $\sum_{n\leq 0} t^n$ has an infinite principal part and lies in no Laurent class. This is exactly the extra hypothesis that Theorem 7.24 (c) adds on $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$.

Monomial finiteness without block finiteness. Let $S_r = tL^r$, $r \geq 0$, in $\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$. Each monomial tL^r occurs in exactly one member, yet $\nu_t S_r = 1$ for every r, the family is not summable, and the would-be block $\sum_{r\geq 0} L^r$ is not a polynomial. Note that the union of supports, $\{(1,r) : r \geq 0\}$, has an infinite fiber and so violates the Hahn clause of Theorem 7.24 (b). Requiring only that each *monomial* occur finitely often is therefore not enough, even on $\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$; the well-based clause is doing real work.

Proposition 7.26 (Fubini for summable families). *Let $(S_{i,k})_{(i,k)\in I\times J}$ be formally summable. Then for each i the family $(S_{i,k})_{k\in J}$ is summable with sum T_i; the family $(T_i)_{i\in I}$ is summable; and*

$$\sum_{i\in I} T_i = \sum_{(i,k)\in I\times J} S_{i,k}.$$

The same holds with the roles of I and J exchanged, so iterated summation may be performed in either order.

Proof. Summability of each slice is inherited, since $\{k : \nu_t S_{i,k} < N\}$ injects into $\{(i',k) : \nu_t S_{i',k} < N\}$. If $\nu_t S_{i,k} \geq N$ for all $k \in J$, then every block of T_i below N vanishes by (7.9), so $\nu_t T_i \geq N$; hence $\{i : \nu_t T_i < N\} \subseteq \{i : \exists k,\ \nu_t S_{i,k} < N\}$, which is finite. Finally, for each n,

$$[t^n]\sum_i T_i = \sum_i [t^n]T_i = \sum_i \sum_k [t^n]S_{i,k} = \sum_{(i,k)} [t^n]S_{i,k} = [t^n]\sum_{(i,k)} S_{i,k},$$

all sums being finite, so ordinary finite-sum associativity applies. □

Remark 7.27 (Summability on the other three classes). Theorem 7.21 and Theorem 7.22 were stated for $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ but use only the valuation laws and the fact that each block lies in a fixed additive group; they hold verbatim in $E[[t]]$ and $E((t))$ for every commutative ring E — in particular in $\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$, $\mathscr{PL}_{\mathrm{rat},\mathbb{K}}$ and $\mathscr{PL}_{\mathrm{itLaur},\mathbb{K}}$. The domain hypothesis on E was used only for the equalities $\nu_t(HS_i) = \nu_t H + \nu_t S_i$ and $\nu_t(S_i T_k) = \nu_t S_i + \nu_t T_k$ in parts (v) and (vi) of that theorem, and there the inequality $\geq$ — valid over any commutative ring — is all that the finiteness arguments need, the displayed set identities becoming inclusions. In $\mathscr{PL}_{\mathrm{itLaur},\mathbb{K}}$, however, (7.9) requires an additional check: the finitely many contributing blocks are Laurent series in L^{-1}, and their sum is again one, so nothing is lost. What does *not* transfer is the *monomialwise* version: in $\mathscr{PL}_{\mathrm{itLaur},\mathbb{K}}$ a single block already involves infinitely many monomials, so summability must be tested on blocks, never on monomials.

7.8 The three O-roles

Never infer one O-role from another

Three unrelated meanings share the letter O in this subject, and the corpus keeps them typographically distinct.

Printed form	Means	Does *not* mean
$\mathcal{O}_t^{\mathrm{form}}(t^N)$	formal valuation tail: $\nu_t(F - G) \geq N$ (Theorem 7.7)	any numerical bound, rate, uniformity in L, convergence, or the existence of a function
$\mathcal{O}_{X\to+\infty,\ X\in S}(A(X))$	analytic estimate: $\lvert R(X)\rvert \leq C\,\lvert A(X)\rvert$ on a stated late part of a stated domain, with branches fixed	anything formal; and it is void unless the regime and domain are printed
$\mathcal{O}(N^2)$	algorithmic cost in a declared model (coefficient-ring operations, bit operations, ...)	a statement about the size or accuracy of the computed object

Remark 7.28 (The bridge, and its price). Passing from the first column to the second is a theorem, never a notation change. It needs, at minimum: a function f on a stated domain; a proof that f has the blockwise expansion in the sense of Theorem 7.10; and, for each cutoff N, an exponent M_N bounding the logarithmic degree of the discarded material. A bare $\mathcal{O}$ with no printed regime is never acceptable for an asymptotic claim.

Section summary

The formal ambient is $\mathscr{PL}_{\mathrm{poly},\mathbb{K}} = \mathbb{K}[L]((t))$: supports are lattice sets with finite fibers and lower-bounded projection, hence reverse well ordered of order type at most ω. The outer valuation ν_t is the computational filtration; the rank-two valuation $v(t^n L^j) = (n, -j)$ is the monomial comparison, and $\mathscr{PL}_{\mathrm{itLaur},\mathbb{K}}$ is exactly the Hahn field of its value group. $\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ is complete and separated and $t^N \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ is a genuine ideal there; in $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ the same set is only a filtration tail, because t is a unit. All infinite formulas in the volume are legitimated by one criterion: a family is summable when $\nu_t(S_i) \to +\infty$, which on the Laurent ring is strictly stronger than block-local finiteness. Leading block, leading monomial and leading scalar are three different objects; log-degree bounds are always hypotheses or proved invariants; and a formal tail carries no size information.

Part III

Arithmetic and differential calculus

Chapter 8

Multiplication: convolution of logarithmic blocks

Multiplication is the operation on which every later construction rests: reciprocals, powers, composition, and reversion are all built from it by triangular recurrences. Its whole content is that the Cauchy rule in the outer variable t is *coefficientwise finite*, so that each output block is an exact finite expression in finitely many input blocks. Nothing here is an analytic rearrangement, and nothing here requires convergence.

Throughout this chapter $\mathbb{K}$ is a field of characteristic zero, X is the large variable, the regime is $X \to +\infty$ on the positive ray unless a different domain is announced, and

$$t := X^{-1}, \qquad L := \log X. \tag{8.1}$$

In the formal algebra t and L are independent generators; their analytic coupling is recorded only by the distinguished derivation, which plays no role in this chapter. The four ambient classes are

class	coefficient ring	elements
$\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$	$\mathbb{K}[L]$	$\sum_{n\geq 0} p_n(L)t^n$
$\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$	$\mathbb{K}[L]$	$\sum_{n\geq n_0} p_n(L)t^n$, $n_0 \in \mathbb{Z}$
$\mathscr{P}\mathscr{L}_{\mathrm{rat},\mathbb{K}}$	$\mathbb{K}(L)$	$\sum_{n\geq n_0} p_n(L)t^n$, p_n rational in L
$\mathscr{P}\mathscr{L}_{\mathrm{itLaur},\mathbb{K}}$	$\mathbb{K}((L^{-1}))$	$\sum_{n\geq n_0} t^n \sum_{j\geq j_n} a_{n,j}L^{-j}$

All truncations below are *exclusive*: $\mathrm{Trunc}_{<N} F$ keeps every logarithmic monomial in every block t^n with $n < N$, and discards the rest.

Remark 8.1 (Reconciliation of the source conventions). Two of the merged reports use an inclusive outer cutoff $[F]_{\leq N}$. The translation is $[F]_{\leq N} = \mathrm{Trunc}_{<N+1} F$, and every dependency range, algorithm bound and error exponent quoted from those reports has been shifted by one block accordingly. Likewise, one report writes the coefficient bracket as $[F]_M$; here the series always stands to the right of the bracket, $[t^n]F = p_n(L)$ and $[t^nL^j]F = a_{n,j}$.

8.1 The block Cauchy product

It is economical to prove the algebra of all four classes at once. What the three coefficient rings $\mathbb{K}[L]$, $\mathbb{K}(L)$, $\mathbb{K}((L^{-1}))$ have in common is a degree function in L together with a coefficient-extraction map obeying the usual convolution rule.

Definition 8.2 (Logarithmically graded domain)**.** A *logarithmically graded domain over* $\mathbb{K}$ is a commutative $\mathbb{K}$-algebra R together with $\mathbb{K}$-linear maps $\kappa_d \colon R \to \mathbb{K}$, one for each $d \in \mathbb{Z}$, such that for every nonzero $f \in R$ the set $\{d : \kappa_d(f) \neq 0\}$ is nonempty and bounded above, and:

(G1) $f = 0$ if and only if $\kappa_d(f) = 0$ for every d;

(G2) $\kappa_d(fg) = \sum_{i+j=d} \kappa_i(f)\kappa_j(g)$ for all $f, g \in R$ and all $d \in \mathbb{Z}$, the sum having only finitely many nonzero terms;

(G3) $\kappa_0(1) = 1$ and $\kappa_d(1) = 0$ for $d \neq 0$.

For $f \neq 0$ put $\deg_{\mathrm{L}} f := \max\{d : \kappa_d(f) \neq 0\}$ and $\mathrm{lc}_L(f) := \kappa_{\deg_{\mathrm{L}} f}(f) \in \mathbb{K}^\times$; set $\deg_{\mathrm{L}} 0 := -\infty$ and $\mathrm{lc}_L(0) := 0$.

Here lc_L is the leading coefficient *inside one block*; it is not the same object as the leading scalar lc of a series, defined in Theorem 8.18 below, although the two agree on the leading block.

Lemma 8.3 (Elementary consequences). *Let R be a logarithmically graded domain over $\mathbb{K}$. For all $f, g \in R$:*

$$\deg_{\mathrm{L}}(fg) = \deg_{\mathrm{L}} f + \deg_{\mathrm{L}} g, \qquad \mathrm{lc}_L(fg) = \mathrm{lc}_L(f)\,\mathrm{lc}_L(g), \qquad \deg_{\mathrm{L}}(f+g) \leq \max(\deg_{\mathrm{L}} f, \deg_{\mathrm{L}} g). \tag{8.2}$$

Here $-\infty + c = -\infty$ and $\max(-\infty, c) = c$. In particular R is an integral domain. Moreover, if $\deg_{\mathrm{L}} f \leq d$ and $\deg_{\mathrm{L}} g \leq e$, then $\kappa_{d+e}(fg) = \kappa_d(f)\kappa_e(g)$.

Proof. If $f = 0$ or $g = 0$ every assertion is trivial under the stated conventions. The third assertion in (8.2) is immediate from $\mathbb{K}$-linearity of each κ_d. For the first two, assume $f, g \neq 0$ and set $d = \deg_{\mathrm{L}} f$, $e = \deg_{\mathrm{L}} g$. If $c > d + e$ and $i + j = c$, then $i > d$ or $j > e$, so every term of (G2) vanishes and $\kappa_c(fg) = 0$. If $c = d + e$, the only possibly nonvanishing term is $i = d$, $j = e$, whence

$$\kappa_{d+e}(fg) = \kappa_d(f)\kappa_e(g) = \mathrm{lc}_L(f)\,\mathrm{lc}_L(g) \neq 0,$$

because $\mathbb{K}$ is a field and both factors are nonzero. This proves $\deg_{\mathrm{L}}(fg) = d + e$, $\mathrm{lc}_L(fg) = \mathrm{lc}_L(f)\,\mathrm{lc}_L(g)$, and, by (G1), $fg \neq 0$; so R is a domain. The last sentence is the same computation with d, e merely upper bounds, all other terms of (G2) again vanishing. ☐

Lemma 8.4 (The three coefficient rings). *Each of*

$$\mathbb{K}[L], \qquad \mathbb{K}(L), \qquad \mathbb{K}((L^{-1}))$$

is a logarithmically graded domain over $\mathbb{K}$, with $\deg_{\mathrm{L}}$ equal respectively to the polynomial degree, to $\deg_{\mathrm{L}}(p/q) = \deg p - \deg q$, and to the largest exponent occurring in the Laurent expansion. There are injective $\mathbb{K}$-algebra homomorphisms

$$\mathbb{K}[L] \hookrightarrow \mathbb{K}(L) \hookrightarrow \mathbb{K}((L^{-1})) \tag{8.3}$$

that preserve $\deg_{\mathrm{L}}$ and lc_L.

Proof. For $\mathbb{K}[L]$ take $\kappa_d(f)$ to be the coefficient of L^d in f for $d \geq 0$ and $\kappa_d = 0$ for $d < 0$; (G1)–(G3) are the definition of polynomial multiplication. For $\mathbb{K}((L^{-1}))$, an element is $f = \sum_{j \leq d} a_j L^j$ with only finitely many $j > 0$ allowed but arbitrarily negative j permitted; put $\kappa_j(f) = a_j$. Then (G1) and (G3) are clear, and in (G2) the constraint $i + j = c$ with $i \leq \deg_{\mathrm{L}} f$, $j \leq \deg_{\mathrm{L}} g$ forces $i \geq c - \deg_{\mathrm{L}} g$, so the sum is finite; it is the definition of the Laurent product. The support of f is bounded above by construction.

The inclusion $\mathbb{K}[L] \subset \mathbb{K}((L^{-1}))$ is a $\mathbb{K}$-algebra map, and every nonzero element of $\mathbb{K}((L^{-1}))$ is invertible: writing $f \neq 0$ as $f = a_d L^d(1 + u)$ with $d = \deg_{\mathrm{L}} f$ and $u \in L^{-1}\mathbb{K}[[L^{-1}]]$, the family $\big((-u)^r\big)_{r \geq 0}$ is L^{-1}-adically summable and its sum inverts $1 + u$, so $f^{-1} = a_d^{-1} L^{-d} \sum_{r \geq 0} (-u)^r$. Hence every nonzero polynomial becomes invertible, and the universal property of the field of fractions supplies a unique $\mathbb{K}$-algebra homomorphism $\iota\colon \mathbb{K}(L) \to \mathbb{K}((L^{-1}))$ extending the inclusion; it is injective because $\mathbb{K}(L)$ is a field and $\iota(1) = 1$. Pulling the maps κ_d back along ι makes $\mathbb{K}(L)$ a logarithmically graded domain, and Theorem 8.3 applied to $\iota(p) = \iota(q)\iota(p/q)$ gives $\deg_{\mathrm{L}}(p/q) = \deg p - \deg q$ and $\mathrm{lc}_L(p/q) = \mathrm{lc}_L(p)\,\mathrm{lc}_L(q)^{-1}$. Both maps in (8.3) preserve κ_d by construction, hence preserve $\deg_{\mathrm{L}}$ and lc_L. □

Definition 8.5 (The block Cauchy product)**.** Let R be a logarithmically graded domain over $\mathbb{K}$ and let

$$F = \sum_{n \geq a} p_n\, t^n, \qquad G = \sum_{m \geq b} q_m\, t^m, \qquad p_n, q_m \in R, \tag{8.4}$$

be elements of $R((t))$, the ring of formal Laurent series in t over R. Their product is

$$FG := \sum_{N \geq a+b} C_N\, t^N, \qquad C_N := \sum_{\substack{n+m=N \\ n \geq a,\ m \geq b}} p_n q_m = \sum_{n=a}^{N-b} p_n\, q_{N-n}. \tag{8.5}$$

Lemma 8.6 (Finiteness of every fiber)**.** *For fixed N the index set in* (8.5) *is $\{n \in \mathbb{Z} : a \leq n \leq N - b\}$, of cardinality $\max(N - a - b + 1,\, 0)$. In particular it is finite, and it is empty exactly when $N < a + b$.*

Proof. The constraints $n \geq a$ and $m = N - n \geq b$ are equivalent to $a \leq n \leq N - b$. An interval of integers $[a, N - b]$ has $N - b - a + 1$ elements when $N - b \geq a$ and is empty otherwise. □

This is the entire mechanism. There is no limit, no rearrangement, and no hypothesis of numerical convergence: each C_N is a finite sum of products in R.

Theorem 8.7 (Closure of the four ambient classes)**.** *Let R be a logarithmically graded domain over $\mathbb{K}$. Then* (8.5) *makes $R((t))$ a commutative $\mathbb{K}$-algebra with unit 1, containing $R[[t]]$ as a subalgebra. Consequently each of*

$$\mathscr{PL}_{\mathrm{pow},\mathbb{K}} = \mathbb{K}[L][[t]], \qquad \mathscr{PL}_{\mathrm{poly},\mathbb{K}} = \mathbb{K}[L]((t)),$$
$$\mathscr{PL}_{\mathrm{rat},\mathbb{K}} = \mathbb{K}(L)((t)), \qquad \mathscr{PL}_{\mathrm{itLaur},\mathbb{K}} = \mathbb{K}((L^{-1}))((t))$$

is closed under multiplication, and the chain of inclusions

$$\mathscr{PL}_{\mathrm{pow},\mathbb{K}} \subset \mathscr{PL}_{\mathrm{poly},\mathbb{K}} \subset \mathscr{PL}_{\mathrm{rat},\mathbb{K}} \subset \mathscr{PL}_{\mathrm{itLaur},\mathbb{K}} \tag{8.6}$$

consists of injective $\mathbb{K}$-algebra homomorphisms.

Proof. By Theorem 8.6 each C_N is a well-defined element of R, and $C_N = 0$ for $N < a + b$; so FG again has support bounded below and lies in $R((t))$. Commutativity is the symmetry of the index set under $(n, m) \mapsto (m, n)$; bilinearity over $\mathbb{K}$ is clear; and $1 \cdot F = F$ because the only admissible pair for $1 = 1 \cdot t^0$ at level N is $(0, N)$. For associativity let $H = \sum_{k \geq c} h_k t^k$. Then

$$[t^N]\big((FG)H\big) = \sum_{r+k=N} \Big(\sum_{n+m=r} p_n q_m \Big) h_k = \sum_{n+m+k=N} p_n q_m h_k,$$

the last sum ranging over the finite set of triples with $n \geq a$, $m \geq b$, $k \geq c$: indeed $n + m + k = N$ with the three lower bounds confines each index to a finite interval, so the regrouping is a finite rearrangement. The same computation with the other bracketing gives the same finite sum, whence $(FG)H = F(GH)$. Distributivity is likewise a finite regrouping.

If $\nu_t F \geq 0$ and $\nu_t G \geq 0$, then $C_N = 0$ for $N < 0$, so $R[[t]]$ is closed under the product; it is a subalgebra. Taking $R = \mathbb{K}[L]$ gives the first two classes, $R = \mathbb{K}(L)$ the third, $R = \mathbb{K}((L^{-1}))$ the fourth, using Theorem 8.4. Finally, a $\mathbb{K}$-algebra homomorphism $\varphi\colon R \to R'$ induces a $\mathbb{K}$-algebra homomorphism $R((t)) \to R'((t))$ by $\sum p_n t^n \mapsto \sum \varphi(p_n) t^n$, because φ commutes with the finite sums (8.5); injectivity is inherited blockwise. Applying this to (8.3) yields (8.6). □

Proposition 8.8 (The monomial-level convolution and the support bound). *Let $F = \sum_{n,j} a_{n,j} t^n L^j$ and $G = \sum_{m,k} b_{m,k} t^m L^k$ lie in $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ (so $j, k \in \mathbb{N}_0$, and $\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ is the case $\nu_t \geq 0$) or in $\mathscr{PL}_{\mathrm{itLaur},\mathbb{K}}$ (so $j, k \in \mathbb{Z}$, bounded above at each fixed outer level). Then*

$$[t^N L^d](FG) = \sum_{n+m=N} \sum_{i+j=d} a_{n,i} b_{m,j}, \tag{8.7}$$

a finite sum, and

$$\mathrm{supp}_{\mathrm{coeff}}(FG) \subseteq \mathrm{supp}_{\mathrm{coeff}}(F) + \mathrm{supp}_{\mathrm{coeff}}(G)\,, \tag{8.8}$$

the Minkowski sum in $\mathbb{Z}^2$.

Proof. Formula (8.7) is (G2) applied to (8.5). Finiteness holds because the outer sum is finite by Theorem 8.6 and, for each of its terms, the inner sum is finite by (G2). For (8.8): if $(N, d) \notin \mathrm{supp}_{\mathrm{coeff}}(F) + \mathrm{supp}_{\mathrm{coeff}}(G)$, then every summand of (8.7) has $a_{n,i} = 0$ or $b_{m,j} = 0$, so $[t^N L^d](FG) = 0$. □

Remark 8.9 ($\mathscr{PL}_{\mathrm{rat},\mathbb{K}}$ has no monomial support of its own). Theorem 8.8 is deliberately not stated for $\mathscr{PL}_{\mathrm{rat},\mathbb{K}}$. An element of $\mathbb{K}(L)((t))$ has coefficient blocks that are rational functions of L, and a rational function is not a (finite or infinite) combination of powers of L until one chooses an expansion point. The coefficient support $\mathrm{supp}_{\mathrm{coeff}}(\cdot)$ becomes defined only after applying the embedding $\mathscr{PL}_{\mathrm{rat},\mathbb{K}} \hookrightarrow \mathscr{PL}_{\mathrm{itLaur},\mathbb{K}}$ of Theorem 8.7, which expands each block at $L = \infty$ and turns the exact coefficient $1/(L+1)$ into the infinite series $L^{-1} - L^{-2} + L^{-3} - \cdots$. Exact rational dependence on L and a monomial support in L are different pieces of data; only the second is what (8.8) constrains.

Remark 8.10 (Why multiplication is a two-level, not a flat, operation). Although (8.7) is a genuine two-dimensional convolution, it should not be implemented as one. The outer index is the asymptotic hierarchy and the inner index is not: at fixed N the inner sum is bounded, whereas a cutoff imposed jointly on $|N| + |d|$ does not respect dominance. See Theorem 8.17.

Remark 8.11 (Hahn supports). Nothing in Theorem 8.6 uses that the exponents are integers, only that the two supports are well ordered in the direction of increasing smallness. If $S, T \subseteq \Gamma$ are reverse-well-ordered subsets of an ordered abelian group, then $S + T$ is reverse-well-ordered and each $\gamma \in S + T$ has only finitely many decompositions $\gamma = s + t$ with $s \in S$, $t \in T$; this is the Hahn support theorem [19, 13]. Everything in this section that does not mention the specific value group $\mathbb{Z}$ transfers verbatim to the Hahn power-log extension $\sum_{\alpha \in S} p_\alpha(L) t^\alpha$ and, in particular, to the Puiseux-log extensions $\mathbb{K}[L]((t^{1/s}))$ used later for fractional powers. We do not repeat the statements.

8.2 Finite determination and precision bookkeeping

The content of Theorem 8.6 deserves to be recorded as a dependency statement, because that is the form in which it is used: it says exactly how much of the input must be known in order to know a given part of the output, and it says that no more is needed and no less will do.

Theorem 8.12 (Finite determination, with exact dependency range). *Let R be a logarithmically graded domain over $\mathbb{K}$, let $a, b, N \in \mathbb{Z}$, and let $F, G \in R((t))$ satisfy $\nu_t F \geq a$ and $\nu_t G \geq b$. Then*

$$[t^N](FG) = \sum_{n=a}^{N-b} p_n\, q_{N-n}, \tag{8.9}$$

so that $[t^N](FG)$ is a $\mathbb{K}$-bilinear function of the two finite tuples

$$(p_a, p_{a+1}, \ldots, p_{N-b}) \qquad \text{and} \qquad (q_b, q_{b+1}, \ldots, q_{N-a}),$$

equivalently of $\mathrm{Trunc}_{<N-b+1} F$ and $\mathrm{Trunc}_{<N-a+1} G$. Both ranges are exact: for every n_0 with $a \leq n_0 \leq N - b$ there are $F, F' \in R((t))$ with $\nu_t \geq a$ that differ only in the block of index n_0, and a G with $\nu_t G \geq b$, such that $[t^N](FG) \neq [t^N](F'G)$; and symmetrically in the second argument.

Proof. Equation (8.9) and the bilinearity are Theorem 8.5 together with Theorem 8.6; the blocks listed are precisely those occurring in the sum. For exactness fix $n_0 \in [a, N-b]$ and put $m_0 = N - n_0$, so that $m_0 \geq b$. Take $G = t^{m_0}$, which has $\nu_t G = m_0 \geq b$. If $n_0 > a$ set $F = t^a + t^{n_0}$ and $F' = t^a$; if $n_0 = a$ set $F = t^a$ and $F' = 2t^a$ (here $2 \neq 0$ because char $\mathbb{K} = 0$). In both cases $\nu_t F = \nu_t F' = a$, the two series agree outside index n_0, and $[t^N](FG) - [t^N](F'G)$ equals 1 respectively -1. The symmetric statement follows by commutativity. ☐

Structural checkpoint

The precision rule. If F is known through $\operatorname{Trunc}_{<P} F$ and G through $\operatorname{Trunc}_{<Q} G$, and $\nu_t F = a$, $\nu_t G = b$, then FG is known through

$$\operatorname{Trunc}_{<\min(P+b,\, Q+a)} FG$$

and, in the worst case, no further. The binding entry is contributed by whichever factor is paired with the *more singular* partner: a factor known to P blocks is degraded by the valuation b of the other factor. In particular, multiplying by a series with a pole of order r, that is with valuation $-r$, costs exactly r blocks of precision.

Corollary 8.13 (Truncated product over a general coefficient ring, at arbitrary sufficient cutoffs). *Let $F, G \in R((t))$ be nonzero with $a = \nu_t F$, $b = \nu_t G$, and let $N \in \mathbb{Z}$. Then for all integers $P \geq N - b$ and $Q \geq N - a$,*

$$\operatorname{Trunc}_{<N} FG = \operatorname{Trunc}_{<N}\big(\operatorname{Trunc}_{<P} F\big)\big(\operatorname{Trunc}_{<Q} G\big); \tag{8.10}$$

and if $N > a + b$ the pair $(P, Q) = (N-b,\ N-a)$ is optimal: neither entry can be lowered by one, that is, there exist F, G with these valuations for which $P = N - b - 1$, respectively $Q = N - a - 1$, falsifies (8.10). *If $F, G \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ and $N \geq 0$, then $(P, Q) = (N, N)$ is admissible, so the cruder law*

$$\operatorname{Trunc}_{<N} FG = \operatorname{Trunc}_{<N}\big(\operatorname{Trunc}_{<N} F\big)\big(\operatorname{Trunc}_{<N} G\big) \qquad (N \geq 0) \tag{8.11}$$

holds there.

Proof. Write $F = \operatorname{Trunc}_{<P} F + E$ and $G = \operatorname{Trunc}_{<Q} G + H$, so that $\nu_t E \geq P$ and $\nu_t H \geq Q$, while $\nu_t \operatorname{Trunc}_{<P} F \geq a$ and $\nu_t \operatorname{Trunc}_{<Q} G \geq b$. Then

$$FG - \big(\operatorname{Trunc}_{<P} F\big)\big(\operatorname{Trunc}_{<Q} G\big) = E \operatorname{Trunc}_{<Q} G + \operatorname{Trunc}_{<P} F\, H + EH,$$

whose ν_t is at least $P + b \geq N$, $a + Q \geq N$ and $P + b \geq N$ respectively; for the third term note that $H = G - \operatorname{Trunc}_{<Q} G$ has $\nu_t H \geq b$, every block of H being a block of G, so $\nu_t(EH) \geq P + b$. Hence the two products agree below t^N, which is (8.10).

For the optimality and the sharpness, assume $N > a + b$, so that $N - 1 \geq a + b$. By Theorem 8.12 the block $[t^{N-1}](FG)$ already depends on p_{N-1-b} and on q_{N-1-a}, so no pair below $(N-b, N-a)$ can serve for all F, G; the following pair of series shows that neither entry may be lowered by one. If $N - b - 1 > a$ put $F = t^a + t^{N-b-1}$; if $N - b - 1 = a$ put $F = t^a$. In both cases take $G = t^b$. Then $[t^{N-1}](FG) = 1$, whereas $\operatorname{Trunc}_{<N-b-1} F$ omits the block t^{N-b-1}, so $\operatorname{Trunc}_{<N}(\operatorname{Trunc}_{<N-b-1} F)G$ has zero block at t^{N-1}. (In the first case $N - 1 > a + b$, so the surviving term t^{a+b} does not sit at level $N - 1$.) The second cutoff is symmetric.

For (8.11), note that $a, b \geq 0$ gives $N \geq N - b$ and $N \geq N - a$, so $(P, Q) = (N, N)$ is admissible. ☐

A common trap

The crude law (8.11) is *false* on the Laurent ring. Take $F = t^{-1}$, $G = 1 + t$, $N = 1$. Then $FG = t^{-1} + 1$, so $\operatorname{Trunc}_{<1} FG = t^{-1} + 1$, whereas $\operatorname{Trunc}_{<1} F \operatorname{Trunc}_{<1} G = t^{-1} \cdot 1 = t^{-1}$. The correct cutoff for G is $N - \nu_t F = 2$, and indeed $t^{-1}(1 + t) = t^{-1} + 1$. Implementations that "normalize every series to start at t^0" silently commit this error whenever a genuine principal part is present.

Corollary 8.14 (Error rule for approximate factors)**.** *Let $F, G \in \mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ with $\nu_t F = a$, $\nu_t G = b$, and suppose $\widetilde{F}, \widetilde{G} \in \mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ satisfy*

$$F = \widetilde{F} + \mathcal{O}_t^{\mathrm{form}}(t^p)\,, \qquad G = \widetilde{G} + \mathcal{O}_t^{\mathrm{form}}(t^q)\,.$$

Then

$$FG = \widetilde{F}\widetilde{G} + \mathcal{O}_t^{\mathrm{form}}\Big(t^{\min(p+b,\; q+a,\; p+q)}\Big)\,. \tag{8.12}$$

If in addition $p \geq a$ and $q \geq b$ — that is, if neither approximation is coarser than the leading order of the quantity it approximates — then the third entry is redundant and

$$FG = \widetilde{F}\widetilde{G} + \mathcal{O}_t^{\mathrm{form}}\Big(t^{\min(p+b,\; q+a)}\Big)\,. \tag{8.13}$$

Proof. Write $E = F - \widetilde{F}$ and $H = G - \widetilde{G}$, so $\nu_t E \geq p$ and $\nu_t H \geq q$. Expanding,

$$FG - \widetilde{F}\widetilde{G} = E\widetilde{G} + \widetilde{F}H + EH. \tag{8.14}$$

Now $\widetilde{G} = G - H$ gives $\nu_t \widetilde{G} \geq \min(\nu_t G, \nu_t H) \geq \min(b, q)$, and likewise $\nu_t \widetilde{F} \geq \min(a, p)$. Since ν_t is additive on products (Theorem 8.21 below, or directly Theorem 8.6) and satisfies $\nu_t(A + B) \geq \min(\nu_t A, \nu_t B)$,

$$\nu_t\big(FG - \widetilde{F}\widetilde{G}\big) \;\geq\; \min\big(p + \min(b, q),\; \min(a, p) + q,\; p + q\big) \;=\; \min(p + b,\; q + a,\; p + q),$$

which is (8.12). If $p \geq a$ and $q \geq b$, then $p + q \geq a + q$ and $p + q \geq p + b$, so $p + q \geq \min(p + b, q + a)$ and the third entry may be dropped. ☐

Remark 8.15 (What (8.12) does and does not say). The symbol $\mathcal{O}_t^{\mathrm{form}}(t^N)$ asserts coefficient agreement below outer order N and nothing else: no numerical bound, no convergence, no uniformity in L. It is also not a statement about a quotient ring. In $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ the element t is a unit, so the ideal generated by t^N is the whole ring; the subset $t^N \mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ is an additive piece of the valuation filtration, not an ideal. Congruences of the form (8.12) must therefore be read as valuation inequalities $\nu_t(F - G) \geq N$, or equivalently as equalities of exclusive truncations. Quotient-algebra language becomes legitimate only after the finite principal part has been removed and one works inside $\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$, where (t^N) is a genuine ideal.

The corresponding algorithm is the schoolbook convolution restricted to the window supplied by Theorem 8.12.

Listing 8.1: Truncated block product

```
# F, G sparse maps  exponent -> coefficient in R;  a = ord(F), b = ord(G)
# returns the blocks of Trunc_{<N}(F*G)
function block_product(F, G, a, b, N):
    C := empty map
    for (n, Pn) in F with a <= n <= N-b-1:
        for (m, Qm) in G with b <= m <= N-n-1:
            C[n+m] := C.get(n+m, 0) + Pn*Qm
    normalize every coefficient of C          # collect in L before zero-testing
    delete the blocks of C that are exactly 0
    return C
```

Remark 8.16 (Cost, and what actually dominates it). With dense consecutive support the loop performs $\mathcal{O}\big((N-a-b)^2\big)$ coefficient products; with sparse supports of sizes s_F, s_G it performs at most $s_F s_G$. If the retained blocks have logarithmic degree $\mathcal{O}(d)$, each coefficient product costs $\mathcal{O}(d^2)$ scalar operations by schoolbook polynomial arithmetic, so the total is $\mathcal{O}\big((N-a-b)^2 d^2\big)$; fast convolution in t and fast polynomial multiplication in L reduce both factors. These are complexity $\mathcal{O}$'s and carry no analytic meaning. In practice, at the moderate orders typical of hand asymptotics, exact coefficient normalization — expansion, collection, and gcd reduction if the coefficients are rational functions of L — costs more than the convolution itself, and the decisive implementation choice is to never form a product whose outer order exceeds the requested cutoff.

Remark 8.17 (A rectangular cutoff is not a truncation). Discarding all monomials $t^n L^j$ with $|n|+|j|$ or $|j|$ above a threshold does not respect dominance. With a cap $|j| \le 10$ one deletes $t^2 L^{100}$ while retaining $t^3 L^{-100}$, although $t^3 L^{-100} \prec t^2 L^{100}$ by an enormous margin. Bounding $\deg_{\mathrm{L}} p_n$ is a second, independent approximation; it is legitimate only when accompanied by a proved degree bound such as Theorem 8.29.

8.3 Leading data and the leading-term law

Definition 8.18 (Leading data over a logarithmically graded domain)**.** Let R be a logarithmically graded domain over $\mathbb{K}$ and let $F = \sum_{n \ge n_0} p_n t^n \in R((t))$ be nonzero with $\nu_t F = \min\{n : p_n \ne 0\}$; set $\nu_t 0 = +\infty$. The *leading block* of F is the whole coefficient $p_{\nu_t F} \in R$, the *leading monomial* is

$$\operatorname{lm}(F) := t^{\nu_t F} L^{\deg_{\mathrm{L}} p_{\nu_t F}}, \tag{8.15}$$

and the *leading scalar* is $\operatorname{lc}(F) := \operatorname{lc}_L\big(p_{\nu_t F}\big) \in \mathbb{K}^\times$. The *rank-two valuation* of F is

$$v(F) := \big(\nu_t F,\ -\deg_{\mathrm{L}} p_{\nu_t F}\big) \in \mathbb{Z}^2, \tag{8.16}$$

with $\mathbb{Z}^2$ lexicographically ordered and $v(0) = +\infty$. These are three different objects and none is determined by the coarse integer $\nu_t F$ alone.

Remark 8.19 (These are the objects of the ring chapter, over a general coefficient ring). Theorem 8.18 introduces no new object. It repeats Theorem 6.4 and Theorem 6.6 with the three coefficient rings $\mathbb{K}[L]$, $\mathbb{K}(L)$, $\mathbb{K}((L^{-1}))$ replaced by an arbitrary logarithmically graded domain R, so that the leading-term law can be proved once for all four ambient classes simultaneously; on those four classes the two readings agree. The same applies to Theorem 8.24 against Theorem 6.11 — whose codomain is $\mathbb{N}_0 \cup \{-\infty\}$ because there the blocks are polynomials, whereas over $\mathbb{K}(L)$ and $\mathbb{K}((L^{-1}))$ the value $\deg_{\mathrm{L}} p_n$ may be negative — and to Theorem 8.13 against Theorem 7.4, which the corollary strengthens to arbitrary sufficient cutoffs (P, Q) and supplements with optimality and sharpness. Nothing below re-proves a ring-chapter result for its own sake; the restatements exist only to make the general coefficient ring available.

Lemma 8.20 (Dominance is growth)**.** *On the positive ray, for integers n, m and j, k,*

$$t^n L^j \succ t^m L^k \quad \Longleftrightarrow \quad \lim_{X \to +\infty} \frac{X^{-n}(\log X)^j}{X^{-m}(\log X)^k} = +\infty \quad \Longleftrightarrow \quad n < m,\ \textit{or}\ (n = m\ \textit{and}\ j > k).$$

Consequently, for F in $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ or in $\mathscr{PL}_{\mathrm{itLaur},\mathbb{K}}$, $\operatorname{lm}(F)$ is the $\succ$-largest element of $\operatorname{supp}_{\mathrm{coeff}}(F)$. (For $\mathscr{PL}_{\mathrm{rat},\mathbb{K}}$ the statement is meaningful only after the expansion of Theorem 8.9.)

Proof. By Theorem 5.12 the relation $\succ$ is *defined* by the right-hand condition, so the first equivalence is a definition and only the second requires an argument; it is Theorem 5.13 specialized to integer exponents, and we repeat the two lines. The ratio equals $X^{m-n}(\log X)^{j-k}$. If $m > n$ it tends to $+\infty$ because $\log X = o_{X\to+\infty}(X^\varepsilon)$ for every $\varepsilon > 0$; if $m = n$ it is $(\log X)^{j-k}$, which tends to $+\infty$ exactly when $j > k$; if $m < n$ it tends to 0 for the same reason as the first case. For the last sentence: a monomial $t^n L^j$ in the support has $n \geq \nu_t F$, with equality attained; among those with $n = \nu_t F$ the largest j with $\kappa_j(p_{\nu_t F}) \neq 0$ is by definition $\deg_{\mathrm{L}} p_{\nu_t F}$. □

Theorem 8.21 (Leading-term law). *Let R be a logarithmically graded domain over $\mathbb{K}$ and let $F, G \in R((t))$ be nonzero. Then $FG \neq 0$ and*

$$\nu_t(FG) = \nu_t F + \nu_t G, \tag{8.17}$$

$$\mathrm{lm}(FG) = \mathrm{lm}(F)\ \mathrm{lm}(G), \tag{8.18}$$

$$\mathrm{lc}(FG) = \mathrm{lc}(F)\ \mathrm{lc}(G), \tag{8.19}$$

$$v(FG) = v(F) + v(G). \tag{8.20}$$

Moreover $\nu_t(F+G) \geq \min(\nu_t F, \nu_t G)$ and $v(F+G) \geq \min\bigl(v(F), v(G)\bigr)$ in the lexicographic order. Each is an equality unless cancellation occurs at the common level: whole leading blocks cancel in the first case, top logarithmic coefficients in the second.

Proof. Put $a = \nu_t F$, $b = \nu_t G$. By Theorem 8.6 the coefficient C_N of FG vanishes for $N < a + b$, and for $N = a + b$ the index set $[a, N - b] = \{a\}$ is a single point, so $C_{a+b} = p_a q_b$. Since R is a domain (Theorem 8.3) and $p_a, q_b \neq 0$, we get $C_{a+b} \neq 0$; hence $FG \neq 0$ and (8.17) holds. Applying (8.2) to $C_{a+b} = p_a q_b$ gives

$$\deg_{\mathrm{L}} C_{a+b} = \deg_{\mathrm{L}} p_a + \deg_{\mathrm{L}} q_b, \qquad \mathrm{lc}_L(C_{a+b}) = \mathrm{lc}_L(p_a)\, \mathrm{lc}_L(q_b),$$

which are exactly (8.18) and (8.19); (8.20) is the same pair of identities rewritten in the notation (8.16).

For the sum, let $c = \min(a, b)$. Every block of $F + G$ of index $< c$ vanishes, so $\nu_t(F + G) \geq c$. If $a < b$ then $[t^c](F + G) = p_a \neq 0$ and the leading data of $F + G$ and F coincide, so $v(F + G) = v(F) = \min(v(F), v(G))$ — note that $a < b$ makes $v(F) < v(G)$ lexicographically. If $a = b = c$ and $p_c + q_c \neq 0$, then $\nu_t(F + G) = c$ and $\deg_{\mathrm{L}}(p_c + q_c) \leq \max(\deg_{\mathrm{L}} p_c, \deg_{\mathrm{L}} q_c)$ by (8.2), i.e. $-\deg_{\mathrm{L}}(p_c + q_c) \geq \min(-\deg_{\mathrm{L}} p_c, -\deg_{\mathrm{L}} q_c)$, which is $v(F+G) \geq \min(v(F), v(G))$. If $p_c + q_c = 0$, then $\nu_t(F + G) > c$ and $v(F + G) > \min(v(F), v(G))$ outright. □

Corollary 8.22 (Integrality and fields). *Each of $\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$, $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$, $\mathscr{PL}_{\mathrm{rat},\mathbb{K}}$, $\mathscr{PL}_{\mathrm{itLaur},\mathbb{K}}$ is an integral domain. The map v of (8.16) is a rank-two valuation on $\mathscr{PL}_{\mathrm{rat},\mathbb{K}}$ and on $\mathscr{PL}_{\mathrm{itLaur},\mathbb{K}}$, with value group $\mathbb{Z}^2$ lexicographically ordered; on $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ it takes values in the sub-semigroup $\mathbb{Z} \times \mathbb{Z}_{\leq 0}$. The classes $\mathscr{PL}_{\mathrm{rat},\mathbb{K}}$ and $\mathscr{PL}_{\mathrm{itLaur},\mathbb{K}}$ are fields; $\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ and $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ are not.*

Proof. Integrality is Theorem 8.21, and the valuation axioms are its last two displays together with $v(F) = +\infty$ iff $F = 0$. Surjectivity onto $\mathbb{Z}^2$ for $\mathscr{PL}_{\mathrm{rat},\mathbb{K}}$ and $\mathscr{PL}_{\mathrm{itLaur},\mathbb{K}}$ follows from $v(t^n L^{-j}) = (n, j)$, legitimate there because L^{-j} lies in $\mathbb{K}(L)$ and in $\mathbb{K}((L^{-1}))$; the group $\mathbb{Z}^2$ with the lexicographic order has exactly one proper nontrivial convex subgroup, $\{0\} \times \mathbb{Z}$, so the valuation has rank two. On $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ the second coordinate is $-\deg_{\mathrm{L}} p_{\nu_t F} \leq 0$ because polynomial degrees are nonnegative.

A Laurent-series ring $R((t))$ over a *field* R is a field, which covers $\mathscr{PL}_{\mathrm{rat},\mathbb{K}}$ and $\mathscr{PL}_{\mathrm{itLaur},\mathbb{K}}$: given $F \neq 0$ with $\nu = \nu_t F$ and leading block $p_\nu \in R^\times$, write $F = p_\nu t^\nu (1 + U)$ with $U = \sum_{n>\nu} (p_n / p_\nu) t^{n-\nu} \in tR[[t]]$; the family $\bigl((-U)^r\bigr)_{r \geq 0}$ is summable because $\nu_t(-U)^r \geq r$, and by Theorem 8.34(2) its sum V satisfies

$$(1 + U)V = \sum_{r \geq 0} (-U)^r - \sum_{r \geq 0} (-U)^{r+1} = (-U)^0 = 1,$$

so $F^{-1} = p_\nu^{-1} t^{-\nu} V$. (Theorem 8.34 is proved in Section 8.5 from Theorem 8.21 alone, so there is no circularity.) Finally $L \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ is not invertible in $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$: if $LH = 1$ then (8.18) gives $L \cdot \mathrm{lm}(H) = 1$, forcing $\deg_{\mathrm{L}}$ of the leading block of H to be -1, impossible for a polynomial. Since $L \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}} \subseteq \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$, an inverse of L inside $\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ would in particular be an inverse inside $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$; so L is not invertible in $\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ either, and neither class is a field. □

Corollary 8.23 (Necessity half of the unit criterion)**.** *If $F \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ is invertible in $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$, then its leading block is a nonzero scalar: $p_{\nu_t F} \in \mathbb{K}^\times$. The same statement holds in $\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ with the additional conclusion $\nu_t F = 0$.*

Proof. If $FH = 1$ then (8.17) gives $\nu_t F + \nu_t H = 0$ and (8.18) gives $\deg_{\mathrm{L}} p_{\nu_t F} + \deg_{\mathrm{L}} h_{\nu_t H} = 0$. Both degrees are nonnegative integers, hence both vanish, so $p_{\nu_t F} \in \mathbb{K}^\times$. In $\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ both orders are ≥ 0 and sum to 0, so both are 0. □

The converse — that a scalar leading block suffices — is a division statement and is proved with the reciprocal recurrence; it is not a multiplication fact and is not repeated here.

8.4 Degree and support bounds

Definition 8.24 (Log-degree profile on the four classes)**.** For $F = \sum_n p_n t^n$ in one of the four classes, the *log-degree profile* is the function $d_F \colon \mathbb{Z} \to \mathbb{Z} \cup \{-\infty\}$,

$$d_F(n) := \begin{cases} \deg_{\mathrm{L}} p_n, & p_n \neq 0, \\ -\infty, & p_n = 0. \end{cases} \tag{8.21}$$

The profile is data additional to $\nu_t F$; it is not recoverable from it. For two profiles define the *max-plus convolution*

$$\big(d_F * d_G\big)(N) := \max_{n+m=N} \big(d_F(n) + d_G(m)\big), \tag{8.22}$$

with the conventions $-\infty + c = -\infty$ and $\max \emptyset = -\infty$. Only the finitely many pairs of Theorem 8.6 can contribute a value other than $-\infty$, so the maximum is effectively over a finite set.

Theorem 8.25 (Degree bound and exact criterion for attainment)**.** *Let $F, G \in R((t))$ with R a logarithmically graded domain over $\mathbb{K}$, let $N \in \mathbb{Z}$, and set $M := (d_F * d_G)(N)$. Let*

$$P_N := \{(n,m) : n+m = N,\ p_n \neq 0,\ q_m \neq 0,\ d_F(n) + d_G(m) = M\}$$

be the (finite, possibly empty) set of maximizing pairs. Then

$$d_{FG}(N) \leq M, \tag{8.23}$$

If $P_N = \emptyset$, equivalently $M = -\infty$, then $[t^N](FG) = 0$ and (8.23) *holds with equality. If $P_N \neq \emptyset$, so that $M \in \mathbb{Z}$, then the coefficient of L^M in the block $[t^N](FG)$ is exactly*

$$\kappa_M\big([t^N](FG)\big) = \sum_{(n,m)\in P_N} \mathrm{lc}_L(p_n)\ \mathrm{lc}_L(q_m), \tag{8.24}$$

and equality holds in (8.23) *if and only if the sum* (8.24) *is nonzero.*

Proof. Each summand of $C_N = \sum_{n+m=N} p_n q_m$ has $\deg_{\mathrm{L}}(p_n q_m) = d_F(n) + d_G(m) \leq M$ by (8.2), with the convention that a vanishing factor gives $-\infty$. A finite sum of elements of $\deg_{\mathrm{L}} \leq M$ has $\deg_{\mathrm{L}} \leq M$, which is (8.23). If $M = -\infty$ then every summand $p_n q_m$ vanishes, so $[t^N](FG) = 0$ and both sides of (8.23) equal $-\infty$; assume from now on $P_N \neq \emptyset$, so that $M \in \mathbb{Z}$. By $\mathbb{K}$-linearity of κ_M,

$$\kappa_M(C_N) = \sum_{n+m=N} \kappa_M(p_n q_m).$$

If $(n,m) \notin P_N$ then either a factor vanishes or $\deg_{\mathrm{L}}(p_n q_m) < M$, and in both cases $\kappa_M(p_n q_m) = 0$. If $(n,m) \in P_N$ then $\deg_{\mathrm{L}}(p_n q_m) = M$ and $\kappa_M(p_n q_m) = \mathrm{lc}_L(p_n q_m) = \mathrm{lc}_L(p_n)\,\mathrm{lc}_L(q_m)$ by Theorem 8.3. This proves (8.24), and equality in (8.23) means precisely $\kappa_M(C_N) \neq 0$. □

Corollary 8.26 (Three sufficient conditions for equality)**.** *In the situation of Theorem 8.25, equality holds in* (8.23) *in each of the following cases.*

1. *P_N is a singleton.*
2. *$N = \nu_t F + \nu_t G$ (both factors nonzero); then P_N is the singleton $\{(\nu_t F, \nu_t G)\}$ and one recovers* (8.18).
3. *$P_N \neq \emptyset$, $\mathbb{K}$ is an ordered field, and $\mathrm{lc}_L(p_n) > 0$, $\mathrm{lc}_L(q_m) > 0$ for every $(n, m) \in P_N$.*

Proof. (1) The sum (8.24) is a single product of two elements of $\mathbb{K}^\times$, hence nonzero. (2) By Theorem 8.6 the only admissible pair at $N = \nu_t F + \nu_t G$ is $(\nu_t F, \nu_t G)$, and both blocks are nonzero, so P_N is that singleton; apply (1). (3) The sum is a nonempty sum of strictly positive elements of an ordered field. □

Example 8.27 (The bound is attained, at every order and by every pair)**.** Over $\mathbb{K} = \mathbb{Q}$ let

$$F = G = \sum_{n \geq 0} L^n t^n \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}, \qquad \text{so} \qquad d_F(n) = d_G(n) = n.$$

Then $(d_F * d_G)(N) = \max_{n+m=N}(n + m) = N$, and *every* one of the $N + 1$ admissible pairs is maximizing, so P_N is as large as it can be and Theorem 8.26(1) does not apply. Nevertheless

$$[t^N](FG) = \sum_{n=0}^{N} L^n \cdot L^{N-n} = (N + 1)L^N,$$

so $d_{FG}(N) = N$ for every N: the bound (8.23) is attained at every order. Formula (8.24) predicts exactly the top coefficient $N + 1 = \sum_{(n,m)\in P_N} 1 \cdot 1$.

A common trap

Theorem 8.27 is the place where characteristic zero is used. Over a field of characteristic $p > 0$ the same computation gives $[t^{p-1}](FG) = p\,L^{p-1} = 0$, so the block vanishes entirely and $d_{FG}(p - 1) = -\infty$ although the bound is $p - 1$. Every statement in this section is asserted over a field of characteristic zero, and this is not a formality.

Example 8.28 (The bound can fail by an arbitrary amount)**.** Let $F = 1 + L^d t$ and $G = 1 - L^d t$ in $\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$, $d \geq 1$. Then $(d_F * d_G)(1) = \max(0 + d,\ d + 0) = d$, while $[t^1](FG) = -L^d + L^d = 0$, so $d_{FG}(1) = -\infty$. Here $P_1 = \{(0, 1), (1, 0)\}$ and the sum (8.24) is $1 \cdot (-1) + 1 \cdot 1 = 0$, as it must be. A drop by exactly one occurs for $F = 1 + (L^2 + L)t$, $G = 1 - (L^2 - L)t$, where the bound at $N = 1$ is 2 while $[t^1](FG) = 2L$ has degree 1: here the top coefficients cancel but the next ones do not.

Proposition 8.29 (Affine profiles and the profile subalgebra)**.** *Let $\alpha, \alpha' \geq 0$ and $\beta, \gamma \in \mathbb{R}$. Let $F, G \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ be nonzero with $a = \nu_t F$, $b = \nu_t G$ and*

$$d_F(n) \leq \alpha n + \beta \ \ (n \geq a), \qquad d_G(m) \leq \alpha' m + \gamma \ \ (m \geq b).$$

Put $\mu = \max(\alpha, \alpha')$. Then for all N,

$$d_{FG}(N) \leq \mu N + \beta + \gamma - (\mu - \alpha)a - (\mu - \alpha')b. \tag{8.25}$$

In particular, if $a, b \geq 0$ then $d_{FG}(N) \leq \mu N + \beta + \gamma$. Consequently, for each $\alpha \geq 0$ the set

$$\mathfrak{A}_\alpha := \{F \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}} :\ \exists \beta \in \mathbb{R},\ d_F(n) \leq \alpha n + \beta \text{ for all } n\}$$

is a $\mathbb{K}$-subspace with $\mathfrak{A}_\alpha \cdot \mathfrak{A}_{\alpha'} \subseteq \mathfrak{A}_{\max(\alpha,\alpha')}$, and $\mathfrak{A} := \bigcup_{\alpha \geq 0} \mathfrak{A}_\alpha$ is a $\mathbb{K}$-subalgebra of $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ containing $\mathbb{K}[L][t, t^{-1}]$.

Proof. Fix N and $n+m=N$ with $n \geq a$, $m \geq b$. Then

$$\alpha n + \alpha' m = \mu(n+m) - (\mu-\alpha)n - (\mu-\alpha')m \leq \mu N - (\mu-\alpha)a - (\mu-\alpha')b,$$

because $\mu - \alpha \geq 0$, $\mu - \alpha' \geq 0$, $n \geq a$ and $m \geq b$. Adding $\beta + \gamma$ and taking the maximum over the finite index set gives (8.25) via (8.23). If $a, b \geq 0$ then the two subtracted terms are ≥ 0, so they may be dropped.

For the algebra statements: $\mathfrak{A}_\alpha$ is a subspace because $d_{F+G}(n) \leq \max(d_F(n), d_G(n)) \leq \alpha n + \max(\beta, \gamma)$ and scalars do not change profiles. The product statement is (8.25), whose right-hand side is $\mu N + \text{const}$. For $\alpha \leq \alpha''$ one has $\mathfrak{A}_\alpha \subseteq \mathfrak{A}_{\alpha''}$. Indeed, let $F \neq 0$ satisfy $d_F(n) \leq \alpha n + \beta$ for all n, and put $a = \nu_t F$. For $n \geq 0$ we have $\alpha n + \beta \leq \alpha'' n + \beta$; for $a \leq n < 0$,

$$\alpha n + \beta = \alpha'' n + (\alpha'' - \alpha)(-n) + \beta \leq \alpha'' n + \beta + (\alpha'' - \alpha)\max(0, -a);$$

and $d_F(n) = -\infty$ for $n < a$. Hence $\beta'' = \beta + (\alpha'' - \alpha)\max(0, -a)$ witnesses $F \in \mathfrak{A}_{\alpha''}$, and $0 \in \mathfrak{A}_{\alpha''}$ trivially. Consequently the union $\mathfrak{A}$ is closed under addition and multiplication and contains 1; every Laurent polynomial $\sum_{n=n_1}^{n_2} p_n(L)t^n$ lies in $\mathfrak{A}_0$ with $\beta = \max_n \deg_{\mathrm{L}} p_n$. $\square$

Remark 8.30 ($\mathfrak{A}$ is not t-adically closed). The affine-profile condition is preserved by finite sums and products but not by t-adic limits. Take $F_r = t^r L^{r^2}$ for $r \geq 1$; each F_r lies in $\mathfrak{A}_0$, and $\nu_t F_r = r \to \infty$, so (F_r) is summable, yet the sum $S = \sum_{r \geq 1} t^r L^{r^2}$ has $d_S(r) = r^2$, which is bounded by no affine function. Hence $S \notin \mathfrak{A}$. A degree bound must therefore be re-derived, not inherited, whenever an infinite construction is performed.

Proposition 8.31 (Profile of a power). *Let $F \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ with $d_F(n) \leq \alpha n + \beta$ for all $n \geq 0$, where $\alpha \geq 0$ and $\beta \geq 0$. Then for every integer $k \geq 0$,*

$$d_{F^k}(N) \leq \alpha N + k\beta \qquad (N \geq 0). \tag{8.26}$$

Proof. Induction on k. For $k=0$ we have $F^0 = 1$, so $d_{F^0}(0) = 0 \leq 0$ and $d_{F^0}(N) = -\infty$ for $N > 0$. Assume the bound for k. By (8.23) applied to $F^{k+1} = F^k \cdot F$,

$$d_{F^{k+1}}(N) \leq \max_{n+m=N,\ n,m \geq 0} \big((\alpha n + k\beta) + (\alpha m + \beta)\big) = \alpha N + (k+1)\beta.$$

$\square$

Remark 8.32 (The Lambert profile, corrected). Two of the sources record the degree law of the canonical Lambert family as $\deg \mathsf{L}_n^{\mathrm{Lam}} = n$ without restriction, and use it to assert that products of Lambert-type expansions have profile $\leq N$. The law $\deg \mathsf{L}_n^{\mathrm{Lam}} = n$ holds only for $n \geq 1$: the normalization $\mathsf{L}_0^{\mathrm{Lam}}(u) = -u$ has degree 1, not 0. The correct hypothesis for a series $F = \sum_{n \geq 0} \mathsf{L}_n^{\mathrm{Lam}}(L)t^n$ is therefore $d_F(n) \leq n+1$ for all $n \geq 0$, i.e. $\alpha = 1$, $\beta = 1$ in Theorem 8.29; a product of two such series has $d_{FG}(N) \leq N+2$, and a k-th power has $d_{F^k}(N) \leq N + k$ by Theorem 8.31. The unrestricted claim $d_{FG}(N) \leq N$ is false already at $N = 0$, where $[t^0](FG) = \mathsf{L}_0^{\mathrm{Lam}}(L)^2 = L^2$ has degree 2.

8.5 Powers and binomial powers

Infinite sums enter through powers, so we first record the exact sense in which they are legitimate.

Definition 8.33 (Summable family over a logarithmically graded domain). Let R be a logarithmically graded domain over $\mathbb{K}$. A family $(S_i)_{i \in I}$ in $R((t))$ is *summable* if for every $N \in \mathbb{Z}$ the set $\{i \in I : \nu_t S_i < N\}$ is finite. Its sum is defined blockwise: $[t^n]\sum_i S_i := \sum_i [t^n] S_i$, a finite sum in the coefficient ring.

Lemma 8.34 (Summable families are well behaved). *Let $(S_i)_{i \in I}$ and $(T_j)_{j \in J}$ be summable families in $R((t))$.*

1. $\inf_i \nu_t S_i > -\infty$, *so* $\sum_i S_i$ *is a genuine element of* $R((t))$ *(its support is bounded below).*
2. *The family* $(S_i T_j)_{(i,j)\in I\times J}$ *is summable and* $\bigl(\sum_i S_i\bigr)\bigl(\sum_j T_j\bigr) = \sum_{(i,j)} S_i T_j$.

Proof. (1) Taking $N = 0$ in Theorem 8.33, only finitely many i have $\nu_t S_i < 0$, and each of those has $\nu_t S_i \in \mathbb{Z}$ (a member of that finite set is in particular nonzero). Let λ be the minimum of that finite set of integers, or $\lambda = 0$ if the set is empty. Then $\nu_t S_i \geq \min(0, \lambda)$ for all i, and every block of $\sum_i S_i$ below that bound vanishes.

(2) Let $\alpha = \inf_i \nu_t S_i$ and $\beta = \inf_j \nu_t T_j$, both $> -\infty$ by (1). If $\alpha = +\infty$ or $\beta = +\infty$ then every S_i, respectively every T_j, is zero, so every $S_i T_j$ is zero and both assertions are trivial; assume therefore $\alpha, \beta \in \mathbb{Z}$. Fix N. If $\nu_t(S_i T_j) = \nu_t S_i + \nu_t T_j < N$ then $\nu_t S_i < N - \beta$ and $\nu_t T_j < N - \alpha$, and each of those conditions holds for only finitely many indices; so the family is summable. For the identity, fix n and compute in the coefficient ring, all sums having only finitely many nonzero terms:

$$[t^n]\Bigl(\sum_i S_i \sum_j T_j\Bigr) = \sum_{p+q=n} \Bigl(\sum_i [t^p] S_i\Bigr)\Bigl(\sum_j [t^q] T_j\Bigr) = \sum_{i,j} \sum_{p+q=n} [t^p] S_i \, [t^q] T_j = \sum_{i,j} [t^n](S_i T_j).$$

□

Remark 8.35. Part (1) is worth isolating. For a *sequence* in a Laurent ring, t-adic Cauchyness alone does not guarantee a limit with a finite principal part; a common eventual lower support bound must be imposed. For a *summable family* no such extra hypothesis is needed, because the finiteness condition at $N = 0$ already forces a common lower bound.

Proposition 8.36 (Coefficients of an integer power)**.** *Let* $F = \sum_{n\geq 0} p_n t^n \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ *and* $k \geq 0$. *Then*

$$[t^N] F^k = \sum_{\substack{n_1,\ldots,n_k \geq 0 \\ n_1+\cdots+n_k = N}} p_{n_1} \cdots p_{n_k} := C_{N,k}(p_0, p_1, \ldots), \tag{8.27}$$

with $C_{0,0} = 1$ *and* $C_{N,0} = 0$ *for* $N > 0$. *These* convolution polynomials *satisfy*

$$C_{N,k+1} = \sum_{n=0}^{N} p_n \, C_{N-n,k}, \tag{8.28}$$

which computes the whole triangular table in $\mathcal{O}(N^2 k)$ *coefficient products. If* $p_0 = 0$, *then* $C_{N,k} = 0$ *for* $N < k$, *and, writing* $F = \sum_{n\geq 1} p_n t^n$,

$$[t^N] F^k = \frac{k!}{N!} \mathcal{B}^{\exp}_{N,k}\bigl(1!\,p_1,\ 2!\,p_2, \ldots, (N-k+1)!\,p_{N-k+1}\bigr). \tag{8.29}$$

Proof. Formula (8.27) follows by induction on k from Theorem 8.5: the case $k = 0$ is the convention, and

$$[t^N] F^{k+1} = \sum_{n+m=N} p_n \, [t^m] F^k = \sum_{n=0}^{N} p_n \sum_{\substack{n_1,\ldots,n_k \geq 0 \\ n_1+\cdots+n_k = N-n}} p_{n_1} \cdots p_{n_k},$$

which is both (8.28) and the composition sum for $k+1$. All sums are finite because every $n_i \geq 0$ and they sum to N. If $p_0 = 0$, every n_i in a nonvanishing term is ≥ 1, so $N \geq k$.

For (8.29), recall the defining generating identity of the exponential partial Bell polynomials [4, 3]: in $R[[z]]$ over a commutative $\mathbb{Q}$-algebra R,

$$\frac{1}{k!}\Bigl(\sum_{j\geq 1} x_j \frac{z^j}{j!}\Bigr)^k = \sum_{N\geq k} \mathcal{B}^{\exp}_{N,k}(x_1, x_2, \ldots) \frac{z^N}{N!}.$$

Apply it with $R = \mathbb{K}[L]$ (a $\mathbb{Q}$-algebra, as $\operatorname{char} \mathbb{K} = 0$), $z = t$ and $x_j = j!\,p_j$, so that $\sum_{j\geq 1} x_j z^j/j! = F$. Comparing coefficients of t^N gives $\frac{1}{k!}[t^N]F^k = \frac{1}{N!}\mathcal{B}^{\exp}_{N,k}(1!p_1, 2!p_2, \ldots)$, which is (8.29); only the arguments $x_1, \ldots, x_{N-k+1}$ occur because a term of $\mathcal{B}^{\exp}_{N,k}$ is a product of k variables with index sum N, so no index can exceed $N - k + 1$. □

Theorem 8.37 (Binomial powers and their finite determination). *Let $U \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ with $\nu_t U \geq e$ for an integer $e \geq 1$, write $u_n = [t^n]U$, and let $\sigma \in \mathbb{K}$. Put $\binom{\sigma}{r} = \sigma(\sigma-1)\cdots(\sigma-r+1)/r! \in \mathbb{K}$, with $\binom{\sigma}{0} = 1$. Then:*

1. *The family $\left(\binom{\sigma}{r}U^r\right)_{r\geq 0}$ is summable, so*

$$(1+U)^\sigma := \sum_{r\geq 0}\binom{\sigma}{r}U^r \tag{8.30}$$

is a well-defined element of $\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ with constant block 1.

2. (Finite determination.) *For $N \geq 0$,*

$$[t^N](1+U)^\sigma = \sum_{r=0}^{\lfloor N/e\rfloor}\binom{\sigma}{r}[t^N]U^r, \tag{8.31}$$

and this depends only on the blocks $u_e, u_{e+1}, \ldots, u_N$, that is, only on $\mathrm{Trunc}_{<N+1}U$. The range of r is exact: for $U = t^e$ with $e \mid N$ the term $r = N/e$ contributes $\binom{\sigma}{N/e}$, nonzero whenever $\sigma \notin \{0, 1, \ldots, N/e-1\}$. The top block enters exactly linearly: for $N \geq 1$,

$$[t^N](1+U)^\sigma = \sigma u_N + \Psi_N(u_e, \ldots, u_{N-e}), \tag{8.32}$$

where Ψ_N is a universal polynomial with coefficients in $\mathbb{Q}[\sigma]$, with the convention that $\Psi_N = 0$ when the argument list is empty. Hence for $\sigma \neq 0$ the block u_N cannot be omitted from the dependency range.

3. (Exponent law.) *For $\sigma, \tau \in \mathbb{K}$,*

$$(1+U)^\sigma(1+U)^\tau = (1+U)^{\sigma+\tau}, \qquad (1+U)^0 = 1, \tag{8.33}$$

and for $k \in \mathbb{N}_0$ the series $(1+U)^k$ agrees with the k-fold product. Consequently $(1+U)^\sigma$ is a unit of $\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ with inverse $(1+U)^{-\sigma}$, and $\sigma \mapsto (1+U)^\sigma$ is a homomorphism from $(\mathbb{K}, +)$ to the units of $\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$.

4. (Profile.) *If $d_U(n) \leq \alpha n + \beta$ for all $n \geq e$, with $\alpha \geq 0$, then, writing $\beta^+ = \max(\beta, 0)$,*

$$d_{(1+U)^\sigma}(N) \leq \Big(\alpha + \frac{\beta^+}{e}\Big)N \qquad (N \geq 0). \tag{8.34}$$

In particular $\mathfrak{A} \cap \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ is stable under $U \mapsto (1+U)^\sigma$ whenever $\nu_t U \geq 1$.

Proof. (1) By (8.17), $\nu_t(U^r) \geq re$, which tends to $+\infty$; so for each N only the finitely many $r \leq N/e$ can have $\nu_t(\binom{\sigma}{r}U^r) < N$. Theorem 8.34(1) gives a well-defined sum, and $\nu_t U \geq e \geq 1$ makes the $r = 0$ term the only contributor to t^0.

(2) Blockwise, $[t^N](1+U)^\sigma = \sum_{r\geq 0}\binom{\sigma}{r}[t^N]U^r$, and $[t^N]U^r = 0$ once $re > N$; this is (8.31). By Theorem 8.12 applied $r-1$ times, $[t^N]U^r$ depends only on the blocks u_n with $e \leq n \leq N-(r-1)e$, and for $r \geq 1$ this range is contained in $[e, N]$. For the sharpness of the r-range take $U = t^e$: then $U^r = t^{re}$ and the only contribution to t^N comes from $r = N/e$, with value $\binom{\sigma}{N/e}$, which vanishes exactly when σ is one of $0, 1, \ldots, N/e-1$. For (8.32), the term $r = 0$ contributes nothing to t^N for $N \geq 1$, the term $r = 1$ contributes σu_N, and each term with $r \geq 2$ involves only u_n with $n \leq N-(r-1)e \leq N-e$.

(3) By Theorem 8.34(2),

$$(1+U)^{\sigma}(1+U)^{\tau} = \sum_{r,s\geq 0} \binom{\sigma}{r}\binom{\tau}{s} U^{r+s} = \sum_{m\geq 0} \Big(\sum_{r+s=m} \binom{\sigma}{r}\binom{\tau}{s} \Big) U^{m},$$

the regrouping being legitimate because for each fixed t-block only finitely many pairs (r, s) contribute. Vandermonde's convolution [18]

$$\sum_{r+s=m} \binom{\sigma}{r}\binom{\tau}{s} = \binom{\sigma+\tau}{m}$$

holds for all $\sigma, \tau \in \mathbb{K}$: both sides are polynomial expressions in (σ, τ) with coefficients in $\mathbb{Q}$; they agree for all pairs of nonnegative integers by comparing coefficients in $(1+z)^{\sigma}(1+z)^{\tau} = (1+z)^{\sigma+\tau}$ in $\mathbb{Q}[z]$; a polynomial in two variables over the infinite domain $\mathbb{Q}$ vanishing on $\mathbb{N}_0^2$ is the zero polynomial; and $\mathbb{Q} \subseteq \mathbb{K}$ because $\operatorname{char} \mathbb{K} = 0$, so the identity persists in $\mathbb{K}$. This proves (8.33); $(1+U)^0 = 1$ is immediate. For $k \in \mathbb{N}_0$ we have $\binom{k}{r} = 0$ for $r > k$, so (8.30) is the finite binomial expansion of the k-fold product in the commutative ring $\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$. Invertibility follows by taking $\tau = -\sigma$.

(4) By (8.31) it suffices to bound $\deg_{\mathrm{L}}[t^N]U^r$ for $1 \leq r \leq \lfloor N/e \rfloor$ (the term $r = 0$ contributes only at $N = 0$, where the bound reads $0 \leq 0$). By (8.27) and (8.2),

$$\deg_{\mathrm{L}}[t^N]U^r \leq \max_{\substack{n_1+\cdots+n_r=N \\ n_i \geq e}} \sum_{i=1}^{r} (\alpha n_i + \beta) = \alpha N + r\beta \leq \alpha N + \lfloor N/e \rfloor \beta^{+} \leq \Big(\alpha + \frac{\beta^{+}}{e}\Big) N,$$

where the second inequality uses $r \leq \lfloor N/e \rfloor$ when $\beta \geq 0$ and $r\beta \leq 0 \leq \lfloor N/e \rfloor \beta^{+}$ when $\beta < 0$. Taking the maximum over r gives (8.34). □

Corollary 8.38 (Formal roots). *With U as above and $p, q \in \mathbb{Z}$, $q \geq 1$,*

$$\big((1+U)^{p/q}\big)^{q} = (1+U)^{p},$$

the right-hand side being the ordinary p-th power (or the reciprocal of the $|p|$-th power if $p < 0$). In particular $(1+U)^{1/q}$ is a q-th root of $1+U$ in $\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$, and it is the unique one with constant block 1.

Proof. Iterating (8.33) $q-1$ times gives $((1+U)^{p/q})^q = (1+U)^{qp/q} = (1+U)^p$, and Theorem 8.37(3) identifies the right-hand side with the ordinary power. For uniqueness, suppose $V, W \in \mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ have constant block 1 and $V^q = W^q = 1+U$. Then $(V-W)\sum_{i=0}^{q-1} V^i W^{q-1-i} = 0$; the second factor has constant block $q \neq 0$ (characteristic zero), hence is nonzero, and $\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ is a domain by Theorem 8.22, so $V = W$. □

Remark 8.39 (Infinite products). If $(U_r)_{r\geq 1}$ lies in $\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ with $\nu_t U_r \to +\infty$, the partial products $P_R = \prod_{r\leq R}(1+U_r)$ converge t-adically. Indeed, given N choose R_N with $\nu_t U_r \geq N$ for $r > R_N$; for $R > R_N$,

$$P_R - P_{R_N} = P_{R_N}\Big(\prod_{R_N < r \leq R} (1+U_r) - 1 \Big),$$

and expanding the finite product shows that the bracket is a sum of products of at least one U_r with $r > R_N$, hence of $\nu_t \geq N$; since $\nu_t P_{R_N} \geq 0$ we get $\nu_t(P_R - P_{R_N}) \geq N$. Thus (P_R) is t-adically Cauchy in the complete ring $\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ and $\prod_{r\geq 1}(1+U_r)$ is well defined. On $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ the same argument applies once one notes that $\nu_t U_r \to +\infty$ forces all but finitely many factors to lie in $\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$, so the partial products have a common lower support bound.

8.6 Three worked computations

Example 8.40 (A Laurent product through four blocks, with degree bookkeeping)**.** Over $\mathbb{K} = \mathbb{Q}$ take

$$\begin{aligned} F &= 3t^{-2} + (L-1)t^{-1} + (L^2+2) + 2L\,t + (L^2-L)t^2 + \mathcal{O}_t^{\text{form}}\big(t^3\big)\,, \\ G &= t^{-1} + (2-L) + (L+1)t + L^2t^2 + \mathcal{O}_t^{\text{form}}\big(t^3\big)\,. \end{aligned}$$

Here $\nu_t F = -2$ and $\nu_t G = -1$; F is known through $\operatorname{Trunc}_{<3} F$ and G through $\operatorname{Trunc}_{<3} G$. By the precision rule the product is known through

$$\operatorname{Trunc}_{<\min(3+(-1),\,3+(-2))} FG = \operatorname{Trunc}_{<1} FG,$$

that is, through the block t^0 inclusive — four blocks in all, starting at t^{-3}. The extra block $[t^2]F$ is wasted: it can influence only $[t^1](FG)$ and beyond, for which $[t^3]G$ would also be needed.

Writing $A_n = [t^n]F$ and $B_m = [t^m]G$, the convolution (8.5) gives

$$\begin{aligned} C_{-3} &= A_{-2}B_{-1} = 3, \\ C_{-2} &= A_{-2}B_0 + A_{-1}B_{-1} = 3(2-L) + (L-1) = 5 - 2L, \\ C_{-1} &= A_{-2}B_1 + A_{-1}B_0 + A_0B_{-1} \\ &= 3(L+1) + (-L^2+3L-2) + (L^2+2) = 6L+3, \\ C_0 &= A_{-2}B_2 + A_{-1}B_1 + A_0B_0 + A_1B_{-1} \\ &= 3L^2 + (L^2-1) + (-L^3+2L^2-2L+4) + 2L = -L^3+6L^2+3. \end{aligned}$$

Hence

$$FG = 3t^{-3} + (5-2L)t^{-2} + (6L+3)t^{-1} + (-L^3+6L^2+3) + \mathcal{O}_t^{\text{form}}\big(t^1\big)\,. \tag{8.35}$$

Leading data: $\operatorname{lm}(F) = t^{-2}$ with $\operatorname{lc}(F) = 3$, $\operatorname{lm}(G) = t^{-1}$ with $\operatorname{lc}(G) = 1$; Theorem 8.21 predicts $\operatorname{lm}(FG) = t^{-3}$ and $\operatorname{lc}(FG) = 3$, confirmed by $C_{-3} = 3$.

The degree bookkeeping is instructive. The profiles are $d_F(-2) = 0$, $d_F(-1) = 1$, $d_F(0) = 2$, $d_F(1) = 1$ and $d_G(-1) = 0$, $d_G(0) = 1$, $d_G(1) = 1$, $d_G(2) = 2$. Then:

N	maximizing pairs P_N	bound M	top coefficient (8.24)	$d_{FG}(N)$
-3	$(-2,-1)$	0	$3\cdot 1 = 3$	0
-2	$(-2,0),\ (-1,-1)$	1	$3(-1) + 1\cdot 1 = -2$	1
-1	$(-1,0),\ (0,-1)$	2	$1\cdot(-1) + 1\cdot 1 = 0$	$1 < 2$
0	$(0,0)$	3	$1\cdot(-1) = -1$	3

At $N = -2$ two pairs compete and their contributions -3 and $+1$ do not cancel, so the bound is attained, with top coefficient -2: indeed $C_{-2} = -2L+5$. At $N = -1$ two pairs compete and cancel exactly, so the degree drops from the bound 2 to the actual value 1: indeed $C_{-1} = 6L+3$. At $N = 0$ the maximizing pair is unique, so Theorem 8.26(1) forces attainment, with top coefficient -1: indeed $C_0 = -L^3 + \cdots$. This is exactly the phenomenon Theorem 8.25 isolates; no inspection of the finished polynomial is needed to predict it.

Example 8.41 (A square root through four blocks, verified by squaring)**.** Let

$$U = 2Lt + (L^2+2)t^2 + 4L\,t^3 + \mathcal{O}_t^{\text{form}}\big(t^4\big) \in t\mathscr{P}\mathscr{L}_{\text{pow},\mathbb{K}},$$

so $e = 1$ in Theorem 8.37, and take $\sigma = \frac{1}{2}$, with

$$\binom{1/2}{0} = 1, \quad \binom{1/2}{1} = \tfrac{1}{2}, \quad \binom{1/2}{2} = -\tfrac{1}{8}, \quad \binom{1/2}{3} = \tfrac{1}{16}.$$

By (8.31) only $r \le N$ contributes to t^N, so through t^3 we need

$$U^2 = 4L^2t^2 + (4L^3 + 8L)t^3 + \mathcal{O}_t^{\mathrm{form}}\big(t^4\big)\,, \qquad U^3 = 8L^3t^3 + \mathcal{O}_t^{\mathrm{form}}\big(t^4\big)\,.$$

Block by block:

$$\begin{aligned}
[t^1](1+U)^{1/2} &= \tfrac{1}{2}(2L) = L,\\
[t^2](1+U)^{1/2} &= \tfrac{1}{2}(L^2+2) - \tfrac{1}{8}(4L^2) = \tfrac{L^2}{2} + 1 - \tfrac{L^2}{2} = 1,\\
[t^3](1+U)^{1/2} &= \tfrac{1}{2}(4L) - \tfrac{1}{8}(4L^3+8L) + \tfrac{1}{16}(8L^3) = 2L - \tfrac{L^3}{2} - L + \tfrac{L^3}{2} = L.
\end{aligned}$$

Hence

$$(1+U)^{1/2} = 1 + L\,t + t^2 + L\,t^3 + \mathcal{O}_t^{\mathrm{form}}\big(t^4\big)\,. \tag{8.36}$$

Two logarithmic cancellations occur, one at each of t^2 and t^3; the degree bound (8.34) with $\alpha = 1$, $\beta^+ = 0$, $e = 1$ predicts only $d_{(1+U)^{1/2}}(N) \le N$, and the actual profile $0, 1, 0, 1$ is strictly smaller at $N = 2, 3$. A degree bound is an upper bound; it is not a prediction.

Verification. Theorem 8.38 says the square of (8.36) must reproduce $1 + U$ through t^3. Squaring by (8.5):

$$[t^0] = 1, \quad [t^1] = 2L, \quad [t^2] = 2\cdot 1 + L^2 = L^2 + 2, \quad [t^3] = 2L + 2L = 4L,$$

which are exactly the blocks $1, u_1, u_2, u_3$ of $1 + U$. Note the precision accounting: squaring a series known through $\mathrm{Trunc}_{<4}\cdot$ with $\nu_t = 0$ returns a result known through $\mathrm{Trunc}_{<4}\cdot$, so the check is available at every block computed and at none beyond.

Example 8.42 (A product in the iterated field)**.** In $\mathscr{PL}_{\mathrm{itLaur},\mathbb{K}}$ the coefficient convolution is finite at the inner level too. Take the two blocks

$$f = \frac{1}{L+1} = L^{-1} - L^{-2} + L^{-3} - \cdots\,, \qquad g = \frac{1}{L-1} = L^{-1} + L^{-2} + L^{-3} + \cdots$$

in $\mathbb{K}((L^{-1}))$. For $r \ge 2$, the coefficient of L^{-r} in fg is a sum over the finitely many decompositions $r = i + j$ with $i, j \ge 1$:

$$\kappa_{-r}(fg) = \sum_{i=1}^{r-1} (-1)^{i+1}\cdot 1 = \begin{cases} 1, & r \text{ even},\\ 0, & r \text{ odd},\end{cases}$$

so $fg = L^{-2} + L^{-4} + L^{-6} + \cdots = (L^2 - 1)^{-1}$, as it must be. The degree law is visible: $\deg_{\mathrm{L}} f = \deg_{\mathrm{L}} g = -1$, $\deg_{\mathrm{L}}(fg) = -2$, and $\mathrm{lc}_L(f) = \mathrm{lc}_L(g) = \mathrm{lc}_L(fg) = 1$. Consequently, for $F = f\,t^{-1} + \cdots$ and $G = g + \cdots$ in $\mathscr{PL}_{\mathrm{itLaur},\mathbb{K}}$ one gets $\mathrm{lm}(FG) = t^{-1}L^{-2}$ and $\mathrm{lc}(FG) = 1$ by Theorem 8.21, with no need to compute any further block. Observe that lm here carries a *negative* power of L: this is legitimate in $\mathscr{PL}_{\mathrm{rat},\mathbb{K}}$ and $\mathscr{PL}_{\mathrm{itLaur},\mathbb{K}}$ and impossible in $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$, which is exactly the content of Theorem 8.22.

8.7 The analytic product rule

Everything above is formal algebra. We now record the exact circumstances under which the block convolution computes the asymptotic expansion of an actual product of functions. Nothing in this subsection asserts convergence: an asymptotic expansion is a statement about remainders, and the formal product theorem is used as an input, not replaced.

Definition 8.43 (Polynomial–logarithmic asymptotic expansion)**.** Let $\mathbb{K} \subseteq \mathbb{C}$, let $X_0 > 1$, and let $S = [X_0, +\infty)$ with the real branch $L = \log X$. A function $f\colon S \to \mathbb{C}$ is *PL-asymptotic* to $F \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$, written $f \approx F$, if for every $N \in \mathbb{Z}$ there exist $M_N \in \mathbb{N}_0$, $C_N > 0$ and $X_N \ge X_0$ such that

$$\big|f(X) - \big(\mathrm{Trunc}_{<N} F\big)(X)\big| \le C_N\, X^{-N} (\log X)^{M_N} \qquad (X \ge X_N), \tag{8.37}$$

where $\big(\mathrm{Trunc}_{<N} F\big)(X) = \sum_{\nu_t F \le n < N} p_n(\log X) X^{-n}$ is the actual finite sum obtained by substituting the generators. Equivalently, $f - \mathrm{Trunc}_{<N} F = \mathcal{O}_{X\to+\infty,\ X\in S}\big(X^{-N}(\log X)^{M_N}\big)$.

Lemma 8.44 (Basic properties of the expansion relation $\approx$)**.** *Let $f, g \colon S \to \mathbb{C}$ and $F, G \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$.*

1. *If* (8.37) *holds for arbitrarily large N, it holds for every N; so it suffices to verify it along a cofinal set of cutoffs.*
2. *$f \approx F$ and $f \approx F'$ imply $F = F'$.*
3. *If $f \approx F$ and $g \approx G$, then $f + g \approx F + G$ and $\lambda f \approx \lambda F$ for $\lambda \in \mathbb{K}$.*
4. *$f \approx 0$ if and only if $f = \mathcal{O}_{X\to+\infty}(X^{-N})$ for every N.*

Proof. (1) Let $N' < N$ and suppose (8.37) holds at N. Then

$$f - \operatorname{Trunc}_{<N'} F = \left(f - \operatorname{Trunc}_{<N} F\right) + \sum_{N' \le n < N} p_n(\log X)X^{-n},$$

a finite sum; the first term is $\mathcal{O}_{X\to+\infty}\left(X^{-N}(\log X)^{M_N}\right) \subseteq \mathcal{O}_{X\to+\infty}\left(X^{-N'}(\log X)^{M_N}\right)$ since $N \ge N'$, and each term of the sum is $\mathcal{O}_{X\to+\infty}\left(X^{-N'}(\log X)^{\deg_{\mathrm{L}} p_n}\right)$.

(2) Suppose $D = F - F' \neq 0$, with $n_1 = \nu_t D$ and leading block p of degree d and leading coefficient $c \neq 0$. Take $N = n_1 + 1$ and subtract the two instances of (8.37):

$$\left(\operatorname{Trunc}_{<n_1+1} D\right)(X) = p(\log X)X^{-n_1} = \mathcal{O}_{X\to+\infty}\left(X^{-n_1-1}(\log X)^{M}\right)$$

for some M. But $|p(\log X)| \sim |c|(\log X)^d$, so the left side divided by $X^{-n_1-1}(\log X)^M$ is $\sim |c|X(\log X)^{d-M} \to +\infty$, a contradiction.

(3) Immediate from $\operatorname{Trunc}_{<N} F + G = \operatorname{Trunc}_{<N} F + \operatorname{Trunc}_{<N} G$ and the triangle inequality.

(4) If $f \approx 0$ then $\operatorname{Trunc}_{<N} 0 = 0$ and $f = \mathcal{O}_{X\to+\infty}\left(X^{-N}(\log X)^{M_N}\right)$ for every N; since $(\log X)^{M_N} = o_{X\to+\infty}(X)$, this gives $f = \mathcal{O}_{X\to+\infty}\left(X^{-N+1}\right)$ for every N, hence $f = \mathcal{O}_{X\to+\infty}\left(X^{-N}\right)$ for every N. The converse is trivial. □

Theorem 8.45 (Analytic product rule)**.** *Let $\mathbb{K} \subseteq \mathbb{C}$, $S = [X_0, +\infty)$ with the real logarithm, and let $f, g \colon S \to \mathbb{C}$ with $f \approx F$ and $g \approx G$ for $F, G \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$. Then*

$$fg \approx FG,$$

where FG is the formal Cauchy product (8.5)*. Consequently the set $\mathcal{A}(S) = \{f : \exists F \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}},\ f \approx F\}$ is a $\mathbb{K}$-algebra, the map $f \mapsto F$ is a well-defined $\mathbb{K}$-algebra homomorphism $\mathcal{A}(S) \to \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$, and its kernel is the ideal of functions that are $\mathcal{O}_{X\to+\infty}\left(X^{-N}\right)$ for every N.*

Proof. First dispose of the degenerate case. Any g with $g \approx G$ satisfies $g = \mathcal{O}_{X\to+\infty}(X^c(\log X)^m)$ for some $c \in \mathbb{Z}$ and $m \in \mathbb{N}_0$: take $N = 1$ in (8.37), note that $\operatorname{Trunc}_{<1} G$ is a finite sum of monomials $t^n L^j$, and set $c = \max(-\nu_t G, -1)$ (with $c = -1$ if $G = 0$). Hence, if $F = 0$, then Theorem 8.44(4) gives $f = \mathcal{O}_{X\to+\infty}\left(X^{-P}\right)$ for every P, so $fg = \mathcal{O}_{X\to+\infty}\left(X^{-P+c}(\log X)^m\right)$ for every P, whence $fg \approx 0 = FG$ by Theorem 8.44(4) again; the case $G = 0$ is symmetric. Assume from now on $F, G \neq 0$ and set $a = \nu_t F$, $b = \nu_t G$.

Fix $N \in \mathbb{Z}$ and put

$$P = \max(N - b,\ a), \qquad Q = \max(N - a,\ b).$$

Write $f_P = (\operatorname{Trunc}_{<P} F)(X)$, $r = f - f_P$, $g_Q = (\operatorname{Trunc}_{<Q} G)(X)$, $s = g - g_Q$. By hypothesis $r = \mathcal{O}_{X\to+\infty}\left(X^{-P}(\log X)^{m_1}\right)$ and $s = \mathcal{O}_{X\to+\infty}\left(X^{-Q}(\log X)^{m_2}\right)$ for suitable exponents. Since $\operatorname{Trunc}_{<P} F$ is a finite sum of blocks with outer exponents $\ge a$,

$$f_P = \mathcal{O}_{X\to+\infty}\left(X^{-a}(\log X)^{m_3}\right), \qquad g_Q = \mathcal{O}_{X\to+\infty}\left(X^{-b}(\log X)^{m_4}\right)$$

with $m_3 = \max_{a \le n < P} \deg_{\mathrm{L}} p_n$ and similarly for m_4 (the estimates being vacuously true if the sums are empty). Therefore

$$fg - f_P g_Q = f_P s + r g_Q + rs$$

is, term by term,

$$\mathcal{O}_{X\to+\infty}\left(X^{-(a+Q)}(\log X)^{m_3+m_2}\right) + \mathcal{O}_{X\to+\infty}\left(X^{-(P+b)}(\log X)^{m_1+m_4}\right) + \mathcal{O}_{X\to+\infty}\left(X^{-(P+Q)}(\log X)^{m_1+m_2}\right).$$

Now $a+Q \ge a+(N-a) = N$, $P+b \ge (N-b)+b = N$, and $P+Q \ge a+(N-a) = N$; hence

$$fg - f_P g_Q = \mathcal{O}_{X\to+\infty}\left(X^{-N}(\log X)^{M'}\right) \tag{8.38}$$

for some M'.

Next, $f_P g_Q$ is the value at X of the finite polynomial–logarithmic expression $\left(\mathrm{Trunc}_{<P} F\right)\left(\mathrm{Trunc}_{<Q} G\right) \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$, because substitution of the generators $t = X^{-1}$, $L = \log X$ into a finite sum of monomials is a ring homomorphism into functions on S. Since $P \ge N - b$ and $Q \ge N - a$, Theorem 8.13 gives

$$\mathrm{Trunc}_{<N}\left(\mathrm{Trunc}_{<P} F\right)\left(\mathrm{Trunc}_{<Q} G\right) = \mathrm{Trunc}_{<N} FG.$$

Hence $\left(\mathrm{Trunc}_{<P} F\right)\left(\mathrm{Trunc}_{<Q} G\right) - \mathrm{Trunc}_{<N} FG$ is a *finite* sum of monomials $c\,t^n L^j$ with $n \ge N$, so its value at X is $\mathcal{O}_{X\to+\infty}\left(X^{-N}(\log X)^{M''}\right)$ with M'' the largest j occurring. Combining with (8.38),

$$fg - \left(\mathrm{Trunc}_{<N} FG\right)(X) = \mathcal{O}_{X\to+\infty}\left(X^{-N}(\log X)^{\max(M',M'')}\right),$$

which is (8.37) for fg and FG. As N was arbitrary, $fg \approx FG$.

The algebra statements now follow: $\mathcal{A}(S)$ is closed under $+$, scalars and $\cdot$ by Theorem 8.44(3) and the above; the assignment $f \mapsto F$ is well defined by Theorem 8.44(2) and is a homomorphism by the same two statements; its kernel is described by Theorem 8.44(4). □

Remark 8.46 (Which hypotheses are actually used, and the sectorial version). The proof uses only three things: that f and g have remainders bounded by X^{-N} times *some* power of $\log X$; that finitely many blocks are involved at each stage; and the formal finite-determination theorem Theorem 8.12, which is what allows the two truncation cutoffs P and Q to be chosen independently of the shape of F and G. No convergence and no uniformity in L is used or obtained.

The same proof works verbatim on an unbounded set $S \subseteq \{z : |\arg z| \le \theta\}$ with $0 < \theta < \pi$ and the principal logarithm, with X^{-N} replaced by $|X|^{-N}$ and $(\log X)^M$ by $(\log|X|)^M$; the only additional input is $|\log X| \le \log|X| + \theta \le 2\log|X|$ for $|X| \ge \mathrm{e}^{\theta}$, so that a bound in $|\log X|$ and a bound in $\log|X|$ differ only by a constant factor. Uniformity claims on a sector must be stated for a *closed* subsector; the branch of $\log$ is part of the statement and cannot be recovered from the formal data.

A common trap

Three cautions, each corresponding to an overstatement found in the sources.

1. *A formal identity is not an analytic theorem.* Theorem 8.45 presupposes that f and g *already have* expansions; it does not produce one. The homomorphism $f \mapsto F$ has a large kernel — $\mathrm{e}^{-X} \approx 0$ — so a formal computation determines an actual function at best up to a flat error.
2. *The formal $\mathcal{O}_t^{\mathrm{form}}(t^N)$ carries no uniformity in L.* Asserting $F = G + \mathcal{O}_t^{\mathrm{form}}(t^N)$ says only that the blocks agree below outer order N. In particular the constant C_N and the exponent M_N of (8.37) are not supplied by any formal statement and must be produced separately.
3. *The three $\mathcal{O}$-roles are distinct.* The formal tail $\mathcal{O}_t^{\mathrm{form}}(t^N)$, the analytic estimate

$\mathcal{O}_{X\to+\infty}\big(X^{-N}(\log X)^M\big)$ of (8.37), and the complexity bound $\mathcal{O}\big((N-a-b)^2d^2\big)$ of Theorem 8.16 are three different assertions and none follows from the others. A bare $\mathcal{O}$ for an asymptotic claim, with the regime left to context, is never acceptable.

Section summary

Multiplication is Cauchy convolution in $t = X^{-1}$ with exact coefficient arithmetic in $L = \log X$. Every output block below a cutoff is a finite bilinear expression in finitely many input blocks, with the exact range $a \le n \le N-b$, $b \le m \le N-a$; the resulting precision rule is $\mathrm{Trunc}_{<\min(P+b,\,Q+a)}\,FG$ and it cannot be improved. All four ambient classes are closed and are integral domains, the leading monomial and leading scalar are multiplicative, and the rank-two valuation $v(F) = (\nu_t F, -\deg_{\mathrm{L}} p_{\nu_t F})$ is additive. The log-degree profile obeys the max-plus bound $d_{FG} \le d_F * d_G$, with equality exactly when the maximizing pairs do not cancel; affine profiles are stable under products, powers, and binomial powers, but not under t-adic limits. Binomial powers $(1+U)^\sigma$ are defined for every $\sigma \in \mathbb{K}$ by a summable family, are finitely determined at every order, and satisfy the exponent law. Under printed remainder hypotheses on a stated domain and branch, the formal product computes the asymptotic expansion of the analytic product — and nothing more.

Chapter 9

Units, reciprocals, and division

Addition and multiplication never leave $\mathscr{PL}_{\mathrm{poly},\mathbb{K}} = \mathbb{K}[L]((t))$. Division is the first operation that does. This chapter settles exactly when it does not – the unit criterion – and then follows the three escape routes that the criterion forces: an accidental exact quotient inside $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$, the rational-log field $\mathscr{PL}_{\mathrm{rat},\mathbb{K}} = \mathbb{K}(L)((t))$, and the iterated Laurent-log field $\mathscr{PL}_{\mathrm{itLaur},\mathbb{K}} = \mathbb{K}((L^{-1}))((t))$. Each of the three is a genuinely different object, and conflating them is the most frequent error in informal polynomial–logarithmic computation.

Throughout, $\mathbb{K}$ is a field of characteristic zero, $t := X^{-1}$ and $L := \log X$ are independent formal generators, and a nonzero element of $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ is written

$$F = \sum_{n \geq n_0} p_n(L)t^n, \qquad p_n \in \mathbb{K}[L], \quad p_{n_0} \neq 0, \tag{9.1}$$

so that $\nu_t F = n_0$. We call $p_{\nu_t F}$ the *leading block* of F; it is a polynomial, not a scalar, and it must be distinguished from the leading monomial $\operatorname{lm} F = t^{n_0} L^d$ and from the scalar leading coefficient $\operatorname{lc} F \in \mathbb{K}^{\times}$, where $d = \deg_{\mathrm{L}} p_{n_0}$. Truncation is always the exclusive operator $\operatorname{Trunc}_{<N} F = \sum_{n<N} p_n(L)t^n$, and $F = G + \mathcal{O}_t^{\mathrm{form}}(t^N)$ abbreviates $\nu_t(F - G) \geq N$; it carries no analytic content whatever.

9.1 The exact unit criterion

Two elementary facts do all the work. Both are standard, but the second is the step that three of the six sources compress into a single clause, and it is precisely the step that decides the nonunit direction of the criterion.

Lemma 9.1 (Units of the coefficient ring). *Let $\mathbb{K}$ be a field. Then $\mathbb{K}[L]^{\times} = \mathbb{K}^{\times}$: a polynomial is invertible in $\mathbb{K}[L]$ if and only if it is a nonzero constant.*

Proof. A nonzero constant c has inverse c^{-1}. Conversely, suppose $pq = 1$ with $p, q \in \mathbb{K}[L]$. Since $\mathbb{K}$ is a field, $\mathbb{K}[L]$ is an integral domain and $\deg_{\mathrm{L}}(pq) = \deg_{\mathrm{L}} p + \deg_{\mathrm{L}} q$. Hence $\deg_{\mathrm{L}} p + \deg_{\mathrm{L}} q = \deg_{\mathrm{L}} 1 = 0$, and as both degrees are nonnegative integers, $\deg_{\mathrm{L}} p = \deg_{\mathrm{L}} q = 0$. So $p \in \mathbb{K}$, and $p \neq 0$ because $pq = 1$. $\square$

Lemma 9.2 (The Laurent ring is a domain and ν_t is additive)**.** *Let $F, G \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ be nonzero, with leading blocks $p_{\nu_t F}$ and $q_{\nu_t G}$. Then $FG \neq 0$,*

$$\nu_t(FG) = \nu_t F + \nu_t G, \tag{9.2}$$

and the leading block of FG is $p_{\nu_t F}\, q_{\nu_t G}$. In particular $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ is an integral domain, and the same statements hold in $\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$, in $\mathbb{K}(L)((t))$, and in $E((t))$ for any integral domain E.

Proof. Write $m = \nu_t F$, $r = \nu_t G$. The Cauchy convolution gives $[t^k](FG) = \sum_{i+j=k} p_i q_j$, a finite sum because $i \geq m$ and $j \geq r$ force $m \leq i \leq k-r$. For $k < m+r$ every term has $i < m$ or $j < r$ and therefore vanishes. For $k = m+r$ the only surviving pair is $(i,j) = (m,r)$, so $[t^{m+r}](FG) = p_m q_r$, which is nonzero because $\mathbb{K}[L]$ is an integral domain. Hence $FG \neq 0$ and both assertions hold. □

Theorem 9.3 (Unit criterion). *Let $F \in \mathscr{P}\mathscr{L}_{\text{poly},\mathbb{K}}$ be nonzero, written as in Equation* (9.1)*. The following are equivalent.*

1. *F is a unit of $\mathscr{P}\mathscr{L}_{\text{poly},\mathbb{K}}$.*
2. *The leading block p_{n_0} is a nonzero constant, that is $p_{n_0} \in \mathbb{K}^\times$.*
3. *$F = c\, t^{n_0}(1+U)$ for some $c \in \mathbb{K}^\times$ and some $U \in t\mathscr{P}\mathscr{L}_{\text{pow},\mathbb{K}}$.*

For $F \in \mathscr{P}\mathscr{L}_{\text{pow},\mathbb{K}}$ with $n_0 = 0$, these are equivalent to $p_0 \in \mathbb{K}^\times$, and F^{-1} then lies in $\mathscr{P}\mathscr{L}_{\text{pow},\mathbb{K}}$.

Proof. (2) $\Leftrightarrow$ (3). If $p_{n_0} = c \in \mathbb{K}^\times$, put $U \coloneqq c^{-1}t^{-n_0}F - 1$; every block of $c^{-1}t^{-n_0}F$ of index ≥ 1 is a polynomial and the index-0 block is 1, so $U \in t\mathscr{P}\mathscr{L}_{\text{pow},\mathbb{K}}$. Conversely, in $F = ct^{n_0}(1+U)$ with $U \in t\mathscr{P}\mathscr{L}_{\text{pow},\mathbb{K}}$ the block of index n_0 is c.

(3) $\Rightarrow$ (1). Set

$$G \coloneqq c^{-1}t^{-n_0}\sum_{k\geq 0}(-U)^k. \tag{9.3}$$

Because $\nu_t(U^k) \geq k$, for each N only the terms with $k < N$ contribute to blocks of index $< N$; the family is therefore formally summable and $G \in \mathscr{P}\mathscr{L}_{\text{poly},\mathbb{K}}$, with polynomial blocks because $\mathscr{P}\mathscr{L}_{\text{pow},\mathbb{K}}$ is closed under multiplication. For every $N \geq 0$,

$$\begin{aligned}\mathrm{Trunc}_{<N}\Big((1+U)\sum_{k\geq 0}(-U)^k\Big) &= \mathrm{Trunc}_{<N}\Big((1+U)\sum_{k<N}(-U)^k\Big)\\ &= \mathrm{Trunc}_{<N}\big(1-(-U)^N\big) = \mathrm{Trunc}_{<N}\,1,\end{aligned}$$

using the telescoping identity $(1+U)\sum_{k<N}(-U)^k = 1-(-U)^N$ and $\nu_t((-U)^N) \geq N$. Since two elements of $\mathscr{P}\mathscr{L}_{\text{poly},\mathbb{K}}$ with equal exclusive truncations at every order are equal, $(1+U)\sum_{k\geq 0}(-U)^k = 1$, whence $FG = 1$.

(1) $\Rightarrow$ (2). Suppose $FG = 1$ with $G \in \mathscr{P}\mathscr{L}_{\text{poly},\mathbb{K}}$. Then $G \neq 0$, and Theorem 9.2 gives $\nu_t F + \nu_t G = \nu_t 1 = 0$ together with $p_{\nu_t F}\, q_{\nu_t G} = 1$ in $\mathbb{K}[L]$, where $q_{\nu_t G}$ is the leading block of G. By Theorem 9.1, $p_{\nu_t F} \in \mathbb{K}^\times$.

The statement for $\mathscr{P}\mathscr{L}_{\text{pow},\mathbb{K}}$ is the case $n_0 = 0$ of the above, together with the observation that the G constructed in Equation (9.3) lies in $\mathscr{P}\mathscr{L}_{\text{pow},\mathbb{K}}$ when $n_0 = 0$. □

Corollary 9.4 (The unit group). *The group of units of $\mathscr{P}\mathscr{L}_{\text{poly},\mathbb{K}}$ is*

$$\mathscr{P}\mathscr{L}^\times_{\text{poly},\mathbb{K}} = \big\{c\,t^m(1+U) : m \in \mathbb{Z},\ c \in \mathbb{K}^\times,\ U \in t\mathscr{P}\mathscr{L}_{\text{pow},\mathbb{K}}\big\}, \tag{9.4}$$

and the factorisation is unique, so that $\mathscr{P}\mathscr{L}^\times_{\text{poly},\mathbb{K}} \cong \mathbb{Z} \times \mathbb{K}^\times \times (1 + t\mathscr{P}\mathscr{L}_{\text{pow},\mathbb{K}})$ as abelian groups. Moreover $\mathscr{P}\mathscr{L}^\times_{\text{pow},\mathbb{K}} = \mathbb{K}^\times \cdot (1 + t\mathscr{P}\mathscr{L}_{\text{pow},\mathbb{K}})$.

Proof. The description of the set is Theorem 9.3. Uniqueness: in $F = ct^m(1+U)$ one has $m = \nu_t F$ by Theorem 9.2, then c is the leading block, then $U = c^{-1}t^{-m}F - 1$. The set $1 + t\mathscr{P}\mathscr{L}_{\text{pow},\mathbb{K}}$ is a subgroup of $\mathscr{P}\mathscr{L}^\times_{\text{poly},\mathbb{K}}$: it is closed under multiplication, and by Theorem 9.3 the inverse of $1+U$ is a unit of $\mathscr{P}\mathscr{L}_{\text{pow},\mathbb{K}}$ with leading block 1, hence again of the form $1+U'$ with $U' \in t\mathscr{P}\mathscr{L}_{\text{pow},\mathbb{K}}$. Since $\mathscr{P}\mathscr{L}^\times_{\text{poly},\mathbb{K}}$ is abelian,

$$\mu : \mathbb{Z} \times \mathbb{K}^\times \times (1 + t\mathscr{P}\mathscr{L}_{\text{pow},\mathbb{K}}) \longrightarrow \mathscr{P}\mathscr{L}^\times_{\text{poly},\mathbb{K}}, \qquad \mu\big(m, c, 1+U\big) = c\,t^m(1+U),$$

is a group homomorphism. It is surjective by Theorem 9.3 and injective because $\mu(m, c, 1 + U) = 1 = 1 \cdot t^0 \cdot (1 + 0)$ forces $m = 0$, $c = 1$ and $U = 0$ by the uniqueness just proved. Hence μ is an isomorphism. $\square$

Example 9.5 (A nonzero nonunit)**.** The element $L + 1 \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ is nonzero, but it is not a unit of $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$. Indeed $\nu_t(L + 1) = 0$ and its leading block is $L + 1$, which is not constant; Theorem 9.3 applies. Unwinding the proof gives the direct argument: if $(L + 1)G = 1$ with $G = \sum_{n\geq n_1} q_n(L)t^n$ nonzero, then $\nu_t G = 0$ and $(L + 1)q_0 = 1$ in $\mathbb{K}[L]$, which is impossible because $\deg_{\mathrm{L}}\big((L + 1)q_0\big) = 1 + \deg_{\mathrm{L}} q_0 \geq 1$. The same argument shows that L, $L + t$, $L^2 + t^3$ and $t^{-5}(L - 7)$ are nonzero nonunits. Their reciprocals exist in $\mathscr{PL}_{\mathrm{rat},\mathbb{K}}$:

$$\frac{1}{L+t} = \frac{1}{L} - \frac{t}{L^2} + \frac{t^2}{L^3} - \cdots \in \mathscr{PL}_{\mathrm{rat},\mathbb{K}} \setminus \mathscr{PL}_{\mathrm{poly},\mathbb{K}}. \tag{9.5}$$

Remark 9.6 (The dominant-monomial fallacy)*.* Factoring out the dominant monomial is *not* enough to invert. Writing $L + t = L(1 + t/L)$ and expanding the second factor geometrically already commits one to the coefficient ring $\mathbb{K}(L)$ or $\mathbb{K}((L^{-1}))$: the step that leaves $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ is the inversion of the scalar-free factor L, not the infinite t-expansion. Equation (9.5) makes the point in one line – the failure is visible already in the block of index 0.

9.2 Existence, uniqueness, and the reciprocal recurrence

Equation (9.3) proves existence but is a poor algorithm. The practical construction is triangular.

Theorem 9.7 (Reciprocal of a unit)**.** *Let $B = \sum_{n\geq 0} B_n(L)t^n \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ with $B_0 = c \in \mathbb{K}^\times$. Then there is exactly one $C \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ with $BC = 1$, namely $C = B^{-1} = \sum_{n\geq 0} C_n(L)t^n$ with*

$$C_0 = c^{-1}, \tag{9.6}$$

$$C_n = -c^{-1} \sum_{j=1}^{n} B_j\, C_{n-j} \qquad (n \geq 1), \tag{9.7}$$

and every C_n lies in $\mathbb{K}[L]$. Moreover C is the unique reciprocal of B in $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$, and for a Laurent unit $F = t^m B$ one has $F^{-1} = t^{-m}C$.

Proof. Existence. Define $C_n \in \mathbb{K}[L]$ by Equations (9.6) and (9.7); the right-hand side of Equation (9.7) involves only $C_0, \ldots, C_{n-1}$, so the definition is legitimate by strong induction, and each C_n is a $\mathbb{K}$-linear combination of products of polynomials, hence a polynomial. Put $C := \sum_{n\geq 0} C_n t^n \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$. For $n \geq 1$,

$$[t^n](BC) = \sum_{j=0}^{n} B_j C_{n-j} = cC_n + \sum_{j=1}^{n} B_j C_{n-j} = 0$$

by Equation (9.7), and $[t^0](BC) = cC_0 = 1$. Hence $BC = 1$.

Uniqueness. If $BC = B\widetilde{C} = 1$ with $C, \widetilde{C} \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$, then $C = C(B\widetilde{C}) = (CB)\widetilde{C} = \widetilde{C}$. This argument uses only associativity and commutativity, so uniqueness holds in $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ – and in any commutative ring – not merely in $\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$.

Laurent case. t^m is a unit with inverse t^{-m}, so $(t^m B)(t^{-m}C) = BC = 1$.

Generality. Nothing above uses more about the coefficient ring than that it is commutative and that B_0 is invertible in it. Hence for any commutative ring R and any $B \in R[[t]]$ with $B_0 \in R^\times$, the same recurrence produces the unique reciprocal of B in $R[[t]]$, and that reciprocal remains the unique one in $R((t))$. $\square$

Corollary 9.8 (Quotient recurrence). *Let E be a field, $A = \sum_{n\ge0} A_n t^n$ and $B = \sum_{n\ge0} B_n t^n$ in $E[[t]]$ with $B_0 \neq 0$. Then $Q = A/B = \sum_{n\ge0} Q_n t^n \in E[[t]]$ is the unique solution of $BQ = A$, and*

$$Q_n = B_0^{-1}\Big(A_n - \sum_{j=1}^{n} B_j Q_{n-j}\Big), \qquad n \ge 0, \tag{9.8}$$

the sum being empty when $n = 0$. When $E = \mathbb{K}[L]$ is used instead of a field, Equation (9.8) *still produces polynomial blocks provided $B_0 \in \mathbb{K}^\times$, and then $Q \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$.*

Proof. Existence and uniqueness follow from Theorem 9.7 applied over the coefficient ring E (or $\mathbb{K}[L]$, where $B_0 \in \mathbb{K}^\times$ is invertible): $Q := A\,B^{-1}$ solves $BQ = A$, and if $BQ = BQ'$ then $Q = Q'$ after multiplying by B^{-1}. Comparing coefficients of t^n in $BQ = A$ gives $B_0 Q_n + \sum_{j=1}^{n} B_j Q_{n-j} = A_n$, which is Equation (9.8). Since Equation (9.8) determines Q_n from $Q_0, \dots, Q_{n-1}$, it also proves uniqueness directly. □

The next statement is the one an implementation actually needs: it says that a finite computation with truncated inputs returns exactly the truncation of the true quotient, and that the truncated quotient is characterised without reference to the recurrence at all.

Theorem 9.9 (Correctness of truncated division). *Let $B \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ with $B_0 \in \mathbb{K}^\times$, let $A \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$, and let $N \ge 0$. Let $Q_0, \dots, Q_{N-1}$ be produced by Equation* (9.8) *and set $P := \sum_{n<N} Q_n t^n$. Then*

$$BP = A + \mathcal{O}_t^{\mathrm{form}}\left(t^N\right), \tag{9.9}$$

and P is the only *element of $\mathbb{K}[L][t]$ of t-degree $< N$ satisfying Equation* (9.9)*. Consequently $P = \mathrm{Trunc}_{<N}(A/B)$.*

Proof. For $n < N$ the coefficient $[t^n](BP) = \sum_{j=0}^{n} B_j Q_{n-j}$ involves only $Q_0, \dots, Q_n$, all of which are among the computed blocks, and it equals A_n by Equation (9.8). This is Equation (9.9).

Uniqueness: if P' is another such polynomial, put $D = P - P'$, a polynomial in t of degree $< N$ with $\nu_t(BD) \ge N$. If $D \neq 0$ then Theorem 9.2 gives $\nu_t(BD) = \nu_t B + \nu_t D = \nu_t D < N$, a contradiction; so $D = 0$.

Finally, $B\big(A/B - P\big) = A - BP$ has $\nu_t \ge N$, and $\nu_t B = 0$, so $\nu_t(A/B - P) \ge N$ by Theorem 9.2. As P has t-degree $< N$, this says exactly $P = \mathrm{Trunc}_{<N}(A/B)$. □

Remark 9.10 (Truncation convention). Report S6 states Theorem 9.9 in the inclusive form

$$B \sum_{n\le N} Q_n t^n = A + \mathcal{O}_t^{\mathrm{form}}\left(t^{N+1}\right).$$

Because $[F]_{\le N} = \mathrm{Trunc}_{<N+1} F$, that statement is the case $N \mapsto N+1$ of the one above. Anyone transferring a precision budget between the two conventions must account for the one-block shift; it is the commonest silent off-by-one in this subject.

Proposition 9.11 (Exact input requirement: relative precision is preserved). *Let E be a field, and let $A, B \in E((t))$ with $A \neq 0 \neq B$ (the case $A = 0$ being trivial). Put $a = \nu_t A$ and $b = \nu_t B$, so that $\nu_t(A/B) = a - b$. For every $M \ge 0$, the first M blocks of A/B, namely the coefficients of $t^{a-b}, \dots, t^{a-b+M-1}$, are determined by the first M blocks of A and the first M blocks of B, and by nothing else. Equivalently, in absolute cutoffs,*

$$\mathrm{Trunc}_{<N}(A/B) \quad \textit{is determined by} \quad \mathrm{Trunc}_{<\,N+b}\, A \textit{ and } \mathrm{Trunc}_{<\,N+2b-a}\, B. \tag{9.10}$$

Proof. Write $\widetilde{B} = t^{-b}B$, so $\nu_t\widetilde{B} = 0$ and $\widetilde{B}_j = B_{j+b}$. Then $A/B = t^{-b}(A/\widetilde{B})$, and $G := A/\widetilde{B}$ has $\nu_t G = a$. Applying Equation (9.8) to G after shifting indices by a, the block G_{a+i} is a function of $A_a, \ldots, A_{a+i}$ and $\widetilde{B}_0, \ldots, \widetilde{B}_i$ alone. Taking $i < M$ gives the first assertion, since $\widetilde{B}_0, \ldots, \widetilde{B}_{M-1}$ are the first M blocks of B. For Equation (9.10), note first that $\mathrm{Trunc}_{<N}(A/B) = 0$ when $N \leq a - b$, so assume $N > a - b$. The requested blocks of A/B are then those of index $< N$, that is the first $M = N - (a - b)$ blocks; these need A through index $a + M - 1 = N + b - 1$ and B through index $b + M - 1 = N + 2b - a - 1$. □

Remark 9.12 (Guard precision). Theorem 9.11 is the precise form of the informal rule that a quotient must be computed "beyond the requested block". The guard is $b = \nu_t B$ blocks on the numerator and $2b - a$ on the denominator; it is zero exactly when $a = b = 0$, and it is what a symbolic system must propagate through an expression tree instead of a single global cutoff. A negative guard means that fewer input blocks suffice. On the numerator this happens exactly when $b < 0$, that is when the denominator has a Laurent principal part; on the denominator it happens exactly when $a > 2b$, for instance when a numerator of high t-order is divided by a unit.

Example 9.13 (A reciprocal computed to five blocks)**.** Let

$$B = 1 + (L+1)t + L^2t^2 + (L-2)t^3 \in \mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}. \tag{9.11}$$

Here $B_0 = 1 \in \mathbb{K}^\times$ and $B_n = 0$ for $n \geq 4$. Equations (9.6) and (9.7) give

$$\begin{aligned}
C_0 &= 1,\\
C_1 &= -(L+1),\\
C_2 &= -\big((L+1)C_1 + L^2C_0\big) = 2L + 1,\\
C_3 &= -\big((L+1)C_2 + L^2C_1 + (L-2)C_0\big) = L^3 - L^2 - 4L + 1,\\
C_4 &= -\big((L+1)C_3 + L^2C_2 + (L-2)C_1\big) = -L^4 - 2L^3 + 5L^2 + 2L - 3,
\end{aligned}$$

so that

$$B^{-1} = 1 - (L+1)t + (2L+1)t^2 + (L^3 - L^2 - 4L + 1)t^3 + (-L^4 - 2L^3 + 5L^2 + 2L - 3)t^4 + \mathcal{O}_t^{\mathrm{form}}\left(t^5\right). \tag{9.12}$$

Multiplying Equation (9.11) by Equation (9.12) and truncating at t^5 returns 1; by Theorem 9.9 this residual check certifies the five blocks independently of the method used to produce them.

Example 9.14 (A quotient with polynomial blocks)**.** With

$$A = 1 + (L+1)t + (2L^2 - L)t^2 + \mathcal{O}_t^{\mathrm{form}}\left(t^3\right), \qquad B = 1 + (2-L)t + (L^2+3)t^2 + \mathcal{O}_t^{\mathrm{form}}\left(t^3\right),$$

Equation (9.8) gives $Q_0 = 1$, $Q_1 = (L+1) - (2-L) = 2L - 1$, and

$$Q_2 = (2L^2 - L) - (2-L)(2L-1) - (L^2+3) = 3L^2 - 6L - 1,$$

so $A/B = 1 + (2L-1)t + (3L^2 - 6L - 1)t^2 + \mathcal{O}_t^{\mathrm{form}}\left(t^3\right)$. The block Q_3 is *not* determined by the displayed data: Theorem 9.11 with $a = b = 0$ says three input blocks determine exactly three output blocks. If one additionally posits $A_3 = B_3 = 0$, then $Q_3 = L^3 - 11L^2 + 5L + 5$.

Do not read a block that the data does not contain

An input written with a formal tail $\mathcal{O}_t^{\mathrm{form}}\left(t^3\right)$ determines the quotient only through t^2. Continuing the recurrence past that point silently substitutes the assumption "the omitted blocks are zero". Both S1 and S6 print one extra block of a worked example on exactly this unstated convention; when it is intended it must be said.

Proposition 9.15 (Degree profile of a reciprocal and of a quotient). *Let $\alpha \geq 0$ and $\beta \geq 0$ be real. Suppose $B \in \mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ has $B_0 \in \mathbb{K}^\times$ and $\deg_{\mathrm{L}} B_n \leq \alpha n$ for all $n \geq 1$. Then the blocks of B^{-1} satisfy*

$$\deg_{\mathrm{L}} C_n \leq \alpha n \qquad (n \geq 0). \tag{9.13}$$

If moreover $A \in \mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ has $\deg_{\mathrm{L}} A_n \leq \alpha n + \beta$, then the quotient blocks satisfy $\deg_{\mathrm{L}} Q_n \leq \alpha n + \beta$.

Proof. Strong induction on n. For $n = 0$, $C_0 = B_0^{-1} \in \mathbb{K}$ has degree $0 \leq 0$. For $n \geq 1$, each summand of Equation (9.7) obeys $\deg_{\mathrm{L}}(B_j C_{n-j}) \leq \alpha j + \alpha(n-j) = \alpha n$ by the inductive hypothesis and the additivity of $\deg_{\mathrm{L}}$ on products in $\mathbb{K}[L]$; a finite sum cannot raise the bound, and multiplying by the scalar $-B_0^{-1}$ changes nothing. The quotient case is identical, with the extra term A_n of degree $\leq \alpha n + \beta$ and $\deg_{\mathrm{L}}(B_j Q_{n-j}) \leq \alpha j + \alpha(n-j) + \beta$. □

Remark 9.16. The bound in Equation (9.13) may be strict: cancellation can lower a block's degree, and it does so in Theorem 9.13, where $\deg_{\mathrm{L}} B_2 = 2$ and $\deg_{\mathrm{L}} C_2 = 1$. An affine profile $d_F(n) \leq \alpha n + \beta$ is a hypothesis or a proved invariant, never part of the definition of the ring; the Lambert expansion is the archetype with $\alpha = 1$, $\beta = 0$.

9.3 Geometric inversion and two-level summability

The geometric formula Equation (9.3) is the conceptual construction, and in $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ its justification is purely t-adic: $\nu_t(U^k) \geq k \to \infty$. Over a completed logarithmic coefficient field this reason is no longer available, and a second level of the argument is needed. Several sources normalise a divisor in $\mathscr{P}\mathscr{L}_{\mathrm{itLaur},\mathbb{K}}$ as $B = b\, t^\nu L^\mu (1 + S)$ with $S \prec 1$ and then assert summability of $\sum_k (-S)^k$ "by the support conditions". That is true but not for the t-adic reason: S may have $\nu_t S = 0$, so $\nu_t(S^k) = 0$ for every k. The correct statement is the following.

Lemma 9.17 (Two-level summability). *For $s \in \mathbb{K}((L^{-1}))$ let $\omega(s)$ denote the L^{-1}-adic order, so that $s \in L^{-\omega(s)}\mathbb{K}[[L^{-1}]]$ and $\omega(0) = +\infty$; ω is additive on products (Theorem 9.2 with L^{-1} in place of t and $E = \mathbb{K}$) and satisfies $\omega(s + s') \geq \min\{\omega(s), \omega(s')\}$. Let $S = \sum_{n \geq 0} s_n t^n \in \mathscr{P}\mathscr{L}_{\mathrm{itLaur},\mathbb{K}}$ satisfy*

$$\nu_t S \geq 0 \qquad \text{and} \qquad \omega(s_0) \geq 1, \tag{9.14}$$

which is exactly the condition $S \prec 1$ in the dominance order. Fix $n \geq 0$ and put

$$c_n := \max\{0,\ -\omega(s_1),\ \ldots,\ -\omega(s_n)\} \in \mathbb{N}_0.$$

Then for every $k \geq 0$,

$$\omega\big([t^n]S^k\big) \geq k - n\,(1 + c_n). \tag{9.15}$$

Consequently, for each fixed monomial $t^n L^{-j}$ only the finitely many indices $k \leq j + n(1 + c_n)$ contribute to $\sum_{k \geq 0} (-S)^k$; the family is summable in $\mathscr{P}\mathscr{L}_{\mathrm{itLaur},\mathbb{K}}$, and $(1 + S)\sum_{k \geq 0}(-S)^k = 1$.

Proof. Because $\nu_t S \geq 0$,

$$[t^n]S^k = \sum_{\substack{(n_1,\ldots,n_k) \in \mathbb{N}_0^k \\ n_1 + \cdots + n_k = n}} s_{n_1} \cdots s_{n_k},$$

a finite sum in which every index satisfies $n_i \leq n$. Fix one tuple and let r be the number of indices with $n_i \geq 1$; then $r \leq n$, since each such index contributes at least 1 to the total n. The $k - r$ factors with $n_i = 0$ have $\omega(s_0) \geq 1$, and the r remaining factors have $\omega(s_{n_i}) \geq -c_n$ by definition of c_n. Additivity of ω on products gives

$$\omega\big(s_{n_1} \cdots s_{n_k}\big) \geq (k - r) \cdot 1 + r \cdot (-c_n) \geq k - n - nc_n,$$

using $r \le n$ and $c_n \ge 0$. Taking the minimum over the finitely many tuples yields Equation (9.15).

A power L^{-j} can occur in an element of order ω only when $j \ge \omega$; hence $[t^n L^{-j}]S^k = 0$ as soon as $k - n(1 + c_n) > j$, which proves the finiteness assertion.

Summability in $\mathscr{PL}_{\mathrm{itLaur},\mathbb{K}}$ needs one more line. The outer support of every S^k lies in $n \ge 0$ because $\nu_t S \ge 0$, so the outer support of the sum is bounded below. For a fixed n, Equation (9.15) with $k \ge 0$ gives $\omega\big([t^n]S^k\big) \ge -n(1 + c_n)$ for *every* k, so each contribution to the n-th block is supported in $j \ge -n(1 + c_n)$; that block is therefore a well-defined element of $\mathbb{K}((L^{-1}))$, and the sum lies in $\mathscr{PL}_{\mathrm{itLaur},\mathbb{K}}$.

For the identity, fix (n, j) and choose K larger than $j + n(1 + c_n)$. Telescoping gives $(1 + S)\sum_{k\le K}(-S)^k = 1-(-S)^{K+1}$, and by the finiteness just proved the coefficient of $t^n L^{-j}$ in $(-S)^{K+1}$ vanishes and the coefficient of $t^n L^{-j}$ in $(1+S)\sum_{k\le K}(-S)^k$ already equals that in $(1+S)\sum_{k\ge 0}(-S)^k$. Hence the two sides agree at every monomial. □

Remark 9.18. Theorem 9.17 is why the catalogue insists that a finite representation in $\mathscr{PL}_{\mathrm{itLaur},\mathbb{K}}$ needs an outer t-cutoff *and* an inner L^{-1}-cutoff, or one explicitly declared staircase. A single outer cutoff does not bound the work: the coefficient of $t^0 L^{-j}$ still receives contributions from $j + 1$ powers of S.

9.4 Why unrestricted division needs the rational-log field

Theorem 9.19 (Laurent series over a field)**.** *Let E be a field. Then $E((t))$ is a field. In particular $\mathscr{PL}_{\mathrm{rat},\mathbb{K}} = \mathbb{K}(L)((t))$ and $\mathscr{PL}_{\mathrm{itLaur},\mathbb{K}} = \mathbb{K}((L^{-1}))((t))$ are fields.*

Proof. $E((t))$ is a commutative ring with 1, and it is an integral domain by Theorem 9.2. Let $F \ne 0$ with $m = \nu_t F$ and leading coefficient $f_m \in E^\times$. Put $U := f_m^{-1}t^{-m}F - 1 \in tE[[t]]$. Exactly as in the proof of Theorem 9.3, $\nu_t(U^k) \ge k$ makes $\sum_{k\ge 0}(-U)^k$ formally summable in $E[[t]]$, and telescoping gives $(1 + U)\sum_{k\ge 0}(-U)^k = 1$. Hence $F \cdot f_m^{-1}t^{-m}\sum_{k\ge 0}(-U)^k = 1$. Finally $\mathbb{K}(L)$ and $\mathbb{K}((L^{-1}))$ are fields: the first is the fraction field of the integral domain $\mathbb{K}[L]$; the second is the Laurent series field over $\mathbb{K}$ in the variable L^{-1}, a field by the case already proved. □

The next theorem says that $\mathscr{PL}_{\mathrm{rat},\mathbb{K}}$ is not merely *a* field containing $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$, but the smallest one of the shape "Laurent series in t over a coefficient ring between $\mathbb{K}[L]$ and $\mathbb{K}(L)$". This is the precise sense in which unrestricted division of *coefficient blocks* forces $\mathbb{K}(L)$.

Theorem 9.20 (Minimality of the rational-log coefficient field)**.** *Let E be a subring of $\mathbb{K}(L)$ with $\mathbb{K}[L] \subseteq E$. Then $E((t))$ is a field if and only if $E = \mathbb{K}(L)$. Consequently $\mathscr{PL}_{\mathrm{rat},\mathbb{K}}$ is the unique smallest field of the form $E((t))$ containing $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ with $E \subseteq \mathbb{K}(L)$.*

Proof. If $E = \mathbb{K}(L)$, apply Theorem 9.19. Conversely suppose $E((t))$ is a field and let $e \in E$, $e \ne 0$. Its inverse $G = \sum_n G_n t^n \in E((t))$ satisfies $eG = 1$, i.e. $eG_0 = 1$ and $eG_n = 0$ for $n \ne 0$. Since $E \subseteq \mathbb{K}(L)$ is an integral domain and $e \ne 0$, we get $G_n = 0$ for $n \ne 0$ and $G_0 = e^{-1} \in E$. Hence E is a field. A subfield of $\mathbb{K}(L)$ containing $\mathbb{K}[L]$ contains p^{-1} for every nonzero $p \in \mathbb{K}[L]$ and hence every quotient p/q, so $E = \mathbb{K}(L)$. □

9.5 An integral domain that is not a field, made precise

"$\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ is an integral domain but not a field" is the standard slogan. It is worth saying exactly how far from a field it is, and – a point none of the six sources records – that its field of fractions is *strictly smaller* than $\mathscr{PL}_{\mathrm{rat},\mathbb{K}}$. Passing to $\mathscr{PL}_{\mathrm{rat},\mathbb{K}}$ is therefore not the same operation as "forming the fraction field".

Lemma 9.21 (Denominator lemma). *Let $A, B \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ with $B \neq 0$, and let $b_0 \in \mathbb{K}[L]$ be the leading block of B. Then, computed in $\mathscr{PL}_{\mathrm{rat},\mathbb{K}}$,*

$$\frac{A}{B} \in \mathbb{K}[L][b_0^{-1}]\,((t)), \tag{9.16}$$

where $\mathbb{K}[L][b_0^{-1}] = \{p/b_0^k : p \in \mathbb{K}[L],\ k \geq 0\} \subseteq \mathbb{K}(L)$. In particular every element of the fraction field of $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ has all of its coefficient blocks in a single *localisation $\mathbb{K}[L][q^{-1}]$ of $\mathbb{K}[L]$ at one nonzero polynomial q.*

Proof. Write $m = \nu_t B$ and $S := \mathbb{K}[L][b_0^{-1}]$, a subring of $\mathbb{K}(L)$. Then $t^{-m}B = \sum_{k\geq 0} b_k t^k$ with $b_k \in \mathbb{K}[L]$, so $b_0^{-1}t^{-m}B = 1 + U$ with $U := \sum_{k\geq 1}(b_k/b_0)t^k \in tS[[t]]$. By the proof of Theorem 9.19 applied to the ring S – only summability and telescoping are used, not that S is a field – $(1+U)^{-1} = \sum_{k\geq 0}(-U)^k \in S[[t]]$. Hence $B^{-1} = t^{-m}b_0^{-1}(1+U)^{-1} \in S((t))$ and, since $A \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}} \subseteq S((t))$, also $A/B \in S((t))$. The last sentence is this statement with $q = b_0$. □

Theorem 9.22 (The fraction field of $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ is strictly smaller than $\mathscr{PL}_{\mathrm{rat},\mathbb{K}}$). *Let $\mathbb{K}$ have characteristic zero. Then*

$$\mathscr{PL}_{\mathrm{poly},\mathbb{K}} \subsetneq \mathrm{Frac}(\mathscr{PL}_{\mathrm{poly},\mathbb{K}}) \subsetneq \mathscr{PL}_{\mathrm{rat},\mathbb{K}}. \tag{9.17}$$

An explicit witness for the second strict inclusion is

$$F := \sum_{n\geq 0} \frac{t^n}{L-n} \in \mathscr{PL}_{\mathrm{rat},\mathbb{K}}. \tag{9.18}$$

Proof. The first inclusion is strict because L is a nonzero nonunit (Theorem 9.5). For the second, $\mathrm{Frac}(\mathscr{PL}_{\mathrm{poly},\mathbb{K}})$ embeds in the field $\mathscr{PL}_{\mathrm{rat},\mathbb{K}}$ because $\mathscr{PL}_{\mathrm{rat},\mathbb{K}}$ contains $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$, so we must only show $F \notin \mathrm{Frac}(\mathscr{PL}_{\mathrm{poly},\mathbb{K}})$. Certainly $F \in \mathscr{PL}_{\mathrm{rat},\mathbb{K}}$: its blocks $(L-n)^{-1}$ lie in $\mathbb{K}(L)$ and its support is bounded below by 0.

Suppose $F = A/B$ with $A, B \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$, $B \neq 0$. By Theorem 9.21 there is a nonzero $q \in \mathbb{K}[L]$ with every block of F in $\mathbb{K}[L][q^{-1}]$. Fix $n \geq 0$. Then $(L-n)^{-1} = p/q^k$ for some $p \in \mathbb{K}[L]$ and $k \geq 0$, i.e. $q^k = (L-n)p$. Since $\mathbb{K}$ has characteristic zero, the elements $0, 1, 2, \ldots$ of $\mathbb{K}$ are pairwise distinct, so $L-n$ is a nonassociate irreducible of the unique factorisation domain $\mathbb{K}[L]$ for each n, and $(L-n) \mid q^k$ forces $(L-n) \mid q$. As this holds for every $n \geq 0$, the polynomial q has infinitely many roots, hence $q = 0$, a contradiction. □

Corollary 9.23 (The precise structure). *$\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ is an integral domain whose unit group is Equation* (9.4). *It is not a field, and its nonunits do not form an ideal, so it is not a local ring: L and $1-L$ are nonunits whose sum is the unit 1. The obstruction to division is therefore not concentrated at one maximal ideal but is spread over the primes of $\mathbb{K}[L]$: for each monic irreducible $\pi \in \mathbb{K}[L]$, the set $\pi\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ is a proper nonzero ideal, and $\bigcap_\pi \pi\mathscr{PL}_{\mathrm{poly},\mathbb{K}} = 0$.*

Proof. The first two sentences have been proved. For the last, $\pi\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ is an ideal; it is nonzero and proper because π is a nonzero nonunit (Theorem 9.3). If $F \neq 0$ lay in $\pi\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ for every π, its leading block $p_{\nu_t F}$ would be divisible in $\mathbb{K}[L]$ by every monic irreducible – by Theorem 9.2 the leading block of πG is π times the leading block of G – and $\mathbb{K}[L]$ has infinitely many pairwise nonassociate irreducibles (Euclid's argument applies verbatim), forcing $p_{\nu_t F} = 0$. □

9.6 Conditional divisibility

A nonunit divisor still divides some elements exactly. The following proposition makes the test explicit; it also shows that "being divisible" is a genuinely global condition, so that no unconditional division theorem in $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ is available.

Proposition 9.24 (Coefficientwise divisibility test)**.** *Let $B \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ be nonzero with $\nu_t B = 0$ and leading block $B_0 \in \mathbb{K}[L]$, and let $A \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$. Let $Q = A/B \in \mathbb{K}(L)[[t]]$ have blocks Q_n given by Equation* (9.8) *over $E = \mathbb{K}(L)$. Then:*

1. *$A/B \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ if and only if $Q_n \in \mathbb{K}[L]$ for every $n \geq 0$;*
2. *given $Q_0, \ldots, Q_{n-1} \in \mathbb{K}[L]$, the block Q_n lies in $\mathbb{K}[L]$ if and only if B_0 divides $A_n - \sum_{j=1}^{n} B_j Q_{n-j}$ in $\mathbb{K}[L]$;*
3. *the set $\{A \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}} : A/B \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}\}$ is exactly the principal ideal $B\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$, which is nonzero, and proper as soon as B is not a unit.*

Proof. (1) is the definition of $\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ as the set of series with polynomial blocks, together with the uniqueness in Theorem 9.8: the blocks of A/B computed over $\mathbb{K}(L)$ are the only candidates. (2) is immediate from Equation (9.8), since $Q_n = B_0^{-1} R_n$ with $R_n = A_n - \sum_{j=1}^{n} B_j Q_{n-j} \in \mathbb{K}[L]$ under the stated hypothesis. For (3): $A/B \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ means $A = B \cdot (A/B)$ with $A/B \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$, i.e. $A \in B\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$; conversely $A = BC$ with $C \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ gives $A/B = C$. The ideal $B\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ contains $B \neq 0$, and it equals $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ exactly when $1 \in B\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$, i.e. when B is a unit. □

Example 9.25 (Exact division by a nonunit, and a near miss)**.** Take $B = L + t$, a nonzero nonunit. Then

$$\frac{L^2 - t^2}{L + t} = L - t \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}, \tag{9.19}$$

verified by the single multiplication $(L+t)(L-t) = L^2 - t^2$. So a nonunit can divide exactly. Change one sign and the divisibility fails at the third step: with $A = L^2 + t^2$, the test of Theorem 9.24 (2) gives

$$Q_0 = \frac{L^2}{L} = L, \qquad Q_1 = \frac{0 - 1 \cdot L}{L} = -1, \qquad Q_2 = \frac{1 - 1 \cdot (-1)}{L} = \frac{2}{L} \notin \mathbb{K}[L],$$

consistently with the exact identity $\dfrac{L^2 + t^2}{L + t} = L - t + \dfrac{2t^2}{L + t}$. Two successful steps prove nothing about the third.

Remark 9.26 (The test is a semi-decision procedure). Theorem 9.24 (2) refutes divisibility in finite time whenever it fails at some finite order. It never confirms divisibility in finite time on its own, because the condition quantifies over all n. A positive answer needs a finite certificate: either a candidate quotient $C \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ with only finitely many blocks – so that $BC = A$ is a finite identity, as in Equation (9.19) – or an a priori bound reducing the infinitely many conditions to finitely many, or a structural argument. Absent such a certificate, a computation that "keeps coming out polynomial" is evidence, not proof.

Structural checkpoint

Three closure questions must be kept apart.

1. Does the quotient exist in *some* field? Always, in $\mathscr{PL}_{\mathrm{rat},\mathbb{K}}$.
2. Does it have rational blocks and nothing worse? Always, again in $\mathscr{PL}_{\mathrm{rat},\mathbb{K}}$; this is Theorem 9.19.
3. Does it have polynomial blocks? Only under the unit criterion, or by a certified accident as in Theorem 9.25.

Answering (1) and reporting it as an answer to (3) is the standard failure mode.

9.7 Newton iteration and the exact precision-doubling law

The reciprocal recurrence produces one block per step. Newton's method for the equation $C^{-1} - B = 0$ produces the map

$$C \longmapsto C(2 - BC), \tag{9.20}$$

which doubles the number of correct blocks. Every source states Equation (9.20) and the identity $1 - B\,C(2 - BC) = (1 - BC)^2$; none states which truncations of B and of C suffice to compute the doubled output, which is precisely what makes the scheme an algorithm rather than an identity. Part (4) below supplies that statement.

Theorem 9.27 (Precision doubling)**.** *Let $B \in \mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ with $B_0 \in \mathbb{K}^\times$, let $C \in \mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$, let $N \geq 1$, and suppose*

$$\nu_t(1 - BC) \geq N. \tag{9.21}$$

Put $C' \coloneqq C(2 - BC)$. Then:

1. *$1 - BC' = (1 - BC)^2$, hence $\nu_t(1 - BC') \geq 2N$;*
2. *$\nu_t\big(B^{-1} - C'\big) = \nu_t(1 - BC') \geq 2N$, and more precisely $\nu_t(B^{-1} - C) = \nu_t(1 - BC)$;*
3. *$\nu_t\big(1 - B\,\mathrm{Trunc}_{<2N}\,C'\big) \geq 2N$, so truncating the output at order $2N$ preserves the invariant;*
4. *$\mathrm{Trunc}_{<2N}\,C'$ depends on B and C only through $\mathrm{Trunc}_{<2N}\,B$ and $\mathrm{Trunc}_{<N}\,C$. Explicitly, if $\overline{B} \in \mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ and $\overline{C} \in \mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ satisfy $\nu_t(B - \overline{B}) \geq 2N$ and $\nu_t(C - \overline{C}) \geq N$, then $\nu_t\big(C' - \overline{C}(2 - \overline{B}\,\overline{C})\big) \geq 2N$.*

Proof. (1) $BC' = BC(2 - BC) = 2BC - (BC)^2$, so $1 - BC' = 1 - 2BC + (BC)^2 = (1 - BC)^2$. Valuations add (Theorem 9.2), so $\nu_t(1 - BC') \geq 2N$.

(2) B is a unit by Theorem 9.3 and $\nu_t(B^{-1}) = -\nu_t B = 0$. From $B^{-1} - C' = B^{-1}\,(1 - BC')$ and additivity of ν_t we get $\nu_t(B^{-1} - C') = \nu_t(1 - BC')$. The same computation with C in place of C' gives the second identity.

(3) $C' - \mathrm{Trunc}_{<2N}\,C' \in t^{2N}\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$, so $B(C' - \mathrm{Trunc}_{<2N}\,C') \in t^{2N}\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ because $\nu_t B = 0$; adding this to (1) gives $\nu_t(1 - B\,\mathrm{Trunc}_{<2N}\,C') \geq 2N$.

(4) Put $\delta = C - \overline{C}$ and $\varepsilon = B - \overline{B}$, so $\nu_t\delta \geq N$ and $\nu_t\varepsilon \geq 2N$. Since $C' = 2C - BC^2$ and $\overline{C}(2 - \overline{B}\,\overline{C}) = 2\overline{C} - \overline{B}\,\overline{C}^2$,

$$C' - \overline{C}(2 - \overline{B}\,\overline{C}) = 2\delta - \big(BC^2 - \overline{B}\,\overline{C}^2\big) = 2\delta - \varepsilon C^2 - \overline{B}\,(C + \overline{C})\,\delta = \delta\big(2 - \overline{B}(C + \overline{C})\big) - \varepsilon C^2,$$

using $BC^2 - \overline{B}\,\overline{C}^2 = \varepsilon C^2 + \overline{B}(C^2 - \overline{C}^2)$ and $C^2 - \overline{C}^2 = (C + \overline{C})\delta$. The second term has $\nu_t \geq 2N$. For the first,

$$2 - \overline{B}(C + \overline{C}) = \big(1 - \overline{B}C\big) + \big(1 - \overline{B}\,\overline{C}\big),$$

and $1 - \overline{B}C = (1 - BC) + \varepsilon C$ has $\nu_t \geq N$, while $1 - \overline{B}\,\overline{C} = (1 - BC) + \varepsilon C + B\delta - \varepsilon\delta$ also has $\nu_t \geq N$. Hence $\nu_t\big(\delta(2 - \overline{B}(C + \overline{C}))\big) \geq N + N = 2N$, and the whole difference has $\nu_t \geq 2N$. ☐

Algorithm 9.28 (Newton reciprocal to exclusive order N)**.** Given an integer $N \geq 1$ and $B \in \mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ with $B_0 \in \mathbb{K}^\times$, known through order t^{N-1}:

```
function newton_reciprocal(B, N):
    require B[0] is a nonzero scalar in K
    C := 1/B[0]                      # exact through order t^0
    m := 1
    while m < N:
        m2 := min(2*m, N)
        E  := Trunc(1 - B*C, m2)    # valuation >= m by the invariant
        C  := Trunc(C*(1 + E), m2)
```

```
        m   := m2
    return C                          # equals Trunc_{<N} of the reciprocal
```

Corollary 9.29 (The iterates are exact truncations). *Let*

$$C^{(0)} = B_0^{-1}, \qquad C^{(k+1)} = \mathrm{Trunc}_{<2^{k+1}}\big(C^{(k)}(2 - BC^{(k)})\big).$$

Then for every $k \geq 0$,

$$C^{(k)} = \mathrm{Trunc}_{<2^k} B^{-1}. \tag{9.22}$$

In particular Theorem 9.28 returns $\mathrm{Trunc}_{<N} B^{-1}$, and the intermediate steps require no blocks of B beyond index $2^{k+1} - 1$ at step k.

Proof. Induction on k. For $k = 0$, $1 - BC^{(0)} = 1 - B_0^{-1}B$ has zero constant block, so $\nu_t(1 - BC^{(0)}) \geq 1$, and by Theorem 9.27 (2) $\nu_t(B^{-1} - C^{(0)}) \geq 1$; as $C^{(0)}$ has t-degree $0 < 1$, it equals $\mathrm{Trunc}_{<1} B^{-1}$. Assume $C^{(k)} = \mathrm{Trunc}_{<2^k} B^{-1}$; then $\nu_t(B^{-1} - C^{(k)}) \geq 2^k$, so $\nu_t(1 - BC^{(k)}) \geq 2^k$ by Theorem 9.27 (2). Parts (1) and (3) give $\nu_t(1 - BC^{(k+1)}) \geq 2^{k+1}$, hence $\nu_t(B^{-1} - C^{(k+1)}) \geq 2^{k+1}$ by (2) again; since $C^{(k+1)}$ has t-degree $< 2^{k+1}$, it is $\mathrm{Trunc}_{<2^{k+1}} B^{-1}$.

The capped step. In Theorem 9.28 the new precision is $m' = \min\{2m, N\} \leq 2m$. Suppose $C = \mathrm{Trunc}_{<m} B^{-1}$ at the top of the loop, so $\nu_t(B^{-1} - C) \geq m$ and hence $\nu_t(1 - BC) \geq m$ by Theorem 9.27 (2). Part (1) gives $\nu_t(1 - BC') \geq 2m \geq m'$ for $C' = C(2 - BC)$, and the computation in the proof of part (3), run with m' in place of $2N$, gives $\nu_t\big(1 - B\,\mathrm{Trunc}_{<m'} C'\big) \geq m'$; by (2) once more $\nu_t\big(B^{-1} - \mathrm{Trunc}_{<m'} C'\big) \geq m'$, and $\mathrm{Trunc}_{<m'} C'$ has t-degree $< m'$, so it equals $\mathrm{Trunc}_{<m'} B^{-1}$. The invariant holds initially with $m = 1$ and the loop terminates with $m = N$, so the returned value is $\mathrm{Trunc}_{<N} B^{-1}$ for every $N \geq 1$, not only for N a power of two. The statement about required blocks of B is Theorem 9.27 (4). □

Remark 9.30 (A wording to avoid). Writing $E = 1 - BC$, the step $\nu_t(E^2) \geq 2N$ holds because ν_t is *additive* on products (Theorem 9.2), not because it is multiplicative; ν_t is a homomorphism from the multiplicative monoid of nonzero elements to $(\mathbb{Z}, +)$.

Remark 9.31 (Where the iteration lives). Theorem 9.27 used only that ν_t is additive and that B has an invertible leading block; it therefore holds verbatim over the coefficient rings $\mathbb{K}(L)$ and $\mathbb{K}((L^{-1}))$, and in $\mathscr{PL}_{\mathrm{itLaur},\mathbb{K}}$ after normalising the leading Laurent-log coefficient. It is an exact algebraic identity, not an approximation: unlike numerical Newton iteration, there is no convergence hypothesis and no loss of significance – by Theorem 9.29 each iterate is literally a truncation of the answer, so the logarithmic degrees appearing in the iterates are exactly those of B^{-1} and cannot swell beyond them, provided one truncates at every step.

9.8 Cost: the recurrence against the doubling scheme

We count multiplications in the coefficient ring $\mathbb{K}[L]$: every product of two coefficient blocks costs one *block multiplication*, including those products one of whose factors happens to be the scalar block $C_0 = B_0^{-1}$. Additions are free, and so is the single outer normalisation by B_0^{-1} that Equations (9.7) and (9.8) apply once per step. Let $M(N)$ denote the number of block multiplications used by whatever routine computes $\mathrm{Trunc}_{<N}(PQ)$ from $\mathrm{Trunc}_{<N} P$ and $\mathrm{Trunc}_{<N} Q$.

Proposition 9.32 (Cost of the recurrence). *Computing $\mathrm{Trunc}_{<N} B^{-1}$ by Equations (9.6) and (9.7) uses exactly*

$$\binom{N}{2} = \frac{N(N-1)}{2} \tag{9.23}$$

block multiplications, and computing $\mathrm{Trunc}_{<N}(A/B)$ by Equation (9.8) uses the same number.

Proof. Step n forms the products B_jC_{n-j} for $j = 1, \dots, n$, that is n block multiplications, for $n = 0, 1, \dots, N-1$. The total is $\sum_{n=0}^{N-1} n = N(N-1)/2$. □

Proposition 9.33 (Cost of the doubling scheme). *Let $N = 2^K$.*

1. *If M is nondecreasing and satisfies $2M(n) \le M(2n)$, then Theorem 9.28 uses at most $4M(N)$ block multiplications.*
2. *With schoolbook truncated multiplication, and exploiting the invariant $\nu_t(1 - BC) \ge m$ so that the first m blocks of BC are known in advance, Theorem 9.28 uses exactly*

$$\frac{N^2 + N - 2}{2} \tag{9.24}$$

block multiplications, which exceeds Equation (9.23) *by exactly $N - 1$.*

Proof. (1) The step from precision m to $2m$ forms two truncated products at order $2m$, so costs at most $2M(2m)$. Summing over $m = 1, 2, 4, \dots, 2^{K-1}$ gives at most $\sum_{j=1}^{K} 2M(2^j)$. The hypothesis $2M(n) \le M(2n)$ gives $M(2^j) \le M(2^K)/2^{K-j}$, so the sum is at most $2M(N)\sum_{j=1}^{K} 2^{j-K} \le 4M(N)$.

(2) Consider the step from m to $2m$, with $C = \mathrm{Trunc}_{<m} B^{-1}$ supported on indices $0, \dots, m-1$. *Forming $E = 1 - BC$ modulo t^{2m}.* By the invariant, the blocks of BC of index $< m$ are known to be $1, 0, \dots, 0$; only indices $m, \dots, 2m-1$ must be computed. The needed products are B_jC_i with $0 \le i < m$ and $m \le i + j < 2m$, and for each of the m values of i there are exactly m admissible j. Cost: m^2. *Forming $C(1 + E)$ modulo t^{2m}.* E is supported on indices $m, \dots, 2m-1$, and C on $0, \dots, m-1$; the products C_iE_j with $i+j < 2m$ number $\sum_{j=m}^{2m-1}(2m - j) = \sum_{s=1}^{m} s = m(m+1)/2$. Total for the step: $m^2 + m(m+1)/2 = (3m^2 + m)/2$. Summing over $m = 2^k$, $k = 0, \dots, K-1$,

$$\sum_{k=0}^{K-1} \frac{3 \cdot 4^k + 2^k}{2} = \frac{3}{2} \cdot \frac{4^K - 1}{3} + \frac{2^K - 1}{2} = \frac{N^2 - 1}{2} + \frac{N - 1}{2} = \frac{N^2 + N - 2}{2}.$$

Subtracting Equation (9.23) gives $(2N - 2)/2 = N - 1$.

How fragile the gap is. Suppose instead that a product one of whose factors is the scalar $C_0 = B_0^{-1}$ is charged nothing. That removes one product per step $n = 1, \dots, N-1$ from the recurrence, hence $N - 1$ in all, and $2m$ from the doubling step at precision m (the m products B_jC_0 and the m products C_0E_j), hence $\sum_{k=0}^{K-1} 2 \cdot 2^k = 2N - 2$ in all. Both counts then drop to $\binom{N-1}{2}$ and the two schemes cost *exactly the same.* The $N - 1$ gap of part (2) is therefore an artefact of the accounting; what is robust in the comparison is part (1). □

Corollary 9.34 (When doubling pays). *Which of the two schemes wins is decided by the growth of M, and the two counts settle the two extremes. Precisely: if $M(N)/N^2 \to 0$ as $N \to \infty$ (with M as in Theorem 9.33 (1)), then the doubling scheme uses $o(N^2)$ block multiplications against $\Theta(N^2)$ for the recurrence; whereas with schoolbook block multiplication it uses exactly $N - 1$* more *block multiplications than the recurrence in the accounting fixed at the head of Section 9.8, and exactly as many under the alternative accounting described at the end of the proof of Theorem 9.33. Under neither accounting is it faster.*

Proof. Combine Theorems 9.32 and 9.33; the first claim is part (1) with the stated growth hypothesis, the second is part (2). □

Remark 9.35 (Scalar cost under an affine degree profile). Block multiplications are not unit-cost. Assume the hypothesis of Theorem 9.15 with $\alpha > 0$ an integer, so $\deg_{\mathrm{L}} B_n \le \alpha n$ and $\deg_{\mathrm{L}} C_n \le \alpha n$, and count $\mathbb{K}$-multiplications with schoolbook polynomial arithmetic in L. The recurrence then costs at most

$$\sum_{n=1}^{N-1}\sum_{j=1}^{n}(\alpha j+1)\big(\alpha(n-j)+1\big)=\Theta\big(\alpha^2N^4\big),\qquad \text{with leading term } \frac{\alpha^2N^4}{24},$$

because $\sum_{j=1}^{n} j(n-j) = \binom{n+1}{3} \sim n^3/6$ and $\sum_{n<N}\binom{n+1}{3} = \binom{N+1}{4} \sim N^4/24$. Quoting a bound $\mathcal{O}(N^2d^2)$ with a fixed d therefore understates the cost whenever the log-degree profile grows; substituting the honest $d \asymp \alpha N$ recovers α^2N^4. This quartic behaviour, not the quadratic block count, is what makes truncating after *every* multiplication the decisive implementation discipline; see [33] for the relaxed and fast-multiplication variants.

9.9 The embedding $\mathscr{PL}_{\mathrm{rat},\mathbb{K}} \hookrightarrow \mathscr{PL}_{\mathrm{itLaur},\mathbb{K}}$

The third response to a nonconstant leading block is to expand each rational coefficient at $L=\infty$. This resolves a finer asymptotic scale, and it is the natural home of a multiseries implementation [29]; it is also the step most likely to be taken without noticing that the object has changed.

Definition 9.36 (Iterated Laurent-log field). $\mathbb{K}((L^{-1}))$ is the field of formal Laurent series in the variable L^{-1}: elements $\sum_{j\geq j_0} a_jL^{-j}$ with $j_0\in\mathbb{Z}$ and $a_j\in\mathbb{K}$, so that only finitely many positive powers of L occur but arbitrarily many negative ones may. Then $\mathscr{PL}_{\mathrm{itLaur},\mathbb{K}} := \mathbb{K}((L^{-1}))((t))$, whose elements are $\sum_{n\geq n_0} t^n \sum_{j\geq j_n} a_{n,j}L^{-j}$.

Theorem 9.37 (The expansion map). *There is exactly one $\mathbb{K}$-algebra homomorphism $\iota_0 : \mathbb{K}(L)\to\mathbb{K}((L^{-1}))$ with $\iota_0(L)=L$, namely Laurent expansion at $L=\infty$, and it is injective. Applying it blockwise defines an injective ring homomorphism $\iota : \mathscr{PL}_{\mathrm{rat},\mathbb{K}}\to\mathscr{PL}_{\mathrm{itLaur},\mathbb{K}}$ that restricts to the identity on $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$. The map ι is* not *surjective; its image is therefore a proper subfield of $\mathscr{PL}_{\mathrm{itLaur},\mathbb{K}}$.*

Proof. Existence on coefficients. A polynomial $\sum_{i\leq d} c_iL^i$ already lies in $\mathbb{K}((L^{-1}))$ (as $\sum_{j\geq -d} c_{-j}L^{-j}$), giving an injective ring map $\mathbb{K}[L]\hookrightarrow\mathbb{K}((L^{-1}))$. Since $\mathbb{K}((L^{-1}))$ is a field (Theorem 9.19), every nonzero polynomial maps to a unit, so by the universal property of the localisation $\mathbb{K}(L)=\operatorname{Frac}\mathbb{K}[L]$ the map extends uniquely to a ring homomorphism $\iota_0:\mathbb{K}(L)\to\mathbb{K}((L^{-1}))$; a ring homomorphism out of a field is injective. Concretely, for $q=q_dL^d(1+w)$ with $q_d\in\mathbb{K}^\times$ and $w\in L^{-1}\mathbb{K}[L^{-1}]$, $\iota_0(1/q)=q_d^{-1}L^{-d}\sum_{k\geq 0}(-w)^k$, the series being summable because w has positive L^{-1}-order.

Uniqueness of ι_0 is the uniqueness clause in that universal property: a $\mathbb{K}$-algebra map on $\mathbb{K}(L)$ is determined by the image of L.

Extension to Laurent series. Define $\iota\big(\sum_{n\geq n_0} f_nt^n\big)=\sum_{n\geq n_0}\iota_0(f_n)t^n$. The support stays bounded below, so the image is in $\mathscr{PL}_{\mathrm{itLaur},\mathbb{K}}$. Applying a ring homomorphism blockwise to Laurent series is again a ring homomorphism, because each block of a sum or a Cauchy product is a finite algebraic expression in the blocks of the arguments; and ι is injective because ι_0 is. On $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ every block is a polynomial and ι_0 is the inclusion, so ι is the identity there.

Not surjective. Since $\operatorname{char}\mathbb{K}=0$, the binomial series $\sigma:=\sum_{k\geq 0}\binom{1/2}{k}L^{-k}\in\mathbb{K}[[L^{-1}]]$ is defined and satisfies $\sigma^2=1+L^{-1}$, the identity holding coefficientwise because it holds in $\mathbb{Q}[[u]]$ for $u=L^{-1}$. Suppose $\sigma=\iota_0(f)$ for some $f\in\mathbb{K}(L)$. Then $f^2=(L+1)/L$, so $\big(Lf\big)^2=L(L+1)$ with $Lf\in\mathbb{K}(L)$. Writing $Lf=u/v$ in lowest terms gives $u^2=L(L+1)v^2$; comparing multiplicities of the irreducible L on the two sides of this equation in the unique factorisation domain $\mathbb{K}[L]$ yields an even number on the left and an odd one on the right, a contradiction. Hence σ is not in the image of ι_0, and σ regarded as a t^0-block is not in the image of ι. □

Example 9.38 (The exact coefficient becomes an infinite series).

$$\iota\Big(\frac{1}{L+1}\Big) = L^{-1} - L^{-2} + L^{-3} - \cdots = \sum_{j\geq 1}(-1)^{j-1}L^{-j}, \qquad \iota\Big(\frac{1}{L+t}\Big) = \sum_{n\geq 0}(-1)^n L^{-n-1}t^n. \tag{9.25}$$

The left-hand sides are single finite data; the right-hand sides are infinite.

Proposition 9.39 (The iteration order is part of the type). *Regard both $\mathscr{P}\mathscr{L}_{\mathrm{itLaur},\mathbb{K}} = \mathbb{K}((L^{-1}))((t))$ and $\mathbb{K}((t))((L^{-1}))$ as sets of families $(a_{n,j})$ of scalars indexed by $(n,j) \in \mathbb{Z}^2$, representing $\sum a_{n,j}t^n L^{-j}$. For the first, the support condition is: the set of n carrying a nonzero entry is bounded below, and for each such n the set of j with $a_{n,j} \neq 0$ is bounded below. For the second, the roles of n and j are exchanged. Then neither set contains the other. Witnesses:*

$$\sum_{n\geq 0} L^n t^n \in \mathbb{K}((L^{-1}))((t)) \setminus \mathbb{K}((t))((L^{-1})), \qquad \sum_{j\geq 0} L^{-j}t^{-j} \in \mathbb{K}((t))((L^{-1})) \setminus \mathbb{K}((L^{-1}))((t)). \tag{9.26}$$

Proof. The first element has $a_{n,-n} = 1$ for $n \geq 0$ and all other entries zero. Its outer index n is bounded below by 0, and for each n exactly one j occurs, so the first support condition holds; indeed the element lies already in $\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$. But for the second condition one must read the family by inner index: the set of j carrying a nonzero entry is $\{0,-1,-2,\dots\}$, which is not bounded below, so the element is not in $\mathbb{K}((t))((L^{-1}))$.

The second element has $a_{-j,j} = 1$ for $j \geq 0$ and all other entries zero. Its inner index j is bounded below by 0, and for each j exactly one n occurs, so it lies in $\mathbb{K}((t))((L^{-1}))$. But the set of n carrying a nonzero entry is $\{0,-1,-2,\dots\}$, so it has infinitely many positive powers of X and is excluded from $\mathbb{K}((L^{-1}))((t))$. □

Expanding at $L = \infty$ changes the object

Three distinct things go wrong if ι is applied silently.

1. *Finiteness.* An exact block such as $1/(L+1)$ becomes an infinite series. A finite representation in $\mathscr{P}\mathscr{L}_{\mathrm{itLaur},\mathbb{K}}$ therefore needs an outer t-cutoff *and* an inner L^{-1}-cutoff, or one declared staircase in the lexicographic monomial order. An outer cutoff alone does not make the data finite.

2. *Algebra.* ι is injective but not surjective (Theorem 9.37), so $\mathscr{P}\mathscr{L}_{\mathrm{rat},\mathbb{K}}$ sits inside $\mathscr{P}\mathscr{L}_{\mathrm{itLaur},\mathbb{K}}$ as a proper subfield: an identity established in $\mathscr{P}\mathscr{L}_{\mathrm{itLaur},\mathbb{K}}$ need not descend to $\mathscr{P}\mathscr{L}_{\mathrm{rat},\mathbb{K}}$, and an element of $\mathscr{P}\mathscr{L}_{\mathrm{itLaur},\mathbb{K}}$ need not be a rational function of L at all. Moreover the leading data change meaning – in $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ each block has a finite $\deg_{\mathrm{L}}$, while in $\mathscr{P}\mathscr{L}_{\mathrm{itLaur},\mathbb{K}}$ the relevant invariant is the L^{-1}-adic order.

3. *Asymptotics.* Let $\mathbb{K} \subseteq \mathbb{C}$ and let $f \in \mathbb{K}(L)$ have largest pole modulus $\rho(f)$, with $\rho(f) = 0$ when f is a polynomial and $\iota_0(f)$ a finite sum. The numerical series $\iota_0(f)$ converges for $|L| > \rho(f)$, that is for $X > \mathrm{e}^{\rho(f)}$ on the positive ray, and for nonpolynomial f diverges for $|L| < \rho(f)$. There is in general *no* threshold valid for all blocks at once: for $F = \sum_{n\geq 0} t^n/(L-n) \in \mathscr{P}\mathscr{L}_{\mathrm{rat},\mathbb{K}}$ the n-th block requires $X > \mathrm{e}^n$. Hence “F equals its L^{-1}-expansion for large X” is not a theorem about F; it is a statement about one block at a time, and any uniform claim needs its own proof with the regime printed as $\mathcal{O}_{X\to+\infty}(\cdot)$.

Remark 9.40 (When to expand and when not to). Keeping $1/(L+1)$ unexpanded is exact, finite, and preserves cancellation; expanding it resolves the finer scale $t \prec \cdots \prec L^{-2} \prec L^{-1} \prec 1$ and is required as soon as one must compare two blocks of the same t-order. The decision is a modelling decision and should be recorded, not inferred: a system that expands rational coefficients automatically will destroy exact cancellations and swell expressions, while one that never expands cannot answer questions about the inner scale. The same tension appears in every multiseries implementation [29, 30].

Section summary

An element of $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ is invertible there exactly when its leading coefficient block is a nonzero scalar; the leading *block* is a polynomial, and factoring out the dominant monomial does not by itself invert. Reciprocals of units exist, are unique, and are computed by a triangular recurrence whose truncated output $P = \mathrm{Trunc}_{<N}(A/B)$ is certified by the residual $BP = A + \mathcal{O}_t^{\mathrm{form}}(t^N)$ alone; relative precision is exactly preserved, so M leading input blocks give M leading output blocks. Newton's map $C \mapsto C(2 - BC)$ doubles the correct order and, with the input requirement of Theorem 9.27 (4), yields exact truncations of the reciprocal; it beats the recurrence asymptotically when block multiplication is subquadratic, and does not beat it at all when multiplication is schoolbook. Unrestricted division of coefficient blocks forces $\mathscr{PL}_{\mathrm{rat},\mathbb{K}}$, the smallest Laurent field over an intermediate coefficient ring, which strictly contains the fraction field of $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$. Expanding each rational coefficient at $L = \infty$ embeds $\mathscr{PL}_{\mathrm{rat},\mathbb{K}}$ into $\mathscr{PL}_{\mathrm{itLaur},\mathbb{K}}$ injectively but not surjectively, and converts one exact datum into an infinite series with no uniform analytic threshold.

Chapter 10

Powers, logarithms, exponentials, and the closure boundary

Addition, multiplication and division were settled by finite coefficient algebra. The operations of this chapter – powers with an arbitrary exponent, logarithms, exponentials, and more generally the substitution of a series into a scalar power series – are the first for which the answer to "does the result stay in the class?" is neither always yes nor always no. The purpose of the section is to replace the informal closure tables of the sources by statements with hypotheses and proofs, and in particular to prove the two negative results that fix the boundary of the polynomial–logarithmic scale: a logarithm leaves the scale exactly when the leading coefficient block is nonconstant, and an exponential leaves *every* power–log scale exactly when its argument is not of nonnegative outer order with an affine t^0-block.

Throughout, $\mathbb{K}$ is a field of characteristic zero, $t := X^{-1}$ and $L := \log X$ are independent formal generators, the default analytic regime is $X \to +\infty$ on the positive ray, and a nonzero $F \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}} = \mathbb{K}[L]((t))$ is written

$$F = \sum_{n \geq n_0} p_n(L)t^n, \qquad p_n \in \mathbb{K}[L], \quad p_{n_0} \neq 0, \quad n_0 = \nu_t F. \tag{10.1}$$

We write $p_{\nu_t F}$ for the leading block, $\deg_{\mathrm{L}} p_{\nu_t F}$ for its degree, and $t\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ for the elements of $\mathscr{PL}_{\mathrm{pow},\mathbb{K}} = \mathbb{K}[L][[t]]$ of strictly positive outer order. Truncation is always the exclusive operator $\mathrm{Trunc}_{<N} F = \sum_{n<N} p_n(L)t^n$, and $F = G + \mathcal{O}_t^{\mathrm{form}}(t^N)$ abbreviates $\nu_t(F - G) \geq N$, a statement about coefficients only.

Two facts proved elsewhere in the volume are used freely: the unit criterion (Theorem 9.3), namely that F is a unit of $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ if and only if its leading block is a nonzero scalar; and the expansion embedding $\iota\colon \mathscr{PL}_{\mathrm{rat},\mathbb{K}} \hookrightarrow \mathscr{PL}_{\mathrm{itLaur},\mathbb{K}}$ (Theorem 9.37), Laurent expansion of each coefficient at $L = \infty$. On $\mathbb{K}((L^{-1}))$ we use the L^{-1}-adic order ω of Theorem 9.17: $\omega(\sum_{j \geq j_0} a_j L^{-j}) = \min\{j : a_j \neq 0\}$, so that $\omega(p) = -\deg_{\mathrm{L}} p$ for a nonzero polynomial p, ω is additive on products, and $\omega(h + h') \geq \min\{\omega(h), \omega(h')\}$.

The distinguished derivation $D_X = t\partial_L - t^2\partial_t$ appears in the hypotheses of several statements below; it is constructed, and all of its properties used here are proved, in Part 11, which is independent of the present chapter. The only facts borrowed are the block formula $D_X(t^n p) = t^{n+1}(p' - np)$, the resulting bound $\nu_t(D_X F) \geq \nu_t F + 1$, and $\ker D_X = \mathbb{K}$.

10.1 Substituting a small series into a scalar power series

Every construction in this section rests on one mechanism: a series of positive outer order may be substituted into an arbitrary formal power series over $\mathbb{K}$, because the outer order of the r-th power grows with r. All six sources state this; we record it once, in the form that also delivers the finite-determination statement that a computation actually needs.

Lemma 10.1 (Summable sequences and blockwise assembly). *Let $(F_r)_{r\geq 0}$ be a family in $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ with $\nu_t F_r \to +\infty$ and $\inf_r \nu_t F_r > -\infty$. Then for each $n \in \mathbb{Z}$ only finitely many r have $[t^n]F_r \neq 0$, the element*

$$\sum_{r\geq 0} F_r := \sum_{n\in\mathbb{Z}} \Big(\sum_{r\geq 0} [t^n]F_r\Big) t^n$$

is a well-defined member of $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$, and the assignment is $\mathbb{K}$-linear and compatible with multiplication by a fixed element: $G\sum_r F_r = \sum_r GF_r$. The same statements hold in $\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$, in $\mathscr{PL}_{\mathrm{rat},\mathbb{K}}$ and in $\mathscr{PL}_{\mathrm{itLaur},\mathbb{K}}$, with the coefficient ring changed accordingly.

Proof. Fix n. Since $\nu_t F_r \to +\infty$, the set $R_n = \{r : \nu_t F_r \leq n\}$ is finite, and $[t^n]F_r = 0$ for $r \notin R_n$; so the inner sum is a finite sum in the coefficient ring. The common lower bound $m = \inf_r \nu_t F_r$ is finite and gives $[t^n]F_r = 0$ for all $n < m$, so the support of the result is bounded below and the result lies in $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$. Linearity is immediate. For the last claim, $[t^n](G\sum_r F_r) = \sum_{i+j=n}[t^i]G\sum_r [t^j]F_r$, a finite double sum that may be reordered. No step used that the coefficients are polynomials. □

Proposition 10.2 (Substitution of a small series into a scalar power series). *Let $U \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ with $q := \nu_t U \geq 1$. For $\Phi = \sum_{r\geq 0} c_r z^r \in \mathbb{K}[[z]]$ put*

$$\Phi(U) := \sum_{r\geq 0} c_r U^r. \tag{10.2}$$

Then the sum is defined by Theorem 10.1, $\Phi(U) \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ with $\nu_t \Phi(U) \geq q \cdot \mathrm{ord}_z \Phi$, and

$$\Phi \longmapsto \Phi(U)$$

is a $\mathbb{K}$-algebra homomorphism $\mathbb{K}[[z]] \to \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ sending z to U. It is the unique such homomorphism that is continuous for the z-adic and t-adic topologies. The same holds with $\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ replaced by $\mathbb{K}(L)[[t]]$ or $\mathbb{K}((L^{-1}))[[t]]$.

Proof. $\nu_t(U^r) \geq rq \to +\infty$ and all orders are ≥ 0, so Theorem 10.1 applies and gives $\Phi(U) \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$; if $c_r = 0$ for $r < r_0$ then every surviving term has order $\geq r_0 q$. Additivity is clear. For multiplicativity, let $\Psi = \sum_s d_s z^s$. Then

$$\Phi(U)\Psi(U) = \sum_r c_r U^r \Psi(U) = \sum_r \sum_s c_r d_s U^{r+s},$$

where the first step is the last clause of Theorem 10.1 and the second applies it again inside each term; the resulting double family is summable because $\nu_t(U^{r+s}) \geq (r+s)q$, so it may be summed by the antidiagonals $r + s = m$, giving $\sum_m (\sum_{r+s=m} c_r d_s) U^m = (\Phi\Psi)(U)$. Uniqueness: a continuous homomorphism is determined on the dense subring $\mathbb{K}[z]$, where it is determined by the image of z. □

The next statement is what a computation uses. The sources give only its first half – that few powers contribute – and leave implicit which blocks of U must be known; both halves are needed to certify a truncated result.

Theorem 10.3 (Finite determination of a substituted scalar series). *Let $U \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ with $\nu_t U = q \geq 1$ and $\Phi = \sum_r c_r z^r \in \mathbb{K}[[z]]$. For $N \leq 0$, $\mathrm{Trunc}_{<N} \Phi(U) = 0$. For $N \geq 1$, put $m = \lfloor (N-1)/q \rfloor$. Then*

$$\mathrm{Trunc}_{<N} \Phi(U) = \mathrm{Trunc}_{<N} \sum_{r=0}^{m} c_r \big(\mathrm{Trunc}_{<N} U\big)^r. \tag{10.3}$$

In particular $\mathrm{Trunc}_{<N} \Phi(U)$ depends only on the $m+1$ scalars $c_0, \ldots, c_m$ and on the blocks $[t^n]U$ with $n < N$, and $[t^N]\Phi(U)$ receives contributions only from the powers U^r with $r \leq \lfloor N/q \rfloor$.

Proof. Both sides lie in $\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$, so the case $N \le 0$ is trivial. Let $N \ge 1$. If $r > m$ then $r \ge m+1 > (N-1)/q$, hence $rq > N-1$ and $rq \ge N$; since $\nu_t(c_r U^r) \ge rq$, such terms do not affect $\mathrm{Trunc}_{<N} \cdot$, which proves that only $r \le m$ matter. For the replacement of U by $\mathrm{Trunc}_{<N} U$, write $E = U - \mathrm{Trunc}_{<N} U$, so $\nu_t E \ge N$. For $r \ge 1$,

$$U^r - \bigl(\mathrm{Trunc}_{<N} U\bigr)^r = E \sum_{i=0}^{r-1} U^i \bigl(\mathrm{Trunc}_{<N} U\bigr)^{r-1-i}$$

has outer order $\ge N$, because every factor in the sum has nonnegative order. Hence the two truncations agree. The last clause is the same computation with N replaced by $N+1$: $rq > N$ forces $[t^N](c_r U^r) = 0$. □

Corollary 10.4 (The elementary series). *Let $U \in t\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ and $\alpha \in \mathbb{K}$. Then*

$$(1+U)^\alpha := \sum_{r\ge 0} \binom{\alpha}{r} U^r \in 1 + t\mathscr{PL}_{\mathrm{pow},\mathbb{K}}, \tag{10.4}$$

$$\log(1+U) := \sum_{r\ge 1} \frac{(-1)^{r+1}}{r} U^r \in t\mathscr{PL}_{\mathrm{pow},\mathbb{K}}, \tag{10.5}$$

$$\exp U := \sum_{r\ge 0} \frac{U^r}{r!} \in 1 + t\mathscr{PL}_{\mathrm{pow},\mathbb{K}} \tag{10.6}$$

are well defined, and every identity between formal power series over $\mathbb{K}$ in one or two variables that involves only these operations transfers. In particular, for $U, V \in t\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$,

$$(1+U)^\alpha (1+U)^\beta = (1+U)^{\alpha+\beta}, \qquad \bigl((1+U)^\alpha\bigr)^n = (1+U)^{n\alpha}\ (n \in \mathbb{Z}), \tag{10.7}$$

$$\exp(U+V) = (\exp U)(\exp V), \qquad \exp\bigl(\log(1+U)\bigr) = 1+U, \qquad \log(\exp U) = U, \tag{10.8}$$

$$\log\bigl((1+U)(1+V)\bigr) = \log(1+U) + \log(1+V), \qquad (1+U)^\alpha = \exp\bigl(\alpha \log(1+U)\bigr). \tag{10.9}$$

Proof. Each of Equations (10.4) to (10.6) is $\Phi(U)$ for a $\Phi \in \mathbb{K}[[z]]$ (here $\operatorname{char} \mathbb{K} = 0$ is used to divide by r and $r!$), so Theorem 10.2 applies; the order statements come from $\mathrm{ord}_z \Phi$. For the identities, note that a one-variable identity $\Phi_1 = \Phi_2$ in $\mathbb{K}[[z]]$ is carried to $\Phi_1(U) = \Phi_2(U)$ by the homomorphism. For the two-variable identities, the same argument applies to the unique continuous $\mathbb{K}$-algebra homomorphism $\mathbb{K}[[z_1, z_2]] \to \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ with $z_i \mapsto U, V$, which exists by the proof of Theorem 10.2 verbatim; the required identities $\exp(z_1+z_2) = \exp z_1 \exp z_2$ and so on are classical in $\mathbb{Q}[[z_1, z_2]]$. The two identities in Equation (10.8) that involve a composition use one further step: if $\Psi \in \mathbb{K}[[z]]$ has $\Psi(0) = 0$, then $(\Phi \circ \Psi)(U) = \Phi(\Psi(U))$ for every Φ. Indeed both $\Phi \mapsto (\Phi \circ \Psi)(U)$ and $\Phi \mapsto \Phi(\Psi(U))$ are continuous $\mathbb{K}$-algebra homomorphisms $\mathbb{K}[[z]] \to \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ – the first as the composite of $\Phi \mapsto \Phi \circ \Psi$, which is Theorem 10.2 applied inside $\mathbb{K}[[z]]$, with evaluation at U; the second directly, since $\nu_t \Psi(U) \ge 1$ – and both send $z \mapsto \Psi(U)$, so they agree by the uniqueness clause. Applying this to the classical identities $\exp(\log(1+z)) = 1+z$ and $\log(\exp z) = z$ in $\mathbb{Q}[[z]]$ gives the two displayed identities. □

For explicit coefficients there are two standard devices, a recurrence and a closed form; each is preferable in a different situation.

Proposition 10.5 (Coefficients of a substituted series). *Let $U = \sum_{n\ge 1} u_n t^n \in t\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ and $\Phi = \sum_{r\ge 0} c_r z^r \in \mathbb{K}[[z]]$. Then $[t^0]\Phi(U) = c_0$ and, for $N \ge 1$,*

$$[t^N]\Phi(U) = \sum_{k=1}^{N} c_k \frac{k!}{N!} \mathcal{B}^{\exp}_{N,k}\bigl(1!\,u_1, 2!\,u_2, \ldots, (N-k+1)!\,u_{N-k+1}\bigr). \tag{10.10}$$

In particular, with $E = \exp U$ and $M = \log(1+U)$,

$$[t^N]E = \frac{1}{N!} \mathcal{B}^{\exp}_{N}\bigl(1!\,u_1, \ldots, N!\,u_N\bigr), \tag{10.11}$$

$$[t^N]M = \frac{1}{N!}\sum_{k=1}^{N}(-1)^{k-1}(k-1)!\,\mathcal{B}^{\exp}_{N,k}\big(1!\,u_1,\dots,(N-k+1)!\,u_{N-k+1}\big). \tag{10.12}$$

Moreover E and M satisfy the recurrences

$$N\,[t^N]E = \sum_{m=1}^{N} m\,u_m\,[t^{N-m}]E, \qquad N\,[t^N]M = N\,u_N - \sum_{m=1}^{N-1} m\,[t^m]M\;u_{N-m}, \tag{10.13}$$

which compute all blocks through t^N with $\mathcal{O}(N^2)$ coefficient multiplications.

Proof. The partial exponential Bell polynomials are defined [4] by the identity

$$\frac{1}{k!}\Big(\sum_{m\ge 1} a_m\frac{z^m}{m!}\Big)^k = \sum_{n\ge k}\mathcal{B}^{\exp}_{n,k}(a_1,a_2,\dots)\frac{z^n}{n!}$$

in $\mathbb{Q}[a_1,a_2,\dots][[z]]$; since it is an identity with universal coefficients it holds after any specialisation into a commutative $\mathbb{Q}$-algebra. Specialise $a_m = m!\,u_m \in \mathbb{K}[L]$ and $z=t$; then $\sum_m a_m z^m/m! = U$, and comparing blocks gives $[t^n]U^k = (k!/n!)\mathcal{B}^{\exp}_{n,k}(1!u_1, 2!u_2,\dots)$, the arguments beyond index $n-k+1$ being irrelevant. Summing over k and using $[t^N]U^k = 0$ for $k>N$ gives Equation (10.10). Equations (10.11) and (10.12) are the cases $c_k = 1/k!$ and $c_k = (-1)^{k-1}/k$, together with $\mathcal{B}^{\exp}_N = \sum_{k=1}^N \mathcal{B}^{\exp}_{N,k}$ for $N\ge 1$.

For the recurrences, let ∂_t be the formal partial derivative holding L fixed. It is a derivation of $\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ with $\nu_t(\partial_t F)\ge \nu_t F - 1$, hence commutes with summable sums, so applying it to Equation (10.6) termwise gives $\partial_t E = (\partial_t U)E$. Comparing the coefficient of t^{N-1} and using $[t^{N-1}]\partial_t E = N[t^N]E$ and $[t^{m-1}]\partial_t U = m u_m$ yields the first recurrence. Likewise $\partial_t M = (\partial_t U)(1+U)^{-1}$, that is $(1+U)\partial_t M = \partial_t U$; comparing coefficients of t^{N-1} gives $N[t^N]M + \sum_{m=1}^{N-1} m[t^m]M\,u_{N-m} = N u_N$. Each recurrence performs $\mathcal{O}(N)$ coefficient multiplications per block. □

Example 10.6 (A finite logarithm)**.** Let $U = (L+1)t + L^2t^2$, so $q=1$. By Theorem 10.3 with $N=4$, only $r\le 3$ contribute below t^4, and

$$\begin{aligned}\log(1+U) &= U - \tfrac12 U^2 + \tfrac13 U^3 + \mathcal{O}^{\mathrm{form}}_t\big(t^4\big)\\ &= (L+1)t + \Big(L^2 - \tfrac{(L+1)^2}{2}\Big)t^2 + \Big(\tfrac{(L+1)^3}{3} - L^2(L+1)\Big)t^3 + \mathcal{O}^{\mathrm{form}}_t\big(t^4\big)\,.\end{aligned}$$

Every step is a finite polynomial computation, and the certificate $\exp(\log(1+U)) - (1+U) = \mathcal{O}^{\mathrm{form}}_t\big(t^4\big)$ checks it.

10.2 The ambient exponential extension

The exponential of an element of $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ need not lie in $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$, so a statement such as "$\exp F$ leaves the class" is meaningless until an ambient object has been named in which $\exp F$ exists. We therefore fix one once and for all. Nothing below depends on which ambient is chosen: every negative result is proved from the axioms alone, so it holds in all of them simultaneously.

Definition 10.7 (Exponential extension)**.** An *exponential extension* of $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ is a triple $(\mathcal{T},\partial,\exp)$ in which

1. $\mathcal{T}$ is an integral domain containing $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ as a subring, and ∂ is a derivation of $\mathcal{T}$ restricting to D_X on $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$;
2. $\exp\colon (\mathcal{T},+)\to(\mathcal{T}^\times,\cdot)$ is a group homomorphism with $\partial(\exp F) = (\partial F)\exp F$ for all $F\in\mathcal{T}$;
3. (normalisation) $\exp L = t^{-1}$ and $\exp\big(t\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}\big)\subseteq 1 + t\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$.

We write $\mathrm{Const}(\mathcal{T}) = \ker\partial$ for the constants, say that $G \in \mathcal{T}$ is *a logarithm of* F when $\exp G = F$, and for $\alpha \in \mathbb{K}$ and F admitting a logarithm G we set $F^\alpha := \exp(\alpha G)$, a value that depends on the choice of G.

Remark 10.8 (Realisations). For $\mathbb{K} = \mathbb{R}$ the field $\mathbb{T}$ of logarithmic–exponential transseries is such an extension, with ∂ the transseries derivation and $\exp$ its exponential [1, 33, 13]; so is any Hardy field containing the germs at $+\infty$ of $X \mapsto X$ and $X \mapsto \log X$ and closed under $\exp$, for instance the germs of the $\log$–$\exp$-definable functions [32, 20]. Condition (iii) is a normalisation, not a restriction: it merely says that the ambient exponential extends the relation $L = \log X$ and does not multiply small exponentials by a stray constant. Theorem 10.9 shows that (iii) already forces $\exp$ to agree on $t\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ with the series of Equation (10.6).

Lemma 10.9 (The ambient exponential is the series exponential)**.** *Let $(\mathcal{T}, \partial, \exp)$ be an exponential extension. For every $U \in t\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$,*

$$\exp U = \sum_{r\geq 0} \frac{U^r}{r!},$$

the right-hand side being Equation (10.6)*. Consequently, for $V \in 1 + t\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ the element $\log V$ of Equation* (10.5) *is a logarithm of V in $\mathcal{T}$.*

Proof. Write $S = \sum_{r\geq 0} U^r/r! \in 1 + t\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$, a unit of $\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$. Since D_X is a derivation with $\nu_t(D_X F) \geq \nu_t F + 1$, it commutes with summable sums, so $D_X S = \sum_{r\geq 1} U^{r-1}(D_X U)/(r-1)! = (D_X U)S$. Hence $Z := (\exp U)S^{-1}$ lies in $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ by (iii), and

$$\partial Z = (\partial \exp U)S^{-1} - (\exp U)S^{-2} D_X S = (D_X U)Z - (D_X U)Z = 0.$$

Thus $Z \in \ker D_X \cap \mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}} = \mathbb{K}$ (Theorem 11.12), while $Z \in 1 + t\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ by (iii); the only scalar in $1 + t\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ is 1. For the last sentence, apply the first with $U = \log V$ and use Equation (10.8). □

Lemma 10.10 (Logarithmic derivative)**.** *Let $(\mathcal{T}, \partial, \exp)$ be an exponential extension and let $F \in \mathcal{T}$ admit a logarithm G. Then $F \in \mathcal{T}^\times$ and*

$$\partial G = \frac{\partial F}{F}. \tag{10.14}$$

Conversely, if G_1, G_2 are logarithms of F then $G_1 - G_2 \in \mathrm{Const}(\mathcal{T})$, and if $H \in \mathcal{T}$ satisfies $\partial H = \partial F/F$ then $H - G \in \mathrm{Const}(\mathcal{T})$.

Proof. $F = \exp G$ is a unit by (ii). Differentiating $F = \exp G$ gives $\partial F = (\partial G)F$, which is Equation (10.14). If $\exp G_1 = \exp G_2$ then $\exp(G_1 - G_2) = 1$, so $\partial(G_1 - G_2) = \partial(1)/1 = 0$. The last claim is immediate from Equation (10.14). □

10.3 Power–log scales

To say that an operation leaves the polynomial–logarithmic class *and every reasonable enlargement of it* we must quantify over the enlargements. The following definition collects them: fractional or real exponents of X, rational or Laurent coefficients in L, and any enlargement of the scalars.

Definition 10.11 (Power–log scale)**.** Let $\mathbb{K}'$ be a field of characteristic zero containing $\mathbb{K}$, let Γ be an ordered additive subgroup of $\mathbb{R}$ with $\mathbb{Z} \subseteq \Gamma$, fixed together with an embedding $\Gamma \hookrightarrow \mathbb{K}'$ of additive groups extending $\mathbb{Z} \subseteq \mathbb{K}'$ (so that the products γp_γ below are defined), and let C be one of $\mathbb{K}'[L]$, $\mathbb{K}'(L)$, $\mathbb{K}'((L^{-1}))$, each equipped with the derivation ∂_L. The associated *power–log scale* is the Hahn series ring

$$\mathcal{H}(C, \Gamma) := \Big\{\, F = \sum_{\gamma\in S} p_\gamma t^\gamma \;:\; p_\gamma \in C,\ S \subseteq \Gamma \text{ well ordered} \Big\},$$

with the Hahn product [19], the valuation $\nu_t F = \min\{\gamma : p_\gamma \neq 0\}$, and the operator

$$D_X F \coloneqq \sum_{\gamma \in S} t^{\gamma+1}\big(\partial_L p_\gamma - \gamma p_\gamma\big). \tag{10.15}$$

Thus $\mathscr{PL}_{\mathrm{poly},\mathbb{K}} = \mathcal{H}(\mathbb{K}[L],\mathbb{Z})$, $\mathscr{PL}_{\mathrm{rat},\mathbb{K}} = \mathcal{H}(\mathbb{K}(L),\mathbb{Z})$, $\mathscr{PL}_{\mathrm{itLaur},\mathbb{K}} = \mathcal{H}(\mathbb{K}((L^{-1})),\mathbb{Z})$, and the Puiseux-log ring $\mathbb{K}[L]((t^{1/s}))$ is $\mathcal{H}(\mathbb{K}[L], s^{-1}\mathbb{Z})$.

Lemma 10.12 (Basic properties of a power–log scale). *Let $\mathcal{H} = \mathcal{H}(C,\Gamma)$ be a power–log scale.*

1. *$\mathcal{H}$ is an integral domain, ν_t is additive on nonzero products, and D_X is a derivation of $\mathcal{H}$.*
2. *$\omega(\partial_L h) \geq \omega(h) + 1$ for every $h \in C$, where ω denotes the L^{-1}-adic order computed in $\mathbb{K}'((L^{-1}))$ after the expansion embedding (Theorem 9.37). Consequently, for $h \neq 0$, $\omega(\partial_L h/h) \geq 1$, and $\partial_L h - \gamma h \neq 0$ for every $\gamma \in \Gamma \setminus \{0\}$.*
3. *$\nu_t(D_X F) \geq \nu_t F + 1$ for every $F \in \mathcal{H}$, and $\ker D_X = \mathbb{K}'$.*
4. *If $F \in \mathcal{H}$ is nonzero and F^{-1} exists in the fraction field of $\mathcal{H}$, then $\nu_t(D_X F/F) \geq 1$, and*

$$[t^1]\Big(\frac{D_X F}{F}\Big) = \frac{\partial_L b}{b} - e, \qquad e = \nu_t F, \quad b = [t^e]F. \tag{10.16}$$

Proof. (i) Hahn multiplication is well defined and makes $\mathcal{H}$ a domain when C is a domain, and ν_t is additive, because the minimum of the Minkowski sum of two well-ordered sets is the sum of the minima and the corresponding coefficient is the product of the leading coefficients [19]. That D_X is additive is clear; for the Leibniz rule it suffices, by Theorem 10.1 applied in the Hahn setting, to check it on monomials:

$$D_X\big(t^\alpha p \cdot t^\beta q\big) = t^{\alpha+\beta+1}\big((pq)' - (\alpha+\beta)pq\big) = \big(D_X(t^\alpha p)\big)t^\beta q + t^\alpha p\, D_X(t^\beta q).$$

(ii) For $h = \sum_{j \geq j_0} a_j L^{-j}$ with $a_{j_0} \neq 0$, we have $\partial_L h = \sum_j (-j) a_j L^{-j-1}$, whose terms all have order $\geq j_0 + 1$. The case $C = \mathbb{K}'(L)$ reduces to this one because the expansion embedding ι_0 of Theorem 9.37 commutes with ∂_L: it is a ring homomorphism fixing L, hence fixing polynomials and their derivatives, and applying it to $\partial_L(u/v) = (u'v - uv')/v^2$ for $u, v \in \mathbb{K}'[L]$ gives $\iota_0(\partial_L(u/v)) = \partial_L(\iota_0(u/v))$ by the quotient rule in $\mathbb{K}'((L^{-1}))$. Then $\omega(\partial_L h/h) = \omega(\partial_L h) - \omega(h) \geq 1$. If $\gamma \neq 0$ then $\omega(\gamma h) = \omega(h) < \omega(\partial_L h)$, so no cancellation is possible and $\partial_L h - \gamma h \neq 0$.

(iii) The bound is immediate from Equation (10.15). If $D_X F = 0$ then $\partial_L p_\gamma = \gamma p_\gamma$ for every γ; by (ii) this forces $p_\gamma = 0$ for $\gamma \neq 0$. For $\gamma = 0$ we get $\partial_L p_0 = 0$; writing $p_0 = \sum_{j \geq j_0} a_j L^{-j}$ in $\mathbb{K}'((L^{-1}))$, this says $-j a_j = 0$ for all j, hence $a_j = 0$ for $j \neq 0$ and $p_0 = a_0 \in \mathbb{K}'$.

(iv) $\nu_t(D_X F/F) = \nu_t(D_X F) - \nu_t F \geq 1$ by (iii) and additivity. Write $F = t^e b(1+S)$ with $\nu_t S > 0$, which is possible in the fraction field. Then, D_X being a derivation,

$$\frac{D_X F}{F} = \frac{D_X(t^e)}{t^e} + \frac{D_X b}{b} + \frac{D_X S}{1+S} = -et + t\frac{\partial_L b}{b} + \frac{D_X S}{1+S},$$

using $D_X(t^e) = -et^{e+1}$ and $D_X b = t\,\partial_L b$. The last term has order $\geq \nu_t S + 1 > 1$, so it does not contribute to the block at t^1. ☐

Remark 10.13 (A common ambient scale). Given two power–log scales $\mathcal{H}(C_1,\Gamma_1)$ and $\mathcal{H}(C_2,\Gamma_2)$ inside one exponential extension, both embed D_X-compatibly in $\mathcal{H}(\mathbb{K}'''((L^{-1})), \Gamma_1 + \Gamma_2)$, where $\mathbb{K}'''$ is the compositum of the two scalar fields: on coefficients this is Theorem 9.37, and on exponents it is the inclusion of well-ordered supports. Hence in the theorems below one may always assume that the scale containing the input also contains the output.

10.4 Exponentials: the exact boundary

The mechanism is the logarithmic derivative. If $E = \exp F$ lies in a power–log scale, then $D_X F = D_X E/E$ is a logarithmic derivative of an element of that scale, and Theorem 10.12 constrains such an object twice: its outer order is at least 1, and its block at t^1 has positive L^{-1}-order. Those two constraints are the whole boundary.

Proposition 10.14 (Necessary conditions)**.** *Let $(\mathcal{T}, \partial, \exp)$ be an exponential extension, let $\mathcal{H} = \mathcal{H}(C, \Gamma) \subseteq \mathcal{T}$ be a power–log scale, and let $F \in \mathcal{H}$ be such that $E := \exp F$ lies in $\mathcal{H}$. Put $e = \nu_t E \in \Gamma$ and $b = [t^e]E \neq 0$. Then*

$$\nu_t F \geq 0 \qquad \textit{and} \qquad \partial_L\big([t^0]F\big) + e = \frac{\partial_L b}{b}. \tag{10.17}$$

In particular $\omega\big(\partial_L([t^0]F) + e\big) \geq 1$; and if $[t^0]F$ is a polynomial then

$$\partial_L\big([t^0]F\big) = -e, \qquad \textit{that is} \qquad [t^0]F = a - eL \ \ \textit{with } a \in \mathbb{K}'. \tag{10.18}$$

Proof. E is nonzero and invertible in the fraction field of $\mathcal{H}$, to which ν_t extends with additivity. By Theorem 10.10, $D_X F = D_X E/E$, so Theorem 10.12(iv) gives $\nu_t(D_X F) \geq 1$ and

$$[t^1]D_X F = \frac{\partial_L b}{b} - e.$$

Write $F = \sum_\gamma p_\gamma t^\gamma$. By Equation (10.15), $\nu_t(D_X F) \geq 1$ forces $\partial_L p_\gamma - \gamma p_\gamma = 0$ for every $\gamma < 0$, and Theorem 10.12(ii) then gives $p_\gamma = 0$ for $\gamma < 0$; that is, $\nu_t F \geq 0$. Again by Equation (10.15), the block of $D_X F$ at t^1 comes only from $\gamma = 0$ and equals $\partial_L p_0$, which proves Equation (10.17). Its right-hand side has $\omega \geq 1$ by Theorem 10.12(ii). If p_0 is a polynomial then $\partial_L p_0 + e$ is a polynomial, and a nonzero polynomial h has $\omega(h) = -\deg_{\mathrm{L}} h \leq 0$; so $\partial_L p_0 + e = 0$, which is Equation (10.18). □

Corollary 10.15 (The three ways out)**.** *Let $F \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$.*

1. *If $\nu_t F < 0$ then $\exp F$ lies in* no *power–log scale whatsoever.*
2. *If $\nu_t F \geq 0$ and $\deg_{\mathrm{L}}[t^0]F \geq 2$ then $\exp F$ lies in no power–log scale.*
3. *If $\nu_t F \geq 0$ and $[t^0]F = a + mL$, then $\exp F$ lies in no power–log scale whose exponent group omits m; in particular it lies in no Puiseux-log ring at all when m is irrational.*

Proof. By Theorem 10.13 we may work inside a single scale $\mathcal{H}(C, \Gamma)$ containing F, and $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ has polynomial coefficients, so Equation (10.18) applies. (i) If $\nu_t F < 0$ this contradicts the first half of Equation (10.17). (ii) Here $\partial_L[t^0]F$ is a nonzero polynomial of degree $\deg_{\mathrm{L}}[t^0]F - 1 \geq 1$, so it is not the constant $-e$, contradicting Equation (10.18). (iii) Now $\partial_L[t^0]F = m$, so Equation (10.18) forces $m = -e \in \Gamma$. □

Theorem 10.16 (Exponential closure boundary)**.** *Let $(\mathcal{T}, \partial, \exp)$ be an exponential extension and $F \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$. Then $\exp F \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ if and only if*

$$\nu_t F \geq 0, \qquad [t^0]F = a + mL \quad \textit{with } m \in \mathbb{Z},\ a \in \mathbb{K},\ \exp a \in \mathbb{K}. \tag{10.19}$$

In that case

$$\exp F = (\exp a)\, t^{-m}\, \exp\big(F - [t^0]F\big), \qquad \exp\big(F - [t^0]F\big) \in 1 + t\mathscr{PL}_{\mathrm{pow},\mathbb{K}}. \tag{10.20}$$

Proof. Sufficiency. Put $V = F - [t^0]F \in t\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$. Since $\exp$ is a homomorphism and $\exp L = t^{-1}$, for $m \in \mathbb{Z}$ we get $\exp(mL) = (\exp L)^m = t^{-m}$, so $\exp F = (\exp a)t^{-m} \exp V$; and $\exp V \in 1 + t\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ by Theorem 10.9. All three factors lie in $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$.

Necessity. Theorem 10.14 with $\mathcal{H} = \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$, $C = \mathbb{K}[L]$, $\Gamma = \mathbb{Z}$ gives $\nu_t F \geq 0$ and $[t^0]F = a - eL$ with $a \in \mathbb{K}$ and $e = \nu_t(\exp F) \in \mathbb{Z}$; put $m = -e$. Then Equation (10.20) holds by the sufficiency computation, and solving it for $\exp a$ exhibits $\exp a$ as a product of elements of $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$, so $\exp a \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$. Finally $\partial(\exp a) = (\partial a)\exp a = 0$, so $\exp a \in \ker D_X \cap \mathscr{PL}_{\mathrm{poly},\mathbb{K}} = \mathbb{K}$ by Theorem 11.12. □

Corollary 10.17 (Fractional and irrational slopes). *Let $\mathcal{H} = \mathcal{H}(\mathbb{K}'[L], \Gamma)$ be a power–log scale with polynomial coefficients on which the ambient exponential is normalised, in the sense that $\exp V \in 1 + \{W \in \mathcal{H} : \nu_t W > 0\}$ whenever $V \in \mathcal{H}$ has $\nu_t V > 0$. For $F \in \mathcal{H}$, $\exp F \in \mathcal{H}$ if and only if $\nu_t F \geq 0$, $[t^0]F = a + mL$ with $m \in \Gamma$, and the constant $\kappa := \exp(a + mL)\, t^m$ lies in $\mathbb{K}'$. Thus $\exp(\frac{1}{2}L) = X^{1/2}$ lies in the Puiseux-log ring $\mathbb{K}[L]((t^{1/2}))$ but not in $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$, and $\exp(\pi L) = X^{\pi}$ requires a Hahn scale with $\pi \in \Gamma$.*

Proof. That κ is a constant is the computation $\partial\kappa = mt\kappa - mt\kappa = 0$, using $D_X(t^m) = -mt^{m+1}$. Sufficiency: $\exp F = \kappa\, t^{-m} \exp(F - [t^0]F)$ with all factors in $\mathcal{H}$. Necessity: Equation (10.18) gives $\nu_t F \geq 0$ and $[t^0]F = a + mL$ with $m = -e \in \Gamma$, and then $\kappa = \exp(F)\exp(-(F - [t^0]F))t^m \in \mathcal{H} \cap \mathrm{Const}(\mathcal{T}) = \mathbb{K}'$ by Theorem 10.12(iii). □

Remark 10.18 (Laurent coefficients admit more exponentials). The affine criterion Equation (10.18) is a statement about *polynomial* coefficients. It genuinely fails for Laurent coefficients: $F = L^{-1}$ has $\nu_t F = 0$ and a t^0-block that is not affine in L, yet $\exp(L^{-1}) = \sum_{k\geq 0} L^{-k}/k! \in \mathbb{K}((L^{-1})) \subseteq \mathscr{PL}_{\mathrm{itLaur},\mathbb{K}}$. There is no contradiction: Equation (10.17) is satisfied with $e = 0$ and $b = \exp(L^{-1})$, because $\partial_L b/b = -L^{-2}$ does have positive L^{-1}-order. The exact criterion in a scale with Laurent coefficients is precisely Equation (10.17): the t^0-block of F, shifted by $-eL$, must be a logarithm inside C of an element of C.

Example 10.19 (The three witnesses). $\exp(t^{-1}) = \mathrm{e}^X$ and $\exp(t^{-1}L) = X^X$ fail (i); $\exp(L^2) = X^{\log X}$ fails (ii); $\exp(\frac{1}{2}L) = X^{1/2}$ fails (iii) for $\Gamma = \mathbb{Z}$ but succeeds in the Puiseux-log ring. By contrast $\exp(3L + t) = X^3\mathrm{e}^{1/X} \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$, and $\exp(Lt) = 1 + Lt + \frac{1}{2}L^2t^2 + \cdots \in 1 + t\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$: smallness is measured by the dominance order, not by the size of the coefficient polynomials.

Remark 10.20 (The exponential hierarchy). Theorem 10.15(i) is the structural reason for the transseries hierarchy. The elements of negative outer order are exactly the *purely large* elements of the scale, and their exponentials are new transmonomials that no enlargement of the exponent group or of the coefficient field can express: enlarging Γ adds monomials t^γ, all of which have logarithmic derivative $-\gamma t$ of outer order 1, whereas e^X has logarithmic derivative 1. Closing under $\exp$ and $\log$ therefore requires a monomial group built by transfinite iteration, which is precisely the construction of the logarithmic–exponential transseries [1, 33, 13, 8]. The polynomial–logarithmic scale is the fragment of that field met after a dominant logarithmic change of variable, and its virtue is that all closure questions have the finite answers proved here.

Corollary 10.21 (Which elements are exponentials). *Let $\mathbb{K} = \mathbb{R}$ and let $\mathcal{T}$ be an exponential extension with $\exp(\mathbb{R}) \subseteq \mathbb{R}$ equal to the real exponential. Then*

$$\mathcal{E} := \{F \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}} : \exp F \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}\} = \mathbb{R} \oplus \mathbb{Z}L \oplus t\mathscr{PL}_{\mathrm{pow},\mathbb{K}},$$

and $\exp$ restricts to a group isomorphism from $(\mathcal{E}, +)$ onto the subgroup of $\mathscr{PL}^{\times}_{\mathrm{poly},\mathbb{K}}$ consisting of the units whose scalar leading coefficient is positive.

Proof. The description of $\mathcal{E}$ is Theorem 10.16 with $\mathbb{K} = \mathbb{R}$, since $\exp a \in \mathbb{R}$ for every $a \in \mathbb{R}$. Equation (10.20) shows that $\exp$ maps $\mathcal{E}$ into the units with positive leading coefficient e^a. It is onto: a unit is $c\,t^{-m}(1+S)$ with $c > 0, m \in \mathbb{Z}, S \in t\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ (Theorem 9.3), and $F = \log c + mL + \log(1+S) \in \mathcal{E}$ has $\exp F = c\,t^{-m}(1 + S)$ by Equation (10.8) and Theorem 10.9. It is injective: if $\exp F = 1$ with $F \in \mathcal{E}$ then $D_X F = 0$ by Theorem 10.10, so $F \in \mathbb{K} = \mathbb{R}$ by Theorem 11.12, and $\mathrm{e}^F = 1$ gives $F = 0$. □

10.5 Logarithms: the leading block decides

We now prove the counterpart for logarithms. One elementary observation about Laurent coefficients does all the work in the negative direction, and it will reappear in Part 11 as the antiderivative obstruction; we state it here in the form needed.

Definition 10.22 (Logarithmic residue at $L = \infty$)**.** For $h = \sum_{j \ge j_0} a_j L^{-j} \in \mathbb{K}'((L^{-1}))$ put $\operatorname{res}(h) := a_1$, the coefficient of L^{-1}. For $h \in \mathbb{K}'(L)$, $\operatorname{res}(h)$ means res of its expansion at $L = \infty$; for $h \in \mathbb{K}'[L]$, $\operatorname{res}(h) = 0$. In the classical convention $\operatorname{Res}_{L=\infty}(h \, \mathrm{d}L) = -\operatorname{res}(h)$.

Lemma 10.23 (Derivatives have no residue)**.** $\operatorname{res}(\partial_L h) = 0$ *for every* $h \in \mathbb{K}'((L^{-1}))$.

Proof. $\partial_L\left(\sum_{j \ge j_0} a_j L^{-j}\right) = \sum_{j \ge j_0} (-j) a_j L^{-j-1}$, and the exponent $-j-1$ equals -1 only for $j = 0$, whose coefficient $-0 \cdot a_0$ is 0. □

Lemma 10.24 (Image and kernel of ∂_L)**.** *Let* $g \in \mathbb{K}'((L^{-1}))$*. There exists* $h \in \mathbb{K}'((L^{-1}))$ *with* $\partial_L h = g$ *if and only if* $\operatorname{res}(g) = 0$*, and in that case* h *is unique up to an additive constant. Equivalently,*

$$\ker \partial_L = \mathbb{K}', \qquad \operatorname{im} \partial_L = \left\{ g \in \mathbb{K}'((L^{-1})) : \operatorname{res}(g) = 0 \right\},$$

so res *induces an isomorphism* $\mathbb{K}'((L^{-1}))/\operatorname{im} \partial_L \cong \mathbb{K}'$*.*

Proof. Write $h = \sum_{j \ge j_0} a_j L^{-j}$ and $g = \sum_{k \ge k_0} b_k L^{-k}$. Then $\partial_L h = \sum_{j \ge j_0} (-j) a_j L^{-j-1}$, so $\partial_L h = g$ holds if and only if

$$-j\, a_j = b_{j+1} \qquad \text{for every } j. \tag{10.21}$$

For $j \neq 0$ this determines $a_j = -b_{j+1}/j$ uniquely, and characteristic zero is what makes the division legitimate. For $j = 0$ the left side is 0, so (10.21) is solvable exactly when $b_1 = \operatorname{res}(g) = 0$, and then a_0 is unconstrained. This gives both the criterion and uniqueness up to the additive constant a_0.

The candidate h so defined is a legitimate element of $\mathbb{K}'((L^{-1}))$: if $b_k = 0$ for $k < k_0$ then $a_j = 0$ for $j < k_0 - 1$, so the support of h is bounded below by $k_0 - 1$. Finally $\ker \partial_L = \mathbb{K}'$ is the case $g = 0$ of the same computation. □

Corollary 10.25 (Residue of a logarithmic derivative)**.** *For nonzero* $h \in \mathbb{K}'((L^{-1}))$, $\operatorname{res}(\partial_L h/h) = -\omega(h)$*. In particular, for a nonzero polynomial* p, $\operatorname{res}(\partial_L p/p) = \deg_{\mathrm{L}} p$*, and for* $p = a \prod_i (L - \lambda_i)^{m_i}$ *over* $\overline{\mathbb{K}'}$ *the residue of* $\partial_L p/p$ *at the finite point* λ_i *is* $m_i \neq 0$*.*

Proof. Write $h = a_{j_0} L^{-j_0}(1 + w)$ with $\omega(w) \ge 1$ and $j_0 = \omega(h)$. Then $\partial_L h/h = -j_0 L^{-1} + \partial_L w/(1 + w)$ by the Leibniz rule, and $\omega(\partial_L w/(1 + w)) \ge \omega(w) + 1 \ge 2$ by Theorem 10.12(ii); so the coefficient of L^{-1} is $-j_0$. For a polynomial, $\omega(p) = -\deg_{\mathrm{L}} p$. The last claim is the partial-fraction identity $\partial_L p/p = \sum_i m_i/(L - \lambda_i)$, and $m_i \neq 0$ in $\mathbb{K}'$ because $\operatorname{char} \mathbb{K}' = 0$. □

Theorem 10.26 (Logarithm boundary)**.** *Let* $(\mathcal{T}, \partial, \exp)$ *be an exponential extension, let* $F \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ *be nonzero with* $n_0 = \nu_t F$*, leading block* $p = p_{n_0}$*, scalar leading coefficient* $a \in \mathbb{K}^\times$ *and* $d = \deg_{\mathrm{L}} p$*, and write* $F = t^{n_0} p\,(1 + S)$ *with* $S \in t\mathbb{K}(L)[[t]]$*. Assume* F *has a logarithm* G *in* $\mathcal{T}$ *and that* L *has a logarithm* Λ *in* $\mathcal{T}$*. Then:*

1. *G is determined modulo* $\operatorname{Const}(\mathcal{T})$ *by*

$$G \equiv -n_0 L + d\,\Lambda + \log\!\left(\frac{p}{aL^d}\right) + \log(1 + S) \pmod{\operatorname{Const}(\mathcal{T})}, \tag{10.22}$$

where $\log(p/(aL^d)) = \sum_{r \ge 1} \frac{(-1)^{r+1}}{r} w^r$ *with* $w = p/(aL^d) - 1 \in L^{-1}\mathbb{K}[L^{-1}]$*, and* $\log(1 + S)$ *is the series Equation* (10.5) *with coefficients in* $\mathbb{K}(L)$*. All the displayed terms except* $d\Lambda$ *lie in* $\mathscr{PL}_{\mathrm{itLaur},\mathbb{K}}$*.*

2. *G lies in some power–log scale if and only if $d = 0$, that is (Theorem 9.3) if and only if F is a unit of $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$. When $d = 0$ one may take $G = c - n_0 L + \log(1 + S) \in \mathbb{K}(c)[L]((t))$ for any $c \in \mathcal{T}$ with $\exp c = a$.*

Proof. (i) Call the right-hand side of Equation (10.22) H. By Theorem 10.10 it suffices to prove $\partial H = D_X F/F$, for then $G - H$ is a constant. Now $\partial\Lambda = D_X L/L = t/L$ by Theorem 10.10 applied to L, and D_X applied to the two logarithmic series termwise (legitimate by Theorem 10.1, the families being summable in the L^{-1}-adic and t-adic order respectively) gives $D_X \log(1 + w) = D_X w/(1 + w) = t\,\partial_L w/(1 + w)$ and $D_X \log(1 + S) = D_X S/(1 + S)$. Hence

$$\partial H = -n_0 t + \frac{d\,t}{L} + t\,\frac{\partial_L w}{1 + w} + \frac{D_X S}{1 + S} = -n_0 t + t\,\frac{\partial_L p}{p} + \frac{D_X S}{1 + S},$$

because $p = aL^d(1 + w)$ gives $\partial_L p/p = d/L + \partial_L w/(1 + w)$. On the other side, $F = t^{n_0} p(1 + S)$ and the Leibniz rule give exactly the same expression for $D_X F/F$. The membership claims are clear: $w \in L^{-1}\mathbb{K}[L^{-1}]$ makes $\log(1+w) \in L^{-1}\mathbb{K}[[L^{-1}]]$, and $\log(1+S) \in t\mathbb{K}(L)[[t]] \subseteq \mathscr{PL}_{\mathrm{itLaur},\mathbb{K}}$ after Theorem 9.37.

(ii) *Sufficiency.* If $d = 0$ then $p = a$, $w = 0$, and Equation (10.22) exhibits a logarithm in $\mathbb{K}(c)[L]((t))$ once the constant is fixed as c; note $S \in t\mathbb{K}[L][[t]]$ in this case, since $F = at^{n_0}(1 + S)$ with $F \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$.

Necessity. Suppose $G \in \mathcal{H}(C, \Gamma)$ and $d \geq 1$. By Theorem 10.10 and the computation in (i),

$$D_X G = \frac{D_X F}{F}, \qquad [t^1] D_X G = \frac{\partial_L p}{p} - n_0.$$

On the other hand Equation (10.15) gives $[t^1]D_X G = \partial_L g_0$ with $g_0 = [t^0]G \in C$. Hence $\partial_L(g_0 + n_0 L) = \partial_L p/p$ with $g_0 + n_0 L \in C$. Apply res: the left side vanishes by Theorem 10.23, the right side equals $d \neq 0$ by Theorem 10.25 – a contradiction. The argument covers all three coefficient rings at once, since each embeds ∂_L- compatibly in $\mathbb{K}'((L^{-1}))$ (Theorem 10.12(ii)); for $C = \mathbb{K}'[L]$ one may argue even more directly, since $\partial_L p/p$ has a pole when $d \geq 1$ and is therefore not a polynomial. □

Example 10.27 (A forced nested logarithm). Let $F = L^2 t^{-3}\big(1 + (L + 1)t\big)$, so $n_0 = -3$, $p = L^2$, $a = 1$, $d = 2$, $w = 0$. Then

$$\log F \equiv 3L + 2\Lambda + (L + 1)t - \tfrac{1}{2}(L + 1)^2 t^2 + \tfrac{1}{3}(L + 1)^3 t^3 + \mathcal{O}_t^{\mathrm{form}}\big(t^4\big)\,.$$

The new generator $\Lambda = \log\log X$ is forced by the factor L^2, not by the small correction, and it enters with coefficient exactly $d = 2$.

Corollary 10.28 (Logarithms inside the class). *Let $F \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ be nonzero. Then F has a logarithm in $\mathbb{K}'[L]((t))$ for some field $\mathbb{K}' \supseteq \mathbb{K}$ if and only if F is a unit of $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$; one may take $\mathbb{K}' = \mathbb{K}(c)$ with $\exp c$ the scalar leading coefficient of F. If F is not a unit, a logarithm exists in $\mathscr{PL}_{\mathrm{itLaur},\mathbb{K}} \oplus \mathbb{K}\Lambda \oplus \mathrm{Const}(\mathcal{T})$ and in no power–log scale.*

Proof. Immediate from Theorem 10.26; the second sentence is Equation (10.22), whose terms other than $d\Lambda$ lie in $\mathscr{PL}_{\mathrm{itLaur},\mathbb{K}}$. □

10.6 Powers

Proposition 10.29 (General powers). *Let $F \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ be a unit, $F = a\,t^{n_0}(1 + S)$ with $a \in \mathbb{K}^\times$, $n_0 \in \mathbb{Z}$, $S \in t\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$, let $\alpha \in \mathbb{K}$, fix $c \in \mathcal{T}$ with $\exp c = a$, and let $G = c - n_0 L + \log(1 + S)$ be the corresponding logarithm of F (Theorem 10.26). Then*

$$F^\alpha = \exp(\alpha G) = \mu\, t^{\alpha n_0}\,(1 + S)^\alpha, \qquad (1 + S)^\alpha = \sum_{r \geq 0} \binom{\alpha}{r} S^r \in 1 + t\mathscr{PL}_{\mathrm{pow},\mathbb{K}}, \tag{10.23}$$

where $\mu := \exp(\alpha c - \alpha n_0 L)\, t^{\alpha n_0}$ *is a constant of* $\mathcal{T}$*, equal to* $\exp(\alpha c)$ *whenever* $\alpha n_0 \in \mathbb{Z}$*. Consequently:*

1. *if* $\alpha n_0 \in \mathbb{Z}$ *and* $\exp(\alpha c) \in \mathbb{K}'$*, then* $F^\alpha \in \mathbb{K}'[L]((t))$*;*

2. *if* $\alpha n_0 \in s^{-1}\mathbb{Z}\setminus\mathbb{Z}$*, then* F^α *lies in the Puiseux-log ring* $\mathbb{K}'[L]((t^{1/s}))$ *and in no scale with* $\alpha n_0 \notin \Gamma$*; for* αn_0 *irrational a Hahn scale is required;*

3. *if* F *is not a unit, with leading block* p *of degree* $d \geq 1$*, then* F^α *lies in a power–log scale only if* $\alpha d \in \mathbb{Z}$*.*

Proof. Since exp is a homomorphism, $\exp(\alpha G) = \exp(\alpha c - \alpha n_0 L)\exp(\alpha \log(1+S))$, and the second factor is $(1+S)^\alpha$ by the last identity of Equation (10.9) together with Theorem 10.9; inserting $t^{\alpha n_0} t^{-\alpha n_0}$ gives Equation (10.23). That μ is a constant is the computation of Theorem 10.17, and for $\alpha n_0 \in \mathbb{Z}$ the normalisation $\exp L = t^{-1}$ gives $\exp(-\alpha n_0 L) = t^{\alpha n_0}$, whence $\mu = \exp(\alpha c)$. Items (i) and (ii) follow, the negative half of (ii) from Theorem 10.15(iii) applied to αG. For (iii), Equation (10.22) shows that αG contains the term $\alpha d\,\Lambda$; equivalently, $D_X(F^\alpha)/F^\alpha = \alpha D_X F/F$ has res $= \alpha d$ by Theorem 10.25, while by that same corollary the logarithmic derivative of a nonzero element of any power–log scale has residue $-\omega(b) \in \mathbb{Z}$ when the coefficients are polynomial, and in general an integer whenever the leading coefficient lies in $\mathbb{K}'((L^{-1}))$. Hence $\alpha d \in \mathbb{Z}$. ☐

Remark 10.30 (Branches). Over $\mathbb{C}$ the scalars $\exp c = a^1$ and $\exp \alpha c$ both depend on the choice of c, and different choices differ by $\exp(2\pi i k \alpha)$; over $\mathbb{R}$ with $a > 0$ the principal choice $c = \operatorname{Log} a$ is available. A formula for F^α is incomplete until the branch is named, and the choice propagates into every later block. This is a genuine mathematical datum, not a notational nuisance: the two square roots of $X^2(1+t)$ are $\pm X(1 + \frac{1}{2}t - \frac{1}{8}t^2 + \cdots)$, and only the choice of sign makes the expansion an asymptotic expansion of a given branch as $X \to +\infty$.

10.7 The closure boundary

We can now state the closure boundary as a theorem rather than a table. The first part says that the polynomial–logarithmic class is generated from its two generators by one construction, Laurent assembly; the second says that a long list of further operations does not enlarge it; the third exhibits, for each operation that does enlarge it, a witness and the theorem that proves non-membership.

Theorem 10.31 (The generated class and its boundary)**.** *Let* $(\mathcal{T}, \partial, \exp)$ *be an exponential extension.*

1. $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ *is the smallest subring* $\mathcal{A} \subseteq \mathcal{T}$ *that contains* $\mathbb{K} \cup \{t, t^{-1}, L\}$ *and is closed under Laurent assembly: whenever* $p_n \in \mathbb{K}[L]$ *for* $n \geq n_0$ *and every* $p_n t^n$ *lies in* $\mathcal{A}$*, the element* $\sum_{n \geq n_0} p_n t^n \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ *lies in* $\mathcal{A}$*.*

2. $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ *is closed under: the ring operations; inversion of its units;* $\Phi(U)$ *for* $\Phi \in \mathbb{K}[[z]]$ *and* $\nu_t U \geq 1$*; the derivation* D_X *and the Euler derivation* ϑ*; antidifferentiation; the exponential of any element of* $\mathbb{K}_0 \oplus \mathbb{Z}L \oplus t\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$*, where* $\mathbb{K}_0 = \{a \in \mathbb{K} : \exp a \in \mathbb{K}\}$*; and the logarithm of any unit whose scalar leading coefficient has a logarithm in* $\mathbb{K}$*.*

3. *Each of the following leaves* $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$*, and none of the enlargements it forces can be avoided: reciprocal of a nonunit; a power with* $\alpha n_0 \notin \mathbb{Z}$*; the logarithm of a nonunit; the exponential of an element of negative outer order or with* $\deg_{\mathrm{L}}[t^0]F \geq 2$*. Witnesses and minimal enlargements are given in the proof and collected in Table 10.1.*

Proof. (i) The intersection of any family of subrings closed under Laurent assembly is again one, so a smallest $\mathcal{A}$ exists. $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ is a subring of $\mathcal{T}$ containing the generators and is closed under Laurent assembly by definition, so $\mathcal{A} \subseteq \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$. Conversely $\mathcal{A}$ contains $\mathbb{K}[L][t, t^{-1}]$, hence every $p_n t^n$, hence by assembly every element of $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$.

(ii) Ring operations and inversion of units are Theorem 9.3; $\Phi(U)$ is Theorem 10.2; D_X and ϑ are Theorem 11.13; antidifferentiation is Theorem 11.18; the exponential and logarithm clauses are Theorem 10.16 and Theorem 10.28.

(iii) The witnesses and their proofs are: $1/(L+1) \notin \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ (Theorem 9.22), with minimal enlargement $\mathscr{PL}_{\mathrm{rat},\mathbb{K}}$, or $\mathscr{PL}_{\mathrm{itLaur},\mathbb{K}}$ after expansion; $t^{1/2} = X^{-1/2}$ for $F = t$, $\alpha = 1/2$ (Theorem 10.29), with minimal enlargement a Puiseux-log or Hahn scale; $\log L = \Lambda$ for $F = L$ (Theorem 10.26), with enlargement the adjunction of the new generator Λ; e^X and $X^{\log X}$ (Theorem 10.15), with enlargement a genuinely exponential transseries field. In each case the cited statement proves non-membership in every power–log scale of the relevant type, so no enlargement of the coefficient field or of the exponent group alone suffices. □

Table 10.1: Closure of the polynomial–logarithmic scale under the elementary operations. "Scale" means any power–log scale $\mathcal{H}(C,\Gamma)$ in the sense of the definition in this section.

Operation	Hypothesis	Result
$F+G$, FG	none	$\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$
$1/F$	leading block in $\mathbb{K}^\times$	$\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$
$1/F$	leading block of degree ≥ 1	$\mathscr{PL}_{\mathrm{rat},\mathbb{K}}$; after expansion $\mathscr{PL}_{\mathrm{itLaur},\mathbb{K}}$
$\Phi(U)$, $\Phi \in \mathbb{K}[[z]]$	$\nu_t U \geq 1$	$\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ (Theorem 10.2)
$D_X F$, ϑF	none	same class (Theorem 11.13)
antiderivative	none	$\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ (Theorem 11.18)
antiderivative in $\mathscr{PL}_{\mathrm{itLaur},\mathbb{K}}$	$\operatorname{res} F = 0$	$\mathscr{PL}_{\mathrm{itLaur},\mathbb{K}}$; otherwise adjoin Λ (Theorem 11.21)
F^α	F a unit, $\alpha\nu_t F \in \mathbb{Z}$, $a^\alpha \in \mathbb{K}$	$\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$
F^α	$\alpha\nu_t F \notin \mathbb{Z}$	Puiseux-log or Hahn scale
$\log F$	F a unit, $\log a \in \mathbb{K}$	$\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ (Theorem 10.28)
$\log F$	leading block of degree $d \geq 1$	$\mathscr{PL}_{\mathrm{itLaur},\mathbb{K}} \oplus \mathbb{K}\Lambda$; no scale (Theorem 10.26)
$\exp F$	$\nu_t F \geq 0$, $[t^0]F = a + mL$, $m \in \mathbb{Z}$, $\mathrm{e}^a \in \mathbb{K}$	$\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ (Theorem 10.16)
$\exp F$	$\nu_t F \geq 0$, $[t^0]F = a + mL$, $m \notin \mathbb{Z}$	Hahn scale with $m \in \Gamma$
$\exp F$	$\deg_{\mathrm{L}}[t^0]F \geq 2$, or $\nu_t F < 0$	no scale; exponential transseries (Theorem 10.15)

A common trap

Three errors recur in the sources and in informal practice.

- *"Small" is not about the coefficients.* Lt is small and L^{-1} is not, in the sense that decides summability of $\sum_r c_r U^r$: the criterion is $\nu_t U \geq 1$, not the numerical size of the blocks. For $\nu_t U = 0$ the series $\log(1+U)$ need not be summable at all, and a normalisation must precede the expansion.

- *A closure table is not a theorem.* "$\exp F$ may require new monomials" is not a statement one can act on. What is true is Theorem 10.15: for $\nu_t F < 0$ no enlargement of the coefficient field or of the exponent group ever suffices.

- *Formal identities are not analytic ones.* Every statement in this section is an identity between formal objects. That $\exp(\log(1+U)) = 1+U$ holds coefficientwise says nothing about the convergence of either series, and an equality $F = G + \mathcal{O}_t^{\mathrm{form}}(t^N)$ carries no bound of the form $\mathcal{O}_{X\to+\infty}(X^{-N})$ without a separate realisation theorem.

Section summary

Substitution of a series of positive outer order into a formal power series is unconditional and finitely determined: $\operatorname{Trunc}_{<N} \Phi(U)$ depends on $c_0, \ldots, c_{\lfloor (N-1)/q \rfloor}$ and on $\operatorname{Trunc}_{<N} U$ alone. Hence $(1+U)^\alpha$, $\log(1+U)$ and $\exp U$ stay in $\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$. Beyond that, the leading data decide. A logarithm stays in the scale precisely when the leading block is a scalar; otherwise the generator $\Lambda = \log L$ appears with coefficient $\deg_{\mathrm{L}} p$, and the coefficients must be expanded at $L = \infty$. An exponential stays in a power–log scale precisely when $\nu_t F \geq 0$ and the t^0-block is affine in L with slope in the exponent group; otherwise the object is a new transmonomial and the exponential hierarchy begins.

Chapter 11

Differentiation and antiderivatives

Differentiation is the operation for which the polynomial–logarithmic scale behaves best: it is unconditionally closed, it raises the outer order by exactly one except in one identifiable case, and – unlike almost every other restricted asymptotic scale – so does its inverse. This chapter proves those statements, in the sharp form, and then determines exactly which elements of each ambient class have an antiderivative inside that class. The answer is a residue: for $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ there is no obstruction at all, for $\mathscr{P}\mathscr{L}_{\mathrm{itLaur},\mathbb{K}}$ the obstruction is a single linear functional, and for $\mathscr{P}\mathscr{L}_{\mathrm{rat},\mathbb{K}}$ there is one linear condition at every pole of every coefficient block.

11.1 The distinguished derivation

In the formal algebra t and L are independent generators. The analytic relation $t = X^{-1}$, $L = \log X$ enters the algebra in exactly one place: it selects a derivation.

Theorem 11.1 (Existence and uniqueness)**.** *Let ∂_L and ∂_t be the formal partial derivatives of $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$, each holding the other generator fixed.*

1. *The operator*
$$D_X \coloneqq t\,\partial_L - t^2\,\partial_t \tag{11.1}$$
is a $\mathbb{K}$-derivation of $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ with $D_X t = -t^2$ and $D_X L = t$, and it maps $\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ into $t\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$.

2. *D_X is the unique $\mathbb{K}$-derivation δ of $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ with $\delta t = -t^2$, $\delta L = t$ and $\delta(\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}) \subseteq \mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$. The last hypothesis may be replaced by t-adic continuity of δ.*

3. *Without any such hypothesis, uniqueness still holds on the subring $\mathbb{K}[L][t, t^{-1}]$.*

Proof. (i) Both ∂_L and ∂_t are $\mathbb{K}$-derivations of $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$. Indeed each is defined blockwise or termwise, $\partial_L\big(\sum p_n t^n\big) = \sum p_n' t^n$ and $\partial_t\big(\sum p_n t^n\big) = \sum n p_n t^{n-1}$, both supports being bounded below; and the Leibniz rule holds because every block of a product is the finite sum $[t^k](FG) = \sum_{i+j=k} p_i q_j$, to which the Leibniz rule for polynomials, respectively the classical computation $k\sum_{i+j=k} p_i q_j = \sum_{i+j=k}(i p_i q_j + j p_i q_j)$, applies. A left $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$-linear combination of derivations is a derivation, so D_X is one. The values on the generators are immediate, and the block formula Equation (11.2) below gives $D_X(\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}) \subseteq t\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$.

(ii) Let δ be as stated and put $\eta = \delta - D_X$, a $\mathbb{K}$-derivation with $\eta t = \eta L = 0$, hence $\eta(t^{-1}) = -t^{-2}\eta(t) = 0$ and $\eta = 0$ on $\mathbb{K}[L][t, t^{-1}]$. Let $F \in \mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ and $N > \nu_t F$. Write $F = \operatorname{Trunc}_{<N} F + t^N R$ with $\operatorname{Trunc}_{<N} F \in \mathbb{K}[L][t, t^{-1}]$ and $R \in \mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$. Then

$$\eta F = \eta\big(t^N R\big) = N t^{N-1}\eta(t)R + t^N \eta R = t^N \eta R.$$

By hypothesis $\delta R \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ and $D_X R \in t\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$, so $\eta R \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ and $\nu_t(\eta F) \geq N$. As N was arbitrary, $\eta F = 0$. If instead η is continuous, choose N_0 with $\nu_t(\eta G) \geq 0$ whenever $\nu_t G \geq N_0$; applying the display with $N + N_0$ in place of N gives $\eta F = t^N \eta(t^{N_0} R)$ of order $\geq N$, and the same conclusion.

(iii) Contained in the first two sentences of (ii). □

Remark 11.2 (The hypothesis is not decorative). The sources, and the notation catalogue, call D_X *the* unique derivation with $D_X t = -t^2$ and $D_X L = t$. That is correct on the dense subring $\mathbb{K}[L][t, t^{-1}]$ and correct on $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ under the boundedness hypothesis of (ii), which every derivation arising in practice satisfies; but the passage to the completion genuinely uses a hypothesis, since a derivation of $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ is not a priori bounded on $\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$. We have stated the hypothesis rather than suppressing it.

11.2 One block, and the Euler derivation

Proposition 11.3 (Block formula). *For $n \in \mathbb{Z}$ and $p \in \mathbb{K}[L]$,*

$$D_X\big(t^n p(L)\big) = t^{n+1}\big(\partial_L - n\big)p(L), \tag{11.2}$$

and hence, for $F = \sum_{n \geq n_0} p_n t^n \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$,

$$D_X F = \sum_{n \geq n_0} t^{n+1}\big(p_n' - np_n\big), \qquad \text{that is} \qquad [t^{n+1}]D_X F = \big(\partial_L - n\big)[t^n]F. \tag{11.3}$$

The same formula holds with p in $\mathbb{K}(L)$ or $\mathbb{K}((L^{-1}))$ and with n replaced by any exponent α of a Puiseux-log or Hahn scale: $D_X(t^\alpha p) = t^{\alpha+1}(p' - \alpha p)$. Each output block receives a contribution from exactly one input block.

Proof. $\partial_L(t^n p) = t^n p'$ and $\partial_t(t^n p) = nt^{n-1}p$, so $D_X(t^n p) = t^{n+1}p' - nt^{n+1}p$. The series form follows because D_X is additive and commutes with summable sums, being continuous by Equation (11.2). For the general scale this is Equation (10.15), which was verified to be a derivation in Theorem 10.12(i). □

Remark 11.4 (Formal crosswalk). The displayed equation (11.2) is Exact in `TransseriesDifferentialBlock.lean`, while Theorem 11.3 is Partial overall. The module's nine-theorem surface is `Fabius.derivation_pow_t`, `Fabius.derivation_block`, `Fabius.exists_block_primitive`, `Fabius.derivation_block_zero`, and `Fabius.exists_block_primitive_resonant`, together with `Fabius.derivation_val_inv`, `Fabius.derivation_pow_inv`, `Fabius.derivation_zpow_t`, and `Fabius.derivation_block_zpow`. In an abstract differential algebra with $d(t) = -t^2$, $d(L) = t$, and unit t, the last theorem gives exactly (11.2) for $n \in \mathbb{Z}$; the natural-power form needs no unit hypothesis. The two primitive theorems give nonresonant primitives for input exponents at least two in characteristic zero (more generally, when the relevant natural-number cast is nonzero) and the resonant exponent-one primitive. The printed proposition also includes the Laurent-series coefficient formula, rational and inverse-log coefficient fields, arbitrary scale exponents, and the one-input-block assertion. Those extensions require a concrete Laurent ambient ring and are not present. In particular the abstract evaluation $p \mapsto p(L)$ need not be injective, so it cannot by itself yield the manuscript's ambient uniqueness and degree claims.

Proposition 11.5 (The Euler derivation). *Put $\vartheta := t^{-1}D_X$, the formal counterpart of $X\,\mathrm{d}/\mathrm{d}X$. Then ϑ is a $\mathbb{K}$-derivation of $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ with $\vartheta t = -t$ and $\vartheta L = 1$, it preserves each outer block,*

$$\vartheta\big(t^n p(L)\big) = t^n\big(\partial_L - n\big)p(L), \tag{11.4}$$

and therefore $\nu_t(\vartheta F) \geq \nu_t F$ for all F. On the block $t^n\mathbb{K}[L]$ the operator ϑ acts as $\partial_L - n$, whose only eigenvalue is $-n$; it is invertible on that block if and only if $n \neq 0$.

Proof. A derivation multiplied by a ring element is again a derivation, and $t^{-1} \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$; the values on the generators and Equation (11.4) follow from Equation (11.2). On $t^n\mathbb{K}[L]$, the operator $\partial_L - n$ is, in the basis $1, L, \ldots, L^D$ of polynomials of degree $\leq D$, triangular with every diagonal entry $-n$, because $\partial_L L^i = iL^{i-1}$ lowers the degree. Hence its determinant on that space is $(-n)^{D+1}$, which vanishes exactly when $n = 0$; and ∂_L itself is not injective on $\mathbb{K}[L]$. □

Remark 11.6 (Resonance is a zero eigenvalue). Theorem 11.5 is the conceptual source of everything that follows. The scale decomposes into the ϑ-invariant blocks $t^n\mathbb{K}[L]$; on block n the Euler derivation is the scalar $-n$ plus a nilpotent, so it is invertible exactly when $n \neq 0$. The block $n = 0$ is *resonant*: there ϑ is the nilpotent ∂_L, with kernel $\mathbb{K}$ and cokernel 0 on $\mathbb{K}[L]$ – but with a one-dimensional cokernel on $\mathbb{K}((L^{-1}))$. Every statement below about constants of integration, logarithms and residues is a statement about that single block.

11.3 Repeated derivatives

Definition 11.7 (Repeated-derivative operator)**.** For n in the exponent group and $k \in \mathbb{N}_0$ put

$$\Delta_{n,k} \coloneqq \prod_{j=0}^{k-1} \big(\partial_L - n - j\big), \qquad \Delta_{n,0} \coloneqq 1. \tag{11.5}$$

The $k = 0$ product is empty and acts as the identity; the factors commute, each being a polynomial in ∂_L.

Two of the sources abbreviate the single factor $\partial_L - n$ by Δ_n. In the notation of this volume that operator is $\Delta_{n,1}$: the second index counts derivatives and is never optional, since $\Delta_{n,k}$ is *not* the k-th power of $\Delta_{n,1}$.

Theorem 11.8 (Repeated derivative formula)**.** *For every $k \in \mathbb{N}_0$, every n, and every coefficient p,*

$$D_X^k\big(t^n p(L)\big) = t^{n+k}\,\Delta_{n,k}p(L), \tag{11.6}$$

and $\Delta_{n,k+1} = \Delta_{n+1,k} \circ (\partial_L - n) = (\partial_L - n - k) \circ \Delta_{n,k}$.

Proof. Induction on k. For $k = 0$ both sides are $t^n p$, the empty product being the identity – the convention that the sources state and then, in Equation (11.6)'s applications, occasionally forget. Assume Equation (11.6) for k. Applying Equation (11.2) at outer exponent $n + k$ to the coefficient $\Delta_{n,k}p$ gives

$$D_X^{k+1}\big(t^n p\big) = t^{n+k+1}\big(\partial_L - n - k\big)\Delta_{n,k}p = t^{n+k+1}\Delta_{n,k+1}p,$$

the last step because the factors of Equation (11.5) commute. □

Corollary 11.9 (Two specialisations)**.** *For $k \geq 0$:*

$$D_X^k\big(t^n\big) = (-1)^k\,(n)^{\overline{k}}\;t^{n+k}, \tag{11.7}$$

with $(n)^{\overline{k}} = n(n+1)\cdots(n+k-1)$ and $(n)^{\overline{0}} = 1$; and, for $p \in \mathbb{K}[L]$,

$$\vartheta^k p(L) = t^{-k}D_X^k p(L) = \Delta_{0,k}p = \sum_{j=0}^{k} s(k,j)\;\partial_L^j p, \tag{11.8}$$

where $s(k,j)$ are the signed Stirling numbers of the first kind. More generally,

$$\Delta_{n,k} = \sum_{j=0}^{k} s(k,j)\;(\partial_L - n)^j.$$

Proof. For Equation (11.7), $\Delta_{n,k}1 = \prod_{j=0}^{k-1}(-n-j) = (-1)^k \prod_{j=0}^{k-1}(n+j)$. For Equation (11.8), the generating identity of the signed Stirling numbers of the first kind is $(z)^{\underline{k}} = \prod_{j=0}^{k-1}(z-j) = \sum_j s(k,j)\, z^j$ [4, 18]; substituting the commuting operator $z = \partial_L$, respectively $z = \partial_L - n$, gives both displays, and $\vartheta^k = t^{-k} D_X^k$ holds on the block $n = 0$ by Equation (11.4). □

Corollary 11.10 (Derivative budget). *$\nu_t(D_X^k F) \geq \nu_t F + k$ for every $F \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ and every $k \geq 0$. Consequently, in a Taylor-type family $\sum_{k\geq 0} H^k D_X^k F/k!$ with $\nu_t H \geq r \geq 0$, the k-th term has outer order at least $\nu_t F + (r+1)k$; the family is summable as soon as $r \geq 0$, and the k-th term is invisible below t^N once $(r+1)k > N - 1 - \nu_t F$.*

Proof. Iterate Theorem 10.12(iii), or read off Equation (11.6). □

11.4 The valuation of a derivative, exactly

The sources state $\nu_t(D_X F) \geq \nu_t F + 1$ and then describe the failure of equality by three cases. The three cases are correct, but they leave the actual value of $\nu_t(D_X F)$ undetermined in the third. It is determined, and the sharp statement is no harder.

Theorem 11.11 (Sharp valuation). *Let $F \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ and write $a_{0,0} = [t^0 L^0]F \in \mathbb{K}$ for its constant term.*

1. *$\nu_t(D_X F) \geq \nu_t F + 1$ always.*
2. *If $F \notin \mathbb{K}$, then*
$$\nu_t\big(D_X F\big) = \nu_t\big(F - a_{0,0}\big) + 1. \tag{11.9}$$
3. *Equality $\nu_t(D_X F) = \nu_t F + 1$ holds if and only if F is not of the form $a_{0,0} + \widetilde{F}$ with $a_{0,0} \neq 0$ and $\nu_t \widetilde{F} \geq 1$. Explicitly, with $n_0 = \nu_t F$, $p = p_{n_0}$, $d = \deg_{\mathrm{L}} p$ and a the scalar leading coefficient:*
   - *if $n_0 \neq 0$, the leading term of $D_X F$ is $-n_0 a\, t^{n_0+1} L^d$ and equality holds;*
   - *if $n_0 = 0$ and $d > 0$, the leading term of $D_X F$ is $da\, t L^{d-1}$ and equality holds;*
   - *if $n_0 = d = 0$, the leading block is annihilated and $\nu_t(D_X F) = \nu_t(F - a) + 1 \geq 2$.*

The same statements hold in $\mathscr{PL}_{\mathrm{rat},\mathbb{K}}$ and $\mathscr{PL}_{\mathrm{itLaur},\mathbb{K}}$ with d replaced by $-\omega(p_{n_0})$, and in any power–log scale, except that in the third case the bound reads $\nu_t(D_X F) > 1$ when the exponent group is not $\mathbb{Z}$.

Proof. (i) is Theorem 10.12(iii). For (iii), apply Equation (11.2) to the leading block: the coefficient of t^{n_0+1} in $D_X F$ is $(\partial_L - n_0)p$. If $n_0 \neq 0$ this is a polynomial of degree d with leading coefficient $-n_0 a \neq 0$, so it is nonzero. If $n_0 = 0$ it is p', of degree $d-1$ with leading coefficient da, nonzero exactly when $d > 0$. If $n_0 = d = 0$ then $p = a$ and $p' = 0$, so the block vanishes.

For (ii), let $\widetilde{F} = F - a_{0,0}$, which is nonzero since $F \notin \mathbb{K}$, and note $D_X F = D_X \widetilde{F}$. It suffices to show that $\widetilde{F}$ falls under one of the first two cases of (iii). Let $m = \nu_t \widetilde{F}$. If $m \neq 0$ we are in the first case. If $m = 0$ then $[t^0]\widetilde{F} = [t^0]F - a_{0,0}$ is a nonzero polynomial with vanishing constant term, hence of degree ≥ 1, and we are in the second case. In both cases $\nu_t(D_X \widetilde{F}) = m + 1$. The version for the larger classes is the same computation with ω in place of $-\deg_{\mathrm{L}}$, using Theorem 10.12(ii). □

Corollary 11.12 (Kernel). $\ker D_X = \mathbb{K}$ *in $\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$, $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$, $\mathscr{PL}_{\mathrm{rat},\mathbb{K}}$, $\mathscr{PL}_{\mathrm{itLaur},\mathbb{K}}$, and* $\ker D_X = \mathbb{K}'$ *in every power–log scale $\mathcal{H}(C,\Gamma)$. Consequently* $\ker \vartheta = \mathbb{K}$ *as well.*

Proof. Theorem 10.12(iii); alternatively, if $F \notin \mathbb{K}$ then Equation (11.9) gives $\nu_t(D_X F) < +\infty$, so $D_X F \neq 0$. For $\vartheta = t^{-1} D_X$, multiplication by the unit t^{-1} does not change the kernel. □

11.5 Closure of the ambient classes

Proposition 11.13 (Differential closure). *D_X and ϑ map each of $\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$, $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$, $\mathscr{P}\mathscr{L}_{\mathrm{rat},\mathbb{K}}$, $\mathscr{P}\mathscr{L}_{\mathrm{itLaur},\mathbb{K}}$, $\mathbb{K}[L]((t^{1/s}))$ and every power–log scale into itself. Moreover $D_X(\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}) = t\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ and $D_X(t\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}) = t^2\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$, so $\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ and its powers $t^k\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ form a filtration by D_X-stable subgroups, and each of $\mathscr{P}\mathscr{L}_{\mathrm{rat},\mathbb{K}}$, $\mathscr{P}\mathscr{L}_{\mathrm{itLaur},\mathbb{K}}$ is a differential field.*

Proof. Closure under D_X follows from Theorem 11.3, because the operator $\partial_L - n$ maps each of the three coefficient rings into itself. For ϑ multiply by t^{-1}, a unit of every class listed except $\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$. The two inclusions

$$D_X(\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}) \subseteq t\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}, \qquad D_X(t\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}) \subseteq t^2\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$$

are immediate from Equation (11.2). Conversely, run the construction of Theorem 11.18 below on an F with $\nu_t F \geq 1$: it produces blocks only at indices $n \geq 0$, so the antiderivative lies in $\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$. On an F with $\nu_t F \geq 2$ it meets only nonresonant indices $n \geq 1$, so the antiderivative lies in $t\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$. □

Remark 11.14 (Iterated logarithms are not polynomially closed). Set $L_1 = L$ and $L_{j+1} = \log L_j$. Then, by Theorem 10.10,

$$D_X L_j = \frac{D_X L_{j-1}}{L_{j-1}}, \qquad \text{hence} \qquad D_X L_j = \frac{t}{L_1 L_2 \cdots L_{j-1}} \quad (j \geq 1), \tag{11.10}$$

by induction from $D_X L_1 = t$. Consequently a polynomial ring $\mathbb{K}[L_1, \ldots, L_r]((t))$ with $r \geq 2$ is *not* closed under D_X: already $D_X L_2 = t/L_1$ leaves it. The smallest natural differential extension replaces the coefficient ring by $\mathbb{K}(L_1, \ldots, L_r)$, or by iterated Laurent coefficients, and the chain rule then reads

$$D_X\big(a(L_1, \ldots, L_r)t^n\big) = t^{n+1}\Big(\partial_{L_1} a + \tfrac{1}{L_1}\partial_{L_2} a + \cdots + \tfrac{1}{L_1 \cdots L_{r-1}}\partial_{L_r} a - na\Big).$$

This is the exact sense in which one logarithmic level is special: it is the only depth at which polynomial coefficients are differentially closed.

A common trap

The formal partial derivative ∂_t, which extracts coefficients, is not the derivative with respect to X. A system that treats L as a symbol independent of X will compute $D_X(p(L)t^n) = -np(L)t^{n+1}$, losing the term $t^{n+1}p'(L)$ – exactly the term that carries all logarithmic information. Conversely, differentiating after the substitution $L = -\log t$ is a different operation again. Print the differentiated variable.

A separate trap is analytic. Termwise differentiation of an *asymptotic* expansion is not automatic: a remainder that is $o_{X\to+\infty}(X^{-N})$ may have a derivative that is not $o_{X\to+\infty}(X^{-N-1})$. Formal closure of $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ under D_X is an algebraic fact and licenses no analytic conclusion without a realisation theorem – analyticity in a sector, or monotonicity plus a derivative estimate [25, 20].

11.6 The block equation and the resonant block

Antidifferentiation is, by Equation (11.3), a block-by-block problem: $D_X G = F$ with $G = \sum q_n t^n$ is the system

$$\big(\partial_L - n\big)q_n = p_{n+1}, \qquad n \in \mathbb{Z}, \tag{11.11}$$

one independent equation per block, with the outer index shifted down by one. Everything therefore reduces to the solvability of one linear ordinary equation in one polynomial or coefficient unknown.

Lemma 11.15 (Nonresonant blocks are triangular). *Let $a \in \mathbb{K} \setminus \{0\}$ and $q \in \mathbb{K}[L]$ of degree D. Then $\big(\partial_L - a\big)p = q$ has the unique polynomial solution*

$$p = -\sum_{j=0}^{D} \frac{q^{(j)}}{a^{j+1}}, \tag{11.12}$$

a finite sum, and $\deg_{\mathrm{L}} p = D$ with leading coefficient $-a^{-1}$ times that of q. The same formula solves the equation uniquely in $\mathbb{K}((L^{-1}))$ for any $q \in \mathbb{K}((L^{-1}))$, the sum then being convergent in the L^{-1}-adic order; hence $\partial_L - a$ is bijective on $\mathbb{K}[L]$ and on $\mathbb{K}((L^{-1}))$.

Proof. In the basis $1, L, \dots, L^D$, the operator $\partial_L - a$ is triangular with diagonal $-a \neq 0$, hence bijective on polynomials of degree $\leq D$; this proves existence, uniqueness and the degree statement. For the closed form, apply $\partial_L - a$ to the right-hand side of Equation (11.12):

$$\big(\partial_L - a\big)\Big(-\sum_{j\geq 0} \frac{q^{(j)}}{a^{j+1}}\Big) = -\sum_{j\geq 0} \frac{q^{(j+1)}}{a^{j+1}} + \sum_{j\geq 0} \frac{q^{(j)}}{a^{j}} = q,$$

the sums telescoping and $q^{(j)} = 0$ for $j > D$. In $\mathbb{K}((L^{-1}))$ the same computation is valid because $\omega(q^{(j)}) \geq \omega(q) + j \to +\infty$ by Theorem 10.12(ii), so the family is summable and may be differentiated termwise; injectivity is Theorem 10.12(ii). □

Lemma 11.16 (The resonant block). *On $\mathbb{K}[L]$, ∂_L is surjective with kernel $\mathbb{K}$: every $q \in \mathbb{K}[L]$ of degree D has the antiderivatives $p = c + \sum_{i=0}^{D} \frac{q_i}{i+1} L^{i+1}$, of degree $D+1$, one for each $c \in \mathbb{K}$. On $\mathbb{K}((L^{-1}))$, ∂_L has kernel $\mathbb{K}$ and image $\{h : \mathrm{res}(h) = 0\}$, a subspace of codimension one.*

Proof. The polynomial statement uses $\operatorname{char}\mathbb{K} = 0$ to divide by $i+1$; that $\deg_{\mathrm{L}} p = D+1$ is clear from the leading term, and $p' = 0$ forces $p \in \mathbb{K}$. The statement for $\mathbb{K}((L^{-1}))$ is Theorem 10.24, and res is onto $\mathbb{K}$ because $\mathrm{res}(L^{-1}) = 1$. □

Remark 11.17 (The same dichotomy as in the motivation part). Theorem 11.15 and Theorem 11.16 are the nonresonant and resonant halves of Theorem 4.16, sharpened in the two ways the antiderivative theorem needs: the nonresonant inverse is given in closed form by Equation (11.12), and both halves are proved on $\mathbb{K}((L^{-1}))$ as well as on $\mathbb{K}[L]$, where the resonant image acquires the codimension-one residue obstruction. Nothing about the mechanism is new: the unique resonant outer level is still $n = 0$ for $\partial_L - n$, equivalently the level $n = 1$ of the series being antidifferentiated, and it is still the only level at which the logarithmic degree can rise.

11.7 Antiderivatives in the polynomial–logarithmic class

Theorem 11.18 (D_X is onto $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$). *The sequence*

$$0 \longrightarrow \mathbb{K} \longrightarrow \mathscr{PL}_{\mathrm{poly},\mathbb{K}} \xrightarrow{D_X} \mathscr{PL}_{\mathrm{poly},\mathbb{K}} \longrightarrow 0 \tag{11.13}$$

is exact. That is: every $F \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ has an antiderivative $G \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$, and G is unique up to an additive constant of $\mathbb{K}$. Explicitly, with $F = \sum_{n\geq n_0} p_n t^n$, the antiderivatives are

$$G = c + \int p_1 \, \mathrm{d}L \; + \sum_{\substack{n\geq n_0-1\\ n\neq 0}} t^n \Big(-\sum_{j\geq 0} \frac{p_{n+1}^{(j)}}{n^{j+1}}\Big), \qquad c \in \mathbb{K}, \tag{11.14}$$

where $\int p_1 \, \mathrm{d}L$ is the polynomial antiderivative with zero constant term. The log-degree profile of G is

$$d_G(n) = d_F(n+1) \ (n \neq 0), \qquad d_G(0) = d_F(1) + 1 \tag{11.15}$$

for the choice $c = 0$, with the convention $-\infty + 1 = -\infty$. The same holds in every Puiseux-log ring and in every power–log scale with polynomial coefficients.

Proof. Exactness at the left and middle is Theorem 11.12. For surjectivity, solve Equation (11.11) block by block: for $n \neq 0$ use Theorem 11.15 with $a = n$, obtaining the unique $q_n \in \mathbb{K}[L]$ displayed in Equation (11.14); for $n = 0$ use Theorem 11.16, obtaining $q_0 = c + \int p_1 \, \mathrm{d}L$. The resulting family $(q_n t^n)$ has support bounded below by $n_0 - 1$, so Theorem 10.1 assembles it into an element $G \in \mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$, and Equation (11.3) gives $D_X G = F$. Uniqueness up to $\mathbb{K}$ is again Theorem 11.12. The profile statement is the degree bookkeeping of Theorems 11.15 and 11.16: the nonresonant solution has the same degree as its right-hand side, the resonant one has degree one higher. Nothing in the argument used $\Gamma = \mathbb{Z}$; for a general scale with polynomial coefficients the only resonant exponent is still $\gamma = 0$. ☐

Example 11.19 (Both blocks)**.** Take $F = L^2 t^3 + L^2 t$. The first term is nonresonant with $n = 2$: Equation (11.12) with $a = 2$, $q = L^2$ gives $q_2 = -\frac{1}{2}L^2 - \frac{1}{2}L - \frac{1}{4}$. The second is the resonant block $n = 0$: $q_0 = \frac{1}{3}L^3 + c$. Hence

$$\int \Big(\frac{(\log X)^2}{X^3} + \frac{(\log X)^2}{X} \Big) \, \mathrm{d}X = -\frac{2L^2 + 2L + 1}{4X^2} + \frac{L^3}{3} + c.$$

Differentiation checks both blocks. Note the degree growth in the resonant block only, as Equation (11.15) predicts.

Remark 11.20 (Why the classical obstruction is invisible here). Theorem 11.18 may look too strong, since $\int \mathrm{d}X/(X \log X) = \log\log X$ is the standard example of an integral that leaves an elementary scale. There is no conflict: the integrand is t/L, which is not in $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ at all. The obstruction is invisible in $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ precisely because a polynomial has no poles and no residue; it becomes visible, and is then exactly one linear condition, as soon as the coefficient ring is enlarged. That is the content of the next subsection.

11.8 The exact obstruction in the larger coefficient classes

Theorem 11.21 (Antiderivatives in $\mathscr{P}\mathscr{L}_{\mathrm{itLaur},\mathbb{K}}$)**.** *Define* $\mathrm{res}\colon \mathscr{P}\mathscr{L}_{\mathrm{itLaur},\mathbb{K}} \to \mathbb{K}$ *by*

$$\mathrm{res}(F) := [t^1 L^{-1}]F = \mathrm{res}\big([t^1]F\big), \tag{11.16}$$

the coefficient of the single monomial tL^{-1}. Then the sequence

$$0 \longrightarrow \mathbb{K} \longrightarrow \mathscr{P}\mathscr{L}_{\mathrm{itLaur},\mathbb{K}} \xrightarrow{D_X} \mathscr{P}\mathscr{L}_{\mathrm{itLaur},\mathbb{K}} \xrightarrow{\mathrm{res}} \mathbb{K} \longrightarrow 0 \tag{11.17}$$

is exact. That is: $F \in \mathscr{P}\mathscr{L}_{\mathrm{itLaur},\mathbb{K}}$ has an antiderivative in $\mathscr{P}\mathscr{L}_{\mathrm{itLaur},\mathbb{K}}$ if and only if $\mathrm{res}(F) = 0$; the antiderivative is then unique up to $\mathbb{K}$. In general, if Λ is a logarithm of L in an exponential extension, then

$$F - \mathrm{res}(F)\,\frac{t}{L} \in D_X\big(\mathscr{P}\mathscr{L}_{\mathrm{itLaur},\mathbb{K}}\big), \tag{11.18}$$

so that every $F \in \mathscr{P}\mathscr{L}_{\mathrm{itLaur},\mathbb{K}}$ has an antiderivative in $\mathscr{P}\mathscr{L}_{\mathrm{itLaur},\mathbb{K}} \oplus \mathbb{K}\Lambda$.

Proof. Exactness at $\mathbb{K}$ and at the first $\mathscr{P}\mathscr{L}_{\mathrm{itLaur},\mathbb{K}}$ is Theorem 11.12. The system Equation (11.11) has, by Theorem 11.15, a unique solution $q_n \in \mathbb{K}((L^{-1}))$ for every $n \neq 0$, and by Theorem 11.16 a solution q_0 if and only if $\mathrm{res}(p_1) = 0$, in which case it is unique up to $\mathbb{K}$. Since $\mathrm{res}(p_1) = \mathrm{res}(F)$ by definition, and since the assembled family has support bounded below by $\nu_t F - 1$ (Theorem 10.1), this proves exactness at the second $\mathscr{P}\mathscr{L}_{\mathrm{itLaur},\mathbb{K}}$. Surjectivity of res follows from $\mathrm{res}(t/L) = 1$. For Equation (11.18), subtracting $\mathrm{res}(F)t/L$ makes the residue vanish, and $D_X\Lambda = D_X L/L = t/L$ by Theorem 10.10. ☐

Corollary 11.22 (Where the second logarithm comes from)**.** *$F = t/L = 1/(X\log X)$ has $\operatorname{res}(F) = 1$, so it has no antiderivative in $\mathscr{P}\mathscr{L}_{\mathrm{itLaur},\mathbb{K}}$, hence none in $\mathscr{P}\mathscr{L}_{\mathrm{rat},\mathbb{K}}$ or $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$; it has the antiderivative $\Lambda = \log\log X$. More generally, by Theorem 10.25, for nonzero F in $\mathscr{P}\mathscr{L}_{\mathrm{itLaur},\mathbb{K}}$ with leading coefficient b,*

$$\operatorname{res}\Big(\frac{D_X F}{F}\Big) = -\omega(b) = \deg_{\mathrm{L}} b \quad \textit{when } b \textit{ is a polynomial}, \tag{11.19}$$

so the logarithmic derivative of F has an antiderivative in $\mathscr{P}\mathscr{L}_{\mathrm{itLaur},\mathbb{K}}$ if and only if $\omega(b) = 0$. This is Theorem 10.26(ii) again: $\log F = \int D_X F/F$, and the two obstructions are the same functional.

Proof. Combine Theorem 11.21 with Theorem 10.12(iv) and Theorem 10.25: $[t^1](D_X F/F) = \partial_L b/b - e$, whose residue is $-\omega(b)$, the constant e contributing nothing. □

The rational-coefficient field behaves differently. The sources record, all of them correctly, that $\mathscr{P}\mathscr{L}_{\mathrm{rat},\mathbb{K}}$ is closed under differentiation, and they prove closure under antiderivatives for the polynomial class only; none of them remarks that the antiderivative statement fails after the enlargement to $\mathbb{K}(L)$ – an inference their closure tables invite. It does fail, and not only at the resonant block: the obstruction is a linear condition at every pole of every coefficient block.

Lemma 11.23 (Twisted residues)**.** *Let $a \in \mathbb{K}$ and $q \in \mathbb{K}(L)$. For $\lambda \in \overline{\mathbb{K}}$ define the* twisted residue

$$\rho_{a,\lambda}(q) := \operatorname{Res}_{L=\lambda}\Big(\mathrm{e}^{-a(L-\lambda)}q\Big) = \sum_{k\ge 0}\frac{(-a)^k}{k!}\operatorname{Res}_{L=\lambda}\big((L-\lambda)^k q\big), \tag{11.20}$$

a finite sum, and note $\rho_{0,\lambda} = \operatorname{Res}_\lambda$. Then

$$\big(\partial_L - a\big)\mathbb{K}(L) = \big\{q \in \mathbb{K}(L) : \rho_{a,\lambda}(q) = 0 \textit{ for every } \lambda \in \overline{\mathbb{K}}\big\}\,. \tag{11.21}$$

In particular $\mathbb{K}[L] \subseteq (\partial_L - a)\mathbb{K}(L)$, while for $a \neq 0$ neither $1/L$ nor $1/L^2$ lies in the image.

Proof. Work first over $\overline{\mathbb{K}}$ and fix λ; write $u = L - \lambda$ and expand in $\overline{\mathbb{K}}((u))$. The multiplier $\mathrm{e}^{-au} = \sum_k (-au)^k/k!$ is a unit of $\overline{\mathbb{K}}[[u]]$, the displayed sum is finite because $u^k q$ is regular for k at least the pole order, and

$$\mathrm{e}^{-au}\big(\partial_L - a\big)p = \partial_L\big(\mathrm{e}^{-au}p\big)$$

for every $p \in \overline{\mathbb{K}}((u))$. Since the residue of a derivative of a Laurent series vanishes, $\rho_{a,\lambda}$ kills the image; this proves the inclusion $\subseteq$, the value at a point where q is regular being 0.

For $\supseteq$, decompose $q = Q + \sum_\lambda q_\lambda$ into a polynomial and its polar parts over $\overline{\mathbb{K}}$; $\rho_{a,\lambda}(q) = \rho_{a,\lambda}(q_\lambda)$ because the other summands are regular at λ. The polynomial part is in the image by Theorem 11.15 (for $a \neq 0$) or Theorem 11.16 (for $a = 0$). Fix λ and let M be the pole order of q_λ. If p has a pole of order $m \ge 1$ at λ, then $(\partial_L - a)p$ has a pole of order exactly $m+1$ there, since the top coefficient is $-mc_m \neq 0$. Hence the linear map sending polar parts of order $\le M-1$ to polar parts of order $\le M$ is injective, so its image has codimension one in the M-dimensional space of polar parts of order $\le M$; the functional $\rho_{a,\lambda}$ vanishes on that image and is not identically zero on the target (it takes the value d_1 on $d_1 u^{-1}$). Therefore the image is exactly $\ker \rho_{a,\lambda}$, and q_λ lies in it. Summing the local solutions and the polynomial solution gives $p \in \overline{\mathbb{K}}(L)$ with $(\partial_L - a)p = q$.

Finally, descend to $\mathbb{K}$. If $a \neq 0$, p is unique by Theorem 11.15, hence fixed by every automorphism of $\overline{\mathbb{K}}$ over $\mathbb{K}$ (which fixes q, a and L), so $p \in \mathbb{K}(L)$. If $a = 0$, p is unique up to a constant; choosing a finite Galois extension $E/\mathbb{K}$ with $p \in E(L)$ and replacing p by $[E:\mathbb{K}]^{-1}\sum_{\sigma\in\mathrm{Gal}(E/\mathbb{K})}\sigma(p)$, which is again an antiderivative because each $\sigma(p)$ is one, gives an element of $\mathbb{K}(L)$. The examples: $1/L$ has $\rho_{a,0} = 1$; $1/L^2$ has $\rho_{a,0} = -a$. □

Theorem 11.24 (Antiderivatives in $\mathscr{P}\mathscr{L}_{\mathrm{rat},\mathbb{K}}$)**.** *Let $F = \sum_{n\ge n_0} p_n t^n \in \mathscr{P}\mathscr{L}_{\mathrm{rat},\mathbb{K}}$. Then F has an antiderivative in $\mathscr{P}\mathscr{L}_{\mathrm{rat},\mathbb{K}}$ if and only if*

$$\rho_{n-1,\,\lambda}\big(p_n\big) = 0 \qquad \textit{for every } n \in \mathbb{Z} \textit{ and every } \lambda \in \overline{\mathbb{K}}, \tag{11.22}$$

and the antiderivative is then unique up to $\mathbb{K}$. The condition at $n=1$ is the vanishing of all residues of $[t^1]F$; the conditions at $n \neq 1$ are the twisted conditions with weight $\mathrm{e}^{-(n-1)(L-\lambda)}$. Consequently:

1. *every $F \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ satisfies Equation* (11.22) *(polynomials have no poles), recovering Theorem 11.18;*
2. *$t^2/L = 1/(X^2 \log X)$ has no antiderivative in $\mathscr{PL}_{\mathrm{rat},\mathbb{K}}$; it does have one in $\mathscr{PL}_{\mathrm{itLaur},\mathbb{K}}$, namely $t\sum_{j\geq 0}(-1)^{j+1} j!\, L^{-j-1}$;*
3. *$\mathscr{PL}_{\mathrm{rat},\mathbb{K}}$ is a differential field that is not closed under antidifferentiation, whereas $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ is a differential ring, not a field, that is closed under antidifferentiation.*

Proof. The system Equation (11.11) is solvable in $\mathbb{K}(L)$ block by block if and only if $p_{n+1} \in (\partial_L - n)\mathbb{K}(L)$ for every n, which by Theorem 11.23 is Equation (11.22) after the index shift $n \mapsto n-1$. Solutions assemble by Theorem 10.1 because the support is bounded below, and the ambiguity is $\ker D_X = \mathbb{K}$ by Theorem 11.12. (i) is immediate. For (ii), $n=2$ and $\lambda = 0$ give $\rho_{1,0}(1/L) = 1 \neq 0$, so the criterion fails; the element displayed is the unique solution of $(\partial_L - 1)q_1 = L^{-1}$ provided by Theorem 11.15 in $\mathbb{K}((L^{-1}))$, namely $q_1 = -\sum_{j\geq 0}(L^{-1})^{(j)} = -\sum_{j\geq 0}(-1)^j j!\, L^{-j-1}$. (iii) collects Theorem 11.13, Theorem 11.18 and part (ii); that $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ is not a field is Theorem 9.23. □

Remark 11.25 (A divergent formal antiderivative). The series $t\sum_{j\geq 0}(-1)^{j+1} j! L^{-j-1}$ of Theorem 11.24(ii) is a perfectly good element of $\mathscr{PL}_{\mathrm{itLaur},\mathbb{K}}$: its defining sum is L^{-1}-adically summable, and its blocks are determined by finitely many operations each. It is not, and does not claim to be, a convergent series; the factorial growth of its coefficients is the algebraic shadow of the fact that $\int \mathrm{d}X/(X^2 \log X)$ is not elementary, and the analytic content of the expansion is a matter of summation theory rather than of the formal algebra [12, 8]. Nothing in Theorems 11.21 and 11.24 asserts convergence; the statements are exact statements about coefficients.

Table 11.1: The differential calculus of the four ambient classes. Here $\operatorname{res} F = [t^1 L^{-1}]F$ and $\rho_{a,\lambda}$ are the twisted residues defined in this section.

Class	D_X-closed	$\ker D_X$	antiderivative exists in the class
$\mathscr{PL}_{\mathrm{pow},\mathbb{K}} = \mathbb{K}[L][[t]]$	yes, into $t\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$	$\mathbb{K}$	iff $\nu_t F \geq 1$ (Theorem 11.13)
$\mathscr{PL}_{\mathrm{poly},\mathbb{K}} = \mathbb{K}[L]((t))$	yes	$\mathbb{K}$	always (Theorem 11.18)
$\mathscr{PL}_{\mathrm{rat},\mathbb{K}} = \mathbb{K}(L)((t))$	yes	$\mathbb{K}$	iff all $\rho_{n-1,\lambda}([t^n]F) = 0$ (Theorem 11.24)
$\mathscr{PL}_{\mathrm{itLaur},\mathbb{K}} = \mathbb{K}((L^{-1}))((t))$	yes	$\mathbb{K}$	iff $\operatorname{res} F = 0$ (Theorem 11.21)

Structural checkpoint

The whole of Part 11 turns on one line: D_X sends the outer block n to the block $n+1$ by the operator $\partial_L - n$. Everything else is a statement about that operator. It is injective for $n \neq 0$ and has kernel $\mathbb{K}$ for $n = 0$; it is surjective on $\mathbb{K}[L]$ for every n, surjective on $\mathbb{K}((L^{-1}))$ for $n \neq 0$ and of corank one for $n = 0$, and of infinite corank on $\mathbb{K}(L)$ for every n as soon as poles are admitted. Reading the four ambient classes off from that single operator is the fastest route through the material.

Section summary

The distinguished derivation $D_X = t\partial_L - t^2\partial_t$ is the unique bounded $\mathbb{K}$-derivation with $D_X t = -t^2$ and $D_X L = t$; on one block it is $D_X(t^n p) = t^{n+1}(\partial_L - n)p$, and k derivatives give $t^{n+k}\Delta_{n,k} p$

with the empty-product convention. The Euler derivation $\vartheta = t^{-1}D_X$ preserves each block and acts there as $\partial_L - n$, whose invertibility fails exactly at the resonant block $n = 0$. Valuation rises by exactly one except when the leading block is a nonzero constant, and then $\nu_t(D_X F) = \nu_t(F - a_{0,0}) + 1$. All four ambient classes are closed under D_X. For antiderivatives the answer is a residue: D_X is onto $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ with kernel $\mathbb{K}$; on $\mathscr{P}\mathscr{L}_{\mathrm{itLaur},\mathbb{K}}$ the single functional $\operatorname{res} F = [t^1 L^{-1}]F$ is the entire obstruction, and it is repaired by adjoining $\Lambda = \log L$; on $\mathscr{P}\mathscr{L}_{\mathrm{rat},\mathbb{K}}$ there is one twisted-residue condition at every pole of every block, so exact rational coefficients are the one enlargement that costs closure under integration.

Part IV

Composition

Chapter 12

Admissible composition and substitution

Throughout this part $\mathbb{K}$ is a field of characteristic zero, X is the large variable, and

$$t := X^{-1}, \qquad L := \log X$$

are the two generators. Unless a sector is named, the analytic regime is the positive real ray $X \to +\infty$ and every logarithm is the real logarithm there. We write $\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ for $\mathbb{K}[L][[t]]$, $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ for $\mathbb{K}[L]((t))$, ν_t for the outer valuation, $\mathrm{Trunc}_{<N} F$ for the *exclusive* truncation $\sum_{n<N} p_n(L)t^n$, and $D_X = t\partial_L - t^2\partial_t$ for the distinguished derivation. Recall the repeated-derivative operator

$$\Delta_{n,k} := \prod_{j=0}^{k-1}(\partial_L - n - j), \qquad \Delta_{n,0} = 1, \qquad D_X^k\big(t^n p(L)\big) = t^{n+k}\Delta_{n,k}p(L). \tag{12.1}$$

The empty product at $k = 0$ is the identity operator; this convention is used without further comment.

Composition is the first operation for which the analytic relation between the two generators is essential. Addition and multiplication hold L fixed and merely recombine supports that are already present. Substitution of $X \mapsto G(X)$ moves both generators at once, and it can move them out of the ring: fractional powers, a second logarithmic level, or exponential monomials may appear. The whole content of this chapter is a set of hypotheses under which the substitution is defined, closed, associative, and computable block by block, together with a careful separation of that formal statement from the analytic statement that it computes the asymptotics of an actual composite function.

12.1 What substitution has to mean

For a polynomial in t and L, substitution is finite algebra and needs no theory. For an infinite series $F = \sum_{n\geq n_0} p_n(L)t^n$ one wants to declare

$$F \circ G := \sum_{n\geq n_0} p_n(L\circ G)\,(t\circ G)^n,$$

and three separate things must be checked.

1. *The generators must land in a named ring.* Both $t \circ G$ and $L \circ G$ must be expansible in the chosen monomial scale.
2. *The outer sum must be summable.* For each output monomial, only finitely many indices n may contribute; otherwise a coefficient is an undefined infinite sum in $\mathbb{K}$.

3. *The result must have admissible support.* In a Laurent setting this means finitely many negative outer exponents; in a Hahn setting it means a well-ordered exponent set.

Each requirement fails for some perfectly meaningful G.

Example 12.1 (Four failures of naive substitution). 1. $F = t$, $G = X^{1/2}$: here $t \circ G = t^{1/2} \notin \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$, and a Puiseux extension is forced.

2. $F = t$, $G = \log X$: here $t \circ G = L^{-1} \notin \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$, and negative powers of L are forced, that is, a Laurent-log coefficient field.

3. $F = L$, $G = X \log X$: here $L \circ G = L + \log L$, and a second logarithmic level is forced.

4. $F = L$, $G = \mathrm{e}^X$: here $t \circ G = \mathrm{e}^{-X}$, smaller than every power of t, and an exponential monomial is forced.

Failure of summability is separate and more insidious. Let $F = \sum_{n\geq 0} t^n$ and let $G = c$ be a nonzero constant, so that $t \circ G = c^{-1}$ has outer valuation 0. Every summand c^{-n} then lands in the block t^0, and the coefficient of t^0 in the putative composite is $\sum_{n\geq 0} c^{-n}$, which is not an element of $\mathbb{K}$. The same collapse occurs for any G whose leading exponent is 0; this is why the hypothesis $\rho > 0$ below is not decorative.

Remark 12.2. Nothing in Theorem 12.1 says that the composite function does not exist. Closure is a statement about a chosen representation class, not about existence. When a substitution leaves the class, the correct response is to name the enlargement it demands, not to suppress the offending symbol.

We record the summability criterion once, in the generality needed later. Let $\Gamma \subseteq \mathbb{R}$ be an ordered additive subgroup and let

$$\mathbb{K}[L][[t^\Gamma]] := \left\{\textstyle\sum_{\alpha\in S} p_\alpha(L) t^\alpha :\ p_\alpha \in \mathbb{K}[L],\ S \subseteq \Gamma \text{ well ordered}\right\}$$

be the associated polynomial-logarithmic Hahn ring. For $F \neq 0$ write $E(F) := \{\alpha \in \Gamma : [t^\alpha]F \neq 0\}$ for its exponent support, so that $\nu_t(F) = \min E(F)$ and $\mathrm{supp}_{\mathrm{coeff}}(F)$ is the finer set of pairs (α, j) with $a_{\alpha,j} \neq 0$. For $\Gamma = \mathbb{Z}$ this ring is exactly $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$, and for $\Gamma = s^{-1}\mathbb{Z}$ it is the Puiseux-log ring $\mathbb{K}[L]((t^{1/s}))$.

Lemma 12.3 (Summable families in a polynomial–logarithmic Hahn ring). *Let $\mathcal{A} = \mathbb{K}[L][[t^\Gamma]]$ and let $(S_i)_{i\in I}$ be a family in $\mathcal{A}$ such that*

$$\{i \in I :\ \nu_t(S_i) \leq \beta\} \quad \textit{is finite for every } \beta \in \Gamma. \tag{12.2}$$

Then $\bigcup_i E(S_i)$ is a well-ordered subset of Γ, every exponent occurs in only finitely many S_i, and

$$\sum_{i\in I} S_i := \sum_\alpha \Big(\sum_i [t^\alpha] S_i\Big) t^\alpha$$

is a well-defined element of $\mathcal{A}$. The sum is unchanged by any regrouping or reindexing of I; in particular a doubly indexed summable family may be summed in either order. Finally, call a $\mathbb{K}$-linear map $\Psi : \mathcal{A} \to \mathcal{B}$ between two such rings compatible with summable families *when there is a non-decreasing unbounded φ with $\nu_t(\Psi a) \geq \varphi(\nu_t a)$ for all a and $\Psi\big(\sum_i S_i\big) = \sum_i \Psi S_i$ for every summable family; the first condition alone already forces (ΨS_i) to satisfy* (12.2)*, and a composite of two compatible maps is compatible.*

Proof. Let $T \subseteq \bigcup_i E(S_i)$ be nonempty and pick $\beta \in T$. By (12.2) the index set $I_\beta = \{i : \nu_t(S_i) \leq \beta\}$ is finite, and every element of T that is $\leq \beta$ lies in $\bigcup_{i\in I_\beta} E(S_i)$, a finite union of well-ordered sets and hence well ordered. Therefore $T \cap (-\infty, \beta]$ has a least element, which is the least element of T. So the union is well ordered. Taking $\beta = \alpha$ shows that only the finitely many $i \in I_\alpha$ can contribute to the exponent α, so each inner sum is finite and the displayed series has well-ordered support. Regrouping and reindexing change none of the finitely many terms contributing to a fixed α. For the last assertion, let φ be non-decreasing and unbounded with $\nu_t(\Psi a) \geq \varphi(\nu_t a)$, and fix β. Choose $\beta_0 \in \Gamma$ with $\varphi(\gamma) > \beta$ for all $\gamma > \beta_0$. Then $\{i : \nu_t(\Psi S_i) \leq \beta\} \subseteq \{i : \nu_t(S_i) \leq \beta_0\}$, which is finite. ☐

Remark 12.4. Condition (12.2) at $\beta = 0$ already forces a common lower support bound: only finitely many members have valuation ≤ 0, and each of those has finite valuation. A t-adically Cauchy *sequence* in a Laurent ring carries no such guarantee, which is why the family formulation is the one used throughout.

12.2 Formal exponential and logarithm of a one-unit

Write $\mathfrak{m} := t\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ for the maximal ideal of $\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$; in a Hahn ring $\mathfrak{m}$ denotes the ideal of elements of strictly positive valuation. For $Z \in \mathfrak{m}$ put

$$\mathrm{e}^Z := \sum_{k\geq 0} \frac{Z^k}{k!}, \qquad \log(1+Z) := \sum_{r\geq 1} \frac{(-1)^{r-1}}{r} Z^r. \tag{12.3}$$

Both families satisfy (12.2) because $\nu_t(Z^k) \geq k\nu_t(Z) \to +\infty$.

Lemma 12.5 (Formal exponential and logarithm)**.** *The maps* (12.3) *are mutually inverse bijections between* $\mathfrak{m}$ *and* $1+\mathfrak{m}$*, and*

$$\mathrm{e}^{Z_1+Z_2} = \mathrm{e}^{Z_1}\mathrm{e}^{Z_2}, \qquad \log(P_1P_2) = \log P_1 + \log P_2, \qquad \log(P^n) = n\log P$$

for $Z_1, Z_2 \in \mathfrak{m}$*,* $P, P_1, P_2 \in 1+\mathfrak{m}$ *and* $n \in \mathbb{Z}$*. Moreover, if* Ψ *is a* $\mathbb{K}$*-algebra homomorphism into another such ring, compatible with summable families and carrying elements of positive valuation to elements of positive valuation, then* $\Psi(\mathrm{e}^Z) = \mathrm{e}^{\Psi Z}$ *and* $\Psi(\log(1+Z)) = \log(1+\Psi Z)$*.*

Proof. In $\mathbb{Q}[[z_1, z_2]]$ one has the classical identities $\exp(z_1+z_2) = \exp z_1 \exp z_2$, $\log((1+z_1)(1+z_2)) = \log(1+z_1) + \log(1+z_2)$, $\log\exp z_1 = z_1$ and $\exp\log(1+z_1) = 1+z_1$. Because $\nu_t(Z_i) > 0$, the monomial family $\big(a_{k_1k_2}Z_1^{k_1}Z_2^{k_2}\big)_{k_1,k_2}$ satisfies (12.2) for every choice of coefficients $a_{k_1k_2} \in \mathbb{Q}$, so $z_i \mapsto Z_i$ extends to a $\mathbb{Q}$-algebra homomorphism $\mathbb{Q}[[z_1, z_2]] \to \mathcal{A}$ that is compatible with summable families (Theorem 12.3); applying it to the four identities gives the first three assertions for $n \geq 0$, and $\log(P^{-1}) = -\log P$ follows from $\log(PP^{-1}) = \log 1 = 0$. The last sentence is termwise application of Ψ to (12.3). ☐

12.3 Monomial-leading inner arguments and the substitution homomorphism

Definition 12.6 (Admissible monomial-leading argument)**.** An *admissible monomial-leading inner argument* over $\mathbb{K}$ is a quadruple $(c, \log c, \rho, U)$ consisting of

$$c \in \mathbb{K}^\times, \qquad \text{a chosen value } \log c \in \mathbb{K}, \qquad \rho \in \mathbb{N}_{>0}, \qquad U \in t\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}},$$

presented as the inner argument

$$G(X) = cX^\rho\big(1+U(X)\big) = c\,t^{-\rho}(1+U) \in \mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}. \tag{12.4}$$

We put

$$V := \log(1+U) \in t\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}. \tag{12.5}$$

On the real ray we additionally require $c > 0$ and take $\log c$ real, so that $G(X) \to +\infty$.

The datum $\log c$ is part of the inner argument, not a consequence of it. If $\mathbb{K}$ does not already contain a logarithm of c, it must first be enlarged; and if $\mathbb{K} \subseteq \mathbb{C}$ contains $2\pi\mathrm{i}$, then c admits several admissible logarithms, which give genuinely different substitutions (Theorem 12.9).

Theorem 12.7 (Substitution homomorphism of a monomial-leading argument)**.** *Let G be as in Theorem 12.6. For $F = \sum_{n\geq n_0} p_n(L)t^n \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ the family*

$$\big(\, p_n(\rho L + \log c + V)\, c^{-n} t^{\rho n}(1+U)^{-n}\,\big)_{n\geq n_0} \tag{12.6}$$

is summable in $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$, so that

$$F \circ G := \sum_{n\geq n_0} p_n\!\big(\rho L + \log c + V\big)\, c^{-n}\, t^{\rho n}\, (1+U)^{-n} \tag{12.7}$$

is a well-defined element of $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$. The resulting map $\Phi_G : F \mapsto F \circ G$ is

1. *a $\mathbb{K}$-algebra homomorphism $\mathscr{PL}_{\mathrm{poly},\mathbb{K}} \to \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$, compatible with summable families;*
2. *determined by its values on the generators,*
$$\Phi_G(t) = c^{-1}t^{\rho}(1+U)^{-1}, \qquad \Phi_G(L) = \rho L + \log c + V; \tag{12.8}$$
3. *exactly valuation-multiplying: $\nu_t(F\circ G) = \rho\, \nu_t(F)$ for all F. In particular Φ_G is injective, and $\Phi_G(\mathscr{PL}_{\mathrm{pow},\mathbb{K}}) \subseteq \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$.*

Proof. Since $\nu_t(U) \geq 1$, Theorem 12.5 gives $V \in t\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$, and the binomial family $(\binom{-n}{r}U^r)_{r\geq 0}$ is summable for every $n \in \mathbb{Z}$, so $(1+U)^{-n} \in 1 + t\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$. Because p_n is a polynomial of some finite degree d_n,

$$p_n(\rho L + \log c + V) = \sum_{j=0}^{d_n} \frac{p_n^{(j)}(\rho L + \log c)}{j!}\, V^j \tag{12.9}$$

is a *finite* sum, each summand lying in $\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$; here $p_n^{(j)}$ is the j-th derivative of p_n in its own variable and the polynomial is evaluated at $\rho L + \log c$. Hence the n-th member of (12.6) lies in $t^{\rho n}\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$, and its valuation is at least ρn. Since $\rho \geq 1$ and $n \geq n_0$, condition (12.2) holds, and Theorem 12.3 gives the sum (12.7).

For the algebra structure, note that (12.7) is exactly "substitute (12.8) into the expansion of F". Concretely, let $\Psi_0 : \mathbb{K}[L][t,t^{-1}] \to \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ be the unique $\mathbb{K}$-algebra homomorphism with $\Psi_0(L) = \rho L + \log c + V$ and $\Psi_0(t) = c^{-1}t^{\rho}(1+U)^{-1}$; it exists because L and t are algebraically independent over $\mathbb{K}$ and $\Psi_0(t)$ is a unit of $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$, the required inverse being $c\,t^{-\rho}(1+U)$. Then $\Phi_G(p_n(L)t^n) = \Psi_0(p_n(L)t^n)$ for every n and every p_n, and $\Phi_G(F) = \sum_n \Psi_0(p_n(L)t^n)$. Additivity of Φ_G is clear. For multiplicativity, let $F = \sum_n p_n t^n$ and $F' = \sum_m q_m t^m$. The double family $\big(\Psi_0(p_n t^n)\Psi_0(q_m t^m)\big)_{n,m}$ satisfies (12.2), because its (n,m) member has valuation at least $\rho(n+m)$; summing it in either order (Theorem 12.3) gives

$$\Phi_G(F)\Phi_G(F') = \sum_{n,m} \Psi_0\big(p_n q_m t^{n+m}\big) = \sum_r \Psi_0\Big(\sum_{n+m=r} p_n q_m\, t^r\Big) = \Phi_G(FF').$$

It remains to check compatibility with summable families in the sense of Theorem 12.3. The valuation half holds with $\varphi(x) = \rho x$, which is non-decreasing and unbounded. For the interchange, let $(S_i)_{i\in I}$ be summable with sum S and write $S_i = \sum_n p_{i,n}(L)t^n$. By Theorem 12.4 there is n_1 with $\nu_t(S_i) \geq n_1$ for all i, so $p_{i,n} = 0$ for $n < n_1$, and for each fixed n only the finitely many i with $\nu_t(S_i) \leq n$ can have $p_{i,n} \neq 0$; consequently $[t^n]S = \sum_i p_{i,n}$ is a finite sum. The doubly indexed family $\big(\Psi_0(p_{i,n}t^n)\big)_{i,n}$ has (i,n) member of valuation at least $\rho n \geq \rho n_1$ and therefore satisfies (12.2). Summing it with i

innermost gives $\sum_n \Psi_0\big([t^n]S \cdot t^n\big) = \Phi_G(S)$, and summing it with n innermost gives $\sum_i \Phi_G(S_i)$; the two agree by Theorem 12.3. This proves (i) and (ii).

For (iii), let $n_0 = \nu_t(F)$. Every summand with $n > n_0$ has valuation at least $\rho n > \rho n_0$, and the summand with $n = n_0$ equals $c^{-n_0} t^{\rho n_0}\big(p_{n_0}(\rho L + \log c) + W\big)$ with $\nu_t(W) \geq 1$, by (12.9) and $(1+U)^{-n_0} \in 1 + t\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$. Since $\rho \neq 0$, the substitution $L \mapsto \rho L + \log c$ is a $\mathbb{K}$-algebra automorphism of $\mathbb{K}[L]$, so $p_{n_0}(\rho L + \log c) \neq 0$. Hence $[t^{\rho n_0}](F \circ G) = c^{-n_0} p_{n_0}(\rho L + \log c) \neq 0$ and $\nu_t(F \circ G) = \rho n_0$. Injectivity and preservation of $\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ follow at once. □

Corollary 12.8 (Scaling and pure power substitution)**.** *With $U = 0$,*

$$F(cX^\rho) = \sum_{n \geq n_0} c^{-n} t^{\rho n}\, p_n(\rho L + \log c).$$

Thus a pure scaling acts on coefficient blocks by the affine change of logarithmic variable $L \mapsto \rho L + \log c$ and on outer exponents by $n \mapsto \rho n$; no new monomials are created.

Proof. Immediate from (12.7) with $U = V = 0$. □

Remark 12.9 (The choice of logarithm is part of the data). Take $\mathbb{K} = \mathbb{C}$, $c = 1$, $\rho = 1$, $U = 0$, so that $G = X$ is the identity map. Choosing $\log c = 0$ gives $\Phi_G = \mathrm{id}$. Choosing $\log c = 2\pi\mathrm{i}$ gives the *different* $\mathbb{K}$-algebra automorphism $t \mapsto t$, $L \mapsto L + 2\pi\mathrm{i}$, which is (12.7) for the very same G. Therefore "$F \circ G$" is shorthand for $\Phi_{(c,\log c,\rho,U)}(F)$, and a symbolic implementation must carry $\log c$ as data rather than replace it by a principal value. All statements below that compose two inner arguments carry an explicit compatibility hypothesis on the chosen logarithms.

12.4 Finite determination

The practical content of Theorem 12.7 is that each output block is a finite algebraic expression in finitely many input blocks, with an explicit and *triangular* precision budget.

Theorem 12.10 (Finite determination of a composition)**.** *Let G be as in Theorem 12.6, let $F = \sum_{n \geq a} p_n(L) t^n \in \mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ with $a = \nu_t(F)$, and let $N \in \mathbb{Z}$. Then*

$$[t^N](F \circ G) = \sum_{\substack{a \leq n \\ \rho n \leq N}} c^{-n} \sum_{\substack{j \geq 0,\ r \geq 0 \\ j \leq \deg p_n}} \frac{\binom{-n}{r}}{j!}\, [t^{N-\rho n}]\big(V^j U^r\big)\, p_n^{(j)}(\rho L + \log c), \tag{12.10}$$

and in the inner sum only the pairs with $j + r \leq N - \rho n$ contribute. Both index ranges are finite, so every output block below any cutoff is a finite sum. Consequently $\mathrm{Trunc}_{<N} F \circ G$ *is a finite algebraic expression in $c^{\pm 1}$,* $\log c$*, ρ, the blocks p_n with $a \leq n < N/\rho$, and the blocks of U below outer order $N - \rho a$.*

Proof. Insert (12.9) and the binomial expansion $(1+U)^{-n} = \sum_{r \geq 0} \binom{-n}{r} U^r$ into (12.7) and extract the coefficient of t^N; the triple family is summable because its (n, j, r) member has valuation at least $\rho n + j + r$, so Theorem 12.3 licenses the rearrangement. The outer index is restricted by $a \leq n$ and $\rho n \leq N$, that is $n \leq \lfloor N/\rho \rfloor$, and the inner indices by $j + r \leq N - \rho n$, because $\nu_t(V^j U^r) \geq j + r$; both are finite ranges. Finally, $\mathrm{Trunc}_{<N} F \circ G$ collects (12.10) for all $N' < N$; the constraint $\rho n < N$ is $n < N/\rho$, and every occurrence of U or V sits inside a factor $t^{\rho n}$ with $\rho n \geq \rho a$, so only the blocks of U below outer order $N - \rho a$ can be reached. □

The last clause is worth isolating, because it is the difference between a usable and an unusable implementation: the required precision of the inner data is not uniform across outer blocks.

Corollary 12.11 (Triangular precision budget). *In* (12.7) *the n-th summand needs V and $(1+U)^{-n}$ only through outer order $N-\rho n-1$. Computing all of them to the single global precision $N-\rho a-1$ is correct but wasteful.*

Proof. The n-th summand carries the prefactor $t^{\rho n}$, so anything of outer order $\geq N-\rho n$ inside it contributes only to $\mathcal{O}_t^{\mathrm{form}}(t^N)$. □

Finite determination also yields the two stability estimates that make composition continuous in each argument. They are used repeatedly in the reversion theory.

Proposition 12.12 (Stability in the outer and inner arguments). *Let G, G_1, G_2 be admissible monomial-leading arguments with the same c, $\log c$, ρ and with $U_1, U_2 \in t\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$.*

1. *If $\nu_t(F_1-F_2) \geq M$ then $\nu_t(F_1\circ G - F_2\circ G) \geq \rho M$.*
2. *If $\nu_t(U_1-U_2) \geq M \geq 1$ and $a=\nu_t(F)$, then $\nu_t(F\circ G_1 - F\circ G_2) \geq \rho a + M$.*

Proof. (i) is Theorem 12.7(iii) applied to F_1-F_2 together with linearity of Φ_G.

For (ii) write $V_i = \log(1+U_i)$ and $D = U_1-U_2$. From $U_1^r - U_2^r = D\sum_{i=0}^{r-1} U_1^i U_2^{r-1-i}$ and $\nu_t(U_i)\geq 1$ we get $\nu_t(U_1^r-U_2^r) \geq M+r-1$, whence $\nu_t(V_1-V_2)\geq M$ by (12.3) and Theorem 12.3. The same computation applied to $(1+U_i)^{-n} = \sum_r \binom{-n}{r} U_i^r$ gives $\nu_t\big((1+U_1)^{-n}-(1+U_2)^{-n}\big) \geq M$. For a polynomial $p\in\mathbb{K}[L]$ there is a two-variable polynomial q over $\mathbb{K}$ with $p(y_1)-p(y_2) = (y_1-y_2)q(y_1,y_2)$; substituting $y_i = \rho L + \log c + V_i$, whose values have nonnegative valuation, gives $\nu_t\big(p_n(\rho L+\log c+V_1) - p_n(\rho L + \log c + V_2)\big) \geq M$. Writing the n-th summands of (12.7) as $A_1B_1 - A_2B_2 = (A_1-A_2)B_1 + A_2(B_1-B_2)$ with $A_i = p_n(\rho L+\log c+V_i)$ and $B_i = c^{-n}t^{\rho n}(1+U_i)^{-n}$, all factors having valuation at least 0 resp. ρn, we obtain a valuation at least $\rho n + M \geq \rho a + M$ for every n. □

12.5 The shifted-Taylor form of the composition formula

Formula (12.7) separates the logarithmic shift from the power prefactor. The two can be packaged into a single shifted operator, which is both more compact and better adapted to coefficient recurrences.

Lemma 12.13 (Shifted-Taylor identity). *Let $p\in\mathbb{K}[L]$, let $n\in\mathbb{Z}$, and let W be an element of positive valuation in a polynomial-logarithmic Hahn ring. Then*

$$\mathrm{e}^{-nW}\, p(z+W) = \sum_{k\geq 0} \frac{W^k}{k!}\,\big((\partial_z - n)^k p\big)(z), \tag{12.11}$$

where z is any element of the ring, ∂_z denotes differentiation of the polynomial p in its own variable, and the operator is applied to p before *evaluation at z. The family on the right is summable; for $p\neq 0$ it terminates exactly when $n=0$, since for $n\neq 0$ the polynomial $(\partial_z-n)^k p$ has the same degree as p and leading coefficient $(-n)^k$ times that of p.*

Proof. Both sides are $\mathbb{K}$-linear in p, so it suffices to expand. On the left,

$$\mathrm{e}^{-nW} p(z+W) = \Big(\sum_{i\geq 0}\frac{(-n)^i W^i}{i!}\Big)\Big(\sum_{j\geq 0}\frac{W^j}{j!}p^{(j)}(z)\Big),$$

the second factor being the finite Taylor expansion of the polynomial p. Collecting the total power $k=i+j$ gives the coefficient $\sum_{j=0}^k \frac{(-n)^{k-j}}{(k-j)!\,j!}p^{(j)}(z)$. On the right, the binomial theorem for the commuting operators ∂_z and $-n$ gives

$$\frac{1}{k!}(\partial_z-n)^k p = \frac{1}{k!}\sum_{j=0}^{k}\binom{k}{j}(-n)^{k-j}p^{(j)} = \sum_{j=0}^{k}\frac{(-n)^{k-j}}{(k-j)!\,j!}p^{(j)}.$$

The two coefficients agree. Summability holds because $\nu_t(W^k) \geq k\nu_t(W) \to +\infty$, and the rearrangement of the double family is licensed by Theorem 12.3. □

Corollary 12.14 (Operator form of the composition formula). *With G as in Theorem 12.6,*

$$F \circ G = \sum_{n \geq n_0} c^{-n} t^{\rho n} E_n, \qquad E_n := \sum_{k \geq 0} \frac{V^k}{k!} \big((\partial_z - n)^k p_n\big)(\rho L + \log c). \tag{12.12}$$

Proof. By Theorem 12.6 and Theorem 12.5, $(1+U)^{-n} = \mathrm{e}^{-nV}$. Apply Theorem 12.13 with $W = V$ and $z = \rho L + \log c$, and substitute into (12.7). □

A common trap

Two ordering traps. First, *normal ordering*. The operator in (12.12) may be written suggestively as $:\mathrm{e}^{V(\partial_z - n)}:$, the colons meaning that every power of V stands to the left and is never differentiated. It is *not* the operator exponential $\exp\big(M_V(\partial_L - n)\big)$, where M_V is multiplication by V: iterating $M_V \partial_L$ would let ∂_L act on V, which depends on L. The two agree only when $\partial_L V = 0$.
Second, *which variable is differentiated.* In (12.12) the operator ∂_z acts on the coefficient polynomial p_n in its own variable, and the result is evaluated at $\rho L + \log c$ afterwards. Writing ∂_L instead is harmless only when $\rho = 1$. Indeed the chain rule gives

$$(\partial_L - n)^k \big[p_n(\rho L + \log c)\big] = \big((\rho\,\partial_z - n)^k p_n\big)(\rho L + \log c),$$

so the misprint replaces the operator $(\partial_z - n)^k$ by $(\rho\,\partial_z - n)^k$. These two agree only for $\rho = 1$, and they are not even proportional: the factor ρ^k relates them in the single case $n = 0$. For $\rho = 2$, $n = 1$, $\log c = 0$, $p_n(z) = z$ and $k = 1$, the correct coefficient of V is $1 - 2L$ whereas the misprint produces $2 - 2L$.

The operator form makes the block coefficients of a single outer level satisfy an exact first-order recurrence. We state it for arbitrary ρ; the case $\rho = 1$ is the form found in the sources.

Proposition 12.15 (Exact block recurrence). *Assume $\rho \in \mathbb{K}^\times$. With E_n as in* (12.12)*, write $E_n = \sum_{r \geq 0} E_{n,r}(L) t^r$ and*

$$\theta := \frac{\partial_t V}{\rho + \partial_L V} = \sum_{q \geq 0} \theta_q(L) t^q, \tag{12.13}$$

the denominator being a unit of $\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$. Then

$$\partial_t E_n = \theta\,(\partial_L - n\rho) E_n, \tag{12.14}$$

and therefore

$$E_{n,0} = p_n(\rho L + \log c), \qquad r\,E_{n,r} = \sum_{q=0}^{r-1} \theta_q\,(\partial_L - n\rho) E_{n,r-1-q} \quad (r \geq 1). \tag{12.15}$$

Proof. By Theorem 12.14 and Theorem 12.13, $E_n = \mathrm{e}^{-nV} p_n(\rho L + \log c + V)$. Write $P = p_n$ and $w = \rho L + \log c + V$. Since ∂_t and ∂_L are formal partial derivatives and $\partial_t w = \partial_t V$, $\partial_L w = \rho + \partial_L V$,

$$\begin{aligned} \partial_t E_n &= \mathrm{e}^{-nV}\,\partial_t V\,\big(P' - nP\big)(w), \\ \partial_L E_n &= \mathrm{e}^{-nV}\Big[-n(\partial_L V)P(w) + (\rho + \partial_L V)P'(w)\Big]. \end{aligned}$$

Subtracting $n\rho E_n = \mathrm{e}^{-nV} n\rho P(w)$ from the second line gives

$$(\partial_L - n\rho)E_n = \mathrm{e}^{-nV}\,(\rho + \partial_L V)\,\big(P' - nP\big)(w),$$

because $-n(\partial_L V) - n\rho = -n(\rho + \partial_L V)$. Dividing the two displays yields (12.14); the denominator $\rho + \partial_L V$ is a unit because $\rho \in \mathbb{K}^\times$ and $\partial_L V \in t\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$. Comparing coefficients of t^{r-1} in (12.14) gives (12.15); and $E_{n,0} = [t^0]E_n = p_n(\rho L + \log c)$ because $\nu_t(V^k) \geq k$, so only $k = 0$ contributes to the block t^0 in (12.12). □

Remark 12.16. The naive substitute $\theta = \partial_t V$ is wrong. It ignores that the logarithmic shift itself depends on L, and it already fails at $r = 2$ whenever $\partial_L V \neq 0$. The correction factor $(\rho + \partial_L V)^{-1}$ is exactly the Jacobian of the change of logarithmic coordinate.

12.6 Rational and real leading exponents

Nothing in the proof of Theorem 12.7 used integrality of ρ except to keep the output in $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$.

Proposition 12.17 (Puiseux-log and Hahn extensions)**.** *Let $F = \sum_{n \geq a} p_n(L)t^n \in \mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$, let $c \in \mathbb{K}^\times$ with a chosen $\log c \in \mathbb{K}$, let $U \in t\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$, and let $\rho > 0$ be real with $\rho \in \mathbb{K}$. Define $G = cX^\rho(1 + U)$ and $F \circ G$ by* (12.7)*.*

1. *If $\rho = r/s$ with $r, s \in \mathbb{N}_{>0}$, then $F \circ G \in \mathbb{K}[L]((t^{1/s}))$, the Puiseux-log ring obtained by adjoining $t^{1/s}$.*
2. *For arbitrary real $\rho > 0$, the family* (12.6) *is summable in $\mathbb{K}[L][[t^\Gamma]]$ with $\Gamma = \mathbb{Z} + \rho\mathbb{Z}$, and*
$$\mathrm{supp}_{\mathrm{coeff}}(F \circ G) \subseteq \{(\rho n + m,\ j) :\ n \geq a,\ m \in \mathbb{N}_0,\ j \in \mathbb{N}_0\}\,.$$
The exponent set $\{\rho n + m : n \geq a,\ m \geq 0\}$ is locally finite: only finitely many of its elements lie below any real bound.
3. *In both cases the conclusions of Theorem 12.7 and Theorem 12.10 hold with $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ replaced as target by $\mathbb{K}[L]((t^{1/s}))$ resp. $\mathbb{K}[L][[t^\Gamma]]$, with N ranging over Γ and $\lfloor N/\rho \rfloor$ read as the largest admissible n.*

Proof. Only the support statement needs an argument. Fix a real bound B. Since $\rho > 0$, the inequality $\rho n + m \leq B$ with $n \geq a$, $m \geq 0$ forces $a \leq n \leq B/\rho$, finitely many values, and for each such n it forces $0 \leq m \leq B - \rho n$, again finitely many. Hence the exponent set is locally finite, in particular well ordered, and it is contained in the group Γ. In particular the n-th member of (12.6) has valuation at least ρn and $\{n \geq a : \rho n \leq \beta\}$ is finite for every real β, which is (12.2); so Theorem 12.3 applies and every proof above goes through word for word. When $\rho = r/s$ we have $\Gamma \subseteq s^{-1}\mathbb{Z}$ and local finiteness upgrades to "finitely many negative exponents", which is the Puiseux-log condition. □

Remark 12.18. The union over all s of the rings $\mathbb{K}[L]((t^{1/s}))$ admits rational exponents but no single denominator bound; it is a field, but a fixed element of it does not carry a uniform s. Real exponents genuinely require the Hahn setting, where the support condition, not the exponent group, does the work. Complex exponents require in addition a declared tie-breaking order, since ordering by real part alone is not a total order [19, 33].

Example 12.19 (Where the one-logarithm class ceases to be closed)**.** The following table is a diagnostic; each row is proved by writing down $t \circ G$ and $L \circ G$.

Leading behavior of G	New object created	Minimal enlargement
cX^ρ, $\rho \in \mathbb{N}_{>0}$	none	$\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$
$cX^{r/s}$	$t^{1/s}$	$\mathbb{K}[L]((t^{1/s}))$
cX^ρ, ρ real	t^ρ	$\mathbb{K}[L][[t^\Gamma]]$
$cX^\rho L^\sigma$, $\sigma \neq 0$	$\log L$	one further iterated-log level
cL^σ, $\sigma > 0$	$L^{-\sigma}$, and $\log L$ if some $p_n \notin \mathbb{K}$	completion order reversed
e^X	e^{-nX}	logarithmic-exponential transseries

For the fourth row, $\log G = \rho L + \sigma \log L + \log c + \log(1+U)$, so a single outer coefficient of positive degree in its argument already produces $\log\log X$. The fifth row is the most drastic: $t \circ G = c^{-1}L^{-\sigma}$ has outer valuation 0, so the composite is not a series in t at all and the roles of the two completions are exchanged; since $\mathscr{PL}_{\mathrm{itLaur},\mathbb{K}} = \mathbb{K}((L^{-1}))((t)) \neq \mathbb{K}((t))((L^{-1}))$, naming the target field is part of the statement. For the sixth, $t \circ \mathrm{e}^X = \mathrm{e}^{-X}$, which is smaller than every power of t; such height-changing substitutions belong to the field of logarithmic-exponential transseries [13, 33, 1].

Example 12.20 (A scale-changing substitution computed exactly). Let $F(Y) = Y + \log Y + Y^{-1}$ and $G(X) = 3X^2L^{-1}$ on the ray $X \to +\infty$. The inner argument is *not* admissible in the sense of Theorem 12.6, because of the factor L^{-1}; this is exactly why a new scale appears. Substituting the generators,

$$F \circ G = 3X^2L^{-1} + \big(\log 3 + 2L - \log L\big) + \tfrac{1}{3}Lt^2.$$

The second logarithmic level appears exactly, not as a remainder; the result lies in a two-logarithm class and in no smaller one.

12.7 Generator substitution for a near-identity argument

The case $c = 1$, $\log c = 0$, $\rho = 1$ is the one that governs reversion.

Definition 12.21 (Near-identity argument). For $H \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ set

$$G = X + H = X(1 + tH),$$

so that $U = tH \in t\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$, $c = 1$, $\log c = 0$, $\rho = 1$. We write $\mathcal{G}_{\mathrm{poly},\mathbb{K}} = \{X + H : H \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}\}$.

Proposition 12.22 (Exact generator substitution). *For $G = X + H$ with $H \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$,*

$$t \circ G = \frac{t}{1 + tH}, \qquad L \circ G = L + \log(1 + tH), \tag{12.16}$$

and consequently, for $F = \sum_{n \geq n_0} p_n(L)t^n$,

$$F \circ (X + H) = \sum_{n \geq n_0} t^n(1 + tH)^{-n}\, p_n\big(L + \log(1 + tH)\big). \tag{12.17}$$

Moreover $\nu_t(F \circ (X + H)) = \nu_t(F)$, so Φ_{X+H} preserves the valuation exactly and maps $\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ injectively into $\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$.

Proof. Specialize (12.8) and (12.7) to $c = 1$, $\log c = 0$, $\rho = 1$, $U = tH$; the valuation statement is Theorem 12.7(iii) with $\rho = 1$. □

A common trap

Both generators move, and neither moves the way it looks. Let $H \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ and $G = X + H$.

Wrong rule 1: $t \mapsto t + H$. The symbol t denotes X^{-1}, not X. Adding H to t models $X^{-1} \mapsto X^{-1} + H$, which is not $(X + H)^{-1}$. The mistake is visible at once from valuations: for $H \notin t\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ one has $\nu_t(t + H) = 0$, whereas $\nu_t(1/G) = 1$ because $G \to \infty$.

Wrong rule 2: update L, hold t fixed. Take $F = tL = \log X/X$ and $H = L$, so $G = X + \log X$. The correct value, from (12.17), is

$$F \circ G = \big(t - Lt^2 + L^2t^3 - \cdots\big)\big(L + Lt - \tfrac{1}{2}L^2t^2 + \cdots\big) = tL + (L - L^2)t^2 + \Big(L^3 - \tfrac{3}{2}L^2\Big)t^3 + \mathcal{O}_t^{\mathrm{form}}(t^4)\,.$$

Substituting only $L \mapsto L + \log(1 + Lt)$ returns $tL + Lt^2 - \frac{1}{2}L^2t^3 + \mathcal{O}_t^{\mathrm{form}}(t^4)$, which misses the term $-L^2t^2$ entirely: the answer is already wrong in the first corrected block. Substituting only $t \mapsto t/(1 + Lt)$ returns $tL - L^2t^2 + L^3t^3 + \mathcal{O}_t^{\mathrm{form}}(t^4)$ and misses $+Lt^2$. A correct implementation substitutes the two generators *simultaneously*: the replacement for L contains t, and the replacement for t contains L, so no sequential substitution is legitimate.

Proposition 12.23 (The near-identity monoid)**.** *For $A, B \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$,*

$$(X + A) \circ (X + B) = X + B + A \circ (X + B), \tag{12.18}$$

and $A \circ (X + B) \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$. Hence $\mathcal{G}_{\mathrm{poly},\mathbb{K}}$ is a monoid under composition, with identity X, and the map $G \mapsto \Phi_G$ is an anti-homomorphism of monoids (Theorem 12.27).

Proof. Apply Φ_{X+B}, a $\mathbb{K}$-algebra homomorphism, to $X + A = t^{-1} + A$: $\Phi_{X+B}(t^{-1}) = (\Phi_{X+B}t)^{-1} = X + B$, and $\Phi_{X+B}(A) = A \circ (X + B)$, which lies in $\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ by Theorem 12.22. □

Proposition 12.24 (Filtration continuity, with a one-block gain on the right)**.** *Let $F, F_1, F_2 \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ and $H, K \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$. Then*

$$\nu_t\big(F_1 \circ (X + H) - F_2 \circ (X + H)\big) = \nu_t(F_1 - F_2), \tag{12.19}$$

$$\nu_t\big(F \circ (X + H) - F \circ (X + K)\big) \geq \nu_t(F) + 1 + \nu_t(H - K). \tag{12.20}$$

Proof. (12.19) is Theorem 12.22 applied to $F_1 - F_2$. For (12.20) use the Taylor formula (13.4), proved independently in Part 13: the difference equals $\sum_{k\geq 1} \frac{H^k - K^k}{k!} D_X^k F$. Since $H^k - K^k = (H - K)\sum_{i=0}^{k-1} H^i K^{k-1-i}$ and all factors lie in $\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$, $\nu_t(H^k - K^k) \geq \nu_t(H - K)$; and $\nu_t(D_X^k F) \geq \nu_t(F) + k \geq \nu_t(F) + 1$ for $k \geq 1$. □

Example 12.25 (A worked near-identity composition)**.** Let

$$F = L^2 + (L + 1)t + (L^2 - L)t^2 + (2L + 1)t^3, \qquad G = X + L.$$

Here $U = Lt$ and, through outer order 3,

$$t \circ G = t - Lt^2 + L^2t^3 + \mathcal{O}_t^{\mathrm{form}}(t^4), \qquad L \circ G = L + Lt - \tfrac{1}{2}L^2t^2 + \tfrac{1}{3}L^3t^3 + \mathcal{O}_t^{\mathrm{form}}(t^4).$$

Simultaneous substitution and collection of blocks give

$$F \circ G = L^2 + (2L^2 + L + 1)t + (-L^3 + L^2 - L)t^2 + \Big(\tfrac{2}{3}L^4 - 2L^3 + \tfrac{7}{2}L^2 + L + 1\Big)t^3 + \mathcal{O}_t^{\mathrm{form}}(t^4). \tag{12.21}$$

Example 12.26 (Increasing logarithmic degree is not an artifact)**.** Let $F = L + L^2t$ and $G = X + L$. Then

$$F \circ G = L + (L^2 + L)t + \Big(-L^3 + \tfrac{3}{2}L^2\Big)t^2 + \Big(L^4 - \tfrac{8}{3}L^3 + L^2\Big)t^3 + \Big(-L^5 + \tfrac{41}{12}L^4 - 2L^3\Big)t^4 + \mathcal{O}_t^{\mathrm{form}}(t^5).$$

The log-degree profile $d_{F\circ G}$ grows by one per block: every extra power of $\log(1 + Lt)$ supplies another factor L. No uniform bound on $\deg_{\mathrm{L}}$ survives composition, which is why the ring is defined without one.

12.8 The associativity domain

Associativity is a theorem, not a formality: it holds on the nose only when the composite inner argument is itself admissible *and* the chosen logarithms are compatible.

Theorem 12.27 (Associativity). *Let*

$$G = c_G X^\rho(1+U), \qquad H = c_H X^\sigma(1+W)$$

be admissible monomial-leading arguments with chosen logarithms $\log c_G, \log c_H \in \mathbb{K}$, *with* $\rho, \sigma \in \mathbb{N}_{>0}$ *and* $U, W \in t\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$. *Put*

$$c := c_G c_H^\rho, \qquad \log c := \log c_G + \rho \log c_H, \qquad 1 + U^* := (1+W)^\rho\big(1 + \Phi_H U\big). \tag{12.22}$$

Then $U^* \in t\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$, *the quadruple* $(c, \log c, \rho\sigma, U^*)$ *is an admissible monomial-leading argument representing* $G \circ H$, *and for every* $F \in \mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$

$$(F \circ G) \circ H = F \circ (G \circ H), \qquad \textit{equivalently} \qquad \Phi_H \circ \Phi_G = \Phi_{G\circ H}. \tag{12.23}$$

The identity X *is a two-sided unit, and* $\mathcal{G}_{\mathrm{poly},\mathbb{K}}$ *is closed under composition with the canonical choices* $c = 1$, $\log c = 0$.

Proof. First, admissibility. Φ_H is a $\mathbb{K}$-algebra homomorphism compatible with summable families (Theorem 12.7), and $\nu_t(\Phi_H U) = \sigma\, \nu_t(U) \geq \sigma \geq 1$; also $\nu_t\big((1+W)^\rho - 1\big) \geq 1$, since ρ is a positive integer and $(1+W)^\rho - 1$ is a sum of positive powers of W. Hence $U^* = (1+W)^\rho(1+\Phi_H U) - 1 \in t\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$. Furthermore

$$G \circ H = \Phi_H(G) = c_G\,(\Phi_H t)^{-\rho}\,(1+\Phi_H U) = c_G c_H^\rho\, t^{-\rho\sigma}(1+W)^\rho(1+\Phi_H U) = c\, t^{-\rho\sigma}(1+U^*),$$

which is (12.22).

Second, the identity. Both $\Phi_H \circ \Phi_G$ and $\Phi_{G\circ H}$ are $\mathbb{K}$-algebra homomorphisms $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}} \to \mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ compatible with summable families, the first because a composite of two such maps is one (Theorem 12.3). By the definition (12.7) of Φ, a homomorphism with this compatibility is determined by its values on t and L: indeed it maps $p_n(L)t^n$ to $p_n(\text{image of } L) \cdot (\text{image of } t)^n$ and then sums. So it suffices to compare the generators.

For t:

$$\Phi_H\big(\Phi_G t\big) = \Phi_H\big(c_G^{-1} t^\rho (1+U)^{-1}\big) = c_G^{-1}\big(c_H^{-1} t^\sigma (1+W)^{-1}\big)^\rho \big(1+\Phi_H U\big)^{-1} = c^{-1} t^{\rho\sigma}(1+U^*)^{-1} = \Phi_{G\circ H}(t).$$

For L, using Theorem 12.5 twice,

$$\begin{aligned}
\Phi_H\big(\Phi_G L\big) &= \Phi_H\big(\rho L + \log c_G + \log(1+U)\big)\\
&= \rho\big(\sigma L + \log c_H + \log(1+W)\big) + \log c_G + \log\big(1+\Phi_H U\big)\\
&= \rho\sigma L + \big(\log c_G + \rho \log c_H\big) + \log\Big((1+W)^\rho\big(1+\Phi_H U\big)\Big)\\
&= \rho\sigma L + \log c + \log(1+U^*) = \Phi_{G\circ H}(L).
\end{aligned}$$

Hence the two homomorphisms agree. Taking $G = X$ with $c = 1$, $\log c = 0$, $\rho = 1$, $U = 0$ gives $\Phi_X = \mathrm{id}$, which is the two-sided identity statement; closure of $\mathcal{G}_{\mathrm{poly},\mathbb{K}}$ is Theorem 12.23. ☐

Remark 12.28 (Right substitution reverses the apparent order). Equation (12.23) reads $\Phi_H \circ \Phi_G = \Phi_{G\circ H}$: the operator that acts *second* is the one indexed by the inner argument that acts *first*. Substitution on the right is a contravariant operation, so $G \mapsto \Phi_G$ is an anti-homomorphism. Implementations of nested composition should test exactly this identity; reversing it silently produces correct-looking output for commuting pairs and wrong output otherwise.

Remark 12.29 (The compatibility hypothesis on logarithms cannot be dropped). Suppose $\mathbb{K} = \mathbb{C}$ and one chooses for $G \circ H$ the logarithm $\log c + 2\pi\mathrm{i}$ instead of $\log c_G + \rho \log c_H$. Then $\Phi_{G\circ H}(L)$ differs from $\Phi_H(\Phi_G L)$ by $2\pi\mathrm{i}$, and (12.23) fails for every F with a coefficient block of positive degree. Associativity is a statement about admissible *data*, not about the underlying maps alone.

Remark 12.30 (Rational and real exponents). Theorem 12.27 was stated for $\rho, \sigma \in \mathbb{N}_{>0}$ so that every displayed object stays in $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$. For $\rho, \sigma > 0$ lying in $\mathbb{K}$, the same proof works verbatim inside the Hahn ring $\mathbb{K}[L][[t^\Gamma]]$ with $\Gamma = \mathbb{Z} + \rho\mathbb{Z} + \sigma\mathbb{Z} + \rho\sigma\mathbb{Z}$, provided one *defines* $(1+W)^\rho := \mathrm{e}^{\rho\log(1+W)}$, which is legitimate by Theorem 12.5 and agrees with the integer power when $\rho \in \mathbb{Z}$. The exponent of the composite is $\rho\sigma$, so the class of admissible exponents must be closed under multiplication; $\mathbb{N}_{>0}$ and $\mathbb{Q}_{>0}$ are, a fixed denominator lattice $s^{-1}\mathbb{Z}$ is not.

A common trap

Formal closure is not a license. Theorem 12.27 asserts $(F \circ G) \circ H = F \circ (G \circ H)$ only when every displayed composition exists in the selected support class. A syntactically meaningful nested expression in a larger Hahn field need not have summable support; support theorems for general transseries composition are genuinely harder and carry their own hypotheses [14, 33].

12.9 The chain rule

Theorem 12.31 (Chain rule). *Let G be an admissible monomial-leading argument with $\rho > 0$. Then for every $F \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$,*

$$D_X\big(F \circ G\big) = \big(D_X F\big) \circ G \cdot D_X G. \tag{12.24}$$

Proof. Set $\delta_1(F) = D_X(\Phi_G F)$ and $\delta_2(F) = \Phi_G(D_X F) \cdot D_X G$. Both are $\mathbb{K}$-linear, both are compatible with summable families (each is a composite of maps that transform valuations by a non-decreasing unbounded function, so Theorem 12.3 applies), and both satisfy the twisted Leibniz rule $\delta(F_1F_2) = \delta(F_1)\Phi_G(F_2) + \Phi_G(F_1)\delta(F_2)$: for δ_1 because D_X is a derivation and Φ_G a homomorphism, for δ_2 for the same reason. It therefore suffices to check (12.24) for $F = t$ and $F = L$; the twisted Leibniz rule then propagates it to $\mathbb{K}[L][t]$, the relation $\delta(t \cdot t^{-1}) = 0$ propagates it to t^{-1}, and compatibility with summable families finishes.

Write $G = ct^{-\rho}(1+U)$. From $D_X(t^\alpha p(L)) = t^{\alpha+1}(p' - \alpha p)$ we get $D_X(t^{-\rho}) = \rho t^{1-\rho}$, so

$$\frac{D_X G}{G} = \frac{c\rho t^{1-\rho}(1+U) + c\,t^{-\rho} D_X U}{c\,t^{-\rho}(1+U)} = \rho t + \frac{D_X U}{1+U}. \tag{12.25}$$

For $F = t$: $\delta_1(t) = D_X(G^{-1}) = -G^{-2}D_X G$, while $\delta_2(t) = \Phi_G(-t^2)D_X G = -G^{-2}D_X G$. For $F = L$: $\delta_2(L) = \Phi_G(t)D_X G = D_X G/G$, while

$$\delta_1(L) = D_X\big(\rho L + \log c + \log(1+U)\big) = \rho t + \sum_{r\geq 1}(-1)^{r-1}U^{r-1}D_X U = \rho t + \frac{D_X U}{1+U},$$

termwise differentiation being legitimate by Theorem 12.3. By (12.25) the two agree. □

12.10 Analytic admissibility

Everything above is an identity between formal objects. It becomes a statement about functions only under additional hypotheses, which we now make explicit. Take $\mathbb{K} \subseteq \mathbb{C}$.

Definition 12.32 (Asymptotic realization). Let S be a ray $[X_0, \infty)$ or a closed subsector of a sector at infinity on which a branch of log has been fixed. A function $f : S \to \mathbb{C}$ *realizes* $F = \sum_{n\geq a} p_n(L)t^n \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ on S, written $f \sim_S F$, if for every $N \in \mathbb{Z}$

$$f(X) - \big(\mathrm{Trunc}_{<N} F\big)(X) = o_{X\to\infty,\ X\in S}\big(X^{1-N}\big)\,, \tag{12.26}$$

where $\big(\mathrm{Trunc}_{<N} F\big)(X) = \sum_{a\leq n<N} p_n(\log X)X^{-n}$ is a finite sum evaluated with the fixed branch.

Lemma 12.33 (Equivalent forms and uniqueness). *$f \sim_S F$ holds if and only if for every N and every $\varepsilon > 0$, $f - \operatorname{Trunc}_{<N} F = \mathcal{O}_{X\to\infty,\, X\in S}(X^{-N+\varepsilon})$. A function realizes at most one F. The stronger relation $f - \operatorname{Trunc}_{<N} F = \mathcal{O}_{X\to\infty,\, X\in S}(X^{-N})$* fails *for a realization whenever $\deg_{\mathrm{L}} p_N > 0$, since the first omitted block is then not $\mathcal{O}_{X\to\infty}(X^{-N})$; the ε may not be dropped.*

Proof. If (12.26) holds for all N, apply it at $N+1$ and add $p_N(\log X)X^{-N} = \mathcal{O}_{X\to\infty}(X^{-N+\varepsilon})$. Conversely the ε-form at N with $\varepsilon < 1$ implies (12.26). For uniqueness, if $F \neq \tilde{F}$ first differ in block M by a nonzero $R \in \mathbb{K}[L]$, subtracting the two relations (12.26) at $N = M+1$ gives $R(\log X)X^{-M} = o_{X\to\infty}(X^{-M})$, so $R(\log X) \to 0$; a nonzero polynomial does not tend to 0 along $\log X \to \infty$. The last sentence is the observation that $X^{-N}(\log X)^d$ is not $\mathcal{O}_{X\to\infty}(X^{-N})$ for $d > 0$. □

Lemma 12.34 (Realizations form a ring). *If $f \sim_S F$ and $g \sim_S \tilde{F}$ then $f+g \sim_S F+\tilde{F}$ and $fg \sim_S F\tilde{F}$.*

Proof. Addition is clear. For the product put $a = \nu_t F$, $\tilde{a} = \nu_t \tilde{F}$ and fix N and $\varepsilon > 0$. Since $\operatorname{Trunc}_{<N'} F\tilde{F} - \operatorname{Trunc}_{<N} F\tilde{F}$ is, for $N' \geq N$, a finite sum of terms $q(\log X)X^{-m}$ with $m \geq N$ and hence $\mathcal{O}_{X\to\infty}(X^{-N+\varepsilon})$, the assertion at a cutoff N' implies the assertion at every smaller cutoff; we may therefore assume $N \geq a + \tilde{a}$. Write $f = \operatorname{Trunc}_{<N-\tilde{a}} F + R$, $g = \operatorname{Trunc}_{<N-a} \tilde{F} + \tilde{R}$. By Theorem 12.33, $R = \mathcal{O}_{X\to\infty}(X^{-N+\tilde{a}+\varepsilon/4})$ and $\tilde{R} = \mathcal{O}_{X\to\infty}(X^{-N+a+\varepsilon/4})$, while $\operatorname{Trunc}_{<N-\tilde{a}} F = \mathcal{O}_{X\to\infty}(X^{-a+\varepsilon/4})$ and similarly for $\tilde{F}$. Hence each of the three cross terms is $\mathcal{O}_{X\to\infty}(X^{-N+\varepsilon})$; for the term $R\tilde{R}$, whose exponent is $-2N+a+\tilde{a}+\varepsilon/2$, this is where $N \geq a+\tilde{a}$ is used. The remaining product of two finite sums is a finite sum of terms $q(\log X)X^{-m}$; those with $m < N$ constitute $\operatorname{Trunc}_{<N} F\tilde{F}$ by the definition of the Cauchy product, and those with $m \geq N$ are $\mathcal{O}_{X\to\infty}(X^{-N+\varepsilon})$. □

Theorem 12.35 (Analytic transfer under composition). *Let $S' = [X_1, \infty)$ and $S = [Y_0, \infty)$ be rays, let $f \sim_S F$ with $F = \sum_{n\geq a} p_n(L)t^n$, and let $g : S' \to \mathbb{R}$ satisfy the following three conditions.*

(A1) Regime. *$g(X) \to +\infty$ and $g(X) \in S$ for all sufficiently large $X \in S'$.*

(A2) Monomial-leading realization. *There are $c > 0$, $\rho > 0$ real and $U \in t\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ with real coefficients such that $u := c^{-1}X^{-\rho}g(X) - 1$ satisfies $u \sim_{S'} U$ and $u(X) \to 0$.*

(A3) Branch compatibility. *$\log g(X) = \rho \log X + \log c + \log(1+u(X))$ for all large $X \in S'$, with the principal real logarithm on both sides.*

Then $f \circ g$ realizes $F \circ G$ on S', where $G = cX^\rho(1+U)$ and $F \circ G$ is (12.7) in the Hahn ring of Theorem 12.17, whose support is locally finite so that $\operatorname{Trunc}_{<\beta} \cdot$ at a real cutoff β is again a finite sum; that is, for every real β,

$$f(g(X)) - \big(\operatorname{Trunc}_{<\beta} F \circ G\big)(X) = o_{X\to+\infty}\Big(X^{-\beta+1}\Big).$$

Proof. Fix β and $\varepsilon \in (0,1)$. Choose an integer $N > (\beta+1)/\rho$ with $N > a$, and split $f = \operatorname{Trunc}_{<N} F + R_N$. By Theorem 12.33, $R_N(Y) = \mathcal{O}_{Y\to+\infty}(Y^{-N+\varepsilon})$. By (A1) and (A2), $g(X) = cX^\rho(1+u(X))$ with $u \to 0$, so $g(X) \asymp X^\rho$ and

$$R_N\big(g(X)\big) = \mathcal{O}_{X\to+\infty}\Big(X^{-\rho(N-\varepsilon)}\Big) = o_{X\to+\infty}\Big(X^{-\beta}\Big) \tag{12.27}$$

for ε small, since $\rho N > \beta + 1$. This is the *remainder transport* step; note that it consumes ρ^{-1} times as many outer orders of the outer expansion as it produces.

It remains to expand the finite sum $\operatorname{Trunc}_{<N} F \circ g$. By (A3) and Theorem 12.33, $\log(1+u(X))$ realizes $V = \log(1+U)$ on S': indeed for any M, $\log(1+u) = \sum_{r=1}^{M} \frac{(-1)^{r-1}}{r} u^r + \mathcal{O}_{X\to+\infty}\Big(|u|^{M+1}\Big)$ by Taylor's theorem with remainder for $\log(1+\cdot)$ at 0, applicable because $u \to 0$; each u^r realizes U^r by Theorem 12.34, and $|u|^{M+1} = \mathcal{O}_{X\to+\infty}(X^{-M-1+\varepsilon})$. The same argument with the binomial series shows that $(1+u(X))^{-n}$ realizes $(1+U)^{-n}$ for each n. Consequently, for each $n < N$,

$$p_n\big(\log g(X)\big)g(X)^{-n} = p_n\big(\rho \log X + \log c + \log(1+u(X))\big)\, c^{-n}X^{-\rho n}(1+u(X))^{-n},$$

which by (A3), the finite expansion (12.9) and Theorem 12.34 realizes $p_n(\rho L+\log c+V)c^{-n}t^{\rho n}(1+U)^{-n}$ on S'. Summing the finitely many $n < N$ and using (12.27) gives the claim, because by Theorem 12.10 the blocks of $F \circ G$ of exponent $< \beta$ come only from those n with $\rho n < \beta \le \rho N$. □

Example 12.36 (Each hypothesis is used)**.**

1. *(A1)/(A2) with $\rho > 0$ are not decorative.* Let $f(Y) = \mathrm{e}^{-Y}$. Then $f \sim_S 0$ on any ray: the zero series is a correct realization to all orders. Substituting $g(X) = \log X$, which has no positive power of X as leading monomial, gives $f(g(X)) = X^{-1}$, which realizes $t \neq 0$. Substituting $g(X) = X^{-1} \to 0$, which violates (A1), gives $f(g(X)) \to 1$, which realizes the series $1 \neq 0$. So composition is not even well defined on realizations without (A1)–(A2); what makes the theorem work is that $\rho > 0$ transports flatness to flatness.

2. *(A3) is a hypothesis, not a consequence.* On a complex sector containing $\arg X$ close to π, the principal logarithm satisfies $\operatorname{Log} g(X) = \rho \operatorname{Log} X + \log c + \operatorname{Log}(1+u)$ only while $|\rho \arg X + \arg(1+u)| < \pi$; crossing that threshold changes the answer by $2\pi\mathrm{i}$ in every coefficient block of positive degree. On the positive real ray with $c > 0$, (A3) is automatic.

3. *Precision transport is lossy for $\rho < 1$.* By (12.27), an output valid through exponent β requires the outer expansion through outer order $N > \beta/\rho$. For $\rho = 1/2$ one must double the outer precision; for $\rho = 2$ one gains.

A common trap

Three separate obligations. A formal identity $F \circ G$ established by Theorem 12.7, Theorem 12.10 or Theorem 12.27 carries no analytic content. To claim an asymptotic statement one must additionally exhibit (A1) a domain on which the inner map lands in the outer expansion's regime, (A2) a monomial-leading realization of the inner map with a positive exponent, and (A3) a branch identity for every logarithm and fractional power used. Differentiating a realization requires yet more: a Poincaré expansion cannot in general be differentiated termwise without sectorial analyticity or direct derivative estimates [25, 10]. The formal derivation D_X is always defined; its analytic interpretation is not.

Chapter 13

The formal Taylor formula

For near-identity inner arguments there is a second description of composition that looks exactly like the classical Taylor expansion but is an exact identity of formal series. We prove it from scratch, because the summability and the multiplicativity of the Taylor operator are both nontrivial and are the two points at which the sources are thin.

13.1 The normal-ordered Taylor operator

Definition 13.1 (Taylor operator)**.** For $H \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ define, for $F \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$,

$$\mathcal{T}_H(F) := \sum_{k\geq 0} \frac{H^k}{k!}\, D_X^k F. \tag{13.1}$$

Lemma 13.2 (Summability, with an explicit derivative budget)**.** *Let $H \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ and $F \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ with $a = \nu_t(F)$. Then*

$$\nu_t\Big(\frac{H^k}{k!} D_X^k F\Big) \geq a + k + k\,\nu_t(H) \;\geq\; a + k, \tag{13.2}$$

so the family in (13.1) *is summable and $\mathcal{T}_H(F) \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ with $\nu_t(\mathcal{T}_H F) \geq a$. Consequently, for every $N \in \mathbb{Z}$,*

$$\mathrm{Trunc}_{<N}\, \mathcal{T}_H(F) = \mathrm{Trunc}_{<N}\Big(\sum_{k=0}^{N-a-1} \frac{H^k}{k!} D_X^k F\Big), \tag{13.3}$$

that is, only the indices $k \leq N-a-1$ can affect any block below t^N, and only $k \leq N-a$ can affect the block t^N itself. If moreover $\nu_t(H) \geq 1$, the budget halves: only $k \leq (N-a)/2$ can affect t^N.

Proof. Write $F = \sum_{n\geq a} p_n(L)t^n$. By (12.1) and Theorem 12.3 applied to the continuous operator D_X, $D_X^k F = \sum_{n\geq a} t^{n+k}\Delta_{n,k}p_n$, so $\nu_t(D_X^k F) \geq a+k$; the inequality may be strict, and is so exactly when the operators $\Delta_{n,k}$ annihilate the leading blocks, but the bound is all we need. Multiplying by H^k, whose valuation is $k\,\nu_t(H) \geq 0$, gives (13.2). The right-hand side tends to $+\infty$ with k, which is (12.2), so Theorem 12.3 applies. A summand contributes to the block t^N only if $a+k \leq N$, and to a block strictly below t^N only if $a+k \leq N-1$. If $\nu_t(H) \geq 1$, (13.2) improves to $a+2k \leq N$. ☐

Lemma 13.3 (The Taylor operator is a homomorphism)**.** *For every $H \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$, $\mathcal{T}_H : \mathscr{PL}_{\mathrm{poly},\mathbb{K}} \to \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ is a $\mathbb{K}$-algebra homomorphism compatible with summable families.*

Proof. $\mathbb{K}$-linearity is clear from (13.1), and $\mathcal{T}_H(1) = 1$ because $D_X 1 = 0$.

For compatibility with summable families, the valuation half holds with $\varphi = \mathrm{id}$, since $\nu_t(\mathcal{T}_H F) \geq \nu_t(F)$ by (13.2). For the interchange, observe first that D_X itself is compatible with summable families:

it is $\mathbb{K}$-linear, raises valuation by at least 1, and by (12.1) the block $[t^{n+1}]D_X F$ depends $\mathbb{K}$-linearly on the single block $[t^n]F$, so $D_X\big(\sum_i S_i\big) = \sum_i D_X S_i$ for every summable family; the same then holds for each D_X^k. Now let $(S_i)_{i\in I}$ be summable with sum S. By Theorem 12.4 the valuations $\nu_t(S_i)$ are bounded below, so the doubly indexed family $\big(\frac{H^k}{k!}D_X^k S_i\big)_{i,k}$, whose (i,k) member has valuation at least $\nu_t(S_i)+k$ by (13.2), satisfies (12.2). Summing it with i innermost gives $\mathcal{T}_H(S)$ and with k innermost gives $\sum_i \mathcal{T}_H(S_i)$.

For multiplicativity, first record the Leibniz rule for the derivation D_X: by induction on m, $D_X^m(F_1F_2) = \sum_{i=0}^m \binom{m}{i} D_X^i F_1\, D_X^{m-i}F_2$. Hence

$$\mathcal{T}_H(F_1F_2) = \sum_{m\ge 0}\frac{H^m}{m!}\sum_{i=0}^m\binom{m}{i}D_X^iF_1\,D_X^{m-i}F_2 = \sum_{m\ge0}\sum_{i=0}^m \frac{H^iD_X^iF_1}{i!}\cdot\frac{H^{m-i}D_X^{m-i}F_2}{(m-i)!}.$$

The doubly indexed family $\big(\frac{H^iD_X^iF_1}{i!}\cdot\frac{H^jD_X^jF_2}{j!}\big)_{i,j\ge0}$ has $\nu_t \ge \nu_t(F_1)+\nu_t(F_2)+i+j$ by (13.2), hence is summable; regrouping it by $i+j=m$ recovers the display, and regrouping it as a product of two summable families gives $\mathcal{T}_H(F_1)\mathcal{T}_H(F_2)$. Both regroupings are licensed by Theorem 12.3. ☐

Remark 13.4. Theorem 13.3 is not a formality. The operator $\mathcal{T}_H$ is the normal-ordered exponential $:\mathrm{e}^{HD_X}:$, in which every power of H is placed to the left and is never differentiated. It is *not* the operator exponential $\exp(M_HD_X)$, since $(M_HD_X)^2 = M_HM_{D_XH}D_X + M_{H^2}D_X^2$. The latter is also an automorphism, but a different one. What makes (13.1) multiplicative is the Leibniz rule combined with the elementary identity $\binom{m}{i}/m! = 1/(i!\,(m-i)!)$, not any operator exponential formalism.

13.2 The formal Taylor formula

Lemma 13.5 (Generators determine such a homomorphism)**.** *Let $\mathcal{A}$ be a polynomial-logarithmic Hahn ring and let $\Psi_1,\Psi_2 : \mathscr{PL}_{\mathrm{poly},\mathbb{K}} \to \mathcal{A}$ be $\mathbb{K}$-algebra homomorphisms compatible with summable families. If $\Psi_1(t)=\Psi_2(t)$ and $\Psi_1(L)=\Psi_2(L)$, then $\Psi_1=\Psi_2$.*

Proof. Both fix $\mathbb{K}$ and agree on L, hence agree on $\mathbb{K}[L]$; they agree on t, hence on t^n for $n\ge0$, and on t^{-1} because $\Psi_i(t)\Psi_i(t^{-1})=1$ forces $\Psi_i(t^{-1})=\Psi_i(t)^{-1}$. So they agree on every $p_n(L)t^n$. For $F=\sum_n p_nt^n$ the family $(p_n(L)t^n)_n$ is summable, so $\Psi_i(F)=\sum_n\Psi_i(p_nt^n)$, and the two sums coincide. ☐

Theorem 13.6 (Formal Taylor formula)**.** *For every $F\in\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ and every $H\in\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$,*

$$F\circ(X+H) = \sum_{k\ge0}\frac{H^k}{k!}\,D_X^kF. \tag{13.4}$$

This is an exact identity in $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$, not an approximation: both sides have the same coefficient in every block. If $a=\nu_t(F)$, the coefficient of t^N on the right receives contributions only from $k\le N-a$.

Proof. By Theorem 12.22 the map Φ_{X+H} is a $\mathbb{K}$-algebra homomorphism compatible with summable families, and by Theorem 13.3 so is $\mathcal{T}_H$. By Theorem 13.5 it suffices to compare them on t and L.

For $F=t$: from $D_Xt=-t^2$ and (12.1) with $n=1$, $p=1$, one gets $D_X^kt = t^{1+k}\prod_{j=0}^{k-1}(-1-j) = (-1)^kk!\,t^{k+1}$. Hence

$$\mathcal{T}_H(t) = \sum_{k\ge0}\frac{H^k}{k!}(-1)^kk!\,t^{k+1} = t\sum_{k\ge0}(-tH)^k = \frac{t}{1+tH} = \Phi_{X+H}(t),$$

the geometric family being summable because $\nu_t(tH)\ge1$.

For $F = L$: $D_X L = t$, so $D_X^k L = D_X^{k-1} t = (-1)^{k-1}(k-1)!\, t^k$ for $k \geq 1$. Hence

$$\mathcal{T}_H(L) = L + \sum_{k\geq 1} \frac{H^k}{k!}(-1)^{k-1}(k-1)!\, t^k = L + \sum_{k\geq 1} \frac{(-1)^{k-1}}{k}(tH)^k = L + \log(1+tH) = \Phi_{X+H}(L),$$

by (12.3). The derivative budget is Theorem 13.2. □

Remark 13.7 (The small parameter is H/X, not H). No smallness of H is assumed, and none is available: H may be L^{100}, which tends to infinity. What the proof uses is $H/X = tH \prec 1$, which is automatic for $H \in \mathscr{PL}_{\text{pow},\mathbb{K}}$ because every polynomial in $\log X$ is $o_{X\to+\infty}(X)$. The mechanism is that D_X raises the outer order by one while multiplication by H does not lower it. The base point is the asymptotic variable itself, and the conclusion is an identity, not an estimate; there is no analytic differentiability hypothesis anywhere.

Remark 13.8 (Where the mechanism fails). In a larger transseries field, "Δ small compared with X" is not enough. Take $F(X) = \mathrm{e}^{\mathrm{e}^X}$ and $\Delta = \mathrm{e}^{-X/2}$, so $\Delta \prec X$. Then

$$\Delta\, D_X F = \mathrm{e}^{-X/2}\mathrm{e}^X \mathrm{e}^{\mathrm{e}^X} = \mathrm{e}^{X/2} F \succ F,$$

so the $k = 1$ Taylor term dominates the $k = 0$ term and the family is not summable. The quantity that must be small is not Δ but $\Delta \cdot D_X \mathfrak{b}/\mathfrak{b}$, where $\mathfrak{b}$ is the dominant monomial of F: the increment multiplied by the logarithmic derivative of the outer scale. In the polynomial-logarithmic class this is automatically small, because for $\mathfrak{b} = t^n L^j$ one has $D_X \mathfrak{b}/\mathfrak{b} = t\big(jL^{-1} - n\big) \prec 1$, whereas for $\mathfrak{b} = \mathrm{e}^{\mathrm{e}^X}$ one has $D_X \mathfrak{b}/\mathfrak{b} = \mathrm{e}^X \succ 1$. Sharper Taylor approximation theorems for larger classes require exactly such a logarithmic-derivative hypothesis [14, 22].

Remark 13.9 (On a tempting but circular proof). One sometimes sees the Taylor formula deduced by introducing a scalar parameter s, forming $\Phi(s) = F(X + sH)$ "with H held constant", and expanding Φ in powers of s at $s = 0$. This is not a proof: the symbol $F(X + sH)$ has no meaning prior to the composition one is trying to compute, and the claim $\partial_s^k \Phi(0) = H^k D_X^k F$ is the chain rule for that same composition. One can rescue the argument by working in $\mathscr{PL}_{\text{poly},\mathbb{K}}[[s]]$ and *defining* $\Phi(s) = \Phi_{X+sH}(F)$, but then one must still prove that $\partial_s \Phi = H \cdot \Phi_{X+sH}(D_X F)$, which is Theorem 12.31. The route through Theorem 13.3 and Theorem 13.5 is shorter and self-contained.

13.3 The finite coefficient formula

Theorem 13.10 (Finite coefficient formula). *Let $F = \sum_{n\geq a} p_n(L)t^n \in \mathscr{PL}_{\text{poly},\mathbb{K}}$ and $H \in \mathscr{PL}_{\text{pow},\mathbb{K}}$, and write*

$$H^k = \sum_{m\geq 0} h_{k,m}(L)\, t^m, \qquad h_{0,0} = 1, \quad h_{0,m} = 0\ (m > 0). \tag{13.5}$$

Then for every $r \in \mathbb{Z}$

$$[t^r]\big(F \circ (X + H)\big) = \sum_{\substack{n\geq a,\ k\geq 0,\ m\geq 0 \\ n+k+m=r}} \frac{h_{k,m}(L)}{k!}\, \Delta_{n,k} p_n(L), \tag{13.6}$$

a sum with exactly $\binom{r-a+2}{2}$ index triples when $r \geq a$, and empty when $r < a$. If $\nu_t(H) \geq 1$, the constraint $m \geq k$ applies as well, so $n + 2k \leq r$ and roughly half the triples drop out.

Proof. By Theorem 13.6 and (12.1),

$$F \circ (X + H) = \sum_{k\geq 0} \frac{H^k}{k!} \sum_{n\geq a} t^{n+k} \Delta_{n,k} p_n = \sum_{k\geq 0}\sum_{n\geq a}\sum_{m\geq 0} \frac{h_{k,m}}{k!}\, \Delta_{n,k} p_n\, t^{n+k+m},$$

the triple family being summable because its (n, k, m) member has valuation at least $n + k + m$. Extracting $[t^r]$ gives (13.6). The affine constraint $n + k + m = r$ with $n \geq a$ and $k, m \geq 0$ has exactly $\binom{r-a+2}{2}$ solutions. If $\nu_t(H) \geq 1$ then $\nu_t(H^k) \geq k$, so $h_{k,m} = 0$ for $m < k$. $\square$

Remark 13.11 (Three separable pieces of work). Formula (13.6) factors the computation into (a) the powers H^k, computed once by truncated multiplication and reused for every outer series; (b) the shifted differential operators $\Delta_{n,k}$ acting on coefficient polynomials, which depend only on F; and (c) a finite index convolution. It is therefore the formula of choice when one inner correction H is composed with many outer series, and the generator form (12.17) is the formula of choice when a single sparse F is composed once.

13.4 Convolution coefficients

The blocks $h_{k,m}$ of (13.5) deserve a name and a recurrence. Write

$$H = \sum_{j \geq 0} h_j(L)\, t^j, \qquad h_j \in \mathbb{K}[L]. \tag{13.7}$$

Definition 13.12 (Convolution coefficients)**.** For $m, r \in \mathbb{N}_0$ put

$$C_{m,r} := \sum_{\substack{j_1,\dots,j_r \geq 0 \\ j_1+\cdots+j_r = m}} h_{j_1} \cdots h_{j_r} \;\in \mathbb{K}[L]. \tag{13.8}$$

The value at $r = 0$ is an empty product summed over the empty composition, so

$$C_{0,0} = 1, \qquad C_{m,0} = 0 \quad (m > 0). \tag{13.9}$$

A common trap

Index order. With (13.5) and (13.8) one has

$$h_{k,m} = C_{m,k} :$$

in $h_{k,m}$ the *first* index is the exponent of H and the second is the outer order; in $C_{m,r}$ it is the other way round. Both notations occur in the literature. Any transfer between them must transpose.

Proposition 13.13 (Structure of the convolution coefficients)**.** *With the conventions* (13.9)*:*

1. $H^r = \sum_{m \geq 0} C_{m,r}\, t^m$ *for every* $r \in \mathbb{N}_0$.
2. $C_{m,1} = h_m$*, and for all* $m, r \geq 0$

$$C_{m,r+1} = \sum_{j=0}^{m} h_j\, C_{m-j,r}. \tag{13.10}$$

3. *If* $h_0 = 0$*, then* $C_{m,r} = 0$ *whenever* $m < r$*, and* $C_{r,r} = h_1^r$.
4. *If* $h_0 = 0$*, then for* $m \geq r \geq 1$

$$C_{m,r} = \frac{r!}{m!}\, \mathcal{B}^{\exp}_{m,r}\big(1!\,h_1,\ 2!\,h_2, \dots, (m-r+1)!\,h_{m-r+1}\big). \tag{13.11}$$

Proof. (i) Expanding $H^r = \big(\sum_j h_j t^j\big)^r$ and collecting the total outer order m is exactly (13.8); for $r = 0$ the statement is $H^0 = 1$, which is (13.9). The rearrangement is legitimate because the family indexed by $(j_1, \dots, j_r)$ has valuation $j_1 + \cdots + j_r$ and is summable.

(ii) Take $r = 1$ in (i). For the recurrence, write $H^{r+1} = H \cdot H^r$ and read off the coefficient of t^m using (i); the inner index is bounded by m because $C_{m',r}$ is 0 for $m' < 0$. At $r = 0$ the recurrence returns $C_{m,1} = h_m C_{0,0} = h_m$, consistent with (13.9).

(iii) If $h_0 = 0$ then $\nu_t(H) \geq 1$, so $\nu_t(H^r) \geq r$ and (i) forces $C_{m,r} = 0$ for $m < r$. For $m = r$ the only composition of r into r positive parts is $(1, \dots, 1)$, giving h_1^r.

(iv) With $h_0 = 0$, set $\hat{h}_j = j!\, h_j$. The partial exponential Bell polynomials are defined by the generating identity

$$\sum_{m \geq r} \mathcal{B}^{\exp}_{m,r}(\hat{h}_1, \hat{h}_2, \dots) \frac{z^m}{m!} = \frac{1}{r!}\Big(\sum_{j \geq 1} \hat{h}_j \frac{z^j}{j!}\Big)^r = \frac{1}{r!}\Big(\sum_{j \geq 1} h_j z^j\Big)^r$$

[4]. Substituting $z \mapsto t$, which is legitimate coefficientwise since both sides are formal power series in one variable with coefficients in $\mathbb{K}[L]$, and comparing with (i) gives $\frac{C_{m,r}}{r!} = \frac{1}{m!}\mathcal{B}^{\exp}_{m,r}(\hat{h})$, which is (13.11). □

Example 13.14 (The first convolution coefficients)**.** For $H = h_0 + h_1 t + h_2 t^2 + h_3 t^3 + \mathcal{O}^{\text{form}}_t(t^4)$:

	$m = 0$	$m = 1$	$m = 2$	$m = 3$
$r = 0$	1	0	0	0
$r = 1$	h_0	h_1	h_2	h_3
$r = 2$	h_0^2	$2h_0h_1$	$h_1^2 + 2h_0h_2$	$2h_0h_3 + 2h_1h_2$
$r = 3$	h_0^3	$3h_0^2h_1$	$3h_0^2h_2 + 3h_0h_1^2$	$h_1^3 + 6h_0h_1h_2 + 3h_0^2h_3$

Each entry is obtained from the row above by (13.10). Setting $h_0 = 0$ empties the lower-left triangle, as Theorem 13.13(iii) predicts.

13.5 Low-order composition laws

We now write out (13.6) for the case that governs reversion: both maps tangent to the identity, through four outer blocks.

Proposition 13.15 (Composition through four blocks)**.** *Let*

$$F = X + f_0(L) + f_1(L)t + f_2(L)t^2 + f_3(L)t^3 + \mathcal{O}^{\text{form}}_t(t^4)\,, \tag{13.12}$$
$$G = X + g_0(L) + g_1(L)t + g_2(L)t^2 + g_3(L)t^3 + \mathcal{O}^{\text{form}}_t(t^4)\,, \tag{13.13}$$

with all $f_j, g_j \in \mathbb{K}[L]$, and write $F \circ G = X + h_0 + h_1 t + h_2 t^2 + h_3 t^3 + \mathcal{O}^{\text{form}}_t(t^4)$. Primes denote ∂_L. Then

$$h_0 = f_0 + g_0, \tag{13.14}$$
$$h_1 = f_1 + g_1 + g_0 f_0', \tag{13.15}$$
$$h_2 = f_2 + g_2 + g_1 f_0' + g_0\big(f_1' - f_1\big) + \frac{g_0^2}{2}\big(f_0'' - f_0'\big), \tag{13.16}$$
$$\begin{aligned} h_3 = {} & f_3 + g_3 + g_2 f_0' + g_1\big(f_1' - f_1\big) + g_0\big(f_2' - 2f_2\big) + g_0 g_1\big(f_0'' - f_0'\big) \\ & + \frac{g_0^2}{2}\big(f_1'' - 3f_1' + 2f_1\big) + \frac{g_0^3}{6}\big(f_0''' - 3f_0'' + 2f_0'\big). \end{aligned} \tag{13.17}$$

Proof. Put $H = G - X = g_0 + g_1 t + g_2 t^2 + g_3 t^3 + \mathcal{O}^{\text{form}}_t(t^4)$, so $H \in \mathscr{PL}_{\text{pow},\mathbb{K}}$ and $\nu_t(F) = -1$. By Theorem 13.2 only $k \leq 3 - (-1) = 4$ can reach the block t^3; but $D_X^k X = 0$ for $k \geq 2$ and $\nu_t(D_X^k(F - X)) \geq k$, so in fact only $k \leq 3$ matter. We need the first three derivatives of F, which (12.1) supplies blockwise with $\Delta_{n,k} = \prod_{j=0}^{k-1}(\partial_L - n - j)$:

$$D_X F = 1 + t f_0' + t^2\big(f_1' - f_1\big) + t^3\big(f_2' - 2f_2\big) + \mathcal{O}^{\text{form}}_t(t^4)\,,$$

$$
\begin{aligned}
D_X^2 F &= t^2\big(f_0'' - f_0'\big) + t^3\big(f_1'' - 3f_1' + 2f_1\big) + \mathcal{O}_t^{\mathrm{form}}\big(t^4\big)\,,\\
D_X^3 F &= t^3\big(f_0''' - 3f_0'' + 2f_0'\big) + \mathcal{O}_t^{\mathrm{form}}\big(t^4\big)\,,
\end{aligned}
$$

using $\Delta_{0,1} = \partial_L$, $\Delta_{1,1} = \partial_L - 1$, $\Delta_{2,1} = \partial_L - 2$, $\Delta_{0,2} = \partial_L(\partial_L - 1)$, $\Delta_{1,2} = (\partial_L - 1)(\partial_L - 2)$ and $\Delta_{0,3} = \partial_L(\partial_L - 1)(\partial_L - 2)$. Also $H^2 = g_0^2 + 2g_0g_1t + \mathcal{O}_t^{\mathrm{form}}\big(t^2\big)$ and $H^3 = g_0^3 + \mathcal{O}_t^{\mathrm{form}}(t)$. Substituting into (13.4),

$$
\begin{aligned}
F \circ G = {} & \big(X + f_0 + f_1 t + f_2 t^2 + f_3 t^3\big)\\
& + \big(g_0 + g_1 t + g_2 t^2 + g_3 t^3\big)\big(1 + t f_0' + t^2(f_1' - f_1) + t^3(f_2' - 2f_2)\big)\\
& + \tfrac{1}{2}\big(g_0^2 + 2g_0g_1t\big)\big(t^2(f_0'' - f_0') + t^3(f_1'' - 3f_1' + 2f_1)\big)\\
& + \tfrac{1}{6} g_0^3\, t^3\big(f_0''' - 3f_0'' + 2f_0'\big) + \mathcal{O}_t^{\mathrm{form}}\big(t^4\big)\,.
\end{aligned}
$$

Collecting the blocks $t^0, \ldots, t^3$ gives (13.14)–(13.17). □

Remark 13.16 (Which derivatives can occur, and with which coefficients). The formulas are triangular: derivatives fall only on the *outer* coefficients f_j, never on the inner g_j, because it is the outer logarithmic argument that is Taylor-expanded while the inner coefficients only supply the increment. At block t^r the highest derivative of f_n that occurs is of order $r - n$, and the operator applied to f_n at Taylor order k is exactly $\Delta_{n,k}$. The integers appearing in (13.16)–(13.17) are therefore signed Stirling numbers of the first kind: since $\Delta_{n,k} = \prod_{j=0}^{k-1}\big((\partial_L - n) - j\big) = ((\partial_L - n))^{\underline{k}}$,

$$
\Delta_{n,k} = \sum_{i=0}^{k} s(k,i)\ (\partial_L - n)^i. \tag{13.18}
$$

For $n = 0, k = 2$ this gives $\partial_L^2 - \partial_L$; for $n = 1, k = 2$, $\partial_L^2 - 3\partial_L + 2$; for $n = 0, k = 3$, $\partial_L^3 - 3\partial_L^2 + 2\partial_L$, which are precisely the three brackets displayed above.

Example 13.17 (Consistency check: a constant shift)**.** Take $G = X + c$ with $c \in \mathbb{K}$, so $g_0 = c$ and $g_1 = g_2 = g_3 = 0$. Then (13.14)–(13.16) reduce to

$$
h_0 = f_0 + c, \qquad h_1 = f_1 + c f_0', \qquad h_2 = f_2 + c\big(f_1' - f_1\big) + \frac{c^2}{2}\big(f_0'' - f_0'\big),
$$

which is (13.4) read off directly, since $D_X f_0(L) = t f_0'$ and $D_X^2 f_0(L) = t^2(f_0'' - f_0')$.

Corollary 13.18 (First commutator block)**.** *Let $F = X + f(L)$ and $G = X + g(L)$ with $f, g \in \mathbb{K}[L]$. Then*

$$
F \circ G - G \circ F = \big(g f' - f g'\big)\, t + \mathcal{O}_t^{\mathrm{form}}\big(t^2\big)\,. \tag{13.19}
$$

In particular composition is associative (Theorem 12.27) but not commutative, and the leading obstruction is the Lie bracket of the one-dimensional vector fields $f\partial_L$ and $g\partial_L$.

Proof. Apply (13.14)–(13.15) with $f_0 = f$, $g_0 = g$ and all other blocks zero: $h_0 = f + g$ is symmetric and $h_1 = g f'$; exchanging the roles of F and G gives $f g'$. □

Example 13.19 (Explicit noncommutativity)**.** For $F = X + L$ and $G = X + L^2$,

$$
\begin{aligned}
F \circ G &= X + L^2 + L + L^2 t - \tfrac{1}{2} L^4 t^2 + \mathcal{O}_t^{\mathrm{form}}\big(t^3\big)\,,\\
G \circ F &= X + L + L^2 + 2L^2 t + \big(L^2 - L^3\big) t^2 + \mathcal{O}_t^{\mathrm{form}}\big(t^3\big)\,,
\end{aligned}
$$

so $F \circ G - G \circ F = -L^2 t + \mathcal{O}_t^{\mathrm{form}}\big(t^2\big)$, in agreement with (13.19): $g f' - f g' = L^2 - 2L^2 = -L^2$.

Example 13.20 (A four-block composition with nontrivial coefficients)**.** Let F and G be the *exact* finite series

$$F(Y) = Y + \big((\log Y)^2 - 1\big) + \frac{2\log Y}{Y} + \frac{1}{Y^2}, \qquad G(X) = X + (1-L) + \frac{L^2}{X},$$

so that $f_0 = L^2 - 1$, $f_1 = 2L$, $f_2 = 1$, $f_3 = 0$, and $g_0 = 1 - L$, $g_1 = L^2$, $g_2 = g_3 = 0$. Then Theorem 13.15 gives

$$h_0 = L^2 - L, \quad h_1 = -L^2 + 4L, \quad h_2 = L^3 + 5L^2 - 7L + 4, \quad h_3 = \tfrac{4}{3}L^4 - L^3 - 8L^2 + \tfrac{41}{3}L - 6,$$

that is,

$$F\circ G = X + \big(L^2 - L\big) + \frac{-L^2+4L}{X} + \frac{L^3+5L^2-7L+4}{X^2} + \frac{\frac{4}{3}L^4 - L^3 - 8L^2 + \frac{41}{3}L - 6}{X^3} + \mathcal{O}_t^{\mathrm{form}}\big(t^4\big)\,.$$

Had F and G been specified only modulo Y^{-3} and t^2, the block h_3 would not have been determined, since it contains $f_3 + g_3 + g_2 f_0'$.

Example 13.21 (One step of a reversion)**.** Let $F(Y) = Y + \log Y$ and $G(X) = X - L + LX^{-1}$, so $f_0 = L$, $f_1 = f_2 = f_3 = 0$, $g_0 = -L$, $g_1 = L$, $g_2 = g_3 = 0$. Then $h_0 = -L + L = 0$ and $h_1 = f_1 + g_1 + g_0 f_0' = 0 + L - L = 0$, while

$$h_2 = f_2 + g_2 + g_1 f_0' + g_0(f_1' - f_1) + \frac{g_0^2}{2}(f_0'' - f_0') = 0 + 0 + L + 0 + \frac{L^2}{2}(0-1) = L - \frac{L^2}{2}.$$

Adding $\mathsf{L}_2^{\mathrm{Lam}}(L) = \frac{1}{2}L^2 - L$ to the X^{-2} block of G cancels h_2, because that block enters h_2 additively through g_2. The canonical Lambert polynomial $\mathsf{L}_2^{\mathrm{Lam}}$ is thus recovered as the unique second-order correction, exactly as the reversion theory predicts.

13.6 Composing with an outer analytic function

Two further compositions occur constantly and are governed by Bell polynomials: substituting a polynomial-logarithmic series of positive valuation into a scalar power series, and differentiating a composite repeatedly.

Remark 13.22 (Substitution into a scalar power series: where it is proved). Let $\varphi(z) = \sum_{k\ge 0} \frac{\varphi_k}{k!} z^k \in \mathbb{K}[[z]]$ and let $W = \sum_{j\ge 1} w_j(L)t^j \in t\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$, so that $\nu_t W \ge 1$. Everything used below about $\varphi(W) := \sum_{k\ge 0} \frac{\varphi_k}{k!} W^k$ is proved in the elementary-series part, applied to $\Phi = \sum_{r\ge 0} c_r z^r$ with $c_r = \varphi_r/r!$ and $U = W$. The family is summable, $\varphi(W) \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$, and $\varphi \mapsto \varphi(W)$ is the unique continuous $\mathbb{K}$-algebra homomorphism $\mathbb{K}[[z]] \to \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ sending z to W (Theorem 10.2); only $\varphi_0, \dots, \varphi_r$ are needed for the block t^r (Theorem 10.3); and

$$[t^0]\varphi(W) = \varphi_0, \qquad [t^r]\varphi(W) = \frac{1}{r!}\sum_{k=1}^{r} \varphi_k\, \mathcal{B}_{r,k}^{\exp}\big(1!\,w_1, 2!\,w_2, \dots, (r-k+1)!\,w_{r-k+1}\big) \quad (r \ge 1),$$

which is (10.10) of Theorem 10.5 after that normalisation. Two additions to what is proved there. First, if φ is analytic at z_0 with Taylor coefficients in $\mathbb{K}$ and $W - z_0 \in t\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$, then the same three statements hold after translation: apply them to $\psi(z) := \varphi(z_0 + z) \in \mathbb{K}[[z]]$ and to $W - z_0$. Second, the coefficient formula also follows from the convolution structure established in this part, without the generating identity for the partial Bell polynomials: Theorem 13.13(iv), applied to W in place of H (whose block w_0 vanishes), gives $[t^r]W^k = \frac{k!}{r!}\mathcal{B}_{r,k}^{\exp}(1!w_1, \dots)$ for $r \ge k \ge 1$ and 0 for $r < k$; multiplying by $\varphi_k/k!$ and summing over $k \le r$ reproduces the display.

Example 13.23. Taking $\varphi(z) = (1+z)^{-n}$, $\varphi(z) = \log(1+z)$ and $\varphi(z) = \mathrm{e}^z$ in Theorem 13.22 recovers, respectively, the binomial factor $(1+U)^{-n}$, the logarithmic increment $V = \log(1+U)$, and $\mathrm{e}^V = 1+U$ used throughout Part 12. For the exponential one may replace the partial by the complete Bell polynomial: $\mathrm{e}^W = \sum_{r\geq 0} \frac{1}{r!}\mathcal{B}^{\exp}_r(1!w_1,\ldots,r!w_r)\,t^r$, with $\mathcal{B}^{\exp}_0 = 1$.

For repeated differentiation of a composite the relevant identity is Faà di Bruno's. We prove it in the exact generality needed, from a recurrence for the partial Bell polynomials which we also prove.

Lemma 13.24 (Shift recurrence for partial Bell polynomials)**.** *Let $\mathcal{D} = \sum_{i\geq 1} x_{i+1}\partial_{x_i}$ act on polynomials in the indeterminates $x_1, x_2, \ldots$ over $\mathbb{Q}$. Then for all $n \geq 0$ and all $k \geq 1$, with the convention $\mathcal{B}^{\exp}_{n,k} = 0$ for $k > n$,*

$$\mathcal{B}^{\exp}_{n+1,k} = \mathcal{D}\,\mathcal{B}^{\exp}_{n,k} + x_1\,\mathcal{B}^{\exp}_{n,k-1}, \tag{13.20}$$

with $\mathcal{B}^{\exp}_{n,0} = 0$ for $n \geq 1$ and $\mathcal{B}^{\exp}_{0,0} = 1$.

Proof. Let $\Xi(z) = \sum_{m\geq 1} x_m z^m/m!$, so that $\sum_{n\geq k}\mathcal{B}^{\exp}_{n,k} z^n/n! = \Xi^k/k!$ is the defining generating identity [4]. Applying $\mathcal{D}$ coefficientwise, $\mathcal{D}\Xi = \sum_{i\geq 1} x_{i+1}z^i/i! = \Xi'(z) - x_1$, because $\Xi'(z) = \sum_{i\geq 0} x_{i+1}z^i/i!$. Since $\mathcal{D}$ is a derivation that does not act on z,

$$\mathcal{D}\Big(\frac{\Xi^k}{k!}\Big) = \frac{\Xi^{k-1}}{(k-1)!}\,\mathcal{D}\Xi = \frac{\Xi^{k-1}}{(k-1)!}\Xi' - x_1\frac{\Xi^{k-1}}{(k-1)!} = \frac{\mathrm{d}}{\mathrm{d}z}\Big(\frac{\Xi^k}{k!}\Big) - x_1\frac{\Xi^{k-1}}{(k-1)!}.$$

Comparing coefficients of $z^n/n!$ and using $\frac{\mathrm{d}}{\mathrm{d}z}\sum_n a_n z^n/n! = \sum_n a_{n+1}z^n/n!$ gives $\mathcal{D}\mathcal{B}^{\exp}_{n,k} = \mathcal{B}^{\exp}_{n+1,k} - x_1\mathcal{B}^{\exp}_{n,k-1}$ for all $n \geq 0$ and $k \geq 1$; when $k > n+1$ both sides vanish, and when $k = n+1$ the identity reads $0 = x_1^{n+1} - x_1 \cdot x_1^n$. ☐

Proposition 13.25 (Faà di Bruno for admissible composition)**.** *Let G be an admissible monomial-leading argument with $\rho > 0$, and let $\gamma \coloneqq D_X G$. Then for every $F \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ and every $n \geq 1$,*

$$D^n_X\big(F\circ G\big) = \sum_{k=1}^{n}\big(D^k_X F\big)\circ G\;\mathcal{B}^{\exp}_{n,k}\big(\gamma, D_X\gamma,\ldots,D^{n-k}_X\gamma\big). \tag{13.21}$$

Proof. Induction on n. For $n = 1$ this is Theorem 12.31, since $\mathcal{B}^{\exp}_{1,1} = x_1$. Assume (13.21) for n. Apply D_X, using the product rule, the chain rule (12.24) in the form $D_X\big((D^k_X F)\circ G\big) = (D^{k+1}_X F)\circ G\cdot\gamma$, and the fact that $D_X\big[\mathcal{B}^{\exp}_{n,k}(\gamma, D_X\gamma,\ldots)\big] = \big(\mathcal{D}\mathcal{B}^{\exp}_{n,k}\big)(\gamma, D_X\gamma,\ldots)$, which is the ordinary chain rule for a polynomial in the entries $x_i = D^{i-1}_X\gamma$. This gives

$$D^{n+1}_X(F\circ G) = \sum_{k\geq 1}\big(D^k_X F\big)\circ G\,\Big[\gamma\,\mathcal{B}^{\exp}_{n,k-1} + \mathcal{D}\,\mathcal{B}^{\exp}_{n,k}\Big]\big(\gamma, D_X\gamma,\ldots\big),$$

after re-indexing the first sum. By Theorem 13.24 with $x_1 = \gamma$ the bracket equals $\mathcal{B}^{\exp}_{n+1,k}$, completing the induction. All rearrangements involve finitely many terms for each n. ☐

Remark 13.26 (Valuation bookkeeping). For a near-identity $G = X+H = t^{-1}+H$ one has $D_X(t^{-1}) = t^0(0+1) = 1$, hence $\gamma = D_X G = 1 + D_X H$ with $\nu_t(D_X H) \geq 1$, and $D^i_X\gamma = D^{i+1}_X H$ has valuation at least $i+1$ for $i \geq 1$. Hence $\mathcal{B}^{\exp}_{n,k}(\gamma, D_X\gamma,\ldots)$ has valuation at least $n-k$, and the k-th term of (13.21) has valuation at least $\nu_t(F)+k+(n-k) = \nu_t(F)+n$, consistent with $\nu_t(D^n_X(F\circ G)) \geq \nu_t(F)+n$. In this class the size of (13.21) is a combinatorial, not an analytic, difficulty.

13.7 Which engine to use

The generator form (12.17) and the Taylor form (13.4) have identical formal content, by Theorem 13.6, and different computational profiles.

	Generator substitution	Taylor composition
Core data	$(1+U)^{-1}$ and $\log(1+U)$	$D_X^k F$ and $H^k/k!$
Reused across	many outer blocks of one F	many outer series with one H
Truncation index	powers of U	derivative order $k \le N - \nu_t F$
Valid for	any admissible $\rho > 0$	$\rho = 1$, $c = 1$ only
Typical error	updating one generator only	applying it when $H \notin \mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$

Structural checkpoint

Before composing, verify in order: (1) the inner argument has a leading monomial cX^ρ with $\rho > 0$ and a chosen $\log c$ in the coefficient field; (2) the residual factor is $1+U$ with $U \prec 1$; (3) the images $1/G$ and $\log G$ stay in the declared exponent group and logarithmic depth; (4) both generators are updated simultaneously; (5) the precision of U and V is set blockwise by Theorem 12.11, not globally; (6) if an asymptotic statement is intended, hypotheses (A1)–(A3) of Theorem 12.35 are separately verified.

Section summary

For an admissible monomial-leading $G = cX^\rho(1+U)$ with a chosen $\log c \in \mathbb{K}$, composition is exact simultaneous substitution of the two generators, $t \mapsto c^{-1}t^\rho(1+U)^{-1}$ and $L \mapsto \rho L + \log c + \log(1+U)$. It is a $\mathbb{K}$-algebra homomorphism, multiplies valuations by ρ, determines every output block by a finite sum with a triangular precision budget, and is associative on the nose once the chosen logarithms are compatible; right substitution reverses the apparent operator order. For $\rho = 1$, $c = 1$ it coincides with the exact formal Taylor formula $F \circ (X+H) = \sum_{k\ge 0} H^k D_X^k F/k!$, whose summability comes from the fact that D_X raises outer order while multiplication by $H \in \mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ does not lower it, so that only $k \le N - \nu_t(F)$ reach the block t^N. None of this is an analytic statement; asymptotic validity requires the regime, branch and remainder-transport hypotheses separately.

Chapter 14

The near-identity composition group

Throughout this chapter $\mathbb{K}$ is a field of characteristic zero, the large variable is X, and

$$t := X^{-1}, \qquad L := \log X, \qquad \mathscr{PL}_{\mathrm{pow},\mathbb{K}} = \mathbb{K}[L][[t]], \qquad \mathscr{PL}_{\mathrm{poly},\mathbb{K}} = \mathbb{K}[L]((t)).$$

The regime is $X \to +\infty$ along the positive real ray; in a complex sector a branch of $\log$ must be fixed once and for all, and every statement below is then read with that branch. The outer valuation is ν_t, the distinguished derivation is $D_X = t\partial_L - t^2\partial_t$, and $\Delta_{n,k} = \prod_{j=0}^{k-1}(\partial_L - n - j)$ with $\Delta_{n,0} = 1$, so that

$$D_X^k\big(t^n p(L)\big) = t^{n+k}\,\Delta_{n,k}p(L), \qquad n \in \mathbb{Z},\ p \in \mathbb{K}[L]. \tag{14.1}$$

All truncations are the exclusive ones, $\mathrm{Trunc}_{<N} F = \sum_{n<N} p_n(L)t^n$.

14.1 The set, and why its correction term is unrestricted

Definition 14.1 (The near-identity class)**.**

$$\mathcal{G}_{\mathrm{poly},\mathbb{K}} := \{X + A \;:\; A \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}\} \subseteq \mathscr{PL}_{\mathrm{poly},\mathbb{K}}.$$

Membership imposes no smallness on A itself: the correction may be L^{100}, or any polynomial block of any degree. What is small is the *relative* correction

$$\frac{X + A}{X} = 1 + tA \in 1 + t\mathscr{PL}_{\mathrm{pow},\mathbb{K}}, \tag{14.2}$$

and every element of $1 + t\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ arises exactly once this way. This is the whole content of the phrase “tangent to the identity” in the present class: $tA \prec 1$ because $t^n L^j \prec 1$ for every $n \geq 1$ and every $j \geq 0$ in the dominance order of the positive ray.

A common trap

$1 + tA = 1 + \mathcal{O}_t^{\mathrm{form}}(t)$ is a statement about coefficient support only. It asserts no numerical bound, no uniformity in L, and no analytic realization. The corresponding analytic statement, $G(X)/X \to 1$ as $X \to +\infty$, is a separate hypothesis about a chosen realization of the series.

We shall use the formal exponential and logarithm on $1 + t\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ constantly.

Remark 14.2 (Formal exponential and logarithm: where they are proved). The assignments $u \mapsto \exp u$ and $1 + u \mapsto \log(1+u)$, given by the usual series, are mutually inverse bijections between $t\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ and $1 + t\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$, and they carry products to sums. This is Theorem 12.5 applied to the Hahn ring $\mathcal{A} = \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ with $\Gamma = \mathbb{Z}$ and $\mathfrak{m} = t\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$; every reference to the present remark below is a reference to that lemma, whose extra clauses — additivity of $\exp$, $\log(P^n) = n\log P$ for $n \in \mathbb{Z}$, and commutation with a homomorphism compatible with summable families — are used here as well.

14.2 Substitution, and the exact Taylor formula

Definition 14.3 (Substitution by a near-identity map). Let $G = X + A \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}$ and put

$$R := (1+tA)^{-1} \in 1 + t\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}, \qquad V := \log(1+tA) \in t\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}.$$

For $F = \sum_{n\geq n_0} p_n(L)t^n \in \mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ define

$$\Phi_G(F) = F\circ G := \sum_{n\geq n_0} t^n R^n\, p_n(L+V). \tag{14.3}$$

The two generator images encoded in (14.3) are

$$t\circ G = \frac{t}{1+tA}, \qquad L\circ G = L + \log(1+tA). \tag{14.4}$$

A common trap

The substitution is $t\mapsto t(1+tA)^{-1}$, never $t\mapsto t+A$: the symbol t denotes X^{-1}, not X. Updating L while leaving t fixed, or conversely, is equally wrong; the two generators must be replaced simultaneously.

Proposition 14.4 (Substitution operator). *Let $G = X + A \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}$. Then*

1. *the family in* (14.3) *is formally summable, so $\Phi_G : \mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}} \to \mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ is well defined, and* $\mathrm{Trunc}_{<N}(F\circ G)$ *depends only on* $\mathrm{Trunc}_{<N} F$ *and on* $\mathrm{Trunc}_{<N-n_0} A$*;*
2. *Φ_G is a $\mathbb{K}$-algebra endomorphism of $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$;*
3. *$\Phi_G(t) = t(1+tA)^{-1}$, $\Phi_G(L) = L + \log(1+tA)$, and $\Phi_G(X) = G$;*
4. *$\nu_t\big(\Phi_G(F)\big) = \nu_t(F)$ for every F, and $\Phi_G(t^N\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}) \subseteq t^N\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ for every $N\in\mathbb{Z}$; in particular Φ_G is injective and t-adically continuous;*
5. *$\Phi_G(\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}) \subseteq \mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ and $\Phi_G\big(\log(1+u)\big) = \log\big(1+\Phi_G(u)\big)$ for $u\in t\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$.*

Proof. (1) Since $R\in 1+t\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ and $V\in t\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$, and since p_n is a polynomial, $p_n(L+V) = \sum_{j\leq \deg_L p_n} \frac{p_n^{(j)}(L)}{j!}V^j$ is a *finite* sum lying in $\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$; likewise $R^n\in\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ for $n\geq 0$ and $R^n = (1+tA)^{-n}\in\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ for $n<0$. The n-th summand of (14.3) therefore lies in $t^n\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$, and ν_t of the n-th summand tends to $+\infty$; the family is summable and n_0 bounds the support below. Only the summands with $n<N$ meet outer order $<N$, and within such a summand only R and V through order t^{N-n-1} are needed, whence the stated dependency.

(2) Additivity is clear. For multiplicativity write $F = \sum_n p_n t^n$, $\widetilde{F} = \sum_m q_m t^m$. Evaluation $p\mapsto p(L+V)$ is a $\mathbb{K}$-algebra homomorphism $\mathbb{K}[L]\to\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$, so

$$\Phi_G(F\widetilde{F}) = \sum_r t^r R^r\Big(\sum_{n+m=r} p_n q_m\Big)(L+V) = \sum_{n,m} t^{n+m}R^{n+m}p_n(L+V)q_m(L+V) = \Phi_G(F)\Phi_G(\widetilde{F}),$$

all rearrangements being legitimate because the families are summable.

(3) Take $F=t$, $F=L$, $F=t^{-1}$ in (14.3); the last gives $t^{-1}R^{-1} = t^{-1}(1+tA) = X+A = G$.

(4) The coefficient of t^{n_0} in $\Phi_G(F)$ is $p_{n_0}(L)$, because $R^{n_0}\in 1+t\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ and $p_{n_0}(L+V)\in p_{n_0}(L)+t\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$; no other summand reaches outer order n_0. Hence $\nu_t(\Phi_G F) = n_0 = \nu_t(F)$, and the displayed inclusion is immediate from (14.3).

(5) The inclusion is the case $n_0\geq 0$. For the last identity, Φ_G is multiplicative and continuous, and $\Phi_G(u)\in t\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ by (4), so $\Phi_G(\log(1+u)) = \sum_{k\geq 1}\frac{(-1)^{k+1}}{k}\Phi_G(u)^k = \log(1+\Phi_G(u))$. $\square$

Remark 14.5 (Exact formal Taylor formula). For every $F \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ and every $A \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ the family $(A^k D_X^k F/k!)_{k\geq 0}$ is formally summable and $F \circ (X+A) = \sum_{k\geq 0} A^k D_X^k F/k!$; if $a = \nu_t(F)$, only the indices $k \leq N-a$ affect the coefficient of t^N. This is Theorem 13.6, equation (13.4), proved there; it is an identity of formal series and carries no analytic content. Every citation of the present remark is a citation of that theorem. Two features of its proof are used repeatedly below: the operator $T_A = \sum_{k\geq 0} \frac{A^k}{k!} D_X^k$ is a *continuous* $\mathbb{K}$-algebra endomorphism of $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ (Theorem 13.3), and two such endomorphisms agreeing on t and L are equal (Theorem 13.5).

The single most useful consequence is a gain of one outer block whenever a near-identity argument is perturbed.

Corollary 14.6 (Filtered gain, and the Lipschitz contraction)**.** *Let $A, B, C \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$. Then*

$$\nu_t\big(A\circ(X+B) - A\circ(X+C)\big) \;\geq\; \nu_t(A) + \nu_t(B-C) + 1, \tag{14.5}$$
$$\nu_t\big(A\circ(X+B) - A\big) \;\geq\; \nu_t(A) + \nu_t(B) + 1, \tag{14.6}$$

and, more precisely,

$$A\circ(X+B) - A \;\equiv\; B\,D_X A \pmod{t^{\nu_t(A)+2\nu_t(B)+2}\mathscr{PL}_{\mathrm{pow},\mathbb{K}}}. \tag{14.7}$$

Proof. By Theorem 14.5, $A\circ(X+B) - A\circ(X+C) = \sum_{k\geq 1} \frac{B^k - C^k}{k!} D_X^k A$. Since $B^k - C^k = (B-C)\sum_{i=0}^{k-1} B^i C^{k-1-i}$ and all factors have nonnegative valuation, $\nu_t(B^k - C^k) \geq \nu_t(B-C)$; and $\nu_t(D_X^k A) \geq \nu_t(A) + k$. Hence the k-th summand has $\nu_t \geq \nu_t(B-C) + \nu_t(A) + k$, and the minimum over $k \geq 1$ is attained at $k=1$; this gives (14.5). Taking $C = 0$ and using $\nu_t(B^k) \geq k\,\nu_t(B)$ gives, for the k-th term, the bound $\nu_t(A) + k(1+\nu_t(B))$, which is $\geq \nu_t(A) + \nu_t(B) + 1$ for $k \geq 1$ and $\geq \nu_t(A) + 2\nu_t(B) + 2$ for $k \geq 2$; this is (14.6) and (14.7). □

14.3 The group theorem

Theorem 14.7 (The near-identity composition group)**.** *$\mathcal{G}_{\mathrm{poly},\mathbb{K}}$ is a group under composition. Explicitly:*

1. (Closure) *for $F = X + A$ and $G = X + B$ in $\mathcal{G}_{\mathrm{poly},\mathbb{K}}$,*
$$F\circ G = X + B + A\circ G \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}; \tag{14.8}$$
2. (Associativity) *$\Phi_H \circ \Phi_G = \Phi_{G\circ H}$ for $G, H \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}$; consequently $(F\circ G)\circ H = F\circ(G\circ H)$ for every $F \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ and $G, H \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}$;*
3. (Identity) *$X \circ G = G$ and $F \circ X = F$;*
4. (Inverses) *every $F = X + A \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}$ has a unique two-sided compositional inverse $F^{\langle -1\rangle_\circ} = X + B \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}$, and $\nu_t(B) = \nu_t(A)$.*

The group is not abelian.

Proof. (1) $\Phi_G(X+A) = \Phi_G(X) + \Phi_G(A) = G + A\circ G$ by Theorem 14.4(2),(3), and $A\circ G \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ by part (5) of the same proposition; so $F\circ G = X + \big(B + A\circ G\big)$ with correction in $\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$.

(2) Both $\Phi_H \circ \Phi_G$ and $\Phi_{G\circ H}$ are continuous $\mathbb{K}$-algebra endomorphisms of $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$, so by Theorem 13.5 it suffices to check them on t and L. Write $G = X + A$, $H = X + B$. Since Φ_H is a ring homomorphism with $\Phi_H(X) = H$ and $\Phi_H(A) = A\circ H$,

$$\Phi_H(G) = H + A\circ H = G\circ H. \tag{14.9}$$

Hence $\Phi_H(\Phi_G(t)) = \Phi_H(G^{-1}) = \Phi_H(G)^{-1} = (G\circ H)^{-1} = \Phi_{G\circ H}(t)$. For L, using Theorem 14.4(5) and Theorem 14.2,

$$\Phi_H\big(\Phi_G(L)\big) = \Phi_H\big(L + \log(1+tA)\big) = L + \log(1+tB) + \log\Big(1 + \tfrac{t}{1+tB}(A\circ H)\Big)$$

$$= L + \log\Big((1+tB)\Big(1 + \tfrac{t(A\circ H)}{1+tB}\Big)\Big) = L + \log\big(1 + t(B + A\circ H)\big) = \Phi_{G\circ H}(L),$$

the last equality because the correction of $G \circ H$ is $B + A \circ H$ by (14.8). Applying the resulting operator identity to F gives $(F \circ G) \circ H = F \circ (G \circ H)$.

(3) $\Phi_G(X) = G$ is Theorem 14.4(3), and $\Phi_X = \mathrm{id}$ because $A = 0$ makes $R = 1$, $V = 0$ in (14.3).

(4) By (1)–(3), $(\mathcal{G}_{\mathrm{poly},\mathbb{K}}, \circ)$ is a monoid with identity X, and by (14.8) an element $G = X + B$ of $\mathcal{G}_{\mathrm{poly},\mathbb{K}}$ satisfies $F \circ G = X$ if and only if

$$B = -A \circ (X + B). \tag{14.10}$$

Theorem 16.11 — proved from items (1)–(3) above, Theorem 14.6, and the t-adic completeness of $\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$, so that the present item may cite it — says that (14.10) has exactly one solution $B \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$, that the resulting $G = X + B$ is the unique element of $\mathcal{G}_{\mathrm{poly},\mathbb{K}}$ with $F \circ G = X$, that it satisfies $G \circ F = X$ as well and is the unique element of $\mathcal{G}_{\mathrm{poly},\mathbb{K}}$ doing so, and that $\nu_t(B) = \nu_t(A)$ with $[t^{\nu_t(A)}]B = -[t^{\nu_t(A)}]A$. Writing $F^{\langle -1\rangle_\circ} = F^{\langle -1\rangle_\circ} = G$ gives (4).

Noncommutativity is Theorem 14.8. □

Example 14.8 (Two maps that do not commute)**.** Let $F = X + L$ and $G = X + 1$. Then $G \circ F = F + 1 = X + L + 1$, whereas

$$F\circ G = G + L\circ G = X+1+\log(X+1) = X+1+L+\log(1+t) = X+1+L+t-\tfrac{1}{2}t^2+\mathcal{O}_t^{\mathrm{form}}\big(t^3\big)\,.$$

Hence $F \circ G - G \circ F = t - \frac{1}{2}t^2 + \mathcal{O}_t^{\mathrm{form}}\big(t^3\big) \neq 0$.

Remark 14.9 (Direction of the dependency on Part IV). Items (1)–(3) of Theorem 14.7 are proved here from Theorem 14.4 and Theorem 13.5 alone, and Theorem 14.6 needs nothing beyond Theorem 14.5. Item (4) is different: the existence, uniqueness, two-sidedness and leading data of $F^{\langle -1\rangle_\circ}$ are proved once in the volume, in Theorem 16.11, and item (4) cites that theorem. The dependency runs in one direction only. Part IV takes items (1)–(3) and Theorem 14.6 as given — they are the imports recorded in Theorem 16.3 — and from them constructs the solution of (14.10) by Picard iteration with the sharp rate $\nu_t(B_{j+1} - B_j) \geq (j+1)\alpha + j$, $\alpha = \nu_t(A)$; it adds the first-order form $F^{\langle -1\rangle_\circ} = X - A + \mathcal{O}_t^{\mathrm{form}}\big(t^{2\alpha+1}\big)$ (Theorem 16.12), the residual certificates, the Lagrange–Bürmann coefficient formula and the algorithms that compute it. A reader who wants only the group law therefore needs, beyond this section, the existence half of Part IV and nothing else.

Remark 14.10 (Substitution reverses order). By Theorem 14.7(2), $G \mapsto \Phi_G$ is an injective *anti*-homomorphism from $\mathcal{G}_{\mathrm{poly},\mathbb{K}}$ into the group of continuous $\mathbb{K}$-algebra automorphisms of $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$; injectivity holds because $\Phi_G(X) = G$. Every sign discrepancy between group-theoretic and vector-field formulas below can be traced to this order reversal.

14.4 What the group is not

A common trap

$\mathcal{G}_{\mathrm{poly},\mathbb{K}}$ is *not* a multiplicative group of coefficient series. Four distinct statements are involved, and conflating them is the commonest error in this subject.

- No element of $\mathcal{G}_{\mathrm{poly},\mathbb{K}}$ lies in $\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$: each has a t^{-1} block. The elements of $\mathcal{G}_{\mathrm{poly},\mathbb{K}}$ are maps, not coefficient series.
- The bijection $X + A \mapsto 1 + tA$ of (14.2) onto the multiplicative group $1 + t\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ is *not* a group homomorphism. Indeed $1 + t\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ is abelian and, by Theorem 14.8, $\mathcal{G}_{\mathrm{poly},\mathbb{K}}$ is not.
- $F^{\langle -1\rangle_\circ} \neq F^{-1}$. For $F = X + L$, $F^{-1} = t(1+tL)^{-1} = t - Lt^2 + L^2t^3 - \cdots$, whereas

$F^{\langle -1\rangle_\circ} = X - L + Lt + \mathcal{O}_t^{\mathrm{form}}(t^2)$.

- $\mathcal{G}_{\mathrm{poly},\mathbb{K}}$ is not the additive group of corrections either: $F\circ G = X + A + B + (A\circ G - A)$, and the last bracket vanishes only in the associated graded object (Theorem 14.15).

Note also that $t^N \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ is not an ideal of $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$, because t is a unit there; all filtration statements below are statements about the valuation ν_t, not about quotient rings.

14.5 The valuation filtration and its normal subgroups

Definition 14.11 (Contact filtration). For $r \in \mathbb{N}_0$ put

$$\mathcal{G}^{(r)}_{\mathrm{poly},\mathbb{K}} := \{X + A \in \mathcal{G}_{\mathrm{poly},\mathbb{K}} :\ \nu_t(A) \geq r\}, \qquad \mathcal{G}^{(0)}_{\mathrm{poly},\mathbb{K}} = \mathcal{G}_{\mathrm{poly},\mathbb{K}}.$$

Write $F \equiv_r F'$ when the corrections of $F, F' \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}$ differ by an element of $t^r \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$, i.e. when $F - F' = \mathcal{O}_t^{\mathrm{form}}(t^r)$; one says that F and F' have *contact of order* r.

Proposition 14.12 (The filtration is by normal subgroups). *For every $r \geq 0$, $\equiv_r$ is a congruence for composition: if $F \equiv_r F'$ and $G \equiv_r G'$ then $F\circ G \equiv_r F'\circ G'$. Consequently $\mathcal{G}_{\mathrm{poly},\mathbb{K}}/\equiv_r$ is a group, the quotient map is a surjective homomorphism, and $\mathcal{G}^{(r)}_{\mathrm{poly},\mathbb{K}}$ is a normal subgroup of $\mathcal{G}_{\mathrm{poly},\mathbb{K}}$ with $\mathcal{G}_{\mathrm{poly},\mathbb{K}}/\mathcal{G}^{(r)}_{\mathrm{poly},\mathbb{K}} \cong \mathcal{G}_{\mathrm{poly},\mathbb{K}}/\equiv_r$.*

Proof. Write $F = X + A$, $F' = X + A'$, $G = X + B$, $G' = X + B'$. Then

$$F\circ G - F'\circ G = (A - A')\circ G,$$

which has $\nu_t = \nu_t(A - A') \geq r$ by Theorem 14.4(4). Next

$$F'\circ G - F'\circ G' = (B - B') + \big(A'\circ G - A'\circ G'\big),$$

whose first term has $\nu_t \geq r$ and whose second has $\nu_t \geq r+1$ by (14.5). Adding gives $F\circ G \equiv_r F'\circ G'$. Since $\equiv_r$ is a congruence on a group, the quotient is a group and the class of X, which is exactly $\mathcal{G}^{(r)}_{\mathrm{poly},\mathbb{K}}$, is a normal subgroup. ☐

Proposition 14.13 (Separation and completeness). $\bigcap_{r\geq 0} \mathcal{G}^{(r)}_{\mathrm{poly},\mathbb{K}} = \{X\}$*, and the natural map*

$$\mathcal{G}_{\mathrm{poly},\mathbb{K}} \longrightarrow \varprojlim_r \mathcal{G}_{\mathrm{poly},\mathbb{K}}/\mathcal{G}^{(r)}_{\mathrm{poly},\mathbb{K}}$$

is an isomorphism of groups.

Proof. If $X + A \in \mathcal{G}^{(r)}_{\mathrm{poly},\mathbb{K}}$ for all r then $\nu_t(A) = +\infty$, so $A = 0$; this is injectivity as well. For surjectivity, let $(F_r\mathcal{G}^{(r)}_{\mathrm{poly},\mathbb{K}})_r$ be a compatible family. Compatibility says $F_{r+1} \equiv_r F_r$, so the coefficient blocks of the corrections stabilise: $[t^n]A_{F_r}$ is independent of r once $r > n$. Define A by $[t^n]A := [t^n]A_{F_{n+1}}$; then $A \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ and $X + A \equiv_r F_r$ for every r. ☐

Proposition 14.14 (Commutators drop one further block). *For $r, s \geq 0$,*

$$[\mathcal{G}^{(r)}_{\mathrm{poly},\mathbb{K}}, \mathcal{G}^{(s)}_{\mathrm{poly},\mathbb{K}}] \subseteq \mathcal{G}^{(r+s+1)}_{\mathrm{poly},\mathbb{K}}, \qquad \textit{where } [F,G] = F^{\langle -1\rangle_\circ}\circ G^{\langle -1\rangle_\circ}\circ F\circ G.$$

More precisely, if $F = X + A \in \mathcal{G}^{(r)}_{\mathrm{poly},\mathbb{K}}$ and $G = X + B \in \mathcal{G}^{(s)}_{\mathrm{poly},\mathbb{K}}$ then the correction of $[F,G]$ satisfies

$$[F,G] - X \ \equiv\ B\,D_X A - A\,D_X B \pmod{t^{r+s+2}\mathscr{PL}_{\mathrm{pow},\mathbb{K}}}. \tag{14.11}$$

Consequently the lower central series obeys $\gamma_k(\mathcal{G}_{\mathrm{poly},\mathbb{K}}) \subseteq \mathcal{G}^{(k-1)}_{\mathrm{poly},\mathbb{K}}$, each quotient $\mathcal{G}_{\mathrm{poly},\mathbb{K}}/\mathcal{G}^{(r)}_{\mathrm{poly},\mathbb{K}}$ is nilpotent of class at most r, and $\mathcal{G}_{\mathrm{poly},\mathbb{K}}$ is residually nilpotent.

Proof. Put $P = F \circ G = X + A_P$ and $Q = G \circ F = X + A_Q$. By (14.8) and (14.7),

$$A_P = B + A \circ G \equiv A + B + B\,D_X A, \qquad A_Q = A + B \circ F \equiv A + B + A\,D_X B$$

modulo $t^{r+s+2}\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$; note that the error terms in (14.7) are of order t^{r+2s+2} and t^{s+2r+2}, both at least t^{r+s+2}. Hence

$$A_P - A_Q \equiv B\,D_X A - A\,D_X B \pmod{t^{r+s+2}\mathscr{PL}_{\mathrm{pow},\mathbb{K}}}, \qquad \nu_t(A_P - A_Q) \geq r+s+1. \quad (14.12)$$

Now $[F,G] = Q^{\langle -1\rangle\circ} \circ P$. Writing $Q^{\langle -1\rangle\circ} = X + B_Q$ and using $X = Q^{\langle -1\rangle\circ} \circ Q = X + A_Q + B_Q \circ Q$, i.e. $B_Q \circ Q = -A_Q$, we get

$$[F,G] = P + B_Q \circ P = X + \big(A_P - A_Q\big) + \big(B_Q \circ P - B_Q \circ Q\big).$$

The last bracket has $\nu_t \geq \nu_t(A_P - A_Q) + 1 \geq r + s + 2$ by (14.5). With (14.12) this proves both displayed claims. The statement about γ_k follows by induction: $\gamma_1 = \mathcal{G}_{\mathrm{poly},\mathbb{K}} = \mathcal{G}^{(0)}_{\mathrm{poly},\mathbb{K}}$ and $\gamma_{k+1} = [\gamma_k, \mathcal{G}_{\mathrm{poly},\mathbb{K}}] \subseteq [\mathcal{G}^{(k-1)}_{\mathrm{poly},\mathbb{K}}, \mathcal{G}^{(0)}_{\mathrm{poly},\mathbb{K}}] \subseteq \mathcal{G}^{(k)}_{\mathrm{poly},\mathbb{K}}$. ◻

14.6 The associated graded object

Proposition 14.15 (Graded pieces)**.** *For every $r \geq 0$ the map*

$$\mathcal{G}^{(r)}_{\mathrm{poly},\mathbb{K}}/\mathcal{G}^{(r+1)}_{\mathrm{poly},\mathbb{K}} \longrightarrow (\mathbb{K}[L], +), \qquad (X + A)\mathcal{G}^{(r+1)}_{\mathrm{poly},\mathbb{K}} \longmapsto [t^r]A,$$

is a group isomorphism. In particular each graded piece is abelian and uniquely divisible.

Proof. If $F = X + A$ and $G = X + B$ lie in $\mathcal{G}^{(r)}_{\mathrm{poly},\mathbb{K}}$, then by (14.8) and (14.6) $F \circ G = X + A + B + (A \circ G - A)$ with $\nu_t(A \circ G - A) \geq r + r + 1 \geq r + 1$; hence $[t^r](A_{F\circ G}) = [t^r]A + [t^r]B$, so the map is a homomorphism. It is surjective (take $A = p(L)t^r$) and its kernel is $\mathcal{G}^{(r+1)}_{\mathrm{poly},\mathbb{K}}$. ◻

Theorem 14.16 (The graded Lie algebra)**.** *Let*

$$\mathcal{V} := \mathbb{K}[L][t]\,D_X,$$

the $\mathbb{K}$-span of the derivations $a(L)t^r D_X$ with $a \in \mathbb{K}[L]$ and $r \in \mathbb{N}_0$, a Lie algebra under $[\alpha D_X, \beta D_X] = (\alpha D_X\beta - \beta D_X\alpha)D_X$, graded by $\deg(a(L)t^r D_X) := r + 1$; explicitly

$$\big[a\,t^r D_X,\ b\,t^s D_X\big] = t^{r+s+1}\Big(a\big(b' - sb\big) - b\big(a' - ra\big)\Big)D_X, \qquad {}' = \partial_L. \quad (14.13)$$

Then the associated graded group $\operatorname{gr}\mathcal{G}_{\mathrm{poly},\mathbb{K}} = \bigoplus_{r\geq 0}\mathcal{G}^{(r)}_{\mathrm{poly},\mathbb{K}}/\mathcal{G}^{(r+1)}_{\mathrm{poly},\mathbb{K}}$, equipped with the bracket induced by the group commutator, is a graded Lie algebra, and

$$\theta : \operatorname{gr}\mathcal{G}_{\mathrm{poly},\mathbb{K}} \longrightarrow \mathcal{V}, \qquad \theta\big((X + A)\mathcal{G}^{(r+1)}_{\mathrm{poly},\mathbb{K}}\big) = -\big([t^r]A\big)t^r D_X$$

is an isomorphism of graded Lie algebras.

Proof. That $\mathcal{V}$ is a Lie algebra needs no verification beyond closure: for $\alpha, \beta \in \mathbb{K}[L][t]$, $\alpha D_X\beta - \beta D_X\alpha \in \mathbb{K}[L][t]$ by (14.1), and the bracket of two derivations of an associative algebra is a derivation, so the Jacobi identity is inherited. Formula (14.13) is (14.1) applied twice, and it shows that the bracket adds the degrees: $(r + 1) + (s + 1) = (r + s + 1) + 1$.

By Theorem 14.14 the group commutator induces a well-defined biadditive map $\mathcal{G}^{(r)}_{\mathrm{poly},\mathbb{K}}/\mathcal{G}^{(r+1)}_{\mathrm{poly},\mathbb{K}} \times \mathcal{G}^{(s)}_{\mathrm{poly},\mathbb{K}}/\mathcal{G}^{(s+1)}_{\mathrm{poly},\mathbb{K}} \to \mathcal{G}^{(r+s+1)}_{\mathrm{poly},\mathbb{K}}/\mathcal{G}^{(r+s+2)}_{\mathrm{poly},\mathbb{K}}$: replacing A by $A + \widetilde{A}$ with $\nu_t\widetilde{A} \geq r + 1$ changes $BD_XA -$

AD_XB by an element of $t^{r+s+2}\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$, and biadditivity follows from (14.11) together with Theorem 14.15. Writing $A = a(L)t^r + \cdots$ and $B = b(L)t^s + \cdots$, (14.1) gives

$$[t^{r+s+1}]\big(B\,D_XA - A\,D_XB\big) = b\,(a' - ra) - a\,(b' - sb) = (ba' - ab') + (s-r)ab.$$

On the other hand (14.13) gives

$$\big[-a\,t^rD_X,\ -b\,t^sD_X\big] = t^{r+s+1}\Big(a(b'-sb)-b(a'-ra)\Big)D_X = -t^{r+s+1}\Big((ba'-ab')+(s-r)ab\Big)D_X,$$

which is exactly θ applied to the previous display. Thus θ intertwines the two brackets; it is bijective and degree preserving by Theorem 14.15. Since $\mathcal{V}$ satisfies the Jacobi identity, so does $\operatorname{gr}\mathcal{G}_{\mathrm{poly},\mathbb{K}}$. □

Remark 14.17. The sign in θ is forced by Theorem 14.10: composition and operator composition run in opposite directions, so the induced bracket is the opposite of the vector-field bracket. For $r = s = 0$, that is for $F = X + f(L)$, $G = X + g(L)$, (14.11) reads $F \circ G - G \circ F = t\,(gf' - fg') + \mathcal{O}_t^{\mathrm{form}}(t^2)$, the classical bracket of one-dimensional vector fields in the coordinate $L = \log X$.

14.7 The exponential of a vector field, and iteration

The Taylor operator $T_A = \sum_k \frac{A^k}{k!}D_X^k$ of Theorem 14.5 is *normally ordered*: every power of A stands to the left of every D_X. It must not be confused with the operator exponential $\exp(AD_X)$, in which D_X also differentiates A. Both are automorphisms, and both are substitutions — but by different maps.

Theorem 14.18 (Exponential correspondence)**.** *For $A \in \mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ the operator $\exp(AD_X) = \sum_{k\geq 0}\frac{1}{k!}(AD_X)^k$ is a well-defined continuous $\mathbb{K}$-algebra automorphism of $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$, and*

$$\exp(AD_X) = \Phi_{\mathcal{E}(A)}, \qquad \mathcal{E}(A) := \exp(AD_X)(X) = X + A + \tfrac{1}{2}A\,D_XA + \cdots \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}. \quad (14.14)$$

The map $\mathcal{E} : \mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}} \to \mathcal{G}_{\mathrm{poly},\mathbb{K}}$ is a bijection, and $\mathcal{E}(A) \equiv X + A \pmod{t^{2\nu_t(A)+1}\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}}$.

Proof. Well-definedness. $\delta := AD_X$ is a derivation with $\delta(t^n\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}) \subseteq t^{n+1}\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$, so for $F \in t^{-m}\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ the family $(\delta^kF/k!)_k$ is summable and $\exp(\delta)F \in t^{-m}\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$; continuity and $\mathbb{K}$-linearity are clear. Multiplicativity follows from the Leibniz rule exactly as for T_A, and $\exp(-\delta)$ is a two-sided inverse because the identity $\sum_k \frac{x^k}{k!}\sum_j \frac{(-x)^j}{j!} = 1$ holds coefficientwise.

Identification with a substitution. Let $\Psi_s = \exp(s\delta)$; for each n, $[t^n]\Psi_s(F)$ is a polynomial in s with coefficients in $\mathbb{K}[L]$, because $\nu_t(\delta^kF) \geq \nu_t(F) + k$. Put

$$u(s) = \Psi_s(L), \qquad v(s) = \log\big(\Psi_s(X)\big) := L + \log\big(1 + t(\Psi_s(X) - X)\big),$$

the latter legitimate because $\Psi_s(X) - X \in \mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$, so that $t(\Psi_s(X) - X) \in t\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ and Theorem 14.2 applies. Differentiating in s coefficientwise, and using $\frac{\mathrm{d}}{\mathrm{d}s}\Psi_s = \Psi_s \circ \delta$,

$$u'(s) = \Psi_s(A\,D_XL) = \Psi_s(At), \qquad v'(s) = \frac{\Psi_s(A\,D_XX)}{\Psi_s(X)} = \Psi_s(A)\,\Psi_s(t) = \Psi_s(At),$$

where $\Psi_s(X)^{-1} = \Psi_s(t)$ because Ψ_s is a ring automorphism. Since $u(0) = v(0) = L$ and $\mathbb{K}$ has characteristic zero, $u = v$. Setting $s = 1$ and writing $G = \mathcal{E}(A) = \Psi_1(X)$, we get $\Psi_1(L) = \log G = \Phi_G(L)$ and $\Psi_1(t) = \Psi_1(X)^{-1} = G^{-1} = \Phi_G(t)$; both maps are continuous algebra homomorphisms, so $\Psi_1 = \Phi_G$.

Bijectivity. Write $\mathcal{E}(A) = X + A + \mathcal{R}(A)$ with $\mathcal{R}(A) = \sum_{k\geq 2}\frac{1}{k!}(AD_X)^{k-1}A$. Each term of $\mathcal{R}(A)$ is a sum of products of k factors drawn from A and its D_X-derivatives, carrying $k-1$ derivatives in total, so $\nu_t \geq k\,\nu_t(A) + k - 1 \geq 2\nu_t(A) + 1$; this proves the last assertion. Replacing A by A' one factor at a time in each such product shows, for $A, A' \in \mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$,

$$\nu_t\big(\mathcal{R}(A) - \mathcal{R}(A')\big) \geq \nu_t(A - A') + 1,$$

because each single-factor difference contributes $\nu_t(A - A')$ and the remaining $k - 1 \geq 1$ derivatives contribute at least 1. Hence $\mathcal{E}(A) - \mathcal{E}(A') \equiv A - A'$ modulo $t^{\nu_t(A-A')+1}\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$. Injectivity is immediate. For surjectivity, let $G = X + A_* \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}$; construct $A^{(r)}$ with $\mathcal{E}(A^{(r)}) \equiv_r G$ by setting $A^{(0)} = 0$ and $A^{(r+1)} = A^{(r)} - e_r t^r$, where $e_r t^r$ is the outer block of order r of $\mathcal{E}(A^{(r)}) - G$. The sequence is Cauchy and its limit A satisfies $\mathcal{E}(A) = G$ by continuity. ☐

Corollary 14.19 (Unique divisibility and continuous iteration). *For $F \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}$ let $A = \mathcal{E}^{-1}(F)$ and define*

$$F^{[s]} := \exp(sAD_X)(X) = \mathcal{E}(sA), \qquad s \in \mathbb{K}.$$

Then $F^{[1]} = F$, $F^{[0]} = X$, $F^{[s]} \circ F^{[s']} = F^{[s+s']}$, $F^{[n]}$ is the n-fold composite for $n \in \mathbb{N}_{>0}$, and $F^{[-1]} = F^{\langle -1\rangle_\circ}$. In particular $\mathcal{G}_{\mathrm{poly},\mathbb{K}}$ is uniquely divisible: for each $n \in \mathbb{N}_{>0}$ there is exactly one $H \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}$ with $H^{[n]} = F$, namely $H = \mathcal{E}(A/n)$. Moreover $F^{[s]} \equiv X + sA \pmod{t^{2\nu_t(A)+1}\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}}$.

Proof. By Theorem 14.18 and Theorem 14.7(2), $\Phi_{F^{[s']}\circ F^{[s]}} = \Phi_{F^{[s]}}\Phi_{F^{[s']}} = \exp(sAD_X)\exp(s'AD_X) = \exp((s+s')AD_X) = \Phi_{F^{[s+s']}}$; apply both sides to X and use injectivity of $G \mapsto \Phi_G$. The remaining assertions are immediate, uniqueness of n-th roots because $H^{[n]} = \mathcal{E}(nB)$ for $H = \mathcal{E}(B)$ and $\mathcal{E}$ is injective. ☐

Remark 14.20 (Normal ordering versus the flow). $T_A \neq \exp(AD_X)$ in general. $T_A = \Phi_{X+A}$ substitutes the map $X + A$; $\exp(AD_X) = \Phi_{\mathcal{E}(A)}$ is the time-one flow of the vector field AD_X, and $\mathcal{E}(A) = X + A + \frac{1}{2}AD_X A + \cdots$. For $A = L$, say, $\mathcal{E}(L) = X + L + \frac{1}{2}Lt + \mathcal{O}_t^{\mathrm{form}}(t^2) \neq X + L$. The two operators agree only when $AD_X A = 0$ to all relevant orders, in particular when $A \in \mathbb{K}$. Confusing them turns an exact composition formula into an approximate one.

Section summary

$\mathcal{G}_{\mathrm{poly},\mathbb{K}} = X + \mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ is a group under composition, with identity X and with inverses produced by a fixed-point iteration that gains one outer block per step. Its contact filtration $(\mathcal{G}^{(r)}_{\mathrm{poly},\mathbb{K}})$ consists of normal subgroups with $[\mathcal{G}^{(r)}_{\mathrm{poly},\mathbb{K}}, \mathcal{G}^{(s)}_{\mathrm{poly},\mathbb{K}}] \subseteq \mathcal{G}^{(r+s+1)}_{\mathrm{poly},\mathbb{K}}$; the quotients are nilpotent, the group is their inverse limit, and the associated graded object is the graded Lie algebra $\mathbb{K}[L][t]D_X$ of polynomial vector fields. Exponentiating those vector fields makes $\mathcal{G}_{\mathrm{poly},\mathbb{K}}$ uniquely divisible and gives a continuous iteration $F^{[s]}$. None of this is a statement about multiplication of coefficient series.

Chapter 15

Scale-changing composition and deeper logarithms

Part 14 treated inner maps that perturb the scale. This chapter treats inner maps that *change* it. The governing principle is the same in both cases and is worth isolating before any case analysis.

15.1 The substitution principle

Every $F \in \mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ is built from the two generators $t = X^{-1}$ and $L = \log X$ by sums, products and t-adic limits. Therefore a substitution $X \mapsto G$ is completely determined by the two images

$$t \circ G = G^{-1}, \qquad L \circ G = \log G, \tag{15.1}$$

and the only question is which ambient class contains them and is closed enough to receive the resulting sums. Answering that question *first* is the difference between a correct computation and a silently truncated one: computing in a class that lacks a monomial forced by (15.1) does not make that monomial disappear, it makes the retained terms wrong.

Definition 15.1 (Nested logarithms, monomial groups, Hahn ambients)**.** Put

$$L_0 \coloneqq X, \qquad L_{j+1} \coloneqq \log L_j,$$

so that $L_1 = L = \log X$ and $L_2 = \log\log X$. For $r \geq 1$ and an ordered subgroup $\Gamma \subseteq \mathbb{R}$ let

$$\mathcal{M}_r(\Gamma) \coloneqq \{L_0^{\alpha_0} L_1^{\alpha_1} \cdots L_r^{\alpha_r} :\ \alpha_0, \ldots, \alpha_r \in \Gamma\},$$

ordered by declaring one monomial dominant over another when, at the least index j at which the exponents differ, the first exponent is the larger. This is a totally ordered abelian group, and we write $\mathcal{H}_r(\mathbb{K}, \Gamma)$ for the Hahn field of series $\sum_{\mathfrak{m} \in S} c_{\mathfrak{m}} \mathfrak{m}$ with $c_{\mathfrak{m}} \in \mathbb{K}$ and $S \subseteq \mathcal{M}_r(\Gamma)$ well ordered in the direction of increasing smallness [19, 13]. On the positive ray this order is the asymptotic order as $X \to +\infty$. One has

$$\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}} \subset \mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}} \subset \mathscr{P}\mathscr{L}_{\mathrm{rat},\mathbb{K}} \subset \mathscr{P}\mathscr{L}_{\mathrm{itLaur},\mathbb{K}} \subset \mathcal{H}_1(\mathbb{K}, \mathbb{Z}), \qquad \mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}} \subset \mathbb{K}[L]\big((t^{1/s})\big) \subset \mathcal{H}_1\big(\mathbb{K}, \tfrac{1}{s}\mathbb{Z}\big).$$

A common trap

$\mathcal{H}_r(\mathbb{K}, \Gamma)$ is a bookkeeping device, not a licence. Membership is a statement about support; it says nothing about convergence, and for non-integer exponents it presupposes a branch of X^{α_0} and of every logarithm involved. All statements below are for $X \to +\infty$ on the positive real ray with the real positive branches; in a sector, each power and each logarithm must be assigned a branch before the classification is applied.

15.2 Power–logarithm inner maps: the classification

Theorem 15.2 (Classification of power–log scale changes). *Let $\rho \in \mathbb{R}_{>0}$, $\sigma \in \mathbb{R}$, $c \in \mathbb{K}^\times$ with a chosen $\log c \in \mathbb{K}$, and $U \in t\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$. Put*

$$G = c\, X^\rho L^\sigma (1+U),$$

and let $\Gamma_0 = \rho\mathbb{Z} + \mathbb{Z}$, $\Gamma_1 = \sigma\mathbb{Z} + \mathbb{Z}$. Then for every $F = \sum_{n \geq n_0} p_n(L) t^n \in \mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ the family

$$F \circ G = \sum_{n \geq n_0} c^{-n}\, t^{\rho n} L^{-\sigma n} (1+U)^{-n}\, p_n\big(\rho L + \sigma L_2 + \log c + \log(1+U)\big) \tag{15.2}$$

is well based and locally finite, so $F \circ G$ is a well-defined element of $\mathcal{H}_2(\mathbb{K}, \Gamma_0 + \Gamma_1)$, with L_0-exponents in $-\Gamma_0$, L_1-exponents in $-\sigma\mathbb{Z} + \mathbb{N}_0$ and L_2-exponents in $\mathbb{N}_0$; and $\Phi_G : F \mapsto F \circ G$ is an injective $\mathbb{K}$-algebra homomorphism. Moreover:

1. *if $\sigma = 0$ and $\rho \in \mathbb{N}_{>0}$, then $\Phi_G(\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}) \subseteq \mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$;*
2. *if $\sigma = 0$ and $\rho = r/s \in \mathbb{Q}_{>0}$ in lowest terms, then $\Phi_G(\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}) \subseteq \mathbb{K}[L]((t^{1/s}))$; for $s > 1$ this is sharp, in that $\Phi_G(t) \notin \mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ and the subgroup of $\mathbb{R}$ generated by the L_0-exponents occurring in $\Phi_G(\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}})$ is exactly $\frac{1}{s}\mathbb{Z}$;*
3. *if $\sigma = 0$ and $\rho \notin \mathbb{Q}$, then no Puiseux-log extension $\mathbb{K}[L]((t^{1/s}))$ contains $\Phi_G(t)$, and Γ_0 is dense in $\mathbb{R}$;*
4. *if $\sigma \neq 0$, then L_2 occurs in $F \circ G$ if and only if $p_n \notin \mathbb{K}$ for some n; and if $p_n \neq 0$ with $\sigma n \notin -\mathbb{N}_0$ then an L_1-exponent outside $\mathbb{N}_0$ occurs. In particular $\Phi_G(\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}})$ is contained in no depth-one class as soon as some coefficient block of some F is nonconstant.*

Proof. Formula (15.2) is (15.1) substituted into $F = \sum p_n(L) t^n$, using

$$G^{-1} = c^{-1} t^\rho L^{-\sigma} (1+U)^{-1}, \qquad \log G = \log c + \rho L_1 + \sigma L_2 + \log(1+U)$$

with the chosen branches, together with the finite Taylor expansion $p_n(a+w) = \sum_{j \leq \deg_{\mathrm{L}} p_n} \frac{p_n^{(j)}(a)}{j!} w^j$ at $a = \rho L + \sigma L_2 + \log c$, $w = \log(1+U) \in t\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$.

Support. The n-th summand has L_0-exponents in $-\rho n - \mathbb{N}_0$, since $(1+U)^{-n}$ and $\log(1+U)$ lie in $\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ and p_n is a polynomial. Because $\rho > 0$, for each real M only finitely many pairs $(n,k) \in \mathbb{Z}_{\geq n_0} \times \mathbb{N}_0$ satisfy $\rho n + k \leq M$. Hence the set of L_0-exponents occurring is well ordered in the direction of increasing smallness, each is reached by only finitely many pairs, and each contribution carries only finitely many monomials in L_1, L_2. The family is therefore summable in $\mathcal{H}_2(\mathbb{K}, \Gamma_0 + \Gamma_1)$, and the stated exponent ranges are read off the display.

Homomorphism and injectivity. Multiplicativity is proved exactly as in Theorem 14.4(2), evaluation of polynomials being a ring homomorphism. For injectivity let $F \neq 0$ with $\nu_t F = n_0$. Since $\rho \neq 0$, the map $Y \mapsto \rho L + \sigma L_2 + \log c$ induces an injection $\mathbb{K}[Y] \hookrightarrow \mathbb{K}[L_1, L_2]$, so $p_{n_0}(\rho L + \sigma L_2 + \log c) \neq 0$; its dominant monomial times $c^{-n_0} X^{-\rho n_0} L^{-\sigma n_0}$ is the dominant monomial of the whole sum, because every other summand and every U-correction has strictly smaller L_0-exponent. Hence $F \circ G \neq 0$.

(1) With $\sigma = 0$ and $\rho \in \mathbb{N}_{>0}$ all L_0-exponents lie in $\mathbb{Z}$, all L_1-exponents in $\mathbb{N}_0$, no L_2 occurs, and only finitely many L_0-exponents exceed $-\rho n_0$; this is exactly membership of $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$.

(2) All exponents lie in $\frac{1}{s}\mathbb{Z}$. Taking $U = 0$ and $F = t^k$ gives $\Phi_G(t^k) = c^{-k} t^{kr/s}$; as k ranges over $\mathbb{Z}$ these exponents generate $\frac{r}{s}\mathbb{Z} + \mathbb{Z} = \frac{1}{s}\mathbb{Z}$, because $\gcd(r,s) = 1$. For $s > 1$ we have $\rho \notin \mathbb{Z}$, so $\Phi_G(t)$ has L_0-exponent $-\rho \notin \mathbb{Z}$ and cannot lie in $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$, all of whose L_0-exponents are integers.

(3) $\Phi_G(t)$ has L_0-exponent $-\rho \notin \mathbb{Q}$, while every element of $\mathbb{K}[L]((t^{1/s}))$ has L_0-exponents in $\frac{1}{s}\mathbb{Z} \subseteq \mathbb{Q}$. Density of $\rho\mathbb{Z} + \mathbb{Z}$ for irrational ρ is the classical equidistribution-free statement that a nondiscrete subgroup of $\mathbb{R}$ is dense.

(4) If every p_n lies in $\mathbb{K}$, then (15.2) reduces to $\sum_n c^{-n} p_n\, t^{\rho n} L^{-\sigma n}(1+U)^{-n}$, in which L_2 does not occur. Conversely, let n_1 be least with $\deg_{\mathrm{L}} p_{n_1} \geq 1$. Summands with $n < n_1$ have constant p_n and contribute no L_2, so the coefficient of $X^{-\rho n_1}$ receives L_2 only from $n = n_1$, where it equals $c^{-n_1} L^{-\sigma n_1} p_{n_1}(\rho L + \sigma L_2 + \log c)$; its L_2-degree is $\deg_{\mathrm{L}} p_{n_1} \geq 1$, with leading coefficient $\sigma^{\deg_{\mathrm{L}} p_{n_1}}$ times that of p_{n_1}, hence nonzero because $\sigma \neq 0$. The claim about L_1-exponents is read off the same summand, whose L_1-exponents are $-\sigma n + \mathbb{N}_0$. ☐

Corollary 15.3 (The three closure regimes for a pure power). *For $G = X^\rho$ with $\rho > 0$ one has $t \circ G = t^\rho$, $L \circ G = \rho L$, and*

$$\Big(\sum_{n \geq n_0} p_n(L) t^n\Big) \circ X^\rho = \sum_{n \geq n_0} p_n(\rho L)\, t^{\rho n}. \tag{15.3}$$

The composite stays in $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ for $\rho \in \mathbb{N}_{>0}$; requires exactly the Puiseux-log extension $\mathbb{K}[L]((t^{1/s}))$ for $\rho = r/s \in \mathbb{Q}_{>0}$ in lowest terms; and requires a Hahn exponent group containing $\rho\mathbb{Z} + \mathbb{Z}$ for irrational ρ. For $\rho \in \mathbb{N}_{>0}$, Φ_{X^ρ} is an automorphism of $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ if and only if $\rho = 1$.

Proof. All but the last sentence is Theorem 15.2(1)–(3) with $c = 1$, $\sigma = 0$, $U = 0$. For $\rho \geq 2$, (15.3) shows every L_0-exponent of the image lies in $\rho\mathbb{Z}$, so t is not in the image; for $\rho = 1$, Φ_X is the identity. ☐

Proposition 15.4 (Dilations and the semidirect structure). *Let $C \subseteq \mathbb{K}^\times$ be a subgroup carrying a fixed group homomorphism* $\lg : C \to (\mathbb{K}, +)$, *a* multiplicative branch choice*; an analytic realisation additionally requires $\mathrm{e}^{\lg(c)} = c$. For $c \in C$ write $m_c(X) = cX$. Then*

$$t \circ m_c = c^{-1} t, \qquad L \circ m_c = L + \lg(c),$$

Φ_{m_c} is an automorphism of $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ preserving $\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$, the dilations normalise $\mathcal{G}_{\mathrm{poly},\mathbb{K}}$ by

$$m_c^{\langle -1 \rangle \circ} \circ (X + A) \circ m_c = X + c^{-1}\big(A \circ m_c\big) \in \mathcal{G}_{\mathrm{poly},\mathbb{K}},$$

and $\mathcal{G}_{\mathrm{lin}} = \{cX + A :\ c \in C,\ A \in \mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}\}$ is a group under composition; it is the semidirect product of C acting on the normal subgroup $\mathcal{G}_{\mathrm{poly},\mathbb{K}}$ by the displayed conjugation.

Proof. The generator images are immediate from (15.1), and they show that Φ_{m_c} maps $\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ into $\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ with two-sided inverse $\Phi_{m_{c^{-1}}}$, using $\lg(c^{-1}) = -\lg(c)$. Next, $(X+A) \circ m_c = cX + A \circ m_c$, and composing with $m_{c^{-1}}$ on the outside multiplies by c^{-1}; since $A \circ m_c \in \mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$, the conjugate lies in $\mathcal{G}_{\mathrm{poly},\mathbb{K}}$. Finally $cX + A = m_c \circ (X + c^{-1}A)$ writes $\mathcal{G}_{\mathrm{lin}}$ as the product of the subgroup $\{m_c : c \in C\} \cong C$ with the normal subgroup $\mathcal{G}_{\mathrm{poly},\mathbb{K}}$, the two intersecting trivially; the group axioms follow from Theorem 14.7 together with the displayed normalisation. ☐

Remark 15.5 (Why the branch constant must be carried as data). For $\mathbb{K} = \mathbb{R}$ and $C = \mathbb{R}_{>0}$ the ordinary logarithm is a homomorphism and Theorem 15.4 applies verbatim. For $\mathbb{K} = \mathbb{C}$ no multiplicative branch choice exists on all of $\mathbb{C}^\times$: a homomorphism $\lg : \mathbb{C}^\times \to (\mathbb{C}, +)$ splitting $\exp$ would exhibit $\ker \exp = 2\pi\mathrm{i}\mathbb{Z}$ as a direct summand of $(\mathbb{C}, +)$; but the latter is a $\mathbb{Q}$-vector space, hence divisible, every direct summand of a divisible group is divisible, and $\mathbb{Z}$ is not. Consequently a symbolic implementation must carry the chosen value of $\log c$ as data attached to the composition rather than recompute it from a principal branch: changing $\operatorname{Log} c$ by $2\pi\mathrm{i}j$ changes the constant block of $L \circ G$ and can move the result onto a different branch of a multivalued inverse.

15.3 Two boundary cases: a logarithmic shift and a logarithmic factor

The maps $X \mapsto X + c\log X$ and $X \mapsto X\log X$ look similar and behave completely differently.

Proposition 15.6 (A logarithmic shift stays inside, at a cost). *Let $c \in \mathbb{K}^\times$ and $G = X + cL$. Then $G \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}$; in particular $\Phi_G(\mathscr{PL}_{\mathrm{poly},\mathbb{K}}) \subseteq \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ and G is invertible in $\mathcal{G}_{\mathrm{poly},\mathbb{K}}$. The logarithmic degrees of the composite are nevertheless unbounded:*

$$t \circ G = \frac{t}{1 + cLt} = \sum_{n\geq 1}(-c)^{n-1}L^{n-1}t^n, \qquad L \circ G = L + \sum_{n\geq 1}\frac{(-1)^{n+1}c^n}{n}L^n t^n,$$

so that, in the log-degree profile notation, $d_{(t\circ G)}(n) = n-1$ and $d_{(L\circ G)}(n) = n$ for $n \geq 1$.

Proof. $cL \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$, so $G \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}$ and Theorem 14.4 and Theorem 14.7 apply. The two displays are the geometric and logarithmic series for $u = cLt \in t\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$, legitimate by Theorem 14.2; each coefficient block is a single monomial in L, so its degree is read off directly. □

Remark 15.7. Theorem 15.6 is the reason why the definition of $\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ imposes no uniform bound on $\deg_{\mathrm{L}} p_n$: the smallest class closed under composition by $X + cL$ must already contain blocks whose degree grows linearly in n. A linear bound $d_F(n) \leq an + b$ is therefore a hypothesis or a proved invariant, never part of the ring definition. The compositional inverse of $X + cL$ is the Wright-omega object whose coefficient blocks are the Lambert polynomials $\mathsf{L}_n^{\mathrm{Lam}}$; see Theorem 16.11 and the surrounding material in Part IV.

Proposition 15.8 (A logarithmic factor forces depth two). *Let $G = XL = X\log X$. Then $t\circ G = tL^{-1}$, $L \circ G = L + L_2$, and for $F = \sum_{n\geq n_0} p_n(L)t^n \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$,*

$$F \circ (XL) = \sum_{n\geq n_0} t^n L^{-n}\, p_n(L + L_2) \;\in\; \mathbb{K}[L_2]\big[L, L^{-1}\big]((t)). \tag{15.4}$$

The symbol L_2 occurs if and only if some p_n is nonconstant; a negative power of L occurs if and only if $p_n \neq 0$ for some $n > 0$. Hence

$$F \circ (XL) \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}} \quad\Longleftrightarrow\quad F \in \mathbb{K}[X] = \mathbb{K}[t^{-1}].$$

Proof. Apply Theorem 15.2 with $c = 1$, $\rho = 1$, $\sigma = 1$, $U = 0$; (15.2) becomes (15.4). The L_2 criterion is part (4) of that theorem. For the powers of L: the coefficient of t^n is $L^{-n}p_n(L + L_2)$, whose L-exponents run over $\{-n, \dots, -n + \deg_{\mathrm{L}} p_n\}$, so a negative exponent occurs precisely when some $n > 0$ has $p_n \neq 0$ – note that the top L_2-degree part of $p_n(L + L_2)$ has L-degree 0, so no cancellation can remove the exponent $-n$. Both defects are excluded in $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$, whose L-exponents lie in $\mathbb{N}_0$ and which contains no L_2. Membership in $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ therefore forces every p_n to be a constant and $p_n = 0$ for $n > 0$, that is, $F \in \mathbb{K}[t^{-1}] = \mathbb{K}[X]$; conversely $X^k \circ (XL) = t^{-k}L^k \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ for $k \geq 0$. □

Example 15.9 (Both enlargements at once). Let $F(X) = X + \log X + X^{-1}$ and $G(X) = 3X^2L^{-1}$, with $\log 3 \in \mathbb{K}$. Then

$$F \circ G = 3X^2L^{-1} + \log\big(3X^2L^{-1}\big) + \frac{L}{3X^2} = 3X^2L^{-1} + 2L - L_2 + \log 3 + \tfrac{1}{3}L\,t^2.$$

Here $\rho = 2$ and $\sigma = -1$: the leading block carries L^{-1} and the second logarithm L_2 appears *exactly*, not as a remainder. The ambient predicted by Theorem 15.2 with $\Gamma_0 = \Gamma_1 = \mathbb{Z}$ is $\mathbb{K}[L_2][L, L^{-1}]((t))$, which is where the answer lies.

15.4 The two extreme scale changes: logarithm and exponential

Proposition 15.10 (Composition with log is a level shift). *Let $\Sigma(F) := F \circ \log$. Then*

$$t \circ \log = L_1^{-1}, \qquad L \circ \log = L_2, \qquad \Sigma\Big(\sum_{n \geq n_0} p_n(L) t^n\Big) = \sum_{n \geq n_0} p_n(L_2)\, L_1^{-n},$$

and Σ is an isomorphism of ordered rings from $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}} = \mathbb{K}[L_1]((L_0^{-1}))$ onto $\mathbb{K}[L_2]((L_1^{-1}))$, carrying the valuation ν_t to the L_1^{-1}-adic valuation and the dominance order to the dominance order. It is a differential isomorphism for the derivation D_{L_1} of the target, i.e. $\Sigma \circ D_X = D_{L_1} \circ \Sigma$; it is not *compatible with D_X on the target, where instead*

$$D_X \circ \Sigma = t\left(\Sigma \circ D_X\right). \tag{15.5}$$

In particular $\mathbb{K}[L_2]((L_1^{-1}))$ is not stable under D_X.

Proof. The generator images are (15.1) with $G = \log X$. Substituting them into $F = \sum p_n(L)t^n$ gives the displayed formula, which is a relabelling of indeterminates: it sends the topological generators $t^{\pm 1}, L_1$ of the source bijectively to the topological generators $L_1^{\mp 1}, L_2$ of the target, preserving sums, products, limits, the lower-bounded integer support, and the lexicographic dominance order. Hence Σ is an isomorphism of ordered rings, and $\Sigma \circ D_X = D_{L_1} \circ \Sigma$ because both sides are continuous $\mathbb{K}$-linear maps satisfying the same Leibniz rule and agreeing on t and L_1. Since $D_X = \frac{\mathrm{d}L_1}{\mathrm{d}X}\, D_{L_1} = t\, D_{L_1}$ on the target, formula (15.5) follows. Finally $D_X(L_1^{-1}) = -tL_1^{-2}$ involves t, which does not occur in $\mathbb{K}[L_2]((L_1^{-1}))$. $\square$

Proposition 15.11 (Composition with exp leaves every logarithmic class). *One has $t \circ \mathrm{e}^X = \mathrm{e}^{-X}$ and $L \circ \mathrm{e}^X = X$, so that*

$$\Big(\sum_{n \geq n_0} p_n(L)t^n\Big) \circ \mathrm{e}^X = \sum_{n \geq n_0} p_n(X)\, \mathrm{e}^{-nX}. \tag{15.6}$$

The monomial e^{-X} belongs to no $\mathcal{M}_r(\Gamma)$ of Theorem 15.1: for $\mathfrak{m} = \prod_{j \leq r} L_j^{\alpha_j}$ one has $\log \mathfrak{m} = \mathcal{O}_{X \to +\infty}(L_1)$, whereas $\log \mathrm{e}^{-X} = -X$ and $X/L_1 \to +\infty$. Moreover the loss is total on the analytic side: if f is a function on some $[X_1, +\infty)$ admitting F as an asymptotic expansion in the scale $\left\{X^{-n}(\log X)^j\right\}$ with $a = \nu_t F \geq 1$ and $d = \deg_{\mathrm{L}} p_a$, then

$$f\left(\mathrm{e}^X\right) = \mathcal{O}_{X \to +\infty}\Big(\mathrm{e}^{-aX} X^d\Big) = o_{X \to +\infty}\left(X^{-N}\right) \quad \text{for every } N, \tag{15.7}$$

so the polynomial–logarithmic expansion of $X \mapsto f(\mathrm{e}^X)$ is identically zero although the function itself need not be.

Proof. The generator images are (15.1) with $G = \mathrm{e}^X$, and (15.6) is their substitution. For the monomials, $\log \mathfrak{m} = \alpha_0 L_1 + \alpha_1 L_2 + \cdots + \alpha_{r-1} L_r + \alpha_r \log L_r$, each summand being $\mathcal{O}_{X \to +\infty}(L_1)$. For the last claim, the hypothesis gives $f(Y) = p_a(\log Y) Y^{-a} + \mathcal{O}_{Y \to +\infty}\Big(Y^{-a-1}(\log Y)^{d'}\Big)$ for a suitable d', hence $|f(Y)| \leq CY^{-a}(\log Y)^d$ for Y large; put $Y = \mathrm{e}^X$. Since $a \geq 1$, $\mathrm{e}^{-aX} X^d$ is smaller than every power X^{-N}. $\square$

Remark 15.12. (15.6) is the reason a formal exponential change of scale must be met by enlarging the *monomial* group, not the coefficient ring: the objects $\mathrm{e}^{-nX} X^j$ are new transmonomials, and the resulting field is the logarithmic–exponential transseries field of positive exponential height [1, 13, 33]. The polynomial–logarithmic class is exactly its height-zero, depth-one corner.

15.5 The table of scale changes

Table 15.1: Scale changes and the resulting ambient class. Throughout $A \in \mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$, $U \in t\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$, $c \in \mathbb{K}^\times$ with a chosen $\log c \in \mathbb{K}$, and the regime is $X \to +\infty$ on the positive ray.

Inner map G	$t \circ G$; $L \circ G$	Smallest ambient class for $F \circ G$
$X + A$	$t(1+tA)^{-1}$; $L + \log(1+tA)$	$\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$; a group (Theorem 14.7)
cX	$c^{-1}t$; $L + \log c$	$\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ (Theorem 15.4)
$X + cL$	$t(1+cLt)^{-1}$; $L + \log(1+cLt)$	$\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$; $\deg_{\mathrm{L}}$ grows linearly (Theorem 15.6)
$cX^\rho(1+U)$, $\rho \in \mathbb{N}_{>0}$	$c^{-1}t^\rho(1+U)^{-1}$; $\rho L + \log c + \log(1+U)$	$\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ (Theorem 15.2(1))
$X^{r/s}$, $\gcd(r,s) = 1$, $s > 1$	$t^{r/s}$; $(r/s)L$	Puiseux-log $\mathbb{K}[L]((t^{1/s}))$; not $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$
X^ρ, $\rho > 0$ irrational	t^ρ; ρL	Hahn $\mathcal{H}_1(\mathbb{K}, \rho\mathbb{Z} + \mathbb{Z})$; no Puiseux extension suffices
$XL^\sigma, \sigma \neq 0$	$tL^{-\sigma}$; $L + \sigma L_2$	depth two: $\mathcal{H}_2$; for $\sigma \in \mathbb{Z}$, $\mathbb{K}[L_2][L, L^{-1}]((t))$
$cX^\rho L^\sigma(1+U)$, $\rho > 0$, $\sigma \neq 0$	$c^{-1}t^\rho L^{-\sigma}(1+U)^{-1}$; $\rho L + \sigma L_2 + \log c + \log(1+U)$	$\mathcal{H}_2(\mathbb{K}, \Gamma_0 + \Gamma_1)$ (Theorem 15.2)
$\log X$	L^{-1}; L_2	$\mathbb{K}[L_2]((L^{-1}))$: the level shift Σ (Theorem 15.10)
e^X	e^{-X}; X	outside every $\mathcal{H}_r$: logarithmic–exponential transseries (Theorem 15.11)
$cX^\rho, \rho \leq 0$	—	inadmissible: G does not tend to $+\infty$
cX, $c < 0$, $\mathbb{K} \subseteq \mathbb{R}$	—	inadmissible on the positive ray: $\log c \notin \mathbb{R}$; extend $\mathbb{K}$ and fix a sector
$X = 1/x$ (regime change)	$t \mapsto x$; $L \mapsto -\log x$	$\mathbb{K}[\Lambda]((x))$, $\Lambda = -\log x$: an isomorphism, not an endomorphism (Theorem 15.18)

A common trap

The last three rows are not failures of a formula but failures of a *hypothesis*. An expression may be perfectly meaningful while lying outside $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$; and conversely a formally computed composite is meaningless if the inner map does not carry the outer expansion's regime. Closure is a statement about a chosen representation class, never about existence of the composed function. In particular the whole table is directional: that $F \circ G$ is defined and lies in a given class says nothing about $G \circ F$.

15.6 The nested logarithmic scale

Proposition 15.13 (Derivation of the logarithmic tower). *With $L_0 = X$ and $L_{j+1} = \log L_j$,*

$$D_X L_j = \frac{t}{L_1 L_2 \cdots L_{j-1}} \qquad (j \geq 1), \tag{15.8}$$

the product being empty for $j = 1$. Put

$$\mathcal{D}_r := \partial_{L_1} + \sum_{j=2}^{r} \frac{1}{L_1 L_2 \cdots L_{j-1}} \partial_{L_j}, \qquad \mathcal{L}_r := \mathbb{K}\big[L_1^{\pm 1}, \ldots, L_{r-1}^{\pm 1}, L_r\big].$$

Then $\mathcal{D}_r(\mathcal{L}_r) \subseteq \mathcal{L}_r$, and for $n \in \mathbb{Z}$, $p \in \mathcal{L}_r$,

$$D_X\big(t^n p\big) = t^{n+1}\big(\mathcal{D}_r - n\big)p. \tag{15.9}$$

By contrast, for $r \geq 2$ the polynomial ring $\mathbb{K}[L_1, \ldots, L_r]$ is not stable under $\mathcal{D}_r$: $\mathcal{D}_r L_2 = L_1^{-1}$.

Proof. (15.8) holds for $j = 1$ since $D_X L_1 = t$, and inductively $D_X L_{j+1} = (D_X L_j)/L_j$. Stability: ∂_{L_1} maps $\mathcal{L}_r$ into $\mathcal{L}_r$ because the ring is Laurent in L_1; for $2 \le j \le r$, ∂_{L_j} lowers the L_j-degree and the prefactor $(L_1 \cdots L_{j-1})^{-1}$ lies in $\mathcal{L}_r$ — here it matters that only $L_1, \dots, L_{r-1}$ need be inverted, since ∂_{L_r} is multiplied by $(L_1 \cdots L_{r-1})^{-1}$. Formula (15.9) follows from $D_X t^n = -nt^{n+1}$ and (15.8) by the Leibniz rule. □

Theorem 15.14 (The near-identity group at logarithmic depth r)**.** *Let $r \ge 1$ and*

$$\mathcal{N}_r := \{X + A : \ A \in \mathcal{L}_r[[t]]\}\,, \qquad \mathcal{N}_1 = \mathcal{G}_{\mathrm{poly},\mathbb{K}}.$$

Then $\mathcal{N}_r$ is a group under composition, and every statement of Part 14 holds verbatim with $\mathbb{K}[L]$ replaced by $\mathcal{L}_r$, ∂_L by $\mathcal{D}_r$, and $\Delta_{n,k}$ by $\prod_{j=0}^{k-1}(\mathcal{D}_r - n - j)$. The corresponding statement for $\{X + A : A \in \mathbb{K}[L_1, \dots, L_r][[t]]\}$ is false for $r \ge 2$.

Proof. Fix $G = X + A$ with $A \in \mathcal{L}_r[[t]]$ and define Φ_G on the topological generators $t^{\pm 1}, L_1^{\pm 1}, \dots, L_{r-1}^{\pm 1}, L_r$ of $\mathcal{L}_r((t))$ by $\Phi_G(t) = t(1 + tA)^{-1}$, $\Phi_G(L_{j+1}) = \log\big(\Phi_G(L_j)\big)$, starting from $\Phi_G(L_0) = X(1 + tA)$. Set $w_0 = tA$ and, for $j \ge 1$,

$$w_j := L_j^{-1} \log(1 + w_{j-1}). \tag{15.10}$$

An induction using Theorem 14.2 gives $w_j \in t\mathcal{L}_r[[t]]$ and

$$\Phi_G(L_j) = L_j\big(1 + w_j\big) \qquad (0 \le j \le r-1), \qquad \Phi_G(L_r) = L_r + \log(1 + w_{r-1}),$$

since $\log\big(L_j(1 + w_j)\big) = L_{j+1} + \log(1 + w_j) = L_{j+1}\big(1 + L_{j+1}^{-1} \log(1 + w_j)\big)$. The construction inverts exactly $L_1, \dots, L_{r-1}$ and never L_r; this is what forces the coefficient ring $\mathcal{L}_r$ and no smaller one. Extending Φ_G multiplicatively and t-adically reproduces Theorem 14.4 word for word, and (15.9) reproduces $D_X(\mathcal{L}_r[[t]]) \subseteq t\mathcal{L}_r[[t]]$, which is the only property of D_X used in Theorem 14.5 and Theorem 14.6.

Two points genuinely need rechecking, and both go through. First, the *gain estimate*: if $\nu_t(B - C) = m$, then comparing (15.10) for $G = X + B$ and $G' = X + C$ gives $\nu_t(w_0 - w_0') \ge m + 1$ and, inductively, $\nu_t(w_j - w_j') \ge m + 1$, because $\log(1 + w) - \log(1 + w') = \log\big(1 + \frac{w - w'}{1 + w'}\big)$ and L_j^{-1} is a unit of $\mathcal{L}_r$ for $j \le r - 1$. Hence every generator except X itself has Φ_G- and $\Phi_{G'}$-images differing by an element of $t^{m+1}\mathcal{L}_r[[t]]$, and expanding $\Phi_G(a_n t^n) - \Phi_{G'}(a_n t^n)$ by $PQ - P'Q' = (P - P')Q + P'(Q - Q')$ yields $\nu_t\big(A \circ G - A \circ G'\big) \ge m + 1$. Second, *associativity*: $\Phi_H(\Phi_G(L_0)) = \Phi_H(G) = G \circ H = \Phi_{G \circ H}(L_0)$ as in (14.9), and if the identity $\Phi_H \Phi_G(L_j) = \Phi_{G \circ H}(L_j)$ holds then applying the continuous algebra homomorphism Φ_H to $\Phi_G(L_{j+1}) = L_{j+1} + \log(1 + w_j)$ and using $\Phi_H(\log(1 + u)) = \log(1 + \Phi_H(u))$ together with $\log(uv) = \log u + \log v$ gives the identity at level $j + 1$; the case $j + 1 = 1$ is the computation displayed in the proof of Theorem 14.7(2). With these, the fixed-point construction of inverses (Theorem 16.11, whose proof uses only the gain estimate Theorem 14.6 and t-adic completeness, both available verbatim in $\mathcal{L}_r[[t]]$) and the monoid argument are unchanged.

For the final assertion take $r = 2$, $G = X + 1 \in \mathcal{N}_2$ and $F = L_2 \in \mathbb{K}[L_1, L_2]$. Then

$$F \circ G = \log\log(X + 1) = L_2 + \log\Big(1 + L_1^{-1} \log(1 + t)\Big) = L_2 + L_1^{-1} t + \mathcal{O}_t^{\mathrm{form}}\big(t^2\big)\,,$$

whose block of outer order 1 is $L_1^{-1} \notin \mathbb{K}[L_1, L_2]$. Hence $\{X + A : A \in \mathbb{K}[L_1, L_2][[t]]\}$ is not closed under composition. □

Proposition 15.15 (The unit criterion changes shape with depth)**.** $\mathcal{L}_r^\times = \big\{c\, L_1^{a_1} \cdots L_{r-1}^{a_{r-1}} : \ c \in \mathbb{K}^\times,\ a_i \in \mathbb{Z}\big\}$. *Consequently a nonzero $F = \sum_{n \ge n_0} p_n t^n \in \mathcal{L}_r((t))$ is a unit if and only if its leading block p_{n_0} is a nonzero scalar times a Laurent monomial in $L_1, \dots, L_{r-1}$. For $r = 1$ this is the classical criterion "$p_{n_0} \in \mathbb{K}^\times$"; for $r \ge 2$ strictly more elements are invertible, for instance $L_1^{-1} + \mathcal{O}_t^{\mathrm{form}}(t)$.*

Proof. Write $\mathcal{L}_r = R[L_r]$ with $R = \mathbb{K}[L_1^{\pm 1}, \ldots, L_{r-1}^{\pm 1}]$. Since R is an integral domain, $(R[L_r])^\times = R^\times$; and by induction on the number of variables, the units of a Laurent polynomial ring over a domain are the units of the base times monomials. The criterion for $\mathcal{L}_r((t))$ is the standard one for a Laurent series ring over a commutative ring: if p_{n_0} is a unit, $F = t^{n_0} p_{n_0}(1+v)$ with $v \in t\mathcal{L}_r[[t]]$, and $(1+v)^{-1}$ is given by the geometric series; conversely $FF^{-1} = 1$ forces $p_{n_0}\,[t^{-n_0}]F^{-1} = 1$. □

15.7 Lifting a one-logarithm result, and where the lift fails

Proposition 15.16 (Transport by the level shift)**.** *Let $\mathcal{S}$ be any statement about the structure $\big(\mathscr{PL}_{\mathrm{poly},\mathbb{K}};\ t,\ L,\ \nu_t,\ \preccurlyeq,\ D_X\big)$ expressed purely in that language. Then $\mathcal{S}$ holds if and only if the corresponding statement holds for $\big(\mathbb{K}[L_2]((L_1^{-1}));\ L_1^{-1},\ L_2,\ \nu_{L_1^{-1}},\ \preccurlyeq,\ D_{L_1}\big)$. In particular*

$$\{L_1 + B :\ B \in \mathbb{K}[L_2][[L_1^{-1}]]\}$$

is a group under composition in the variable L_1, with all the structure of Theorem 14.7, Theorem 14.12 and Theorem 14.16.

Proof. Theorem 15.10 exhibits Σ as an isomorphism of the two structures. □

Remark 15.17 (The correct fast variable). This is the precise content of the practical rule that a nested expansion should be read in the right variable. A series $\sum_{n\geq 0} p_n(\log\log X)(\log X)^{-n}$, which is a depth-two object in X, is a *one-logarithm* object in the fast variable $T = L_1 = \log X$, with $\Lambda = L_2 = \log T$. Choosing the wrong fast variable makes such a problem look as though it fell outside the theory.

A common trap

Three things do *not* lift.

- *Statements about D_X.* By (15.5), Σ intertwines D_X with D_{L_1}, not with D_X; and $\mathbb{K}[L_2]((L_1^{-1}))$ is not even stable under D_X. A result whose statement mentions the derivation of the *original* variable must be re-derived, not transported.
- *Genuinely mixed depth.* Σ reaches only the shifted copy. A series such as $\sum_n p_n(L_1, L_2)t^n$ with genuinely bivariate blocks lies in neither $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ nor $\Sigma(\mathscr{PL}_{\mathrm{poly},\mathbb{K}})$, and no change of fast variable makes it single-logarithmic; one must work in $\mathcal{L}_r((t))$ and accept Theorem 15.13.
- *The unit criterion.* Its shape changes at depth ≥ 2 (Theorem 15.15), so any argument that used "leading block a nonzero scalar" must be restated as "leading block a unit of the coefficient ring".

15.8 From infinity to a finite point

Proposition 15.18 (The infinity–zero dictionary)**.** *Put $x = X^{-1}$ and*

$$\Lambda := \log\frac{1}{x} = -\log x = \log X = L.$$

Then $F \mapsto F(1/x)$ is an isomorphism of ordered rings

$$\iota:\ \mathbb{K}[L]((t)) \longrightarrow \mathbb{K}[\Lambda]((x)), \qquad t \longmapsto x, \quad L \longmapsto \Lambda,$$

matching the regime $X \to +\infty$ with $x \to 0^+$ and the dominance order with the order of decreasing size as $x \to 0^+$. On the differential side

$$\frac{\mathrm{d}}{\mathrm{d}x} = -X^2 D_X, \qquad \frac{\mathrm{d}}{\mathrm{d}x} = \partial_x - \frac{1}{x}\,\partial_\Lambda, \qquad \frac{\mathrm{d}}{\mathrm{d}x}\big(x^n P(\Lambda)\big) = x^{n-1}\big(nP - P'\big), \tag{15.11}$$

the exact mirror of (14.1)*. Under ι the group $\mathcal{G}_{\mathrm{poly},\mathbb{K}}$ corresponds to the group of germs*

$$\left\{ g(x) = \frac{x}{1 + x\,a(x)} \;:\; a \in \mathbb{K}[\Lambda][[x]] \right\}$$

with the composition law of $\mathcal{G}_{\mathrm{poly},\mathbb{K}}$, the isomorphism being $G \mapsto j \circ G \circ j$ with $j(x) = 1/x$.

Proof. The map ι is the relabelling $t \mapsto x$, $L \mapsto \Lambda$; it is a ring isomorphism, and it preserves the order because $t^n L^j \succ t^m L^k$ and $x^n \Lambda^j \succ x^m \Lambda^k$ are governed by the same inequalities on (n, j) and (m, k). From $x = X^{-1}$, $\frac{\mathrm{d}x}{\mathrm{d}X} = -X^{-2}$, so $D_X = -x^2 \frac{\mathrm{d}}{\mathrm{d}x}$, which is the first identity; the second is the chain rule with $\partial_x \Lambda = -1/x$; the third follows: $\frac{\mathrm{d}}{\mathrm{d}x}(x^n P) = n x^{n-1} P - x^n \frac{1}{x} P'$. Finally j is an involution, so $G \mapsto j \circ G \circ j$ is a group isomorphism, and for $G = X + A$, $j \circ G \circ j = \big(x^{-1} + A(1/x)\big)^{-1} = x\big(1 + x\,\iota(A)\big)^{-1}$. $\square$

A common trap

The dictionary is $L = -\log x$, never $L = \log x$. Substituting the wrong sign reverses ∂_L and turns the block operator $\Delta_{n,k} = \prod_{j=0}^{k-1}(\partial_L - n - j)$ into a different operator; all reversion coefficients then come out with wrong signs while remaining formally consistent, which makes the error hard to detect.

Remark 15.19 (What is deferred). Theorem 15.18 is only the integer-exponent dictionary. Local dynamics at a finite point uses *Dulac* classes: real or well-ordered exponent sets x^α, the generator $\ell = -1/\log x$ in place of Λ, and their iterated versions, for which composition and normal forms require the machinery of [23, 24, 27, 26]. That development, including the change between Λ and ℓ and the consequences of admitting real exponents, is carried out systematically in Part 26. Note also that the other natural passage to a finite point, $X = -\log x$, is the exponential change of scale of Theorem 15.11 and leaves the class.

Section summary

A substitution is decided by the two generator images G^{-1} and $\log G$, and by nothing else. For $G = cX^\rho L^\sigma(1 + U)$ with $\rho > 0$: integral ρ and $\sigma = 0$ keep the composite in $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$; rational $\rho = r/s$ forces exactly the Puiseux-log extension $\mathbb{K}[L]((t^{1/s}))$; irrational ρ forces a Hahn exponent group; $\sigma \neq 0$ forces the second logarithm L_2 precisely when some coefficient block is nonconstant, together with L-exponents outside $\mathbb{N}_0$. Composition with $\log$ is an isomorphism onto the level-shifted copy — the formal statement of the "correct fast variable" rule — while composition with $\exp$ leaves every logarithmic class and annihilates the polynomial–logarithmic expansion entirely. At depth r the near-identity group survives, but only over the coefficient ring $\mathbb{K}[L_1^{\pm 1}, \dots, L_{r-1}^{\pm 1}, L_r]$ that the derivation and the generator substitutions force.

Part V

Series reversal at infinity

Chapter 16

Existence and uniqueness of the near-identity inverse

16.1 Conventions, and what is imported from the composition part

Throughout this part $\mathbb{K}$ is a field of characteristic zero, the large variable is X with

$$t := X^{-1}, \qquad L := \log X,$$

and the default analytic regime, whenever one is invoked at all, is $X \to +\infty$ on the positive real ray. We work in

$$\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}} = \mathbb{K}[L][[t]], \qquad \mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}} = \mathbb{K}[L]((t)), \qquad \mathscr{P}\mathscr{L}_{\mathrm{rat},\mathbb{K}} = \mathbb{K}(L)((t)),$$

with the outer valuation ν_t and the exclusive truncation $\operatorname{Trunc}_{<N} F = \sum_{n<N} p_n(L)t^n$. The near-identity group is

$$\mathcal{G}_{\mathrm{poly},\mathbb{K}} := \{X + A : A \in \mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}\} = \left\{F \in \mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}} : \ \nu_t(F) \geq -1 \text{ and } [t^{-1}]F = 1\right\}. \quad (16.1)$$

Definition 16.1 (Truncation and blocks of a near-identity map). For $G = X + B \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}$ with $B = \sum_{n\geq 0} b_n(L)t^n$ we call b_n the *n-th inverse block* of G and set

$$\operatorname{Trunc}_{<N} G := X + \operatorname{Trunc}_{<N} B = X + \sum_{n<N} b_n(L)t^n.$$

For $F \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}$, the phrase "$G$ is correct through outer block N" means

$$\operatorname{Trunc}_{<N+1} G = \operatorname{Trunc}_{<N+1} F^{\langle -1\rangle_\circ},$$

that is, agreement in the $N+1$ blocks $b_0, \ldots, b_N$.

Remark 16.2 (Inclusive versus exclusive cutoffs). Report 6 writes its truncations inclusively as $[F]_{\leq N}$. The translation is $[F]_{\leq N} = \operatorname{Trunc}_{<N+1} F$, so every cutoff index borrowed from that source has been raised by one below. The place where this matters most is the finite certificate (Theorem 16.16): correctness *through* block N is the exclusive cutoff $< N+1$, and the certifying residual is a $\mathcal{O}_t^{\mathrm{form}}\big(t^{N+1}\big)$, not a $\mathcal{O}_t^{\mathrm{form}}\big(t^{N}\big)$.

Remark 16.3 (Standing facts imported from the composition part). Four facts about $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$, D_X and $\circ$ are used constantly. They are proved elsewhere in this volume; we record them in the exact form needed here and prove nothing about them except the completeness statement, whose precise shape is used below.

(I1) Block derivative (Theorem 11.8). For $n \in \mathbb{Z}$ and $p \in \mathbb{K}[L]$,

$$D_X\big(t^n p(L)\big) = t^{n+1}\big(\partial_L - n\big)p(L), \qquad D_X^k\big(t^n p(L)\big) = t^{n+k}\Delta_{n,k}p(L), \tag{16.2}$$

where $\Delta_{n,k} = \prod_{j=0}^{k-1}(\partial_L - n - j)$ and $\Delta_{n,0} = 1$ is the empty product.

(I2) Exact formal Taylor formula (Theorem 13.6). For $S \in \mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ and $P \in \mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ the family $\big(P^k D_X^k S/k!\big)_{k\geq 0}$ is formally summable and

$$S \circ (X + P) = \sum_{k\geq 0} \frac{P^k}{k!}\, D_X^k S. \tag{16.3}$$

This is an identity of formal series; it carries no analytic content.

(I3) Closure, associativity and chain rule (Theorem 14.7 (1)–(3), Theorem 12.31). For $F = X + A$ and $G = X + B$ in $\mathcal{G}_{\mathrm{poly},\mathbb{K}}$,

$$F \circ G = X + B + A \circ G \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}; \tag{16.4}$$

$(S \circ H_1) \circ H_2 = S \circ (H_1 \circ H_2)$ whenever all three displayed composites exist in the ambient class; this covers near-identity inner arguments and monomial inner arguments cX for which a value of $\log c$ lies in $\mathbb{K}$. The series X is a two-sided identity for $\circ$, and

$$D_X(S \circ H) = \big((D_X S) \circ H\big) \cdot D_X H. \tag{16.5}$$

(I4) Completeness (Theorem 7.18). $\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ is complete for the t-adic filtration: if (F_j) is Cauchy then for each n the block $[t^n]F_j \in \mathbb{K}[L]$ is eventually constant, say p_n; the series $F = \sum_{n\geq 0} p_n t^n$ lies in $\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ — no uniform bound on $\deg_L p_n$ is required — and $F_j \to F$. On $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ the same argument needs the extra hypothesis that the $\nu_t(F_j)$ are bounded below, without which a Cauchy sequence need not define a Laurent series with a finite principal part.

Lemma 16.4 (Valuation gain of the derivation). *For every $S \in \mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ and every $k \in \mathbb{N}_0$, $\nu_t(D_X^k S) \geq \nu_t(S) + k$.*

Proof. By (16.2), D_X^k sends the block $t^n p$ to $t^{n+k}\Delta_{n,k}p$, hence maps $t^m \mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ into $t^{m+k}\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ for every $m \in \mathbb{Z}$. Apply this with $m = \nu_t(S)$. Equality may fail, since $\Delta_{n,k}p$ can vanish; for instance $D_X(t^0 \cdot 1) = 0$. ☐

Lemma 16.5 (Right composition with a near-identity preserves leading data). *Let $S \in \mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$, $P \in \mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$, and $m = \nu_t(S)$. Then*

$$\nu_t\big(S \circ (X + P)\big) = m, \qquad [t^m]\big(S \circ (X + P)\big) = [t^m]S.$$

In particular $S \circ (X + P) = 0$ if and only if $S = 0$, and right composition by an element of $\mathcal{G}_{\mathrm{poly},\mathbb{K}}$ carries units of $\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ to units of $\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$.

Proof. In (16.3) the term $k = 0$ is S. For $k \geq 1$, $\nu_t\big(P^k D_X^k S/k!\big) \geq k\,\nu_t(P) + m + k \geq m + k > m$, using Theorem 16.4 and $\nu_t(P) \geq 0$. So no earlier block is created and the block at level m is unchanged. ☐

The next lemma re-expands a composition around a shifted base point without inverting anything. It may therefore be used inside the existence proof, and it is the engine of every estimate in this part.

Lemma 16.6 (Two-point Taylor expansion). *Let $S \in \mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ and $P, Q \in \mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$. Then the family $\big(Q^j (D_X^j S) \circ (X + P)/j!\big)_{j\geq 0}$ is formally summable and*

$$S \circ \big(X + P + Q\big) = \sum_{j\geq 0} \frac{Q^j}{j!}\, \big(D_X^j S\big) \circ (X + P). \tag{16.6}$$

Proof. Summability: by Theorems 16.4 and 16.5, $\nu_t\big(Q^j(D_X^j S)\circ(X+P)/j!\big) \geq j\,\nu_t(Q)+\nu_t(S)+j \to \infty$.

Apply (16.3) with $P+Q \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ in place of P:

$$S\circ(X+P+Q) = \sum_{k\geq 0}\frac{(P+Q)^k}{k!}D_X^k S = \sum_{k\geq 0}\sum_{i+j=k}\frac{P^i}{i!}\frac{Q^j}{j!}D_X^{i+j}S.$$

Each term has $\nu_t \geq \nu_t(S)+i+j$, so the double family is summable and may be summed in either order. Grouping by j and applying (16.3) once more, now to $D_X^j S$,

$$\sum_{j\geq 0}\frac{Q^j}{j!}\sum_{i\geq 0}\frac{P^i}{i!}D_X^i\big(D_X^j S\big) = \sum_{j\geq 0}\frac{Q^j}{j!}\big(D_X^j S\big)\circ(X+P),$$

which is (16.6). ☐

16.2 The inverse equation as a fixed point

Proposition 16.7 (The reversion equation). *Let $F = X + A \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}$ and $G = X + B \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}$. Then $F\circ G = X$ if and only if $B = -A\circ(X+B)$.*

Proof. By (16.4), $F\circ G = X + B + A\circ G$, which equals X exactly when $B + A\circ(X+B) = 0$. ☐

Thus B is a fixed point of

$$\Phi\colon \mathscr{PL}_{\mathrm{pow},\mathbb{K}} \to \mathscr{PL}_{\mathrm{pow},\mathbb{K}}, \qquad \Phi(B) := -A\circ(X+B). \tag{16.7}$$

The map Φ is neither linear nor defined on all of $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$: its argument must be an admissible near-identity displacement.

Remark 16.8 (Variable names). Reports 3 and 6 write the reversion equation in a second letter, $B(Y) = -A(Y+B(Y))$, to separate visually the argument of the inverse from the argument of F. This is a typographic device only: X is a free generator of the formal algebra and the renaming is the identity map. We use X throughout, in agreement with the notation catalogue.

Remark 16.9 (Filtered Lipschitz gain: the fifth standing import). Let $A \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ with $\alpha := \nu_t(A)$, and let $B, C \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$. Then

$$\nu_t\big(A\circ(X+B) - A\circ(X+C)\big) \;\geq\; \alpha + \nu_t(B-C)+1, \tag{16.8}$$

which is (14.5) of Theorem 14.6, proved there. In particular, for the map Φ of (16.7), $\nu_t\big(\Phi(B)-\Phi(C)\big) \geq \alpha+\nu_t(B-C)+1$. The reader who prefers an argument internal to this part may instead put $S = A$, $P = C$, $Q = B - C$ in Theorem 16.6 and bound the j-th term by $\alpha + j(\nu_t(Q)+1)$ using Theorems 16.4 and 16.5; the minimum over $j \geq 1$ is at $j = 1$.

Remark 16.10 (What “a gain of one block” means, precisely). All six reports describe (16.8) as a contraction. Three points must be kept apart.

1. *It is a statement about ν_t alone.* If B and C agree in the blocks $0,\ldots,r-1$, then $\Phi(B)$ and $\Phi(C)$ agree in the blocks $0,\ldots,r$: one further block of the output is pinned down. Nothing whatever is asserted about the size, the coefficients, or the logarithmic degree of the blocks that still differ; $\deg_L$ is not contracted and in general grows.

2. *It is a genuine metric contraction for the standard t-adic absolute value.* Setting $|F|_t = 2^{-\nu_t(F)}$ and $|0|_t = 0$, inequality (16.8) says exactly that Φ is Lipschitz with constant $2^{-1-\alpha} \leq \frac{1}{2}$. The space $(\mathscr{PL}_{\mathrm{pow},\mathbb{K}}, |\cdot|_t)$ is complete by (I4), so the Banach fixed-point theorem applies verbatim. We nevertheless give the coefficient-stabilization proof below, because that is what an implementation performs and because it yields the sharp iteration rate.

3. *The gain is at least one, and in general exactly one.* Take $A = L$, $B = 0$ and $C = ct^r$ with $c \in \mathbb{K}^\times$, $r \geq 0$. Then $A \circ (X + C) - A \circ (X + B) = \log(1 + ct^{r+1}) = ct^{r+1} + \mathcal{O}_t^{\mathrm{form}}\big(t^{r+2}\big)$, and (16.8) is an equality. The summand $\alpha = \nu_t(A)$ in (16.8) does not appear in any of the six sources; it is what makes the sharp Newton estimate of Theorem 17.11 come out.

16.3 Existence, uniqueness, and the two-sided property

Theorem 16.11 (Near-identity inverse theorem). *Let $\mathbb{K}$ have characteristic zero and let $F = X + A \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}$ with $\alpha = \nu_t(A)$. Then:*

1. *the Picard iterates $B_0 = 0$, $B_{j+1} = \Phi(B_j) = -A \circ (X + B_j)$ satisfy*
$$\nu_t\big(B_{j+1} - B_j\big) \;\geq\; (j+1)\alpha + j \qquad (j \geq 0), \tag{16.9}$$
and converge t-adically to some $B \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ with
$$\nu_t(B - B_j) \;\geq\; (j+1)\alpha + j; \tag{16.10}$$
2. *B is the unique element of $\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ with $B = -A \circ (X + B)$, and $G := X + B$ is the unique element of $\mathcal{G}_{\mathrm{poly},\mathbb{K}}$ with $F \circ G = X$;*
3. *$G \circ F = X$ as well, and G is also the unique element of $\mathcal{G}_{\mathrm{poly},\mathbb{K}}$ with that property; we write $G = F^{\langle -1 \rangle_\circ}$;*
4. *$\nu_t(B) = \alpha$ and $[t^\alpha]B = -[t^\alpha]A$.*

Proof. *(1)* For $j = 0$, $B_1 - B_0 = -A \circ X = -A$ has $\nu_t \geq \alpha = (0+1)\alpha + 0$. If (16.9) holds for $j - 1$, then by Theorem 16.9
$$\nu_t(B_{j+1} - B_j) = \nu_t\big(\Phi(B_j) - \Phi(B_{j-1})\big) \geq \alpha + \big(j\alpha + j - 1\big) + 1 = (j+1)\alpha + j.$$
The bound tends to $+\infty$, so (B_j) is Cauchy and converges to some $B \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ by (I4); moreover $\nu_t(B - B_j) \geq \min_{i \geq j} \nu_t(B_{i+1} - B_i) \geq (j+1)\alpha + j$, which is (16.10).

That B solves $B = \Phi(B)$ needs no separate continuity argument. For every j, $B - \Phi(B) = (B - B_{j+1}) + \big(\Phi(B_j) - \Phi(B)\big)$. The first summand has $\nu_t \geq (j+2)\alpha + j + 1$ by (16.10); the second has $\nu_t \geq \alpha + \nu_t(B_j - B) + 1 \geq (j+2)\alpha + j + 1$ by Theorem 16.9. Hence $\nu_t\big(B - \Phi(B)\big) \geq (j+2)\alpha + j + 1$ for every j, which exceeds every fixed integer as $j \to \infty$; so $B = \Phi(B)$.

(2) If B and $\widetilde{B}$ are fixed points with $r = \nu_t(B - \widetilde{B}) < \infty$, then Theorem 16.9 gives $r = \nu_t\big(\Phi(B) - \Phi(\widetilde{B})\big) \geq \alpha + r + 1 > r$, a contradiction; so $B = \widetilde{B}$. By Theorem 16.7, $G = X + B$ is then the unique right inverse of F inside $\mathcal{G}_{\mathrm{poly},\mathbb{K}}$.

(3) Apply (1)–(2) to $G \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}$: there is $H \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}$ with $G \circ H = X$. Using (I3),
$$F = F \circ X = F \circ (G \circ H) = (F \circ G) \circ H = X \circ H = H,$$
so $G \circ F = G \circ H = X$. If $G' \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}$ also satisfies $G' \circ F = X$, then $G' = G' \circ (F \circ G) = (G' \circ F) \circ G = X \circ G = G$.

(4) From $B = -A \circ (X + B)$ and Theorem 16.5, $\nu_t(B) = \nu_t(A) = \alpha$ and $[t^\alpha]B = -[t^\alpha]A$. □

Together with closure and associativity, which are (I3), part (3) of Theorem 16.11 supplies the missing clause of Theorem 14.7: $(\mathcal{G}_{\mathrm{poly},\mathbb{K}}, \circ)$ is a group with identity X, and $F \mapsto F^{\langle -1 \rangle_\circ}$ is its inversion map. This is the only place in the volume where the inverse is constructed.

Corollary 16.12 (First-order form of the inverse). *For $F = X + A \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}$ with $\alpha = \nu_t(A)$, $F^{\langle -1 \rangle_\circ} = X - A + \mathcal{O}_t^{\mathrm{form}}\big(t^{2\alpha+1}\big)$.*

Proof. Take $j = 1$ in (16.10): $B_1 = -A$ and $\nu_t(B - B_1) \geq 2\alpha + 1$. □

Remark 16.13 (A second proof that a one-sided inverse is two-sided). Report 1 argues differently, and the argument is worth keeping because it localizes the only arithmetic input. Let G be the right inverse and put $J = G \circ F = X + R \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}$. By (I3), $J \circ J = G \circ (F \circ G) \circ F = G \circ F = J$, so J is a composition idempotent. On the other hand, by (16.3),

$$J \circ J = (X + R) \circ (X + R) = X + R + R \circ (X + R) = X + 2R + \sum_{k \geq 1} \frac{R^k}{k!} D_X^k R,$$

and the last sum has $\nu_t \geq 2\nu_t(R) + 1$ by Theorem 16.4. If $R \neq 0$ with $\rho = \nu_t(R)$, comparing the t^ρ blocks in $J \circ J = J$ gives $2[t^\rho]R = [t^\rho]R$, hence $[t^\rho]R = 0$, a contradiction. So $R = 0$. This proof does not construct a right inverse for G, but it does compare two different integer multiples of one block; the monoid argument in Theorem 16.11(3) uses nothing beyond associativity.

Section summary

For $F = X + A \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}$ the equation $B = -A \circ (X + B)$ has exactly one solution, because $D_X A$ has strictly positive outer valuation: changing the guess in block r changes $-A \circ (X + \cdot)$ only from block $r + 1 + \nu_t(A)$ onward. Iteration from $B_0 = 0$ therefore freezes at least one further block per step, and t-adic completeness turns that gain into existence. Uniqueness, the group property of $\mathcal{G}_{\mathrm{poly},\mathbb{K}}$, and two-sidedness of the inverse all follow.

16.4 The reversal residual and the finite certificate

Definition 16.14 (Reversal residual)**.** For $F, H \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}$ define

$$\mathrm{Err}_{\mathrm{rev}}(F, H) := F \circ H - X \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}. \tag{16.11}$$

We call $\mathrm{Err}_{\mathrm{rev}}(F, H)$ the *right residual* of the candidate inverse H for the map F, and $H \circ F - X$ the *left residual*.

A common trap

The residual has *two* arguments and is not determined by the candidate H alone. Several source reports abbreviate it by a one-argument symbol in which the ambient map F is pushed into a subscript or into the surrounding prose. In a document that reverses more than one map — for instance $X + \log X$ and its perturbation $X + \log X + c/X$, both of which occur below — such a symbol is ambiguous, and the ambiguity is invisible in the typeset formula. Always print both arguments, as in $\mathrm{Err}_{\mathrm{rev}}(F, H)$. The same warning applies to the left residual, which is a different function of the same pair.

The next proposition is what makes residuals usable as certificates. It is stated in none of the six sources, although several of them silently pass from "the residual has order m" to "m blocks are correct".

Proposition 16.15 (The residual measures the error exactly, on both sides)**.** *Let $F \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}$, $G = F^{\langle -1 \rangle_\circ}$, $H \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}$, and $E = H - G \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$. Then*

$$\nu_t\big(\mathrm{Err}_{\mathrm{rev}}(F, H)\big) = \nu_t\big(H \circ F - X\big) = \nu_t(E), \tag{16.12}$$

and, if $E \neq 0$ with $e = \nu_t(E)$, all three series have the same *leading coefficient block:*

$$[t^e]\,\mathrm{Err}_{\mathrm{rev}}(F, H) = [t^e]\big(H \circ F - X\big) = [t^e]E.$$

Proof. Right residual. Write $F = X + A$, $G = X + B$, $H = X + B + E$, and apply Theorem 16.6 with $S = F$, $P = B$, $Q = E$:

$$F \circ H = F \circ G + \big((D_X F) \circ G\big)E + \sum_{j\geq 2} \frac{E^j}{j!}\big(D_X^j F\big) \circ G.$$

Here $F \circ G = X$. Since $D_X F = 1 + D_X A$ with $\nu_t(D_X A) \geq 1$, Theorem 16.5 gives, for $J := (D_X F) \circ G$, that $\nu_t(J) = 0$ and $[t^0]J = 1$. For $j \geq 2$, $D_X^j F = D_X^j A$ has $\nu_t \geq j$, so the j-th tail term has $\nu_t \geq je + j \geq 2e + 2$. Therefore

$$\mathrm{Err}_{\mathrm{rev}}(F, H) = J\,E + \mathcal{O}_t^{\mathrm{form}}\big(t^{2e+2}\big), \qquad \nu_t(J) = 0, \quad [t^0]J = 1. \tag{16.13}$$

Since $e \geq 0$ we have $2e + 2 > e$, while $\nu_t(JE) = e$ and $[t^e](JE) = [t^0]J \cdot [t^e]E = [t^e]E$.

Left residual. Right composition by F is substitution, hence a ring homomorphism, so, using $G \circ F = X$,

$$H \circ F - X = H \circ F - G \circ F = (H - G) \circ F = E \circ F,$$

and Theorem 16.5 gives $\nu_t(E \circ F) = e$ with $[t^e](E \circ F) = [t^e]E$. □

Corollary 16.16 (Finite certificate). *Let $F \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}$, $N \geq 0$, and $G_N = X + \sum_{j=0}^{N} b_j(L)t^j$ with $b_j \in \mathbb{K}[L]$. The following are equivalent.*

1. $G_N = \mathrm{Trunc}_{<N+1} F^{\langle -1\rangle_\circ}$*, that is, G_N is correct through outer block N.*
2. $\mathrm{Err}_{\mathrm{rev}}(F, G_N) = F \circ G_N - X = \mathcal{O}_t^{\mathrm{form}}\big(t^{N+1}\big)$ (right certificate).
3. $G_N \circ F - X = \mathcal{O}_t^{\mathrm{form}}\big(t^{N+1}\big)$ (left certificate).

Proof. By Theorem 16.15, each of (2) and (3) is equivalent to $\nu_t(G_N - F^{\langle -1\rangle_\circ}) \geq N + 1$. Since G_N has no block beyond t^N, that is exactly $\mathrm{Trunc}_{<N+1} G_N = \mathrm{Trunc}_{<N+1} F^{\langle -1\rangle_\circ}$, i.e. (1). □

Proposition 16.17 (The residual reads off the next inverse block). *Let $F \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}$, write $F^{\langle -1\rangle_\circ} = X + \sum_{n\geq 0} b_n(L)t^n$, and let $G_{N-1} = \mathrm{Trunc}_{<N} F^{\langle -1\rangle_\circ}$ for $N \geq 0$. Then*

$$\mathrm{Err}_{\mathrm{rev}}(F, G_{N-1}) = -b_N(L)\,t^N + \mathcal{O}_t^{\mathrm{form}}\big(t^{N+1}\big). \tag{16.14}$$

Proof. Here $E = G_{N-1} - F^{\langle -1\rangle_\circ} = -\sum_{n\geq N} b_n t^n$, so $\nu_t(E) \geq N$ and $[t^N]E = -b_N$. By (16.13) the residual has $\nu_t \geq N$ and $[t^N]\,\mathrm{Err}_{\mathrm{rev}}(F, G_{N-1}) = [t^N]E = -b_N$. □

Remark 16.18 (Why both certificates are computed, and what they really test). Theorem 16.15 gives more than the equivalence of the two certificates: for an exact $G_N \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}$ the right and left residuals have the same valuation *and* the same leading coefficient block. Report 1's claim that at finite order the two "need not have identical first omitted blocks" is therefore incorrect as stated. What is true, and is the genuine reason to compute both, is this.

- The two are computed along different code paths: $F \circ G_N$ substitutes the candidate into the map, $G_N \circ F$ substitutes the map into the candidate. A transposed argument order, a wrong update of $L \circ G = L + \log\big(1 + t(G - X)\big)$, or a routine that substitutes only the t-part of the inner series will usually break exactly one of them.
- As soon as the inputs are themselves truncated — F known only to some block, or G_N stored with different tails in the two calls — the two computed residuals are the residuals of *different* exact elements, and then their leading blocks may indeed differ. That is a statement about inconsistent precision management, not about the algebra of $\mathcal{G}_{\mathrm{poly},\mathbb{K}}$, and it is quantified by Theorem 17.14 and Theorem 17.15.

Proposition 16.19 (The derivative certificate is one level weaker)**.** *Let $F, H \in \mathcal{G}_{\text{poly},\mathbb{K}}$, $G = F^{\langle -1 \rangle \circ}$, $E = H - G$, $e = \nu_t(E)$. Then*

$$\nu_t\Big(D_X H - \big((D_X F) \circ H\big)^{-1}\Big) \ \geq \ e + 1, \tag{16.15}$$

with equality whenever $e \geq 1$. For $e = 0$ the bound need not be attained: if $E = c \in \mathbb{K}^\times$ then $D_X E = 0$ and the left-hand side has valuation at least 2.

Proof. Differentiating $F \circ G = X$ by the chain rule (16.5) gives $\big((D_X F) \circ G\big) D_X G = 1$, so $D_X G = \big((D_X F) \circ G\big)^{-1}$. Hence

$$D_X H - \big((D_X F) \circ H\big)^{-1} = D_X E + \Big(\big((D_X F) \circ G\big)^{-1} - \big((D_X F) \circ H\big)^{-1}\Big).$$

By Theorem 16.6 applied to $S = D_X F$, $(D_X F) \circ H - (D_X F) \circ G = \sum_{j \geq 1} \frac{E^j}{j!} (D_X^{j+1} A) \circ G$, of valuation $\geq e + 2$ because $\nu_t(D_X^{j+1} A) \geq j + 1 \geq 2$. Both reciprocals have valuation 0, so the bracket has valuation $\geq e + 2$. Finally $\nu_t(D_X E) \geq e + 1$, with equality when $e \geq 1$: if $[t^e]E = c \neq 0$ then $[t^{e+1}] D_X E = (\partial_L - e)c$, and $(\partial_L - e)c = 0$ forces $c' = ec$, impossible for a nonzero polynomial when $e \neq 0$. For $e = 0$ the operator ∂_L annihilates constants. □

Remark 16.20. Report 1 recommends the derivative identity $D_X G = \big((D_X F) \circ G\big)^{-1}$ as a check that "often detects a wrong block one order earlier". Theorem 16.19 shows the opposite: the derivative residual first becomes nonzero at least one level *later* than $\mathrm{Err}_{\text{rev}}(F, H)$, and it under-reports by two levels an error whose leading block is a nonzero constant at level 0. It remains a useful *independent* check — it exercises D_X and the reciprocal rather than the composition engine — but as a detector of the first wrong block it is strictly weaker.

16.5 What the theorem does not say

A common trap

Theorem 16.11 is a theorem about formal series. It asserts nothing about the existence of an analytic inverse branch, about convergence of $\sum_n b_n(L) t^n$ for any value of X, or about any constant uniform in L. A formal certificate $\mathrm{Err}_{\text{rev}}(F, G_N) = \mathcal{O}_t^{\text{form}}(t^{N+1})$ means coefficient agreement below outer order $N + 1$ and nothing more; in particular it does not imply $f(g_N(X)) - X = \mathcal{O}_{X \to +\infty}(X^{-N-1})$ for any realization f, g_N, since the discarded blocks carry unbounded powers of $\log X$.

Proposition 16.21 (Analytic residual transfer)**.** *Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$ be continuously differentiable and injective, let $y \in f(I)$, and let $x_* = f^{\langle -1 \rangle \circ}(y)$. Let $x_N \in I$ and suppose $|f'(x)| \geq m > 0$ for all x between x_N and x_*. Then*

$$|x_N - x_*| \leq \frac{|f(x_N) - y|}{m}.$$

Proof. By the mean value theorem there is ξ between x_N and x_* with $f(x_N) - y = f(x_N) - f(x_*) = f'(\xi)(x_N - x_*)$, whence $|f(x_N) - y| \geq m\,|x_N - x_*|$. □

Remark 16.22. Both hypotheses are essential and neither is supplied by the formal theory. For $f(x) = x + \log x$ on $(1, \infty)$ one may take $m = 1$, because $f'(x) = 1 + 1/x > 1$, so there a numerical residual does control the inverse error — but that is a statement about one realization on one ray, proved analytically, and it is not a corollary of Theorem 16.16. Transferring a *formal* residual to an asymptotic estimate additionally requires a realization theorem for the class, treated separately in this volume.

Chapter 17

Triangular reversal, Newton iteration, and precision doubling

17.1 The triangular correction

Lemma 17.1 (Block perturbation). *Let $F \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}$, $G \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}$, $N \in \mathbb{N}_0$, $c \in \mathbb{K}[L]$. Then*

$$F \circ \left(G + c(L)t^N\right) = F \circ G + c(L)t^N + \mathcal{O}_t^{\mathrm{form}}\left(t^{N+1}\right). \tag{17.1}$$

Proof. Apply Theorem 16.6 with $S = F$, $P = G - X$, $Q = \delta := ct^N$:

$$F \circ (G + \delta) - F \circ G = \big((D_X F) \circ G\big)\delta + \sum_{j \geq 2} \frac{\delta^j}{j!}\left(D_X^j F\right) \circ G.$$

By Theorem 16.5, $(D_X F) \circ G = 1 + \mathcal{O}_t^{\mathrm{form}}(t)$, so the linear term is $ct^N + \mathcal{O}_t^{\mathrm{form}}\left(t^{N+1}\right)$; and for $j \geq 2$ the j-th term has $\nu_t \geq jN + j \geq 2N + 2 > N$ since $N \geq 0$. □

Theorem 17.2 (Triangular reversal). *Let $F \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}$. Put $G_{-1} = X$ and, for $n = 0, 1, 2, \ldots$,*

$$r_n(L) := [t^n]\,\mathrm{Err}_{\mathrm{rev}}(F, G_{n-1}), \qquad G_n := G_{n-1} - r_n(L)t^n. \tag{17.2}$$

Then for every $n \geq -1$,

$$\mathrm{Err}_{\mathrm{rev}}(F, G_n) = \mathcal{O}_t^{\mathrm{form}}\left(t^{n+1}\right), \qquad G_n = \mathrm{Trunc}_{<n+1} F^{\langle -1 \rangle \circ},$$

and $r_n = -b_n$, where $F^{\langle -1 \rangle \circ} = X + \sum_{j \geq 0} b_j(L)t^j$.

Proof. Induction on n. For $n = -1$, $\mathrm{Err}_{\mathrm{rev}}(F, X) = F - X = A$ has $\nu_t \geq 0$, which is the assertion $\mathcal{O}_t^{\mathrm{form}}\left(t^0\right)$, and $\mathrm{Trunc}_{<0} F^{\langle -1 \rangle \circ} = X$. Assume $\mathrm{Err}_{\mathrm{rev}}(F, G_{n-1}) = \mathcal{O}_t^{\mathrm{form}}(t^n)$. Then r_n is its coefficient at level n, and by Theorem 17.1 with $c = -r_n$,

$$\mathrm{Err}_{\mathrm{rev}}(F, G_n) = \mathrm{Err}_{\mathrm{rev}}(F, G_{n-1}) - r_n t^n + \mathcal{O}_t^{\mathrm{form}}\left(t^{n+1}\right) = \mathcal{O}_t^{\mathrm{form}}\left(t^{n+1}\right),$$

since the inherited residual and the correction contribute $r_n t^n$ and $-r_n t^n$ at level n and nothing below. As G_n has no block beyond t^n, Theorem 16.16 identifies $G_n = \mathrm{Trunc}_{<n+1} F^{\langle -1 \rangle \circ}$. Finally Theorem 16.17, applied to $G_{n-1} = \mathrm{Trunc}_{<n} F^{\langle -1 \rangle \circ}$, gives $r_n = -b_n$. □

Algorithm 17.3. Residual-cancellation reversal. *Input:* $F = X + A \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}$ with A known below t^{N+1}; a target block $N \geq 0$. *Output:* $\mathrm{Trunc}_{<N+1} F^{\langle -1 \rangle \circ}$.

1. $G \leftarrow X$.

2. For $n = 0, 1, \ldots, N$: compute $R \leftarrow \operatorname{Trunc}_{<N+1}\big(F \circ G - X\big)$; set $r_n \leftarrow [t^n]R$; set $G \leftarrow G - r_n(L)t^n$.
3. Return G together with the certificate $\operatorname{Trunc}_{<N+1}\big(F \circ G - X\big) = 0$.

Every intermediate composition may be truncated below t^{N+1}: by Theorem 16.9 a change of G at order $\geq N+1$ changes $F \circ G$ only at order $\geq N+1$, so a discarded block never returns.

Step (17.2) divides the residual block by the leading block of $D_X F$, which is 1 for $F \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}$. Outside $\mathcal{G}_{\mathrm{poly},\mathbb{K}}$ this is exactly where the unit criterion of the coefficient ring enters, and the sources treat it too briefly.

Proposition 17.4 (Reversal and triangular correction at a general leading slope)**.** *Let $a \in \mathbb{K}^\times$ and fix a value* $\log a \in \mathbb{K}$. *Let θ be the $\mathbb{K}$-algebra automorphism of $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ induced by the substitution $X \mapsto a^{-1}X$, that is $\theta(t) = at$ and $\theta(L) = L - \log a$; equivalently $\theta(S) = S \circ (a^{-1}X)$. Let $A \in \mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$, $F = aX + A$, $G_0 \coloneqq X + a^{-1}A \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}$, and $\mathcal{G}_a \coloneqq a^{-1}X + \mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}} = \theta(\mathcal{G}_{\mathrm{poly},\mathbb{K}})$. Then:*

1. *θ preserves ν_t, maps $\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ onto $\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$, and satisfies $\theta(S \circ H) = S \circ \theta(H)$ whenever the composites exist;*
2. *$F = a\,G_0$ (a product, not a composite), and F has exactly one compositional inverse in $\mathcal{G}_a$, namely $F^{\langle -1\rangle\circ} = \theta\big(G_0^{\langle -1\rangle\circ}\big)$, which is two-sided;*
3. *for every $G \in \mathcal{G}_a$, $c \in \mathbb{K}[L]$ and $N \geq 0$,*

$$F \circ \big(G + c(L)t^N\big) = F \circ G + a\,c(L)\,t^N + \mathcal{O}_t^{\mathrm{form}}\big(t^{N+1}\big)\,, \tag{17.3}$$

so a residual block r_N is cancelled by the correction $-r_N/a$, which stays in $\mathbb{K}[L]$.

Proof. (1) $\theta(t^n p(L)) = a^n t^n p(L - \log a)$, so θ is a bijective ring homomorphism preserving ν_t and $\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$; it is well defined because $\log a \in \mathbb{K}$. The intertwining relation is (I3): $\theta(S \circ H) = (S \circ H) \circ (a^{-1}X) = S \circ \big(H \circ (a^{-1}X)\big) = S \circ \theta(H)$.

(2) $F = aX + A = a(X + a^{-1}A) = a\,G_0$. Put $H = G_0^{\langle -1\rangle\circ}$. Then, by (1), $F \circ \theta(H) = \theta(F \circ H) = \theta\big(a(G_0 \circ H)\big) = \theta(aX) = a \cdot a^{-1}X = X$. On the other side, $\theta(H) \circ F = H \circ (a^{-1}X) \circ F = H \circ (a^{-1}F) = H \circ G_0 = X$. For uniqueness, let $G' \in \mathcal{G}_a$ with $F \circ G' = X$ and write $G' = \theta(\widetilde{G})$ with $\widetilde{G} \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}$. Then $\theta(F \circ \widetilde{G}) = X$, so $F \circ \widetilde{G} = \theta^{-1}(X) = aX$, i.e. $G_0 \circ \widetilde{G} = X$, so $\widetilde{G} = G_0^{\langle -1\rangle\circ}$ by Theorem 16.11(2).

(3) Write $G = \theta(\widetilde{G})$ with $\widetilde{G} \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}$ and $\delta = ct^N$, $\eta = \theta^{-1}(\delta) = a^{-N}c(L + \log a)t^N \in \mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$. Since θ is additive, $G + \delta = \theta(\widetilde{G} + \eta)$, so by (1) and $F = aG_0$,

$$F \circ (G + \delta) = \theta\big(F \circ (\widetilde{G} + \eta)\big) = \theta\big(a\,G_0 \circ (\widetilde{G} + \eta)\big).$$

By Theorem 17.1, $G_0 \circ (\widetilde{G} + \eta) = G_0 \circ \widetilde{G} + \eta + \mathcal{O}_t^{\mathrm{form}}\big(t^{N+1}\big)$. Applying $a\theta(\cdot)$ and using $\theta(\eta) = \delta$ together with the fact that θ preserves ν_t gives (17.3). ☐

Remark 17.5 (Two repairs to the general-slope rule). *(i) The scaling reduction needs* $\log a \in \mathbb{K}$. Report 3 performs the reduction $F = a\,G_0$, $F^{\langle -1\rangle\circ}(X) = G_0^{\langle -1\rangle\circ}(X/a)$ and notes in passing that the substitution “shifts the logarithm by $-\log c$”; report 4 states the division rule with no hypothesis at all. As Theorem 17.4(1) makes explicit, the substitution $X \mapsto a^{-1}X$ is an automorphism of $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ only if a value of $\log a$ lies in $\mathbb{K}$; otherwise $\mathbb{K}$ must first be extended, and the inverse of a map with coefficients in $\mathbb{K}$ acquires coefficients in $\mathbb{K}(\log a)$. Composition with F itself is likewise undefined without that value.

(ii) The divisor must be a unit of the coefficient ring, not merely nonzero. What is really being inverted in (17.3) is $\lambda \coloneqq [t^0](D_X F)$, the leading block of the derivative. For $F = aX + A$ it equals $a \in \mathbb{K}^\times$ and all is well. But λ can be a nonconstant polynomial: for $F = X \log X = t^{-1}L$ one

has $D_X F = (\partial_L + 1)L = L + 1$, so $\lambda = L + 1$, which is not a unit of $\mathbb{K}[L]$. Division by λ then leaves $\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ for $\mathscr{PL}_{\mathrm{rat},\mathbb{K}}$. This is not a removable inconvenience: $X \log X$ has no compositional inverse in $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ at all. Indeed, for any $G \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}$ the composite $(X + B)\log(X + B)$ has $\nu_t = -1$ with t^{-1}-block $L \neq 1$, so it cannot equal X; and $\log(X \log X) = L + \log L$ forces the second logarithm into the scale. A leading slope that is not a unit is a reliable symptom that the inverse leaves the one-logarithm class.

17.2 Coefficient stabilization by functional iteration

Algorithm 17.6. Coefficient-stabilization reversal. *Input:* $F = X + A \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}$, target block $N \geq 0$. *Output:* $\operatorname{Trunc}_{<N+1} F^{\langle -1 \rangle \circ}$.

1. $B \leftarrow 0$.
2. Repeat $N + 1$ times: $B \leftarrow \operatorname{Trunc}_{<N+1}\big(-A \circ (X + B)\big)$.
3. Return $X + B$ with the certificate $\operatorname{Trunc}_{<N+1}\big(F \circ (X + B) - X\big) = 0$.

One may stop as soon as two consecutive iterates agree below t^{N+1}. By Theorem 16.11(1) that happens after at most $N + 1$ steps, and after at most $\lceil (N + 1 - \alpha)/(\alpha + 1) \rceil$ steps when $\alpha = \nu_t(A)$.

Proposition 17.7 (Truncation is safe)**.** *Let $A \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$, $M \geq 0$, and $B, \widetilde{B} \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ with $\nu_t(B - \widetilde{B}) \geq M$. Then $\operatorname{Trunc}_{<M} \Phi(B) = \operatorname{Trunc}_{<M} \Phi(\widetilde{B})$. Consequently, truncating the iterate below t^M after every step of Theorem 17.6 alters no block below t^M of any later iterate, and the algorithm returns the same answer as its untruncated form.*

Proof. Theorem 16.9 gives $\nu_t\big(\Phi(B) - \Phi(\widetilde{B})\big) \geq \nu_t(A) + M + 1 > M$, which is the first claim. For the second, let B_j be the exact iterates and $\widehat{B}_j$ the truncated ones, and suppose $\nu_t(B_j - \widehat{B}_j) \geq M$. Then $\nu_t\big(\Phi(B_j) - \Phi(\widehat{B}_j)\big) \geq M + 1$, and truncating $\Phi(\widehat{B}_j)$ below t^M introduces an error of order $\geq M$; hence $\nu_t(B_{j+1} - \widehat{B}_{j+1}) \geq M$, and the invariant persists. ☐

Remark 17.8 (An algebraically equivalent iteration need not be order-improving). An existence proof by iteration depends on the chosen normal form, not only on the equation. For $F = X + \log X$ the inverse $\omega = F^{\langle -1 \rangle \circ}$ satisfies $\omega + \log \omega = X$. The rearrangement

$$\omega = X - \log \omega \tag{17.4}$$

is order-improving: if $\omega_1, \omega_2 \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}$ and $\omega_1 - \omega_2 = \mathcal{O}_t^{\mathrm{form}}(t^r)$, then $\omega_1 \omega_2^{-1} = 1 + \mathcal{O}_t^{\mathrm{form}}\big(t^{r+1}\big)$, hence $\log \omega_1 - \log \omega_2 = \mathcal{O}_t^{\mathrm{form}}\big(t^{r+1}\big)$; this is Theorem 16.9 again. The algebraically equivalent rearrangement

$$\omega = \mathrm{e}^{X - \omega} \tag{17.5}$$

fails for a sharper reason than slow convergence: e^X is not an element of $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$, so the right-hand side of (17.5) does not define a map of the class and no truncation of it is meaningful. Report 6 makes the same point with an exponential rearrangement of the Lambert equation, attributing the observation to Edgar [13]: a fixed-point scheme must increase asymptotic precision, and algebraic equivalence is no evidence that it does. The test is supplied by Theorem 16.9: the iteration map must send a difference of valuation r to one of valuation $> r$, uniformly in r.

17.3 Newton iteration

Lemma 17.9 (The Newton denominator is a unit)**.** *Let $F = X + A \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}$ and $G \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}$. Then $J := (D_X F) \circ G$ satisfies $J = 1 + \mathcal{O}_t^{\mathrm{form}}(t)$ and is a unit of $\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$, with $J^{-1} = \sum_{k \geq 0} (1 - J)^k$.*

Proof. $D_X F = 1 + D_X A$ and $\nu_t(D_X A) \geq \nu_t(A) + 1 \geq 1$ by Theorem 16.4. By Theorem 16.5, $\nu_t(J) = 0$ and $[t^0]J = 1$, so $W := 1 - J$ has $\nu_t(W) \geq 1$; the family $(W^k)_{k\geq 0}$ is summable and $J \sum_{k\geq 0} W^k = (1 - W) \sum_{k\geq 0} W^k = 1$. (This is the unit criterion in $\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$: U is a unit exactly when $[t^0]U \in \mathbb{K}^\times$.) $\square$

Definition 17.10 (Newton reversion map). For $F = X + A \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}$ and $G \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}$ put

$$\mathcal{N}_F(G) := G - \frac{\mathrm{Err}_{\mathrm{rev}}(F, G)}{(D_X F) \circ G} \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}. \tag{17.6}$$

Theorem 17.11 (Precision doubling, exact arithmetic). *Let $F = X + A \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}$, $\alpha = \nu_t(A)$, $G \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}$, and $m = \nu_t\big(\mathrm{Err}_{\mathrm{rev}}(F, G)\big)$. Then*

$$\nu_t\Big(\mathrm{Err}_{\mathrm{rev}}(F, \mathcal{N}_F(G))\Big) \geq 2m + 2 + \alpha. \tag{17.7}$$

Equivalently, by Theorem 16.15: if G is correct through outer block $m - 1$, then $\mathcal{N}_F(G)$ is correct through outer block $2m + 1 + \alpha$; the number of correct blocks passes from m to $2m + 2 + \alpha$.

Proof. Put $R = \mathrm{Err}_{\mathrm{rev}}(F, G)$, $J = (D_X F) \circ G$, and $C = -RJ^{-1}$, so $\mathcal{N}_F(G) = G + C$. By Theorem 17.9, $\nu_t(C) = \nu_t(R) = m$. By Theorem 16.6 with $S = F$, $P = G - X$, $Q = C$,

$$\mathrm{Err}_{\mathrm{rev}}(F, G + C) = \underbrace{R + JC}_{=0} + \sum_{j\geq 2} \frac{C^j}{j!} \big(D_X^j F\big) \circ G.$$

For $j \geq 2$, $D_X^j F = D_X^j A$, and Theorems 16.4 and 16.5 give $\nu_t\big((D_X^j A) \circ G\big) \geq \alpha + j$. Hence the j-th term has $\nu_t \geq jm + \alpha + j = \alpha + j(m + 1)$, minimal at $j = 2$ with value $2m + 2 + \alpha$. $\square$

Remark 17.12 (Diagnosis of the six sources). The reports state the Newton estimate in five different ways.

- Reports 1, 5 and 6 assert the exponent $2m + 2$, which is (17.7) with the term α omitted; their proofs are correct as far as they go.
- Report 2 states $2N$ under the extra hypothesis $N \geq 1$. The hypothesis is never used and is not needed; the case $N = 0$ is legitimate and is precisely the situation at the first Newton step from $G = X$, where the residual is A itself. The argument given there in fact yields $2N + 2$.
- Report 3 states $2m$ and adds only that "in this particular scale the order is often even higher". The improvement is not occasional but automatic and equals $2m + 2 + \nu_t(A)$.
- Report 4 is internally inconsistent: it announces t^{2N+1} in the text and proves t^{2N+2} two lines later, bridging the gap with the phrase "allowing coarse bounds and Laurent shifts". There are no Laurent shifts here: every object in (17.6) lies in $\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$, and the estimate is exact.

A second and subtler slide occurs in reports 1 and 5, which pass from "the residual has order $2m+2$" to "the number of correct blocks doubles" without argument. That step is legitimate, but only because of Theorem 16.15, which identifies the residual valuation with $\nu_t(G - F^{\langle -1\rangle\circ})$; no source proves or even states that identification. Without it, a small residual is a priori compatible with a large coefficient error — exactly as it would be for a map whose derivative were not a unit.

Remark 17.13 (Sharpness). The exponent in (17.7) cannot be improved in general. For $F = X + \log X$ (so $\alpha = 0$) and $G = X - L(1 + t)^{-1}$ one has $m = 2$, and after one exact Newton step the residual is $-\frac{1}{8}L^4 t^6 + \mathcal{O}_t^{\mathrm{form}}(t^7)$, of order precisely $2m + 2 = 6$. See Theorem 17.17.

17.4 Precision doubling with truncated intermediates

In an implementation none of R, J or the quotient is known exactly, and the exponent $2m+2+\alpha$ is attained only if the truncations are deep enough. Reports 1, 2, 4, 5 and 6 omit the hypothesis; report 3 gestures at it ("computed consistently to sufficient precision"). The following two results make it exact.

Theorem 17.14 (Newton step with an inexact correction). *Let $F = X + A \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}$, $\alpha = \nu_t(A)$, $G \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}$, $R = \mathrm{Err}_{\mathrm{rev}}(F,G)$, $m = \nu_t(R)$, $J = (D_X F) \circ G$, and let $C = -RJ^{-1}$ be the exact Newton correction. Let $\widehat{C} \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ satisfy $\nu_t(\widehat{C} - C) \geq \sigma$ with $\sigma \geq m$. Then*

$$\nu_t\Big(\mathrm{Err}_{\mathrm{rev}}\Big(F, G + \widehat{C}\Big)\Big) \;\geq\; \min\{\, 2m + 2 + \alpha,\ \sigma \,\}. \tag{17.8}$$

Proof. Since $\nu_t(C) = m$ and $\nu_t(\widehat{C} - C) \geq \sigma \geq m$, we get $\nu_t(\widehat{C}) \geq m$. By Theorem 16.6 with $S = F$, $P = G - X$, $Q = \widehat{C}$,

$$\mathrm{Err}_{\mathrm{rev}}\Big(F, G + \widehat{C}\Big) = R + J\widehat{C} + \sum_{j \geq 2} \frac{\widehat{C}^{\,j}}{j!}\big(D_X^j A\big) \circ G.$$

The tail has $\nu_t \geq 2m + 2 + \alpha$ exactly as in Theorem 17.11. For the head, $R + JC = 0$ gives $R + J\widehat{C} = J(\widehat{C} - C)$, and $\nu_t(J) = 0$, so $\nu_t\big(R + J\widehat{C}\big) = \nu_t(\widehat{C} - C) \geq \sigma$. ☐

Lemma 17.15 (Precision propagation through the Newton step). *Keep the notation of Theorem 17.14 and let $\widehat{R}, \widehat{K} \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ and $q \in \mathbb{N}_0$ satisfy*

$$\nu_t\big(\widehat{R} - R\big) \geq \sigma_R \geq m, \qquad \nu_t\big(\widehat{K} - J^{-1}\big) \geq \sigma_K \geq 0.$$

Put $\widehat{C} = -\operatorname{Trunc}_{<q}\big(\widehat{R}\,\widehat{K}\big)$. Then $\nu_t(\widehat{C} - C) \geq \min\{q,\ \sigma_R,\ m + \sigma_K\}$. Consequently, to certify $\nu_t\big(\mathrm{Err}_{\mathrm{rev}}\big(F, G + \widehat{C}\big)\big) \geq q$ for a target $q \leq 2m + 2 + \alpha$, it suffices to take $\sigma_R \geq q$ and $\sigma_K \geq q - m$. Moreover:

1. *$\widehat{K}$ with $\sigma_K = s$ is obtained from $\operatorname{Trunc}_{<s} J$, since for units $J, \widehat{J}$ with $[t^0]J = [t^0]\widehat{J} = 1$ one has $\widehat{J}^{-1} - J^{-1} = (J - \widehat{J})\widehat{J}^{-1}J^{-1}$;*
2. *$\operatorname{Trunc}_{<s} J$ is determined by $\operatorname{Trunc}_{<s-1} A$ and $\operatorname{Trunc}_{<s-2} G$, because changing G by $\mathcal{O}_t^{\mathrm{form}}(t^u)$ changes J only by $\mathcal{O}_t^{\mathrm{form}}\big(t^{u+2}\big)$;*
3. *$\operatorname{Trunc}_{<q} R$ is determined by $\operatorname{Trunc}_{<q} A$ and $\operatorname{Trunc}_{<q} G$, and no coarser knowledge of G suffices: for $\epsilon \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$,*

   $$\mathrm{Err}_{\mathrm{rev}}(F, G + \epsilon) - \mathrm{Err}_{\mathrm{rev}}(F, G) = J\epsilon + \mathcal{O}_t^{\mathrm{form}}\Big(t^{2\nu_t(\epsilon)+2}\Big),$$

   whose valuation is exactly $\nu_t(\epsilon)$. Thus a block of G at level $u < q$ moves R at level u, and if G is stored with fewer than q blocks its zero padding is part of the definition of the element being corrected and must be used consistently throughout the step.

Proof. Write $\widehat{C} - C = -\big(\operatorname{Trunc}_{<q}(\widehat{R}\widehat{K}) - \widehat{R}\widehat{K}\big) - \big(\widehat{R}\widehat{K} - RJ^{-1}\big)$. The first bracket has $\nu_t \geq q$. For the second,

$$\widehat{R}\widehat{K} - RJ^{-1} = \big(\widehat{R} - R\big)\widehat{K} + R\big(\widehat{K} - J^{-1}\big),$$

and $\nu_t(\widehat{K}) \geq \min\{0, \sigma_K\} = 0$ because $\nu_t(J^{-1}) = 0$; so its valuation is at least $\min\{\sigma_R,\ m + \sigma_K\}$. With $\sigma_R \geq q$, $\sigma_K \geq q - m$ the bound becomes q, and Theorem 17.14 applies since $q \geq m$.

For (1), the displayed identity gives $\nu_t\big(\widehat{J}^{-1} - J^{-1}\big) = \nu_t(J - \widehat{J}) \geq s$. For (2), Theorem 16.6 applied to $S = D_X F$ gives $(D_X F) \circ (G + \epsilon) - (D_X F) \circ G = \sum_{j \geq 1} \frac{\epsilon^j}{j!}(D_X^{j+1} A) \circ G$, whose valuation is at least $\nu_t(\epsilon) + 2$ because $\nu_t(D_X^{j+1} A) \geq j + 1 \geq 2$. For (3), Theorem 16.6 with $S = F$, $P = G - X$, $Q = \epsilon$ gives $\mathrm{Err}_{\mathrm{rev}}(F, G + \epsilon) - \mathrm{Err}_{\mathrm{rev}}(F, G) = J\epsilon + \sum_{j \geq 2} \frac{\epsilon^j}{j!}(D_X^j A) \circ G$, and the tail has valuation at least $2\nu_t(\epsilon) + 2 > \nu_t(\epsilon)$ while $\nu_t(J\epsilon) = \nu_t(\epsilon)$. ☐

Corollary 17.16 (Doubling schedule). *Let $F = X + A \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}$ and let $N \geq 0$ be a target block. Put $H_0 = X$, $q_0 = \nu_t(A)$, and for $i \geq 0$*

$$q_{i+1} = \min\{N+1,\ 2q_i + 2\}, \qquad H_{i+1} = \mathrm{Trunc}_{<q_{i+1}}\Big(H_i - \mathrm{Trunc}_{<q_{i+1}} R_i \cdot \mathrm{Trunc}_{<\,q_{i+1}-q_i} J_i^{-1}\Big),$$

where $R_i = \mathrm{Err}_{\mathrm{rev}}(F, H_i)$ and $J_i = (D_X F) \circ H_i$. Then $\nu_t\big(\mathrm{Err}_{\mathrm{rev}}(F, H_i)\big) \geq q_i$ for every i, and $H_i = \mathrm{Trunc}_{<N+1} F^{\langle -1 \rangle \circ}$ as soon as $q_i = N+1$. The number of steps is $\mathcal{O}(\log N)$.

Proof. Induction on i. The case $i = 0$ is $\mathrm{Err}_{\mathrm{rev}}(F, X) = A$. Assume $\nu_t(R_i) \geq q_i$. Apply Theorem 17.15 with $m = q_i$, $q = q_{i+1}$, $\sigma_R = q_{i+1}$ and $\sigma_K = q_{i+1} - q_i$, which is ≥ 1 because $q_{i+1} \geq \min\{N+1, 2q_i+2\} > q_i$ while $q_i < N+1$; the final outer truncation $\mathrm{Trunc}_{<q_{i+1}} \cdot$ contributes an error of valuation $\geq q_{i+1}$ and is absorbed in the same bound. Then Theorem 17.14 gives $\nu_t(R_{i+1}) \geq \min\{2q_i + 2 + \nu_t(A),\ q_{i+1}\} = q_{i+1}$. The identification with $\mathrm{Trunc}_{<N+1} F^{\langle -1 \rangle \circ}$ is Theorem 16.16. Finally $q_{i+1} + 1 \geq 2(q_i + 1)$ until the cap is reached, so $q_i + 1 \geq 2^i$. □

A common trap

A Newton step at target q treats the stored G as an exact element of $\mathcal{G}_{\mathrm{poly},\mathbb{K}}$ — zero-padded beyond its last stored block — and must form $R = \mathrm{Err}_{\mathrm{rev}}(F, G)$ below t^q from *that same* element, even though only $m < q$ of its blocks are known to be correct. The blocks between levels m and $q-1$ are the guard blocks: they are not required to be correct, but they are required to be fixed and used identically in the residual, in the Jacobian factor and in the update. Computing J to the final cutoff while computing R only to the current accuracy m, or recomputing R from a differently truncated copy of G, destroys the doubling: by (17.8) the guaranteed order collapses to σ, and the collapse is invisible in the printed series.

Example 17.17 (Doubling, and its failure, for $F = X + \log X$). Let $F = X + \log X = X + L$, so $A = L$, $\alpha = 0$, and write $\omega = F^{\langle -1 \rangle \circ}$. Here $D_X F = 1 + t$, and for $G = X + B$

$$\mathrm{Err}_{\mathrm{rev}}(F, G) = B + L + \log(1 + Bt), \qquad J = (D_X F) \circ G = 1 + G^{-1} = 1 + \frac{t}{1 + Bt}.$$

First step. From $H_0 = X$: $R_0 = L$, $m = 0$, $J_0 = 1 + t$, so

$$H_1 = X - \frac{L}{1+t} = X - L + Lt - Lt^2 + Lt^3 - \cdots.$$

Theorem 17.11 predicts $\nu_t(R_1) \geq 2$, and indeed

$$R_1 = -\tfrac{1}{2}L^2 t^2 + \big(L^2 - \tfrac{1}{3}L^3\big)t^3 + \big(-\tfrac{3}{2}L^2 + L^3 - \tfrac{1}{4}L^4\big)t^4 + \mathcal{O}_t^{\mathrm{form}}(t^5),$$

so $m = 2$ exactly. By Theorem 16.15 the error must have the same leading block, and $H_1 - \omega = -\frac{1}{2}L^2 t^2 + \mathcal{O}_t^{\mathrm{form}}(t^3)$.

Exact second step. With $q = 2m + 2 = 6$ and all data carried below t^6,

$$\mathrm{Err}_{\mathrm{rev}}(F, \mathcal{N}_F(H_1)) = -\tfrac{1}{8}L^4 t^6 + \mathcal{O}_t^{\mathrm{form}}(t^7),$$

of order exactly 6: blocks $0, \ldots, 5$ of ω are correct, six blocks in place of two.

Guard-starved second step. Suppose the residual is truncated one doubling too early, $\widehat{R} = \mathrm{Trunc}_{<4} R_1$, while J and the division are still carried below t^6. Then $\sigma_R = 4$ and Theorem 17.15 guarantees only order 4 — and the bound is attained. With $\rho_4 := [t^4]R_1 = -\frac{1}{4}L^4 + L^3 - \frac{3}{2}L^2$,

$$\widehat{C} - C = \big(R_1 - \widehat{R}\big)J^{-1} = \rho_4 t^4 + \mathcal{O}_t^{\mathrm{form}}(t^5),$$

hence $\mathrm{Err}_{\mathrm{rev}}\Big(F, H_1 + \widehat{C}\Big) = \rho_4 t^4 + \mathcal{O}_t^{\mathrm{form}}(t^5)$ and, by Theorem 16.15, the computed inverse first deviates from ω in block 4, by exactly ρ_4. Four correct blocks in place of six: the loss equals the two blocks of residual that were discarded, and no amount of extra precision in J or in the division repairs it.

17.5 Four reversal strategies compared

Write $C(n)$ for the cost of one composition $F \circ G$ truncated below t^n, $M(n)$ for a truncated multiplication, and $I(n)$ for the reciprocal of a unit of $\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$. The counts below are counts of these operations, not bit complexities: each block is a polynomial in L whose degree grows with the block index — linearly when $\deg_L[t^0]A = 1$, and like dn when $\deg_L[t^0]A = d$ — so a further degree-dependent factor is hidden inside C, M and I.

Strategy	Operation count to block N	Strength	Failure mode
Triangular, Theorem 17.2	$N+1$ residual evaluations; naively $\sum_{n\le N} C(n{+}1)$, less with incremental residual updating	one exact new block per step, each of them printable; no division beyond the leading slope; the residual is itself the certificate	fails when the leading block of $D_X F$ is not a unit (Theorem 17.5); strictly sequential, so it cannot exploit fast arithmetic
Functional iteration, Theorem 17.6	$N+1$ compositions below t^{N+1}: $(N{+}1)\,C(N{+}1)$	shortest existence proof; needs composition only; insensitive to the initial guess	gains only $\nu_t(A)+1$ blocks per step and recomputes low blocks each time; an algebraically equivalent rearrangement may gain nothing (Theorem 17.8)
Newton, Theorem 17.16	$\mathcal{O}(\log N)$ steps at precisions q_i with $q_i{+}1 \ge 2^i$; total $\sum_i \big(2C(q_i) + M(q_i) + I(q_i)\big)$	residual valuation more than doubles each step; fastest at high order	needs $D_X F$, a reciprocal and a composition per step; silently loses blocks when the guard precisions of Theorem 17.15 are not respected
Lagrange–Bürmann, Part 18	powers A^r, $r \le N+1$: N multiplications, plus $\mathcal{O}(N^2)$ applications of $\Delta_{n,k}$	closed universal formula; no recursion; the natural route to coefficient identities and degree laws	expression swell in A^r and in the repeated derivatives; logarithmic degrees reach $\deg_L \approx d(N{+}1)$ although only N blocks are wanted

Remark 17.18 (On the cost claims). The assertion that Newton is asymptotically superior, made in reports 1, 3, 4 and 6, is a statement about the cost model: it holds when C, M and I are superadditive, so that the geometric schedule $q_0 < q_1 < \cdots < N+1$ sums to a constant multiple of the cost at the final precision. It is not a theorem about the polynomial–logarithmic class, and it can fail if a composition routine carries a large fixed overhead, or if the coefficient degrees rather than the block count dominate the work. We assert only what is proved here: the number of Newton steps is $\mathcal{O}(\log N)$, and every step is subject to Theorem 17.15. Practical schemes seed with two or three triangular blocks, switch to Newton, and certify with an independent Lagrange–Bürmann coefficient; algorithms for the wider exp–log setting are in [29, 30, 33].

17.6 Two worked reversions with certificates

Example 17.19 (The inverse of $X+\log X$, through block 4)**.** Take $F = X+\log X = X+L \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}$, so $A = L$ and $\alpha = 0$. For $G = X + B$,

$$\mathrm{Err}_{\mathrm{rev}}(F, G) = B + L + \log\bigl(1 + Bt\bigr), \tag{17.9}$$

since $F \circ G = G + \log G = X + B + L + \log(1 + Bt)$. Run Theorem 17.2.

Stage -1. $G_{-1} = X$, $B = 0$, so $\mathrm{Err}_{\mathrm{rev}}(F, G_{-1}) = L$; $r_0 = L$ and $b_0 = -L$.

Stage 0. $G_0 = X - L$. Now $B + L = 0$ and (17.9) collapses to a closed form,

$$\mathrm{Err}_{\mathrm{rev}}(F, G_0) = \log(1 - Lt) = -Lt - \tfrac{1}{2}L^2t^2 - \tfrac{1}{3}L^3t^3 - \tfrac{1}{4}L^4t^4 - \cdots ,$$

so $r_1 = -L$ and $b_1 = L$.

Stage 1. $G_1 = X - L + Lt$, and

$$\mathrm{Err}_{\mathrm{rev}}(F, G_1) = \bigl(L - \tfrac{1}{2}L^2\bigr)t^2 + \bigl(L^2 - \tfrac{1}{3}L^3\bigr)t^3 + \bigl(-\tfrac{1}{2}L^2 + L^3 - \tfrac{1}{4}L^4\bigr)t^4 + \mathcal{O}_t^{\mathrm{form}}\bigl(t^5\bigr) ,$$

so $r_2 = L - \frac{1}{2}L^2$ and $b_2 = \frac{1}{2}L^2 - L$.

Stage 2. $G_2 = G_1 + \bigl(\frac{1}{2}L^2 - L\bigr)t^2$, and

$$\mathrm{Err}_{\mathrm{rev}}(F, G_2) = \bigl(-\tfrac{1}{3}L^3 + \tfrac{3}{2}L^2 - L\bigr)t^3 + \bigl(-\tfrac{1}{4}L^4 + \tfrac{3}{2}L^3 - \tfrac{3}{2}L^2\bigr)t^4 + \mathcal{O}_t^{\mathrm{form}}\bigl(t^5\bigr) ,$$

so $b_3 = \frac{1}{3}L^3 - \frac{3}{2}L^2 + L$.

Stage 3. $G_3 = G_2 + \bigl(\frac{1}{3}L^3 - \frac{3}{2}L^2 + L\bigr)t^3$, and

$$\mathrm{Err}_{\mathrm{rev}}(F, G_3) = \bigl(-\tfrac{1}{4}L^4 + \tfrac{11}{6}L^3 - 3L^2 + L\bigr)t^4 + \mathcal{O}_t^{\mathrm{form}}\bigl(t^5\bigr) ,$$

so $b_4 = \frac{1}{4}L^4 - \frac{11}{6}L^3 + 3L^2 - L$.

Stage 4. $G_4 = G_3 + b_4t^4$ and $\mathrm{Err}_{\mathrm{rev}}(F, G_4) = \mathcal{O}_t^{\mathrm{form}}\bigl(t^5\bigr)$, which is the certificate of Theorem 16.16 for $N = 4$. Collecting,

$$(X+\log X)^{\langle -1\rangle\circ} = X-L+Lt+\Bigl(\frac{L^2}{2}-L\Bigr)t^2+\Bigl(\frac{L^3}{3}-\frac{3L^2}{2}+L\Bigr)t^3+\Bigl(\frac{L^4}{4}-\frac{11L^3}{6}+3L^2-L\Bigr)t^4+\mathcal{O}_t^{\mathrm{form}}\bigl(t^5\bigr) . \tag{17.10}$$

The blocks are the canonical zero-based Lambert polynomials, $b_n = \mathsf{L}_n^{\mathrm{Lam}}(L)$ for $0 \le n \le 4$, with $\mathsf{L}_0^{\mathrm{Lam}}(u) = -u$, $\mathsf{L}_1^{\mathrm{Lam}}(u) = u$, $\mathsf{L}_2^{\mathrm{Lam}}(u) = \frac{1}{2}u^2 - u$, $\mathsf{L}_3^{\mathrm{Lam}}(u) = \frac{1}{3}u^3 - \frac{3}{2}u^2 + u$ and $\mathsf{L}_4^{\mathrm{Lam}}(u) = \frac{1}{4}u^4 - \frac{11}{6}u^3 + 3u^2 - u$. The map $(X + \log X)^{\langle -1\rangle\circ}$ is Wright's omega function ω [5, 7]; its polynomials, the degree law $\deg \mathsf{L}_n^{\mathrm{Lam}} = n$ for $n \ge 1$ — which fails at $n = 0$, where $\deg \mathsf{L}_0^{\mathrm{Lam}} = 1$ — and the Stirling-number formula are developed in the Lambert part of this volume.

Remark 17.20 (The residual is minus the next Lambert polynomial). Every residual displayed in Theorem 17.19 has leading block $-\mathsf{L}_n^{\mathrm{Lam}}(L)$ at level n. This is not an accident of the computation but an instance of Theorem 16.17: for $G_{n-1} = \mathrm{Trunc}_{<n}\,\omega$,

$$\mathrm{Err}_{\mathrm{rev}}(X + \log X,\ \mathrm{Trunc}_{<n}\,\omega) = -\mathsf{L}_n^{\mathrm{Lam}}(L)\, t^n + \mathcal{O}_t^{\mathrm{form}}\bigl(t^{n+1}\bigr) .$$

This is a one-line certificate for any published table of Lambert polynomials, and it is independent of whatever recurrence generated the table.

Example 17.21 (A two-parameter logarithmic-rational perturbation, through block 4)**.** Let $a, b \in \mathbb{K}$ and

$$F(X) = X + a\log X + \frac{b}{X} = X + aL + bt \ \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}.$$

For $G = X + B$,

$$\mathrm{Err}_{\mathrm{rev}}(F, G) = B + aL + a\log(1 + Bt) + \frac{bt}{1 + Bt}. \tag{17.11}$$

Running Theorem 17.2 on (17.11) gives $r_n = -b_n$ with

$$b_0 = -aL, \tag{17.12}$$
$$b_1 = a^2 L - b, \tag{17.13}$$
$$b_2 = a^3\Big(\frac{L^2}{2} - L\Big) + ab\,(1 - L), \tag{17.14}$$
$$b_3 = a^4\Big(\frac{L^3}{3} - \frac{3L^2}{2} + L\Big) + a^2 b\,\big(-L^2 + 3L - 1\big) - b^2, \tag{17.15}$$
$$b_4 = a^5\Big(\frac{L^4}{4} - \frac{11L^3}{6} + 3L^2 - L\Big) + a^3 b\Big(-L^3 + \frac{11L^2}{2} - 6L + 1\Big) + ab^2\Big(\frac{5}{2} - 3L\Big). \tag{17.16}$$

The certificates exhibited at the successive stages are

$$\begin{aligned}
\mathrm{Err}_{\mathrm{rev}}(F, G_{-1}) &= aL + bt,\\
\mathrm{Err}_{\mathrm{rev}}(F, G_0) &= \big(b - a^2 L\big)t + \mathcal{O}_t^{\mathrm{form}}\big(t^2\big)\,,\\
\mathrm{Err}_{\mathrm{rev}}(F, G_1) &= \Big[a^3\Big(L - \frac{L^2}{2}\Big) + ab\,(L - 1)\Big]t^2 + \mathcal{O}_t^{\mathrm{form}}\big(t^3\big)\,,\\
\mathrm{Err}_{\mathrm{rev}}(F, G_2) &= -b_3\, t^3 + \mathcal{O}_t^{\mathrm{form}}\big(t^4\big)\,,\\
\mathrm{Err}_{\mathrm{rev}}(F, G_3) &= -b_4\, t^4 + \mathcal{O}_t^{\mathrm{form}}\big(t^5\big)\,,\\
\mathrm{Err}_{\mathrm{rev}}(F, G_4) &= \mathcal{O}_t^{\mathrm{form}}\big(t^5\big)\,,
\end{aligned}$$

their valuations being $0, 1, 2, 3, 4$ and ≥ 5 respectively (for $a \neq 0$). Three checks come free.

- $b = 0$. Then $b_n = a^{n+1}\mathsf{L}_n^{\mathrm{Lam}}(L)$ for $0 \leq n \leq 4$; at $a = 1$ this reproduces (17.10). The pattern persists for every n, since in the Lagrange–Bürmann formula of Part 18 only the index $r = n+1$ contributes to block n when $A = aL$, giving $a^{n+1}\frac{(-1)^{n+1}}{(n+1)!}\Delta_{0,n}L^{n+1} = a^{n+1}\mathsf{L}_n^{\mathrm{Lam}}(L)$.
- $a = 0$. Then $b_0 = b_2 = b_4 = 0$ and $b_1 = -b$, $b_3 = -b^2$. The vanishing of the even blocks is forced by parity: $F = X + bX^{-1}$ satisfies $F(-X) = -F(X)$, so by the uniqueness in Theorem 16.11 its inverse is odd as well, and $F^{\langle -1\rangle\circ}(-X) = -F^{\langle -1\rangle\circ}(X)$ reads $b_n(-1)^n = -b_n$. See Theorem 17.22.
- *Degrees.* For $a \neq 0$, $\deg_L b_n = n$ for $1 \leq n \leq 4$ while $\deg_L b_0 = 1$, matching the Lambert degree law and its exception at $n = 0$.

Formulas (17.12)–(17.15) agree with the specializations recorded in reports 1, 2, 3, 4, 5 and 6, which we have checked term by term against each other and against the triangular recursion; (17.16) appears in none of them.

Example 17.22 (An exact benchmark: the Catalan inverse)**.** The case $a = 0$ of Theorem 17.21 has a closed form and is therefore the sharpest available test of signs and of the branch convention. Let $c \in \mathbb{K}^\times$ and $F = X + ct$. Since $\mathbb{K}$ has characteristic zero,

$$S := \big(1 - 4ct^2\big)^{1/2} = \sum_{k\geq 0}\binom{1/2}{k}(-4c)^k t^{2k} \in \mathbb{K}[[t]]$$

is well defined, and $\binom{1/2}{k}(-4)^k = -\frac{2}{k}\binom{2k-2}{k-1}$ for $k \geq 1$, i.e. -2 times the Catalan number C_{k-1}. Put $G = \frac{1}{2}X(1 + S)$. Then $\nu_t(G - X) \geq 1$, so $G \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}$, and

$$G^2 - XG = \tfrac{1}{4}X^2(1 + S)\big[(1 + S) - 2\big] = \tfrac{1}{4}X^2\big(S^2 - 1\big) = -cX^2t^2 = -c,$$

so $G^2 - XG + c = 0$. Dividing by G, a unit of $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ because $\nu_t(G) = -1$ with leading block 1, gives $G + cG^{-1} = X$, that is $F \circ G = X$. By Theorem 16.11 this G is $F^{\langle -1\rangle_\circ}$, and expanding,

$$\left(X + cX^{-1}\right)^{\langle -1\rangle_\circ} = X - \sum_{k\geq 1} C_{k-1} c^k t^{2k-1} = X - \frac{c}{X} - \frac{c^2}{X^3} - \frac{2c^3}{X^5} - \frac{5c^4}{X^7} - \cdots ,$$

with Catalan coefficients $1, 1, 2, 5, 14, \ldots$ [4, 18]. Blocks 0, 1 and 3 reproduce (17.12), (17.13) and (17.15) at $a = 0$. Note that the second root $\frac{1}{2}X(1 - S)$ has $\nu_t = 1$ and does not lie in $\mathcal{G}_{\mathrm{poly},\mathbb{K}}$: the near-identity normalization is exactly what selects the branch, and no coefficient comparison is needed to discard the other one.

Structural checkpoint

Before accepting a computed reversal, record five items: the coefficient field $\mathbb{K}$, including any logarithm of a scaling constant that had to be adjoined; the exclusive cutoff $N + 1$ of the answer; the guard precisions actually used for R, for J and for the quotient; the valuation of the right residual; and the valuation of the left residual. A result whose residual valuation is not *strictly* greater than the advertised block index is not certified, however plausible its coefficients look.

17.7 Exercises

Exercise 17.23. Let $F = X + A \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}$ with $\nu_t(A) = \alpha$. Show that Theorem 17.6 reaches block N after at most $\lceil (N + 1 - \alpha)/(\alpha + 1)\rceil$ iterations, and exhibit an A with $\alpha = 0$ for which $N + 1$ iterations are necessary.

Exercise 17.24. Carry out the guard-starved Newton step of Theorem 17.17 in the other direction: keep $q = \sigma_R = 6$ but under-resolve the Jacobian, taking $\sigma_K \in \{2, 3, 4\}$. Predict the resulting residual order from Theorem 17.15, and verify that in each case the bound $\min\{q, \sigma_R, m + \sigma_K\}$ is attained exactly.

Exercise 17.25. Let $F = X \log X$. Show that $\nu_t(D_X F) = 0$ with leading block $L + 1$, and that the first triangular correction already requires $(L + 1)^{-1}$. Show that F has no compositional inverse in $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$, identify the leading monomial of the true inverse, and locate the first block in which $\log \log X$ appears.

Exercise 17.26. Let $F = X + A \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}$ and $\alpha = \nu_t(A)$. Using Theorem 16.15 and Theorem 16.12, show that $H \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}$ satisfies $\mathrm{Err}_{\mathrm{rev}}(F, H) = \mathcal{O}_t^{\mathrm{form}}\left(t^{2\alpha+1}\right)$ if and only if $H = X - A + \mathcal{O}_t^{\mathrm{form}}\left(t^{2\alpha+1}\right)$. Deduce that $X - \mathrm{Trunc}_{<2\alpha+1} A$ is the unique element of $\mathcal{G}_{\mathrm{poly},\mathbb{K}}$ whose correction is supported below $t^{2\alpha+1}$ and whose residual has valuation at least $2\alpha + 1$.

Chapter 18

The Lagrange–Bürmann formula at infinity

Fixed-point and Newton reversion produce the inverse of a near-identity map one filtration block at a time. The Lagrange–Bürmann formula produces all blocks at once, as a single explicit family of differential monomials in the perturbation. This chapter states that formula, proves it *formally* – no analytic Lagrange inversion theorem is invoked and no convergence is used or claimed – derives the finite coefficient formula in the polynomial–logarithmic basis together with a sharp determination of its index range, proves the transport formula for a composed observable, and establishes the degree law that governs the growth of logarithmic degrees under reversion.

Throughout, $\mathbb{K}$ is a field of characteristic zero, X is the large variable, and

$$t := X^{-1}, \qquad L := \log X.$$

The default analytic regime, whenever one is mentioned at all, is $X \to +\infty$ along the positive real ray. Truncation is always *exclusive*: $\operatorname{Trunc}_{<N} F = \sum_{n<N} p_n(L)t^n$. Coefficient extraction is written with the series to the right of the bracket, $[t^n]F = p_n(L)$ and $[t^n L^j]F = a_{n,j}$.

18.1 Reversion frames

Every argument below uses only four features of $\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$: it is a characteristic-zero algebra, it carries a valuation, it is complete for that valuation, and its distinguished derivation strictly increases the valuation. Isolating these makes the theorems of this section available verbatim in the Puiseux–log, iterated Laurent-log and nested-logarithmic enlargements that Part 19 is forced to introduce. The cost of the abstraction is one definition.

Definition 18.1 (Reversion frame)**.** Let $\mathbb{K}$ be a field of characteristic zero and let $\Gamma \subseteq \mathbb{R}$ be an ordered subgroup with $1 \in \Gamma$. A *reversion frame* over $\mathbb{K}$ with value group Γ is a quadruple $(\mathfrak{A}, \nu, \partial, X)$ in which $\mathfrak{A}$ is a commutative $\mathbb{K}$-algebra, ∂ is a $\mathbb{K}$-derivation of $\mathfrak{A}$, $X \in \mathfrak{A}$, and $\nu\colon \mathfrak{A} \to \Gamma \cup \{+\infty\}$ satisfies the following.

(F1) $\nu(f) = +\infty$ exactly when $f = 0$; $\nu(f+g) \geq \min\{\nu(f), \nu(g)\}$; $\nu(fg) = \nu(f) + \nu(g)$; and $\nu(\lambda) = 0$ for every $\lambda \in \mathbb{K}^\times$.

(F2) Call a family $(f_i)_{i\in I}$ in $\mathfrak{A}$ *summable* when $\{i \in I : \nu(f_i) < \gamma\}$ is finite for every $\gamma \in \Gamma$. Every summable family has a sum $\sum_i f_i \in \mathfrak{A}$; summation is $\mathbb{K}$-linear, satisfies $\nu\big(\sum_i f_i\big) \geq \inf_i \nu(f_i)$, is unchanged by regrouping the index set into disjoint blocks, and satisfies $\big(\sum_i f_i\big)\big(\sum_j g_j\big) = \sum_{i,j} f_i g_j$ for summable (f_i) and (g_j).

(F3) $\nu(\partial f) \geq \nu(f) + 1$ for every $f \in \mathfrak{A}$.

(F4) $\partial X = 1$.

We write

$$\mathfrak{A}_{>-1} := \{f \in \mathfrak{A} : \nu(f) > -1\},$$

the set of *admissible perturbations*. It is closed under addition and under multiplication by $\{\nu \geq 0\}$, and contains 0; it is not in general a subring, since $\nu(f^2) = 2\nu(f)$ may drop below -1.

Remark 18.2. The threshold is -1, not 0, because the summability of every family below is governed by the *gain*

$$\delta := 1 + \nu(A) > 0 \tag{18.1}$$

of one derivative against one factor of the perturbation. In the polynomial–logarithmic frame the value group is $\mathbb{Z}$, so $\nu_t(A) > -1$ is the same as $A \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ and $\mathcal{G}(\mathscr{PL}_{\mathrm{poly},\mathbb{K}}) = \mathcal{G}_{\mathrm{poly},\mathbb{K}}$; the distinction matters only in the Puiseux–log frames of Part 19, where a perturbation of outer order $-\frac{1}{2}$ is admissible and is exactly what the reduction of a ramified reversion produces (Theorem 19.11).

Condition (F3) forces ∂ to be ν-continuous, so it commutes with sums of summable families: if (f_i) is summable then so is (∂f_i), and $\partial \sum_i f_i = \sum_i \partial f_i$, because the two sides agree modulo $\{\nu \geq \gamma\}$ for every γ after discarding the finitely many indices with $\nu(f_i) < \gamma$.

Lemma 18.3 (The polynomial–logarithmic frame). $\left(\mathscr{PL}_{\mathrm{poly},\mathbb{K}}, \nu_t, D_X, t^{-1}\right)$ *is a reversion frame over* $\mathbb{K}$ *with* $\Gamma = \mathbb{Z}$.

Proof. $\mathscr{PL}_{\mathrm{poly},\mathbb{K}} = \mathbb{K}[L]((t))$ is a domain because $\mathbb{K}[L]$ is, and (F1) is the standard outer valuation together with multiplicativity of leading blocks. For (F2), let (f_i) be summable. Taking $\gamma = n+1$ shows that only finitely many i have $\nu_t(f_i) \leq n$, so $[t^n]\sum_i f_i := \sum_i [t^n] f_i$ is a finite sum of elements of $\mathbb{K}[L]$; taking $\gamma = 0$ shows that $\inf_i \nu_t(f_i)$ is finite, so the resulting series has only finitely many negative powers of t and lies in $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$. Linearity, the valuation bound, regrouping and the product rule are then finite-sum statements at each fixed n. For (F3), the block formula $D_X(t^n p(L)) = t^{n+1}(\partial_L - n)p(L)$ gives $\nu_t(D_X f) \geq \nu_t(f) + 1$. Finally $D_X(t^{-1}) = D_X X = 1$, which is (F4). ☐

Remark 18.4. The same verification, word for word, applies to the Puiseux–log ring $\mathbb{K}[L]((t^{1/s}))$ with $\Gamma = s^{-1}\mathbb{Z}$, to the iterated Laurent-log field $\mathscr{PL}_{\mathrm{itLaur},\mathbb{K}}$ with $\Gamma = \mathbb{Z}$, and to the nested-logarithmic ring $\mathbb{K}((L^{-1}))[\log L]((t))$, on which $D_X(t^n p) = t^{n+1}\left(\partial_L p + L^{-1}\partial_{\log L} p - np\right)$. All four are used in Part 19. What is *not* covered is a value group of rank two, such as the refinement of ν_t by logarithmic degree; see Theorem 19.15.

Lemma 18.5 (Substitution calculus). *Let* $(\mathfrak{A}, \nu, \partial, X)$ *be a reversion frame and let* $H \in \mathfrak{A}_{>-1}$. *Define*

$$T_H(f) := \sum_{k \geq 0} \frac{H^k}{k!}\, \partial^k f, \qquad f \in \mathfrak{A}, \tag{18.2}$$

and write $f \circ (X + H) := T_H(f)$. *Then:*

1. *the family in Equation* (18.2) *is summable, its* k*-th member having* $\nu \geq \nu(f) + k\big(1 + \nu(H)\big)$, *and* $\nu(T_H f) \geq \nu(f)$;
2. T_H *is a* $\mathbb{K}$*-algebra endomorphism of* $\mathfrak{A}$, *continuous for* ν, *with* $T_0 = \mathrm{id}$ *and* $T_H(X) = X + H$;
3. $\partial \circ T_H = (1 + \partial H) \cdot (T_H \circ \partial)$;
4. *for* $H, H' \in \mathfrak{A}_{>-1}$, $T_{H'} \circ T_H = T_{H''}$ *with* $H'' = H' + T_{H'}(H) \in \mathfrak{A}_{>-1}$;
5. $\mathcal{G}(\mathfrak{A}) := \{X + H : H \in \mathfrak{A}_{>-1}\}$ *is a group under* $(X + H) \circ (X + H') := T_{H'}(X + H) = X + H' + T_{H'}(H)$, *and* T *is an anti-homomorphism from it to the monoid of endomorphisms of* $\mathfrak{A}$. *In the polynomial–logarithmic frame,* $\mathcal{G}(\mathscr{PL}_{\mathrm{poly},\mathbb{K}}) = \mathcal{G}_{\mathrm{poly},\mathbb{K}}$.

Proof. (1) By (F1) and (F3), $\nu\big(H^k\partial^k f/k!\big) \geq k\nu(H) + \nu(f) + k = \nu(f) + k\big(1+\nu(H)\big)$, which tends to $+\infty$ because $1+\nu(H) > 0$; summability and the valuation bound follow from (F2), the minimum over $k \geq 0$ being attained at $k = 0$.

(2) Linearity is clear. For multiplicativity, expand $\partial^k(fg)$ by Leibniz and regroup, which (F2) permits:

$$T_H(fg) = \sum_{k\geq 0} \frac{H^k}{k!} \sum_{i+j=k} \binom{k}{i} \partial^i f\, \partial^j g = \sum_{i,j\geq 0} \frac{H^i\partial^i f}{i!} \cdot \frac{H^j\partial^j g}{j!} = T_H(f)\, T_H(g).$$

$T_H(1) = 1$ and $T_H(\lambda) = \lambda$. $T_H(X) = X + H$ because $\partial X = 1$ and $\partial^k X = 0$ for $k \geq 2$. Continuity follows from $\nu(T_H f) \geq \nu(f)$ and linearity.

(3) Differentiate Equation (18.2) term by term, which is legitimate because ∂ commutes with summable sums:

$$\partial T_H(f) = \sum_{k\geq 1} \frac{kH^{k-1}(\partial H)}{k!}\partial^k f + \sum_{k\geq 0} \frac{H^k}{k!}\partial^{k+1} f = (\partial H)\, T_H(\partial f) + T_H(\partial f).$$

(4) Let $\mathfrak{A}\langle z\rangle_a$ denote the set of formal series $P = \sum_{m\geq 0} P_m z^m$ with $P_m \in \mathfrak{A}$ and $\nu(P_m) \geq m - a$, and $\mathfrak{A}\langle z\rangle = \bigcup_{a\in\Gamma} \mathfrak{A}\langle z\rangle_a$. By (F1) and (F2) this is a subring of $\mathfrak{A}[[z]]$ with $\mathfrak{A}\langle z\rangle_a \mathfrak{A}\langle z\rangle_b \subseteq \mathfrak{A}\langle z\rangle_{a+b}$, and for each $g \in \mathfrak{A}_{>-1}$ the evaluation $\mathrm{ev}_g(P) = \sum_m P_m g^m$ is a well-defined $\mathbb{K}$-algebra homomorphism $\mathfrak{A}\langle z\rangle \to \mathfrak{A}$, because $\nu(P_m g^m) \geq m\big(1+\nu(g)\big) - a \to +\infty$ makes the family summable and (F2) makes the product rule a rearrangement of a summable double family. Put $\tau(f) = \sum_k (\partial^k f) z^k/k!$; by the computation in (1), $\tau(f) \in \mathfrak{A}\langle z\rangle_{-\nu(f)}$, by the Leibniz argument in (2) τ is a ring homomorphism, and $\partial_z \tau = \tau\partial$, $\mathrm{ev}_0\tau = \mathrm{id}$, $T_g = \mathrm{ev}_g \circ \tau$.

Fix $f \in \mathfrak{A}$ and consider the doubly indexed family

$$w_{j,k} := \frac{\partial^k f}{j!\,(k-j)!}\, \tau(H)^j\, z^{k-j} \in \mathfrak{A}\langle z\rangle, \qquad 0 \leq j \leq k.$$

Its coefficient of z^m is nonzero only when $k - j \leq m$, and $\nu\big([z^m]w_{j,k}\big) \geq (\nu(f)+k) + \big(j\nu(H) + m - k + j\big) = \nu(f) + m + j\big(1+\nu(H)\big)$. Since $k \leq m+j$ and $1+\nu(H) > 0$, for each fixed m and each γ only finitely many pairs (j,k) give valuation below γ; the family is therefore summable in every z-degree, and may be regrouped freely. Summing first over k and then over j gives $\sum_j \tau(H)^j \partial_z^j \tau(f)/j!$, which by $\partial_z^j\tau = \tau\partial^j$ and multiplicativity of τ equals $\tau(T_H f)$; summing first over j and then over k, and using the binomial theorem, gives $\sum_k \big(\partial^k f/k!\big)(z+\tau(H))^k$. Hence

$$\tau\big(T_H f\big) = \sum_{k\geq 0} \frac{\partial^k f}{k!}\big(z + \tau(H)\big)^k.$$

Applying the ring homomorphism $\mathrm{ev}_{H'}$, which likewise commutes with the regrouping of this summable family, and using $\mathrm{ev}_{H'}\tau(H) = T_{H'}(H)$ and $\mathrm{ev}_{H'}(z) = H'$, gives $T_{H'}(T_H f) = \sum_k (\partial^k f)(H' + T_{H'}H)^k/k! = T_{H''}(f)$. Finally $\nu(H'') \geq \min\{\nu(H'), \nu(T_{H'}H)\} \geq \min\{\nu(H'), \nu(H)\} > -1$.

(5) Part (4) says exactly that $\circ$ is associative on $\mathcal{G}(\mathfrak{A})$ and that $T_{F\circ G} = T_G \circ T_F$; the element X is a two-sided identity. By Theorem 18.6 below every element has a right inverse, and a monoid in which every element has a right inverse is a group: if $ab = X$ and $bc = X$ then $ba = ba(bc) = b(ab)c = bc = X$. There is no circularity, because the proof of Theorem 18.6 uses only parts (1) and (2). □

Proposition 18.6 (Right inverse: existence, uniqueness, characterisation)**.** *Let* $(\mathfrak{A}, \nu, \partial, X)$ *be a reversion frame,* $A \in \mathfrak{A}_{>-1}$*, and* $F = X + A$*. There is exactly one* $B \in \mathfrak{A}_{>-1}$ *with*

$$F \circ (X+B) = X, \qquad \textit{equivalently} \qquad B = -T_B(A) = -A \circ (X+B). \tag{18.3}$$

It satisfies $\nu(B) \geq \nu(A)$*. We write* $F^{\langle -1\rangle\circ} = X + B$*.*

Proof. The two displayed conditions are the same because $F\circ(X+B) = T_B(X+A) = X+B+T_B(A)$.

A contraction estimate. Let $B, B' \in \mathfrak{A}_{>-1}$ and put $m = \min\{\nu(B), \nu(B')\} > -1$. Then

$$\nu\big(T_{B'}(A) - T_B(A)\big) \;\geq\; \nu(B'-B) + \delta, \qquad \delta = 1+\nu(A) > 0. \tag{18.4}$$

Indeed $T_{B'}(A) - T_B(A) = \sum_{k\geq 1}\big(B'^k - B^k\big)\partial^k A/k!$, the $k=0$ terms cancelling, and for $k \geq 1$ $B'^k - B^k = (B'-B)\sum_{i=0}^{k-1} B'^i B^{k-1-i}$ has valuation at least $\nu(B'-B)+(k-1)m$, while $\nu(\partial^k A) \geq \nu(A)+k$. The k-th term therefore has valuation at least $\nu(B'-B)+\nu(A)+k(1+m)-m$, which is increasing in k because $1+m>0$; its value at $k=1$ is $\nu(B'-B)+\nu(A)+1$.

Uniqueness. If B, B' both solve Equation (18.3), then Equation (18.4) gives $\nu(B-B') \geq \nu(B'-B)+\delta$ with $\delta > 0$, forcing $\nu(B-B') = +\infty$.

Existence. Set $B_0 = 0$ and $B_{n+1} = -T_{B_n}(A)$; by Theorem 18.5(1), $\nu(B_{n+1}) \geq \nu(A) > -1$, so every B_n lies in $\mathfrak{A}_{>-1}$ and Equation (18.4) applies with $m \geq \min\{0, \nu(A)\}$. Hence $\nu(B_{n+1}-B_n) \geq \nu(B_n - B_{n-1})+\delta$, so $\nu(B_{n+1}-B_n) \geq \nu(A)+n\delta$. The family $(B_{n+1}-B_n)_{n\geq 0}$ is summable, and $B := \sum_{n\geq 0}(B_{n+1}-B_n)$ satisfies $\nu(B-B_n) \geq \nu(A)+n\delta$. Applying Equation (18.4) once more, $\nu\big(T_B(A) - T_{B_n}(A)\big) \geq \nu(A)+(n+1)\delta$, so $\nu\big(B+T_B(A)\big) \geq \min\{\nu(B-B_{n+1}), \nu(T_{B_n}A - T_B A)\} \to +\infty$ and $B = -T_B(A)$. Finally $\nu(B) = \nu(T_B(A)) \geq \nu(A)$. □

Remark 18.7. The characterisation Equation (18.3) is the only property of $F^{\langle -1\rangle\circ}$ used below; the two-sidedness supplied by Theorem 18.5(5) is never needed in the proof of the Lagrange–Bürmann formula and is recorded only so that “the” inverse is unambiguous.

18.2 Formal summability

Lemma 18.8 (Order of the r-th Lagrange term). *Let $(\mathfrak{A}, \nu, \partial, X)$ be a reversion frame and $A \in \mathfrak{A}$. For every $r \geq 1$,*

$$\nu\Big(\partial^{r-1}(A^r)\Big) \;\geq\; r\,\nu(A) + r - 1 \;=\; r\delta - 1, \qquad \delta = 1+\nu(A). \tag{18.5}$$

If $A \in \mathfrak{A}_{>-1}$, so that $\delta > 0$, the family $\big((-1)^r\partial^{r-1}(A^r)/r!\big)_{r\geq 1}$ is summable, and its sum S satisfies $\nu(S) \geq \nu(A) > -1$, whence $S \in \mathfrak{A}_{>-1}$.

Proof. $\nu(A^r) = r\nu(A)$ by (F1) and $r-1$ applications of (F3) add at least $r-1$. If $\delta > 0$ the right-hand side of Equation (18.5) tends to $+\infty$, so the family is summable by (F2); its minimum over $r \geq 1$ is attained at $r = 1$ and equals $\nu(A)$, which bounds $\nu(S)$. □

The hypothesis $\nu(A) > -1$ is not merely convenient: it is exactly the summability condition, in the polynomial–logarithmic frame and in its Puiseux–log enlargements alike.

Proposition 18.9 (Sharp summability criterion). *Fix $s \in \mathbb{N}_{>0}$ and let $A \in \mathbb{K}[L]((t^{1/s}))$ be nonzero, with $\alpha := \nu_t(A) \in s^{-1}\mathbb{Z}$. The family $\big((-1)^r D_X^{r-1}(A^r)/r!\big)_{r\geq 1}$ is summable if and only if $\alpha > -1$. Indeed, if $\alpha \leq -1$ then*

$$\nu_t\Big(D_X^{r-1}(A^r)\Big) = r(\alpha+1) - 1 \qquad \textit{for every } r \geq 1, \tag{18.6}$$

which is nonincreasing in r and hence never tends to $+\infty$. For $s = 1$, that is in $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$, the value group is $\mathbb{Z}$ and the criterion reads $\nu_t(A) \geq 0$, i.e. $A \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$.

Proof. Sufficiency is Theorem 18.8. For necessity assume $\alpha \leq -1$ and write $A = t^\alpha p(L) + A_1$ with $p \neq 0$ and $\nu_t(A_1) > \alpha$. Then $A^r = t^{r\alpha}p^r + (\text{terms of } \nu_t > r\alpha)$. Applying the repeated-derivative formula $D_X^{r-1}\big(t^n p(L)\big) = t^{n+r-1}\Delta_{n,r-1}p(L)$ with $n = r\alpha$,

$$D_X^{r-1}\big(A^r\big) = t^{r\alpha+r-1}\prod_{j=0}^{r-2}\big(\partial_L - r\alpha - j\big)p(L)^r + \big(\text{terms of } \nu_t > r\alpha + r - 1\big).$$

For $0 \le j \le r-2$ one has $-r\alpha - j \ge r-(r-2) = 2 > 0$, so each factor $\partial_L - (r\alpha + j)$ multiplies the leading coefficient of a nonzero polynomial by the nonzero scalar $-r\alpha - j$ and in particular is injective on $\mathbb{K}[L] \setminus \{0\}$. Hence the displayed leading block is nonzero and Equation (18.6) holds. Since $\alpha + 1 \le 0$, the exponent $r(\alpha+1)-1$ is at most -1 for all r, so $\{r : \nu_t < 0\}$ is infinite and the family is not summable. ☐

Remark 18.10. Theorem 18.9 makes precise what "near identity" means here, and it is *not* that A is bounded. For $A = L = \log X$ one has $A \to +\infty$ yet $\nu_t(A) = 0 > -1$, and the theory applies; for $A = X/L$ one has $A = o_{X\to+\infty}(X)$ yet $\nu_t(A) = -1$, and it does not. The decisive quantity is the outer order measured against the threshold $-1 = \nu_t(X)$, equivalently the dominance comparison $A \prec X$ *refined to the outer filtration*: logarithmic gains are invisible to ν_t, and a perturbation of the same outer order as X itself is never admissible however many logarithms separate it from X.

18.3 The parameter identity

The classical Lagrange–Bürmann theorem is a statement about a formal parameter. We prove the version we need – the *shifted*, or reversion, form – from scratch, over an arbitrary commutative $\mathbb{Q}$-algebra and with no invertibility hypothesis whatsoever.

Lemma 18.11 (Lagrange reversion identity). *Let C be a commutative $\mathbb{Q}$-algebra and let $\psi \in C[[z]]$. For $f \in C[[z]]$ and $w \in \varepsilon\, C[[z]][[\varepsilon]]$ write $f_{[w]} := \sum_{k\ge0}(\partial_z^k f)w^k/k!$. There is exactly one $u \in \varepsilon\, C[[z]][[\varepsilon]]$ with*

$$u = \varepsilon\, \psi_{[u]}, \tag{18.7}$$

and for every $\gamma \in C[[z]]$ and every $m \ge 1$,

$$[\varepsilon^m]\, \gamma_{[u]} = \frac{1}{m!}\, \partial_z^{m-1}\big(\gamma'\psi^m\big), \qquad \gamma' := \partial_z\gamma. \tag{18.8}$$

Proof. Existence and uniqueness of u: comparing coefficients of ε^m in Equation (18.7) determines $[\varepsilon^m]u$ from $[\varepsilon^1]u, \dots, [\varepsilon^{m-1}]u$, because w^k contributes to ε^m only for $k \le m$ and then only through those coefficients.

For fixed u, the map $f \mapsto f_{[u]}$ is a C-algebra homomorphism of $C[[z]]$ into $C[[z]][[\varepsilon]]$ (Leibniz, exactly as in Theorem 18.5(2)), and differentiating its defining series term by term gives the two chain rules

$$\partial_\varepsilon\big(f_{[u]}\big) = f'_{[u]}\, \partial_\varepsilon u, \qquad \partial_z\big(f_{[u]}\big) = f'_{[u]}\big(1 + \partial_z u\big). \tag{18.9}$$

Applying Equation (18.9) to Equation (18.7),

$$\partial_\varepsilon u = \psi_{[u]} + \varepsilon\psi'_{[u]}\partial_\varepsilon u, \qquad \partial_z u = \varepsilon\psi'_{[u]}\big(1 + \partial_z u\big).$$

Since $\varepsilon\psi'_{[u]} \in \varepsilon C[[z]][[\varepsilon]]$, the element $1 - \varepsilon\psi'_{[u]}$ is a unit; solving the second relation gives $1 + \partial_z u = \big(1 - \varepsilon\psi'_{[u]}\big)^{-1}$, and substituting into the first gives the key identity

$$\partial_\varepsilon u = \psi_{[u]}\big(1 + \partial_z u\big). \tag{18.10}$$

Now let $\gamma \in C[[z]]$ be arbitrary and let $\Gamma \in C[[z]]$ be the formal antiderivative of $\gamma'\psi$ with $\Gamma(0) = 0$, which exists because C is a $\mathbb{Q}$-algebra. Using Equation (18.9), Equation (18.10) and multiplicativity of $f \mapsto f_{[u]}$,

$$\partial_\varepsilon\big(\gamma_{[u]}\big) = \gamma'_{[u]}\psi_{[u]}\big(1 + \partial_z u\big) = \big(\gamma'\psi\big)_{[u]}\big(1 + \partial_z u\big) = \Gamma'_{[u]}\big(1 + \partial_z u\big) = \partial_z\big(\Gamma_{[u]}\big).$$

Extracting $[\varepsilon^{m-1}]$ from both ends and using $[\varepsilon^{m-1}]\partial_\varepsilon = m\,[\varepsilon^m]$,

$$m\,[\varepsilon^m]\gamma_{[u]} = \partial_z\Big([\varepsilon^{m-1}]\Gamma_{[u]}\Big), \qquad m \ge 1. \tag{18.11}$$

We prove Equation (18.8) by induction on m, the statement being quantified over all $\gamma \in C[[z]]$. For $m = 1$, $[\varepsilon^0]\Gamma_{[u]} = \Gamma$ because $u \in \varepsilon C[[z]][[\varepsilon]]$, so Equation (18.11) gives $[\varepsilon^1]\gamma_{[u]} = \Gamma' = \gamma'\psi$, which is Equation (18.8) with $m = 1$. For $m \geq 2$, the inductive hypothesis applied to Γ gives $[\varepsilon^{m-1}]\Gamma_{[u]} = \partial_z^{m-2}\big(\Gamma'\psi^{m-1}\big)/(m-1)! = \partial_z^{m-2}\big(\gamma'\psi^m\big)/(m-1)!$, and Equation (18.11) yields $m[\varepsilon^m]\gamma_{[u]} = \partial_z^{m-1}(\gamma'\psi^m)/(m-1)!$. □

Remark 18.12. A second argument runs through formal residues. Writing $\operatorname{res}_\varepsilon$ for the coefficient of ε^{-1} in a formal Laurent series, one extracts $[\varepsilon^m]$ as a residue and substitutes $\varepsilon = z/\psi(z)$. That change of variables is legitimate only when $\psi(0)$ is invertible, so the general case must be recovered by performing the computation in the universal polynomial ring generated by the coefficients of ψ with the constant coefficient inverted, observing that both sides are polynomial in finitely many of those generators at each fixed m, and specialising. The route is instructive, but the ∂_ε–∂_z exchange Equation (18.10) used above needs no localisation, no residue theory, and no hypothesis on $\psi(0)$. Classical accounts of formal Lagrange inversion are in [4, 18].

18.4 Specialising the bookkeeping parameter

Every source we merge performs the step "now set $\varepsilon = 1$" and justifies it by the sentence that only finitely many terms contribute below any fixed outer order. That sentence is true but it is not a proof: evaluation at $\varepsilon = 1$ has to be a *ring homomorphism* defined on the subalgebra in which the computation takes place, and the fixed-point equation Equation (18.7) contains an infinite sum that must be shown to survive it. The following lemma supplies the missing object.

Lemma 18.13 (Weighted deformation algebra and evaluation at one). *Let $(\mathfrak{A}, \nu, \partial, X)$ be a reversion frame and fix a weight $\delta \in \Gamma_{>0}$. For $a \in \Gamma$ put*

$$\mathfrak{A}\langle\varepsilon\rangle_a := \Big\{\, u = \sum_{m\geq 0} u_m\varepsilon^m \in \mathfrak{A}[[\varepsilon]] \;:\; \nu(u_m) \geq m\delta - a \text{ for all } m \,\Big\}, \qquad \mathfrak{A}\langle\varepsilon\rangle = \bigcup_{a\in\Gamma} \mathfrak{A}\langle\varepsilon\rangle_a.$$

Then:

1. *$\mathfrak{A}\langle\varepsilon\rangle$ is a subring of $\mathfrak{A}[[\varepsilon]]$ containing $\mathfrak{A}$ and ε, with $\mathfrak{A}\langle\varepsilon\rangle_a\,\mathfrak{A}\langle\varepsilon\rangle_b \subseteq \mathfrak{A}\langle\varepsilon\rangle_{a+b}$, and it is stable under the coefficientwise extension of ∂;*

2. *$\operatorname{ev}\colon \mathfrak{A}\langle\varepsilon\rangle \to \mathfrak{A}$, $\operatorname{ev}(u) = \sum_{m\geq 0} u_m$, is a well-defined $\mathbb{K}$-algebra homomorphism with $\operatorname{ev}\circ\partial = \partial\circ\operatorname{ev}$, $\operatorname{ev}(\varepsilon) = 1$, and $\nu(\operatorname{ev}(u)) \geq -a$ for $u \in \mathfrak{A}\langle\varepsilon\rangle_a$;*

3. *if $\big(f^{(k)}\big)_{k\geq 0}$ is a family in $\mathfrak{A}\langle\varepsilon\rangle_a$ with $f^{(k)} \in \varepsilon^k\mathfrak{A}[[\varepsilon]]$, then $\sum_k f^{(k)} \in \mathfrak{A}\langle\varepsilon\rangle_a$, the family $\big(\operatorname{ev} f^{(k)}\big)_k$ is summable in $\mathfrak{A}$, and $\operatorname{ev}\big(\sum_k f^{(k)}\big) = \sum_k \operatorname{ev}\big(f^{(k)}\big)$.*

Proof. (1) If $\nu(u_m) \geq m\delta - a$ and $\nu(v_m) \geq m\delta - b$ then $\nu\big(\sum_{i+j=m} u_iv_j\big) \geq \min_{i+j=m}(i\delta - a + j\delta - b) = m\delta - a - b$. Stability under ∂ follows from (F3), which gives $\nu(\partial u_m) \geq m\delta - (a-1)$.

(2) For $u \in \mathfrak{A}\langle\varepsilon\rangle_a$ the family $(u_m)_{m\geq 0}$ is summable, since $\nu(u_m) \geq m\delta - a \to +\infty$; the sum exists by (F2) and has $\nu \geq \inf_m(m\delta - a) = -a$. Additivity is clear, and multiplicativity is the product rule of (F2): $\operatorname{ev}(uv) = \sum_m \sum_{i+j=m} u_iv_j = \sum_{i,j} u_iv_j = \big(\sum_i u_i\big)\big(\sum_j v_j\big)$, the regrouping being licensed because $\nu(u_iv_j) \geq (i+j)\delta - a - b$ makes the double family summable. Commutation with ∂ is the continuity of ∂.

(3) Since $f^{(k)} \in \varepsilon^k\mathfrak{A}[[\varepsilon]]$, for each m only the indices $k \leq m$ contribute to $[\varepsilon^m]\sum_k f^{(k)}$, so the sum is defined coefficientwise and lies in $\mathfrak{A}\langle\varepsilon\rangle_a$. The doubly indexed family $\big([\varepsilon^m]f^{(k)}\big)_{k,m}$ is summable: it vanishes for $k > m$, and $\nu\big([\varepsilon^m]f^{(k)}\big) \geq m\delta - a$, so $\{(k,m) : \nu < \gamma\} \subseteq \{(k,m) : m\delta < \gamma + a,\ k \leq m\}$ is finite. Both iterated sums therefore equal the unordered sum, by the regrouping clause of (F2). □

18.5 The formula and the transport formula

Theorem 18.14 (Lagrange–Bürmann reversion at infinity). *Let $(\mathfrak{A}, \nu, \partial, X)$ be a reversion frame and let $A \in \mathfrak{A}_{>-1}$, $F = X + A$. Then the family $\big((-1)^r \partial^{r-1}(A^r)/r!\big)_{r\geq 1}$ is summable and*

$$F^{\langle -1\rangle_\circ} = X + \sum_{r\geq 1} \frac{(-1)^r}{r!}\, \partial^{r-1}\big(A^r\big). \tag{18.12}$$

In the polynomial–logarithmic frame this reads

$$(X+A)^{\langle -1\rangle_\circ}(X) = X + \sum_{r\geq 1} \frac{(-1)^r}{r!}\, D_X^{r-1}\big(A(X)^r\big), \qquad A \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}, \tag{18.13}$$

and the r-th summand has $\nu_t \geq r-1$.

Proof. Summability is Theorem 18.8; call the sum S, so $\nu(S) \geq \nu(A) > -1$. By Theorem 18.6 it suffices to prove $S = -T_S(A)$. Write $\delta = 1 + \nu(A) > 0$ and use this δ as the weight in Theorem 18.13.

Take $C = \mathfrak{A}$ in Theorem 18.11 and let $\tau\colon \mathfrak{A} \to \mathfrak{A}[[z]]$, $\tau(f) = \sum_k (\partial^k f) z^k/k!$, be the Taylor-shift homomorphism of Theorem 18.5(4); recall $\partial_z \tau = \tau\partial$ and $\mathrm{ev}_{z=0}\tau = \mathrm{id}$. Put

$$\psi := \tau(-A) \in \mathfrak{A}[[z]],$$

and let $u \in \varepsilon\mathfrak{A}[[z]][[\varepsilon]]$ be the unique solution of Equation (18.7). Apply Equation (18.8) with $\gamma = \tau(X) = X + z$, so that $\gamma' = 1 = \tau(1)$:

$$[\varepsilon^m]\,\gamma_{[u]} = \frac{1}{m!}\partial_z^{m-1}\big(\psi^m\big) = \frac{(-1)^m}{m!}\,\partial_z^{m-1}\tau\big(A^m\big) = \frac{(-1)^m}{m!}\,\tau\Big(\partial^{m-1}\big(A^m\big)\Big),$$

using that τ is a ring homomorphism intertwining ∂_z with ∂. Let $\mathrm{ev}_{z=0}\colon \mathfrak{A}[[z]][[\varepsilon]] \to \mathfrak{A}[[\varepsilon]]$ act coefficientwise in ε; it is a ring homomorphism. Setting $U := \mathrm{ev}_{z=0}(u) \in \varepsilon\mathfrak{A}[[\varepsilon]]$ and using $\gamma_{[u]} = \sum_k (\partial_z^k\gamma)u^k/k! = (X+z) + u$, we obtain

$$U = \sum_{m\geq 1} \frac{(-1)^m}{m!}\,\partial^{m-1}\big(A^m\big)\,\varepsilon^m. \tag{18.14}$$

Likewise, applying $\mathrm{ev}_{z=0}$ to Equation (18.7) and using $\psi_{[u]} = \sum_k (\partial_z^k\psi)u^k/k!$ together with $\mathrm{ev}_{z=0}\partial_z^k\tau(-A) = \partial^k(-A)$,

$$U = -\varepsilon \sum_{k\geq 0} \frac{\partial^k A}{k!}\, U^k. \tag{18.15}$$

Now specialise. By Equation (18.14) and Theorem 18.8, $\nu\big([\varepsilon^m]U\big) \geq m\delta - 1$, so $U \in \mathfrak{A}\langle\varepsilon\rangle_1$ and $\mathrm{ev}(U) = S$ by construction. Since U has zero constant term, $[\varepsilon^m](U^k) = 0$ for $m < k$ and, splitting $m = m_1 + \cdots + m_k$ with every $m_i \geq 1$, $\nu\big([\varepsilon^m](U^k)\big) \geq \sum_i (m_i\delta - 1) = m\delta - k$; with $\nu(\partial^k A) \geq \nu(A) + k \geq k - 1 + \delta \geq k\cdot\min\{1,\delta\}$ – more simply, with $\nu(\partial^k A) \geq \nu(A) + k$ – this gives

$$\nu\Big([\varepsilon^m]\Big(\tfrac{\partial^k A}{k!}U^k\Big)\Big) \;\geq\; \big(\nu(A) + k\big) + \big(m\delta - k\big) = m\delta + \nu(A), \qquad \tfrac{\partial^k A}{k!}U^k \in \varepsilon^k\mathfrak{A}[[\varepsilon]].$$

So Theorem 18.13(3) applies to $f^{(k)} = \partial^k A\, U^k/k!$ with $a = -\nu(A)$, and applying ev to Equation (18.15) gives

$$S = \mathrm{ev}(U) = -\sum_{k\geq 0} \frac{\partial^k A}{k!}\,\mathrm{ev}(U)^k = -\sum_{k\geq 0} \frac{S^k}{k!}\partial^k A = -T_S(A).$$

By Theorem 18.6, $X + S = F^{\langle -1\rangle_\circ}$. □

The same construction, run with a general observable in place of X, produces the expansion of any transseries along the inverse map without first building the inverse.

Theorem 18.15 (Transport formula for a composed observable)**.** *Let $(\mathfrak{A}, \nu, \partial, X)$ be a reversion frame, let $A \in \mathfrak{A}_{\geq 0}$, $F = X + A$, and let $\varphi \in \mathfrak{A}$ be arbitrary. Then the family $\big((-1)^r \partial^{r-1}\big((\partial\varphi)A^r\big)/r!\big)_{r\geq 1}$ is summable, the r-th member having*

$$\nu\Big(\partial^{r-1}\big((\partial\varphi)A^r\big)\Big) \;\geq\; \nu(\varphi) + r\big(1 + \nu(A)\big) \;\geq\; \nu(\varphi) + r, \tag{18.16}$$

and

$$\varphi \circ F^{\langle -1 \rangle_\circ} = \varphi + \sum_{r\geq 1} \frac{(-1)^r}{r!}\, \partial^{r-1}\big((\partial\varphi)\, A^r\big). \tag{18.17}$$

In particular $[t^N]\big(\varphi \circ F^{\langle -1 \rangle_\circ}\big)$ in the polynomial–logarithmic frame receives contributions only from $r \leq N - \nu_t(\varphi)$.

Proof. For Equation (18.16): $\nu(\partial\varphi) \geq \nu(\varphi) + 1$ by (F3), $\nu(A^r) = r\nu(A)$ by (F1), and $r - 1$ further derivatives add at least $r - 1$; the total is $\nu(\varphi) + 1 + r\nu(A) + r - 1 = \nu(\varphi) + r(1 + \nu(A))$, which is at least $\nu(\varphi) + r \to +\infty$. Summability follows from (F2).

Run the proof of Theorem 18.14 with $\gamma := \tau(\varphi)$ in Equation (18.8). Since $\gamma' = \partial_z \tau(\varphi) = \tau(\partial\varphi)$ and τ is a ring homomorphism intertwining ∂_z with ∂,

$$[\varepsilon^m]\gamma_{[u]} = \frac{1}{m!}\partial_z^{m-1}\big(\tau(\partial\varphi)\,\tau(-A)^m\big) = \frac{(-1)^m}{m!}\,\tau\Big(\partial^{m-1}\big((\partial\varphi)A^m\big)\Big).$$

Applying $\mathrm{ev}_{z=0}$ and writing $c_m := (-1)^m \partial^{m-1}\big((\partial\varphi)A^m\big)/m!$,

$$\sum_{k\geq 0} \frac{\partial^k\varphi}{k!}\, U^k = \varphi + \sum_{m\geq 1} c_m\, \varepsilon^m \qquad \text{in } \mathfrak{A}[[\varepsilon]], \tag{18.18}$$

where U is as in Equation (18.14). By Equation (18.16), $\nu(c_m) \geq \nu(\varphi) + m$, so the right-hand side of Equation (18.18) lies in $\mathfrak{A}\langle\varepsilon\rangle_{-\nu(\varphi)}$. On the left, $\nu\big([\varepsilon^m](\partial^k\varphi\, U^k/k!)\big) \geq \big(\nu(\varphi) + k\big) + (m - k) = \nu(\varphi) + m$ and $\partial^k\varphi\, U^k/k! \in \varepsilon^k\mathfrak{A}[[\varepsilon]]$, so Theorem 18.13(3) applies with $a = -\nu(\varphi)$. Applying ev to Equation (18.18) and using $\mathrm{ev}(U) = S$ with $X + S = F^{\langle -1 \rangle_\circ}$ gives $T_S(\varphi) = \varphi + \sum_{m\geq 1} c_m$, which is Equation (18.17). □

Corollary 18.16. *With $F = X + A$, $A \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$:*

$$F^{\langle -1 \rangle_\circ} = X + \sum_{r\geq 1} \frac{(-1)^r}{r!} D_X^{r-1}\big(A^r\big) \qquad (\varphi = X), \tag{18.19}$$

$$\log F^{\langle -1 \rangle_\circ} = L + \sum_{r\geq 1} \frac{(-1)^r}{r!} D_X^{r-1}\big(t\, A^r\big) \qquad (\varphi = L), \tag{18.20}$$

$$\big(F^{\langle -1 \rangle_\circ}\big)^{-q} = t^q - q\sum_{r\geq 1} \frac{(-1)^r}{r!} D_X^{r-1}\big(t^{q+1}A^r\big) \qquad (\varphi = t^q,\ q \in \mathbb{Z}). \tag{18.21}$$

Proof. Substitute $\varphi = X = t^{-1}$, $\varphi = L$ and $\varphi = t^q$ in Equation (18.17), using $D_X X = 1$, $D_X L = t$ and $D_X t^q = -q\, t^{q+1}$. In Equation (18.20) the left-hand side means the observable L evaluated along $F^{\langle -1 \rangle_\circ}$, that is $L \circ F^{\langle -1 \rangle_\circ} = \log\big(F^{\langle -1 \rangle_\circ}\big)$ in the sense of the generator substitution $L \circ (X + B) = L + \log(1 + tB)$. □

Example 18.17. For $A = L$, Equation (18.20) gives, with $\omega = (X + L)^{\langle -1 \rangle_\circ}$,

$$\log\omega = L - tL + t^2\Big(L - \frac{L^2}{2}\Big) + t^3\Big(-\frac{L^3}{3} + \frac{3}{2}L^2 - L\Big) + \mathcal{O}_t^{\mathrm{form}}\big(t^4\big)\,,$$

which is exactly $X - \omega$ read off from the defining relation $\omega + \log\omega = X$, and therefore a nontrivial check of both Theorem 18.15 and the coefficient table for $\mathsf{L}_n^{\mathrm{Lam}}$.

18.6 The finite coefficient formula and its exact index range

Theorem 18.18 (Coefficient formula in the polynomial–logarithmic basis). *Let $A \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$, $F = X + A$, and write*

$$A = \sum_{m\geq 0} a_m(L)\,t^m, \qquad A^r = \sum_{m\geq 0} A_{r,m}(L)\,t^m, \qquad F^{\langle -1\rangle\circ} = X + \sum_{N\geq 0} b_N(L)\,t^N.$$

Then for every $N \in \mathbb{N}_0$,

$$b_N(L) = \sum_{r=1}^{N+1} \frac{(-1)^r}{r!} \left[\prod_{s=N-r+1}^{N-1} \big(\partial_L - s\big) \right] A_{r,\,N-r+1}(L), \tag{18.22}$$

where a product whose lower bound exceeds its upper bound is empty and acts as the identity; this occurs exactly at $r = 1$.

Proof. By Theorem 18.14 and the summability of the family, the coefficient of t^N may be computed term by term. For fixed $r \geq 1$ and $m \geq 0$, the repeated-derivative formula gives

$$D_X^{r-1}\big(A_{r,m}(L)t^m\big) = t^{m+r-1}\,\Delta_{m,r-1}A_{r,m}(L), \qquad \Delta_{m,r-1} = \prod_{j=0}^{r-2}\big(\partial_L - m - j\big).$$

This contributes to t^N exactly when $m = N - r + 1$, and then the parameters $m + j$ with $0 \leq j \leq r - 2$ run over the integers $N - r + 1, N - r + 2, \ldots, N - 1$, so $\Delta_{N-r+1,r-1} = \prod_{s=N-r+1}^{N-1}(\partial_L - s)$. The constraint $m \geq 0$ is $r \leq N + 1$. For $r = 1$ the operator is $\Delta_{N,0} = 1$, the empty product. □

Proposition 18.19 (The range $r \leq N + 1$ is exactly right). *Fix $N \in \mathbb{N}_0$ and let $A \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ be nonzero.*

1. *No summand with $r > N + 1$ contributes to b_N: indeed $\nu_t\big(D_X^{r-1}(A^r)\big) \geq r - 1 > N$.*
2. *The bound cannot be lowered. For $N \geq 1$, the summand $r = N + 1$ contributes a nonzero polynomial to Equation* (18.22) *if and only if $\deg_{\mathrm{L}} a_0 \geq 1$; for $N = 0$ the only summand is $r = 1$ and $b_0 = -a_0 \neq 0$.*

In particular, for $A = L$ every $N \geq 0$ attains the bound, so no uniform improvement of the range is possible.

Proof. (1) is Theorem 18.8. For (2), the $r = N+1$ summand of Equation (18.22) is $\frac{(-1)^{N+1}}{(N+1)!}\big[\prod_{s=0}^{N-1}(\partial_L - s)\big]a_0^{N+1}$, because $A_{N+1,0} = a_0^{N+1}$. Write $q = a_0^{N+1}$; since $\mathbb{K}[L]$ is a domain, $q \neq 0$ and $\deg_{\mathrm{L}} q = (N + 1)\deg_{\mathrm{L}} a_0$. If $\deg_{\mathrm{L}} a_0 = 0$ then q is a nonzero constant and $\partial_L q = 0$, so the product vanishes. If $\deg_{\mathrm{L}} a_0 \geq 1$ then $\partial_L q \neq 0$, and for $s \geq 1$ the operator $\partial_L - s$ multiplies the leading coefficient of any nonzero polynomial by $-s \neq 0$, hence is injective; so the whole product is nonzero. For $A = L$ one has $a_0 = L$ and $\deg_{\mathrm{L}} a_0 = 1$. □

Remark 18.20. Theorem 18.19 is the exact stopping rule for computation: $\mathrm{Trunc}_{<N+1} F^{\langle -1\rangle\circ}$, that is, the inverse through outer order t^N inclusive, needs A^r only for $r \leq N + 1$ and only through outer order t^{N-r+1}. In the inclusive notation of report 6 this is $\big[F^{\langle -1\rangle\circ}\big]_{\leq N} = \mathrm{Trunc}_{<N+1} F^{\langle -1\rangle\circ}$; the one-block difference between the two conventions is a frequent source of off-by-one errors when tables are transferred between reports.

18.7 The degree law

Reversion is finite at every outer block, but the *logarithmic* degrees inside those blocks grow. Recall the log-degree profile $d_F(n) = \deg_{\mathrm{L}} p_n$, with $d_F(n) = -\infty$ when $p_n = 0$.

Theorem 18.21 (Degree law for linearly bounded profiles). *Let $A \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$, $F = X + A$, $F^{\langle -1\rangle_\circ} = X + B$. Suppose there are real $\alpha, \beta \geq 0$ with*

$$d_A(n) \leq \alpha n + \beta \qquad \textit{for all } n \in \mathbb{N}_0. \tag{18.23}$$

Then

$$d_B(N) \;\leq\; N \max\{\alpha, \beta\} + \beta \qquad \textit{for all } N \in \mathbb{N}_0. \tag{18.24}$$

More precisely, $d_B(0) \leq \beta$ and, for $N \geq 1$,

$$d_B(N) \;\leq\; \max\Big\{ (N+1)\beta - 1,\; \max_{1\leq r\leq N} \big[\alpha(N-r+1) + r\beta\big]\Big\}. \tag{18.25}$$

In particular the class of perturbations with linearly bounded log-degree profile is stable under reversion.

Proof. Every operator $\partial_L - s$ satisfies $\deg_{\mathrm{L}}\big((\partial_L - s)p\big) \leq \deg_{\mathrm{L}} p$, with the strict bound $\deg_{\mathrm{L}}(\partial_L p) \leq \deg_{\mathrm{L}} p - 1$ when $s = 0$. In Equation (18.22) the operator attached to index r is $\prod_{s=N-r+1}^{N-1}(\partial_L - s)$; the value $s = 0$ occurs in that range precisely when $N - r + 1 \leq 0 \leq N - 1$, that is (given $1 \leq r \leq N+1$) when $N \geq 1$ and $r = N + 1$.

Next, $A_{r,m} = \sum_{m_1+\cdots+m_r=m} a_{m_1}\cdots a_{m_r}$, so Equation (18.23) gives $\deg_{\mathrm{L}} A_{r,m} \leq \sum_i(\alpha m_i + \beta) = \alpha m + r\beta$. Hence the r-th summand of Equation (18.22) has degree at most $\alpha(N-r+1)+r\beta$, improved by one when $r = N+1$ and $N \geq 1$, where $m = 0$ and the bound is $(N+1)\beta - 1$. Taking the maximum over $1 \leq r \leq N+1$ gives Equation (18.25), and $d_B(0) = \deg_{\mathrm{L}}(-a_0) \leq \beta$.

For Equation (18.24), note $\alpha(N - r + 1) + r\beta = \alpha N + \alpha + r(\beta - \alpha)$ is monotone in r, so on $1 \leq r \leq N$ it is at most $\max\{\alpha N + \beta,\ \alpha + N\beta\} \leq N\max\{\alpha, \beta\} + \beta$, while $(N+1)\beta - 1 < N\beta + \beta \leq N\max\{\alpha,\beta\} + \beta$. □

The bound is attained, with an exact leading coefficient, in the situation the sources treat.

Theorem 18.22 (Exact degree and leading coefficient). *Let $A = \sum_{n\geq 0} a_n(L)t^n \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ with*

$$\deg_{\mathrm{L}} a_0 = d \geq 1, \qquad a_d := [L^d]a_0 \neq 0, \qquad \deg_{\mathrm{L}} a_n \leq d - 1 \;\textit{ for all } n \geq 1.$$

Then $b_0 = -a_0$ has degree d, and for every $N \geq 1$

$$\deg_{\mathrm{L}} b_N = d(N+1) - 1, \qquad \big[L^{\,d(N+1)-1}\big]b_N = \frac{d}{N}\, a_d^{N+1}. \tag{18.26}$$

Thus $d_B(N) = dN + (d-1)$ for $N \geq 1$: the logarithmic degree of the leading block of the perturbation becomes the slope *of the inverse's log-degree profile.*

Proof. Consider the summands of Equation (18.22) for $N \geq 1$.

The summand $r = N+1$. It equals $\frac{(-1)^{N+1}}{(N+1)!}\,\partial_L(\partial_L - 1)\cdots(\partial_L - N + 1)\,a_0^{N+1}$. The polynomial a_0^{N+1} has degree $d(N+1)$ and leading coefficient a_d^{N+1}. The factor ∂_L lowers the degree by one and multiplies the leading coefficient by $d(N+1)$; each factor $\partial_L - s$ with $1 \leq s \leq N - 1$ preserves the degree and multiplies the leading coefficient by $-s$. Hence this summand has degree $d(N+1) - 1$ and leading coefficient

$$\frac{(-1)^{N+1}}{(N+1)!}\, d(N+1)\, a_d^{N+1}\,(-1)^{N-1}(N-1)! = \frac{d(N+1)(N-1)!}{(N+1)!}\, a_d^{N+1} = \frac{d}{N}\, a_d^{N+1},$$

which is nonzero because $\mathbb{K}$ has characteristic zero.

The summands $1 \le r \le N$. Here $m = N - r + 1 \ge 1$, so in every product $a_{m_1} \cdots a_{m_r}$ occurring in $A_{r,m}$ at least one index m_i is ≥ 1 and contributes degree at most $d-1$, the remaining $r-1$ factors contributing at most d each. Therefore $\deg_{\mathrm{L}} A_{r,N-r+1} \le rd - 1 \le Nd - 1 < d(N+1) - 1$, and the attached operator does not raise degrees.

Adding the summands, the term of degree $d(N+1)-1$ survives with the stated coefficient. □

Corollary 18.23 (Pure polynomial–logarithmic perturbations). *Let $P \in \mathbb{K}[L]$ with $\deg_{\mathrm{L}} P = d \ge 1$ and leading coefficient a_d, and let $F = X + P(L)$. Then only the summand $r = N+1$ contributes to b_N, so*

$$F^{\langle -1\rangle\circ} = X - P(L) + \sum_{N\ge 1} b_N(L) t^N, \qquad b_N(L) = \frac{(-1)^{N+1}}{(N+1)!}\left[\prod_{s=0}^{N-1} (\partial_L - s)\right] P(L)^{N+1}, \tag{18.27}$$

and $\deg_{\mathrm{L}} b_N = d(N+1) - 1$ *with* $\left[L^{d(N+1)-1}\right] b_N = (d/N) a_d^{N+1}$ *for* $N \ge 1$*. For* $P = aL$ *this specialises to* $b_N = a^{N+1} \mathsf{L}_N^{\mathrm{Lam}}(L)$*, so*

$$(X + a\log X)^{\langle -1\rangle\circ}(Y) = Y + \sum_{N\ge 0} a^{N+1} \mathsf{L}_N^{\mathrm{Lam}}(\log Y)\, Y^{-N}, \qquad a \in \mathbb{K}. \tag{18.28}$$

Proof. For $A = P(L)$ one has $A_{r,m} = P^r \delta_{m,0}$, so Equation (18.22) retains only $r = N+1$, giving Equation (18.27); the degree statement is Theorem 18.22 with $a_n = 0$ for $n \ge 1$. For $P = aL$, Equation (18.27) reads $b_N = a^{N+1} \frac{(-1)^{N+1}}{(N+1)!} \left[\prod_{s=0}^{N-1} (\partial_L - s)\right] L^{N+1}$, which is the defining formula of the canonical zero-based Lambert polynomial $\mathsf{L}_N^{\mathrm{Lam}}$; the case $N = 0$ gives $\mathsf{L}_0^{\mathrm{Lam}}(L) = -L$, so the displayed sum may start at $N = 0$. □

Remark 18.24 (Expression swell, stated correctly). Theorem 18.22 is a statement about *sizes of exact symbolic objects*, not about an algorithm. It says that computing the inverse through outer order t^N requires manipulating polynomials in L of degree about dN; it says nothing about the number of arithmetic operations, and nothing whatever about the analytic size of the discarded tail. The three big-O roles must be kept apart: the tail statement $F^{\langle -1\rangle\circ} - \mathrm{Trunc}_{<N} F^{\langle -1\rangle\circ} = \mathcal{O}_t^{\mathrm{form}}\left(t^N\right)$ is a formal valuation statement with no analytic content.

18.8 Worked extraction of the first coefficients

Writing a dot for ∂_L, Equation (18.22) gives the first four universal coefficients of the inverse of $F = X + \sum_{n\ge 0} a_n(L) t^n$ directly. Grouping by the Lagrange index r:

$$b_0 = -\, a_0, \tag{18.29}$$

$$b_1 = -\, a_1 + a_0 \dot{a}_0, \tag{18.30}$$

$$b_2 = -\, a_2 + \left[\dot{a}_0 a_1 + a_0 \dot{a}_1 - a_0 a_1\right] + \left[-a_0 \dot{a}_0^2 + \tfrac{1}{2} a_0^2 \dot{a}_0 - \tfrac{1}{2} a_0^2 \ddot{a}_0\right], \tag{18.31}$$

$$\begin{aligned} b_3 = &-\, a_3 + \left[\dot{a}_0 a_2 + a_0 \dot{a}_2 + a_1 \dot{a}_1 - 2 a_0 a_2 - a_1^2\right] \\ &+ \left[3 a_0 \dot{a}_0 a_1 - a_0^2 a_1 - \dot{a}_0^2 a_1 - a_0 \ddot{a}_0 a_1 - 2 a_0 \dot{a}_0 \dot{a}_1 + \tfrac{3}{2} a_0^2 \dot{a}_1 - \tfrac{1}{2} a_0^2 \ddot{a}_1\right] \\ &+ \left[a_0 \dot{a}_0^3 - \tfrac{3}{2} a_0^2 \dot{a}_0^2 + \tfrac{1}{3} a_0^3 \dot{a}_0 + \tfrac{3}{2} a_0^2 \dot{a}_0 \ddot{a}_0 - \tfrac{1}{2} a_0^3 \ddot{a}_0 + \tfrac{1}{6} a_0^3 \dddot{a}_0\right]. \end{aligned} \tag{18.32}$$

In each line the successive bracketed groups are the contributions of $r = 2, 3, 4$; the leading $-a_N$ is the contribution of $r = 1$.

Derivation. Apply Equation (18.22). For $N=0$ only $r=1$ occurs and the operator is empty. For $N=1$, $r=1$ gives $-A_{1,1}=-a_1$ and $r=2$ gives $\frac{1}{2}\partial_L(a_0^2)=a_0\dot a_0$. For $N=2$: $r=1$ gives $-a_2$; $r=2$ gives $\frac{1}{2}(\partial_L-1)A_{2,1}=\frac{1}{2}(\partial_L-1)(2a_0a_1)$, which is the first bracket; $r=3$ gives $-\frac{1}{6}\partial_L(\partial_L-1)a_0^3$, and with $\partial_L(\partial_L-1)q=\ddot q-\dot q$ applied to $q=a_0^3$, $\dot q=3a_0^2\dot a_0$ and $\ddot q=6a_0\dot a_0^2+3a_0^2\ddot a_0$, this is the second bracket. For $N=3$: $r=1$ gives $-a_3$; $r=2$ gives $\frac{1}{2}(\partial_L-2)A_{2,2}$ with $A_{2,2}=2a_0a_2+a_1^2$; $r=3$ gives $-\frac{1}{6}(\partial_L-1)(\partial_L-2)A_{3,1}$ with $A_{3,1}=3a_0^2a_1$ and $(\partial_L-1)(\partial_L-2)q=\ddot q-3\dot q+2q$; $r=4$ gives $\frac{1}{24}\partial_L(\partial_L-1)(\partial_L-2)a_0^4$ with $\partial_L(\partial_L-1)(\partial_L-2)q=\dddot q-3\ddot q+2\dot q$, and $\dot q=4a_0^3\dot a_0$, $\ddot q=12a_0^2\dot a_0^2+4a_0^3\ddot a_0$, $\dddot q=24a_0\dot a_0^3+36a_0^2\dot a_0\ddot a_0+4a_0^3\dddot a_0$. Collecting gives the displayed brackets. □

Example 18.25 (A fully mixed perturbation). Let

$$F(X)=X+2\log X+1+\frac{3\log X-1}{X}+\frac{(\log X)^2+2}{X^2},$$

so $a_0=2L+1$, $a_1=3L-1$, $a_2=L^2+2$, $a_n=0$ for $n\geq 3$. Then Equation (18.29)–Equation (18.32) give

$$\begin{aligned}F^{\langle -1\rangle\circ}(Y)&=Y-2M-1+\frac{M+3}{Y}+\frac{-3M^2+7M-3}{Y^2}\\&\quad+\frac{-\frac{32}{3}M^3+25M^2-6M-\frac{59}{6}}{Y^3}+\mathcal{O}_t^{\mathrm{form}}\big(t^4\big),\qquad M=\log Y.\end{aligned}\tag{18.33}$$

Grouped by Lagrange index, the contributions to the block t^3 are $-4M^3-5M^2+9M-4$ from $r=2$, $-12M^3+46M^2-11M-\frac{17}{2}$ from $r=3$, and $\frac{16}{3}M^3-16M^2-4M+\frac{8}{3}$ from $r=4$; nothing comes from $r=1$. A summand that is "later" in the Lagrange index therefore still reaches the same outer block: the stopping rule is the filtration bound of Theorem 18.19, never the number of displayed summands.

Example 18.26 (A logarithmic–rational family). For $F(X)=X+a\log X+bX^{-1}+cX^{-2}$, that is $a_0=aL$, $a_1=b$, $a_2=c$,

$$\begin{aligned}F^{\langle -1\rangle\circ}(Y)&=Y-aM+\frac{a^2M-b}{Y}+\frac{a^3\big(\frac{1}{2}M^2-M\big)-abM+ab-c}{Y^2}\\&\quad+\frac{a^4\big(\frac{1}{3}M^3-\frac{3}{2}M^2+M\big)+a^2b\big(-M^2+3M-1\big)-2acM+ac-b^2}{Y^3}+\mathcal{O}_t^{\mathrm{form}}\big(t^4\big).\end{aligned}\tag{18.34}$$

Setting $b=c=0$ recovers Equation (18.28); setting $a=c=0$ recovers Theorem 18.27 with $\kappa=b$. The parameter c, which enters F at outer order t^2, first perturbs the inverse at the same order, and the product b^2 first appears at t^3, where two rational perturbations can meet.

Example 18.27 (An exact check: Catalan numbers). For $F(X)=X+\kappa X^{-1}$, Equation (18.13) with $A=\kappa t$ and $D_X^{r-1}(t^r)=t^{2r-1}\prod_{j=0}^{r-2}(-r-j)=(-1)^{r-1}\frac{(2r-2)!}{(r-1)!}t^{2r-1}$ gives

$$F^{\langle -1\rangle\circ}(Y)=Y-\sum_{r\geq 1}\frac{(2r-2)!}{r!\,(r-1)!}\kappa^r\,Y^{-(2r-1)}=Y-\frac{\kappa}{Y}-\frac{\kappa^2}{Y^3}-\frac{2\kappa^3}{Y^5}-\frac{5\kappa^4}{Y^7}-\cdots,$$

the coefficients $(2r-2)!/\big(r!\,(r-1)!\big)$ being the Catalan numbers. This matches the branch $\frac{1}{2}\big(Y+\sqrt{Y^2-4\kappa}\big)$ of $Z+\kappa Z^{-1}=Y$ that is asymptotic to Y, and simultaneously checks the signs, the derivative order $r-1$, and the branch selection implicit in the near-identity normalisation.

The coefficients produced by Theorem 18.18 are certified by a finite computation, and the certificate is the one already established in the reversion part: for $F=X+A\in\mathcal{G}_{\mathrm{poly},\mathbb{K}}$ and $G_N=X+\sum_{j=0}^{N}g_j(L)t^j$, the three conditions $G_N=\mathrm{Trunc}_{<N+1}F^{\langle -1\rangle\circ}$, $\mathrm{Err}_{\mathrm{rev}}(F,G_N)=\mathcal{O}_t^{\mathrm{form}}\big(t^{N+1}\big)$ and

$G_N \circ F - X = \mathcal{O}_t^{\mathrm{form}}\big(t^{N+1}\big)$ are equivalent (Theorem 16.16); moreover the order of either residual equals $\nu_t(G_N - F^{\langle -1\rangle\circ})$ exactly, with the same leading block (Theorem 16.15). Nothing in that argument uses the Lagrange–Bürmann representation, so it certifies coefficients obtained by any of the routes of this part.

Example 18.28. Substituting Equation (18.33) back into the F of Theorem 18.25, using the exact generator substitutions $t \circ G = t/(1+U)$ and $L \circ G = L + \log(1+U)$ with $U = t(G - Y)$, gives

$$F \circ G - Y = \Big(28M^4 - \tfrac{211}{3}M^3 - 15M^2 + 91M - \tfrac{59}{3}\Big)t^4 + \mathcal{O}_t^{\mathrm{form}}\big(t^5\big)\,.$$

The vanishing through t^3 certifies every coefficient claimed in Equation (18.33); the nonzero t^4 block is expected, being the first block not represented in G. Computing one residual block beyond the target is the cheapest available check, and it detects exactly the errors that a second run of the same code would not.

What the formula does and does not say

For $F = X + A$ with $A \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$, $F^{\langle -1\rangle\circ} = X + \sum_{r\geq 1} \frac{(-1)^r}{r!} D_X^{r-1}(A^r)$, and the block t^N is a finite sum over $1 \leq r \leq N+1$. Summability holds precisely because $\nu_t(A) \geq 0$, and *only* then (Theorem 18.9). The same construction transports any observable $\varphi \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ along the inverse without building it (Theorem 18.15). Logarithmic degrees grow linearly, with slope $\deg_{\mathrm{L}} a_0$ (Theorem 18.22). None of this asserts convergence, an analytic inverse branch, or any bound uniform in L.

Chapter 19

Reversion beyond the identity scale

Part 18 inverts $F = X + A$ with $\nu_t(A) \geq 0$. Two things can go wrong outside that class: the leading exponent of F need not be 1, and the leading coefficient block need not be a scalar. This chapter shows that these are the *only* two obstructions, proves a criterion that decides in advance which enlargement of $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ the inverse will require, turns dominant balance into an algorithm with a proof attached, and exhibits the way a careless scale change silently corrupts an otherwise correct near-identity computation.

Throughout this chapter, composition means composition of maps: whenever we write $F \circ G$ for germs at $+\infty$, associativity and the functorial identity

$$(F \circ G)^{\langle -1 \rangle_\circ} = G^{\langle -1 \rangle_\circ} \circ F^{\langle -1 \rangle_\circ} \tag{19.1}$$

are the ordinary group-theoretic facts, valid in any group of invertible maps, and are used without further comment for genuine germs. For formal series they hold in the near-identity group of any reversion frame by Theorem 18.5(5), and more generally whenever all displayed composites are admissible in the chosen support class.

19.1 Monomial-leading maps and the closure criterion

Definition 19.1 (Monomial-leading map). Let $\Gamma \subseteq \mathbb{Q}$ be an exponent group. A series G is *monomial-leading of exponent* μ if

$$G = dX^\mu\bigl(1 + V\bigr), \qquad d \in \mathbb{K}^\times, \quad \mu \in \Gamma_{>0}, \quad \nu_t(V) > 0, \tag{19.2}$$

with $\log d \in \mathbb{K}$. This is the class for which the generator substitutions

$$t \circ G = d^{-1} t^\mu (1 + V)^{-1}, \qquad L \circ G = \mu L + \log d + \log(1 + V) \tag{19.3}$$

stay inside the ambient coefficient ring, and hence the class for which composition with an outer polynomial–logarithmic series is defined.

Remark 19.2. The scalar hypothesis $d \in \mathbb{K}^\times$ cannot be relaxed to a polynomial leading block. If $G = X^\mu q(L)(1 + V)$ with $q \in \mathbb{K}[L]$ of degree $e \geq 1$ and leading coefficient q_e, then $L \circ G = \mu L + \log q(L) + \log(1 + V)$ and $\log q(L) = e \log L + \log q_e + \mathcal{O}_{X\to+\infty}\bigl(L^{-1}\bigr)$, so a new generator $\log L$ appears. That generator is genuinely new: if $\log L = p(L) + R$ with $p \in \mathbb{K}[L]$ and $R \to 0$ as $X \to +\infty$, then $\log L - p(L) \to 0$, which is impossible, since for constant p the difference tends to $+\infty$ and for $\deg_{\mathrm{L}} p \geq 1$ it tends to $\pm\infty$. This is the same argument used in Theorem 19.4.

Theorem 19.3 (Closure criterion for reversion in the one-logarithm ring)**.** *Let $\Gamma \subseteq \mathbb{Q}$ be an exponent group containing $\mathbb{Z}$, let $\mathfrak{R}$ be the corresponding Puiseux–log ring (so $\mathfrak{R} = \mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ when $\Gamma = \mathbb{Z}$, and $\mathfrak{R} = \mathbb{K}[L]((t^{1/s}))$ when $\Gamma = s^{-1}\mathbb{Z}$), and let*

$$F = cX^\rho L^\sigma\bigl(1 + U\bigr), \qquad c \in \mathbb{K}^\times, \quad \rho \in \Gamma_{>0}, \quad \sigma \in \mathbb{N}_0, \quad \nu_t(U) > 0. \tag{19.4}$$

Then F admits a monomial-leading right inverse G in the sense of Theorem 19.1, over $\mathfrak{R}$ or over any extension of $\mathfrak{R}$ obtained by enlarging the coefficient field alone, if and only if

$$\rho^{-1} \in \Gamma \qquad \text{and} \qquad \sigma = 0. \tag{19.5}$$

When Equation (19.5) holds, the exponent of G is $\mu = \rho^{-1}$ and its leading scalar is any d with $cd^\rho = 1$, so the construction requires a field $\mathbb{K}' \supseteq \mathbb{K}$ containing such a d together with $\log d$. In particular:

1. *in $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ itself $(\Gamma = \mathbb{Z})$, the criterion is $\rho = 1$ and $\sigma = 0$, that is, $c^{-1}F \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}(\mathscr{PL}_{\mathrm{pow},\mathbb{K}})$;*
2. *enlarging the exponent group to $\rho^{-1}\mathbb{Z}$ removes the exponent obstruction but never the obstruction $\sigma \geq 1$.*

Proof. Write $F(Y) = \sum_{n \geq -\rho} p_n(\log Y)Y^{-n}$ with $p_{-\rho}(\lambda) = c\lambda^\sigma$ and $\nu_t(F - cX^\rho L^\sigma) > -\rho$, and let $G = dX^\mu(1+V)$ be monomial-leading. By Equation (19.3) and the monomial-leading composition formula,

$$F \circ G = \sum_{n \geq -\rho} p_n\big(\mu L + \log d + \log(1+V)\big) d^{-n} t^{\mu n}(1+V)^{-n}.$$

Since $\nu_t(V) > 0$ and $\nu_t \log(1+V) > 0$, the term $n = -\rho$ contributes

$$c\, d^\rho \left(\mu L + \log d\right)^\sigma t^{-\mu\rho} + \big(\text{terms of } \nu_t > -\mu\rho\big),$$

and every term with $n > -\rho$ has $\nu_t \geq -\mu n > -\mu\rho$. Hence $\nu_t(F \circ G) = -\mu\rho$ and the leading block of $F \circ G$ is $cd^\rho(\mu L + \log d)^\sigma$.

Requiring $F \circ G = X = t^{-1}$ forces $\mu\rho = 1$, so $\mu = \rho^{-1}$ must lie in Γ, and forces

$$c\, d^\rho \left(\mu L + \log d\right)^\sigma = 1. \tag{19.6}$$

If $\sigma \geq 1$, the left side of Equation (19.6) is a polynomial in L of degree $\sigma \geq 1$ with leading coefficient $cd^\rho \mu^\sigma \neq 0$, so it cannot be the constant 1; hence $\sigma = 0$, and then Equation (19.6) reads $cd^\rho = 1$.

Conversely, suppose $\rho^{-1} \in \Gamma$ and $\sigma = 0$. Choose d with $cd^\rho = 1$ in a finite extension $\mathbb{K}' \supseteq \mathbb{K}$ also containing $\log d$, and set $F_0 = cX^\rho$, $F_0^{\langle -1 \rangle \circ} = dX^{1/\rho}$. Then $\Phi := F \circ F_0^{\langle -1 \rangle \circ}$ satisfies $\Phi = cd^\rho X\big(1 + U \circ F_0^{\langle -1 \rangle \circ}\big) = X\big(1 + \widetilde{U}\big)$ with $\nu_t(\widetilde{U}) > 0$, because $\nu_t\big(U \circ F_0^{\langle -1 \rangle \circ}\big) = \rho^{-1}\nu_t(U) > 0$. Hence $\Phi \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}$ of the Puiseux–log frame $\mathbb{K}'[L]((t^{1/s}))$, s a common denominator of ρ^{-1} and the exponents of $\widetilde{U}$, and Theorem 18.14 produces $\Phi^{\langle -1 \rangle \circ}$. By Equation (19.1), $F^{\langle -1 \rangle \circ} = F_0^{\langle -1 \rangle \circ} \circ \Phi^{\langle -1 \rangle \circ}$, which is monomial-leading of exponent ρ^{-1}. ☐

Proposition 19.4 (A leading logarithm forces a nested logarithm). *Let $F = cX^\rho L^\sigma(1+U)$ as in Equation (19.4) with $\sigma \geq 1$, and let G be a germ at $+\infty$ with $F(G(Y)) = Y$ for all large Y. Put $S = \log Y$ and $u = \log G(Y)$. Then*

$$u = \frac{1}{\rho}\Big(S - \sigma \log S + \sigma \log \rho - \log c\Big) + \mathcal{O}_{S \to +\infty}\Big(\tfrac{\log S}{S}\Big), \tag{19.7}$$

so $\log G$ contains the term $-(\sigma/\rho)\log\log Y$ with nonzero coefficient. Consequently G lies in no Puiseux–log ring $\mathbb{K}'[S]((Y^{-1/s}))$ with $\mathbb{K}' \supseteq \mathbb{K}$ and $s \in \mathbb{N}_{>0}$: a second logarithm is unavoidable.

Proof. Taking logarithms in $F(G(Y)) = Y$ and writing $u = \log G(Y)$,

$$S = \log c + \rho u + \sigma \log u + \log\big(1 + U(G(Y))\big). \tag{19.8}$$

Since $\nu_t(U) > 0$ there is $\delta > 0$ with $U(X) = \mathcal{O}_{X \to +\infty}\big(X^{-\delta}\big)$, so the last term of Equation (19.8) is $\mathcal{O}_{Y \to +\infty}\big(G(Y)^{-\delta}\big)$; since $G(Y) \to +\infty$ like a positive power of Y, it is smaller than every power of $1/S$. From Equation (19.8), $\rho u \sim S$, hence $\log u = \log S - \log \rho + o_{S \to +\infty}(1)$; substituting back, $\rho u = S - \log c - \sigma \log S + \sigma \log \rho + o_{S \to +\infty}(1)$, which is Equation (19.7).

Suppose $G \in \mathbb{K}'[S]((Y^{-1/s}))$. Then $G = q(S)Y^{-\lambda}\big(1 + o_{Y\to+\infty}(1)\big)$ for some $\lambda \in s^{-1}\mathbb{Z}$ and some nonzero $q \in \mathbb{K}'[S]$, so

$$u = \log G = -\lambda S + \log q(S) + o_{Y\to+\infty}(1) = -\lambda S + (\deg q)\log S + \log(\operatorname{lc} q) + o_{Y\to+\infty}(1)\,.$$

Comparing with Equation (19.7), the coefficients of S give $\lambda = -1/\rho$ and those of $\log S$ give $\deg q = -\sigma/\rho < 0$, which is impossible for a polynomial. □

Corollary 19.5 (Which enlargement is needed)**.** *For F as in Equation* (19.4)*:*

leading data	*obstruction*	*smallest ambient for $F^{\langle -1\rangle\circ}$*
$\rho = 1,\ \sigma = 0$	*none*	$\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ *over* $\mathbb{K}(c^{-1}, \log c)$
$\rho \in \mathbb{Q}_{>0},\ \sigma = 0$	*exponent*	$\mathbb{K}'[L]((t^{1/s}))$, s *a denominator of* ρ^{-1}
$\rho \notin \mathbb{Q},\ \sigma = 0$	*exponent*	*Hahn power-log with* $\rho^{-1}\mathbb{Z} \subseteq \Gamma$
$\sigma \geq 1$	*logarithm*	*nested-logarithmic;* $\log\log X$ *required*

Proof. Rows one to three are Theorem 19.3 together with the explicit construction in its converse half; row four is Theorem 19.4 together with Theorem 19.7 below, which produces the inverse in the nested-logarithmic frame. □

19.2 Dominant balance as a theorem

The workflow "guess the leading behaviour, normalise, iterate" becomes a theorem once one names the object that does the normalising.

Theorem 19.6 (Reduction to the near-identity case)**.** *Let $\mathfrak{G}$ be a group of invertible germs at $+\infty$ (or of formal maps in a fixed support class, with all displayed composites admissible), let $F \in \mathfrak{G}$, and let $F_0 \in \mathfrak{G}$ be any* leading model*, that is, any element whose inverse is known. Put*

$$\Phi := F \circ F_0^{\langle -1\rangle\circ}. \tag{19.9}$$

Then $F = \Phi \circ F_0$ and

$$F^{\langle -1\rangle\circ} = F_0^{\langle -1\rangle\circ} \circ \Phi^{\langle -1\rangle\circ}. \tag{19.10}$$

If moreover $\Phi = X + A$ with A in the nonnegative part of some reversion frame, then $\Phi^{\langle -1\rangle\circ}$ is given by Equation (18.12)*, and*

$$F^{\langle -1\rangle\circ} = F_0^{\langle -1\rangle\circ} \circ \Big(X + \sum_{r\geq 1} \frac{(-1)^r}{r!}\partial^{r-1}\big(A^r\big)\Big). \tag{19.11}$$

The dual normalisation $\Psi := F_0^{\langle -1\rangle\circ} \circ F$ gives $F = F_0 \circ \Psi$ and $F^{\langle -1\rangle\circ} = \Psi^{\langle -1\rangle\circ} \circ F_0^{\langle -1\rangle\circ}$.

Proof. $\Phi \circ F_0 = F \circ F_0^{\langle -1\rangle\circ} \circ F_0 = F$. Applying Equation (19.1) to $F = \Phi \circ F_0$ gives $F^{\langle -1\rangle\circ} = F_0^{\langle -1\rangle\circ} \circ \Phi^{\langle -1\rangle\circ}$. The last two statements are the same computation with the factors exchanged, and Equation (19.11) is Theorem 18.14 substituted into Equation (19.10). □

Theorem 19.6 is content-free until a leading model is produced. For a monomial-leading F the model is supplied by one canonical change of scale, and this is where dominant balance actually happens.

Theorem 19.7 (Logarithmic normal form of a power–log leading term)**.** *Let $c \in \mathbb{K}^\times$ with $\log c \in \mathbb{K}$, let $\rho \in \mathbb{Q}_{>0}$ with $\log\rho \in \mathbb{K}$, let $\sigma \in \mathbb{N}_0$, and set $F_0(X) = cX^\rho(\log X)^\sigma$. Write $E(X) = \mathrm{e}^X$ and $m_\rho(X) = \rho X$. Then*

$$G := m_\rho^{\langle -1\rangle\circ} \circ \log \circ F_0 \circ E \;=\; u + \frac{\sigma}{\rho}\log u + \frac{\log c}{\rho} \tag{19.12}$$

as a map in the large variable u; in particular G lies in the near-identity group of the reversion frame $\mathbb{K}[\log u][[u^{-1}]]$ with perturbation $A(u) = (\sigma/\rho)\log u + (\log c)/\rho$, and

$$F_0^{\langle -1\rangle\circ} = E \circ G^{\langle -1\rangle\circ} \circ m_{1/\rho} \circ \log . \tag{19.13}$$

Consequently, with $S = \log Y$ and $\Lambda = \log S$,

$$\begin{aligned}\log F_0^{\langle -1\rangle\circ}(Y) = \frac{1}{\rho}\Big[\, & S - \sigma\Lambda + \sigma\log\rho - \log c \\ & + \frac{\sigma^2\Lambda + \sigma\log c - \sigma^2\log\rho}{S} + \mathcal{O}^{\mathrm{form}}_{S^{-1}}\big(S^{-2}\big)\Big],\end{aligned} \tag{19.14}$$

$$F_0^{\langle -1\rangle\circ}(Y) = c^{-1/\rho}\rho^{\sigma/\rho}Y^{1/\rho}S^{-\sigma/\rho}\Big[1 + \frac{\sigma^2\Lambda + \sigma\log c - \sigma^2\log\rho}{\rho S} + \mathcal{O}^{\mathrm{form}}_{S^{-1}}\big(S^{-2}\big)\Big]. \tag{19.15}$$

Proof. For $X = \mathrm{e}^u$, $\log F_0(\mathrm{e}^u) = \log c + \rho u + \sigma\log u$. Post-composing with $m_\rho^{\langle -1\rangle\circ} = m_{1/\rho}$ divides by ρ and gives Equation (19.12). Since $\log u \in \mathbb{K}[\log u][[u^{-1}]]$ has zero outer order in the variable u, the perturbation A has $\nu(A) = 0$, so $G \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}$ of that frame. Reading Equation (19.12) as $G = m_{1/\rho} \circ \log \circ F_0 \circ E$ and inverting with Equation (19.1), using $E^{\langle -1\rangle\circ} = \log$ and $(\log)^{\langle -1\rangle\circ} = E$,

$$G^{\langle -1\rangle\circ} = E^{\langle -1\rangle\circ} \circ F_0^{\langle -1\rangle\circ} \circ (\log)^{\langle -1\rangle\circ} \circ (m_{1/\rho})^{\langle -1\rangle\circ} = \log \circ F_0^{\langle -1\rangle\circ} \circ E \circ m_\rho,$$

which rearranges to Equation (19.13).

For the expansions, apply Equation (18.29) and Equation (18.30) to $A(u) = a_0(\log u)$ with $a_0(\lambda) = (\sigma/\rho)\lambda + (\log c)/\rho$ and $a_n = 0$ for $n \geq 1$:

$$b_0 = -a_0, \qquad b_1 = a_0\dot{a}_0 = \frac{\sigma}{\rho}\Big(\frac{\sigma}{\rho}\lambda + \frac{\log c}{\rho}\Big) = \frac{\sigma}{\rho^2}\big(\sigma\lambda + \log c\big).$$

Hence, writing v for the large variable of $G^{\langle -1\rangle\circ}$ and $\lambda = \log v$,

$$G^{\langle -1\rangle\circ}(v) = v - \frac{\sigma\lambda + \log c}{\rho} + \frac{\sigma(\sigma\lambda + \log c)}{\rho^2 v} + \mathcal{O}^{\mathrm{form}}_{v^{-1}}\big(v^{-2}\big)\,.$$

By Equation (19.13), $\log F_0^{\langle -1\rangle\circ}(Y) = G^{\langle -1\rangle\circ}(S/\rho)$. Substituting $v = S/\rho$, so $\lambda = \Lambda - \log\rho$ and $v^{-1} = \rho/S$,

$$\log F_0^{\langle -1\rangle\circ}(Y) = \frac{S}{\rho} - \frac{\sigma(\Lambda - \log\rho) + \log c}{\rho} + \frac{\sigma\big(\sigma(\Lambda - \log\rho) + \log c\big)}{\rho S} + \mathcal{O}^{\mathrm{form}}_{S^{-1}}\big(S^{-2}\big)\,,$$

which is Equation (19.14). Exponentiating and using $\mathrm{e}^{S/\rho} = Y^{1/\rho}$, $\mathrm{e}^{-\sigma\Lambda/\rho} = S^{-\sigma/\rho}$, and $\mathrm{e}^{(\sigma\log\rho - \log c)/\rho} = \rho^{\sigma/\rho}c^{-1/\rho}$, gives Equation (19.15), the exponential of the S^{-1} block being expanded as $1 + w + \mathcal{O}^{\mathrm{form}}_{S^{-1}}\big(S^{-2}\big)$. □

Remark 19.8. Equation (19.12) explains the "one level down" phenomenon: after conjugation by e, the expansion parameter is $1/\log Y$ and the coefficients are polynomials in $\log\log Y$. The polynomial–logarithmic calculus is applied in the frame of the variable $u = \log X$, not of X. Note also that Equation (19.12) is exact only for $U = 0$; for $F = F_0\cdot(1+U)$ with $\nu_t(U) > 0$ the conjugated map carries an extra term $\rho^{-1}\log(1 + U(\mathrm{e}^u))$, which is smaller than every power of u^{-1} and therefore invisible in the frame $\mathbb{K}[\log u][[u^{-1}]]$. That term is not negligible after exponentiating back, and it is precisely what the second stage Equation (19.9) of Theorem 19.6 handles: with $\Phi = F \circ F_0^{\langle -1\rangle\circ} = X(1 + U \circ F_0^{\langle -1\rangle\circ})$, the two scales $1/\log Y$ and $Y^{-1/\rho}$ are separated rather than mixed.

Algorithm 19.9 (Reversion with a nonidentity leading term)**.** Input: a germ F at $+\infty$ presented as $F = cX^\rho L^\sigma(1 + U)$ with $c \in \mathbb{K}^\times$, $\rho \in \mathbb{Q}_{>0}$, $\sigma \in \mathbb{N}_0$, $\nu_t(U) > 0$; a target outer order N.

1. **Extend the coefficient field.** Replace $\mathbb{K}$ by a field containing $\log c$, $\log \rho$ and $c^{-1/\rho}$. Record the branch of every root and logarithm chosen.

2. **Fix the ambient.** If $\sigma = 0$ and $\rho = 1$, the ambient is $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$. If $\sigma = 0$ and $\rho = p/q$ in lowest terms, the ambient is $\mathbb{K}[L]((t^{1/p}))$. If $\sigma \geq 1$, the ambient is the nested-logarithmic frame of Theorem 19.7. (Theorem 19.3 and Theorem 19.5.)

3. **Invert the leading model.** Set $F_0 = cX^\rho L^\sigma$. If $\sigma = 0$, $F_0^{\langle -1\rangle\circ} = c^{-1/\rho}X^{1/\rho}$. If $\sigma \geq 1$, obtain $F_0^{\langle -1\rangle\circ}$ from Equation (19.13): revert $G(u) = u + (\sigma/\rho)\log u + (\log c)/\rho$ by Theorem 18.14 in the u-frame to the required order, then evaluate at $u = S/\rho$ and exponentiate.

4. **Normalise.** Form $\Phi = F \circ F_0^{\langle -1\rangle\circ}$. By construction $\Phi = X\bigl(1 + U \circ F_0^{\langle -1\rangle\circ}\bigr)$, which is near-identity in the ambient of step 2.

5. **Revert the near-identity factor.** Apply Theorem 18.14 or the coefficient formula Equation (18.22) to Φ, through the order fixed by Theorem 18.19.

6. **Recompose.** Return $F^{\langle -1\rangle\circ} = F_0^{\langle -1\rangle\circ} \circ \Phi^{\langle -1\rangle\circ}$, as in Equation (19.10). *Both* generators must be substituted: t and L change together, by Equation (19.3).

7. **Certify.** Verify $F \circ F^{\langle -1\rangle\circ} - X = \mathcal{O}_t^{\mathrm{form}}\bigl(t^{N+1}\bigr)$ and, independently, $F^{\langle -1\rangle\circ} \circ F - X = \mathcal{O}_t^{\mathrm{form}}\bigl(t^{N+1}\bigr)$, as in Theorem 16.16. The two residuals detect different substitution-order errors.

19.3 Worked examples

Example 19.10 (Leading behaviour $X + c\log X$: inside the theory)**.** The map $F(X) = X + a\log X + b$ with $a \neq 0$ has $A = aL + b \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$, so $\nu_t(A) = 0$ and no enlargement is needed, even though $A \to +\infty$. By Theorem 18.23 with $P = aL + b$, the inverse is

$$F^{\langle -1\rangle\circ}(Y) = Y - aM - b + \frac{a(aM+b)}{Y} + \frac{a(aM+b)(aM+b-2a)}{2Y^2} + \frac{a(aM+b)}{6Y^3}\bigl(2a^2M^2 - 9a^2M + 4abM + 6a^2 - 9ab + 2b^2\bigr) + \mathcal{O}_t^{\mathrm{form}}\bigl(t^4\bigr)\,, \tag{19.16}$$

with $M = \log Y$, and $\deg_{\mathrm{L}} b_N = 2(N+1) - 1$ when $\deg P = 1\ldots$ no: with $d = \deg_{\mathrm{L}} P = 1$, Equation (18.26) gives $\deg_{\mathrm{L}} b_N = N$ and $\bigl[M^N\bigr]b_N = a^{N+1}/N$ for $N \geq 1$. The family also has an exact closed form: $Y = X + a\log X + b$ is equivalent to $\frac{X}{a}\mathrm{e}^{X/a} = \frac{1}{a}\mathrm{e}^{(Y-b)/a}$, whence

$$F^{\langle -1\rangle\circ}(Y) = a\,W_k\Bigl(\tfrac{1}{a}\mathrm{e}^{(Y-b)/a}\Bigr), \tag{19.17}$$

with the branch k selected by the intended real interval or sector. Equation (19.16) is the large-Y expansion of Equation (19.17); the formal calculation does not choose k [5, 11, 7].

Example 19.11 (Leading behaviour cX^ρ: Puiseux)**.** Let $F(X) = X^2 + X$. Here $\rho = 2$, $\sigma = 0$, $c = 1$, so Theorem 19.3 forbids an inverse in $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ and Theorem 19.5 prescribes $\mathbb{K}[L]((t^{1/2}))$. Theorem 19.9 gives $F_0 = X^2$, $F_0^{\langle -1\rangle\circ} = X^{1/2}$, $\Phi = F \circ F_0^{\langle -1\rangle\circ} = X + X^{1/2}$, so $A = X^{1/2} = t^{-1/2}$ has $\nu_t(A) = -\frac{1}{2} < 0$ and step 4 fails: the normalisation must be redone with $F_0 = X^2 + X$ treated exactly, or, more simply, with the substitution $X = Z - \frac{1}{2}$. Doing the latter, $F = Z^2 - \frac{1}{4}$, and

$$F^{\langle -1\rangle\circ}(Y) = \Bigl(Y + \tfrac{1}{4}\Bigr)^{1/2} - \tfrac{1}{2} = Y^{1/2} - \frac{1}{2} + \frac{1}{8}Y^{-1/2} - \frac{1}{128}Y^{-3/2} + \cdots\,,$$

which is the branch $\frac{1}{2}\bigl(-1 + \sqrt{1+4Y}\bigr)$ asymptotic to $Y^{1/2}$. No integer-power ansatz can reproduce it.

Remark 19.12. Theorem 19.11 shows that the normalisation step of Theorem 19.9 must use the *full* leading monomial $cX^\rho L^\sigma$ of F, not merely a convenient power. In Theorem 19.3 the identity $\Phi = X\big(1 + U \circ F_0^{\langle -1\rangle\circ}\big)$ holds because $F = F_0(1+U)$ exactly; if $F - F_0$ is not $o_{X\to+\infty}(F_0)$, the step produces a Φ outside $\mathcal{G}_{\mathrm{poly},\mathbb{K}}$ and Theorem 18.9 then says that Equation (18.12) is not available.

Example 19.13 (Leading behaviour $X \log X$: nested logarithm)**.** Let $F(X) = X\log X$, so $\rho = 1$, $\sigma = 1$, $c = 1$. By Theorem 19.3 there is no monomial-leading inverse in $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$, and by Theorem 19.4 none in any Puiseux–log ring. Theorem 19.7 gives the leading model in logarithmic coordinates: $G(u) = u + \log u$, whose inverse is Wright omega, so

$$F^{\langle -1\rangle\circ}(Y) = \mathrm{e}^{\omega(S)} = \frac{Y}{W(Y)}, \qquad S = \log Y, \tag{19.18}$$

the second equality because $Z\log Z = Y$ with $Z = \mathrm{e}^w$ gives $w\mathrm{e}^w = Y$. Expanding $W(Y) = S - \Lambda + \sum_{n\ge 1} \mathsf{L}_n^{\mathrm{Lam}}(\Lambda)S^{-n}$ with $\Lambda = \log S$ and inverting the resulting series,

$$F^{\langle -1\rangle\circ}(Y) = \frac{Y}{S}\left(1 + \frac{\Lambda}{S} + \frac{\Lambda^2 - \Lambda}{S^2} + \mathcal{O}_{Y\to+\infty}\Big(\tfrac{\Lambda^3}{S^3}\Big)\right). \tag{19.19}$$

Every coefficient is a polynomial in $\Lambda = \log\log Y$; a one-logarithm ansatz, however many terms it carries, cannot represent Equation (19.19). The same mechanism produces depth two in the inverses of Dulac transseries with a logarithm in the leading monomial [24, 27].

Example 19.14 (General power–log monomial)**.** For $Y = cX^\rho(\log X)^\sigma$ with $c, \rho > 0$, Equation (19.15) gives, in fully explicit form,

$$X = c^{-1/\rho}\rho^{\sigma/\rho}Y^{1/\rho}(\log Y)^{-\sigma/\rho}\left[1 + \frac{\sigma^2\log\log Y + \sigma\log c - \sigma^2\log\rho}{\rho\log Y} + \mathcal{O}_{Y\to+\infty}\Big(\tfrac{(\log\log Y)^2}{(\log Y)^2}\Big)\right].$$

An exact Lambert form is also available when $\sigma \neq 0$: putting $s = \log X$, the equation becomes $(Y/c)^{1/\sigma} = s\mathrm{e}^{(\rho/\sigma)s}$, so

$$X = \exp\left[\frac{\sigma}{\rho}W_k\left(\frac{\rho}{\sigma}\Big(\frac{Y}{c}\Big)^{1/\sigma}\right)\right]. \tag{19.20}$$

Because the argument of W contains a power of Y, the expansion of Equation (19.20) contains $\log Y$ and $\log\log Y$: a second logarithm is forced whenever $\sigma \neq 0$, in agreement with Theorem 19.4.

A scale change that silently corrupts a correct series

Let $F(X) = 2X + \log X$. This is not near-identity, but it is affine in the leading scale, so Theorem 19.6 applies with the model $F_0 = m_2$, $m_2(Z) = 2Z$. Writing $G(X) = X + \frac{1}{2}\log X$, one has $F = m_2 \circ G$ and therefore

$$F^{\langle -1\rangle\circ}(Y) = G^{\langle -1\rangle\circ}\big(Y/2\big). \tag{19.21}$$

By Equation (18.28) with $a = \frac{1}{2}$, $G^{\langle -1\rangle\circ}(Z) = Z + \sum_{N\ge 0} 2^{-(N+1)}\mathsf{L}_N^{\mathrm{Lam}}(\log Z)Z^{-N}$. Substituting $Z = Y/2$ changes *both* generators: $Z^{-N} = 2^N Y^{-N}$ *and* $\log Z = \log Y - \log 2$. Hence

$$F^{\langle -1\rangle\circ}(Y) = \frac{Y}{2} + \frac{1}{2}\sum_{N\ge 0}\mathsf{L}_N^{\mathrm{Lam}}\big(\log Y - \log 2\big)Y^{-N}. \tag{19.22}$$

The silent failure is to substitute only the power, leaving $\log Z$ as $\log Y$. The result, $\frac{Y}{2} + \frac{1}{2}\sum_N \mathsf{L}_N^{\mathrm{Lam}}(\log Y)Y^{-N}$, is a perfectly well formed element of $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$; every block has the right shape, the right degree, and the right sign pattern, and no formal check inside the near-identity calculation detects anything. It is wrong by a constant. Numerically, with $Y = 20 + \log 10 =$

22.302585 . . ., for which $F^{\langle -1\rangle\circ}(Y) = 10$ exactly, the truncations through Y^{-3} give

$$\text{Equation (19.22)}: \ 10.000004\ldots, \qquad \text{naive substitution}: \ 9.67026\ldots,$$

the leading discrepancy being $\frac{1}{2}\log 2 = 0.34657\ldots$. Two further observations belong with this example.

- The coefficient field must contain $\log 2$. Over $\mathbb{K} = \mathbb{Q}$ the identity Equation (19.22) is not even a statement about $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$; the field has to be extended first (Theorem 19.9, step 1).
- The residual certificate of Theorem 16.16 does catch the error, because it is computed against F, not against G. This is why step 7 of Theorem 19.9 certifies against the original map and never against the normalised one.

Remark 19.15 (Sufficient, not necessary). Theorem 18.9 settles summability *for the outer valuation*. It does not say that Equation (18.12) fails whenever $\nu_t(A) < 0$; it says the theorem as proved does not apply. Take $F(X) = X + X/L = X(1 + 1/L)$, so $A = t^{-1}L^{-1} \in \mathscr{PL}_{\mathrm{itLaur},\mathbb{K}}$ with $\nu_t(A) = -1$, while $A \prec X$. Here every Lagrange summand contributes to the single outer block t^{-1}:

$$-A = -XL^{-1}, \quad \tfrac{1}{2}D_X(A^2) = XL^{-2} - XL^{-3}, \quad -\tfrac{1}{6}D_X^2(A^3) = -XL^{-3} + \tfrac{5}{2}XL^{-4} - 2XL^{-5},$$

and so on, so no finite number of summands determines any block. The family is nevertheless summable for the L^{-1}-adic topology of the coefficient field $\mathbb{K}((L^{-1}))$ at each fixed outer level, and the resulting series $Y\bigl(1 - M^{-1} + M^{-2} - 2M^{-3} + \cdots\bigr)$, $M = \log Y$, does satisfy $F \circ F^{\langle -1\rangle\circ} - Y = \mathcal{O}_{Y\to+\infty}\bigl(YM^{-4}\bigr)$, as direct substitution confirms. A theorem covering this case needs a rank-two value group and Hahn-style summability, in which $\{i : \nu(f_i) < \gamma\}$ is replaced by a well-ordered-support condition. We have not carried that out, and we do not claim it: the statement "Equation (18.12) holds whenever $A \prec X$ in the rank-two order, with summation in the refined filtration" is left as an open problem.

Remark 19.16 (What does not belong here). Reversion at a critical point is a different problem in a different regime. If $f(x) - f(x_0) = c_m(x - x_0)^m + \cdots$ with $c_m \neq 0$ and $m \geq 2$, the local inverse begins $x - x_0 \sim c_m^{-1/m}(y - f(x_0))^{1/m}$ and has m branches; Lambert W near $-\mathrm{e}^{-1}$ is the case $m = 2$ [5]. This is a Puiseux expansion at a finite point, not an expansion in the scale $X \to +\infty$, and none of the machinery of this section applies to it. In the at-infinity setting the derivative never vanishes on the relevant class: for $F = X + A$ with $A \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$, $D_X F = 1 + \mathcal{O}_t^{\mathrm{form}}(t)$ is a unit of $\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$.

Beyond the identity scale

A monomial-leading $F = cX^\rho L^\sigma(1 + U)$ has a monomial-leading inverse inside a Puiseux–log ring exactly when ρ^{-1} lies in the exponent group and $\sigma = 0$ (Theorem 19.3). A noninteger ρ is repaired by enlarging the exponent group; $\sigma \geq 1$ cannot be repaired that way and forces $\log\log X$ (Theorem 19.4). In every case reversion factors as $F^{\langle -1\rangle\circ} = F_0^{\langle -1\rangle\circ} \circ \Phi^{\langle -1\rangle\circ}$ with $\Phi = F \circ F_0^{\langle -1\rangle\circ}$ near-identity (Theorem 19.6), and the leading model F_0 is inverted by conjugating with e and dividing by ρ (Theorem 19.7). Every scale change substitutes *both* generators; substituting only the power silently produces a plausible and wrong series (Section 19.3).

Part VI

Wright omega, the Lambert polynomials, and Lambert W

Chapter 20

Wright omega: the canonical logarithmic inverse

Every result of Parts I–IV was stated for a general polynomial–logarithmic series. This part specialises all of it to a single map,

$$F(Y) = Y + \log Y,$$

and shows that this one reversion carries the whole Lambert circle of problems. The correction blocks of $F^{\langle -1\rangle\circ}$ are the Lambert polynomials $\mathsf{L}_n^{\mathrm{Lam}}$; they are defined and studied in Part 21, and the present chapter explains where they come from and exactly what is being asserted when one writes them down.

20.1 Standing conventions for this part

Throughout, $\mathbb{K}$ is a field of characteristic zero, so that $\mathbb{Q} \subseteq \mathbb{K}$. On the formal side we work in the rings of Theorem 5.3 with the large variable X, the small generator $t = X^{-1}$, and the logarithmic generator $L = \log X$; the near-identity class is $\mathcal{G}_{\mathrm{poly},\mathbb{K}} = X + \mathbb{K}[L][[t]]$ of Theorem 14.1, the derivation is $D_X = t\,\partial_L - t^2\,\partial_t$, and truncation is always the *exclusive* $\mathrm{Trunc}_{<N} F = \mathrm{Trunc}_{<N} F$.

On the analytic side the regime is $x \to +\infty$ along the positive real ray, with the real logarithm; every deviation from that regime — a complex sector, a nonprincipal branch, a limit at a finite point — is announced explicitly.

Two symbols carry a variable role and must be read locally. In this section L is the generator $\log X$ of the coefficient ring. In Part 21 the Lambert polynomials are treated as polynomials in an indeterminate u, and s is the formal variable of their generating series; in Theorem 21.14 the letter s is, instead, a complex parameter of an honest integral. The two roles of s are unrelated and never appear in the same statement.

A common trap

Three inverses meet in this part and must not be conflated. For a series F, the multiplicative inverse is F^{-1} (Theorem 9.7) and the compositional inverse is $F^{\langle -1\rangle\circ}$ (Theorem 16.11); for the Lambert function, the subscript in W_{-1} is a *branch label* and is neither of the two. The glyph -1 is the same in all three cases and the objects are different.

20.2 The real bijection and its inverse

Proposition 20.1 (The real logarithmic bijection). *The map*

$$\phi\colon (0,\infty) \longrightarrow \mathbb{R}, \qquad \phi(Y) = Y + \log Y,$$

is a strictly increasing real-analytic bijection. Its inverse $\omega\colon \mathbb{R} \to (0,\infty)$ *is strictly increasing and real-analytic, and satisfies, for every* $x \in \mathbb{R}$,

$$\omega(x) + \log\omega(x) = x, \qquad \omega'(x) = \frac{\omega(x)}{1+\omega(x)}, \qquad \omega(x) = W_0(\mathrm{e}^x). \tag{20.1}$$

Here W_0 *is the branch of the inverse of* $w \mapsto w\mathrm{e}^w$ *taking values in* $[-1,\infty)$*. Moreover* $\omega(1) = 1$*, and for every* x *with* $\omega(x) > 1$,

$$x - \log x < \omega(x) < x. \tag{20.2}$$

Proof. On $(0,\infty)$ the function ϕ is real-analytic with $\phi'(Y) = 1 + Y^{-1} > 1 > 0$, so it is strictly increasing and injective. As $Y \to 0^+$ we have $\log Y \to -\infty$ and $Y \to 0$, so $\phi(Y) \to -\infty$; as $Y \to +\infty$, $\phi(Y) \geq Y \to +\infty$. By the intermediate value theorem ϕ is onto $\mathbb{R}$. Hence ϕ is a bijection and $\omega := \phi^{\langle -1\rangle\circ}$ is defined on all of $\mathbb{R}$, strictly increasing, and real-analytic by the inverse function theorem, ϕ' being nowhere zero.

Differentiating $\omega + \log\omega = x$ gives $\omega'(1+\omega^{-1}) = 1$, i.e. the displayed derivative formula. For the Lambert identity fix $x \in \mathbb{R}$ and put $w = \omega(x) > 0$. Then $\log(w\mathrm{e}^w) = \log w + w = x$, so $w\mathrm{e}^w = \mathrm{e}^x$. The map $w \mapsto w\mathrm{e}^w$ is strictly increasing on $[-1,\infty)$ with value $-\mathrm{e}^{-1}$ at $w = -1$, hence injective there; since $w > 0 > -1$, w is the value of the principal branch, i.e. $w = W_0(\mathrm{e}^x)$.

Finally $\phi(1) = 1 + \log 1 = 1$, so $\omega(1) = 1$. If $Y = \omega(x) > 1$ then $\log Y > 0$, so $Y = x - \log Y < x$; and then $\log Y < \log x$, whence $Y = x - \log Y > x - \log x$. □

Remark 20.2 (What is not repeated here). The elementary two-sided envelope for ω and the sharper two-order statement were established in Theorem 4.1 and Theorem 4.3; Equation (20.2) is recorded only because the proof of Theorem 20.17 uses the crude lower bound $\omega(x) > x - \log x$ directly. In particular $\omega(x) \to +\infty$ and $\omega(x)/x \to 1$ as $x \to +\infty$.

Remark 20.3 (Totality versus asymptotics). ω is a genuine function on all of $\mathbb{R}$; it is not defined only near $+\infty$. The polynomial–logarithmic expansion studied below describes it *only* as $x \to +\infty$, and says nothing about, say, $x \leq 0$, where $\log x$ is not even real. Several of the six source drafts introduce ω with the phrase "for x sufficiently large"; that restriction belongs to the expansion, not to the function.

20.3 Every Lambert problem is this one reversion

The point of isolating ω is that a one-parameter family of logarithmic inversion problems collapses onto it exactly, with no new coefficients. We give the analytic reduction first and the formal one second.

Lemma 20.4 (Exact scaling reduction, analytic form). *Let* $a > 0$ *be real. Then* $y \mapsto y + a\log y$ *is a strictly increasing bijection of* $(0,\infty)$ *onto* $\mathbb{R}$*, and its inverse is*

$$y = a\,\omega\Big(\frac{x}{a} - \log a\Big), \qquad x \in \mathbb{R}. \tag{20.3}$$

Equivalently, for $\alpha > 0$ *and* $X > 0$*, the unique* $y > 0$ *with* $y^\alpha \mathrm{e}^y = X$ *is* $y = \alpha\,\omega\big(\alpha^{-1}\log X - \log\alpha\big)$.

Proof. Write $y = az$ with $z > 0$. Then $y + a\log y = az + a\log a + a\log z = a\big(z + \log z\big) + a\log a$, so $y + a\log y = x$ is equivalent to $z + \log z = x/a - \log a$, that is to $z = \omega(x/a - \log a)$ by Theorem 20.1. Monotonicity and bijectivity are inherited from ϕ through the same substitution. For the second form, take logarithms in $y^\alpha\mathrm{e}^y = X$: since $y > 0$ and $X > 0$, this is equivalent to $y + \alpha\log y = \log X$, and (20.3) applies with $a = \alpha$, $x = \log X$. □

The formal counterpart is stronger: it holds over any characteristic-zero $\mathbb{K}$, for every scalar a, and it exhibits the *same* polynomials with only a scalar reweighting.

Theorem 20.5 (The one-parameter family of logarithmic reversions). *Let $a \in \mathbb{K}$ and let $F_a := X + aL \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}$. Then $F_a^{\langle -1\rangle\circ} \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}$ exists, is unique, and equals*

$$F_a^{\langle -1\rangle\circ}(X) = X + \sum_{n\geq 0} a^{n+1}\mathsf{L}_n^{\mathrm{Lam}}(L)\, t^n = X - aL + \sum_{n\geq 1} a^{n+1}\mathsf{L}_n^{\mathrm{Lam}}(L)\, t^n, \tag{20.4}$$

with $\mathsf{L}_n^{\mathrm{Lam}}$ as in Theorem 21.1. In particular $F_1^{\langle -1\rangle\circ} = \omega$ and

$$(X-L)^{\langle -1\rangle\circ}(X) = X + L + \sum_{n\geq 1}(-1)^{n+1}\mathsf{L}_n^{\mathrm{Lam}}(L)\, t^n. \tag{20.5}$$

Proof. Existence and uniqueness are Theorem 16.11, since $aL \in \mathbb{K}[L][[t]]$. For the formula, write $\mathcal{P}(t,L) := \sum_{n\geq 0}\mathsf{L}_n^{\mathrm{Lam}}(L)t^n \in \mathbb{K}[L][[t]]$. By Theorem 20.9,

$$\mathcal{P} + L + \log(1 + t\mathcal{P}) = 0. \tag{20.6}$$

Substituting $t \mapsto at$ into (20.6) — a legitimate $\mathbb{K}$-algebra endomorphism of $\mathbb{K}[L][[t]]$, continuous for the t-adic filtration, hence commuting with the formal logarithm — gives

$$\mathcal{P}(at,L) + L + \log\big(1 + at\,\mathcal{P}(at,L)\big) = 0.$$

Put $g := a\,\mathcal{P}(at,L) \in \mathbb{K}[L][[t]]$ and multiply by a:

$$g + aL + a\log(1+tg) = 0. \tag{20.7}$$

Now let $G = X + g$. By generator substitution (Theorem 12.22), $L \circ G = L + \log(1+tg)$, so

$$F_a \circ G = G + a\,(L\circ G) = X + g + aL + a\log(1+tg) = X$$

by (20.7). Thus G is a right inverse of F_a in $\mathcal{G}_{\mathrm{poly},\mathbb{K}}$, hence the two-sided inverse by uniqueness (Theorem 16.11(2),(3)). Finally $g = a\sum_{n\geq 0}\mathsf{L}_n^{\mathrm{Lam}}(L)a^nt^n = \sum_{n\geq 0}a^{n+1}\mathsf{L}_n^{\mathrm{Lam}}(L)t^n$, which is (20.4); and $a^{n+1}\big|_{a=-1} = (-1)^{n+1}$, with $(-1)^1\mathsf{L}_0^{\mathrm{Lam}}(L) = L$, gives (20.5). □

Remark 20.6 (Why no "second Lambert family" exists). The inverse of $X - \log X$ is often presented as a separate expansion with its own alternating coefficients. Equation (20.5) settles the matter: the blocks are $(-1)^{n+1}\mathsf{L}_n^{\mathrm{Lam}}$, the same polynomials with a sign. The frequently quoted weight $(-1)^n$ is correct for $-(X-L)^{\langle -1\rangle\circ}$, not for $(X-L)^{\langle -1\rangle\circ}$ itself; that sign flip is exactly the substitution $w \mapsto -w$ used to reach the lower real Lambert branch, and it is the source of a recurring off-by-one-sign error in the source drafts. Concretely, for $x \to 0^-$, putting $T = \log(1/(-x))$ and $\ell = \log T$, the substitution $W_{-1}(x) = -v$ turns $we^w = x$ into $v - \log v = T$, so (20.5) gives the block pattern

$$W_{-1}(x)\text{ blocks } = -T - \ell + \sum_{n\geq 1}(-1)^n\mathsf{L}_n^{\mathrm{Lam}}(\ell)T^{-n}.$$

The formal identity is (20.5); the asymptotic assertion about W_{-1} needs the analytic transfer of Section 20.6 applied on the branch in question and is not proved here.

Remark 20.7 (Branch coordinates). Taking a compatible logarithm in $W(z)e^{W(z)} = z$ gives $W_k(z) + \log W_k(z) = \xi_k$ with $\xi_k = \operatorname{Log} z + 2\pi i k$. Thus every large branch of W is the *same* reversion evaluated at a different large variable, and the formal expansion of W_k is obtained from Theorem 20.10 by the substitutions $X \mapsto \xi_k$, $L \mapsto \ell_k = \log_{\mathcal{B}}\xi_k$ for a declared branch $\mathcal{B}$. The polynomials $\mathsf{L}_n^{\mathrm{Lam}}$ are therefore branch-independent; the coordinates ξ_k, ℓ_k, the branch $\mathcal{B}$, the sector, and the resulting analytic estimates are not. We do not develop the complex branch analysis here.

20.4 The formal expansion

The reversion machinery of Part IV applies verbatim, and in this special case it degenerates in a useful way: each Lagrange–Bürmann summand produces exactly one block, so no cross-order collection is ever needed. We first record the operator identity that makes this visible.

Lemma 20.8 (Repeated derivative of a coefficient block). *Let $n \in \mathbb{Z}$, $k \in \mathbb{N}_0$ and $p \in \mathbb{K}[L]$. Then*

$$D_X^k\big(t^n p(L)\big) = t^{n+k}\,\Delta_{n,k}\,p(L), \qquad \Delta_{n,k} = \prod_{j=0}^{k-1}\big(\partial_L - n - j\big), \tag{20.8}$$

the empty product at $k = 0$ acting as the identity.

Proof. Induction on k. The case $k = 0$ is trivial. Assume (20.8) for k and write $q = \Delta_{n,k}p \in \mathbb{K}[L]$. Since $D_X t = -t^2$ and $D_X L = t$, Leibniz and the chain rule for the derivation D_X give

$$D_X\big(t^{n+k}q(L)\big) = (n+k)t^{n+k-1}(-t^2)q(L) + t^{n+k}q'(L)\,t = t^{n+k+1}\big(\partial_L - (n+k)\big)q(L).$$

The factors of $\Delta_{n,k}$ are polynomials in ∂_L and therefore commute, so $\big(\partial_L - (n+k)\big)\Delta_{n,k} = \Delta_{n,k+1}$, which is the statement for $k+1$. □

Proposition 20.9 (The implicit equation for the correction). *Let $\mathcal{T} : \mathbb{K}[L][[t]] \to \mathbb{K}[L][[t]]$ be $\mathcal{T}(U) = -L - \log(1+tU)$, the logarithm being the formal one of Theorem 12.5. Then $\mathcal{T}$ has a unique fixed point $\mathcal{P} \in \mathbb{K}[L][[t]]$; equivalently, the equation*

$$\mathcal{P} + L + \log(1 + t\mathcal{P}) = 0 \tag{20.9}$$

has exactly one solution in $\mathbb{K}[L][[t]]$. That solution lies in $-L + t\mathbb{K}[L][[t]]$, satisfies $\mathcal{P}(t,0) = 0$, and

$$\omega = (X+L)^{\langle -1\rangle\circ} = X + \mathcal{P}. \tag{20.10}$$

Proof. For $U \in \mathbb{K}[L][[t]]$ we have $tU \in t\mathbb{K}[L][[t]]$, so the family $\big((-1)^{j-1}(tU)^j/j\big)_{j\geq 1}$ is formally summable and $\log(1+tU) \in t\mathbb{K}[L][[t]]$; hence $\mathcal{T}$ is well defined and maps into $-L + t\mathbb{K}[L][[t]]$. If $\nu_t(U - V) \geq N$, then, $1 + tV$ being a unit of $\mathbb{K}[L][[t]]$,

$$\mathcal{T}(U) - \mathcal{T}(V) = -\log\frac{1+tU}{1+tV} = -\log\left(1 + \frac{t(U-V)}{1+tV}\right) \in t^{N+1}\mathbb{K}[L][[t]],$$

so $\nu_t\big(\mathcal{T}(U) - \mathcal{T}(V)\big) \geq \nu_t(U-V) + 1$. Thus $\mathcal{T}$ is a strict contraction for the t-adic filtration on the whole of $\mathbb{K}[L][[t]]$, which is separated and complete (Theorem 14.13). Two fixed points would satisfy $\nu_t(U-V) \geq \nu_t(U-V)+1$, forcing $\nu_t(U-V) = +\infty$ and $U = V$; and the iterates $U_0 = 0$, $U_{r+1} = \mathcal{T}(U_r)$ form a Cauchy sequence whose limit is a fixed point. Its membership in $-L + t\mathbb{K}[L][[t]]$ is the image statement above.

Setting $L = 0$ in (20.9) — that is, applying the $\mathbb{K}$-algebra map $\mathbb{K}[L][[t]] \to \mathbb{K}[[t]]$, $L \mapsto 0$, which is t-adically continuous and hence commutes with the formal logarithm — shows that $V := \mathcal{P}(t,0)$ solves $V + \log(1+tV) = 0$ in $\mathbb{K}[[t]]$. The same contraction argument applied to $V \mapsto -\log(1+tV)$ gives a unique solution, and $V = 0$ is one; hence $\mathcal{P}(t,0) = 0$.

Finally put $G = X + \mathcal{P} \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}$. By Theorem 12.22, $(X+L)\circ G = G + L + \log(1+t\mathcal{P}) = X$ by (20.9), so G is a right inverse of $X + L$, hence $G = (X+L)^{\langle -1\rangle\circ}$ by Theorem 16.11(2),(3). That $(X+L)^{\langle -1\rangle\circ}$ is what the notation ω denotes formally is Theorem 20.1 together with Theorem 20.17 below; at this point (20.10) is a definition of the symbol on the formal side. □

Theorem 20.10 (Polynomial–logarithmic expansion of ω). *In $\mathcal{G}_{\mathrm{poly},\mathbb{K}}$,*

$$\omega = X + \sum_{n\geq 0} \mathsf{L}_n^{\mathrm{Lam}}(L)\, t^n = X - L + \sum_{n\geq 1} \mathsf{L}_n^{\mathrm{Lam}}(L)\, t^n, \tag{20.11}$$

where $\mathsf{L}_n^{\mathrm{Lam}}$ is the Lambert polynomial of Theorem 21.1. More precisely, the r-th Lagrange–Bürmann summand for $F = X + L$ is a single *block:*

$$\frac{(-1)^r}{r!}\, D_X^{r-1}\big(L^r\big) = \mathsf{L}_{r-1}^{\mathrm{Lam}}(L)\, t^{r-1}, \qquad r \geq 1. \tag{20.12}$$

Proof. Apply the Lagrange–Bürmann formula Theorem 18.14, in the form (18.13), to $F = X + A$ with $A = L$:

$$F^{\langle -1\rangle_\circ}(X) = X + \sum_{r\geq 1} \frac{(-1)^r}{r!} D_X^{r-1}\big(L^r\big),$$

the family being formally summable because the r-th summand has t-order at least $r-1$. By Theorem 20.8 with $n = 0$, $k = r-1$, $p = L^r$,

$$D_X^{r-1}\big(L^r\big) = t^{r-1}\Delta_{0,r-1}L^r = t^{r-1}\left[\prod_{j=0}^{r-2}(\partial_L - j)\right] L^r.$$

Setting $n = r-1$, the scalar prefactor is $\frac{(-1)^r}{r!} = \frac{(-1)^{n+1}}{(n+1)!}$ and the operator is $\prod_{j=0}^{n-1}(\partial_L - j)$ applied to L^{n+1}; by Theorem 21.1 the product of the two is exactly $\mathsf{L}_n^{\mathrm{Lam}}(L)$. This is (20.12). Since the summand of index r has t-order exactly $r-1$ and no other summand meets that order, summing over $r \geq 1$ gives (20.11) with no collection of contributions; the $r = 1$ term is $-L = \mathsf{L}_0^{\mathrm{Lam}}(L)$. □

Remark 20.11 (Two normalisations of the same expansion). The two displays in (20.11) are the same series. The zero-based one is canonical: it is the form in which (20.12) says "one Lagrange index per block", and the form in which Theorem 20.5 needs no case distinction. The one-based form, which displays $-L$ separately, is the classical way of writing the Lambert expansion and is the one the source drafts mostly use. The price of the zero-based normalisation is that $\deg \mathsf{L}_0^{\mathrm{Lam}} = 1 \neq 0$: the degree law $\deg \mathsf{L}_n^{\mathrm{Lam}} = n$ holds for $n \geq 1$ only (Theorem 21.7).

20.5 The truncation residual

Definition 20.12 (Omega truncations). For $N \in \mathbb{N}_0$ set

$$\omega_N := X + \mathrm{Trunc}_{<N+1}\,\mathcal{P} = X - L + \sum_{n=1}^{N} \mathsf{L}_n^{\mathrm{Lam}}(L)\, t^n \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}. \tag{20.13}$$

In the inclusive notation of source 6 this is $X + [\mathcal{P}]_{\leq N}$; the one-block difference $[F]_{\leq N} = \mathrm{Trunc}_{<N+1} F$ is the only discrepancy between the two conventions and is a frequent source of apparent off-by-one disagreements between the drafts.

Theorem 20.13 (Exact leading residual). *For every $N \in \mathbb{N}_0$, in $\mathbb{K}[L][[t]]$,*

$$\mathrm{Err}_{\mathrm{rev}}(X + L, \omega_N) = \omega_N + \log\omega_N - X = -\mathsf{L}_{N+1}^{\mathrm{Lam}}(L)\, t^{N+1} + \mathcal{O}_t^{\mathrm{form}}\big(t^{N+2}\big)\,. \tag{20.14}$$

In particular $\nu_t\big(\mathrm{Err}_{\mathrm{rev}}(X + L, \omega_N)\big) = N + 1$ exactly, for every $N \geq 0$.

Proof. Write $\mathcal{P}_N = \operatorname{Trunc}_{<N+1}\mathcal{P}$ and $\Delta = \mathcal{P} - \mathcal{P}_N = \sum_{n\geq N+1} \mathsf{L}_n^{\mathrm{Lam}}(L)t^n$, so that $\nu_t\Delta = N+1$ and $[t^{N+1}]\Delta = \mathsf{L}_{N+1}^{\mathrm{Lam}}(L)$. By generator substitution (Theorem 12.22),

$$\operatorname{Err}_{\mathrm{rev}}(X+L,\omega_N) = \mathcal{P}_N + L + \log(1+t\mathcal{P}_N).$$

Using $\mathcal{P}_N = \mathcal{P} - \Delta$, the identity $1 + t\mathcal{P}_N = (1+t\mathcal{P})\Big(1 - \dfrac{t\Delta}{1+t\mathcal{P}}\Big)$, and additivity of the formal logarithm on $1 + t\mathbb{K}[L][[t]]$ (Theorem 12.5), we get

$$\operatorname{Err}_{\mathrm{rev}}(X+L,\omega_N) = \underbrace{\mathcal{P} + L + \log(1+t\mathcal{P})}_{=0\text{ by (20.9)}} - \Delta + \log\left(1 - \frac{t\Delta}{1+t\mathcal{P}}\right).$$

Since $1+t\mathcal{P}$ is a unit of $\mathbb{K}[L][[t]]$ and $\nu_t\Delta = N+1$, the argument of the last logarithm lies in $1 + t^{N+2}\mathbb{K}[L][[t]]$, so that logarithm lies in $t^{N+2}\mathbb{K}[L][[t]]$. Hence

$$\operatorname{Err}_{\mathrm{rev}}(X+L,\omega_N) = -\Delta + \mathcal{O}_t^{\mathrm{form}}\big(t^{N+2}\big) = -\mathsf{L}_{N+1}^{\mathrm{Lam}}(L)t^{N+1} + \mathcal{O}_t^{\mathrm{form}}\big(t^{N+2}\big).$$

Exactness of the order follows because $\mathsf{L}_{N+1}^{\mathrm{Lam}} \neq 0$ for every $N \geq 0$, by Theorem 21.7. □

Corollary 20.14 (Finite certificate for the Lambert polynomials). *Let $p_0,\dots,p_N \in \mathbb{K}[L]$ and set $H = X + \sum_{n=0}^N p_n(L)t^n$. Then $p_n = \mathsf{L}_n^{\mathrm{Lam}}$ for $0 \leq n \leq N$ if and only if* $\operatorname{Err}_{\mathrm{rev}}(X+L,H) = \mathcal{O}_t^{\mathrm{form}}\big(t^{N+1}\big)$.

Proof. Necessity is Theorem 20.13. Conversely, if the residual has t-order at least $N+1$ then H is a right inverse of $X+L$ through block N in the sense of Theorem 16.16, so $\operatorname{Trunc}_{<N+1} H - X = \operatorname{Trunc}_{<N+1}\mathcal{P}$, i.e. $p_n = \mathsf{L}_n^{\mathrm{Lam}}$ for $n \leq N$. □

Example 20.15 (The residual at $N=2$). With $\mathcal{P}_2 = -L + Lt + \big(\frac{1}{2}L^2 - L\big)t^2$,

$$\mathcal{P}_2 + L + \log(1+t\mathcal{P}_2) = -\left(\tfrac{1}{3}L^3 - \tfrac{3}{2}L^2 + L\right)t^3 + \mathcal{O}_t^{\mathrm{form}}\big(t^4\big) = -\mathsf{L}_3^{\mathrm{Lam}}(L)t^3 + \mathcal{O}_t^{\mathrm{form}}\big(t^4\big),$$

which one checks by expanding $\log(1+t\mathcal{P}_2) = t\mathcal{P}_2 - \frac{1}{2}t^2\mathcal{P}_2^2 + \frac{1}{3}t^3\mathcal{P}_2^3 - \cdots$ and collecting t^0, t^1, t^2, t^3. This single computation certifies $\mathsf{L}_0^{\mathrm{Lam}}, \mathsf{L}_1^{\mathrm{Lam}}, \mathsf{L}_2^{\mathrm{Lam}}$ simultaneously and predicts $\mathsf{L}_3^{\mathrm{Lam}}$; it is the cheapest available unit test for an implementation of the recurrence.

Example 20.16 (Every truncation is exact at $x=1$). Because $\mathsf{L}_n^{\mathrm{Lam}}(0) = 0$ for all n (Theorem 21.2), substituting $x=1$, where $L = \log 1 = 0$, gives $\omega_N(1) = 1 - 0 + 0 = 1 = \omega(1)$ for every N. The same phenomenon in the Lambert coordinates is the exactness of the branch expansion at $z = \mathrm{e}$, where $\xi_0 = 1$ and $\ell_0 = 0$. It is a consequence of the vanishing $\mathsf{L}_n^{\mathrm{Lam}}(0) = 0$, not of any convergence property.

20.6 Formal statement, analytic statement

Theorem 20.13 is an identity in $\mathbb{K}[L][[t]]$. It asserts coefficient agreement below a cutoff and nothing else: no numerical bound, no convergence, no uniformity in L, no claim that ω exists as a function. An analytic statement about the real function of Theorem 20.1 needs four further ingredients, none of which is formal algebra:

1. a *regime*: here the ray $x \to +\infty$ with the real logarithm, so that $L = \log x$ is a real quantity that *grows with* x;
2. an *analytic remainder* for the truncated logarithm, replacing the formal $\log(1+t\mathcal{P}_N)$ by a finite sum plus an estimated tail;
3. a *lower bound on the derivative* of ϕ between the true and the approximate root, to convert a residual bound into an error bound;

4. an *admissibility check* that the approximation stays inside the domain of ϕ, here $\omega_N(x) > 0$.

The following theorem supplies all four. Note that the estimate is *not* uniform in N, and that the exponent of $\log x$ in the remainder is quadratic in N, not linear; this is an honest feature of the argument, not an artefact.

Theorem 20.17 (Asymptotic validity on the positive real ray). *Fix $N \in \mathbb{N}_0$ and put $K = (N+2)^2$. Then, with $L = \log x$,*

$$\omega_N(x) + \log \omega_N(x) - x = -\frac{\mathsf{L}^{\mathrm{Lam}}_{N+1}(L)}{x^{N+1}} + \mathcal{O}_{x\to+\infty}\left(\frac{L^K}{x^{N+2}}\right) \tag{20.15}$$

and

$$\omega(x) - \omega_N(x) = \frac{\mathsf{L}^{\mathrm{Lam}}_{N+1}(L)}{x^{N+1}} + \mathcal{O}_{x\to+\infty}\left(\frac{L^K}{x^{N+2}}\right). \tag{20.16}$$

Consequently

$$\omega(x) - \omega_N(x) \sim \frac{(\log x)^{N+1}}{(N+1)\, x^{N+1}}, \qquad x \to +\infty, \tag{20.17}$$

so ω_N approximates ω to order exactly $L^{N+1}x^{-(N+1)}$, and $X + \sum_{n\ge 0} \mathsf{L}^{\mathrm{Lam}}_n(L) t^n$ is an asymptotic expansion of ω on the scale $\{(\log x)^m x^{-n}\}$ in the sense of Theorem 8.43.

Proof. Write $t = 1/x$, $L = \log x$, and $p = p(x) := -L + \sum_{n=1}^N \mathsf{L}^{\mathrm{Lam}}_n(L) x^{-n}$, so $\omega_N(x) = x + p = x(1+\varepsilon)$ with $\varepsilon = p/x$.

Step 1: ε is small. For $x \ge \mathrm{e}$ we have $L \ge 1$, and $\left|\mathsf{L}^{\mathrm{Lam}}_n(L)\right| \le c_n L^n$ with $c_n = \sum_m \left|[u^m]\mathsf{L}^{\mathrm{Lam}}_n\right|$. Hence, once $x \ge L$, $\sum_{n=1}^N \left|\mathsf{L}^{\mathrm{Lam}}_n(L)\right| x^{-n} \le \sum_{n=1}^N c_n (L/x)^n \le C_N\, L/x$. Therefore $|p| \le L + C_N L/x \le 2L$ and $|\varepsilon| \le 2L/x$ for all x beyond some x_N, which we enlarge so that $|\varepsilon| \le \frac{1}{2}$. In particular $\omega_N(x) \ge x/2 > 0$, settling item (4) above, and $\log \omega_N(x) = L + \log(1+\varepsilon)$ with the real logarithm.

Step 2: an analytic remainder for the logarithm. For $|\varepsilon| \le \frac{1}{2}$,

$$\log(1+\varepsilon) = \sum_{j=1}^{N+1} \frac{(-1)^{j-1}}{j}\varepsilon^j + \theta, \qquad |\theta| \le \sum_{j>N+1} \frac{|\varepsilon|^j}{j} \le 2\,|\varepsilon|^{N+2} \le 2\left(\frac{2L}{x}\right)^{N+2}.$$

Consequently

$$\omega_N(x) + \log\omega_N(x) - x = \Pi(1/x, \log x) + \theta, \qquad \Pi := \mathcal{P}_N + L + \sum_{j=1}^{N+1} \frac{(-1)^{j-1}}{j}\left(t\mathcal{P}_N\right)^j, \tag{20.18}$$

where $\Pi \in \mathbb{K}[L][t]$ is a *polynomial* in t and L, evaluated at the real numbers $1/x$ and $\log x$.

Step 3: identifying Π. In $\mathbb{K}[L][[t]]$ we have $\nu_t(t\mathcal{P}_N) \ge 1$, so $\log(1+t\mathcal{P}_N) - \sum_{j\le N+1} \frac{(-1)^{j-1}}{j}(t\mathcal{P}_N)^j \in t^{N+2}\mathbb{K}[L][[t]]$. Combining this with Theorem 20.13,

$$\Pi \equiv -\mathsf{L}^{\mathrm{Lam}}_{N+1}(L)\, t^{N+1} \pmod{t^{N+2}\mathbb{K}[L][t]}.$$

Now $\mathcal{P}_N$ has t-degree $\le N$ and L-degree $\le \max(N,1)$, so Π has t-degree $\le (N+1)^2$ and L-degree $\le (N+1)\max(N,1) \le K$. Writing $\Pi = -\mathsf{L}^{\mathrm{Lam}}_{N+1}(L) t^{N+1} + \sum_{i=N+2}^{(N+1)^2} c_i(L) t^i$ with $\deg c_i \le K$, we obtain, for $x \ge x_N$,

$$\left|\Pi(1/x, \log x) + \mathsf{L}^{\mathrm{Lam}}_{N+1}(L) x^{-(N+1)}\right| \le C'_N\, \frac{L^K}{x^{N+2}}.$$

Since also $|\theta| \le 2^{N+3} L^{N+2} x^{-(N+2)}$ and $N+2 \le K$, (20.15) follows from (20.18).

Step 4: residual to error. Denote the left side of (20.15) by $R_N(x)$, so $\phi(\omega_N(x)) - \phi(\omega(x)) = R_N(x)$ with $\phi(Y) = Y + \log Y$. Both $\omega_N(x)$ and $\omega(x)$ lie in $(0, \infty)$, and by Equation (20.2) and Step 1 both exceed $x/2$ for $x \geq x_N$ (enlarging x_N again). By the mean value theorem there is ζ between them, hence $\zeta > x/2$, with

$$\omega_N(x) - \omega(x) = \frac{R_N(x)}{\phi'(\zeta)} = \frac{R_N(x)}{1+\zeta^{-1}}, \qquad \frac{1}{1+\zeta^{-1}} = 1 + \mathcal{O}_{x\to+\infty}\big(x^{-1}\big)\,.$$

This is the general residual-transfer principle of Theorem 16.21 in the concrete case at hand, with derivative lower bound $m = 1$. Multiplying (20.15) by $-(1+\zeta^{-1})^{-1}$ and absorbing $\big|\mathsf{L}^{\mathrm{Lam}}_{N+1}(L)\big|\,x^{-(N+1)} \cdot \mathcal{O}_{x\to+\infty}\big(x^{-1}\big) = \mathcal{O}_{x\to+\infty}\big(L^{N+1}x^{-(N+2)}\big)$ into the remainder gives (20.16).

Step 5. By Theorem 21.7, $\mathsf{L}^{\mathrm{Lam}}_{N+1}(L) = L^{N+1}/(N+1) + \mathcal{O}_{L\to+\infty}\big(L^N\big)$, and $L^K x^{-(N+2)} = o_{x\to+\infty}\big(L^{N+1}x^{-(N+1)}\big)$ because $L^{K-N-1}/x \to 0$. Hence (20.17). Since (20.17) holds for every N, the partial sums are an asymptotic expansion on the stated scale. □

Remark 20.18 (The rate in (20.17) is only logarithmic). Statement (20.16) is the useful one: the error of ω_N equals the *next block* with a relative error $\mathcal{O}_{x\to+\infty}\big(L^K/x\big)$. The simplified form (20.17) discards the lower-degree part of $\mathsf{L}^{\mathrm{Lam}}_{N+1}$, whose largest term is $-H_N L^N$; the relative error of (20.17) is therefore only $\mathcal{O}_{L\to\infty}(1/L)$, i.e. $\mathcal{O}_{x\to+\infty}(1/\log x)$. For $N = 4$ and $x = 5 \cdot 10^5$ the ratio of the true error to $\mathsf{L}^{\mathrm{Lam}}_5(L)x^{-5}$ is $1.0000148\ldots$, while its ratio to $L^5/(5x^5)$ is still far from 1. Quoting (20.17) where (20.16) is meant is a standard way to make a correct theorem look numerically wrong.

The next proposition isolates the one convergence statement that the formal theory does yield, and makes clear why it is not the statement one wants.

Proposition 20.19 (Convergence at frozen logarithm)**.** *For each $u_0 \in \mathbb{C}$ there are $r > 0$ and $\rho > 0$ such that the series $\sum_{n\geq 1} \mathsf{L}^{\mathrm{Lam}}_n(u)s^n$ converges absolutely for $|s| < r$, uniformly for $|u - u_0| \leq \rho$, and its sum $Q(s,u)$ is the unique holomorphic solution near $(0, u_0)$ of*

$$Q = -\log(1 - su + sQ), \qquad Q(0,u) = 0. \tag{20.19}$$

Proof. Let $\Phi(s,u,Q) = Q + \mathrm{Log}(1-su+sQ)$, holomorphic on a neighbourhood of $\{(0,u,0) : u \in \mathbb{C}\}$ because $1 - su + sQ = 1$ there. We have $\Phi(0,u,0) = 0$ and $\partial_Q\Phi(0,u,0) = 1 + \frac{s}{1-su+sQ}\big|_{s=0} = 1 \neq 0$. The holomorphic implicit function theorem gives $r, \rho > 0$ and a unique holomorphic Q on $|s| < r$, $|u - u_0| < \rho$, with $Q(0,u) = 0$ and $\Phi = 0$. Its Taylor coefficients in s satisfy the same triangular recursion as those of the formal solution of (20.19), which by Theorem 21.11 is $\sum_{n\geq 1}\mathsf{L}^{\mathrm{Lam}}_n(u)s^n$; by uniqueness of Taylor coefficients the two agree. □

Remark 20.20 (What is *not* proved here). Theorem 20.19 freezes u and lets $s \to 0$. The asymptotic problem does the opposite: it couples them by $u = \log(1/s)$, so that $u \to \infty$ as $s \to 0$, and Theorem 20.19 says nothing about it, because r is not claimed to be bounded below as $|u_0| \to \infty$. In fact the implicit function theorem fails on the locus $\partial_Q\Phi = 0$, which after eliminating Q from (20.19) is $u = s^{-1} + 1 - \log(-s)$; this suggests $r \asymp 1/|u|$ for large $|u|$, which would be compatible with genuine convergence of $\sum_n \mathsf{L}^{\mathrm{Lam}}_n(\log x)x^{-n}$ for large x. We record that as a heuristic only. The literature asserts absolute convergence of the associated double series on explicit regions [5, 21, 11]; no proof of any such convergence statement is given here, and none follows from anything in Parts I–IV. Likewise, uniformity of (20.16) in a complex sector, in the branch index k of Theorem 20.7, or in N, is neither claimed nor proved.

Chapter 21

The Lambert polynomials

This chapter is self-contained: the polynomials are defined by an operator formula, and every property — the recurrence, the closed Stirling form, the coefficient structure, the integrality, the generating equation, the Laplace transform — is derived from that definition without using Part 20. The two sections meet in Theorem 21.11, which reproves Theorem 20.10 without the reversion machinery.

Throughout, u is an indeterminate, ∂_u is $\mathrm{d}/\,\mathrm{d}u$, and $\Delta_{0,n} = \prod_{j=0}^{n-1}(\partial_u - j)$ is the falling-factorial operator $\partial_u^{\underline{n}}$, with the empty product at $n = 0$ acting as the identity.

21.1 Definition and first values

Definition 21.1 (Canonical zero-based Lambert polynomials)**.** For $n \in \mathbb{N}_0$ put

$$\mathsf{L}_n^{\mathrm{Lam}}(u) := \frac{(-1)^{n+1}}{(n+1)!}\left[\prod_{j=0}^{n-1}(\partial_u - j)\right] u^{n+1} \;\in\; \mathbb{Q}[u], \tag{21.1}$$

the product being empty at $n = 0$. Since $\operatorname{char}\mathbb{K} = 0$ we regard $\mathsf{L}_n^{\mathrm{Lam}} \in \mathbb{K}[u]$ whenever convenient.

At $n = 0$ the operator is the identity and the prefactor is $(-1)^1/1! = -1$, so $\mathsf{L}_0^{\mathrm{Lam}}(u) = -u$. The next values are

$$\mathsf{L}_1^{\mathrm{Lam}}(u) = u, \qquad \mathsf{L}_2^{\mathrm{Lam}}(u) = \frac{u^2}{2} - u, \qquad \mathsf{L}_3^{\mathrm{Lam}}(u) = \frac{u^3}{3} - \frac{3}{2}u^2 + u, \tag{21.2}$$

computed as follows. For $n = 1$, $\partial_u u^2 = 2u$ and $(-1)^2/2! = \frac{1}{2}$, giving u. For $n = 2$, $(\partial_u - 1)\partial_u u^3 = (\partial_u - 1)(3u^2) = 6u - 3u^2$ and $(-1)^3/3! = -\frac{1}{6}$, giving $\frac{1}{2}u^2 - u$. For $n = 3$, $(\partial_u - 2)(\partial_u - 1)\partial_u u^4 = (\partial_u - 2)(\partial_u - 1)(4u^3) = (\partial_u - 2)(12u^2 - 4u^3) = 24u - 12u^2 - 24u^2 + 8u^3$, and $(-1)^4/4! = \frac{1}{24}$, giving $\frac{1}{3}u^3 - \frac{3}{2}u^2 + u$.

A common trap

The degree law $\deg \mathsf{L}_n^{\mathrm{Lam}} = n$ is asserted for $n \geq 1$ only. The zero-index member $\mathsf{L}_0^{\mathrm{Lam}}(u) = -u$ has degree 1. Any statement quantified over $n \geq 0$ that mentions the degree, the leading coefficient, or a coefficient indexed relative to the top must exclude $n = 0$ explicitly.

21.2 The recurrence

Theorem 21.2 (Differential and integral recurrence)**.** *For every* $n \in \mathbb{N}_0$,

$$\mathsf{L}_n^{\mathrm{Lam}}(0) = 0, \tag{21.3}$$

$$\frac{\mathrm{d}}{\mathrm{d}u}\mathsf{L}^{\mathrm{Lam}}_{n+1}(u) = n\,\mathsf{L}^{\mathrm{Lam}}_{n}(u) - \frac{\mathrm{d}}{\mathrm{d}u}\mathsf{L}^{\mathrm{Lam}}_{n}(u), \tag{21.4}$$

$$\mathsf{L}^{\mathrm{Lam}}_{n+1}(u) = -\mathsf{L}^{\mathrm{Lam}}_{n}(u) + n\int_0^u \mathsf{L}^{\mathrm{Lam}}_{n}(v)\;\mathrm{d}v. \tag{21.5}$$

Moreover, (21.5) *for all* $n \geq 0$ *is equivalent to the conjunction of* (21.4) *for all* $n \geq 0$ *with* (21.3) *for all* $n \geq 0$*; and either version, together with* $\mathsf{L}^{\mathrm{Lam}}_0(u) = -u$*, determines the whole family uniquely.*

Proof. Boundary value. Expanding the operator in (21.1) as a $\mathbb{Q}$-linear combination of ∂_u^k, $0 \leq k \leq n$, and applying it to u^{n+1} produces a linear combination of the monomials u^{n+1-k} with $n+1-k \geq 1$. Hence $\mathsf{L}^{\mathrm{Lam}}_n$ has zero constant term, which is (21.3).

Differential recurrence. Write $A_n = \Delta_{0,n}$, so $A_{n+1} = (\partial_u - n)A_n$ and A_n commutes with ∂_u, all factors being polynomials in ∂_u. From (21.1),

$$\mathsf{L}^{\mathrm{Lam}}_{n+1}(u) = \frac{(-1)^n}{(n+2)!}\,(\partial_u - n)A_n\,u^{n+2}.$$

Differentiate and commute ∂_u past $(\partial_u - n)A_n$:

$$\frac{\mathrm{d}}{\mathrm{d}u}\mathsf{L}^{\mathrm{Lam}}_{n+1}(u) = \frac{(-1)^n}{(n+2)!}(\partial_u - n)A_n\,\partial_u u^{n+2} = \frac{(-1)^n(n+2)}{(n+2)!}(\partial_u - n)A_n\,u^{n+1} = \frac{(-1)^n}{(n+1)!}(\partial_u - n)A_n u^{n+1}.$$

By (21.1) again, $A_n u^{n+1} = (-1)^{n+1}(n+1)!\,\mathsf{L}^{\mathrm{Lam}}_n(u)$, so

$$\frac{\mathrm{d}}{\mathrm{d}u}\mathsf{L}^{\mathrm{Lam}}_{n+1}(u) = \frac{(-1)^n(-1)^{n+1}(n+1)!}{(n+1)!}\,(\partial_u - n)\mathsf{L}^{\mathrm{Lam}}_n(u) = -(\partial_u - n)\mathsf{L}^{\mathrm{Lam}}_n(u),$$

which is (21.4).

Equivalence of the two forms. Given (21.4) and (21.3), integrate (21.4) from 0 to u and use $\mathsf{L}^{\mathrm{Lam}}_{n+1}(0) = \mathsf{L}^{\mathrm{Lam}}_n(0) = 0$ to get (21.5). Conversely, differentiating (21.5) returns (21.4), and evaluating (21.5) at $u = 0$ gives $\mathsf{L}^{\mathrm{Lam}}_{n+1}(0) = -\mathsf{L}^{\mathrm{Lam}}_n(0)$; since $\mathsf{L}^{\mathrm{Lam}}_0(0) = 0$, induction yields (21.3) for all n.

Uniqueness. (21.5) computes $\mathsf{L}^{\mathrm{Lam}}_{n+1}$ from $\mathsf{L}^{\mathrm{Lam}}_n$ by operations with no free constant, so the family is determined by $\mathsf{L}^{\mathrm{Lam}}_0 = -u$. ☐

Lean status: Exact for the definition, the boundary value and the differential recurrence; the integral form and the uniqueness clause are not formalized. `LambertCoefficientPolynomials.lean` defines `Fabius.lambertPoly` by the operator formula (21.1) verbatim, with the shifted derivative `Fabius.lambertShiftOp` and the falling operator `Fabius.lambertFallingOp`; `Fabius.lambertPoly_zero` and `Fabius.lambertPoly_one` are $\mathsf{L}^{\mathrm{Lam}}_0 = -u$, $\mathsf{L}^{\mathrm{Lam}}_1 = u$. The boundary value (21.3) is `Fabius.lambertPoly_eval_zero`, proved by the order-of-vanishing bookkeeping of the proof above (`Fabius.coeff_eq_zero_lambertFalling`), and the differential recurrence (21.4) is `Fabius.derivative_lambertPoly_succ`, with the commutation of the operator with ∂_u as `Fabius.derivative_lambertFalling`. The integral form (21.5) and the uniqueness clause need polynomial antidifferentiation and are not formalized.

Remark 21.3 (Reconciliation with the one-based drafts). Sources 3 and 6 state the recurrence as starting from $\mathsf{L}^{\mathrm{Lam}}_1(u) = u$ with $\mathsf{L}^{\mathrm{Lam}}_n(0) = 0$ imposed. The zero-based normalisation subsumes this: with $\mathsf{L}^{\mathrm{Lam}}_0 = -u$, (21.5) at $n = 0$ reads $\mathsf{L}^{\mathrm{Lam}}_1 = -\mathsf{L}^{\mathrm{Lam}}_0 + 0 = u$. Source 4 derives (21.4) by "equating coefficients" in the first-order equation $T\mathcal{L}\,\partial_T\mathcal{L} + (1+T)\partial_T\mathcal{L} + \mathcal{L} = 0$ satisfied by the generating series; that equation is *quadratic* in $\mathcal{L}$ and equating coefficients in it yields a convolution recurrence, not the two-term linear one. The correct generating-function route runs through the *linear* equation (21.15) below, and is given in Theorem 21.11.

21.3 The closed Stirling form

We use the signed and unsigned Stirling numbers of the first kind in the catalogue normalisation:

$$(z)^{\underline{n}} = z(z-1)\cdots(z-n+1) = \sum_{k=0}^{n} s(n,k)\, z^k, \qquad \begin{bmatrix} n \\ k \end{bmatrix} = (-1)^{n-k} s(n,k)\,, \tag{21.6}$$

with $s(0,0) = 1$ and $s(n,k) = 0$ for $k < 0$ or $k > n$.

Theorem 21.4 (Stirling-number formula)**.** *For every $n \geq 1$,*

$$\mathsf{L}_n^{\mathrm{Lam}}(u) = \sum_{m=1}^{n} (-1)^{n-m} \begin{bmatrix} n \\ n-m+1 \end{bmatrix} \frac{u^m}{m!}. \tag{21.7}$$

Equivalently, writing $\mathsf{L}_n^{\mathrm{Lam}}(u) = \sum_{m\geq 0} \lambda_{n,m} u^m$,

$$\lambda_{n,m} = \frac{(-1)^{n-m}}{m!} \begin{bmatrix} n \\ n-m+1 \end{bmatrix} = \frac{(-1)^{n+1}}{m!} s(n, n-m+1)\,, \qquad 1 \leq m \leq n, \tag{21.8}$$

and $\lambda_{n,m} = 0$ for $m = 0$ and for $m > n$. If the upper limit in (21.7) *is raised from n to $n+1$, the formula holds for $n = 0$ as well and returns $\mathsf{L}_0^{\mathrm{Lam}}(u) = -u$.*

Proof. Substituting $z \mapsto \partial_u$ in (21.6) — a substitution of a commuting operator into a polynomial identity, hence valid — gives $\Delta_{0,n} = \sum_{k=0}^{n} s(n,k)\, \partial_u^k$. Since $\partial_u^k u^{n+1} = \frac{(n+1)!}{(n+1-k)!} u^{n+1-k}$ for $0 \leq k \leq n+1$, (21.1) gives

$$\mathsf{L}_n^{\mathrm{Lam}}(u) = \frac{(-1)^{n+1}}{(n+1)!} \sum_{k=0}^{n} s(n,k)\, \frac{(n+1)!}{(n+1-k)!} u^{n+1-k} = (-1)^{n+1} \sum_{k=0}^{n} s(n,k)\, \frac{u^{n+1-k}}{(n+1-k)!}.$$

Put $m = n+1-k$, so $k = n-m+1$ and m runs from 1 to $n+1$:

$$\mathsf{L}_n^{\mathrm{Lam}}(u) = (-1)^{n+1} \sum_{m=1}^{n+1} s(n, n-m+1)\, \frac{u^m}{m!}.$$

This is the second equality of (21.8), and the first follows from $s(n, n-m+1) = (-1)^{n-(n-m+1)} \begin{bmatrix} n \\ n-m+1 \end{bmatrix} = (-1)^{m-1} \begin{bmatrix} n \\ n-m+1 \end{bmatrix}$, so that $(-1)^{n+1}(-1)^{m-1} = (-1)^{n+m} = (-1)^{n-m}$. For $n \geq 1$, the term $m = n+1$ carries $s(n,0) = 0$ and may be dropped, giving (21.7). For $n = 0$ the only term is $m = 1$, with $s(0,0) = 1$ and prefactor $(-1)^1$, giving $-u$. □

Remark 21.5 (A second proof, and where the Stirling numbers come from). Formula (21.7) can also be obtained from Theorem 21.2 alone, and that route explains the appearance of Stirling numbers structurally. Put $\sigma_{n,m} := m!\, \lambda_{n,m}$. The integral recurrence (21.5) reads, coefficientwise, $\lambda_{n+1,m} = -\lambda_{n,m} + \frac{n}{m} \lambda_{n,m-1}$, which in the σ normalisation becomes

$$\sigma_{n+1,m} = -\sigma_{n,m} + n\, \sigma_{n,m-1}, \qquad \sigma_{0,1} = -1. \tag{21.9}$$

Setting $\sigma_{n,m} = (-1)^{n-m} \begin{bmatrix} n \\ n-m+1 \end{bmatrix}$ and $k = n-m+2$ turns (21.9) into $\begin{bmatrix} n+1 \\ k \end{bmatrix} = \begin{bmatrix} n \\ k-1 \end{bmatrix} + n \begin{bmatrix} n \\ k \end{bmatrix}$, the defining recurrence of the unsigned Stirling numbers. Thus the Lambert recurrence *is* the Stirling recurrence, transported by the normalisation $u^m \mapsto u^m/m!$. Source 4 uses this induction; it is valid, but it presupposes the recurrence, whereas the proof above needs only the definition.

21.4 Coefficient structure

The four "edge" coefficients below all come from a single generating polynomial, so we prove the Stirling evaluations rather than quote them.

Lean status: Exact. `LambertPolynomialStirling.lean` proves the signed coefficient form of (21.8) as `Fabius.coeff_lambertPoly`, for every $m \le n+1$; the unsigned form (21.7) is `Fabius.coeff_lambertPoly_eq_unsigned` in `LambertPolynomialCoefficients.lean`, for $1 \le m \le n$, and the vanishing outside that range is `Fabius.coeff_lambertPoly_of_lt` together with `Fabius.coeff_lambertPoly_succ_self`. The route is the one above made literal: `Fabius.lambertFallingOp_eq_derivEval` says the falling operator is the falling factorial evaluated at ∂_u, the evaluation being `Polynomial.aeval` into the endomorphism algebra of $\mathbb{Q}[u]$, and `Fabius.descPochhammer_eq_sum_monomial_signedStirlingFirst` supplies (21.6).

Lemma 21.6 (Four Stirling evaluations). *For $n \ge 1$ let $H_r = \sum_{j=1}^{r} j^{-1}$ and $H_r^{(2)} = \sum_{j=1}^{r} j^{-2}$, with $H_0 = H_0^{(2)} = 0$. Then*

$$\begin{bmatrix} n \\ 1 \end{bmatrix} = (n-1)!, \qquad \begin{bmatrix} n \\ 2 \end{bmatrix} = (n-1)!\,H_{n-1}, \qquad \begin{bmatrix} n \\ 3 \end{bmatrix} = \frac{(n-1)!}{2}\Big(H_{n-1}^2 - H_{n-1}^{(2)}\Big),$$

and for $n \ge 2$, $\left[{n \atop n-1}\right] = \binom{n}{2}$, while $\left[{n \atop n}\right] = 1$.

Proof. Replacing z by $-z$ in (21.6) gives the unsigned form $(z)^{\overline{n}} = \sum_k \left[{n \atop k}\right] z^k$, where $(z)^{\overline{n}} = z(z+1)\cdots(z+n-1)$. Since $\left[{n \atop 0}\right] = 0$ for $n \ge 1$, divide by z and set

$$f_n(z) := \prod_{j=1}^{n-1}(z+j) = \sum_{k\ge 1} \begin{bmatrix} n \\ k \end{bmatrix} z^{k-1}.$$

Then $\left[{n \atop 1}\right] = f_n(0) = (n-1)!$; f_n is monic of degree $n-1$, so $\left[{n \atop n}\right] = 1$; and the coefficient of z^{n-2} in f_n is $\sum_{j=1}^{n-1} j = \binom{n}{2}$, i.e. $\left[{n \atop n-1}\right] = \binom{n}{2}$. For the remaining two, logarithmic differentiation gives $f_n' = f_n \sum_{j=1}^{n-1}(z+j)^{-1}$ and $f_n'' = f_n\big[(\sum_j (z+j)^{-1})^2 - \sum_j (z+j)^{-2}\big]$; evaluating at $z=0$, $\left[{n \atop 2}\right] = f_n'(0) = (n-1)!H_{n-1}$ and $\left[{n \atop 3}\right] = \frac{1}{2}f_n''(0) = \frac{1}{2}(n-1)!(H_{n-1}^2 - H_{n-1}^{(2)})$. □

Corollary 21.7 (Coefficient structure of $\mathsf{L}_n^{\mathrm{Lam}}$). *Let $n \ge 1$. Then:*

(i) $\deg \mathsf{L}_n^{\mathrm{Lam}} = n$*, the constant term is* 0*, and* $u \mid \mathsf{L}_n^{\mathrm{Lam}}$*;*

(ii) *all n coefficients $\lambda_{n,1}, \dots, \lambda_{n,n}$ are nonzero and strictly alternate:* $\operatorname{sgn} \lambda_{n,m} = (-1)^{n-m}$*;*

(iii) $[u^n]\mathsf{L}_n^{\mathrm{Lam}} = 1/n$*;*

(iv) $[u^{n-1}]\mathsf{L}_n^{\mathrm{Lam}} = -H_{n-1}$ *for $n \ge 2$;*

(v) $[u]\mathsf{L}_n^{\mathrm{Lam}} = (-1)^{n-1}$*, equivalently* $\mathsf{L}_n^{\mathrm{Lam}\prime}(0) = (-1)^{n-1}$*;*

(vi) $[u^{n-2}]\mathsf{L}_n^{\mathrm{Lam}} = \dfrac{n-1}{2}\Big(H_{n-1}^2 - H_{n-1}^{(2)}\Big)$ *for $n \ge 3$;*

(vii) $[u^2]\mathsf{L}_n^{\mathrm{Lam}} = (-1)^n \dfrac{n(n-1)}{4}$ *for $n \ge 2$.*

In particular $\mathsf{L}_n^{\mathrm{Lam}}(u) = \dfrac{u^n}{n} - H_{n-1}u^{n-1} + \cdots + (-1)^{n-1}u.$

Proof. All statements read off (21.8) together with Theorem 21.6. (i) and (ii): for $1 \le m \le n$ the index $k = n-m+1$ satisfies $1 \le k \le n$, so $\left[{n \atop k}\right] > 0$ (it counts the permutations of n letters with k cycles, and there is at least one); hence $\lambda_{n,m} \ne 0$ with sign $(-1)^{n-m}$, and $\lambda_{n,m} = 0$ outside $1 \le m \le n$.

(iii): $m = n$ gives $k = 1$ and $\lambda_{n,n} = \left[{n \atop 1}\right]/n! = (n-1)!/n! = 1/n$. (iv): $m = n-1$ gives $k = 2$ and $\lambda_{n,n-1} = -\left[{n \atop 2}\right]/(n-1)! = -H_{n-1}$. (v): $m = 1$ gives $k = n$ and $\lambda_{n,1} = (-1)^{n-1}\left[{n \atop n}\right] = (-1)^{n-1}$. (vi): $m = n-2$ gives $k = 3$ and $\lambda_{n,n-2} = \left[{n \atop 3}\right]/(n-2)! = \frac{(n-1)!}{2(n-2)!}(H_{n-1}^2 - H_{n-1}^{(2)})$. (vii): $m = 2$ gives $k = n-1$ and $\lambda_{n,2} = (-1)^{n-2}\binom{n}{2}/2! = (-1)^n n(n-1)/4$. $\square$

Remark 21.8 (The formulas overlap consistently). At small n several of the seven statements describe the same coefficient, and they agree. At $n = 3$, (iv) and (vii) both give $[u^2]\mathsf{L}_3^{\mathrm{Lam}} = -\frac{3}{2}$, and (v) and (vi) both give $[u]\mathsf{L}_3^{\mathrm{Lam}} = 1$. At $n = 4$, (vi) and (vii) both give $[u^2]\mathsf{L}_4^{\mathrm{Lam}} = 3$, since $\frac{3}{2}(H_3^2 - H_3^{(2)}) = \frac{3}{2}\left(\frac{121}{36} - \frac{49}{36}\right) = 3$. Formula (vi) also returns the correct value 0 at $n = 2$, where $H_1^2 - H_1^{(2)} = 0$ and $[u^0]\mathsf{L}_2^{\mathrm{Lam}} = 0$; it is stated for $n \geq 3$ only because u^{n-2} is not a monomial for $n < 2$.

21.5 Integrality and the scaled family

Lean status: partial; (i), (ii), (iii), (v), (vii) Exact, (iv) and (vi) not formalized. In `LambertPolynomialCoefficients.lean`: (i) is `Fabius.natDegree_lambertPoly` and `Fabius.X_dvd_lambertPoly`, with the constant term `Fabius.lambertPoly_eval_zero`; (ii) is `Fabius.neg_one_pow_mul_coeff_lambertPoly_pos`, that $(-1)^{n-m}\lambda_{n,m} > 0$ for $1 \leq m \leq n$, together with `Fabius.coeff_lambertPoly_ne_zero`, resting on the positivity `Fabius.stirlingFirst_pos` of the unsigned numbers; (iii) is `Fabius.coeff_lambertPoly_self`; (v) is `Fabius.coeff_lambertPoly_one`; (vii) is `Fabius.coeff_lambertPoly_two`. Items (iv) and (vi) need the harmonic-number values of Theorem 21.6, which are not in Mathlib and not yet in the corpus; they will be cited once formalized.

Corollary 21.9 (Integrality). *For $n \geq 1$ and $0 \leq m \leq n$,*

$$\sigma_{n,m} := m!\,[u^m]\mathsf{L}_n^{\mathrm{Lam}}(u) = (-1)^{n-m}\left[{n \atop n-m+1}\right] \in \mathbb{Z}. \tag{21.10}$$

Consequently the scaled family

$$\mathsf{L}_n^{\mathrm{Lam,sc}}(u) := n!\,\mathsf{L}_n^{\mathrm{Lam}}(u) = \sum_{m=1}^{n} \frac{n!}{m!}\,\sigma_{n,m}u^m \tag{21.11}$$

lies in $\mathbb{Z}[u]$, is divisible by u, has degree n, leading coefficient $(n-1)!$ and linear coefficient $(-1)^{n-1}n!$, and satisfies

$$\mathsf{L}_{n+1}^{\mathrm{Lam,sc}}(u) = -(n+1)\mathsf{L}_n^{\mathrm{Lam,sc}}(u) + n(n+1)\int_0^u \mathsf{L}_n^{\mathrm{Lam,sc}}(v)\,\mathrm{d}v. \tag{21.12}$$

The convention $\mathsf{L}_0^{\mathrm{Lam,sc}}(u) = 0!\,\mathsf{L}_0^{\mathrm{Lam}}(u) = -u$ makes (21.12) valid at $n = 0$ as well.

Proof. (21.10) is (21.8) rewritten, and $\left[{n \atop k}\right] \in \mathbb{Z}$. Then $n!\lambda_{n,m} = \frac{n!}{m!}\sigma_{n,m}$ is an integer because $m \leq n$. The listed data are Theorem 21.7(i), (iii), (v) multiplied by $n!$. Multiplying (21.5) by $(n+1)!$ and using $(n+1)! = (n+1)\cdot n!$ gives (21.12); at $n = 0$ both sides equal u. $\square$

A common trap

(21.12) is an identity in $\mathbb{Q}[u]$ and does *not* manifestly preserve $\mathbb{Z}[u]$: the antiderivative divides the coefficient of u^m by $m+1$, and $n(n+1)/(m+1)$ need not be an integer. Integrality of $\mathsf{L}_n^{\mathrm{Lam,sc}}$ is a theorem (Theorem 21.9), proved through the Stirling form, not a visible feature of the recurrence. An implementation that wants to stay inside $\mathbb{Z}$ should iterate (21.9) on the integers $\sigma_{n,m}$ — equivalently, build the Stirling triangle — and divide by $m!$ only at output time.

The first scaled polynomials are

$$\mathsf{L}_0^{\mathrm{Lam,sc}}(u) = -u, \qquad \mathsf{L}_1^{\mathrm{Lam,sc}}(u) = u,$$

$$\mathsf{L}_2^{\mathrm{Lam,sc}}(u) = u^2 - 2u, \qquad \mathsf{L}_3^{\mathrm{Lam,sc}}(u) = 2u^3 - 9u^2 + 6u,$$
$$\mathsf{L}_4^{\mathrm{Lam,sc}}(u) = 6u^4 - 44u^3 + 72u^2 - 24u, \quad \mathsf{L}_5^{\mathrm{Lam,sc}}(u) = 24u^5 - 250u^4 + 700u^3 - 600u^2 + 120u, \tag{21.13}$$

$$\mathsf{L}_6^{\mathrm{Lam,sc}}(u) = 120u^6 - 1644u^5 + 6750u^4 - 10200u^3 + 5400u^2 - 720u. \tag{21.14}$$

Further values, and the corresponding $\mathsf{L}_n^{\mathrm{Lam}}$, are tabulated in Part D.

21.6 The generating equation

Lemma 21.10 (The linear generating equation). *Let* $\mathcal{P}(s,u) := \sum_{n\geq 0} \mathsf{L}_n^{\mathrm{Lam}}(u)s^n \in \mathbb{K}[u][[s]]$. *Then*

$$s^2\,\partial_s\mathcal{P} = (1+s)\,\partial_u\mathcal{P} + 1. \tag{21.15}$$

Proof. Multiply (21.4) by s^{n+1} and sum over $n \geq 0$. The right-hand side gives $\sum_{n\geq 0} n\mathsf{L}_n^{\mathrm{Lam}}(u)s^{n+1} = s \cdot s\,\partial_s\mathcal{P} = s^2\partial_s\mathcal{P}$. The left-hand side gives

$$\sum_{n\geq 0}\Big(\mathsf{L}_{n+1}^{\mathrm{Lam}\prime}(u) + \mathsf{L}_n^{\mathrm{Lam}\prime}(u)\Big)s^{n+1} = \big(\partial_u\mathcal{P} - \mathsf{L}_0^{\mathrm{Lam}\prime}(u)\big) + s\,\partial_u\mathcal{P} = (1+s)\partial_u\mathcal{P} + 1,$$

using $\mathsf{L}_0^{\mathrm{Lam}\prime}(u) = -1$. □

Theorem 21.11 (Implicit generating equation and formal uniqueness). *Let* $Q(s,u) := \sum_{n\geq 1} \mathsf{L}_n^{\mathrm{Lam}}(u)s^n \in \mathbb{K}[u][[s]]$. *Then* $Q(0,u) = 0$ *and*

$$Q = -\log\big(1 - su + sQ\big), \tag{21.16}$$

the logarithm being the formal one on $1 + s\mathbb{K}[u][[s]]$. *Conversely,* (21.16) *together with* $Q(0,u) = 0$ *has exactly one solution in* $\mathbb{K}[u][[s]]$*; hence it determines the family* $(\mathsf{L}_n^{\mathrm{Lam}})_{n\geq 1}$*, and therefore also* $\mathsf{L}_0^{\mathrm{Lam}} = -u$ *through* (21.5)*, uniquely.*

Proof. Write $\mathcal{P} = -u + Q$ as in Theorem 21.10. Since $s\mathcal{P} \in s\mathbb{K}[u][[s]]$, the element $E := \mathcal{P} + u + \log(1 + s\mathcal{P}) \in \mathbb{K}[u][[s]]$ is defined, and (21.16) is exactly $E = 0$.

Step 1: E satisfies the homogeneous equation. Put $D = 1 + s\mathcal{P}$, a unit of $\mathbb{K}[u][[s]]$. Then

$$\partial_u E = \partial_u\mathcal{P} + 1 + \frac{s\,\partial_u\mathcal{P}}{D}, \qquad \partial_s E = \partial_s\mathcal{P} + \frac{\mathcal{P} + s\,\partial_s\mathcal{P}}{D}.$$

Using $s^2\partial_s\mathcal{P} = (1+s)\partial_u\mathcal{P} + 1$ from (21.15), hence $s^3\partial_s\mathcal{P} = s(1+s)\partial_u\mathcal{P} + s$, we get

$$s^2\partial_s E - (1+s)\partial_u E = \big[(1+s)\partial_u\mathcal{P} + 1\big] - (1+s)\partial_u\mathcal{P} - (1+s) + \frac{s^2\mathcal{P} + s(1+s)\partial_u\mathcal{P} + s - s(1+s)\partial_u\mathcal{P}}{D}.$$

The fraction is $\dfrac{s^2\mathcal{P} + s}{D} = \dfrac{s(1+s\mathcal{P})}{D} = s$, so the whole expression equals $1 - (1+s) + s = 0$. Thus

$$s^2\partial_s E = (1+s)\partial_u E. \tag{21.17}$$

Step 2: E vanishes on $u = 0$. By (21.3), $\mathcal{P}(s,0) = \sum_n \mathsf{L}_n^{\mathrm{Lam}}(0)s^n = 0$. Applying the continuous $\mathbb{K}$-algebra map $u \mapsto 0$ to E gives $E(s,0) = 0 + 0 + \log(1+0) = 0$.

Step 3. Write $E = \sum_{n\geq 0} e_n(u)s^n$. Comparing coefficients of s^0 and s^{n+1} in (21.17) gives $e_0' = 0$ and $e_{n+1}' = n\,e_n - e_n'$; Step 2 gives $e_n(0) = 0$ for all n. Hence e_0 is a constant with $e_0(0) = 0$, so $e_0 = 0$; and inductively, if $e_n = 0$ then $e_{n+1}' = 0$ and $e_{n+1}(0) = 0$, so $e_{n+1} = 0$. Thus $E = 0$, which is (21.16).

Uniqueness. Let $\mathcal{S}(Q) = -\log(1 - su + sQ)$. For $Q \in \mathbb{K}[u][[s]]$ the argument lies in $1 + s\mathbb{K}[u][[s]]$, so $\mathcal{S}(Q) \in s\mathbb{K}[u][[s]]$ and in particular $\mathcal{S}(Q)(0, u) = 0$. If $Q_1 - Q_2 \in s^N\mathbb{K}[u][[s]]$ then, $1 - su + sQ_2$ being a unit,

$$\mathcal{S}(Q_1) - \mathcal{S}(Q_2) = -\log\left(1 + \frac{s(Q_1 - Q_2)}{1 - su + sQ_2}\right) \in s^{N+1}\mathbb{K}[u][[s]].$$

So $\mathcal{S}$ is a strict contraction for the s-adic filtration on $\mathbb{K}[u][[s]]$, which is separated and complete; it therefore has exactly one fixed point, and that fixed point automatically satisfies $Q(0, u) = 0$. □

Corollary 21.12 (Reversion-free proof of Theorem 20.10)**.** *The generating equation* (21.16) *determines the expansion of* ω *on its own:*

$$(X + L)^{\langle -1\rangle_\circ} = X + \mathcal{P}(t, L), \qquad \mathcal{P}(t, L) = \sum_{n\geq 0} \mathsf{L}_n^{\mathrm{Lam}}(L)t^n,$$

so Theorem 20.10 holds without appeal to Lagrange–Bürmann and without Theorem 20.8.

Proof. Substitute $s \mapsto t$ and $u \mapsto L$ in (21.16), and add $u = L$ to both sides of $\mathcal{P} = -u + Q$. The result is exactly the assertion that $\mathcal{P}(t, L)$ satisfies the implicit equation (20.9). By the uniqueness half of Theorem 20.9, the solution of that equation in $\mathcal{G}_{\mathrm{poly},\mathbb{K}}$ is unique, so $(X + L)^{\langle -1\rangle_\circ} = X + \mathcal{P}(t, L)$.

The two ingredients used here are the existence and uniqueness of the fixed point of the generating equation and the uniqueness half of Theorem 20.9; neither depends on Theorem 18.14 or on the repeated-derivative computation, which is what makes this route independent. □

Remark 21.13 (Which route to use). Three independent characterisations of the same family are now available: the operator formula (21.1), the recurrence (21.5), and the generating equation (21.16). The recurrence is the cheapest generator (one exact polynomial integration and one subtraction per index); the operator formula is the right tool for proving identities, since it makes the falling factorial visible; the generating equation is the right tool for uniqueness statements and for transporting the family to a new inversion problem, as in Theorem 20.5. Agreement of two of them on a computed table is a genuine check only if the two were implemented independently; (20.14) of Theorem 20.13 is a stronger test, because it exercises the composition code as well.

21.7 The Laplace transform and the exponential moments

The remaining result is of a different type: it is an assertion about convergent integrals of the real polynomial functions $\mathsf{L}_n^{\mathrm{Lam}}$, not a formal identity, and its hypotheses have real content.

Theorem 21.14 (Laplace transform of $\mathsf{L}_n^{\mathrm{Lam}}$)**.** *Let* $n \geq 1$ *and let* $s \in \mathbb{C}$ *with* $\Re s > 0$*. Then the integral below converges absolutely and*

$$\int_0^\infty \mathrm{e}^{-su}\,\mathsf{L}_n^{\mathrm{Lam}}(u)\,\mathrm{d}u = \frac{1}{s^{n+1}}\prod_{j=1}^{n-1}(j - s) = \frac{(1 - s)^{\overline{n-1}}}{s^{n+1}}, \tag{21.18}$$

the product being empty (and equal to 1*) when* $n = 1$*.*

Proof. Absolute convergence is clear: $\mathsf{L}_n^{\mathrm{Lam}}$ is a polynomial and $|\mathrm{e}^{-su}| = \mathrm{e}^{-u\Re s}$ with $\Re s > 0$, so the integrand is $\mathcal{O}_{u\to+\infty}(\mathrm{e}^{-u\Re s/2})$. Write $I_n(s) = \int_0^\infty \mathrm{e}^{-su}\mathsf{L}_n^{\mathrm{Lam}}(u)\,\mathrm{d}u$.

The integration by parts. For $0 < b < \infty$,

$$\int_0^b \mathrm{e}^{-su}\mathsf{L}_n^{\mathrm{Lam}\prime}(u)\,\mathrm{d}u = \left[\mathrm{e}^{-su}\mathsf{L}_n^{\mathrm{Lam}}(u)\right]_0^b + s\int_0^b \mathrm{e}^{-su}\mathsf{L}_n^{\mathrm{Lam}}(u)\,\mathrm{d}u.$$

The boundary term at $u = 0$ vanishes because $\mathsf{L}_n^{\mathrm{Lam}}(0) = 0$ (Theorem 21.2); the boundary term at $u = b$ tends to 0 as $b \to \infty$ because $\Re s > 0$ and $\mathsf{L}_n^{\mathrm{Lam}}$ is a polynomial. Both hypotheses are used, and neither can be dropped: without $\Re s > 0$ the integral does not converge, and without $\mathsf{L}_n^{\mathrm{Lam}}(0) = 0$ an extra term $-\mathsf{L}_n^{\mathrm{Lam}}(0)$ would survive. Letting $b \to \infty$,

$$\int_0^\infty \mathrm{e}^{-su} \mathsf{L}_n^{\mathrm{Lam}\prime}(u)\ \mathrm{d}u = s\, I_n(s). \tag{21.19}$$

The recursion. Apply (21.19) to both sides of (21.4), which is an identity of polynomials and may therefore be integrated against e^{-su}:

$$s\, I_{n+1}(s) = n\, I_n(s) - s\, I_n(s), \qquad \text{i.e.} \qquad I_{n+1}(s) = \frac{n-s}{s}\, I_n(s) \quad (n \geq 1).$$

The base case is $I_1(s) = \int_0^\infty \mathrm{e}^{-su} u\ \mathrm{d}u = s^{-2}$. Iterating, $I_n(s) = s^{-2} \prod_{j=1}^{n-1} \frac{j-s}{s} = s^{-(n+1)} \prod_{j=1}^{n-1}(j - s)$. Finally $\prod_{j=1}^{n-1}(j-s) = (1-s)(2-s)\cdots(n-1-s) = \overline{(1-s)}^{\,n-1}$. ☐

Corollary 21.15 (Exponential moment cancellations)**.** *Let $n \geq 1$. For every integer r with $1 \leq r \leq n-1$,*

$$\int_0^\infty \mathrm{e}^{-ru}\, \mathsf{L}_n^{\mathrm{Lam}}(u)\ \mathrm{d}u = 0. \tag{21.20}$$

The range is empty for $n = 1$, so the first genuine cancellation is $\int_0^\infty \mathrm{e}^{-u} \mathsf{L}_2^{\mathrm{Lam}}(u)\ \mathrm{d}u = 0$.

Proof. Such an r is a positive real number, so $\Re r > 0$ and Theorem 21.14 applies with $s = r$. The factor $j = r$ occurs in $\prod_{j=1}^{n-1}(j-s)$ precisely because $1 \leq r \leq n-1$, and it vanishes at $s = r$. ☐

Remark 21.16 (The transform is the falling factorial again). Theorem 21.14 and Theorem 21.4 are equivalent, and the equivalence explains the cancellations of Theorem 21.15. Indeed $(s)^{\underline{n}} = s \prod_{j=1}^{n-1}(s - j) = (-1)^{n-1} s \prod_{j=1}^{n-1}(j-s)$, so by (21.6)

$$\prod_{j=1}^{n-1}(j-s) = (-1)^{n-1} \sum_{k \geq 1} s(n,k)\, s^{k-1} = \sum_{k \geq 1} (-1)^{k-1} \begin{bmatrix} n \\ k \end{bmatrix} s^{k-1},$$

whence $I_n(s) = \sum_{k=1}^n (-1)^{k-1} \left[{n \atop k}\right] s^{-(n-k+2)}$. Comparing with the termwise transform $I_n(s) = \sum_{m=1}^n \lambda_{n,m}\, m!\, s^{-(m+1)}$ and matching powers via $m = n - k + 1$ returns exactly (21.8). Thus (21.20) says that the falling factorial $(s)^{\underline{n}}$ vanishes at $s = 1, \ldots, n-1$ — the cancellations are the roots of the very operator that defines $\mathsf{L}_n^{\mathrm{Lam}}$ in (21.1).

A common trap

The letter s carries two unrelated meanings in this section. In Theorem 21.10 and Theorem 21.11 it is a formal indeterminate and no analytic content is asserted. In Theorem 21.14 it is a complex number constrained by $\Re s > 0$, and the identity is an equality of convergent integrals with analytic functions of s on the half-plane. A "Laplace transform of the generating series" mixing the two would be a different object and is not claimed anywhere here.

Chapter 22

Lambert W: branches and expansions

Every result of Parts I–IV was formal: an identity between elements of $\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$, $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$, $\mathscr{PL}_{\mathrm{rat},\mathbb{K}}$ or $\mathscr{PL}_{\mathrm{itLaur},\mathbb{K}}$, or between elements of the near-identity group $\mathcal{G}_{\mathrm{poly},\mathbb{K}}$. Lambert's W is where that formal theory meets a genuine multivalued analytic function, and it is therefore the place where the boundary between an algebraic identity and an analytic theorem has to be drawn with care. The six source reports draw it in six different places, and none of them draws it entirely correctly; the present section states exactly what is proved, proves it, and marks what is not.

The organising principle is a single substitution. The equation $we^w = z$ is not itself a polynomial–logarithmic reversion problem: its natural variable z is exponentially larger than its solution. After one logarithm it becomes the Wright omega problem $w + \log w = \xi$, which is exactly the reversion of $X + L$ in $\mathcal{G}_{\mathrm{poly},\mathbb{K}}$, already solved in Part 20. All of W — every branch, the lower real branch near zero, and the generalized Lambert families of Part 23 — is that one reversion evaluated at different arguments. What changes from branch to branch is the argument, not the reversion.

22.1 The defining equation and the real branches

Definition 22.1 (Branches of W). For $k \in \mathbb{Z}$, a *branch* W_k of the Lambert function is a determination of the inverse of

$$g\colon \mathbb{C} \longrightarrow \mathbb{C}, \qquad g(w) \coloneqq w\mathrm{e}^w,$$

so that

$$W_k(z)\mathrm{e}^{W_k(z)} = z. \tag{22.1}$$

The map g is entire and infinite-to-one, so Equation (22.1) alone determines nothing: a branch is the pair consisting of a solution and a rule for selecting it. Following the notation catalogue, W_0 is the *principal* branch, real and increasing on $[-\mathrm{e}^{-1}, \infty)$ with $W_0(0) = 0$, analytic at 0, and cut along $(-\infty, -\mathrm{e}^{-1}]$; W_{-1} is the *lower real* branch, real on $[-\mathrm{e}^{-1}, 0)$ with values in $(-\infty, -1]$. A branchless symbol $W(z)$ is admissible only where the domain forces the real principal branch.

Proposition 22.2 (The two real branches). *Let $g(w) = w\mathrm{e}^w$ on $\mathbb{R}$. Then:*

1. *g is strictly decreasing on $(-\infty, -1]$, with image $[-\mathrm{e}^{-1}, 0)$, and strictly increasing on $[-1, \infty)$, with image $[-\mathrm{e}^{-1}, \infty)$;*
2. *consequently g restricted to $[-1, \infty)$ has a continuous strictly increasing inverse $W_0\colon [-\mathrm{e}^{-1}, \infty) \to [-1, \infty)$, and g restricted to $(-\infty, -1]$ has a continuous strictly decreasing inverse $W_{-1}\colon [-\mathrm{e}^{-1}, 0) \to (-\infty, -1]$;*
3. *for $x \in (-\mathrm{e}^{-1}, 0)$ the equation $g(w) = x$ has exactly the two real solutions $W_0(x) \in (-1, 0)$ and $W_{-1}(x) \in (-\infty, -1)$; for $x = -\mathrm{e}^{-1}$ it has the single solution -1; for $x \geq 0$ it has exactly one; and for $x < -\mathrm{e}^{-1}$ none;*

4. $W_{-1}(x) \to -\infty$ *as* $x \uparrow 0$.

Proof. We have $g'(w) = (1+w)\mathrm{e}^w$, which is negative for $w < -1$, zero at $w = -1$ and positive for $w > -1$; thus g is strictly decreasing on $(-\infty, -1]$ and strictly increasing on $[-1, \infty)$, with the single minimum $g(-1) = -\mathrm{e}^{-1}$. On $(-\infty, -1]$ we have $g(w) < 0$ for $w < 0$ and $g(w) \to 0^-$ as $w \to -\infty$, because $|w|\,\mathrm{e}^w \to 0$; hence g maps $(-\infty, -1]$ bijectively onto $[-\mathrm{e}^{-1}, 0)$, decreasingly. On $[-1, \infty)$, $g(-1) = -\mathrm{e}^{-1}$ and $g(w) \to +\infty$, so g maps $[-1, \infty)$ bijectively onto $[-\mathrm{e}^{-1}, \infty)$, increasingly. Items (i)–(iii) follow, and (iv) is the statement that the decreasing inverse of $g|_{(-\infty,-1]}$ tends to $-\infty$ as its argument tends to the endpoint 0^- of the image. □

Lemma 22.3 (Derivative of a branch). *Let* $w = W_k(z)$ *be analytic near* z_0 *with* $z_0 \neq 0$ *and* $w \neq -1$. *Then*

$$\frac{\mathrm{d}W_k}{\mathrm{d}z}(z) = \frac{W_k(z)}{z\bigl(1 + W_k(z)\bigr)}. \tag{22.2}$$

Proof. Differentiating $w\mathrm{e}^w = z$ gives $(1+w)\mathrm{e}^w w' = 1$. Since $\mathrm{e}^w = z/w$ whenever $w \neq 0$, this is $(1+w)(z/w)w' = 1$, which is Equation (22.2). The excluded cases $z = 0$ and $w = -1$ are exactly the points where the substitution $\mathrm{e}^w = z/w$ or the division by $1 + w$ is illegitimate; $w = -1$ is the branch point studied in Section 22.7. □

22.2 Branch coordinates and the substitution to the omega problem

Definition 22.4 (Branch coordinates). Let $z \in \mathbb{C} \setminus \{0\}$ and $k \in \mathbb{Z}$. The *outer branch coordinate* is

$$\xi_k := \operatorname{Log} z + 2\pi \mathrm{i} k, \tag{22.3}$$

with Log the principal logarithm, so that $\mathrm{e}^{\xi_k} = z$. Given in addition $\xi_k \neq 0$, an *inner branch coordinate* is any $\ell_k \in \mathbb{C}$ with

$$\mathrm{e}^{\ell_k} = \xi_k, \tag{22.4}$$

written $\ell_k = \log_{\mathcal{B}_k} \xi_k$ when a particular determination $\mathcal{B}_k$ has been fixed. The pair (ξ_k, ℓ_k) is the *branch datum*. A *coordinate sector* is a connected open $S_k \subseteq \mathbb{C} \setminus \{0\}$ on which $z \mapsto \xi_k$ and $\xi_k \mapsto \ell_k$ admit single-valued analytic determinations and on which $|\xi_k| \to \infty$ along the limits considered.

Three data must be recorded and are routinely conflated: the choice of $\operatorname{Log} z$ (equivalently, of the strip in which $\operatorname{Arg} z$ is taken), the integer k, and the determination $\mathcal{B}_k$ of the second logarithm. Reports S1, S2 and S4 all write $\ell_k = \operatorname{Log} \xi_k$, i.e. they silently force $\mathcal{B}_k$ to be principal. That is a legitimate normalisation and we use it in all numerical statements, but it is a hypothesis, not a convention-free fact: Theorem 22.16 shows exactly where it is used.

Proposition 22.5 (Reduction to the omega problem). *Let* $z \neq 0$, $k \in \mathbb{Z}$, *and let* $w \in \mathbb{C} \setminus \{0\}$. *The following are equivalent.*

1. $w\mathrm{e}^w = z$ *and there is a logarithm* λ *of* w *(that is,* $\mathrm{e}^\lambda = w$*) with* $w + \lambda = \xi_k$.

2. $w + \lambda = \xi_k$ *for some* λ *with* $\mathrm{e}^\lambda = w$.

Moreover, if $w\mathrm{e}^w = z$ *and* λ *is any logarithm of* w, *then* $w + \lambda = \xi_j$ *for exactly one* $j \in \mathbb{Z}$, *and every* j *arises from a suitable choice of* λ.

Proof. Assume (ii). Then $w\mathrm{e}^w = \mathrm{e}^\lambda \mathrm{e}^w = \mathrm{e}^{w+\lambda} = \mathrm{e}^{\xi_k} = z$, which is (i). Conversely (i) contains (ii).

For the last statement, let $w\mathrm{e}^w = z$ and $\mathrm{e}^\lambda = w$. Then $\mathrm{e}^{w+\lambda} = w\mathrm{e}^w = z = \mathrm{e}^{\operatorname{Log} z}$, so $w + \lambda - \operatorname{Log} z \in 2\pi\mathrm{i}\mathbb{Z}$, that is $w + \lambda = \xi_j$ for a unique j. Replacing λ by $\lambda + 2\pi\mathrm{i}m$ (still a logarithm of w) replaces j by $j + m$, so every integer occurs. □

Remark 22.6 (What the substitution does and does not say). Theorem 22.5 is the exact form of the informal step "take a logarithm of $we^w = z$". Two things must be noticed. First, the branch index k is *defined* by the relation $w + \lambda = \xi_k$; it is not an independent piece of data that can be chosen after w and λ. Second, the equivalence is with the *additive* equation $w + \lambda = \xi_k$, in which λ is a logarithm of the unknown w. Whether λ is the principal logarithm of w is a separate question, settled for large $|\xi_k|$ by Theorem 22.16. The discrepancy $\lambda - \operatorname{Log} w$, divided by $2\pi\mathrm{i}$, is the *unwinding number*; the identity $\omega(s) = W(\mathrm{e}^s)$ asserted without qualification in S1 and S3 holds only where that unwinding number vanishes.

22.3 The formal expansion and branch independence

We first record the formal statement, which is an identity in the near-identity group and involves no analysis whatsoever. Throughout, $\mathbb{K}$ is a field of characteristic zero, X is the large variable, $t = X^{-1}$, $L = \log X$, and $\mathcal{G}_{\mathrm{poly},\mathbb{K}} = X + \mathbb{K}[L][[t]]$.

Theorem 22.7 (Formal affine-logarithmic reversion). *Let $a \in \mathbb{K}$ and let $F = X + aL \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}$. Then the compositional inverse of F in $\mathcal{G}_{\mathrm{poly},\mathbb{K}}$ is*

$$F^{\langle -1\rangle\circ} = X + a\sum_{n\geq 0} a^n \mathsf{L}_n^{\mathrm{Lam}}(L)\, t^n = X - aL + \sum_{n\geq 1} a^{n+1}\mathsf{L}_n^{\mathrm{Lam}}(L)\, t^n. \tag{22.5}$$

In particular the coefficient of t^n in $F^{\langle -1\rangle\circ}$ is the fixed polynomial $\mathsf{L}_n^{\mathrm{Lam}}$ multiplied by the scalar a^{n+1}; no new polynomial family occurs for any value of a, and $a = 1$ recovers the Wright omega expansion of Theorem 20.10.

Proof. Let $\mathcal{Q}(s,u) = \sum_{n\geq 1} \mathsf{L}_n^{\mathrm{Lam}}(u)s^n \in \mathbb{K}[u][[s]]$ be the generating series of Theorem 21.11, characterised there as the unique element of $s\mathbb{K}[u][[s]]$ with

$$\mathcal{Q} = -\log\bigl(1 - su + s\mathcal{Q}\bigr). \tag{22.6}$$

Because at has positive t-order, the assignment $s \mapsto at$, $u \mapsto L$ extends to a continuous $\mathbb{K}$-algebra homomorphism $\mathbb{K}[u][[s]] \to \mathbb{K}[L][[t]]$ for the respective adic topologies. Write

$$\mathcal{Q}_a \coloneqq \mathcal{Q}(at, L) = \sum_{n\geq 1} a^n \mathsf{L}_n^{\mathrm{Lam}}(L)\, t^n \in t\mathbb{K}[L][[t]].$$

Applying the homomorphism to Equation (22.6) gives, in $\mathbb{K}[L][[t]]$,

$$\mathcal{Q}_a = -\log\bigl(1 - atL + at\,\mathcal{Q}_a\bigr), \tag{22.7}$$

the logarithm being the formal logarithm of Theorem 12.5, legitimate because the argument lies in $1 + t\mathbb{K}[L][[t]]$.

Put $G \coloneqq X - aL + a\mathcal{Q}_a$. Since $-aL + a\mathcal{Q}_a \in \mathbb{K}[L][[t]]$, we have $G \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}$, and

$$G = X\Bigl(1 - at\,(L - \mathcal{Q}_a)\Bigr),$$

whose second factor lies in $1 + t\mathbb{K}[L][[t]]$. By the exact generator substitution rule of Theorem 12.22,

$$L \circ G = \log G = L + \log\bigl(1 - atL + at\,\mathcal{Q}_a\bigr) = L - \mathcal{Q}_a,$$

the last equality being Equation (22.7). Hence

$$F \circ G = G + a\,(L \circ G) = X - aL + a\mathcal{Q}_a + aL - a\mathcal{Q}_a = X.$$

So G is a right inverse of F inside $\mathcal{G}_{\mathrm{poly},\mathbb{K}}$. By Theorem 16.11 the inverse in $\mathcal{G}_{\mathrm{poly},\mathbb{K}}$ exists and is unique and two-sided, so $G = F^{\langle -1\rangle\circ}$. Finally $a\mathsf{L}_0^{\mathrm{Lam}}(L) = -aL$, which converts the second display of Equation (22.5) into the first. $\square$

Remark 22.8 (Two proofs in the sources, and why this one is preferred). Reports S2 and S4 obtain Equation (22.5) by inspecting the Lagrange–Bürmann series and observing that its m-th term carries a factor a^m. That argument is correct but needs the summability of the Lagrange family (Theorem 18.9) and a separate identification of the m-th term with $\mathsf{L}_m^{\mathrm{Lam}}$. The proof above verifies a closed-form candidate against the defining equation and appeals only to uniqueness in $\mathcal{G}_{\mathrm{poly},\mathbb{K}}$; it is shorter, it is the same proof for every a including $a=-1$, and it is the one that transports without change to the analytic statement of Theorem 22.13.

Theorem 22.9 (Formal branch expansion)**.** *Let $k\in\mathbb{Z}$ and let (ξ_k,ℓ_k) be a branch datum in the sense of Theorem 22.4. Substituting $X\mapsto\xi_k$, $L\mapsto\ell_k$, $t\mapsto\xi_k^{-1}$ in Equation* (22.5) *with $a=1$ produces the formal series*

$$\xi_k+\sum_{n\geq 0}\frac{\mathsf{L}_n^{\mathrm{Lam}}(\ell_k)}{\xi_k^n}=\xi_k-\ell_k+\sum_{n\geq 1}\frac{\mathsf{L}_n^{\mathrm{Lam}}(\ell_k)}{\xi_k^n},\tag{22.8}$$

equivalently, by Theorem 21.4,

$$\xi_k-\ell_k+\sum_{n\geq 1}\frac{(-1)^n}{\xi_k^n}\sum_{m=1}^{n}\begin{bmatrix}n\\ n-m+1\end{bmatrix}\frac{(-\ell_k)^m}{m!}.\tag{22.9}$$

The coefficient polynomials $\mathsf{L}_n^{\mathrm{Lam}}$ do not depend on k, on the determination $\mathcal{B}_k$, or on the sector.

Proof. The first display is Equation (22.5) with $a=1$ after the stated substitution, which is a substitution of symbols and requires no justification beyond $\xi_k\neq 0$. For the second, insert the closed coefficient formula $\mathsf{L}_n^{\mathrm{Lam}}(u)=\sum_{m=1}^{n}(-1)^{n-m}\left[{n\atop n-m+1}\right]u^m/m!$ of Theorem 21.4 and use $(-1)^{n-m}=(-1)^n(-1)^m$, which converts u^m into $(-u)^m$ with an overall factor $(-1)^n$. Branch independence is the observation that Theorem 22.7 was proved once, in $\mathbb{K}[L][[t]]$, with no branch data present. $\square$

The next theorem is the precise form of the catalogue's assertion that "the polynomial family is branch-independent; the coordinates ξ_k,ℓ_k, their branches, and the analytic domain are not". Its first half is the uniqueness of the coefficients, and its second half exhibits the genuine k-dependence.

Theorem 22.10 (Branch independence of the coefficients, dependence of the coordinates)**.**

1. (Universality.) *There is exactly one sequence $(p_n)_{n\geq 0}$ in $\mathbb{K}[u]$ such that for every $a\in\mathbb{K}$ the map $X+aL\in\mathcal{G}_{\mathrm{poly},\mathbb{K}}$ has inverse $X+a\sum_{n\geq 0}a^np_n(L)t^n$; namely $p_n=\mathsf{L}_n^{\mathrm{Lam}}$. A fortiori, for each fixed k, the expansion Equation* (22.8) *determines the polynomials $\mathsf{L}_n^{\mathrm{Lam}}$ uniquely, and the answer is the same for all k.*
2. (Uniqueness against the coordinates.) *Fix k and a coordinate sector S_k on which $|\xi_k|\to\infty$ and $\ell_k=\operatorname{Log}\xi_k$. If $(q_n)_{n\geq 0}$ in $\mathbb{C}[u]$ satisfies $\sum_{n=0}^{N}q_n(\ell_k)\xi_k^{-n}=\mathcal{O}_{|\xi_k|\to\infty,\ \xi_k\in S_k}\left(\ell_k^{N+1}/\xi_k^{N+1}\right)$ for every N, then $q_n=0$ for all n.*
3. (The coordinates really differ.) *For $j\neq k$ and z in a common domain, $\xi_j-\xi_k=2\pi\mathrm{i}(j-k)\neq 0$, and the corresponding solutions w_j,w_k of Equation* (22.1) *supplied by Theorem 22.13 satisfy $w_j-w_k\to 2\pi\mathrm{i}(j-k)$ as $|z|\to\infty$ with $\operatorname{Arg}z$ bounded. In particular they are distinct functions.*

Proof. (i) Existence is Theorem 22.7. For uniqueness take $a=1$: if $X+\sum_{n\geq 0}p_n(L)t^n$ and $X+\sum_{n\geq 0}\mathsf{L}_n^{\mathrm{Lam}}(L)t^n$ are both inverses of $X+L$ in $\mathcal{G}_{\mathrm{poly},\mathbb{K}}$, they are equal by the uniqueness clause of Theorem 16.11, and equality in $\mathbb{K}[L][[t]]$ is coefficientwise by Theorem 6.9. (Uniqueness already follows from the single value $a=1$; no compatibility across different a is needed.)

(ii) Induct on n. Suppose $q_0=\cdots=q_{n-1}=0$ (vacuous for $n=0$). Taking $N=n$ in the hypothesis and multiplying by ξ_k^n gives $q_n(\ell_k)=\mathcal{O}_{|\xi_k|\to\infty,\ \xi_k\in S_k}\left(\ell_k^{n+1}/\xi_k\right)$. Suppose $q_n\neq 0$, say of degree $d\geq 0$ with leading coefficient $c\neq 0$. Since $\ell_k=\operatorname{Log}\xi_k$ we have $|\ell_k|\leq\log|\xi_k|+\pi$ and $|\ell_k|\geq\log|\xi_k|$, so $|\ell_k|\to\infty$ and $|q_n(\ell_k)|\geq\frac{1}{2}|c|\,|\ell_k|^d$ for $|\xi_k|$ large, while the right-hand side is at most $C(\log|\xi_k| + \pi)^{n+1}/|\xi_k| \to 0$. For $d \geq 1$ the left side tends to ∞ and for $d = 0$ it is the nonzero constant $|c|$; either way we contradict the bound. Hence $q_n = 0$.

(iii) The identity $\xi_j - \xi_k = 2\pi\mathrm{i}(j-k)$ is immediate from Equation (22.3). Since $\operatorname{Arg} z$ stays bounded, both $|\xi_j|$ and $|\xi_k|$ tend to infinity while $|\ell_j|, |\ell_k|$ grow only logarithmically, so for $|z|$ large both branch data satisfy $|\xi| \geq 10(1+|\ell|)$; we restrict to that range. By Theorem 22.13 applied with base point ξ_j, respectively ξ_k, we have $w_j = \xi_j - \ell_j + \mathcal{Y}_j$ and $w_k = \xi_k - \ell_k + \mathcal{Y}_k$, where Theorem 22.14 with $N = 0$ bounds $|\mathcal{Y}_j|, |\mathcal{Y}_k|$ by $10(1+|\ell|)/|\xi|$, which tends to 0. Hence

$$w_j - w_k - 2\pi\mathrm{i}(j-k) = -(\ell_j - \ell_k) + \mathcal{Y}_j - \mathcal{Y}_k.$$

Now $\ell_j - \ell_k = \operatorname{Log}(\xi_j) - \operatorname{Log}(\xi_k) \to 0$, because $\xi_j/\xi_k = 1 + 2\pi\mathrm{i}(j-k)/\xi_k \to 1$ and both $\operatorname{Arg}\xi_j$ and $\operatorname{Arg}\xi_k$ tend to 0 when $\operatorname{Arg} z$ stays bounded. All three terms on the right tend to 0, so $w_j - w_k \to 2\pi\mathrm{i}(j-k) \neq 0$. □

Universality is a statement about one problem, not about all of them

Theorem 22.10 (i) says that the *same* polynomials serve every a and every branch. It does not say that $\mathsf{L}_n^{\mathrm{Lam}}$ is the coefficient family of every logarithmic reversion. Even inside the affine-logarithmic family the scalar a^{n+1} is present, and as soon as the perturbation ceases to be aL — for instance for $X + aL^2$ with $a \neq 0$, whose t^0-coefficient is $-aL^2$ and whose t^n-coefficient for $n \geq 1$ has degree $2n+1$ in L and leading coefficient $2a^{n+1}/n$ by Theorem 18.22 — the coefficients are new polynomials. The universality of $\mathsf{L}_n^{\mathrm{Lam}}$ is exactly the universality of the single map $X + L$.

Remark 22.11 (An unexplained Catalan pattern in the $X + aL^2$ family). Reversing $X + aL^2$ in exact rational arithmetic through $n = 9$, the *lowest*-degree term of the t^n-coefficient is

$$(-1)^n\, \mathsf{C}_{n+1}^{\mathrm{Cat}}\, a^{n+1} L^{n+2}, \qquad 1 \leq n \leq 9,$$

where $\mathsf{C}_m^{\mathrm{Cat}} = \binom{2m}{m}/(m+1)$: the coefficients are $2, -5, 14, -42, 132, -429, 1430, -4862, 16796$. The top-degree end of the same coefficient is governed by Theorem 18.22, which gives degree $2n+1$ and leading coefficient $2a^{n+1}/n$; the bottom-degree end has no such explanation here. This is an observation, not a theorem. Nine confirmations are evidence and nothing more, and no proof is offered: the Lagrange–Bürmann coefficient formula of Theorem 18.18 controls the top of each block, because the highest power of L comes from the term of largest r, while the bottom of the block collects contributions from every r at once and is not visibly a single Catalan-counted sum. A proof would presumably identify the L^{n+2} coefficient with a lattice-path or tree count already attached to the quadratic-logarithm reversion; we have not found that identification. It is recorded because a pattern this clean is either a theorem with a short proof or a coincidence that ends at $n = 10$, and both outcomes are worth knowing.

22.4 The analytic status of the expansion

Theorem 22.9 is a theorem about symbols. It asserts nothing about the function W_k. The sources at this point diverge sharply. S1 and S3 assert Poincaré validity “in an admissible sector” without defining the sector; S6 asserts convergence by an implicit-function argument that quantifies incorrectly (see Theorem 22.18); S1, S3 and S5 all repeat from the literature the much stronger claim that for real z the series converges for $z \geq \mathrm{e}$. We prove here a self-contained analytic theorem which is strong enough for every use made of the expansion in this volume — it gives absolute convergence, an explicit region, and an explicit uniform remainder — and we mark the literature claim as unverified.

The whole analytic content sits in one two-variable lemma. Its point is that the two small quantities in the problem are $1/\xi_k$ and ℓ_k/ξ_k, and that the solution is jointly analytic in them near the origin. Separating them is what makes the estimate uniform in ℓ_k, which is precisely the step that no source performs.

Lemma 22.12 (The two-variable solution)**.** *Let*

$$D := \{(\sigma,\tau) \in \mathbb{C}^2 : |\sigma| + |\tau| < \tfrac{1}{2}\}.$$

For $(\sigma,\tau) \in D$ *the equation*

$$y = -\operatorname{Log}\bigl(1 - \tau + \sigma y\bigr) \tag{22.10}$$

has a unique solution $y = \mathcal{Y}(\sigma,\tau)$ *in the closed unit disc* $|y| \le 1$*, and it satisfies* $|\mathcal{Y}(\sigma,\tau)| \le \log 2$*. The function* $\mathcal{Y}$ *is holomorphic on* D*; its Taylor series* $\mathcal{Y} = \sum_{a,b\ge 0} c_{ab}\sigma^a\tau^b$ *converges absolutely at every point of* D*; the coefficients satisfy*

$$|c_{ab}| \le 5^{\,a+b} \qquad (a,b \ge 0); \tag{22.11}$$

and for every $u \in \mathbb{C}$ *and every* $n \ge 0$*,*

$$\sum_{a+b=n} c_{ab}u^b = \begin{cases} 0, & n = 0, \\ \mathsf{L}_n^{\mathrm{Lam}}(u), & n \ge 1. \end{cases} \tag{22.12}$$

Consequently $c_{ab} = \lambda_{a+b,b}$ *for* $a + b \ge 1$*, where* $\mathsf{L}_n^{\mathrm{Lam}}(u) = \sum_m \lambda_{n,m}u^m$*, and for all* $(\sigma,\tau) \in D$ *and* $n \ge 1$*,*

$$\Bigl|\textstyle\sum_{a+b=n} c_{ab}\sigma^a\tau^b\Bigr| \le \bigl(5(|\sigma| + |\tau|)\bigr)^n. \tag{22.13}$$

Proof. Existence, uniqueness and the bound. Fix $(\sigma,\tau) \in D$ and put $\theta = |\sigma| + |\tau| < \frac{1}{2}$. For $|y| \le 1$ we have $|\sigma y - \tau| \le |\sigma| + |\tau| = \theta < \frac{1}{2}$, so $\zeta := 1 - \tau + \sigma y$ lies in the open disc of radius $\frac{1}{2}$ about 1, where the principal logarithm is holomorphic and, by the Maclaurin series $\operatorname{Log}(1+\nu) = \sum_{m\ge 1}(-1)^{m+1}\nu^m/m$,

$$|\operatorname{Log}(1+\nu)| \le \sum_{m\ge 1} \frac{|\nu|^m}{m} = -\log\bigl(1 - |\nu|\bigr) \qquad (|\nu| < 1). \tag{22.14}$$

Hence the map $\Phi(y) = -\operatorname{Log}(1 - \tau + \sigma y)$ sends the closed unit disc into the closed disc of radius $-\log(1-\theta) \le \log 2 < 1$. Moreover

$$\bigl|\Phi'(y)\bigr| = \left|\frac{\sigma}{1 - \tau + \sigma y}\right| \le \frac{|\sigma|}{1-\theta} \le \frac{\theta}{1-\theta} < 1,$$

so Φ is a contraction of the complete metric space $\{|y| \le 1\}$. Banach's fixed point theorem gives a unique fixed point $\mathcal{Y}(\sigma,\tau)$, and it lies in the image, so $|\mathcal{Y}| \le -\log(1-\theta) \le \log 2$.

Holomorphy. Let $y_0 \equiv 0$ and $y_{j+1} = \Phi(y_j)$. Each y_j is holomorphic on D as a function of (σ,τ), by induction: $1 - \tau + \sigma y_j$ is holomorphic with values in $\{|\zeta - 1| < \frac{1}{2}\}$, on which Log is holomorphic. On a compact $E \subset D$ we have $\sup_E \theta = \theta_E < \frac{1}{2}$, hence $|y_{j+1} - y_j| \le \bigl(\theta_E/(1-\theta_E)\bigr)^j\,|y_1 - y_0|$ with $\theta_E/(1-\theta_E) < 1$, so (y_j) converges uniformly on E to $\mathcal{Y}$. A locally uniform limit of holomorphic functions is holomorphic.

Taylor expansion and Cauchy bounds. Let $(\sigma_0,\tau_0) \in D$ and choose $\rho_1 > |\sigma_0|$, $\rho_2 > |\tau_0|$ with $\rho_1 + \rho_2 < \frac{1}{2}$. The closed polydisc of polyradius (ρ_1,ρ_2) is contained in D and $|\mathcal{Y}| \le 1$ there, so the Cauchy integral formula on the distinguished boundary yields the absolutely and uniformly convergent expansion $\mathcal{Y} = \sum_{a,b} c_{ab}\sigma^a\tau^b$ on the open polydisc, with $|c_{ab}| \le \rho_1^{-a}\rho_2^{-b}$. Since (σ_0,τ_0) was arbitrary in D, the series converges absolutely at every point of D, and the coefficients c_{ab} do not depend on the polydisc. Taking $\rho_1 = \rho_2 = \frac{1}{5}$, which satisfies $\rho_1 + \rho_2 = \frac{2}{5} < \frac{1}{2}$, gives Equation (22.11).

Identification of the diagonal sums. Fix $u \in \mathbb{C}$ and restrict to the line $\tau = \sigma u$; the restriction lies in D for $|\sigma| < \bigl(2(1+|u|)\bigr)^{-1}$. On that disc, $s \mapsto \mathcal{Y}(s, su)$ is holomorphic, vanishes at $s = 0$, and satisfies $\mathcal{Y} = -\operatorname{Log}(1 - su + s\mathcal{Y})$; hence its Taylor coefficients form the unique element of $s\mathbb{C}[[s]]$ satisfying Equation (22.6) at the given u. By the formal uniqueness clause of Theorem 21.11, those coefficients are $\mathsf{L}_n^{\mathrm{Lam}}(u)$. On the other hand, absolute convergence permits the rearrangement $\mathcal{Y}(s, su) = \sum_{n\geq 0}\bigl(\sum_{a+b=n} c_{ab}u^b\bigr)s^n$. Comparing coefficients gives Equation (22.12) for the given u; as u ranges over $\mathbb{C}$, two polynomials agreeing everywhere are equal, and comparing coefficients of u^b gives $c_{ab} = \lambda_{a+b,b}$.

The block bound. For $x, y \geq 0$, $\sum_{a+b=n} x^a y^b \leq \sum_{a+b=n} \binom{n}{a} x^a y^b = (x+y)^n$. Combining this with Equation (22.11) gives Equation (22.13). □

Theorem 22.13 (Convergent branch expansion from an arbitrary base point). *Let $\xi \in \mathbb{C}$. Let $K \in \mathbb{C} \setminus \{0\}$, let $\log K$ be any logarithm of K, and put*

$$M := K + \log K - \xi. \tag{22.15}$$

Assume the seating condition

$$2\bigl(1 + |M|\bigr) < |K|\,. \tag{22.16}$$

Then:

1. *the series $\sum_{n\geq 0} \mathsf{L}_n^{\mathrm{Lam}}(M)K^{-n}$ converges absolutely;*
2. *the number*
$$w := K + \sum_{n\geq 0} \frac{\mathsf{L}_n^{\mathrm{Lam}}(M)}{K^n} = K - M + \sum_{n\geq 1} \frac{\mathsf{L}_n^{\mathrm{Lam}}(M)}{K^n}$$
satisfies $|w - (K - M)| \leq \log 2$ and $w \neq 0$;
3. *with $\log_* w := \log K + \mathrm{Log}(w/K)$, which is a logarithm of w,*
$$w + \log_* w = \xi, \qquad \textit{hence} \qquad w\mathrm{e}^w = \mathrm{e}^\xi.$$

Proof. Set $\sigma = 1/K$ and $\tau = M/K$. Then $|\sigma| + |\tau| = (1 + |M|)/\,|K| < \frac{1}{2}$ by Equation (22.16), so $(\sigma, \tau) \in D$. Let $\mathcal{Y} = \mathcal{Y}(\sigma, \tau)$ be as in Theorem 22.12. Applying Equation (22.12) with $u = M$ and using $\tau = \sigma M$,

$$\mathcal{Y} = \sum_{n\geq 1} \mathsf{L}_n^{\mathrm{Lam}}(M)\sigma^n = \sum_{n\geq 1} \frac{\mathsf{L}_n^{\mathrm{Lam}}(M)}{K^n},$$

the series converging absolutely by Theorem 22.12. Since $\mathsf{L}_0^{\mathrm{Lam}}(M) = -M$, this is (i), and $w = K - M + \mathcal{Y}$ with $|\mathcal{Y}| \leq \log 2$, which is the first half of (ii). For $w \neq 0$: $|w| \geq |K| - |M| - \log 2 > 2 + |M| - \log 2 > 1$, using Equation (22.16).

For (iii), compute

$$\frac{w}{K} = \frac{K - M + \mathcal{Y}}{K} = 1 - \tau + \sigma\mathcal{Y},$$

which lies in the disc of radius $\frac{1}{2}$ about 1 by the proof of Theorem 22.12; hence $\mathrm{Log}(w/K)$ is defined and, by Equation (22.10), $\mathrm{Log}(w/K) = -\mathcal{Y}$. Therefore $\log_* w = \log K - \mathcal{Y}$, and $\mathrm{e}^{\log_* w} = K \cdot (w/K) = w$, so $\log_* w$ is indeed a logarithm of w. Finally

$$w + \log_* w = (K - M + \mathcal{Y}) + (\log K - \mathcal{Y}) = K + \log K - M = \xi$$

by Equation (22.15), and exponentiating gives $w\mathrm{e}^w = \mathrm{e}^{w+\log_* w} = \mathrm{e}^\xi$. □

Theorem 22.14 (Explicit uniform remainder). *In the situation of Theorem 22.13, put*

$$\theta := \frac{5\bigl(1 + |M|\bigr)}{|K|} \tag{22.17}$$

and assume $\theta \le \frac{1}{2}$, that is $|K| \ge 10(1+|M|)$. Then for every $N \in \mathbb{N}_0$,

$$\left| w - K - \sum_{n=0}^{N} \frac{\mathsf{L}_n^{\mathrm{Lam}}(M)}{K^n} \right| \le 2\,\theta^{N+1} = 2\left(\frac{5(1+|M|)}{|K|}\right)^{N+1}. \tag{22.18}$$

The bound is uniform: its constants 5 and 2 are absolute, and no hypothesis beyond $|K| \ge 10(1+|M|)$ is used.

Proof. With $\sigma = 1/K, \tau = M/K$ as above, $5(|\sigma|+|\tau|) = \theta$, so Equation (22.13) reads $\left|\mathsf{L}_n^{\mathrm{Lam}}(M)K^{-n}\right| \le \theta^n$ for $n \ge 1$. Since $\theta \le \frac{1}{2}$,

$$\left| \sum_{n>N} \frac{\mathsf{L}_n^{\mathrm{Lam}}(M)}{K^n} \right| \le \sum_{n>N} \theta^n = \frac{\theta^{N+1}}{1-\theta} \le 2\theta^{N+1}. \qquad \square$$

Specialising to the canonical seating $K = \xi_k$ gives the statement the notation catalogue asks for, now with every hypothesis printed and every constant explicit.

Corollary 22.15 (Canonical branch expansion)**.** *Let $z \ne 0$, $k \in \mathbb{Z}$, and let (ξ_k, ℓ_k) be a branch datum with $\xi_k \ne 0$. Take $K = \xi_k$ and $\log K = \ell_k$ in Theorem 22.13; then $M = \ell_k$. Consequently:*

1. *if $2(1+|\ell_k|) < |\xi_k|$, the series*

$$w = \xi_k - \ell_k + \sum_{n\ge1} \frac{\mathsf{L}_n^{\mathrm{Lam}}(\ell_k)}{\xi_k^n}$$

converges absolutely and defines a solution of $w\mathrm{e}^w = z$ with $\log_* w = \xi_k - w$;

2. *if moreover $|\xi_k| \ge 10(1+|\ell_k|)$, then for every N,*

$$\left| w - \xi_k + \ell_k - \sum_{n=1}^{N} \frac{\mathsf{L}_n^{\mathrm{Lam}}(\ell_k)}{\xi_k^n} \right| \le 2\left(\frac{5(1+|\ell_k|)}{|\xi_k|}\right)^{N+1} = \mathcal{O}_{|\xi_k|\to\infty,\ |\ell_k|\ge1}\left(\ell_k^{N+1}/\xi_k^{N+1}\right);$$

3. *in particular, if $\ell_k = \operatorname{Log}\xi_k$, both hypotheses hold as soon as $|\xi_k| \ge 100$, and for real $z \ge \mathrm{e}^6$ with $k = 0$ hypothesis* (i) *holds while for real $z \ge \mathrm{e}^{50}$ hypothesis* (ii) *holds.*

Proof. (i) and (ii) are Theorems 22.13 and 22.14 with $M = \xi_k + \ell_k - \xi_k = \ell_k$; the $\mathcal{O}.(\cdot)$ form of the bound in (ii) follows because $(1+|\ell_k|) \le 2\,|\ell_k|$ once $|\ell_k| \ge 1$, so the displayed bound is at most $2 \cdot 10^{N+1}\,|\ell_k/\xi_k|^{N+1}$.

(iii) If $\ell_k = \operatorname{Log}\xi_k$ then $|\ell_k| \le \log|\xi_k| + \pi$. Put $R = |\xi_k|$. Hypothesis (ii) holds if $R \ge 10(1+\pi+\log R)$; the function $R \mapsto R - 10(1+\pi+\log R)$ has derivative $1 - 10/R > 0$ for $R > 10$ and is positive at $R = 100$ (indeed $10(1+\pi+\log 100) < 87.5$), hence for all $R \ge 100$; and hypothesis (i) is weaker. For real $z > 1$ and $k = 0$, $\xi_0 = \log z > 0$ and $\ell_0 = \log\log z$ are real, so (i) requires $2(1+|\log\xi_0|) < \xi_0$; at $\xi_0 = 6$ one has $2(1+\log 6) < 5.59 < 6$, and $\xi_0 \mapsto \xi_0 - 2 - 2\log\xi_0$ is increasing for $\xi_0 > 2$. Similarly (ii) requires $\xi_0 \ge 10(1+\log\xi_0)$, which holds at $\xi_0 = 50$ ($10(1+\log 50) < 49.2$) and hence for all $\xi_0 \ge 50$ since $\xi_0 \mapsto \xi_0 - 10 - 10\log\xi_0$ is increasing for $\xi_0 > 10$. $\square$

Proposition 22.16 (Identification of the branch label)**.** *Let $z \ne 0$, $k \in \mathbb{Z}$, and let (ξ_k, ℓ_k) be a branch datum with $2(1+|\ell_k|) < |\xi_k|$. Let $w = w_k(z)$ be the solution produced by Theorem 22.15 (i), and assume in addition*

$$|\operatorname{Im}\ell_k| \le \pi - \tfrac{3}{4}, \tag{22.19}$$

that is, ξ_k is at angular distance at least $\frac{3}{4}$ from the cut of the chosen determination of the second logarithm. Then $\log_* w = \operatorname{Log} w$, so

$$w + \operatorname{Log} w = \xi_k, \qquad \operatorname{Im} w \in \left[\, 2\pi k + \operatorname{Arg} z - \pi,\ 2\pi k + \operatorname{Arg} z + \pi\,\right). \tag{22.20}$$

In particular distinct k produce solutions in disjoint horizontal strips, so $k \mapsto w_k(z)$ is injective. Independently of Equation (22.19), $w_k(z)$ is the unique solution of $we^w = z$ in the closed disc $|v - (\xi_k - \ell_k)| \le 1$.

Proof. By the proof of Theorem 22.13, $\log_* w = \ell_k - \mathcal{Y}$ with $|\mathcal{Y}| \le \log 2 < \frac{3}{4}$. Hence $|\operatorname{Im} \log_* w| \le |\operatorname{Im} \ell_k| + \log 2 < (\pi - \frac{3}{4}) + \frac{3}{4} = \pi$. A logarithm of w whose imaginary part lies in $(-\pi, \pi)$ is the principal one, so $\log_* w = \operatorname{Log} w$. Then Theorem 22.13 (iii) reads $w + \operatorname{Log} w = \xi_k$; taking imaginary parts and using $\operatorname{Im} \operatorname{Log} w = \operatorname{Arg} w \in (-\pi, \pi]$ and $\operatorname{Im} \xi_k = \operatorname{Arg} z + 2\pi k$ gives $\operatorname{Im} w = \operatorname{Arg} z + 2\pi k - \operatorname{Arg} w$, which lies in the stated interval. Those intervals for distinct k are disjoint half-open intervals of length 2π, which gives injectivity.

For the disc uniqueness, let $ve^v = z$ with $|v - (\xi_k - \ell_k)| \le 1$, and put $\sigma = 1/\xi_k$, $\tau = \ell_k/\xi_k$, so $(\sigma, \tau) \in D$ by the seating condition. Then $|v - \xi_k| \le |\ell_k| + 1 \le \frac{1}{2}|\xi_k|$, so $|v/\xi_k - 1| \le \frac{1}{2}$ and $y := -\operatorname{Log}(v/\xi_k)$ is defined with $|y| \le \log 2$ by Equation (22.14). Now $\lambda := \ell_k - y$ satisfies $e^\lambda = \xi_k \cdot (v/\xi_k) = v$, so Theorem 22.5 gives $v + \lambda = \xi_k + 2\pi i m$ for some $m \in \mathbb{Z}$, that is $v = \xi_k - \ell_k + y + 2\pi i m$. Since $|v - (\xi_k - \ell_k)| \le 1$ and $|y| \le \log 2$, we get $2\pi |m| \le 1 + \log 2 < 2\pi$, hence $m = 0$ and $v = \xi_k - \ell_k + y$. Dividing by ξ_k gives $v/\xi_k = 1 - \tau + \sigma y$, so $y = -\operatorname{Log}(1 - \tau + \sigma y)$: y is a fixed point of Equation (22.10) in $|y| \le 1$. By the uniqueness in Theorem 22.12, $y = \mathcal{Y}(\sigma, \tau)$, whence $v = w_k(z)$. □

Remark 22.17 (No sector is needed for the estimate). The remainder bound Equation (22.18) holds pointwise on the explicitly described region $\{|\xi_k| \ge 10(1 + |\ell_k|)\}$, with no sector hypothesis at all. The sector enters for two other purposes, and it is worth separating them:

1. to make $z \mapsto \xi_k$ and $\xi_k \mapsto \ell_k$ single-valued analytic functions, so that w_k is an analytic function of z rather than a value attached to a branch datum;
2. to identify w_k with the function that the standard references call W_k; this is where the distance-to-the-cut hypothesis Equation (22.19) is used.

Sources that write "in an admissible sector" without saying which are conflating these two roles with the estimate itself. With Theorems 22.15 and 22.16 one may take, for instance,

$$S_k = \left\{ z : |\operatorname{Arg} z| < \pi,\ |\xi_k| > 100,\ |\operatorname{Arg} \xi_k| < \pi - \tfrac{3}{4} \right\},$$

on which ξ_k and $\ell_k = \operatorname{Log} \xi_k$ are analytic, w_k is analytic, and all hypotheses hold.

Remark 22.18 (The gap in the convergence argument of S6). S6 argues: the implicit function theorem shows that for each fixed u the series $\sum_n \mathsf{L}_n^{\mathrm{Lam}}(u) s^n$ converges for $|s|$ small; along the Lambert path $s = 1/\xi$, $u = \ell$, one has $|su| \to 0$; hence the series converges. The conclusion does not follow from the premises. The radius of convergence in s supplied by the implicit function theorem depends on u and shrinks as $|u| \to \infty$, so a statement about $|su|$ is not a statement about $|s|$ lying inside that radius. What repairs the argument is exactly the separation of variables in Theorem 22.12: one must apply the implicit function theorem in the two variables $(\sigma, \tau) = (1/\xi, \ell/\xi)$, *both* of which are small, rather than in s alone with u frozen. The frozen-u statement is true — it is Theorem 20.19 — but it is not sufficient.

What is proved here, and what is only reported

Theorem 22.15 proves absolute convergence of the canonical series for real $z \ge e^6 \approx 403$, and the explicit uniform remainder for $z \ge e^{50}$. Reports S1, S3 and S5 all state, citing [11, 5], that for the principal branch and real z the series converges for every $z \ge e$. That claim is far stronger than anything proved here and we have not verified it; it is recorded as a literature report, not as a theorem of this volume. Note that the check offered by S5 — "at $z = e$ one has $\ell_0 = 0$, so every term vanishes" — is no evidence for it at all: it shows only that the series is trivially convergent at the single point where all its terms are zero (Theorem 22.20). Similarly, S5's report of the convergence domains $X > (\alpha e)^\alpha$ for $\alpha \ge 1$ and $X > e$ for $\alpha < 1$ in the generalized problem [21] is repeated in Part 23 as attribution only.

22.5 The principal real specialization

Corollary 22.19 (Principal real branch at infinity). *Let $z > 1$ be real, and take $k = 0$, so that $\xi_0 = \log z > 0$ and $\ell_0 = \log\log z$ are real. Write $L_1 = \log z$ and $L_2 = \log\log z$. Then for $z \geq \mathrm{e}^{50}$,*

$$\begin{aligned} W_0(z) = L_1 - L_2 + \frac{L_2}{L_1} + \frac{L_2(L_2-2)}{2L_1^2} + \frac{L_2(2L_2^2 - 9L_2 + 6)}{6L_1^3} \\ + \frac{L_2(3L_2^3 - 22L_2^2 + 36L_2 - 12)}{12L_1^4} \\ + \frac{L_2(12L_2^4 - 125L_2^3 + 350L_2^2 - 300L_2 + 60)}{60L_1^5} + \varrho_5, \end{aligned} \tag{22.21}$$

with the explicit bound $|\varrho_5| \leq 2\big(5(1+L_2)/L_1\big)^6$, and the infinite series converges absolutely to $W_0(z)$ already for $z \geq \mathrm{e}^6$.

Proof. For real $z > 1$ the datum $(\xi_0, \ell_0) = (\log z, \log\log z)$ satisfies Equation (22.19) with $\operatorname{Im} \ell_0 = 0$, so Theorem 22.16 applies and identifies the constructed solution with the real principal branch: $w_0(z)$ is real (the whole construction is real for real data, because $\mathcal{Y}(\sigma,\tau)$ is real for real $(\sigma,\tau) \in D$ by uniqueness of the fixed point in $\{|y| \leq 1\}$) and $w_0 > 1$, hence $w_0 = W_0(z)$ by Theorem 22.2 (iii). The five displayed corrections are $\mathsf{L}_1^{\mathrm{Lam}}, \ldots, \mathsf{L}_5^{\mathrm{Lam}}$ evaluated at L_2, in the factored forms of Part D, and the bound on ϱ_5 is Theorem 22.15 (ii) with $N = 5$. Convergence for $z \geq \mathrm{e}^6$ is Theorem 22.15 (iii). ☐

Proposition 22.20 (Exactness at $z = \mathrm{e}$, and what it does not mean). *At $z = \mathrm{e}$ one has $L_1 = 1$, $L_2 = 0$, and every truncation $\xi_0 - \ell_0 + \sum_{n=1}^{N} \mathsf{L}_n^{\mathrm{Lam}}(\ell_0)\xi_0^{-n}$ equals 1, which is the exact value $W_0(\mathrm{e}) = 1$. The reason is that $\mathsf{L}_n^{\mathrm{Lam}}(0) = 0$ for every $n \in \mathbb{N}_0$. This gives no information whatever about convergence, because at that point every term of the series is 0; indeed the seating condition Equation* (22.16) *reads $2 < 1$ there and fails.*

Proof. $W_0(\mathrm{e}) = 1$ because $1 \cdot \mathrm{e}^1 = \mathrm{e}$ and $1 > -1$, so Theorem 22.2 identifies the branch. For $n \geq 1$, Theorem 21.7 expresses $\mathsf{L}_n^{\mathrm{Lam}}$ as a sum over $1 \leq m \leq n$ of multiples of u^m, so $\mathsf{L}_n^{\mathrm{Lam}}(0) = 0$; and $\mathsf{L}_0^{\mathrm{Lam}}(u) = -u$ gives $\mathsf{L}_0^{\mathrm{Lam}}(0) = 0$. Hence every truncation is $1 - 0 + 0 = 1$. For the last sentence, $|K| = \xi_0 = 1$ and $|M| = |\ell_0| = 0$, so Equation (22.16) asserts $2 < 1$. ☐

22.6 The lower real branch near zero

The catalogue reserves the symbol $\varepsilon_{\mathrm{br}}$ for the positive small argument of the lower branch; it is not an error tolerance and not an inverse iterated logarithm. We write ε for it in this subsection, where no other ε occurs.

Proposition 22.21 (The lower-branch equation). *The map $h(u) = u - \log u$ is a strictly increasing bijection from $(1,\infty)$ onto $(1,\infty)$. For $0 < \varepsilon < \mathrm{e}^{-1}$, put*

$$S := \log(1/\varepsilon) > 1, \qquad L := \log S. \tag{22.22}$$

Then $u := -W_{-1}(-\varepsilon)$ is the unique solution in $(1,\infty)$ of

$$u - \log u = S. \tag{22.23}$$

Proof. $h'(u) = 1 - 1/u > 0$ on $(1,\infty)$, $h(1) = 1$, and $h(u) \to \infty$; so h is a strictly increasing bijection onto $(1,\infty)$. By Theorem 22.2, $W_{-1}(-\varepsilon) \in (-\infty,-1)$ for $0 < \varepsilon < \mathrm{e}^{-1}$, so $u > 1$. Writing $w = -u$ in $w\mathrm{e}^w = -\varepsilon$ gives $u\mathrm{e}^{-u} = \varepsilon$; taking the real logarithm, $\log u - u = \log\varepsilon = -S$, which is Equation (22.23). Uniqueness is the injectivity of h. ☐

Theorem 22.22 (Expansion of the lower real branch)**.** *With S, L as in Equation (22.22), and provided $2(1+|L|) < S$, the series below converges absolutely and*

$$W_{-1}(-\varepsilon) = -S - L + \sum_{n\geq 1} (-1)^n \frac{\mathsf{L}_n^{\mathrm{Lam}}(L)}{S^n}, \tag{22.24}$$

equivalently, in the zero-based normalisation and with $\mathsf{L}_0^{\mathrm{Lam}}(L) = -L$,

$$W_{-1}(-\varepsilon) = -S + \sum_{n\geq 0} (-1)^n \frac{\mathsf{L}_n^{\mathrm{Lam}}(L)}{S^n}. \tag{22.25}$$

If in addition $S \geq 10(1+|L|)$, then for every N,

$$\left| W_{-1}(-\varepsilon) + S + L - \sum_{n=1}^{N} (-1)^n \frac{\mathsf{L}_n^{\mathrm{Lam}}(L)}{S^n} \right| \leq 2 \left(\frac{5(1+|L|)}{S} \right)^{N+1}. \tag{22.26}$$

Explicitly,

$$\begin{aligned} W_{-1}(-\varepsilon) = \; & -S - L - \frac{L}{S} + \frac{L(L-2)}{2S^2} - \frac{L(2L^2-9L+6)}{6S^3} \\ & + \frac{L(3L^3-22L^2+36L-12)}{12S^4} + \cdots . \end{aligned} \tag{22.27}$$

Proof. This is the analytic affine-logarithmic theorem with $a = -1$. Concretely, apply Theorem 22.12 with $\sigma = -1/S$ and $\tau = -L/S$; then $|\sigma| + |\tau| = (1+|L|)/S < \frac{1}{2}$ by hypothesis, so $(\sigma, \tau) \in D$, and $\tau = \sigma L$, so Equation (22.12) gives

$$\mathcal{Y} = \sum_{n\geq 1} \mathsf{L}_n^{\mathrm{Lam}}(L)\sigma^n = \sum_{n\geq 1} (-1)^n \frac{\mathsf{L}_n^{\mathrm{Lam}}(L)}{S^n},$$

absolutely convergent. Put $v := S + L - \mathcal{Y}$. Then

$$\frac{v}{S} = 1 + \frac{L}{S} - \frac{\mathcal{Y}}{S} = 1 - \tau + \sigma\mathcal{Y},$$

which lies in the disc of radius $\frac{1}{2}$ about 1; hence $v > 0$ and $\operatorname{Log}(v/S) = -\mathcal{Y}$ by Equation (22.10). All the data being real, $\mathcal{Y}$ is real and $\log(v/S) = -\mathcal{Y}$ is the real logarithm. Therefore

$$v - \log v = v - \log S - \log(v/S) = S + L - \mathcal{Y} - L + \mathcal{Y} = S.$$

Also $v > 1$: the hypothesis $2(1+|L|) < S$ gives $|L| < S/2 - 1$ and $S > 2$, hence $v \geq S + L - |\mathcal{Y}| \geq S - |L| - \log 2 > S/2 + 1 - \log 2 > 2 - \log 2 > 1$. By the uniqueness in Theorem 22.21, $v = u = -W_{-1}(-\varepsilon)$, which gives Equation (22.25) after using $\mathsf{L}_0^{\mathrm{Lam}}(L) = -L$. The remainder bound is Equation (22.13) exactly as in Theorem 22.14, with $5(|\sigma|+|\tau|) = 5(1+|L|)/S \leq \frac{1}{2}$. The displayed first terms are $\mathsf{L}_1^{\mathrm{Lam}}, \ldots, \mathsf{L}_4^{\mathrm{Lam}}$ with the alternating signs. □

The alternating signs do not define a second Lambert family

Every source displays Equation (22.27) and remarks that "the same polynomials reappear with alternating block signs". The statement deserves to be made exactly, because the alternation is sometimes absorbed into the polynomials, producing a spurious second family $\widetilde{\mathsf{L}_n^{\mathrm{Lam}}} = (-1)^n \mathsf{L}_n^{\mathrm{Lam}}$ with its own tables and its own recurrence. There is no second family. By Theorem 22.7, the coefficient of X^{-n} in the inverse of $X + aL$ is $a^{n+1}\mathsf{L}_n^{\mathrm{Lam}}(L)$, a *scalar* multiple of the one canonical polynomial; the lower real branch is the single value $a = -1$ of that scalar, so the coefficient of S^{-n} is $(-1)^{n+1}\mathsf{L}_n^{\mathrm{Lam}}(L)$ in the equation $u - \log u = S$ and $(-1)^n\mathsf{L}_n^{\mathrm{Lam}}(L)$ after the reflection $w = -u$. The sign is data about the argument, not about the polynomial. Concretely: the recurrence $\mathsf{L}_{n+1}^{\mathrm{Lam}} = -\mathsf{L}_n^{\mathrm{Lam}} + n\int_0^u \mathsf{L}_n^{\mathrm{Lam}}$ of Theorem 21.2, the Stirling formula of Theorem 21.4 and the tables of Part D are used unchanged for the lower branch; only the evaluation point and the sign of the outer variable change.

Remark 22.23 (Why the lower branch is not literally an ω value). Setting $y = aY$ in $y + a\log y = X$ turns it into $Y + \log Y = X/a - \log a$, so for $a > 0$ the affine-logarithmic problem is a rescaled Wright omega evaluation. For $a = -1$ this substitution requires $\log(-1)$, and the resulting omega argument is $-X \mp \pi\mathrm{i}$: the real lower branch corresponds to a *complex* point of ω. This is why Theorem 22.22 is proved directly from Theorem 22.12 rather than by quoting the omega expansion, and it is the precise content of the catalogue's warning that $\varepsilon_{\mathrm{br}}$ is a branch-point-independent parameter for a problem of its own.

22.7 The square-root branch point

Lemma 22.24 (The critical point of $w\mathrm{e}^w$). *Let $g(w) = w\mathrm{e}^w$ and $w = -1 + q$. Then*

$$1 + \mathrm{e}\, g(-1+q) = \sum_{n\geq 2} \frac{(n-1)\, q^n}{n!} = \frac{q^2}{2} + \frac{q^3}{3} + \frac{q^4}{8} + \frac{q^5}{30} + \frac{q^6}{144} + \cdots, \tag{22.28}$$

an entire function of q. In particular $g(-1) = -\mathrm{e}^{-1}$, $g'(-1) = 0$ and $g''(-1) = \mathrm{e}^{-1} \neq 0$, so -1 is a critical point of g of order exactly two.

Proof. $\mathrm{e}\, g(-1+q) = \mathrm{e}(-1+q)\mathrm{e}^{-1+q} = (q-1)\mathrm{e}^q$. Now

$$(q-1)\mathrm{e}^q = \sum_{n\geq 0} \frac{q^{n+1}}{n!} - \sum_{n\geq 0} \frac{q^n}{n!} = -1 + \sum_{n\geq 1} q^n \left(\frac{1}{(n-1)!} - \frac{1}{n!}\right) = -1 + \sum_{n\geq 1} \frac{(n-1)q^n}{n!},$$

and the $n = 1$ summand vanishes. Adding 1 gives Equation (22.28). Comparing coefficients, $\mathrm{e}\, g'(-1) = 0$ and $\mathrm{e}\, g''(-1)/2! = \frac{1}{2}$, i.e. $g''(-1) = \mathrm{e}^{-1}$. ☐

Theorem 22.25 (Puiseux expansion at the branch point). *Put*

$$p := \sqrt{2(1 + \mathrm{e}z)}. \tag{22.29}$$

There are $r > 0$ and a function φ holomorphic on $|p| < r$ with $\varphi(0) = 0$, $\varphi'(0) = 1$, such that for $|1 + \mathrm{e}z| < r^2/2$ the two solutions of $w\mathrm{e}^w = z$ near $w = -1$ are exactly

$$w = -1 + \varphi(p) \qquad \text{and} \qquad w = -1 + \varphi(-p), \tag{22.30}$$

for either determination of the square root in Equation (22.29)*. Its expansion begins*

$$\varphi(p) = p - \frac{p^2}{3} + \frac{11p^3}{72} - \frac{43p^4}{540} + \frac{769p^5}{17280} - \frac{221p^6}{8505} + \cdots. \tag{22.31}$$

In particular the leading behaviour is

$$w + 1 = \pm\sqrt{2(1 + \mathrm{e}z)} + \mathcal{O}_{z\to-\mathrm{e}^{-1}}(|1 + \mathrm{e}z|). \tag{22.32}$$

For real z slightly larger than $-\mathrm{e}^{-1}$ and $p > 0$, the branch $\varphi(p)$ is W_0 and $\varphi(-p)$ is W_{-1}; the two are exchanged by $p \mapsto -p$.

Proof. Write Equation (22.28) as $1 + \mathrm{e}z = \frac{1}{2}q^2 h(q)$ with

$$h(q) := \sum_{n\geq 2} \frac{2(n-1)}{n!} q^{n-2} = 1 + \frac{2q}{3} + \frac{q^2}{4} + \frac{q^3}{15} + \frac{q^4}{72} + \cdots,$$

entire with $h(0) = 1$. Choose $r_0 > 0$ with $h \neq 0$ on $|q| < r_0$ and let $h^{1/2}$ be the holomorphic square root there with $h^{1/2}(0) = 1$. Put $\psi(q) = q\, h^{1/2}(q)$, so $\psi(0) = 0$, $\psi'(0) = 1$ and

$$\psi(q)^2 = q^2 h(q) = 2(1 + \mathrm{e}z) \qquad \text{whenever } z = g(-1+q). \tag{22.33}$$

By the holomorphic inverse function theorem ψ is a biholomorphism of a disc $|q| < r_1 \leq r_0$ onto a neighbourhood of 0 containing a disc $|p| < r$; let $\varphi = \psi^{-1}$ on that disc, so $\varphi(0) = 0$ and $\varphi'(0) = 1/\psi'(0) = 1$.

Now fix z with $|1 + \mathrm{e}z| < r^2/2$ and let p be either square root of $2(1 + \mathrm{e}z)$, so $|p| < r$. If $w = -1 + q$ with $|q| < r_1$ solves $g(w) = z$, then Equation (22.33) gives $\psi(q)^2 = p^2$, hence $\psi(q) = \pm p$ and $q = \varphi(\pm p)$; conversely each of $q = \varphi(\pm p)$ satisfies $\frac{1}{2}q^2 h(q) = \frac{1}{2}\psi(q)^2 = \frac{1}{2}p^2 = 1 + \mathrm{e}z$, i.e. $g(-1+q) = z$ by Equation (22.28). Shrinking r so that $|\varphi| < r_1$ on $|p| < r$ makes this a bijection between the two square roots and the two nearby solutions; they are distinct for $p \neq 0$ because $\varphi(p) - \varphi(-p) = 2p + \mathcal{O}_{p\to 0}(p^3)$.

For the coefficients, write $\varphi(p) = \sum_{i\geq 1} a_i p^i$ with $a_1 = 1$ and determine $a_2, a_3, \ldots$ by substituting into $\psi(\varphi(p)) = p$, where $\psi(q) = q + \frac{1}{3}q^2 + \frac{5}{72}q^3 + \frac{11}{1080}q^4 + \frac{59}{51840}q^5 + \cdots$ is obtained by expanding $h^{1/2}$ from the displayed h. Comparing coefficients of $p^2, \ldots, p^6$ gives in turn $a_2 = -\frac{1}{3}$, $a_3 = \frac{11}{72}$, $a_4 = -\frac{43}{540}$, $a_5 = \frac{769}{17280}$, $a_6 = -\frac{221}{8505}$, which is Equation (22.31). Since $\varphi(p) = p + \mathcal{O}_{p\to 0}(p^2)$ and $p^2 = 2(1 + \mathrm{e}z)$, we get Equation (22.32).

Finally let $z \in (-\mathrm{e}^{-1}, -\mathrm{e}^{-1} + \delta)$ be real and take $p = +\sqrt{2(1 + \mathrm{e}z)} > 0$ small. Then $\varphi(p) > 0$ and $\varphi(-p) < 0$, so $-1 + \varphi(p) > -1$ and $-1 + \varphi(-p) < -1$; by Theorem 22.2 (iii) these are $W_0(z)$ and $W_{-1}(z)$ respectively. □

Corollary 22.26 (No power–logarithmic expansion at the branch point)**.** *Let* $z_0 = -\mathrm{e}^{-1}$.

1. *There is no single-valued holomorphic function* F *on a punctured disc* $0 < |z - z_0| < \rho$ *with* $g(F(z)) = z$ *and* $F(z) \to -1$.
2. *The local monodromy of the two-valued germ at* z_0 *has order exactly two.*
3. *Consequently no determination of the inverse near* -1 *has, on a slit disc, a representation*

$$F(z) = \sum_{j=0}^{d} A_j(z) \big(\log(z - z_0)\big)^j$$

with $d \geq 1$*, the* A_j *single-valued holomorphic on the punctured disc, and* $A_d \not\equiv 0$*. That is, no logarithm of* $z - z_0$ *occurs at any power; the correct local scale is the Puiseux scale* $\{(z - z_0)^{m/2}\}$*, on which Theorem 22.25 is an expansion.*

Proof. (ii) Continuing $p = \sqrt{2\mathrm{e}(z - z_0)}$ once counterclockwise around z_0 replaces p by $-p$, hence exchanges the two determinations $-1 + \varphi(\pm p)$ of Theorem 22.25; continuing twice restores them. The exchange is nontrivial because the two determinations differ for $p \neq 0$. So the monodromy permutation has order two.

(i) A single-valued F would be invariant under one continuation, but the only two candidate values are exchanged; contradiction.

(iii) Write $\Lambda = \log(z - z_0)$ and suppose $F = \sum_{j=0}^{d} A_j \Lambda^j$ with $d \geq 1$ and $A_d \not\equiv 0$. Continuation m times around z_0 replaces Λ by $\Lambda + 2\pi\mathrm{i}m$ and hence F by $F_m = \sum_j A_j(\Lambda + 2\pi\mathrm{i}m)^j$. Expanding, F_m is again of the same shape with top coefficient A_d and next coefficient $A_{d-1} + 2\pi\mathrm{i}m\, d\, A_d$. Since the representation of a germ in the form $\sum_{j\le d} A_j\Lambda^j$ with single-valued A_j is unique — the functions $1, \Lambda, \dots, \Lambda^d$ are linearly independent over the field of single-valued meromorphic germs, because Λ is transcendental over it — the germs F_m, $m \in \mathbb{Z}$, are pairwise distinct. So the germ would have infinitely many determinations, contradicting (ii). If instead F were given by a convergent series in integer powers of $z - z_0$, it would be single-valued, contradicting (i). $\square$

Remark 22.27 (The general principle). Theorem 22.26 is the local instance of a rule used repeatedly in Part 23: the fractional-power scale of an inverse is dictated by the order of vanishing of the derivative of the direct map. A simple zero of g' — equivalently, a critical point of order two — forces the exponent $1/2$ and monodromy of order two. A zero of order $m-1$ would force exponent $1/m$; an essential degeneracy would force something else again. Logarithms in an inverse come from a different mechanism entirely, namely a logarithm in the direct map (Theorem 19.4), and the two mechanisms never substitute for one another.

22.8 Residual correction and one Newton step in branch coordinates

The formal theory of Part 16 measures the quality of a truncated inverse by the reversal residual $\mathrm{Err}_{\mathrm{rev}}(F, H) = F \circ H - X$, and Theorem 16.15 shows that its t-order is exactly the number of correct blocks. In branch coordinates the analytic counterpart is the *logarithmic residual*, and it is a much better numerical object than the multiplicative residual $v\mathrm{e}^v - z$, which overflows as soon as z is large.

Definition 22.28 (Truncations and the logarithmic residual). Let (ξ, ℓ) be a branch datum with $|\xi| \ge 10(1 + |\ell|)$. For $N \in \mathbb{N}_0$ put

$$w_N := \xi - \ell + \sum_{n=1}^{N} \frac{\mathsf{L}_n^{\mathrm{Lam}}(\ell)}{\xi^n}, \tag{22.34}$$

so that w_N is the value of the exclusive truncation $\mathrm{Trunc}_{<N+1}(X + L)^{\langle -1\rangle_\circ}$ at $(X, L) = (\xi, \ell)$. On the closed disc $\mathcal{D} := \{v : |v - (\xi - \ell)| \le 1\}$ define $\log_* v := \ell + \mathrm{Log}(v/\xi)$ and $F(v) := v + \log_* v$; set

$$R_N := F(w_N) - \xi = w_N + \log_* w_N - \xi, \qquad \delta_N := -\frac{R_N}{1 + 1/w_N}. \tag{22.35}$$

Proposition 22.29 (The disc is admissible). *Under the hypothesis of Theorem 22.28: $\mathcal{D}$ is contained in the region where $\log_*$ is defined and holomorphic; for $v \in \mathcal{D}$ one has $|v| \ge \frac{9}{10}|\xi| \ge 9$, $\frac{8}{9} \le |F'(v)| \le \frac{10}{9}$, and $|F''(v)| \le \frac{5}{4}|\xi|^{-2}$; and both w_N (for every N) and the exact solution w of Theorem 22.15 lie in $\mathcal{D}$.*

Proof. Write $\theta = 5(1 + |\ell|)/|\xi| \le \frac{1}{2}$. For $v \in \mathcal{D}$, $|v - \xi| \le |\ell| + 1 \le |\xi|/10$, so $|v/\xi - 1| \le \frac{1}{10} < 1$ and $\mathrm{Log}(v/\xi)$ is defined and holomorphic there; also $|v| \ge \frac{9}{10}|\xi|$, and $|\xi| \ge 10(1 + |\ell|) \ge 10$ gives $|v| \ge 9$. Since $F'(v) = 1 + 1/v$ and $F''(v) = -1/v^2$, we get $|F'(v) - 1| \le \frac{1}{9}$ and $|F''(v)| \le (10/9)^2|\xi|^{-2} \le \frac{5}{4}|\xi|^{-2}$. Finally $|w_N - (\xi - \ell)| \le \sum_{n\ge1}\theta^n = \theta/(1-\theta) \le 1$ by Equation (22.13), and $|w - (\xi - \ell)| = |\mathcal{Y}| \le \log 2 \le 1$. $\square$

Theorem 22.30 (Residual order, residual-to-error transfer, and one Newton step). *In the situation of Theorem 22.28, put $\theta = 5(1 + |\ell|)/|\xi| \le \frac{1}{2}$ and let w be the exact solution of Theorem 22.15. Then, with $e := w - w_N$:*

1. $|e| \le 2\theta^{N+1}$ *and* $|R_N| \le \frac{20}{9}\theta^{N+1}$*; in particular*

$$R_N = \mathcal{O}_{|\xi|\to\infty,\ |\ell|\ge1}\left(\ell^{\,N+1}/\xi^{\,N+1}\right),$$

the same order as the error itself;

2. $|e| \le \frac{5}{4}|R_N|$ *(residual-to-error transfer, with an explicit constant);*

3. $|e - \delta_N| \le \dfrac{3}{2}\dfrac{|R_N|^2}{|\xi|^2}$ (*one Newton step is quadratically better*).

Proof. (i) The bound on e is Theorem 22.14. The segment $[w_N, w]$ lies in the convex set $\mathcal{D}$ by Theorem 22.29, and $F(w) = \xi$ by Theorem 22.13 (iii) — note that on $\mathcal{D}$ the function $\log_*$ of Theorem 22.28 agrees with the $\log_*$ of Theorem 22.13 — so $|R_N| = |F(w_N) - F(w)| \le \sup_{\mathcal{D}} |F'| \cdot |e| \le \frac{10}{9} \cdot 2\theta^{N+1}$. The $\mathcal{O}.(\cdot)$ form follows as in Theorem 22.15 (ii).

(ii) and (iii). Taylor's formula with integral remainder along $[w_N, w] \subset \mathcal{D}$ gives

$$0 = F(w) - \xi = R_N + F'(w_N)e + E, \qquad E = \int_0^1 (1-s)F''\big(w_N + se\big)e^2 \,\mathrm{d}s,$$

so $|E| \le \frac{1}{2} \cdot \frac{5}{4} |\xi|^{-2} |e|^2 = \frac{5}{8} |e|^2 |\xi|^{-2}$. Since both points lie in $\mathcal{D}$ we have $|e| \le 2$, and $|\xi| \ge 10$, so $|E| \le \frac{5}{8} \cdot 2 |e| / 100 = |e| / 80$. Using $|F'(w_N)| \ge \frac{8}{9}$,

$$\tfrac{8}{9} |e| \le |R_N| + \tfrac{1}{80} |e|, \qquad \text{hence} \qquad |e| \le \frac{|R_N|}{8/9 - 1/80} \le \tfrac{5}{4} |R_N|,$$

which is (ii). Dividing the Taylor identity by $F'(w_N) = 1 + 1/w_N$ gives $e = \delta_N - E/F'(w_N)$, so

$$|e - \delta_N| \le \frac{9}{8} |E| \le \frac{9}{8} \cdot \frac{5}{8} \cdot \frac{|e|^2}{|\xi|^2} \le \frac{45}{64} \cdot \frac{25}{16} \frac{|R_N|^2}{|\xi|^2} \le \frac{3}{2} \frac{|R_N|^2}{|\xi|^2}. \qquad \square$$

Remark 22.31 (Where the transfer fails). Every constant above degrades as $|v|$ approaches 0, and the argument collapses entirely at $v = -1$, where $F'(v) = 1 + 1/v = 0$. That is exactly the branch point of Section 22.7: the residual-to-error transfer is a statement about a nonvanishing derivative, and at a critical point a small residual is compatible with an error of the order of its square root. This is the analytic shadow of the algebraic fact recorded in Theorem 17.9: Newton reversion needs the denominator to be a unit.

Remark 22.32 (Exact identities as independent checks). Two exact relations cost nothing and catch branch mistakes that coefficient checks do not. On any branch, $\log_* W_k(z) = \xi_k - W_k(z)$ and $\mathrm{e}^{W_k(z)} = z/W_k(z)$. The first lets one expand $\log W_k$ without composing a logarithm with a series, and the second lets one check $W_k \mathrm{e}^{W_k} = z$ by a division rather than an exponentiation. A truncation that satisfies the coefficient recurrences but violates the first identity has a branch mismatch, not a coefficient error.

22.9 Numerical behaviour of the truncations

All six sources tabulate truncation errors, and the tables disagree — not in their numbers but in what their column headings mean. S1 and S4 tabulate against $x = \log z$, S3 and S5 against z. Both families of numbers are correct, and each is a subset of the other's grid; we tabulate once, in the intrinsic variable $\xi_0 = \log z$, and record the concordance.

Remark 22.33 (Concordance of the source tables). For the reader of a particular source: the S1 columns headed $L = 10, 50, 100$ and the S4 columns headed $x = 10, 25, 100$ are columns of Table 22.1 (S4's index N counts from the zero-based family, $\omega_N(x) = x + \sum_{n=0}^{N} \mathsf{L}_n^{\mathrm{Lam}}(\log x)x^{-n}$, which is our w_N). The S3 and S5 columns headed $z = 10, 100, 10^6$ correspond to $\xi_0 = 2.3026,\ 4.6052,\ 13.8155$. For the first two the seating condition Equation (22.16) *fails* ($2(1+0.8340) = 3.67 \not< 2.30$ and $2(1+1.5272) = 5.05 \not< 4.61$), so nothing proved here applies to them; for the third it holds ($2(1 + 2.6259) = 7.25 < 13.82$), so the series converges there, but the remainder hypothesis $|\xi_0| \ge 10(1 + |\ell_0|) = 36.3$ still fails and the explicit bound Equation (22.18) is unavailable. Those columns are therefore empirical illustrations, not instances of a theorem. Every numerical value in all four source tables was recomputed to 60 digits and found correct.

Table 22.1: Absolute error $\left|w_N - W_0(\mathrm{e}^{\xi_0})\right|$ of the truncation Equation (22.34) on the principal real branch, $\ell_0 = \log \xi_0$. Recomputed here to 60 significant digits and in agreement with the tables of S1, S3, S4 and S5 wherever those overlap.

N	$\xi_0 = 10$	$\xi_0 = 25$	$\xi_0 = 50$	$\xi_0 = 100$
0	$2.3201 \cdot 10^{-1}$	$1.3180 \cdot 10^{-1}$	$7.9742 \cdot 10^{-2}$	$4.6657 \cdot 10^{-2}$
1	$1.7467 \cdot 10^{-3}$	$3.0442 \cdot 10^{-3}$	$1.5017 \cdot 10^{-3}$	$6.0513 \cdot 10^{-4}$
2	$1.7370 \cdot 10^{-3}$	$9.4510 \cdot 10^{-5}$	$5.7352 \cdot 10^{-6}$	$5.2671 \cdot 10^{-6}$
3	$1.5606 \cdot 10^{-4}$	$1.7340 \cdot 10^{-5}$	$1.5654 \cdot 10^{-6}$	$8.1538 \cdot 10^{-8}$
5	$5.1336 \cdot 10^{-6}$	$2.7114 \cdot 10^{-8}$	$2.2625 \cdot 10^{-9}$	$1.5423 \cdot 10^{-10}$
8	$1.4208 \cdot 10^{-8}$	$1.0628 \cdot 10^{-11}$	$3.9030 \cdot 10^{-13}$	$2.9383 \cdot 10^{-15}$
12	$1.9269 \cdot 10^{-11}$	$2.7265 \cdot 10^{-15}$	$1.8339 \cdot 10^{-18}$	$2.1347 \cdot 10^{-21}$

Table 22.2: Absolute error of the truncation of Equation (22.24) for $W_{-1}(-\varepsilon)$, with $S = \log(1/\varepsilon)$, $L = \log S$.

N	$\varepsilon = 10^{-3}$ ($S = 6.908$)	$\varepsilon = 10^{-6}$ ($S = 13.816$)	$\varepsilon = 10^{-20}$ ($S = 46.052$)
0	$2.7761 \cdot 10^{-1}$	$1.8521 \cdot 10^{-1}$	$8.1518 \cdot 10^{-2}$
1	$2.1725 \cdot 10^{-3}$	$4.8547 \cdot 10^{-3}$	$1.6446 \cdot 10^{-3}$
2	$3.5365 \cdot 10^{-3}$	$5.5019 \cdot 10^{-4}$	$7.5505 \cdot 10^{-6}$
3	$2.9763 \cdot 10^{-4}$	$8.7527 \cdot 10^{-5}$	$1.8886 \cdot 10^{-6}$
5	$1.8100 \cdot 10^{-5}$	$1.0098 \cdot 10^{-6}$	$3.0375 \cdot 10^{-9}$
8	$1.8025 \cdot 10^{-8}$	$2.4269 \cdot 10^{-9}$	$6.2816 \cdot 10^{-13}$
12	$3.8085 \cdot 10^{-10}$	$2.5958 \cdot 10^{-13}$	$3.3774 \cdot 10^{-18}$

Remark 22.34 (Non-monotonicity is not a contradiction). At $\xi_0 = 10$ the error at $N = 2$ is essentially the same as at $N = 1$, and at $\varepsilon = 10^{-3}$ the error at $N = 2$ is *larger* than at $N = 1$. Nothing is wrong: Theorem 22.14 bounds the tail by $2\theta^{N+1}$ with $\theta = 5(1 + |\ell|)/|\xi|$, and at $\xi_0 = 10$ one has $\theta \approx 1.65 > 1$, so the theorem says nothing there at all. Even inside the proved region the bound is a geometric envelope, not a term-by-term ordering: individual $\mathsf{L}_n^{\mathrm{Lam}}(\ell)$ change sign (Theorem 21.7), so two consecutive corrections can nearly cancel. The residual R_N of Theorem 22.28, not the size of the last retained block, is the quantity that should decide where to truncate; by Theorem 22.30 (ii) it controls the error with an explicit constant.

Algorithm 22.35 (Branch-aware evaluation).

```
input : z != 0, branch index k, order N, tolerance tol
1  xi := Log(z) + 2*pi*i*k                # principal Log; record the choice
2  if xi = 0: fail (the coordinates are undefined)
3  ell := Log(xi)                         # record the determination B_k
4  if |xi| < 10*(1 + |ell|):
       warn: outside the region of Theorem lw-uniform-remainder
5  w := xi - ell
6  for n := 1 to N:  w := w + P_n(ell) / xi^n     # tables, or the recurrence
7  R := w + ell + Log(w/xi) - xi          # the same branch data as line 3
8  while |R| > tol:
       w := w - R / (1 + 1/w)             # Newton in branch coordinates
       R := w + ell + Log(w/xi) - xi
9  return w, R
```

Line 7 evaluates $\log_* w$ as $\ell + \mathrm{Log}(w/\xi)$, never as $\mathrm{Log}\, w$: by Theorem 22.16 the two agree only when the distance-to-the-cut hypothesis Equation (22.19) holds, and an implementation that silently uses

$\operatorname{Log} w$ will report a spurious residual of size 2π near the cut. The multiplicative residual $we^w - z$ must not be substituted for line 7: it overflows for $|z|$ large, whereas R never forms e^w.

Section summary

Lambert W is one reversion evaluated at many arguments. The substitution $we^w = z \rightsquigarrow w + \log_* w = \xi_k$ (Theorem 22.5) turns each branch into the reversion of $X + L$, so the coefficient polynomials $\mathsf{L}_n^{\mathrm{Lam}}$ are the same for every branch (Theorem 22.10) while the coordinates ξ_k, ℓ_k carry all the branch dependence. The formal expansion is a theorem in $\mathcal{G}_{\mathrm{poly},\mathbb{K}}$ with no hypotheses; its analytic validity is a separate theorem with explicit ones (Theorems 22.13 and 22.14), and the frequently repeated claim of convergence for real $z \geq e$ is not among them. The lower real branch is the single scalar $a = -1$ of the affine-logarithmic family and introduces no new polynomials. At $z = -e^{-1}$ the derivative of we^w vanishes, the local monodromy has order two, and the correct scale is a square root, not a logarithm.

Exercise 22.36. Show that the seating condition Equation (22.16) holds for $K = \xi_0 = \log z$ and $M = \ell_0$ whenever $z \geq e^6$, but fails for every $z \leq e^5$; and find the exact threshold to four decimal places.

Exercise 22.37. Take $K = \xi_k - \ell_k$ in Theorem 22.13 and show that $M = \log(1 - \ell_k/\xi_k)$, so that $|M| = \mathcal{O}_{|\xi_k|\to\infty}(|\ell_k/\xi_k|)$. Deduce that the seating condition for this base point is satisfied on a strictly larger region than for $K = \xi_k$, and compare with Theorem 23.21.

Exercise 22.38. Using $\log_* W = \xi - W$ and the division rules of Part 9, expand $W_k(z)/(1 + W_k(z))$ through ξ^{-3} and verify against Equation (22.2).

Exercise 22.39. Prove that the two determinations of Theorem 22.25 satisfy $\varphi(p) + \varphi(-p) = -\frac{2}{3}p^2 - \frac{43}{270}p^4 + \mathcal{O}_{p\to 0}(p^6)$, and explain why this sum is an even function of p, hence a single-valued holomorphic function of $1 + ez$ near 0.

Chapter 23

Generalized Lambert equations and reusable inverse templates

Part 22 solved one equation, $w + \log w = \xi$, and obtained every branch of W from it. This chapter asks how far that leverage extends. The answer has three layers. First, an explicitly delimited family of equations reduces to $w + \log w = \xi$ by an exact algebraic change of variable, so that no new coefficient family is needed at all: this is the class $y^\alpha e^{\beta y} = x$ and its relatives (Section 23.2). Second, a much larger class does not reduce exactly but has the same *shape* of dominant balance — a monomial times a power of a logarithm — and for that class there is a mechanical template producing the polynomial–logarithmic inverse, with the Lambert polynomials appearing as the leading part and computable corrections beyond (Section 23.3). Third, and equally important, there are equations that look similar and are not in the class; the last worked example is one of them, and the point of including it is that the reader should be able to recognise the boundary before spending a page computing on the wrong side of it.

23.1 The affine-logarithmic core, analytically

Theorem 22.7 is the formal statement. Its analytic counterpart is proved by exactly the argument of Theorem 22.13, with the pair (σ, τ) chosen to carry the parameter a.

Theorem 23.1 (Analytic affine-logarithmic reversion). *Let $a \in \mathbb{C} \setminus \{0\}$, $X \in \mathbb{C} \setminus \{0\}$, and let ℓ satisfy $e^\ell = X$. Assume*

$$2\,|a|\,\big(1 + |\ell|\big) < |X|\,. \tag{23.1}$$

Then the series below converges absolutely; the number

$$y := X + a \sum_{n \geq 0} \left(\frac{a}{X}\right)^n \mathsf{L}_n^{\mathrm{Lam}}(\ell) = X - a\ell + \sum_{n \geq 1} a^{n+1} \frac{\mathsf{L}_n^{\mathrm{Lam}}(\ell)}{X^n} \tag{23.2}$$

satisfies $|y/X - 1| \leq \frac{1}{2}$, hence $y \neq 0$; and with $\log_* y := \ell + \mathrm{Log}(y/X)$, which is a logarithm of y,

$$y + a \log_* y = X. \tag{23.3}$$

If moreover $10\,|a|\,(1 + |\ell|) \leq |X|$, then with $\theta_a := 5\,|a|\,(1 + |\ell|)/\,|X| \leq \frac{1}{2}$ and every $N \in \mathbb{N}_0$,

$$\left| y - X - a \sum_{n=0}^{N} \left(\frac{a}{X}\right)^n \mathsf{L}_n^{\mathrm{Lam}}(\ell) \right| \leq 2\,|a|\;\theta_a^{N+1}. \tag{23.4}$$

All data being real, y is real.

Proof. Put $\sigma = a/X$ and $\tau = a\ell/X$, so that $\tau = \sigma\ell$ and $|\sigma| + |\tau| = |a|\,(1 + |\ell|)/\,|X| < \frac{1}{2}$ by Equation (23.1); thus $(\sigma, \tau) \in D$. Let $\mathcal{Y} = \mathcal{Y}(\sigma, \tau)$ be as in Theorem 22.12. By Equation (22.12) with $u = \ell$,

$$\mathcal{Y} = \sum_{n\geq 1} \mathsf{L}_n^{\mathrm{Lam}}(\ell)\sigma^n = \sum_{n\geq 1} a^n \frac{\mathsf{L}_n^{\mathrm{Lam}}(\ell)}{X^n},$$

absolutely convergent, and since $\mathsf{L}_0^{\mathrm{Lam}}(\ell) = -\ell$ the number y of Equation (23.2) equals $X - a\ell + a\mathcal{Y}$. Then

$$\frac{y}{X} = 1 - \frac{a\ell}{X} + \frac{a\mathcal{Y}}{X} = 1 - \tau + \sigma\mathcal{Y},$$

which lies in the disc of radius $\frac{1}{2}$ about 1, so $|y/X - 1| \leq \frac{1}{2}$, $y \neq 0$, $\mathrm{Log}(y/X) = -\mathcal{Y}$ by Equation (22.10), and $\mathrm{e}^{\log_* y} = X \cdot (y/X) = y$. Therefore

$$y + a\log_* y = (X - a\ell + a\mathcal{Y}) + a(\ell - \mathcal{Y}) = X.$$

The remainder bound is Equation (22.13) with $5(|\sigma| + |\tau|) = \theta_a$, multiplied by the outer factor $|a|$. Reality follows from uniqueness of the fixed point in Theorem 22.12: for real (σ, τ) the contraction Φ preserves the real interval $[-1, 1]$, so its fixed point is real. □

23.2 The class $y^\alpha \mathrm{e}^{\beta y} = x$

Proposition 23.2 (Exact reduction, with the branch bookkeeping made explicit)**.** *Let $\alpha, \beta \in \mathbb{C} \setminus \{0\}$ and $x \in \mathbb{C} \setminus \{0\}$; fix a logarithm ξ of x and a logarithm γ of α/β. Then the map*

$$(y, \lambda) \longmapsto (u, \mu) := \left(\frac{\beta}{\alpha} y,\ \lambda - \gamma\right) \tag{23.5}$$

is a bijection between

$$\left\{(y, \lambda) : \mathrm{e}^\lambda = y,\ \beta y + \alpha\lambda = \xi\right\} \quad \textit{and} \quad \{(u, \mu) : \mathrm{e}^\mu = u,\ u + \mu = \zeta\}, \qquad \zeta := \frac{\xi}{\alpha} - \gamma.$$

Consequently, writing $x^{1/\alpha} := \mathrm{e}^{\xi/\alpha}$, every solution of $y^\alpha \mathrm{e}^{\beta y} = x$ compatible with the chosen logarithms has the form

$$y = \frac{\alpha}{\beta} u, \qquad u\mathrm{e}^u = \mathrm{e}^\zeta = \frac{\beta}{\alpha} x^{1/\alpha}, \tag{23.6}$$

that is, $y = (\alpha/\beta)W_k\big((\beta/\alpha)x^{1/\alpha}\big)$ for the branch label k determined by Theorem 22.5.

Proof. Let $\mathrm{e}^\lambda = y$ and $\beta y + \alpha\lambda = \xi$; set $u = (\beta/\alpha)y$ and $\mu = \lambda - \gamma$. Then $\mathrm{e}^\mu = \mathrm{e}^\lambda \mathrm{e}^{-\gamma} = y \cdot (\beta/\alpha) = u$, and $u + \mu = (\beta/\alpha)y + \lambda - \gamma = \alpha^{-1}(\beta y + \alpha\lambda) - \gamma = \xi/\alpha - \gamma = \zeta$. The inverse map is $(u, \mu) \mapsto ((\alpha/\beta)u, \mu + \gamma)$, so the correspondence is bijective. For the last statement, $\mathrm{e}^\mu = u$ and $u + \mu = \zeta$ give $u\mathrm{e}^u = \mathrm{e}^{u+\mu} = \mathrm{e}^\zeta$ by Theorem 22.5, and $\mathrm{e}^\zeta = \mathrm{e}^{\xi/\alpha}\mathrm{e}^{-\gamma} = (\beta/\alpha)x^{1/\alpha}$. Finally $y^\alpha \mathrm{e}^{\beta y}$ means $\mathrm{e}^{\alpha\lambda + \beta y}$ once the logarithm λ of y has been chosen, so the equation $y^\alpha \mathrm{e}^{\beta y} = x$ is exactly $\beta y + \alpha\lambda = \xi$ modulo $2\pi\mathrm{i}\mathbb{Z}$, i.e. exactly the left-hand set after adjusting ξ by a multiple of $2\pi\mathrm{i}$. □

"With branches chosen consistently" is three choices, not a disclaimer

Reports S3, S4, S5 and S6 all display $y = (\alpha/\beta)W_k((\beta/\alpha)x^{1/\alpha})$ followed by a phrase such as "with powers and branches chosen consistently". Theorem 23.2 says what the phrase must mean: one chooses *(a)* a logarithm ξ of x, which fixes the meaning of $x^{1/\alpha}$ as $\mathrm{e}^{\xi/\alpha}$ and is *not* determined by x alone when $\alpha \notin \mathbb{Z}$; *(b)* a logarithm γ of α/β; *(c)* the branch label k, which by Theorem 22.5 is then determined by the solution, not free. Different choices at (a) or (b) give genuinely different solutions y of the same equation, not different names for one.

Theorem 23.3 (Generalized Lambert expansion). *In the situation of Theorem 23.2, put*

$$a := \frac{\alpha}{\beta}, \qquad X := \frac{\xi}{\beta}, \tag{23.7}$$

and let ℓ satisfy $\mathrm{e}^{\ell} = X$. If $2\,|a|\,(1+|\ell|) < |X|$, then

$$y = X + a\sum_{n\ge 0}\left(\frac{a}{X}\right)^n \mathsf{L}^{\mathrm{Lam}}_n(\ell) = \frac{\xi}{\beta} - \frac{\alpha}{\beta}\ell + \sum_{n\ge 1}\frac{\alpha^{n+1}}{\beta}\,\frac{\mathsf{L}^{\mathrm{Lam}}_n(\ell)}{\xi^n} \tag{23.8}$$

converges absolutely and solves $y^{\alpha}\mathrm{e}^{\beta y} = x$ with $\lambda = \ell + \mathrm{Log}(y/X)$ as the logarithm of y; the remainder after N terms is bounded by Equation (23.4)*. In particular, for $\beta = 1$, real $x \to +\infty$, $L_1 = \log x$ and $L_2 = \log\log x$,*

$$y = L_1 - \alpha L_2 + \sum_{n\ge 1}\alpha^{n+1}\frac{\mathsf{L}^{\mathrm{Lam}}_n(L_2)}{L_1^n}. \tag{23.9}$$

Proof. By Theorem 23.2 the equation is $\beta y + \alpha\lambda = \xi$ with $\mathrm{e}^{\lambda} = y$; dividing by β gives $y + a\lambda = X$. Theorem 23.1 with this a and X produces y as in Equation (23.2), which is the first equality in Equation (23.8); the second is $a(a/X)^n = \alpha^{n+1}\beta^{-n-1}\cdot\beta^n\xi^{-n} = \alpha^{n+1}\beta^{-1}\xi^{-n}$. For $\beta = 1$ we have $X = \xi = L_1$, $\ell = L_2$, $a = \alpha$, and $a\cdot a^n\mathsf{L}^{\mathrm{Lam}}_n(L_2)L_1^{-n} = \alpha^{n+1}\mathsf{L}^{\mathrm{Lam}}_n(L_2)L_1^{-n}$. $\square$

Remark 23.4 (Comparison with the sources, and with the literature). Equation (23.9) is S5's generalized Lambert expansion and S2's α-expansion; both are correct and both are proved there only formally. What is added here is the analytic half: a printed seating condition and an explicit remainder. The inversion of $y^{\alpha}\mathrm{e}^{y}$ is classical: Comtet treated it together with $y\log y$ [3], and Jeffrey, Corless, Hare and Knuth expressed the coefficients through associated Stirling numbers and studied the convergence of the resulting series [21]. S5 reports their sufficient domains as $x > (\alpha\mathrm{e})^{\alpha}$ for $\alpha \ge 1$ and $x > \mathrm{e}$ for $\alpha < 1$. Those domains are much larger than the region $2\,|\alpha|\,(1+|L_2|) < L_1$ proved here; they are recorded as attribution and are not theorems of this volume. See Section 23.5 for the structural reason why a different regrouping converges on a larger domain, and for what this volume can prove about it.

Proposition 23.5 (The r-Lambert equation: the same expansion to all orders). *Fix $r \in \mathbb{R}$. There is $z_r > 0$ such that for real $z > z_r$ the equation*

$$w\mathrm{e}^{w} + rw = z \tag{23.10}$$

has a unique large positive solution $w = \mathcal{W}_r(z)$, it tends to $+\infty$, and

$$\left|\mathcal{W}_r(z) - W_0(z)\right| \le \frac{3\,|r|\log z}{z} \qquad (z \to +\infty). \tag{23.11}$$

Consequently $\mathcal{W}_r$ and W_0 have the same *polynomial–logarithmic asymptotic expansion to all orders in the scale $\left\{\ell_0^j\xi_0^{-n}\right\}$: the parameter r is invisible beyond all orders.*

Proof. Let $f(w) = w\mathrm{e}^{w} + rw$. Then $f'(w) = (1+w)\mathrm{e}^{w} + r$, and $(1+w)\mathrm{e}^{w} \to \infty$, so there is $w_r \ge 1$ with $f' > 0$ on $[w_r,\infty)$; since $f \to +\infty$, f is a strictly increasing bijection from $[w_r,\infty)$ onto $[f(w_r),\infty)$. Put $z_r = \max\left\{f(w_r),\mathrm{e}^2\right\}$; for $z > z_r$ let $w > w_r$ be the unique preimage. As $z \to \infty$, $w \to \infty$. Each of the finitely many smallness conditions used below holds for all sufficiently large z; we enlarge z_r once, at the end of the proof, so that all of them hold for $z > z_r$.

From Equation (23.10), $w\mathrm{e}^{w} = z - rw$, so

$$w + \log w = \log(z - rw) = \xi_0 + R, \qquad R := \log\left(1 - \frac{rw}{z}\right), \qquad \xi_0 = \log z, \tag{23.12}$$

valid once $z - rw > 0$. To see that this holds for z large, note first that $we^w = z - rw \le z + |r|\,w$; since $w \to \infty$ we may assume $e^w \ge 2\,|r|$, and then $2\,|r|\,w \le we^w \le z + |r|\,w$ forces $|r|\,w \le z$ and hence $we^w \le 2z$. Since $W_0(y) \le \log y$ for $y \ge e$, this gives $w \le W_0(2z) \le \log(2z) \le \frac{3}{2}\log z$ for $z \ge 4$. For z large, $|rw/z| \le \frac{1}{2}$ and therefore $|R| \le 2\,|r|\,w/z \le 3\,|r|\log z/z$, using $|\log(1-\nu)| \le 2\,|\nu|$ for $|\nu| \le \frac{1}{2}$ and $w \le \log(2z) \le \frac{3}{2}\log z$.

Let $F(v) = v + \log v$ on $(0,\infty)$; then $F'(v) = 1 + 1/v > 1$, so $|F(v_1) - F(v_2)| \ge |v_1 - v_2|$. The principal branch satisfies $F(W_0(z)) = \xi_0$ (take the real logarithm of $W_0 e^{W_0} = z$, legitimate since $W_0(z) > 0$), and $F(w) = \xi_0 + R$; hence $|w - W_0(z)| \le |R|$, which is Equation (23.11).

Finally $\log z/z = \xi_0 e^{-\xi_0}$, which is $o_{\xi_0\to\infty}\left(\xi_0^{-n}\right)$ for every n; so subtracting $\mathcal{W}_r$ from W_0 changes no coefficient of the expansion. □

Remark 23.6 (Why this proposition is here). None of the six sources treats the r-Lambert family, so nothing is consolidated in Theorem 23.5; it is included because the family is the standard first generalisation of W in the literature and because the answer is instructive. It exhibits, in the simplest possible setting, the limitation that every result of this volume shares: a polynomial–logarithmic expansion sees a function only modulo quantities that are smaller than every power of the large variable, and a one-parameter family of genuinely different functions can share one expansion. The same phenomenon is revisited in Theorem 23.18.

23.3 The template

Definition 23.7 (Monomial–logarithmic inverse datum)**.** Let $\mathbb{K}$ be a field of characteristic zero. An inverse problem $\Phi(y) = x$ is *of monomial–logarithmic type* with data (ρ, μ, β, B) if, after a logarithm ξ of x and a logarithm Z of y have been chosen, it is equivalent to

$$\rho Z + \mu \log Z + \beta + B(Z) = \xi, \tag{23.13}$$

where $\rho \in \mathbb{K}^\times$, $\mu, \beta \in \mathbb{K}$, and, writing $t = Z^{-1}$ and $L = \log Z$ for the generators of the logarithmic chart,

$$B = \sum_{m\ge1} b_m(L)\, t^m \in t\mathbb{K}[L][[t]], \qquad b_m \in \mathbb{K}[L]. \tag{23.14}$$

Exponentiating, this says $\Phi(y) = e^\beta y^\rho (\log y)^\mu e^{B(\log y)}$, all powers being defined through the chosen logarithms; the leading balance is the monomial $e^\beta y^\rho (\log y)^\mu$.

The three hypotheses are worth naming, because each fails in a recognisable way.

(H1) $\rho \ne 0$: there is a genuine monomial in the balance. If $\rho = 0$ the inverse changes exponential level and leaves every polynomial–logarithmic class (Theorem 15.11); see Theorem 23.17.

(H2) μ is a *constant*, so that the logarithmic part of the balance is $\mu \log Z$ and not something with a variable exponent.

(H3) B has strictly positive t-order and coefficients *polynomial* in L. If B has t-order 0 or its coefficients are not polynomial, the reversion leaves $\mathcal{G}_{\mathrm{poly},\mathbb{K}}$; see Theorem 23.15.

Theorem 23.8 (Reusable inverse template)**.** *Let $\Phi(y) = x$ be of monomial–logarithmic type with data (ρ, μ, β, B) as in Theorem 23.7. Put*

$$a := \frac{\mu}{\rho}, \qquad X := \frac{\xi - \beta}{\rho}, \qquad t = X^{-1}, \qquad \ell = \log X. \tag{23.15}$$

Then:

1. *$F := X + aL + \rho^{-1}B$ lies in $\mathcal{G}_{\mathrm{poly},\mathbb{K}}$, and Equation* (23.13) *is equivalent to $F(Z) = X$; hence there is a unique $Z \in X + \mathbb{K}[\ell][[t]]$ solving it, namely $Z = F^{\langle -1\rangle_\circ}$ evaluated at X;*

2. *writing* $F^{\langle -1\rangle_\circ} = X + \sum_{n\geq 0} q_n(L)t^n$*, one has* $q_0 = -aL$*, and for every* $N \geq 0$ *the polynomial* q_N *is a polynomial function of* a *and of the coefficients of* $b_1, \dots, b_N$ *only, given explicitly by the coefficient formula of Theorem 18.18;*

3. $q_N = a^{N+1}\mathsf{L}_N^{\mathrm{Lam}} + E_N$*, where* E_N *depends only on* $a, b_1, \dots, b_N$ *and vanishes identically when* $b_1 = \cdots = b_N = 0$*; in particular* $q_N = a^{N+1}\mathsf{L}_N^{\mathrm{Lam}}$ *for all* N *when* $B = 0$*;*

4. *re-exponentiating,*

$$y = \mathrm{e}^Z = \left(\frac{x}{\mathrm{e}^\beta}\right)^{1/\rho} X^{-\mu/\rho} \exp\left(\sum_{n\geq 1} q_n(\ell)\, X^{-n}\right), \tag{23.16}$$

with $x^{1/\rho} := \mathrm{e}^{\xi/\rho}$*; the last factor lies in* $1 + t\mathbb{K}[\ell][[t]]$*. In particular the leading behaviour is*

$$y \sim \mathrm{e}^{-\beta/\rho}\, x^{1/\rho} \left(\frac{\xi}{\rho}\right)^{-\mu/\rho}. \tag{23.17}$$

When $1/\rho$ *or* μ/ρ *is not an integer, Equation* (23.16) *lives in the ramified Puiseux–logarithmic tower of Theorem 5.25, not in* $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$*.*

Proof. (i) Dividing Equation (23.13) by ρ and moving β/ρ to the right gives $Z + a\log Z + \rho^{-1}B(Z) = X$. The perturbation $A := aL + \rho^{-1}B$ lies in $\mathbb{K}[L][[t]]$ by Equation (23.14), so $F = X + A \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}$ (Theorem 14.1), and Theorem 16.11 gives a unique inverse in $\mathcal{G}_{\mathrm{poly},\mathbb{K}}$.

(ii) Write $A^r = \sum_{m\geq 0} A_{r,m}(L)t^m$. By Theorem 18.18,

$$q_N(L) = \sum_{r=1}^{N+1} \frac{(-1)^r}{r!} \left[\prod_{s=N-r+1}^{N-1} (\partial_L - s)\right] A_{r,N-r+1}(L),$$

with the empty product acting as the identity. Now $A_{r,m}$ is a sum of products of r blocks $A_{j_1}, \dots, A_{j_r}$ with $j_1 + \cdots + j_r = m$, so it depends only on $A_0, \dots, A_m$; since $m = N - r + 1 \leq N$, the whole formula involves only $A_0 = aL$ and $A_1 = \rho^{-1}b_1, \dots, A_N = \rho^{-1}b_N$, polynomially. The block q_0 is the $r = 1, m = 0$ term, namely $-A_{1,0} = -A_0 = -aL$.

(iii) Setting $b_1 = \cdots = b_N = 0$ in the formula of (ii) leaves the coefficients of $(X + aL)^{\langle -1\rangle_\circ}$, which are $a^{N+1}\mathsf{L}_N^{\mathrm{Lam}}$ by Theorem 22.7. Define E_N as the difference; by (ii) it is a polynomial function of $a, b_1, \dots, b_N$ vanishing when the b's do.

(iv) From $Z = X + q_0(\ell) + \sum_{n\geq 1} q_n(\ell)X^{-n}$ and $q_0(\ell) = -a\ell$,

$$\mathrm{e}^Z = \mathrm{e}^X \cdot \mathrm{e}^{-a\ell} \cdot \exp\Big(\sum_{n\geq 1} q_n(\ell)X^{-n}\Big).$$

Here $\mathrm{e}^{-a\ell} = (\mathrm{e}^\ell)^{-a} = X^{-a} = X^{-\mu/\rho}$ and $\mathrm{e}^X = \mathrm{e}^{(\xi-\beta)/\rho} = \mathrm{e}^{\xi/\rho}\mathrm{e}^{-\beta/\rho} = x^{1/\rho}\mathrm{e}^{-\beta/\rho}$. The exponential of an element of $t\mathbb{K}[\ell][[t]]$ lies in $1 + t\mathbb{K}[\ell][[t]]$ by Theorem 12.5, which also gives Equation (23.17) after replacing $X = (\xi - \beta)/\rho$ by ξ/ρ, a substitution absorbed by the $1 + t\mathbb{K}[\ell][[t]]$ factor. The final sentence is Theorem 12.17 together with Theorem 5.25. □

Corollary 23.9 (Closed form when $B = 0$)**.** *If* $B = 0$*, then*

$$Z = \frac{\xi - \beta}{\rho} + \frac{\mu}{\rho}\sum_{n\geq 0}\left(\frac{\mu}{\xi - \beta}\right)^n \mathsf{L}_n^{\mathrm{Lam}}(\ell), \qquad \ell = \log\frac{\xi - \beta}{\rho}, \tag{23.18}$$

and the series converges absolutely, with the remainder bound Equation (23.4)*, as soon as* $2\,|\mu|\,(1 + |\ell|) < |\xi - \beta|$.

Proof. Theorem 23.1 with $a = \mu/\rho$ and $X = (\xi - \beta)/\rho$; note $a/X = \mu/(\xi - \beta)$ and $a \cdot (a/X)^n$ has the prefactor μ/ρ shown. The seating condition $2\,|a|\,(1 + |\ell|) < |X|$ is $2\,|\mu|\,(1 + |\ell|) < |\xi - \beta|$ after multiplying by $|\rho|$. □

Note the shape of Equation (23.18): the exponent ρ of the monomial appears only in the leading term, in the overall prefactor μ/ρ and inside ℓ — never in the ratio $a/X = \mu/(\xi - \beta)$ that drives the series — while the exponent μ of the logarithm is the expansion parameter. This is why the generalized Lambert family is a one-parameter family and not a two-parameter one.

Algorithm 23.10 (The template, mechanically)**.**

```
input : the direct map Phi, the target x, an order N
1  choose logarithms: xi with e^xi = x; Z will be a logarithm of y
2  rewrite Phi(y) = x as  rho*Z + mu*log Z + beta + B(Z) = xi
     # if this is impossible with constant rho != 0 and constant mu,
     # or with B of positive t-order and polynomial coefficients,
     # the template does not apply: see Example lw-failure
3  a := mu/rho ;  X := (xi - beta)/rho ;  ell := log X ;  t := 1/X
4  A := a*L + B/rho                      # an element of K[L][[t]]
5  Z := reverse(X + A) to order N        # triangular or Newton reversion
     # equivalently  q_n = a^{n+1} P_n(ell) + E_n(b_1..b_n)
6  y := exp(Z)                           # = (x e^-beta)^(1/rho) * X^(-a) * unit
7  residual := (rho*Z + mu*log Z + beta + B(Z) - xi)   # must be O(t^{N+1})
```

Step 5 is where the whole of Part 17 is reused unchanged; step 7 is the finite certificate of Theorem 16.16. Step 6 is the only place where the answer can leave $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$, and it does so exactly when $1/\rho$ or μ/ρ is not an integer.

23.4 Three worked examples

Example 23.11 (A fractional power forcing a Puiseux–logarithmic extension)**.** Solve $Y = X^2 \log X$ for X as $Y \to +\infty$.

Here $\rho = 2$, $\mu = 1$, $\beta = 0$, $B = 0$, and $a = \mu/\rho = 1/2$. With $\xi = \log Y$, the template coordinates are $X_* = \xi/2$ and $\ell = \log(\xi/2)$, and $a/X_* = \mu/\xi = 1/\xi$. Theorem 23.9 gives

$$\log X = \frac{\xi}{2} + \frac{1}{2}\sum_{n\ge 0}\frac{\mathsf{L}_n^{\mathrm{Lam}}(\ell)}{\xi^n} = \frac{\xi}{2} - \frac{\ell}{2} + \frac{\ell}{2\xi} + \frac{\ell^2/2 - \ell}{2\xi^2} + \cdots, \tag{23.19}$$

convergent as soon as $2(1 + |\ell|) < \xi$. Exponentiating,

$$X = \sqrt{\frac{2Y}{\log Y}}\left(1 + \frac{\ell}{2\xi} + \frac{3\ell^2/8 - \ell/2}{\xi^2} + \mathcal{O}_{Y\to+\infty}\big(\ell^3/\xi^3\big)\right), \quad \xi = \log Y,\ \ell = \log\frac{\log Y}{2}. \tag{23.20}$$

Two features are structural rather than accidental. First, the answer contains $Y^{1/2}$: by Theorem 23.8 the exponent is $1/\rho$, so any ρ with $1/\rho \notin \mathbb{Z}$ takes the answer out of $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ into the ramified tower of Theorem 5.25. Second, the answer contains $\log\log Y$ through ℓ, although the problem contained only one logarithm; this is Theorem 19.4, and it is why the coefficient ring has to be $\mathbb{K}[\ell]$ and not $\mathbb{K}$. The relative errors of the three displayed truncations at $Y = 10^{20}$ are $3.4 \cdot 10^{-2}$, $9.8 \cdot 10^{-4}$ and $1.3 \cdot 10^{-5}$.

The general case $Y = cX^\rho(\log X)^\mu$ is Equation (23.17): $X \sim c^{-1/\rho}\rho^{\mu/\rho}Y^{1/\rho}(\log Y)^{-\mu/\rho}$, which is the leading form quoted in S3 and S6.

Example 23.12 (An iterated logarithm, and where the coefficient field breaks)**.** Solve $Y = F(X) = X + \alpha\log X + \gamma\log\log X$ for X as $Y \to +\infty$, with $\alpha, \gamma \in \mathbb{K}$.

This is *not* covered by Theorem 23.8 as stated. In the template's variables the equation is $Z + \alpha \log Z + \gamma \log\log Z = Y$, so besides the permitted block $\mu \log Z$ with $\mu = \alpha$ there is a *second* block of t-order 0, namely $\gamma \log\log Z$, whose coefficient $M = \log L$ is not in $\mathbb{K}[L]$: hypothesis (H3) fails. Equivalently, in the reversion chart the perturbation is $A = \alpha L + \gamma M$ with $L = \log Y$, $M = \log L$, and $A \notin \mathbb{K}[L][[t]]$. The example is covered by the depth-two version: by Theorem 15.14 the near-identity group at logarithmic depth 2 is a group over the coefficient ring generated by L and M, and the proof of Theorem 23.8 goes through verbatim once $\mathbb{K}[L]$ is replaced by that ring. Writing

$$A = \alpha L + \gamma M, \qquad \varpi := \alpha + \frac{\gamma}{L}, \qquad \frac{\mathrm{d}A}{\mathrm{d}Y} = \frac{\varpi}{Y}, \qquad \frac{\mathrm{d}^2 A}{\mathrm{d}Y^2} = -\frac{1}{Y^2}\left(\alpha + \frac{\gamma}{L} + \frac{\gamma}{L^2}\right), \tag{23.21}$$

the first three Lagrange–Bürmann terms of Theorem 18.14 give

$$F^{\langle -1 \rangle_\circ}(Y) = Y - A + \frac{A\varpi}{Y} + \frac{1}{Y^2}\left[-A\varpi^2 + \frac{A^2}{2}\left(\alpha + \frac{\gamma}{L} + \frac{\gamma}{L^2}\right)\right] + \mathcal{O}_t^{\mathrm{form}}\left(t^3\right), \qquad t = Y^{-1}. \tag{23.22}$$

The remainder here is the *formal* tail of Theorem 18.14, not an analytic estimate: the three displayed blocks are produced by a computation in the depth-two reversion chart and say nothing by themselves about the function $F^{\langle -1 \rangle_\circ}$. They do become an asymptotic statement, with remainder $\mathcal{O}_{Y\to+\infty}\left(L^4/Y^3\right)$, once one adds that F realises its expansion in the sense of Theorem 12.32 — which it does, being a sum of three elementary functions — and invokes the transfer of Theorem 16.21; that step is analysis, and it is not supplied by the template. The instructive point of the example is elsewhere: it is where the coefficient ring first fails to be polynomial.

Proposition 23.13 (The first non-polynomial coefficient)**.** *In Equation* (23.22)*, expand the block of* Y^{-1}*:*

$$A\varpi = (\alpha L + \gamma M)\left(\alpha + \frac{\gamma}{L}\right) = \alpha^2 L + \alpha\gamma M + \alpha\gamma + \frac{\gamma^2 M}{L}. \tag{23.23}$$

Hence for every $\gamma \neq 0$ *the coefficient of* Y^{-1} *already fails to lie in* $\mathbb{K}[L, M]$ *— one block earlier than S6 records — and the natural coefficient ring is* $\mathbb{K}((L^{-1}))[M]$*. For* $\gamma = 0$ *the problem degenerates to the affine-logarithmic case and every block is in* $\mathbb{K}[L]$*.*

Proof. The displayed product is immediate. The functions $L = \log Y$ and $M = \log\log Y$ are algebraically independent over $\mathbb{K}$: if $P \in \mathbb{K}[U, V]$ were nonzero with $P(L, M) \equiv 0$ for large Y, then among the monomials $L^i M^j$ occurring in P there would be a unique dominant one by Theorem 4.7 (the order being lexicographic in (i, j), since $M = o_{Y\to+\infty}(L^\epsilon)$ for every $\epsilon > 0$), and a dominant term cannot cancel. Hence $\mathbb{K}[L, M]$ is a polynomial ring in two indeterminates, in which L is prime and does not divide M; therefore $M/L \notin \mathbb{K}[L, M]$. Since $\alpha^2 L + \alpha\gamma M + \alpha\gamma \in \mathbb{K}[L, M]$ and $\gamma^2 \neq 0$, the sum Equation (23.23) lies outside $\mathbb{K}[L, M]$. For $\gamma = 0$, $A = \alpha L$ and Theorem 22.7 applies, giving $q_n = \alpha^{n+1} \mathsf{L}_n^{\mathrm{Lam}}(L) \in \mathbb{K}[L]$. □

Remark 23.14 (Numerical check). With $\alpha = \gamma = 1$ and $Y = 10^{10}$, the absolute errors of the truncations of Equation (23.22) at orders Y^0, Y^{-1}, Y^{-2} are $2.73 \cdot 10^{-9}$, $3.29 \cdot 10^{-18}$ and $5.16 \cdot 10^{-27}$; the ratio of consecutive errors is about $Y^{-1}L$, as Equation (23.22) predicts.

Example 23.15 (An equation that fails the hypotheses)**.** Solve

$$Y = X + \frac{X}{\log X} \tag{23.24}$$

for X as $Y \to +\infty$.

The map looks like a perturbation of the identity, and it is one analytically: $X/\log X = o_{X\to+\infty}(X)$. It is not one *in the sense of* $\mathcal{G}_{\mathrm{poly},\mathbb{K}}$. Writing $t = X^{-1}$ and $L = \log X$, the perturbation is $L^{-1}t^{-1}$; read inside $\mathscr{PL}_{\mathrm{itLaur},\mathbb{K}}$ it has $\nu_t = -1$, the same outer order as X itself, and its coefficient L^{-1} is not in $\mathbb{K}[L]$. So hypothesis (H3) of Theorem 23.7 fails twice over — wrong outer order and wrong coefficient ring — and Theorem 16.11 does not apply.

The correct diagnosis is that the problem is a near-identity problem one level down. Put $\lambda = \log Y$ and $Z = \log X$; taking logarithms of Equation (23.24),

$$\lambda = Z + \log\left(1 + \frac{1}{Z}\right) = Z + \frac{1}{Z} - \frac{1}{2Z^2} + \frac{1}{3Z^3} - \cdots, \tag{23.25}$$

which *is* of monomial–logarithmic type with $\rho = 1$, $\mu = 0$, $\beta = 0$ and $B(Z) = \log(1 + 1/Z) \in Z^{-1}\mathbb{K}[[Z^{-1}]]$. Theorem 23.8 therefore applies in the logarithmic chart and gives, with all q_n constants because $a = \mu/\rho = 0$ and every b_m is constant,

$$Z = \lambda - \frac{1}{\lambda} + \frac{1}{2\lambda^2} - \frac{4}{3\lambda^3} + \frac{7}{4\lambda^4} - \frac{121}{30\lambda^5} + \cdots. \tag{23.26}$$

Re-exponentiating, or equivalently using $X = Y\big(1 - 1/(Z+1)\big)$,

$$X = Y\left(1 - \frac{1}{\lambda} + \frac{1}{\lambda^2} - \frac{2}{\lambda^3} + \frac{7}{2\lambda^4} - \frac{22}{3\lambda^5} + \frac{179}{12\lambda^6} - \cdots\right), \qquad \lambda = \log Y. \tag{23.27}$$

The relative errors of the first four truncations at $Y = 10^{40}$ are $1.09 \cdot 10^{-2}$, $1.17 \cdot 10^{-4}$, $2.54 \cdot 10^{-6}$, $4.81 \cdot 10^{-8}$.

Proposition 23.16 (The answer is not a polynomial–logarithmic series). *Let $X = X(Y)$ be the solution of Equation* (23.24) *for $Y \to +\infty$. There is no $G = \sum_{n \geq n_0} p_n(L)t^n \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ — with $t = Y^{-1}$, $L = \log Y$, coefficients $p_n \in \mathbb{K}[L]$ — that is an asymptotic expansion of X in the sense of Theorem 8.43. The answer Equation* (23.27) *does lie in $\mathscr{PL}_{\mathrm{itLaur},\mathbb{K}} = \mathbb{K}((L^{-1}))((t))$.*

Proof. Suppose such a G existed. Since $X/Y \to 1$, the outer order is $n_0 = -1$, and Theorem 8.43 implies in particular that, for some $d \in \mathbb{N}_0$ and all large Y,

$$X = p_{-1}(L)\, Y + \mathcal{O}_{Y\to+\infty}\left(L^d\right), \tag{23.28}$$

the error absorbing the block $p_0(L)$ together with the tail. The two leading terms of X/Y are obtained directly from Equation (23.24), with no appeal to Equation (23.27): writing $Z = \log X$, the equation $Y = X(1 + 1/Z)$ gives

$$\frac{X}{Y} = \frac{Z}{Z+1} = 1 - \frac{1}{Z+1}, \qquad L = \log Y = Z + \log\Big(1 + \frac{1}{Z}\Big) = Z + \mathcal{O}_{Y\to+\infty}\big(Z^{-1}\big),$$

so $Z = L + \mathcal{O}_{Y\to+\infty}\big(L^{-1}\big)$ and hence

$$\frac{X}{Y} = 1 - \frac{1}{L} + \mathcal{O}_{Y\to+\infty}\big(L^{-2}\big),$$

whereas Equation (23.28) gives $X/Y = p_{-1}(L) + \mathcal{O}_{Y\to+\infty}\big(L^d/Y\big)$. Subtracting,

$$p_{-1}(L) - 1 + \frac{1}{L} = \mathcal{O}_{Y\to+\infty}\big(L^{-2}\big) + \mathcal{O}_{Y\to+\infty}\Big(L^d/Y\Big).$$

As $Y \to +\infty$ both right-hand terms tend to 0 (the second because $L^d/Y \to 0$ for every fixed d), while $1/L \to 0$; hence $p_{-1}(L) \to 1$ along $L \to +\infty$, which for a polynomial forces $p_{-1} \equiv 1$. But then the left side is exactly $1/L$, which is not $\mathcal{O}_{Y\to+\infty}\big(L^{-2}\big) + \mathcal{O}_{Y\to+\infty}\big(L^d/Y\big)$, because $L \cdot (1/L) = 1 \not\to 0$. Contradiction.

For the positive half, Equation (23.27) exhibits X as $c(L)\, t^{-1}$ with $c(L) = 1 - L^{-1} + L^{-2} - 2L^{-3} + \cdots \in \mathbb{K}((L^{-1}))$, which is one block of an element of $\mathscr{PL}_{\mathrm{itLaur},\mathbb{K}}$ (Theorem 5.3); the inclusion $\mathscr{PL}_{\mathrm{poly},\mathbb{K}} \subsetneq \mathscr{PL}_{\mathrm{itLaur},\mathbb{K}}$ and its strictness are Theorem 5.7. $\square$

Remark 23.17 (The other way to fail: no monomial at all). Hypothesis (H1) fails for $Y = (\log X)^2$, where Equation (23.13) reads $0 \cdot Z + 2\log Z = \xi$. The solution is $X = \exp(\mathrm{e}^{\xi/2}) = \exp(\sqrt{Y})$ — an exponential of a power of the large variable, hence outside every polynomial–logarithmic class by Theorem 15.11. No amount of coefficient-ring enlargement recovers it: the failure is a change of exponential level, not of coefficient field. The rule of thumb is that (H1) controls the *level*, (H2) the shape of the leading logarithm, and (H3) the coefficient field; a violation of each is repaired, if at all, in a different direction.

Remark 23.18 (Beyond all orders). A last failure mode passes every hypothesis and still leaves the answer undetermined. For $Y = X + \mathrm{e}^{-X}$, the perturbation e^{-X} is not in $\mathbb{K}[L][[t]]$, but it is $o_{X\to+\infty}(t^n)$ for every n; the template applied to the truncated data returns $X = Y$, and every polynomial–logarithmic truncation of the true inverse is indeed Y. The expansion is correct and useless: it does not see e^{-Y}. The same phenomenon appears in Theorem 23.5, where an entire one-parameter family of functions shares one expansion. Recovering the invisible terms requires transseries with exponential monomials [12, 8, 1], which are outside the classes of Part 5.

Remark 23.19 (Two classical instances, recorded for reference). The template covers, with no new work, two examples that recur across the sources.

1. $Y = X\log X$. Put $u = \log X$; then $u\mathrm{e}^u = Y$, so $X = \mathrm{e}^{W(Y)} = Y/W(Y)$, and dividing by the branch expansion, using the reciprocal recurrence of Theorem 9.7,

$$X = \frac{Y}{L_1}\left(1 + \frac{L_2}{L_1} + \frac{L_2^2 - L_2}{L_1^2} + \frac{L_2^3 - \frac{5}{2}L_2^2 + L_2}{L_1^3} + \mathcal{O}_{Y\to+\infty}\big(L_2^4/L_1^4\big)\right),$$

   with $L_1 = \log Y$, $L_2 = \log\log Y$. This is the form given in S3 and S6, and it is correct.

2. $X = y/\log y$. Setting $v = y/X$ turns this into $v - \log v = \log X$, which is the lower-branch equation of Theorem 22.21 with $S = T$, $T := \log X$; hence $y = -XW_{-1}(-1/X)$ and, with $\Lambda = \log T$,

$$y = X\left(T + \Lambda + \sum_{n\geq 1}(-1)^{n+1}\frac{\mathsf{L}_n^{\mathrm{Lam}}(\Lambda)}{T^n}\right),$$

   convergent whenever $2(1+|\Lambda|) < T$ by Theorem 22.22. Note the sign: it is $(-1)^{n+1}$, not $(-1)^n$, because the reflection $y = Xv$ has already been undone.

Remark 23.20 (Reductions that are exact, and reductions that are not). It is worth separating two things the sources sometimes present together. Theorem 23.2 and Theorem 23.19 are *exact* reductions: an algebraic change of variable turns the equation into $u + \log u = \zeta$ with no error term, so the whole of Part 22 applies verbatim, convergence statements included. Theorem 23.8 is not of that kind: it produces a formal inverse in $\mathcal{G}_{\mathrm{poly},\mathbb{K}}$ whose analytic status must be argued separately, either by Theorem 12.35 (if the direct map realises its expansion) or by an inverse-function estimate as in Theorem 16.21. S6's example $Y = X + \alpha\log X + \beta + (\gamma\log X + \delta)/X + \cdots$, whose inverse is

$$X = Y - \alpha L - \beta + \frac{(\alpha^2-\gamma)L + \alpha\beta - \delta}{Y} + \mathcal{O}_{Y\to+\infty}\big(L^2/Y^2\big), \qquad L = \log Y,$$

is of the second kind. Its displayed coefficients are correct, and they are the case $\rho = 1$, $\mu = \alpha$, $b_1 = \gamma\ell + \delta$ of Theorem 23.8: the template shifts the constant away, working in $X = Y - \beta$ and $\ell = \log(Y - \beta)$, where $q_1(\ell) = (\alpha^2-\gamma)\ell - \delta$ with no constant term, and the extra $\alpha\beta$ in the display above is produced when ℓ is re-expanded as $L - \beta/Y + \mathcal{O}_{Y\to+\infty}(Y^{-2})$. The error term, however, is legitimate only under a hypothesis on the direct map, not merely on its formal expansion.

23.5 Reseating the expansion, and the associated Stirling numbers

Theorem 22.13 was stated with an arbitrary base point K for a reason: the choice $K = \xi_k$ is convenient but not optimal, and the literature on Lambert W contains a whole *sequence* of series obtained by other choices [6], as well as rearrangements whose coefficients are associated Stirling numbers rather than ordinary ones [21, 3]. This subsection isolates the mechanism, which is entirely elementary once the two-variable picture of Theorem 22.12 is available: the coefficient array (c_{ab}) can be summed along its diagonals $a + b = n$, which gives the Lambert polynomials, or along its rows $a =$ const, which gives something else — and the something else turns out to be the same family again, evaluated elsewhere.

Theorem 23.21 (Transverse resummation)**.** *In $\mathbb{Q}[[\sigma, \tau]]$, let $\mathcal{Y}$ be the unique solution with $\mathcal{Y}(0, 0) = 0$ of $\mathcal{Y} = -\log(1 - \tau + \sigma\mathcal{Y})$. Then*

$$\mathcal{Y}(\sigma, \tau) = \sum_{a\geq 0} \mathsf{L}_a^{\mathrm{Lam}}\big(\log(1-\tau)\big) \left(\frac{\sigma}{1-\tau}\right)^a, \tag{23.29}$$

where $\log(1 - \tau) = -\sum_{m\geq 1} \tau^m/m \in \tau\mathbb{Q}[[\tau]]$. Equivalently, writing $\mathsf{L}_n^{\mathrm{Lam}}(u) = \sum_m \lambda_{n,m}u^m$, for all $a, b \geq 0$ with $a + b \geq 1$,

$$\lambda_{a+b,b} = [\tau^b] \left\{ \frac{\mathsf{L}_a^{\mathrm{Lam}}\big(\log(1-\tau)\big)}{(1-\tau)^a} \right\}. \tag{23.30}$$

Proof. Existence and uniqueness of $\mathcal{Y}$ in $\mathbb{Q}[[\sigma, \tau]]$ with $\mathcal{Y}(0, 0) = 0$ follow from the (σ, τ)-adic contraction argument, the map $y \mapsto -\log(1 - \tau + \sigma y)$ raising the order by one on the maximal ideal; the analytic realisation is Theorem 22.12. Put $\zeta = 1 - \tau$, $\Lambda = -\log\zeta \in \tau\mathbb{Q}[[\tau]]$ and $\widehat{\sigma} = \sigma/\zeta$. Then

$$\mathcal{Y} = -\log(\zeta + \sigma\mathcal{Y}) = -\log\zeta - \log\big(1 + \widehat{\sigma}\mathcal{Y}\big) = \Lambda - \log\big(1 + \widehat{\sigma}\mathcal{Y}\big).$$

Set $\mathcal{Q} := \mathcal{Y} - \Lambda$. Substituting $\mathcal{Y} = \Lambda + \mathcal{Q}$,

$$\mathcal{Q} = -\log\big(1 + \widehat{\sigma}\Lambda + \widehat{\sigma}\mathcal{Q}\big) = -\log\big(1 - \widehat{\sigma}\,(-\Lambda) + \widehat{\sigma}\mathcal{Q}\big),$$

which is precisely the generating equation Equation (22.6) with $s = \widehat{\sigma}$ and $u = -\Lambda$. Moreover $\mathcal{Q}$ has positive order in $\widehat{\sigma}$. By the uniqueness clause of Theorem 21.11, transported along the continuous $\mathbb{Q}$-algebra homomorphism $\mathbb{Q}[u][[s]] \to \mathbb{Q}[[\sigma, \tau]]$ sending $s \mapsto \widehat{\sigma}$ and $u \mapsto -\Lambda$,

$$\mathcal{Q} = \sum_{a\geq 1} \mathsf{L}_a^{\mathrm{Lam}}(-\Lambda)\,\widehat{\sigma}^{\,a}.$$

Since $\mathsf{L}_0^{\mathrm{Lam}}(-\Lambda) = \Lambda$, adding Λ gives $\mathcal{Y} = \sum_{a\geq 0} \mathsf{L}_a^{\mathrm{Lam}}(-\Lambda)\widehat{\sigma}^{\,a}$, which is Equation (23.29) because $-\Lambda = \log(1 - \tau)$.

For Equation (23.30), compare with the diagonal grouping. Restricting Equation (22.12) to $\tau = \sigma u$ identified $\sum_{a+b=n} c_{ab}u^b = \mathsf{L}_n^{\mathrm{Lam}}(u)$, i.e. $c_{ab} = \lambda_{a+b,b}$ for $a + b \geq 1$. Extracting $\sigma^a\tau^b$ from Equation (23.29) gives $c_{ab} = [\tau^b]\{\mathsf{L}_a^{\mathrm{Lam}}(\log(1 - \tau))(1 - \tau)^{-a}\}$. □

Corollary 23.22 (The edge coefficients of the Lambert polynomials, twice)**.** *Taking $a = 0$ and $a = 1$ in Equation* (23.30)*:*

$$\lambda_{b,b} = [\tau^b]\big(-\log(1-\tau)\big) = \frac{1}{b} \qquad (b \geq 1), \tag{23.31}$$

$$\lambda_{b+1,b} = [\tau^b]\frac{\log(1-\tau)}{1-\tau} = -H_b \qquad (b \geq 0), \tag{23.32}$$

where $H_b = \sum_{j=1}^{b} 1/j$. These recover the leading coefficient $1/n$ and the next coefficient $-H_{n-1}$ of $\mathsf{L}_n^{\mathrm{Lam}}$ recorded in Theorem 21.7, now as the Taylor coefficients of two elementary functions. Continuing to $a = 2$,

$$\frac{\mathsf{L}_2^{\mathrm{Lam}}\big(\log(1-\tau)\big)}{(1-\tau)^2} = \tau + 3\tau^2 + \tfrac{35}{6}\tau^3 + \tfrac{75}{8}\tau^4 + \tfrac{203}{15}\tau^5 + \cdots,$$

whose coefficients are $\lambda_{3,1}, \lambda_{4,2}, \lambda_{5,3}, \lambda_{6,4}, \lambda_{7,5}$.

Proof. For $a = 0$, $\mathsf{L}_0^{\mathrm{Lam}}(u) = -u$ gives $-\log(1-\tau) = \sum_{b\geq 1}\tau^b/b$. For $a = 1$, $\mathsf{L}_1^{\mathrm{Lam}}(u) = u$ gives $\log(1-\tau)/(1-\tau)$, and $\sum_{b\geq 1} H_b\tau^b = -\log(1-\tau)/(1-\tau)$ is the classical Cauchy-product identity $\big(\sum_{m\geq 1}\tau^m/m\big)\big(\sum_{j\geq 0}\tau^j\big) = \sum_{b\geq 1} H_b\tau^b$, with $H_0 = 0$. The listed values for $a = 2$ are obtained by the same expansion. □

Corollary 23.23 (Reseating at the two-term approximation)**.** *In Theorem 22.13, take the base point $\widetilde{K} := K - M$ with $\log\widetilde{K} := \log K + \operatorname{Log}(1 - M/K)$, assuming $|M/K| < 1$. Then the associated parameter is*

$$\widetilde{M} = \operatorname{Log}\left(1 - \frac{M}{K}\right), \qquad \left|\widetilde{M}\right| \leq -\log\left(1 - \frac{|M|}{|K|}\right), \tag{23.33}$$

so that $\widetilde{M} \to 0$ when $M/K \to 0$. If $2(1 + \left|\widetilde{M}\right|) < \left|\widetilde{K}\right|$, then the series $\widetilde{w} := \widetilde{K} + \sum_{a\geq 0}\mathsf{L}_a^{\mathrm{Lam}}(\widetilde{M})\widetilde{K}^{-a}$ converges absolutely, with the remainder bound Equation (22.18) *in the reseated data, and $\widetilde{w}\mathrm{e}^{\widetilde{w}} = \mathrm{e}^{\xi}$. In the canonical case $K = \xi_k$, $M = \ell_k$, assume in addition to the hypotheses of Theorem 22.16 that*

$$4\big(1 + |\ell_k|\big) \leq |\xi_k|\,. \tag{23.34}$$

Then the reseated seating condition holds automatically, $\widetilde{w} = W_k(z)$, and

$$W_k(z) = \big(\xi_k - \ell_k\big) + \sum_{a\geq 0}\frac{\mathsf{L}_a^{\mathrm{Lam}}\big(\operatorname{Log}(1 - \ell_k/\xi_k)\big)}{(\xi_k - \ell_k)^a}. \tag{23.35}$$

For real $z > 1$ and $k = 0$ the identification $\widetilde{w} = W_0(z)$ needs neither Equation (23.34) *nor Theorem 22.16: all the reseated data are then real, hence so is $\widetilde{w}$, and a positive real argument has exactly one real preimage under $w\mathrm{e}^w$ by Theorem 22.2* (iii).

Proof. By Equation (22.15), the parameter attached to a base point $\widetilde{K}$ with a chosen logarithm is $\widetilde{M} = \widetilde{K} + \log\widetilde{K} - \xi$. With $\widetilde{K} = K - M$ and the stated logarithm,

$$\widetilde{M} = (K - M) + \log K + \operatorname{Log}(1 - M/K) - \xi = (K + \log K - \xi) - M + \operatorname{Log}(1 - M/K),$$

and the bracket is M, leaving $\widetilde{M} = \operatorname{Log}(1 - M/K)$. The stated modulus bound is Equation (22.14). Note $\mathrm{e}^{\log\widetilde{K}} = K(1 - M/K) = K - M = \widetilde{K}$, so the chosen logarithm is legitimate. The convergence statement, the remainder bound and $\widetilde{w}\mathrm{e}^{\widetilde{w}} = \mathrm{e}^{\xi}$ are then Theorems 22.13 and 22.14 applied verbatim to the reseated data $(\widetilde{K}, \widetilde{M})$.

It remains to identify $\widetilde{w}$ in the canonical case. Assume Equation (23.34), so $|\ell_k| \leq \frac{1}{4}|\xi_k| - 1$ and in particular $|\ell_k| / |\xi_k| \leq \frac{1}{4}$ and $|\xi_k| \geq 4$. By Equation (23.33), $\left|\widetilde{M}\right| \leq -\log(1 - \frac{1}{4}) = \log\frac{4}{3} < 0.2877$, while $\left|\widetilde{K}\right| \geq |\xi_k| - |\ell_k| \geq \frac{3}{4}|\xi_k| + 1 \geq 4$; since $2(1 + \left|\widetilde{M}\right|) < 2.576 < 4$, the reseated seating condition holds automatically. Theorem 22.13 (ii) applied to the reseated data gives $\left|\widetilde{w} - (\widetilde{K} - \widetilde{M})\right| \leq \log 2$, hence

$$\left|\widetilde{w} - (\xi_k - \ell_k)\right| \leq \left|\widetilde{M}\right| + \log 2 < 0.2877 + 0.6932 < 1.$$

So $\widetilde{w}$ is a solution of $w\mathrm{e}^w = z$ lying in the closed disc $|v - (\xi_k - \ell_k)| \leq 1$; by the uniqueness clause of Theorem 22.16 — whose hypothesis $2(1 + |\ell_k|) < |\xi_k|$ follows from Equation (23.34) — it equals $w_k(z)$, which under the distance-to-the-cut hypothesis Equation (22.19) is $W_k(z)$. This is Equation (23.35).

For the real case, let $z > 1$ be real and $k = 0$. Then $\xi_0 = \log z > 0$ and $\ell_0 = \log\xi_0 < \xi_0$, so $\widetilde{K} = \xi_0 - \ell_0 > 0$ and $\widetilde{M} = \log(1 - \ell_0/\xi_0)$ are real; the fixed point $\mathcal{Y}(\sigma,\tau)$ of Theorem 22.12 is real for real $(\sigma,\tau) \in D$, because the contraction Φ then maps $[-1,1]$ into itself and its fixed point there must be the unique fixed point in $|y| \leq 1$. Hence $\widetilde{w}$ is real and satisfies $\widetilde{w}\mathrm{e}^{\widetilde{w}} = z > 0$, and Theorem 22.2 (iii) leaves it no choice but $W_0(z)$. □

Remark 23.24 (What this buys, quantitatively). The reseated parameter $\widetilde{M} = \operatorname{Log}(1 - \ell_k/\xi_k)$ is $\mathcal{O}_{|\xi_k|\to\infty}(\ell_k/\xi_k)$, whereas $M = \ell_k \to \infty$. The seating condition Equation (22.16) therefore holds for the reseated series on a strictly larger region: for real z it becomes $2(1 + |\log(1 - \ell_0/\xi_0)|) < \xi_0 - \ell_0$, whose threshold is $\xi_0 = 4.2849\ldots$, so that it holds already at $\xi_0 = 5$, whereas the canonical condition $2(1 + \log \xi_0) < \xi_0$ needs $\xi_0 > 5.3567\ldots$ (the root of $\xi = 2 + 2\log\xi$, namely $5.356693980\ldots$). Numerically, with $K = \xi_0$ and $\widetilde{K} = \xi_0 - \ell_0$:

ξ_0	base point	M	$N = 2$	$N = 5$	$N = 10$
5	ξ_0	$+1.6094$	$6.4 \cdot 10^{-3}$	$3.0 \cdot 10^{-5}$	$1.4 \cdot 10^{-7}$
	$\xi_0 - \ell_0$	-0.3884	$1.1 \cdot 10^{-2}$	$1.1 \cdot 10^{-3}$	$4.9 \cdot 10^{-5}$
10	ξ_0	$+2.3026$	$1.7 \cdot 10^{-3}$	$5.1 \cdot 10^{-6}$	$6.4 \cdot 10^{-10}$
	$\xi_0 - \ell_0$	-0.2617	$6.9 \cdot 10^{-4}$	$4.3 \cdot 10^{-6}$	$1.7 \cdot 10^{-9}$
30	ξ_0	$+3.4012$	$4.1 \cdot 10^{-5}$	$1.5 \cdot 10^{-9}$	$8.2 \cdot 10^{-14}$
	$\xi_0 - \ell_0$	-0.1203	$7.3 \cdot 10^{-6}$	$6.9 \cdot 10^{-10}$	$2.2 \cdot 10^{-16}$

The honest reading: reseating enlarges the region on which convergence is *proved*, and it is numerically better at $\xi_0 = 30$ throughout the displayed range, and at $\xi_0 = 10$ at the two lower orders — but not at $N = 10$, where the canonical series is already ahead by a factor of about three; at $\xi_0 = 5$ the canonical series is better at every displayed order, even though nothing here proves that it converges at all. A larger proved domain and a smaller error at a given point are different goods, and the table shows that neither implies the other.

Remark 23.25 (Associated Stirling numbers). Theorem 23.21 is the structural reason why a second family of combinatorial numbers appears in this problem. Summing the coefficient array (c_{ab}) along diagonals $a+b = n$ gives $\mathsf{L}_n^{\mathrm{Lam}}$, whose coefficients are ordinary Stirling cycle numbers (Theorem 21.4). Summing along rows $a = \text{const}$ gives the functions $\mathsf{L}_a^{\mathrm{Lam}}(\log(1-\tau))(1-\tau)^{-a}$, whose Taylor coefficients are a different array. Comtet inverted $y^\alpha \mathrm{e}^y$ and $y \log y$ [3], and Jeffrey, Corless, Hare and Knuth identified the coefficients of the row-type rearrangements with *associated* Stirling numbers — those counting permutations all of whose cycles have length at least two — and used them to obtain series converging on larger domains [21]; Corless, Jeffrey and Knuth developed the resulting sequence of series systematically [6]. S5's observation that the two small parameters are $\sigma = \alpha/L_1$ and $\tau = \alpha L_2/L_1$, and that associated Stirling numbers arise from "a different small-parameter geometry", is correct and is made precise by Theorems 23.21 and 23.23: the different geometry is the base-point change $K \rightsquigarrow K - M$. This volume does not prove the identification of the row coefficients with associated Stirling numbers; that identification is cited, not derived, and Equation (23.30) is what is proved here.

Section summary

Three layers. *Exact reduction*: $y^\alpha \mathrm{e}^{\beta y} = x$ and its relatives become $u + \log u = \zeta$ by an algebraic substitution, so Part 22 transfers verbatim, convergence included (Theorems 23.2 and 23.3); the only new content is bookkeeping, and $\mathsf{L}_n^{\mathrm{Lam}}$ reappears with the scalar a^{n+1}, $a = \mu/\rho$. *Template*: any equation whose dominant balance is a monomial times a power of a logarithm, perturbed by a series of positive outer order with polynomial logarithmic coefficients, has a mechanically computable polynomial–logarithmic inverse (Theorem 23.8), whose leading blocks are the Lambert polynomials and whose later blocks differ from them by explicitly computable corrections. *Boundary*: the hypotheses are not decorative. A zero monomial exponent changes exponential level (Theorem 23.17); a perturbation of outer order zero forces the iterated Laurent coefficient field $\mathscr{P}\mathscr{L}_{\mathrm{itLaur},\mathbb{K}}$ (Theorem 23.16); an exponentially small perturbation is invisible to the whole apparatus (Theorem 23.18).

Exercise 23.26. Carry Equation (23.8) to $n = 3$ for $\alpha = \frac{1}{2}, \beta = 1$ and verify the seating condition at $x = \mathrm{e}^{40}$.

Exercise 23.27. Apply Theorem 23.8 to $Y = X + \alpha \log X + \beta + (\gamma \log X + \delta)/X$, in the shifted chart $X_* = Y - \beta$, $\ell = \log(Y - \beta)$, and check from Theorem 18.18 that $q_1(\ell) = (\alpha^2 - \gamma)\ell - \delta$. Identify the part $a^2 \mathsf{L}_1^{\mathrm{Lam}}(\ell) = \alpha^2 \ell$ and the correction $E_1 = -\gamma\ell - \delta$. Then re-expand ℓ in $L = \log Y$ and recover the block $\big((\alpha^2 - \gamma)L + \alpha\beta - \delta\big)/Y$ displayed above.

Exercise 23.28. Show that $\mathsf{L}_3^{\mathrm{Lam}}(\log(1-\tau))(1-\tau)^{-3}$ has Taylor coefficients $0, -1, -5, -\frac{85}{6}, -\frac{245}{8}, \ldots$, and identify them among the coefficients of $\mathsf{L}_3^{\mathrm{Lam}}, \mathsf{L}_4^{\mathrm{Lam}}, \mathsf{L}_5^{\mathrm{Lam}}, \mathsf{L}_6^{\mathrm{Lam}}, \mathsf{L}_7^{\mathrm{Lam}}$.

Exercise 23.29. Prove that the logarithmic integral satisfies $\mathrm{li}(x) \sim x/\log x \cdot \sum_{j \ge 0} j!(\log x)^{-j}$ and that formal inversion of this relation is an instance of Theorem 23.8 with $\rho = 1$, $\mu = -1$; deduce the shape of the first three blocks of the inverse, and explain why an analytic theorem about li — not the template — is what is needed to pass from the formal inverse to an asymptotic formula for the nth prime.

Part VII

From formal transseries to analytic asymptotics

Chapter 24

Poincaré asymptotics on the polynomial–logarithmic scale

Everything proved so far about $\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$, $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$, $\mathscr{PL}_{\mathrm{rat},\mathbb{K}}$ and $\mathscr{PL}_{\mathrm{itLaur},\mathbb{K}}$ is algebra: statements about coefficient arrays, the valuation ν_t, and the t-adic filtration. None of it mentions a function, a limit, or an inequality. This part supplies the bridge, and states its toll. The toll is real: four of the six sources assert asymptotic validity, uniformity in a parameter, or termwise differentiability on the strength of a formal identity alone, and each such assertion is marked with a `% ed.` comment and repaired where it occurs below.

Two relations between a function and a series have already been used in this volume: $f \approx F$ of Theorem 8.43, introduced with a $\mathcal{O}.(\cdot)$ remainder carrying an unspecified power of $\log X$, and $f \sim_S F$ of Theorem 12.32, introduced with a $o.(\cdot)$ remainder. They were introduced independently, in different parts, for different purposes. The first order of business is to prove that they are the same relation (Theorem 24.6), so that results proved about one may be applied to the other without comment. The second is to determine exactly how much of a function the relation remembers (Theorem 24.13 and Theorem 24.14); the answer is: everything except a flat perturbation, and nothing less.

Throughout, $\mathbb{K} \subseteq \mathbb{C}$ is a field of characteristic zero, the regime is $X \to +\infty$ along the real ray $S = [X_0, +\infty)$ with $X_0 > 1$ and the real logarithm $L = \log X$, and $t = X^{-1}$. Where a statement holds verbatim on a closed subsector of a sector at infinity with a fixed branch of the logarithm, this is said explicitly.

24.1 What may be called a scale here

Poincaré's definition of an asymptotic expansion presupposes an *enumerated* scale: a sequence $\phi_0, \phi_1, \ldots$ of positive functions with $\phi_{k+1} = o_{X\to+\infty}(\phi_k)$. Two of the sources assert that the polynomial–logarithmic monomials $X^{-n}(\log X)^j$ form such a scale under the lexicographic dominance order of Theorem 5.12. They do not, and the reason is structural rather than pedantic.

Proposition 24.1 (The full monomial family admits no decreasing enumeration)**.** *Let*

$$\Gamma = \left\{ X^{-n}(\log X)^j \;:\; n \in \mathbb{Z},\ j \in \mathbb{N}_0 \right\},$$

ordered by dominance $\prec$ as in Theorem 5.12. Then for every $n \in \mathbb{Z}$ the set $\{\mu \in \Gamma : \mu \prec X^{-n}\}$ has no $\prec$-greatest element. Consequently there is no sequence $(\phi_k)_{k\in\mathbb{N}_0}$ in Γ with $\phi_{k+1} \prec \phi_k$ whose terms exhaust Γ.

Proof. Fix n. A monomial $X^{-m}(\log X)^j$ is $\prec X^{-n} = X^{-n}(\log X)^0$ if and only if $m > n$, since for $m = n$ the only monomial below $(\log X)^0$ in the block would need a negative logarithmic exponent, and there is none in Γ. Given such a monomial, $X^{-m}(\log X)^{j+1}$ is again $\prec X^{-n}$ (still $m > n$) and strictly dominates it. So the set has no greatest element.

For the consequence, suppose (ϕ_k) were a strictly decreasing enumeration of Γ. The monomial X^{-n} occurs as some ϕ_k, and then $\phi_{k+1} \prec X^{-n}$ while every element of Γ below X^{-n} must occur later in the sequence, hence be $\preccurlyeq \phi_{k+1}$. That makes ϕ_{k+1} a greatest element of a set which has none. □

Definition 24.2 (Admissible enumerated scale)**.** Let $n_0 \in \mathbb{Z}$ and let $\delta\colon \mathbb{Z}_{\geq n_0} \to \mathbb{N}_0$ be any function. The *scale of profile* δ is the family

$$\Sigma_\delta = \left\{X^{-n}(\log X)^j \;:\; n \geq n_0,\ 0 \leq j \leq \delta(n)\right\},$$

enumerated by increasing n and, inside each block, by decreasing j.

Lemma 24.3 (A profile makes a genuine Poincaré scale)**.** *The enumeration of Theorem 24.2 is strictly decreasing for dominance, and consecutive members satisfy $\phi_{k+1} = o_{X\to+\infty}(\phi_k)$.*

Proof. Each block contributes the finitely many indices $j = \delta(n), \dots, 0$, so the enumeration is well defined and of order type ω. Inside a block, $X^{-n}(\log X)^{j-1}/X^{-n}(\log X)^j = (\log X)^{-1} \to 0$. At a block boundary, the successor of X^{-n} is $X^{-n-1}(\log X)^{\delta(n+1)}$ and the ratio is $(\log X)^{\delta(n+1)}/X \to 0$. □

24.2 The expansion relation, and the equivalent remainder conditions

Theorem 8.43 defines $f \approx F$ for $F \in \mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$. We record the extension to rational logarithmic blocks, which is forced on us later by division (Theorem 24.16) and is already needed to say anything at all about so simple a function as $1/\log X$ (Theorem 24.10).

Definition 24.4 (Polynomial–logarithmic asymptotic expansion over the rational-log field)**.** Let $F = \sum_{n\geq n_0} p_n(L)t^n$ lie in $\mathscr{P}\mathscr{L}_{\mathrm{rat},\mathbb{K}}$, so that every block p_n is a rational function of L over $\mathbb{K}$. Fix $N \in \mathbb{Z}$. Only the finitely many blocks with $n_0 \leq n < N$ are involved below, and each has finitely many poles, so for all sufficiently large X none of them has a pole at $\log X$; for such X put

$$\big(\mathrm{Trunc}_{<N} F\big)(X) = \sum_{n_0\leq n<N} p_n(\log X)\, X^{-n},$$

a finite sum of complex numbers. A function $f\colon S \to \mathbb{C}$ is *PL-asymptotic* to F, written $f \approx F$, if for every $N \in \mathbb{Z}$ there are $M_N \in \mathbb{N}_0$, $C_N > 0$ and $X_N \geq X_0$ with

$$\big|f(X) - \big(\mathrm{Trunc}_{<N} F\big)(X)\big| \leq C_N\, X^{-N}(\log X)^{M_N} \qquad (X \geq X_N). \tag{24.1}$$

For $F \in \mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ this is exactly Theorem 8.43.

Remark 24.5 (Why the ambient is not $\mathscr{P}\mathscr{L}_{\mathrm{itLaur},\mathbb{K}}$). The definition does not extend to $F \in \mathscr{P}\mathscr{L}_{\mathrm{itLaur},\mathbb{K}}$ as it stands: there a block p_n lies in $\mathbb{K}((L^{-1}))$, so it is an infinite series and not a function of $\log X$, and $\big(\mathrm{Trunc}_{<N} F\big)(X)$ is not a number. The correct object there is a *doubly* indexed condition, in which one truncates first in t and then, inside each retained block, in L^{-1}; the two limits do not commute, since t is smaller than every power of L^{-1} (Theorem 5.22). Two of the sources use $\mathscr{P}\mathscr{L}_{\mathrm{itLaur},\mathbb{K}}$ coefficients and an undecorated one-parameter remainder in the same formula. We do not develop the iterated notion here and no statement below uses it; it is recorded as an obligation in Part G.

Theorem 24.6 (Equivalent forms of the remainder condition)**.** *Let $F = \sum_{n\geq n_0} p_n(L)t^n \in \mathscr{P}\mathscr{L}_{\mathrm{rat},\mathbb{K}}$ and let $f\colon S \to \mathbb{C}$. Write $k_n \in \mathbb{Z}$ for the degree of p_n as a rational function, so that $p_n(\ell) \sim c_n\ell^{k_n}$ with $c_n \neq 0$ when $p_n \neq 0$. The following are equivalent.*

(i) *For every N, $f - \mathrm{Trunc}_{<N} F = o_{X\to+\infty}\big(X^{1-N}\big)$.*

(ii) *For every N and every $\varepsilon > 0$, $f - \mathrm{Trunc}_{<N} F = \mathcal{O}_{X\to+\infty}\big(X^{-N+\varepsilon}\big)$.*

(iii) *$f \approx F$, that is,* (24.1) *holds for every N.*

(iv) *For every N, $f - \mathrm{Trunc}_{<N} F - p_N(\log X)X^{-N} = o_{X\to+\infty}\big(X^{-N}\big)$.*

(v) *There are $M_N \in \mathbb{N}_0$ with $f - \mathrm{Trunc}_{<N+1} F = o_{X\to+\infty}\big(X^{-N}(\log X)^{M_N}\big)$ for every N.*

If moreover $F \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$, these are further equivalent to:

(vi) *for some (equivalently, every) profile δ with $\delta(n) \geq \deg_{\mathrm{L}} p_n$ for all n, the function f has the Poincaré expansion $\sum_k c_k\phi_k$ relative to the scale Σ_δ of Theorem 24.2, where c_k is the coefficient in F of the monomial ϕ_k; that is, $f - \sum_{i\leq k} c_i\phi_i = o_{X\to+\infty}(\phi_k)$ for every k.*

The equivalence is between the families *of conditions, and cannot be localised to one cutoff. Cutoff by cutoff only the implications* (iv)⇒(iii)⇒(ii) ⇒(i) *hold; the reverse implications need the whole family. For instance, with $f(X) = X^{-N+1/2}$ and $F = 0$, condition* (i) *holds at that N while* (iii) *fails at that N for every M_N.*

Proof. Throughout we use that for any fixed $m \in \mathbb{Z}$, $d \geq 0$ and $\varepsilon > 0$,

$$X^{-m}(\log X)^d = o_{X\to+\infty}\big(X^{-m+\varepsilon}\big)\,, \qquad X^{-m-1}(\log X)^d = o_{X\to+\infty}\big(X^{-m}\big)\,, \tag{24.2}$$

both because $(\log X)^d/X^\varepsilon \to 0$. We also use that a nonzero $p \in \mathbb{K}(L)$ satisfies $p(\log X) = \mathcal{O}_{X\to+\infty}\big((\log X)^{\max(k,0)}\big)$ with k its degree, so each individual block of F obeys $p_n(\log X)X^{-n} = \mathcal{O}_{X\to+\infty}\big(X^{-n}(\log X)^{\max(k_n,0)}\big)$.

(iii)⇒(ii)⇒(i). The first is (24.2); the second follows by taking $\varepsilon < 1$.

(i)⇒(iv). Condition (i) at the cutoff $N+1$ reads $f - \mathrm{Trunc}_{<N+1} F = o_{X\to+\infty}\big(X^{-N}\big)$, and one has the identity $\mathrm{Trunc}_{<N+1} F = \mathrm{Trunc}_{<N} F + p_N(\log X)X^{-N}$.

(iv)⇒(iii). Given N, (iv) at that N gives $f - \mathrm{Trunc}_{<N} F = p_N(\log X)X^{-N} + o_{X\to+\infty}\big(X^{-N}\big)$, which is $\mathcal{O}_{X\to+\infty}\big(X^{-N}(\log X)^{M_N}\big)$ with $M_N = \max(k_N, 0)$.

(iv)⇒(v) with $M_N = 0$, since (iv) at N is precisely $f - \mathrm{Trunc}_{<N+1} F = o_{X\to+\infty}\big(X^{-N}\big)$.

(v)⇒(i). Apply (v) at index $N+1$: $f - \mathrm{Trunc}_{<N+2} F = o_{X\to+\infty}\big(X^{-N-1}(\log X)^{M_{N+1}}\big)$, which by (24.2) is $o_{X\to+\infty}\big(X^{-N}\big)$. Adding the single block $p_{N+1}(\log X)X^{-N-1} = o_{X\to+\infty}\big(X^{-N}\big)$ gives $f - \mathrm{Trunc}_{<N+1} F = o_{X\to+\infty}\big(X^{-N}\big)$, which is (i) at cutoff $N+1$. As N is arbitrary, (i) holds at every cutoff.

(iii)⇒(vi), *for $F \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$.* Fix an admissible profile δ, and let $\phi_k = X^{-n}(\log X)^j$. The partial sum $\sum_{i\leq k} c_i\phi_i$ equals $\mathrm{Trunc}_{<n} F(X) + X^{-n}\sum_{i\geq j} a_{n,i}(\log X)^i$, where $p_n = \sum_i a_{n,i}L^i$. Hence

$$f - \sum_{i\leq k} c_i\phi_i = \Big(f - \mathrm{Trunc}_{<n+1} F\Big) + X^{-n}\sum_{i<j} a_{n,i}(\log X)^i.$$

By (iii) at cutoff $n+1$ and (24.2), the first bracket is $o_{X\to+\infty}(X^{-n})$, hence $o_{X\to+\infty}(\phi_k)$ because $\phi_k \geq X^{-n}$; the second is $\mathcal{O}_{X\to+\infty}\big(X^{-n}(\log X)^{j-1}\big) = o_{X\to+\infty}(\phi_k)$ (empty, hence zero, if $j = 0$).

(vi)⇒(i). Take $\phi_k = X^{-n}$, the last member of block n; the corresponding partial sum is exactly $\mathrm{Trunc}_{<n+1} F(X)$, so (vi) gives $f - \mathrm{Trunc}_{<n+1} F = o_{X\to+\infty}(X^{-n})$ for every $n \geq n_0$, which is (i) at every cutoff $> n_0$; at cutoffs $\leq n_0$ the truncation is 0 and the estimate follows from the one at $n_0 + 1$. That the choice of admissible δ is immaterial is now clear, since the conclusion (i) does not mention δ.

Finally, the last assertion: for $f = X^{-N+1/2}$ and $F = 0$ one has $f = o_{X\to+\infty}\big(X^{1-N}\big)$, while $X^{-N+1/2}/(X^{-N}(\log X)^M) = X^{1/2}(\log X)^{-M} \to +\infty$ for every M. □

Remark 24.7 (Concordance with the six sources and with this volume). Theorem 24.6 reconciles every convention in play. Form (iii) is Theorem 8.43; form (i) is Theorem 12.32; form (iv), written with the inclusive cutoff $\sum_{n\leq N}$, is the definition used by S4 and S5; form (v), again inclusive and with a logarithmic weight on the right, is the definition used by S6. All of them say the same thing, and consequently Theorems 12.33 to 12.35 may be used interchangeably with Theorems 8.44 and 8.45. What is *not* interchangeable is a single truncation order: the last clause of the theorem shows that a one-cutoff statement in one convention is strictly weaker or stronger than the corresponding one-cutoff statement in another, which is why a printed asymptotic formula must always say which convention and which cutoff it uses.

Proposition 24.8 (Uniqueness, with rational blocks)**.** *A function $f\colon S \to \mathbb{C}$ is PL-asymptotic to at most one $F \in \mathscr{PL}_{\mathrm{rat},\mathbb{K}}$. The same holds on a closed subsector $\{z : |z| \geq R,\ |\arg z - \theta_0| \leq \theta\}$ of a sector at infinity carrying a fixed analytic branch of the logarithm.*

Proof. Suppose $f \approx F$ and $f \approx \widetilde{F}$ with $D = F - \widetilde{F} \neq 0$. Put $n_1 = \nu_t D$ and let $q \in \mathbb{K}(L)$ be its leading block, of degree $k \in \mathbb{Z}$ and leading coefficient $c \neq 0$, so $q(\ell)/\ell^k \to c$ as $|\ell| \to \infty$. Subtracting the two instances of (24.1) at $N = n_1 + 1$ and using $\mathrm{Trunc}_{<n_1+1} D = q(L)t^{n_1}$ gives

$$q(\log X)\, X^{-n_1} = \mathcal{O}_{X\to+\infty}\big(X^{-n_1-1}(\log X)^M\big)$$

for some $M \in \mathbb{N}_0$. Dividing, $|q(\log X)|\ X\ (\log X)^{-M}$ would have to stay bounded; but it is asymptotic to $|c|\, X(\log X)^{k-M} \to +\infty$, a contradiction. On a subsector, $\log z = \log|z| + \mathrm{i}\arg z$ has $|\arg z|$ bounded, so $|\log z| \sim \log|z|$ and the same computation applies with $|z|$ in place of X. □

Remark 24.9 (Exactly what uniqueness gives, and what it does not). Theorem 24.8 says the map $f \mapsto F$ is well defined where it is defined. It does *not* say:

1. that f is determined by F. It is not, by Theorem 24.14; the fibre over F is an entire coset of the flat ideal.
2. that F determines f even up to a null set, or on any finite interval. The relation is a statement about germs at $+\infty$ only.
3. that F converges anywhere. Convergence is a separate theorem about one series and one domain; see Theorem 20.19 and Theorem 20.20 for the only case in this volume where the question is settled at all, and there only in a form that does not answer the asymptotic question.
4. that f is differentiable, or that $f' \approx D_X F$ if it is (Theorem 24.20).
5. that f exists. That every F is realized is a theorem (Theorem 24.13), and its proof constructs f by hand; no part of the formal algebra supplies it.
6. anything about the coefficients being computable from finitely many values of f. Each p_n is a limit of quotients along the whole ray.

Example 24.10 (A smooth positive function with no $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ expansion)**.** Let $f(X) = 1/\log X$ on $[\mathrm{e}, \infty)$. If $f \approx F$ with $F \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$, then $\log X \cdot f(X) = 1$ and $\log X \approx L$, so Theorem 8.45 gives $LF = 1$ in $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$. But L is not a unit of $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ by Theorem 9.3, since its leading block L is not a unit of $\mathbb{K}[L]$; contradiction. On the other hand $f \approx L^{-1} \in \mathscr{PL}_{\mathrm{rat},\mathbb{K}}$ trivially, the remainder being zero at every cutoff $N \geq 1$. So the enlargement of Theorem 24.4 from $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ to $\mathscr{PL}_{\mathrm{rat},\mathbb{K}}$ is forced by an example this elementary, and is not a convenience.

24.3 The algebra of expansions and the flat ideal

Definition 24.11 (The expansion algebra and its flat ideal)**.** Let $\mathcal{A}$ be the set of germs at $+\infty$ of functions $f\colon [X_f, \infty) \to \mathbb{C}$ for which there exists $F \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ with $f \approx F$, and let

$$\mathcal{N} = \{f \in \mathcal{A} : f \approx 0\} = \big\{f :\ f = \mathcal{O}_{X\to+\infty}\big(X^{-N}\big) \text{ for every } N\big\}\,.$$

Members of $\mathcal{N}$ are called *flat.* Write $\mathsf{exp}\colon \mathcal{A} \to \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$, $\mathsf{exp}(f) = F$, for the expansion map, well defined by Theorem 24.8.

The second description of $\mathcal{N}$ is Theorem 8.44(4), and that $\mathcal{N}$ is an ideal of the $\mathbb{K}$-algebra $\mathcal{A}$ is part of Theorem 8.45: $\mathcal{N} = \ker \mathsf{exp}$ and exp is a $\mathbb{K}$-algebra homomorphism. The next proposition records the three finer structural facts, which are not in the sources.

Proposition 24.12 (Structure of the flat ideal)**.** *$\mathcal{N}$ is a proper ideal of $\mathcal{A}$, and*

1. *$\mathcal{N}$ is prime but not maximal;*
2. *$\mathcal{N}$ is not stable under differentiation, even on the subalgebra of germs of entire functions;*
3. *$\mathcal{N}$ contains no minimal nonzero scale: if $f \in \mathcal{N}$ is eventually nonvanishing then $f^2 \in \mathcal{N}$ and $f^2 = o_{X\to+\infty}(f)$.*

Proof. Properness: $1 \approx 1 \neq 0$, so $1 \notin \mathcal{N}$.

(1) By Theorem 24.13 below, exp is onto, so $\mathcal{A}/\mathcal{N} \cong \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$. That ring is an integral domain (Theorem 8.22) and is not a field (Theorem 9.23); hence $\mathcal{N}$ is prime and not maximal. A witness for non-maximality is the germ of $\log X$, whose class is nonzero and not invertible: an inverse would force L to be a unit of $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$, refuted in Theorem 24.10.

(2) See Theorem 24.20, where $f \in \mathcal{N}$ is entire and $f' \notin \mathcal{A}$, a fortiori $f' \notin \mathcal{N}$.

(3) $\mathcal{N}$ is an ideal and $f \in \mathcal{N} \subseteq \mathcal{A}$, so $f^2 \in \mathcal{N}$. Taking $N = 1$ in the definition of $\mathcal{N}$ gives $f(X) \to 0$, so $f^2/f = f \to 0$ wherever $f \neq 0$, that is $f^2 = o_{X\to+\infty}(f)$. ☐

Theorem 24.13 (Borel–Ritt on the polynomial–logarithmic scale)**.** *For every $F \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ there is a C^∞ function $f\colon (0,\infty) \to \mathbb{C}$, real valued if $\mathbb{K} \subseteq \mathbb{R}$, with $f \approx F$. Consequently*

$$0 \longrightarrow \mathcal{N} \longrightarrow \mathcal{A} \xrightarrow{\mathsf{exp}} \mathscr{PL}_{\mathrm{poly},\mathbb{K}} \longrightarrow 0$$

is an exact sequence of $\mathbb{K}$-algebras and $\mathcal{A}/\mathcal{N} \cong \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$.

Proof. Write $F = \sum_{n\geq n_0} p_n(L)t^n$. Fix once and for all a C^∞ function $\chi\colon \mathbb{R} \to [0,1]$ with $\chi = 0$ on $(-\infty, 0]$ and $\chi = 1$ on $[1,\infty)$. Choose real numbers $X_{n_0} < X_{n_0+1} < \cdots$ with $X_n \geq \max(\mathrm{e}, X_{n-1}+1)$, $X_n \to +\infty$, and

$$|p_n(\log X)| \leq 2^{-|n|}X \qquad \text{for all } X \geq X_n; \tag{24.3}$$

this is possible because $p_n(\log X)/X \to 0$ for each fixed n. Set

$$f(X) = \sum_{n\geq n_0} \chi(X - X_n)\, p_n(\log X)\, X^{-n}.$$

On any bounded interval only the finitely many n with $X_n <$ sup of that interval contribute, so the sum is locally finite and $f \in C^\infty((0,\infty))$.

By Theorem 8.44(1) it suffices to establish (24.1) for every $N > n_0$. Fix such an N and split

$$f - \mathrm{Trunc}_{<N} F = \underbrace{\sum_{n_0\leq n<N} \big(\chi(X - X_n) - 1\big)p_n(\log X)X^{-n}}_{=:\,\Sigma_1} + \underbrace{\sum_{n\geq N} \chi(X - X_n)p_n(\log X)X^{-n}}_{=:\,\Sigma_2}.$$

Σ_1 is a finite sum whose n-th term vanishes once $X \geq X_n+1$; hence $\Sigma_1 = 0$ for $X \geq X_{N-1}+1$. In Σ_2, the term $n = N$ is bounded by $|p_N(\log X)|\, X^{-N} \leq CX^{-N}(\log X)^{M_N}$ with $M_N = \max(\deg_{\mathrm{L}} p_N, 0)$. For $n \geq N+1$ the n-th term vanishes unless $X \geq X_n$, and where it does not vanish (24.3) and $\chi \leq 1$ bound it by $2^{-|n|}X^{1-n} \leq 2^{-|n|}X^{-N}$ for $X \geq 1$; summing $\sum_{n\in\mathbb{Z}} 2^{-|n|} = 3$,

$$|\Sigma_2| \;\leq\; CX^{-N}(\log X)^{M_N} + 3X^{-N}.$$

This is (24.1) at the cutoff N.

Exactness at $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ is the surjectivity just proved, exactness at $\mathcal{A}$ is $\mathcal{N} = \ker \mathsf{exp}$ (Theorem 8.45), and exp is a $\mathbb{K}$-algebra map by the same theorem. ☐

Corollary 24.14 (The converse fails, and by exactly one coset)**.** *For $F \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ the set of $f \in \mathcal{A}$ with $f \approx F$ is nonempty, and equals $f_0 + \mathcal{N}$ for any one of its members f_0. The ideal $\mathcal{N}$ is nonzero, since it contains e^{-X}, $\mathrm{e}^{-\sqrt{X}}$ and $X^{-\log X}$; so infinitely many pairwise distinct germs share every polynomial–logarithmic expansion, and no property of a function that is not invariant under adding a flat germ can be read off from its expansion.*

Proof. Nonemptiness is Theorem 24.13; the fibre description is the statement that exp is additive with kernel $\mathcal{N}$. For membership in $\mathcal{N}$: $\mathrm{e}^{-X}X^N \to 0$, $\mathrm{e}^{-\sqrt{X}}X^N \to 0$ and $X^{-\log X}X^N = \mathrm{e}^{-(\log X)^2 + N\log X} \to 0$ for every N. ☐

Example 24.15 (Flatness has no bottom)**.** Extending the dominance symbol of Theorem 5.12 from monomials to arbitrary positive germs by declaring $\phi \succ \psi$ to mean $\psi = o_{X\to+\infty}(\phi)$, the germs

$$X^{-\log X} \succ \mathrm{e}^{-\sqrt{X}} \succ \mathrm{e}^{-X} \succ \mathrm{e}^{-\mathrm{e}^X}$$

all lie in $\mathcal{N}$; iterating Theorem 24.12(3) produces from any one of them an infinite strictly decreasing chain inside $\mathcal{N}$. The expansion map cannot distinguish any of them from 0. This is the precise sense in which a polynomial–logarithmic expansion "remembers the whole power–log sector and nothing beyond it"; adjoining exponential monomials is the standard remedy and takes one out of the class studied here (Theorem 10.16).

Theorem 24.16 (Quotients)**.** *Let $f \approx F$ and $g \approx G$ with $F, G \in \mathscr{PL}_{\mathrm{rat},\mathbb{K}}$ and $G \neq 0$. Put $m = \nu_t G$ and let $q \in \mathbb{K}(L)$ be the leading block of G. Then g is eventually nonvanishing, and*

$$\frac{f}{g} \approx \frac{F}{G} \in \mathscr{PL}_{\mathrm{rat},\mathbb{K}}.$$

If $F, G \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ and $q \in \mathbb{K}^\times$, then $F/G \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$; if $q \notin \mathbb{K}$, then F/G need not lie in $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$, the quotient $F = 1$, $G = L$ being the smallest example (Theorem 24.10).

Proof. Write k for the degree of q and $c \neq 0$ for its leading coefficient. Applying (24.1) to g at cutoff $m+1$,

$$g(X) = X^{-m}\Big(q(\log X) + \mathcal{O}_{X\to+\infty}\big(X^{-1}(\log X)^M\big)\Big),$$

and $|q(\log X)| \sim |c|\,(\log X)^k$ while $X^{-1}(\log X)^M = o_{X\to+\infty}\big((\log X)^k\big)$. Hence for large X

$$|g(X)| \;\geq\; \tfrac{1}{2}\,|c|\; X^{-m}(\log X)^k > 0, \tag{24.4}$$

so g is eventually nonvanishing and $|1/g(X)| \leq 2\,|c|^{-1}\,X^m(\log X)^{-k}$.

Since $\mathscr{PL}_{\mathrm{rat},\mathbb{K}} = \mathbb{K}(L)((t))$ is the field of Laurent series over the field $\mathbb{K}(L)$ (Theorem 9.19), $H := F/G \in \mathscr{PL}_{\mathrm{rat},\mathbb{K}}$ exists and $F = GH$. Fix N. The truncation $\mathrm{Trunc}_{<N} H$ is a finite polynomial–logarithmic expression, hence realizes itself: $\mathrm{Trunc}_{<N} H \approx \mathrm{Trunc}_{<N} H$, because for any cutoff N' the difference $\mathrm{Trunc}_{<N} H - \mathrm{Trunc}_{<N'} \mathrm{Trunc}_{<N} H$ is a finite sum of blocks of index $\geq N'$. By Theorem 8.45 and linearity (Theorem 8.44(3)),

$$\phi := f - g \cdot \mathrm{Trunc}_{<N} H \;\approx\; E := F - G\ \mathrm{Trunc}_{<N} H = G\big(H - \mathrm{Trunc}_{<N} H\big).$$

By additivity of ν_t (Theorem 6.2), $\nu_t E \geq m + N$, so $\mathrm{Trunc}_{<m+N} E = 0$ and (24.1) applied to ϕ at cutoff $m+N$ gives $\phi = \mathcal{O}_{X\to+\infty}\Big(X^{-(m+N)}(\log X)^{M'}\Big)$. Dividing by g and using the lower bound (24.4),

$$\frac{f}{g} - \mathrm{Trunc}_{<N} H = \frac{\phi}{g} = \mathcal{O}_{X\to+\infty}\Big(X^{-N}(\log X)^{\max(M'-k,\,0)}\Big).$$

As N was arbitrary, $f/g \approx H$. For the last sentence: F/G lies in $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ for every $F \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ if and only if $t^{-m}G$ is a unit of $\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$, which by Theorem 9.3 happens if and only if its leading block q lies in $\mathbb{K}^\times$, and then $1/G$ is given by the recurrence of Theorem 9.7. ☐

Proposition 24.17 (Substitution into a function holomorphic at the origin). *Let $u\colon S \to \mathbb{C}$ satisfy $u \approx U$ with $U \in \mathscr{PL}_{\mathrm{rat},\mathbb{K}}$, $\nu_t U \geq 1$. Let $h(z) = \sum_{k\geq 0} h_k z^k$ be holomorphic on $|z| < r$ with coefficients in $\mathbb{K}$. Then $u(X) \to 0$, $h \circ u$ is defined for large X, and*

$$h \circ u \;\approx\; \sum_{k\geq 0} h_k U^k,$$

the right side being the formally summable substitution of Theorem 10.2, applied in $\mathbb{K}(L)[[t]]$.

Proof. Since $\nu_t U \geq 1$, $\mathrm{Trunc}_{<1} U = 0$, so (24.1) at cutoff 1 gives $u = \mathcal{O}_{X\to+\infty}\big(X^{-1}(\log X)^{M_1}\big)$, whence $u \to 0$ and, for each $k \geq 0$, $u^k = \mathcal{O}_{X\to+\infty}\big(X^{-k}(\log X)^{kM_1}\big)$. In particular $|u| \leq r/2$ for large X, so $h \circ u$ is defined there.

Fix N and choose an integer $J \geq \max(N, 0)$. Write $h(z) = \sum_{k<J} h_k z^k + z^J \widetilde{h}(z)$ with $\widetilde{h}$ holomorphic on $|z| < r$; let $C_J = \max_{|z|\leq r/2} \left|\widetilde{h}(z)\right|$. Then

$$h(u) - \sum_{k<J} h_k u^k = u^J \widetilde{h}(u) = \mathcal{O}_{X\to+\infty}\big(X^{-J}(\log X)^{JM_1}\big) = \mathcal{O}_{X\to+\infty}\big(X^{-N}(\log X)^{JM_1}\big)\,.$$

By Theorem 8.45 and linearity, $\sum_{k<J} h_k u^k \approx \sum_{k<J} h_k U^k$. Finally $\nu_t(U^k) \geq k \geq J \geq N$ for $k \geq J$, so $\mathrm{Trunc}_{<N} \sum_{k<J} h_k U^k = \mathrm{Trunc}_{<N} \sum_{k\geq 0} h_k U^k$. Combining the last three displays gives (24.1) for $h \circ u$ at cutoff N. □

Corollary 24.18 (The three small-argument operations). *Let $u \approx U$ with $\nu_t U \geq 1$, and let $\alpha \in \mathbb{K}$. Then, with the real logarithm and the principal branch of the power for large real X,*

$$\log(1+u) \approx \log(1+U), \qquad (1+u)^\alpha \approx (1+U)^\alpha, \qquad \mathrm{e}^u \approx \exp U,$$

where the right-hand sides are the scalar substitutions of Theorem 10.2 in $\mathbb{K}(L)[[t]]$; when $U \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ they are the formal operations of Theorem 12.5 and Theorem 8.37. In particular the formal geometric, logarithmic and binomial series may be substituted termwise, with no hypothesis beyond $\nu_t U \geq 1$.

Proof. Apply Theorem 24.17 to $h(z) = \log(1+z)$ and $h(z) = (1+z)^\alpha$, both holomorphic on $|z| < 1$, and to $h(z) = \mathrm{e}^z$, entire; $u \to 0$ puts the argument in the disc for large X, and on the positive real ray the principal branches agree with the power series. That the formal sums are the substitutions named, and that for $U \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ they coincide with the operations of Theorem 12.5 and Theorem 8.37, is Theorem 10.2 together with the uniqueness of the continuous $\mathbb{K}$-algebra homomorphism sending z to U. □

Remark 24.19 (Composition: the hypotheses are geometric, not algebraic). Composition with a general inner function is *not* covered by Theorem 24.17 and is not free. The statement to use is Theorem 12.35, whose three hypotheses (A1) regime, (A2) monomial-leading realization with a positive exponent, and (A3) branch compatibility are each shown to be necessary in Theorem 12.36. All five sources that discuss composition list "uniformity of the outer remainder over the range of the inner map" as a requirement; that is (A1) together with the remainder transport step (12.27), and it costs ρ^{-1} outer orders per output order. None of the sources records that cost.

24.4 Termwise differentiation is not automatic

Example 24.20 (An entire flat function whose derivative has no expansion). Let

$$f(X) = \mathrm{e}^{-X}\sin\big(\mathrm{e}^X\big), \qquad X \geq 1.$$

Then f is the restriction to the ray of an entire function, $|f(X)| \le \mathrm{e}^{-X}$, so $f \in \mathcal{N}$ and $f \approx 0$; the formal derivative of 0 is 0. But

$$f'(X) = \cos\bigl(\mathrm{e}^X\bigr) - \mathrm{e}^{-X}\sin\bigl(\mathrm{e}^X\bigr),$$

and f' is not PL-asymptotic to *any* $F \in \mathscr{PL}_{\mathrm{rat},\mathbb{K}}$. Indeed $|f'| \le 1 + \mathrm{e}^{-1}$ is bounded, so if $f' \approx F$ then $\nu_t F \ge 0$ — otherwise (24.1) at cutoff $\nu_t F + 1$ would force $|f'| \sim |p_{\nu_t F}(\log X)|\, X^{-\nu_t F} \to \infty$ — and then (24.1) at cutoff 1 gives $f'(X) - p_0(\log X) \to 0$. Since $f'(X) = \cos(\mathrm{e}^X) + o_{X\to+\infty}(1)$, this forces $p_0(\log X) \to 1$ along $X_k = \log(2k\pi)$ and $p_0(\log X) \to -1$ along $Y_k = \log((2k+1)\pi)$. Both sequences tend to $+\infty$, and a rational function has a single limit in $\mathbb{K} \cup \{\infty\}$ at $+\infty$; contradiction.

An expansion says nothing about the derivative

The example is entire, bounded, and flat. Real-analyticity is therefore not a sufficient hypothesis for termwise differentiation, nor is boundedness, nor is flatness. In particular the frequently seen inference

“$f(X) = X + \mathcal{O}_{X\to+\infty}(\log X)$, hence $f'(X) = 1 + o_{X\to+\infty}(1)$”

is invalid: $f(X) = X + \sin(\mathrm{e}^X)$ satisfies the hypothesis and has $f'(X) = 1 + \mathrm{e}^X\cos(\mathrm{e}^X)$, which is unbounded. Two sufficient hypotheses that do work are given in Theorem 24.25 and Theorem 24.27.

24.5 Termwise integration is unconditional

The asymmetry between the two operations is complete: differentiation needs an extra hypothesis and integration needs none. The reason is visible in Theorem 24.20: differentiation amplifies fast oscillation, while integration averages it away.

Lemma 24.21 (The elementary tail integral)**.** *Let $N \ge 2$ and $M \in \mathbb{N}_0$. For $X \ge \mathrm{e}$,*

$$\int_X^\infty s^{-N}(\log s)^M \,\mathrm{d}s \;\le\; C_{N,M}\, X^{1-N}(\log X)^M, \qquad C_{N,M} = \frac{1}{N-1}\sum_{i=0}^{M}\frac{M!}{(M-i)!\,(N-1)^i}.$$

Proof. Let $I_M(X)$ denote the integral. Integrating by parts with $u = (\log s)^M$ and $\mathrm{d}v = s^{-N}\,\mathrm{d}s$, and using $s^{1-N}(\log s)^M \to 0$,

$$I_M(X) = \frac{X^{1-N}(\log X)^M}{N-1} + \frac{M}{N-1} I_{M-1}(X), \qquad I_0(X) = \frac{X^{1-N}}{N-1}.$$

Unwinding the recursion gives $I_M(X) = \frac{X^{1-N}}{N-1}\sum_{i=0}^{M}\frac{M!}{(M-i)!(N-1)^i}(\log X)^{M-i}$, and $(\log X)^{M-i} \le (\log X)^M$ for $X \ge \mathrm{e}$. ☐

Theorem 24.22 (Termwise integration)**.** *Let $g\colon [X_0,\infty) \to \mathbb{C}$ be locally integrable with $g \approx G \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$, and let $A \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ be any antiderivative of G, that is $D_X A = G$; such an A exists and is unique up to an additive constant of $\mathbb{K}$ by Theorem 11.18 and Theorem 11.12. Then there is a constant $c \in \mathbb{C}$ with*

$$\Bigl(\int_{X_0}^X g(s)\,\mathrm{d}s\Bigr) - c \;\approx\; A,$$

equivalently $\int_{X_0}^X g \approx A + c$ once the coefficient field is enlarged from $\mathbb{K}$ to $\mathbb{K}(c)$. That constant is the only ambiguity: no hypothesis beyond $g \approx G$ is needed, and none of g beyond local integrability.

Proof. Write $G = \sum_n g_n(L)t^n$ and $A = \sum_n a_n(L)t^n$, and put $h(X) = \int_{X_0}^X g$. We first record the exact formal identity

$$\frac{\mathrm{d}}{\mathrm{d}X}\Big[\big(\mathrm{Trunc}_{<N-1} A\big)(X)\Big] = \big(\mathrm{Trunc}_{<N} D_X A\big)(X) = \big(\mathrm{Trunc}_{<N} G\big)(X), \tag{24.5}$$

valid for every N. Indeed by Theorem 11.3 the block of $D_X A$ at index m is $(\partial_L - (m-1))a_{m-1}$, so the blocks of $D_X A$ of index $< N$ are exactly the images of the blocks of A of index $< N-1$; differentiating the finite sum $\sum_{n<N-1} a_n(\log X)X^{-n}$ term by term reproduces them.

Fix $N \geq 2$ and set $\psi_N = h - \mathrm{Trunc}_{<N-1} A$. By (24.5), $\psi_N' = g - \mathrm{Trunc}_{<N} G =: r_N$, and by hypothesis $|r_N(X)| \leq C_N X^{-N}(\log X)^{M_N}$ for $X \geq X_N$. By Theorem 24.21 this is integrable on $[X_N, \infty)$, so $c_N := \lim_{X\to\infty} \psi_N(X)$ exists and

$$|\psi_N(X) - c_N| = \Big|\int_X^\infty r_N\Big| \leq C_N C_{N,M_N}\, X^{1-N}(\log X)^{M_N}. \tag{24.6}$$

Moreover for $N \geq 2$,

$$c_{N+1} - c_N = \lim_{X\to\infty}\big(\mathrm{Trunc}_{<N-1} A - \mathrm{Trunc}_{<N} A\big)(X) = -\lim_{X\to\infty} a_{N-1}(\log X)\, X^{1-N} = 0,$$

since $N - 1 \geq 1$. So all c_N with $N \geq 2$ coincide; call the common value c.

Then (24.6) says $h - c - \mathrm{Trunc}_{<N-1} A = \mathcal{O}_{X\to+\infty}\big(X^{-(N-1)}(\log X)^{M_N}\big)$ for every $N \geq 2$, that is, (24.1) holds for $h - c$ against A at every cutoff ≥ 1. By Theorem 8.44(1) a cofinal family of cutoffs suffices, so $h - c \approx A$. □

Corollary 24.23 (The tail integral of an integrable expansion). *If $g\colon [X_0, \infty) \to \mathbb{C}$ is locally integrable and $g \approx G$ with $\nu_t G \geq 2$, then g is integrable near $+\infty$ and*

$$\int_X^\infty g(s)\,\mathrm{d}s \;\approx\; -A,$$

where A is the unique antiderivative of G in $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ with $\nu_t A \geq 1$.

Proof. Existence and uniqueness of A with $\nu_t A \geq 1$: writing $G = \sum_n g_n(L)t^n$ and $A = \sum_n a_n(L)t^n$, by Theorem 11.3 the block of A at index $n \geq 1$ must satisfy $(\partial_L - n)a_n = g_{n+1}$, which for $n \neq 0$ has a unique solution in $\mathbb{K}[L]$ by Theorem 11.15, and $\nu_t G \geq 2$ means only indices $n \geq 1$ occur; the resonant index $n = 0$, the one carrying the constant of Theorem 11.12 and the extra logarithmic degree of (11.15), is not reached. Now apply Theorem 24.22: with $h(X) = \int_{X_0}^X g$ there is $c \in \mathbb{C}$ with $h - c \approx A$. Since $\nu_t A \geq 1$ we have $\big(\mathrm{Trunc}_{<1} A\big)(X) = 0$, so (24.1) at cutoff 1 gives $h(X) - c \to 0$; hence $\int_{X_0}^\infty g$ converges, with value c, and $\int_X^\infty g = c - h(X) = -(h-c) \approx -A$. □

Remark 24.24 (Where the resonant block bites). If $\nu_t G = 1$, the antiderivative acquires a block at index 0 of logarithmic degree one higher than that of g_1 ((11.15)): integrating $p(\log s)s^{-1}$ produces a polynomial in $\log X$, not a constant. This is the analytic shadow of the resonance of Theorem 11.16, and it is the reason Theorem 24.23 needs $\nu_t G \geq 2$ and not merely $\nu_t G \geq 1$. In $\mathscr{P}\mathscr{L}_{\mathrm{rat},\mathbb{K}}$ the same mechanism at the block $L^{-1}t$ produces $\log\log X$ and leaves the class altogether (Theorems 11.22 and 11.24).

Theorem 24.25 (Termwise differentiation is rigid where it is possible). *Let f be differentiable on a ray with $f \approx F \in \mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$. If $f' \approx G$ for* some *$G \in \mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$, then $G = D_X F$.*

Proof. First, f' is locally integrable on a tail. It is Borel measurable, being a pointwise limit of continuous difference quotients, and $f' \approx G$ applied at the cutoff $\nu_t G + 1$ exhibits f' on some $[X_1, \infty)$ as a continuous function plus a remainder bounded on compacts; hence f' is bounded on compact subsets of $[X_1, \infty)$. A function that is differentiable everywhere on a compact interval with Lebesgue integrable derivative satisfies the fundamental theorem of calculus there, so $f(X) - f(X_1) = \int_{X_1}^{X} f'$ for $X \geq X_1$.

Now Theorem 24.22 applied to $g = f'$ gives, for any $A \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ with $D_X A = G$, a constant $c \in \mathbb{C}$ with $f - \gamma \approx A$, where $\gamma = c + f(X_1)$. Also $f \approx F$, so $f - \gamma \approx F - \gamma$. Both A and $F - \gamma$ lie in $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ over the field $\mathbb{K}(\gamma) \subseteq \mathbb{C}$, so Theorem 24.8, applied over that field, gives $F - \gamma = A$. Applying D_X and using $\ker D_X = \mathbb{K}(\gamma)$ (Theorem 11.12) gives $D_X F = D_X A = G$. □

Remark 24.26 (The exact shape of the extra hypothesis). Theorem 24.25 isolates the content of the folklore statement "an asymptotic expansion may not be differentiated termwise". The formal answer is never wrong; what can fail is that f' has *no* expansion at all. So the extra hypothesis needed for $f \approx F \Rightarrow f' \approx D_X F$ is exactly:

f' admits some polynomial–logarithmic asymptotic expansion.

This is a hypothesis about f', not about F, and no property of F implies it. Theorem 24.27 gives the standard checkable sufficient condition; a Hardy-field or tame-germ hypothesis is another [31, 32, 1].

Theorem 24.27 (Sectorial expansions may be differentiated termwise)**.** *Let $0 < \theta \leq \pi$ and $R > 1$, let $S(\theta, R) = \{z \in \mathbb{C} : |z| \geq R,\ |\arg z| < \theta\}$ with the principal logarithm, and let f be holomorphic on $S(\theta, R)$. Suppose $F \in \mathscr{PL}_{\mathrm{rat},\mathbb{K}}$ and that for every $\theta' < \theta$ and every N there are C, M, R' with*

$$\big|f(z) - \big(\mathrm{Trunc}_{<N} F\big)(z)\big| \leq C\,|z|^{-N}\,\big(\log|z|\big)^{M} \qquad \big(|z| \geq R',\ |\arg z| \leq \theta'\big). \tag{24.7}$$

Then f' satisfies (24.7) *with F replaced by $D_X F$: the expansion may be differentiated termwise, on every closed subsector.*

Proof. Fix $\theta' < \theta$ and N. Choose θ'' with $\theta' < \theta'' < \theta$ and put $\delta = \min\big(\frac{1}{2}, \sin(\theta'' - \theta')\big)$. For z with $|\arg z| \leq \theta'$, every w with $|w - z| \leq \delta\,|z|$ satisfies $|\arg w - \arg z| \leq \arcsin\delta \leq \theta'' - \theta'$, hence $|\arg w| \leq \theta''$, and $(1-\delta)\,|z| \leq |w| \leq (1+\delta)\,|z|$.

Apply (24.7) on the subsector θ'' at cutoff $N+1$: setting $G = f - \mathrm{Trunc}_{<N+1} F$, which is holomorphic there, there are C, M, R'' with

$$|G(w)| \leq C\,|w|^{-N-1}\,\big(\log|w|\big)^{M} \qquad \big(|\arg w| \leq \theta'',\ |w| \geq R''\big).$$

For $|z| \geq 2R''$ with $|\arg z| \leq \theta'$, the circle $|w - z| = \delta\,|z|$ lies in that region, so Cauchy's estimate gives, with $\kappa = \max\big((1-\delta)^{-N-1}, (1+\delta)^{-N-1}\big)$,

$$\big|G'(z)\big| \leq \frac{1}{\delta\,|z|} \max_{|w-z|=\delta|z|} |G(w)| \leq \frac{C\,\kappa\,\big(\log((1+\delta)\,|z|)\big)^{M}}{\delta\,|z|^{N+2}} \leq C'\,|z|^{-(N+2)}\,\big(\log|z|\big)^{M}$$

for $|z|$ large. Finally, exactly as in (24.5), the derivative of $\mathrm{Trunc}_{<N+1} F$ with respect to z equals $\mathrm{Trunc}_{<N+2} D_X F$, so that

$$G' = f' - \mathrm{Trunc}_{<N+2} D_X F.$$

The display above is therefore (24.7) for f' against $D_X F$, at the cutoff $N+2$. Since N was arbitrary and such cutoffs are cofinal, the conclusion holds at every cutoff by Theorem 8.44(1). □

Remark 24.28 (The hypothesis is sectorial, not real-analytic). Theorem 24.20 shows why (24.7) must be required on a genuine two-dimensional sector. The function $\mathrm{e}^{-z}\sin\mathrm{e}^{z}$ is entire, and its restriction to the ray is flat; but on any sector $|\arg z| \leq \theta'$ with $\theta' > 0$ it is unbounded, so it fails (24.7) against $F = 0$ at every cutoff. "Analytic" alone is worth nothing here; "analytic with a uniform remainder bound on a sector of positive opening" is worth Cauchy's estimate, which is the whole of the proof.

Section summary

On the polynomial–logarithmic scale, the four elementary transfers behave as follows. *Sums and products*: unconditional (Theorem 8.44, Theorem 8.45). *Quotients*: unconditional once the divisor's leading block is nonzero, with the quotient living in $\mathscr{P}\mathscr{L}_{\mathrm{rat},\mathbb{K}}$ unless that block is a constant (Theorem 24.16). *Substitution into a function holomorphic at* 0: unconditional (Theorem 24.17). *Composition with a general inner map*: three geometric hypotheses, each necessary (Theorem 12.35). *Integration*: unconditional (Theorem 24.22). *Differentiation*: not transferable at all without an extra hypothesis, but rigid once f' is known to have any expansion (Theorem 24.25, Theorem 24.27). Every function with an expansion is a member of a coset of the flat ideal, and every formal series is realized (Theorem 24.13, Theorem 24.14).

Chapter 25

Residual-to-error transfer

A formal reversion produces a candidate G_N and a certificate: the residual $\mathrm{Err}_{\mathrm{rev}}(F, G_N)$ has t-order at least N (Theorem 16.16). A user of the answer wants a number: how far is $G_N(x)$ from the true solution of $F(y) = x$? The passage from the first to the second is the subject of this chapter. It has three steps, and only the first is trivial:

1. convert a residual *value* into an error bound (calculus);
2. convert the formal residual *order* into an analytic residual *size* (Theorem 25.5, which needs an analytic hypothesis on the map being reversed);
3. verify that the true solution exists, is unique, and lies where the derivative bound holds (Theorem 25.2).

Steps (2) and (3) are exactly what the sources omit.

25.1 The transfer inequality and its enclosure form

Remark 25.1 (What Theorem 16.21 does and does not assume). Theorem 16.21 states: if f is C^1 and injective on an interval I, $y \in f(I)$, x_* is the unique point of I with $f(x_*) = y$, $x_N \in I$ and $|f'| \geq m > 0$ between x_N and x_*, then

$$|x_N - x_*| \leq \frac{|f(x_N) - y|}{m}. \tag{25.1}$$

Two observations about the hypotheses, neither of them in the sources. First, injectivity is used only to *name* x_*; the mean-value argument itself needs only differentiability on the segment joining x_N to x_*. Second, and more seriously, the derivative bound is required on an interval whose right endpoint is the unknown. In an actual computation one knows x_N and $f(x_N) - y$ and does not know x_*; a bound "between x_N and x_*" is therefore not directly verifiable, and using (25.1) as stated is circular unless a separate argument locates x_*. Theorem 25.2 removes the circularity by hypothesising the derivative bound on a *chosen* interval around x_N and deducing the existence of x_* inside it.

Theorem 25.2 (Residual enclosure: existence, uniqueness, and two-sided error)**.** *Let $\rho > 0$, let $I = [x_N - \rho, x_N + \rho] \subseteq \mathbb{R}$, and let $f\colon I \to \mathbb{R}$ be continuously differentiable with*

$$m := \min_{x \in I} \left|f'(x)\right| > 0, \qquad M := \max_{x \in I} \left|f'(x)\right|.$$

Let $y \in \mathbb{R}$ satisfy

$$|f(x_N) - y| \;\leq\; m\rho. \tag{25.2}$$

Then f is strictly monotone on I; there is exactly one $x_* \in I$ with $f(x_*) = y$; and

$$\frac{|f(x_N) - y|}{M} \;\leq\; |x_N - x_*| \;\leq\; \frac{|f(x_N) - y|}{m}. \tag{25.3}$$

No hypothesis of injectivity is imposed; it is a conclusion.

Proof. Since f' is continuous on the interval I and never zero, the intermediate value theorem applied to f' shows that f' has constant sign; replacing f by $-f$ and y by $-y$ if necessary, assume $f' \geq m > 0$. Then f is strictly increasing, hence injective.

By the mean value theorem, $f(x_N + \rho) - f(x_N) \geq m\rho \geq y - f(x_N)$ and $f(x_N) - f(x_N - \rho) \geq m\rho \geq f(x_N) - y$, using (25.2) in both. Hence $f(x_N - \rho) \leq y \leq f(x_N + \rho)$, and the intermediate value theorem applied to f produces $x_* \in I$ with $f(x_*) = y$; it is unique by injectivity.

Finally the mean value theorem gives ξ strictly between x_N and x_* (or $x_N = x_*$, in which case (25.3) is trivial) with $f(x_N) - y = f(x_N) - f(x_*) = f'(\xi)(x_N - x_*)$. Since $m \leq |f'(\xi)| \leq M$, both inequalities of (25.3) follow. □

Theorem 25.3 (Complex enclosure by Rouché). *Let f be holomorphic on a neighbourhood of the closed disc $\overline{D} = \{z : |z - x_N| \leq \rho\} \subseteq \mathbb{C}$, put $\mu = |f'(x_N)|$, assume $\mu > 0$, and suppose*

$$\left|f'(z) - f'(x_N)\right| \leq \frac{\mu}{4} \quad (z \in \overline{D}), \qquad |f(x_N) - y| < \frac{\mu\rho}{4}. \tag{25.4}$$

Then f is injective on $\overline{D}$, the equation $f(z) = y$ has exactly one solution x_* in D, and

$$\frac{4\,|f(x_N) - y|}{5\mu} \;\leq\; |x_N - x_*| \;\leq\; \frac{4\,|f(x_N) - y|}{3\mu}. \tag{25.5}$$

Proof. Write $g(z) = f(z) - y$ and let $h(z) = \big(f(x_N) - y\big) + f'(x_N)(z - x_N)$ be its linearisation at x_N. For $z \in \overline{D}$, integrating $f' - f'(x_N)$ along the segment from x_N to z, which lies in the convex set $\overline{D}$, gives

$$|g(z) - h(z)| = \left|\int_0^1 \big(f'(x_N + s(z - x_N)) - f'(x_N)\big)(z - x_N)\,\mathrm{d}s\right| \leq \frac{\mu}{4}\,|z - x_N|\,. \tag{25.6}$$

On the circle $|z - x_N| = \rho$ this is $\leq \mu\rho/4$, while $|h(z)| \geq \mu\rho - |f(x_N) - y| > \mu\rho - \mu\rho/4 = 3\mu\rho/4$. Thus $|g - h| < |h|$ on the circle, and Rouché's theorem gives that g and h have equally many zeros in D, counted with multiplicity. The only zero of h is $z_0 = x_N - (f(x_N) - y)/f'(x_N)$, and $|z_0 - x_N| < (\mu\rho/4)/\mu = \rho/4 < \rho$, so $z_0 \in D$ and g has exactly one zero $x_* \in D$.

For injectivity and the error bounds, note that for $z_1, z_2 \in \overline{D}$,

$$f(z_1) - f(z_2) = (z_1 - z_2)\int_0^1 f'\big(z_2 + s(z_1 - z_2)\big)\,\mathrm{d}s,$$

and the integral differs from $f'(x_N)$ by at most $\mu/4$ by (25.4); hence its modulus lies in $[\,3\mu/4,\ 5\mu/4\,]$ because $|f'(x_N)| = \mu$. Injectivity follows from the lower end. Taking $z_1 = x_N$ and $z_2 = x_*$,

$$\tfrac{3\mu}{4}\,|x_N - x_*| \;\leq\; |f(x_N) - y| \;\leq\; \tfrac{5\mu}{4}\,|x_N - x_*|\,,$$

which is (25.5). □

Remark 25.4 (Using a lower bound for $|f'(x_N)|$). If μ is only known to satisfy $0 < \mu \leq |f'(x_N)|$, the proof of Theorem 25.3 still gives injectivity, existence and uniqueness of $x_* \in D$, and the upper bound $|x_N - x_*| \leq \frac{4}{3\mu}\,|f(x_N) - y|$; only the left-hand inequality of (25.5) uses $|f'(x_N)| = \mu$. This is the asymmetry that produces the constant 5 on the left and 3 on the right.

A formal residual is not an inequality

The statement $\mathrm{Err}_{\mathrm{rev}}(F, G_N) = \mathcal{O}_t^{\mathrm{form}}\big(t^N\big)$ is a membership in a filtration step of $\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ (Theorem 7.7); the statement $F(G_N(x)) - x = \mathcal{O}_{x\to+\infty}\big(x^{-N}(\log x)^M\big)$ is an inequality between real numbers valid beyond some x_N. Neither implies the other without a theorem. The direction one needs is supplied by Theorem 25.5, and its hypothesis is that the map being reversed *has* a polynomial–logarithmic asymptotic expansion — an analytic fact about a function, of the kind that Theorem 24.13 shows cannot be recovered from the coefficients.

25.2 From the formal residual order to an analytic residual size

Theorem 25.5 (Analytic size of a residual). *Let $\mathbb{K} \subseteq \mathbb{R}$, let $Y_0 > 1$, let $\Phi = X + A \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}$, and let $\varphi\colon [Y_0, \infty) \to \mathbb{R}$ satisfy $\varphi \approx \Phi$ in the sense of Theorem 24.4, with the real logarithm and the regime $Y \to +\infty$. Let $H = X + B \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}$ with $B \in \mathbb{K}[L][t]$ a* finite *sum of blocks, and let $h(x) = x + B(\log x, 1/x)$ be its evaluation for large real x. Put $r = \nu_t\big(\mathrm{Err}_{\mathrm{rev}}(\Phi, H)\big) \in \mathbb{N}_0 \cup \{+\infty\}$. Then $h(x) \to +\infty$, $h(x)/x \to 1$, and for every finite $N \le r$ there are M, C, x_1 with*

$$\big|\varphi\big(h(x)\big) - x\big| \;\le\; C\, x^{-N} (\log x)^M \qquad (x \ge x_1). \tag{25.7}$$

Proof. Since $\nu_t B \ge 0$ and B is a finite sum of blocks, $B(\log x, 1/x) = \mathcal{O}_{x\to+\infty}\big((\log x)^d\big)$ with $d = \max_n \deg_{\mathrm{L}} b_n$; hence $h(x) = x\big(1 + u(x)\big)$ with $u(x) = B(\log x, 1/x)/x \to 0$, and $h(x) \to +\infty$, $h(x)/x \to 1$. In particular $h(x) \ge Y_0$ for large x, so $\varphi \circ h$ is defined there.

We verify the three hypotheses of Theorem 12.35 for the outer function φ and the inner function h. (A1) is the previous paragraph. (A2) holds with $c = 1$, $\rho = 1$ and $U = tB \in t\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$, whose coefficients are real because $\mathbb{K} \subseteq \mathbb{R}$, as that hypothesis of Theorem 12.35 requires: the function u is the evaluation of the finite sum tB, so $u \approx U$ exactly (its remainder vanishes at every cutoff beyond the t-degree of tB), and $u \to 0$. (A3) holds automatically on the positive real ray with $c = 1 > 0$. By Theorem 24.6 the relation $\approx$ of Theorem 24.4 and the relation $\sim_S$ used in Theorem 12.35 coincide, so the theorem applies and yields

$$\varphi \circ h \;\approx\; \Phi \circ H.$$

The identity function realizes the series X, so by Theorem 8.44(3) the germ $x \mapsto \varphi(h(x)) - x$ is PL-asymptotic to $\Phi \circ H - X = \mathrm{Err}_{\mathrm{rev}}(\Phi, H)$. Let $N \le r$ be finite. Since $\nu_t \,\mathrm{Err}_{\mathrm{rev}}(\Phi, H) = r \ge N$, we have $\mathrm{Trunc}_{<N} \mathrm{Err}_{\mathrm{rev}}(\Phi, H) = 0$, and (24.1) at cutoff N is exactly (25.7). $\square$

Theorem 25.6 (Transfer of a formal inverse expansion to the analytic inverse). *Let $\mathbb{K} \subseteq \mathbb{R}$ and assume:*

(H1) Regime and derivative bound. *$\varphi\colon [Y_0, \infty) \to \mathbb{R}$ is C^1 with $\varphi'(Y) \ge m_0 > 0$ for all $Y \ge Y_0$, and $\varphi(Y) \to +\infty$. (Then φ is a strictly increasing bijection onto $[\varphi(Y_0), \infty)$; write ψ for its inverse.)*

(H2) Analytic expansion of the map. *$\varphi \approx \Phi$ for some $\Phi = X + A \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}$, with the real logarithm as $Y \to +\infty$.*

Let $G = \Phi^{\langle -1\rangle_\circ} = X + B \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}$ be the formal compositional inverse supplied by Theorem 16.11, and put $g_N(x) = x + \big(\mathrm{Trunc}_{<N} B\big)(x)$. Then for every $N \in \mathbb{N}_0$ there are $M_N \in \mathbb{N}_0$, $C_N > 0$ and a threshold τ_N with

$$|\psi(x) - g_N(x)| \;\le\; C_N\, x^{-N} (\log x)^{M_N} \qquad (x \ge \tau_N), \tag{25.8}$$

and consequently $\psi \approx G$: the formal compositional inverse is the asymptotic expansion of the analytic inverse.

Proof. (H1) makes ψ well defined. $\varphi' \ge m_0 > 0$ forces φ strictly increasing, hence injective, and $\varphi(Y) \to +\infty$ with the intermediate value theorem makes it onto $[\varphi(Y_0), \infty)$; ψ is its inverse, $\psi(x) \ge Y_0$, and $\psi(x) \to +\infty$.

Formal order of the residual. Fix N and put $G_N = X + \mathrm{Trunc}_{<N} B \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}$, a finite sum of blocks. Then $G - G_N = B - \mathrm{Trunc}_{<N} B$ has $\nu_t \ge N$, so by Theorem 16.15

$$\nu_t\big(\mathrm{Err}_{\mathrm{rev}}(\Phi, G_N)\big) = \nu_t(G_N - G) \;\ge\; N.$$

Analytic size of the residual. By (H2) and Theorem 25.5 applied with $H = G_N$, there are M_N, C, x_1 with $|\varphi(g_N(x)) - x| \le Cx^{-N}(\log x)^{M_N}$ for $x \ge x_1$.

Transfer. For x large, both $g_N(x)$ and $\psi(x)$ lie in the interval $[Y_0, \infty)$, on all of which $|\varphi'| \ge m_0$. Applying Theorem 16.21 on that interval with $f = \varphi$, argument value $y = x$, approximation $g_N(x)$, exact solution $x_* = \psi(x)$ and $m = m_0$ gives

$$|\psi(x) - g_N(x)| \le \frac{|\varphi(g_N(x)) - x|}{m_0} \le \frac{C}{m_0}\, x^{-N}(\log x)^{M_N},$$

which is (25.8). Since $g_N(x) = \big(\operatorname{Trunc}_{<N} G\big)(x)$ for every $N \ge 0$, this is (24.1) at every cutoff $N \ge 0$, and cutoffs $N < 0$ follow from Theorem 8.44(1). Hence $\psi \approx G$. □

Remark 25.7 (Which hypothesis does which job). (H1) is the only source of the constant m_0, and it must be verified on the function, not deduced from an expansion: by Theorem 24.20 the expansion $\Phi = X + A$ does not entail $\varphi' = 1 + o_{Y\to+\infty}(1)$, or indeed any statement about φ' whatever. (H2) is what turns a t-adic order into an inequality; it is an analytic hypothesis and no algebraic theorem supplies it. A third condition, the admissibility check $g_N(x) \ge Y_0$, is listed as a hypothesis in several of the sources and in the four-item list preceding Theorem 20.17; it is *not* a hypothesis of Theorem 25.6, because $g_N(x) = x + \mathcal{O}_{x\to+\infty}\big((\log x)^d\big) \to +\infty$ makes it automatic in the limit. It is nonetheless indispensable at any *specific* x at which one wants to use the resulting inequality: a truncated formal inverse can take values outside $[Y_0, \infty)$ for moderate x, and then $\varphi(g_N(x))$ is not even defined, so the numerical bound is vacuous while the asymptotic theorem remains true. Item (2) of the list preceding Theorem 20.17 — an analytic remainder for the truncated logarithm — is not (H2) but what Theorem 25.5 *deduces* from (H2); for $\varphi(Y) = Y + \log Y$ the hypothesis (H2) itself is trivial, since $\Phi = X + L$ is realized by φ with zero remainder at every cutoff.

25.3 Explicit constants for the Wright omega truncations

For the archetype $\varphi(Y) = Y + \log Y$ the constants of Theorem 25.2 can be written down exactly, with no asymptotic hypothesis at all. This is the strongest form in which a residual bound is available anywhere in this volume, and it is available because $\varphi'(Y) = 1 + Y^{-1}$ is monotone and explicitly bounded on $(0, \infty)$.

Proposition 25.8 (A computable two-sided error bound, valid for every $x > 1$). *Let $x > 1$, let $N \in \mathbb{N}_0$, let $\omega_N(x) = x - \log x + \sum_{n=1}^{N} \mathsf{L}_n^{\mathrm{Lam}}(\log x)x^{-n}$ be the evaluation of the truncation of Theorem 20.12, and suppose $\omega_N(x) > 0$. Put*

$$R_N(x) = \omega_N(x) + \log \omega_N(x) - x, \qquad \mu_N(x) = \min\{\omega_N(x),\ x - \log x\}.$$

Then $\mu_N(x) > 0$ and

$$\frac{|R_N(x)|}{1 + \mu_N(x)^{-1}} \ \le\ |\omega(x) - \omega_N(x)| \ \le\ |R_N(x)|, \tag{25.9}$$

and $\omega(x) - \omega_N(x)$ has the sign of $-R_N(x)$. Both bounds are computable from x and N alone; neither involves an implied constant, an unspecified threshold, or a limit.

Proof. By Theorem 20.1, $\varphi(Y) = Y + \log Y$ is a strictly increasing bijection $(0, \infty) \to \mathbb{R}$ with $\varphi'(Y) = 1 + Y^{-1}$; the same proposition gives $\omega(1) = 1$ and ω strictly increasing, so $\omega(x) > 1$ for $x > 1$, which is the hypothesis under which the envelope (20.2) gives $\omega(x) > x - \log x$. Also $x - \log x \ge 1 > 0$ for every $x > 0$, because $x \mapsto x - \log x$ has its minimum at $x = 1$. Hence both $\omega(x)$ and $\omega_N(x)$ lie in $[\mu_N(x), \infty)$, and so does the whole segment between them.

On $[\mu_N(x), \infty)$ the derivative satisfies $1 < \varphi'(Y) \le 1 + \mu_N(x)^{-1}$. Since $\varphi(\omega(x)) = x$, the mean value theorem gives ζ in that segment with

$$R_N(x) = \varphi\big(\omega_N(x)\big) - \varphi\big(\omega(x)\big) = \varphi'(\zeta)\big(\omega_N(x) - \omega(x)\big),$$

whence the sign statement and, dividing by $\varphi'(\zeta) \in (1, 1+\mu_N(x)^{-1}]$, the two inequalities (25.9). $\square$

Proposition 25.9 (The two-term approximation, in closed form). *For every $x \geq 1$, with $L = \log x$,*

$$R_0(x) = (x-L) + \log(x-L) - x = \log\Big(1 - \frac{L}{x}\Big) \;\leq\; 0, \tag{25.10}$$

and

$$\frac{L}{x} \;\leq\; |R_0(x)| \;\leq\; \frac{L}{x} + \frac{L^2}{x^2}. \tag{25.11}$$

Consequently $x - \log x$ always underestimates $\omega(x)$, and

$$\frac{L}{x} \cdot \frac{x-L}{x-L+1} \;\leq\; \omega(x) - (x - \log x) \;\leq\; \frac{L}{x} + \frac{L^2}{x^2}. \tag{25.12}$$

Proof. For $x \geq 1$ we have $L \geq 0$ and $x - L \geq 1 > 0$, so $\log(x-L)$ is defined and $(x-L) + \log(x-L) - x = \log(x-L) - L = \log\frac{x-L}{x} = \log(1 - L/x)$, which is ≤ 0. Set $u = L/x$. The function $x \mapsto (\log x)/x$ attains its maximum e^{-1} at $x = \mathrm{e}$, so $0 \leq u \leq \mathrm{e}^{-1} < \frac{1}{2}$. Then

$$-\log(1-u) = \sum_{j\geq 1} \frac{u^j}{j} \;\geq\; u, \qquad -\log(1-u) \leq u + \frac{1}{2}\sum_{j\geq 2} u^j = u + \frac{u^2}{2(1-u)} \leq u + u^2,$$

the last step because $u \leq \frac{1}{2}$. This is (25.11). At $x = 1$ all three quantities in (25.12) vanish, since $L = 0$ and $\omega(1) = 1 = x - \log x$, so assume $x > 1$. Now apply Theorem 25.8 with $N = 0$, where $\omega_0(x) = x - L$ and hence $\mu_0(x) = x - L$: the sign of $-R_0 \geq 0$ gives $\omega(x) \geq x - L$, the upper bound of (25.9) combined with the right half of (25.11) gives the right half of (25.12), and the lower bound of (25.9) combined with the left half of (25.11) gives $|R_0|/(1 + (x-L)^{-1}) \geq (L/x)(x-L)/(x-L+1)$. $\square$

Remark 25.10 (What is *not* obtained). Theorem 25.8 is uniform in N only because it does not evaluate R_N; the closed form (25.10) is available at $N = 0$ and, by the same computation, at $N = 1$, but a constant C with $|R_N(x)| \leq C\left|\mathsf{L}^{\mathrm{Lam}}_{N+1}(\log x)\right| x^{-N-1}$ uniformly in N is *not* obtained here and is not obtained in Theorem 20.17 either: the constant there depends on N, and the logarithmic exponent $K = (N+2)^2$ in its remainder is quadratic in N. Any statement in the literature of the form "the error is bounded by the first omitted term" for this expansion, with a constant independent of N, is stronger than anything proved here. We do not assert it.

25.4 A worked bound, to digits

Example 25.11 (The enclosure at $x = 100$, $N = 3$). Take $x = 100$, so that

$$L = \log 100 = 4.605170185988091368\ldots,$$

and use the Lambert polynomials of Theorem 21.1,

$$\mathsf{L}^{\mathrm{Lam}}_1(u) = u, \qquad \mathsf{L}^{\mathrm{Lam}}_2(u) = \tfrac{1}{2}u^2 - u, \qquad \mathsf{L}^{\mathrm{Lam}}_3(u) = \tfrac{1}{3}u^3 - \tfrac{3}{2}u^2 + u.$$

The three correction terms are

$$\begin{aligned}
\mathsf{L}^{\mathrm{Lam}}_1(L)\, x^{-1} &= 4.60517018598809137 \times 10^{-2},\\
\mathsf{L}^{\mathrm{Lam}}_2(L)\, x^{-2} &= 5.99862603496870465 \times 10^{-4},\\
\mathsf{L}^{\mathrm{Lam}}_3(L)\, x^{-3} &= 5.34863899981332245 \times 10^{-6},
\end{aligned}$$

so that

$$\omega_3(100) = 95.4414867271142862294\ldots$$

Substituting into $\varphi(Y) = Y + \log Y$ gives the residual

$$R_3(100) = \omega_3(100) + \log \omega_3(100) - 100 = 8.23927836460666826 \times 10^{-8}.$$

Here $x - \log x = 95.39482981401190863\ldots$ and $\omega_3(100) > x - \log x$, so $\mu_3 = 95.394829814\ldots$ and $1 + \mu_3^{-1} = 1.010482748404\ldots$. Theorem 25.8 therefore yields the rigorous enclosure

$$8.153804088 \times 10^{-8} \;\le\; |\omega(100) - \omega_3(100)| \;\le\; 8.239278365 \times 10^{-8},$$

with $\omega(100) < \omega_3(100)$ because $R_3 > 0$. The true value, computed independently to 40 digits from $\omega(100) = W_0(\mathrm{e}^{100}) = 95.44148664557583184017\ldots$, is

$$|\omega(100) - \omega_3(100)| = 8.153845439 \times 10^{-8},$$

inside the enclosure and within 5×10^{-6} relative of its lower end. The lower bound is close to sharp because $\varphi'(\zeta) = 1 + \zeta^{-1}$ is close to 1; the whole enclosure has relative width $(1+\mu_3)^{-1} = 1.037\%$, and in particular at most $\mu_3^{-1} \approx 1.05\%$.

For comparison, Theorem 20.17 predicts the error $\mathsf{L}_4^{\mathrm{Lam}}(L)x^{-4} = -7.593614686 \times 10^{-8}$ up to a relative correction $\mathcal{O}_{x\to+\infty}\big(L^K/x\big)$; the predicted and true errors agree to about 6.9%, which is the size of the ratio $\mathsf{L}_5^{\mathrm{Lam}}(L)x^{-5}/\mathsf{L}_4^{\mathrm{Lam}}(L)x^{-4} = 7.2 \times 10^{-2}$. The enclosure of Theorem 25.8 has width 8.55×10^{-10}, about seven times smaller than the 5.60×10^{-9} by which the asymptotic prediction misses; and unlike the asymptotic statement it is a theorem about this one value of x, with no implied constant and no threshold.

Table 25.1: Rigorous enclosure of $|\omega(100) - \omega_N(100)|$ by Theorem 25.8. All entries computed at 40 significant digits and rounded; $\mu_N = x - \log x = 95.394829814\ldots$ in every row.

N	$\lvert R_N(100)\rvert$	lower bound	true error	upper bound
0	$4.714580382 \times 10^{-2}$	$4.665671324 \times 10^{-2}$	$4.665683156 \times 10^{-2}$	$4.714580382 \times 10^{-2}$
1	$6.114700456 \times 10^{-4}$	$6.051266551 \times 10^{-4}$	$6.051297040 \times 10^{-4}$	$6.114700456 \times 10^{-4}$
2	$5.322287245 \times 10^{-6}$	$5.267073836 \times 10^{-6}$	$5.267100545 \times 10^{-6}$	$5.322287245 \times 10^{-6}$
3	$8.239278365 \times 10^{-8}$	$8.153804088 \times 10^{-8}$	$8.153845439 \times 10^{-8}$	$8.239278365 \times 10^{-8}$
4	$5.661006395 \times 10^{-9}$	$5.602279113 \times 10^{-9}$	$5.602307524 \times 10^{-9}$	$5.661006395 \times 10^{-9}$
5	$1.558418770 \times 10^{-10}$	$1.542251733 \times 10^{-10}$	$1.542259554 \times 10^{-10}$	$1.558418770 \times 10^{-10}$

Remark 25.12 (Cancellation in the residual, with the digit count). Table 25.1 is a table of *differences of numbers of size* 95. At $N = 5$ the residual is 1.56×10^{-10}, so forming it loses about $\log_{10}(95.44/1.56 \times 10^{-10}) \approx 11.8$ decimal digits. A double-precision evaluation of $R_N(x)$ carries roughly 16 digits and is therefore already worthless at $N = 6$, where $R_6 \approx 2.05 \times 10^{-12}$ and the loss is about 13.7 digits. The bound of Theorem 25.8 is exact mathematics and useless numerics unless the residual is formed in extended precision. This is a property of the bound, not of the expansion: the expansion itself is evaluated by a stable Horner recurrence. One source advises "sufficient guard precision" when evaluating the coordinates; none quantifies the cancellation in the residual itself, although it is that cancellation, and not the coordinates, that decides how many digits are needed.

Remark 25.13 (The error is not monotone in N). At $x = 100$ the errors in Table 25.1 decrease at every step, but this is a feature of that value of x, not a theorem. At $x = 10$, where $\omega(10) = 7.92942009501969734856 2\ldots$, the successive errors $\omega(10) - \omega_N(10)$ are

$$\begin{aligned} N = 4:&\quad 1.9026 \times 10^{-5}, & N = 5:&\quad 5.1336 \times 10^{-6},\\ N = 6:&\quad 3.8679 \times 10^{-8}, & N = 7:&\quad -1.2836 \times 10^{-7},\\ N = 11:&\quad 4.8696 \times 10^{-12}, & N = 12:&\quad -1.9269 \times 10^{-11}, \end{aligned}$$

so the absolute error increases from $N = 6$ to $N = 7$ and again from $N = 11$ to $N = 12$. This is the reason to truncate by residual size rather than by formal order when a numerical answer is wanted.

25.5 An example where the hypotheses fail

Example 25.14 (The Lambert branch point: a residual of 10^{-8} with an error of 2.3×10^{-4})**.** Let $\varphi(Y) = Y\mathrm{e}^Y$ on $\mathbb{R}$, so that $\varphi'(Y) = (1+Y)\mathrm{e}^Y$ and $\varphi'(-1) = 0$; the two real solutions of $\varphi(Y) = z$ for $-\mathrm{e}^{-1} < z < 0$ are $W_0(z) > -1$ and $W_{-1}(z) < -1$ [5, 11]. Take

$$\eta = 10^{-8}, \qquad z = -\mathrm{e}^{-1} + \eta = -0.36787943117144232159552\ldots, \qquad Y_{\text{app}} = -1.$$

The residual is exactly

$$\varphi(Y_{\text{app}}) - z = -\mathrm{e}^{-1} - z = -\eta = -10^{-8},$$

as small as one could wish. The two true solutions are

$$W_0(z) = -0.99976685372178274818\ldots, \qquad W_{-1}(z) = -1.00023318252197543531\ldots,$$

so that the actual error of Y_{app} is $2.3314627822 \times 10^{-4}$ or $2.3318252198 \times 10^{-4}$ according to which root is meant — larger than the residual by a factor of about 2.33×10^4.

Every hypothesis fails, and each failure is instructive.

1. *No admissible* m. On the interval joining $Y_{\text{app}} = -1$ to either root, $\inf|\varphi'| = 0$, attained at the endpoint -1. So there is no $m > 0$ satisfying the hypothesis of Theorem 16.21, and (25.1) is not merely weak, it is unavailable.

2. *No enclosure.* In Theorem 25.2 one needs $m = \min_I |\varphi'| > 0$ on an interval I around Y_{app}; any such I contains -1, where φ' vanishes, so $m = 0$ and (25.2) cannot be satisfied for any ρ.

3. *No uniqueness.* The residual cannot distinguish the two roots: both lie within 2.34×10^{-4} of Y_{app}, and the single number -10^{-8} certifies neither. A residual bound is a statement about *a* root, and at a critical point "a root" is not "the root".

4. *The true law is a square root.* Since $\varphi(-1) = -\mathrm{e}^{-1}$, $\varphi'(-1) = 0$ and $\varphi''(-1) = \mathrm{e}^{-1}$, Taylor's theorem gives $\varphi(Y) + \mathrm{e}^{-1} = \frac{1}{2}\mathrm{e}^{-1}(Y+1)^2 + \mathcal{O}_{Y\to-1}\big((Y+1)^3\big)$, so the error obeys $|Y+1| \approx \sqrt{2\mathrm{e}\,|R|}$. With $|R| = \eta = 10^{-8}$ this predicts $\sqrt{2\mathrm{e}\eta} = 2.3316439816 \times 10^{-4}$, which matches both true errors to four significant figures. A linear residual bound with any finite constant is false here for all small η, since $\sqrt{\eta}/\eta \to \infty$.

The repair is a change of chart, not a better constant: in the variable $p = \sqrt{2\mathrm{e}(z + \mathrm{e}^{-1})}$ the two branches are analytic and the transfer machinery applies again. That is the analytic reason the branch point carries a Puiseux, not a polynomial–logarithmic, expansion.

Remark 25.15 (The three ways the transfer fails, in general). Theorem 25.14 exhibits the first and worst of three failure modes.

1. *A critical point of the map.* φ' vanishes between the approximation and the root. The residual then controls only a fractional power of the error, with the exponent $1/k$ at a zero of φ' of order $k-1$. No choice of constants rescues the linear bound.

2. *A small derivative without a critical point.* φ' is bounded below by $m > 0$, but m is tiny. Then (25.3) is true and useless: the error bound is inflated by m^{-1}, and the enclosure hypothesis (25.2) demands a residual smaller than $m\rho$. This is the situation for W_{-1} at moderate arguments, where φ' is small but nonzero.

3. *An unverifiable analytic residual.* (H2) of Theorem 25.6 fails or is not checked: the formal residual has high t-order and nothing is known about the size of $\varphi(g_N(x)) - x$. This mode is silent — no computation goes wrong, no inequality is violated, and the resulting statement is simply unproved. It is the mode that occurs in the sources.

Section summary

A residual is converted into an error bound by the mean value theorem, and that step is elementary. Everything else is not. The derivative lower bound must hold on an interval that contains the unknown root, which is circular unless one instead *encloses* the root (Theorem 25.2, and Theorem 25.3 by Rouché in the complex case). The analytic size of the residual does not follow from its formal t-order; it follows from the formal order *together with* an analytic expansion of the map being reversed (Theorem 25.5), and with that in hand the formal inverse is the asymptotic expansion of the analytic inverse (Theorem 25.6). For $\varphi(Y) = Y + \log Y$ all constants are explicit and the enclosure (25.9) holds at every single $x > 1$, with relative width at most $1/(x - \log x)$; at $x = 100$, $N = 3$ it brackets the true error 8.153845×10^{-8} between 8.153804×10^{-8} and 8.239278×10^{-8}. Where the derivative vanishes — the Lambert branch point — a residual of 10^{-8} coexists with an error of 2.3×10^{-4}, and no constant repairs the bound.

Chapter 26

Power–log series near a finite point and the Dulac conventions

Every result proved so far has been stated for $X \to +\infty$. The literature that motivates the class — local dynamics of planar vector fields, normal forms of parabolic and hyperbolic germs, Dulac's non-accumulation problem — works instead at a finite point, almost always $x \to 0^+$, and with generators, signs, orderings and truncation conventions that differ from ours in every one of those four respects. The purpose of this chapter is to make the passage exact.

Theorem 15.18 already recorded the integer-exponent dictionary $x = X^{-1}$, $\Lambda = \log(1/x) = L$, and Theorem 15.10 recorded the level shift $F \mapsto F \circ \log$. Those two propositions are the input here; the remark closing Part 15 deferred to the present chapter three things they do not supply:

(a) a statement of *which* class each of the two natural substitutions produces, and a proof that the two are not interchangeable;

(b) the Dulac conventions themselves — what a Dulac series is, what a Dulac germ is, how the support condition compares with the one that defines $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$, and where the two genuinely differ — together with the reason the first-return map of a hyperbolic polycycle lands in the class;

(c) the enlargements a finite point forces — real exponents in x, real exponents in the logarithm, and the two distinct mechanisms by which finiteness of the logarithmic degree fails.

The contribution of this chapter is precision rather than machinery. The sources are thin on proofs here and thick on convention: S2 devotes a chapter to the small-variable dictionary and states its results without proof; S1 and S6 each give half a page. Worse, S1, S2 and S6 do not use the same letter for the same object, and the discrepancy is invisible in any single formula. We therefore state definitions, prove transport theorems, and record as *checkable* propositions the eight ways a transported statement goes wrong.

26.1 Zero-side generators, and the two substitutions

Definition 26.1 (Local coordinate and zero-side generators)**.** Let $a \in \mathbb{R}$. A *local coordinate at a from the right* is $x = z - a$; from the left it is $x = a - z$. In either case the regime is $x \to 0^+$, and we work on an interval $0 < x < x_0$ with $x_0 < \mathrm{e}^{-1}$, so that the two zero-side generators

$$x, \qquad \Lambda := \log\frac{1}{x} = -\log x \tag{26.1}$$

satisfy $0 < x < \mathrm{e}^{-1} < 1 < \Lambda$ and $\Lambda \to +\infty$. We write $\Lambda_2 := \log \Lambda$ for the second zero-side logarithm and

$$\ell := \Lambda^{-1} = -\frac{1}{\log x} \in (0,1), \qquad \ell \to 0^+, \tag{26.2}$$

for the *small* logarithmic generator used in the normal-form literature. In the formal algebra x and Λ are independent indeterminates, and the four zero-side classes are

$$\mathbb{K}[\Lambda][[x]], \qquad \mathbb{K}[\Lambda]((x)), \qquad \mathbb{K}(\Lambda)((x)), \qquad \mathbb{K}((\Lambda^{-1}))((x)) = \mathbb{K}((\ell))((x)),$$

carrying the x-adic valuation ν_x, the exclusive truncation $\mathrm{Trunc}_{<N} f = \sum_{n<N} p_n(\Lambda)x^n$, and the derivation determined by

$$\frac{\mathrm{d}x}{\mathrm{d}x} = 1, \qquad \frac{\mathrm{d}\Lambda}{\mathrm{d}x} = -\frac{1}{x}. \tag{26.3}$$

The last three of these classes are stable under $\mathrm{d}/\,\mathrm{d}x$; the first is not, since $\mathrm{d}\Lambda/\,\mathrm{d}x = -x^{-1} \notin \mathbb{K}[\Lambda][[x]]$. This is the first asymmetry with the situation at infinity, where $\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ is D_X-stable, and it recurs as trap (T6) of Theorem 26.14. For a complex germ one replaces the interval by a sector $\{0 < |x| < x_0,\ |\arg x| < \theta\}$ and fixes a branch of log; every statement below that mentions positivity or a real branch is then a hypothesis, not a consequence.

The letter ℓ means two reciprocal things

Source S2 writes $\ell = \log(1/x)$; sources S1 and S6, following [23, 24, 27, 26], write $\ell = -1/\log x$. These are reciprocal: $\ell_{\mathrm{S2}} = \Lambda \to +\infty$ while $\ell_{\mathrm{S1,S6}} = \Lambda^{-1} \to 0^+$. A monomial $x^\alpha \ell^m$ therefore means $x^\alpha \Lambda^m$ in one convention and $x^\alpha \Lambda^{-m}$ in the other; the exponent of the logarithm changes sign, and with it the direction in which the logarithmic part of the dominance order runs. Throughout this volume the large zero-side logarithm is Λ and the small one is written Λ^{-1} or, where the comparison with the normal-form literature is the point, ℓ with (26.2) recalled on the spot.

Definition 26.2 (The reciprocal and the exponential substitution)**.** Let $\mathcal{M}_2(\mathbb{K},\mathbb{R})$ and $\mathcal{H}_2(\mathbb{K},\mathbb{R})$ be the depth-two monomial group and Hahn field of Theorem 15.1. Write $\mathcal{Z} := \iota\big(\mathcal{H}_2(\mathbb{K},\mathbb{R})\big)$ for its zero-side copy under the relabelling

$$\iota: \quad L_0 = X \longmapsto x^{-1}, \qquad L_1 \longmapsto \Lambda, \qquad L_2 \longmapsto \Lambda_2, \tag{26.4}$$

so that a monomial of $\mathcal{Z}$ is $x^a\Lambda^b\Lambda_2^c$ with $(a,b,c) \in \mathbb{R}^3$, and

$$x^a\Lambda^b\Lambda_2^c \succ x^{a'}\Lambda^{b'}\Lambda_2^{c'} \iff a < a', \text{ or } (a = a',\ b > b'), \text{ or } (a = a',\ b = b',\ c > c'). \tag{26.5}$$

This is (26.4) applied to the order of Theorem 15.1, since the exponent of $L_0 = x^{-1}$ is $-a$. The two substitutions are the $\mathbb{K}$-algebra embeddings of $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ into $\mathcal{Z}$ determined on generators by

$$\sigma: \quad t \longmapsto x, \qquad L \longmapsto \Lambda \qquad (\textit{reciprocal}\text{: } X = 1/x), \tag{26.6}$$
$$\tau: \quad t \longmapsto \Lambda^{-1}, \qquad L \longmapsto \Lambda_2 \qquad (\textit{exponential}\text{: } X = \log(1/x), \text{ i.e. } x = \mathrm{e}^{-X}). \tag{26.7}$$

On germs, σ is $F \mapsto F(1/x)$, which is ι restricted to $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$; and τ is $F \mapsto F(\log(1/x))$, which is $\iota \circ \Sigma$ with $\Sigma = (\,\cdot\,) \circ \log$ the level shift of Theorem 15.10. Throughout this section σ and τ denote these two substitutions and nothing else; elsewhere in the volume σ is a logarithmic exponent (Theorem 15.2) and τ is the Taylor-shift homomorphism used in the proof of Theorem 18.14, whose role here is played by θ.

Theorem 26.3 (What each substitution does to each class)**.** *Let $\mathbb{K}$ have characteristic zero.*

(i) *σ is an isomorphism of ordered rings onto its image and restricts to isomorphisms*

$$\mathscr{PL}_{\mathrm{pow},\mathbb{K}} \to \mathbb{K}[\Lambda][[x]], \qquad \mathscr{PL}_{\mathrm{poly},\mathbb{K}} \to \mathbb{K}[\Lambda]((x)),$$
$$\mathscr{PL}_{\mathrm{rat},\mathbb{K}} \to \mathbb{K}(\Lambda)((x)), \qquad \mathscr{PL}_{\mathrm{itLaur},\mathbb{K}} \to \mathbb{K}((\Lambda^{-1}))((x)),$$

and, for every $s \in \mathbb{N}_{>0}$ and every ordered subgroup $\Gamma \subseteq \mathbb{R}$, $\mathbb{K}[L]((t^{1/s})) \to \mathbb{K}[\Lambda]((x^{1/s}))$ and $\mathcal{H}_1(\mathbb{K},\Gamma) \to \iota\big(\mathcal{H}_1(\mathbb{K},\Gamma)\big)$. It carries ν_t to ν_x, $\mathrm{Trunc}_{<N}\cdot$ to $\mathrm{Trunc}_{<N}\cdot$, $\deg_{\mathrm{L}}$ to $\deg_\Lambda$, and the dominance order of Theorem 5.12 to the order of decreasing eventual size as $x \to 0^+$. It does not *carry D_X to $\mathrm{d}/\mathrm{d}x$; instead*

$$\sigma \circ D_X = \Big(-x^2 \frac{\mathrm{d}}{\mathrm{d}x}\Big) \circ \sigma. \tag{26.8}$$

(ii) *τ is an isomorphism of ordered rings onto $\mathbb{K}[\Lambda_2]((\Lambda^{-1}))$, carrying ν_t to the Λ^{-1}-adic valuation and dominance to dominance. It intertwines D_X with the derivation D_Λ of the target determined by $D_\Lambda \Lambda = 1$, $D_\Lambda \Lambda_2 = \Lambda^{-1}$; it does* not *intertwine D_X with $\mathrm{d}/\mathrm{d}x$, and $\mathbb{K}[\Lambda_2]((\Lambda^{-1}))$ is not stable under $\mathrm{d}/\mathrm{d}x$, because*

$$\frac{\mathrm{d}}{\mathrm{d}x} = -\frac{1}{x}\, D_\Lambda, \qquad \frac{\mathrm{d}}{\mathrm{d}x}\Lambda^{-1} = \frac{1}{x\Lambda^2}. \tag{26.9}$$

(iii) *Inside $\mathcal{Z}$,*

$$\sigma(\mathscr{PL}_{\mathrm{poly},\mathbb{K}}) \cap \tau(\mathscr{PL}_{\mathrm{poly},\mathbb{K}}) = \mathbb{K}[\Lambda]. \tag{26.10}$$

More precisely, in the exponent coordinates (a,b,c) of (26.5)*, $\sigma(\mathscr{PL}_{\mathrm{poly},\mathbb{K}})$ consists of the elements of $\mathcal{Z}$ whose support lies in $\mathbb{Z} \times \mathbb{N}_0 \times \{0\}$ with bounded-below first coordinate, while $\tau(\mathscr{PL}_{\mathrm{poly},\mathbb{K}})$ consists of those whose support lies in $\{0\} \times \mathbb{Z} \times \mathbb{N}_0$ with bounded-above second coordinate.*

Proof. (i) The map σ is the relabelling $t \mapsto x$, $L \mapsto \Lambda$ of two algebraically independent generators, extended continuously; it is therefore a ring isomorphism onto the correspondingly relabelled ring, and each of the six displayed restrictions is the same relabelling applied to the same support condition. Since σ does not move the exponent pair (n,j) of a monomial, it commutes with the operators $\nu_t \mapsto \nu_x$, $\mathrm{Trunc}_{<N}\cdot$, $\deg_{\mathrm{L}} \mapsto \deg_\Lambda$, which are defined from that pair alone; and it preserves the order because (5.10) and its zero-side counterpart are governed by the same inequalities on (n,j). That the zero-side order is the order of decreasing eventual size is Theorem 5.13 composed with $X = 1/x$. Finally (26.8) is (15.11), restated as an identity of operators.

(ii) Composing Theorem 15.10 with the relabelling (26.4) gives at once that τ is an isomorphism of ordered rings of $\mathscr{PL}_{\mathrm{poly},\mathbb{K}} = \mathbb{K}[L_1]((L_0^{-1}))$ onto $\mathbb{K}[L_2]((L_1^{-1})) = \mathbb{K}[\Lambda_2]((\Lambda^{-1}))$ with the stated transport of valuation and order, and $\Sigma \circ D_X = D_{L_1} \circ \Sigma$ becomes $\tau \circ D_X = D_\Lambda \circ \tau$. For the negative statement, the chain rule and (26.3) give $\mathrm{d}/\mathrm{d}x = (\mathrm{d}\Lambda/\mathrm{d}x) D_\Lambda = -x^{-1} D_\Lambda$, whence $\mathrm{d}\Lambda^{-1}/\mathrm{d}x = -x^{-1}\cdot(-\Lambda^{-2}) = (x\Lambda^2)^{-1}$, which involves x and therefore does not lie in $\mathbb{K}[\Lambda_2]((\Lambda^{-1}))$; this is (15.5) written at zero.

(iii) Both images lie in $\mathcal{Z}$ and Hahn representations are unique, so it suffices to identify supports, which we record in the coordinates (a,b,c) of (26.5). An element of $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ is $\sum_{n \geq n_0} p_n(L) t^n$; applying σ gives support $\{(n,j,0)\}$ with $n \geq n_0$ and $0 \leq j \leq \deg p_n$, while applying τ gives support $\{(0,-n,j)\}$ with the same ranges. An element of the intersection has support in $\{0\} \times \mathbb{Z} \times \{0\}$, i.e. it is a series in Λ alone; lying in $\sigma(\mathscr{PL}_{\mathrm{poly},\mathbb{K}})$ makes it a *polynomial* in Λ, and every polynomial in Λ does lie in $\tau(\mathscr{PL}_{\mathrm{poly},\mathbb{K}}) = \mathbb{K}[\Lambda_2]((\Lambda^{-1}))$, since that ring admits finitely many negative powers of its outer generator Λ^{-1}. Hence the intersection is exactly $\mathbb{K}[\Lambda]$. ☐

Remark 26.4 (The transport theorem is deliberately trivial as algebra). Item (i) is a relabelling and nothing more; this is the point. All of the difficulty in passing between infinity and a finite point is concentrated in the two places where the relabelling is *not* available: the derivation, by (26.8), and the analytic meaning of a monomial, by Theorem 8.43 versus Theorem 26.9 below. A transported statement that mentions neither is automatically correct; a transported statement that mentions either must be re-derived. The eight traps of Theorem 26.14 are all instances of this one sentence.

The two substitutions are not alternatives between which one may choose freely; they produce classes that are, apart from the polynomials in Λ, disjoint. The next proposition is the precise obstruction, and it also supplies the practical criterion.

Proposition 26.5 (The two scales at a finite point are incomparable)**.** *Work on $0 < x < \mathrm{e}^{-1}$ with the real branches, so that $\Lambda > 1$ and $\Lambda_2 > 0$.*

(i) *Let $F \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ be nonzero, with leading monomial $c\,t^n L^j$, and let f be a function on $(0, x_0)$ that realizes $\tau(F)$ at leading order, that is, $f \sim c\,\Lambda^{-n}\Lambda_2^j$ as $x \to 0^+$; this holds in particular whenever $f \approx \tau(F)$ in the sense of Theorem 8.43 read in the variable $X = \Lambda$. Then*

$$\log|f(x)| = \mathcal{O}_{x\to 0^+}(\Lambda_2) = o_{x\to 0^+}(\Lambda)\,;$$

in particular, for every $\varepsilon > 0$, $x^\varepsilon \prec f \prec x^{-\varepsilon}$.

(ii) *$\Lambda_2 = \log\log(1/x)$ is not asymptotically equivalent to $c\,x^\alpha\Lambda^\beta$ for any $c \neq 0$ and any $\alpha, \beta \in \mathbb{R}$. Consequently Λ_2 admits no asymptotic expansion in the scale $\{x^\alpha\Lambda^\beta : \alpha, \beta \in \mathbb{R}\}$, and in particular lies in none of the four σ-images, nor in $\sigma\big(\mathcal{H}_1(\mathbb{K}, \mathbb{R})\big)$ realized on that scale.*

(iii) *Symmetrically, x is not asymptotically equivalent to $c\,\Lambda^\beta\Lambda_2^\gamma$ for any $c \neq 0$ and any $\beta, \gamma \in \mathbb{R}$; so x has no expansion in the scale produced by τ.*

Proof. (i) By hypothesis $f/(c\,\Lambda^{-n}\Lambda_2^j) \to 1$. That $f \approx \tau(F)$ implies this is the estimate (8.37) at the cutoff $N = n+1$, which reads $|f - p_n(\Lambda_2)\Lambda^{-n}| \le C\Lambda^{-(n+1)}\Lambda_2^M$, combined with $p_n(\Lambda_2) \sim c\,\Lambda_2^j$ and $\Lambda^{-1}\Lambda_2^{M-j} \to 0$. Hence

$$\log|f| = -n\Lambda_2 + j\log\Lambda_2 + \mathcal{O}_{x\to 0^+}(1) = \mathcal{O}_{x\to 0^+}(\Lambda_2)\,,$$

and $\Lambda_2 = o_{x\to 0^+}(\Lambda)$ with $\log x = -\Lambda$ gives $\log|f| \,/\, \log x \to 0$, which is the second assertion.

(ii) Suppose $\Lambda_2 \sim c\,x^\alpha\Lambda^\beta$. Taking logarithms,

$$\log\Lambda_2 = \log|c| - \alpha\Lambda + \beta\Lambda_2 + o_{x\to 0^+}(1)\,.$$

The left side is $o_{x\to 0^+}(\Lambda_2)$, hence $o_{x\to 0^+}(\Lambda)$. If $\alpha \neq 0$ the right side is $\asymp \Lambda$, a contradiction; so $\alpha = 0$. Then the right side is $\beta\Lambda_2 + \mathcal{O}_{x\to 0^+}(1)$ while the left side is $o_{x\to 0^+}(\Lambda_2)$, forcing $\beta = 0$; but then the right side is bounded and the left side tends to $+\infty$. For the consequence: if Λ_2 had an expansion in that scale with a nonzero leading term, that term would be such a $c\,x^\alpha\Lambda^\beta$, and if the expansion were 0 then $\Lambda_2 = \mathcal{O}_{x\to 0^+}\big(x^N\big)$ for some $N > 0$, which is false.

(iii) Suppose $x \sim c\,\Lambda^\beta\Lambda_2^\gamma$. Taking logarithms, $-\Lambda = \log|c| + \beta\Lambda_2 + \gamma\log\Lambda_2 + o_{x\to 0^+}(1)$. The right side is $\mathcal{O}_{x\to 0^+}(\Lambda_2) = o_{x\to 0^+}(\Lambda)$ while the left side is $-\Lambda$, a contradiction. $\square$

Remark 26.6 (Which substitution to use). Theorem 26.5 turns the choice into a measurement. Given a germ f at 0^+, form $\log|f(x)| \,/\, \log x$.

- If it tends to a finite limit α and the corrections to f/x^α are powers of x times polynomials in Λ, the reciprocal substitution σ is the right one: f is a power–log germ and $f(1/X)$ is a candidate element of $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ or one of its enlargements.
- If it tends to 0 while f itself is unbounded or tends to 0, the germ lives on the logarithmic scale and σ is useless: by Theorem 26.5(ii) even the depth-one Hahn class in x and Λ cannot express Λ_2. Use τ, i.e. take $X = \log(1/x)$ as the large variable.
- If it tends to $\pm\infty$, neither substitution applies: the germ is beyond every logarithmic class, by Theorem 15.11 read through ι.

Theorem 26.7 is the standard germ for which the second case occurs, and for which the first case fails in an instructive way.

Example 26.7 (The lower Lambert branch at a finite point)**.** For $0 < x < \mathrm{e}^{-1}$ the branch $W_{-1}(-x)$ is the unique solution $w < -1$ of $w\mathrm{e}^w = -x$. Writing $w = -U$ with $U > 1$, the equation becomes $U\mathrm{e}^{-U} = x$, i.e.

$$U - \log U = \Lambda. \tag{26.11}$$

The map $g(u) = u - \log u$ has $g'(u) = 1 - 1/u > 0$ on $(1, +\infty)$, with $g(1) = 1$ and $g(u) \to +\infty$; it is therefore an increasing bijection $(1, +\infty) \to (1, +\infty)$, and $U = g^{\langle -1 \rangle \circ}(\Lambda)$ is well defined for $\Lambda > 1$, i.e. for $0 < x < \mathrm{e}^{-1}$.

Now apply τ. In the variables $X = \Lambda$, $L = \Lambda_2$, the equation (26.11) is exactly the reversion of $X \mapsto X - L$, which is the member $a = -1$ of the family of Theorem 20.5. Substituting (20.5) gives the expansion — with $\approx$ the relation of Theorem 8.43 read in the large variable $X = \Lambda$, not the zero-side relation $\approx_0$ of Theorem 26.9, whose scale is powers of x —

$$W_{-1}(-x) \;\approx\; -\Lambda - \Lambda_2 - \sum_{n \geq 1} (-1)^{n+1} \frac{\mathsf{L}_n^{\mathrm{Lam}}(\Lambda_2)}{\Lambda^n} = -\Lambda - \Lambda_2 - \frac{\Lambda_2}{\Lambda} + \frac{\Lambda_2^2/2 - \Lambda_2}{\Lambda^2} - \cdots, \tag{26.12}$$

whose analytic validity is Theorem 26.8. Two points deserve emphasis. First, the expansion uses *no* power of x: it lies entirely in $\tau(\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}})$, consistently with $\log|W_{-1}(-x)| / \log x \to 0$. Second, the reciprocal substitution genuinely fails here: by Theorem 26.5(ii) the term Λ_2 already lies outside every σ-image, so no amount of enlarging the coefficient field in Λ will produce (26.12). A second logarithm at infinity would be needed — and that is precisely the level shift that τ performs once and for all.

Proposition 26.8 (Asymptotic validity of the lower-branch expansion)**.** *For $N \in \mathbb{N}_0$ put*

$$V_N(\Lambda) \coloneqq \Lambda + \Lambda_2 + \sum_{n=1}^{N} (-1)^{n+1} \frac{\mathsf{L}_n^{\mathrm{Lam}}(\Lambda_2)}{\Lambda^n}.$$

Then there are $K_N \in \mathbb{N}_0$, $C_N > 0$ and $\Lambda_N > 1$ such that, with $U = -W_{-1}(-x)$ and $\Lambda = \log(1/x)$,

$$|U - V_N(\Lambda)| \leq C_N \frac{\Lambda_2^{K_N}}{\Lambda^{N+1}} \qquad (\Lambda \geq \Lambda_N). \tag{26.13}$$

In particular (26.12) *holds in the sense of Theorem 8.43 read in the large variable $X = \Lambda$, with $L = \Lambda_2$: the estimate* (26.13) *is* (8.37) *at the cutoff $N+1$, and the cutoffs $N+1$, $N \in \mathbb{N}_0$, are cofinal, so Theorem 8.44(1) supplies the remaining ones. Concretely,*

$$W_{-1}(-x) = -\Lambda - \Lambda_2 - \frac{\Lambda_2}{\Lambda} + \frac{\Lambda_2^2/2 - \Lambda_2}{\Lambda^2} + \mathcal{O}_{x \to 0^+}\left(\frac{\Lambda_2^3}{\Lambda^3}\right).$$

Proof. Throughout, $g(u) = u - \log u$ and $U = g^{\langle -1 \rangle \circ}(\Lambda)$ as in Theorem 26.7.

Step 1: an a priori bound. From $U = \Lambda + \log U$ and $U > 1$ we get $U > \Lambda$. For $u \geq 8$ one has $\log u \leq u/2$, hence $g(u) \geq u/2$; so if $\Lambda \geq 4$ and $u > 2\Lambda \geq 8$ then $g(u) > \Lambda$, and monotonicity of g gives $U \leq 2\Lambda$. Therefore $\Lambda < U \leq 2\Lambda$ and $\Lambda_2 < \log U \leq \Lambda_2 + \log 2$ for $\Lambda \geq 4$.

Step 2: the formal identity. Work in $\mathbb{K}[L][[t]]$ and put $\Psi \coloneqq L + \sum_{n \geq 1} (-1)^{n+1} \mathsf{L}_n^{\mathrm{Lam}}(L) t^n$, so that (20.5) reads $(X - L)^{\langle -1 \rangle \circ} = X + \Psi$. Applying $X - L$ to $X + \Psi$ and using generator substitution (Theorem 12.22), $L \circ (X + \Psi) = L + \log(1 + t\Psi)$, the defining relation becomes

$$\Psi = L + \log(1 + t\Psi). \tag{26.14}$$

Let $\Psi_N \coloneqq \mathrm{Trunc}_{<N+1} \Psi$. Since $\nu_t(\Psi - \Psi_N) \geq N+1$ and $\log(1 + t\,\cdot\,)$ raises the outer order by at least one, $\nu_t\big(\log(1 + t\Psi) - \log(1 + t\Psi_N)\big) \geq N + 2$; subtracting (26.14) therefore gives

$$\nu_t\Big(\Psi_N - L - \log(1 + t\Psi_N)\Big) \geq N + 1. \tag{26.15}$$

Step 3: the analytic residual. Put $\rho_N(\Lambda) = \sum_{n=1}^{N} (-1)^{n+1} \mathsf{L}_n^{\mathrm{Lam}}(\Lambda_2)\Lambda^{-n}$, so $V_N = \Lambda + \Lambda_2 + \rho_N = \Lambda(1+w)$ with $w = (\Lambda_2 + \rho_N)/\Lambda$. For Λ large, $|w| \le 2\Lambda_2/\Lambda \le \frac{1}{2}$, so $V_N > \Lambda/2 > 1$ and

$$R_N \coloneqq g(V_N) - \Lambda = V_N - \log V_N - \Lambda = \rho_N - \log(1+w).$$

Expand $\log(1+w) = \sum_{k=1}^{N+1} \frac{(-1)^{k-1}}{k} w^k + \theta$ with $|\theta| \le 2\,|w|^{N+2} \le 2^{N+3}\Lambda_2^{N+2}\Lambda^{-(N+2)}$, the bound on the tail being $\sum_{k \ge N+2} |w|^k / k \le |w|^{N+2} / \big((N+2)(1-|w|)\big)$. What remains,

$$P \coloneqq \rho_N - \sum_{k=1}^{N+1} \frac{(-1)^{k-1}}{k} w^k,$$

is the value at $t = \Lambda^{-1}$, $L = \Lambda_2$ of the *polynomial* $\mathcal{P} = (\Psi_N - L) - \sum_{k=1}^{N+1} \frac{(-1)^{k-1}}{k} (t\Psi_N)^k \in \mathbb{K}[L][t]$, because both are the same finite algebraic combination of t, L and Ψ_N. Now

$$\mathcal{P} = \underbrace{\Big(\Psi_N - L - \log(1 + t\Psi_N)\Big)}_{\nu_t \ge N+1 \text{ by (26.15)}} + \underbrace{\sum_{k \ge N+2} \frac{(-1)^{k-1}}{k} (t\Psi_N)^k}_{\nu_t \ge N+2},$$

so $\nu_t \mathcal{P} \ge N+1$, i.e. $\mathcal{P} = t^{N+1} Q$ with $Q \in \mathbb{K}[L][t]$. Writing $K_N = \deg_{\mathrm{L}} Q$ and evaluating at $t = \Lambda^{-1} \le 1$, $L = \Lambda_2 \ge 1$ gives $|P| \le C \Lambda_2^{K_N} \Lambda^{-(N+1)}$; enlarging K_N to absorb θ, $|R_N| \le C_N' \Lambda_2^{K_N} \Lambda^{-(N+1)}$.

Step 4: inversion. We have $g(U) = \Lambda$ and $g(V_N) = \Lambda + R_N$. Both U and V_N exceed $\Lambda/2 \ge 2$ for Λ large, so on the interval between them $g'(u) = 1 - 1/u \ge 1/2$. The mean value theorem gives $|U - V_N| \le 2\,|R_N|$, which is (26.13).

For the last display, apply (26.13) with $N = 3$ and add back the $n = 3$ term: $U - V_2 = \mathsf{L}_3^{\mathrm{Lam}}(\Lambda_2)\Lambda^{-3} + \mathcal{O}_{x \to 0^+}\Big(\Lambda_2^{K_3}\Lambda^{-4}\Big) = \mathcal{O}_{x \to 0^+}\big(\Lambda_2^3 \Lambda^{-3}\big)$, since $\deg \mathsf{L}_3^{\mathrm{Lam}} = 3$ and $\Lambda_2^K \Lambda^{-4} = o_{x \to 0^+}\big(\Lambda_2^3\Lambda^{-3}\big)$ for every fixed K. □

26.2 The transported dictionary, and the repeated derivative

The purely algebraic half of the dictionary is σ; the analytic half needs its own definition, because Theorem 8.43 names the regime $X \to +\infty$ explicitly.

Definition 26.9 (Zero-side asymptotic expansion)**.** Let $\mathbb{K} \subseteq \mathbb{C}$, let $0 < x_0 < \mathrm{e}^{-1}$, and let $f\colon (0, x_0) \to \mathbb{C}$. Let $\hat{f}$ be a formal series in the zero-side generators with left-finite x-support (in the sense of Theorem 30.1(b)) and polynomial coefficient blocks. We write $f \approx_0 \hat{f}$ if for every $N \in \mathbb{R}$ there are $M_N \in \mathbb{N}_0$, $C_N > 0$ and $x_N \in (0, x_0]$ with

$$\Big| f(x) - \big(\mathrm{Trunc}_{<N} \hat{f}\big)(x) \Big| \le C_N\, x^N \Lambda^{M_N} \qquad (0 < x < x_N), \tag{26.16}$$

where $\big(\mathrm{Trunc}_{<N} \hat{f}\big)(x)$ is the finite sum obtained by substituting the generators.

Proposition 26.10 (The two asymptotic relations correspond)**.** *Let $F \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ and let f be a function on $(0, x_0)$. Put $\tilde{f}(X) \coloneqq f(1/X)$ on $(1/x_0, +\infty)$. Then $f \approx_0 \sigma(F)$ if and only if $\tilde{f} \approx F$ in the sense of Theorem 8.43. The same equivalence holds for the Puiseux-log and Hahn enlargements, with N ranging over the relevant exponent group.*

Proof. Substituting $X = 1/x$ turns $\big(\mathrm{Trunc}_{<N} F\big)(X)$ into $\big(\mathrm{Trunc}_{<N} \sigma F\big)(x)$ monomial by monomial, because σ commutes with $\mathrm{Trunc}_{<N} \cdot$ (Theorem 26.3(i)); it turns $\log X$ into Λ and $X^{-N}(\log X)^M$ into $x^N \Lambda^M$; and it is a bijection between $(1/x_0, +\infty)$ and $(0, x_0)$ matching $X \ge X_N$ with $0 < x \le 1/X_N$. Thus (8.37) and (26.16) are the same inequality read in two coordinates. □

We can now transport a genuinely differential statement, and we do it in both directions, on the theorem whose zero-side form the Dulac literature uses most: the repeated-derivative formula. Recall $\Delta_{n,k} = \prod_{j=0}^{k-1}(\partial_L - n - j)$, with the empty product equal to the identity.

Theorem 26.11 (Repeated derivative at a finite point)**.** *Let $p \in \mathbb{K}[\Lambda]$, $n \in \mathbb{Z}$ and $k \in \mathbb{N}_0$. Then, in $\mathbb{K}[\Lambda]((x))$,*

$$\frac{\mathrm{d}^k}{\mathrm{d}x^k}\big(x^n p(\Lambda)\big) = (-1)^k\, x^{n-k}\, \Delta_{n-k+1,k}\, p(\Lambda) = x^{n-k} \prod_{j=0}^{k-1} \big(n - j - \partial_\Lambda\big)\, p(\Lambda), \tag{26.17}$$

the factors of the product commuting. In particular $\mathrm{d}(x^n p)/\,\mathrm{d}x = x^{n-1}(np - p')$, and $\nu_x\big(\mathrm{d}f/\,\mathrm{d}x\big) \geq \nu_x(f) - 1$ for every $f \in \mathbb{K}[\Lambda]((x))$, with equality unless $\nu_x(f) = 0$ and the block $[x^0]f$ is a nonzero constant.

Proof. By (26.8), $\mathrm{d}/\,\mathrm{d}x = -x^{-2}\,\sigma D_X \sigma^{-1}$ as operators on $\mathbb{K}[\Lambda]((x))$. Iterating,

$$\frac{\mathrm{d}^k}{\mathrm{d}x^k} = \big(-x^{-2}\,\sigma D_X \sigma^{-1}\big)^k,$$

which is not a power of $\sigma D_X \sigma^{-1}$ because x^{-2} and $\sigma D_X \sigma^{-1}$ do not commute; so we argue directly by induction on k, using the case $k = 1$ supplied by (15.11), namely $\mathrm{d}(x^m q)/\,\mathrm{d}x = x^{m-1}(mq - q')$.

For $k = 0$ both sides of (26.17) are $x^n p$, the empty product being the identity. Assume (26.17) for k, in the second form. Applying the case $k = 1$ at outer exponent $n - k$ to the coefficient $q = \prod_{j=0}^{k-1}(n - j - \partial_\Lambda)p$ gives

$$\frac{\mathrm{d}^{k+1}}{\mathrm{d}x^{k+1}}(x^n p) = x^{n-k-1}\big((n-k)q - q'\big) = x^{n-(k+1)}(n - k - \partial_\Lambda)q = x^{n-(k+1)} \prod_{j=0}^{k}(n - j - \partial_\Lambda)p,$$

which is the case $k + 1$; the factors commute because they are polynomials in the single operator ∂_Λ. Finally $\prod_{j=0}^{k-1}(n - j - \partial_\Lambda) = (-1)^k \prod_{j=0}^{k-1}(\partial_\Lambda - n + j)$, and as j runs from 0 to $k - 1$ the shifts $n - j$ run over $n, n-1, \ldots, n-k+1$, the same set as the shifts $(n - k + 1) + i$ for $0 \leq i \leq k - 1$; hence the product is $(-1)^k \Delta_{n-k+1,k}$, which is the first form.

For the valuation statement, let $n_0 = \nu_x(f)$ and $p = [x^{n_0}]f$. The coefficient of $x^{n_0 - 1}$ in $\mathrm{d}f/\,\mathrm{d}x$ is $n_0 p - p'$. If $n_0 \neq 0$ this has the same Λ-degree as p and leading coefficient n_0 times that of p, hence is nonzero; if $n_0 = 0$ it is $-p'$, nonzero exactly when $\deg_\Lambda p > 0$. ☐

The shift index moves, and only by k

The infinity-side formula (11.6) reads $D_X^k(t^n p) = t^{n+k}\Delta_{n,k}p$: the outer exponent moves *up* by k and the shift index stays at n. The finite-point formula (26.17) reads $\mathrm{d}^k(x^n p)/\,\mathrm{d}x^k = (-1)^k x^{n-k}\Delta_{n-k+1,k}p$: the outer exponent moves *down* by k, a sign $(-1)^k$ appears, and the shift index moves to $n - k + 1$. Writing $\Delta_{n,k}$ on the zero side is correct for $k \leq 1$ and wrong for every $k \geq 2$.

Transporting in the other direction is not a formality either, because the statement the sources make at a finite point is stated for a *real* exponent, as Dulac exponents require, whereas Theorem 11.8 is stated over $\mathbb{Z}$. Carrying it back therefore yields a strengthening.

Proposition 26.12 (The return trip, and a real-exponent extension)**.** *Let $\Gamma \subseteq \mathbb{R}$ be an ordered subgroup and let $\alpha \in \Gamma$, $p \in \mathbb{K}[L]$, $k \in \mathbb{N}_0$. In the Hahn power–log ring $\mathbb{K}[L][[t^\Gamma]]$, with the derivation determined by $D_X t^\alpha = -\alpha t^{\alpha+1}$ and $D_X L = t$,*

$$D_X^k\big(t^\alpha p(L)\big) = t^{\alpha+k}\,\Delta_{\alpha,k}\, p(L), \qquad \Delta_{\alpha,k} = \prod_{j=0}^{k-1}(\partial_L - \alpha - j), \tag{26.18}$$

and applying σ returns (26.17) *with n replaced by α. In particular the two transports are mutually inverse: starting from* (11.6)*, carrying it to the finite point by Theorem 26.11 and carrying the result back by* (26.8) *reproduces* (11.6) *verbatim, with the index substitutions $n \mapsto n$ and $n \mapsto n-k+1$ composing to the identity on the pair* (*outer exponent*, *shift index*).

Proof. The block rule $D_X(t^\alpha p) = t^{\alpha+1}(p' - \alpha p)$ holds for real α by the Leibniz rule applied to $D_X t^\alpha = -\alpha t^{\alpha+1}$ and $D_X L = t$; the induction proving Theorem 11.8 uses nothing about n beyond this rule and the commutation of the factors, so it goes through verbatim with n replaced by α. This is (26.18).

Applying σ to (26.18) and using (26.8) k times gives $\left(-x^2\,\mathrm{d}/\,\mathrm{d}x\right)^k(x^\alpha p(\Lambda)) = x^{\alpha+k}\Delta_{\alpha,k}p(\Lambda)$; and the induction of Theorem 26.11, which again uses only the case $k=1$, gives $\mathrm{d}^k(x^\alpha p)/\,\mathrm{d}x^k = (-1)^k x^{\alpha-k}\Delta_{\alpha-k+1,k}p$. These two are consistent. Indeed, one application of $-x^2\,\mathrm{d}/\,\mathrm{d}x$ sends $x^m q$ to

$$-x^2 \cdot x^{m-1}\big(mq - q'\big) = x^{m+1}\big(\partial_\Lambda - m\big)q,$$

so starting from $x^\alpha p$ the $(j+1)$-st application acts at outer exponent $\alpha + j$ and contributes the factor $\partial_\Lambda - \alpha - j$; after k steps the result is $x^{\alpha+k}\prod_{j=0}^{k-1}(\partial_\Lambda - \alpha - j)p = x^{\alpha+k}\Delta_{\alpha,k}p$, which is exactly the σ-image of (26.18). Reading the same computation backwards recovers (11.6), the shift index of the composite being α again. □

Remark 26.13 (What the sources say). S2 displays (26.17) in the second, product form, for real λ and with no proof and no empty-product convention. The formula is correct; what is missing is the identification of the operator, which is where the $k \geq 2$ index shift of Section 26.2 becomes visible, and the observation that the real-exponent case is a strengthening of Theorem 11.8 rather than an instance of it. S6 gives only the case $k = 1$.

26.3 Eight traps, each as a checkable statement

Proposition 26.14 (Transport traps). *Let $F = \sum_{n \geq n_0} p_n(L)t^n \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ and $f = \sigma F$. Each of the following is an identity or an equivalence, and each is the correct form of a step that is routinely taken in the reverse or the wrong direction.*

(T1) Index sign. *Writing F in powers of the variable, $F = \sum_{m \leq -n_0} q_m(L)X^m$ with $q_{-n} = p_n$, one has $f = \sum_{m \leq -n_0} q_m(\Lambda)x^{-m}$. Thus $\deg_X F = -\nu_t F = -\nu_x f$: the outer index used at infinity and the one used at zero are negatives of each other, while ν_t and ν_x agree.*

(T2) Dominance against the variable. *$X^a \succ X^b$ if and only if $a > b$, whereas $x^a \succ x^b$ if and only if $a < b$. The dominance order read on the exponent of the variable therefore reverses; read on the exponent of t, respectively x, it is preserved.*

(T3) Leading term. *The leading monomial of F is the one of largest X-exponent and that of f the one of smallest x-exponent, and σ matches them. Consequently a series written in increasing powers of X has its leading term last, while a series written in increasing powers of x has it first.*

(T4) Truncation bound. $\mathrm{Trunc}_{<N} F$ *retains exactly the monomials with X-exponent $> -N$ — a* lower *bound — and* $\mathrm{Trunc}_{<N} f$ *retains exactly those with x-exponent $< N$ — an* upper *bound. The remainder $F - \mathrm{Trunc}_{<N} F$ consists of the monomials of* small *X-exponent and of* large *x-exponent.*

(T5) The derivation is not transported. *$\sigma D_X \sigma^{-1} = -x^2\,\mathrm{d}/\,\mathrm{d}x$, not $\mathrm{d}/\,\mathrm{d}x$. On blocks, $\partial_L - n$ becomes $n - \partial_\Lambda$, and for k derivatives the shift index moves as in Section 26.2.*

(T6) Differentiation changes the remainder in opposite directions. *$\nu_t(D_X F) \geq \nu_t F + 1$ (Theorem 11.11), whereas $\nu_x(\mathrm{d}f/\,\mathrm{d}x) \geq \nu_x(f) - 1$ (Theorem 26.11). Hence $\mathcal{O}_t^{\mathrm{form}}\big(t^N\big)$ differentiates* *to $\mathcal{O}_t^{\mathrm{form}}(t^{N+1})$ at infinity and $\mathcal{O}_x^{\mathrm{form}}(x^N)$ differentiates to $\mathcal{O}_x^{\mathrm{form}}(x^{N-1})$ at a finite point: at infinity a derivative improves a formal tail, at a finite point it costs one order.*

(T7) Sign of the logarithm. $\Lambda = -\log x$. *A block written as a polynomial P in $\log x$ equals $p(\Lambda)$ with $p(u) = P(-u)$; degrees agree, and every coefficient of odd degree changes sign. Writing $\log x$ for Λ therefore leaves every formula formally consistent while reversing the sign of half the coefficients.*

(T8) Reciprocal logarithmic generator. *With $\ell = \Lambda^{-1}$, a monomial $x^\alpha \ell^\beta$ is $x^\alpha \Lambda^{-\beta}$, i.e. $X^{-\alpha} L^{-\beta}$. The exponent of the logarithm changes sign; a family listed by increasing powers of ℓ is listed by* decreasing *size in the logarithmic direction, the opposite of a family listed by increasing powers of Λ.*

Proof. (T1) $X^m = t^{-m}$ and $\sigma(t^{-m}) = x^{-m}$; the rest is the definition of ν_t and ν_x.

(T2) For $a > b$, $X^a/X^b = X^{a-b} \to +\infty$ as $X \to +\infty$, so $X^a \succ X^b$; and $x^a/x^b = x^{a-b} \to 0$ as $x \to 0^+$, so $x^a \prec x^b$, i.e. $x^a \succ x^b$ exactly when $a < b$. On the exponent of t, respectively x, (5.10) is literally the same inequality on both sides, which is Theorem 26.3(i).

(T3) Immediate from (T1) and (T2).

(T4) By Theorem 7.1, $\mathrm{Trunc}_{<N} F = \sum_{n<N} p_n t^n$; the monomial t^n is X^{-n}, and $n < N$ is $-n > -N$. On the zero side t^n becomes x^n and the condition stays $n < N$. The statement about the remainder is the negation of the two conditions.

(T5) (26.8) and Theorem 26.11. The two operators are genuinely different: $\sigma D_X \sigma^{-1}$ raises ν_x by one and $\mathrm{d}/\mathrm{d}x$ lowers it by one.

(T6) The two valuation statements are Theorem 11.11(i) and the last clause of Theorem 26.11. The consequences for formal tails follow because $\mathcal{O}_t^{\mathrm{form}}(t^N)$ means $\nu_t \geq N$ (Theorem 7.7) and likewise at zero.

(T7) $P(\log x) = P(-\Lambda) = p(\Lambda)$ with $p(u) = P(-u)$; the coefficient of u^j in p is $(-1)^j$ times that of P.

(T8) $\ell^\beta = \Lambda^{-\beta}$ by (26.2), and $\Lambda = L$, $x = t$ by (26.6). Since $\ell \to 0^+$ and $\Lambda \to +\infty$, $\ell^\beta \prec \ell^{\beta'}$ exactly when $\beta > \beta'$, the reverse of (5.10) read in Λ. ☐

Three ways a transported theorem is wrong while looking right

The dangerous traps are (T5)–(T8), because each of them leaves a formula *formally consistent*. A calculation performed at a finite point with $\Delta_{n,k}$ in place of $\Delta_{n-k+1,k}$, or with $\log x$ in place of Λ, or with the reciprocal ℓ in place of Λ, produces a self-consistent array of coefficients that satisfies every internal check and is simply not the expansion of the intended function. The cheapest detection is a two-sided residual: substitute the computed inverse back and verify that the residual has the predicted order *and the predicted sign*, as in Theorem 16.15. The second cheapest is a numerical spot check at one value of x, which $\Lambda \leftrightarrow \ell$ confusion fails immediately.

26.4 A statement that looks transportable and is not: reversion at a finite point

All three of S1, S2 and S6 assert that near-identity reversion and the Lagrange–Bürmann formula “have exactly the same form” at a finite point. The formulas are indeed the same; the reason is not that they are transports of one another, and asserting that they are is a mistake with consequences, because the summability mechanism runs in the opposite direction. The following proposition is the obstruction, stated so that it can be checked.

Proposition 26.15 (The reversion-frame axioms are not σ-invariant)**.** *Let $\mathfrak{B} := \mathbb{K}[\Lambda]((x))$ and $\partial := \mathrm{d}/\mathrm{d}x$.*

(i) *σ carries the polynomial–logarithmic reversion frame* $\big(\mathscr{PL}_{\mathrm{poly},\mathbb{K}}, \nu_t, D_X, t^{-1}\big)$ *of Theorem 18.3 to the frame* $\big(\mathfrak{B}, \nu_x, -x^2\partial,\ x^{-1}\big)$. *Its derivation is* $-x^2\partial$, *not* ∂, *and its distinguished element is* x^{-1}, *not* x.

(ii) *The quadruple* $\big(\mathfrak{B}, \nu_x, \partial, x\big)$ *— the one that governs reversion of germs at* 0^+ *— satisfies* (F1), (F2) *and* (F4) *of Theorem 18.1 with* $\Gamma = \mathbb{Z}$, *but* not (F3)*: instead*

$$\nu_x(\partial f) \geq \nu_x(f) - 1 \qquad (f \in \mathfrak{B}), \tag{26.19}$$

and (F3) *fails already at* $f = x$, *where* $\nu_x(\partial x) = 0 < \nu_x(x) + 1 = 2$.

(iii) *No rescaling repairs this: for* $c \in \mathbb{R}^\times$ *the map* $c\,\nu_x$ *satisfies* (F3) *for* ∂ *only if* $c \leq -1$, *and for* $c < 0$ *it violates* (F1), *as witnessed by* $f = x$, $g = -x + x^5$, *for which* $c\nu_x(f+g) = 5c < c = \min\{c\nu_x f, c\nu_x g\}$.

Consequently Theorem 18.14 does not apply to $\big(\mathfrak{B}, \nu_x, \partial, x\big)$, *and the near-zero Lagrange–Bürmann formula is a second theorem, with its own proof.*

Proof. (i) σ is a ring isomorphism carrying ν_t to ν_x (Theorem 26.3(i)), D_X to $-x^2\partial$ ((26.8)) and $X = t^{-1}$ to x^{-1}. Since the frame axioms are formulated in terms of these four data, the image is a frame.

(ii) (F1): $\mathbb{K}[\Lambda]$ is a domain, so ν_x is a valuation with $\nu_x(\lambda) = 0$ for $\lambda \in \mathbb{K}^\times$. (F2): a family with $\{i : \nu_x(f_i) < \gamma\}$ finite for all γ has coefficientwise finite sums at each level and only finitely many levels below any bound, exactly as in the proof of Theorem 18.3. (F4): $\partial x = 1$. Inequality (26.19) is the last clause of Theorem 26.11, and $\partial x = 1$ gives the stated failure of (F3).

(iii) If $\nu = c\nu_x$ satisfies (F3), then applying it to $f = x^n$ with $\partial^n x^n = n!$ gives $0 = \nu(n!) \geq \nu(x^n) + n = n(c+1)$ for every $n \geq 1$, so $c \leq -1$; in particular $c < 0$. The displayed witness then contradicts the ultrametric inequality in (F1). ☐

The correct zero-side objects are these.

Definition 26.16 (The near-identity class at a finite point)**.** Put

$$\mathfrak{m}_0 := x^2\mathbb{K}[\Lambda][[x]] = \{h \in \mathbb{K}[\Lambda]((x)) : \nu_x(h) \geq 2\}, \qquad \mathcal{G}_0 := \{x + h : h \in \mathfrak{m}_0\}.$$

For $B \in \mathfrak{m}_0$ define the *shift operator*

$$T_B(f) := \sum_{k \geq 0} \frac{B^k}{k!}\frac{\mathrm{d}^k f}{\mathrm{d}x^k}, \qquad f \in \mathbb{K}[\Lambda]((x)), \tag{26.20}$$

and write $f \circ (x + B) := T_B(f)$.

Lemma 26.17 (The shift operator is substitution of $x + B$)**.** *Let* $B \in \mathfrak{m}_0$. *Then:*

(i) *the family in* (26.20) *is summable, its* k*-th member having* $\nu_x \geq \nu_x(f) + k$; *hence* $\nu_x(T_B f) \geq \nu_x(f)$;

(ii) T_B *is a* $\mathbb{K}$*-algebra endomorphism of* $\mathbb{K}[\Lambda]((x))$, *continuous for* ν_x, *with* $T_0 = \mathrm{id}$;

(iii) *on generators,*

$$T_B(x) = x + B, \qquad T_B(\Lambda) = \Lambda - \log\Big(1 + \frac{B}{x}\Big) = \log\frac{1}{x + B}; \tag{26.21}$$

(iv) $T_{B'} \circ T_B = T_{B''}$ *with* $B'' = B' + T_{B'}(B) \in \mathfrak{m}_0$; *consequently* $\mathcal{G}_0$ *is a monoid under* $(x + B) \circ (x + B') := T_{B'}(x + B) = x + B''$, *with neutral element* x, *and* T *is an anti-homomorphism from it into the monoid of endomorphisms of* $\mathbb{K}[\Lambda]((x))$.

Proof. (i) $\nu_x(B^k) \geq 2k$ and, by (26.19), $\nu_x(\mathrm{d}^k f/\,\mathrm{d}x^k) \geq \nu_x(f) - k$; the product has $\nu_x \geq \nu_x(f) + k \to +\infty$, so (F2) for $(\mathfrak{B}, \nu_x)$ — established in Theorem 26.15(ii) — gives the sum, and the bound is the minimum at $k = 0$.

(ii) Linearity is clear. For multiplicativity, expand $\mathrm{d}^k(fg)/\,\mathrm{d}x^k$ by Leibniz and regroup; the double family $\big(B^i\partial^i f/i!\big)\big(B^j\partial^j g/j!\big)$ has $\nu_x \geq \nu_x(f) + \nu_x(g) + i + j$, so it is summable and the regrouping clause of (F2) applies:

$$T_B(fg) = \sum_k \frac{B^k}{k!} \sum_{i+j=k} \binom{k}{i} \partial^i f\, \partial^j g = \sum_{i,j} \frac{B^i \partial^i f}{i!} \cdot \frac{B^j \partial^j g}{j!} = T_B(f)\, T_B(g).$$

Continuity follows from (i).

(iii) $\partial x = 1$ and $\partial^k x = 0$ for $k \geq 2$, giving $T_B(x) = x + B$. For Λ: by (26.3), $\partial\Lambda = -x^{-1}$ and hence $\partial^k\Lambda = (-1)^k(k-1)!\, x^{-k}$ for $k \geq 1$, so

$$T_B(\Lambda) = \Lambda + \sum_{k\geq 1} \frac{B^k}{k!}(-1)^k(k-1)!\, x^{-k} = \Lambda + \sum_{k\geq 1} \frac{(-1)^k}{k}\Big(\frac{B}{x}\Big)^k = \Lambda - \log\Big(1 + \frac{B}{x}\Big),$$

the formal logarithm being defined because $\nu_x(B/x) \geq 1$. Since $\Lambda = \log(1/x)$ and $\log\big(1 + B/x\big) = \log\big((x+B)/x\big)$, this is $\log\big(1/(x+B)\big)$.

(iv) Both $T_{B'} \circ T_B$ and $T_{B''}$ are continuous $\mathbb{K}$-algebra endomorphisms, and $\mathbb{K}[\Lambda]((x))$ is topologically generated by $x^{\pm 1}$ and Λ; so it suffices to compare them there. On x: $T_{B'}(T_B x) = T_{B'}(x + B) = x + B' + T_{B'}(B) = x + B''$, and $T_{B''}(x) = x + B''$. On x^{-1}: a ring homomorphism sends the inverse of a unit to the inverse of its image, and $x + B'$ is a unit of $\mathbb{K}[\Lambda]((x))$, its leading block being the scalar 1 (Theorem 9.3, transported by Theorem 26.3(i)); so the values on x^{-1} are determined by those on x. On Λ: $T_{B'}$ is a continuous ring homomorphism, hence $T_{B'}\big(\log(1+w)\big) = \log\big(1 + T_{B'}(w)\big)$ whenever $\nu_x(w) > 0$, and $T_{B'}(B/x) = T_{B'}(B)/(x + B')$. Therefore, by (iii),

$$\begin{aligned} T_{B'}\big(T_B\Lambda\big) &= \Lambda - \log\Big(1 + \frac{B'}{x}\Big) - \log\Big(1 + \frac{T_{B'}(B)}{x + B'}\Big) \\ &= \Lambda - \log\Big(\frac{x + B'}{x} \cdot \frac{x + B' + T_{B'}(B)}{x + B'}\Big) = \Lambda - \log\Big(1 + \frac{B''}{x}\Big), \end{aligned}$$

which is $T_{B''}(\Lambda)$. Finally $\nu_x(B'') \geq \min\{\nu_x(B'), \nu_x(B)\} \geq 2$ by (i), so $B'' \in \mathfrak{m}_0$ and $\mathcal{G}_0$ is closed; neutrality of x is $T_0 = \mathrm{id}$, and associativity is (iv) itself. $\square$

Theorem 26.18 (Lagrange–Bürmann reversion at a finite point)**.** *Let $h \in \mathfrak{m}_0$, i.e. $h \in \mathbb{K}[\Lambda][[x]]$ with $\nu_x(h) \geq 2$, and put $\delta := \nu_x(h) - 1 \geq 1$. Then the family*

$$\Big(B_r\Big)_{r\geq 1}, \qquad B_r := \frac{(-1)^r}{r!} \frac{\mathrm{d}^{\,r-1}}{\mathrm{d}x^{\,r-1}}\big(h^r\big),$$

is summable, with $\nu_x(B_r) \geq r\delta + 1$; its sum B lies in $\mathfrak{m}_0$; and

$$(x+h)^{\langle -1\rangle_\circ} = x + \sum_{r\geq 1} \frac{(-1)^r}{r!} \frac{\mathrm{d}^{\,r-1}}{\mathrm{d}x^{\,r-1}}\big(h^r\big) \tag{26.22}$$

is the two-sided compositional inverse of $x + h$ in $\mathcal{G}_0$. In particular $\mathcal{G}_0$ is a group. The same statement holds in $\mathbb{K}[\Lambda]((x^{1/s}))$ and in the Hahn enlargements, with $\mathfrak{m}_0$ replaced by $\{h : \nu_x(h) > 1\}$; the threshold $\nu_x(h) > 1$ is exactly the mirror of $\nu_t(A) > -1$ of Theorem 18.1, and $1 = \nu_x(x)$ is the mirror of $-1 = \nu_t(X)$.

Proof. Write $\partial = \mathrm{d}/\,\mathrm{d}x$ and $\mathfrak{B} = \mathbb{K}[\Lambda]((x))$.

Step 1: summability. $\nu_x(h^r) \geq r\nu_x(h)$ and ∂^{r-1} costs at most $r-1$ by (26.19), so $\nu_x(B_r) \geq r\nu_x(h) - (r-1) = r\delta + 1$, which tends to $+\infty$ since $\delta > 0$. Hence (B_r) is summable and $\nu_x(B) \geq \delta + 1 = \nu_x(h) \geq 2$; also $B \in \mathbb{K}[\Lambda][[x]]$, so $B \in \mathfrak{m}_0$ and $x + B \in \mathcal{G}_0$.

Step 2: what has to be proved. By Theorem 26.17(iv), $(x+h)\circ(x+B) = T_B(x+h) = x+B+T_B(h)$. So $x+B$ is a right inverse of $x+h$ in $\mathcal{G}_0$ if and only if

$$B = -T_B(h). \tag{26.23}$$

Granting (26.23), every element of the monoid $\mathcal{G}_0$ has a right inverse; and in a monoid in which every element has a right inverse, every element is invertible: if $ab = x$ and $bc = x$ then $a = a\circ(b\circ c) = (a\circ b)\circ c = c$, whence $ba = bc = x$. So $\mathcal{G}_0$ is a group and $x + B = (x+h)^{\langle -1\rangle_\circ}$ is the two-sided inverse. It remains to prove (26.23).

Step 3: the weighted deformation algebra. Define $\mathfrak{B}\langle\varepsilon\rangle_a$ and $\mathfrak{B}\langle\varepsilon\rangle$ exactly as in Theorem 18.13, with the present ν_x, the present ∂ and the weight δ. The proof of Theorem 18.13 uses (F3) in one place only, to see that ∂ maps $\mathfrak{B}\langle\varepsilon\rangle_a$ into some $\mathfrak{B}\langle\varepsilon\rangle_{a'}$; by (26.19) that holds here with $a' = a+1$ in place of $a-1$. Everything else in that proof uses only (F1) and (F2), which hold by Theorem 26.15(ii). Thus all three conclusions of Theorem 18.13 are available: $\mathfrak{B}\langle\varepsilon\rangle$ is a ∂-stable subring of $\mathfrak{B}[[\varepsilon]]$; evaluation $\mathrm{ev}(u) = \sum_m u_m$ is a $\mathbb{K}$-algebra homomorphism with $\mathrm{ev}(\varepsilon) = 1$; and ev commutes with sums of families $\big(f^{(k)}\big)$ lying in a common $\mathfrak{B}\langle\varepsilon\rangle_a$ with $f^{(k)} \in \varepsilon^k\mathfrak{B}[[\varepsilon]]$.

Step 4: the parameter identity. Theorem 18.11 is stated for an arbitrary commutative $\mathbb{Q}$-algebra and uses no frame; take $C = \mathfrak{B}$. Let

$$\theta\colon \mathfrak{B} \to \mathfrak{B}[[z]], \qquad \theta(f) = \sum_{k\geq 0}\big(\partial^k f\big)\frac{z^k}{k!}.$$

This is a $\mathbb{K}$-algebra homomorphism (Leibniz, as in Theorem 26.17(ii), with no summability needed because $\mathfrak{B}[[z]]$ is a formal power-series ring), it satisfies $\partial_z\theta = \theta\partial$ by a term shift, and $\mathrm{ev}_{z=0}\theta = \mathrm{id}$. Put $\psi = \theta(-h)$ and $\gamma = \theta(x) = x + z$, so $\partial_z\gamma = 1$. Let $u \in \varepsilon\mathfrak{B}[[z]][[\varepsilon]]$ be the unique solution of $u = \varepsilon\psi_{[u]}$. Since $\gamma_{[u]} = x + z + u$, formula (18.8) gives, for $m \geq 1$,

$$[\varepsilon^m]u = \frac{1}{m!}\partial_z^{m-1}\big(\psi^m\big) = \frac{(-1)^m}{m!}\partial_z^{m-1}\theta\big(h^m\big) = \theta\big(B_m\big). \tag{26.24}$$

Step 5: evaluation at $z = 0$. The map $\mathrm{ev}_{z=0}$, applied coefficientwise in ε, is a $\mathbb{K}$-algebra homomorphism $\mathfrak{B}[[z]][[\varepsilon]] \to \mathfrak{B}[[\varepsilon]]$; each ε-coefficient of $\psi_{[u]} = \sum_k(\partial_z^k\psi)u^k/k!$ is a finite sum because $u \in \varepsilon(\cdot)$, so applying it to $u = \varepsilon\psi_{[u]}$ is legitimate. Write $\tilde{u} := \mathrm{ev}_{z=0}(u) = \sum_{m\geq 1}B_m\varepsilon^m$ by (26.24), and note $\mathrm{ev}_{z=0}\big(\partial_z^k\psi\big) = \mathrm{ev}_{z=0}\theta\big(\partial^k(-h)\big) = -\partial^k h$. Hence

$$\tilde{u} = \varepsilon\sum_{k\geq 0}\frac{-\partial^k h}{k!}\,\tilde{u}^k \quad \text{in } \mathfrak{B}[[\varepsilon]]. \tag{26.25}$$

By Step 1, $\nu_x(B_m) \geq m\delta + 1$, so $\tilde{u} \in \mathfrak{B}\langle\varepsilon\rangle_0$ and $\mathrm{ev}(\tilde{u}) = \sum_m B_m = B$.

Step 6: evaluation at $\varepsilon = 1$. Put $f^{(k)} := \varepsilon\,\frac{-\partial^k h}{k!}\tilde{u}^k \in \varepsilon^{k+1}\mathfrak{B}[[\varepsilon]]$. Its ε^m coefficient is a finite sum of terms $\frac{-\partial^k h}{k!}B_{m_1}\cdots B_{m_k}$ with $m_1 + \cdots + m_k = m - 1$ and every $m_i \geq 1$, so

$$\nu_x\big([\varepsilon^m]f^{(k)}\big) \geq \big(\nu_x(h) - k\big) + \sum_{i=1}^{k}\big(m_i\delta + 1\big) = \nu_x(h) + \delta(m-1) = \delta m + 1 \geq \delta m,$$

the bound being uniform in k. Thus every $f^{(k)}$ lies in the *same* $\mathfrak{B}\langle\varepsilon\rangle_0$, and Theorem 18.13(3) applies to (26.25):

$$B = \mathrm{ev}(\tilde{u}) = \sum_{k\geq 0}\mathrm{ev}\big(f^{(k)}\big) = \sum_{k\geq 0}\frac{-\partial^k h}{k!}\,\mathrm{ev}(\tilde{u})^k = -\sum_{k\geq 0}\frac{B^k}{k!}\,\partial^k h = -T_B(h),$$

using $\mathrm{ev}(\varepsilon) = 1$ and multiplicativity. This is (26.23).

For the Puiseux-log and Hahn versions, every step used only $\delta = \nu_x(h) - 1 > 0$ and the properties (F1), (F2) of ν_x, which hold verbatim over the enlarged exponent group; and Theorem 26.17, whose proof uses $\nu_x(B) \geq 2$ only through the estimate $\nu_x\big(B^k \partial^k f/k!\big) \geq \nu_x(f) + k\big(\nu_x(B) - 1\big)$ and the membership $\nu_x(B/x) > 0$, holds there with $\mathfrak{m}_0$ replaced by $\{B : \nu_x(B) > 1\}$. □

Proposition 26.19 (Logarithmic degree of a finite-point reversion). *Let $h = \sum_{n\geq 2} c_n(\Lambda)x^n \in \mathfrak{m}_0$ with $\deg_\Lambda c_n \leq d$ for all n. Then, for every $n \geq 2$,*

$$\deg_\Lambda \big[x^n\big](x + h)^{\langle -1\rangle\circ} \leq (n-1)\, d, \tag{26.26}$$

and the bound is attained for $h = ax^2\Lambda$ with $a \neq 0$, $d = 1$.

Proof. Differentiation does not raise the logarithmic degree: by Theorem 26.11 one has

$$\frac{\mathrm{d}}{\mathrm{d}x}(x^m p) = x^{m-1}(mp - p'), \qquad \deg_\Lambda(mp - p') \leq \deg_\Lambda p.$$

Multiplication adds degrees, so every block of h^r has $\deg_\Lambda \leq rd$, and hence so does every block of $\partial^{r-1}(h^r)$. By Step 1 of Theorem 26.18 with $\delta \geq 1$, $\nu_x(B_r) \geq r+1$, so only the terms with $r \leq n-1$ contribute to $[x^n](x+h)^{\langle -1\rangle\circ}$; the largest degree among them is at most $(n-1)d$. Attainment is Theorem 26.20 below, where the x^n block has degree exactly $n-1$ for $n = 2, 3, 4$; in general the $r = n-1$ term of $h = ax^2\Lambda$ contributes $\frac{(-1)^{n-1}}{(n-1)!}\partial^{n-2}\big(a^{n-1}x^{2n-2}\Lambda^{n-1}\big)$, whose Λ^{n-1} coefficient is $\frac{(-1)^{n-1}a^{n-1}}{(n-1)!}\prod_{j=0}^{n-3}(2n-2-j) \neq 0$ by (26.17) — the pure top-degree part of $\prod_j(m - j - \partial_\Lambda)\Lambda^{n-1}$ being $\prod_j(m-j)\Lambda^{n-1}$ with $m = 2n-2$ — and no other r contributes to Λ^{n-1} at level x^n. □

Example 26.20 (A Dulac reversion computed at zero and checked at infinity). Let $a \in \mathbb{K}^\times$ and $f(x) = x + ax^2\Lambda$, a tangent-to-identity Dulac germ. Here $h = ax^2\Lambda$, $\nu_x(h) = 2$, $\delta = 1$. Using (26.17) at each step:

$$\begin{aligned}
B_1 &= -h = -ax^2\Lambda,\\
B_2 &= \tfrac{1}{2}\partial\big(a^2x^4\Lambda^2\big) = \tfrac{a^2}{2}x^3\big(4\Lambda^2 - 2\Lambda\big) = a^2x^3\big(2\Lambda^2 - \Lambda\big),\\
B_3 &= -\tfrac{1}{6}\partial^2\big(a^3x^6\Lambda^3\big) = -\tfrac{a^3}{6}x^4(\partial_\Lambda - 5)(\partial_\Lambda - 6)\Lambda^3 = -\tfrac{a^3}{6}x^4\big(30\Lambda^3 - 33\Lambda^2 + 6\Lambda\big),
\end{aligned}$$

and $\nu_x(B_4) \geq 5$. Hence

$$f^{\langle -1\rangle\circ}(x) = x - ax^2\Lambda + a^2x^3\big(2\Lambda^2 - \Lambda\big) + a^3x^4\Big(-5\Lambda^3 + \tfrac{11}{2}\Lambda^2 - \Lambda\Big) + \mathcal{O}_x^{\mathrm{form}}\big(x^5\big)\,. \tag{26.27}$$

Direct verification. Write $g = x + B$ with $B = -a\Lambda x^2 + a^2(2\Lambda^2 - \Lambda)x^3 + \beta_4x^4$, $w = B/x$, and use $\Lambda \circ g = \Lambda - \log(1+w)$ from (26.21). Since g^2 has $\nu_x = 2$, only two terms of $\log(1+w)$ are needed:

$$\begin{aligned}
\log(1+w) &= -a\Lambda x + a^2\big(\tfrac{3}{2}\Lambda^2 - \Lambda\big)x^2 + \mathcal{O}_x^{\mathrm{form}}\big(x^3\big)\,,\\
g^2 &= x^2 - 2a\Lambda x^3 + a^2(5\Lambda^2 - 2\Lambda)x^4 + \mathcal{O}_x^{\mathrm{form}}\big(x^5\big)\,.
\end{aligned}$$

Multiplying,

$$a\,g^2\,(\Lambda \circ g) = a\Lambda x^2 + a^2\big(\Lambda - 2\Lambda^2\big)x^3 + a^3\big(5\Lambda^3 - \tfrac{11}{2}\Lambda^2 + \Lambda\big)x^4 + \mathcal{O}_x^{\mathrm{form}}\big(x^5\big)\,,$$

so that $f \circ g = g + ag^2(\Lambda \circ g) = x$ through order x^4 exactly when $\beta_4 = a^3\big(-5\Lambda^3 + \frac{11}{2}\Lambda^2 - \Lambda\big)$, which is (26.27).

The same germ at infinity. Conjugating by $j(x) = 1/x$ as in Theorem 15.18 and writing $L = \Lambda$, the germ $F := j \circ f \circ j$ is $F(X) = 1/f(1/X)$:

$$F(X) = \frac{1}{X^{-1} + aX^{-2}L} = \frac{X}{1 + atL} = X - aL + a^2L^2t - a^3L^3t^2 + \cdots \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}.$$

Thus the single zero-side block $ax^2\Lambda$ becomes an *infinite* perturbation $A = -aL + a^2L^2t - \cdots \in \mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ at infinity: the map conjugation $f \mapsto j \circ f \circ j$ is not the coefficientwise dictionary σ, even though σ is a term-by-term relabelling on series. Reversion at infinity now proceeds by (18.13), with D_X and the shift operators $\Delta_{n,k}$ – a different computation with a different summability count, whose answer is $j \circ f^{\langle -1\rangle\circ} \circ j$. This is the concrete form of Theorem 26.15.

Remark 26.21 (The source formula for tangent-to-identity Dulac germs). S1 computes, for $f(x) = x + ax^\alpha\ell^m$ with $\ell = -1/\log x = \Lambda^{-1}$ and $\alpha > 1$,

$$f^{\langle -1\rangle\circ}(x) = x - ax^\alpha\ell^m + a^2x^{2\alpha-1}\big(\alpha\ell^{2m} + m\ell^{2m+1}\big) + \cdots .$$

Here $h = ax^\alpha\ell^m$ has $\nu_x(h) = \alpha > 1$ but is not in $\mathbb{K}[\Lambda][[x]]$ when $m > 0$; the applicable form of Theorem 26.18 is therefore its last clause, in the Hahn enlargement with real x-exponents and Laurent Λ-coefficients. Granting that, the formula agrees with (26.22) at $r \le 2$: from $\mathrm{d}\ell/\mathrm{d}x = \ell^2/x$ one gets $h' = ax^{\alpha-1}(\alpha\ell^m + m\ell^{m+1})$ and $\frac{1}{2}\partial(h^2) = hh'$. In our generator the same statement reads $f^{\langle -1\rangle\circ} = x - ax^\alpha\Lambda^{-m} + a^2x^{2\alpha-1}\big(\alpha\Lambda^{-2m} + m\Lambda^{-2m-1}\big) + \cdots$: the exponents of the logarithm are the *negatives* of S1's. Note also that the dominant logarithmic term of the $x^{2\alpha-1}$ block is $\alpha\ell^{2m}$ when $m > 0$, since $\ell \to 0$; in Λ it is $\alpha\Lambda^{-2m}$, again the dominant one, so the ordering statement survives the translation even though every exponent changes sign. This is Theorem 26.14 (T8) in action.

26.5 Dulac series, Dulac germs, and how the support condition differs

Two distinct conventions travel under the name "power–log series near zero". The older one, of Dulac and of the finiteness literature, keeps *polynomial* coefficients in the large logarithm and enlarges the exponents of x to a set of reals increasing to $+\infty$. The newer one, of the normal-form literature [23, 24, 27, 26], keeps x-exponents real and takes real exponents of the *small* logarithm $\ell = \Lambda^{-1}$. They are not the same class, they are not nested, and their intersection is small. We define both and prove the comparison.

Definition 26.22 (Dulac series)**.** A set $S \subseteq \mathbb{R}$ is *left-finite* if $S \cap (-\infty, M)$ is finite for every $M \in \mathbb{R}$; equivalently (Theorem 30.6(i)), S is finite or can be enumerated as a strictly increasing sequence $\alpha_0 < \alpha_1 < \cdots \to +\infty$. A *Dulac series* over $\mathbb{K}$ is a formal expression

$$\hat f = \sum_{\alpha\in S} x^\alpha p_\alpha(\Lambda), \qquad S \subseteq \mathbb{R} \text{ left-finite}, \quad p_\alpha \in \mathbb{K}[\Lambda] \setminus \{0\}, \tag{26.28}$$

and $\mathfrak{D}(\mathbb{K})$ denotes the set of these. For $\hat f \neq 0$ and $N \in \mathbb{R}$ we write

$$\nu_x(\hat f) = \min S, \qquad \mathrm{Trunc}_{<N}\,\hat f = \sum_{\alpha<N} x^\alpha p_\alpha(\Lambda);$$

both are meaningful because S is left-finite. For a sub-semigroup $\Gamma \subseteq (\mathbb{R}, +)$ and $\alpha_0 \in \mathbb{R}$ we say that $\hat f$ is *carried by the grid* $\alpha_0 + \Gamma$ if $S \subseteq \alpha_0 + \Gamma$; this is Theorem 30.1(c). Finally $\mathfrak{D}_{>0}(\mathbb{K}) := \Big\{\hat f \in \mathfrak{D}(\mathbb{K}) : \nu_x(\hat f) > 0\Big\}$.

Lemma 26.23 (Finitely generated positive semigroups are left-finite)**.** *Let $g_1, \ldots, g_k \in \mathbb{R}_{>0}$ and let $\Gamma = \{n_1g_1 + \cdots + n_kg_k : n_i \in \mathbb{N}_0\}$. Then Γ is left-finite, and every $\gamma \in \Gamma$ has only finitely many representations. Consequently every grid $\alpha_0 + \Gamma$ is left-finite, and a sum of two such grids is again one.*

Proof. Put $g_{\min} = \min_i g_i > 0$. If $\sum n_ig_i \le M$ then $g_{\min}\sum n_i \le M$, so $\sum n_i \le M/g_{\min}$; there are only finitely many tuples $(n_1, \ldots, n_k) \in \mathbb{N}_0^k$ with bounded sum, which proves both assertions at once. Translation preserves left-finiteness, and $(\alpha_0 + \Gamma) + (\alpha_0' + \Gamma') = (\alpha_0 + \alpha_0') + \Gamma''$ with Γ'' generated by the union of the two generating sets. ☐

Proposition 26.24 ($\mathfrak{D}(\mathbb{K})$ is a complete valued domain)**.** $\mathfrak{D}(\mathbb{K})$ *is a commutative* $\mathbb{K}$*-algebra under the operations*

$$\hat{f}+\hat{g}=\sum_{\alpha}x^{\alpha}\big(p_\alpha+q_\alpha\big),\qquad \hat{f}\hat{g}=\sum_{\gamma}x^{\gamma}\sum_{\alpha+\beta=\gamma}p_\alpha q_\beta,$$

both well defined; it is an integral domain; ν_x *is a valuation on it with value group* $\mathbb{R}$*; it is complete and separated for the filtration by* $\{\nu_x\ge N\}$*; and*

$$\hat{f}\in\mathfrak{D}(\mathbb{K})^{\times}\iff p_{\nu_x(\hat{f})}\in\mathbb{K}^{\times},\tag{26.29}$$

i.e. exactly when the leading coefficient block is a nonzero scalar — the verbatim mirror of Theorem 9.3.

Proof. Well-definedness of the sum is clear (a union of two left-finite sets is left-finite). For the product, Theorem 30.6(iii) gives that $S+T$ is left-finite and that every fiber $\{(\alpha,\beta)\in S\times T:\alpha+\beta=\gamma\}$ is finite, so each inner sum is a finite sum in $\mathbb{K}[\Lambda]$. Associativity, distributivity and the $\mathbb{K}$-algebra axioms are then verified fiberwise.

Leading blocks multiply: $[x^{\nu_x\hat{f}+\nu_x\hat{g}}](\hat{f}\hat{g})=p_{\nu_x\hat{f}}q_{\nu_x\hat{g}}\ne0$ because $\mathbb{K}[\Lambda]$ is a domain. This gives both integrality and $\nu_x(\hat{f}\hat{g})=\nu_x\hat{f}+\nu_x\hat{g}$; the ultrametric inequality is immediate.

Separatedness is clear. For completeness, let $(\hat{f}_k)$ be Cauchy: for every M there is k_M with $\nu_x(\hat{f}_k-\hat{f}_l)\ge M$ for $k,l\ge k_M$. Then for each M the truncations $\operatorname{Trunc}_{<M}\hat{f}_k$ are eventually constant; define $\hat{f}$ by these stable truncations. Its support below any M equals the support of $\operatorname{Trunc}_{<M}\hat{f}_{k_M}$, which is finite, so the support of $\hat{f}$ is left-finite and $\hat{f}\in\mathfrak{D}(\mathbb{K})$, with $\hat{f}_k\to\hat{f}$.

For (26.29): if $\hat{f}\hat{g}=1$ then the leading blocks multiply to 1 in $\mathbb{K}[\Lambda]$, whose units are $\mathbb{K}^{\times}$. Conversely if $p_{\alpha_0}=c\in\mathbb{K}^{\times}$ with $\alpha_0=\nu_x\hat{f}$, write $\hat{f}=cx^{\alpha_0}(1+u)$ with $\nu_x(u)>0$; the support of u is left-finite and contained in $\mathbb{R}_{>0}$, so by Theorem 30.6(iv) the semigroup it generates is left-finite, the family $(-u)^k$ is summable, and $\hat{f}^{-1}=c^{-1}x^{-\alpha_0}\sum_{k\ge0}(-u)^k\in\mathfrak{D}(\mathbb{K})$. □

Theorem 26.25 (Dulac series against the classes of this volume)**.** *Inside* $\mathfrak{D}(\mathbb{K})$*, with* σ *as in* (26.6)*:*

(i) $\sigma(\mathscr{PL}_{\mathrm{poly},\mathbb{K}})=\Big\{\hat{f}\in\mathfrak{D}(\mathbb{K}):S\subseteq\mathbb{Z}\Big\}$ *and* $\sigma\big(\mathbb{K}[L]((t^{1/s}))\big)=\Big\{\hat{f}:S\subseteq s^{-1}\mathbb{Z}\Big\}$*; the inclusions*

$$\sigma(\mathscr{PL}_{\mathrm{poly},\mathbb{K}})\subsetneq\bigcup_{s\ge1}\sigma\big(\mathbb{K}[L]((t^{1/s}))\big)\subsetneq\mathfrak{D}(\mathbb{K})$$

are strict, the second witnessed by $x^{\sqrt2}$*.*

(ii) $\mathfrak{D}(\mathbb{K})$ *is* strictly *contained in the set of elements of* $\sigma\big(\mathcal{H}_1(\mathbb{K},\mathbb{R})\big)$ *all of whose coefficient blocks are polynomials in* Λ*: the Hahn condition on such an element is well-basedness of its* x*-support, which is strictly weaker than left-finiteness once the exponent group is dense. The series* $\sum_{k\ge1}x^{1-1/k}$ *has well-based* x*-support of order type* ω *and is not a Dulac series.*

(iii) *For* $S\subseteq\mathbb{Z}$ *— and more generally for* $S\subseteq\Gamma$ *with* $\Gamma\subseteq\mathbb{R}$ *a discrete subgroup — the three conditions "bounded below", "left-finite" and "well-based" coincide, which is Theorem 5.15. This is the exact sense in which the support condition defining* $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ *and the one defining* $\mathfrak{D}(\mathbb{K})$ *agree, and* (ii) *is the exact sense in which they differ.*

Proof. (i) By Theorem 26.3(i), σ sends $\sum_{n\ge n_0}p_n(L)t^n$ to $\sum_{n\ge n_0}p_n(\Lambda)x^n$, whose support is a subset of $\mathbb{Z}$ bounded below, hence left-finite; conversely a left-finite $S\subseteq\mathbb{Z}$ is bounded below and $\sum_{\alpha\in S}x^\alpha p_\alpha$ is the σ-image of the corresponding element of $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$. The same argument with $s^{-1}\mathbb{Z}$ gives the Puiseux case. Strictness of the first inclusion: $x^{1/2}$. Of the second: $x^{\sqrt2}$ lies in $\mathfrak{D}(\mathbb{K})$ but its exponent lies in no $s^{-1}\mathbb{Z}$.

(ii) The displayed series has support $\{1-1/k:k\ge1\}$, which is well ordered of order type ω and therefore admissible in a Hahn ring, but its intersection with $(-\infty,1)$ is infinite, so it is not left-finite;

this is Theorem 30.6(ii). Every left-finite set is well based, since a strictly decreasing sequence would have infinitely many terms below its first, so the inclusion holds and is strict.

(iii) A subset of a discrete subgroup $\Gamma \subseteq \mathbb{R}$ is bounded below if and only if it is left-finite, because Γ itself is left-finite above any bound; and bounded below is equivalent to well based in a discrete group for the same reason. $\square$

"Left-finite" is the Dulac condition, "well-based" is not

The condition $\alpha_j \to +\infty$ that every source imposes on a Dulac series is left-finiteness, not well-basedness. The distinction is invisible in $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$, where the two coincide by Theorem 26.25(iii), and is essential the moment real exponents enter: a Hahn series such as $\sum_{k\geq 1} x^{1-1/k}$ has a perfectly legitimate support, a leading term, and a well-defined product, but its truncation $\mathrm{Trunc}_{<N} \cdot$ is an infinite sum for every $N \geq 1$, and already $\mathrm{Trunc}_{<1} \cdot$ is the whole series — so every algorithm of Part 27, every finite-determination theorem, and the very notion of "computing the expansion to order N" fails for it. Dulac theory needs left-finiteness precisely because it needs those.

Definition 26.26 (Dulac germ)**.** Let $\mathbb{K} = \mathbb{R}$. A germ $f\colon (0, x_0) \to \mathbb{R}$ is a *Dulac germ* if there is $\hat{f} \in \mathfrak{D}_{>0}(\mathbb{R})$ with $f \approx_0 \hat{f}$ in the sense of Theorem 26.9. We call $\hat{f}$ the *Dulac series of* f; it is unique when it exists, and we write $\nu_x(f) := \nu_x(\hat{f}) > 0$ for its leading exponent.

Proposition 26.27 (Uniqueness, and the flat kernel)**.** *Let f, g be germs at 0^+ and $\hat{f}, \hat{g} \in \mathfrak{D}(\mathbb{R})$.*

(i) *If $f \approx_0 \hat{f}$ and $f \approx_0 \hat{g}$ then $\hat{f} = \hat{g}$.*

(ii) *$f \approx_0 0$ if and only if $f(x) = \mathcal{O}_{x\to 0^+}(x^N)$ for every N. Such germs exist and are nonzero:* $\mathrm{e}^{-1/x} \approx_0 0$.

(iii) *If $f \approx_0 \hat{f}$ and $g \approx_0 \hat{g}$ then $f + g \approx_0 \hat{f} + \hat{g}$ and $fg \approx_0 \hat{f}\hat{g}$. Consequently the germs admitting a Dulac series form a ring, the map $f \mapsto \hat{f}$ is a ring homomorphism* into $\mathfrak{D}(\mathbb{R})$, *and by* (ii) *its kernel — the flat germs — is a nonzero ideal. The map is therefore* not *injective. Whether it is surjective — a Borel–Ritt theorem for left-finite real exponent sets, of which Theorem 24.13 is the integer-exponent case at infinity — is not decided here and is used nowhere below.*

Proof. (i) Let $\hat{h} = \hat{f} - \hat{g} \neq 0$, with $\alpha_0 = \nu_x(\hat{h})$ and leading block $p \neq 0$. Because the support of $\hat{h}$ is left-finite, there is $\epsilon > 0$ with no support element in $(\alpha_0, \alpha_0 + \epsilon)$; put $N = \alpha_0 + \epsilon$, so that $\mathrm{Trunc}_{<N} \hat{h} = x^{\alpha_0} p(\Lambda)$. Subtracting the two estimates (26.16) at this N and using linearity of $\mathrm{Trunc}_{<N} \cdot$ gives $\left|\mathrm{Trunc}_{<N} \hat{h}(x)\right| \leq C x^N \Lambda^M$. But $|x^{\alpha_0} p(\Lambda)| \geq c\, x^{\alpha_0} \Lambda^{\deg p}$ for small x, with $c > 0$ half the absolute value of the leading coefficient of p, and $x^{\alpha_0}\Lambda^{\deg p}/(x^{\alpha_0+\epsilon}\Lambda^M) = x^{-\epsilon}\Lambda^{\deg p - M} \to +\infty$. Contradiction.

(ii) If $\hat{f} = 0$ then $\mathrm{Trunc}_{<N} \hat{f} = 0$ and (26.16) is exactly $|f| \leq C_N x^N \Lambda^{M_N}$, which for every N' is $\mathcal{O}_{x\to 0^+}\left(x^{N'}\right)$ upon taking $N = N' + 1$; the converse is immediate. Finally $\mathrm{e}^{-1/x} x^{-N} \to 0$ for every N.

(iii) Additivity is the triangle inequality applied to (26.16), since $\mathrm{Trunc}_{<N} \cdot$ is linear: take C_N the sum and M_N the maximum of the two data.

For multiplicativity, first note that if $\hat{h} \in \mathfrak{D}(\mathbb{R})$ is nonzero with $b = \nu_x(\hat{h})$ and $h \approx_0 \hat{h}$, then $\mathrm{Trunc}_{<N'} \hat{h}$ is, for each N', a finite sum $\sum_{b\leq\gamma<N'} x^\gamma q_\gamma(\Lambda)$, whence

$$\left|\left(\mathrm{Trunc}_{<N'} \hat{h}\right)(x)\right| \leq C\, x^b \Lambda^M, \qquad |h(x)| \leq C'\, x^b \Lambda^{M'} \tag{26.30}$$

for small x, the second by adding the remainder at $N' = b + 1$. Now let $a = \nu_x(\hat{f})$, $b = \nu_x(\hat{g})$, both finite, fix $N \in \mathbb{R}$ and put $N_1 = N - b$, $N_2 = N - a$. Then

$$fg - \left(\mathrm{Trunc}_{<N_1} \hat{f}\right)\left(\mathrm{Trunc}_{<N_2} \hat{g}\right) = \left(f - \mathrm{Trunc}_{<N_1} \hat{f}\right)g + \left(\mathrm{Trunc}_{<N_1} \hat{f}\right)\left(g - \mathrm{Trunc}_{<N_2} \hat{g}\right),$$

and by (26.16) together with (26.30) the two summands are $\mathcal{O}_{x\to 0^+}\big(x^{N_1+b}\Lambda^*\big)$ and $\mathcal{O}_{x\to 0^+}\big(x^{a+N_2}\Lambda^*\big)$, both $\mathcal{O}_{x\to 0^+}\big(x^N\Lambda^*\big)$. Finally $\big(\mathrm{Trunc}_{<N_1}\hat{f}\big)\big(\mathrm{Trunc}_{<N_2}\hat{g}\big)-\mathrm{Trunc}_{<N}\hat{f}\hat{g}$ is a finite sum of monomials $x^{\alpha+\beta}$ with $\alpha+\beta\geq N$: a pair (α,β) with $\alpha+\beta<N$, $\alpha\geq a$, $\beta\geq b$ automatically has $\alpha<N-b=N_1$ and $\beta<N-a=N_2$, so no term of $\mathrm{Trunc}_{<N}\hat{f}\hat{g}$ is missing from the product of truncations. Hence this difference is $\mathcal{O}_{x\to 0^+}\big(x^N\Lambda^*\big)$ as well, and $fg\approx_0\hat{f}\hat{g}$. If $\hat{f}=0$, then $f=\mathcal{O}_{x\to 0^+}\big(x^N\big)$ for every N by (ii) and $|g|\leq C'x^b\Lambda^{M'}$ by (26.30), so $fg\approx_0 0=\hat{f}\hat{g}$. □

A Dulac series is not a Dulac germ

In $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ the series *is* the object. At a finite point this is false in both directions. A Dulac series is typically divergent, so it defines no germ; and by Theorem 26.27(ii) a germ is not determined by its Dulac series, since exponentially flat germs are invisible to it. That the germs *arising from analytic vector fields* are nevertheless determined by their Dulac series — quasi-analyticity of the corresponding class — is a deep theorem, equivalent in strength to the non-accumulation of limit cycles on a polycycle, and is the content of the resolution of Dulac's problem by Écalle [12] and, independently, by Il'yashenko; see [27, 24] for the modern account and for the associated ε-neighbourhood invariants. *No* formal statement in this section, and no amount of algebra in $\mathfrak{D}(\mathbb{K})$, implies it.

Theorem 26.28 (Composition of Dulac series)**.** *Let* $\mathbb{K}=\mathbb{R}$. *Let* $\hat{f}=\sum_{\alpha\in S}x^\alpha p_\alpha(\Lambda)\in\mathfrak{D}(\mathbb{K})$ *and let*

$$\hat{g}=c\,x^\beta\big(1+u\big),\qquad c>0,\quad \beta>0,\quad u\in\mathfrak{D}(\mathbb{K}),\quad \nu_x(u)>0.$$

Define

$$\hat{f}\circ\hat{g}:=\sum_{\alpha\in S}c^\alpha x^{\alpha\beta}(1+u)^\alpha\,p_\alpha\big(\beta\Lambda-\log c-\log(1+u)\big),\tag{26.31}$$

where $(1+u)^\alpha=\sum_{k\geq 0}\binom{\alpha}{k}u^k$ *and* $\log(1+u)=\sum_{k\geq 1}\frac{(-1)^{k-1}}{k}u^k$. *Then:*

(i) *the family in* (26.31) *is summable and* $\hat{f}\circ\hat{g}\in\mathfrak{D}(\mathbb{K})$*, the set of* x*-exponents occurring in it satisfying*

$$\Big\{x\text{-exponents of }\hat{f}\circ\hat{g}\Big\}\subseteq\beta S+\Gamma_u,\tag{26.32}$$

Γ_u *denoting the sub-semigroup of* $\mathbb{R}_{\geq 0}$ *generated by the* x*-support of* u*;*

(ii) *if* $S\subseteq\alpha_0+\Gamma_1$ *and the support of* u *lies in a finitely generated* $\Gamma_2\subseteq\mathbb{R}_{>0}$*, with* Γ_1 *finitely generated, then* $\hat{f}\circ\hat{g}$ *is carried by the grid* $\alpha_0\beta+\Gamma_3$*, where* Γ_3 *is generated by* $\beta\cdot$ *(generators of* Γ_1*) together with the generators of* Γ_2*; in particular* Γ_3 *is finitely generated;*

(iii) $\nu_x(\hat{f}\circ\hat{g})=\beta\,\nu_x(\hat{f})$*, the leading block is* $c^{\alpha_0}p_{\alpha_0}(\beta\Lambda-\log c)$ *with* $\alpha_0=\nu_x(\hat{f})$*, and* $\deg_\Lambda$ *of the leading block equals* $\deg p_{\alpha_0}$*;*

(iv) $\hat{f}\mapsto\hat{f}\circ\hat{g}$ *is an injective* $\mathbb{K}$*-algebra homomorphism of* $\mathfrak{D}(\mathbb{K})$ *into itself, continuous for the filtration.*

Proof. Write T for the support of u; it is left-finite and contained in $\mathbb{R}_{>0}$, so by Theorem 30.6(iv) the semigroup $\Gamma_u=T^*$ is left-finite and every element of it has only finitely many representations as a sum of elements of T.

The individual summands. Since $\nu_x(u)>0$, the families $(\binom{\alpha}{k}u^k)_{k\geq 0}$ and $(\frac{(-1)^{k-1}}{k}u^k)_{k\geq 1}$ are summable in $\mathfrak{D}(\mathbb{K})$ — $\nu_x(u^k)\geq k\nu_x(u)\to+\infty$ — so $(1+u)^\alpha$ and $\log(1+u)$ are defined, with supports contained in Γ_u. Substituting the element $\beta\Lambda-\log c-\log(1+u)$ into the polynomial p_α is a finite algebraic operation, so $p_\alpha(\beta\Lambda-\log c-\log(1+u))$ lies in $\mathfrak{D}(\mathbb{K})$ with support in Γ_u and with Λ-degree $\deg p_\alpha$ at the exponent 0. Hence the α-th summand of (26.31) has support in $\alpha\beta+\Gamma_u$.

Summability. Fix $M\in\mathbb{R}$. A summand can contribute below M only if $\alpha\beta\leq M$, i.e. $\alpha\leq M/\beta$; since S is left-finite and $\beta>0$, only finitely many α qualify. Each of those contributes a left-finite set of exponents. Therefore the total support meets $(-\infty, M)$ in a finite set, the sum is defined coefficientwise, and (26.32) holds; the result lies in $\mathfrak{D}(\mathbb{K})$.

(ii) $\beta S + \Gamma_u \subseteq \beta\alpha_0 + \beta\Gamma_1 + \Gamma_2$, and $\beta\Gamma_1 + \Gamma_2 = \Gamma_3$ is generated by the displayed finite set; Theorem 26.23 makes it left-finite.

(iii) The exponent $\alpha_0\beta$ is attained only by the summand $\alpha = \alpha_0$ and, within it, only by the Γ_u-degree-zero part, which is $c^{\alpha_0} p_{\alpha_0}(\beta\Lambda - \log c)$; this is nonzero of the same Λ-degree as p_{α_0} because $\beta \neq 0$.

(iv) Additivity is clear from (26.31). For multiplicativity, write $\hat{g}^\alpha := c^\alpha x^{\alpha\beta}(1+u)^\alpha$ and $\Lambda \circ \hat{g} := \beta\Lambda - \log c - \log(1+u)$, so that (26.31) is the $\mathbb{K}$-linear extension of

$$x^\alpha \Lambda^j \longmapsto \hat{g}^\alpha \left(\Lambda \circ \hat{g}\right)^j$$

over the (summable) monomial decomposition of $\hat{f}$. This assignment is multiplicative on monomials: $c^{\alpha+\alpha'} = c^\alpha c^{\alpha'}$, and $(1+u)^\alpha(1+u)^{\alpha'} = (1+u)^{\alpha+\alpha'}$ because the coefficient of u^k in the product is $\sum_{i+j=k} \binom{\alpha}{i}\binom{\alpha'}{j} = \binom{\alpha+\alpha'}{k}$ by the Chu–Vandermonde identity, which is a polynomial identity in α, α'. Multiplicativity on all of $\mathfrak{D}(\mathbb{K})$ then follows by expanding both sides as summable double families and regrouping, which the finiteness established above licenses. Continuity and injectivity follow from (iii), which gives $\nu_x(\hat{f} \circ \hat{g}) = \beta\nu_x(\hat{f})$. □

Corollary 26.29 (The class does not depend on the analytic chart)**.** *Let φ be a germ at 0 of a real-analytic diffeomorphism with $\varphi(0) = 0$ and $\varphi'(0) > 0$. Then $\hat{f} \mapsto \hat{f} \circ \hat{\varphi}$ maps $\mathfrak{D}(\mathbb{R})$ into itself, preserves ν_x and the leading Λ-degree, and carries a grid $\alpha_0 + \Gamma$ to the grid $\alpha_0 + \Gamma'$ with Γ' generated by the generators of Γ together with 1. Consequently the notions “Dulac series”, “leading exponent” and “leading logarithmic degree” do not depend on the analytic parametrisation of the transversal used to define them; the corresponding statement for Dulac* germs *follows from the analytic composition clause quoted in Theorem 26.32.*

Proof. Write $\varphi(x) = x\psi(x)$ with ψ analytic and $c := \psi(0) = \varphi'(0) > 0$. Then $\hat{\varphi} = c\, x^1 \big(1+u\big)$ with $u = \psi(x)/c - 1 \in x\mathbb{R}[[x]]$, whose support lies in $\mathbb{N}_{>0}$; so $\beta = 1$ and Γ_2 is the semigroup generated by 1, and Theorem 26.28 applies, giving all the assertions. The last sentence follows because two analytic parametrisations of the same transversal differ by such a φ, possibly after replacing φ by $\varphi^{\langle -1 \rangle \circ}$, which is again analytic with positive derivative. □

Proposition 26.30 (The two logarithmic conventions)**.** *Recall $\ell = \Lambda^{-1}$. Then, inside the Hahn ambient $\mathcal{Z}$:*

(i) $\mathbb{K}[\Lambda] \cap \mathbb{K}[[\ell]] = \mathbb{K}$*; the Dulac coefficient ring $\mathbb{K}[\Lambda]$ and the normal-form coefficient ring $\mathbb{K}[[\ell]]$ meet only in the constants.*

(ii) *The smallest of the four classes of Theorem 26.1 containing both is $\mathbb{K}((\ell))((x)) = \sigma(\mathscr{PL}_{\mathrm{itLaur},\mathbb{K}})$, the transported iterated Laurent-log field: its coefficient ring $\mathbb{K}((\Lambda^{-1}))$ contains $\mathbb{K}[\Lambda]$ and $\mathbb{K}[[\ell]]$, and any ring containing both contains their sums.*

(iii) *Admitting real exponents of the logarithm, i.e. monomials $x^\alpha\ell^\gamma = x^\alpha\Lambda^{-\gamma}$ with $\gamma \in \mathbb{R}$ and well-based γ-support, produces the rank-two Hahn ring $\sigma\big(\mathcal{H}_1(\mathbb{K}, \mathbb{R})\big)$ refined by real logarithmic exponents; both conventions embed in it.*

(iv) *In the dominance order (26.5), $\ell^\gamma \prec \ell^{\gamma'}$ if and only if $\gamma > \gamma'$. A list of monomials $x^\alpha\ell^m$ written by increasing m is therefore written by* decreasing *size, while a list of $x^\alpha\Lambda^j$ written by increasing j is written by increasing size.*

Proof. (i) In $\mathcal{Z}$ a polynomial in Λ has support in $\{0\} \times \mathbb{N}_0 \times \{0\}$ and an element of $\mathbb{K}[[\ell]]$ has support in $\{0\} \times (-\mathbb{N}_0) \times \{0\}$; the intersection of the supports is $\{(0, 0, 0)\}$, and uniqueness of the Hahn representation gives the claim.

(ii) $\mathbb{K}((\Lambda^{-1}))$ consists of the series $\sum_{j\geq j_0} a_j\Lambda^{-j}$ with $j_0 \in \mathbb{Z}$; taking $j_0 \leq 0$ gives every polynomial in Λ, and taking $j_0 = 0$ gives $\mathbb{K}[[\ell]]$. Among $\mathbb{K}[\Lambda]$, $\mathbb{K}(\Lambda)$ and $\mathbb{K}((\Lambda^{-1}))$, the first does not contain ℓ, and the second does not contain $\sum_{j\geq 0} j!\,\ell^j$. Indeed, the Laurent expansion at $\Lambda = \infty$ of a rational function P/Q has coefficients satisfying the linear recurrence with constant coefficients determined by Q, whereas $(j!)$ satisfies none: if $\sum_{i=0}^{d} c_i(j+i)! = 0$ for all large j, division by $j!$ gives $\sum_{i=0}^{d} c_i(j+1)(j+2)\cdots(j+i) = 0$ for all large j, a polynomial identity in j whose leading coefficient is c_d, so $c_d = 0$ and, inductively, every $c_i = 0$. So $\mathbb{K}((\Lambda^{-1}))$ is the smallest of the three.

(iii) Immediate from (26.2).

(iv) $\ell^\gamma = \Lambda^{-\gamma}$, and by (26.5) $\Lambda^b \succ \Lambda^{b'}$ iff $b > b'$; put $b = -\gamma$. □

26.6 The transition map of a hyperbolic sector

The reason the class of Theorem 26.22 is the right one is a theorem about planar vector fields. We state it precisely, we prove the model case, and we mark clearly which part is quoted.

Example 26.31 (The linear saddle: an exact power law, and where the logarithm hides)**.** Consider $\dot u = \lambda_u u$, $\dot v = -\lambda_s v$ on $\mathbb{R}^2$ with $\lambda_u, \lambda_s > 0$, and put

$$r := \frac{\lambda_s}{\lambda_u} > 0$$

(the *hyperbolicity ratio*). Fix $\varepsilon > 0$ and take the transversals $\Sigma^{\mathrm{in}} = \{v = \varepsilon,\ 0 < u < \varepsilon\}$ with coordinate $x = u$ and $\Sigma^{\mathrm{out}} = \{u = \varepsilon,\ 0 < v < \varepsilon\}$ with coordinate $y = v$. The orbit through (x, ε) is $u(\theta) = x\mathrm{e}^{\lambda_u\theta}$, $v(\theta) = \varepsilon\mathrm{e}^{-\lambda_s\theta}$; it meets Σ^{out} at the transit time

$$T(x) = \frac{1}{\lambda_u}\log\frac{\varepsilon}{x} = \frac{1}{\lambda_u}\big(\Lambda + \log\varepsilon\big), \tag{26.33}$$

and the transition map is

$$D(x) = \varepsilon\mathrm{e}^{-\lambda_s T(x)} = \varepsilon^{1-r}x^r, \tag{26.34}$$

an exact power law with no logarithm. The logarithm is nevertheless present: by (26.33) the time spent near the saddle is exactly Λ/λ_u up to a constant, and it is the exact exponential cancellation in (26.34) that removes it. For an analytic saddle that is not linearisable the cancellation is only approximate, and the surviving discrepancy is what produces the polynomials p_j in Theorem 26.32.

Theorem 26.32 (Dulac; quoted, not proved here)**.** *Let $\mathcal{X}$ be a real-analytic vector field on a neighbourhood of $0 \in \mathbb{R}^2$ with a hyperbolic saddle at 0, of eigenvalues $\lambda_u > 0 > -\lambda_s$ and hyperbolicity ratio $r = \lambda_s/\lambda_u > 0$. Let Σ^{in} and Σ^{out} be analytic arcs transverse to the stable and to the unstable separatrix, analytically parametrised by $x \geq 0$ and $y \geq 0$ with the separatrix at the parameter value 0. Then the transition map $D\colon (0,\varepsilon) \to (0,\infty)$ obtained by following the flow from Σ^{in} to Σ^{out} is a Dulac germ in the sense of Theorem 26.26: there are $c > 0$, a left-finite set $\{r = \alpha_0 < \alpha_1 < \cdots\} \subseteq \mathbb{R}_{>0}$ contained in a finitely generated sub-semigroup of $\mathbb{R}_{>0}$, and polynomials $p_j \in \mathbb{R}[\Lambda]$ with $p_0 = c$ constant, such that*

$$D \approx_0 \hat D, \qquad \hat D(x) = c\,x^r + \sum_{j\geq 1} x^{\alpha_j} p_j(\Lambda). \tag{26.35}$$

Moreover the composition of two Dulac germs whose leading exponents are positive is again a Dulac germ, with Dulac series the formal composite (26.31).

Remark 26.33 (Provenance, and what we have and have not verified). Theorem 26.32 is classical: it originates with Dulac and is proved in the modern finiteness literature; the accounts we have used are [27, 24], with the normal-form machinery that makes the exponent grid explicit in [23, 26]. Three points of hygiene.

(a) We have *proved* the model case (Theorem 26.31), the closure of the formal class under the composition (26.31) (Theorem 26.28), and the chart-independence (Theorem 26.29). What is quoted is the existence of the asymptotic expansion for an analytic saddle and the analytic half of the composition statement, which requires uniform remainder control of the kind Theorem 12.35 supplies at infinity and which we have not carried out at a finite point.

(b) The sources record further structure that we quote but do not use: that the exponents may be taken in the sub-semigroup of $\mathbb{R}_{>0}$ generated by 1 and r, and that in the non-resonant case $r \notin \mathbb{Q}$ every p_j is constant, so that the expansion has no logarithms at all. We do not assert these; only the left-finiteness and the finite generation of the grid are used below, and both are consequences of the displayed statement.

(c) The polynomials are usually printed as $P_j(\ln x)$ in that literature. The translation to our normalisation is $p_j(u) = P_j(-u)$, Theorem 26.14 (T7): the degrees agree and every odd-degree coefficient changes sign.

Corollary 26.34 (The first-return map of a hyperbolic polycycle). *Let γ be a hyperbolic polycycle of a real-analytic planar vector field: a cyclically ordered finite family of hyperbolic saddles $s_1, \dots, s_k$, with ratios $r_1, \dots, r_k > 0$, joined by regular orbit arcs. Let Σ be an analytic transversal at a regular point of γ, analytically parametrised by $x \geq 0$, and let P be the first-return map on the side from which the orbits approach γ. Then P is a Dulac germ. Its Dulac series $\hat{P}$ is the formal composite*

$$\hat{P} = \hat{R}_k \circ \hat{D}_k \circ \cdots \circ \hat{R}_1 \circ \hat{D}_1,$$

carried by a finitely generated grid, with

$$\nu_x(\hat{P}) = r := \prod_{i=1}^{k} r_i, \qquad \big[x^r\big]\hat{P} \in \mathbb{R}_{>0} \text{ a constant}, \tag{26.36}$$

and a change of the analytic parametrisation of Σ conjugates $\hat{P}$ by an analytic chart change φ as in Theorem 26.29, which changes neither membership in $\mathfrak{D}(\mathbb{R})$, nor $\nu_x(\hat{P}) = r$, nor the leading logarithmic degree, nor the finite generation of the grid. In particular, if $r \neq 1$ then $\hat{P}(x) - x$ has leading term $\big([x^r]\hat{P}\big)x^r$ when $r < 1$ and $-x$ when $r > 1$; if $r = 1$ the leading block of $\hat{P} - x$ is not determined by the ratios alone and may carry a logarithm.

Proof. Each corner map D_i is a Dulac germ with leading exponent $r_i > 0$, by Theorem 26.32; each regular transition R_i is a germ of an analytic diffeomorphism fixing 0 with positive derivative, hence a Dulac germ with $\hat{R}_i \in x\mathbb{R}[[x]]$, leading exponent 1 and support in $\mathbb{N}_{>0}$ (a convergent Taylor series is $\approx_0$ itself). That P is the displayed composite of germs is the definition of the return map. By the last clause of Theorem 26.32 the composite is a Dulac germ with Dulac series the formal composite; by Theorem 26.28(i),(ii) the formal composite lies in $\mathfrak{D}(\mathbb{R})$ and is carried by a finitely generated grid, since each factor is; by Theorem 26.28(iii) applied $2k$ times the leading exponent multiplies, giving $\prod_i r_i \cdot 1^k = r$. The leading block is a positive constant by the same induction: at each stage the outer leading block is a positive constant κ and the inner leading coefficient a positive constant c, and Theorem 26.28(iii) gives the composite leading block $c^{\alpha_0}\kappa > 0$. Independence of the chart is Theorem 26.29. The final sentence compares $\nu_x(\hat{P}) = r$ with $\nu_x(x) = 1$ in the dominance order (26.5). □

Remark 26.35 (Dulac's problem, and the exact place formal algebra stops). Limit cycles accumulating on γ correspond to zeros of $P(x) - x$ accumulating at 0. If $\hat{P}(x) \neq x$, the formal algebra of Theorem 26.24 shows that $\hat{P}(x) - x$ has a leading block $x^\alpha p(\Lambda)$ with $p \neq 0$, and hence — *if* P were determined by $\hat{P}$ — that $P(x) - x$ has finitely many zeros near 0. The gap is exactly Theorem 26.27: a germ whose Dulac series is x may still oscillate, and nothing in $\mathfrak{D}(\mathbb{R})$ forbids it. Closing that gap for germs of analytic origin is Dulac's problem, solved by Écalle [12] and by Il'yashenko. The algebraic content of this section — closure of the class, the grid, the leading data — is what makes the statement of the problem precise; it is not, and cannot be, an ingredient of its solution. Compare the o-minimal route: the unary germs definable in $\mathbb{R}_{\mathrm{an,exp}}$ form a Hardy field whose asymptotic expansions live in the field of logarithmic–exponential series [31, 32], so for a germ known to be definable the quasi-analyticity is automatic; we make no claim about the definability of a general Dulac germ.

26.7 Enlargements forced at a finite point

At infinity the enlargements of Part 15 were forced by composition. At a finite point three further mechanisms operate, and the third is the one the sources treat least carefully.

Proposition 26.36 (Real and ramified exponents)**.** *Let $r > 0$ be the hyperbolicity ratio of Theorem 26.31 and let $D(x) = cx^r$ be the model transition map.*

(i) *If $r \in \mathbb{N}_{>0}$ then $\hat{D} \in \sigma(\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}})$.*

(ii) *If $r = p/q \in \mathbb{Q}_{>0}$ in lowest terms, then $\hat{D} \in \sigma\big(\mathbb{K}[L]((t^{1/q}))\big)$ and in no $\sigma\big(\mathbb{K}[L]((t^{1/s}))\big)$ with $q \nmid s$; the Puiseux-log ramification index is exactly q.*

(iii) *If $r \notin \mathbb{Q}$ then $\hat{D}$ lies in no Puiseux-log ring at all. The grid produced by composing $\hat{D}$ with analytic charts on either side is contained, by Theorem 26.29 and Theorem 26.28(ii), in the semigroup generated by r and 1; it is left-finite by Theorem 26.23, but the group $\mathbb{Z} + r\mathbb{Z}$ it generates is dense in $\mathbb{R}$, so no ramification index bounds it.*

Proof. (i) and the membership in (ii) are Theorem 26.25(i). For the sharpness in (ii): the exponent p/q lies in $s^{-1}\mathbb{Z}$ if and only if $sp/q \in \mathbb{Z}$, i.e. if and only if $q \mid sp$, i.e. — since $\gcd(p, q) = 1$ — if and only if $q \mid s$. For (iii), an irrational r lies in no $s^{-1}\mathbb{Z}$; and $\mathbb{Z} + r\mathbb{Z}$ is a non-cyclic subgroup of $\mathbb{R}$, hence dense. This is Theorem 15.2(2),(3) read through σ. ☐

Theorem 26.37 (Two ways the logarithmic degree stops being finite)**.** *Work in a Hahn ambient admitting monomials $x^a\Lambda^b$ with $a, b \in \mathbb{R}$ and well-based supports.*

(i) (Division.) $\big(x(\Lambda + 1)\big)^{-1} = x^{-1}\sum_{k\geq 0}(-1)^k\Lambda^{-k-1}$*. Its coefficient block is an infinite series in Λ^{-1}; it lies in $\sigma(\mathscr{P}\mathscr{L}_{\mathrm{itLaur},\mathbb{K}})$ and in $\sigma(\mathscr{P}\mathscr{L}_{\mathrm{rat},\mathbb{K}})$, but in no class with polynomial blocks. The* block *has ceased to be finite while every exponent is still an integer.*

(ii) (Non-integer powers.) *Let $\rho \in \mathbb{R} \setminus \mathbb{N}_0$. Then*

$$(\Lambda + 1)^\rho = \Lambda^\rho \sum_{k\geq 0} \binom{\rho}{k} \Lambda^{-k}, \qquad \binom{\rho}{k} \neq 0 \text{ for every } k \geq 0.$$

Consequently $\big(x(\Lambda+1)\big)^\rho = x^\rho\Lambda^\rho\sum_k \binom{\rho}{k}\Lambda^{-k}$ has infinitely many nonzero logarithmic coefficients, hence lies in no class with polynomial coefficient blocks: not in $\sigma(\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}})$, not in a Puiseux-log ring $\sigma\big(\mathbb{K}[L]((t^{1/s}))\big)$, and not in $\mathfrak{D}(\mathbb{K})$. If moreover $\rho \notin \mathbb{Z}$, its leading logarithmic exponent ρ is not an integer, and it then lies in neither $\sigma(\mathscr{P}\mathscr{L}_{\mathrm{rat},\mathbb{K}})$ nor $\sigma(\mathscr{P}\mathscr{L}_{\mathrm{itLaur},\mathbb{K}})$, every monomial of those two classes having logarithmic exponent in $\mathbb{Z}$; for $\rho \in \mathbb{Z}_{<0}$, by contrast, it does lie in both, as (i) *shows for $\rho = -1$. In the non-integer case the* degree *has ceased to be a natural number.*

(iii) *Neither failure is exotic. Failure* (i) *is forced by a single division, since $\Lambda + 1$ is not a unit of $\mathbb{K}[\Lambda]$ (Theorem 9.3 transported). Failure* (ii) *is forced by a single square root: the element $x(\Lambda + 1)$, a member of $\sigma(\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}})$, has by* (ii) *with $\rho = 1/2$ the square root $x^{1/2}\Lambda^{1/2}\sum_{k\geq 0}\binom{1/2}{k}\Lambda^{-k}$, which lies in no Dulac, Puiseux-log or iterated Laurent-log class.*

(iv) *Within $\sigma(\mathscr{PL}_{\mathrm{poly},\mathbb{K}})$ itself, where every block is a polynomial, the degrees need not be bounded: $\sum_{n\geq 1} x^n\Lambda^n$ has $\deg_\Lambda[x^n] = n$. No statement in this volume may assume a uniform logarithmic degree.*

Proof. (i) $(\Lambda+1)^{-1} = \Lambda^{-1}(1+\Lambda^{-1})^{-1} = \sum_{k\geq 0}(-1)^k\Lambda^{-k-1}$, the geometric series being summable in the Λ^{-1}-adic topology. All coefficients are $\pm 1 \neq 0$, so the support is infinite; membership in $\sigma(\mathscr{PL}_{\mathrm{itLaur},\mathbb{K}})$ is the definition of $\mathbb{K}((\Lambda^{-1}))$, and membership in $\sigma(\mathscr{PL}_{\mathrm{rat},\mathbb{K}})$ holds because $(\Lambda+1)^{-1} \in \mathbb{K}(\Lambda)$.

(ii) Write $\Lambda + 1 = \Lambda(1+\Lambda^{-1})$; the binomial series converges Λ^{-1}-adically and $\binom{\rho}{k} = \rho(\rho-1)\cdots(\rho-k+1)/k!$ vanishes only if $\rho \in \{0,1,\ldots,k-1\}$, which is excluded by $\rho \notin \mathbb{N}_0$. Since all these coefficients are nonzero, the single x-block at the exponent ρ is an infinite series in Λ, so the element is not in any class whose blocks are polynomials in Λ, that is, not in $\sigma(\mathscr{PL}_{\mathrm{poly},\mathbb{K}})$, not in $\sigma\big(\mathbb{K}[L]((t^{1/s}))\big)$ and not in $\mathfrak{D}(\mathbb{K})$; Hahn representations are unique, so no rewriting can repair this. The leading monomial is $x^\rho\Lambda^\rho$; if $\rho \notin \mathbb{Z}$ its logarithmic exponent is not an integer, whereas every monomial of $\sigma(\mathscr{PL}_{\mathrm{rat},\mathbb{K}})$ and of $\sigma(\mathscr{PL}_{\mathrm{itLaur},\mathbb{K}})$ has logarithmic exponent in $\mathbb{Z}$, which excludes those two classes as well. For $\rho \in \mathbb{Z}_{<0}$ no such exclusion is available, and indeed (i) places $\rho = -1$ inside both.

(iii) The reciprocal of $x(\Lambda+1)$ is not in $\sigma(\mathscr{PL}_{\mathrm{poly},\mathbb{K}})$ because $\Lambda+1$ is not a unit of $\mathbb{K}[\Lambda]$; this is Theorem 9.3. The square-root assertion is (ii) with $\rho = 1/2$: the displayed element has leading monomial $x^{1/2}\Lambda^{1/2}$, whose logarithmic exponent $1/2$ is not an integer, and its logarithmic support is infinite because $\binom{1/2}{k} \neq 0$ for all k.

(iv) Immediate; the series lies in $\mathbb{K}[\Lambda][[x]]$ because each block is a single monomial. □

Proposition 26.38 (A logarithm in the inner monomial forces Λ_2)**.** *Let $\hat g = c\,x^\beta\Lambda^\gamma(1+u)$ with $c>0$, $\beta>0$, $\gamma\in\mathbb{R}$, $\nu_x(u)>0$. Then*

$$\Lambda\circ\hat g = \log\frac{1}{\hat g} = \beta\Lambda - \gamma\Lambda_2 - \log c - \log(1+u), \tag{26.37}$$

so that for $\hat f = \sum_\alpha x^\alpha p_\alpha(\Lambda)$:

(i) *if $\gamma = 0$, Theorem 26.28 applies and $\hat f\circ\hat g$ stays in $\mathfrak{D}(\mathbb{K})$;*

(ii) *if $\gamma \neq 0$, the second logarithm Λ_2 occurs in $\hat f\circ\hat g$ if and only if $p_\alpha \notin \mathbb{K}$ for some α; and if $\alpha_0 = \nu_x(\hat f) = \min S$ satisfies $\gamma\alpha_0 \notin \mathbb{N}_0$, then a Λ-exponent outside $\mathbb{N}_0$ occurs. In the first case $\hat f\circ\hat g$ lies in no class of depth one at a finite point, that is, in no class whose monomials are $x^a\Lambda^b$; in the second it lies in no depth-one class whose logarithmic exponents are natural numbers — $\mathbb{K}[\Lambda][[x]]$, $\mathbb{K}[\Lambda]((x))$, the Puiseux-log rings $\mathbb{K}[\Lambda]((x^{1/s}))$ and $\mathfrak{D}(\mathbb{K})$ — and, if moreover $\gamma\alpha_0 \notin \mathbb{Z}$, in none of the four classes of Theorem 26.1.*

Proof. Formula (26.37) is $-\log$ of the product, using $\log x = -\Lambda$ and $\log\Lambda = \Lambda_2$. Item (i) is Theorem 26.28.

For (ii), substitute the generator images into $\hat f = \sum_\alpha x^\alpha p_\alpha(\Lambda)$:

$$\hat f\circ\hat g = \sum_{\alpha\in S} c^\alpha x^{\alpha\beta}\Lambda^{\gamma\alpha}(1+u)^\alpha\, p_\alpha\big(\beta\Lambda - \gamma\Lambda_2 - \log c - \log(1+u)\big),$$

which is summable by the argument of Theorem 26.28 verbatim, since the extra factor $\Lambda^{\gamma\alpha}$ affects no x-exponent. The α-th summand has x-support in $\alpha\beta + \Gamma_u \subseteq [\alpha\beta, +\infty)$, so the x-exponent $\alpha\beta$ receives a contribution only from summands with $\alpha' \leq \alpha$, and from the summand α itself only through the Γ_u-degree-zero part, in which the factors $(1+u)^\alpha$ and $\log(1+u)$ contribute only their constant terms 1 and 0. That part is

$$c^\alpha\Lambda^{\gamma\alpha}\, p_\alpha\big(\beta\Lambda - \gamma\Lambda_2 - \log c\big). \tag{26.38}$$

Writing $d = \deg p_\alpha$ and $a_d \neq 0$ for its leading coefficient, the coefficient of Λ_2^d in (26.38) is $c^\alpha \Lambda^{\gamma\alpha} a_d(-\gamma)^d$, which is nonzero when $d \geq 1$ because $\gamma \neq 0$; and it is 0 for every α when every p_α is a constant, in which case no summand contains Λ_2 at all, since $(1+u)^\alpha$ and $\log(1+u)$ do not. That settles the "only if". For the "if", let $\alpha_* = \min\{\alpha \in S : p_\alpha \notin \mathbb{K}\}$, which exists because S is left-finite. Every summand $\alpha' < \alpha_*$ has $p_{\alpha'} \in \mathbb{K}$ and therefore contributes no Λ_2 anywhere, while the summand α_* contributes at the x-exponent $\alpha_*\beta$ the nonzero Λ_2^d-coefficient just computed; nothing can cancel it, so Λ_2 occurs.

For the second assertion take $\alpha = \alpha_0 = \min S$. No summand $\alpha' < \alpha_0$ exists, so the block of $\hat{f} \circ \hat{g}$ at the x-exponent $\alpha_0\beta$ is exactly (26.38). Its Λ-exponents form the set $\gamma\alpha_0 + \{0, 1, \ldots, \deg p_{\alpha_0}\}$, and the exponent $\gamma\alpha_0$ itself occurs with coefficient $c^{\alpha_0} p_{\alpha_0}(-\gamma\Lambda_2 - \log c)$, a nonzero polynomial in Λ_2 because $\gamma \neq 0$ and $p_{\alpha_0} \neq 0$; so if $\gamma\alpha_0 \notin \mathbb{N}_0$ a Λ-exponent outside $\mathbb{N}_0$ occurs, and if $\gamma\alpha_0 \notin \mathbb{Z}$ one outside $\mathbb{Z}$ occurs. Finally, depth one at a finite point means support in $\{(a, b, 0)\}$ in the coordinates of (26.5), which a nonzero Λ_2-exponent contradicts; and a non-natural Λ-exponent excludes every class with $b \in \mathbb{N}_0$. Read through σ^{-1}, this is Theorem 15.2(4). □

Table 26.1: The enlargement ladder at a finite point. Every row names the smallest class containing the object produced, and the statement that forces it.

what forces it	smallest zero-side class	reference
nothing (integer exponents)	$\mathbb{K}[\Lambda]((x)) = \sigma(\mathscr{PL}_{\mathrm{poly},\mathbb{K}})$	Theorem 26.3
division by a nonconstant block	$\mathbb{K}(\Lambda)((x))$, then $\mathbb{K}((\ell))((x))$	Theorem 26.37(i)
ratio $r = p/q \in \mathbb{Q}$	$\mathbb{K}[\Lambda]((x^{1/q}))$	Theorem 26.36(ii)
ratio $r \notin \mathbb{Q}$	$\mathfrak{D}(\mathbb{K})$, grid in $\mathbb{N}_0 + r\mathbb{N}_0$	Theorem 26.36(iii)
non-integer power of a block	real Λ-exponents, rank-two Hahn	Theorem 26.37(ii)
inner monomial with Λ^γ, $\gamma \neq 0$	depth two: Λ_2 appears	Theorem 26.38
a logarithmic leading scale	the τ-image $\mathbb{K}[\Lambda_2]((\ell))$	Theorem 26.3(ii)
an exponentially flat germ	none: the expansion is 0	Theorem 26.27(ii)

Remark 26.39 (What is not claimed). Three questions are left open here, and none of them is answered by anything above.

(a) Whether the logarithmic degrees $\deg p_j$ of the Dulac series of a hyperbolic-polycycle return map grow at most linearly in j. Theorem 26.19 gives a linear bound for reversion of a perturbation of uniformly bounded logarithmic degree, and Theorem 18.21 gives the corresponding statement at infinity; neither applies to a composite of transition maps, whose logarithmic degrees are not supplied by Theorem 26.32 in any form we have verified.

(b) Whether the analytic half of Theorem 26.32 — that a composition of Dulac germs is a Dulac germ with the formally composed series — can be proved by transporting Theorem 12.35 through Theorem 26.10. The obstruction is that the latter is stated for integer exponents, and the uniform control needed for the outer expansion at real exponents has not been established here.

(c) Whether every Dulac germ arising from an analytic saddle is definable in an o-minimal expansion of the real field; see Theorem 26.35.

Section summary

At a finite point there are two substitutions, not one. The reciprocal substitution $\sigma\colon X = 1/x$ is a relabelling of $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ onto $\mathbb{K}[\Lambda]((x))$ with $\Lambda = \log(1/x)$; it preserves valuation, truncation, leading data and dominance, and carries D_X to $-x^2\,\mathrm{d}/\mathrm{d}x$, not to $\mathrm{d}/\mathrm{d}x$. The exponential substitution $\tau\colon X = \log(1/x)$ lands on the disjoint logarithmic scale $\mathbb{K}[\Lambda_2]((\ell))$ and is the only one that reaches germs such as $W_{-1}(-x)$. Every trap in transporting a theorem is an instance of one sentence: a statement mentioning neither the derivation nor the analytic meaning of a monomial transports automatically, and one mentioning either must be re-derived — as the near-zero Lagrange–Bürmann formula must, since the reversion-frame axiom $\nu(\partial f) \geq \nu(f) + 1$ becomes $\nu_x(\partial f) \geq \nu_x(f) - 1$. A Dulac series is an element of the Hahn ring in x and Λ whose x-support is *left-finite*, strictly more than the well-basedness a Hahn ring requires and exactly what finite truncation needs; over $\mathbb{Z}$ the two conditions coincide, which is why the distinction is invisible in $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$. The Dulac class is closed under composition with a positive-leading-exponent argument and under analytic changes of chart, which is what puts the first-return map of a hyperbolic polycycle in it; but the passage from a germ to its Dulac series has a nonzero kernel, and closing that gap is Dulac's problem, not algebra.

Part VIII

Algorithms, certificates, and diagnostics

Chapter 27

Data structures and finite-order algorithms

Parts I–IV established what the polynomial–logarithmic calculus *is*: which sums, products, quotients, compositions and inverses exist, and which finitely many input blocks determine a given output block. This chapter turns those finiteness theorems into routines. Nothing here is a new closure result; everything here is a contract. The mathematical content is of three kinds: the exact statement of what a truncated object denotes, the exact statement of how much precision each primitive delivers, and the exact statement of what each primitive costs.

Two conventions are in force throughout, and both are consequences of the normative catalogue rather than choices of this chapter.

- Truncation is *exclusive*: $\operatorname{Trunc}_{<N} F = \sum_{n<N} p_n(L)t^n$. A source that writes $[F]_{\le N}$ is using the inclusive marker, and $[F]_{\le N} = \operatorname{Trunc}_{<N+1} F$. Every index bound below has been converted; the one-block discrepancy this conversion repairs is the subject of Part 29.
- The three O-roles are never conflated. In this chapter $\mathcal{O}_t^{\mathrm{form}}\left(t^N\right)$ is a statement about coefficients, $\mathcal{O}_{X\to+\infty}(\cdot)$ is a statement about functions, and $\mathcal{O}(\cdot)$ counts machine operations. A complexity bound is not an error bound and an error bound is not a complexity bound.

A common trap

The tail $t^N \mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ is an additive subgroup of $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ and a filtration step, *not* an ideal of $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$: in $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}} = \mathbb{K}[L]((t))$ the element t is a unit, so the ideal generated by t^N is the whole ring. Every statement below about "working modulo t^N" is therefore a statement about cosets of an additive subgroup, and quotient-ring language is illegitimate on the Laurent ring. Reduction modulo (t^N) is a genuine ring quotient only after one restricts to $\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$.

27.1 What a computation stores

Definition 27.1 (Sparse block representation)**.** Let R be a commutative $\mathbb{K}$-algebra with a decidable zero test and a distinguished canonical form; the cases of interest are $R = \mathbb{K}[L]$, $R = \mathbb{K}(L)$, and $R = \mathbb{K}((L^{-1}))$ truncated at a declared inner cutoff, together with the tower of Theorem 9.36. A *truncated representative* over R is a pair

$$\mathcal{T} = \left(v, N\right), \qquad N \in \mathbb{Z} \cup \{+\infty\},$$

in which v is a finitely supported map

$$v\colon \mathbb{Z} \longrightarrow R, \qquad v(n) = 0 \text{ for } n \ge N,$$

stored as the strictly increasing list of the pairs $(n, v(n))$ with $v(n) \neq 0$. The outer index n is the primary key and the internal representation of $v(n)$ — for $R = \mathbb{K}[L]$, a polynomial stored by logarithmic degree — is subordinate to it. The representative additionally carries the name of R and, when the ground field contains chosen values of logarithms or fractional powers, the branch data selecting them.

The two-level key is forced by the dominance order of Theorem 4.7: $t^n L^j \succ t^m L^k$ compares n first and j only within a fixed n. Any storage scheme that flattens (n, j) into a single total degree destroys the hierarchy, because $L^D \succ 1$ while L^D has the larger total degree; truncating by total degree therefore discards *dominant* monomials. Blocks are finite (Theorem 4.8) but their degrees are unbounded across blocks (Theorem 8.24), so the second level must be a variable-length object, never a fixed second array dimension.

Definition 27.2 (Denotation)**.** For a truncated representative $\mathcal{T} = (v, N)$ over $R = \mathbb{K}[L]$ put

$$\mathrm{Den}(\mathcal{T}) := \{F \in \mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}} : \mathrm{Trunc}_{<N} F = v\}.$$

Lemma 27.3 (The denotation is a coset of a filtration step)**.** *Let $\mathcal{T} = (v, N)$ with N finite. Then*

$$\mathrm{Den}(\mathcal{T}) = v + t^N \mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}},$$

and every $F \in \mathrm{Den}(\mathcal{T})$ satisfies $\nu_t(F) \geq m(\mathcal{T})$, where

$$m(\mathcal{T}) := \begin{cases} \nu_t(v), & v \neq 0, \\ N, & v = 0. \end{cases}$$

The bound $m(\mathcal{T})$ is attained: it equals $\min\{\nu_t(F) : F \in \mathrm{Den}(\mathcal{T})\}$.

Proof. If $\mathrm{Trunc}_{<N} F = v$ then every block of $F - v$ with index $< N$ vanishes, so $F - v \in t^N \mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$; conversely if $F = v + w$ with $w \in t^N \mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ then $\mathrm{Trunc}_{<N} F = \mathrm{Trunc}_{<N} v = v$ because v is supported below N. This proves the displayed equality. For the valuation, $\nu_t(F) \geq \min\{\nu_t(v), N\}$, and $\nu_t(v) \leq N - 1 < N$ when $v \neq 0$, which gives $\nu_t(F) \geq \nu_t(v) = m(\mathcal{T})$ with equality for $F = v$. When $v = 0$ the set is $t^N \mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$, whose minimum valuation is N, attained at t^N. □

Remark 27.4 (The order bound is derived, not stored). Theorem 27.3 shows that the valuation bound $m(\mathcal{T})$ is determined by (v, N). An implementation must therefore recompute it from the canonical form and must not carry it as an independent mutable field: a retained zero block at the nominal least exponent makes a stored bound stale, and a stale bound falsifies exactly the precision law of Theorem 27.10, where m occurs on the right-hand side.

Definition 27.5 (Relative precision)**.** The *absolute precision* of $\mathcal{T} = (v, N)$ is N; its *relative precision* is

$$\rho(\mathcal{T}) := N - m(\mathcal{T}) \in \mathbb{N}_0 \cup \{+\infty\}.$$

Thus ρ counts the number of outer blocks known from the leading one onwards, and $\rho(\mathcal{T}) = 0$ exactly when $v = 0$.

Definition 27.6 (Soundness and sharpness)**.** Let $\varphi\colon \mathcal{D} \to \mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ be a partial operation on k-tuples. A routine R that maps truncated representatives $\mathcal{T}_1, \dots, \mathcal{T}_k$ to a truncated representative $\mathsf{R}(\mathcal{T}_1, \dots, \mathcal{T}_k) = (w, N_{\mathrm{out}})$ is *sound* for φ if

$$\varphi\big(\mathrm{Den}(\mathcal{T}_1) \times \cdots \times \mathrm{Den}(\mathcal{T}_k)\big) \subseteq \mathrm{Den}\big(\mathsf{R}(\mathcal{T}_1, \dots, \mathcal{T}_k)\big)$$

whenever every tuple in the product lies in $\mathcal{D}$. It is *sharp* if in addition no larger value of N_{out} has this property, that is, if there exist two admissible tuples whose φ-images differ in the block of index N_{out}.

Soundness is the invariant that every routine must preserve. It is worth isolating the reason that it is enough to check it one primitive at a time.

Proposition 27.7 (Soundness composes)**.** *Let* R_1 *be sound for* φ_1 *and* R_2 *sound for* φ_2*, and suppose the declared output type of* R_1 *is an admissible input type of* R_2*. Then* $\mathsf{R}_2 \circ \mathsf{R}_1$ *is sound for* $\varphi_2 \circ \varphi_1$ *on the locus where the latter is defined.*

Proof. Let $F \in \mathrm{Den}(\mathcal{T})$ with $\varphi_2(\varphi_1(F))$ defined. Soundness of R_1 gives $\varphi_1(F) \in \mathrm{Den}(\mathsf{R}_1(\mathcal{T}))$. Applying soundness of R_2 to the representative $\mathsf{R}_1(\mathcal{T})$ gives $\varphi_2(\varphi_1(F)) \in \mathrm{Den}(\mathsf{R}_2(\mathsf{R}_1(\mathcal{T})))$, which is the assertion. The argument uses nothing about R_1 except that its output denotation contains $\varphi_1(F)$; in particular it is indifferent to how many blocks R_1 discarded. □

Corollary 27.8 (Truncating early is free)**.** *If every primitive in a straight-line computation is sound, then truncating each intermediate result to its own declared precision changes no block of the final result that the final declared precision advertises.*

Proof. Immediate from Theorem 27.7: the truncation of an intermediate is the passage from an element of a denotation to the representative of that denotation, and soundness of the next routine is stated for the whole denotation. □

Theorem 27.8 is the theorem behind the universal advice of all six sources to truncate after every operation. The advice is not a heuristic trade of accuracy against speed: at the declared precision the answer is bit-for-bit the same.

Definition 27.9 (The routine invariant)**.** A routine is *conforming* when it guarantees all of:

(P1) *Soundness* in the sense of Theorem 27.6, with the declared N_{out}.

(P2) *Canonicity*: the returned map has strictly increasing keys, no key $\geq N_{\mathrm{out}}$, no zero value, and every value in the canonical form of R.

(P3) *Order honesty*: the order bound is recomputed from the returned canonical form (Theorem 27.4).

(P4) *Ring honesty*: if the computation left the declared R — for instance by dividing by a nonunit block, Theorem 9.3 — the enlarged R is part of the returned type and not a silent simplification.

27.2 The precision calculus

The following theorem is the arithmetic that (P1) requires. All six sources state fragments of it; none states it sharply, and the missing sharpness is the source of the guard-block failures catalogued in Part 29.

Theorem 27.10 (Precision calculus, with sharpness)**.** *Let* $\mathcal{T}_i = (v_i, N_i)$ *be truncated representatives with finite* N_i*, order bounds* $m_i = m(\mathcal{T}_i)$ *and relative precisions* $\rho_i = N_i - m_i$*. Then the following output precisions are sound, and each is sharp in the sense of Theorem 27.6 — for* every *instance in cases 1–5, and for every instance satisfying the nondegeneracy hypothesis stated there in case 6.*

1. Sum. $N_{\mathrm{out}} = \min\{N_1, N_2\}$.
2. Product. $N_{\mathrm{out}} = \min\{N_1 + m_2,\ N_2 + m_1\}$*, with output order bound* $m_1 + m_2$.
3. Reciprocal. *If every* $F \in \mathrm{Den}(\mathcal{T}_1)$ *is a unit of* $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$*, then* $N_{\mathrm{out}} = N_1 - 2m_1$*, with output order bound* $-m_1$.
4. Quotient. *If every* $G \in \mathrm{Den}(\mathcal{T}_2)$ *is a unit, then* $N_{\mathrm{out}} = \min\{N_1 - m_2,\ N_2 + m_1 - 2m_2\}$*, with output order bound* $m_1 - m_2$.

5. Derivation. $N_{\text{out}} = N_1 + 1$ *for* D_X*, and this is sharp without exception. The output order bound* $m_1 + 1$ *is always sound; for* $v_1 \neq 0$ *it is attained if and only if* $(\partial_L - m_1)v_1(m_1) \neq 0$*, that is, unless* $m_1 = 0$ *and* $v_1(0) \in \mathbb{K}^\times$*, while for* $v_1 = 0$ *it is attained trivially, both sides being* $N_1 + 1$*; in the exceptional case the true order is strictly larger and must be recomputed from the returned canonical form (Theorem 27.4).*

6. Near-identity composition. *Let* $\mathcal{T}_1$ *represent the outer series* $F \in \mathscr{PL}_{\text{poly},\mathbb{K}}$ *and let* $\mathcal{T}_2 = (u, N_2)$ *represent* $H \in \mathscr{PL}_{\text{pow},\mathbb{K}}$*, so that the inner map is* $X + H \in \mathcal{G}_{\text{poly},\mathbb{K}}$*. Then*

$$N_{\text{out}} = \min\{N_1,\ N_2 + m_1 + 1\},$$

with output order bound m_1*. Sharpness holds for every instance in which* $N_1 \leq N_2 + m_1 + 1$*, and, when* $N_2 + m_1 + 1 < N_1$*, for every instance with* $\nu_t(D_X v_1) = m_1 + 1$*. The excluded case is real: for* $v_1 = c \in \mathbb{K}^\times$ *every admissible pair gives* $F \circ (X + H) = c + a \circ (X + H)$ *with* $\nu_t(a) \geq N_1$*, so the composite is in fact determined to the larger cutoff* N_1*, and the declared* $N_{\text{out}} = N_2 + 1$ *is sound but pessimistic.*

Proof. Write $F = v_1 + a$ and $G = v_2 + b$ with $\nu_t(a) \geq N_1$, $\nu_t(b) \geq N_2$; by Theorem 27.3 this parametrises the denotations, and $\nu_t(v_i) \geq m_i$, $N_i \geq m_i$.

1. $(F+G) - (v_1+v_2) = a+b$ has valuation at least $\min\{N_1, N_2\}$. For sharpness assume $N_1 \leq N_2$ (otherwise exchange the two arguments) and compare the admissible pairs (v_1, v_2) and $(v_1 + t^{N_1}, v_2)$: their sums differ by t^{N_1}, a nonzero block of index exactly $N_1 = N_{\text{out}}$. This witness exists for every instance, not only for $v_1 = v_2 = 0$.

2. $FG - v_1v_2 = v_1b + av_2 + ab$. The three summands have valuations at least $m_1 + N_2$, $N_1 + m_2$ and $N_1 + N_2$; since $N_i \geq m_i$ the third is not smaller than the first two, so $\nu_t(FG - v_1v_2) \geq N_{\text{out}}$. Additivity of ν_t (Theorem 9.2) gives the order bound. For sharpness one must exhibit, for the given instance, two admissible pairs whose products differ in the block of index *exactly* N_{out}. *Case* $N_1 + m_2 \leq N_2 + m_1$ *with* $v_2 \neq 0$. The admissible pairs (v_1, v_2) and $(v_1 + t^{N_1}, v_2)$ have products differing by $t^{N_1}v_2$, of valuation exactly $N_1 + \nu_t(v_2) = N_1 + m_2 = N_{\text{out}}$. *Case* $N_2 + m_1 \leq N_1 + m_2$ *with* $v_1 \neq 0$. Symmetrically the pairs (v_1, v_2) and $(v_1, v_2 + t^{N_2})$ have products differing by $t^{N_2}v_1$, of valuation exactly $N_2 + m_1 = N_{\text{out}}$. *Remaining cases.* If $v_2 = 0$ and $v_1 \neq 0$ then $m_2 = N_2$ while $m_1 \leq N_1 - 1$, so $N_2 + m_1 < N_1 + N_2 = N_1 + m_2$ and the second case applies; symmetrically if $v_1 = 0$ and $v_2 \neq 0$. If $v_1 = v_2 = 0$ then $N_{\text{out}} = N_1 + N_2$ and the pairs $(0,0)$, (t^{N_1}, t^{N_2}) have products 0 and $t^{N_1+N_2}$. Every instance is covered.

3. Here $v_1 \neq 0$ and $m_1 = \nu_t(v_1) = \nu_t(F)$. From $F^{-1} - v_1^{-1} = (v_1 - F)/(Fv_1) = -a/(Fv_1)$ and additivity of ν_t,

$$\nu_t\bigl(F^{-1} - v_1^{-1}\bigr) = \nu_t(a) - 2m_1 \geq N_1 - 2m_1.$$

Equality holds for $a = t^{N_1}$, which proves sharpness.

4. With $A = v_1 + a$ and $B = v_2 + b$,

$$\frac{A}{B} - \frac{v_1}{v_2} = \frac{av_2 - bv_1}{Bv_2},$$

whose valuation is at least $\min\{N_1 + m_2, N_2 + m_1\} - 2m_2$, which is the assertion. For sharpness note that $\nu_t(v_2) = m_2$ and $N_2 > m_2$, so $\nu_t\bigl(v_2(v_2 + t^{N_2})\bigr) = 2m_2$ exactly. Perturbing the numerator from v_1 to $v_1 + t^{N_1}$ at fixed denominator v_2 changes the quotient by t^{N_1}/v_2, of valuation exactly $N_1 - m_2$. Perturbing the denominator from v_2 to $v_2 + t^{N_2}$ at a fixed admissible numerator A changes it by $-At^{N_2}/\bigl(v_2(v_2 + t^{N_2})\bigr)$, of valuation exactly $\nu_t(A) + N_2 - 2m_2$; take $A = v_1$ if $v_1 \neq 0$ and $A = t^{N_1}$ if $v_1 = 0$, so that $\nu_t(A) = m_1$ in both cases and the valuation is $m_1 + N_2 - 2m_2$. Whichever of the two competing bounds is the smaller, the corresponding witness differs in the block of index exactly N_{out}.

5. $D_X(F - v_1) = D_X a$ and $\nu_t(D_X a) \geq \nu_t(a) + 1 \geq N_1 + 1$ (Theorem 16.4). For sharpness take $a = t^{N_1}L$, which is admissible because $\nu_t(a) = N_1$. Then $D_X a = t^{N_1+1}(\partial_L - N_1)L = t^{N_1+1}(1 - N_1 L) \neq 0$

is a nonzero block of index exactly $N_1 + 1 = N_{\text{out}}$, for every $N_1 \in \mathbb{Z}$, including $N_1 = 0$. For the order clause, the block of index $m_1 + 1$ of $D_X v_1$ is $(\partial_L - m_1)v_1(m_1)$ by the block formula; the operator $\partial_L - m_1$ is injective on $\mathbb{K}[L]$ for $m_1 \neq 0$ and has kernel exactly $\mathbb{K}$ for $m_1 = 0$, while $v_1(m_1) \neq 0$ whenever $v_1 \neq 0$. Hence that block vanishes precisely when $m_1 = 0$ and $v_1(0) \in \mathbb{K}^\times$. This — and not any failure of the cutoff law — is the algorithmic shadow of $\ker D_X = \mathbb{K}$ (Theorem 11.12).

6. Write $G = X + H$. Perturbing the outer argument by Φ with $\nu_t(\Phi) \geq N_1$ changes the value by $\Phi \circ G$, and

$$\nu_t(\Phi \circ G) = \nu_t(\Phi) \qquad (\Phi \in \mathscr{PL}_{\text{poly},\mathbb{K}},\ G \in \mathcal{G}_{\text{poly},\mathbb{K}}), \tag{27.1}$$

because the exact Taylor formula Theorem 14.5 gives $\Phi \circ G = \sum_{k \geq 0} H^k D_X^k \Phi / k!$, whose $k = 0$ term is Φ and whose $k \geq 1$ terms have valuation at least $\nu_t(\Phi) + k > \nu_t(\Phi)$; this is Theorem 16.5. Hence the outer argument contributes precision N_1, sharply, by taking $\Phi = t^{N_1}$. Perturbing the inner correction by η with $\nu_t(\eta) \geq N_2$ changes the value by

$$F \circ (X + H + \eta) - F \circ (X + H) = \sum_{k \geq 1} \frac{(H + \eta)^k - H^k}{k!} D_X^k F.$$

Since $\nu_t(H) \geq 0$ we have $\nu_t((H + \eta)^k - H^k) \geq \nu_t(\eta) \geq N_2$, while $\nu_t(D_X^k F) \geq m_1 + k$; so the k-th summand has valuation at least $N_2 + m_1 + k$, and the minimum over $k \geq 1$ is $N_2 + m_1 + 1$. For sharpness in the regime $N_2 + m_1 + 1 < N_1$, assume $\nu_t(D_X v_1) = m_1 + 1$ and take $F = v_1$, $\eta = t^{N_2}$. The $k = 1$ summand of the displayed difference is exactly $\eta\, D_X F = t^{N_2} D_X v_1$, of valuation exactly $N_2 + m_1 + 1$; for $k \geq 2$ one has $\nu_t\big((H + \eta)^k - H^k\big) \geq N_2$ and $\nu_t(D_X^k F) \geq \nu_t(D_X F) + (k - 1) = m_1 + k \geq m_1 + 2$, so all later summands have valuation at least $N_2 + m_1 + 2$. The admissible inner data H and $H + \eta$ therefore give composites differing in the block of index exactly N_{out}. Finally the output order bound is m_1 by Equation (27.1). ☐

Corollary 27.11 (Relative precision is the conserved quantity)**.** *In the notation of Theorem 27.10, and writing $h = m_2 \geq 0$ for the inner order in part 6,*

$$\rho(FG) = \min\{\rho_1, \rho_2\}, \qquad \rho\big(F^{-1}\big) = \rho_1,$$
$$\rho(A/B) = \min\{\rho_1, \rho_2\}, \qquad \rho\big(F \circ (X + H)\big) = \min\{\rho_1,\ \rho_2 + h + 1\}.$$

For the derivation, $\rho(D_X F) \leq \rho_1$; for $v_1 \neq 0$, equality holds if and only if $(\partial_L - m_1)v_1(m_1) \neq 0$, and in the exceptional case the drop can be total, as it is for $v_1 = c \in \mathbb{K}^\times$; for $v_1 = 0$ both sides are 0. For the sum, $\rho \leq \max\{\rho_1, \rho_2\}$ always, but no lower bound in terms of ρ_1, ρ_2 holds: ρ can drop to 0.

Proof. Each identity is $N_{\text{out}} - m_{\text{out}}$ computed from Theorem 27.10; for the product, the reciprocal, the quotient and the composition respectively,

$$\begin{aligned}
\min\{N_1 + m_2, N_2 + m_1\} - (m_1 + m_2) &= \min\{N_1 - m_1,\ N_2 - m_2\}, \\
(N_1 - 2m_1) - (-m_1) &= N_1 - m_1, \\
\min\{N_1 - m_2, N_2 + m_1 - 2m_2\} - (m_1 - m_2) &= \min\{N_1 - m_1,\ N_2 - m_2\}, \\
\min\{N_1, N_2 + m_1 + 1\} - m_1 &= \min\{\rho_1,\ (N_2 - h) + h + 1\}.
\end{aligned}$$

For the derivation, the output cutoff is $N_1 + 1$ and the output order is $m_{\text{out}} = \nu_t(D_X v_1) \geq m_1 + 1$, so $\rho(D_X F) = N_1 + 1 - m_{\text{out}} \leq N_1 - m_1 = \rho_1$, with equality exactly when $m_{\text{out}} = m_1 + 1$, which for $v_1 \neq 0$ is by Theorem 27.105 the condition $(\partial_L - m_1)v_1(m_1) \neq 0$; for $v_1 = c \in \mathbb{K}^\times$ the returned representative is $(0, N_1 + 1)$, of relative precision 0; and for $v_1 = 0$ one has $m_1 = N_1$ and $m_{\text{out}} = N_1 + 1 = m_1 + 1$, so $\rho = 0 = \rho_1$. For the sum, the returned representative is $\big(v_1 + v_2, \min\{N_1, N_2\}\big)$. If $v_1 + v_2 = 0$ then $\rho = 0$. Otherwise $m_{\text{out}} = \nu_t(v_1 + v_2) \geq \min\{m_1, m_2\}$, where the right-hand side is read as m_2 when $v_1 = 0$ and as m_1 when $v_2 = 0$; hence $\rho \leq \min\{N_1, N_2\} - \min\{m_1, m_2\}$, and assuming without loss of generality $N_1 \leq N_2$ this last quantity equals $N_1 - m_1 = \rho_1$ when $m_1 \leq m_2$

and is at most $N_2 - m_2 = \rho_2$ when $m_2 < m_1$. In either case $\rho \le \max\{\rho_1, \rho_2\}$. That no lower bound holds is shown by

$$F = \sum_{n=-1}^{N-1} t^n, \qquad G = -\sum_{n=-1}^{N-2} t^n,$$

both with $m = -1$ and $N_i = N$, hence $\rho_i = N + 1$; their sum is t^{N-1}, with order bound $N - 1$ and absolute precision N, so $\rho = 1$; and by $G = -F$, for which the sum is the zero representative $(0, N)$ and $\rho = 0$. □

Remark 27.12 (Addition, not multiplication, is the dangerous primitive). Theorem 27.11 inverts the usual folklore. Products, quotients and compositions transport relative precision unchanged or improved; the two primitives that can destroy it are addition, which destroys it exactly when leading blocks cancel — the formal counterpart of catastrophic cancellation — and, in the single exceptional case $(\partial_L - m_1)v_1(m_1) = 0$, the derivation, which destroys it for the same reason one block further in. A routine that reports only absolute cutoffs cannot see this loss; a routine that reports ρ sees it immediately, because ρ drops at the subtraction that caused it.

Example 27.13 (The one-block guard)**.** Let $F = X = t^{-1}$, known exactly, and let $G \in \mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ be known to exclusive precision N with $m_2 = 0$. Then Theorem 27.102 gives

$$N_{\mathrm{out}} = \min\{\infty + 0,\ N + (-1)\} = N - 1 :$$

the product is determined only one block less far than the factor. The two admissible completions $G_1 = 1$ and $G_2 = 1 + t^N$ satisfy

$$\mathrm{Trunc}_{<N} G_1 = \mathrm{Trunc}_{<N} G_2 = 1, \qquad FG_1 = t^{-1}, \qquad FG_2 = t^{-1} + t^{N-1},$$

so the two products differ in the block of index $N - 1$. Conversely, to obtain the product through block $N - 1$ inclusively one must request G through block N, that is, one guard block beyond the naive expectation, because the block of index n of FG is the block of index $n + 1$ of G.

Remark 27.14 (Why the sharp law matters). The sound-but-blunt rule "both inputs modulo t^N give the output modulo t^N" is correct only when both order bounds are ≥ 0. On the Laurent ring it is false, by Theorem 27.13, and the failure is silent: the routine returns a block it has no right to return. This is failure mode 29.2 of Part 29.

27.3 Cost models

Definition 27.15 (Two cost models)**.** In the *coefficient-operation model* one counts multiplications, additions, and inversions of units in the coefficient ring R, each at unit cost. In the *scalar model* one counts multiplications and additions in $\mathbb{K}$. Complexity statements below use $\mathcal{O}(\cdot)$ in the sense of operation counting only; the notation carries no analytic content.

The two models are not interchangeable, and the discrepancy between them is governed by the log-degree profile d_F of Theorem 8.24, which is data attached to a series and is *not* bounded by any function of the outer cutoff.

Lemma 27.16 (Scalar cost of a block-bilinear routine)**.** *Let a routine over $R = \mathbb{K}[L]$ perform a multiset $\mathcal{M}$ of coefficient multiplications $p \cdot q$, each carried out by schoolbook polynomial multiplication. Then it performs exactly*

$$\sum_{(p,q)\in\mathcal{M}} (\deg p + 1)(\deg q + 1)$$

scalar multiplications in $\mathbb{K}$, with the convention that a zero polynomial contributes the factor 0.

Proof. Schoolbook multiplication of polynomials of degrees d and e forms exactly the $(d+1)(e+1)$ products of pairs of coefficients. Summing over $\mathcal{M}$ gives the count. ☐

Proposition 27.17 (Exact scalar cost of truncated block convolution). *Let $F, G \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ and $N \geq 1$. The schoolbook algorithm computing $\operatorname{Trunc}_{<N} FG$ as $\sum_{i+j<N} p_i q_j$ performs exactly*

$$\sum_{\substack{i,j\geq 0\\ i+j<N}} \big(d_F(i)+1\big)\big(d_G(j)+1\big)$$

scalar multiplications, where $d_F(n)+1$ is read as 0 when the block p_n vanishes. In the coefficient-operation model the count is at most $\binom{N+1}{2}$ multiplications in $\mathbb{K}[L]$.

Proof. Immediate from Theorem 27.16 together with Theorem 8.5: the pairs contributing to blocks below N are exactly those with $i, j \geq 0$ and $i+j<N$, of which there are $\binom{N+1}{2}$. ☐

Corollary 27.18 (The scalar cost is not $\mathcal{O}(N^2)$).

1. *If $d_F(n), d_G(n) \leq D$ for all $n<N$, the scalar cost is at most $(D+1)^2\binom{N+1}{2}$.*
2. *If $d_F(n) = d_G(n) = n$ for all $n<N$, the scalar cost is exactly $\binom{N+3}{4}$.*
3. *No bound of the form $\mathcal{O}(N^c)$, with an absolute implied constant, holds uniformly over $\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$: already for $N=1$ the cost of multiplying $F=G=L^M$ is $(M+1)^2$, unbounded in M.*

Proof. Part (i) is immediate. For part (ii),

$$\sum_{i+j=r} (i+1)(j+1) = \sum_{k=1}^{r+1} k\,(r+2-k) = (r+2)\frac{(r+1)(r+2)}{2} - \frac{(r+1)(r+2)(2r+3)}{6} = \binom{r+3}{3},$$

using $\sum_{k=1}^{M} k = M(M+1)/2$ and $\sum_{k=1}^{M} k^2 = M(M+1)(2M+1)/6$ with $M=r+1$. Summing over $0 \leq r \leq N-1$ and applying the hockey-stick identity $\sum_{j=3}^{N+2}\binom{j}{3} = \binom{N+3}{4}$ gives the stated closed form. Part (iii) is the displayed example. ☐

A common trap

The profile of case (ii) is not artificial: up to the single block of index 0 it is the profile of the correction $(X+L)^{\langle -1\rangle_\circ} - X$, by Theorem 29.4 below, for which the exact scalar count is $\binom{N+3}{4} + N^2 + N + 1$. So the natural output of the reversion algorithms already costs $N^4/24 + \mathcal{O}(N^3)$ scalar multiplications per truncated product: the exponent is four, not the naively expected two. Every complexity statement in this section is therefore stated in the coefficient-operation model and must be multiplied by the profile factor before it describes a machine.

27.4 Multiplication

Algorithm 27.19 (Truncated block product). **Input.** Truncated representatives $\mathcal{T}_1 = (v_1, N_1)$, $\mathcal{T}_2 = (v_2, N_2)$ over R, with derived order bounds m_1, m_2.
Precondition. None beyond Theorem 27.9.
Output. (w, N) with $N = \min\{N_1+m_2, N_2+m_1\}$ and $w = \operatorname{Trunc}_{<N} v_1 v_2$.
Postcondition. $FG \in \operatorname{Den}(w, N)$ for every $F \in \operatorname{Den}(\mathcal{T}_1)$, $G \in \operatorname{Den}(\mathcal{T}_2)$, and N is the largest cutoff with this property.
Steps.

1. Compute m_1, m_2 from the canonical forms and set $N \leftarrow \min\{N_1+m_2, N_2+m_1\}$. If $N \leq m_1+m_2$ return the zero representative $(0, N)$.

2. Initialise an empty ordered map w.
3. For each stored pair $(i, v_1(i))$ and each stored pair $(j, v_2(j))$ with $i + j < N$, accumulate $w(i+j) \leftarrow w(i+j) + v_1(i)v_2(j)$.
4. Normalise every touched value in R and delete the zero values.

Cost. At most $\binom{N-m_1-m_2+1}{2}$ multiplications in R, and exactly $s_1 s_2$ minus the number of skipped pairs when the supports have s_1, s_2 stored blocks. In the scalar model the count is that of Theorem 27.17.

Correctness. Soundness and sharpness of the declared N are Theorem 27.102. That the loop computes $\mathrm{Trunc}_{<N} v_1 v_2$ is Theorem 8.5 together with the finiteness of every fiber (Theorem 8.6); no pair with $i+j \geq N$ can contribute to a retained block, so skipping such pairs is exact rather than approximate. □

```
function multiply(T1, T2):
    m1 := order(T1);  m2 := order(T2)
    N  := min(T1.cutoff + m2, T2.cutoff + m1)
    w  := empty ordered map
    for (i, a) in T1.blocks:
        if i >= N - m2: break          # keys are increasing
        for (j, b) in T2.blocks:
            if i + j >= N: break
            w[i+j] := normalize(w[i+j] + a*b)
    return canonicalize(w, N)
```

The two `break` statements are legitimate exactly because the keys are stored in increasing order; with a hash map they would be wrong, and the routine would also lose the determinism required in Section 28.5.

27.5 Reciprocal, division, and powers

Algorithm 27.20 (Reciprocal by the linear recurrence)**. Input.** $\mathcal{T} = (v, N_1)$ over R, with $m_1 = m(\mathcal{T})$.
Precondition. $v \neq 0$ and the leading block $v(m_1)$ is a unit of R. For $R = \mathbb{K}[L]$ this means $v(m_1) \in \mathbb{K}^\times$, by Theorems 9.1 and 9.3; if the test fails the routine must not proceed but must report the obstruction, per (P4) of Theorem 27.9.
Output. $(w, N_1 - 2m_1)$ with $w = \mathrm{Trunc}_{<N_1-2m_1} v^{-1}$.
Steps. Factor $v = t^{m_1}\widetilde{v}$ with $\widetilde{v} = b_0 + b_1 t + \cdots$, b_0 a unit; set $c_0 \leftarrow b_0^{-1}$ and, for $n = 1, \ldots, \rho - 1$ where $\rho = N_1 - m_1$,

$$c_n \leftarrow -b_0^{-1} \sum_{j=1}^{n} b_j c_{n-j};$$

return $t^{-m_1} \sum_{n<\rho} c_n t^n$.
Cost. Exactly $\binom{\rho}{2}$ products $b_j c_{n-j}$ in R, plus $\rho - 1$ multiplications by the precomputed b_0^{-1} and one inversion. In the scalar model, $\sum_{n=1}^{\rho-1} \sum_{j=1}^{n} (\deg b_j + 1)(\deg c_{n-j} + 1)$.

Correctness is Theorem 9.7; the precision statement is Theorem 27.103, and the coefficient-model count restates Theorem 9.32 in the present bookkeeping.

Algorithm 27.21 (Reciprocal by Newton doubling)**. Input.** As in Theorem 27.20.
Precondition. As in Theorem 27.20.
Output. The same representative as Theorem 27.20.
Steps. With $\widetilde{v}$ as above, set $C \leftarrow b_0^{-1}$, $\mu \leftarrow 1$; while $\mu < \rho$, set $\mu' \leftarrow \min\{2\mu, \rho\}$ and

$$C \leftarrow \mathrm{Trunc}_{<\mu'} C\,(2 - \widetilde{v}C), \qquad \mu \leftarrow \mu'.$$

Return $t^{-m_1}C$.
Cost. $\lceil \log_2 \rho \rceil$ iterations, the i-th using two truncated products at cutoff 2^i.

Correctness. The update squares the residual: with $E = 1 - \widetilde{v}C$,

$$1 - \widetilde{v}\,C(2 - \widetilde{v}C) = 1 - 2\widetilde{v}C + (\widetilde{v}C)^2 = E^2.$$

Hence $\nu_t(E) \geq \mu$ implies $\nu_t(E^2) \geq 2\mu$, and by Theorem 27.103 the reciprocal is then known to precision 2μ. That the iterates are exactly the truncations of the reciprocal, so that no spurious blocks are produced, is Theorem 9.29; the precision schedule and the crossover with Theorem 27.20 are Theorems 9.27, 9.33 and 9.34. □

A common trap

Newton doubling is stated for the *reciprocal*, an operation on the multiplicative structure. It has nothing to do with the compositional inverse beyond a formal analogy, and the two must not share a symbol: this volume writes F^{-1} and $F^{\langle -1\rangle_\circ}$ for the two operations and never a bare superscript -1.

Algorithm 27.22 (Truncated division). **Input.** $\mathcal{T}_1 = (v_1, N_1)$ (numerator) and $\mathcal{T}_2 = (v_2, N_2)$ (denominator).
Precondition. The leading block of v_2 is a unit of R.
Output. (w, N) with $N = \min\{N_1 - m_2,\ N_2 + m_1 - 2m_2\}$, equivalently the representative of relative precision $\min\{\rho_1, \rho_2\}$ and order $m_1 - m_2$.
Steps. Either compose Theorem 27.20 or Theorem 27.21 with Theorem 27.19, or run the direct recurrence of Theorem 9.8 on the normalised operands, which saves one temporary and ρ coefficient products.
Cost. $\binom{\rho}{2} + \mathcal{O}(\rho)$ coefficient multiplications for the direct recurrence, where $\rho = \min\{\rho_1, \rho_2\}$.

Correctness of the truncated recurrence is Theorem 9.9; that the declared relative precision is exactly right, and in particular that the numerator need not be known to a larger absolute cutoff than the denominator, is Theorem 9.11 together with Theorem 27.11. The profile of the answer is Theorem 9.15; it is in general *not* bounded by the profiles of the inputs, which is why the scalar cost of the recurrence cannot be read off from the inputs alone.

For powers, two routines are available and they are not interchangeable.

Proposition 27.23 (Power recurrence). *Let $\mathbb{K}$ have characteristic zero, let $B = \sum_{n\geq 0} b_n t^n \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ with b_0 a unit of R, and let $\alpha \in \mathbb{K}$. Define $C = B^\alpha$ by Theorem 8.37, that is, $C = b_0^\alpha(1+U)^\alpha$ with $U = b_0^{-1}B - 1$ of positive order and a chosen value $b_0^\alpha \in R$. Then $C = \sum_{n\geq 0} c_n t^n$ satisfies $c_0 = b_0^\alpha$ and, for $n \geq 1$,*

$$n\, b_0 c_n = \sum_{j=1}^{n} \big((\alpha+1)j - n\big) b_j c_{n-j}. \tag{27.2}$$

Proof. Let ∂_t be the formal partial derivative in t with L held fixed; it is an R-linear derivation of $\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$, and it is continuous for the t-adic filtration because $\partial_t(t^N \mathscr{PL}_{\mathrm{pow},\mathbb{K}}) \subseteq t^{N-1}\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$. Write $C = b_0^\alpha(1+U)^\alpha = b_0^\alpha \sum_{k\geq 0}\binom{\alpha}{k}U^k$, a summable family since $\nu_t(U^k) \geq k$ (Theorem 8.34). Applying ∂_t termwise, which is legitimate for a summable family by continuity, and using $k\binom{\alpha}{k} = \alpha\binom{\alpha-1}{k-1}$,

$$\partial_t C = b_0^\alpha \sum_{k\geq 1} k\binom{\alpha}{k} U^{k-1}\partial_t U = \alpha\, b_0^\alpha(1+U)^{\alpha-1}\partial_t U.$$

Multiplying by $B = b_0(1+U)$ and using $\partial_t B = b_0 \partial_t U$ gives the differential relation

$$B\,\partial_t C = \alpha\, C\,\partial_t B. \tag{27.3}$$

Extracting the coefficient of t^{n-1} from Equation (27.3) yields

$$\sum_{j=0}^{n}(n-j)\,b_j c_{n-j} = \alpha\sum_{j=1}^{n} j\,b_j c_{n-j},$$

that is, $nb_0c_n + \sum_{j=1}^{n}(n-j)b_jc_{n-j} = \alpha\sum_{j=1}^{n} jb_jc_{n-j}$, which rearranges to Equation (27.2). Division by n uses characteristic zero. □

Algorithm 27.24 (Powers). **Input.** $\mathcal{T} = (v, N_1)$, an exponent α.
Precondition. For $\alpha \in \mathbb{N}_0$: none. For general $\alpha \in \mathbb{K}$: $m_1 = 0$, b_0 a unit of R, and a chosen value $b_0^\alpha \in R$ recorded as branch data.
Output. Relative precision ρ_1, order αm_1 (an integer when $\alpha \in \mathbb{Z}$; otherwise the computation lives in a Puiseux extension and must be typed accordingly).
Steps. For $\alpha \in \mathbb{N}_0$ either binary powering by Theorem 27.19 or Equation (27.2). For general α, Equation (27.2) only.
Cost. Binary powering: $\mathcal{O}(\log\alpha)$ truncated products, hence $\mathcal{O}(\rho^2\log\alpha)$ coefficient multiplications. Recurrence Equation (27.2): exactly $\binom{\rho}{2}$ products b_jc_{n-j}, independent of α.

Remark 27.25. Binary powering is preferable only for very small α; the recurrence is independent of α altogether and is the *only* option for $\alpha \notin \mathbb{N}_0$, including the formal roots of Theorem 8.38. The profile of the answer is Theorem 8.31.

27.6 Logarithm and exponential of a positive-order argument

Proposition 27.26 (Coefficient recurrences for log and exp). *Let $\mathbb{K}$ have characteristic zero and let $U = \sum_{n\geq 1} u_nt^n \in t\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$.*

1. *If $\Lambda = \log(1+U) = \sum_{n\geq 1}\lambda_nt^n$, then for $n \geq 1$*

$$\lambda_n = u_n - \frac{1}{n}\sum_{j=1}^{n-1} j\,\lambda_j\,u_{n-j}. \tag{27.4}$$

2. *If $E = \exp(U) = \sum_{n\geq 0} e_nt^n$, then $e_0 = 1$ and for $n \geq 1$*

$$e_n = \frac{1}{n}\sum_{j=1}^{n} j\,u_j\,e_{n-j}. \tag{27.5}$$

Both series lie in $\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$, with $\nu_t(\Lambda) = \nu_t(U)$ and $E \in 1 + t\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$, by Theorem 12.5.

Proof. Both families $\sum_{m\geq 1}(-1)^{m+1}U^m/m$ and $\sum_{k\geq 0}U^k/k!$ are summable, since $\nu_t(U^m) \geq m$ (Theorem 8.34), so their sums are defined and ∂_t may be applied termwise by the continuity noted in the proof of Theorem 27.23. Termwise differentiation gives

$$\partial_t\Lambda = \Big(\sum_{m\geq 1}(-1)^{m+1}U^{m-1}\Big)\partial_tU = \frac{\partial_tU}{1+U}, \qquad \partial_tE = E\,\partial_tU,$$

the first because $\sum_{m\geq 1}(-1)^{m+1}U^{m-1}$ is the reciprocal of $1+U$ (Theorem 9.7). Rewrite the first as $(1+U)\partial_t\Lambda = \partial_tU$ and extract the coefficient of t^{n-1}:

$$n\lambda_n + \sum_{\substack{i\geq 1,\ j\geq 1\\ i+j=n}} u_i\,j\lambda_j = n\,u_n,$$

which is Equation (27.4) after writing $i = n-j$. From the second, the coefficient of t^{n-1} is $ne_n = \sum_{j=1}^{n} j\,u_j\,e_{n-j}$, which is Equation (27.5). Division by n uses characteristic zero. □

Algorithm 27.27 (Logarithm and exponential)**. Input.** $\mathcal{T} = (u, N)$ representing U.
Precondition. $m(\mathcal{T}) \geq 1$. The routine must *refuse* an argument of order 0: exp of a nonconstant block leaves the ring ($\exp(L) = X$), and log of a unit $b_0(1 + U)$ with $b_0 \notin \mathbb{K}$, or of an element of nonzero order, produces $\log b_0$ or a multiple of L and must be handled by the generator formulas of Theorem 12.22, not by these recurrences.
Output. Precision N in both cases; order ≥ 1 for log, order 0 for exp.
Steps. Equation (27.4) or Equation (27.5) for $n = 1, \ldots, N - 1$.
Cost. Exactly $\binom{N-1}{2}$ coefficient multiplications $\lambda_j u_{n-j}$ for Equation (27.4) and exactly $\binom{N}{2}$ coefficient multiplications $u_j e_{n-j}$ for Equation (27.5), against $\mathcal{O}(N)$ truncated products, hence $\mathcal{O}(N^3)$ coefficient multiplications, for the naive summation of the defining series.

Precision. Perturb U by η with $\nu_t(\eta) \geq N$. Then $\log(1+U+\eta) - \log(1+U) = \log\bigl(1+\eta(1+U)^{-1}\bigr)$ has order $\nu_t(\eta) \geq N$, and $\exp(U+\eta) - \exp(U) = \exp(U)\bigl(\exp(\eta) - 1\bigr)$ likewise has order $\nu_t(\eta) \geq N$. Both are sharp, as $\eta = t^N$ shows. □

27.7 The derivation and its iterates

Algorithm 27.28 (The distinguished derivation)**. Input.** $\mathcal{T} = (v, N)$ over $R = \mathbb{K}[L]$.
Precondition. R supports ∂_L.
Output. $(w, N+1)$ with $w(n+1) = (\partial_L - n)v(n)$ for every stored key n, and order bound $m + 1$.
Cost. One ∂_L and one scalar multiple per stored block: $\mathcal{O}(N)$ coefficient operations, $\mathcal{O}\bigl(\sum_n (d_F(n) + 1)\bigr)$ scalar operations. No coefficient *multiplication* is required.

Correctness is the block formula $D_X(t^n p) = t^{n+1}(\partial_L - n)p$; the precision gain and its single exception are Theorem 27.105. Iterates use the shifted operator $\Delta_{n,k} = \prod_{j=0}^{k-1}(\partial_L - n - j)$, with $\Delta_{n,0}$ the empty product and hence the identity:

$$D_X^k\bigl(t^n p(L)\bigr) = t^{n+k}\Delta_{n,k}p(L).$$

Proposition 27.29 (Cost of iterated differentiation)**.** *Computing $D_X^k F$ from F by k applications of Theorem 27.28 costs $\mathcal{O}(kN)$ coefficient operations, whereas applying $\Delta_{n,k}$ directly to a* single *block costs $\mathcal{O}(k)$. Moreover, for $p \in \mathbb{K}[L]$ and $k \geq 1$,*

$$\Delta_{n,k}p = 0 \iff p = 0, \quad \textit{or} \quad p \in \mathbb{K} \textit{ and } n \leq 0 \leq n + k - 1,$$

and $\deg \Delta_{n,k}p = \deg p$ *whenever* $n, n+1, \ldots, n+k-1$ *are all nonzero, while* $\deg \Delta_{n,k}p = \deg p - 1$ *when one of them vanishes and* $\deg p \geq 1$.

Proof. The operation counts are immediate. Each factor $\partial_L - s$ with $s \neq 0$ maps a polynomial of degree $d \geq 0$ to one of degree d, its leading coefficient multiplied by $-s$; in particular it is injective, and being degree-preserving and $\mathbb{K}$-linear on each finite-dimensional space of polynomials of degree $\leq d$ it is bijective on $\mathbb{K}[L]$. The factor ∂_L lowers the degree by one and has kernel $\mathbb{K}$. The factors commute, each being a polynomial in ∂_L. If none of $n, \ldots, n+k-1$ vanishes, $\Delta_{n,k}$ is a composite of bijections, whence injective and degree-preserving. If exactly one of them vanishes — they are distinct consecutive integers, so at most one can — write $\Delta_{n,k} = \partial_L \circ Q$ with Q the product of the remaining factors, a degree-preserving bijection mapping $\mathbb{K}$ onto $\mathbb{K}$. Then $\Delta_{n,k}p = 0$ if and only if $Qp \in \mathbb{K}$, that is, if and only if $p \in \mathbb{K}$; and for $\deg p \geq 1$ the degree of $\Delta_{n,k}p$ is $\deg Qp - 1 = \deg p - 1$. □

Antidifferentiation is the inverse problem and is needed both by integration-based algorithms and by the diagnosis of Section 29.6.

Corollary 27.30 (Antiderivative)**.** *The map $D_X : \mathscr{PL}_{\mathrm{pow},\mathbb{K}} \to t\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ is surjective and $\mathbb{K}$-linear with kernel $\mathbb{K}$. Concretely, given $G = \sum_{m\geq 1} g_m t^m$, the solutions $Y = \sum_{n\geq 0} y_n t^n$ of $D_X Y = G$ are* *obtained blockwise from* $(\partial_L - n)y_n = g_{n+1}$*; for* $n \geq 1$ *the solution is unique, and for* $n = 0$ *it is* $y_0 = \int_0^L g_1 + c$ *with* $c \in \mathbb{K}$ *arbitrary.*

Proof. The blockwise reduction is the block formula for D_X. For $n \neq 0$ the operator $\partial_L - n$ is bijective on $\mathbb{K}[L]$: it preserves the filtration by degree and acts on the $(d+1)$-dimensional space of polynomials of degree $\leq d$ by a triangular matrix with all diagonal entries $-n \neq 0$. For $n = 0$ it is ∂_L, which is surjective on $\mathbb{K}[L]$ in characteristic zero with kernel $\mathbb{K}$. Assembling the blocks gives surjectivity onto $t\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ and kernel $\mathbb{K}$. This is the ring-level form of Theorem 4.16. □

A common trap

Theorem 27.30 says that an antiderivative routine is *not* a function of its input. A routine that returns one antiderivative without recording how the resonant constant was chosen is nondeterministic in the sense of Theorem 28.24, and two correct runs may disagree by an element of $\mathbb{K}$, which is asymptotically larger than every block of positive order.

27.8 Composition

Two engines are available. The Taylor engine of Theorem 14.5 applies to a near-identity inner map; the generator engine of Theorem 12.22 applies to every admissible monomial-leading inner argument of Theorem 12.6. They compute the same thing (Theorem 12.7) and are the canonical pair of independent implementations for the differential test of Section 28.4.

Proposition 27.31 (Sharp Taylor budget, exclusive convention). *Let* $F \in \mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ *with* $\nu_t(F) \geq a$*, let* $H \in \mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ *with* $\nu_t(H) \geq h \geq 0$*, and let* $N \in \mathbb{Z}$*. Then*

$$\mathrm{Trunc}_{<N} F \circ (X + H) = \mathrm{Trunc}_{<N} \sum_{k=0}^{K-1} \frac{H^k}{k!} D_X^k F\,, \qquad K = \left\lceil \frac{N-a}{h+1} \right\rceil,$$

and no smaller K *works for every such pair* F *and* H*. Equivalently, the last retained index is* $k = \lfloor (N-1-a)/(h+1) \rfloor$*.*

Proof. By Theorem 14.5 the identity $F \circ (X+H) = \sum_{k\geq 0} H^k D_X^k F/k!$ is exact, and the family is summable because $\nu_t(H^k D_X^k F) \geq kh + a + k = a + k(h+1)$ by Theorem 16.4. A summand can influence a block of index $< N$ only if $a + k(h+1) < N$, that is, $k < (N-a)/(h+1)$. For $N > a$ the number of nonnegative integers k satisfying this is $\lceil (N-a)/(h+1) \rceil$ and the largest is $\lfloor (N-1-a)/(h+1) \rfloor$. For $N \leq a$ there is no such k, the stated K is nonpositive, the sum is empty, and both sides of the displayed identity vanish because $\nu_t\big(F \circ (X+H)\big) = \nu_t(F) \geq a \geq N$ by Equation (27.1). Sharpness: take $F = t^a L$ and $H = t^h$, so that $\nu_t F = a$ and $\nu_t H = h$. Then $D_X^k F = t^{a+k} \Delta_{a,k} L$, and $\Delta_{a,k} L \neq 0$ for every $k \geq 0$ because $L \notin \mathbb{K}$ (Theorem 27.29). Hence $H^k D_X^k F/k!$ is a nonzero multiple of $t^{a+k(h+1)}$ for every k, and the summand with $k = K-1$ contributes a nonzero block of index $a + (K-1)(h+1) < N$, while all summands with $k \geq K$ have valuation at least N. Deleting the term $k = K-1$ therefore changes the truncation. □

Algorithm 27.32 (Near-identity composition, Taylor engine). **Input.** $\mathcal{T}_F = (v_F, N_F)$, $\mathcal{T}_H = (v_H, N_H)$ with $m_H \geq 0$.
Precondition. $m_H \geq 0$, so that $X + H \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}$. An inner argument of negative order is *not* near-identity and must be routed to Theorem 27.34 after the dominant monomial has been normalised.
Output. Precision $N = \min\{N_F,\ N_H + m_F + 1\}$, order m_F.
Steps. Set $S \leftarrow 0$, $P \leftarrow 1$, $D \leftarrow v_F$, $k \leftarrow 0$. While $\nu_t(P) + \nu_t(D) < N$: set $S \leftarrow \mathrm{Trunc}_{<N} S + PD/k!$; then $D \leftarrow D_X D$, $P \leftarrow \mathrm{Trunc}_{<N-\nu_t(D)} PH$, $k \leftarrow k+1$. Return S.
Cost. $K = \lceil (N - m_F)/(m_H + 1) \rceil$ iterations, each one truncated product and one derivative: $\mathcal{O}\big(K\rho^2\big) = \mathcal{O}\big(\rho^3/(m_H+1)\big)$ coefficient multiplications, with $\rho = N - m_F$.

Precision and sharpness are Theorem 27.106; the stopping test is Theorem 27.31, implemented as the dynamic test $\nu_t(P) + \nu_t(D) \geq N$, which is never weaker than the static bound and is often much stronger because D_X may annihilate blocks. The cutoff $N - \nu_t(D)$ carried by the running power is the exact one demanded by Theorem 27.102: the discarded part of P has valuation at least $N - \nu_t(D)$, so its product with D has valuation at least N and cannot reach a retained block. For $m_F \geq 0$ this cutoff is at most N and the naive choice N is merely wasteful; for $m_F < 0$, which is the normal case when the outer series is Laurent, it is strictly larger than N and the naive choice is wrong.

For the generator engine one needs the elementary but useful observation that substituting into the logarithmic generator is a *finite* Taylor shift.

Proposition 27.33 (Polynomial shift). *Let $p \in \mathbb{K}[L]$ of degree d and let $V \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$. Then, in $\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$,*

$$p(L+V) = \sum_{k=0}^{d} \frac{(\partial_L^k p)(L)}{k!}\, V^k, \tag{27.6}$$

a sum of $d+1$ terms.

Proof. Both sides are $\mathbb{K}$-linear in p, so it suffices to take $p = L^e$ with $e \leq d$. The binomial theorem gives $p(L+V) = \sum_{k=0}^{e} \binom{e}{k} L^{e-k} V^k$, and $\partial_L^k L^e/k! = \binom{e}{k} L^{e-k}$. Terms with $k > e$ vanish on both sides. Characteristic zero is used only to divide by $k!$. □

Algorithm 27.34 (Composition by generator images). **Input.** $\mathcal{T}_F = (v_F, N_F)$; an inner argument $G = cX^{\varrho}(1+U)$ with $c \in \mathbb{K}^{\times}$, $\varrho \in \mathbb{N}_{>0}$, $U \in t\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ given by $\mathcal{T}_U = (v_U, N_U)$; a chosen value $\log c \in \mathbb{K}$.
Precondition. G admissible in the sense of Theorem 12.6; $\log c \in \mathbb{K}$, failing which $\mathbb{K}$ must first be enlarged; $\varrho \in \mathbb{N}_{>0}$, failing which the output lives in a Puiseux or Hahn extension (Theorem 12.17) and the type must say so.
Output. $\mathrm{Trunc}_{<N} F \circ G$ with

$$N = \min\{\varrho N_F,\ \varrho m_F + N_U\},$$

which is sound by Theorem 12.12: perturbing the outer argument below its cutoff moves the value by at least ϱN_F, and perturbing U below N_U moves it by at least $\varrho m_F + N_U$. The internal budget for the auxiliary powers is Theorem 12.11.
Steps.

1. $V \leftarrow \log(1+U)$ by Theorem 27.27; $R \leftarrow (1+U)^{-1}$ by Theorem 27.20 or Theorem 27.21.
2. Let $D = \max_n \deg v_F(n)$ over retained blocks and let m_F be the order bound of $\mathcal{T}_F$. Precompute $V^k/k!$ for $1 \leq k \leq \min\{D,\ N - \varrho m_F - 1\}$. The second entry of the minimum is exactly the budget of Theorem 12.11, and it cannot be lowered: the summand carrying the outer index n is $c^{-n} t^{\varrho n} R^n\, (\partial_L^k p_n)(\ldots) V^k/k!$, of valuation at least $\varrho n + k$, so it can reach a block of index $< N$ only for $k < N - \varrho n$, and $n \geq m_F$ with $\varrho \geq 1$ makes $N - \varrho m_F - 1$ the largest useful k. For $m_F \geq 0$ this cap is at most $N-1$; for $m_F < 0$ it is strictly larger, and the value $N-1$ is *wrong*, not merely wasteful.
3. Precompute R^n incrementally for the retained outer indices n (for negative n, $R^n = (1+U)^{|n|}$, an ordinary positive power).
4. Accumulate $\sum_n c^{-n} t^{\varrho n} R^n\, p_n(\varrho L + \log c + V)$ with the inner evaluation performed by Equation (27.6) applied to $p_n(\varrho L + \log c + \cdot)$, truncating after every product.

Cost. At most $\min\{D,\ N - \varrho m_F - 1\}$ truncated products for the powers of V, at most $N - m_F$ for the powers of R, and $\sum_n(\deg v_F(n) + 1)$ polynomial-by-series products for the shifts: in the coefficient-operation model $\mathcal{O}\big((N+D)N^2\big)$.

Remark 27.35 (Why the powers of V are shared). Theorem 27.33 converts every one of the $\mathcal{O}(N)$ polynomial evaluations into a fixed linear combination of the *same* powers $V^k/k!$. This is the precise content of the caching advice given by all six sources. Horner evaluation in $L+V$, by contrast, performs $\deg p_n$ truncated products *per block* and does not share work between blocks; it is asymptotically worse by a factor N.

A common trap

Composition changes *both* generators. For $G = X + H = X(1 + tH)$,

$$t \circ G = \frac{t}{1 + tH}, \qquad L \circ G = L + \log(1 + tH),$$

by Theorem 12.22. Substituting $t \mapsto t + H$, or updating one generator and not the other, is not an approximation but a different and wrong operation; see Section 29.3.

27.9 Reversion by the four strategies

Throughout this subsection $F = X + A \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}$ with $A \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$, $h = \nu_t(A) \geq 0$, and $G = F^{\langle -1 \rangle \circ} = X + B$. All four strategies compute the same B, by uniqueness (Theorem 16.11). All four are precision-neutral in the following sense, so none of them needs guard blocks in the outer argument.

Proposition 27.36 (Reversion neither gains nor loses precision). *If $F_1 = X + A_1$ and $F_2 = X + A_2$ lie in $\mathcal{G}_{\mathrm{poly},\mathbb{K}}$ with $\nu_t(A_1 - A_2) \geq N$, then $\nu_t(F_1^{\langle -1 \rangle \circ} - F_2^{\langle -1 \rangle \circ}) \geq N$. Consequently a routine that receives A at precision N may return B at precision N, and no larger cutoff is sound.*

Proof. Put $G_2 = F_2^{\langle -1 \rangle \circ}$. Then $F_1 \circ G_2 - X = (F_2 \circ G_2 - X) + (A_1 - A_2) \circ G_2 = (A_1 - A_2) \circ G_2$, whose order equals $\nu_t(A_1 - A_2) \geq N$ by Equation (27.1). By Theorem 28.5 below, applied to F_1 and the candidate G_2, $\nu_t(G_2 - F_1^{\langle -1 \rangle \circ}) = \nu_t(F_1 \circ G_2 - X) \geq N$. Sharpness: take $A_1 = A_2 + t^N$; the same computation gives $\nu_t(G_2 - F_1^{\langle -1 \rangle \circ}) = N$ exactly. The statement is the block form of Theorem 17.7. □

Algorithm 27.37 (Strategy 1: Picard iteration)**. Input.** A at precision N.
Precondition. $\nu_t(A) \geq 0$.
Output. B at precision N.
Steps. $B_0 \leftarrow 0$; $B_{r+1} \leftarrow \mathrm{Trunc}_{<r+1} - A \circ (X + B_r)$ for $r = 0, \dots, N-1$; return B_N.
Cost. N compositions, the r-th at cutoff $r + 1$.

Correctness is the fixed-point equation $B = -A \circ (X + B)$ of Theorem 16.7 together with the one-block gain of Theorem 16.9, which gives $\nu_t(B - B_r) \geq r$; the limit exists by completeness (Theorem 14.13). Picard is the cheapest strategy to implement and the most expensive to run; its virtue is that it needs nothing but composition.

Algorithm 27.38 (Strategy 2: triangular block solve)**. Input.** A at precision N.
Precondition. $\nu_t(A) \geq 0$; the leading slope of F is 1. For leading slope $c \in \mathbb{K}^\times$ divide each correction by c (Theorem 17.4).
Output. B at precision N.
Steps. $G \leftarrow X$; for $n = 0, \dots, N-1$: compute the block of index n of $\mathrm{Err}_{\mathrm{rev}}(F, G)$, call it $r_n(L)$, and set $G \leftarrow G - r_n(L)t^n$.
Cost. N residual evaluations; with the block-perturbation update of Theorem 17.1 each step touches only the blocks that the new correction can reach, and the total is one composition-sized computation rather than N.

Correctness is Theorem 17.2; that the residual block reads off the next inverse block is Theorem 16.17. This is the strategy of choice when only a few blocks are wanted, when the answer is required in expanded differential-polynomial form, or when a lazy interface must produce blocks on demand.

Algorithm 27.39 (Strategy 3: Newton reversion with doubling)**. Input.** A at precision N; an initial G correct to precision $\mu_0 \geq 1$.
Precondition. $\nu_t(A) \geq 0$; $D_X F \circ G$ is a unit, which holds automatically since it equals $1 + \mathcal{O}_t^{\text{form}}(t)$ (Theorem 17.9).
Output. B at precision N.
Steps. Set $\mu \leftarrow \mu_0$. While $\mu < N$: set $\mu' \leftarrow \min\{2\mu, N\}$ and

$$G \leftarrow \operatorname{Trunc}_{<\mu'} G - \frac{F \circ G - X}{(D_X F) \circ G}, \qquad \mu \leftarrow \mu'.$$

Recompute $\nu_t(\operatorname{Err}_{\text{rev}}(F, G))$ after each step and abort if it is smaller than μ.
Cost. $\lceil \log_2(N/\mu_0) \rceil$ stages; each stage one composition, one derivative-composition and one division at the stage cutoff.

Correctness in exact arithmetic is Theorem 17.11; the behaviour with inexact data, and hence the guard precision each stage must request, is Theorem 17.14, and the schedule is Theorem 17.16.

Algorithm 27.40 (Strategy 4: Lagrange–Bürmann streaming)**. Input.** A at precision N.
Precondition. $\nu_t(A) \geq h \geq 0$.
Output. B at precision N.
Steps. $B \leftarrow 0$, $P \leftarrow 1$; for $r = 1, 2, \ldots$: set $P \leftarrow \operatorname{Trunc}_{<N-r+1} PA$, form $Q \leftarrow D_X^{r-1} P$ by $r - 1$ applications of Theorem 27.28, and accumulate $B \leftarrow \operatorname{Trunc}_{<N} B + (-1)^r Q/r!$. Stop when $r(h+1) - 1 \geq N$, that is, after $R = \lceil (N+1)/(h+1) \rceil - 1$ terms.
Cost. R truncated products and $\mathcal{O}(R^2)$ derivative passes; memory $\mathcal{O}(N)$, since only the current power is retained.

Stopping rule. By Theorem 18.14, $B = \sum_{r \geq 1} ((-1)^r/r!) D_X^{r-1}(A^r)$. Since $\nu_t(A^r) \geq rh$ and each D_X raises the order by at least one (Theorem 16.4), $\nu_t(D_X^{r-1}(A^r)) \geq rh + r - 1 = r(h+1) - 1$. A summand can meet a block of index $< N$ only if $r(h+1) - 1 < N$, i.e. $r < (N+1)/(h+1)$, which is the stated bound. For $h = 0$ it reads $r \leq N$; for the inclusive statement "through block N" it reads $r \leq N + 1$, which is the source of the apparent one-block discrepancy discussed in Section 29.1. □

Algorithm 27.41 (Hybrid driver)**. Input.** A map F whose dominant part need not be X; a target N.
Steps.

1. Normalise: factor the dominant monomial of F and invert it exactly. If the leading exponent is not a positive integer, or if the leading derivative vanishes, stop: the problem is not near-identity reversion and the class must be changed (Section 29.9).
2. Run Theorem 27.38 for two or three blocks. This exposes the coefficient ring actually required and the correct leading slope before any expensive stage.
3. Run Theorem 27.39 with the doubling schedule to the target cutoff.
4. Compute *both* residuals of Theorem 28.5 and compare their orders.
5. If either order falls short of the target, enlarge the working cutoff and repeat from step 3; do not adjust the claimed precision instead.

Proposition 27.42 (Comparison of the four strategies)**.** *Let $C(\nu)$ denote the cost of one composition at cutoff ν in the coefficient-operation model, and assume C is nondecreasing and satisfies $C(2\nu) \geq 2C(\nu)$. Then, to reach precision N from A of order $h \geq 0$:*

1. *Picard costs* $\sum_{r\leq N} C(r)$ *and uses only composition;*
2. *the triangular solve costs* $\mathcal{O}(C(N))$ *when the block-perturbation update is used, and* $\sum_{n\leq N} C(n)$ *when it is not;*
3. *Newton costs* $\sum_{i\leq \log_2 N}\big(2C(2^i)+\mathcal{O}(4^i)\big) \leq 4C(N)+\mathcal{O}(N^2)$*;*
4. *Lagrange–Bürmann costs* $R = \lceil (N+1)/(h+1)\rceil - 1$ *truncated products and* $\mathcal{O}\big((N/(h+1))^2\big)$ *derivative passes, and uses no composition at all.*

Proof. Parts (i)–(iii) are the step counts stated with the algorithms; the estimate in (iii) uses

$$\sum_{i\leq \log_2 N} C(2^i) \leq 2C(N),$$

which follows by summing the geometric majorisation implied by $C(2\nu) \geq 2C(\nu)$, and the division at stage i costs $\mathcal{O}(4^i)$ by Theorem 27.22. Part (iv) is the stopping rule of Theorem 27.40: the r-th term needs one product to update A^r and $r-1$ derivative passes, and $\sum_{r\leq R}(r-1) = \mathcal{O}(R^2)$. ☐

Remark 27.43 (Which to use). Lagrange–Bürmann is the only strategy that never composes, and it is therefore the strategy of choice for theoretical derivations and for maps whose powers and derivatives simplify — the Lambert polynomials being the model case, as Theorem 29.4 shows. Its weakness is that its cost is insensitive to cancellation: it forms A^r for every r up to the stopping index even when most terms cancel. Newton dominates at high N, provided composition is itself efficient; Picard and the triangular solve dominate at low N and are the only ones that give blocks lazily.

Section summary

Every routine in this section returns a coset, not an element. Its contract is the triple (order bound, absolute cutoff, coefficient ring), its correctness obligation is soundness at the declared cutoff, and its cost must be quoted in a named model. Relative precision, not absolute cutoff, is the quantity that products, quotients and compositions conserve. Addition destroys it whenever leading blocks cancel; differentiation gains one block of absolute cutoff and conserves the relative precision, except in the single annihilating case $(\partial_L - m_1)v_1(m_1) = 0$, where it destroys that too.

Chapter 28

Residual certificates and reproducible verification

Part 27 says what each routine must deliver. This chapter says how one finds out whether it did. The mechanism is uniform: every operation of this volume is characterised by an equation that its output satisfies exactly, so a computed candidate can be substituted back into that equation and the valuation of the result compared with the declared cutoff. The mathematics needed to make this a proof rather than a habit is small but it is not empty, and it is exactly the following three points, each of which is made precise below.

- A residual certifies a *coset*, and only when the residual equation is *faithful* — when the valuation of the residual determines the valuation of the error. Every primitive of Part 27 that inverts a cheaper one — reciprocal, quotient, root, logarithm, exponential, composition, reversion — has a faithful equation with an explicit offset (Table 28.1), with exactly one exception, the antiderivative, and the exception is the resonant constant (Theorem 28.18). The three primitives that invert nothing — addition, multiplication, the derivation — have no entry in that table: they are certified by independent recomputation, or by the unfaithful derivative tests of Theorem 28.19.
- A residual test is a statement about coefficients in R, decided by comparison of canonical forms. Substituting numbers for L does not decide it (Theorem 28.3).
- A residual computed by the same code path that produced the answer, and in particular the residual that *drove* an iteration, certifies nothing at all (Theorem 28.7). This is the precise reason for computing both residuals of a reversion, and the reason is not that one of them is more accurate: at finite precision they certify exactly as far as each other (Theorem 28.53).

28.1 Faithful defining equations

Definition 28.1 (Defining equation, faithfulness, gauge)**.** Let φ be a partial operation on the ambient class, defined on a set $\mathcal{D}$ of input tuples with values in a declared solution class $\mathcal{Y}$. A *defining equation* for φ is a map E that assigns to an input tuple $z \in \mathcal{D}$ and a candidate $Y \in \mathcal{Y}$ an element $E(Y;z)$ of the ambient class, computable from Y and z by finitely many conforming routines, such that

$$E(Y;z) = 0 \iff Y = \varphi(z).$$

The equation is *faithful with gauge* $\kappa \in \mathbb{Z}$ if moreover

$$\nu_t E(Y;z) = \nu_t\big(Y - \varphi(z)\big) + \kappa \qquad \text{for all } z \in \mathcal{D},\ Y \in \mathcal{Y}, \tag{28.1}$$

with the convention $\nu_t 0 = +\infty$. A *residual certificate* for a computed Y at cutoff M is the pair consisting of the computed residual $E(Y;z)$ and the verified assertion $\nu_t E(Y;z) \geq M + \kappa$.

Proposition 28.2 (What a faithful certificate certifies). *Let E be faithful with gauge κ and let $M \in \mathbb{Z}$. Then, for every $Y \in \mathcal{Y}$,*

$$\nu_t E(Y;z) \geq M + \kappa \iff \nu_t\big(Y - \varphi(z)\big) \geq M \iff \operatorname{Trunc}_{<M} Y = \operatorname{Trunc}_{<M} \varphi(z).$$

In particular the certificate is an equivalence, not an implication: a candidate that passes is correct below M, and a candidate that is correct below M passes.

Proof. The first equivalence is (28.1). The second is Theorem 27.3: $\nu_t(Y - \varphi(z)) \geq M$ says exactly that $Y - \varphi(z) \in t^M \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$, which says that Y and $\varphi(z)$ have the same blocks of index $< M$. □

The gauge is not a decoration: it is the number of blocks by which the residual test must be run *beyond* the cutoff one wants to certify, and it is the commonest source of an apparently failing test on a correct answer. For the reciprocal of a series of order m it equals m, which is negative exactly when the series has a pole.

Proposition 28.3 (Exact zero testing). *Let $\mathbb{K}$ be infinite and let $R = \mathbb{K}[L]$. Let $s \geq 1$ and let $\lambda_1, \dots, \lambda_s \in \mathbb{K}$ be any finite list of sample points. Then there is a nonzero $p \in R$ of degree s with $p(\lambda_i) = 0$ for all i, namely $p = \prod_{i=1}^{s}(L - \lambda_i)$. Consequently no test that evaluates residual blocks at finitely many values of L can decide whether a residual block vanishes, and a residual test is sound only when it compares canonical forms in R.*

Proof. The displayed polynomial is monic of degree s, hence nonzero, and vanishes at each λ_i. A test that inspects only the values $p(\lambda_1), \dots, p(\lambda_s)$ cannot distinguish it from 0. □

Remark 28.4 (The three coefficient rings test differently). For $R = \mathbb{K}[L]$ the canonical form is the coefficient vector and the test is a comparison with the zero vector. For $R = \mathbb{K}(L)$ one must normalise the fraction and test the numerator, since an unnormalised numerator and denominator may share a factor; testing a rational function at sample points is subject to Theorem 28.3 twice over, once for the numerator and once for a spurious pole. For $R = \mathbb{K}((L^{-1}))$, a block is stored only to a declared *inner* cutoff, so the test decides vanishing only below that cutoff, and the inner cutoff is a second precision parameter that the type must carry and the certificate must quote (Theorem 5.22). A test that silently compares a polynomial block with a truncated Laurent block is comparing objects of two different types.

A common trap

A certificate is a statement about the pair (candidate, defining equation). It is not a statement about the routine that produced the candidate, and it is not a statement about the routine that computes the residual. If the residual is computed by the very operation whose correctness is in question, a passing test is consistent with that operation being arbitrarily wrong; see Theorem 28.7 and the engine of Theorem 28.8, which passes its own residual test identically while getting every block after the first one wrong.

28.2 The two residuals of a reversion

Throughout this subsection $F = X + A \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}$ with $A \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$, and $G \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}$ is a candidate for $F^{\langle -1\rangle\circ}$. The two residuals are the right residual $\mathrm{Err}_{\mathrm{rev}}(F,G) = F \circ G - X$ and the left residual $G \circ F - X$ of Theorem 16.14.

Theorem 28.5 (Two-sided residual certificate). *Let $F, G \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}$ and put $\Delta = G - F^{\langle -1\rangle\circ} \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$; the letter E is reserved for defining equations, as in Theorem 28.1.*

1. Both equations are faithful with gauge 0:

$$\nu_t\big(\mathrm{Err}_{\mathrm{rev}}(F,G)\big) = \nu_t\big(G \circ F - X\big) = \nu_t(\Delta),$$

and when $\Delta \neq 0$ all three series have the same leading block.

2. Certification. *For every $M \in \mathbb{N}_0$, each of $\nu_t(\mathrm{Err}_{\mathrm{rev}}(F,G)) \geq M$ and $\nu_t(G \circ F - X) \geq M$ is equivalent to $\mathrm{Trunc}_{<M}\, G - X = \mathrm{Trunc}_{<M}\, F^{\langle -1 \rangle_\circ} - X$.*

3. Attainable certification cutoff. *Suppose F and G are given by conforming representatives of the corrections A and $G - X$, with finite cutoffs N_A and N_B. Then each of the two residuals is determined by that data exactly to the cutoff $\min\{N_A, N_B\}$, and to no larger cutoff. Hence neither residual can certify G beyond block $\min\{N_A, N_B\} - 1$, and the two residuals certify exactly as far as each other.*

Proof. Part 1 is Theorem 16.15, recorded here in the form the certificate discipline uses; in the language of Theorem 28.1 it says that $E(G;F) = \mathrm{Err}_{\mathrm{rev}}(F,G)$ and $E'(G;F) = G \circ F - X$ are defining equations for $\varphi = \cdot^{\langle -1 \rangle_\circ}$ on $\mathcal{Y} = \mathcal{G}_{\mathrm{poly},\mathbb{K}}$, faithful with gauge $\kappa = 0$. Part 2 is then Theorem 28.2; it contains Theorem 16.16, which is the special case in which $G - X$ is a polynomial in t of degree $M - 1$.

For part 3, apply Theorem 27.106 to $F \circ G$: the outer argument is F, whose representative has order bound $m_F = -1$ — the block $X = t^{-1}$ — and cutoff N_A; the inner correction is $G - X$, with cutoff N_B. The declared output cutoff is therefore

$$\min\{N_A,\ N_B + m_F + 1\} = \min\{N_A, N_B\}\,,$$

and subtracting the exactly known X changes neither cutoff nor determinacy. Sharpness in the outer-limited regime is automatic; in the inner-limited regime it needs $\nu_t(D_X v_F) = m_F + 1 = 0$, which holds for every admissible F because $D_X X = 1$ and every other block of v_F has index ≥ 0, so $\nu_t(D_X v_F) = 0$. The same computation with the roles of F and G exchanged gives $\min\{N_B, N_A\}$ for the left residual. Since certification below M requires the residual to be known to be zero on all blocks of index $< M$, and blocks of index $\geq \min\{N_A, N_B\}$ are not determined by the data, no certificate can reach beyond $\min\{N_A, N_B\} - 1$. □

Remark 28.6 (Why compute both, then). Theorem 28.5 removes two folklore reasons for computing both residuals and leaves one. It is *not* true that one residual is sharper than the other: by 1 they have the same order and even the same leading block, and by 3 they are determined to the same cutoff by the same data. It is *not* true that the left residual detects an error the right one cannot: by 1 either one detects an error exactly when there is one. What remains, and what is decisive in practice, is independence: the two residuals evaluate the composition routine at arguments in the opposite order, so a composition implemented correctly in one order and incorrectly in the other is caught, and — far more importantly — an iteration driven by one of them cannot be certified by that one (Theorem 28.7).

Proposition 28.7 (A driven residual certifies nothing)**.** *Let $\odot$ be any binary operation on $\mathcal{G}_{\mathrm{poly},\mathbb{K}}$-representatives whatsoever, and suppose the triangular loop of Theorem 27.38 is run with $\odot$ in place of $\circ$, the block r_n at each step being read from $F \odot G - X$. If the loop terminates having driven the blocks of $F \odot G - X$ of index $< N$ to zero, then the assertion "$\nu_t(F \odot G - X) \geq N$" holds by construction and is independent of whether $\odot = \circ$. Consequently it carries no information about G, and in particular it does not certify $\mathrm{Trunc}_{<N}\, G - X = \mathrm{Trunc}_{<N}\, F^{\langle -1 \rangle_\circ} - X$.*

Proof. The loop's exit condition is precisely the assertion in question: the loop terminates only when the blocks of $F \odot G - X$ of index $< N$ have been made to vanish, and each step changes G so as to make one more of them vanish. Nothing in the argument used any property of $\odot$. Hence the assertion is a theorem about the loop, not about G, and it holds equally for an operation $\odot \neq \circ$, for which the conclusion $\mathrm{Trunc}_{<N}\, G - X = \mathrm{Trunc}_{<N}\, F^{\langle -1 \rangle_\circ} - X$ is false; an explicit such $\odot$ is exhibited in Theorem 28.8. □

Example 28.8 (An engine that passes its own test and is wrong from the second block on)**.** Define $\Psi_G\colon \mathscr{PL}_{\mathrm{poly},\mathbb{K}} \to \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ as the $\mathbb{K}[L]$-algebra homomorphism that substitutes the outer generator and leaves the logarithmic generator alone:

$$\Psi_G\Bigl(\sum_n p_n(L)t^n\Bigr) := \sum_n p_n(L)\bigl(G^{-1}\bigr)^n, \qquad G = X + B \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}.$$

This is well defined: $G^{-1} = t/(1+tB)$ has $\nu_t = 1$, so the family is summable (Theorem 8.34). It is the "substitute t, forget L" engine of Section 29.3. Take $F = X + L$, the map whose inverse is ω (Theorem 20.10). Then

$$F \odot G := \Psi_G(F) = \Psi_G(t^{-1}) + \Psi_G(L) = G + L,$$

so the triangular loop driven by $F \odot G - X = B + L$ sets $b_0 = -L$ and every later block to 0, terminating with

$$\widetilde{G} = X - L \qquad \text{and} \qquad F \odot \widetilde{G} - X = 0 \text{ identically.}$$

The true inverse is $\omega = X - L + Lt + (\frac{1}{2}L^2 - L)t^2 + \cdots$, so $\widetilde{G}$ is correct in the block of index 0 and wrong in every later block. The true right residual detects this at once:

$$\mathrm{Err}_{\mathrm{rev}}\Bigl(F, \widetilde{G}\Bigr) = (X - L) + \log(X - L) - X = \log(1 - Lt) = -Lt - \tfrac{1}{2}L^2t^2 - \cdots,$$

of order 1, and the true left residual is

$$\widetilde{G} \circ F - X = (X + L) - \log(X + L) - X = -\log(1 + Lt) = -Lt + \tfrac{1}{2}L^2t^2 - \cdots,$$

also of order 1, with the same leading block $-L$ as predicted by Theorem 28.51.

The engine of Theorem 28.8 is worth one more moment, because it shows which structural tests can and cannot see a wrong generator image.

Proposition 28.9 (The ring tests are blind to the generator; the chain rule is not)**.** *Let $G = X + B \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}$ with $B \neq 0$, $h = \nu_t(B) \geq 0$, and let Ψ_G be as in Theorem 28.8.*

1. *Ψ_G is a $\mathbb{K}[L]$-algebra homomorphism of $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$, continuous for the t-adic filtration. Hence it satisfies every identity built from addition, multiplication, reciprocals and $\mathbb{K}[L]$-linearity — in particular $\Psi_G(F_1F_2) = \Psi_G(F_1)\Psi_G(F_2)$ and $\Psi_G(F^{-1}) = \Psi_G(F)^{-1}$ — so no test built from those operations distinguishes it from composition.*

2. *Ψ_G fails the chain rule already on the single generator L:*

$$D_X\bigl(\Psi_G(L)\bigr) - \Psi_G(D_XL)\, D_XG = t - G^{-1}\, D_XG,$$

and this element has valuation exactly $h+2$, with block $-(\partial_L - h - 1)b_h$ there, where b_h is the leading block of B.

Proof. (i) Ψ_G is defined on generators by $t \mapsto G^{-1}$, $L \mapsto L$, and extended by the summable family in the definition; summability, since $\nu_t\bigl((G^{-1})^n\bigr) = n$, makes the extension a continuous $\mathbb{K}[L]$-algebra map, exactly as in Theorem 12.7 with the logarithmic generator held fixed. Multiplicativity and compatibility with reciprocals are formal consequences.

(ii) $\Psi_G(L) = L$ and $D_XL = t$, so the left side is $t - \Psi_G(t)D_XG = t - G^{-1}D_XG$. Now $G^{-1} = t(1+tB)^{-1}$ and $D_XG = 1 + D_XB$, so

$$G^{-1}\, D_XG = t\,(1 + D_XB)\bigl(1 - tB + t^2B^2 - \cdots\bigr) = t\Bigl[1 + (D_XB - tB) + W\Bigr],$$

where W collects the products of at least two of the small terms D_XB, tB, each of valuation $\geq h+1$; hence $\nu_tW \geq 2h+2 \geq h+2$. By the block formula the block of index $h+1$ of D_XB is $(\partial_L - h)b_h$ and that of tB is b_h, so $D_X B - tB$ has block $(\partial_L - h - 1)b_h$ at index $h+1$. Since $h+1 \neq 0$, the operator $\partial_L - h - 1$ is injective on $\mathbb{K}[L]$ (Theorem 27.29), so that block is nonzero and $\nu_t(D_X B - tB) = h+1 < h+2 \leq \nu_t W$. Therefore $t - G^{-1}D_X G = -t(D_X B - tB) - tW$ has valuation exactly $h+2$ and block $-(\partial_L - h - 1)b_h$ there. □

Remark 28.10. Theorem 28.9 explains why the chain-rule test is listed as mandatory in Section 28.4 even though it looks redundant next to the composition residual: the composition residual computed with a wrong engine is the object of Theorem 28.7, whereas the chain rule couples the engine to the derivation, which is an independent primitive (Theorem 27.28) with an independent certificate.

28.3 Certificates for the algebraic primitives

Proposition 28.11 (Quotient and reciprocal certificates)**.** *Let $B \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ be a unit, $m = \nu_t(B)$, and $A \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$. For every candidate $Q \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$,*

$$\nu_t(BQ - A) = \nu_t\Big(Q - \frac{A}{B}\Big) + m.$$

Thus $E(Q) = BQ - A$ is a defining equation for the quotient, faithful with gauge m; the case $A = 1$ is the reciprocal, with the same gauge.

Proof. $BQ - A = B\big(Q - A/B\big)$, and ν_t is additive on $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ (Theorem 9.2), with $\nu_t(B) = m$ finite because $B \neq 0$. □

Remark 28.12 (The gauge is where the sources' division test goes wrong). Two of the six sources state the division certificate in the form "residual $\mathcal{O}_t^{\mathrm{form}}\big(t^N\big)$ certifies the quotient through block $N-1$". By Theorems 28.2 and 28.11 the correct statement is that such a residual certifies the quotient through block $N - m - 1$, where $m = \nu_t(B)$. The two agree exactly when the denominator is normalised to order 0, which is the case the sources have in mind and the case an implementation should enforce before testing — but the enforcement is then part of the contract and must be checked, not assumed.

Proposition 28.13 (Logarithm and exponential certificates)**.** *Let $U \in t\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ and let ∂_t be the formal derivative in t at fixed L.*

1. *For every candidate $\Lambda \in t\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$, $\nu_t\big(\exp\Lambda - (1+U)\big) = \nu_t\big(\Lambda - \log(1+U)\big)$: gauge 0.*
2. *For every candidate $\Lambda \in t\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$, $\nu_t\big((1+U)\partial_t\Lambda - \partial_t U\big) = \nu_t\big(\Lambda - \log(1+U)\big) - 1$: gauge -1.*
3. *For every candidate $\mathcal{E} \in 1 + t\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$, $\nu_t\big(\log\mathcal{E} - U\big) = \nu_t\big(\mathcal{E} - \exp U\big)$: gauge 0.*

Proof. Write $\Delta = \Lambda - \log(1+U) \in t\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$.

(i) By Theorem 12.5, $\exp\Lambda = \exp\big(\log(1+U)\big)\exp\Delta = (1+U)\exp\Delta$, so $\exp\Lambda - (1+U) = (1+U)(\exp\Delta - 1)$. Now $\exp\Delta - 1 = \sum_{k\geq 1}\Delta^k/k!$ and $\nu_t(\Delta^k) \geq k\nu_t\Delta > \nu_t\Delta$ for $k \geq 2$ because $\nu_t\Delta \geq 1$; hence $\nu_t(\exp\Delta - 1) = \nu_t\Delta$. Multiplying by the unit $1+U$ of order 0 preserves the order.

(ii) Termwise differentiation of the summable family defining the logarithm gives $(1+U)\partial_t\log(1+U) = \partial_t U$ (proof of Theorem 27.26), so $(1+U)\partial_t\Lambda - \partial_t U = (1+U)\partial_t\Delta$. If $\Delta \neq 0$ has $\nu_t\Delta = n \geq 1$ and leading block $\delta_n \neq 0$, then $\partial_t\Delta$ has block $n\delta_n \neq 0$ at index $n-1$, using characteristic zero, and no block below; so $\nu_t(\partial_t\Delta) = n - 1$. Multiplying by $1+U$ preserves it.

(iii) With $\Theta = \mathcal{E} - \exp U$, one has $\log\mathcal{E} - U = \log\big(\mathcal{E}(\exp U)^{-1}\big) = \log\big(1 + \Theta(\exp U)^{-1}\big)$, and $\exp U$ is a unit of order 0, so the argument of the logarithm lies in $1 + t\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ and $\nu_t\log(1+\Xi) = \nu_t\Xi$ for $\Xi \in t\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ by Theorem 12.5. Hence the order equals $\nu_t\Theta$. □

Remark 28.14 (The differential certificate is cheaper, not stronger). Part (ii) of Theorem 28.13 detects an error one block earlier than part (i), and it costs one truncated product and two derivatives instead of an exponential. It does not, however, certify further. If the data are known to cutoff N, the residual of (i) is known to cutoff N and certifies $\nu_t\Delta \geq M$ for $M \leq N$; the residual of (ii) is known only to cutoff $N-1$, because ∂_t shifts the filtration by one, and its threshold is $M-1$, so it too certifies exactly $M \leq N$. The two reach the same block for opposite reasons.

Proposition 28.15 (Power and root certificates). *Let $B \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ whose block of index 0 is a scalar $b_0 \in \mathbb{K}^\times$ — equivalently, by Theorem 9.3, a unit of $\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ — let $p \in \mathbb{Z}$ and $q \in \mathbb{N}_{>0}$, and let $C = B^{p/q}$ be the power determined by a chosen value $b_0^{p/q} \in \mathbb{K}^\times$ satisfying $\big(b_0^{p/q}\big)^q = b_0^p$, the powers being those of Theorem 8.37. Then $C^q = B^p$, and for every candidate $D \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ whose block of index 0 is the scalar $b_0^{p/q}$,*

$$\nu_t\big(D^q - B^p\big) = \nu_t(D - C) :$$

gauge 0. The restriction on the block of index 0 of D cannot be dropped, and the equation cannot replace it: if $\zeta \in \mathbb{K}^\times$ satisfies $\zeta^q = 1$ and $\zeta \neq 1$, then $D = \zeta C$ satisfies $D^q - B^p = 0$ identically while $D \neq C$. The branch datum is part of the specification and is certified only by comparing the blocks of index 0.

Proof. The exponent law $(1+U)^a(1+U)^b = (1+U)^{a+b}$ for binomial powers is the Vandermonde convolution $\sum_k \binom{a}{k}\binom{b}{n-k} = \binom{a+b}{n}$ [18], valid in any $\mathbb{Q}$-algebra; iterating it q times with $a = b = \cdots = p/q$ gives $(1+U)^p$ where $U = b_0^{-1}B - 1$. Combined with the hypothesis $\big(b_0^{p/q}\big)^q = b_0^p$ this yields $C^q = B^p$. Next,

$$D^q - C^q = (D - C)\sum_{i=0}^{q-1} D^i C^{\,q-1-i},$$

and each summand has order 0 with block of index 0 equal to $d_0^i c_0^{q-1-i}$, where $c_0 = b_0^{p/q}$ and d_0 is the block of index 0 of D. Under the hypothesis $d_0 = c_0$ the sum has block $qc_0^{q-1} \neq 0$ there, because $\mathbb{K}$ has characteristic zero and $c_0 \neq 0$; hence the sum has order exactly 0 and the displayed identity follows from additivity of ν_t. For the final assertion, $(\zeta C)^q = \zeta^q C^q = C^q = B^p$ while $\zeta C - C = (\zeta - 1)C \neq 0$. ☐

Proposition 28.16 (A faithful composition certificate). *Let $F \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$, $G \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}$, and let $H \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ be a candidate for $F \circ G$. Then*

$$\nu_t\big(H \circ G^{\langle -1\rangle_\circ} - F\big) = \nu_t\big(H - F \circ G\big) :$$

gauge 0.

Proof. Right composition with $G^{\langle -1\rangle_\circ} \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}$ is a ring homomorphism of $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ (Theorem 12.7) and satisfies $(F \circ G) \circ G^{\langle -1\rangle_\circ} = F \circ (G \circ G^{\langle -1\rangle_\circ}) = F$ by associativity on the admissible domain (Theorem 12.27) and $G \circ G^{\langle -1\rangle_\circ} = X$ (Theorem 14.7). Hence $H \circ G^{\langle -1\rangle_\circ} - F = (H - F \circ G) \circ G^{\langle -1\rangle_\circ}$, whose order equals $\nu_t(H - F \circ G)$ by Equation (27.1). ☐

Remark 28.17 (Composition has one faithful certificate and several cheap unfaithful ones). Theorem 28.16 is the only single-equation certificate for composition in this volume, and it is not free: it requires the inverse of the inner map, hence an independent reversion. The cheap alternatives — the chain rule (Theorem 12.31), the multiplicativity $(F_1F_2) \circ G = (F_1 \circ G)(F_2 \circ G)$, associativity, and the agreement of the two engines of Section 27.8 — are *consistency* tests: each of them is an identity that a wrong engine may or may not satisfy. Theorem 28.9 shows that the multiplicative tests are satisfied by a wrong engine, and the chain rule is not, so among the cheap tests the chain rule is the one to run.

Proposition 28.18 (The antiderivative equation is the one unfaithful defining equation). *Let $G \in t\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$.*

1. *The equation $E(Y) = D_X Y - G$ does not define a single-valued operation on $\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$: its solution set is a coset of $\mathbb{K}$ (Theorem 27.30), and E is not faithful with any gauge, since $E(Y+c) = E(Y)$ while $\nu_t\big((Y+c)-Y\big) = 0$ for $c \in \mathbb{K}^\times$.*

2. *Fix $c \in \mathbb{K}$ and let $\mathcal{Y}_c = \{Y \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}} : (\text{block of index } 0 \text{ of } Y) \text{ takes the value } c \text{ at } L = 0\}$. Then E restricted to $\mathcal{Y}_c$ is a defining equation for a single-valued operation, and it is faithful with gauge 1.*

Proof. (i) is Theorem 27.30 together with the displayed observation. For (ii), let $Y, Y^* \in \mathcal{Y}_c$ with $D_X Y^* = G$ and put $\Delta = Y - Y^*$, so that $E(Y) = D_X\Delta$. The block of index 0 of Δ vanishes at $L = 0$, so it is not a nonzero constant. If $\nu_t\Delta = 0$, its leading block Δ_0 is a nonzero polynomial with $\Delta_0(0) = 0$, hence $\deg \Delta_0 \geq 1$ and $\partial_L \Delta_0 \neq 0$ in characteristic zero; if $\nu_t\Delta = n \geq 1$, then $(\partial_L - n)\Delta_n \neq 0$ because $\partial_L - n$ is injective on $\mathbb{K}[L]$. In both cases the block formula gives $\nu_t(D_X\Delta) = \nu_t(\Delta) + 1$. Uniqueness of the solution in $\mathcal{Y}_c$ is the case $\Delta = 0$. □

Remark 28.19 (Every derivative-based certificate inherits the same defect). The Leibniz test $D_X H - (F\,D_X G + G\,D_X F)$ for a candidate product H, and the chain-rule test for a candidate composition, are residuals of the form $D_X(\text{error})$. By Theorem 27.105 they have gauge 1 *except* when the leading block of the error is a constant sitting at index 0, which they annihilate. A derivative-based test therefore cannot see an error in $\mathbb{K}$, which is exactly the error an antiderivative can commit. Pair them with one gauge-zero test.

Table 28.1: Defining equations, their gauges, and where they are proved. The gauge is the number of blocks by which the residual test must be run beyond the cutoff to be certified (Theorem 28.2).

Operation and candidate	Defining equation E	Gauge	Reference
Quotient Q of A/B	$BQ - A$	$\nu_t B$	Theorem 28.11
Reciprocal C of B^{-1}	$BC - 1$	$\nu_t B$	Theorem 28.11
Rational power D of $B^{p/q}$, $\nu_t B = 0$	$D^q - B^p$	0	Theorem 28.15
Logarithm Λ of $\log(1+U)$	$\exp\Lambda - (1+U)$	0	Theorem 28.13(i)
Logarithm Λ, differential form	$(1+U)\partial_t\Lambda - \partial_t U$	-1	Theorem 28.13(ii)
Exponential $\mathcal{E}$ of $\exp U$	$\log\mathcal{E} - U$	0	Theorem 28.13(iii)
Composition H of $F \circ G$, $G \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}$	$H \circ G^{\langle -1\rangle\circ} - F$	0	Theorem 28.16
Reversion G of $F^{\langle -1\rangle\circ}$, right	$F \circ G - X$	0	Theorem 28.5
Reversion G of $F^{\langle -1\rangle\circ}$, left	$G \circ F - X$	0	Theorem 28.5
Omega truncation ω_N	$\omega_N + \log\omega_N - X$	0	Theorem 20.13
Antiderivative Y with declared constant	$D_X Y - G$	1 on $\mathcal{Y}_c$	Theorem 28.18
Antiderivative Y, constant undeclared	$D_X Y - G$	none	Theorem 28.18(i)

28.4 A specification for a self-checking implementation

The tests below are chosen so that each one fails on a specific error that this volume documents, and so that no test is checked by the code path it tests. They are stated as a specification: an implementation conforms when it runs all of them at every build, over the declared coefficient ring, with exact arithmetic.

Two of them need an object to test against. The Wright omega expansion is the canonical choice, because it is the reversion of the single map $X + L$, its blocks are the Lambert polynomials, and it satisfies four identities that share no code path: the reversion residual (Theorem 20.13), the integral recurrence (21.5), the Stirling formula (Theorem 21.4), and the block recurrence of the differentiated equation, which we now record because it is the only one of the four that exercises D_X and the truncated product rather than ∂_L alone.

Lemma 28.20 (The block recurrence of the omega equation, and the constant it cannot see). *Let $\mathcal{Q} = \sum_{n\geq 0} q_n(L)t^n \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ and $\Omega = X + \mathcal{Q} \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}$.*

1. *If $\Omega + \log\Omega = X + c$ for some $c \in \mathbb{K}$, then*

$$(1 + t + t\mathcal{Q})\, D_X\mathcal{Q} = -t. \tag{28.2}$$

2. *Conversely,* (28.2) *holds if and only if $\partial_L q_0 = -1$ and, for every $n \geq 1$,*

$$(\partial_L - n)q_n = -(\partial_L - n + 1)q_{n-1} - \sum_{j=1}^{n} q_{n-j}\,(\partial_L - j + 1)q_{j-1}. \tag{28.3}$$

   In particular (28.2) *determines q_n uniquely from $q_0, \ldots, q_{n-1}$ for every $n \geq 1$, and determines q_0 only up to an additive constant.*

3. *The solutions of* (28.2) *in $\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ are exactly the corrections $(X + L - c)^{\langle -1\rangle\circ} - X$ for $c \in \mathbb{K}$, one for each value of $q_0(0) = c$; only $c = 0$ solves $\Omega + \log\Omega = X$, and the residual of Theorem 20.13 detects $c \neq 0$ already in the block of index 0.*

Proof. (i) Apply D_X to $\Omega + \log\Omega = X + c$. By Theorem 10.10, $D_X \log\Omega = (D_X\Omega)\Omega^{-1}$, and $D_X(X + c) = 1$, so $(D_X\Omega)(\Omega + 1) = \Omega$. Substituting $\Omega = t^{-1}(1 + t\mathcal{Q})$, $\Omega + 1 = t^{-1}(1 + t + t\mathcal{Q})$ and $D_X\Omega = 1 + D_X\mathcal{Q}$, and multiplying by t, gives $(1 + D_X\mathcal{Q})(1 + t + t\mathcal{Q}) = 1 + t\mathcal{Q}$, which rearranges to (28.2).

(ii) By the block formula, $D_X\mathcal{Q} = \sum_{m\geq 1} d_m t^m$ with $d_m = (\partial_L - m + 1)q_{m-1}$, and $d_0 = 0$. Extracting the coefficient of t^m in (28.2) gives

$$d_m + d_{m-1} + \sum_{j=1}^{m-1} q_{m-1-j} d_j = -\,[m = 1] \qquad (m \geq 1),$$

where $[m = 1]$ is the Iverson bracket and the sum is empty for $m = 1$. The case $m = 1$ is $d_1 = -1$, that is $\partial_L q_0 = -1$; the case $m = n + 1 \geq 2$ is (28.3) after substituting the values of the d_j. Since $\partial_L - n$ is bijective on $\mathbb{K}[L]$ for $n \neq 0$ (Theorem 27.30), q_n is determined; and $\partial_L q_0 = -1$ has solution set $-L + \mathbb{K}$ in characteristic zero.

(iii) For $c \in \mathbb{K}$ the map $F_c = X + L - c$ lies in $\mathcal{G}_{\mathrm{poly},\mathbb{K}}$, so $\Omega_c = F_c^{\langle -1\rangle\circ}$ exists and is unique (Theorem 16.11), and $F_c \circ \Omega_c = X$ reads $\Omega_c + \log\Omega_c = X + c$; by (i) its correction solves (28.2), and its block of index 0 satisfies $q_0 + L = c$, so $q_0(0) = c$. Conversely a solution of (28.2) has $q_0 = -L + c$ for some $c \in \mathbb{K}$ by (ii), and then agrees with the correction of Ω_c block by block, again by the uniqueness in (ii). Finally $\Omega_c + \log\Omega_c - X = c$, which is the block of index 0 of the residual. □

Remark 28.21. Theorem 21.2 records the recurrence for the Lambert polynomials in its intrinsic two-term form, which is the right form for proving things about them. Theorem 28.20 is the form an implementation can run without knowing the family in advance: it is a recurrence for the *blocks of a series*, driven by D_X and one truncated product, and it therefore tests exactly the primitives of Sections 27.4 and 27.7. Its resonance at $n = 0$ is not an artefact of the derivation: it is Theorem 27.30 in the smallest instance anybody actually computes.

Example 28.22 (Hand-checkable test vectors)**.** Every quantity below can be verified with pencil and paper, which is the point of a test vector.

1. *Lambert blocks.* $\mathsf{L}_0^{\mathrm{Lam}} = -L$, $\mathsf{L}_1^{\mathrm{Lam}} = L$, $\mathsf{L}_2^{\mathrm{Lam}} = \frac{1}{2}L^2 - L$, $\mathsf{L}_3^{\mathrm{Lam}} = \frac{1}{3}L^3 - \frac{3}{2}L^2 + L$. At $n = 3$ the recurrence (28.3) reads
$$(\partial_L - 3)\mathsf{L}_3^{\mathrm{Lam}} = -L^3 + \tfrac{11}{2}L^2 - 6L + 1,$$
and its right-hand side,
$$-(\partial_L - 2)\mathsf{L}_2^{\mathrm{Lam}} - \left[\mathsf{L}_2^{\mathrm{Lam}}\,\partial_L\mathsf{L}_0^{\mathrm{Lam}} + \mathsf{L}_1^{\mathrm{Lam}}(\partial_L - 1)\mathsf{L}_1^{\mathrm{Lam}} + \mathsf{L}_0^{\mathrm{Lam}}(\partial_L - 2)\mathsf{L}_2^{\mathrm{Lam}}\right],$$
is the same polynomial; both are computed from the displayed $\mathsf{L}_n^{\mathrm{Lam}}$ in four lines.

2. *Composition through the logarithmic generator.* With $H = 1$, the Taylor engine gives $D_X L = t$, $D_X^2 L = -t^2$, $D_X^3 L = 2t^3$, hence
$$L \circ (X + 1) = L + t - \tfrac{1}{2}t^2 + \tfrac{1}{3}t^3 + \mathcal{O}_t^{\mathrm{form}}\left(t^4\right) = L + \log(1 + t) + \mathcal{O}_t^{\mathrm{form}}\left(t^4\right),$$
in agreement with the generator formula Theorem 12.22. An engine that returns L has substituted into the wrong generator.

3. *Guard block.* $F = X$ exactly, and G known to cutoff 1, its only stored block being the constant 1 at index 0: the sound output cutoff for FG is $\min\{+\infty,\ 1 - 1\} = 0$, so a routine that returns the block of index 0 of FG has returned an undetermined block (Theorem 27.13).

4. *Nonunit denominator.* $(L + t)^{-1} = \sum_{n\geq 0}(-1)^n t^n L^{-n-1}$, an element of $\mathscr{P}\mathscr{L}_{\mathrm{rat},\mathbb{K}}$ and not of $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$; the identity is checked by multiplying out and telescoping. A routine that returns a polynomial answer here has violated (P4) of Theorem 27.9.

5. *Resonance.* $D_X Y = t$ has solution set $L + \mathbb{K}$; a routine that returns L without recording the choice is nondeterministic in the sense of Theorem 28.24.

6. *Sampling is not a zero test.* The residual block $\prod_{i=1}^{s}(L - \lambda_i)$ vanishes at every sample point $\lambda_1, \ldots, \lambda_s$ and is nonzero (Theorem 28.3).

Table 28.2: The conformance suite. Each test is a checked identity; the third column names the error class it is there to catch, with the diagnosis in Part 29.

	Checked identity	Catches
T1	Declared cutoffs of $+, \times, /, \circ, D_X$ equal those of Theorem 27.10 on random inputs, including inputs of negative order	the "modulo t^N in, modulo t^N out" fallacy; Section 29.2
T2	$\mathrm{Trunc}_{<N} F$ retains the block of index $N-1$ and discards the block of index N	inclusive/exclusive confusion; Section 29.1
T3	$BQ - A$ vanishes below $N + \nu_t(B)$ after a quotient at cutoff N	a gauge dropped from the division test; Theorem 28.12

	Checked identity	Catches
T4	$\exp\log(1+U) = 1+U$ and $\log\exp U = U$ to the declared cutoff	off-by-one in the two recurrences of Theorem 27.26
T5	$D_X(F \circ G) = ((D_X F) \circ G)\, D_X G$ on the single input $F = L$	substitution into the wrong generator; Theorem 28.9
T6	Taylor engine and generator engine agree to the declared cutoff	an error in either engine, but not in a wrong convention shared by both
T7	$H \circ G^{\langle -1 \rangle_\circ} = F$ to the declared cutoff	composition errors that the consistency tests miss; Theorem 28.16
T8	*Both* $F \circ G - X$ and $G \circ F - X$ vanish below $\min\{N_A, N_B\}$, and their leading blocks agree	a reversion certified by the residual that drove it; Theorem 28.7
T9	The four reversion strategies of Section 27.9 agree to a common cutoff	strategy-specific indexing errors, in particular the Lagrange stopping rule; Section 29.1
T10	$\omega_N + \log\omega_N - X$ has order exactly $N+1$ and leading block $-\mathsf{L}^{\mathrm{Lam}}_{N+1}$	any error in the Lambert blocks, and a truncation off by one, which shows as order N or $N+2$; Theorem 20.13
T11	The blocks produced by (28.3), by (21.5) and by Theorem 21.4 coincide	an error in D_X, in the truncated product, or in the Stirling coefficients; Theorem 28.20
T12	$D_X Y - G = 0$ *and* the recorded resonant constant is reproduced	a lost constant of integration; Section 29.6
T13	Residual blocks are tested by canonical form, and a planted residual $\prod_i (L - \lambda_i)$ is rejected	numerical sampling used as a zero test; Theorem 28.3
T14	Two runs on the same input, and runs with permuted input orderings, produce byte-identical canonical forms	nondeterminism; Section 28.5

Remark 28.23 (What a passing suite does not establish). Every test above is a formal identity in the coefficient ring. A complete pass establishes that the computed truncations satisfy the defining equations below their declared cutoffs. It establishes nothing about any function: the passage to an analytic statement needs a realization theorem (Theorems 12.32 and 12.35), a regime, and a derivative bound (Theorems 16.21 and 20.17). Nor does a pass establish that the answer is what the user wanted: the class, the branch and the coefficient ring are inputs, and a computation performed in the wrong class can be internally impeccable. Section 29.9 exists for that reason.

28.5 Determinism and reproducibility

Definition 28.24 (Deterministic routine, reproducible computation)**.** A routine is *deterministic* when its returned representative is a function of its input representatives together with the declared branch and normalisation data — and, in particular, is independent of the order in which stored blocks are visited, of any hashing, and of any internally chosen constant. A computation is *reproducible* when its record

$$\big(\text{series},\ R,\ \text{cutoff},\ \text{order bound},\ \text{residual and its order},\ \text{branch and normalisation data}\big)$$

is returned in full, so that a second run, or a different implementation, can be compared with it item by item.

Proposition 28.25 (Soundness pins the answer)**.** *Let φ be single-valued on denotations, and let* R *be sound for φ with declared cutoff N_{out}, returning w. Then for every admissible input tuple $(F_1, \ldots, F_k)$ in* *the denotations,*

$$w = \mathrm{Trunc}_{<N_{\mathrm{out}}}\, \varphi(F_1, \dots, F_k).$$

Consequently any two sound routines for φ that declare the same cutoff return the same value, and if in addition both are canonical in the sense of (P2) of Theorem 27.9, they return the same stored object.

Proof. Write $z = (F_1, \dots, F_k)$. Soundness says $\varphi(z) \in \mathrm{Den}(w, N_{\mathrm{out}})$, which by Theorem 27.3 says $\mathrm{Trunc}_{<N_{\mathrm{out}}}\, \varphi(z) = w$. The right-hand side does not depend on the routine, and by hypothesis it does not depend on which element of the denotations was chosen either. Canonicity makes the stored representation of w unique. □

Corollary 28.26 (Reproducibility is a theorem, and its failure is a diagnosis)**.** *A conforming implementation — sound, canonical, with a declared cutoff that is a function of the declared input data — is deterministic on every operation whose φ is single-valued on denotations. Consequently, if two runs of a conforming implementation disagree, then at least one of the following holds, and these are the only possibilities:*

1. *the operation is not single-valued — an antiderivative (Theorem 27.30), a root or a fractional power (Theorem 28.15), a logarithm of a constant, or a branch choice — and the choice was made internally instead of being an input;*
2. *the declared cutoff is not a function of the declared input data, for instance because it was taken from a stale stored order bound (Theorem 27.4);*
3. *the returned object is not in canonical form, for instance because a zero block was retained or a rational coefficient was left unnormalised;*
4. *the implementation is unsound.*

Proof. The positive statement is Theorem 28.25: with (P1), (P2) and a cutoff determined by the inputs, the returned object is $\mathrm{Trunc}_{<N_{\mathrm{out}}}\, \varphi(\cdot)$ in canonical form, hence the same in both runs. For the list, negate the hypotheses one at a time: the proof used exactly single-valuedness of φ on the denotations, the fact that N_{out} is the same in both runs, canonicity, and soundness. A disagreement therefore falsifies one of them, and each of the four items is the failure of one. □

A common trap

Three habits break Theorem 28.26 in practice and are worth naming. Iterating over a hash map makes the accumulation order depend on the hash seed; with exact arithmetic the mathematical value is unaffected but the canonical form is reached by a different route and any timing-dependent truncation heuristic becomes visible, which is why the ordered map of Theorem 27.1 is part of the contract and not an optimisation. Floating-point coefficients break the exact zero test of Theorem 28.3, hence break canonicity, hence break determinism. And a global "simplification" pass applied to coefficients outside the declared R can silently change the type, which is failure (P4).

Section summary

A certificate is a defining equation plus a checked valuation, and it certifies a coset up to the equation's gauge. Every primitive of Part 27 that inverts a cheaper primitive has a faithful equation with a known gauge, except the antiderivative, whose equation is blind to a constant — which is exactly the resonance; addition, multiplication and the derivation have none, and are certified by recomputation. Both reversion residuals certify the same blocks; the reason to compute both is that a residual which drove the iteration certifies nothing. Zero tests are comparisons of canonical forms, never numerical samples. Reproducibility is then not a matter of discipline but a corollary of soundness, canonicity, and a cutoff that is a function of the inputs.

Chapter 29

Failure modes, diagnostics, and a decision procedure

Polynomial–logarithmic computations are forgiving at leading order and unforgiving after it. Each of the eight failure modes below is one that the six source drafts either commit, warn against without a diagnosis, or warn against with the wrong diagnosis. Each is presented in the same shape: *symptom*, the observable behaviour; *diagnosis*, the theorem that was violated; *minimal reproducing example*, the smallest instance in which the error is visible, chosen so that it can be checked by hand and installed as a regression test; and *repair*, what the contract must say so that the error cannot recur silently. The section closes with a decision procedure: given a problem, which class, which algorithm, which certificate.

29.1 Truncation-convention off-by-one

Symptom. The last requested block is missing, or is present but wrong; two implementations agree in every block but the last; a residual has order one less than predicted, or one more; a loop that should run to R runs to $R-1$ and the answer is correct except at the most expensive block, the one nobody checks by hand.

Diagnosis. Two conventions are in circulation and they differ by one block:

$$\operatorname{Trunc}_{<N} F = \sum_{n<N} p_n(L)t^n \qquad \text{against} \qquad [F]_{\leq N} = \operatorname{Trunc}_{<N+1} F,$$

the first adopted here and the second used by source 6. The conversion is harmless in a statement and fatal in a loop bound, because a loop bound is where the two conventions meet an index. The three places in this volume where the discrepancy is real are the Taylor budget (Theorem 27.31), the Lagrange stopping rule (Theorem 27.40), and the certifying residual, which for correctness *through* block N must be a $\mathcal{O}_t^{\mathrm{form}}\big(t^{N+1}\big)$ and not a $\mathcal{O}_t^{\mathrm{form}}\big(t^{N}\big)$ (Theorem 16.16).

Minimal reproducing example. Reverting $F = X + L$, whose correction has $h = \nu_t(L) = 0$, by Lagrange–Bürmann (Theorem 27.40). Each summand contributes exactly one block: for $r \geq 1$,

$$\frac{(-1)^r}{r!} D_X^{r-1}(L^r) = \mathsf{L}_{r-1}^{\mathrm{Lam}}(L)\, t^{r-1},$$

which is Equation (20.12). Asking for the blocks of index $0, 1, 2$ — that is, for the exclusive cutoff $N = 3$ — therefore requires $r = 1, 2, 3$, which is $R = \lceil (N+1)/(h+1) \rceil - 1 = 3$. A loop written to the inclusive reading of "through block N" but run against an exclusive cutoff stops at $r = 2$ and returns

$$-L + Lt + 0 \cdot t^2 \qquad \text{instead of} \qquad -L + Lt + \Big(\tfrac{1}{2}L^2 - L\Big)t^2,$$

losing exactly $\mathsf{L}_2^{\mathrm{Lam}}$. The right residual then has order 2 instead of the predicted 3, which is the diagnosis printed in one line.

Repair. Carry the cutoff in the type, never in the prose; convert once, at the boundary, with $[F]_{\leq N} = \mathrm{Trunc}_{<N+1} F$; and let the residual order be the arbiter, since by Theorem 28.51 it is exactly the first index at which the answer is wrong.

29.2 Reading a block the input data does not determine

Symptom. A routine returns a block that changes when an input block *beyond* the requested cutoff is changed; two runs with different but equally admissible completions of the same input disagree; a computation is reproducible on every test case in which the inputs happen to have order ≥ 0 and wrong on the first Laurent input.

Diagnosis. The folklore rule "both inputs modulo t^N, output modulo t^N" is the product law Theorem 27.102 specialised to $m_1 = m_2 = 0$. The law is $N_{\mathrm{out}} = \min\{N_1 + m_2, N_2 + m_1\}$, which is *better* than the folklore rule when a factor has positive order and *worse* as soon as a factor has a pole. Worse means: the routine returns a block it has no right to return, and the failure is silent, because the returned block is a perfectly good function of the stored data — it is just not a function of the mathematical object.

Minimal reproducing example. $F = X = t^{-1}$, known exactly; G known to exclusive cutoff 1, its only stored block being 1 at index 0. The sound cutoff for FG is $\min\{+\infty,\ 1 + (-1)\} = 0$: only the block of index -1 is determined. Indeed $G_1 = 1$ and $G_2 = 1 + t$ have the same representative and $FG_1 = t^{-1}$, $FG_2 = t^{-1} + 1$ differ at index 0. A routine that declares cutoff 1 here has invented the block 0.

Repair. Compute N_{out} from the derived order bounds (Theorem 27.4), never from the input cutoffs alone; report the relative precision ρ alongside the cutoff, since by Theorem 27.11 it is ρ, not N, that the multiplicative operations conserve; and run test T1 of Table 28.2 with at least one input of negative order, which is the only kind of input that exhibits the bug.

29.3 Substituting into the wrong generator

Symptom. The leading block is right and every later block is wrong, so the answer looks nearly correct and passes any eyeball test; the residual has order exactly one more than the leading block; ring-based consistency tests all pass.

Diagnosis. Composition moves *both* generators (Theorem 12.22):

$$t \circ G = G^{-1}, \qquad L \circ G = \log G.$$

Only the second is easy to forget, and forgetting it is not an approximation but a different operation — one that is still a $\mathbb{K}[L]$-algebra homomorphism, hence still passes every multiplicative test (Theorem 28.9(i)). A related error is to "substitute $t \mapsto t + H$", which is not an operation on $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ at all: it is the substitution appropriate to a series in a variable that is not the reciprocal of the argument.

Minimal reproducing example. Two, at different depths.

1. One generator, one block. $L \circ (X+1) = L + \log(1+t) = L + t - \frac{1}{2}t^2 + \frac{1}{3}t^3 + \mathcal{O}_t^{\mathrm{form}}(t^4)$ (Theorem 28.22(ii)). An engine that returns L has forgotten the logarithmic generator. The chain-rule test on this single input detects it (Theorem 28.9(ii)).

2. A whole reversion. The engine Ψ_G of Theorem 28.8 reverts $X + L$ to $X - L$, which is the correct block of index 0 of ω and nothing else, while its own residual vanishes identically.

Repair. Implement composition only through the generator images (Theorem 27.34) or the Taylor engine (Theorem 27.32), never by textual substitution; run the chain-rule test T5 and the faithful certificate T7 of Table 28.2; and never certify a reversion with the residual that drove it (Theorem 28.7).

29.4 Treating a formal tail as an analytic bound

Symptom. A numerical error bound of the form CX^{-N} that experiment violates; a claim that a truncation "converges"; a remainder quoted as $\mathcal{O}(X^{-3})$ that in fact contains $X^{-3}(\log X)^{100}$; an identity between formal objects presented as an identity between functions.

Diagnosis. $\mathcal{O}_t^{\mathrm{form}}(t^N)$ is a statement about coefficient support and nothing else (Theorems 7.7 and 7.8). It becomes a statement about a function only through a realization (Theorem 12.32) together with a theorem that transports it (Theorems 12.35, 16.21 and 20.17). Two further confusions travel with this one: formal summability (Theorem 7.21) is a statement about valuations and implies nothing about numerical convergence; and a bare $\mathcal{O}$ with no printed regime silently invites the reader to supply the wrong one.

Proposition 29.1 (Two ways in which the formal tail says nothing analytic)**.**

1. Realizations are not unique. *Let f be a function on a ray $(X_0, +\infty)$ with $f \approx F$ in the sense of Theorem 12.32, and let $g(X) = f(X) + \mathrm{e}^{-X}$. Then $g \approx F$ as well, and $f \neq g$ everywhere. Consequently no formal statement about F determines any value of a realization, and the difference of two realizations is invisible to every block.*
2. Formal summability is not convergence. *The family $\left(n!\,t^n\right)_{n\geq 0}$ is formally summable, with sum $F = \sum_{n\geq 0} n!\,t^n \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$, yet for every real $X > 0$ the numerical series $\sum_{n\geq 0} n!\,X^{-n}$ diverges.*

Proof. (i) For every fixed N, $\mathrm{e}^{-X} = o_{X\to+\infty}\left(X^{-N}\right)$, since $X^N\mathrm{e}^{-X} \to 0$; hence g has the same asymptotic remainders as f at every order, which is the definition of $g \approx F$. And $\mathrm{e}^{-X} > 0$, so $f \neq g$ pointwise.

(ii) Formal summability requires $\nu_t(n!\,t^n) = n \to \infty$ (Theorem 7.21), which holds, so the sum exists in $\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$; each block is a single scalar and no coefficient receives infinitely many contributions. For the divergence, fix $X > 0$; then $n!\,X^{-n} \to \infty$ as $n \to \infty$, because $\left((n+1)!X^{-n-1}\right)/\left(n!X^{-n}\right) = (n+1)/X \geq 2$ for $n \geq 2X$. A series whose terms do not tend to zero diverges. □

Minimal reproducing example. Theorem 29.1(ii) with $N = 3$: the "remainder after three blocks" is formally $\mathcal{O}_t^{\mathrm{form}}(t^3)$, while numerically, at $X = 10$, the terms $n!\,10^{-n}$ decrease only until $n = 9$ and $n = 10$, where both equal $9!/10^9 = 3.63 \times 10^{-4}$, and grow without bound afterwards. The formal tail marker predicts none of this and forbids none of it: it is not a statement about numbers.

Repair. Never write a bare $\mathcal{O}$ for an asymptotic claim. Print the regime, as in

$$\mathcal{O}_{X\to+\infty}\Big((\log X)^d X^{-3}\Big)\,;$$

state which realization theorem is being used, and if none is, say that the statement is formal. Section 20.6 is the model of the four ingredients an analytic statement needs.

29.5 Dividing by a nonunit

Symptom. Coefficients acquire denominators in L; block degrees go negative; a routine that promised polynomial coefficients returns something else, or — worse — silently truncates an infinite series in L^{-1} and returns a polynomial that is wrong in every block; a residual test fails at the leading block for no visible reason.

Diagnosis. The units of $\mathbb{K}[L]$ are $\mathbb{K}^\times$ (Theorem 9.1); hence an element of $\mathscr{P}\mathscr{L}_{\text{poly},\mathbb{K}}$ is a unit if and only if its leading block is a nonzero *scalar* (Theorem 9.3). The reciprocal recurrence divides by the leading block at every step, so a leading block of positive L-degree takes the computation out of $\mathscr{P}\mathscr{L}_{\text{poly},\mathbb{K}}$ and into $\mathscr{P}\mathscr{L}_{\text{rat},\mathbb{K}}$, or after expansion into $\mathscr{P}\mathscr{L}_{\text{itLaur},\mathbb{K}}$. This is not a defect of the algorithm but information about the problem, and (P4) of Theorem 27.9 requires it to appear in the returned type.

Proposition 29.2 (The smallest nonunit, and the two rings it forces).

1. *$B = L + t$ is not a unit of $\mathscr{P}\mathscr{L}_{\text{poly},\mathbb{K}}$.*
2. *In $\mathscr{P}\mathscr{L}_{\text{rat},\mathbb{K}} = \mathbb{K}(L)((t))$ it is a unit, with $B^{-1} = \sum_{n\geq 0}(-1)^n L^{-n-1}t^n$.*
3. *For $B' = L + 1 + t$ the reciprocal is $\sum_{n\geq 0}(-1)^n (L+1)^{-n-1}t^n$, and each of its blocks, when expanded in $\mathbb{K}((L^{-1}))$, is an infinite series; so the passage from $\mathscr{P}\mathscr{L}_{\text{rat},\mathbb{K}}$ to $\mathscr{P}\mathscr{L}_{\text{itLaur},\mathbb{K}}$ introduces a second, inner cutoff, which the type must carry and the certificate must quote.*

Proof. (i) The leading block of B is L, which is not a unit of $\mathbb{K}[L]$ because a product $Lq(L)$ has degree ≥ 1; now apply Theorem 9.3.

(ii) Multiply out and telescope:

$$\begin{aligned}(L+t)\sum_{n\geq 0}(-1)^n L^{-n-1}t^n &= \sum_{n\geq 0}(-1)^n L^{-n}t^n + \sum_{n\geq 0}(-1)^n L^{-n-1}t^{n+1} \\ &= \sum_{n\geq 0}(-1)^n L^{-n}t^n - \sum_{m\geq 1}(-1)^m L^{-m}t^m = 1.\end{aligned}$$

Every block lies in $\mathbb{K}(L)$ and the family is summable, the n-th term having $\nu_t = n$.

(iii) The same telescoping with L replaced by $L+1$ gives the displayed reciprocal. Expanding $(L+1)^{-n-1} = L^{-n-1}(1+L^{-1})^{-n-1} = \sum_{k\geq 0}\binom{-n-1}{k}L^{-n-1-k}$ exhibits infinitely many nonzero inner coefficients, since $\binom{-n-1}{k} = (-1)^k\binom{n+k}{k} \neq 0$ for every k. □

Minimal reproducing example. Theorem 29.2 with the quotient $1/(L+t)$ requested in $\mathscr{P}\mathscr{L}_{\text{poly},\mathbb{K}}$: a conforming routine refuses and reports the obstruction; a nonconforming one returns $L^{-1} - L^{-2}t + \cdots$ typed as a polynomial-coefficient object, and every downstream degree computation is then meaningless.

Repair. Test the leading block for unit-hood before dividing, taking logarithms, or forming a general power; enlarge R explicitly and record it in the returned type; and remember that in $\mathscr{P}\mathscr{L}_{\text{itLaur},\mathbb{K}}$ the log-degree profile of Theorem 8.24 is no longer the relevant invariant — the inner valuation is.

29.6 Losing the constant of integration at the resonant block

Symptom. Two runs differ by an additive constant; an identity that held before an integration fails after it; a derived expansion is a shift of the intended one, so that every block of positive index is right and the answer is still wrong; or, in the rational class, a routine returns nonsense where the true answer needs $\log L$.

Diagnosis. The block equation for $D_X Y = G$ is $(\partial_L - n)y_n = g_{n+1}$. The operator $\partial_L - n$ is bijective on $\mathbb{K}[L]$ for $n \neq 0$ and equals ∂_L for $n = 0$, whose kernel is $\mathbb{K}$ (Theorem 27.30). So the block of index 0 — the resonant block — is determined only up to a constant, and the defining equation $D_X Y - G$ is blind to that constant (Theorem 28.18). In $\mathscr{P}\mathscr{L}_{\text{rat},\mathbb{K}}$ the same block carries an obstruction as well as a freedom: the equation $\partial_L y_0 = g_1$ is solvable in $\mathbb{K}(L)$ only when g_1 has zero residue (Theorems 10.23 and 11.24).

Minimal reproducing example. Two, one for the freedom and one for the obstruction.

1. *Freedom.* Recover the ω expansion from the differentiated equation (28.2) instead of from $\Omega + \log\Omega = X$. By Theorem 28.20(ii)–(iii) the differentiated equation determines every block of index ≥ 1 uniquely and determines the block of index 0 only up to a constant c, and each c gives the genuine reversion of the genuinely different map $X + L - c$. Every block of positive index is then wrong as well, since it is a block of the wrong map: solving (28.3) at $n = 1$ with $q_0 = c - L$ gives $(\partial_L - 1)q_1 = 1 + c - L$, hence $q_1 = L - c$. For $c = 1$ the computed answer begins $1 - L + (L-1)t$ where the intended one begins $-L + Lt$; no block is correct, and the residual $\Omega + \log\Omega - X = c$ exposes it in the block of index 0.

2. *Obstruction.* Antidifferentiate $G = t/L \in \mathscr{P}\mathscr{L}_{\mathrm{rat},\mathbb{K}}$. The block equation at $n = 0$ is $\partial_L y_0 = 1/L$, which has no solution in $\mathbb{K}(L)$, because the derivative of a rational function has zero residue at 0 (Theorem 10.23) while $1/L$ has residue 1. The class must be enlarged to admit $\log L$, which is Theorem 11.22.

Repair. Treat the antiderivative as an operation with an input, not as a function: the resonant constant is part of the specification, it is returned in the record of Theorem 28.24, and the certificate is the pair (T12 of Table 28.2). Before integrating in $\mathscr{P}\mathscr{L}_{\mathrm{rat},\mathbb{K}}$, test the residue.

29.7 Assuming a bound uniform in L

Symptom. A truncation that is provably correct as $X \to +\infty$ behaves badly at every argument one can actually compute with; the "next correction" is larger than the term it corrects; an error constant that was quoted once turns out to depend on the number of blocks retained, or on the log-degrees of the discarded ones.

Diagnosis. The dominance $t^{n+1}L^D \prec t^n$ is a statement about $X \to +\infty$ at *fixed* D (Theorems 4.7 and 5.13). Since $L = \log X$ grows, the crossover beyond which the monomial $L^D X^{-1}$ stays small is, for $D \geq 3$, at $\log X > x^*(D)$, where $x^*(D)$ is the largest root of $x = D\log x$ (Theorem 29.3(iv); for $D \leq 2$ there is no crossover). Moreover $x^*(D) \sim D\log D$ as $D \to \infty$. Indeed, writing $y = \log x^*$, the equation $x^* = D\log x^*$ reads $y - \log y = \log D$; the left side is strictly increasing to $+\infty$ on $y > 1$, and $y > \log D \geq \log 3 > 1$, so $y \to \infty$ with D, whence $\log y = o_{D\to\infty}(y)$, $y \sim \log D$, and $x^* = Dy \sim D\log D$. A profile that grows with the block index therefore pushes the crossover out with it, and no constant covers all blocks at once.

Proposition 29.3 (No constant is uniform in the number of blocks). *Fix an integer $D \geq 1$, take $\mathbb{K} = \mathbb{R}$, and let $F = \sum_{n\geq 0} L^{Dn}t^n \in \mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$, so that $d_F(n) = Dn$. All statements below about numbers are on the positive real ray with the real logarithm.*

1. *On the ray $X > 1$ with $L = \log X$, the numerical series converges exactly when $L^D < X$, with sum $f(X) = X/(X - L^D)$, and $f \approx F$ as $X \to +\infty$.*

2. *For each fixed $N \geq 1$, the remainder is $f(X) - \sum_{n<N} L^{Dn}X^{-n} = \big(L^D X^{-1}\big)^N \big(1 - L^D X^{-1}\big)^{-1}$, which is $o_{X\to+\infty}\big(X^{-N+\varepsilon}\big)$ for every $\varepsilon > 0$ but is* not *$\mathcal{O}_{X\to+\infty}\big(X^{-N}\big)$.*

3. *There is no pair (C, X_0) with $\big|f(X) - \sum_{n<N} L^{Dn}X^{-n}\big| \leq CX^{-N}$ for all $N \geq 0$ and all $X \geq X_0$.*

4. *Suppose $D \geq 3$. Then $x = D\log x$ has exactly two roots $x_-(D) < D < x^*(D)$, and at a given $X > 1$ the blocks decrease in size if and only if $\log X < x_-(D)$ or $\log X > x^*(D)$; thus $x^*(D)$ is the threshold beyond which they decrease for good. For $D = 10$, $x_- = 1.1183\ldots$ and $x^* = 35.7715\ldots$; so at $X = 10^{15}$ the block of index 1 exceeds the block of index 0 by the factor $L^{10}/X = 2.4158\ldots$, and the expansion becomes permanently decreasing only for $X > 3.4 \times 10^{15}$. For $D \leq 2$ there is no threshold at all: the blocks decrease at every $X > 1$.*

Proof. (i) The series is geometric with ratio $L^D X^{-1}$, whence the convergence criterion and the sum

$1/(1 - L^D X^{-1}) = X/(X - L^D)$. For the asymptotic statement, fix $N \geq 0$; by (ii) with $N+1$ in place of N,

$$f(X) - \sum_{n \leq N} L^{Dn} X^{-n} = \frac{L^{D(N+1)} X^{-(N+1)}}{1 - L^D X^{-1}} = o_{X \to +\infty}\left(X^{-N}\right),$$

because $L^{D(N+1)} X^{-1} = (\log X)^{D(N+1)}/X \to 0$ and the denominator tends to 1. This is the level-wise condition of Theorem 12.32, so $f \approx F$.

(ii) Sum the geometric tail. The remainder equals $L^{DN} X^{-N}\left(1 - L^D X^{-1}\right)^{-1} \geq L^{DN} X^{-N}$ for X large, and $L^{DN} \to \infty$, so the remainder divided by X^{-N} is unbounded as $X \to +\infty$; hence it is not $\mathcal{O}_{X \to +\infty}\left(X^{-N}\right)$. The little-o statement is the estimate just made.

(iii) Since $L^D X^{-1} = (\log X)^D / X \to 0$, one may fix an $X \geq \max\left\{X_0, \mathrm{e}^2\right\}$ with $L^D < X$, so that $f(X)$ is defined by (i); then $L = \log X \geq 2$ and in particular $L^D \geq 2$. The remainder at cutoff N is at least $L^{DN} X^{-N}$, so the asserted bound would force $C \geq L^{DN} \geq 2^{DN}$ for every N, which is impossible for a finite C.

(iv) The block of index n evaluates to $\left(L^D X^{-1}\right)^n$, so the blocks decrease if and only if $L^D < X$, that is $D \log L < L$ with $L = \log X$, that is $\psi(L) > 0$ for $\psi(x) = x - D \log x$. On $(0, \infty)$ the function ψ is smooth with $\psi'(x) = 1 - D/x$, so its only critical point is $x = D$, a minimum, with $\psi(D) = D(1 - \log D)$, and $\psi \to +\infty$ at both ends. Hence: for $D \leq 2$ one has $\log D < 1$, so $\psi(D) = D(1 - \log D) > 0$ and therefore $\psi > 0$ everywhere, and the blocks decrease at every $X > 1$; for $D \geq 3$ one has $\psi(D) < 0$, so ψ has exactly one root $x_-(D) \in (0, D)$ and exactly one root $x^*(D) \in (D, \infty)$, and $\psi(L) > 0$ is equivalent to $L < x_-(D)$ or $L > x^*(D)$. For $D = 10$, $x - 10 \log x$ is $+0.147$ at $x = 1.1$ and -0.092 at $x = 1.13$, and -0.553 at $x = 35$ and $+0.165$ at $x = 36$, so $x_- = 1.1183\ldots$, $x^* = 35.7715\ldots$ and $\mathrm{e}^{x^*} = 3.43 \times 10^{15}$. At $X = 10^{15}$ one has $L = 34.5388\ldots$ and $L^{10}/X = 2.4158\ldots$. ☐

Minimal reproducing example. Theorem 29.3(iv) with $D = 10$ at $X = 10^{15}$: a series that is a perfectly valid asymptotic expansion, whose first correction is 2.4 times its leading term.

Repair. Quote the log-degrees with the error term, as in $\mathcal{O}_{X \to +\infty}\left((\log X)^d X^{-N}\right)$; state the constant's dependence on N and on the profile; and, for numerical work, compare the actual magnitude of the first discarded block at the actual argument rather than trusting the ordering of the monomials.

29.8 Using a degree bound that was a hypothesis

Symptom. A fixed-width coefficient array overflows, or silently truncates the top logarithmic coefficients; a cost estimate is off by a factor that grows with N; a proof of a degree law is quoted without its hypothesis.

Diagnosis. Two distinct claims are easily confused. The degree law for a reversion (Theorem 18.21) is a theorem *under the hypothesis* that the profile of the input is linearly bounded, and it bounds the output profile in terms of that hypothesis. What is not true, under any hypothesis, is that the output profile is bounded by the input profile: a single reciprocal already destroys that, and reversion destroys it spectacularly. The log-degree profile is output data (Theorems 6.12, 8.24 and 9.15), to be measured and reported, never an input constant.

Proposition 29.4 (Bounded input, unbounded output; and the exact scalar cost). *1.* One reciprocal suffices. *$B = 1 + Lt$ has $d_B(0) = 0$, $d_B(1) = 1$ and no other nonzero block, while $B^{-1} = \sum_{n \geq 0} (-1)^n L^n t^n$ has $d_{B^{-1}}(n) = n$ for every n.*

2. Reversion. *$F = X + L$ has a single nonzero correction block, of degree 1, and $F^{\langle -1 \rangle_\circ} = \omega$ has correction blocks $\mathsf{L}_n^{\mathrm{Lam}}(L)$ with*

$$d_{F^{\langle -1 \rangle_\circ} - X}(0) = 1, \qquad d_{F^{\langle -1 \rangle_\circ} - X}(n) = n \quad (n \geq 1).$$

3. Cost. *For $N \geq 1$, the schoolbook truncated product of two series with the profile of (ii) costs exactly*

$$\binom{N+3}{4} + N^2 + N + 1$$

scalar multiplications, which exceeds the naive "$\binom{N+1}{2}$ multiplications" count of Theorem 27.17 by a factor $\Theta(N^2)$.

Proof. (i) The geometric series identity $(1 + Lt) \sum_{n \geq 0} (-1)^n L^n t^n = 1$ is immediate by telescoping, and $\deg\big((-1)^n L^n\big) = n$.

(ii) $(X + L)^{\langle -1 \rangle_\circ} = \omega = X + \sum_{n \geq 0} \mathsf{L}_n^{\mathrm{Lam}}(L) t^n$ is Theorem 20.10. For the degrees, $\mathsf{L}_0^{\mathrm{Lam}}(u) = -u$ has degree 1, and for $n \geq 1$, $\mathsf{L}_n^{\mathrm{Lam}} = \big((-1)^{n+1}/(n+1)!\big) \Delta_{0,n} u^{n+1}$ by Theorem 21.1, where $\Delta_{0,n} = \prod_{j=0}^{n-1} (\partial_u - j)$ contains the factor ∂_u exactly once, at $j = 0$; by Theorem 27.29 the degree therefore drops by exactly one, giving $\deg \mathsf{L}_n^{\mathrm{Lam}} = (n+1) - 1 = n$. This is the degree law of Theorem 21.7.

(iii) Write $\mathcal{P} = F^{\langle -1 \rangle_\circ} - X$. By Theorem 27.17 the scalar count is $\sum_{i+j<N} w(i)w(j)$ with $w(n) = d_{\mathcal{P}}(n) + 1$, so $w(0) = 2$ and $w(n) = n + 1$ for $n \geq 1$; no block vanishes, since $\mathsf{L}_n^{\mathrm{Lam}} \neq 0$ for all n. Using the Iverson bracket, which is 1 when the enclosed statement holds and 0 otherwise, we have $w(i) = (i + 1) + [i = 0]$ and hence

$$w(i)w(j) = (i+1)(j+1) + (i+1)[j=0] + (j+1)[i=0] + [i=0][j=0].$$

Summing over $i, j \geq 0$ with $i + j < N$: the first term gives $\binom{N+3}{4}$ by Theorem 27.18(ii); the second gives $\sum_{i=0}^{N-1} (i + 1) = N(N + 1)/2$, and the third the same by symmetry; the fourth gives 1. The total is $\binom{N+3}{4} + N(N+1) + 1 = \binom{N+3}{4} + N^2 + N + 1$. Since $\binom{N+3}{4} = N^4/24 + \mathcal{O}(N^3)$ while $\binom{N+1}{2} = N^2/2 + \mathcal{O}(N)$, the ratio is $\Theta(N^2)$. □

Minimal reproducing example. Theorem 29.4(i): an implementation that stores coefficient polynomials in an array of width $\max_n d_B(n) + 1 = 2$, inferred from the input, loses every logarithmic coefficient above degree 1 from the third block of the reciprocal on.

Repair. Store blocks as variable-length objects (Theorem 27.1); measure the profile of the output and return it; quote costs in the coefficient-operation model and multiply by the measured profile factor, as Theorem 27.18 requires; and when invoking a degree law, print its hypothesis.

29.9 A decision procedure

The eight failure modes have a common shape: a decision was made implicitly. The procedure below makes each of them explicit and in the order in which the information becomes available. Its first three steps decide the class, its next two the algorithm and the precision budget, its last three the certificate and the status of the answer. Nothing in it is optional: skipping step 3 is failure mode Section 29.5, skipping step 4 is Section 29.2, and skipping step 7 is Section 29.4.

Algorithm 29.5 (Decision procedure: class, algorithm, certificate)**. Input.** A problem: an expansion, an implicit equation, an inverse, or a composition; and a target amount of information.
Steps.

1. *Name the regime and the branch.* Which variable tends where, on which ray or sector, with which logarithm branch. For a problem at $x \to 0^+$, transport it to $X \to +\infty$ by the dictionary of Theorem 15.18 before anything else; for a problem at a finite critical point, stop — the scale is ramified, not polynomial–logarithmic.

2. *Solve the leading balance and normalise.* Factor the dominant monomial. If the map is monomial-leading, $cX^{\varrho}(1 + \ldots)$ with $\varrho > 0$ (Theorem 19.1), reduce to the near-identity case by Theorem 19.6; a leading logarithmic factor is first put in normal form by Theorem 19.7. If the leading

derivative degenerates, the problem is not near-identity reversion at all and needs a ramified scale.

3. *Compute the generator images before anything else.* Form $t \circ G = G^{-1}$ and $L \circ G = \log G$ and read the smallest closed class off Table 29.1. Adopt that class; if it is larger than the one the problem was posed in, say so in the output type. This is the generator-first rule, and it is much safer than expanding first and inferring the class from the mess.

4. *Fix the precision budget backwards.* From the target cutoff, propagate the required input cutoffs through Theorem 27.10, remembering that the guard blocks are not a safety margin but part of the law (Theorem 27.13), and that relative precision, not absolute cutoff, is what the multiplicative operations conserve (Theorem 27.11).

5. *Choose the algorithm* from Table 29.2, by how many blocks are wanted, whether they are wanted lazily, and whether composition is available and cheap.

6. *Choose the certificate* from Table 28.1, compute the residual, and compare its order with the declared cutoff *plus the gauge.* If the answer was produced by an iteration, the certificate must not be the residual that drove it (Theorem 28.7).

7. *Declare the status of the answer.* Formal — an identity of coefficients, and nothing more; asymptotic — name the realization and the transfer theorem (Theorems 12.32, 12.35, 16.21 and 20.17); or convergent — name the convergence theorem, and note that this volume proves none for the ω series (Theorem 20.20).

8. *Return the record* of Theorem 28.24: series, ring, cutoff, order bound, residual and its order, branch and normalisation data.

Table 29.1: Step 3 of Theorem 29.5: what the generator images force. The class named is the smallest one in this volume that is closed for the operation; a smaller one can be used only by refusing the operation.

Observed data	Smallest closed class	Authority
Sums, products, derivatives of blocks with polynomial coefficients	$\mathscr{PL}_{\mathrm{poly},\mathbb{K}} = \mathbb{K}[L]((t))$	Theorems 8.7 and 11.1
Inner map $X + H$, $H \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$; reversion of it	$\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$, group $\mathcal{G}_{\mathrm{poly},\mathbb{K}}$	Theorems 14.7 and 16.11
Inner map $cX^{\varrho}(1+U)$, $\varrho \in \mathbb{N}_{>0}$, $\log c \in \mathbb{K}$	$\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$	Theorem 12.22
Division, logarithm or general power whose leading block is a nonconstant polynomial	$\mathscr{PL}_{\mathrm{rat},\mathbb{K}} = \mathbb{K}(L)((t))$	Theorems 9.3 and 9.20
A rational coefficient expanded for large L; a second, inner cutoff appears	$\mathscr{PL}_{\mathrm{itLaur},\mathbb{K}} = \mathbb{K}((L^{-1}))((t))$	Theorems 9.37 and 29.2
$\varrho \in \mathbb{Q}_{>0} \setminus \mathbb{Z}$, or a formal root	Puiseux–log tower	Theorems 5.25, 8.38 and 12.17
ϱ irrational, or a non-cyclic exponent group	Hahn field of Γ	Theorem 5.19
Leading term cX^aL^b with $b \neq 0$; or an inner map with a logarithmic factor	depth two: $\log\log X$	Theorems 15.8, 19.4 and 19.5
Composition with $\log$, or with a map of logarithmic depth r	depth-r tower	Theorems 15.10 and 15.14
An exponential of a block that is not a constant; composition with $\exp$	outside every logarithmic class	Theorems 10.16 and 15.11

Observed data	Smallest closed class	Authority
Antiderivative in $\mathscr{P}\mathscr{L}_{\mathrm{rat},\mathbb{K}}$ of a block with nonzero residue	adjoin $\log L$	Theorems 11.22 and 11.24

Table 29.2: Steps 5 and 6 of Theorem 29.5: which algorithm, and which certificate goes with it. Costs are in the coefficient-operation model of Theorem 27.15 and must be multiplied by the profile factor of Theorem 27.18 to describe a machine.

Goal	Algorithm	Certificate
Product at cutoff N	Theorem 27.19	Leibniz test, plus T1 of Table 28.2
Reciprocal, small N	Theorem 27.20, $\binom{\rho}{2}$ products	$BC - 1$, gauge $\nu_t B$
Reciprocal, large N	Theorem 27.21, $\lceil \log_2 \rho \rceil$ stages	same, recomputed at each stage
Quotient	Theorem 27.22	$BQ - A$, gauge $\nu_t B$
Power, α a small nonnegative integer	binary powering, Theorem 27.24	$D^q - B^p$ with $q = 1$
Power, general α; formal roots	recurrence Equation (27.2)	$D^q - B^p$ and a check of the block of index 0 (Theorem 28.15)
$\log(1+U)$, $\exp U$, $\nu_t U \geq 1$	Theorem 27.27	Theorem 28.13, either form
Composition, inner near-identity	Theorem 27.32	T5, T6, and Theorem 28.16
Composition, inner monomial-leading	Theorem 27.34	the same three
Reversion, a few blocks, or lazily	Theorem 27.38	*left* residual, never the driving one
Reversion, large N, composition cheap	Theorem 27.39 with Theorem 17.16	both residuals, plus the abort test
Reversion, closed form or a map whose powers simplify	Theorem 27.40	both residuals; for $X + L$, T10
Reversion with only a composition primitive	Theorem 27.37	both residuals
Antiderivative	Theorem 27.30, blockwise	$D_X Y - G$ with the constant declared (T12)
Mixed or unknown leading data	Theorem 27.41	as at each stage

Table 29.3: Diagnostic index: an observed symptom and the section that explains it.

Symptom	Diagnosis
Exactly one block, the last, disagrees between two implementations	Section 29.1
A residual has order one less, or one more, than predicted	Section 29.1
An output block changes when an input block beyond the cutoff changes	Section 29.2

Symptom	Diagnosis
Everything is correct until the first input with a pole	Section 29.2
The leading block is right and every later block is wrong	Section 29.3
Multiplicative tests pass and the chain-rule test fails	Section 29.3
A numerical error bound is violated in experiment	Section 29.4 or Section 29.7
Denominators in L appear where polynomials were promised	Section 29.5
Two runs differ by an additive constant	Section 29.6
Every block of positive index is a block of a shifted problem	Section 29.6
The first correction is larger than the term it corrects	Section 29.7
A coefficient array overflows, or a cost estimate is off by a growing factor	Section 29.8
Two runs of the same input differ at all	Theorem 28.26

Section summary

The eight failure modes are eight implicit decisions. Truncation convention and guard blocks are decisions about indices; the generator images and the unit test are decisions about the class; the resonant constant — and with it the branch datum of Theorem 28.15 — is a decision about which of several correct answers is wanted; the formal tail and the uniformity in L are decisions about what kind of statement is being made; and the degree bound is a decision about which hypotheses are in force. Theorem 29.5 orders them so that each is made when its information is available, and Table 28.1 says what to compute afterwards to find out whether the decisions were kept.

Part IX

Extensions and the wider landscape

Chapter 30

Generalized power–log classes, nested logarithms, and multiseries

Parts I–IV worked inside a deliberately small ambient: integer powers of $t = X^{-1}$, polynomial (or rational, or Laurent) coefficients in the single logarithm $L = \log X$, and no exponentials. Every closure failure recorded there — the reciprocal of a nonunit (Theorem 9.3), a fractional leading exponent (Theorem 19.3), a logarithm inside the leading monomial (Theorem 19.4), the exponential of an element of negative outer order (Theorem 10.16) — names a specific enlargement. This chapter makes each enlargement precise and, for each, says exactly which theorems of Parts I–IV survive verbatim, which survive after a hypothesis is changed, and which become false. The failures are proved, because they are the statements a computation can actually violate; a class that is described only by what it contains is a class one will silently misuse.

30.1 Five independent axes

It is useful to separate the enlargements into five axes that can be turned independently. Nothing in this subsection is a theorem; it is the map of the territory, and each entry is made precise below or in the place indicated.

Table 30.1: The five axes of enlargement. The class of Parts I–IV is the corner $\big(\Gamma = \mathbb{Z},\ C = \mathbb{K}[L],\ r = 1,\ \text{height } 0,\ \text{well-based}\big)$.

Axis	What is enlarged	Forced by	Analysed in
Exponent group in t	$\mathbb{Z} \rightsquigarrow s^{-1}\mathbb{Z} \rightsquigarrow \Gamma \subseteq \mathbb{R}$	fractional or irrational leading exponent (Theorem 19.3)	Section 30.3, Section 30.4
Coefficient ring in L	$\mathbb{K}[L] \rightsquigarrow \mathbb{K}(L) \rightsquigarrow \mathbb{K}((L^{-1}))$	division by a nonunit (Theorem 9.3); the axis stops short of $\sqrt{L}$ (Theorem 30.17)	Part 5, Section 30.4
Logarithmic depth r	one logarithm $\rightsquigarrow L_1, \dots, L_r$	a logarithm in the leading monomial (Theorem 19.4); an unmatched residue (Theorem 30.20)	Part 15, Section 30.5
Exponential height	height $0 \rightsquigarrow$ transmonomials $e^{\mathfrak{p}}$	$\exp$ of an element of negative outer order (Theorem 10.16, Theorem 31.1)	Part 31
Support class	grid-based $\rightsquigarrow$ left-finite $\rightsquigarrow$ well-based	infinite summation; completion (Theorem 30.10)	Section 30.2

The axes are not interchangeable

An enlargement along one axis never repairs an obstruction on another. Ramifying the exponent group of t does not produce $\sqrt{L}$ (Theorem 30.17); adding logarithmic depth does not produce e^{-X} (Theorem 31.2); enlarging the coefficient ring does not remove the obstruction $\sigma \geq 1$ of Theorem 19.3; and no amount of exponential height supplies the antiderivative of $1/(L_0 L_1 \cdots L_r)$ (Theorem 30.20). A closure test must therefore be run once per axis, and a diagnosis that does not name the axis is not a diagnosis.

30.2 Supports: what well-ordering does, and what it does not do

Throughout this subsection Γ is a totally ordered abelian group, written additively, and a subset $S \subseteq \Gamma$ is thought of as a set of exponents of t: a *larger* exponent means a *smaller* monomial, so the least element of a support is its dominant monomial, and the $\succ$-greatest element of Theorem 5.15 (c) is the least exponent. We keep the exponent formulation because it makes the group structure visible; the reader who prefers monomials should read every "least" as "dominant".

Definition 30.1 (Three support classes)**.** Let $S \subseteq \Gamma$.

(a) S is *well based* (equivalently, well ordered) if every nonempty subset of S has a least element.

(b) S is *left-finite* if $S \cap (-\infty, M)$ is finite for every $M \in \Gamma$.

(c) S is *grid-based* if there are $\mathfrak{n} \in \Gamma$ and finitely many $\alpha_1, \ldots, \alpha_k \in \Gamma_{>0}$ with

$$S \subseteq \mathfrak{n} + \{n_1\alpha_1 + \cdots + n_k\alpha_k : \; n_1, \ldots, n_k \in \mathbb{N}_0\}.$$

Each of the three classes is closed under passing to subsets, under finite unions, and under translation.

Verification of the closure claims. Subsets and translates are immediate in all three cases; for (c) a translate changes only $\mathfrak{n}$. Finite unions: for (a) and (b) this is clear. For (c), let $S \subseteq \mathfrak{n} + \sum \mathbb{N}_0\alpha_i$ and $T \subseteq \mathfrak{m} + \sum \mathbb{N}_0\beta_j$, put $\mathfrak{p} = \min(\mathfrak{n}, \mathfrak{m})$, and take as generating set the α_i, the β_j, and whichever of $\mathfrak{n} - \mathfrak{p}$, $\mathfrak{m} - \mathfrak{p}$ is nonzero; both of the latter are ≥ 0, so all generators used are > 0, and $S \cup T \subseteq \mathfrak{p} + \sum \mathbb{N}_0(\cdots)$. ☐

The whole content of Theorem 5.15 is that for $\Gamma = \mathbb{Z}$ the three classes coincide with "bounded below". They separate as soon as Γ is dense, or as soon as Γ fails to be archimedean; Theorem 30.8 says exactly how.

30.2.1 Well-ordering and local finiteness

The next lemma is the reason the catalogue calls the support condition "what makes multiplication locally finite". We give the proof in full because the sources assert it and because part (iii), which is what actually makes reciprocals and geometric series legal, is not the same statement as parts (i)–(ii).

Lemma 30.2 (Convolution of well-based supports)**.** *Let Γ be a totally ordered abelian group.*

(i) *If $S, T \subseteq \Gamma$ are well based then so is $S + T = \{\sigma + \tau : \sigma \in S, \; \tau \in T\}$.*

(ii) *If S, T are well based then for every $\gamma \in \Gamma$ the fiber $R_\gamma = \{(\sigma, \tau) \in S \times T : \sigma + \tau = \gamma\}$ is finite.*

(iii) *Let $S \subseteq \Gamma_{>0}$ be well based and, for $k \geq 0$, let $S^{(k)} = S + \cdots + S$ (k summands; $S^{(0)} = \{0\}$). Then every $S^{(k)}$ is well based, and for every k and every γ only finitely many multisets of k elements of S have sum γ. If moreover Γ is archimedean, then $S^* = \bigcup_{k \geq 0} S^{(k)}$ is well based and every $\gamma \in S^*$ has, in total, only finitely many representations as a sum of elements of S.*

Proof. We use twice the following remark: *every sequence in a well-based set has a non-decreasing subsequence.* Indeed, given $(\sigma_k)_{k\geq 1}$ in the well-based set S, the set $\{\sigma_k : k \geq 1\}$ has a least element; let k_1 attain it. Having chosen $k_1 < \cdots < k_i$, let $k_{i+1} > k_i$ attain the least element of the nonempty set $\{\sigma_k : k > k_i\}$. Since $\{\sigma_k : k > k_i\} \subseteq \{\sigma_k : k > k_{i-1}\}$ while σ_{k_i} is least in the larger set, $\sigma_{k_i} \leq \sigma_{k_{i+1}}$.

(ii) Suppose R_γ is infinite. As $\tau = \gamma - \sigma$ is determined by σ, infinitely many distinct σ occur; choose pairwise distinct $\sigma_k \in S$ with $\gamma - \sigma_k \in T$. Passing to a subsequence we may assume $\sigma_{k_1} \leq \sigma_{k_2} \leq \cdots$, and since they are pairwise distinct the inequalities are strict. Then $\gamma - \sigma_{k_i}$ is a strictly decreasing sequence in T, so $\{\gamma - \sigma_{k_i} : i \geq 1\}$ is a nonempty subset of T with no least element, contradicting well-basedness of T.

(i) Suppose $\gamma_1 > \gamma_2 > \cdots$ is strictly decreasing in $S + T$, say $\gamma_k = \sigma_k + \tau_k$. Pass to a subsequence along which (σ_k) is non-decreasing; along it $\tau_k = \gamma_k - \sigma_k$ is strictly decreasing in T, a contradiction as before. Hence $S+T$ has no infinite strictly decreasing sequence; a totally ordered set with that property is well based, since a nonempty subset without a least element would produce one by dependent choice.

(iii) That $S^{(k)}$ is well based follows from (i) by induction on k. For the representation count, induct on k; the case $k \leq 1$ is trivial (the empty multiset for $\gamma = 0$, and at most one singleton otherwise). Let $k \geq 2$ and suppose infinitely many multisets $\{s_1 \leq s_2 \leq \cdots \leq s_k\} \subseteq S$ have sum γ. If the value s_1 took only finitely many values, then for one of them infinitely many multisets of size $k-1$ would sum to $\gamma - s_1$, contradicting the inductive hypothesis. So infinitely many distinct values of s_1 occur; by the remark, choose a strictly increasing sequence $s_1^{(1)} < s_1^{(2)} < \cdots$ of such values. Then $\gamma - s_1^{(i)}$ is a strictly decreasing sequence in the well-based set $S^{(k-1)}$, a contradiction.

Now assume Γ archimedean. Since S is well based and $S \subseteq \Gamma_{>0}$, it has a least element $\delta > 0$ (we may assume $S \neq \emptyset$, else there is nothing to prove). A sum of k elements of S is $\geq k\delta$; by archimedeanity, for each M only finitely many k satisfy $k\delta < M$. Hence for $\emptyset \neq T \subseteq S^*$ and $\tau \in T$ we have $T \cap (-\infty, \tau] \subseteq \bigcup_{k\leq K} S^{(k)}$ for some finite K, a finite union of well-based sets and therefore well based; its least element is least in T. The total representation count of γ is the sum, over the finitely many admissible k, of the counts already bounded. $\square$

Remark 30.3 (Neumann's lemma without archimedeanity). The last clause of Theorem 30.2 (iii) is true for an arbitrary totally ordered abelian group — this is Neumann's lemma, and it is what makes a Hahn series ring over any such group a field when the coefficients form a field [19, 13, 33]. The proof above uses archimedeanity only in the last paragraph, and $\Gamma \subseteq \mathbb{R}$ is archimedean, so every use of the clause in this volume with a *one-dimensional* exponent group is covered by the proof given. The monomial groups $\mathcal{M}_r(\Gamma)$ of Theorem 15.1 are *not* archimedean for $r \geq 1$, and there we quote Neumann's lemma rather than reprove it. Every *argument* carried out in Section 30.5 uses only Theorem 30.2 (i)–(ii); the one further input, used only through the word "field", is that $\mathcal{H}_r(\mathbb{K}, \Gamma)$ is a field, which is Theorem 15.1 together with Neumann's lemma and is quoted, not reproved.

Proposition 30.4 (Well-basedness is the largest admissible support class)**.** *Let Γ be a totally ordered abelian group and let $T \subseteq \Gamma$ fail to be well based. Then there is a well-based $S \subseteq \Gamma$, of order type ω, and a $\gamma \in \Gamma$ whose fiber $R_\gamma \subseteq S \times T$ is infinite. Consequently, if $\mathcal{F}$ is a family of subsets of Γ that contains every well-based set and for which every convolution fiber $R_\gamma \subseteq S \times T$ with $S, T \in \mathcal{F}$ is finite, then $\mathcal{F}$ is exactly the family of well-based subsets of Γ.*

Proof. If T is not well based it contains a strictly decreasing sequence $\tau_1 > \tau_2 > \cdots$. Put $S = \{-\tau_k : k \geq 1\}$; then $-\tau_1 < -\tau_2 < \cdots$, so S is well based of order type ω. With $\gamma = 0$, every pair $(-\tau_k, \tau_k)$ lies in R_0, which is therefore infinite. For the second assertion, a member $T \in \mathcal{F}$ failing to be well based would, together with the well-based $S \in \mathcal{F}$ just produced, violate the finiteness hypothesis. $\square$

Remark 30.5 (What the sharp statement is, and what it is not). Theorem 30.2 says well-basedness suffices, and Theorem 30.4 says it cannot be relaxed *as a condition on a class of supports.* It is not necessary for a single product. Take $\Gamma = \mathbb{Q}$ and

$$T = \left\{1 + \tfrac{1}{k} :\ k \in \mathbb{N}_{>0}\right\} = \left\{2, \tfrac{3}{2}, \tfrac{4}{3}, \tfrac{5}{4}, \dots\right\},$$

which is strictly decreasing and therefore not well based. Every element of $T + T$ lies in $(2, 4]$, and a representation $\gamma = (1 + \frac{1}{k}) + (1 + \frac{1}{l})$ with $k \le l$ forces $\frac{1}{k} + \frac{1}{l} = \gamma - 2$, hence $\frac{1}{k} \ge \frac{\gamma-2}{2}$ and $\frac{1}{k} < \gamma - 2$, so

$$(\gamma - 2)^{-1} < k \le 2(\gamma - 2)^{-1},$$

which leaves finitely many k, and each k determines $l = \left(\gamma - 2 - \frac{1}{k}\right)^{-1}$ uniquely. Thus T is a non-well-based support for which the square of the corresponding formal sum has finite coefficient fibers. What fails is *closure*: by Theorem 30.4 the same T has an infinite fiber against the perfectly legitimate support $\left\{-1 - \frac{1}{k} : k \ge 1\right\}$, whose elements increase to -1. A support condition is a condition on a class, not on an element, and this is why the axiom is stated as well-basedness.

30.2.2 Left-finite supports

Left-finiteness is the exact condition under which the finite bookkeeping of Parts II and VI — truncation, block recursion, sparse representation — survives a change of exponent group. It is Theorem 5.15 (d) with $\mathbb{Z}$ replaced by Γ, strengthened by the cofinality clause of Theorem 30.6 (i), which is automatic over $\mathbb{Z}$ and is not automatic over a dense Γ.

Proposition 30.6 (Left-finite supports). *Let Γ be a totally ordered abelian group with $\Gamma \ne \{0\}$ and let $S \subseteq \Gamma$.*

(i) *S is left-finite if and only if it can be enumerated as a strictly increasing sequence $\gamma_0 < \gamma_1 < \cdots$, finite or of order type ω, which in the infinite case is cofinal in Γ: for every $M \in \Gamma$ some $\gamma_k > M$.*

(ii) *Left-finite $\Rightarrow$ well based, and the converse fails whenever Γ is densely ordered: for $\Gamma = \mathbb{Q}$ the set $S = \left\{1 - \frac{1}{k} : k \in \mathbb{N}_{>0}\right\}$ is well based of order type ω and is not left-finite.*

(iii) *If S, T are left-finite then so are $S \cup T$, $S + T$ and every translate of S, and all convolution fibers are finite.*

(iv) *Assume in addition that Γ is archimedean. If $S \subseteq \Gamma_{>0}$ is left-finite then S^* is left-finite, and every $\gamma \in S^*$ arises from only finitely many multisets of elements of S.*

Proof. (i) Let S be left-finite and nonempty. For $\mu \in S$ the set $S \cap (-\infty, \mu]$ is finite and nonempty, so S has a minimum γ_0; removing it leaves a left-finite set, and iterating produces a strictly increasing enumeration in which $\mu \in S$ is reached after at most $\#\big(S \cap (-\infty, \mu]\big)$ steps, so every element occurs. If the enumeration is infinite and $\gamma_k \le M$ for all k, pick $g \in \Gamma_{>0}$; then $S \cap (-\infty, M + g) = S$ is infinite, contradicting left-finiteness. Conversely, given such an enumeration and $M \in \Gamma$, choose K with $\gamma_K > M$; then $S \cap (-\infty, M) \subseteq \{\gamma_0, \dots, \gamma_{K-1}\}$.

(ii) If $\emptyset \ne T \subseteq S$ with S left-finite, pick $\tau \in T$; then $T \cap (-\infty, \tau]$ is finite and nonempty, so it has a least element, which is least in T. For the counterexample, the displayed S is enumerated by the strictly increasing sequence $1 - \frac{1}{k}$, so every nonempty subset has a least element; but $S \subseteq (-\infty, 1)$, so $S \cap (-\infty, 1) = S$ is infinite. It is exactly the cofinality clause of (i) that fails. In a general densely ordered Γ the same construction works: pick $a < b$ and, recursively, γ_{k+1} strictly between γ_k and b starting from $\gamma_1 \in (a, b)$; then $\{\gamma_k\}$ is well based of order type ω, and it is not left-finite because it is infinite and contained in $(-\infty, b)$.

(iii) Unions and translates are immediate. A nonempty left-finite set has a minimum by (i); write $\sigma_0 = \min S$, $\tau_0 = \min T$. If $\sigma + \tau < M$ with $\sigma \in S$, $\tau \in T$, then $\sigma < M - \tau_0$ and $\tau < M - \sigma_0$, so σ, τ range over finite sets; hence $(S + T) \cap (-\infty, M)$ is finite and every fiber is finite.

(iv) Put $\delta = \min S > 0$. A sum of k elements of S is at least $k\delta$, so a sum $< M$ forces $k\delta < M$, and archimedeanity leaves finitely many such k. Each summand of such a sum is $< M - (k-1)\delta \le M$, so lies in the finite set $S \cap (-\infty, M)$. There are therefore finitely many admissible multisets, which proves both assertions. $\square$

Remark 30.7 (Archimedean is not decoration). Part (iv) fails for non-archimedean Γ. Take $\Gamma = \mathbb{Z} \times \mathbb{Z}$ ordered lexicographically and $S = \{(0,1)\}$, which is left-finite, indeed finite. Then $S^* = \{(0,k) : k \in \mathbb{N}_0\}$ is infinite and contained in $(-\infty, (1,0))$, so it is not left-finite. The failure is precisely the failure of $\mathbb{Z}\delta$ to be cofinal, and it is the same phenomenon that would break separatedness of a filtration indexed by Γ. This is why Theorem 10.11 fixes $\Gamma \subseteq \mathbb{R}$; the restriction cannot be dropped without redoing every summability argument of Parts II–IV. It also means that in the nested-logarithmic setting of Section 30.5, where the monomial group is non-archimedean by construction, "left-finite" is the wrong notion and only well-basedness survives.

Theorem 30.8 (The support hierarchy, and where it collapses)**.** *Let Γ be a totally ordered abelian group.*

(i) *Grid-based $\Rightarrow$ well based, always.*

(ii) *Left-finite $\Rightarrow$ well based, always.*

(iii) *If Γ is archimedean, grid-based $\Rightarrow$ left-finite.*

(iv) *If Γ is not archimedean, grid-based $\not\Rightarrow$ left-finite.*

(v) *If Γ is archimedean and not densely ordered — equivalently, by Hölder's theorem, Γ is cyclic — then all three classes coincide with "bounded below".*

(vi) *For $\Gamma = \mathbb{Q}$ both inclusions in grid-based $\subsetneq$ left-finite $\subsetneq$ well based are strict.*

Proof. (i) Let $S \subseteq \mathfrak{n} + M$ with $M = \{\sum_i n_i\alpha_i : n_i \in \mathbb{N}_0\}$ and $\alpha_i \in \Gamma_{>0}$. It suffices to show M is well based. Suppose $v^{(1)}, v^{(2)}, \ldots \in \mathbb{N}_0^k$ give a strictly decreasing sequence $\sum_i v_i^{(m)}\alpha_i$. By Dickson's lemma the componentwise order on $\mathbb{N}_0^k$ is a well-quasi-order, so there are $m < m'$ with $v^{(m)} \leq v^{(m')}$ componentwise; as all $\alpha_i > 0$ this gives $\sum_i v_i^{(m)}\alpha_i \leq \sum_i v_i^{(m')}\alpha_i$, contradicting strict decrease. So M has no infinite strictly decreasing sequence and is well based; well-basedness passes to subsets and translates.

(ii) Theorem 30.6 (ii).

(iii) With S and M as in (i) and $M' \in \Gamma$ given, an element $\mathfrak{n} + \sum_i n_i\alpha_i < M'$ has $n_i\alpha_i < M' - \mathfrak{n}$ for each i, because the omitted summands are ≥ 0; archimedeanity leaves finitely many n_i for each i, hence finitely many tuples and finitely many values.

(iv) In $\mathbb{Z} \times \mathbb{Z}$ ordered lexicographically the set $M = \{(0,n) : n \in \mathbb{N}_0\}$ is grid-based with $\mathfrak{n} = (0,0)$ and the single generator $\alpha_1 = (0,1) > 0$, and it is not left-finite, as in Theorem 30.7.

(v) An archimedean group is order-isomorphic to a subgroup of $\mathbb{R}$ (Hölder), and a subgroup of $\mathbb{R}$ is either dense or cyclic. If $\Gamma = \mathbb{Z}\delta$ is cyclic, a set bounded below is finite below every bound, hence left-finite; it is then well based by (ii); and it is grid-based with generator δ and base point any lower bound. Conversely each class consists of sets bounded below: a well-based set has a least element if nonempty. This is Theorem 5.15 in the case $\Gamma = \mathbb{Z}$.

(vi) Left-finite $\subsetneq$ well based is Theorem 30.6 (ii). For grid-based $\subsetneq$ left-finite, take

$$S = \left\{k + \tfrac{1}{k} \;:\; k \in \mathbb{N}_{>0}\right\} \subseteq \mathbb{Q}.$$

It is enumerated by a strictly increasing sequence tending to $+\infty$, hence left-finite by Theorem 30.6 (i). If it were grid-based, it would lie in the subgroup of $\mathbb{Q}$ generated by $\mathfrak{n}, \alpha_1, \ldots, \alpha_k$; a finitely generated subgroup of $\mathbb{Q}$ is cyclic, say $N^{-1}\mathbb{Z}$, and $k + \frac{1}{k} \in N^{-1}\mathbb{Z}$ forces $N/k \in \mathbb{Z}$, that is $k \mid N$, which holds for only finitely many k. $\square$

Structural checkpoint

Three different things are being asked of a support, and they are not the same thing. *Well based* is what makes the product defined at all (Theorem 30.2), and by Theorem 30.4 it is the weakest condition that can be imposed on a whole class. *Left-finite* is what makes truncation produce finite data (Theorem 30.10 (ii)). *Grid-based* is what makes the exponent arithmetic itself finitary: exponents are integer vectors on a fixed finite basis. In the integer-exponent world of Parts I–IV all three collapse into "bounded below", which is exactly why the distinction is invisible there and unavoidable here.

30.3 Real exponents: Hahn power–log scales

Theorem 10.11 already introduced the power–log scale $\mathcal{H}(C,\Gamma)$: well-based supports in an ordered group $\mathbb{Z} \subseteq \Gamma \subseteq \mathbb{R}$, blocks in $C \in \{\mathbb{K}'[L], \mathbb{K}'(L), \mathbb{K}'((L^{-1}))\}$, and the derivation (10.15). We now isolate inside it the subring on which the finite bookkeeping of Parts II and VI is still meaningful, and prove that outside that subring it is not.

Definition 30.9 (The left-finite power–log scale)**.** Let $\mathcal{H} = \mathcal{H}(C,\Gamma)$ be a power–log scale in the sense of Theorem 10.11. For $F = \sum_{\gamma\in S} p_\gamma t^\gamma$ put $\operatorname{Exp}(F) = \{\gamma \in \Gamma : p_\gamma \neq 0\}$, the *exponent set* of F — the projection to Γ of $\operatorname{supp}_{\text{coeff}}(F)$. Define

$$\mathcal{H}_{\text{lf}}(C,\Gamma) := \{F \in \mathcal{H}(C,\Gamma) : \ \operatorname{Exp}(F) \text{ is left-finite}\},$$

and write $C[t^\Gamma]$ for the subring of elements with finite exponent set.

Theorem 30.10 (Truncation, completeness, and what $\mathcal{H}_{\text{lf}}$ is)**.** *Let $\mathcal{H} = \mathcal{H}(C,\Gamma)$ be a power–log scale.*

(i) *$\mathcal{H}_{\text{lf}}(C,\Gamma)$ is a subring of $\mathcal{H}$ containing $C[t^\Gamma]$, stable under D_X, and stable under the inversion of those of its elements that are units of $\mathcal{H}$.*

(ii) *For $F \in \mathcal{H}$: every truncation $\operatorname{Trunc}_{<N} F$, $N \in \Gamma$, has finite exponent set if and only if $F \in \mathcal{H}_{\text{lf}}(C,\Gamma)$.*

(iii) *Both $\mathcal{H}$ and $\mathcal{H}_{\text{lf}}(C,\Gamma)$ are separated and complete for the filtration $\mathfrak{F}_N = \{F \in \mathcal{H} : \nu_t F \geq N\}$, $N \in \Gamma$, whose topology is metrizable because $\mathbb{Z}$ is cofinal in Γ.*

(iv) *$\mathcal{H}_{\text{lf}}(C,\Gamma)$ is the closure of $C[t^\Gamma]$ in $\mathcal{H}$; equivalently it is the completion of $C[t^\Gamma]$ for the induced filtration. If Γ is dense in $\mathbb{R}$ then $\mathcal{H}_{\text{lf}}(C,\Gamma) \subsetneq \mathcal{H}(C,\Gamma)$.*

Proof. (i) Closure under sums and products is Theorem 30.6 (iii), the block of a product at t^γ being the finite sum over the fiber, which is finite by Theorem 30.2 (ii). Stability under D_X holds because (10.15) translates the exponent set by $1 \in \Gamma$. For inverses, let $F \in \mathcal{H}_{\text{lf}}$ be a unit of $\mathcal{H}$ and write $e = \nu_t F$, $p_e = [t^e]F$. If $FG = 1$ then $\nu_t G = -e$ and, by the leading-block law (Theorem 10.12(i) and its proof, or directly: the coefficient of t^0 in FG receives exactly one contribution, from the two leading blocks), $p_e\,[t^{-e}]G = 1$, so $p_e \in C^\times$. Hence $F = t^e p_e(1+v)$ with $v = t^{-e}p_e^{-1}F - 1 \in \mathcal{H}_{\text{lf}}$ and $\nu_t v > 0$; then $F^{-1} = t^{-e}p_e^{-1}\sum_{k\geq 0}(-v)^k$. Put $S = \operatorname{Exp}(v) \subseteq \Gamma_{>0}$, which is left-finite. By Theorem 30.6 (iv), S^* is left-finite and each of its elements has finitely many representations, so the family $\left((-v)^k\right)_{k\geq 0}$ is summable and its sum has exponent set inside S^*; translating by $-e$ preserves left-finiteness.

(ii) If $\operatorname{Exp}(F)$ is left-finite then $\operatorname{Exp}(F) \cap (-\infty, N)$, which is $\operatorname{Exp}(\operatorname{Trunc}_{<N} F)$, is finite. Conversely, if every $\operatorname{Trunc}_{<N} F$ has finite exponent set then $\operatorname{Exp}(F) \cap (-\infty, N)$ is finite for every $N \in \Gamma$.

(iii) *Metrizability and separatedness.* Since $1 \in \Gamma \subseteq \mathbb{R}$ and Γ is archimedean, no element of Γ exceeds every integer, so $\mathbb{Z}$ is cofinal in Γ and the countable subfamily $(\mathfrak{F}_N)_{N\in\mathbb{Z}}$ is a neighbourhood basis of 0. If $\nu_t F \geq N$ for every $N \in \mathbb{Z}$ then $F = 0$, for otherwise $\nu_t F \in \Gamma$ would exceed every integer. *Completeness.* Let (F_r) be Cauchy: for each $N \in \mathbb{Z}$ there is $r(N)$ with $\operatorname{Trunc}_{<N} F_r = \operatorname{Trunc}_{<N} F_{r(N)}$

for all $r \geq r(N)$, and we may take $r(N)$ non-decreasing in N. Define F blockwise by $[t^\gamma]F = [t^\gamma]F_{r(N)}$ for any integer $N > \gamma$; the definition is consistent. Then

$$\operatorname{Exp}(F) = \bigcup_{N\in\mathbb{Z}} \Big(\operatorname{Exp}(F_{r(N)}) \cap (-\infty, N)\Big),$$

which is well based: for $\emptyset \neq T \subseteq \operatorname{Exp}(F)$ pick $\tau \in T$ and an integer $N > \tau$; then $T \cap (-\infty, N)$ is a nonempty subset of the well-based set $\operatorname{Exp}(F_{r(N)})$ and its least element is least in T. So $F \in \mathcal{H}$ and $\nu_t(F - F_r) \geq N$ for $r \geq r(N)$. If all F_r lie in $\mathcal{H}_{\mathrm{lf}}$ then so does F, because $\operatorname{Exp}(F) \cap (-\infty, N) = \operatorname{Exp}(F_{r(N)}) \cap (-\infty, N)$ is finite for every integer N, and every $M \in \Gamma$ is below some integer.

(iv) $\mathcal{H}_{\mathrm{lf}}$ is closed in $\mathcal{H}$ by the last sentence of (iii); it contains $C[t^\Gamma]$; and every $F \in \mathcal{H}_{\mathrm{lf}}$ is the limit of the finitely supported truncations $\operatorname{Trunc}_{<N} F$, $N \in \mathbb{Z}$, by (ii). Conversely a limit of finitely supported elements is left-finite, since $\operatorname{Trunc}_{<N} F = \operatorname{Trunc}_{<N} G$ with G finitely supported makes $\operatorname{Trunc}_{<N} F$ finitely supported. Hence $\mathcal{H}_{\mathrm{lf}}$ is the closure of $C[t^\Gamma]$, and a closed subring containing a subring is its completion when the ambient is complete and separated. Finally, if Γ is dense in $\mathbb{R}$ then $\Gamma \cap (0,1)$ is infinite; an infinite bounded subset of $\mathbb{R}$ contains a strictly monotone sequence, and by passing to a subsequence we obtain $\gamma_1 < \gamma_2 < \cdots$ in $\Gamma \cap (0,1)$. Put $F = 1 + t + \sum_{k\geq 1} t^{\gamma_k}$. Its exponent set $\{0\} \cup \{\gamma_k : k \geq 1\} \cup \{1\}$ is well based — a nonempty subset containing 0 has least element 0; one meeting $\{\gamma_k\}$ has as least element the γ_k of least index; and the only remaining subset is $\{1\}$ — so $F \in \mathcal{H}$; it is not left-finite, since $\operatorname{Exp}(F) \cap (-\infty, 1)$ is infinite, so $F \notin \mathcal{H}_{\mathrm{lf}}$. □

Corollary 30.11 (The left-finite scale carries the near-identity theory)**.** *Let $\mathcal{H} = \mathcal{H}(C, \Gamma)$ and let $\mathcal{H}_{\geq 0} = \{A \in \mathcal{H} : \nu_t A \geq 0\}$.*

(i) *Every statement of Part 14 and of Part 16 whose proof uses only* (a) *that D_X raises ν_t by at least* 1*,* (b) *that a family with $\nu_t \to \infty$ may be summed termwise,* (c) *that a continuous $\mathbb{K}$-algebra endomorphism is determined by the image of the monomial t and of L together with the binomial rule for $(1+u)^\gamma$, $\gamma \in \Gamma$, $\nu_t u > 0$, and* (d) *that ν_t is additive, holds in $\mathcal{H}$ with $\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ replaced by $\mathcal{H}_{\geq 0}$ and the integer filtration by the Γ-filtration $\mathfrak{F}_N$. This covers the substitution operator* (Theorem 14.4)*, the Taylor formula* (Theorem 13.6)*, the filtered gain* (Theorem 14.6)*, the group law* (Theorem 14.7)*, separation and completeness of the contact filtration* (Theorem 14.13)*, and existence and uniqueness of the near-identity inverse* (Theorem 16.11).

(ii) *The subset*

$$\{X + A \ : \ A \in \mathcal{H}_{\mathrm{lf}}(C, \Gamma),\ \nu_t A \geq 0\}$$

is a subgroup of the group of (i)*: it is closed under composition and under compositional inversion.*

Proof. (i) Inspect the four inputs. D_X raises ν_t by 1 on monomials by (10.15) and hence on all of $\mathcal{H}$; summability of $(A^k D_X^k F/k!)_{k\geq 0}$ follows from $\nu_t(A^k D_X^k F) \geq \nu_t F + k$, and $\mathbb{Z}$ is cofinal in Γ, so only finitely many k affect any given block; the substitution operator is fixed on a general monomial by $t^\gamma \circ (X + A) = t^\gamma (1 + tA)^{-\gamma}$, the binomial series being summable since $\nu_t(tA) \geq 1$ (Theorem 8.37, Theorem 10.29), and on L by $L \circ (X + A) = L + \log(1 + tA)$; additivity of ν_t is Theorem 10.12(i). Each of the six listed statements uses only these.

(ii) Let $A, B \in \mathcal{H}_{\mathrm{lf}}$ with $\nu_t A, \nu_t B \geq 0$, and put $S_A = \operatorname{Exp}(A)$, $S_B = \operatorname{Exp}(B)$, $S'_B = S_B \cap \Gamma_{>0}$. By the Taylor formula, $A \circ (X + B) = \sum_{k\geq 0} B^k D_X^k A/k!$ and

$$\operatorname{Exp}\big(B^k D_X^k A\big) \ \subseteq\ (S_A + k) + (S'_B)^* \ \subseteq\ (S_A + k) + \Gamma_{\geq 0},$$

because k copies of S_B contribute a sum of elements of S'_B together with some copies of 0. Fix $M \in \Gamma$. A contribution to $(-\infty, M)$ requires $\min S_A + k < M$, so only finitely many k contribute, since $\mathbb{Z}$ is cofinal in the archimedean Γ. For each such k, the set $(S_A + k) + (S'_B)^*$ is left-finite by Theorem 30.6 (iii)–(iv). A finite union of left-finite sets is left-finite, so $A \circ (X + B) \in \mathcal{H}_{\mathrm{lf}}$. Closure under inversion follows: the inverse of $X + A$ is $X + B$ with B the $\mathfrak{F}$-limit of $B_0 = 0$, $B_{n+1} = -A \circ (X + B_n)$

(Theorem 16.11), every B_n lies in $\mathcal{H}_{\mathrm{lf}}$ by what has just been proved and induction, and $\mathcal{H}_{\mathrm{lf}}$ is closed in $\mathcal{H}$ by Theorem 30.10 (iv). $\square$

What clause (i) deliberately omits

The statements about the *associated graded* of the contact filtration — Theorem 14.14, Theorem 14.15, Theorem 14.16, Theorem 14.18 — are formulated using that $r \mapsto r+1$ is the successor in the index group. Write $\mathcal{G}_\gamma$ for the step of the contact filtration of Theorem 14.11 indexed by $\gamma \in \Gamma$. For dense Γ the quotients $\mathcal{G}_\gamma/\mathcal{G}_{\gamma+1}$ remain defined, because the gain is the element $1 \in \Gamma$, but they are no longer single steps of the filtration, and the identification of the graded object with a Lie algebra is not the same statement. We do not claim it here. The same caution applies to any argument that enumerates blocks one at a time; see the entry for Theorem 17.2 in Table 30.2.

Proposition 30.12 (Truncation, and therefore all of Part VI, fails in the general Hahn scale). *Let $\Gamma \subseteq \mathbb{R}$ be dense with $\mathbb{Z} \subseteq \Gamma$, and let $\mathcal{H} = \mathcal{H}(\mathbb{K}[L], \Gamma)$. There is $F \in \mathcal{H}$ with $\nu_t F = 0$ and every block equal to 1 such that:*

(a) $\mathrm{Trunc}_{<N} F$ *has infinite exponent set for every $N \geq 1$ in Γ;*

(b) $\mathrm{supp}_{\mathrm{coeff}}(F)$ *admits no enumeration as a strictly $\succ$-decreasing sequence of order type at most ω exhausting it, so Theorem 5.15 (d) is false in $\mathcal{H}$, although Theorem 5.15 (c) remains true by definition of $\mathcal{H}$;*

(c) *F has no sparse block representation in the sense of Theorem 27.1 at any precision $N \geq 1$; consequently no routine of Part 27 accepts F as input, and the precision calculus of Theorem 27.10 and the cost model of Theorem 27.17 have no meaning for it.*

Proof. Take $F = 1 + t + \sum_{k \geq 1} t^{\gamma_k}$ with $\gamma_k \in \Gamma \cap (0,1)$ strictly increasing, as constructed in the proof of Theorem 30.10 (iv); then $\nu_t F = 0$ and every block is 1. (a) All the γ_k and 0 are $< 1 \leq N$, so $\mathrm{Exp}(\mathrm{Trunc}_{<N} F) \supseteq \{\gamma_k : k \geq 1\}$ is infinite. (b) An enumeration $s_0 \succ s_1 \succ \cdots$ of order type at most ω exhausting $\mathrm{Exp}(F)$ would be an increasing enumeration of a set whose order type is at most ω. But $\mathrm{Exp}(F)$ has order type $\omega + 1$: the element 1 is strictly greater than each of the infinitely many elements $0 < \gamma_1 < \gamma_2 < \cdots$, so in any exhausting increasing enumeration it would have to be preceded by infinitely many terms and could occupy no finite position. Hence Theorem 5.15 (d) fails. By Theorem 30.6 (i) the correct replacement for (d) in a general Γ is not "order type at most ω" but left-finiteness, which additionally demands cofinality of the enumeration. (c) A sparse block representation is a *finitely supported* map v together with a cutoff N, and by Theorem 27.3 its denotation is the coset $\sum_{\gamma<N} v(\gamma)t^\gamma + \mathfrak{F}_N$. If this coset contained F then $\mathrm{Trunc}_{<N} F$ would equal the finitely supported $\sum_{\gamma<N} v(\gamma)t^\gamma$, contradicting (a). $\square$

30.4 Puiseux-log classes, and the union over denominators

The Puiseux-log ring $\mathbb{K}[L]((t^{1/s}))$ of Theorem 5.24 and the ramified tower of Theorem 5.25 are the minimal enlargement forced by a rational leading exponent (Theorem 19.5). At a fixed denominator nothing is lost at all, and the reason is worth stating as a theorem rather than as the folklore "the derivative formula generalizes without change" of S1, S3 and S5: the ring is not merely similar to $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$, it *is* $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ in a different large variable.

Proposition 30.13 (A fixed denominator is a change of variable, not an enlargement). *Fix $s \in \mathbb{N}_{>0}$ and set*

$$Y := X^{1/s}, \qquad \tau := Y^{-1} = t^{1/s}, \qquad \Lambda := \log Y = s^{-1}L.$$

Table 30.2: Transfer of Parts I–IV and VI to a Hahn power–log scale $\mathcal{H}(C,\Gamma)$ with $\Gamma \subseteq \mathbb{R}$ dense. "Verbatim" means the statement and its proof go through with $n \in \mathbb{Z}$ read as $\gamma \in \Gamma$.

Statement	Status	What changes
Domain, valuation laws (Theorem 6.2)	verbatim	—
Unit criterion (Theorem 9.3)	verbatim	—
Reciprocal of a unit (Theorem 9.7)	modified	the block-by-block recurrence presupposes a successor exponent; replace it by the geometric series, summable by Theorem 30.2 (iii)
t-adic completeness (Theorem 7.18)	verbatim	Theorem 30.10 (iii)
Support characterisation, clause (d) (Theorem 5.15)	**false**	Theorem 30.12 (b)
Block and repeated-derivative formulas (Theorem 11.3, Theorem 11.8)	verbatim	the shift is $\partial_L - \gamma$ with $\gamma \in \Gamma$; the successive shifts $\gamma, \gamma+1, \ldots$ stay integrally spaced
Sharp valuation of D_X (Theorem 11.11), kernel	verbatim	—
Antiderivatives (Theorem 11.18)	verbatim	the resonant exponent is still the single value $\gamma = 0$
Exponential boundary (Theorem 10.16)	modified	m may be any element of Γ (Theorem 10.17)
Taylor formula, near-identity group, reversion (Theorem 14.7, Theorem 16.11)	verbatim	Theorem 30.11
Triangular reversal (Theorem 17.2)	modified	"the next block" is undefined when Γ is dense; one must fix a cofinal sequence in Γ and cancel residuals along it
Lagrange–Bürmann index range $r \leq N+1$ (Theorem 18.19)	**open**	the bound is an integrality statement; see Theorem 31.10
Sparse representation, precision calculus, cost model (Theorem 27.1, Theorem 27.10, Theorem 27.17)	**false**	restored verbatim on $\mathcal{H}_{\mathrm{lf}}(C,\Gamma)$

Then the $\mathbb{K}$-linear map sending $t^{r/s}L^j$ to $s^j\tau^r\Lambda^j$ is an isomorphism of ordered $\mathbb{K}$-algebras

$$\Phi_s:\ \mathbb{K}[L]\big((t^{1/s})\big)\ \longrightarrow\ \mathbb{K}[\Lambda]((\tau)),$$

onto the polynomial–logarithmic Laurent ring of Theorem 5.3 built from the generators τ, Λ, and

$$\Phi_s\circ D_X\circ\Phi_s^{-1}=s^{-1}\tau^{\,s-1}\,D_Y,\tag{30.1}$$

where $D_Y=\tau\partial_\Lambda-\tau^2\partial_\tau$ is the distinguished derivation of the target. Consequently every statement of Parts I–IV that mentions only the ring structure, the order, the valuation, the truncation calculus and the derivation up to a unit factor holds verbatim in $\mathbb{K}[L]((t^{1/s}))$. In particular the unit criterion, the reciprocal recurrence, the completeness theorem, the sparse representation and the precision calculus transfer with no change whatever, and

$$D_X\big(t^{\alpha}p(L)\big)=t^{\alpha+1}\big(\partial_Lp-\alpha p\big),\qquad \alpha\in s^{-1}\mathbb{Z}.\tag{30.2}$$

Proof. Φ_s is the relabelling of generators $t^{1/s}\mapsto\tau$, $L\mapsto s\Lambda$; it is bijective on monomials, hence a $\mathbb{K}$-algebra isomorphism, and it preserves the dominance order because it preserves the exponent pair (r,j) and the order is lexicographic in that pair. For (30.1) it suffices to compare on the two generators. On the one hand $D_X\tau=D_Xt^{1/s}=-s^{-1}t^{1/s+1}=-s^{-1}\tau^{\,s+1}$, and $s^{-1}\tau^{\,s-1}D_Y\tau=-s^{-1}\tau^{\,s-1}\tau^2=-s^{-1}\tau^{\,s+1}$. On the other hand $D_X\Lambda=s^{-1}D_XL=s^{-1}t=s^{-1}\tau^{\,s}$, and $s^{-1}\tau^{\,s-1}D_Y\Lambda=s^{-1}\tau^{\,s-1}\tau=s^{-1}\tau^{\,s}$. Both sides of (30.1) are derivations of $\mathbb{K}[\Lambda]((\tau))$ that raise the τ-order, hence are continuous for the τ-adic filtration; their difference is a continuous derivation vanishing on τ and Λ, so by the Leibniz rule it vanishes on $\mathbb{K}[\Lambda][\tau,\tau^{-1}]$ and by continuity and Theorem 7.18 on all of $\mathbb{K}[\Lambda]((\tau))$. Formula (30.2) is (10.15) for $\Gamma=s^{-1}\mathbb{Z}$, or equivalently (30.1) read backwards. ☐

Remark 30.14 (The one thing the change of variable does *not* transport). Φ_s is an isomorphism of ordered differential rings up to a unit, but it does not carry near-identity maps in X to near-identity maps in Y identically: if $F=X+A$ with $A\in\mathbb{K}[L][[t^{1/s}]]$, then in the variable Y the same map reads

$$Y\longmapsto\big(Y^s+A\big)^{1/s}=Y\big(1+\tau^sA\big)^{1/s}=Y+s^{-1}\tau^{\,s-1}A+\mathcal{O}_\tau^{\text{form}}\big(\tau^{2s-1}\big),$$

which is again near-identity but with a perturbation of outer order $s-1$ higher. Reversion therefore transports through the conjugation $F\mapsto j_s\circ F\circ j_s^{\langle -1\rangle\circ}$ with $j_s(X)=X^{1/s}$, not through Φ_s alone. This is the formal counterpart of the fact that tangency to the identity is a property of a map together with a coordinate.

The union over denominators behaves quite differently, and here the sources are silent about the essential point.

Theorem 30.15 (The Puiseux-log union is separated but not complete)**.** *Let*

$$\mathfrak{P}:=\mathbb{K}[L]\big((t^{1/\infty})\big)=\bigcup_{s\in\mathbb{N}_{>0}}\mathbb{K}[L]\big((t^{1/s})\big)$$

be the ramified union of Theorem 5.25, viewed inside the power–log scale $\mathcal{H}(\mathbb{K}[L],\mathbb{Q})$. Then:

(i) *$\mathfrak{P}\subseteq\mathcal{H}_{\mathrm{lf}}(\mathbb{K}[L],\mathbb{Q})$, and $\mathfrak{P}$ contains every element with finite exponent set;*

(ii) *$\mathfrak{P}$ is separated for the $\mathbb{Q}$-indexed filtration, but it is* not *complete: the sequence of partial sums of*

$$F:=\sum_{k\geq 2}t^{\,k-1/k}\tag{30.3}$$

is Cauchy in $\mathfrak{P}$ and has no limit in $\mathfrak{P}$. Hence Theorem 7.18 is false for $\mathfrak{P}$;

(iii) *the closure of* $\mathfrak{P}$ *inside* $\mathcal{H}(\mathbb{K}[L],\mathbb{Q})$ *is exactly* $\mathcal{H}_{\mathrm{lf}}(\mathbb{K}[L],\mathbb{Q})$*; that is, the left-finite Hahn ring over* $\mathbb{Q}$ *is the completion of the Puiseux-log union.*

Proof. (i) An element of $\mathbb{K}[L]((t^{1/s}))$ has exponent set contained in $s^{-1}\mathbb{Z}$ and bounded below, hence left-finite by Theorem 30.8 (v) applied to the cyclic group $s^{-1}\mathbb{Z}$, or directly. Conversely a finite set of rationals has a common denominator s, so any element with finite exponent set lies in $\mathbb{K}[L]((t^{1/s}))$.

(ii) Separatedness is inherited from Theorem 30.10 (iii). Put $F_K = \sum_{k=2}^{K} t^{k-1/k}$; then $F_K \in \mathbb{K}[L]((t^{1/K!})) \subseteq \mathfrak{P}$, and for $K' > K$ we have $\nu_t(F_{K'} - F_K) = (K+1) - \frac{1}{K+1} > K$, so (F_K) is Cauchy. Its limit in the complete ring $\mathcal{H}(\mathbb{K}[L],\mathbb{Q})$ is F of (30.3), whose exponent set $\left\{k - \frac{1}{k} : k \geq 2\right\}$ is strictly increasing and unbounded, hence left-finite, so $F \in \mathcal{H}_{\mathrm{lf}}$. But $F \notin \mathfrak{P}$: if $F \in \mathbb{K}[L]((t^{1/s}))$ then $k - \frac{1}{k} \in s^{-1}\mathbb{Z}$ for all $k \geq 2$, that is $s/k \in \mathbb{Z}$, that is $k \mid s$, which fails for all $k > s$. By separatedness a limit is unique, so (F_K) has no limit in $\mathfrak{P}$.

(iii) By (i), $\mathbb{K}[L][t^{\mathbb{Q}}] \subseteq \mathfrak{P} \subseteq \mathcal{H}_{\mathrm{lf}}$. Taking closures in $\mathcal{H}(\mathbb{K}[L],\mathbb{Q})$ and using Theorem 30.10 (iv), which identifies the closure of $\mathbb{K}[L][t^{\mathbb{Q}}]$ with $\mathcal{H}_{\mathrm{lf}}$, and the fact that $\mathcal{H}_{\mathrm{lf}}$ is closed, all three closures coincide. □

Remark 30.16 (Why this matters operationally). Every individual computation in the Puiseux tower happens at some finite denominator, and Theorem 30.13 says that at a fixed denominator the whole apparatus of Parts I–IV applies unchanged. What Theorem 30.15 adds is that a *limit* of such computations may leave the tower. An implementation that stores a single denominator per object is therefore sound for any finite sequence of operations and unsound as a model of the completed class; the completed class is $\mathcal{H}_{\mathrm{lf}}$, on which Theorem 30.11 guarantees closure under near-identity composition and reversion but no denominator bound. Bounding the denominator growth of an explicit iteration is the content of Theorem 31.11.

30.4.1 An obstruction no exponent group repairs

Proposition 30.17 (No power–log scale contains a square root of L)**.** *Let* $\mathcal{H} = \mathcal{H}(C,\Gamma)$ *be a power–log scale in the sense of Theorem 10.11, so that* $C \in \left\{\mathbb{K}'[L], \mathbb{K}'(L), \mathbb{K}'((L^{-1}))\right\}$*. Then there is no* $F \in \mathcal{H}$ *with* $F^2 = L$*. This holds for every exponent group* Γ*, in particular for* $\Gamma = \mathbb{R}$*.*

Proof. Suppose $F^2 = L$. Since $\mathcal{H}$ is a domain and ν_t is additive (Theorem 10.12(i)), $2\nu_t F = \nu_t L = 0$, so $\nu_t F = 0$; as $F \neq 0$ and its exponent set is well based, the block $p_0 = [t^0]F$ is defined and nonzero, and comparing blocks at t^0 gives $p_0^2 = L$ in C. It therefore suffices to show that L is not a square in any of the three coefficient rings.

For $C = \mathbb{K}'(L)$ (and hence for the subring $\mathbb{K}'[L]$): write $p_0 = u/v$ with $u, v \in \mathbb{K}'[L]$ coprime and $v \neq 0$; then $u^2 = Lv^2$. The polynomial L is irreducible in the unique factorisation domain $\mathbb{K}'[L]$, and its multiplicity in u^2 is even while its multiplicity in Lv^2 is odd, a contradiction.

For $C = \mathbb{K}'((L^{-1}))$: write $p_0 = \sum_{j \leq d} c_j L^j$ with $c_d \neq 0$. Then $p_0^2 = c_d^2 L^{2d} +$ (lower powers), so comparing the top exponent with that of L gives $2d = 1$, impossible for $d \in \mathbb{Z}$. □

Remark 30.18 (Which axis repairs it). $\sqrt{L}$ does exist in the rank-two Hahn field $\mathcal{H}_1(\mathbb{K},\Gamma)$ of Theorem 15.1 whenever $\frac{1}{2} \in \Gamma$, since there $L_1^{1/2}$ is a monomial. The point of Theorem 30.17 is that the four classes of Part I and all their Puiseux and Hahn refinements share one structural feature — the exponent of L inside a block runs over $\mathbb{Z}$, never over a larger group — and that no amount of refinement along the t-exponent axis touches it. The enlargement required is a ramification of the *monomial group* in which L itself carries fractional exponents; algebraically this is the step from a scale with a coefficient *ring* to a genuine rank-two Hahn field. Concretely, $\sqrt{\log X}$ arises the moment one inverts a map with a half-integral logarithmic exponent. For $Y = X(\log X)^{1/2}$ with $X \to +\infty$, taking logarithms gives $\log Y = \log X + \frac{1}{2}\log\log X$; dividing that relation by $\log Y$ and taking logarithms gives $\log\log X - \log\log Y = \log\bigl(1 - \frac{1}{2}\log\log X/\log Y\bigr) = \mathcal{O}_{Y\to+\infty}(\log\log Y/\log Y)$, whence

$$\log X = \log Y - \tfrac{1}{2}\log\log Y + \mathcal{O}_{Y\to+\infty}(\log\log Y/\log Y)\,,$$

and hence

$$X = Y\,(\log X)^{-1/2} = Y\,(\log Y)^{-1/2}\Bigl(1 + \tfrac{1}{4}\frac{\log\log Y}{\log Y} + \mathcal{O}_{Y\to+\infty}\bigl((\log\log Y)^2(\log Y)^{-2}\bigr)\Bigr),$$

whose leading monomial $Y(\log Y)^{-1/2}$ has a half-integral exponent on the first logarithm; the expansion lives in $\mathcal{H}_2\bigl(\mathbb{K}, \frac{1}{2}\mathbb{Z}\bigr)$ and in no power–log scale. The computation is the shape of Theorem 19.7 at $\rho = 1$, $\sigma = \frac{1}{2}$; that theorem is stated for $\sigma \in \mathbb{N}_0$, and we do not widen its hypothesis. The display above is proved where it stands, and it agrees with (19.15) at $c = 1$, $\rho = 1$, $\sigma = \frac{1}{2}$.

30.5 Nested logarithms, and the residue that forces each new level

The nested-logarithm tower $L_0 = X$, $L_{j+1} = \log L_j$, its monomial groups $\mathcal{M}_r(\Gamma)$ and its Hahn ambients $\mathcal{H}_r(\mathbb{K},\Gamma)$ were introduced in Theorem 15.1; the tower derivation $\mathcal{D}_r$, the minimal Laurent coefficient ring $\mathcal{L}_r$, the near-identity group at depth r and the depth-dependent unit criterion are Theorem 15.13, Theorem 15.14 and Theorem 15.15. Two facts remain to be proved, and they are the ones that make the tower a genuine hierarchy rather than a notational convenience: the derivation extends to the full Hahn ambient, and each level is forced by an obstruction of residue type that no enlargement along any other axis removes.

Throughout, $\mathbb{Z} \subseteq \Gamma \subseteq \mathbb{R}$ is an ordered group equipped, as in Theorem 10.11, with an embedding $\Gamma \hookrightarrow \mathbb{K}$ of additive groups. We extend Theorem 15.1 to $r = 0$ by the same formula, so that $\mathcal{M}_0(\Gamma) = \{X^{\alpha_0} : \alpha_0 \in \Gamma\}$ and $\mathcal{H}_0(\mathbb{K},\Gamma)$ is the logarithm-free Hahn field of real powers of X. We write a monomial of $\mathcal{M}_r(\Gamma)$ as

$$\mathfrak{m} = \prod_{j=0}^{r} L_j^{\alpha_j(\mathfrak{m})}, \qquad \alpha_j(\mathfrak{m}) \in \Gamma, \qquad \mathfrak{n}_j := L_0 L_1 \cdots L_j \quad (0 \le j \le r). \tag{30.4}$$

Each $\alpha_j : \mathcal{M}_r(\Gamma) \to \Gamma$ is a group homomorphism, and $\mathfrak{n}_j \succ 1$, so $\mathfrak{n}_j^{-1} \prec 1$; moreover $\mathfrak{n}_0^{-1} = t \succ \mathfrak{n}_1^{-1} \succ \cdots \succ \mathfrak{n}_r^{-1}$.

Lemma 30.19 (The tower derivation on the Hahn ambient). *For $0 \le j \le r$ let ϑ_j act on $F = \sum_{\mathfrak{m}\in S} c_{\mathfrak{m}}\mathfrak{m} \in \mathcal{H}_r(\mathbb{K},\Gamma)$ by $\vartheta_j F = \sum_{\mathfrak{m}\in S} \alpha_j(\mathfrak{m}) c_{\mathfrak{m}}\mathfrak{m}$, and set*

$$\partial_r := \sum_{j=0}^{r} \mathfrak{n}_j^{-1}\vartheta_j. \tag{30.5}$$

Then:

(i) *each ϑ_j is a well-defined $\mathbb{K}$-linear derivation of $\mathcal{H}_r(\mathbb{K},\Gamma)$, and so is ∂_r;*

(ii) *$\partial_r L_j = \mathfrak{n}_{j-1}^{-1}$ for $1 \le j \le r$ and $\partial_r L_0 = 1$, in agreement with Theorem 15.13; and for $\mathfrak{m} \in \mathcal{M}_r(\Gamma)$,*

$$\frac{\partial_r \mathfrak{m}}{\mathfrak{m}} = \sum_{j=0}^{r} \alpha_j(\mathfrak{m})\,\mathfrak{n}_j^{-1}; \tag{30.6}$$

(iii) *on $\mathscr{PL}_{\mathrm{poly},\mathbb{K}} \subseteq \mathcal{H}_1(\mathbb{K},\mathbb{Z})$, ∂_1 is D_X; on a power–log scale $\mathcal{H}(\mathbb{K}[L],\Gamma)$ it is* (10.15)*; and on $\mathcal{L}_r((t))$ it is* (15.9).

Proof. (i) $\vartheta_j F$ has support inside that of F, so it is well defined. It is a derivation: the coefficient of $\mathfrak{p}$ in $\vartheta_j(FG)$ is $\alpha_j(\mathfrak{p})\sum_{\mathfrak{m}\mathfrak{n}=\mathfrak{p}} c_\mathfrak{m} d_\mathfrak{n}$, while the coefficient of $\mathfrak{p}$ in $(\vartheta_j F)G + F(\vartheta_j G)$ is $\sum_{\mathfrak{m}\mathfrak{n}=\mathfrak{p}}\big(\alpha_j(\mathfrak{m}) + \alpha_j(\mathfrak{n})\big)c_\mathfrak{m} d_\mathfrak{n}$; these agree because α_j is a homomorphism, and both sums are finite by Theorem 30.2 (ii). Multiplying a derivation by a fixed ring element again gives a derivation, and a finite sum of derivations is a derivation, so ∂_r is one. Its support lies in $\bigcup_j \mathfrak{n}_j^{-1}S$, a finite union of translates of a well-based set, and each monomial receives at most $r+1$ contributions, one per j, because $\mathfrak{m}\mapsto\mathfrak{m}\mathfrak{n}_j^{-1}$ is injective.

(ii) For $\mathfrak{m} = L_j$ one has $\alpha_i(L_j) = \delta_{ij}$, so $\partial_r L_j = \mathfrak{n}_j^{-1}L_j = (L_0\cdots L_{j-1})^{-1} = \mathfrak{n}_{j-1}^{-1}$ for $j\ge 1$, and $\partial_r L_0 = L_0^{-1}L_0 = 1$. Formula (30.6) is (30.5) applied to the single monomial $\mathfrak{m}$.

(iii) For $\mathfrak{m} = t^\gamma L^k = L_0^{-\gamma}L_1^k$, (30.6) gives $\partial_1\mathfrak{m} = \mathfrak{m}\big(-\gamma L_0^{-1} + k(L_0L_1)^{-1}\big) = t^{\gamma+1}\big(kL^{k-1} - \gamma L^k\big) = t^{\gamma+1}(\partial_L - \gamma)L^k$, which is (10.15), and for $\gamma = n\in\mathbb{Z}$ this is D_X. The identification with (15.9) is the same computation with $\mathfrak{m} = t^n L_1^{a_1}\cdots L_r^{a_r}$. □

The next theorem is the exact reason the tower cannot be truncated. It is the depth-r form of Theorem 11.22, which is the case $r=1$: there the obstruction was the residue $[t^1L^{-1}]$, and $t^1L^{-1} = \mathfrak{n}_1^{-1}$.

Theorem 30.20 (Each level of the tower is forced by an unmatched residue)**.** *Let $r\ge 0$ and let $\mathcal{H}_r = \mathcal{H}_r(\mathbb{K},\Gamma)$ with $\mathbb{Z}\subseteq\Gamma\subseteq\mathbb{R}$. Then*

$$\big[\mathfrak{n}_r^{-1}\big]\,\partial_r F = 0 \qquad \textit{for every } F\in\mathcal{H}_r, \tag{30.7}$$

where $[\mathfrak{n}]G$ denotes the coefficient of the monomial $\mathfrak{n}$ in G. Consequently:

(i) $\mathfrak{n}_r^{-1} = \dfrac{1}{L_0L_1\cdots L_r}$ *has no antiderivative in $\mathcal{H}_r$, so ∂_r is not surjective and Theorem 11.18 fails at every depth once the ambient is the full Hahn field;*

(ii) *if $(\mathcal{T},\partial)$ is any differential field extension of $\mathcal{H}_r$ whose derivation restricts to ∂_r, and $\Lambda\in\mathcal{T}$ satisfies $\partial\Lambda = \mathfrak{n}_r^{-1}$ — as $L_{r+1} = \log L_r$ does — then $\Lambda\notin\mathcal{H}_r$. In particular $L_{r+1}\notin\mathcal{H}_r(\mathbb{K},\Gamma)$ for every Γ, and*

$$\mathcal{H}_0(\mathbb{K},\Gamma)\subsetneq\mathcal{H}_1(\mathbb{K},\Gamma)\subsetneq\mathcal{H}_2(\mathbb{K},\Gamma)\subsetneq\cdots$$

is a strictly increasing chain of differential subfields;

(iii) *no $\mathcal{H}_r(\mathbb{K},\Gamma)$ is closed under the logarithm of its positive elements: $L_r\in\mathcal{H}_r$ but $\log L_r\notin\mathcal{H}_r$.*

Proof. Fix $F = \sum_{\mathfrak{m}\in S} c_\mathfrak{m}\mathfrak{m}\in\mathcal{H}_r$. By (30.5), the monomial $\mathfrak{n}_r^{-1}$ receives, for each index $j\in\{0,\dots,r\}$, exactly one possible contribution, namely from the unique monomial $\mathfrak{m}_j$ with $\mathfrak{m}_j\mathfrak{n}_j^{-1} = \mathfrak{n}_r^{-1}$, that is

$$\mathfrak{m}_j = \mathfrak{n}_j\mathfrak{n}_r^{-1} = \frac{L_0L_1\cdots L_j}{L_0L_1\cdots L_r} = \prod_{i=j+1}^{r} L_i^{-1},$$

the empty product for $j=r$ being 1. Hence

$$\big[\mathfrak{n}_r^{-1}\big]\partial_r F = \sum_{j=0}^{r}\alpha_j(\mathfrak{m}_j)\,c_{\mathfrak{m}_j},$$

a finite sum, in which $c_{\mathfrak{m}_j}$ is read as 0 when $\mathfrak{m}_j\notin S$. But $\mathfrak{m}_j$ involves only the letters $L_{j+1},\dots,L_r$, so its L_j-exponent is $\alpha_j(\mathfrak{m}_j) = 0$ for every j, and every term vanishes. This proves (30.7).

(i) If $\partial_r F = \mathfrak{n}_r^{-1}$ then the left side has $\mathfrak{n}_r^{-1}$-coefficient 0 and the right side has $\mathfrak{n}_r^{-1}$-coefficient 1.

(ii) If $\Lambda\in\mathcal{H}_r$ then $\partial\Lambda = \partial_r\Lambda$ has vanishing $\mathfrak{n}_r^{-1}$-coefficient by (30.7), contradicting $\partial\Lambda = \mathfrak{n}_r^{-1}$. That L_{r+1} has this derivative is Theorem 15.13 at index $r+1$, rewritten with $\mathfrak{n}_r = L_0L_1\cdots L_r$. Inclusion $\mathcal{H}_r\subseteq\mathcal{H}_{r+1}$ holds because $\mathcal{M}_r(\Gamma)$ is the subgroup of $\mathcal{M}_{r+1}(\Gamma)$ with $\alpha_{r+1}=0$, the induced order agrees, and well-basedness and the convolution product are inherited; strictness is the membership

$L_{r+1} \in \mathcal{H}_{r+1} \setminus \mathcal{H}_r$. Each inclusion commutes with the derivations, since ∂_{r+1} restricted to monomials with $\alpha_{r+1} = 0$ is ∂_r.

(iii) Immediate from (ii) with $\Lambda = L_{r+1}$. □

Corollary 30.21 (Depth is not bounded, and the union is the smallest logarithmically closed ambient). *Put $\mathcal{H}_\infty(\mathbb{K}, \Gamma) = \bigcup_{r\geq 0} \mathcal{H}_r(\mathbb{K}, \Gamma)$, a directed union of differential fields, hence a differential field. Then $\mathcal{H}_\infty$ contains L_j for every j, no proper member $\mathcal{H}_r$ of the chain does, and any subfield of an ambient differential field that contains X and is closed under $\log$ on the elements $L_0, L_1, L_2, \ldots$ contains all of them. In particular a bound on logarithmic depth is never a theorem about a class; it is a hypothesis about a particular computation.*

Proof. A directed union of subfields of a fixed field is a subfield, and the derivations agree on overlaps by Theorem 30.20 (ii); $L_j \in \mathcal{H}_j \subseteq \mathcal{H}_\infty$ while $L_{r+1} \notin \mathcal{H}_r$ by the same theorem. The last assertion is an induction on j starting from $L_0 = X$. □

A depth bound is a hypothesis, not a conclusion

Theorem 19.4 shows one logarithm in a leading monomial forcing one new level; Theorem 30.20 shows that no level is ever redundant. Neither result bounds the depth needed by an arbitrary composite of admissible operations, and this volume proves no such bound. What is proved is that the classification of scale changes (Theorem 15.2) raises the depth of a single power–logarithm substitution by at most one; iterating it k times can therefore be guaranteed only within depth $r + k$. Whether that is sharp for every k is Theorem 31.9.

30.6 Multiseries: nesting, the fast variable, and the completion order

A *multiseries* is an expansion whose coefficients are themselves expansions, in a strictly slower scale, iterated to a finite depth. Formally this is the iterated Laurent construction of Theorem 9.36, carried up the logarithmic tower.

Definition 30.22 (Finite-depth multiseries field). For $r \geq 0$ put

$$\mathfrak{J}_r := \mathbb{K}\big((L_r^{-1})\big)\big((L_{r-1}^{-1})\big) \cdots \big((L_1^{-1})\big)\big((L_0^{-1})\big), \tag{30.8}$$

the iterated field of formal Laurent series in $L_0^{-1} = t$ whose coefficients are formal Laurent series in L_1^{-1}, whose coefficients are formal Laurent series in L_2^{-1}, and so on to depth r; the outermost construction carries the *most dominant* scale. Thus $\mathfrak{J}_0 = \mathbb{K}((t))$ and $\mathfrak{J}_1 = \mathscr{PL}_{\mathrm{itLaur},\mathbb{K}}$. An element of $\mathfrak{J}_r$ is a family $(c_m)_{m\in\mathbb{Z}^{r+1}}$ of scalars, $m = (m_0, \ldots, m_r)$ representing the monomial $L_0^{-m_0} L_1^{-m_1} \cdots L_r^{-m_r}$.

Theorem 30.23 (The iterated field is the Hahn field, and only in the matching order). *Let $r \geq 0$.*

(i) *$\mathfrak{J}_r = \mathcal{H}_r(\mathbb{K}, \mathbb{Z})$ as $\mathbb{K}$-algebras, the identification being the identity on monomials. Explicitly, a family $(c_m)_{m\in\mathbb{Z}^{r+1}}$ lies in $\mathfrak{J}_r$ if and only if its support is well ordered in the direction of increasing smallness for the dominance order of Theorem 15.1.*

(ii) *Let π be a permutation of $\{0, 1, \ldots, r\}$ and let $\mathfrak{J}_\pi$ be built as in (30.8) with the levels nested in the order $\pi(0)$ outermost, …, $\pi(r)$ innermost. If $\pi \neq \mathrm{id}$ then neither of $\mathfrak{J}_\pi$, $\mathfrak{J}_r$ contains the other, as subsets of the space of families (c_m). More generally, if $\pi \neq \pi'$ then neither of $\mathfrak{J}_\pi$, $\mathfrak{J}_{\pi'}$ contains the other; so the $(r + 1)!$ nestings give $(r + 1)!$ pairwise incomparable, and in particular pairwise distinct, subsets.*

Proof. (i) Write $\mathfrak{m}(m) = L_0^{-m_0} \cdots L_r^{-m_r}$. By Theorem 15.1, $\mathfrak{m}(m) \succ \mathfrak{m}(m')$ exactly when $m < m'$ lexicographically, so "the support is well ordered in the direction of increasing smallness" means: every nonempty subset of the support, read as a set of vectors $m \in \mathbb{Z}^{r+1}$, has a lexicographically least element. Membership in $\mathfrak{J}_r$, unfolded, says: the set of m_0 occurring is bounded below; for each such m_0 the set of m_1 occurring with that m_0 is bounded below; and so on through m_r. These two conditions are equivalent.

$\Leftarrow$: given a nonempty subset T, let μ_0 be the least m_0 occurring in T, which exists because the set is a nonempty subset of $\mathbb{Z}$ bounded below; restrict T to that value and let μ_1 be the least m_1 occurring, and so on. The resulting vector $(\mu_0, \dots, \mu_r)$ lies in T and is lexicographically least.

$\Rightarrow$: if for some k and some fixed prefix $(\mu_0, \dots, \mu_{k-1})$ occurring in the support the set of admissible m_k were unbounded below, choose $m^{(1)}, m^{(2)}, \dots$ in the support with that prefix and with $m_k^{(i)}$ strictly decreasing; then $m^{(1)} > m^{(2)} > \cdots$ lexicographically, so $\{m^{(i)} : i \geq 1\}$ has no lexicographically least element.

(ii) Since $\pi \neq \mathrm{id}$ there are $a < b$ in $\{0, \dots, r\}$ with $\pi^{-1}(b) < \pi^{-1}(a)$: level b is nested *outside* level a in $\mathfrak{J}_\pi$, while in $\mathfrak{J}_r$ level a is outside level b. Put

$$A = \sum_{n \geq 0} L_a^{-n} L_b^{n}, \qquad B = \sum_{n \geq 0} L_a^{n} L_b^{-n},$$

so that A has $m_a = n$, $m_b = -n$ and all other coordinates 0, and B has $m_a = -n$, $m_b = n$. Read in the order $0, 1, \dots, r$: for A, the coordinates $m_0, \dots, m_{a-1}$ are 0, the set of m_a is $\mathbb{N}_0$, bounded below, and once $m_a = n$ is fixed all later coordinates are determined; hence $A \in \mathfrak{J}_r$ (indeed $A = \sum_n (L_b/L_a)^n$). For B, the set of m_a occurring is $\{0, -1, -2, \dots\}$, unbounded below, so $B \notin \mathfrak{J}_r$. Read in the order π, the roles of a and b are exchanged and the same computation gives $B \in \mathfrak{J}_\pi$ and $A \notin \mathfrak{J}_\pi$.

The argument used only that the two nestings order the pair a, b oppositely, and $\mathfrak{J}_r = \mathfrak{J}_{\mathrm{id}}$. Two distinct total orders on the finite set $\{0, \dots, r\}$ must order some pair oppositely, so for $\pi \neq \pi'$ the same witnesses — built from the two levels on which π and π' disagree — give an element of $\mathfrak{J}_\pi \setminus \mathfrak{J}_{\pi'}$ and an element of $\mathfrak{J}_{\pi'} \setminus \mathfrak{J}_\pi$. □

Remark 30.24 (The completion order, in one sentence). Theorem 30.23 says that iterating the Laurent construction reproduces the Hahn field *exactly when the nesting order is the reverse of the dominance order* — outermost variable most dominant — and produces an incomparable object otherwise. The case $r = 1$ is Theorem 5.22 and Theorem 9.39; the general case shows that the phenomenon is not an artefact of the rank-two situation but is present at every depth, and that there are $(r+1)!$ pairwise distinct iterated fields on the same monomial group, exactly one of which is the asymptotically meaningful one. The catalogue records the same warning as “the iteration order is part of the type”.

Remark 30.25 (Nested truncation: what a finite datum is). Membership in $\mathfrak{J}_r$ does not make an element finitely representable. A single block of $\mathfrak{J}_1 = \mathscr{PL}_{\mathrm{itLaur},\mathbb{K}}$ is already an infinite Laurent series in L^{-1}, so a finite datum needs $r+1$ cutoffs, or one declared staircase in the lexicographic monomial order; an outer cutoff alone bounds nothing. This is the depth-r form of the caution attached to Theorem 9.37, and it is the reason the sparse representation of Theorem 27.1 fixes an inner cutoff for the coefficient ring $\mathbb{K}((L^{-1}))$. A workable contract at depth r is a monotone staircase: a bound $m_0 < N_0$, then $m_1 < N_1(m_0)$, then $m_2 < N_2(m_0, m_1)$, and so on, chosen so that the region is stable under the convolutions to be performed. The choice is part of the specification of the computation, not a detail of its implementation.

Do not expand a rational coefficient before you must

The exact block $(L^2+1)/(L-1) \in \mathbb{K}(L)$ and its expansion $L+1+2L^{-1}+2L^{-2}+\cdots \in \mathbb{K}((L^{-1}))$ are different data about the same object: the first is finite and exact, the second resolves a finer asymptotic scale and is infinite. By Theorem 9.37 the expansion map ι is injective, so an identity proved after expansion *does* descend; but it is not surjective, so a computation carried out after expansion may leave the image and return no exact rational block, and a cancellation that is exact and finite in $\mathbb{K}(L)$ survives expansion only in the limit — at every finite truncation it is invisible. Expand when the requested scale requires it, and record that you did.

30.6.1 Changing the fast variable

The most useful practical observation about multiseries is that a nested expansion is often an ordinary polynomial–logarithmic expansion in the right variable. The mechanism is already a theorem in this volume — Theorem 15.16, the level shift — and the following example shows it operating on the classical case.

Example 30.26 (A logarithmic prefactor, and where the second logarithm sits)**.** Let $\beta \in \mathbb{K}$ and invert $Y = Z(\log Z)^\beta$ for $Z \to +\infty$. Write $S = \log Y$, $\Lambda = \log S$ and $z = \log Z$, so that the relation reads $S = z + \beta \log z$, that is $z = S - \beta \log z$. Substituting $z = S - \beta\Lambda + w$ and expanding $\log z = \Lambda + \log\bigl(1 - \beta\Lambda S^{-1} + wS^{-1}\bigr)$ forces $w = \beta^2 \Lambda S^{-1} + \mathcal{O}^{\mathrm{form}}_{S^{-1}}\bigl(S^{-2}\bigr)$, the reversion being the near-identity one of $\mathbb{K}[\Lambda][[S^{-1}]]$. Since $\mathrm{e}^S = Y$ and $\mathrm{e}^{-\beta\Lambda} = S^{-\beta}$, exponentiating z gives

$$Z = Y\, S^{-\beta}\Bigl(1 + \beta^2 \frac{\Lambda}{S} + \mathcal{O}^{\mathrm{form}}_{S^{-1}}\bigl(S^{-2}\bigr)\Bigr). \tag{30.9}$$

This is the shape of Theorem 19.7 at $c = 1$, $\rho = 1$, $\sigma = \beta$, whose hypothesis $\sigma \in \mathbb{N}_0$ we do not widen; the display is proved where it stands. The right-hand side is *not* an element of any power–log scale in the variable Y, because the second logarithm Λ occurs. But the bracket *is* an element of $\mathbb{K}[\Lambda][[S^{-1}]]$, that is, of the polynomial–logarithmic power-series ring built on the fast variable $S = \log Y$ and its logarithm $\Lambda = \log S$. Under Theorem 15.16 every statement of Parts I–IV therefore applies to the bracket verbatim, with t read as S^{-1} and L read as Λ. The only genuinely new object in (30.9) is the monomial prefactor $Y S^{-\beta}$, which is where the depth increase is confined.

Remark 30.27 (The pattern, and the honest limit of the pattern). Theorem 30.26 is a special case of the normalisation theorem Theorem 19.6: factor out a leading model whose inverse is known, and the remainder is a near-identity problem in a scale that the leading model itself selects. What the pattern does *not* do is remove the depth increase; it localises it in the prefactor. A claim of the form “after the right change of variable the problem is one-logarithmic” is true of the correction and false of the answer, and Theorem 30.20 is the reason it must be false of the answer whenever $\beta \neq 0$.

Chapter 31

Relation with the full transseries field

Everything in this volume happens at *exponential height zero*: no monomial of any class considered so far is an exponential of a large series. This chapter places the polynomial–logarithmic class inside the ambient in which exponentials are allowed, says precisely what that ambient adds, and is careful about the division of labour. Two things are proved here, because they are what a user of the smaller class needs and because they sharpen results already in the volume: an exact criterion, valid at every logarithmic depth, for an element with a prescribed logarithmic derivative to lie in the class (Theorem 31.1); and the observation that the filtered gain which makes all of Parts III–IV work is a property of the logarithmic scale and is destroyed by a single exponential (Theorem 31.4). Everything else — the construction of the transseries field, its model theory, its composition and summability theory — is quoted from the literature and explicitly *not* reproved.

31.1 The exponential boundary at every logarithmic depth

Theorem 10.16 settled which elements of $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ have exponentials inside $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$, and Theorem 10.17 extended this to a power–log scale. Both are statements at logarithmic depth one. The following theorem is the depth-r statement, and it is what makes the boundary structural rather than a feature of the particular class: it says that in $\mathcal{H}_r(\mathbb{K},\Gamma)$ the logarithmic derivative of a nonzero element is never larger than t, and that its t-coefficient is forced to lie in the exponent group.

Theorem 31.1 (Logarithmic derivatives in a nested-logarithmic Hahn field)**.** *Let $r \geq 0$, $\mathbb{Z} \subseteq \Gamma \subseteq \mathbb{R}$, and let $\mathcal{H}_r = \mathcal{H}_r(\mathbb{K},\Gamma)$ with the derivation ∂_r of Theorem 30.19. Let $F \in \mathcal{H}_r$ be nonzero with dominant monomial $\mathfrak{m} = \operatorname{lm}(F)$ and leading scalar $c = \operatorname{lc}(F)$. Then*

$$\operatorname{lm}\bigl(\partial_r F\bigr) \;\preccurlyeq\; t\,\mathfrak{m}, \qquad \bigl[t\,\mathfrak{m}\bigr]\,\partial_r F = \alpha_0(\mathfrak{m})\,c. \tag{31.1}$$

Consequently, if $(\mathcal{T},\partial)$ is a differential ring extension of $\mathcal{H}_r$ whose derivation restricts to ∂_r, if $E \in \mathcal{H}_r$ is nonzero, and if $u \in \mathcal{H}_r$ is nonzero with $\partial E = uE$, then

$$\operatorname{lm}(u) \preccurlyeq t, \qquad \textit{and if } \operatorname{lm}(u) = t \textit{ then } \operatorname{lc}(u) = \alpha_0(\operatorname{lm} E) \in \Gamma. \tag{31.2}$$

Proof. Write $F = \sum_{\mathfrak{n}\in S} c_{\mathfrak{n}}\mathfrak{n}$ with $\mathfrak{m} = \max_{\succ} S$. By (30.5), every monomial occurring in $\partial_r F$ is of the form $\mathfrak{n}\mathfrak{n}_j^{-1}$ with $\mathfrak{n} \in S$ and $0 \leq j \leq r$. Since $\mathfrak{n} \preccurlyeq \mathfrak{m}$ and $\mathfrak{n}_j^{-1} \preccurlyeq \mathfrak{n}_0^{-1} = t$, each such monomial is $\preccurlyeq t\mathfrak{m}$, which is the first half of (31.1). For the second half, the contributions to $t\mathfrak{m} = \mathfrak{m}\mathfrak{n}_0^{-1}$ come from the pairs $(\mathfrak{n}, j)$ with $\mathfrak{n} = \mathfrak{m}\mathfrak{n}_0^{-1}\mathfrak{n}_j = \mathfrak{m}\, L_1 L_2 \cdots L_j$. For $j = 0$ this is $\mathfrak{n} = \mathfrak{m}$, contributing $\alpha_0(\mathfrak{m})c$. For $j \geq 1$ the monomial $\mathfrak{m}L_1 \cdots L_j$ has the same L_0-exponent as $\mathfrak{m}$ and a strictly larger L_1-exponent, hence $\mathfrak{m}L_1 \cdots L_j \succ \mathfrak{m}$, so it does not lie in S and contributes nothing.

For the consequence, note first that $\mathcal{H}_r$ is an integral domain with $\operatorname{lm}(uE) = \operatorname{lm}(u)\operatorname{lm}(E)$ and $\operatorname{lc}(uE) = \operatorname{lc}(u)\operatorname{lc}(E)$: the coefficient of $\operatorname{lm}(u)\operatorname{lm}(E)$ in uE is the finite sum (Theorem 30.2 (ii)) over pairs $\mathfrak{p}\mathfrak{q} = \operatorname{lm}(u)\operatorname{lm}(E)$ with $\mathfrak{p} \in \operatorname{supp}_{\mathrm{coeff}}(u)$, $\mathfrak{q} \in \operatorname{supp}_{\mathrm{coeff}}(E)$, and any such pair other than $(\operatorname{lm} u, \operatorname{lm} E)$ would have $\mathfrak{p} \prec \operatorname{lm}(u)$ or $\mathfrak{q} \prec \operatorname{lm}(E)$ and hence $\mathfrak{p}\mathfrak{q} \prec \operatorname{lm}(u)\operatorname{lm}(E)$. Now put $\mathfrak{m} = \operatorname{lm}(E)$. From $\partial E = \partial_r E = uE$ and (31.1), $\operatorname{lm}(u)\mathfrak{m} = \operatorname{lm}(uE) = \operatorname{lm}(\partial_r E) \preccurlyeq t\mathfrak{m}$, so $\operatorname{lm}(u) \preccurlyeq t$. If $\operatorname{lm}(u) = t$, comparing the coefficients of $t\mathfrak{m}$ gives $\operatorname{lc}(u)\operatorname{lc}(E) = \alpha_0(\mathfrak{m})\operatorname{lc}(E)$, and $\operatorname{lc}(E) \neq 0$. □

The criterion is applied by enlarging the exponent group so that the logarithmic derivative itself becomes an element of the ambient; this costs nothing, because $\mathcal{H}_r(\mathbb{K}, \Gamma) \subseteq \mathcal{H}_r(\mathbb{K}, \Gamma')$ for $\Gamma \subseteq \Gamma'$, while the conclusion $\operatorname{lc}(u) = \alpha_0(\operatorname{lm} E)$ still reads the exponent of an element of $\mathcal{M}_r(\Gamma)$.

Corollary 31.2 (No finite logarithmic depth admits an exponential)**.** *Let $\mathbb{Z} \subseteq \Gamma \subseteq \Gamma' \subseteq \mathbb{R}$, let $(\mathcal{T}, \partial)$ be a differential ring extension of $\mathcal{H}_r(\mathbb{K}, \Gamma')$ whose derivation restricts to ∂_r, and let $E \in \mathcal{T}$ be nonzero with $\partial E = uE$ for some nonzero $u \in \mathcal{H}_r(\mathbb{K}, \Gamma')$. If $E \in \mathcal{H}_r(\mathbb{K}, \Gamma)$ then $\operatorname{lm}(u) \preccurlyeq t$, and if $\operatorname{lm}(u) = t$ then $\operatorname{lc}(u) \in \Gamma$. Consequently, for every $r \geq 0$ and every $\Gamma \subseteq \mathbb{R}$:*

(i) $\mathrm{e}^{cX} \notin \mathcal{H}_r(\mathbb{K}, \Gamma)$ *for $c \in \mathbb{K}^\times$;*

(ii) $\mathrm{e}^{\sqrt{X}} \notin \mathcal{H}_r(\mathbb{K}, \Gamma)$ *and $X^{\log X} = \mathrm{e}^{L^2} \notin \mathcal{H}_r(\mathbb{K}, \Gamma)$;*

(iii) *no nonzero E in an ambient as above satisfies $\partial E = aX^{-1}E$ with $E \in \mathcal{H}_r(\mathbb{K}, \Gamma)$, when $a \in \mathbb{K}$ does not lie in the image of Γ; informally, $X^a \notin \mathcal{H}_r(\mathbb{K}, \Gamma)$ whenever $a \notin \Gamma$;*

(iv) *by contrast $L \in \mathcal{H}_1$ and $\mathrm{e}^{1/X} \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$, whose logarithmic derivatives are $\mathfrak{n}_1^{-1} = t/L$ and $-t^2$: the criterion is not vacuous in either direction.*

Proof. The displayed criterion is (31.2) applied inside $\mathcal{H}_r(\mathbb{K}, \Gamma')$, which contains E and u; the refinement is that $\operatorname{lm}(E) \in \mathcal{M}_r(\Gamma)$, so $\alpha_0(\operatorname{lm} E) \in \Gamma$.

(i) Take $\Gamma' = \Gamma$ and $u = c$; then $\operatorname{lm}(u) = 1 \succ t$. (ii) For $E = \mathrm{e}^{\sqrt{X}}$ take $\Gamma' = \Gamma + \frac{1}{2}\mathbb{Z}$, which is admissible: since $2 \cdot \frac{1}{2} = 1 \in \Gamma$ and $\mathbb{K}$ has characteristic zero, sending $\frac{1}{2}$ to half the image of 1 extends the embedding $\Gamma \hookrightarrow \mathbb{K}$ to Γ', and the extension is injective because $\gamma + \frac{1}{2} \in \Gamma$ would force $\frac{1}{2} \in \Gamma$, in which case $\Gamma' = \Gamma$. Take $u = \frac{1}{2}X^{-1/2}$, so $\operatorname{lm}(u) = L_0^{-1/2} \succ L_0^{-1} = t$ because $-\frac{1}{2} > -1$. For $E = \mathrm{e}^{L^2}$ assume first $r \geq 1$; take $\Gamma' = \Gamma$ and $u = 2LX^{-1}$, so $\operatorname{lm}(u) = L_0^{-1}L_1 \succ L_0^{-1}$: the two monomials agree in the L_0-exponent and the first has the larger L_1-exponent. Both contradict the criterion. For $r = 0$ the second case follows from the case $r = 1$, because $\mathcal{H}_0(\mathbb{K}, \Gamma) \subseteq \mathcal{H}_1(\mathbb{K}, \Gamma)$ by Theorem 30.20 (ii). (iii) Take $\Gamma' = \Gamma$ and $u = aX^{-1} = a\,t$, which lies in $\mathcal{H}_r(\mathbb{K}, \Gamma)$ because $t \in \mathcal{M}_r(\Gamma)$ and a is a scalar; then $\operatorname{lm}(u) = t$ and $\operatorname{lc}(u) = a$, which is not in the image of Γ. (iv) $\operatorname{lm}(t/L) = L_0^{-1}L_1^{-1} \prec L_0^{-1} = t$ and $\operatorname{lm}(-t^2) = L_0^{-2} \prec L_0^{-1}$; both are consistent with the criterion, and both elements do lie in the stated classes. □

Remark 31.3 (Relation to the depth-one boundary). Theorem 31.2 does not supersede Theorem 10.16: the latter is an exact *iff*, listing the elements whose exponential stays in $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$, while the former is a necessary condition of a different shape — a constraint on the logarithmic derivative — whose merit is that it is uniform in the logarithmic depth and therefore rules out an element from *every* member of the tower at once. Read together they say: within the logarithmic world an exponential can only be a power of X times something already there (Theorem 10.21), and enlarging the logarithmic depth never changes that. The first genuinely new monomial is e^X, and it is new at every depth.

31.2 Where the class sits: the logarithmic–exponential field

The field $\mathbb{T}$ of logarithmic–exponential transseries is built by iterating two constructions on an ordered monomial group: adjoining exponentials $\mathrm{e}^{\mathfrak{p}}$ of *purely large* series $\mathfrak{p}$, and adjoining logarithms; and then closing under well-based sums. Two standard variants must be distinguished, and they correspond exactly to the support classes of Theorem 30.1: the *grid-based* transseries of [33], whose supports lie on a finitely generated multiplicative grid, and the *well-based* field of [1, 13], whose supports need only be well ordered. Theorem 30.8 is the elementary shadow of that distinction.

Section summary

Quoted, not proved. The construction of $\mathbb{T}$ as an ordered, exponentially closed, differential field with well-based supports and finite exponential height is [33, 1, 13]. Its monomial group contains every $\prod_{j=0}^{r} L_j^{\alpha_j}$ with $\alpha_j \in \mathbb{R}$, and $\mathbb{T}$ is closed under well-based sums of its own monomials; consequently, for $\mathbb{K} = \mathbb{R}$ and every $\Gamma \subseteq \mathbb{R}$ and $r \geq 0$, the field $\mathcal{H}_r(\mathbb{R}, \Gamma)$ of Theorem 15.1 is an ordered differential subfield of $\mathbb{T}$. The four classes of Theorem 5.3 therefore sit inside $\mathbb{T}$ as the subrings of $\mathcal{H}_1(\mathbb{K}, \mathbb{Z})$ described in Theorem 5.20. We use this placement only for orientation; no theorem of this volume depends on it.

What the exponential hierarchy adds is exactly one thing, and Theorem 31.2 says it cannot be simulated: monomials whose logarithmic derivative is not $\preccurlyeq t$. Everything else — real exponents, arbitrary logarithmic depth, well-based rather than left-finite supports — lives already at height zero and was treated in Part 30. The cost of the addition is severe, and it is worth stating precisely, because it explains why Parts III and IV could be elementary and the corresponding theory in $\mathbb{T}$ cannot.

Proposition 31.4 (The filtered gain is a logarithmic-scale phenomenon)**.** *Let $(\mathcal{T}, \partial)$ be a differential ring containing a constant c and an element E with $\partial E = cE \neq 0$. Then there is no map ν from $\mathcal{T} \setminus \{0\}$ to a partially ordered set that is constant on nonzero constant multiples and satisfies $\nu(\partial G) > \nu(G)$ for every G with $G, \partial G \neq 0$. In particular the one-block gain $\nu_t(D_X A) \geq \nu_t(A) + 1$, which is the decisive hypothesis behind Theorem 14.6, Theorem 16.4, Theorem 16.11 and the termination of every algorithm of Part 27, cannot survive into any ring containing e^X.*

Proof. Apply the hypothesis to $G = E$: $\nu(\partial E) = \nu(cE) = \nu(E)$, while the assumed property gives $\nu(\partial E) > \nu(E)$. For the second sentence, take $c = 1$, $E = \mathrm{e}^X$, and $\nu = \nu_t$, which is constant on nonzero scalar multiples. By contrast, inside $\mathcal{H}_r(\mathbb{K}, \Gamma)$ the gain does hold: Theorem 31.1 gives $\mathrm{lm}(\partial_r G) \preccurlyeq t \, \mathrm{lm}(G) \prec \mathrm{lm}(G)$ for every nonzero G. □

Remark 31.5 (What replaces it). In $\mathbb{T}$ the derivation is not valuation-increasing, and the contraction arguments of Parts III–IV have no direct analogue. Their replacement is the asymptotic-differential-algebra machinery of [1]: the notion of an H-field, the asymptotic couple attached to the valuation and the derivation, the Newton degree of a differential polynomial, and the axioms ω-freeness and newtonianity, which are what make the analogue of "the residual determines the next block" true again — now for differential equations rather than for a single composition. It is this apparatus, and not any refinement of the elementary filtration, that carries the theory across the exponential boundary.

31.3 The Hardy-field perspective

Everything above is formal. The classical counterpart is Hardy's calculus of *orders of infinity* [20]: the logarithmico-exponential functions — germs at $+\infty$ built from the variable and real constants by the field operations, $\exp$ and $\log$ — are eventually of constant sign, eventually monotone, and pairwise comparable, so their germs form a totally ordered differential field, a *Hardy field.* Three points deserve to be kept apart, and the sources of this consolidation do not always keep them apart.

1. A *Hardy field* consists of germs of actual real functions; its order is eventual dominance of real values.
2. A *transseries field* consists of formal objects; its order is the combinatorial dominance order on supports (Theorem 5.12).

3. An *expansion map* relates the two. Its existence for a given germ is a theorem, never a definition, and its injectivity fails in general: e^{-X} and 0 have the same polynomial–logarithmic expansion.

Within this volume, item (2) is what Theorem 5.13 justifies at the level of monomials, and item (3) is what Theorem 12.32 and Theorem 12.35 implement, with hypotheses printed. The non-injectivity in item (3) is the exact reason the class is blind to resurgence; see Section 31.6.

31.4 Model theory of the ambient, quoted

Section summary

Quoted, not proved. The following are stated as citations and used nowhere in this volume.

- Van den Dries, Macintyre and Marker [31] construct the field of logarithmic–exponential power series, a real closed field closed under $\exp$ and $\log$, and identify it as a model of the theory of the real field with restricted analytic functions and exponentiation; in [32] they develop its ordered differential and compositional structure and its relation with Hardy fields of logarithmico-exponential functions.
- Aschenbrenner, van den Dries and van der Hoeven [1] prove that $\mathbb{T}$, as an ordered valued differential field, is an ω-free newtonian Liouville closed H-field, that this axiomatisation is complete and model complete in their language, and develop the accompanying theory of asymptotic differential algebra.

These results are what license the phrase "the ambient field is well behaved". None of them is needed to justify a single computation in Parts I–VI, and none of them is reproved here.

31.5 Composition, recursion, and convergence in the larger setting

Composition is the operation that degrades worst under enlargement, and the reason is visible already in Part 30: substitution moves supports, and the image of a well-based support need not be well based unless the inner argument is controlled. Edgar's treatment [14] isolates the admissibility conditions — in particular that the inner argument be positive and infinitely large — under which composition of transseries is defined, associative, and compatible with differentiation; the corresponding statements in this volume are Theorem 12.6, Theorem 12.7 and Theorem 12.27, proved there in the narrow setting where admissibility is automatic. Fractional iteration and the iterative logarithm — the formal vector field V with $\exp(V)X = F$ — are developed for transseries in [15]; in the present class they are Theorem 14.18 and Theorem 14.19, where the graded Lie algebra of the contact filtration makes the correspondence exact and the divisibility unique.

Two hazards that only appear after enlargement

Both are invisible in $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ and must be checked in any larger class.

1. *Loss of well-basedness under substitution.* Composition substitutes an inner series into infinitely many outer monomials; the union of the resulting supports is well based only under a hypothesis, and Theorem 30.4 shows the hypothesis cannot simply be dropped in favour of a weaker support class.
2. *Failure of Taylor approximation.* The formal Taylor formula Theorem 13.6 is summable because D_X raises the outer order; Theorem 31.4 shows this fails in an exponentially closed field, so the Taylor expansion of a transseries around a displaced argument requires a separate theorem with its own hypotheses. Such theorems exist [22], and they are not the trivial extension of Theorem 13.6.

31.6 Borel summability and resurgence: what this class cannot see

The formal theory of Parts I–IV is indifferent to convergence, and this is not an oversight: it is a division of labour. A divergent polynomial–logarithmic series may be attached to an actual function by Borel–Laplace summation [8], and the ambiguity in that attachment — the Stokes phenomenon, and the exponentially small terms that measure it — is the subject of resurgence [12]. The following is the honest statement of where the present class stands.

Proposition 31.6 (The class is blind to exponentially small corrections). (i) *For every $r \geq 0$, every $\Gamma \subseteq \mathbb{R}$, every $\mu \in \mathbb{K}^\times$ and every ambient as in Theorem 31.2, the element $\mathrm{e}^{-\mu X}$ does not lie in $\mathcal{H}_r(\mathbb{K}, \Gamma)$.*

(ii) *Let $S = [X_0, \infty)$ be a real ray, let $\mu > 0$ and $\lambda \in \mathbb{C}$, and let $f : S \to \mathbb{C}$ realize $F \in \mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ in the sense of Theorem 12.32. Then the function $g(X) = f(X) + \lambda \mathrm{e}^{-\mu X}$ realizes the* same *F, and by Theorem 12.33 no other element of $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$.*

Consequently no truncation, no residual certificate (Theorem 16.14) *and no error estimate of Parts I–VI distinguishes f from g.*

Proof. (i) This is Theorem 31.2 (i) with $c = -\mu$.

(ii) For every $N \in \mathbb{Z}$ and every $\mu > 0$ one has $\mathrm{e}^{-\mu X} = o_{X\to+\infty}\big(X^{1-N}\big)$, since $X^{N-1}\mathrm{e}^{-\mu X} \to 0$. Hence

$$g(X) - \big(\mathrm{Trunc}_{<N} F\big)(X) = \Big(f(X) - \big(\mathrm{Trunc}_{<N} F\big)(X)\Big) + \lambda \mathrm{e}^{-\mu X} = o_{X\to+\infty}\big(X^{1-N}\big)\,,$$

which is the defining remainder condition of Theorem 12.32 for g. Uniqueness of the realized series is Theorem 12.33. ☐

Remark 31.7 (Why the interesting resurgence is not here). Resurgence is a statement about the *relations* between the exponentially small sectors of a transseries and the singularities of its Borel transform. Theorem 31.6 says the polynomial–logarithmic class contains no such sector at all, at any logarithmic depth; the class is therefore too small for the phenomena to be visible, not merely inconvenient for them. What the class *can* contribute is the height-zero datum on which those phenomena act: the formal power–log series that a Borel–Laplace summation is asked to sum, together with the exact residual certificates of Theorem 16.14 and Theorem 16.15, which control the algebraic part of the error without any convergence hypothesis. Where a convergence statement is actually available in this volume it is proved and its scope is printed — Theorem 20.19 for the Lambert generating function at frozen logarithm — and where it is not, it is recorded as unproved in Section 31.8. No claim of Borel summability is made anywhere in this volume.

31.7 Symbolic asymptotics for larger classes

The algorithmic counterpart of this section is the multiseries technology of Salvy and Shackell. Their programme computes asymptotic expansions of exp–log functions by discovering the scale rather than fixing it in advance: implicit functions and functions of two variables [29], and multiseries of inverse functions [30]. The relationship with Part 27 is the following, and it is a genuine division of labour rather than a comparison.

- Their algorithms decide, at each step, *which* scale the next term lives in — an inverse power of X, an inverse power of $\log X$, a new iterated logarithm, a root, or an exponential monomial — and then recurse. Theorem 19.5, Theorem 19.3 and Theorem 30.20 are the corresponding decision content in the present class, stated as theorems about which enlargement is forced.

- Once the scale is fixed, the coefficient arithmetic is exactly the block calculus of Parts I–IV, and the cost model of Theorem 27.15 applies. The sharp scalar cost of truncated block convolution (Theorem 27.17) and the precision calculus (Theorem 27.10) are statements that a general multiseries implementation does not have available, because they use the finiteness of each block.
- The practical rule that follows is the one this volume has argued for throughout: begin in the smallest class that contains the problem, run the closure test for the next operation, and promote explicitly. Promotion hidden inside a simplifier is how a computation acquires an unproved convergence claim.

31.8 Open problems

The following are stated as open, not as theorems and not as conjectures with suppressed hypotheses. Each is precise enough to be attacked, and each is raised, and left unsettled, by material in this volume.

Remark 31.8 (Open problem: convergence of the Lambert expansion with coupled logarithm). Does there exist $x_0 > 0$ such that $\sum_{n\geq 1} \left|\mathsf{L}_n^{\mathrm{Lam}}(\log x)\right| x^{-n} < \infty$ for every real $x > x_0$? Equivalently, is the radius of convergence $r(u)$ of $s \mapsto \sum_{n\geq 1} \mathsf{L}_n^{\mathrm{Lam}}(u)s^n$ bounded below by κ/u for some $\kappa > 0$ and all large real u? Theorem 20.19 proves convergence for each *frozen* u with no lower bound on $r(u)$. Writing $\omega(X) = X + h$ with $h = \sum_{n\geq 0} \mathsf{L}_n^{\mathrm{Lam}}(u)s^n$, $u = L$, $s = t$, the defining relation $\omega + \log\omega = X$ becomes $h + u + \log(1 + sh) = 0$ — this is the implicit generating equation of Theorem 21.11, and the substitution just made rederives it, the value $h(u, 0) = \mathsf{L}_0^{\mathrm{Lam}}(u) = -u$ fixing the branch. Its h-derivative $1 + s/(1 + sh)$ vanishes exactly at $1 + sh = -s$, that is at $h = -1 - s^{-1}$, and eliminating h puts the singularities of $s \mapsto h(u, s)$ on the curve $u = s^{-1} + 1 - \log(-s)$. This suggests $r(u) \asymp 1/u$; nothing in Parts I–IV proves it. A positive answer would upgrade Theorem 20.10 from an asymptotic expansion to a convergent one on a half-line.

Remark 31.9 (Open problem: sharp logarithmic depth of a composite). Fix $r \geq 1$ and let $G_1, \dots, G_k$ be monomial-leading maps, each of logarithmic depth at most r in the sense that its leading monomial lies in $\mathcal{M}_r(\Gamma)$. Theorem 15.2 guarantees that a single such substitution raises depth by at most one, so $G_1 \circ \cdots \circ G_k$ lies at depth at most $r + k$; this volume proves no corresponding bound for its compositional inverse. Is $r + k$ attained by the composite, and is the inverse subject to any depth bound at all? More precisely: give an algorithm that, from the leading data of $G_1, \dots, G_k$, computes the least r' with $(G_1 \circ \cdots \circ G_k)^{\langle -1\rangle\circ} \in \mathcal{H}_{r'}$, together with a proof of minimality. The only lower bound proved here is the single step of Theorem 19.4, and the only obstruction proved here is the residue of Theorem 30.20; neither is known to be complete.

Remark 31.10 (Open problem: a Lagrange–Bürmann formula at dense exponents). Theorem 18.14 expresses the reversion of a near-identity map as $\sum_{r\geq 1} \frac{(-1)^r}{r!}\partial^{r-1}(A^r)$, and Theorem 18.19 shows that the coefficient of t^N is determined by the indices $r \leq N + 1$ and by no fewer. Both statements use that the exponent group is $\mathbb{Z}$. In a Hahn power–log scale $\mathcal{H}(\mathbb{K}[L], \Gamma)$ with Γ dense, is there a summation index set — necessarily not $\mathbb{N}_{>0}$ with the range $r \leq N + 1$ — and an operator identity reducing to Theorem 18.14 at $\Gamma = \mathbb{Z}$? The near-identity inverse exists there by Theorem 30.11; what is missing is a closed formula for it.

Remark 31.11 (Open problem: denominator growth in the Puiseux tower). By Theorem 30.15 the Puiseux-log union $\mathfrak{P}$ is not complete, so an infinite iteration may leave it. Give an explicit bound, or prove that none exists, for the following: if $F = X + A$ with $A \in \mathbb{K}[L][[t^{1/s}]]$, the Newton scheme of Theorem 17.10 started at X produces iterates $H_0, H_1, \dots$; is there a function $s \mapsto \sigma(s)$, independent of A, with $H_k \in \mathbb{K}[L]((t^{1/\sigma(s)}))$ for all k? For the triangular scheme of Theorem 17.2 the answer is yes with $\sigma(s) = s$, since every block of the inverse has exponent in $s^{-1}\mathbb{Z}$; the question is whether the quadratically convergent scheme, which divides by $F' \circ H_k$, preserves the denominator.

Remark 31.12 (Open problem: uniformity of the analytic transfer). Theorem 20.17 and Theorem 12.35 give, for each fixed truncation order N, an asymptotic error estimate on the positive ray with a constant depending on N. Is there an estimate uniform in N — for instance a bound of the form $|\omega(X) - \mathrm{Trunc}_{<N}\,\omega(X)| \leq C^N N!\, X^{-N}(\log X)^N$ valid for all N and all $X \geq X_0$, with C, X_0 absolute? Is there a version uniform in a complex sector and in the branch index k of W_k? None of these is claimed or proved anywhere in this volume; the literature asserts related convergence statements [5, 21, 11] whose proofs the present consolidation has not verified.

Remark 31.13 (Open problem: realisation of left-finite Hahn elements by germs). Theorem 30.10 identifies $\mathcal{H}_{\mathrm{lf}}(\mathbb{K}[L], \Gamma)$ as the completion of the finitely supported power–log series, and Theorem 30.11 shows it carries the whole near-identity theory. Which of its elements are asymptotic expansions of germs at $+\infty$ lying in a common Hardy field, in the sense of Theorem 12.32? For $\Gamma = \mathbb{Z}$ the question is the classical one about power–log series; for $\Gamma = \mathbb{Q}$, where by Theorem 30.15 the class strictly contains the Puiseux tower, we do not know a characterisation, nor an example of an element of $\mathcal{H}_{\mathrm{lf}}(\mathbb{K}[L], \mathbb{Q}) \setminus \mathfrak{P}$ that is provably realised.

Remark 31.14 (Open problem: a resurgent completion with block arithmetic). Theorem 31.6 shows the class cannot see exponentially small corrections. Is there a minimal enlargement — adjoining a single transmonomial $\mathrm{e}^{-\mu X}$, $\mu > 0$, with coefficients in $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$, that is, the ring $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}[\![\mathrm{e}^{-\mu X}]\!]$ with its natural derivation — in which the block arithmetic of Parts I–IV survives verbatim, the unit criterion and the reversion theorem included? Theorem 31.4 shows the valuation gain of D_X fails on the new generator, so a positive answer would need a two-index filtration (order in t, order in $\mathrm{e}^{-\mu X}$) and a proof that the composition operator is continuous for it. We state this as open because the naive filtration does not work and we have not verified any substitute.

Remark 31.15 (Open problem: which of these statements are machine-checkable today). The formalisation register records each audited result's current proof-assistant status. The basic support-theoretic inputs are no longer merely prospective: Theorems 2.4 and 2.5 are Exact in Lean, with `OrderDual` implementing the volume's well-based orientation. The stronger Hahn-series constructions of Section 30.2 are not thereby formalized. Theorem 30.20 remains a finite computation with a monomial-bookkeeping argument, while Theorem 30.10 still needs a formal Hahn-series library with a Γ-indexed filtration. Those remaining entries should be read as a plan, not as claims of coverage.

Section summary

The polynomial–logarithmic class is a proper, well-delimited fragment of the logarithmic–exponential transseries field. What separates it from the ambient is a single structural fact, proved here at every logarithmic depth: inside the class the logarithmic derivative of a nonzero element is never larger than X^{-1} (Theorem 31.1). Everything the class does easily — triangular reversion, terminating algorithms, exact residual certificates — follows from the valuation gain that this fact expresses, and everything it cannot do — exponentially small terms, Stokes data, resurgence — follows from the same fact read as a limitation (Theorem 31.4, Theorem 31.6). The right attitude is neither to inflate the class nor to abandon it, but to promote explicitly, one axis at a time, with the closure test named.

Appendices

Appendix A

Bell polynomials and Faà di Bruno bookkeeping

The exponential partial Bell polynomials appear at four separate places in the body of this volume: in the coefficients of an integer power (Theorem 8.36), in the coefficients of a series substituted into a scalar power series (Theorems 10.5 and 13.22), in the shift recurrence that drives Faà di Bruno for admissible composition (Theorems 13.24 and 13.25), and in the closed form of the convolution polynomials (Theorem 13.13 (iv)). (The second member of that group restates the first in the part on composition, and adds a clause about a base point $z_0 \neq 0$; only the first is proved from scratch.) In each of those places the defining generating identity is quoted from [4]. This appendix pays that debt: it defines the family, proves the generating identity and the recurrences from the definition, proves the two Stirling evaluations — one of which is exactly the arithmetic content of the Lambert polynomials of Part 21 — and proves the form of Faà di Bruno that the volume needs but does not state, namely the one whose outer function is a scalar-analytic function of a polynomial–logarithmic series.

Throughout, $\mathbb{K}$ is a field of characteristic zero, so that $\mathbb{Q} \subseteq \mathbb{K}$ and every factorial occurring below is invertible. The polynomial identities of Sections A.1 and A.2 are identities in $\mathbb{Q}[x_1, x_2, \ldots]$ and are therefore available over every commutative $\mathbb{Q}$-algebra, in particular over $\mathbb{K}[L]$, $\mathbb{K}(L)$, $\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ and $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$.

A.1 Definition, generating identity, and the two normalizations

Definition A.1 (Exponential partial and complete Bell polynomials)**.** Let $x_1, x_2, \ldots$ be independent indeterminates. For $n, k \in \mathbb{N}_0$ put

$$\mathcal{B}^{\exp}_{n,k}(x_1, x_2, \ldots) := \sum_{\substack{m_1, m_2, \ldots \geq 0 \\ \sum_{j\geq 1} m_j = k \\ \sum_{j \geq 1} j m_j = n}} \frac{n!}{\prod_{j\geq 1} m_j!\,(j!)^{m_j}} \prod_{j\geq 1} x_j^{m_j} \;\in\; \mathbb{Q}[x_1, x_2, \ldots], \tag{A.1}$$

the sum ranging over all sequences $(m_j)_{j\geq 1}$ of nonnegative integers with finite support satisfying the two displayed constraints, and the empty product being 1. The *complete* polynomial is

$$\mathcal{B}^{\exp}_{n}(x_1, \ldots, x_n) := \sum_{k=0}^{n} \mathcal{B}^{\exp}_{n,k}(x_1, \ldots, x_{n-k+1}). \tag{A.2}$$

Lean status. Both polynomials of this definition already exist in the corpus, under the earlier combinatorial-calculus campaign rather than as work done for the present volume, and are recorded here because the crosswalk had not cited them. The partial polynomial is `Fabius.partialBell` in

`PartialBellPolynomials.lean`, defined not by the multinomial sum (A.1) but by the block-of-the-largest-element recurrence (A.9), which is therefore *definitional* there (`Fabius.partialBell_succ_succ`, with the binomial-convolution repackaging `Fabius.partialBell_succ_succ_eq_binomialConv`); the boundary values are `Fabius.partialBell_zero_zero`, `Fabius.partialBell_zero_succ` and `Fabius.partialBell_succ_zero`. The coefficient ring is an arbitrary commutative semiring, so the formal statement is more general than the $\mathbb{Q}[x_1, x_2, \ldots]$ of the definition above; the equivalence with the multinomial sum is not itself formalized, the recurrence being taken as primitive and the generating identity (A.5) proved from it. The complete polynomial (A.2) is `Bell.complete`, and that it is the row sum of the partial ones is `Fabius.bell_complete_eq_sum_partialBell`.

The index set in (A.1) is finite: the constraint $\sum_j jm_j = n$ forces $m_j = 0$ for $j > n$ and $m_j \leq n/j$. It is empty — so that $\mathcal{B}^{\exp}_{n,k} = 0$ — whenever no such sequence exists. Two degenerate cases matter and are fixed once and for all by (A.1) itself: for $n = k = 0$ the only sequence is $m \equiv 0$, giving $\mathcal{B}^{\exp}_{0,0} = 1$; and for $n > 0$, $k = 0$ the constraints $\sum m_j = 0$ and $\sum jm_j = n > 0$ are incompatible, so $\mathcal{B}^{\exp}_{n,0} = 0$. Consequently $\mathcal{B}^{\exp}_0 = 1$.

Lemma A.2 (Support, degree, weight, and boundary values). *Fix $n, k \in \mathbb{N}_0$ and write $B := \mathcal{B}^{\exp}_{n,k}$.*

(i) *$B = 0$ unless $0 \leq k \leq n$; moreover $B = 1$ when $n = k = 0$ and $B = 0$ when $n \geq 1$ and $k = 0$. When $1 \leq k \leq n$, only the variables $x_1, \ldots, x_{n-k+1}$ occur in B.*

(ii) *B is homogeneous of degree k in the variables and isobaric of weight n for the weighting $\mathrm{wt}(x_j) = j$. Equivalently, for indeterminates a, b,*

$$\mathcal{B}^{\exp}_{n,k}\big(abx_1, ab^2x_2, ab^3x_3, \ldots\big) = a^k b^n\, \mathcal{B}^{\exp}_{n,k}(x_1, x_2, \ldots). \tag{A.3}$$

(iii) *The boundary rows and columns are*

$$\mathcal{B}^{\exp}_{n,n} = x_1^n, \qquad \mathcal{B}^{\exp}_{n,1} = x_n\ (n \geq 1), \qquad \mathcal{B}^{\exp}_{n,n-1} = \binom{n}{2} x_1^{n-2} x_2\ (n \geq 2). \tag{A.4}$$

Proof. (i) A contributing sequence has $\sum_j m_j = k$ and $\sum_j jm_j = n$; since every $j \geq 1$, the second sum is at least the first, so $n \geq k$. If $k = 0$ the first sum forces $m \equiv 0$ and then $n = 0$. Suppose $1 \leq k \leq n$ and $m_i \geq 1$ for some i. The remaining $k - 1$ parts contribute at least $k - 1$ to $\sum_j jm_j$, whence $i \leq n - (k-1) = n - k + 1$.

(ii) Each monomial $\prod_j x_j^{m_j}$ occurring in (A.1) has total degree $\sum_j m_j = k$ and weight $\sum_j jm_j = n$. Substituting $x_j \mapsto ab^j x_j$ multiplies that monomial by $a^{\sum_j m_j} b^{\sum_j jm_j} = a^k b^n$, which is (A.3).

(iii) For $k = n \geq 1$, $\sum_j m_j = \sum_j jm_j$ forces $m_j = 0$ for $j \geq 2$ and hence $m_1 = n$; the coefficient is $n!/(n!\,(1!)^n) = 1$. For $k = 1$ the only sequence is $m_n = 1$ with all other entries zero; the coefficient is $n!/(1!\,(n!)^1) = 1$. For $k = n - 1$ with $n \geq 2$ the only sequence is $m_1 = n - 2$, $m_2 = 1$; its coefficient is $n!/\big((n-2)!\,1!\,(1!)^{n-2}(2!)^1\big) = n(n-1)/2 = \binom{n}{2}$. □

Lean status: (i) *and* (ii) *formal,* (iii) *partial.* The vanishing in (i) is `Fabius.partialBell_eq_zero_of_lt` together with the three boundary values cited at Theorem A.1; the last clause of (i), that only $x_1, \ldots, x_{n-k+1}$ occur, is not formalized, the corpus having no variable-support predicate for these polynomials. Statement (ii) is formal in exactly the displayed bihomogeneous form (A.3): `Fabius.partialBell_bihomogeneous` in `BellHomogeneity.lean` reads $\mathcal{B}^{\exp}_{n,k}\big(ab^j x_j\big)_j = a^k b^n \mathcal{B}^{\exp}_{n,k}(x)$ for arbitrary ring elements a, b, and its two halves `Fabius.partialBell_mul_left` (homogeneity of degree k) and `Fabius.partialBell_pow_mul` (isobaric of weight n) are the separate statements that (ii) calls equivalent to it. Of the three boundary identities in (A.4) only the first, $\mathcal{B}^{\exp}_{n,n} = x_1^n$, is formal (`Fabius.partialBell_self`); $\mathcal{B}^{\exp}_{n,1} = x_n$ and the $\binom{n}{2} x_1^{n-2} x_2$ row have no counterpart.

Proposition A.3 (The defining generating identity)**.** *Let R be a commutative $\mathbb{Q}$-algebra, let $\xi_1, \xi_2, \ldots \in R$, and put $\Xi(z) := \sum_{j\geq 1} \xi_j z^j/j! \in zR[[z]]$. Then, in $R[[z]]$,*

$$\frac{\Xi(z)^k}{k!} = \sum_{n\geq k} \mathcal{B}^{\exp}_{n,k}(\xi_1, \xi_2, \ldots)\frac{z^n}{n!} \qquad (k \in \mathbb{N}_0), \tag{A.5}$$

and the family $\left(\Xi^k/k!\right)_{k\geq 0}$ is summable in $R[[z]]$ with

$$\exp \Xi(z) := \sum_{k\geq 0} \frac{\Xi(z)^k}{k!} = \sum_{n\geq 0} \mathcal{B}^{\exp}_{n}(\xi_1, \ldots, \xi_n)\frac{z^n}{n!}. \tag{A.6}$$

Conversely, (A.5) *taken over $R = \mathbb{Q}[x_1, x_2, \ldots]$ with $\xi_j = x_j$ determines* (A.1) *uniquely.*

Proof. Fix $k \geq 1$ (the case $k = 0$ is $1 = 1$). By the multinomial theorem in $R[[z]]$, applied to the k-th power of a series with zero constant term,

$$\Xi(z)^k = \sum_{(j_1,\ldots,j_k)\in\mathbb{N}^k_{>0}} \prod_{i=1}^{k} \frac{\xi_{j_i} z^{j_i}}{j_i!} = \sum_{n\geq k} z^n \sum_{\substack{j_1,\ldots,j_k\geq 1\\ j_1+\cdots+j_k=n}} \prod_{i=1}^{k} \frac{\xi_{j_i}}{j_i!},$$

the rearrangement being legitimate because for fixed n only the finitely many tuples with $j_1+\cdots+j_k = n$ contribute, and there are none when $n < k$. Now group the tuples by their multiplicity vector: a tuple $(j_1, \ldots, j_k)$ with $m_j := \#\{i : j_i = j\}$ satisfies $\sum_j m_j = k$ and $\sum_j jm_j = n$, and the number of tuples with a prescribed multiplicity vector is the multinomial coefficient $k!/\prod_j m_j!$. Every tuple in one such class contributes the same term $\prod_j(\xi_j/j!)^{m_j}$. Hence

$$[z^n]\,\Xi^k = \sum_{\substack{\sum_j m_j=k\\ \sum_j jm_j=n}} \frac{k!}{\prod_j m_j!} \prod_j \left(\frac{\xi_j}{j!}\right)^{m_j} = \frac{k!}{n!}\,\mathcal{B}^{\exp}_{n,k}(\xi_1, \xi_2, \ldots),$$

by (A.1). Dividing by $k!$ gives (A.5).

For (A.6): $\Xi^k/k! \in z^k R[[z]]$, so the family is summable for the z-adic filtration and its sum is computed blockwise; by (A.5) and Theorem A.2 (i) the coefficient of $z^n/n!$ in that sum is $\sum_{k=0}^{n} \mathcal{B}^{\exp}_{n,k}$, which is (A.2). Finally, in the universal case $R = \mathbb{Q}[x_1, x_2, \ldots]$ the coefficients of a formal power series are uniquely determined, so (A.5) recovers (A.1). □

Lean status. Formal in `BellGeneratingFunctions.lean`, all three clauses. The column identity (A.5) is `Fabius.bellWeightSeries_pow`: with Ξ the weight series `Fabius.bellWeightSeries` (constant term zero, `Fabius.constantCoeff_bellWeightSeries`), one has $\Xi^k = k! \cdot \mathrm{egf}\big(n \mapsto \mathcal{B}^{\exp}_{n,k}\big)$, which is (A.5) with the $k!$ cleared. The complete identity (A.6) is `Fabius.exp_subst_bellWeightSeries`, $\exp \Xi = \mathrm{egf}\big(\mathcal{B}^{\exp}_n\big)$, stated as a formal substitution and proved by a differential-equation argument rather than by summation of (A.5) — `Fabius.derivative_bellWeightSeries` is the step it turns on. The converse clause, that (A.5) determines the polynomials, is `Fabius.partialBell_eq_of_bellWeightSeries_pow_eq_egf`: a sequence whose column generating function matches $\Xi^k/k!$ over $\mathbb{Q}$ *is* the sequence of partial Bell polynomials. That last statement is the tool used to prove the Stirling and Lah evaluations of Theorem A.9.

Theorem A.3 is precisely the identity quoted without proof in the proofs of Theorem 8.36, Theorem 13.13 (iv) and Theorem 13.24; those citations may now be read as references to (A.5).

The volume uses two different normalizations for the same combinatorial data, and the conversion between them is a constant source of factorial errors. The next lemma states it once.

Lemma A.4 (Ordinary versus exponential normalization)**.** *Let R be a commutative $\mathbb{Q}$-algebra and $\xi_1, \xi_2, \ldots \in R$. Define the* ordinary *partial Bell polynomial by*

$$\mathcal{B}^{\mathrm{ord}}_{n,k}(\xi_1, \xi_2, \ldots) := \sum_{\substack{m_1, m_2, \ldots \geq 0 \\ \sum_j m_j = k \\ \sum_j j m_j = n}} \frac{k!}{\prod_{j \geq 1} m_j!} \prod_{j \geq 1} \xi_j^{m_j}, \tag{A.7}$$

so that $\left(\sum_{j\geq 1} \xi_j z^j\right)^k = \sum_{n \geq k} \mathcal{B}^{\mathrm{ord}}_{n,k}(\xi_1, \ldots) z^n$. *Then for all* $n, k \in \mathbb{N}_0$

$$\mathcal{B}^{\mathrm{ord}}_{n,k}(\xi_1, \xi_2, \ldots) = \frac{k!}{n!} \mathcal{B}^{\mathrm{exp}}_{n,k}\big(1!\, \xi_1, 2!\, \xi_2, 3!\, \xi_3, \ldots\big). \tag{A.8}$$

In particular, for $H = \sum_{j \geq 1} h_j(L) t^j \in t\mathscr{P}\mathscr{L}_{\mathrm{pow}, \mathbb{K}}$ *the convolution polynomials of Theorem 13.12 satisfy* $C_{n,k} = \mathcal{B}^{\mathrm{ord}}_{n,k}(h_1, h_2, \ldots)$*, which is Theorem 13.13 (iv).*

Proof. Apply (A.3) in the substituted form: putting $\hat{\xi}_j = j!\, \xi_j$ in (A.1) gives

$$\mathcal{B}^{\mathrm{exp}}_{n,k}(\hat{\xi}_1, \hat{\xi}_2, \ldots) = \sum_{\substack{\sum_j m_j = k \\ \sum_j j m_j = n}} \frac{n!}{\prod_j m_j!\, (j!)^{m_j}} \prod_j (j!)^{m_j} \xi_j^{m_j} = \frac{n!}{k!} \sum_{\substack{\sum_j m_j = k \\ \sum_j j m_j = n}} \frac{k!}{\prod_j m_j!} \prod_j \xi_j^{m_j},$$

which is (A.8). The generating identity for (A.7) follows by the same multiplicity grouping as in the proof of Theorem A.3, now without the factors $1/j!$. The last sentence is Theorem 13.13 (i) combined with that generating identity. □

Lean status: Exact. The new two-definition/four-theorem module `OrdinaryPartialBell.lean` represents the ordinary family by its equivalent characterizing coefficient definition rather than by the displayed multiplicity sum. Its definitions `Fabius.ordinarySeries` and `Fabius.ordinaryPartialBell`, together with `Fabius.ordinaryPartialBell_pow`, give exactly the ordinary generating identity following (A.7); `Fabius.ordinaryPartialBell_eq_zero_of_lt` records its triangular support. The theorem `Fabius.bellWeightSeries_eq_ordinarySeries` identifies the divided-coefficient ordinary series, and the multiplicatively cleared theorem `Fabius.factorial_mul_ordinaryPartialBell` is precisely the normalization bridge (A.8) over every commutative $\mathbb{Q}$-algebra. Composing that bridge with the coefficient identity in Theorem 13.13 (i) gives the final statement for $C_{n,k}$; no different object from `OrdinaryBellBivariate.lean` or `PowerSumOrdinaryBell.lean` is being substituted. This does not identify the ordinary family with labelled set partitions.

Three two-index arrays, three index orders

The volume carries three arrays whose entries are indexed by a pair of integers and which are easy to confuse.

- $h_{k,m} = [t^m] H^k$ of (13.5): first index the exponent, second the outer order.
- $C_{m,r}$ of Theorem 13.12: first index the outer order, second the exponent; $h_{k,m} = C_{m,k}$.
- $\mathcal{B}^{\mathrm{exp}}_{n,k}$ of Theorem A.1: first index the *weight* n, second the *degree* k; it is an ordinary polynomial in abstract indeterminates, and only after the substitution $x_j \mapsto j!\, h_j$ and the factor $k!/n!$ of (A.8) does it become $C_{n,k}$.

Transferring a formula between any two of them requires transposing the indices, and between the last and either of the first two also requires the factorial conversion. A missing $k!/n!$ is the single commonest error in this circle of identities.

Proposition A.5 (Two computational recurrences). *For all $n \geq 1$ and $k \geq 1$,*

$$\mathcal{B}^{\exp}_{n,k} = \sum_{j=1}^{n-k+1} \binom{n-1}{j-1} x_j \, \mathcal{B}^{\exp}_{n-j,k-1}, \tag{A.9}$$

$$k\, \mathcal{B}^{\exp}_{n,k} = \sum_{j=1}^{n-k+1} \binom{n}{j} x_j \, \mathcal{B}^{\exp}_{n-j,k-1}. \tag{A.10}$$

Either recurrence, together with the boundary values $\mathcal{B}^{\exp}_{0,0} = 1$, $\mathcal{B}^{\exp}_{n,0} = 0$ for $n \geq 1$ and $\mathcal{B}^{\exp}_{0,k} = 0$ for $k \geq 1$, determines the whole array.

Proof. Work in $R[[z]]$ with $R = \mathbb{Q}[x_1, x_2, \ldots]$ and $\Xi(z) = \sum_{j\geq 1} x_j z^j/j!$.

For (A.10), write $\Xi^k/k! = \frac{1}{k}\,\Xi \cdot \Xi^{k-1}/(k-1)!$ and extract $[z^n/n!]$. For two series $P = \sum_n p_n z^n/n!$ and $Q = \sum_n q_n z^n/n!$ one has $[z^n/n!](PQ) = \sum_{j=0}^{n} \binom{n}{j} p_j q_{n-j}$; applying this with $P = \Xi$ (so $p_0 = 0$, $p_j = x_j$) and $Q = \Xi^{k-1}/(k-1)!$ and using (A.5) gives $\frac{1}{k}\sum_{j\geq 1}\binom{n}{j} x_j \mathcal{B}^{\exp}_{n-j,k-1}$. The upper limit is $n-k+1$ because $\mathcal{B}^{\exp}_{n-j,k-1} = 0$ unless $n-j \geq k-1$ (Theorem A.2 (i)).

For (A.9), differentiate (A.5) with respect to z:

$$\frac{\mathrm{d}}{\mathrm{d}z}\frac{\Xi^k}{k!} = \Xi' \, \frac{\Xi^{k-1}}{(k-1)!}, \qquad \Xi'(z) = \sum_{i\geq 0} x_{i+1}\frac{z^i}{i!}.$$

The left side is $\sum_{n\geq 0} \mathcal{B}^{\exp}_{n+1,k} z^n/n!$. Applying the product rule for exponential generating functions to the right side with $p_i = x_{i+1}$ gives $\sum_{i=0}^{n}\binom{n}{i} x_{i+1}\mathcal{B}^{\exp}_{n-i,k-1}$. Replacing n by $n-1$ and setting $j = i+1$ turns this into $\mathcal{B}^{\exp}_{n,k} = \sum_{j\geq 1}\binom{n-1}{j-1} x_j \mathcal{B}^{\exp}_{n-j,k-1}$, which is (A.9); the upper limit is again forced by Theorem A.2 (i).

Both recurrences express $\mathcal{B}^{\exp}_{n,k}$ in terms of entries $\mathcal{B}^{\exp}_{n-j,k-1}$ with $j \geq 1$, hence in terms of strictly earlier rows; with the three stated boundary values — which supply every entry of row 0 and every entry of column 0 — they determine the array by induction on n. □

Remark A.6 (A third recurrence, and where each is used). Theorem 13.24 proves a third recurrence, $\mathcal{B}^{\exp}_{n+1,k} = \mathcal{D}\,\mathcal{B}^{\exp}_{n,k} + x_1 \mathcal{B}^{\exp}_{n,k-1}$ with $\mathcal{D} = \sum_{i\geq 1} x_{i+1}\partial_{x_i}$, which is the one adapted to *differentiating* a Bell polynomial evaluated at a tower of derivatives; it is the engine of every Faà di Bruno induction, including Theorem 13.25 and Theorem A.12 below. By contrast (A.9) and (A.10) are the ones adapted to *tabulating* the array: each computes one entry from $n-k+1$ entries of the previous column, so the whole triangle up to weight N costs $\mathcal{O}(N^3)$ multiplications of one indeterminate x_j into one previously computed entry. That is a count of *entry* operations, not of scalar ones: the entry $\mathcal{B}^{\exp}_{n,k}$ has exactly one monomial for each partition of n into k parts, so row n carries as many monomials as there are partitions of n, and the number of scalar coefficients written while tabulating up to weight N therefore exceeds every polynomial in N. The distinction between the two levels of counting is the one drawn in Theorem 27.18.

Example A.7 (The array through weight six). Table A.1 lists $\mathcal{B}^{\exp}_{n,k}$ for $1 \leq k \leq n \leq 6$; the entries were generated from (A.9) and checked against (A.1). The first column is $\mathcal{B}^{\exp}_{n,1} = x_n$, the diagonal is x_1^n and the subdiagonal is $\binom{n}{2} x_1^{n-2} x_2$, as (A.4) predicts. Each entry is homogeneous of degree k and isobaric of weight n, which is the cheapest available check on a hand computation: in $\mathcal{B}^{\exp}_{6,3} = 15x_1^2 x_4 + 60 x_1 x_2 x_3 + 15 x_2^3$ every monomial has three factors and subscript sum six. A second check is that setting all $x_j = 1$ must return the Stirling number $\left\{ {n \atop k} \right\}$ of Theorem A.9 (ii); for $n = 6$, $k = 3$ this reads $15 + 60 + 15 = 90$.

Table A.1: The exponential partial Bell polynomials $\mathcal{B}^{\exp}_{n,k}(x_1, x_2, \ldots)$ for $1 \le k \le n \le 6$.

n	k	$\mathcal{B}^{\exp}_{n,k}$
1	1	x_1
2	1	x_2
	2	x_1^2
3	1	x_3
	2	$3x_1x_2$
	3	x_1^3
4	1	x_4
	2	$4x_1x_3 + 3x_2^2$
	3	$6x_1^2x_2$
	4	x_1^4
5	1	x_5
	2	$5x_1x_4 + 10x_2x_3$
	3	$10x_1^2x_3 + 15x_1x_2^2$
	4	$10x_1^3x_2$
	5	x_1^5
6	1	x_6
	2	$6x_1x_5 + 15x_2x_4 + 10x_3^2$
	3	$15x_1^2x_4 + 60x_1x_2x_3 + 15x_2^3$
	4	$20x_1^3x_3 + 45x_1^2x_2^2$
	5	$15x_1^4x_2$
	6	x_1^6

A.2 Stirling evaluations, and the link to the Lambert polynomials

The unsigned Stirling numbers of the first kind were fixed in (21.6) by $(z)^{\underline{n}} = \sum_k s(n,k)\, z^k$ and $\left[{n \atop k}\right] = (-1)^{n-k} s(n,k)$. We first record the exponential generating function that converts that definition into a Bell evaluation.

Lemma A.8 (Rising-factorial and Stirling generating identities). *In* $\mathbb{Q}[z][[u]]$,

$$\sum_{n\ge 0} (z)^{\overline{n}}\, \frac{u^n}{n!} = \exp\Big(z \log \frac{1}{1-u}\Big), \qquad (z)^{\overline{n}} = z(z+1)\cdots(z+n-1), \tag{A.11}$$

where $\log \frac{1}{1-u} := \sum_{i\ge 1} u^i/i$ *and* $\exp$ *is the formal exponential of a series with zero constant term. Consequently* $(z)^{\overline{n}} = \sum_{k=0}^{n} \left[{n \atop k}\right] z^k$ *and, for every* $k \in \mathbb{N}_0$,

$$\frac{1}{k!}\Big(\log \frac{1}{1-u}\Big)^k = \sum_{n\ge k} \left[{n \atop k}\right] \frac{u^n}{n!}. \tag{A.12}$$

Proof. The unsigned form of (21.6) is obtained by replacing z with $-z$: since $((-z))^{\underline{n}} = (-z)(-z-1)\cdots(-z-n+1) = (-1)^n\, (z)^{\overline{n}}$, we get $(z)^{\overline{n}} = (-1)^n \sum_k s(n,k)\,(-z)^k = \sum_k (-1)^{n-k} s(n,k)\, z^k = \sum_k \left[{n \atop k}\right] z^k$.

For (A.11) we compare two solutions of one linear differential equation in $\mathbb{Q}[z][[u]]$. A series $W = \sum_{n\ge 0} w_n u^n/n!$ satisfies

$$(1-u)\,\partial_u W = zW, \qquad w_0 = 1, \tag{A.13}$$

if and only if $w_{n+1} = (z+n)w_n$ for every $n \ge 0$; indeed $\partial_u W = \sum_{n\ge 0} w_{n+1}u^n/n!$ and $u\,\partial_u W = \sum_{n\ge 1} n\, w_n u^n/n!$, so comparing coefficients of $u^n/n!$ in (A.13) gives $w_{n+1} - nw_n = zw_n$. In particular (A.13) has exactly one solution, and $w_n = (z)^{\overline{n}}$ is it, because $(z)^{\overline{0}} = 1$ and $(z)^{\overline{n+1}} = (z+n)\,(z)^{\overline{n}}$.

Now put $\ell(u) := \log \frac{1}{1-u} = \sum_{i\geq 1} u^i/i$, so that $\partial_u \ell = \sum_{i\geq 1} u^{i-1} = (1-u)^{-1}$, and $Z := \exp\bigl(z\ell(u)\bigr) = \sum_{k\geq 0} z^k \ell(u)^k/k!$, a summable family in $\mathbb{Q}[z][[u]]$ because $\ell^k \in u^k\mathbb{Q}[[u]]$. Termwise differentiation of that summable family gives $\partial_u Z = z(\partial_u \ell)Z = zZ/(1-u)$, that is $(1-u)\partial_u Z = zZ$, and Z has constant term 1. By the uniqueness just proved, $Z = \sum_{n\geq 0} (z)^{\overline{n}}\, u^n/n!$, which is (A.11).

Finally, compare the coefficient of z^k on the two sides of (A.11): on the left it is $\sum_{n\geq 0} \left[{n \atop k}\right] u^n/n!$ by the expansion $(z)^{\overline{n}} = \sum_k \left[{n \atop k}\right] z^k$ just established, and on the right it is $\ell(u)^k/k!$. This is (A.12); the sum there may be started at $n = k$ because $\left[{n \atop k}\right] = 0$ for $k > n$. □

Theorem A.9 (Stirling evaluations of the Bell polynomials)**.** *For all $n, k \in \mathbb{N}_0$,*

(i) $\mathcal{B}^{\exp}_{n,k}\bigl(0!, 1!, 2!, \ldots\bigr) = \left[{n \atop k}\right]$*, the unsigned Stirling number of the first kind; equivalently* $\mathcal{B}^{\exp}_{n,k}\bigl((j-1)!\bigr)_{j\geq 1} = (-1)^{n-k} s(n,k)$*;*

(ii) $\mathcal{B}^{\exp}_{n,k}(1, 1, 1, \ldots) = \left\{{n \atop k}\right\}$*, the Stirling number of the second kind;*

(iii) $\mathcal{B}^{\exp}_{n}(0!, 1!, 2!, \ldots) = n!$.

Proof. (i) Apply (A.5) over $R = \mathbb{Q}$ with $\xi_j = (j-1)!$. Then $\Xi(u) = \sum_{j\geq 1}(j-1)!\, u^j/j! = \sum_{j\geq 1} u^j/j = \log \frac{1}{1-u}$, so (A.5) reads

$$\frac{1}{k!}\Bigl(\log \frac{1}{1-u}\Bigr)^k = \sum_{n\geq k} \mathcal{B}^{\exp}_{n,k}\bigl(0!, 1!, 2!, \ldots\bigr)\frac{u^n}{n!}.$$

Comparing with (A.12) and using uniqueness of coefficients in $\mathbb{Q}[[u]]$ gives (i). The second form of (i) is then the sign convention (21.6).

(ii) With $\xi_j = 1$ for all j one gets $\Xi(u) = \sum_{j\geq 1} u^j/j! = \mathrm{e}^u - 1$, and (A.5) becomes $\frac{1}{k!}(\mathrm{e}^u - 1)^k = \sum_{n\geq k} \mathcal{B}^{\exp}_{n,k}(1, 1, \ldots)u^n/n!$. The left-hand side is the classical exponential generating function of the Stirling numbers of the second kind, which we verify from the definition $\left\{{n \atop k}\right\} = \frac{1}{k!}\sum_{i=0}^{k}(-1)^{k-i}\binom{k}{i} i^n$. Expanding $(\mathrm{e}^u - 1)^k$ by the binomial theorem gives

$$(\mathrm{e}^u - 1)^k = \sum_{i=0}^{k}(-1)^{k-i}\binom{k}{i}\mathrm{e}^{iu} = \sum_{n\geq 0}\Bigl(\sum_{i=0}^{k}(-1)^{k-i}\binom{k}{i} i^n\Bigr)\frac{u^n}{n!},$$

so $[u^n/n!]\,\frac{1}{k!}(\mathrm{e}^u - 1)^k = \left\{{n \atop k}\right\}$.

(iii) Sum (i) over $0 \leq k \leq n$ and use $\sum_{k=0}^{n} \left[{n \atop k}\right] = (1)^{\overline{n}} = 1 \cdot 2 \cdots n = n!$, which is the polynomial identity $(z)^{\overline{n}} = \sum_k \left[{n \atop k}\right] z^k$ of Theorem A.8 evaluated at $z = 1$. □

Lean status. All three evaluations are formal, over $\mathbb{N}_0$, and each is derived from the generating identity rather than from the multinomial sum. (i), in the unsigned form, is `Fabius.partialBell_factorial_pred` in `BellGeneratingFunctions.lean`: $\mathcal{B}^{\exp}_{n,k}\bigl((j-1)!\bigr)_j = \left[{n \atop k}\right]$, proved by identifying $\bigl(-\log(1-z)\bigr)^k$ with the corresponding column exponential generating function; the signed restatement $(-1)^{n-k}s(n,k)$ offered as an equivalent above is not separately formalized, the corpus working throughout with the unsigned numbers. (ii) is `Fabius.partialBell_one`, and (iii) is `Fabius.bell_complete_factorial_pred` in `BellFactorialRowSum.lean`, obtained from (i) and the row sum $\sum_k \left[{n \atop k}\right] = n!$. Two neighbouring evaluations not stated in this theorem are formal as well and are worth recording here: `Fabius.partialBell_factorial`, that $x_j = j!$ gives the Lah numbers, and `Fabius.bell_complete_one`, that the complete polynomial at $x = (1, 1, 1, \ldots)$ is the n-th Bell number.

Corollary A.10 (The Lambert triangle is a slice of the Bell array)**.** *For $n \geq 1$ and $1 \leq m \leq n$, the coefficients $\lambda_{n,m} = [u^m]\mathsf{L}_n^{\mathrm{Lam}}(u)$ of the Lambert polynomials (Theorems 21.1 and 21.4) are*

$$\begin{aligned} \lambda_{n,m} &= \frac{(-1)^{n-m}}{m!}\, \mathcal{B}^{\exp}_{n,\,n-m+1}\big(0!, 1!, 2!, \ldots\big), \\ \mathsf{L}_n^{\mathrm{Lam}}(u) &= \sum_{m=1}^{n} \frac{(-1)^{n-m}}{m!}\, \mathcal{B}^{\exp}_{n,\,n-m+1}\big(0!, 1!, 2!, \ldots\big)\, u^m. \end{aligned} \tag{A.14}$$

Reading the array by antidiagonals, the n-th Lambert polynomial consumes exactly the n-th row of the Bell array evaluated at the factorial arguments, in reversed column order and with alternating signs.

Proof. Substitute Theorem A.9 (i) with $k = n - m + 1$ into (21.8). □

Remark A.11 (Why the factorial arguments are the right ones). The evaluation $x_j \mapsto (j-1)!$ looks arbitrary until one notices what it encodes. By the proof of Theorem A.9 (i) it is exactly the evaluation for which the generating series Ξ of Theorem A.3 becomes a logarithm. The Lambert polynomials arise from reverting $F = X + L$, that is from a perturbation which *is* the logarithm (Theorem 20.10); the appearance of cycle numbers in (A.14) is therefore not a coincidence of two unrelated triangles but the same statement twice. The table of $\left[{n \atop n-m+1}\right]$ in Table D.2 is simultaneously a table of Bell polynomials at factorial arguments. Note that the correspondence is between the *unsigned* cycle numbers and the Bell array: the signs in (A.14) come from the alternating prefactor of (21.1) and not from the Bell polynomials, which have nonnegative integer coefficients by (A.1).

A.3 Faà di Bruno with a scalar-analytic outer function

Theorem 13.25 computes $D_X^n(F \circ G)$ when F and G are both polynomial–logarithmic and $\circ$ is the composition of Part 12. The volume also needs the complementary case, in which the outer object is not an element of $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ at all but a scalar power series $\varphi \in \mathbb{K}[[z]]$ applied to a polynomial–logarithmic argument through the functional calculus of Theorem 13.22. The two statements have the same proof, and we isolate that proof once.

Lemma A.12 (Abstract Faà di Bruno for a Leibniz tower)**.** *Let R be a commutative $\mathbb{Q}$-algebra, ∂ a derivation of R, $\gamma \in R$, and let $(u_k)_{k\geq 0}$ be a sequence in R satisfying*

$$\partial u_k = u_{k+1}\,\gamma \qquad (k \geq 0). \tag{A.15}$$

Then for every $n \geq 1$,

$$\partial^{\,n} u_0 = \sum_{k=1}^{n} u_k\, \mathcal{B}^{\exp}_{n,k}\big(\gamma, \partial\gamma, \ldots, \partial^{\,n-k}\gamma\big). \tag{A.16}$$

Proof. Induction on n. For $n = 1$, (A.16) reads $\partial u_0 = u_1 \mathcal{B}^{\exp}_{1,1}(\gamma) = u_1\gamma$, which is (A.15) at $k = 0$.

Assume (A.16) for some $n \geq 1$ and abbreviate $\beta_{n,k} := \mathcal{B}^{\exp}_{n,k}(\gamma, \partial\gamma, \ldots)$, the evaluation of the polynomial $\mathcal{B}^{\exp}_{n,k}$ at $x_i = \partial^{\,i-1}\gamma$. Since ∂ is a derivation and $\mathcal{B}^{\exp}_{n,k}$ is a polynomial in finitely many of the x_i, the ordinary chain rule for polynomials gives

$$\partial\beta_{n,k} = \sum_{i\geq 1} \big(\partial_{x_i}\mathcal{B}^{\exp}_{n,k}\big)(\gamma, \partial\gamma, \ldots) \cdot \partial^{\,i}\gamma = \big(\mathcal{D}\mathcal{B}^{\exp}_{n,k}\big)(\gamma, \partial\gamma, \ldots), \qquad \mathcal{D} = \sum_{i\geq 1} x_{i+1}\partial_{x_i},$$

because $\partial^{\,i}\gamma$ is the value of x_{i+1}. Hence, using the Leibniz rule and (A.15),

$$\partial^{\,n+1} u_0 = \sum_{k\geq 1}\Big(u_{k+1}\gamma\,\beta_{n,k} + u_k\big(\mathcal{D}\mathcal{B}^{\exp}_{n,k}\big)(\gamma, \partial\gamma, \ldots)\Big) = \sum_{k\geq 1} u_k \Big[\gamma\,\mathcal{B}^{\exp}_{n,k-1} + \mathcal{D}\,\mathcal{B}^{\exp}_{n,k}\Big](\gamma, \partial\gamma, \ldots),$$

after shifting the index in the first sum (the term $k = 1$ of the shifted sum carries $\mathcal{B}^{\exp}_{n,0} = 0$ for $n \geq 1$, so nothing is lost). By Theorem 13.24 evaluated at $x_1 = \gamma$, the bracket is $\mathcal{B}^{\exp}_{n+1,k}$. The sum truncates at $k = n+1$ by Theorem A.2 (i), and the arguments needed are $x_1, \ldots, x_{n+1-k+1}$, that is $\gamma, \ldots, \partial^{n+1-k}\gamma$. All sums are finite. □

Lean status. Formal in `BellLeibnizTower.lean` as `Fabius.iterate_derivation_eq_sum_partialBell`, with `Fabius.derivationTower` and `Fabius.derivation_partialBell_tower` as the engine. The partial Bell polynomials it is stated against are the corpus family of Theorem A.1, so this lemma and the Bell chapter share one object rather than two definitions that happen to agree.

Remark A.13 (The two Faà di Bruno statements are two towers). Theorem 13.25 is Theorem A.12 in $R = \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ with $\partial = D_X$, $\gamma = D_X G$ and $u_k = \left(D_X^k F\right) \circ G$: the tower condition (A.15) is then exactly the chain rule Theorem 12.31. Theorem A.15 below is the same lemma with $\gamma = D_X F$ and $u_k = \varphi^{(k)}(F)$. Nothing else distinguishes the two; in particular neither is a corollary of the other, since the towers are different.

Before applying Theorem A.12 to a scalar-analytic outer function we must know that the required tower exists, that is, that the chain rule holds for the functional calculus $\varphi \mapsto \varphi(W)$ of Theorem 13.22. For a *polynomial* φ this is the Leibniz rule; for a power series it needs the continuity of D_X.

Lemma A.14 (Chain rule for a scalar-analytic substitution)**.** *Let $z_0 \in \mathbb{K}$ and let the outer function be given* by its expansion about z_0*: that is, let $\varphi_0, \varphi_1, \ldots \in \mathbb{K}$ be arbitrary scalars and write, formally, $\varphi(z) = \sum_{k\geq 0} \frac{\varphi_k}{k!}(z - z_0)^k$, the datum being the pair $\left(z_0, (\varphi_k)_{k\geq 0}\right)$. Let $F \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ satisfy $\nu_t(F - z_0) \geq 1$, write $W := F - z_0 \in t\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$, and put*

$$\varphi(F) := \sum_{k\geq 0} \frac{\varphi_k}{k!} W^k, \qquad \varphi^{(k)}(F) := \sum_{i\geq 0} \frac{\varphi_{k+i}}{i!} W^i \qquad (k \geq 0), \tag{A.17}$$

each well defined by Theorem 13.22 applied to the series $\sum_{k\geq 0}(\varphi_k/k!)z^k \in \mathbb{K}[[z]]$ and to W. When $z_0 = 0$ this is the ordinary reading $\varphi(F) = \varphi(W)$; when $z_0 \neq 0$ the centre must be part of the datum, since an element of $\mathbb{K}[[z]]$ presented at the origin cannot be re-expanded about z_0 inside $\mathbb{K}[[z]]$. Then

$$D_X\left(\varphi(F)\right) = \varphi'(F)\, D_X F. \tag{A.18}$$

If $\varphi \in \mathbb{K}[z]$ is a polynomial, (A.18) *holds for every $F \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$, with $\varphi(F)$ the ordinary polynomial evaluation and no valuation hypothesis. The two readings of $\varphi(F)$ agree when both apply, since the Taylor expansion of a polynomial about z_0 is exact and finite.*

Proof. Each $\varphi^{(k)}(F)$ is defined by (A.17), the family there being summable because $\nu_t(W^i) \geq i$. Write $c_k = \varphi_k/k! \in \mathbb{K}$, so that $\varphi(F) = \sum_{k\geq 0} c_k W^k$ and $\varphi'(F) = \sum_{k\geq 1} k c_k W^{k-1}$, both families being summable because $\nu_t(W^k) \geq k$ (Theorem 6.2, Theorem 10.1). Note $D_X W = D_X F$, because $D_X z_0 = 0$ for a scalar z_0 (Theorem 11.12).

The identity (A.18) is checked one block at a time. Fix $N \geq 0$ and split $\varphi(F) = P_N + T_N$ with $P_N = \sum_{k\leq N+1} c_k W^k$ and $\nu_t(T_N) \geq N + 2$. Since D_X is a $\mathbb{K}$-derivation of $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ with $\nu_t(D_X H) \geq \nu_t H + 1$ (Theorem 11.1, Theorem 11.11 (1)), we get $\nu_t(D_X T_N) \geq N + 3$, so

$$[t^N]\, D_X\left(\varphi(F)\right) = [t^N]\, D_X P_N = \sum_{k=1}^{N+1} k c_k\, [t^N]\left(W^{k-1} D_X W\right),$$

the last step being the Leibniz rule applied to the *finite* sum P_N. On the other side, $\nu_t(W^{k-1} D_X W) \geq (k-1) + 2 = k + 1$, so the terms with $k \geq N + 2$ do not reach the block t^N and

$$[t^N]\left(\varphi'(F)\, D_X F\right) = \sum_{k\geq 1} k c_k\, [t^N]\left(W^{k-1} D_X W\right) = \sum_{k=1}^{N+1} k c_k\, [t^N]\left(W^{k-1} D_X W\right).$$

The two agree for every N, and an element of $\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ is determined by its blocks (Theorem 7.13), which proves (A.18). For a polynomial φ the sums are finite from the start and only the Leibniz rule is used, so no valuation hypothesis is needed and the identity holds in $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$. □

Theorem A.15 (Faà di Bruno for a scalar-analytic outer function)**.** *Let $z_0 \in \mathbb{K}$, let φ be given by its expansion about z_0 as in Theorem A.14, and let $F \in \mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ satisfy $\nu_t(F - z_0) \geq 1$; or let $\varphi \in \mathbb{K}[z]$ be a polynomial and $F \in \mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ arbitrary. Then for every $n \geq 1$*

$$D_X^n\big(\varphi(F)\big) = \sum_{k=1}^{n} \varphi^{(k)}(F)\, \mathcal{B}_{n,k}^{\exp}\big(D_X F, D_X^2 F, \ldots, D_X^{n-k+1} F\big), \tag{A.19}$$

an identity in $\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ in the first case and in $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ in the second.

Proof. Apply Theorem A.12 in $R = \mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ with $\partial = D_X$, $\gamma = D_X F$ and $u_k = \varphi^{(k)}(F)$. The tower condition (A.15) is $D_X\big(\varphi^{(k)}(F)\big) = \varphi^{(k+1)}(F)\, D_X F$, which is Theorem A.14 applied to $\varphi^{(k)}$ in place of φ. The arguments of the Bell polynomial are $x_i = \partial^{\,i-1}\gamma = D_X^i F$, $1 \leq i \leq n-k+1$. Membership in $\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ in the first case follows because $\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ is closed under D_X (Theorem 11.1 (i)) and under the substitution (Theorem 13.22). □

Corollary A.16 (Derivatives of a power, and the Lagrange terms)**.** *For every $A \in \mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$, every $r \in \mathbb{N}_0$ and every $n \geq 1$,*

$$D_X^n\big(A^r\big) = \sum_{k=1}^{\min\{n,r\}} (r)^{\underline{k}}\ A^{r-k}\ \mathcal{B}_{n,k}^{\exp}\big(D_X A, D_X^2 A, \ldots, D_X^{n-k+1} A\big), \tag{A.20}$$

where $(r)^{\underline{k}} = r(r-1)\cdots(r-k+1)$ is the falling factorial, and where the terms with $k > r$ are absent because $(r)^{\underline{k}} = 0$ there.

Proof. Take $\varphi(z) = z^r$ in the polynomial case of Theorem A.15; then $\varphi^{(k)}(z) = (r)^{\underline{k}}\, z^{r-k}$ for $k \leq r$ and $\varphi^{(k)} = 0$ for $k > r$. □

Remark A.17 (Valuation bookkeeping). In the setting of Theorem A.15 with $\nu_t(F - z_0) \geq 1$, one has $\nu_t(D_X^i F) \geq i + 1$ by Theorem 11.11, so a monomial of $\mathcal{B}_{n,k}^{\exp}$ with multiplicity vector (m_j) evaluates to something of order at least $\sum_j m_j(j+1) = n + k$. The k-th summand of (A.19) therefore has $\nu_t \geq n + k \geq n + 1$, matching the bound $\nu_t\big(D_X^n(\varphi(F))\big) \geq n + 1$ that Theorem 11.11 (2) gives directly, since $\varphi(F) - \varphi(z_0) \in t\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$. Consequently only $k \leq N - n$ can affect the block t^N: the combinatorial size of (A.19) at a fixed target block is bounded independently of any analytic consideration.

Appendix B

Closed-form coefficient extraction

Parts II–IV compute coefficients by *recurrences*: the reciprocal by (9.7), the quotient by (9.8), the convolution polynomials by (13.10), the inverse by Theorem 17.2. A recurrence is the right tool when all coefficients up to a cutoff are wanted, and it is what an implementation should run (Part 27). It is the wrong tool when a *single* block is wanted, when the block is wanted as an explicit function of the input blocks, or when one has to prove something about the block — a degree bound, a sign, an integrality statement — for all N at once.

This chapter derives the non-recursive counterparts. By a *closed form* we mean a formula for $[t^N]$ or $[t^N L^d]$ of the result as a finite sum, indexed by an explicitly bounded set, of products of the input blocks and of universal combinatorial constants, with no reference to previously computed output blocks. Every such formula below is derived from the body of the volume and from Part A; none is quoted.

B.1 The multi-index bookkeeping

Two separate finiteness mechanisms are in play, and conflating them is the source of the "infinite sum" complaints that the sources occasionally raise against these formulas.

Lemma B.1 (The two finiteness mechanisms). *Let $F = \sum_{n \geq n_0} p_n(L)t^n \in \mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ and fix $N \in \mathbb{Z}$, $d \in \mathbb{N}_0$.*

(i) (Outer, by valuation.) *If $(S_i)_{i \in I}$ is a summable family in $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$, then $\{i \in I : [t^N]S_i \neq 0\}$ is finite and $[t^N]\sum_i S_i = \sum_i [t^N]S_i$.*

(ii) (Inner, by degree.) *Each block p_n lies in $\mathbb{K}[L]$, hence has finite $\deg_{\mathrm{L}}$, so $[t^N L^d]F = 0$ for $d > d_F(N)$ and the block p_N is a finite sum of monomials.*

(iii) *The two are independent: a summable family may have blocks of unbounded $\deg_{\mathrm{L}}$ at a fixed level of the partial sums — for instance $S_i = t^i L^i$ is summable although $d_{\sum_{i \leq M} S_i}$ is unbounded in M — and conversely a family of bounded $\deg_{\mathrm{L}}$ need not be summable. Only* (i) *is part of the t-adic topology; a bound on $\deg_{\mathrm{L}}$ is an additional approximation and must be stated separately.*

Proof. (i) is Theorem 7.22 together with Theorem 7.13. (ii) is Theorem 6.11 and the definition of $\mathbb{K}[L]$. For the first half of (iii), $\nu_t(t^i L^i) = i \to \infty$, so the family is summable, while d of the M-th partial sum takes the value M at level M; for the second half, the constant family $S_i = 1$ has $\deg_{\mathrm{L}} S_i = 0$ and is not summable. $\square$

The formal tail does not bound a logarithmic degree

The statement $F = G + \mathcal{O}_t^{\mathrm{form}}(t^N)$ says nothing whatever about $\deg_{\mathrm{L}}$ of the discarded blocks (Theorem 7.7); truncation is exclusive in t and *whole* in L (Theorem 7.1). Every closed formula below therefore returns a complete polynomial block, never a polynomial block truncated in L. Discarding a high power of L inside a retained block is a second, independent approximation and requires its own error statement.

B.2 Products

Proposition B.2 (Scalar coefficients of a product). *Let $F, G \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ be nonzero with $a = \nu_t F$, $b = \nu_t G$, and write $a_{n,e} = [t^n L^e]F$, $b_{m,f} = [t^m L^f]G$ in the notation of Theorem 7.12. Then for all $N \in \mathbb{Z}$ and $d \in \mathbb{N}_0$,*

$$[t^N L^d](FG) = \sum_{n=a}^{N-b} \sum_{e=0}^{d} a_{n,e}\, b_{N-n,\, d-e}, \tag{B.1}$$

a sum of at most $(N-a-b+1)(d+1)$ products, empty when $N < a+b$. The nonzero terms satisfy the sharper constraints $e \le d_F(n)$ and $d - e \le d_G(N-n)$; in particular $[t^N L^d](FG) = 0$ whenever $d > (d_F * d_G)(N) = \max_{n+m=N}\big(d_F(n) + d_G(m)\big)$.

Proof. By Theorem 8.5 and Theorem 8.6, $[t^N](FG) = \sum_{n=a}^{N-b} p_n q_{N-n}$, a finite sum in $\mathbb{K}[L]$. Extracting $[L^d]$ of a product of two polynomials is the Cauchy product in L, which gives (B.1); the index e is confined to $[0, d]$ because both factors are polynomials, hence have no negative powers of L. The sharper constraints and the vanishing statement are the degree bound (8.23) of Theorem 8.25, whose $*$ is the $(\max, +)$-convolution displayed above. $\square$

Formula (B.1) is a genuine two-level convolution: outer in t by valuation, inner in L by degree. Its cost, and the fact that the resulting scalar cost is *not* $\mathcal{O}(N^2)$ once the inner level is counted, are analysed in Theorem 27.17 and Theorem 27.18.

B.3 Powers, reciprocals, and the convolution polynomials

The convolution polynomials $C_{m,r}$ of Theorem 13.12 — defined by (13.8) with the empty-product conventions $C_{0,0} = 1$, $C_{m,0} = 0$ for $m > 0$, and computed by the recurrence (13.10) — are the blocks of the powers of a series. Theorem 13.13 (iv) gives their closed form only under the hypothesis $h_0 = 0$. The general case is needed whenever the inner series has a nonzero level-zero block, which is the normal situation for a near-identity map.

Proposition B.3 (Closed form of the convolution polynomials, general case). *Let $H = \sum_{j \ge 0} h_j(L) t^j \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ and let $C_{m,r}$ be as in* (13.8). *Then for all $m, r \in \mathbb{N}_0$,*

$$C_{m,r} = \sum_{k=0}^{\min\{m,r\}} \binom{r}{k} h_0^{r-k} \frac{k!}{m!} \mathcal{B}^{\exp}_{m,k}\big(1!\,h_1, 2!\,h_2, \ldots, (m-k+1)!\,h_{m-k+1}\big), \tag{B.2}$$

where the term $k = 0$ carries $\mathcal{B}^{\exp}_{m,0}$ and is therefore h_0^r for $m = 0$ and 0 for $m > 0$. When $h_0 = 0$ only $k = r$ survives: for $r \ge 1$ formula (B.2) *then reduces to* (13.11)*, and for $r = 0$ it returns the boundary values $C_{m,0} = \delta_{m,0}$ of* (13.8).

Proof. Split $H = h_0 + H_+$ with $H_+ = \sum_{j \ge 1} h_j t^j \in t\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$. Since h_0 and H_+ commute, the binomial theorem gives $H^r = \sum_{k=0}^{r} \binom{r}{k} h_0^{r-k} H_+^k$, a finite sum. By Theorem 13.13 (i) applied to H and by Theorem 8.36 applied to H_+ (whose level-zero block vanishes),

$$C_{m,r} = [t^m]H^r = \sum_{k=0}^{r} \binom{r}{k} h_0^{r-k}\, [t^m]H_+^k, \qquad [t^m]H_+^k = \frac{k!}{m!}\, \mathcal{B}^{\exp}_{m,k}\big(1!h_1,\dots\big) \ \ (k \ge 1),$$

with $[t^m]H_+^k = 0$ for $m < k$ and $[t^m]H_+^0 = \delta_{m,0}$. Truncating the sum at $k = \min\{m, r\}$ is Theorem A.2 (i). If $h_0 = 0$ then, with the convention $0^0 = 1$, $h_0^{r-k} = 0$ unless $k = r$; the surviving term is (13.11) when $r \ge 1$, and is $(0!/m!)\mathcal{B}^{\exp}_{m,0} = \delta_{m,0}$ when $r = 0$. □

Theorem B.4 (Closed form for an arbitrary power of a unit)**.** *Let $W = \sum_{j\ge1} w_j(L)t^j \in t\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ and $\sigma \in \mathbb{K}$, and let $(1+W)^\sigma \in \mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ be the binomial power of Theorem 8.37. Then $[t^0](1+W)^\sigma = 1$ and, for $r \ge 1$,*

$$[t^r](1+W)^\sigma = \frac{1}{r!}\sum_{k=1}^{r} (\sigma)^{\underline{k}}\ \mathcal{B}^{\exp}_{r,k}\big(1!\,w_1, 2!\,w_2, \dots, (r-k+1)!\,w_{r-k+1}\big), \tag{B.3}$$

where $(\sigma)^{\underline{k}} = \sigma(\sigma-1)\cdots(\sigma-k+1)$. The right-hand side depends only on $w_1,\dots,w_r$, and it is $\mathbb{K}$-linear in w_r with coefficient σ.

If $F \in \mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ is a unit, so that $F = ct^{n_0}(1+W)$ with $c \in \mathbb{K}^\times$, $n_0 = \nu_t F$ and $W \in t\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ by Theorem 9.3, then, once a value $c^\sigma \in \mathbb{K}$ is chosen, one defines *$F^\sigma := c^\sigma t^{\sigma n_0}(1+W)^\sigma$; this lies in $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ when $\sigma n_0 \in \mathbb{Z}$, and otherwise in the Puiseux-log or Hahn extension of Theorem 5.25. For $\sigma \in \mathbb{Z}$ the definition returns the ring power of F.*

Proof. By Theorem 8.37 (i)–(ii), $(1+W)^\sigma = \sum_{k\ge0}\binom{\sigma}{k}W^k$ with the family summable, and $[t^r](1+W)^\sigma = \sum_{k=0}^{r}\binom{\sigma}{k}[t^r]W^k$ since $\nu_t(W^k) \ge k$. For $k \ge 1$, Theorem 8.36 (formula (8.29), whose hypothesis $w_0 = 0$ holds) gives $[t^r]W^k = \frac{k!}{r!}\mathcal{B}^{\exp}_{r,k}(1!w_1, 2!w_2, \dots)$; for $k = 0$, $[t^r]W^0 = \delta_{r,0}$, which contributes nothing when $r \ge 1$. Since $\binom{\sigma}{k}k! = (\sigma)^{\underline{k}}$, summing gives (B.3).

Dependence only on $w_1,\dots,w_r$ is Theorem A.2 (i). For the linear dependence on w_r: the variable x_r occurs in $\mathcal{B}^{\exp}_{r,k}$ only for $k = 1$ – by Theorem A.2 (i) it can occur only when $r \le r-k+1$, i.e. $k \le 1$ – and $\mathcal{B}^{\exp}_{r,1} = x_r$ by (A.4), so the coefficient of w_r is $\frac{1}{r!}\,(\sigma)^{\underline{1}}\cdot r! = \sigma$. This is (8.32) made explicit.

The factorization statement is Theorem 9.3 (3); the exponent $t^{\sigma n_0}$ lies in the integer-exponent scale exactly when $\sigma n_0 \in \mathbb{Z}$, and otherwise requires the ramified tower of Theorem 5.25. For $\sigma \in \mathbb{Z}$ the displayed definition returns the ring power, because c^σ and $t^{\sigma n_0}$ are then the ordinary powers and $(1+W)^\sigma$ agrees with the σ-fold product by the exponent law of Theorem 8.37. □

Corollary B.5 (Closed form for a reciprocal)**.** *With W as in Theorem B.4,*

$$[t^r](1+W)^{-1} = \frac{1}{r!}\sum_{k=1}^{r}(-1)^k\,k!\,\mathcal{B}^{\exp}_{r,k}\big(1!\,w_1,\dots,(r-k+1)!\,w_{r-k+1}\big) \qquad (r \ge 1). \tag{B.4}$$

Consequently, for a unit $F = ct^{n_0}(1+W)$ of $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$, $F^{-1} = c^{-1}t^{-n_0}\big(1+\sum_{r\ge1}[t^r](1+W)^{-1}\,t^r\big)$. In particular

$$[t^1](1+W)^{-1} = -w_1, \quad [t^2](1+W)^{-1} = w_1^2 - w_2, \quad [t^3](1+W)^{-1} = -w_1^3 + 2w_1w_2 - w_3.$$

Proof. Put $\sigma = -1$ in (B.3) and use $((-1))^{\underline{k}} = (-1)(-2)\cdots(-k) = (-1)^k k!$. The factorization is Theorem 9.7. The three displayed values follow from Table A.1: for $r = 1$, $-1!\,x_1/1!$ at $x_1 = w_1$; for $r = 2$, $\big(-1!\,x_2 + 2!\,x_1^2\big)/2!$ at $(x_1, x_2) = (w_1, 2w_2)$, giving $-w_2 + w_1^2$; for $r = 3$, $\big(-x_3 + 2\cdot 3x_1x_2 - 6x_1^3\big)/6$ at $(x_1, x_2, x_3) = (w_1, 2w_2, 6w_3)$, giving $-w_3 + 2w_1w_2 - w_1^3$. □

The Bell form (B.4) is closed but its term count grows like the partition function. A second closed form, with a term count that is a single determinant and which is the one to use for structural arguments about signs and integrality, is the following.

Proposition B.6 (Toeplitz–Hessenberg determinant for the reciprocal). *Let $B = \sum_{n\geq 0} B_n(L)t^n \in \mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ with $B_0 = c \in \mathbb{K}^\times$, and let $B^{-1} = \sum_{r\geq 0} C_r(L)t^r$ be its reciprocal (Theorem 9.7). For $r \geq 1$ let M_r be the $r \times r$ matrix over $\mathbb{K}[L]$ with entries*

$$(M_r)_{ij} \coloneqq B_{i-j+1}, \qquad 1 \leq i, j \leq r, \qquad B_m \coloneqq 0 \text{ for } m < 0. \tag{B.5}$$

Thus M_r is lower Hessenberg, with c on the superdiagonal, B_1 on the diagonal, and B_r in the lower-left corner. Then

$$C_r = \frac{(-1)^r}{c^{r+1}} \det M_r \qquad (r \geq 1), \qquad C_0 = c^{-1}. \tag{B.6}$$

Proof. Put $D_0 \coloneqq 1$ and $D_r \coloneqq \det M_r$. We first show

$$D_r = -\sum_{m=1}^{r} (-1)^m B_m\, c^{m-1} D_{r-m} \qquad (r \geq 1). \tag{B.7}$$

Expand $\det M_r$ along its last row, whose j-th entry is B_{r-j+1}: $D_r = \sum_{j=1}^{r} (-1)^{r+j} B_{r-j+1}\, \mu_j$, where μ_j is the minor obtained by deleting row r and column j. Its rows are indexed by $i \in \{1, \ldots, r-1\}$ and its columns by $q \in \{1, \ldots, r\} \setminus \{j\}$, with entry B_{i-q+1}. If $i \leq j-1$ and $q \geq j+1$, then $i - q + 1 \leq j - 1 - j - 1 + 1 < 0$, so that block vanishes; the minor is therefore block lower-triangular with diagonal blocks

- rows $1, \ldots, j-1$ against columns $1, \ldots, j-1$, which is M_{j-1}, of determinant D_{j-1}; and
- rows $j, \ldots, r-1$ against columns $j+1, \ldots, r$. In the local indices $a = i - j + 1$ and $b = q - j$, both running over $1, \ldots, r-j$, its entry is $B_{i-q+1} = B_{a-b}$; it therefore vanishes strictly above the diagonal and carries $B_0 = c$ on the diagonal, so it is lower triangular with determinant c^{r-j}.

Hence $\mu_j = D_{j-1} c^{r-j}$ and $D_r = \sum_{j=1}^{r} (-1)^{r+j} B_{r-j+1} c^{r-j} D_{j-1}$. Substituting $m = r - j + 1$, so that $j - 1 = r - m$, $r - j = m - 1$ and $(-1)^{r+j} = (-1)^{2r-m+1} = (-1)^{m+1}$, gives (B.7).

Now define $\widetilde{C}_0 = c^{-1}$ and $\widetilde{C}_r = (-1)^r c^{-(r+1)} D_r$ for $r \geq 1$. Multiplying (B.7) by $(-1)^r c^{-(r+1)}$ turns it into

$$\begin{aligned}\widetilde{C}_r &= -\sum_{m=1}^{r} (-1)^{r+m} c^{m-1-(r+1)} B_m D_{r-m} = -c^{-1} \sum_{m=1}^{r} B_m\, (-1)^{r-m} c^{-(r-m+1)} D_{r-m} \\ &= -c^{-1} \sum_{m=1}^{r} B_m \widetilde{C}_{r-m},\end{aligned}$$

using $(-1)^{r+m} = (-1)^{r-m}$ and $D_0 = 1$ for the term $m = r$. This is exactly the recurrence (9.7) characterizing the blocks C_r, and $\widetilde{C}_0 = C_0$ by (9.6); by the uniqueness part of Theorem 9.7, $\widetilde{C}_r = C_r$ for all r. □

Remark B.7 (What the determinant form is good for). Theorem B.6 makes two structural facts immediate that are awkward to see from either (9.7) or (B.4). First, $c^{r+1} C_r$ is a polynomial with integer coefficients in $B_0, \ldots, B_r$, so the only denominators in a reciprocal are powers of the leading scalar: reciprocation introduces no new primes beyond those dividing c. Second, for $r \geq 1$,

$$\deg_{\mathrm{L}} C_r \leq \max\Big\{ \textstyle\sum_{i=1}^{s} d_B(m_i) \ : \ s \geq 1,\ m_i \geq 1,\ m_1 + \cdots + m_s = r \Big\},$$

because a signed product in the expansion of $\det M_r$ is $\prod_{i=1}^{r} B_{i-\sigma(i)+1}$ for a permutation σ, whose subscripts sum to $\sum_i (i - \sigma(i) + 1) = r$; the factors with subscript 0 are the scalar c and contribute nothing to $\deg_{\mathrm{L}}$, and the remaining factors are indexed by a composition of r into positive parts. Setting $d_B(m) \leq \alpha m$ makes the right-hand side at most αr, which is the linear bound of Theorem 9.15, now obtained without a recursion. Neither observation requires expanding the determinant.

B.4 Composition

Theorem 13.10 already gives a closed formula, in terms of the convolution polynomials. Substituting Theorem B.3 makes it closed in terms of the *input blocks* alone.

Proposition B.8 (Composition coefficients in Bell form). *Let $F = \sum_{n \ge n_0} p_n(L) t^n \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ and $H = \sum_{j \ge 1} h_j(L) t^j \in t\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ (so $h_0 = 0$). Then for every $N \in \mathbb{Z}$,*

$$[t^N]\big(F \circ (X+H)\big) = \sum_{n=n_0}^{N} \sum_{r=0}^{\lfloor (N-n)/2 \rfloor} \frac{1}{(N-n-r)!} \mathcal{B}^{\exp}_{N-n-r,\,r}\big(1!\,h_1, 2!\,h_2, \ldots\big)\ \Delta_{n,r}\, p_n, \tag{B.8}$$

where $\Delta_{n,r} = \prod_{j=0}^{r-1}(\partial_L - n - j)$ is the operator of Theorem 11.7. The sum is finite, with at most $\lfloor (N-n_0+2)^2/4 \rfloor$ terms, and it involves only $p_{n_0}, \ldots, p_N$ and $h_1, \ldots, h_{N-n_0}$.

Proof. Start from (13.6), in which the summand indexed by (n,k,m) with $n+k+m = N$ is $\frac{h_{k,m}}{k!}\Delta_{n,k} p_n$ and $h_{k,m} = C_{m,k}$ by the index dictionary of Section 13.4. Rename k as r and use $m = N-n-r$. Since $h_0 = 0$, (13.11) gives $C_{m,r} = \frac{r!}{m!}\mathcal{B}^{\exp}_{m,r}(1!h_1, \ldots)$ for $m \ge r \ge 1$, and $C_{m,r} = 0$ for $m < r$; also $C_{0,0} = 1 = \frac{0!}{0!}\mathcal{B}^{\exp}_{0,0}$ and $C_{m,0} = 0 = \mathcal{B}^{\exp}_{m,0}$ for $m > 0$. Hence in every case $C_{m,r}/r! = \mathcal{B}^{\exp}_{m,r}(\cdot)/m!$, which is the displayed summand. The constraint $r \le m = N-n-r$, i.e. $2r \le N-n$, is Theorem A.2 (i). Counting the pairs (n,r) with $n_0 \le n \le N$ and $0 \le r \le \lfloor (N-n)/2 \rfloor$ and writing $s = N-n$ gives $\sum_{s=0}^{N-n_0}(\lfloor s/2 \rfloor + 1) = \lfloor (N-n_0+2)^2/4 \rfloor$. Dependence on the listed blocks only is visible from the displayed ranges: $n \le N$, and the arguments of $\mathcal{B}^{\exp}_{N-n-r,r}$ reach at most $x_{(N-n-r)-r+1}$, so h_j occurs only for $j \le N-n \le N-n_0$. This agrees with, and refines for this special G, the general finite determination of Theorem 12.10. ☐

Corollary B.9 (Composition with a pure level-zero shift). *Let $F = \sum_{n \ge n_0} p_n(L) t^n \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ and let $H = h_0(L) \in \mathbb{K}[L]$ have no positive t-levels, so that $X + H = X + h_0(L)$. Then for every N,*

$$[t^N]\big(F \circ (X+h_0)\big) = \sum_{n=n_0}^{N} \frac{h_0^{N-n}}{(N-n)!}\ \Delta_{n,\,N-n}\, p_n. \tag{B.9}$$

In particular, for $F = X + A$ and $h_0 = aL + b$ with $a, b \in \mathbb{K}$, the whole composite is computed by a single sum with $N - n_0 + 1$ terms and no convolution at all.

Proof. Here $H^k = h_0^k$ is concentrated at level 0, so $h_{k,m} = h_0^k \delta_{m,0}$ in (13.5). Substituting into (13.6) leaves only the triples $(n,k,m) = (n, N-n, 0)$, which is (B.9). ☐

Example B.10 (Composition with $X + L$). Take $F = t = X^{-1}$, so $n_0 = 1$, $p_1 = 1$ and $p_n = 0$ otherwise. Then (B.9) with $h_0 = L$ gives $[t^N](t \circ (X+L)) = \frac{L^{N-1}}{(N-1)!}\Delta_{1,N-1}1$. Since $\Delta_{1,k}1 = \prod_{j=0}^{k-1}(-1-j) = (-1)^k k!$, this is $(-1)^{N-1}L^{N-1}$, so $t \circ (X+L) = \sum_{N \ge 1}(-1)^{N-1}L^{N-1}t^N = t/(1+tL)$, in agreement with the generator rule (12.16).

B.5 Reversion

Two closed forms are available for the inverse, and they answer different questions. The first is closed in the blocks of the perturbation.

Corollary B.11 (Reversion coefficients, closed in the input blocks). *Let $A = \sum_{m \ge 0} a_m(L) t^m \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$, $F = X + A$, and $F^{\langle -1 \rangle_\circ} = X + \sum_{N \ge 0} b_N(L) t^N$. Then for every $N \in \mathbb{N}_0$,*

$$b_N = \sum_{r=1}^{N+1} \frac{(-1)^r}{r!} \left[\prod_{s=N-r+1}^{N-1} (\partial_L - s)\right] C_{N-r+1,\,r}(a_0, a_1, \ldots), \tag{B.10}$$

where $C_{m,r}$ is the convolution polynomial of Theorem 13.12 formed from the blocks of A, given in closed form by (B.2)*, and where a product whose lower bound exceeds its upper bound is empty. Only $a_0, \dots, a_N$ occur, and the range $r \le N+1$ cannot be shortened.*

Proof. Theorem 18.18 is (B.10) with $A_{r,N-r+1}$ in place of $C_{N-r+1,r}$, and Theorem 13.13 (i) identifies $A_{r,m} = [t^m]A^r = C_{m,r}$. Since $C_{m,r}$ involves only $a_0, \dots, a_m$ and $m = N - r + 1 \le N$ here, only $a_0, \dots, a_N$ occur. Sharpness of the range is Theorem 18.19. □

The special case in which the perturbation is a single level-zero block degenerates spectacularly: exactly one Lagrange summand survives per output block, and the whole inverse is given by one operator applied to one power.

Corollary B.12 (Reversion of a pure level-zero perturbation)**.** *Let $P \in \mathbb{K}[L]$ and $F = X + P(L) \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}$. Then*

$$F^{\langle -1\rangle_\circ} = X + \sum_{N\ge 0} \frac{(-1)^{N+1}}{(N+1)!} \Big[\prod_{j=0}^{N-1} (\partial_L - j)\Big] P^{N+1}\, t^N = X + \sum_{N\ge 0} \frac{(-1)^{N+1}}{(N+1)!} \Delta_{0,N}\big(P^{N+1}\big)\, t^N, \tag{B.11}$$

the product being empty at $N = 0$, where the block is $-P$. For $P(L) = aL$ with $a \in \mathbb{K}$ this is $F^{\langle -1\rangle_\circ} = X + \sum_{N\ge 0} a^{N+1} \mathsf{L}^{\mathrm{Lam}}_N(L) t^N$, which is Theorem 20.5; for $a = 1$ it is the expansion of ω (Theorem 20.10). For a constant $P = c \in \mathbb{K}$ it collapses to $(X + c)^{\langle -1\rangle_\circ} = X - c$.

Proof. Here $a_0 = P$ and $a_m = 0$ for $m \ge 1$, so by (13.8) the convolution polynomials are $C_{0,r} = P^r$ and $C_{m,r} = 0$ for $m \ge 1$. In (B.10) the index $m = N - r + 1$ is therefore forced to 0, that is $r = N + 1$, and the operator becomes $\prod_{s=0}^{N-1}(\partial_L - s) = \Delta_{0,N}$. Substituting gives (B.11).

For $P = aL$, $\Delta_{0,N}\big((aL)^{N+1}\big) = a^{N+1}\Delta_{0,N} L^{N+1}$, and comparison with (21.1) identifies the block as $a^{N+1}\mathsf{L}^{\mathrm{Lam}}_N(L)$. For $P = c$ constant and $N \ge 1$, the factor ∂_L at $s = 0$ annihilates the constant c^{N+1}, so every block beyond $N = 0$ vanishes. □

Remark B.13 (A repair to the level-zero reversion formula). Source S3 records (B.11) in the equivalent but shifted form "$[t^n](F^{\langle -1\rangle_\circ} - X + P) = \cdots$". That form is correct for $n \ge 1$, because P sits at level zero and contributes nothing to higher blocks, but it fails at $n = 0$, where the left-hand side is $-P + P = 0$ while the right-hand side is $-P$. The correction is simply to delete the $+P$; (B.11) is then valid for every $N \ge 0$, and the Lambert normalization $\mathsf{L}^{\mathrm{Lam}}_0(u) = -u$ of Theorem 21.1 becomes the $N = 0$ instance rather than a separately stipulated boundary value.

The second closed form is closed in A and its D_X-derivatives. It is what one wants when A is not given blockwise — for instance when it is a symbolic expression, or when one wants a formula valid for a whole family of perturbations at once.

Theorem B.14 (The Lagrange terms as differential polynomials)**.** *Let $A \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ and $F = X + A$. Then for every $r \ge 1$*

$$\frac{(-1)^r}{r!} D_X^{r-1}(A^r) = (-1)^r \sum_{j=1}^{r} \frac{A^j}{j!} \mathcal{B}^{\exp}_{r-1,\,r-j}\big(D_X A, D_X^2 A, \dots, D_X^j A\big), \tag{B.12}$$

and consequently

$$F^{\langle -1\rangle_\circ} = X + \sum_{r\ge 1} (-1)^r \sum_{j=1}^{r} \frac{A^j}{j!} \mathcal{B}^{\exp}_{r-1,\,r-j}\big(D_X A, \dots, D_X^j A\big), \tag{B.13}$$

the outer family being summable. Writing $A' = D_X A$, $A'' = D_X^2 A$ and so on, the first five Lagrange terms are

$$
\begin{aligned}
F^{\langle -1\rangle_\circ} - X = {} & -A \\
& + AA' \\
& - A(A')^2 - \tfrac{1}{2}A^2A'' \\
& + A(A')^3 + \tfrac{3}{2}A^2A'A'' + \tfrac{1}{6}A^3A''' \\
& - A(A')^4 - 3A^2(A')^2A'' - \tfrac{1}{2}A^3(A'')^2 - \tfrac{2}{3}A^3A'A''' - \tfrac{1}{24}A^4A'''' \\
& + \mathcal{O}_t^{\mathrm{form}}\big(t^5\big),
\end{aligned}
\tag{B.14}
$$

each line being one value of r, from $r = 1$ to $r = 5$.

Proof. Apply Theorem A.16 with $n = r - 1$. For $r = 1$ both sides of (B.12) are $-A$ (the left side is $-D_X^0 A$; the right side has only $j = 1$, with $\mathcal{B}_{0,0}^{\exp} = 1$). For $r \geq 2$, (A.20) gives

$$
D_X^{r-1}\big(A^r\big) = \sum_{k=1}^{r-1} (r)^{\underline{k}}\, A^{r-k}\, \mathcal{B}_{r-1,\,k}^{\exp}\big(D_X A, \ldots, D_X^{r-k} A\big),
$$

the index k stopping at $r-1$ because $\mathcal{B}_{r-1,k}^{\exp} = 0$ for $k > r-1$. Divide by $r!$ and use $(r)^{\underline{k}}/r! = 1/(r-k)!$; putting $j = r-k$, whose range is $1 \leq j \leq r-1$, turns the sum into $\sum_j \frac{A^j}{j!}\mathcal{B}_{r-1,r-j}^{\exp}(D_X A, \ldots, D_X^j A)$. The missing term $j = r$ carries $\mathcal{B}_{r-1,0}^{\exp}$, which vanishes for $r \geq 2$, so it may be added without changing the value; this gives (B.12) in the stated uniform range $1 \leq j \leq r$. The argument list is $x_1, \ldots, x_{(r-1)-(r-j)+1} = x_1, \ldots, x_j$, that is $D_X A, \ldots, D_X^j A$.

Formula (B.13) is now (18.13) of Theorem 18.14 term by term, and the summability of the outer family is the same theorem.

For (B.14), evaluate (B.12) for $r \leq 5$ using Table A.1. At $r = 2$: $j = 1$ gives $A\,\mathcal{B}_{1,1}^{\exp}(A') = AA'$ and $j = 2$ gives $\frac{1}{2}A^2\mathcal{B}_{1,0}^{\exp} = 0$; the sign is $(-1)^2$. At $r = 3$: $j = 1$ gives $A\,\mathcal{B}_{2,2}^{\exp}(A') = A(A')^2$, $j = 2$ gives $\frac{1}{2}A^2\mathcal{B}_{2,1}^{\exp}(A', A'') = \frac{1}{2}A^2A''$, and $j = 3$ gives 0; the sign is -1. At $r = 4$: $\mathcal{B}_{3,3}^{\exp} = (A')^3$, $\mathcal{B}_{3,2}^{\exp} = 3A'A''$, $\mathcal{B}_{3,1}^{\exp} = A'''$, giving $A(A')^3 + \frac{1}{2}A^2 \cdot 3A'A'' + \frac{1}{6}A^3A'''$. At $r = 5$: $\mathcal{B}_{4,4}^{\exp} = (A')^4$, $\mathcal{B}_{4,3}^{\exp} = 6(A')^2A''$, $\mathcal{B}_{4,2}^{\exp} = 4A'A''' + 3(A'')^2$, $\mathcal{B}_{4,1}^{\exp} = A''''$, giving $A(A')^4 + \frac{1}{2}A^2 \cdot 6(A')^2A'' + \frac{1}{6}A^3\big(4A'A''' + 3(A'')^2\big) + \frac{1}{24}A^4A''''$, which is the last line with the sign $(-1)^5$. Finally, Theorem 18.8 gives ν_t at least $r - 1 \geq 5$ for every omitted term, which is the recorded formal tail. □

Remark B.15 (Which reversion formula to run). Theorem B.11 and Theorem B.14 are the same series regrouped. Theorem B.11 groups by output block and is the formula to use when a single b_N is wanted, since it needs only A^r for $r \leq N + 1$ and one application of a differential operator per term. Theorem B.14 groups by Lagrange index and is the formula to use for structural statements, since the r-th line of (B.14) is a universal differential polynomial with rational coefficients, independent of A, of weight r in the grading $\mathrm{wt}(D_X^i A) = i + 1$. Neither is competitive as an *algorithm*: for computing all blocks up to a cutoff, triangular reversal (Theorem 17.2) or Newton reversion (Theorem 17.11) is cheaper, as Theorem 27.42 records.

B.6 What differs from the classical Lagrange inversion formula

The classical formula, in the form standard in enumerative combinatorics [17, 16], reads as follows. Let C be a commutative $\mathbb{Q}$-algebra, let $\psi \in C[[z]]$, and let $u \in \varepsilon C[[\varepsilon]]$ be the unique solution of $u = \varepsilon\psi(u)$. Then for every $\eta \in C[[z]]$ and every $m \geq 1$,

$$
[\varepsilon^m]\,\eta(u) = \frac{1}{m}\,[z^{m-1}]\big(\eta'(z)\psi(z)^m\big). \tag{B.15}
$$

Theorem 18.14 is not this statement, and the differences are worth naming precisely, because each of them is a place where a transplanted classical proof breaks.

1. *The output of one term.* In (B.15) the right-hand side is a single scalar, obtained by a coefficient extraction that evaluates a derivative at the origin. In (18.13) the r-th summand $\frac{(-1)^r}{r!}D_X^{r-1}(A^r)$ is an entire element of $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$, contributing to every block t^N with $N \geq r-1$; the passage to one block is a second step, performed by Theorem 18.18. Accordingly the volume's abstract ingredient Theorem 18.11 states $[\varepsilon^m]\gamma_{[u]} = \frac{1}{m!}\partial_z^{m-1}(\gamma'\psi^m)$ as an identity of *series* in z, and specializes z only afterwards. Evaluating at $z = 0$ recovers (B.15), because $\frac{1}{m!}\big(\partial_z^{m-1}(\gamma'\psi^m)\big)\big|_{z=0} = \frac{1}{m}[z^{m-1}](\gamma'\psi^m)$.

2. *The derivation does not annihilate the coefficients.* In (B.15) the operator ∂_z kills C. Here $D_X L = t \neq 0$ (Theorem 11.1), so differentiating a Lagrange term differentiates its coefficient polynomials; this is what produces the operator $\Delta_{n,k}$ of Theorem 11.7 in (B.10) and, ultimately, the Lambert polynomials themselves. Replacing D_X^{r-1} by "extract the coefficient of t^{r-1}" is therefore wrong by an operator, not by a constant.

3. *What makes the family summable.* Classically the grading is by the single variable ε, and the solution is small because $u \in \varepsilon C[[\varepsilon]]$. Here A need not be small at all: $\nu_t(A) = 0$ is the normal case, since a near-identity map may have a full level-zero block $a_0(L)$. Summability comes instead from the pairing of one derivative against one extra factor of A, that is from the axiom $\nu(\partial f) \geq \nu(f) + 1$ of Theorem 18.1; the resulting admissibility threshold is $\nu(A) > -1$, not $\nu(A) > 0$ (Theorem 18.2, Theorem 18.8).

4. *No formal residue is available.* The usual proof of (B.15) runs through the formal residue on $C((z))$ and the identity $\mathrm{Res}_z\,\partial_z(\cdot) = 0$. Theorem B.16 shows that no such functional exists on $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$. The volume's proof of Theorem 18.14 therefore goes through a weighted deformation algebra and an evaluation homomorphism (Theorem 18.13) instead of through residues.

5. *The contributing range is finite and known.* Classically every $m \geq 1$ contributes to its own coefficient. Here only $r \leq N + 1$ contributes to the block N, and that bound is attained (Theorem 18.19); this is what makes (B.10) a finite formula rather than a limit.

6. *No analytic content.* Theorem 18.14 is an identity of formal objects. The transfer to a genuine inverse function of a real or complex variable is a separate theorem with separate hypotheses (Theorem 16.21), exactly as the catalogue's distinction between $\mathcal{O}_t^{\mathrm{form}}(t^N)$ and $\mathcal{O}_{X\to+\infty}(\cdot)$ requires.

Proposition B.16 (No formal residue on the polynomial–logarithmic Laurent ring). *There is no nonzero $\mathbb{K}$-linear functional $\rho\colon \mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}} \to \mathbb{K}$ with $\rho\big(D_X F\big) = 0$ for every $F \in \mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$. The same holds for any $\mathbb{K}$-linear map from $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ to any $\mathbb{K}$-vector space.*

Proof. By Theorem 11.18 the map $D_X : \mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}} \to \mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ is surjective. If $\rho \circ D_X = 0$ then ρ vanishes on $D_X(\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}) = \mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$, so $\rho = 0$. The same argument applies verbatim to a linear map with any target. ☐

Remark B.17 (The residue that does exist is a residue in L). Theorem B.16 does not contradict Theorem 10.22. That definition attaches a residue to a *coefficient* $h \in \mathbb{K}((L^{-1}))$, namely the coefficient of L^{-1}, and Theorem 10.23 shows that it annihilates ∂_L-images. It is a residue in the inner variable, defined only after the coefficient ring has been enlarged to $\mathbb{K}((L^{-1}))$ — where ∂_L is *not* surjective (Theorem 10.24) — and it does not implement extraction of an outer block. The obstruction that a classical residue proof would need lives in the outer variable t, and there is none, because t is a unit of $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ (Theorem 7.11 (i)) and D_X is onto.

B.7 Which formula to use

Table B.1: Closed forms and their recursive counterparts. "Closed" means a finite sum over an explicitly bounded index set in the input blocks alone; "recursive" means a triangular recurrence in the output blocks.

Task	Closed form	Recursive alternative, and when to prefer it
Block of a product	(B.1) (Theorem B.2)	none needed; the Cauchy product (8.5) is already finite. Use Theorem 7.4 to truncate inputs safely.
Block of an integer power	(8.27), or (B.2) in Bell form	the triangular recurrence (8.28); preferable when all powers $r \leq R$ are wanted, since it costs $\mathcal{O}(N^2R)$ block products.
Block of $(1+W)^\sigma$	(B.3) (Theorem B.4)	the power recurrence of Theorem 27.23; preferable for a full table.
Block of a reciprocal	(B.4), or the determinant (B.6)	(9.7) (Theorem 9.7) for a table; Newton doubling Theorem 9.27 when N is large. Use the determinant for structural claims (Theorem B.7).
Block of a quotient	B^{-1} then (B.1)	(9.8) (Theorem 9.8); almost always preferable.
Block of a composition	(13.6), or (B.8) in Bell form	the block recurrence Theorem 12.15; preferable when one inner map is composed with many outer series.
Composition with a level-zero shift	(B.9) (Theorem B.9)	none needed; the closed form is already a single sum.
Block of a reversion	(B.10) (Theorem B.11)	triangular reversal Theorem 17.2 or Newton Theorem 17.11; both cheaper for a full table (Theorem 27.42).
Reversion of a level-zero perturbation $X + P(L)$	(B.11) (Theorem B.12)	none needed; one operator applied to one power gives every block. Contains the Lambert family as the case $P = aL$.
Reversion as a differential polynomial	(B.13), (B.14) (Theorem B.14)	no recursive counterpart; this is the form for symbolic A and for weight-graded structural arguments.

Appendix C

Operational formula sheet

This chapter is a navigation aid, not an exposition. Every line states a result of this volume and cites where that result is proved; nothing is asserted here that is not proved there, and no line carries a hypothesis that its citation does not carry. Read a line together with its reference before using it: the abbreviations below suppress hypotheses that are essential — which coefficient class an element lives in, whether a leading block is a unit, whether an inner argument is admissible, whether a claim is formal or analytic.

Standing conventions for the whole sheet. $\mathbb{K}$ is a field of characteristic zero; X is the large variable and the default regime is $X \to +\infty$ along the positive real ray; $t = X^{-1}$ and $L = \log X$; F, G, H denote series, p_n, q_n, a_n, b_n, h_n their blocks in $\mathbb{K}[L]$, and N, n, m, r, k integers. Truncation is *exclusive* (Theorem 7.1). Three symbols that look alike are kept apart throughout: $\mathcal{O}_t^{\mathrm{form}}(t^N)$ is a formal valuation tail with no analytic content, $\mathcal{O}_{X\to+\infty}(A(X))$ is an analytic estimate on the printed regime, and $\mathcal{O}(N^2)$ is an operation count.

C.1 The four ambient classes

Table C.1: Generators, classes, and their algebraic type.

Statement	Proved at
$t := X^{-1}$, $L := \log X$; independent generators in the formal algebra	Theorem 5.1
$\mathscr{PL}_{\mathrm{pow},\mathbb{K}} = \mathbb{K}[L][[t]]$, $\mathscr{PL}_{\mathrm{poly},\mathbb{K}} = \mathbb{K}[L]((t))$, $\mathscr{PL}_{\mathrm{rat},\mathbb{K}} = \mathbb{K}(L)((t))$, $\mathscr{PL}_{\mathrm{itLaur},\mathbb{K}} = \mathbb{K}((L^{-1}))((t))$	Theorem 5.3
$\mathscr{PL}_{\mathrm{pow},\mathbb{K}} \subsetneq \mathscr{PL}_{\mathrm{poly},\mathbb{K}} \subsetneq \mathscr{PL}_{\mathrm{rat},\mathbb{K}} \subsetneq \mathscr{PL}_{\mathrm{itLaur},\mathbb{K}}$; the first two are domains and not fields, the last two are fields	Theorem 5.5, Theorem 5.6, Theorem 9.19
$\mathscr{PL}_{\mathrm{itLaur},\mathbb{K}}$ is the Hahn field of the lexicographic rank-two value group	Theorem 5.19, Theorem 5.20
The coefficient extensions $\mathbb{K}[L] \rightsquigarrow \mathbb{K}(L) \rightsquigarrow \mathbb{K}((L^{-1}))$ are not localizations of $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$	Theorem 5.8
Fractional outer exponents require the Puiseux-log tower; a general real exponent requires a Hahn exponent group	Theorem 5.24, Theorem 5.25
$\mathcal{G}_{\mathrm{poly},\mathbb{K}} = \{X + A : A \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}\}$ is a group under $\circ$	Theorem 14.1, Theorem 14.7
Every class is t-adically complete and separated	Theorem 7.18

C.2 Valuation, leading data, and the dominance order

Table C.2: Outer valuation, rank-two valuation, leading data, dominance, log-degree profile.

Statement	Proved at
$\nu_t(F) = \min\{n : p_n \neq 0\}$, $\nu_t(0) = +\infty$	Theorem 6.1
$\nu_t(FG) = \nu_t F + \nu_t G$; $\nu_t(F+G) \geq \min\{\nu_t F, \nu_t G\}$, strict exactly when the leading blocks cancel	Theorem 6.2
Leading block $p_{\nu_t F}$, leading monomial $\mathrm{lm}(F) = t^{\nu_t F} L^{\deg_{\mathrm{L}} p_{\nu_t F}}$, leading scalar $\mathrm{lc}(F)$ — three different objects	Theorem 6.4, Theorem 6.5
$\mathrm{lm}(FG) = \mathrm{lm}(F)\,\mathrm{lm}(G)$ and $\mathrm{lc}(FG) = \mathrm{lc}(F)\,\mathrm{lc}(G)$	Theorem 6.7, Theorem 8.21
The block representation of a nonzero element is unique	Theorem 6.9
$t^n L^j \succ t^m L^k$ iff $n < m$, or $n = m$ and $j > k$	Theorem 5.12
Dominance is genuine growth as $X \to +\infty$	Theorem 5.13
$d_F(n) = \deg_{\mathrm{L}} p_n$, with $-\infty$ at absent blocks	Theorem 6.11
$d_{F+G} \leq \max$, $d_{FG} \leq$ the $(\max, +)$-convolution, with an exact attainment criterion	Theorem 6.12, Theorem 8.25

C.3 Truncation, the formal tail, and summability

Table C.3: Exclusive truncation, the formal big-O token, summability.

Statement	Proved at
$\mathrm{Trunc}_{<N} F = \sum_{n<N} p_n(L) t^n$; every retained block is retained whole	Theorem 7.1
Inclusive-to-exclusive dictionary $[F]_{\leq N} = \mathrm{Trunc}_{<N+1} F$	Theorem 7.2, (7.2)
$\mathrm{Trunc}_{<N}(FG) = \mathrm{Trunc}_{<N}\big(\mathrm{Trunc}_{<N-\nu_t G} F \cdot \mathrm{Trunc}_{<N-\nu_t F} G\big)$	Theorem 7.4
$F = G + \mathcal{O}_t^{\mathrm{form}}\big(t^N\big)$ means $\nu_t(F-G) \geq N$, equivalently $\mathrm{Trunc}_{<N} F = \mathrm{Trunc}_{<N} G$	Theorem 7.7, Theorem 7.8
$t^N \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ is a filtration step of $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$, never an ideal, because t is a unit there	Theorem 7.11
$[t^n]F = p_n(L) \in \mathbb{K}[L]$ and $[t^n L^j]F = a_{n,j} \in \mathbb{K}$; extraction is linear and filtration-continuous	Theorem 7.12, Theorem 7.13
A family is summable iff $\nu_t \to +\infty$; sums may then be regrouped and multiplied termwise	Theorem 7.21, Theorem 7.22, Theorem 7.24, Theorem 7.26
A bound on $\deg_{\mathrm{L}}$ is a second, independent approximation, not part of the t-adic topology	Theorem B.1

C.4 The derivation, its powers, and antiderivatives

Table C.4: The distinguished derivation and its calculus.

Statement	Proved at
$D_X = t\,\partial_L - t^2\partial_t$; the unique $\mathbb{K}$-derivation with $D_X t = -t^2$, $D_X L = t$ and $D_X(\mathscr{PL}_{\text{pow},\mathbb{K}}) \subseteq \mathscr{PL}_{\text{pow},\mathbb{K}}$	Theorem 11.1
Block formula $D_X\big(t^n p(L)\big) = t^{n+1}\big(\partial_L - n\big)p(L)$	Theorem 11.3
Euler form $\vartheta = t^{-1}D_X$ preserves each block and acts as $\partial_L - n$ on $t^n\mathbb{K}[L]$	Theorem 11.5
$\Delta_{n,k} = \prod_{j=0}^{k-1}\big(\partial_L - n - j\big)$, $\Delta_{n,0} = 1$; the second index is never optional	Theorem 11.7
$D_X^k\big(t^n p(L)\big) = t^{n+k}\,\Delta_{n,k}p(L)$	Theorem 11.8
$\Delta_{0,k} = \sum_j s(k,j)\,\partial_L^j$, and $D_X^k t^n = (-1)^k\,(n)^{\overline{k}}\,t^{n+k}$	Theorem 11.9
$\nu_t(D_X F) \geq \nu_t F + 1$, with the sharp value and the three cases	Theorem 11.11
$\ker D_X \cap \mathscr{PL}_{\text{poly},\mathbb{K}} = \mathbb{K}$	Theorem 11.12
D_X is onto $\mathscr{PL}_{\text{poly},\mathbb{K}}$; antiderivatives exist and are unique up to a constant of $\mathbb{K}$	Theorem 11.18
Resonance: $(\partial_L - n)p = q$ is triangular for $n \neq 0$; the block $n = 0$ integrates and adds a constant	Theorem 11.15, Theorem 11.16
In $\mathscr{PL}_{\text{itLaur},\mathbb{K}}$ and $\mathscr{PL}_{\text{rat},\mathbb{K}}$ antidifferentiation is obstructed by a residue at $L = \infty$; this is where $\log L$ is forced	Theorem 11.21, Theorem 11.24, Theorem 11.22

C.5 Product, reciprocal, and division

Table C.5: Multiplicative calculus.

Statement	Proved at
$[t^N](FG) = \sum_{n+m=N} p_n q_m$, a finite sum for each N; the four classes are closed under it	Theorem 8.5, Theorem 8.6, Theorem 8.7
$[t^N L^d](FG) = \sum_{n+m=N}\sum_{e+f=d} a_{n,e}b_{m,f}$	Theorem B.2
F is a unit of $\mathscr{PL}_{\text{poly},\mathbb{K}}$ iff its leading block is a nonzero *scalar*, iff $F = c\,t^{n_0}(1+U)$ with $c \in \mathbb{K}^\times, U \in t\mathscr{PL}_{\text{pow},\mathbb{K}}$	Theorem 9.3, Theorem 9.4
$B^{-1} = \sum_n C_n t^n$ with $C_0 = c^{-1}$ and $C_n = -c^{-1}\sum_{j=1}^n B_j C_{n-j}$, where $B_0 = c \in \mathbb{K}^\times$	Theorem 9.7
$Q_n = B_0^{-1}\big(A_n - \sum_{j=1}^n B_j Q_{n-j}\big)$, valid over $\mathbb{K}[L]$ only when $B_0 \in \mathbb{K}^\times$ — a nonzero *scalar*, not merely a nonzero polynomial	Theorem 9.8, Theorem 9.3
Newton reciprocal step $C \mapsto C(2 - BC)$ squares the residual $1 - BC$ and may be truncated at the doubled order	Theorem 9.27, Theorem 9.29
Truncated division: relative precision is exactly preserved	Theorem 9.9, Theorem 9.11
$(1+U)^\sigma = \sum_{r\geq 0}\binom{\sigma}{r}U^r$ for $\nu_t U \geq 1$, with finite determination at each block	Theorem 8.37

Statement	Proved at
Closed forms: $[t^r](1+W)^\sigma = \frac{1}{r!}\sum_{k=1}^{r} (\sigma)^{\underline{k}}\, \mathcal{B}^{\exp}_{r,k}(1!w_1, 2!w_2, \ldots)$; $\sigma = -1$ gives the reciprocal	Theorem B.4, Theorem B.5
$c^{r+1}[t^r]B^{-1}$ is a Toeplitz–Hessenberg determinant in $B_0, \ldots, B_r$	Theorem B.6
$\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ is a domain but not a field; among the rings $E((t))$ with $\mathbb{K}[L] \subseteq E \subseteq \mathbb{K}(L)$, exactly one is a field, namely $\mathscr{PL}_{\mathrm{rat},\mathbb{K}}$	Theorem 8.22, Theorem 9.20, Theorem 9.23

C.6 Composition and the exact Taylor formula

Table C.6: Substitution, the Taylor formula, and the coefficient formula.

Statement	Proved at
Admissible inner argument $G = cX^\rho(1+U)$, $c \in \mathbb{K}^\times$, $\rho > 0$, $U \in t\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$	Theorem 12.6
Generator images $\Phi_G(t) = c^{-1}t^\rho(1+U)^{-1}$, $\Phi_G(L) = \rho L + \log c + \log(1+U)$; Φ_G is a $\mathbb{K}$-algebra homomorphism compatible with summable families	Theorem 12.7
Near identity: $t \circ (X+H) = t/(1+tH)$, $L \circ (X+H) = L + \log(1+tH)$. Substituting $t \mapsto t+H$, or updating L alone, is wrong	Theorem 12.22
$F \circ (X+H) = \sum_{k\geq 0} \frac{H^k}{k!} D_X^k F$, an exact identity, not an approximation	Theorem 13.6
$[t^r]\big(F\circ(X+H)\big) = \sum_{n+k+m=r} \frac{h_{k,m}}{k!} \Delta_{n,k} p_n$, with $h_{k,m} = [t^m]H^k = C_{m,k}$	Theorem 13.10, Theorem 13.12
$C_{0,0} = 1$, $C_{m,0} = 0$ for $m > 0$, $C_{m,r+1} = \sum_{j=0}^{m} h_j C_{m-j,r}$; closed form in (B.2)	Theorem 13.13, Theorem B.3
Bell form when $h_0 = 0$, and the single-sum formula for a level-zero shift $H = h_0(L)$	Theorem B.8, Theorem B.9
Substituting into a scalar power series: $\varphi(W)$ for $W \in t\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$, with Bell coefficients	Theorem 10.5, Theorem 13.22
$D_X^n\big(\varphi(F)\big) = \sum_{k=1}^{n} \varphi^{(k)}(F)\, \mathcal{B}^{\exp}_{n,k}\big(D_X F, \ldots, D_X^{n-k+1}F\big)$	Theorem A.15
$D_X^n(F \circ G) = \sum_{k=1}^{n} \big(D_X^k F\big) \circ G\ \ \mathcal{B}^{\exp}_{n,k}\big(\gamma, \ldots, D_X^{n-k}\gamma\big)$, $\gamma = D_X G$	Theorem 13.25, Theorem A.12
Chain rule $D_X(F \circ G) = \big((D_X F) \circ G\big) D_X G$; associativity holds inside the admissible class	Theorem 12.31, Theorem 12.27
Finite determination and the triangular precision budget	Theorem 12.10, Theorem 12.11

Statement	Proved at
Analytic transfer under composition is a separate theorem with its own hypotheses	Theorem 12.35

C.7 Reversion

Table C.7: The compositional inverse: fixed point, algorithms, closed formulas.

Statement	Proved at
For $F = X + A \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}$ and $G = X + B$: $F \circ G = X$ iff $B = -A \circ (X + B)$	Theorem 16.7
The fixed point exists, is unique, and is a two-sided inverse; the Picard iterates gain one block per step	Theorem 16.11, Theorem 16.9, Theorem 14.7
$\mathrm{Err}_{\mathrm{rev}}(F, H) = F \circ H - X$; its order equals $\nu_t(H - F^{\langle -1\rangle\circ})$ exactly, on both sides, with the same leading block	Theorem 16.14, Theorem 16.15, Theorem 16.16
Triangular reversal: $r_N = [t^N]\,\mathrm{Err}_{\mathrm{rev}}(F, G_{N-1})$, $G_N = G_{N-1} - r_N t^N$, and $G_N = \mathrm{Trunc}_{<N+1}\, F^{\langle -1\rangle\circ}$	Theorem 17.2, Theorem 16.17
At leading slope $a \in \mathbb{K}^\times$ the correction is $-r_N/a$, and the whole problem conjugates to the near-identity one	Theorem 17.4
Truncating the inputs of a reversal step is safe	Theorem 17.7, Theorem 27.8
Newton map $\mathcal{N}_F(G) = G - \dfrac{\mathrm{Err}_{\mathrm{rev}}(F, G)}{(D_X F) \circ G}$; the denominator is a unit	Theorem 17.10, Theorem 17.9
Exact arithmetic: $\nu_t\,\mathrm{Err}_{\mathrm{rev}}(F, \mathcal{N}_F(G)) \ge 2m + 2 + \alpha$ with $m = \nu_t\,\mathrm{Err}_{\mathrm{rev}}(F, G)$, $\alpha = \nu_t A$	Theorem 17.11
Guard requirement. With an inexact correction accurate only to order σ, the new residual order is $\min\{2m + 2 + \alpha,\ \sigma\}$; doubling is lost unless $\sigma \ge 2m + 2 + \alpha$	Theorem 17.14, Theorem 17.15, Theorem 17.16
$(X + A)^{\langle -1\rangle\circ} = X + \sum_{r\ge 1} \dfrac{(-1)^r}{r!} D_X^{r-1}(A^r)$, the r-th summand of order $\ge r - 1$	Theorem 18.14
$b_N = \sum_{r=1}^{N+1} \dfrac{(-1)^r}{r!} \Big[\prod_{s=N-r+1}^{N-1} (\partial_L - s) \Big] A_{r,N-r+1}$, the empty product occurring exactly at $r = 1$	Theorem 18.18, Theorem B.11
The range $r \le N + 1$ is exactly right and is attained	Theorem 18.19
Level-zero perturbation: $(X + P)^{\langle -1\rangle\circ} = X + \sum_{N\ge 0} \dfrac{(-1)^{N+1}}{(N+1)!} \Delta_{0,N}(P^{N+1}) t^N$ for $P \in \mathbb{K}[L]$	Theorem B.12, Theorem B.13
Transport of an observable: $\varphi \circ F^{\langle -1\rangle\circ} = \varphi + \sum_{r\ge 1} \frac{(-1)^r}{r!} \partial^{r-1}\big((\partial\varphi) A^r\big)$	Theorem 18.15, Theorem 18.16
Universal differential-polynomial form, weight-graded by $\mathrm{wt}(D_X^i A) = i + 1$	Theorem B.14
Beyond the identity scale: reduce by the leading monomial; a leading logarithm forces a nested logarithm	Theorem 19.6, Theorem 19.3, Theorem 19.4

Statement	Proved at
Analytic residual transfer needs injectivity and a derivative lower bound	Theorem 16.21

C.8 Wright omega, the Lambert polynomials, and Lambert W

Table C.8: The canonical logarithmic reversion and its coefficient family.

Statement	Proved at
$y \mapsto y + \log y$ is a bijection $(0, \infty) \to \mathbb{R}$; its inverse is ω, with $\omega(x) = W_0(\mathrm{e}^x)$	Theorem 20.1
$\omega = X + \sum_{n\geq 0} \mathsf{L}_n^{\mathrm{Lam}}(L)t^n = X - L + \sum_{n\geq 1} \mathsf{L}_n^{\mathrm{Lam}}(L)t^n$ in $\mathcal{G}_{\mathrm{poly},\mathbb{K}}$; each Lagrange–Bürmann summand is a single block	Theorem 20.10
$\mathsf{L}_n^{\mathrm{Lam}}(u) = \frac{(-1)^{n+1}}{(n+1)!}\Big[\prod_{j=0}^{n-1}(\partial_u - j)\Big]u^{n+1}$; $\mathsf{L}_0^{\mathrm{Lam}}(u) = -u$, $\mathsf{L}_1^{\mathrm{Lam}}(u) = u$, $\mathsf{L}_2^{\mathrm{Lam}}(u) = \frac{u^2}{2} - u$	Theorem 21.1
$\mathsf{L}_n^{\mathrm{Lam}}(0) = 0$; $\mathsf{L}_{n+1}^{\mathrm{Lam}} = -\mathsf{L}_n^{\mathrm{Lam}} + n\int_0^u \mathsf{L}_n^{\mathrm{Lam}}$; equivalently $\mathsf{L}_{n+1}^{\mathrm{Lam}\prime} = n\mathsf{L}_n^{\mathrm{Lam}} - \mathsf{L}_n^{\mathrm{Lam}\prime}$	Theorem 21.2
$\mathsf{L}_n^{\mathrm{Lam}}(u) = \sum_{m=1}^{n}(-1)^{n-m}\left[{n \atop n-m+1}\right]\frac{u^m}{m!}$ $(n \geq 1)$	Theorem 21.4
$\deg \mathsf{L}_n^{\mathrm{Lam}} = n$ for $n \geq 1$ only; $[u^n]\mathsf{L}_n^{\mathrm{Lam}} = 1/n$; $\lambda_{n,1} = (-1)^{n-1}$; $\mathsf{L}_n^{\mathrm{Lam,sc}} = n!\mathsf{L}_n^{\mathrm{Lam}} \in \mathbb{Z}[u]$	Theorem 21.7, Theorem 21.9
$\left[{n \atop n-m+1}\right] = \mathcal{B}^{\exp}_{n,\,n-m+1}(0!, 1!, 2!, \ldots)$: the Lambert triangle is a slice of the Bell array	Theorem A.9, Theorem A.10
$(X + aL)^{\langle -1\rangle_\circ} = X + \sum_{n\geq 0} a^{n+1}\mathsf{L}_n^{\mathrm{Lam}}(L)t^n$; the case $a = -1$ gives $(-1)^{n+1}\mathsf{L}_n^{\mathrm{Lam}}$, *not* $(-1)^n\mathsf{L}_n^{\mathrm{Lam}}$	Theorem 20.5, Theorem 20.6
Exact residual of the truncation: $\omega_N + \log \omega_N - X = -\mathsf{L}_{N+1}^{\mathrm{Lam}}(L)t^{N+1} + \mathcal{O}_t^{\mathrm{form}}\big(t^{N+2}\big)$	Theorem 20.13, Theorem 20.14
Analytic form, $x \to +\infty$ on the positive ray, $K = (N+2)^2$: $\omega(x) - \omega_N(x) = \mathsf{L}_{N+1}^{\mathrm{Lam}}(L)x^{-(N+1)} + \mathcal{O}_{x\to+\infty}\big(L^K x^{-(N+2)}\big)$	Theorem 20.17
Branch coordinates: $W_k(z) + \log W_k(z) = \xi_k$ with $\xi_k = \mathrm{Log}\, z + 2\pi\mathrm{i}k$; the formal expansion of W_k is Theorem 20.10 under $X \mapsto \xi_k$, $L \mapsto \ell_k = \log_{\mathcal{B}} \xi_k$. The polynomials are branch-independent; the coordinates, sector and analytic estimates are not, and the complex analysis is not developed	Theorem 20.7
Numerical tables of $\mathsf{L}_n^{\mathrm{Lam}}$, of $\left[{n \atop n-m+1}\right]$, and of $\mathsf{L}_n^{\mathrm{Lam}}(1)$	Part D, Table D.1, Table D.2, Table D.3

C.9 Bell polynomials and closed coefficient extraction

Table C.9: The combinatorial layer of this appendix.

Statement	Proved at
$\mathcal{B}^{\exp}_{n,k}$ by multiplicity sum; $\mathcal{B}^{\exp}_{0,0} = 1$, $\mathcal{B}^{\exp}_{n,0} = 0$ for $n \geq 1$	Theorem A.1
$\dfrac{\Xi^k}{k!} = \displaystyle\sum_{n \geq k} \mathcal{B}^{\exp}_{n,k} \frac{z^n}{n!}$ for $\Xi = \sum_{j \geq 1} \xi_j z^j / j!$; and the complete version with $\exp \Xi$	Theorem A.3
Degree k, weight n, bihomogeneity, boundary rows	Theorem A.2
$\mathcal{B}^{\mathrm{ord}}_{n,k}(\xi) = \frac{k!}{n!} \mathcal{B}^{\exp}_{n,k}(1!\xi_1, 2!\xi_2, \ldots)$; this is the factor that converts $C_{n,k}$ to the exponential normalization	Theorem A.4
Tabulating recurrences $\mathcal{B}^{\exp}_{n,k} = \sum_j \binom{n-1}{j-1} x_j \mathcal{B}^{\exp}_{n-j,k-1}$ and its $\binom{n}{j}$ variant	Theorem A.5
Differentiating recurrence $\mathcal{B}^{\exp}_{n+1,k} = \mathcal{D}\mathcal{B}^{\exp}_{n,k} + x_1 \mathcal{B}^{\exp}_{n,k-1}$	Theorem 13.24
$\mathcal{B}^{\exp}_{n,k}(0!, 1!, \ldots) = \left[{n \atop k}\right]$; $\mathcal{B}^{\exp}_{n,k}(1, 1, \ldots) = \left\{{n \atop k}\right\}$	Theorem A.9
Abstract Faà di Bruno for a Leibniz tower $\partial u_k = u_{k+1}\gamma$	Theorem A.12
$D^n_X(A^r) = \displaystyle\sum_k (r)^{\underline{k}} A^{r-k} \mathcal{B}^{\exp}_{n,k}(D_X A, \ldots)$	Theorem A.16
No nonzero linear functional on $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ annihilates all D_X-images: there is no formal residue in t	Theorem B.16
Six named differences from the classical Lagrange inversion formula	Section B.6
Which closed form to use for which task	Table B.1

C.10 What this sheet does not assert

Four things the sheet is silent about

1. **Analytic validity.** Every entry above whose citation is a formal statement is an identity of formal series. It becomes an assertion about a function only through an explicit transfer theorem (Theorem 8.45, Theorem 12.35, Theorem 16.21, Theorem 20.17), each with its own hypotheses and its own printed regime. A bare $\mathcal{O}^{\mathrm{form}}$ never implies a bare $\mathcal{O}.(\cdot)$ (Theorem 7.7).

2. **Uniformity in L.** A block-level identity says nothing about the size of the coefficient polynomial at a given x. The analytic entries above therefore carry an explicit power of L in the error term — for instance $K = (N+2)^2$ in Theorem 20.17 — and none of them is uniform in N.

3. **Degree truncation.** Discarding high powers of L inside a retained block is not part of the t-adic calculus and none of the precision statements above covers it (Theorem B.1, Theorem 7.1).

4. **Closure.** Membership in a class is a hypothesis, never a conclusion of a formula. Exponentials, general powers, logarithms of non-units and reversions at a non-identity leading slope leave $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ under conditions recorded in Theorem 10.16, Theorem 10.26, Theorem 10.31 and Theorem 19.3, and only there.

Appendix D

Tables of the Lambert polynomials

All entries below were produced from the operator definition of Theorem 21.1 in exact rational arithmetic and then verified three independent ways: against the Stirling closed form of Theorem 21.4, against the differential recurrence of Theorem 21.2, and — the only check that does not presuppose any of the derived identities — by substituting the resulting series into the defining equation $\omega + \log\omega = X$ and confirming that every coefficient block through outer order t^8 vanishes identically.

D.1 The scaled integer family

By Theorem 21.9, $n!\mathsf{L}_n^{\mathrm{Lam}}(u) \in \mathbb{Z}[u]$ for $n \geq 1$; the table lists that integer polynomial together with the factorial that divides it, which is the storage normalization $\mathsf{L}_n^{\mathrm{Lam,sc}} = n!\mathsf{L}_n^{\mathrm{Lam}}$. Read the row as $\mathsf{L}_n^{\mathrm{Lam}}(u) = \big(\text{entry}\big)/n!$. The sources tabulate these polynomials only through $n = 8$; rows 9–12 are new here.

Table D.1: The Lambert polynomials $\mathsf{L}_n^{\mathrm{Lam}}(u) = \mathsf{L}_n^{\mathrm{Lam,sc}}(u)/n!$.

n	$n!$	$\mathsf{L}_n^{\mathrm{Lam,sc}}(u) = n!\,\mathsf{L}_n^{\mathrm{Lam}}(u)$
0	1	$-u$
1	1	u
2	2	$u^2 - 2u$
3	6	$2u^3 - 9u^2 + 6u$
4	24	$6u^4 - 44u^3 + 72u^2 - 24u$
5	120	$24u^5 - 250u^4 + 700u^3 - 600u^2 + 120u$
6	720	$120u^6 - 1{,}644u^5 + 6{,}750u^4 - 10{,}200u^3 + 5{,}400u^2 - 720u$
7	5,040	$720u^7 - 12{,}348u^6 + 68{,}208u^5 - 154{,}350u^4 + 147{,}000u^3 - 52{,}920u^2 + 5{,}040u$
8	40,320	$5{,}040u^8 - 104{,}544u^7 + 735{,}392u^6 - 2{,}274{,}384u^5 + 3{,}292{,}800u^4 - 2{,}163{,}840u^3 + 564{,}480u^2 - 40{,}320u$
9	362,880	$40{,}320u^9 - 986{,}256u^8 + 8{,}504{,}928u^7 - 33{,}911{,}136u^6 + 67{,}885{,}776u^5 - 68{,}584{,}320u^4 + 33{,}022{,}080u^3 - 6{,}531{,}840u^2 + 362{,}880u$
10	3,628,800	$362{,}880u^{10} - 10{,}265{,}760u^9 + 105{,}543{,}000u^8 - 521{,}049{,}600u^7 + 1{,}357{,}398{,}000u^6 - 1{,}913{,}375{,}520u^5 + 1{,}428{,}840{,}000u^4 - 526{,}176{,}000u^3 + 81{,}648{,}000u^2 - 3{,}628{,}800u$
11	39,916,800	$3{,}628{,}800u^{11} - 116{,}915{,}040u^{10} + 1{,}402{,}893{,}360u^9 - 8{,}325{,}405{,}000u^8 + 27{,}062{,}085{,}600u^7 - 50{,}009{,}929{,}200u^6 + 52{,}481{,}610{,}720u^5 - 30{,}187{,}080{,}000u^4 + 8{,}781{,}696{,}000u^3 - 1{,}097{,}712{,}000u^2 + 39{,}916{,}800u$

n	$n!$	$\mathsf{L}_n^{\mathrm{Lam,sc}}(u) = n!\,\mathsf{L}_n^{\mathrm{Lam}}(u)$
12	479,001,600	$39{,}916{,}800u^{12} - 1{,}446{,}526{,}080u^{11} + 19{,}921{,}172{,}832u^{10} - 138{,}940{,}660{,}320u^9 + 546{,}429{,}272{,}400u^8 - 1{,}267{,}789{,}406{,}400u^7 + 1{,}754{,}714{,}586{,}240u^6 - 1{,}426{,}718{,}240{,}640u^5 + 652{,}040{,}928{,}000u^4 - 153{,}679{,}680{,}000u^3 + 15{,}807{,}052{,}800u^2 - 479{,}001{,}600u$

Remark D.1 (The $n = 0$ row is not an exception). $\mathsf{L}_0^{\mathrm{Lam}}(u) = -u$ has degree 1, not 0. The degree law $\deg \mathsf{L}_n^{\mathrm{Lam}} = n$ of Theorem 21.7 is therefore stated only for $n \geq 1$, and the integrality normalization $\mathsf{L}_n^{\mathrm{Lam,sc}} = n!\mathsf{L}_n^{\mathrm{Lam}}$ is likewise a statement about $n \geq 1$; the table's $n = 0$ row records the value, not an instance of either law.

D.2 The coefficient triangle

Write $\mathsf{L}_n^{\mathrm{Lam}}(u) = \sum_{m=1}^{n} \lambda_{n,m} u^m$ for $n \geq 1$. By Theorem 21.4,

$$\lambda_{n,m} = \frac{(-1)^{n-m}}{m!} \begin{bmatrix} n \\ n-m+1 \end{bmatrix},$$

so the signs alternate along each row and the magnitudes are unsigned Stirling cycle numbers divided by a factorial. The next table lists the integers $m!\,(-1)^{n-m}\lambda_{n,m} = \left[{n \atop n-m+1}\right]$, which is the row of the Stirling triangle read from the right.

Table D.2: $\left[{n \atop n-m+1}\right] = m!\ |\lambda_{n,m}|$, the unsigned coefficient data of $\mathsf{L}_n^{\mathrm{Lam}}$.

$n\backslash m$	1	2	3	4	5	6	7	8	9
1	1								
2	1	1							
3	1	3	2						
4	1	6	11	6					
5	1	10	35	50	24				
6	1	15	85	225	274	120			
7	1	21	175	735	1,624	1,764	720		
8	1	28	322	1,960	6,769	13,132	13,068	5,040	
9	1	36	546	4,536	22,449	67,284	118,124	109,584	40,320

The first column is $\left[{n \atop n}\right] = 1$, which is Theorem 21.7(iv) in the form $\lambda_{n,1} = (-1)^{n-1}$. The last entry of row n is $\left[{n \atop 1}\right] = (n-1)!$, giving the leading coefficient $\lambda_{n,n} = 1/n$. The penultimate entry is $\left[{n \atop 2}\right] = (n-1)!H_{n-1}$, giving $\lambda_{n,n-1} = -H_{n-1}$.

D.3 Numerical check values

The following exact values are convenient unit tests for an implementation of the recurrence; each is $\mathsf{L}_n^{\mathrm{Lam}}(1)$, which by Theorem 21.4 equals $\sum_{m=1}^{n}(-1)^{n-m}\left[{n \atop n-m+1}\right]/m!$.

D.4 An end-to-end numerical validation

The tables above and the identities of Part 21 are algebraic. A coefficient table can be internally consistent and still be wrong in sign, because a sign error propagates through every identity that derives one normalization from another. The following check is independent of all of them: it solves the defining equation numerically and compares.

Table D.3: Exact values $\mathsf{L}_n^{\mathrm{Lam}}(1)$.

n	$\mathsf{L}_n^{\mathrm{Lam}}(1)$	n	$\mathsf{L}_n^{\mathrm{Lam}}(1)$
0	-1	7	$\frac{15}{56}$
1	1	8	$\frac{457}{1260}$
2	$-\frac{1}{2}$	9	$-\frac{49}{90}$
3	$-\frac{1}{6}$	10	$-\frac{391}{2016}$
4	$\frac{5}{12}$	11	$\frac{32213}{36960}$
5	$-\frac{1}{20}$	12	$-\frac{299363}{1425600}$
6	$-\frac{49}{120}$		

Let $\omega_N(x) = x + \sum_{n=0}^{N} \mathsf{L}_n^{\mathrm{Lam}}(L)x^{-n}$ with $L = \log x$, the truncation of Theorem 20.10. Solving $y + \log y = x$ to fifty significant digits by Newton's method and comparing gives the following. The residual column is $\omega_N + \log \omega_N - x$ computed numerically; the predicted column is $-\mathsf{L}_{N+1}^{\mathrm{Lam}}(L)x^{-(N+1)}$, which is what Equation (20.14) asserts the residual must be to leading order.

Table D.4: Observed versus predicted residual of the truncated Wright-omega expansion. The ratio tends to 1 as x grows, which is the content of Equation (20.14); the departure at $x = 50$ is the omitted $\mathcal{O}_t^{\mathrm{form}}(t^{N+2})$ block making itself felt.

x	N	observed residual	predicted	ratio
50	1	-1.534239×10^{-3}	-1.495976×10^{-3}	1.0256
50	2	-5.859441×10^{-6}	-7.300570×10^{-6}	0.8026
50	3	$+1.599261 \times 10^{-6}$	$+1.473299 \times 10^{-6}$	1.0855
50	4	$+9.405048 \times 10^{-8}$	$+8.979402 \times 10^{-8}$	1.0474
200	1	-2.211720×10^{-4}	-2.184442×10^{-4}	1.0125
200	2	-1.606059×10^{-6}	-1.596060×10^{-6}	1.0063
200	3	-1.802138×10^{-9}	-2.030332×10^{-9}	0.8876
200	4	$+2.386204 \times 10^{-10}$	$+2.316867 \times 10^{-10}$	1.0299
1000	1	-1.701321×10^{-5}	-1.695079×10^{-5}	1.0037
1000	2	-4.535166×10^{-8}	-4.520477×10^{-8}	1.0032
1000	3	$-1.013683 \times 10^{-10}$	$-1.011740 \times 10^{-10}$	1.0019
1000	4	$-9.250010 \times 10^{-14}$	$-9.319150 \times 10^{-14}$	0.9926

Two features of the table are worth reading carefully, because both are predictions of the theory rather than accidents.

First, the residual *changes sign* between $N = 2$ and $N = 3$ at $x = 50$ and between $N = 3$ and $N = 4$ at $x = 200$. That is not instability: by Theorem 21.7 the coefficients of $\mathsf{L}_n^{\mathrm{Lam}}$ alternate, so the leading residual $-\mathsf{L}_{N+1}^{\mathrm{Lam}}(L)x^{-(N+1)}$ changes sign with N whenever L is small enough that the top coefficient does not yet dominate. A sign error anywhere in the chain would break this pattern rather than reproduce it.

Second, at $x = 1000$ the error falls by roughly three orders of magnitude per added block:

N	$\lvert\omega_N - \omega\rvert$	N	$\lvert\omega_N - \omega\rvert$
0	6.924751×10^{-3}	4	9.240705×10^{-14}
1	1.699609×10^{-5}	5	7.844485×10^{-16}
2	4.530604×10^{-8}	6	6.759229×10^{-18}
3	1.012664×10^{-10}		

What this table does *not* show

Improvement through $N = 6$ at one value of x is not convergence. The series of Theorem 20.10 diverges for every fixed x: by Theorem 21.7 the leading coefficient of $\mathsf{L}_n^{\mathrm{Lam}}$ is $1/n$ while its degree is n, so for fixed x the terms eventually grow. What the table exhibits is asymptotic behaviour — for each fixed N, the error is $\mathcal{O}_{x\to\infty}\left(L^{N+2}/x^{N+2}\right)$ — and the correct reading of the second table is a statement about N small relative to $\log x$, not a limit in N. Section 24 makes the distinction precise.

Appendix E

Exercises with solutions

The six source reports carry 83 exercises between them, distributed very unevenly: report 1 sets 58, three at the end of each chapter, and supplies no solutions; report 3 sets 13 and solves all of them; report 2 sets 12 and solves all of them; reports 4, 5 and 6 set none but contain worked examples that make good problems. The overlap is heavy — five of the six ask for the reversion of $X + aL$ in one guise or another — and the quality is uneven, ranging from genuine tests of the formalism down to arithmetic drill.

What follows is a single graded set of 56 exercises, harvested from all six sources, deduplicated, corrected, and organised by the part of the volume they exercise. Every exercise carries a complete solution, not a hint. Where a source supplied a solution it was recomputed independently; the corrections are marked in the LaTeX source with `% ed.:` comments and are also summarised in Part G. Where an important technique of the body had no exercise anywhere in the corpus — the sharp summability threshold, the guard-precision rule, the derivative certificate, the analytic transfer — one has been added.

E.1 How to use this appendix

Three conventions are worth restating, because most of the mistakes recorded below come from breaking one of them.

1. *Truncation is exclusive.* $\operatorname{Trunc}_{<N} F$ keeps the blocks of outer order $n < N$ (Theorem 7.1). Sources 1 and 6 use the inclusive cutoff $[F]_{\leq N} = \operatorname{Trunc}_{<N+1} F$; every index taken from them has been shifted by one block. Exercises whose title contains the word *trap* exist precisely because that shift is invisible until it is wrong.
2. *The three $\mathcal{O}_{\cdot}(\cdot)$-roles are distinct.* $\mathcal{O}_t^{\mathrm{form}}(t^N)$ is a statement about a formal valuation and carries no analytic content whatsoever; $\mathcal{O}_{X\to+\infty}(A(X))$ is an inequality between functions on a stated ray; a complexity bound is neither. No exercise below asks you to pass from the first to the second without an analytic hypothesis, and three of them ask you to find where somebody did.
3. *Every statement names its coefficient field and its regime.* The default field is a field $\mathbb{K}$ of characteristic zero; the default regime is $X \to +\infty$ along the positive real ray with the real logarithm.

Exercises marked *(hard)* need either a genuinely new idea or a long exact computation. Two special kinds recur.

- *Counterexample exercises* ask you to refute a plausible strengthening of a theorem of the body, usually one obtained by deleting a hypothesis. These are Theorems E.4, E.5, E.7, E.15, E.20, E.27, E.29, E.36, E.39, E.44 and E.53. The hypotheses the sources kept dropping are exactly the ones these exercises defend.

- *Traps* hand you a plausible-looking computation containing one documented failure mode — a truncation convention read in the wrong direction, a formal tail treated as an analytic estimate, a division by a nonunit, a scale change applied to one generator only — and ask you to locate the error and repair it. These are Theorems E.6, E.16, E.26, E.32, E.41, E.46, E.52 and E.54.

E.2 The scale and its algebra

Exercise E.1 (Valuation of a sum, and how far it can jump)**.** Work in $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$, using only Theorem 6.2.

1. Prove $\nu_t(F+G) \geq \min\{\nu_t F, \nu_t G\}$, with equality whenever $\nu_t F \neq \nu_t G$.
2. For every $k \geq 1$ and every $d \geq 0$ exhibit $F, G \in \mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ with $\nu_t F = \nu_t G = 0$, $\deg_{\mathrm{L}}\big([t^0]F\big) = d$, and $\nu_t(F+G) = k$.
3. Show that strict inequality in (1) forces $\nu_t F = \nu_t G =: n$ and $[t^n]F = -[t^n]G$.

Solution. (1) Let $n = \min\{\nu_t F, \nu_t G\}$. For $m < n$ both $[t^m]F$ and $[t^m]G$ vanish, hence so does $[t^m](F+G)$; this is $\nu_t(F+G) \geq n$. If $\nu_t F < \nu_t G$ then $n = \nu_t F$ and $[t^n](F+G) = [t^n]F \neq 0$, so $\nu_t(F+G) = n$; the case $\nu_t G < \nu_t F$ is symmetric.

(2) Put

$$F = L^d + t^k, \qquad G = -L^d.$$

Both lie in $\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$, both have outer order 0, the t^0-block of F has L-degree d, and $F+G = t^k$ has outer order exactly k.

(3) If $\nu_t F < \nu_t G$ or $\nu_t G < \nu_t F$ then (1) gives equality, so strictness forces $\nu_t F = \nu_t G = n$. Then $0 = [t^n](F+G) = [t^n]F + [t^n]G$, which is the assertion. Note that the converse fails: cancelling leading blocks guarantee only $\nu_t(F+G) \geq n+1$ and say nothing about how much further the order jumps, which is what (2) exploits. ☐

Exercise E.2 (Completeness of the power-series ring, block by block)**.** Let $(F_r)_{r\geq 0}$ be a t-adically Cauchy sequence in $\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$.

1. Show that for each n the block $[t^n]F_r$ is eventually independent of r; call its value $p_n \in \mathbb{K}[L]$.
2. Show $F := \sum_{n\geq 0} p_n t^n$ lies in $\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ and $F_r \to F$.
3. Locate the exact step at which $\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$, rather than $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$, was used, and show that a Cauchy sequence in $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ nevertheless always satisfies $\inf_r \nu_t(F_r) > -\infty$, so that the same three steps do prove completeness of $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ as well.

Solution. (1) Cauchy means: for every N there is r_N with $\nu_t(F_r - F_s) \geq N$ whenever $r, s \geq r_N$. Applying this with $N = n+1$ gives $[t^n]F_r = [t^n]F_s$ for all $r, s \geq r_{n+1}$, which is the assertion; write p_n for the common value.

(2) Each p_n lies in $\mathbb{K}[L]$ and the index set is $\mathbb{N}_0$, so the support of F is bounded below by 0 and $F \in \mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$. Fix N and let $r \geq \max\{r_1, \ldots, r_N\}$. For $n < N$ we have $[t^n]F_r = p_n = [t^n]F$, so $\nu_t(F - F_r) \geq N$. As N was arbitrary, $F_r \to F$. Uniqueness of the limit is separatedness, Theorem 7.18 (i).

(3) The only step that used $\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ is the claim in (2) that $\sum_n p_n t^n$ is an element of the ring: on $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ the blockwise limits p_n could be nonzero for infinitely many $n < 0$, and then the sum is not an element of $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$. This is not idle: the sequence of Theorem 7.19 stabilises coefficientwise to exactly such a symbol. But that sequence is not Cauchy, and no Cauchy sequence can behave that way. Indeed apply the Cauchy condition with $N = 0$: there is r_0 with $\nu_t(F_r - F_{r_0}) \geq 0$ for all $r \geq r_0$,

whence by part (1) of Theorem E.1

$$\nu_t(F_r) \;\geq\; \min\{\nu_t(F_{r_0}),\, 0\} \qquad (r \geq r_0),$$

and the finitely many indices $r < r_0$ contribute a finite minimum. Hence $\mu := \inf_r \nu_t(F_r) > -\infty$, every F_r lies in $t^\mu \mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$, and steps (1) and (2) may be run verbatim inside that subring. This is Theorem 7.18 (iii),(iv). □

Exercise E.3 (Summable powers)**.** Let $U \in \mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$. Prove that the family $(U^n)_{n\geq 0}$ is formally summable in the sense of Theorem 7.21 if and only if $U = 0$ or $\nu_t(U) \geq 1$, and that in that case $\sum_{n\geq 0} U^n = (1-U)^{-1}$.

Solution. Assume $U \neq 0$ and put $q = \nu_t(U)$. Since $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ is a domain and ν_t is additive on products (Theorem 6.2), $\nu_t(U^n) = nq$ for every $n \geq 0$, with $U^0 = 1$ of order 0.

If $q \geq 1$ then $nq \to +\infty$, so for each N the set $\{n : \nu_t(U^n) < N\}$ is contained in $\{n : n < N\}$ and is finite: the family is summable. If $q \leq 0$ then $nq \leq 0$ for all n, so $\{n : \nu_t(U^n) < 1\} = \mathbb{N}_0$ is infinite and the family is not summable.

For the value, let $S = \sum_{n\geq 0} U^n$, which exists by the summability just proved. Multiplication is continuous for the t-adic filtration on a summable family (Theorem 7.22), so

$$(1-U)S = \sum_{n\geq 0} U^n - \sum_{n\geq 0} U^{n+1} = U^0 = 1,$$

the middle step being a telescoping of two summable families. Since $\nu_t(U) \geq 1$ the element $1-U$ is a unit of $\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ by Theorem 9.3, and its reciprocal is unique, so $S = (1-U)^{-1}$. This is Theorem 7.23. □

Exercise E.4 (Counterexample: block-local finiteness is not summability)**.** Theorem 7.24 (c) characterises summability by two conditions: every block receives finitely many contributions, *and* $\inf_i \nu_t(S_i) > -\infty$. Show that the second clause cannot be deleted.

1. Exhibit a family in $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ that is block-locally finite but not summable.
2. Exhibit one that is block-locally finite, not summable, and whose blockwise sum *is* a legitimate element of $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$. Conclude that “the blockwise sum exists in the ring” is not a substitute for summability either.
3. Show that on $\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ the second clause is automatic.

Solution. (1) Take $S_i = t^{-i}$ for $i \geq 1$. The block t^n receives a contribution from S_i only for $i = -n$, so at most one contribution. But $\nu_t(S_i) = -i$, so $\{i : \nu_t(S_i) < 0\}$ is infinite and the family fails Theorem 7.21. Concretely, the partial sums $T_m = \sum_{i=1}^m t^{-i}$ satisfy $\nu_t(T_{m+1} - T_m) = -(m+1)$, so they are not Cauchy, and the blockwise limits would give a symbol with infinitely many negative blocks, which is not an element of $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ (Theorem 5.15 (b)).

(2) Take

$$S_i = t^{-i} - t^{-i-1} \qquad (i \geq 1).$$

The block t^n receives contributions only from $i = -n$ and $i = -n-1$, so at most two; the family is block-locally finite. Blockwise, $[t^{-1}]\sum_i S_i = 1$ and for $n \leq -2$ the two contributions are $+1$ (from $i = -n$) and -1 (from $i = -n-1$), summing to 0. The blockwise sum is therefore $t^{-1} \in \mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$. Nevertheless $\nu_t(S_i) = -i-1 \to -\infty$, so the family is not summable, and indeed the partial sums $\sum_{i=1}^m S_i = t^{-1} - t^{-m-1}$ do not converge: consecutive partial sums differ by $S_{m+1} = t^{-m-1} - t^{-m-2}$, of outer order $-(m+2) \to -\infty$, so the sequence is not Cauchy. So a blockwise sum can exist in the ring and still fail to be the limit of the partial sums in any sense the filtration recognises. Summability is a statement about the family, not about the array of coefficients it produces.

(3) If every $S_i \in \mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ then $\nu_t(S_i) \geq 0$, so $\inf_i \nu_t(S_i) \geq 0 > -\infty$. □

Exercise E.5 (Counterexample: the tail is a filtration, not an ideal)**.** A plausible reading of "compute modulo t^N" is that $\mathfrak{F}_N = t^N \mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ is an ideal of $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ and that $\mathrm{Trunc}_{<N}\cdot$ is the quotient homomorphism. Refute both halves.

1. Show $\mathfrak{F}_N$ is not an ideal of $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ for any N, and that the ideal it generates is all of $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$.
2. Show $\mathrm{Trunc}_{<N}\cdot$ is not multiplicative on $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$, by exhibiting F, G with $\mathrm{Trunc}_{<N}(FG) \neq \mathrm{Trunc}_{<N}\big(\mathrm{Trunc}_{<N} F\, \mathrm{Trunc}_{<N} G\big)$.
3. State what does survive, and prove it.

Solution. (1) t is a unit of $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$, with inverse t^{-1}. Hence $t^{-1}\cdot t^N = t^{N-1}$, which has outer order $N-1 < N$ and so does not lie in $\mathfrak{F}_N$: $\mathfrak{F}_N$ is not closed under multiplication by ring elements. Moreover $t^{-N}\cdot t^N = 1$, so the ideal generated by t^N contains 1 and is all of $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$. In particular $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}/(t^N) = 0$ and there is no quotient ring to map onto. This is Theorem 7.11 (i),(ii).

(2) Take

$$F = G = \frac{t^{-2}}{1-t} = \sum_{n\geq -2} t^n, \qquad N = 1.$$

Then $FG = t^{-4}(1-t)^{-2} = \sum_{j\geq -4}(j+5)t^j$, so

$$\mathrm{Trunc}_{<1}(FG) = t^{-4} + 2t^{-3} + 3t^{-2} + 4t^{-1} + 5.$$

On the other hand $\mathrm{Trunc}_{<1} F = \mathrm{Trunc}_{<1} G = t^{-2}(1+t+t^2)$, and

$$\mathrm{Trunc}_{<1} F\ \mathrm{Trunc}_{<1} G = t^{-4}\big(1+t+t^2\big)^2 = t^{-4} + 2t^{-3} + 3t^{-2} + 2t^{-1} + 1,$$

which is already of t-degree 0, so truncating again changes nothing. The blocks at t^{-1} and t^0 are wrong, by 2 and by 4.

(3) Three statements survive. First, $\mathfrak{F}_N$ is an additive subgroup and a $\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$-submodule of $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$, so $\mathrm{Trunc}_{<N}\cdot$ is $\mathbb{K}[L]$-linear and $\mathrm{Trunc}_{<N}(F+G) = \mathrm{Trunc}_{<N} F + \mathrm{Trunc}_{<N} G$. Second, on $\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ the set $t^N\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ *is* an ideal for $N \geq 0$ and $\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}/t^N\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}} \cong \mathbb{K}[L][t]/(t^N)$; this is where the quotient language is legitimate. Third, the correct multiplicative law on $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ is the asymmetric one of Theorem 7.4: with $m = \nu_t F$ and $r = \nu_t G$,

$$\mathrm{Trunc}_{<N}(FG) = \mathrm{Trunc}_{<N}\big(\mathrm{Trunc}_{<N-r} F \cdot \mathrm{Trunc}_{<N-m} G\big).$$

In the example $m = r = -2$ and $N = 1$, so the required inputs are $\mathrm{Trunc}_{<3} F$ and $\mathrm{Trunc}_{<3} G$, two blocks more than were used. A direct check: $\mathrm{Trunc}_{<3} F = t^{-2}(1+t+t^2+t^3+t^4)$, whose square is $t^{-4}(1+2t+3t^2+4t^3+5t^4+\cdots)$, and truncating at <1 reproduces the correct answer. □

Exercise E.6 (Trap: a symmetric truncation on the Laurent ring)**.** A routine is asked for $\mathrm{Trunc}_{<0}(FG)$ where

$$F = X^2 + X + \frac{L}{X} + \mathcal{O}_t^{\mathrm{form}}\big(t^2\big)\,, \qquad G = X^3 - X^2 L + \mathcal{O}_t^{\mathrm{form}}\big(t^{-1}\big)\,.$$

It reasons: "both factors are known through outer order t^1 respectively t^{-2}; truncating each at <0 and multiplying, $FG = (X^2+X)(X^3-X^2L) + \cdots = X^5 - X^4L + X^4 - X^3L + \cdots$, so $\mathrm{Trunc}_{<0}(FG) = X^5 + (1-L)X^4 - LX^3$." Find the error, say which blocks of the answer are nevertheless correct, and state exactly what precision in F and G the requested output needs.

Solution. The error is that $\mathrm{Trunc}_{<N}\cdot$ is not multiplicative on $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$ (Theorem E.5), and the symmetric rule (7.4) is valid only when both factors have nonnegative outer order. Here $\nu_t F = -2$ and $\nu_t G = -3$, so Theorem 7.4 demands

$$\mathrm{Trunc}_{<0}(FG) = \mathrm{Trunc}_{<0}\big(\mathrm{Trunc}_{<0-(-3)} F \cdot \mathrm{Trunc}_{<0-(-2)} G\big)$$

$$= \operatorname{Trunc}_{<0}\bigl(\operatorname{Trunc}_{<3} F \cdot \operatorname{Trunc}_{<2} G\bigr).$$

That is, F is needed through outer order t^2 and G through t^1; the routine used F through t^{-1} and G through t^{-1}.

Which blocks survive? Write $F_n = [t^n]F$ and $G_n = [t^n]G$, so that $F_{-2} = 1$, $F_{-1} = 1$, $F_0 = 0$, $F_1 = L$ are known and F_n is unknown for $n \geq 2$; while $G_{-3} = 1$, $G_{-2} = -L$ are known and G_n is unknown for $n \geq -1$. Convolving,

$$\begin{aligned}
[t^{-5}](FG) &= F_{-2}G_{-3} = 1,\\
[t^{-4}](FG) &= F_{-2}G_{-2} + F_{-1}G_{-3} = 1 - L,\\
[t^{-3}](FG) &= F_{-2}G_{-1} + F_{-1}G_{-2} + F_0G_{-3} = G_{-1} - L.
\end{aligned}$$

The first two are determined by the data and agree with the routine's X^5 and $(1-L)X^4$. The third is not determined: it needs G_{-1}, which the routine both discarded and never knew. The printed $-LX^3$ is what one gets by setting $G_{-1} = 0$, an assumption nothing supports. The blocks at t^{-2}, t^{-1}, t^0 are likewise undetermined, and the routine did not print them at all.

So the data are *insufficient* for the requested output. Indeed F is known only modulo $\mathcal{O}_t^{\mathrm{form}}\bigl(t^2\bigr)$, which is exactly $\operatorname{Trunc}_{<2} F$, one block short of the $\operatorname{Trunc}_{<3} F$ required; and G is known only modulo $\mathcal{O}_t^{\mathrm{form}}\bigl(t^{-1}\bigr)$, three blocks short of $\operatorname{Trunc}_{<2} G$. The honest output is

$$\operatorname{Trunc}_{<-3}(FG) = X^5 + (1-L)X^4,$$

with the remaining blocks reported as unknown. This is the error rule of Theorem 8.14 read backwards. Two lessons: the symmetric rule (7.4) is a theorem about $\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$, not about $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$; and a Laurent factor makes a routine need *more* input precision than the output it is asked for, never less. $\square$

Exercise E.7 (Counterexample: no monomial prefactor inside the polynomial ring)**.** Several sources write a nonzero $A \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ in the prefactor normal form $A = cX^\rho L^\beta(1+U)$ with $c \in \mathbb{K}^\times$, $\rho \in \mathbb{Z}$, $\beta \in \mathbb{N}_0$ and $U \in t\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$, "whenever available". Show that for $A = L + t$ it is not available, and that it becomes available exactly after passing to $\mathscr{PL}_{\mathrm{rat},\mathbb{K}}$. Deduce that no argument inside $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ may assume the normal form.

Solution. Suppose $L + t = cX^\rho L^\beta(1+U) = c\,t^{-\rho}L^\beta(1+U)$ with $U \in t\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$. Comparing outer orders, $\nu_t(L+t) = 0$ while $\nu_t\bigl(ct^{-\rho}L^\beta(1+U)\bigr) = -\rho$, so $\rho = 0$. Comparing the t^0-blocks then gives $cL^\beta = L$, so $c = 1$ and $\beta = 1$. Comparing the t^1-blocks gives

$$1 = [t^1]\bigl(L(1+U)\bigr) = L \cdot u_1, \qquad u_1 := [t^1]U \in \mathbb{K}[L],$$

whence $u_1 = L^{-1}$, which is not a polynomial. Contradiction.

Over $\mathbb{K}(L)$ the factorisation exists and is the obvious one, $L + t = L\bigl(1 + L^{-1}t\bigr)$ with $L^{-1}t \in t\mathscr{PL}_{\mathrm{rat},\mathbb{K}}$. So the obstruction is not the shape of the leading monomial but the failure of $\mathbb{K}[L]$ to be a field: the general leading-block factorisation Theorem 6.10 produces $A = t^{n_0}p_{n_0}(L)(1+U)$ with U in $t\mathscr{PL}_{\mathrm{rat},\mathbb{K}}$, and the coefficient p_{n_0} can be inverted only in $\mathbb{K}(L)$. Consequently any argument that silently divides by the leading block — for instance one that defines d_A through the prefactor, or that "normalises A to be monic" — has left $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$. $\square$

Exercise E.8 (A prescribed drop in the log-degree profile)**.** Let $d \geq 1$. For each k with $1 \leq k \leq d$ construct a pair $F, G \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ with $\nu_t F = \nu_t G = 0$ and $d_F(1) = d_G(1) = d$ such that $d_{FG}(1) = d - k$; construct one more pair with $d_{FG}(1) = -\infty$; and exhibit a pair with no drop at all. Then say exactly when equality holds in the bound $d_{FG}(N) \leq (d_F * d_G)(N)$ of Theorem 8.25.

Solution. For $1 \leq k \leq d$ put

$$F = 1 + \bigl(L^d + L^{d-k}\bigr)t, \qquad G = 1 - L^d t.$$

Both have $\nu_t = 0$, $d_F(0) = d_G(0) = 0$ and $d_F(1) = d_G(1) = d$, the term L^{d-k} being of strictly lower degree than L^d because $k \geq 1$. Now

$$[t^1](FG) = \left(L^d + L^{d-k}\right) \cdot 1 + 1 \cdot \left(-L^d\right) = L^{d-k},$$

so $d_{FG}(1) = d - k$, while the convolution bound is $(d_F * d_G)(1) = \max\{0 + d,\ d + 0\} = d$. For the degenerate case take $F = 1 + L^d t$, $G = 1 - L^d t$, where $[t^1](FG) = 0$ and $d_{FG}(1) = -\infty$. For no drop take $F = G = 1 + L^d t$, where $[t^1](FG) = 2L^d$ has degree d; this is the generic situation, and it is what makes the drop worth an exercise.

Equality is decided exactly by Theorem 8.25. Let $M = (d_F * d_G)(N)$ and let P_N be the set of pairs (n, m) with $n+m = N$, $[t^n]F \neq 0 \neq [t^m]G$ and $d_F(n)+d_G(m) = M$. If $P_N = \emptyset$ then $[t^N](FG) = 0$ and equality holds trivially with $M = -\infty$. If $P_N \neq \emptyset$ then the coefficient of L^M in $[t^N](FG)$ equals $\sum_{(n,m)\in P_N} \operatorname{lc}\big([t^n]F\big) \operatorname{lc}\big([t^m]G\big)$, and equality holds if and only if that sum is nonzero. In the dropping family $P_1 = \{(0, 1), (1, 0)\}$ and the sum is $1 \cdot (-1) + 1 \cdot 1 = 0$, which is why the degree drops; how far it drops is then decided by the next coefficients, and the parameter k tunes exactly that. In the last pair the same sum is $1 \cdot 1 + 1 \cdot 1 = 2 \neq 0$ and there is no drop. Note that condition (3) of Theorem 8.26 — positive leading coefficients over an ordered field — excludes the dropping family and covers the last one, as it must. ☐

Exercise E.9 (Which enlargement does each expression force?)**.** For each expression name the smallest class in which it can be represented, choosing among $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$, $\mathscr{PL}_{\mathrm{rat},\mathbb{K}}$, $\mathscr{PL}_{\mathrm{itLaur},\mathbb{K}}$, a Puiseux–log ring, a ring with a second logarithmic level, and a logarithmic-exponential transseries field; and justify both membership and minimality.

$$(L+t)^{-1}, \qquad \sum_{n\geq 0} n!\, L^{-n}, \qquad X^{1/2}, \qquad \log(X \log X), \qquad \mathrm{e}^{-X}.$$

Solution. $(L + t)^{-1}$. Geometric expansion in t gives

$$\frac{1}{L+t} = \frac{1}{L} \cdot \frac{1}{1 + L^{-1}t} = \sum_{n\geq 0} (-1)^n L^{-n-1} t^n, \tag{E.1}$$

every coefficient of which lies in $\mathbb{K}[L, L^{-1}] \subseteq \mathbb{K}(L)$. So the element lies in $\mathscr{PL}_{\mathrm{rat},\mathbb{K}}$. It does not lie in $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$, because the reciprocal in $\mathscr{PL}_{\mathrm{rat},\mathbb{K}}$ is unique and (E.1) has non-polynomial coefficients; equivalently, $L + t$ is not a unit of $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ by Theorem 9.3, its leading block L not being a nonzero constant. Minimality among the classes of the volume: Theorem 9.20 says $\mathscr{PL}_{\mathrm{rat},\mathbb{K}}$ is the smallest field $E((t))$ with $\mathbb{K}[L] \subseteq E \subseteq \mathbb{K}(L)$ containing $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$, and this example is one of the witnesses that forces the enlargement.

$\sum_{n\geq 0} n! L^{-n}$. This is a legitimate element of $\mathbb{K}((L^{-1}))$, hence of $\mathscr{PL}_{\mathrm{itLaur},\mathbb{K}}$ (as a t^0-block). It is not in $\mathbb{K}(L)$: a rational function of L has an eventually linearly recurrent sequence of Laurent coefficients at $L = \infty$, and $(n!)_{n\geq 0}$ satisfies no linear recurrence with constant coefficients, since $n!/(n - 1)! = n$ is unbounded. So this element separates $\mathscr{PL}_{\mathrm{itLaur},\mathbb{K}}$ from $\mathscr{PL}_{\mathrm{rat},\mathbb{K}}$, which is Theorem 5.7.

$X^{1/2} = t^{-1/2}$. A Puiseux–log ring with exponent group $\frac{1}{2}\mathbb{Z}$ contains it; no class with exponent group $\mathbb{Z}$ does, since the outer exponents of every element of $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$, $\mathscr{PL}_{\mathrm{rat},\mathbb{K}}$, $\mathscr{PL}_{\mathrm{itLaur},\mathbb{K}}$ are integers. Enlarging the coefficient field cannot help: the exponent group is not touched by that. See Theorem 5.25.

$\log(X \log X) = L + \log L$. The symbol $\log L$ is not in $\mathbb{K}((L^{-1}))$, let alone $\mathbb{K}(L)$: if it were, it would be a Laurent series in L^{-1}, so $\partial_L \log L = L^{-1}$ would lie in the image of ∂_L on $\mathbb{K}((L^{-1}))$, contradicting Theorem 10.24 (the residue at $L = \infty$ of L^{-1} is 1). A second logarithmic level is forced; this is item (3) of Theorem 12.1.

e^{-X}. By Theorem 10.16, $\exp F$ lies in $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ only if $\nu_t F \geq 0$; here $F = -X$ has $\nu_t = -1$. In fact $\mathrm{e}^{-X} \prec t^N$ for every N, so it is not comparable with any monomial of any power–log scale

(Theorem 10.15); an exponential transmonomial, hence a logarithmic-exponential transseries field, is required. $\square$

Exercise E.10 (Admissible supports, and a support that is not one)**.** Decide, with proof, which of the following subsets of $\mathbb{Z} \times \mathbb{N}_0$ are the coefficient support $\operatorname{supp}_{\text{coeff}}(F)$ of some $F \in \mathscr{PL}_{\text{poly},\mathbb{K}}$, using Theorem 5.15:

$$S_1 = \{(n,n) : n \geq 0\}, \qquad S_2 = \{(n,j) : n \geq 0,\ 0 \leq j \leq n\},$$
$$S_3 = \{(-n,0) : n \geq 0\}, \qquad S_4 = \{(n,0) : n \geq 0\} \cup \{(0,j) : j \geq 0\}.$$

For each admissible one, exhibit an F realising it.

Solution. Criterion (b) of Theorem 5.15: the set of outer exponents occurring must be bounded below, and each fibre $S_n = \{j : (n,j) \in S\}$ must be finite.

S_1: outer exponents $\{n \geq 0\}$ bounded below by 0; fibre $S_n = \{n\}$ finite. Admissible; realised by $F = \sum_{n\geq 0} L^n t^n$.

S_2: exponents bounded below by 0; fibre $S_n = \{0, \dots, n\}$ finite. Admissible; realised by $F = \sum_{n\geq 0}\big(\sum_{j=0}^{n} L^j\big)t^n$.

S_3: exponents $\{-n : n \geq 0\}$ are not bounded below. Not admissible. This is the support the divergent family of Theorem E.4 (1) would have produced.

S_4: exponents bounded below by 0, but the fibre at $n = 0$ is $\mathbb{N}_0$, infinite: a block would have to be a power series in L, not a polynomial. Not admissible in $\mathscr{PL}_{\text{poly},\mathbb{K}}$. Nor is it rescued by passing to the Hahn ambient of Theorem 5.19: that ambient still demands a reverse well-ordered support, which is criterion (c), and S_4 fails (c), since $\{(0,j) : j \geq 0\}$ has no $\succ$-greatest element — larger j at fixed n means larger monomial $t^n L^j$. So S_4 is not admissible in any of the four classes. $\square$

E.3 Arithmetic: product, reciprocal, derivative

Exercise E.11 (The block Cauchy product, and what it consumed)**.** Let

$$A = 1 + (L+1)t + (L^2-1)t^2, \qquad B = 1 + (2-L)t + Lt^2,$$

with all undisplayed blocks zero. Compute AB through outer order t^3. Then, without recomputing, say which blocks of A and of B the block $[t^3](AB)$ actually depends on, and check your answer against Theorem 8.12.

Solution. The blocks convolve:

$$\begin{aligned}
[t^0](AB) &= 1,\\
[t^1](AB) &= (L+1) + (2-L) = 3,\\
[t^2](AB) &= (L^2-1) + (L+1)(2-L) + L = L^2 - 1 + (-L^2 + L + 2) + L = 2L+1,\\
[t^3](AB) &= (L+1)\cdot L + (L^2-1)(2-L) = L^2 + L - L^3 + 2L^2 + L - 2\\
&= -L^3 + 3L^2 + 2L - 2.
\end{aligned}$$

Hence

$$AB = 1 + 3t + (2L+1)t^2 + \big(-L^3 + 3L^2 + 2L - 2\big)t^3 + \mathcal{O}_t^{\text{form}}\big(t^4\big),$$

and in fact the product is exact through t^4 as well, both inputs being polynomials in t.

Dependency: since $\nu_t A = \nu_t B = 0$, Theorem 8.12 says $[t^N](AB)$ depends exactly on $[t^n]A$ and $[t^m]B$ for $0 \leq n, m \leq N$. For $N = 3$ that is $A_0, \dots, A_3$ and $B_0, \dots, B_3$; as $A_3 = B_3 = 0$ by hypothesis, four blocks of each are consulted and only three of each are nonzero. Note that the hypothesis "undisplayed blocks vanish" is not decoration: without it A_3 and B_3 are unknown and $[t^3](AB)$ is undetermined. $\square$

Exercise E.12 (Reciprocal of a unit, and where the unit hypothesis enters)**.** Find $C \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ with $\big(1+(L-1)t+Lt^2\big)C = 1$, through outer order t^4. Then explain why C has polynomial blocks, and give the one-line reason that the same recurrence applied to $L+(L-1)t$ does not.

Solution. Write $B = 1+B_1t+B_2t^2$ with $B_1 = L-1$, $B_2 = L$, and $C = \sum_{n\geq 0} C_n t^n$. By Theorem 9.7 with $c = 1$, $C_0 = 1$ and $C_n = -\sum_{j=1}^{n} B_j C_{n-j}$. Hence

$$
\begin{aligned}
C_1 &= -B_1 = 1-L,\\
C_2 &= -(B_1C_1 + B_2C_0) = (L-1)^2 - L = L^2-3L+1,\\
C_3 &= -(B_1C_2 + B_2C_1) = -(L-1)(L^2-3L+1) - L(1-L)\\
&= -\big(L^3-4L^2+4L-1\big) - \big(L-L^2\big) = -L^3+5L^2-5L+1,\\
C_4 &= -(B_1C_3 + B_2C_2)\\
&= -(L-1)\big(-L^3+5L^2-5L+1\big) - L\big(L^2-3L+1\big)\\
&= -\big(-L^4+6L^3-10L^2+6L-1\big) - \big(L^3-3L^2+L\big)\\
&= L^4-7L^3+13L^2-7L+1.
\end{aligned}
$$

So

$$C = 1+(1-L)t+(L^2-3L+1)t^2+(-L^3+5L^2-5L+1)t^3+(L^4-7L^3+13L^2-7L+1)t^4+\mathcal{O}_t^{\mathrm{form}}\big(t^5\big)\,.$$

Every C_n is a polynomial because the recurrence divides only by $c = B_0 = 1$, a unit of $\mathbb{K}$; the whole content of Theorem 9.3 is that this is the only case in which the division stays inside $\mathbb{K}[L]$. For $B = L+(L-1)t$ the recurrence starts $C_0 = L^{-1}$, which already leaves $\mathbb{K}[L]$; B is not a unit of $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ because its leading block L is not a nonzero constant. □

Exercise E.13 (A quotient, and the exact guard precision it needs)**.** Let

$$A = 1+(2L-1)t+(L^2+3)t^2+(L^3-L)t^3+\mathcal{O}_t^{\mathrm{form}}\big(t^4\big)\,, \qquad B = 1+Lt+(L^2-1)t^2+\mathcal{O}_t^{\mathrm{form}}\big(t^3\big)\,.$$

Compute $\mathrm{Trunc}_{<4}(A/B)$, or explain why the data do not determine it, and in either case state the sharp input requirement given by Theorem 9.11.

Solution. Here $a = \nu_t A = 0$ and $b = \nu_t B = 0$, so (9.10) reads: $\mathrm{Trunc}_{<N}(A/B)$ is determined by $\mathrm{Trunc}_{<N} A$ and $\mathrm{Trunc}_{<N} B$. With $N = 4$ we need A and B each through outer order t^3. A is given to that precision; B is given only modulo $\mathcal{O}_t^{\mathrm{form}}\big(t^3\big)$, i.e. only $\mathrm{Trunc}_{<3} B$. So the block $[t^3](A/B)$ is *not* determined: changing B_3 by δ changes the quotient's third block by $-\delta$.

What is determined is $\mathrm{Trunc}_{<3}(A/B)$. Using (9.8), i.e. $Q_n = A_n - \sum_{j\geq 1} B_j Q_{n-j}$ since $B_0 = 1$:

$$
\begin{aligned}
Q_0 &= 1,\\
Q_1 &= A_1 - B_1Q_0 = (2L-1) - L = L-1,\\
Q_2 &= A_2 - B_1Q_1 - B_2Q_0 = (L^2+3) - L(L-1) - (L^2-1) = -L^2+L+4.
\end{aligned}
$$

If in addition $B_3 = 0$ is assumed — which is what the source that supplies this example intends — then

$$Q_3 = A_3 - B_1Q_2 - B_2Q_1 - B_3Q_0 = (L^3-L) - L(-L^2+L+4) - (L^2-1)(L-1) = L^3-4L-1,$$

giving

$$\frac{A}{B} = 1+(L-1)t+(-L^2+L+4)t^2+(L^3-4L-1)t^3+\mathcal{O}_t^{\mathrm{form}}\big(t^4\big)\,.$$

A useful check, and the one the same source recommends, is to multiply the computed quotient by B and compare exclusive truncations at the requested cutoff; by Theorem 9.9 the truncated quotient is the unique polynomial of t-degree < 4 passing that test. □

Exercise E.14 (Integer and generalised binomial powers)**.** Let $F = 1 + Lt \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$.

1. Compute $[t^n]F^k$ for every integer $k \geq 0$.
2. Compute $[t^n]F^\kappa$ for arbitrary $\kappa \in \mathbb{K}$, where $F^\kappa := \exp(\kappa \log F)$ as in Theorem 8.38, and compare with (1).
3. Compute d_{F^κ} and compare with the bound of Theorem 8.31.
4. Expand $\big(1 + Lt + (L^2 - 1)t^2\big)^\kappa$ through t^2.

Solution. (1) $F^k = (1 + Lt)^k = \sum_{n=0}^{k} \binom{k}{n} L^n t^n$, so $[t^n]F^k = \binom{k}{n} L^n$, zero for $n > k$.

(2) $\log F = \log(1+Lt) = \sum_{m\geq 1} \frac{(-1)^{m+1}}{m} L^m t^m$, and exponentiating gives the generalised binomial series $F^\kappa = \sum_{n\geq 0} \binom{\kappa}{n} L^n t^n$ with $\binom{\kappa}{n} = (\kappa)^{\underline{n}}/n!$. So the coefficient formula is *the same*, $[t^n]F^\kappa = \binom{\kappa}{n} L^n$; the only difference is that for $\kappa = k \in \mathbb{N}_0$ the falling factorial $(k)^{\underline{n}}$ vanishes for $n > k$ and the series terminates, while for $\kappa \notin \mathbb{N}_0$ every coefficient is nonzero and the series is genuinely infinite. Both are legitimate elements of $\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$; the support condition of Theorem 5.15 does not care.

(3) $d_{F^\kappa}(n) = n$ for all $n \geq 0$ when $\kappa \notin \mathbb{N}_0$, and for $0 \leq n \leq k$ when $\kappa = k$. The hypothesis of Theorem 8.31 holds with $\alpha = 1, \beta = 0$, and its conclusion $d_{F^k}(N) \leq N$ is attained: the bound is sharp here.

(4) Put $U = Lt + (L^2 - 1)t^2$, so $U^2 = L^2t^2 + \mathcal{O}_t^{\mathrm{form}}(t^3)$ and

$$(1+U)^\kappa = 1 + \kappa U + \binom{\kappa}{2} U^2 + \mathcal{O}_t^{\mathrm{form}}(t^3) = 1 + \kappa L t + \Big(\kappa(L^2-1) + \tfrac{\kappa(\kappa-1)}{2} L^2\Big) t^2 + \mathcal{O}_t^{\mathrm{form}}(t^3),$$

that is, the t^2-block is $\frac{1}{2}\kappa\big((\kappa+1)L^2 - 2\big)$. □

Exercise E.15 (Counterexample: a nonzero leading block is not enough)**.** A plausible strengthening of Theorem 9.3 is: "F is a unit of $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ if and only if its leading block is nonzero". Refute it, locate $(L + t)^{-1}$ exactly, and deduce that $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ is an integral domain that is not a field — so that the four classes of Theorem 5.3 really are four different objects.

Solution. Every nonzero $F \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ has, by definition, a nonzero leading block; if the proposed criterion were correct then every nonzero element would be a unit and $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ would be a field. It is not. Take $F = L + t$. Its leading block is $L \neq 0$. If F were a unit of $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$, with reciprocal $C = \sum_{n\geq 0} C_n t^n \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$, then comparing t^0-blocks in $FC = 1$ gives $LC_0 = 1$, impossible in $\mathbb{K}[L]$ since $\deg_{\mathrm{L}}(LC_0) \geq 1$ for $C_0 \neq 0$. The reciprocal does exist one class up, in $\mathscr{PL}_{\mathrm{rat},\mathbb{K}}$, and is given by (E.1).

That $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ is a domain is Theorem 8.22: outer orders add and leading blocks multiply in $\mathbb{K}[L]$, which is a domain, so a product of nonzero elements is nonzero. A domain that is not a field is exactly what the unit criterion describes: the units are the elements $ct^n(1 + U)$ with $c \in \mathbb{K}^\times$, $n \in \mathbb{Z}$, $U \in t\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$, a proper subgroup of the nonzero elements. The strengthening fails because the condition is not that the leading block be nonzero but that it be a nonzero *constant*. □

Exercise E.16 (Trap: a division that leaves the ring)**.** A student computes $(L + t)^{-1}$ inside $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ as follows. "By Theorem 9.7 the reciprocal of $B = B_0 + B_1 t + \cdots$ has $C_0 = B_0^{-1}$ and $C_n = -B_0^{-1} \sum_{j=1}^{n} B_j C_{n-j}$. Here $B_0 = L$, $B_1 = 1$, so $C_0 = L^{-1}$, $C_1 = -L^{-2}$, $C_2 = L^{-3}$, and in general $C_n = (-1)^n L^{-n-1}$. Since every C_n is a Laurent polynomial in L and Laurent polynomials are polynomials in L and L^{-1}, the answer $\sum_{n\geq 0} (-1)^n L^{-n-1} t^n$ lies in $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$." Find every error, and state precisely what is true.

Solution. There are two errors, one of hypothesis and one of vocabulary.

Hypothesis. Theorem 9.7 requires $B_0 = c \in \mathbb{K}^\times$; the whole proof consists in dividing by c, and c^{-1} has to exist *in the coefficient ring*. Here $B_0 = L$ is not invertible in $\mathbb{K}[L]$, so the theorem does not apply, and the recurrence as written is being run over the larger ring $\mathbb{K}(L)$. That is legitimate — it computes the reciprocal in $\mathscr{PL}_{\mathrm{rat},\mathbb{K}}$ — but it is no longer the theorem quoted.

Vocabulary. The blocks of an element of $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ are *polynomials* in L, not Laurent polynomials. $\mathbb{K}[L]$ is the coefficient ring; $\mathbb{K}[L, L^{-1}]$ is a strictly larger ring, and $L^{-1} \notin \mathbb{K}[L]$. So the concluding sentence is false at its last step.

What is true: the computed series is correct, and it is the reciprocal of $L+t$ in $\mathscr{PL}_{\mathrm{rat},\mathbb{K}}$; all its blocks in fact lie in the localisation $\mathbb{K}[L, L^{-1}]$, so it lies in the intermediate ring $\mathbb{K}[L, L^{-1}]((t)) \subsetneq \mathscr{PL}_{\mathrm{rat},\mathbb{K}}$. It does not lie in $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$, by Theorem E.15. The general statement replacing the misquoted theorem is Theorem 9.24: with $\nu_t B = 0$ and $A \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$, the quotient A/B lies in $\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ if and only if every block of the $\mathbb{K}(L)$-recurrence happens to be a polynomial, which happens if and only if $A \in B\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$. □

Exercise E.17 (Exact divisibility conditions, block by block)**.** Let $B = (L+1) + t$ and let $A = A_0(L) + A_1(L)t + \mathcal{O}_t^{\mathrm{form}}(t^2)$ with $A_0, A_1 \in \mathbb{K}[L]$. Find necessary and sufficient conditions on A_0, A_1 for $\mathrm{Trunc}_{<2}(A/B)$ to have polynomial blocks, and express them as two evaluations at a single point of $\mathbb{K}$.

Solution. Work over $E = \mathbb{K}(L)$, where B is a unit, and let $Q = A/B$ have blocks Q_n. From $BQ = A$, comparing blocks,

$$(L+1)Q_0 = A_0, \qquad (L+1)Q_1 + Q_0 = A_1.$$

Thus $Q_0 \in \mathbb{K}[L]$ if and only if $(L+1) \mid A_0$; and given that, $Q_1 \in \mathbb{K}[L]$ if and only if $(L+1) \mid (A_1 - Q_0)$. This is Theorem 9.24 (2) for the present B.

To turn the two conditions into evaluations, note that for $P \in \mathbb{K}[L]$ one has $(L+1) \mid P$ if and only if $P(-1) = 0$. The first condition is therefore $A_0(-1) = 0$. Writing $A_0 = (L+1)Q_0$ and differentiating, $A_0' = Q_0 + (L+1)Q_0'$, so $Q_0(-1) = A_0'(-1)$. The second condition, $A_1(-1) = Q_0(-1)$, becomes

$$\boxed{A_0(-1) = 0 \qquad \text{and} \qquad A_1(-1) = A_0'(-1).}$$

Both are conditions at the single point $L = -1$, the root of the leading block. Two remarks. First, the conditions are not vacuous: for $A = 1$ the first already fails, which is another proof that B is not a unit. Second, the pattern continues: $Q_n \in \mathbb{K}[L]$ requires $(L+1) \mid (A_n - Q_{n-1})$, so the whole quotient lies in $\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ exactly when $A \in B\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$, which is Theorem 9.24 (3). □

Exercise E.18 (Repeated derivatives of one block, three ways)**.**

1. Differentiate $A = X^2 + L^3 + (L^2 - 1)X^{-2}$ once, using the block formula of Theorem 11.3.
2. Compute $D_X^k(L^3)$ for $k = 1, 2, 3, 4$ using Theorem 11.8.
3. Recompute $D_X^4(L^3)$ from the Stirling expansion $\Delta_{0,k} = \sum_{j=0}^{k} s(k,j)\, \partial_L^j$ of Theorem 11.9, and check the two answers agree.

Solution. (1) Theorem 11.3 is $D_X\big(t^n p(L)\big) = t^{n+1}(p' - np)$. Apply it to the three blocks, with $n = -2, 0, 2$:

$$\begin{aligned} D_X\big(t^{-2}\big) &= t^{-1}(0+2) = 2X, \\ D_X\big(L^3\big) &= t^1\big(3L^2 - 0\big) = 3L^2 t, \\ D_X\big(t^2(L^2-1)\big) &= t^3\big(2L - 2(L^2-1)\big) = \big(-2L^2 + 2L + 2\big)t^3. \end{aligned}$$

Hence $D_X A = 2X + 3L^2 X^{-1} + (-2L^2 + 2L + 2)X^{-3}$.

(2) For a t^0-block, $D_X^k p = t^k \Delta_{0,k} p$ with $\Delta_{0,k} = (\partial_L - k + 1) \cdots (\partial_L - 1)\partial_L$. With $p = L^3$:

$$\Delta_{0,1} L^3 = 3L^2,$$

$$\begin{aligned}
\Delta_{0,2}L^3 &= (\partial_L - 1)(3L^2) = 6L - 3L^2,\\
\Delta_{0,3}L^3 &= (\partial_L - 2)(6L - 3L^2) = 6 - 6L - 12L + 6L^2 = 6L^2 - 18L + 6,\\
\Delta_{0,4}L^3 &= (\partial_L - 3)(6L^2 - 18L + 6) = 12L - 18 - 18L^2 + 54L - 18 = -18L^2 + 66L - 36.
\end{aligned}$$

So $D_X L^3 = 3L^2 t$, $D_X^2 L^3 = (6L - 3L^2)t^2$, $D_X^3 L^3 = (6L^2 - 18L + 6)t^3$, $D_X^4 L^3 = (-18L^2 + 66L - 36)t^4$.

(3) The signed Stirling numbers of the first kind in row 4 are $s(4,0) = 0$, $s(4,1) = -6$, $s(4,2) = 11$, $s(4,3) = -6$, $s(4,4) = 1$. Applying $\sum_j s(4,j)\,\partial_L^j$ to L^3, whose derivatives are $3L^2, 6L, 6, 0$,

$$-6 \cdot 3L^2 + 11 \cdot 6L - 6 \cdot 6 + 1 \cdot 0 = -18L^2 + 66L - 36,$$

which agrees with (2). Equivalently $X^4 \dfrac{\mathrm{d}^4}{\mathrm{d}X^4}(\log X)^3 = \sum_{j=0}^4 s(4,j)\,\big((\log X)^3\big)^{(j)}$; note that the sum must start at $j = 0$, because $s(k,0) = 0$ only for $k \geq 1$ and equals 1 at $k = 0$. □

Exercise E.19 (Antiderivatives of a single block, resonant and not)**.** Find polynomial-logarithmic antiderivatives of

$$F_1 = t^2(L+1) \qquad \text{and} \qquad F_2 = t(L+1),$$

and verify each by differentiating. Explain why the two computations look different, and identify the block index at which the difference occurs.

Solution. By Theorem 11.3, $D_X\big(t^n Q\big) = t^{n+1}(Q' - nQ)$, so an antiderivative of $t^{n+1}P$ is $t^n Q$ with $Q' - nQ = P$.

$F_1 = t^2(L+1)$, *so* $n = 1$. Solve $Q' - Q = L + 1$ in $\mathbb{K}[L]$. With $Q = \alpha L + \beta$: $\alpha - \alpha L - \beta = L + 1$, so $\alpha = -1$ and $\alpha - \beta = 1$, i.e. $\beta = -2$. Thus $Q = -L - 2$ and

$$\int F_1 \,\mathrm{d}X = -t(L+2) + c, \qquad c \in \mathbb{K}.$$

Check: $D_X\big(-t(L+2)\big) = t^2\big(-1 - (-(L+2))\big) = t^2(L+1)$.

$F_2 = t(L+1)$, *so* $n = 0$. Now the equation is $Q' = L + 1$, with solution $Q = \frac{1}{2}L^2 + L$ up to a constant, and

$$\int F_2 \,\mathrm{d}X = \tfrac{1}{2}L^2 + L + c.$$

Check: $D_X\big(\frac{1}{2}L^2 + L\big) = t\big(L + 1 - 0\big) = t(L+1)$.

The difference is the operator being inverted. For $n \neq 0$ the map $Q \mapsto Q' - nQ$ is bijective on $\mathbb{K}[L]$: it is triangular with diagonal $-n \neq 0$ in the monomial basis, and the explicit inverse is the terminating sum $Q = -\sum_{j \geq 0} P^{(j)}/n^{j+1}$ of (11.14). This is the nonresonant case of Theorem 11.15, and it *lowers* the L-degree, or at worst preserves it. For $n = 0$ the map is $Q \mapsto Q'$, which is surjective on $\mathbb{K}[L]$ but has kernel $\mathbb{K}$ and *raises* the degree by one; this is the resonant block of Theorem 11.16, and it is the only place where the log-degree profile of an antiderivative exceeds that of its derivative, as recorded in (11.15). The resonant index is the one whose outer exponent in F is 1, i.e. the block t^1 of the integrand. □

Exercise E.20 (Counterexample: surjectivity of D_X is not inherited)**.** By Theorem 11.18 the derivation D_X maps $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ onto $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$, with kernel $\mathbb{K}$. A plausible strengthening is that it maps each of the four classes onto itself. Refute this. Show that the element

$$F = \frac{t}{L} = \frac{1}{X \log X},$$

which lies in $\mathscr{PL}_{\mathrm{rat},\mathbb{K}}$ and in $\mathscr{PL}_{\mathrm{itLaur},\mathbb{K}}$, has no antiderivative in $\mathscr{PL}_{\mathrm{itLaur},\mathbb{K}}$, and identify the antiderivative it does have.

Solution. Suppose $G = \sum_n g_n(L)t^n \in \mathscr{PL}_{\mathrm{itLaur},\mathbb{K}}$, so each $g_n \in \mathbb{K}((L^{-1}))$, satisfies $D_X G = t/L$. By Theorem 11.3, extended over the coefficient ring $\mathbb{K}((L^{-1}))$, the block of $D_X G$ at t^{n+1} is $g_n' - ng_n$, where $' = \partial_L$. Comparing at t^1, i.e. $n = 0$, gives

$$g_0' = \frac{1}{L}.$$

Now ∂_L maps $\mathbb{K}((L^{-1}))$ into itself, and for $g_0 = \sum_{k\geq k_0} c_k L^{-k}$ one has $g_0' = \sum_{k\geq k_0}(-k)c_k L^{-k-1}$. The coefficient of L^{-1} in g_0' is the one with $-k-1=-1$, i.e. $k=0$, and it equals $-0\cdot c_0 = 0$. So L^{-1} is never in the image: the residue at $L=\infty$, $\operatorname{res}(g) := [L^{-1}]g$, vanishes identically on derivatives (Theorem 10.23), and $\operatorname{res}(t/L) = 1 \neq 0$. Hence no such g_0 exists, and F has no antiderivative in $\mathscr{PL}_{\mathrm{itLaur},\mathbb{K}}$; a fortiori none in $\mathscr{PL}_{\mathrm{rat},\mathbb{K}}$ (Theorem 10.24).

The obstruction is exactly one function, the residue, and it is met by adjoining one new symbol: $\Lambda = \log\log X$ satisfies $D_X\Lambda = t/L$, because $\frac{\mathrm{d}}{\mathrm{d}X}\log\log X = \frac{1}{X\log X}$. This is Theorem 11.22, and it is where the second logarithmic level of the whole theory comes from: not from composition, but from integration. Note that no contradiction with Theorem 11.18 arises, since $t/L \notin \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$. □

Exercise E.21 (Which formal exponentials stay in the class?)**.** Decide, for each F below, whether $\exp F \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$, and if not, name the smallest enlargement in which it lives:

$$L, \qquad 2L+t, \qquad t, \qquad \tfrac{1}{2}L, \qquad L^2, \qquad t^{-1}.$$

State the criterion you used and check it in each case.

Solution. The criterion is Theorem 10.16: $\exp F \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ if and only if $\nu_t F \geq 0$ and $[t^0]F = a + mL$ with $m \in \mathbb{Z}$, $a \in \mathbb{K}$ and $\exp a \in \mathbb{K}$; and then $\exp F = (\exp a)t^{-m}\exp\big(F - [t^0]F\big)$ with the last factor in $1 + t\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$.

$F = L$: $\nu_t = 0$, $[t^0]F = 0 + 1\cdot L$, $m = 1$, $a = 0$. Yes: $\exp L = t^{-1} = X$.

$F = 2L + t$: $\nu_t = 0$, $[t^0]F = 2L$, $m = 2$, $a = 0$. Yes: $\exp F = t^{-2}\exp(t) = X^2\sum_{k\geq 0} t^k/k! \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$.

$F = t$: $\nu_t = 1 \geq 0$, $[t^0]F = 0$, $m = 0$, $a = 0$. Yes, and $\exp t \in 1 + t\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$.

$F = \frac{1}{2}L$: $m = \frac{1}{2} \notin \mathbb{Z}$. No; $\exp F = X^{1/2}$ needs the Puiseux–log ring with exponent group $\frac{1}{2}\mathbb{Z}$ (Theorem 10.17).

$F = L^2$: $[t^0]F = L^2$ is not affine in L. No. Here the failure is not repaired by any coefficient-field or exponent-group enlargement of a power–log scale: $\exp(L^2) = X^{\log X}$ grows faster than every power of X and slower than $\exp(cX)$, so it is a new transmonomial (Theorem 10.15).

$F = t^{-1} = X$: $\nu_t = -1 < 0$. No; e^X is an exponential transmonomial, and the smallest home is a logarithmic-exponential transseries field.

A remark worth extracting: the three failures are of three different kinds — a fractional slope, a nonaffine t^0-block, a negative outer order — and Theorem 10.15 says these are the only three. □

E.4 Composition

Exercise E.22 (Substituting a constant shift into one monomial)**.** Let $a \in \mathbb{K}$ and $G = X + a$. Compute $(tL)\circ G$ through outer order t^3, and then $(tL)\circ(cX)$ exactly, for $c \in \mathbb{K}^\times$ with $\log c \in \mathbb{K}$.

Solution. By Theorem 12.22, with $H = a$,

$$t\circ G = \frac{t}{1+at} = t - at^2 + a^2t^3 + \mathcal{O}_t^{\mathrm{form}}(t^4)\,, \qquad L\circ G = L + \log(1+at) = L + at - \frac{a^2}{2}t^2 + \mathcal{O}_t^{\mathrm{form}}(t^3)\,.$$

Composition is a ring homomorphism (Theorem 12.7), so $(tL)\circ G = (t\circ G)(L\circ G)$, and multiplying,

$$(tL)\circ(X+a) = \big(t - at^2 + a^2t^3\big)\Big(L + at - \tfrac{a^2}{2}t^2\Big) + \mathcal{O}_t^{\mathrm{form}}\big(t^4\big)$$

$$= tL + a(1-L)t^2 + a^2\Big(L - \tfrac{3}{2}\Big)t^3 + \mathcal{O}_t^{\mathrm{form}}\big(t^4\big)\,.$$

For the dilation $G = cX$ the substitution is exact and finite: $t \circ G = c^{-1}t$ and $L \circ G = L + \log c$, so

$$(tL) \circ (cX) = c^{-1}t\,(L + \log c),$$

and more generally $F \circ (cX) = \sum_n c^{-n} p_n(L + \log c)t^n$ for $F = \sum_n p_n(L)t^n$. Two things are visible here that matter later. First, a dilation changes *both* generators, the power by c^{-n} and the logarithm by the shift $\log c$; forgetting the second is the failure mode of Theorem E.41. Second, the identity is a statement about $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ only if $\log c \in \mathbb{K}$; over $\mathbb{K} = \mathbb{Q}$ and $c = 2$ the right-hand side does not live in $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ at all, and the field must be enlarged first. □

Exercise E.23 (A composition through two blocks)**.** Let $F(Y) = Y + (\log Y)^2$ and $G(X) = X - L$. Compute $F \circ G$ through outer order t^2.

Solution. Write $F = t^{-1} + L^2$ in its own generators and $G = X + H$ with $H = -L$, so $H \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ and $G \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}$. By Theorem 12.22,

$$L \circ G = L + \log(1 - Lt) = L - Lt - \tfrac{1}{2}L^2t^2 + \mathcal{O}_t^{\mathrm{form}}\big(t^3\big)\,.$$

Therefore

$$\begin{aligned}(L \circ G)^2 &= \Big(L - Lt - \tfrac{1}{2}L^2t^2\Big)^2 + \mathcal{O}_t^{\mathrm{form}}\big(t^3\big)\\ &= L^2 - 2L^2t + \big(L^2 - L^3\big)t^2 + \mathcal{O}_t^{\mathrm{form}}\big(t^3\big)\,,\end{aligned}$$

the t^2-block being $2 \cdot L \cdot (-\tfrac{1}{2}L^2) + (-L)^2$. Adding $G = X - L$,

$$F \circ G = X + \big(L^2 - L\big) - \frac{2L^2}{X} + \frac{L^2 - L^3}{X^2} + \mathcal{O}_t^{\mathrm{form}}\big(t^3\big)\,.$$

A check on the shape: $\nu_t(F \circ G) = \nu_t F = -1$, as Theorem 12.22 requires, and no block acquired a degree in L larger than 2+ (the number of substituted logarithms), consistent with Theorem 12.26. □

Exercise E.24 (Composition is multiplicative, from the generators)**.** Let $G = X + H$ with $H \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$, so $G \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}$. Prove that

$$(F\Phi) \circ G = (F \circ G)(\Phi \circ G) \qquad \text{for all } F, \Phi \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}},$$

directly from the exact substitution formula (12.17), without invoking Theorem 12.7.

Solution. Let $F = \sum_{n \ge n_0} p_n(L)t^n$ and $\Phi = \sum_{m \ge m_0} q_m(L)t^m$, and let $G = X + H \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}$. Write

$$\tau := t \circ G = \frac{t}{1 + tH}, \qquad \lambda := L \circ G = L + \log(1 + tH),$$

so that (12.17) reads $F \circ G = \sum_n \tau^n p_n(\lambda)$. Two facts are needed.

(a) The families are summable. Since $\nu_t(\tau) = 1$ and $\nu_t(\lambda - L) \ge 1$, the summand $\tau^n p_n(\lambda)$ has outer order exactly n when $p_n \ne 0$, and is zero otherwise (Theorem 12.22); in either case its order is at least n, so $\{n : \nu_t < N\}$ is finite for every N and Theorem 7.21 is satisfied. The same holds for Φ and for the Cauchy product.

(b) Multiplication distributes over summable families. This is Theorem 7.22 (iv): if (S_i) and (T_j) are summable then so is $(S_iT_j)_{(i,j)}$ and $\big(\sum_i S_i\big)\big(\sum_j T_j\big) = \sum_{i,j} S_iT_j$.

Now compute. Let $r_N = \sum_{n+m=N} p_nq_m$ be the blocks of $F\Phi$. By (b),

$$(F \circ G)(\Phi \circ G) = \Big(\sum_n \tau^n p_n(\lambda)\Big)\Big(\sum_m \tau^m q_m(\lambda)\Big) = \sum_{n,m} \tau^{n+m}\, p_n(\lambda)q_m(\lambda).$$

Group by $N = n + m$; the inner sums are finite because for fixed N only finitely many pairs (n, m) with $n \ge n_0$, $m \ge m_0$ satisfy $n + m = N$. Since evaluation of a polynomial at the fixed element $\lambda \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ is a ring homomorphism $\mathbb{K}[L] \to \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$, $\sum_{n+m=N} p_n(\lambda) q_m(\lambda) = r_N(\lambda)$. Hence

$$(F \circ G)(\Phi \circ G) = \sum_N \tau^N r_N(\lambda) = (F\Phi) \circ G,$$

the last equality being (12.17) applied to $F\Phi$. Additivity is the same computation with one index, and $1 \circ G = 1$; so $\Phi_G \colon F \mapsto F \circ G$ is a $\mathbb{K}$-algebra endomorphism of $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$. □

Exercise E.25 (The sharp Taylor budget, in the exclusive convention)**.** Let $F \in \mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ with $\nu_t F \ge a$, let $H \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ with $\nu_t H \ge h \ge 0$, and let $N \in \mathbb{Z}$.

1. Prove that $\mathrm{Trunc}_{<N}\big(F \circ (X + H)\big)$ is unchanged if the Taylor family $\sum_{k \ge 0} H^k D_X^k F/k!$ is truncated after the index $k = \lfloor (N - 1 - a)/(h + 1) \rfloor$.
2. Show the bound cannot be lowered, by exhibiting F, H, N for which the last retained term does change the answer.
3. A source states the same budget as $k \le \lfloor (N - a)/(h + 1) \rfloor$. Explain the discrepancy and say whether it is harmful.

Solution. (1) By Theorem 11.10, $\nu_t(D_X^k F) \ge a + k$, and $\nu_t(H^k) \ge kh$. So the k-th Taylor term has

$$\nu_t\Big(\frac{H^k}{k!} D_X^k F\Big) \;\ge\; a + k + kh = a + k(h + 1).$$

Such a term is invisible to $\mathrm{Trunc}_{<N} \cdot$ as soon as $a + k(h + 1) \ge N$, i.e. $k \ge (N - a)/(h + 1)$. The retained indices are therefore those with $a + k(h + 1) < N$, that is $k(h + 1) < N - a$, that is $k \le \lfloor (N - 1 - a)/(h + 1) \rfloor$ for integers. Equivalently the number of retained terms is $K = \lceil (N - a)/(h + 1) \rceil$, which is Theorem 27.31.

(2) Take $F = L$ (so $a = 0$), $H = 1$ (so $h = 0$) and $N = 3$. The budget retains $k \le 2$. Now $D_X L = t$, $D_X^2 L = D_X(t) = t^2(0 - 1) = -t^2$, $D_X^3 L = D_X(-t^2) = -t^3(0 - 2) = 2t^3$, so the Taylor sum is

$$L + t - \tfrac{1}{2}t^2 + \tfrac{1}{3}t^3 - \cdots,$$

which is exactly $L \circ (X + 1) = L + \log(1 + t)$, as it must be. The term $k = 2$ contributes $-\frac{1}{2}t^2$, a block strictly below the cutoff $N = 3$; dropping it changes $\mathrm{Trunc}_{<3} \cdot$. So the last retained index is genuinely needed, and the bound is sharp.

(3) The source is working with the inclusive cutoff $[F]_{\le N}$. Since $[F]_{\le N} = \mathrm{Trunc}_{<N+1} F$, its N is our $N - 1$, and its budget $k \le \lfloor (N - a)/(h + 1) \rfloor$ becomes, in our normalisation, exactly $k \le \lfloor (N - 1 - a)/(h + 1) \rfloor$: the two agree. Read as an exclusive bound, however, the source's formula retains one term too many whenever $(h + 1) \mid (N - a)$. That is harmless for *correctness* — the extra term is invisible below t^N — but not for cost, and not for a routine that uses the budget in reverse to decide how much of H to compute. This is the discrepancy recorded at Theorem 27.31, whose last line prints the exclusive form $k \le \lfloor (N - 1 - a)/(h + 1) \rfloor$ explicitly for this reason. □

Exercise E.26 (Trap: how much of the inner argument is needed?)**.** A routine must return $\mathrm{Trunc}_{<1}\big(F \circ (X + H)\big)$ for $F = X^2$. It argues: "the output is requested through outer order t^0; composition is filtration-continuous (Theorem 12.24), so H is needed only through outer order t^0, i.e. only $\mathrm{Trunc}_{<1} H$." Compute $\mathrm{Trunc}_{<1}\big(F \circ (X + H)\big)$ for $H = h_0 + h_1 t + h_2 t^2 + \cdots$, find the error, and state the correct input requirement for a general F.

Solution. Here $F = t^{-2}$, and composition with a near-identity map is exact:

$$F \circ (X + H) = (X + H)^2 = t^{-2} + 2t^{-1}H + H^2,$$

so, block by block,

$$[t^{-2}] = 1, \qquad [t^{-1}] = 2h_0, \qquad [t^0] = 2h_1 + h_0^2,$$

and $\mathrm{Trunc}_{<1}\big(F \circ (X+H)\big) = X^2 + 2h_0X + \big(2h_1 + h_0^2\big)$. The block h_1 enters. The routine's answer would omit it, and would be wrong by $2h_1$.

The error is a confusion of absolute with relative precision. Filtration continuity says $\nu_t\big(F \circ (X+B) - F \circ (X+C)\big) \geq \nu_t(F) + \nu_t(B-C) + 1$ (Theorem 14.6), and the summand $\nu_t(F)$ is negative here. Unwinding: a perturbation of H at outer order j moves the composite at outer order $a+j+1$ with $a = \nu_t F$. It is therefore invisible below t^N exactly when $a+j+1 \geq N$, so the required input is

$$\mathrm{Trunc}_{<\,N-a-1}\, H.$$

For $F = X^2$, $a = -2$ and $N = 1$: $\mathrm{Trunc}_{<2}\, H$, one block more than the routine used. Only when $\nu_t F \geq -1$ does $\mathrm{Trunc}_{<N}\, H$ suffice. This is the sharp statement of Theorem 12.10, and the general principle behind it is Theorem 27.11: what a routine conserves is relative, not absolute, precision. □

Exercise E.27 (Counterexample: the near-identity group is not abelian)**.** Let $f, g \in \mathbb{K}[L]$, $F = X + f(L)$, $G = X + g(L)$.

1. Compute the first block of $F \circ G - G \circ F$.
2. Deduce that $\mathcal{G}_{\mathrm{poly},\mathbb{K}}$ is not abelian, and characterise the pairs (f,g) for which F and G commute to first order.
3. Show that vanishing of the first block does *not* imply commuting, by computing the next block for $f = L$, $g = 2L$.

Solution. (1) By Theorem 12.23, $F \circ G = X + g + f \circ (X+g)$, and by Theorem 14.6, (14.7), $f \circ (X+g) - f \equiv g\, D_X f \pmod{t^2 \mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}}$, where $D_X f = f'(L)t$. Hence

$$F \circ G = X + f + g + gf'\,t + \mathcal{O}_t^{\mathrm{form}}\big(t^2\big)\,,$$

and by symmetry $G \circ F = X + f + g + fg'\,t + \mathcal{O}_t^{\mathrm{form}}\big(t^2\big)$. Subtracting,

$$F \circ G - G \circ F = \big(gf' - fg'\big)t + \mathcal{O}_t^{\mathrm{form}}\big(t^2\big)\,, \tag{E.2}$$

which is Theorem 13.18.

(2) Take $f = L$, $g = L^2$: then $gf' - fg' = L^2 - 2L^2 = -L^2 \neq 0$, so $F \circ G \neq G \circ F$ and $\mathcal{G}_{\mathrm{poly},\mathbb{K}}$ is not abelian, although it is associative (Theorem 12.27). The first block vanishes if and only if $gf' = fg'$, i.e. $(f/g)' = 0$ in $\mathbb{K}(L)$ for $g \neq 0$, i.e. f and g are proportional over $\mathbb{K}$. Equivalently, the vector fields $f\partial_L$ and $g\partial_L$ commute.

(3) Take the proportional pair $f = L$, $g = 2L$, for which $gf' - fg' = 2L - 2L = 0$, so the first block of the commutator vanishes. The maps still do not commute. Indeed by Theorem 12.22,

$$(X+L) \circ (X+2L) = X + 3L + 2Lt - 2L^2t^2 + \mathcal{O}_t^{\mathrm{form}}\big(t^3\big)\,,$$
$$(X+2L) \circ (X+L) = X + 3L + 2Lt - L^2t^2 + \mathcal{O}_t^{\mathrm{form}}\big(t^3\big)\,,$$

the two t^2-blocks coming from $\log(1+2Lt)$ and from $2\log(1+Lt)$ respectively. Hence

$$F \circ G - G \circ F = -L^2t^2 + 2L^3t^3 - \tfrac{7}{2}L^4t^4 + \mathcal{O}_t^{\mathrm{form}}\big(t^5\big)\,.$$

So proportionality kills the first block only. Structurally this is exactly what Theorem 14.16 predicts: the graded Lie algebra of the contact filtration is not abelian, and vanishing of the bracket of the leading symbols only pushes the commutator one step down the filtration (Theorem 14.14). □

Exercise E.28 (A scale change that produces a second logarithm exactly)**.** Let $F(X) = X + \log X + X^{-1}$ and $G(X) = 3X^2L^{-1}$. Compute $F \circ G$ exactly, and say which class each summand lives in. Then do the same for $F(Y) = \log Y$, $G(X) = X^2(\log X)^3$, and identify the smallest new symbol forced in each case.

Solution. First pair. $G = 3t^{-2}L^{-1} \in \mathscr{PL}_{\mathrm{rat},\mathbb{K}}$ is monomial-leading with $\rho = 2$, $c = 3$, $\sigma = -1$ and no correction. Composition acts on the generators by $t \mapsto G^{-1}$ and $L \mapsto \log G$, and here both are exact and finite:

$$G^{-1} = \tfrac{1}{3}Lt^2, \qquad \log G = \log 3 + 2L - \log L.$$

Hence

$$F \circ G = 3X^2L^{-1} + 2L - \log L + \log 3 + \tfrac{1}{3}Lt^2,$$

a *finite* expression. The summands $3X^2L^{-1}$ and $\frac{1}{3}Lt^2$ lie in $\mathscr{PL}_{\mathrm{rat},\mathbb{K}}$; $2L$ lies in $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$; $\log 3$ requires $\log 3 \in \mathbb{K}$; and $\log L = \log\log X$ lies in no Puiseux–log ring over $\mathbb{K}[L]$, by the residue argument of Theorem E.20. The point the source makes is worth keeping: the second logarithmic level appears here *exactly*, in a closed-form finite composite, not merely inside a remainder.

Second pair. $\log G = 2L + 3\log L$, so again the new symbol is $L_2 = \log L = \log\log X$, and the output lies in a two-level coefficient class, not in $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$. This is item (3) of Theorem 12.1 and Theorem 15.8.

In both cases the forced symbol is read off from the images of the generators alone: G^{-1} decides whether the exponent group must grow, $\log G$ decides whether the logarithmic depth must grow. That is the whole content of the admissibility checklist of Part 15. □

Exercise E.29 (Counterexample: a monomial-leading inner argument is not enough)**.** Theorem 12.7 composes F with an inner argument $G = cX^{\rho}(1 + U)$ subject to $\rho > 0$. Show that the hypothesis $\rho > 0$ cannot be weakened to $\rho \geq 0$. Precisely: write $\tau = t \circ G$, $\lambda = L \circ G$, and suppose $\nu_t\tau = 0$. Prove that the substitution family $\big(\tau^n p_n(\lambda)\big)_n$ fails to be summable as soon as $p_n(\lambda) \neq 0$ for infinitely many n, whatever the coefficient field; exhibit such an F; and identify the exact failure. Then show that the quantifier cannot be improved to "every F with infinite coefficient support".

Solution. Suppose G is such that $\tau := t \circ G$ has $\nu_t\tau = 0$; the simplest instance is $G = c \in \mathbb{K}^{\times}$, where $\tau = c^{-1}$ and $\lambda = \log c \in \mathbb{K}$. Let $F = \sum_{n \geq n_0} p_n(L)t^n$. The defining formula for the composite is the sum of the family $\big(\tau^n p_n(\lambda)\big)_n$. Since $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ is a domain and $\nu_t(\tau^n) = n\nu_t(\tau) = 0$,

$$\nu_t\big(\tau^n p_n(\lambda)\big) = \nu_t\big(p_n(\lambda)\big) \qquad \text{for every } n.$$

So no member of the family can be pushed to a high outer order by the factor τ^n: the whole family sits at the orders of the substituted coefficients. For the constant inner argument $G = c$ those are scalars, so every nonzero member sits in the single block t^0. Hence if $p_n(\log c) \neq 0$ for infinitely many n — take $F = 1/(1-t)$, where $p_n = 1$ for all $n \geq 0$ — the set $\{n : \nu_t < 1\}$ is infinite, Theorem 7.21 fails, and the t^0-block of the putative composite would be

$$\sum_{n \geq n_0} c^{-n} p_n(\log c),$$

an infinite sum of scalars, which is not an element of $\mathbb{K}$ unless it happens to terminate. No enlargement of the coefficient field repairs this: the obstruction is that infinitely many summands land in one block, which is a statement about the outer filtration alone. The same argument applies to any G with $\rho = 0$, since then $\nu_t(t \circ G) = 0$.

The quantifier. It is not enough that F have infinite coefficient support: what must be infinite is the set of indices surviving the substitution. With $G = c$ take

$$F = \frac{L - \log c}{1 - t} = \sum_{n \geq 0} (L - \log c)t^n,$$

whose support $\{(n,0),(n,1) : n \geq 0\}$ is infinite. Every substituted coefficient $p_n(\lambda) = \log c - \log c$ vanishes, the family is $(0,0,\dots)$, which is vacuously summable, and $F \circ c = 0$ is perfectly well defined. The honest statement is the one proved above, with the hypothesis on $p_n(\lambda)$ and not on F.

For contrast, when $\rho > 0$ one has $\nu_t(\tau) = \rho > 0$ and $\nu_t(\tau^n) = n\rho \to +\infty$, so the family is summable — in a Hahn ambient with exponent group containing $\rho\mathbb{Z}$, which is Theorem 12.17. This is why $\rho > 0$ is a hypothesis and not a normalisation. □

E.5 Reversion

Exercise E.30 (The first-order form of the inverse, and its sharp error)**.** Let $H \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ with $q = \nu_t(H)$. Show

$$(X+H)^{\langle -1\rangle_\circ} = X - H + \mathcal{O}_t^{\mathrm{form}}\big(t^{2q+1}\big)\,,$$

and show that the exponent $2q+1$ is attained for $H = L$.

Solution. By Theorem 16.11, $(X+H)^{\langle -1\rangle_\circ} = X + B$ where B is the t-adic limit of the Picard iterates $B_0 = 0$, $B_{j+1} = -H \circ (X + B_j)$, and $\nu_t(B - B_j) \geq (j+1)q + j$. Taking $j = 1$, so that $B_1 = -H \circ X = -H$,

$$\nu_t\big(B + H\big) \;\geq\; 2q+1,$$

which is the claim.

Sharpness at $H = L$: here $q = 0$ and the claim is $(X+L)^{\langle -1\rangle_\circ} = X - L + \mathcal{O}_t^{\mathrm{form}}(t)$. By Theorem 20.10 the inverse is $X - L + \mathsf{L}_1^{\mathrm{Lam}}(L)t + \cdots = X - L + Lt + \cdots$, so $B + L = Lt + \cdots$ has outer order exactly $1 = 2\cdot 0 + 1$.

Two remarks. First, the estimate is a statement about ν_t only: the correction $-H$ is exact in the leading block by Theorem 16.11 (4), and everything beyond it is error. Second, the same proof gives the general Picard rate: j iterations from $B_0 = 0$ are correct through outer block $(j+1)q + j - 1$, so each further iteration gains $q+1$ blocks. □

Exercise E.31 (The universal inverse blocks through order two)**.** Let $a, b, c \in \mathbb{K}[L]$ and $F = X + a(L) + b(L)t + c(L)t^2 + \mathcal{O}_t^{\mathrm{form}}\big(t^3\big)$, so $F \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}$. Write $F^{\langle -1\rangle_\circ} = X + b_0 + b_1 t + b_2 t^2 + \mathcal{O}_t^{\mathrm{form}}\big(t^3\big)$. Derive b_0, b_1, b_2 as universal differential polynomials in a, b, c.

Solution. Put $G = X + B$ with $B = b_0 + b_1 t + b_2 t^2 + \cdots$ and impose $F \circ G = X$. By Theorem 12.22,

$$L \circ G = L + \log(1+tB) = L + b_0 t + \Big(b_1 - \tfrac12 b_0^2\Big)t^2 + \mathcal{O}_t^{\mathrm{form}}\big(t^3\big)\,, \qquad t \circ G = t\big(1 - b_0 t + \mathcal{O}_t^{\mathrm{form}}\big(t^2\big)\big).$$

Write $\delta := (L \circ G) - L$, of outer order ≥ 1. Expanding each coefficient by Theorem 27.33, a finite Taylor sum:

$$\begin{aligned}
a(L \circ G) &= a + a'\delta + \tfrac12 a''\delta^2 + \mathcal{O}_t^{\mathrm{form}}\big(t^3\big)\\
&= a + a'b_0 t + \Big(a'\big(b_1 - \tfrac12 b_0^2\big) + \tfrac12 a'' b_0^2\Big)t^2 + \mathcal{O}_t^{\mathrm{form}}\big(t^3\big)\,,\\
b(L \circ G)\cdot(t \circ G) &= \big(b + b'b_0 t\big)\cdot t\big(1 - b_0 t\big) + \mathcal{O}_t^{\mathrm{form}}\big(t^3\big)\\
&= bt + \big(b'b_0 - bb_0\big)t^2 + \mathcal{O}_t^{\mathrm{form}}\big(t^3\big)\,,\\
c(L \circ G)\cdot(t \circ G)^2 &= ct^2 + \mathcal{O}_t^{\mathrm{form}}\big(t^3\big)\,.
\end{aligned}$$

Adding these to $G = X + B$ and equating $F \circ G$ to X block by block:

$$\begin{aligned}
t^0 :&\quad b_0 + a = 0,\\
t^1 :&\quad b_1 + a'b_0 + b = 0,\\
t^2 :&\quad b_2 + a'\big(b_1 - \tfrac12 b_0^2\big) + \tfrac12 a'' b_0^2 + b'b_0 - bb_0 + c = 0.
\end{aligned}$$

Solving in order, with $b_0 = -a$ and $b_1 = aa' - b$,

$$b_0 = -a, \qquad b_1 = aa' - b, \qquad b_2 = \frac{a^2a'}{2} - \frac{a^2a''}{2} - a(a')^2 + a'b + ab' - ab - c. \tag{E.3}$$

Three consistency checks. For $b = c = 0$ this is $b_2 = \frac{1}{2}a^2a' - a(a')^2 - \frac{1}{2}a^2a''$, the form used in Theorem E.33. For $a = \alpha L$ with $\alpha \in \mathbb{K}$ constant and $b = c = 0$ it gives $\alpha^3(L^2/2 - L) = \alpha^3 \mathsf{L}_2^{\mathrm{Lam}}(L)$, matching Theorem 18.23. For a, b constants and $c = 0$ it gives $b_2 = -ab$, matching (17.14) at $\deg_{\mathrm{L}} a = 0$. ☐

Exercise E.32 (Trap: an incomplete universal formula)**.** A source records the third universal inverse block of $F = X + a(L) + b(L)t + c(L)t^2$ as

$$b_2 \overset{?}{=} \frac{1}{2}a^2a' - a(a')^2 - ab + a'b,$$

and verifies it on the example $a = 2L$, $b = 3$, $c = 0$, where it is correct. Show that the formula is nevertheless false in general, by producing three independent witnesses — one for each missing term — and state the correct formula. Explain why the source's verification could not have detected the error.

Solution. Comparing with (E.3), the quoted formula omits $-\frac{1}{2}a^2a''$, $+ab'$ and $-c$. Each is detected by a separate witness.

Missing $-\frac{1}{2}a^2a''$. Take $a = L^2$, $b = c = 0$. Then $a' = 2L$, $a'' = 2$, and (E.3) gives

$$b_2 = \frac{L^4 \cdot 2L}{2} - \frac{L^4 \cdot 2}{2} - L^2 \cdot 4L^2 = L^5 - 5L^4,$$

whereas the quoted formula gives $L^5 - 4L^4$. The discrepancy is $-L^4$. The correct value is confirmed independently by Theorem 18.23, which for $P = L^2$ predicts $\deg_{\mathrm{L}} b_2 = d(N+1) - 1 = 5$ with leading coefficient $(d/N)a_d^{N+1} = 2/2 = 1$; both hold for $L^5 - 5L^4$.

Missing $+ab'$. Take $a = 1$, $b = L$, $c = 0$. Then (E.3) gives $b_2 = -ab + ab' = -L + 1$, while the quoted formula gives $-L$.

Missing $-c$. Take $a = b = 0$, $c = 1$. Then $F = X + t^2$, and $F^{\langle -1 \rangle_\circ} = X - t^2 + \mathcal{O}_t^{\mathrm{form}}(t^3)$ directly by Theorem E.30; so $b_2 = -1$, while the quoted formula gives 0.

The source's verification could not detect any of this because its single test case has $a'' = 0$ (its a is affine in L), $b' = 0$ (its b is a constant) and $c = 0$. All three omitted terms vanish simultaneously on that example. This is the standard hazard of validating a universal formula on one instance; the residual certificate of Theorem 16.16 is the correct test, because it is run against F itself and is an identity in $\mathbb{K}[L][[t]]$, not a numerical sample. ☐

Exercise E.33 (Reversion of a quadratic logarithmic perturbation)**.** Reverse $F = X + L^2$ through outer order t^3, and check the result against the degree law of Theorem 18.22 at $N = 1, 2, 3$.

Solution. Use the Lagrange–Bürmann formula in the form Theorem 18.23: for $F = X + P(L)$ only the summand $r = N + 1$ contributes to b_N, so

$$b_N = \frac{(-1)^{N+1}}{(N+1)!}\Big[\prod_{s=0}^{N-1}(\partial_L - s)\Big]P^{N+1}.$$

With $P = L^2$:

$N = 0$: $b_0 = -P = -L^2$.

$N = 1$: $b_1 = \frac{1}{2}\partial_L(P^2) = PP' = L^2 \cdot 2L = 2L^3$.

$N = 2$: $b_2 = -\frac{1}{6}(\partial_L - 1)\partial_L(P^3)$. Now $\partial_L(P^3) = 3P^2P'$ and $(\partial_L - 1)(3P^2P') = 6P(P')^2 + 3P^2P'' - 3P^2P'$, so

$$b_2 = \tfrac{1}{2}P^2P' - P(P')^2 - \tfrac{1}{2}P^2P'' = L^5 - 4L^4 - L^4 = L^5 - 5L^4.$$

$N = 3$: $b_3 = \frac{1}{24}(\partial_L - 2)(\partial_L - 1)\partial_L(P^4)$. With $P^4 = L^8$: $\partial_L L^8 = 8L^7$; $(\partial_L - 1)(8L^7) = 56L^6 - 8L^7$; $(\partial_L - 2)(56L^6 - 8L^7) = 336L^5 - 56L^6 - 112L^6 + 16L^7 = 16L^7 - 168L^6 + 336L^5$. Dividing by 24,

$$b_3 = \tfrac{2}{3}L^7 - 7L^6 + 14L^5.$$

Collecting,

$$(X + L^2)^{\langle -1\rangle\circ} = X - L^2 + 2L^3t + \big(L^5 - 5L^4\big)t^2 + \Big(\tfrac{2}{3}L^7 - 7L^6 + 14L^5\Big)t^3 + \mathcal{O}_t^{\mathrm{form}}\big(t^4\big)\,.$$

Degree law. Here $d = \deg_{\mathrm{L}} P = 2$ and $a_d = 1$. Theorem 18.22 predicts, for $N \geq 1$, $\deg_{\mathrm{L}} b_N = d(N+1) - 1 = 2N + 1$ and $\big[L^{2N+1}\big]b_N = (d/N)a_d^{N+1} = 2/N$. For $N = 1, 2, 3$ that is degrees $3, 5, 7$ with leading coefficients $2, 1, \frac{2}{3}$; all three agree with the computation. Note that $\deg_{\mathrm{L}} b_0 = 2 \neq d \cdot 1 - 1 = 1$: the law excludes $N = 0$, exactly as the Lambert degree law excludes $n = 0$ (Theorem E.44). □

Exercise E.34 (A perturbation with a genuine t-block). Reverse $F = X + L + at$ through outer order t^3, for $a \in \mathbb{K}$, and verify the answer with the residual certificate of Theorem 16.16.

Solution. Set $G = X + B$, $B = b_0 + b_1t + b_2t^2 + b_3t^3 + \cdots$. Since $F = X + L + at$, Theorem 12.22 gives

$$F \circ G = X + B + L + \log(1 + tB) + \frac{at}{1 + tB}.$$

Expand each piece through t^3. Writing $u = tB = b_0t + b_1t^2 + b_2t^3$,

$$\log(1+u) = b_0t + \Big(b_1 - \tfrac{1}{2}b_0^2\Big)t^2 + \Big(b_2 - b_0b_1 + \tfrac{1}{3}b_0^3\Big)t^3 + \mathcal{O}_t^{\mathrm{form}}\big(t^4\big)\,,$$

$$\frac{at}{1+u} = at - ab_0t^2 + a\big(b_0^2 - b_1\big)t^3 + \mathcal{O}_t^{\mathrm{form}}\big(t^4\big)\,.$$

Equating $F \circ G = X$ block by block:

$$\begin{aligned}
&t^0 : \ b_0 + L = 0 && \Rightarrow\ b_0 = -L,\\
&t^1 : \ b_1 + b_0 + a = 0 && \Rightarrow\ b_1 = L - a,\\
&t^2 : \ b_2 + b_1 - \tfrac{1}{2}b_0^2 - ab_0 = 0 && \Rightarrow\ b_2 = \tfrac{1}{2}L^2 - L - aL + a,\\
&t^3 : \ b_3 + b_2 - b_0b_1 + \tfrac{1}{3}b_0^3 + a\big(b_0^2 - b_1\big) = 0.
\end{aligned}$$

Substituting the earlier blocks in the last line, $b_0b_1 = -L(L - a) = -L^2 + aL$, $\frac{1}{3}b_0^3 = -\frac{1}{3}L^3$, $b_0^2 - b_1 = L^2 - L + a$, so

$$b_3 = -b_2 + b_0b_1 - \tfrac{1}{3}b_0^3 - a\big(b_0^2 - b_1\big) = \frac{L^3}{3} - \frac{3L^2}{2} - aL^2 + 3aL + L - a^2 - a.$$

Hence

$$\begin{aligned}
(X + L + at)^{\langle -1\rangle\circ} = X - L + (L - a)t &+ \Big(\frac{L^2}{2} - L - aL + a\Big)t^2\\
&+ \Big(\frac{L^3}{3} - \frac{3L^2}{2} - aL^2 + 3aL + L - a^2 - a\Big)t^3 + \mathcal{O}_t^{\mathrm{form}}\big(t^4\big)\,.
\end{aligned}$$

Certificate. By Theorem 17.2 the residual of the truncation $G_2 = X + b_0 + b_1t + b_2t^2$ is $\mathrm{Err}_{\mathrm{rev}}(F, G_2) = -b_3t^3 + \mathcal{O}_t^{\mathrm{form}}\big(t^4\big)$; substituting the computed b_2 into the t^3-equation above reproduces exactly $-b_3$, which is the check. Two sanity tests: at $a = 0$ the blocks reduce to $-L, L, \frac{1}{2}L^2 - L, \frac{1}{3}L^3 - \frac{3}{2}L^2 + L$, i.e. to $\mathsf{L}_0^{\mathrm{Lam}}, \ldots, \mathsf{L}_3^{\mathrm{Lam}}$ as Theorem 20.10 requires; and the whole answer agrees with (17.12)–(17.15) at $a \mapsto 1$, $b \mapsto a$. □

Exercise E.35 (The residual measures the error, on both sides)**.** Let $F = X + L$ and let $H = X - L + Lt + \frac{1}{2}L^2t^2$, the second Picard iterate. Compute $\mathrm{Err}_{\mathrm{rev}}(F, H)$ and $H \circ F - X$ through outer order t^3, and verify Theorem 16.15 on this pair, including the part that says the two residuals agree only in their *leading* block.

Solution. Write $G = F^{\langle -1 \rangle \circ} = \omega$, whose blocks are the Lambert polynomials (Theorem 20.10). The error is $E = H - G$, with

$$[t^0]E = [t^1]E = 0, \qquad [t^2]E = \tfrac{1}{2}L^2 - \mathsf{L}_2^{\mathrm{Lam}}(L) = \tfrac{1}{2}L^2 - \left(\tfrac{1}{2}L^2 - L\right) = L,$$

so $e \coloneqq \nu_t(E) = 2$ and $[t^2]E = L$.

Right residual. With $B = -L + Lt + \frac{1}{2}L^2t^2$, $\mathrm{Err}_{\mathrm{rev}}(F, H) = B + L + \log(1 + tB)$. Expanding,

$$\mathrm{Err}_{\mathrm{rev}}(F, H) = L\,t^2 - \frac{L^2(2L - 9)}{6}\,t^3 + \mathcal{O}_t^{\mathrm{form}}\left(t^4\right).$$

Left residual. $H \circ F - X = (F - X) + B \circ F = L + B \circ F$, and expanding with (12.17),

$$H \circ F - X = L\,t^2 - \frac{L^2(2L + 3)}{6}\,t^3 + \mathcal{O}_t^{\mathrm{form}}\left(t^4\right).$$

Both residuals have outer order exactly $2 = e$, and both have leading block $L = [t^2]E$, which is Theorem 16.15. Their t^3-blocks differ, by $2L^2$: the proposition asserts equality of the leading blocks only, and this pair shows that no more is true. The practical consequence is that either residual certifies correctness through a requested block (Theorem 16.16), and either reads off the next inverse block (Theorem 16.17), but neither may be used as an approximation to the error beyond its leading block. □

Exercise E.36 (Counterexample: the derivative certificate is strictly weaker)**.** One source asserts that the derivative residual $D_X H - \big((D_X F) \circ H\big)^{-1}$ "often detects a wrong block one order earlier" than the composition residual $\mathrm{Err}_{\mathrm{rev}}(F, H)$. Refute this, and quantify the true relation.

1. Show the derivative residual has outer order at least $e + 1$, where $e = \nu_t(H - F^{\langle -1 \rangle \circ})$: it is one level *later*, never earlier.
2. Show it is blind at $e = 0$: construct F, H with $e = 0$ for which the composition residual has order 0 while the derivative residual has order 2.

Solution. (1) Let $G = F^{\langle -1 \rangle \circ}$ and $E = H - G$. Differentiating $F \circ G = X$ gives $(D_X F) \circ G \cdot D_X G = 1$, so $D_X G = \big((D_X F) \circ G\big)^{-1}$ and the derivative residual is

$$D_X H - \big((D_X F) \circ H\big)^{-1} = D_X E + \Big(\big((D_X F) \circ G\big)^{-1} - \big((D_X F) \circ H\big)^{-1}\Big).$$

The first summand has $\nu_t \geq \nu_t(E) + 1 = e + 1$ by Theorem 11.10. For the second, write $F = X + A$ with $A \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$, so $D_X F = 1 + D_X A$ with $\nu_t(D_X A) \geq \nu_t(A) + 1 \geq 1$. Then $(D_X F) \circ H - (D_X F) \circ G = (D_X A) \circ H - (D_X A) \circ G$ has $\nu_t \geq \nu_t(D_X A) + e + 1 \geq e + 2$ by Theorem 14.6. Both $(D_X F) \circ H$ and $(D_X F) \circ G$ lie in $1 + t\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$, and for units u, v of that form $u^{-1} - v^{-1} = (v - u)(uv)^{-1}$ with $\nu_t(uv) = 0$, so taking reciprocals preserves the order of a difference exactly; the second summand therefore has $\nu_t \geq e + 2$. Hence the derivative residual has $\nu_t \geq e + 1$, whereas $\mathrm{Err}_{\mathrm{rev}}(F, H)$ has ν_t exactly e (Theorem 16.15). It is therefore always one level later, and can never be earlier. This is Theorem 16.19.

(2) Take $F = X + L$, $G = F^{\langle -1 \rangle \circ} = \omega$ and $H = G + 5$, so $E = 5 \in \mathbb{K}^\times$ and $e = 0$. The composition residual has order 0: indeed $\mathrm{Err}_{\mathrm{rev}}(F, H) = 5 + \big(\text{terms of order} \geq 1\big)$, whose t^0-block is $5 \neq 0$. But $D_X E = D_X(5) = 0$, so the first summand in the display above vanishes identically, and a direct computation gives

$$D_X H - \big((D_X F) \circ H\big)^{-1} = -5\,t^2 + \mathcal{O}_t^{\mathrm{form}}\left(t^3\right),$$

of order 2. So at $e = 0$ the derivative certificate is off by two levels, and a routine that used it alone would accept an inverse that is wrong in its very first block by an additive constant. The reason is structural: D_X kills $\mathbb{K}$, so the derivative certificate is blind to exactly the kernel of D_X, which is the block that ν_t calls level 0. □

Exercise E.37 (A near-identity idempotent is the identity)**.** Let $J \in \mathcal{G}_{\mathrm{poly},\mathbb{K}}$ satisfy $J \circ J = J$. Prove $J = X$, first from the group structure and then, without it, by comparing the first nonzero correction block.

Solution. First proof. By Theorem 14.7, $\mathcal{G}_{\mathrm{poly},\mathbb{K}}$ is a group under composition. Composing $J \circ J = J$ on the right with $J^{\langle -1\rangle\circ}$ gives $J = X$.

Second proof, without inverses. Write $J = X + B$ with $B \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ and suppose $B \neq 0$; let $\beta = \nu_t(B) \geq 0$. By Theorem 12.23,

$$J \circ J = X + B + B \circ (X + B),$$

so $J \circ J = J$ forces $B \circ (X + B) = 0$. But Theorem 12.22 says that substitution by a near-identity map preserves the outer valuation exactly: $\nu_t\bigl(B \circ (X + B)\bigr) = \nu_t(B) = \beta < \infty$. A series of finite outer order is nonzero, a contradiction. Hence $B = 0$ and $J = X$.

The second proof is the more informative one: it shows that the conclusion uses only that Φ_J is injective and valuation-preserving, not that inverses exist. Consequently the same argument works in the near-identity monoid at any logarithmic depth (Theorem 15.14), and in Puiseux–log enlargements. □

Exercise E.38 (One Picard step against one Newton step)**.** Let $F = X + \log X$ and take $G_0 = X - \log X$ as the starting approximation.

1. Compute the next Picard iterate $\Phi(G_0)$, where $\Phi(X + B) = X - A \circ (X + B)$ with $A = L$, and say how many blocks of $F^{\langle -1\rangle\circ}$ it gets right.
2. Compute the Newton iterate $\mathcal{N}_F(G_0)$ of Theorem 17.10 far enough to say how many blocks *it* gets right, and compare with Theorem 17.11.

Solution. Throughout, $F^{\langle -1\rangle\circ} = \omega = X - L + Lt + \bigl(\tfrac{1}{2}L^2 - L\bigr)t^2 + \bigl(\tfrac{1}{3}L^3 - \tfrac{3}{2}L^2 + L\bigr)t^3 + \cdots$ (Theorem 17.19), and $G_0 = X - L$ is correct through block 0, so $e_0 := \nu_t\bigl(G_0 - F^{\langle -1\rangle\circ}\bigr) = 1$.

(1) $\Phi(G_0) = X - L \circ (X - L) = X - \bigl(L + \log(1 - Lt)\bigr)$, so

$$\Phi(G_0) = X - L + Lt + \frac{L^2}{2}t^2 + \frac{L^3}{3}t^3 + \cdots .$$

Blocks 0 and 1 are correct ($-L$ and L); block 2 is $\tfrac{1}{2}L^2$ against the true $\tfrac{1}{2}L^2 - L$, so the error has outer order exactly 2. One Picard step gained one block, in agreement with (16.10): with $\alpha = \nu_t(A) = 0$, each step gains $\alpha + 1 = 1$.

(2) $D_X F = 1 + t$, so $\mathcal{N}_F(G_0) = G_0 - \mathrm{Err}_{\mathrm{rev}}(F, G_0) \cdot \bigl((1 + t) \circ G_0\bigr)^{-1}$. Here $\mathrm{Err}_{\mathrm{rev}}(F, G_0) = \log(1 - Lt) = -Lt - \tfrac{1}{2}L^2t^2 - \tfrac{1}{3}L^3t^3 - \cdots$ and $(1 + t) \circ G_0 = 1 + t/(1 - Lt) = 1 + t + Lt^2 + \cdots$, whose reciprocal is $1 - t + (1 - L)t^2 + \cdots$. Multiplying and subtracting,

$$\mathcal{N}_F(G_0) = X - L + Lt + \Bigl(\tfrac{1}{2}L^2 - L\Bigr)t^2 + \Bigl(\tfrac{1}{3}L^3 - \tfrac{3}{2}L^2 + L\Bigr)t^3 + \Bigl(\tfrac{1}{4}L^4 - \tfrac{11}{6}L^3 + \tfrac{5}{2}L^2 - L\Bigr)t^4 + \cdots ,$$

whose blocks $0, 1, 2, 3$ all agree with ω and whose block 4 does not (the true value is $\tfrac{1}{4}L^4 - \tfrac{11}{6}L^3 + 3L^2 - L = \mathsf{L}_4^{\mathrm{Lam}}(L)$, differing by $\tfrac{1}{2}L^2$). So the error has outer order exactly 4.

Theorem 17.11 predicts $\nu_t\bigl(\mathrm{Err}_{\mathrm{rev}}(F, \mathcal{N}_F(G_0))\bigr) \geq 2m + 2 + \alpha$ with $m = \nu_t\bigl(\mathrm{Err}_{\mathrm{rev}}(F, G_0)\bigr) = 1$ and $\alpha = 0$, i.e. at least 4; the computation attains it. Summary: from a start correct through block 0, one Picard step reaches block 1 and one Newton step reaches block 3. The comparison is not, however, a cost comparison: the Newton step performs a composition, a reciprocal and a multiplication, and which scheme is cheaper depends on the cost model (Theorem 27.42). □

Exercise E.39 (Counterexample: "smaller than X" is not the admissibility test (hard))**.** Theorem 18.9 requires $\nu_t(A) > -1$, which on $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ means $A \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$. A plausible substitute is the dominance condition $A \prec X$, i.e. "A is asymptotically smaller than the identity". Refute it.

1. Show that $A = X/L$ satisfies $A = o_{X\to+\infty}(X)$ yet has $\nu_t(A) = -1$.
2. For $A = \alpha X$ with $\alpha \in \mathbb{K}^\times$, compute every Lagrange–Bürmann summand exactly and show they all have the same outer order, so the family is never summable.
3. For this part only, take $\mathbb{K} \subseteq \mathbb{C}$ with the usual absolute value. Show that the formal sum of those summands is a geometric series whose *numerical* value reproduces the correct inverse $\big((1+\alpha)X\big)^{\langle -1\rangle\circ} = X/(1+\alpha)$ when $|\alpha| < 1$ and is divergent when $|\alpha| \geq 1$. Conclude that the threshold is not a technicality.

Solution. (1) $A = X/L = t^{-1}L^{-1}$ lies in $\mathscr{PL}_{\mathrm{rat},\mathbb{K}}$ with $\nu_t(A) = -1$. As a function, $A(X)/X = 1/\log X \to 0$, so $A \prec X$ in the analytic sense of Theorem 5.13. So the dominance test is passed and the valuation test is failed: logarithmic gains are invisible to ν_t. This is the content of Theorem 18.10.

(2) Let $A = \alpha X = \alpha t^{-1}$. By Theorem 11.3, $D_X\big(t^n\big) = t^{n+1}(0-n) = -nt^{n+1}$, so by induction

$$D_X^k\big(t^{-r}\big) = r(r-1)\cdots(r-k+1)\,t^{-r+k},$$

and with $k = r-1$, $D_X^{r-1}\big(t^{-r}\big) = r!\,t^{-1}$. Hence the r-th Lagrange–Bürmann summand is

$$\frac{(-1)^r}{r!}D_X^{r-1}\big(A^r\big) = \frac{(-1)^r\alpha^r}{r!}\,r!\,t^{-1} = (-\alpha)^r\,X. \tag{E.4}$$

Every summand has outer order -1, so $\{r : \nu_t < 0\}$ is infinite and the family fails Theorem 7.21. This matches the exact order formula (18.6): with $\alpha_{\nu_t} = \nu_t(A) = -1$, $\nu_t\big(D_X^{r-1}(A^r)\big) = r(\nu_t(A)+1) - 1 = -1$ for every r.

(3) Here $\mathbb{K} \subseteq \mathbb{C}$. Formally summing (E.4) as a geometric series,

$$X + \sum_{r\geq 1}(-\alpha)^r X \text{ “}=\text{” } X\cdot\frac{1}{1+\alpha} = \frac{X}{1+\alpha},$$

which is indeed $F^{\langle -1\rangle\circ}$ for $F = X + \alpha X = (1+\alpha)X$. For $|\alpha| < 1$ the scalar series converges and the manipulation is a valid computation with numbers — but not with formal series: the family is not summable, so the left-hand side names no element of $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$, and the agreement is a coincidence of the geometric series, not a theorem. For $\alpha = 2$, say, the same manipulation produces $X\sum_{r\geq 0}(-2)^r$, a divergent series of scalars, while the true inverse $(3X)^{\langle -1\rangle\circ} = X/3$ is perfectly well defined. So the recipe is not merely unjustified at the threshold; it breaks.

The moral: $\nu_t(A) > -1$ is exactly the condition under which the summands' orders tend to $+\infty$, and no growth comparison between A and X substitutes for it. The two agree on $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$, whose value group is $\mathbb{Z}$, and separate on the Puiseux–log enlargement, where $\nu_t(A) > -1$ admits fractional orders in $(-1,0)$ that "$A \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$" would exclude. □

Exercise E.40 (The inverse of $X\log X$ needs two logarithms (hard))**.** Let $F(X) = X\log X$ and $Y = F(X) \to +\infty$. Put $S = \log Y$ and $\Lambda = \log S$.

1. Explain why no one-logarithm ansatz can represent $F^{\langle -1\rangle\circ}$, however many terms it carries.
2. Derive the closed form $F^{\langle -1\rangle\circ}(Y) = Y/W(Y)$ and expand it through S^{-4}.

Solution. (1) $F = cX^\rho L^\sigma(1+U)$ with $c=1$, $\rho=1$, $\sigma=1$, $U=0$. Theorem 19.3 says a monomial-leading inverse exists in a Puiseux–log ring if and only if ρ^{-1} is in the exponent group *and* $\sigma = 0$; here $\sigma = 1$, so no enlargement of the exponent group or of the coefficient field helps. Quantitatively, Theorem 19.4 gives $\log F^{\langle -1\rangle\circ}(Y) = S - \Lambda + \mathcal{O}_{S\to+\infty}(\Lambda/S)$, and the coefficient -1 of $\Lambda = \log\log Y$ is nonzero, so $F^{\langle -1\rangle\circ}$ lies in no ring whose coefficients are polynomials in one logarithm.

(2) Put $u = \log X$. Then $Y = X\log X = u\mathrm{e}^u$, so $u = W(Y)$ on the principal branch and

$$F^{\langle -1\rangle\circ}(Y) = X = \mathrm{e}^u = \frac{Y}{u} = \frac{Y}{W(Y)}.$$

Now $W(Y) = \omega(S)$, because $w + \log w = S$ is $w\mathrm{e}^w = Y$. By Theorem 20.10,

$$W(Y) = S - \Lambda + \sum_{n\ge1}\frac{\mathsf{L}_n^{\mathrm{Lam}}(\Lambda)}{S^n} = S\bigl(1+\varepsilon\bigr), \qquad \varepsilon = -\frac{\Lambda}{S} + \sum_{n\ge1}\frac{\mathsf{L}_n^{\mathrm{Lam}}(\Lambda)}{S^{n+1}}.$$

Hence $F^{\langle -1\rangle\circ}(Y) = \dfrac{Y}{S}\dfrac{1}{1+\varepsilon}$, and expanding the reciprocal with $\mathsf{L}_1^{\mathrm{Lam}} = \Lambda$, $\mathsf{L}_2^{\mathrm{Lam}} = \frac12\Lambda^2 - \Lambda$, $\mathsf{L}_3^{\mathrm{Lam}} = \frac13\Lambda^3 - \frac32\Lambda^2 + \Lambda$ gives, collecting by powers of S^{-1},

$$\begin{aligned}F^{\langle -1\rangle\circ}(Y) = \frac{Y}{S}\Bigl[\,1 + \frac{\Lambda}{S} + \frac{\Lambda(\Lambda-1)}{S^2} + \frac{\Lambda(\Lambda-2)(2\Lambda-1)}{2S^3}&\\ + \frac{\Lambda\bigl(6\Lambda^3 - 26\Lambda^2 + 27\Lambda - 6\bigr)}{6S^4} + \mathcal{O}_{Y\to+\infty}\Bigl(\tfrac{\Lambda^5}{S^5}\Bigr)\Bigr].&\end{aligned} \tag{E.5}$$

For the record, the coefficients arise as follows. Writing $\varepsilon = \sum_{k\ge1} e_k S^{-k}$ with $e_1 = -\Lambda$, $e_2 = \Lambda$, $e_3 = \mathsf{L}_2^{\mathrm{Lam}}(\Lambda)$, $e_4 = \mathsf{L}_3^{\mathrm{Lam}}(\Lambda)$, the coefficient of S^{-4} in $(1+\varepsilon)^{-1}$ is

$$-e_4 + \bigl(2e_1e_3 + e_2^2\bigr) - 3e_1^2e_2 + e_1^4 = -\tfrac13\Lambda^3 + \tfrac32\Lambda^2 - \Lambda - \Lambda^3 + 2\Lambda^2 + \Lambda^2 - 3\Lambda^3 + \Lambda^4,$$

which is $\Lambda^4 - \frac{13}{3}\Lambda^3 + \frac92\Lambda^2 - \Lambda$, the printed S^{-4} numerator divided by 6. Every coefficient is a polynomial in Λ, which is (1) again. The body records this expansion through S^{-2} only, at (19.19); the two further blocks are computed here.

One word on the remainder. Everything above is an identity between formal series in S^{-1} with coefficients in $\mathbb{K}[\Lambda]$; by itself that justifies only a $\mathcal{O}^{\mathrm{form}}_{S^{-1}}\bigl(S^{-5}\bigr)$. The analytic $\mathcal{O}_{Y\to+\infty}\bigl(\Lambda^5S^{-5}\bigr)$ printed in (E.5) is legitimate, but it comes from Theorem 22.15 (ii): with $\xi_0 = S$ and $\ell_0 = \Lambda$ that corollary bounds the error of the N-block truncation of $W(Y)$ by $2\bigl(5(1+\Lambda)/S\bigr)^{N+1}$ once $S \ge 10(1+\Lambda)$, and dividing Y by a quantity bounded away from 0 preserves the order. Without that input the display is a formal identity and nothing more. □

Exercise E.41 (Trap: a scale change applied to one generator only)**.** Let $F(X) = 2X + \log X$. A computation runs: "$F = m_2 \circ G$ with $m_2(Z) = 2Z$ and $G(X) = X + \frac12\log X$, so $F^{\langle -1\rangle\circ}(Y) = G^{\langle -1\rangle\circ}(Y/2)$. By Theorem 18.23 with $a = \frac12$, $G^{\langle -1\rangle\circ}(Z) = Z + \sum_{N\ge0} 2^{-(N+1)}\mathsf{L}_N^{\mathrm{Lam}}(\log Z)Z^{-N}$. Substituting $Z = Y/2$, that is $Z^{-N} = 2^NY^{-N}$, gives

$$F^{\langle -1\rangle\circ}(Y) \overset{?}{=} \frac{Y}{2} + \frac12\sum_{N\ge0}\mathsf{L}_N^{\mathrm{Lam}}(\log Y)\,Y^{-N}.$$

" Find the error, give the correct answer, quantify the discrepancy, and say which of the volume's checks would have caught it.

Solution. The error is that $Z = Y/2$ changes *both* generators, not one:

$$Z^{-N} = 2^NY^{-N} \qquad \text{and} \qquad \log Z = \log Y - \log 2.$$

The computation substituted the first and forgot the second. The correct result is

$$F^{\langle -1\rangle\circ}(Y) = \frac{Y}{2} + \frac{1}{2}\sum_{N\geq 0} \mathsf{L}_N^{\mathrm{Lam}}\big(\log Y - \log 2\big)\, Y^{-N},$$

which is (19.22).

The failure is silent. The wrong answer is a perfectly well formed element of $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$: every block has the right outer order, the right L-degree, the right leading coefficient and the right sign pattern. No check internal to the near-identity calculation for G can see it, because the calculation for G was correct.

Size of the error. The leading discrepancy is in the $N = 0$ block: $\frac{1}{2}\mathsf{L}_0^{\mathrm{Lam}}(\log Y - \log 2) - \frac{1}{2}\mathsf{L}_0^{\mathrm{Lam}}(\log Y) = -\frac{1}{2}(\log Y - \log 2) + \frac{1}{2}\log Y = \frac{1}{2}\log 2 = 0.34657\ldots$, an additive constant. Numerically, at $Y = 20 + \log 10 = 22.3025850929\ldots$, where $F^{\langle -1\rangle\circ}(Y) = 10$ exactly, the two truncations through Y^{-3} — that is, over $0 \leq N \leq 3$ — give

$$\text{correct}: \ 10.0000043732\ldots, \qquad \text{naive}: \ 9.6702070480\ldots,$$

a gap of $0.3298\ldots$, close to but not equal to $\frac{1}{2}\log 2$ because the higher blocks differ as well.

What catches it. Two things, and only two. First, the residual certificate of Theorem 16.16 *computed against F itself*: substituting the wrong series into $2X + \log X$ leaves a residual of outer order 0, namely $\log 2$, rather than the $\mathcal{O}_t^{\mathrm{form}}\big(t^{N+1}\big)$ a correct truncation must leave. A certificate run against the normalised map G would pass, which is why step 7 of Theorem 19.9 certifies against the original map. Second, the coefficient field: the correct answer does not lie in $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ over $\mathbb{K} = \mathbb{Q}$ at all, since its blocks involve $\log 2$. A routine that tracks the field would have refused the substitution before performing it, which is the point of Theorem 19.2. □

E.6 Wright omega and the Lambert polynomials

Exercise E.42 (Generating the Lambert polynomials two ways)**.** Starting from $\mathsf{L}_1^{\mathrm{Lam}}(u) = u$, use the integral recurrence (21.5), $\mathsf{L}_{n+1}^{\mathrm{Lam}} = -\mathsf{L}_n^{\mathrm{Lam}} + n\int_0^u \mathsf{L}_n^{\mathrm{Lam}}(v)\,\mathrm{d}v$, to produce $\mathsf{L}_2^{\mathrm{Lam}}, \mathsf{L}_3^{\mathrm{Lam}}, \mathsf{L}_4^{\mathrm{Lam}}$. Check each against the Stirling formula (21.7), using $\left[{4 \atop 4}\right] = 1$, $\left[{4 \atop 3}\right] = 6$, $\left[{4 \atop 2}\right] = 11$, $\left[{4 \atop 1}\right] = 6$.

Solution. Successive integration, the constant of integration being fixed to 0 by $\mathsf{L}_n^{\mathrm{Lam}}(0) = 0$ (Theorem 21.2):

$$\begin{aligned}
\mathsf{L}_2^{\mathrm{Lam}} &= -u + 1\cdot\int_0^u v\ \mathrm{d}v = \frac{u^2}{2} - u,\\
\mathsf{L}_3^{\mathrm{Lam}} &= -\Big(\frac{u^2}{2} - u\Big) + 2\int_0^u \Big(\frac{v^2}{2} - v\Big)\,\mathrm{d}v = -\frac{u^2}{2} + u + \frac{u^3}{3} - u^2 = \frac{u^3}{3} - \frac{3}{2}u^2 + u,\\
\mathsf{L}_4^{\mathrm{Lam}} &= -\Big(\frac{u^3}{3} - \frac{3}{2}u^2 + u\Big) + 3\int_0^u \Big(\frac{v^3}{3} - \frac{3}{2}v^2 + v\Big)\,\mathrm{d}v\\
&= -\frac{u^3}{3} + \frac{3}{2}u^2 - u + \frac{u^4}{4} - \frac{3}{2}u^3 + \frac{3}{2}u^2 = \frac{u^4}{4} - \frac{11}{6}u^3 + 3u^2 - u.
\end{aligned}$$

Stirling check at $n = 4$. Formula (21.7) reads $\mathsf{L}_n^{\mathrm{Lam}}(u) = \sum_{m=1}^{n}(-1)^{n-m}\left[{n \atop n-m+1}\right]u^m/m!$. For $n = 4$:

$$-\begin{bmatrix}4\\4\end{bmatrix}\frac{u}{1!} + \begin{bmatrix}4\\3\end{bmatrix}\frac{u^2}{2!} - \begin{bmatrix}4\\2\end{bmatrix}\frac{u^3}{3!} + \begin{bmatrix}4\\1\end{bmatrix}\frac{u^4}{4!} = -u + 3u^2 - \frac{11}{6}u^3 + \frac{u^4}{4},$$

which agrees. Note that the alternation is governed by $(-1)^{n-m}$, not by $(-1)^m$: the whole row changes sign with the parity of n. A common slip is to fix the sign pattern from a single row — $n = 4$, where the coefficients run $-, +, -, +$ — and then apply it at $n = 3$, where they run $+, -, +$.

Factored forms. $\mathsf{L}_2^{\mathrm{Lam}} = \frac{1}{2}u(u-2)$, $\mathsf{L}_3^{\mathrm{Lam}} = \frac{1}{6}u(2u^2-9u+6)$, $\mathsf{L}_4^{\mathrm{Lam}} = \frac{1}{12}u(3u^3-22u^2+36u-12)$, matching the scaled integer family $\mathsf{L}_n^{\mathrm{Lam,sc}} = n!\mathsf{L}_n^{\mathrm{Lam}}$ of Table D.1. □

Exercise E.43 (The one-parameter family, from Lagrange–Bürmann)**.** Let $a \in \mathbb{K}$ and $F = X + a\log X$. Prove directly from (18.13) that the coefficient of X^{-n} in $F^{\langle -1\rangle\circ}$ is $a^{n+1}\mathsf{L}_n^{\mathrm{Lam}}(L)$ for every $n \geq 0$, and identify which Lagrange index contributes to which block.

Solution. Apply (18.13) with $A = aL \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}$, legitimate since $\nu_t(A) = 0 > -1$ (Theorem 18.9):

$$F^{\langle -1\rangle\circ} = X + \sum_{r\geq 1} \frac{(-1)^r}{r!} D_X^{r-1}\big(a^r L^r\big) = X + \sum_{r\geq 1} \frac{(-1)^r a^r}{r!} D_X^{r-1}\big(L^r\big).$$

By Theorem 11.8 applied to the t^0-block L^r,

$$D_X^{r-1}\big(L^r\big) = t^{r-1}\,\Delta_{0,r-1}L^r = t^{r-1}\Big[\prod_{j=0}^{r-2}(\partial_L - j)\Big]L^r.$$

So the r-th summand is a *single* block, sitting at outer order exactly $r-1$. Setting $n = r-1$, no two summands ever meet, and

$$[t^n]F^{\langle -1\rangle\circ} = a^{n+1}\cdot\frac{(-1)^{n+1}}{(n+1)!}\Big[\prod_{j=0}^{n-1}(\partial_L - j)\Big]L^{n+1} = a^{n+1}\mathsf{L}_n^{\mathrm{Lam}}(L),$$

by Theorem 21.1. Index r contributes to block $n = r-1$ and to no other; this is the "one Lagrange index per block" of (20.12), and it is special to a perturbation that is a *single* t^0-block. At $a = 1$ one recovers ω (Theorem 20.10) and at $a = -1$ the reversion of $X - \log X$ with weights $(-1)^{n+1}$ ((20.5)); at $n = 0$ the value is $a\mathsf{L}_0^{\mathrm{Lam}}(L) = -aL$, which is the leading correction predicted by Theorem 16.11 (4).

A useful sanity check that does not use the formula: substituting $t \mapsto at$ in the implicit equation (20.9) and multiplying by a transports the unique fixed point of $X + L$ to that of $X + aL$, which is the proof of Theorem 20.5. □

Exercise E.44 (Counterexample: the degree law and the sign of the error at $n = 0$)**.** The Lambert polynomials satisfy $\deg \mathsf{L}_n^{\mathrm{Lam}} = n$, $[u^n]\mathsf{L}_n^{\mathrm{Lam}} = 1/n$ and $[u]\mathsf{L}_n^{\mathrm{Lam}} = (-1)^{n-1}$ — for $n \geq 1$.

1. Show that each of the three statements fails, or is meaningless, at $n = 0$, and determine which parts of Theorem 21.7 do survive there.
2. A plausible consequence of $\omega - \omega_N \sim \mathsf{L}_{N+1}^{\mathrm{Lam}}(L)x^{-(N+1)}$ is that every truncation approaches ω from below on $(1,\infty)$. Refute this, and locate the exact threshold for $N = 1$.

Solution. (1) $\mathsf{L}_0^{\mathrm{Lam}}(u) = -u$. Its degree is 1, not 0; the assertion $[u^0]\mathsf{L}_0^{\mathrm{Lam}} = 1/0$ is meaningless; and $[u]\mathsf{L}_0^{\mathrm{Lam}} = -1$ whereas $(-1)^{0-1} = -1$ — so the third statement happens to hold, by accident, and should not be quoted as evidence. What survives at $n = 0$ is the part of Theorem 21.7 that does not mention the degree or a coefficient indexed from the top: the constant term vanishes and $u \mid \mathsf{L}_0^{\mathrm{Lam}}$, i.e. item (i) minus its degree clause. The two bottom-indexed items also evaluate correctly there — (v) as noted, and (vii) as the identity $0 = 0$ — but by accident, each being proved only for $n \geq 1$ and $n \geq 2$ respectively. Every statement indexed relative to the top — (ii) on the number of nonzero coefficients, (iii), (iv), (vi) — fails or is vacuous. The integrality normalisation $\mathsf{L}_n^{\mathrm{Lam,sc}} = n!\mathsf{L}_n^{\mathrm{Lam}}$ is likewise an $n \geq 1$ statement; at $n = 0$ it returns $-u$, which is the value, not an instance of the law.

Why the exception is unavoidable: the zero-based normalisation is the one in which (20.12) says "one Lagrange index per block" and in which Theorem 20.5 needs no case distinction. The price is paid entirely at $n = 0$, and it is the reason Theorem 20.11 exists.

(2) The claim would follow if $\mathsf{L}^{\mathrm{Lam}}_{N+1}(L) > 0$ for all $x > 1$, i.e. for all $L > 0$. It is false. For $N = 1$ the relevant polynomial is $\mathsf{L}^{\mathrm{Lam}}_2(u) = \frac{1}{2}u(u-2)$, which is *negative* for $0 < u < 2$ and positive for $u > 2$. So the leading term $\mathsf{L}^{\mathrm{Lam}}_2(L)x^{-2}$ of the error changes sign at $L = 2$, that is at $x = \mathrm{e}^2 = 7.389056\ldots$, and the truncation ω_1 overshoots on one side of that point and undershoots on the other.

The exact threshold is a different number, and this is precisely the trap the exercise is about. The function $x \mapsto \omega(x) - \omega_1(x)$ is negative on $(1, x_*)$ and positive on (x_*, ∞), with a single zero at

$$x_* = 8.746508640\ldots, \qquad \log x_* = 2.168654608\ldots,$$

not at e^2. The leading term locates the crossing only to the accuracy with which it approximates the error, and $\mathsf{L}^{\mathrm{Lam}}_3(L)x^{-3}$ is still of comparable size at these values of x. What the body's (20.17) asserts is an equivalence as $x \to +\infty$; it does not name a finite crossing point, and nothing here entitles one to read $L = 2$ as such a point.

Numerically: at $x = \mathrm{e}$ (so $L = 1$) one has $\omega(\mathrm{e}) = 2.0167798\ldots$ while $\omega_1(\mathrm{e}) = \mathrm{e} - 1 + \mathrm{e}^{-1} = 2.0861613\ldots$, an overshoot; at $x = 10$ (so $L = 2.302585\ldots$) one has $\omega(10) = 7.9294201\ldots$ and $\omega_1(10) = 7.9276734\ldots$, an undershoot. The two errors are

$$-6.938 \times 10^{-2} \qquad \text{and} \qquad +1.747 \times 10^{-3},$$

against the leading term $\mathsf{L}^{\mathrm{Lam}}_2(L)x^{-2}$, which is -6.767×10^{-2} and $+3.484 \times 10^{-3}$ respectively. The signs agree in both cases, which is all the sign discussion needs. The magnitudes agree only at $x = \mathrm{e}$: at $x = 10$ the leading term is too large by a factor of about two, because the next two blocks, $\mathsf{L}^{\mathrm{Lam}}_3(L)x^{-3} = -1.581 \times 10^{-3}$ and $\mathsf{L}^{\mathrm{Lam}}_4(L)x^{-4} = -1.752 \times 10^{-4}$, cancel most of it. That is again the same lesson: at $L \approx 2.3$ the expansion is nowhere near its asymptotic regime, and (20.17) claims nothing at a finite x.

More generally $[u]\mathsf{L}^{\mathrm{Lam}}_n = (-1)^{n-1}$, so near $u = 0^+$ — that is for x slightly above 1 — the sign of $\mathsf{L}^{\mathrm{Lam}}_n(L)$ is $(-1)^{n-1}$ and the truncations alternate sides. Only for L large, where the leading term $L^n/n > 0$ dominates, do they settle on one side. The correct statement is the one the body makes, (20.17): an asymptotic equivalence as $x \to +\infty$, with no claim at any finite x. □

Exercise E.45 (The lower real branch near zero)**.** Let $x \in (-\mathrm{e}^{-1}, 0)$ and write $x = -\mathrm{e}^{-\eta}$, so $\eta \to +\infty$ as $x \to 0^-$. Put $M = \log \eta$.

1. Show that $W_{-1}(x) = -v$ where v solves $v - \log v = \eta$.
2. Derive the first four correction blocks of v, and hence of W_{-1}.
3. Some sources say "the same $\mathsf{L}^{\mathrm{Lam}}_n$ occur, multiplied by $(-1)^n$". Say precisely which quantity carries which weight.

Solution. (1) $W_{-1}(x) < -1$, so write $W_{-1}(x) = -v$ with $v > 1$. The defining equation $W\mathrm{e}^W = x$ becomes $-v\mathrm{e}^{-v} = x = -\mathrm{e}^{-\eta}$, i.e. $v\mathrm{e}^{-v} = \mathrm{e}^{-\eta}$. Taking the real logarithm of both positive sides, $\log v - v = -\eta$, that is $v - \log v = \eta$.

(2) So $v = (X - \log X)^{\langle -1\rangle_\circ}(\eta)$. By (20.5),

$$\begin{aligned} v &= \eta + M + \sum_{n\ge1}(-1)^{n+1}\mathsf{L}^{\mathrm{Lam}}_n(M)\,\eta^{-n} \\ &= \eta + M + \frac{M}{\eta} - \frac{\frac{1}{2}M^2 - M}{\eta^2} + \frac{\frac{1}{3}M^3 - \frac{3}{2}M^2 + M}{\eta^3} - \frac{\mathsf{L}^{\mathrm{Lam}}_4(M)}{\eta^4} + \cdots, \end{aligned}$$

and therefore

$$W_{-1}\big(-\mathrm{e}^{-\eta}\big) = -\eta - M - \frac{M}{\eta} + \frac{M(M-2)}{2\eta^2} - \frac{M(2M^2 - 9M + 6)}{6\eta^3}$$

$$+ \frac{M(3M^3 - 22M^2 + 36M - 12)}{12\eta^4} + \mathcal{O}_{\eta \to +\infty}\left(\tfrac{M^5}{\eta^5}\right),$$

which is (22.27). Note that the Lambert part of this volume proves more than an asymptotic statement here: by Theorem 22.22 the series $\sum_{n\geq 1}(-1)^n \mathsf{L}_n^{\mathrm{Lam}}(M)\eta^{-n}$ *converges* once $2(1+|M|) < \eta$, with an explicit remainder bound. That is a theorem about this particular family, proved by a Cauchy estimate; it is not a consequence of the formal reversion. The corresponding statement for ω is *not* missing from the volume, contrary to what one might expect from the asymptotic framing of Part 20: Theorem 22.13 applied with $\xi = x$, $K = x$ and $\log K = L$ gives $M = L$, so its seating condition reads $2(1+L) < x$, and under it the series $\sum_{n\geq 1} \mathsf{L}_n^{\mathrm{Lam}}(L)x^{-n}$ converges absolutely to $\omega(x) - x + L$. The two branches are the two signs $\sigma = \pm 1/\cdot$ of one and the same Cauchy estimate, Theorem 22.12.

(3) The two weights belong to two different objects, and conflating them is the error the sources make. The reversion of $X - \log X$ carries $(-1)^{n+1}$: $(X-L)^{\langle -1\rangle\circ} = X + L + \sum_{n\geq 1}(-1)^{n+1}\mathsf{L}_n^{\mathrm{Lam}}(L)t^n$, so its first correction is $+\mathsf{L}_1^{\mathrm{Lam}}(L)/X = +L/X$, with a *plus* sign. Its negative, which is what enters W_{-1}, carries $(-1)^n$: $W_{-1} = -\eta - M + \sum_{n\geq 1}(-1)^n \mathsf{L}_n^{\mathrm{Lam}}(M)\eta^{-n}$. Both displays above are consistent with this. □

Exercise E.46 (Trap: an off-by-one in the truncation index)**.** A source using the inclusive cutoff defines $\Omega_N = X + [\mathcal{P}]_{\leq N}$, where $\mathcal{P} = \sum_{n\geq 0} \mathsf{L}_n^{\mathrm{Lam}}(L)t^n$, and proves

$$\Omega_N + \log \Omega_N - X = -\mathsf{L}_{N+1}^{\mathrm{Lam}}(L)\, t^{N+1} + \mathcal{O}_t^{\mathrm{form}}\left(t^{N+2}\right).$$

A reader transcribing into the normative exclusive convention writes $\Omega_N = X + \operatorname{Trunc}_{<N}\mathcal{P}$ and keeps the conclusion. Show that the conclusion is then false, compute the correct residual, and give the correct transcription.

Solution. The two cutoffs are related by $[F]_{\leq N} = \operatorname{Trunc}_{<N+1} F$, so the correct transcription is $\omega_N = X + \operatorname{Trunc}_{<N+1}\mathcal{P}$, which is Theorem 20.12. Writing $\operatorname{Trunc}_{<N}\mathcal{P}$ instead drops the block $\mathsf{L}_N^{\mathrm{Lam}}(L)t^N$.

Compute the residual of the mistranscribed object. Put $\mathcal{P}'_N = \operatorname{Trunc}_{<N}\mathcal{P}$ and $\Delta = \mathcal{P} - \mathcal{P}'_N = \sum_{n\geq N} \mathsf{L}_n^{\mathrm{Lam}}(L)t^n$, so $\nu_t\Delta = N$ and $[t^N]\Delta = \mathsf{L}_N^{\mathrm{Lam}}(L)$. Exactly the computation in the proof of Theorem 20.13 — generator substitution, then $1 + t\mathcal{P}'_N = (1+t\mathcal{P})\big(1 - t\Delta/(1+t\mathcal{P})\big)$ and additivity of the formal logarithm — gives

$$\big(X + \operatorname{Trunc}_{<N}\mathcal{P}\big) + \log\big(X + \operatorname{Trunc}_{<N}\mathcal{P}\big) - X = -\mathsf{L}_N^{\mathrm{Lam}}(L)\, t^N + \mathcal{O}_t^{\mathrm{form}}\left(t^{N+1}\right).$$

So the residual has outer order exactly N, not $N+1$, and its leading block is $-\mathsf{L}_N^{\mathrm{Lam}}$, not $-\mathsf{L}_{N+1}^{\mathrm{Lam}}$. For $N = 3$, for instance, the mistranscribed truncation leaves $-\big(\tfrac{1}{3}L^3 - \tfrac{3}{2}L^2 + L\big)t^3 + \mathcal{O}_t^{\mathrm{form}}\big(t^4\big)$, whereas the correct ω_3 leaves $-\mathsf{L}_4^{\mathrm{Lam}}(L)t^4 + \mathcal{O}_t^{\mathrm{form}}\big(t^5\big)$.

The practical bite: a user asking for "the expansion correct through x^{-3}" and receiving $X + \operatorname{Trunc}_{<3}\mathcal{P}$ gets an answer correct only through x^{-2}. The error is one block, it is systematic, and it is undetectable by inspecting the answer — every printed block is a correct Lambert polynomial. What detects it is the certificate of Theorem 20.14: the residual order is the definition of how many blocks are right, and it reads N, not $N+1$. □

Exercise E.47 (The Laplace transform and its cancellations)**.**

1. Compute the transform

$$\int_0^\infty \mathrm{e}^{-su}\, \mathsf{L}_n^{\mathrm{Lam}}(u)\, \mathrm{d}u$$

directly for $n = 1, 2, 3$ and $\Re s > 0$, and compare with (21.18).

2. Deduce the exponential moment cancellations
$$\int_0^\infty \mathrm{e}^{-ru}\mathsf{L}_n^{\mathrm{Lam}}(u)\,\mathrm{d}u = 0$$
for all integers r with $1 \le r \le n-1$, and say why the range is empty at $n = 1$.

3. Explain why the identity says nothing about the convergence of $\sum_n \mathsf{L}_n^{\mathrm{Lam}}(u)s^n$.

Solution. (1) Use $\int_0^\infty \mathrm{e}^{-su}u^m\,\mathrm{d}u = m!/s^{m+1}$ for $\Re s > 0$.
$$\mathsf{L}_1^{\mathrm{Lam}} = u: \quad \frac{1}{s^2}; \qquad \mathsf{L}_2^{\mathrm{Lam}} = \frac{u^2}{2} - u: \quad \frac{1}{2}\cdot\frac{2}{s^3} - \frac{1}{s^2} = \frac{1-s}{s^3};$$
$$\mathsf{L}_3^{\mathrm{Lam}} = \frac{u^3}{3} - \frac{3}{2}u^2 + u: \quad \frac{1}{3}\cdot\frac{6}{s^4} - \frac{3}{2}\cdot\frac{2}{s^3} + \frac{1}{s^2} = \frac{2-3s+s^2}{s^4} = \frac{(1-s)(2-s)}{s^4}.$$
Theorem 21.14 predicts $s^{-(n+1)}\prod_{j=1}^{n-1}(j-s)$, the product empty at $n=1$; the three values are $1/s^2$, $(1-s)/s^3$, $(1-s)(2-s)/s^4$, matching.

(2) The right-hand side of (21.18) vanishes exactly at the zeros of $\prod_{j=1}^{n-1}(j-s)$, i.e. at $s = 1, 2, \ldots, n-1$, all of which satisfy $\Re s > 0$ so that the integral converges absolutely. Hence $\int_0^\infty \mathrm{e}^{-ru}\mathsf{L}_n^{\mathrm{Lam}}(u)\,\mathrm{d}u = 0$ for $1 \le r \le n-1$. At $n = 1$ the product is empty and has no zeros, so the range $1 \le r \le 0$ is empty; the first genuine cancellation is $\int_0^\infty \mathrm{e}^{-u}\mathsf{L}_2^{\mathrm{Lam}}(u)\,\mathrm{d}u = 0$, which one can check by hand: $\frac{1}{2}\cdot 2 - 1 = 0$.

(3) The Laplace identity is a statement about one polynomial at a time. It integrates $\mathsf{L}_n^{\mathrm{Lam}}$ against a fixed kernel over $u \in (0,\infty)$; it involves no sum over n, and gives no bound on $\mathsf{L}_n^{\mathrm{Lam}}(u)$ at a fixed u as $n \to \infty$. Convergence of $\sum_n \mathsf{L}_n^{\mathrm{Lam}}(u)s^n$ at frozen u is a separate theorem, Theorem 20.19, proved by the implicit function theorem for (20.19); and neither statement is about the asymptotic expansion of ω, where $u = \log x$ and $s = 1/x$ move together. Keeping the three apart is the discipline Section E.7 is about. □

Exercise E.48 (Reciprocal and logarithm of W (hard))**.** Let z be large and positive, $L_1 = \log z$, $\ell = \log L_1$, $r = L_1^{-1}$, and $W = W(z)$ on the principal branch.

1. Derive $\log W = L_1 - W$ exactly, and use it to expand $\log W$ through r^4 without ever taking the logarithm of a series.

2. Expand $1/W$ through r^6, and compute one block further than the sources tabulate.

3. Deduce a transseries for e^W with no exponentiation of a series.

Solution. (1) From $W\mathrm{e}^W = z$, taking logarithms of the positive quantities, $\log W + W = \log z = L_1$, so $\log W = L_1 - W$ exactly. Since $W = \omega(L_1)$ — because $W + \log W = L_1$ is the defining equation of ω — Theorem 20.10, transported to branch coordinates by Theorem 22.9, gives $W = L_1 - \ell + \sum_{n\ge1}\mathsf{L}_n^{\mathrm{Lam}}(\ell)r^n$. Subtracting from L_1,
$$\begin{aligned}\log W &= \ell - \sum_{n\ge1}\mathsf{L}_n^{\mathrm{Lam}}(\ell)r^n \\ &= \ell - \ell r + \Big(\ell - \frac{\ell^2}{2}\Big)r^2 + \Big(-\frac{\ell^3}{3} + \frac{3}{2}\ell^2 - \ell\Big)r^3 \\ &\quad + \Big(-\frac{\ell^4}{4} + \frac{11}{6}\ell^3 - 3\ell^2 + \ell\Big)r^4 + \mathcal{O}_r^{\mathrm{form}}(r^5)\,.\end{aligned}$$

No series logarithm was computed: the identity $\log W = L_1 - W$ did all the work. This is the same device as Theorem 10.10 used in closed form.

(2) Factor $W = L_1(1+\varepsilon)$ with

$$\begin{aligned}\varepsilon &= -\ell r + \sum_{n\ge 1} \mathsf{L}_n^{\mathrm{Lam}}(\ell) r^{n+1}\\ &= -\ell r + \ell r^2 + \Big(\frac{\ell^2}{2} - \ell\Big) r^3 + \Big(\frac{\ell^3}{3} - \frac{3}{2}\ell^2 + \ell\Big) r^4 + \mathsf{L}_4^{\mathrm{Lam}}(\ell) r^5 + \mathsf{L}_5^{\mathrm{Lam}}(\ell) r^6 + \cdots ,\end{aligned}$$

so $\nu_t\varepsilon = 1$ in the r-adic filtration and the geometric series $\sum_k (-\varepsilon)^k$ is summable (Theorem E.3). Then $1/W = r\sum_{k\ge 0}(-\varepsilon)^k$. Collecting by powers of r, with $e_1 = -\ell$, $e_2 = \ell$, $e_3 = \mathsf{L}_2^{\mathrm{Lam}}(\ell)$, $e_4 = \mathsf{L}_3^{\mathrm{Lam}}(\ell)$, $e_5 = \mathsf{L}_4^{\mathrm{Lam}}(\ell)$, $e_6 = \mathsf{L}_5^{\mathrm{Lam}}(\ell)$ the coefficients of $\varepsilon = \sum_k e_k r^k$:

$$\begin{aligned}\frac{1}{W} &= r + \ell r^2 + \big(\ell^2 - \ell\big) r^3 + \Big(\ell^3 - \frac{5}{2}\ell^2 + \ell\Big) r^4\\ &\quad + \Big(\ell^4 - \frac{13}{3}\ell^3 + \frac{9}{2}\ell^2 - \ell\Big) r^5 + \Big(\ell^5 - \frac{77}{12}\ell^4 + \frac{71}{6}\ell^3 - 7\ell^2 + \ell\Big) r^6\\ &\quad + \Big(\ell^6 - \frac{87}{10}\ell^5 + \frac{145}{6}\ell^4 - \frac{155}{6}\ell^3 + 10\ell^2 - \ell\Big) r^7 + \mathcal{O}_r^{\mathrm{form}}\big(r^8\big) .\end{aligned} \tag{E.6}$$

The blocks through r^6 are tabulated in the sources; the r^7 block is computed here. As a sample of the bookkeeping, the coefficient of r^5 is $-e_4 + (2e_1e_3 + e_2^2) - 3e_1^2e_2 + e_1^4$, which evaluates to $-\frac{1}{3}\ell^3 + \frac{3}{2}\ell^2 - \ell - \ell^3 + 2\ell^2 + \ell^2 - 3\ell^3 + \ell^4 = \ell^4 - \frac{13}{3}\ell^3 + \frac{9}{2}\ell^2 - \ell$.

Three structural checks now follow, each provable in a line and each visible in (E.6). Write

$$\frac{1}{W} = r\sum_{k\ge 0} c_k(\ell)\, r^k,$$

so that c_k is the block of $1/W$ at r^{k+1}.

Zero constant term. Setting $\ell = 0$ kills every $\mathsf{L}_n^{\mathrm{Lam}}$ ((21.3)), hence $\varepsilon|_{\ell=0} = 0$ and $1/W|_{\ell=0} = r$; so $c_k(0) = 0$ for $k \ge 1$.

Monic of degree k. In $\varepsilon = \sum_{k\ge 1} e_k r^k$ the first coefficient has ℓ-degree 1 and, for $k\ge 2$, $e_k = \mathsf{L}_{k-1}^{\mathrm{Lam}}(\ell)$ has ℓ-degree $k-1$. A monomial $e_{i_1}\cdots e_{i_m}$ with $i_1 + \cdots + i_m = k$ therefore has degree at most k minus the number of indices $i_j \ge 2$, which equals k only when every $i_j = 1$. So the top-degree part of c_k comes from $(-e_1)^k = \ell^k$ alone, and $[\ell^k]c_k = 1$.

Linear coefficient. As ε vanishes at $\ell = 0$, the derivative of $(1+\varepsilon)^{-1}$ there is $-\partial_\ell\varepsilon$, and by $\mathsf{L}_n^{\mathrm{Lam}\,\prime}(0) = (-1)^{n-1}$ (Theorem 21.7(v)),

$$-\partial_\ell\varepsilon\big|_{\ell=0} = r + \sum_{n\ge 1}(-1)^n r^{n+1}.$$

Hence $[\ell]c_k = (-1)^{k-1}$ for $k\ge 1$, which is the alternation visible in (E.6).

Finally, (E.6) is, block for block, the bracket in (E.5): $(X\log X)^{\langle -1\rangle_\circ}(Y) = Y/W(Y)$, so the two computations are the same one.

(3) From $W\mathrm{e}^W = z$, $\mathrm{e}^W = z/W = z\cdot(1/W)$, and (E.6) supplies $1/W$ already. No exponential of a series is ever formed, which matters because Theorem 10.16 shows that exponentiating a block of L-degree ≥ 2 leaves every power–log scale. □

Exercise E.49 (The fifth and sixth Lambert polynomials, three ways (hard))**.** Compute $\mathsf{L}_5^{\mathrm{Lam}}$ and $\mathsf{L}_6^{\mathrm{Lam}}$ from the operator definition (21.1), from the integral recurrence, and from the Stirling formula, using

$$\Big(\begin{bmatrix}5\\5\end{bmatrix},\dots,\begin{bmatrix}5\\1\end{bmatrix}\Big) = (1,10,35,50,24),\qquad \Big(\begin{bmatrix}6\\6\end{bmatrix},\dots,\begin{bmatrix}6\\1\end{bmatrix}\Big) = (1,15,85,225,274,120).$$

Then verify the coefficient identities of Theorem 21.7(iii)–(vii) on both.

Solution. Answers.

$$\begin{aligned}\mathsf{L}_5^{\mathrm{Lam}}(u) &= \frac{u^5}{5} - \frac{25}{12}u^4 + \frac{35}{6}u^3 - 5u^2 + u = \frac{24u^5 - 250u^4 + 700u^3 - 600u^2 + 120u}{120},\\ \mathsf{L}_6^{\mathrm{Lam}}(u) &= \frac{u^6}{6} - \frac{137}{60}u^5 + \frac{75}{8}u^4 - \frac{85}{6}u^3 + \frac{15}{2}u^2 - u\\ &= \frac{120u^6 - 1644u^5 + 6750u^4 - 10200u^3 + 5400u^2 - 720u}{720}.\end{aligned}$$

Stirling route. The two rows are the usual cycle numbers [4, 18]. By (21.7),

$$\mathsf{L}_n^{\mathrm{Lam}}(u) = \sum_{m=1}^{n} (-1)^{n-m} \begin{bmatrix} n \\ n-m+1 \end{bmatrix} \frac{u^m}{m!}.$$

For $n = 5$ the signs are $(-1)^{5-m}$, i.e. $+, -, +, -, +$ for $m = 1, \dots, 5$:

$$\mathsf{L}_5^{\mathrm{Lam}} = \begin{bmatrix}5\\5\end{bmatrix} u - \begin{bmatrix}5\\4\end{bmatrix} \frac{u^2}{2} + \begin{bmatrix}5\\3\end{bmatrix} \frac{u^3}{6} - \begin{bmatrix}5\\2\end{bmatrix} \frac{u^4}{24} + \begin{bmatrix}5\\1\end{bmatrix} \frac{u^5}{120} = u - 5u^2 + \frac{35}{6}u^3 - \frac{25}{12}u^4 + \frac{u^5}{5}.$$

For $n = 6$ the signs are $(-1)^{6-m}$, i.e. $-, +, -, +, -, +$:

$$\mathsf{L}_6^{\mathrm{Lam}} = -u + \frac{15}{2}u^2 - \frac{85}{6}u^3 + \frac{225}{24}u^4 - \frac{274}{120}u^5 + \frac{120}{720}u^6,$$

which is the stated answer since $225/24 = 75/8$, $274/120 = 137/60$ and $120/720 = 1/6$.

Recurrence route. From $\mathsf{L}_4^{\mathrm{Lam}}$ of Theorem E.42, $\mathsf{L}_5^{\mathrm{Lam}} = -\mathsf{L}_4^{\mathrm{Lam}} + 4\int_0^u \mathsf{L}_4^{\mathrm{Lam}}$, giving $-\big(\frac{1}{4}u^4 - \frac{11}{6}u^3 + 3u^2 - u\big) + 4\big(\frac{1}{20}u^5 - \frac{11}{24}u^4 + u^3 - \frac{1}{2}u^2\big) = \frac{1}{5}u^5 - \frac{25}{12}u^4 + \frac{35}{6}u^3 - 5u^2 + u$. Then $\mathsf{L}_6^{\mathrm{Lam}} = -\mathsf{L}_5^{\mathrm{Lam}} + 5\int_0^u \mathsf{L}_5^{\mathrm{Lam}}$ gives the second answer by the same arithmetic.

Operator route. $\mathsf{L}_5^{\mathrm{Lam}} = \frac{1}{720}(\partial_u - 4)(\partial_u - 3)(\partial_u - 2)(\partial_u - 1)\partial_u\, u^6$; applying the five factors in turn to $6u^5$ reproduces the same polynomial. This is the only one of the three routes that presupposes none of the derived identities.

Coefficient checks. With $H_4 = \frac{25}{12}$ and $H_5 = \frac{137}{60}$: (iii) $[u^5]\mathsf{L}_5^{\mathrm{Lam}} = \frac{1}{5}$ and $[u^6]\mathsf{L}_6^{\mathrm{Lam}} = \frac{1}{6}$, i.e. $1/n$. (iv) $[u^4]\mathsf{L}_5^{\mathrm{Lam}} = -\frac{25}{12} = -H_4$ and $[u^5]\mathsf{L}_6^{\mathrm{Lam}} = -\frac{137}{60} = -H_5$. (v) $[u]\mathsf{L}_5^{\mathrm{Lam}} = 1 = (-1)^4$ and $[u]\mathsf{L}_6^{\mathrm{Lam}} = -1 = (-1)^5$. (vii) $[u^2]\mathsf{L}_5^{\mathrm{Lam}} = -5 = (-1)^5\frac{5\cdot 4}{4}$ and $[u^2]\mathsf{L}_6^{\mathrm{Lam}} = \frac{15}{2} = (-1)^6\frac{6\cdot 5}{4}$. Finally (vi), which reads $[u^{n-2}]\mathsf{L}_n^{\mathrm{Lam}} = \frac{n-1}{2}\big(H_{n-1}^2 - H_{n-1}^{(2)}\big)$ — an index counted from the top, not the fixed exponent 3, the two agreeing at $n = 5$ alone. At $n = 5$, with $H_4^{(2)} = \frac{205}{144}$,

$$[u^3]\mathsf{L}_5^{\mathrm{Lam}} = \tfrac{35}{6} = 2\Big(\tfrac{625}{144} - \tfrac{205}{144}\Big) = 2 \cdot \tfrac{420}{144} = \tfrac{35}{6};$$

at $n = 6$, with $H_5^{(2)} = \frac{5269}{3600}$,

$$[u^4]\mathsf{L}_6^{\mathrm{Lam}} = \tfrac{75}{8} = \tfrac{5}{2}\Big(\tfrac{18769}{3600} - \tfrac{5269}{3600}\Big) = \tfrac{5}{2} \cdot \tfrac{15}{4} = \tfrac{75}{8}.$$

Both hold. Reading (vi) at the fixed exponent 3 instead would compare $[u^3]\mathsf{L}_6^{\mathrm{Lam}} = -\frac{85}{6}$ with $\frac{75}{8}$ and report a failure that is entirely an indexing error. Multiplying by $n!$ turns every coefficient into an integer, as Theorem 21.9 requires: $\mathsf{L}_5^{\mathrm{Lam,sc}}$ and $\mathsf{L}_6^{\mathrm{Lam,sc}}$ are the numerators displayed above. □

Exercise E.50 (The branch point forces a square root, not a logarithm (hard))**.** At $z = -\mathrm{e}^{-1}$ the two real branches of W meet at $W = -1$.

1. Set $W = -1 + q$ and expand $1 + \mathrm{e}z$ in powers of q through q^4.
2. With $p = \sqrt{2(1 + \mathrm{e}z)}$, derive $W_0(z) = -1 + p - \frac{1}{3}p^2 + \frac{11}{72}p^3 - \frac{43}{540}p^4 + \cdots$ and W_{-1} by $p \mapsto -p$.

3. Explain, from the local structure of $w\mathrm{e}^w$, why a square root rather than an iterated logarithm is forced here, and why the residual-based error estimate of Theorem 16.21 degrades near this point.

Solution. (1) With $w = -1 + q$,

$$\mathrm{e}z = \mathrm{e}\, w\mathrm{e}^w = (-1+q)\mathrm{e}^q,$$

and multiplying $-1+q$ into $\sum_{k\ge 0} q^k/k!$ the coefficient of q^n for $n \ge 2$ is $\frac{1}{(n-1)!} - \frac{1}{n!} = \frac{n-1}{n!}$, while the constant term is -1 and the coefficient of q is $-1+1=0$. Hence

$$1 + \mathrm{e}z = \sum_{n\ge 2} \frac{(n-1)\,q^n}{n!} = \frac{q^2}{2} + \frac{q^3}{3} + \frac{q^4}{8} + \frac{q^5}{30} + \mathcal{O}_{q\to 0}\big(q^6\big)\,, \tag{E.7}$$

an entire function of q; this is Theorem 22.24.

(2) Then $p^2 = 2(1+\mathrm{e}z) = q^2\big(1 + \frac{2}{3}q + \frac{1}{4}q^2 + \frac{1}{15}q^3 + \cdots\big)$, so, choosing the branch with $p \sim q$,

$$p = q\Big(1 + \frac{2}{3}q + \frac{1}{4}q^2 + \frac{1}{15}q^3 + \cdots\Big)^{1/2} = q + \frac{q^2}{3} + \frac{5q^3}{72} + \frac{11q^4}{1080} + \cdots.$$

Reverting this power series (a near-identity reversion in the ordinary one-variable sense — no logarithms are involved) gives

$$q = p - \frac{p^2}{3} + \frac{11}{72}p^3 - \frac{43}{540}p^4 + \frac{769}{17280}p^5 - \frac{221}{8505}p^6 + \cdots,$$

whence $W_0 = -1 + q$ as stated; the coefficients agree with the standard tabulations [5, 11]. Since p enters only through $p^2 = 2(1+\mathrm{e}z)$, replacing p by $-p$ gives the second solution of $w\mathrm{e}^w = z$ near $w = -1$, namely $W_{-1}(z) = -1 - p - \frac{1}{3}p^2 - \frac{11}{72}p^3 - \frac{43}{540}p^4 - \cdots$. That the reverted series has a positive radius of convergence, and that the two displayed branches are exactly the two solutions there, is Theorem 22.25; the formal reversion above supplies the coefficients but not the radius.

(3) The map $w \mapsto w\mathrm{e}^w$ has derivative $(1+w)\mathrm{e}^w$, which vanishes simply at $w=-1$; the second derivative $(2+w)\mathrm{e}^w$ equals $\mathrm{e}^{-1} \ne 0$ there. So $z + \mathrm{e}^{-1}$ has a simple *quadratic* zero in w at $w=-1$, and inverting a quadratic critical point produces a two-valued square-root branch, with the two branches exchanged by $p \mapsto -p$. Nothing logarithmic occurs: the logarithmic scale of Theorem 20.10 comes from the leading behaviour of $w\mathrm{e}^w$ at $w \to +\infty$, an entirely different regime.

Near the branch point the residual-to-error transfer degrades exactly because W' blows up there. Theorem 16.21 bounds $|x_N - x_*|$ by $|f(x_N) - y|\,/m$ with $m = \inf|f'|$ on the interval between; here $f'(w) = (1+w)\mathrm{e}^w \to 0$ as $w \to -1$, so $m \to 0$ and the bound becomes vacuous. Concretely, a small residual no longer certifies a small error: near a quadratic critical point the error is of order the square root of the residual, not of order the residual. □

E.7 Analytic transfer

Exercise E.51 (From residual to error, on a ray)**.** Let $F(y) = y + \log y$ for $y > 0$, let $X \in \mathbb{R}$, and let $y_* > 0$ be the unique solution of $F(y) = X$.

1. Show that every $y_N > 0$ satisfies $|y_N - y_*| \le |y_N + \log y_N - X|$, and identify the general principle at work.

2. Sharpen this to a two-sided estimate: if $y_0 > 0$ and both y_N and y_* lie in $[y_0, \infty)$, then

$$\frac{|F(y_N) - X|}{1 + 1/y_0} \;\le\; |y_N - y_*| \;\le\; |F(y_N) - X|\,.$$

   Deduce that on such a ray the residual is not merely an upper bound for the error but a proxy for it, correct to within the factor $1 + 1/y_0$.

3. Explain why the same argument gives nothing for $F(y) = y - \log y$ on $(0, 1)$.

Solution. F is continuously differentiable on $(0, \infty)$ with $F'(y) = 1 + 1/y > 1$, so F is strictly increasing, hence injective, and $F\big((0, \infty)\big) = \mathbb{R}$, so y_* exists and is unique (Theorem 20.1). By the mean value theorem there is θ between y_N and y_* with

$$F(y_N) - X = F(y_N) - F(y_*) = F'(\theta)\,(y_N - y_*),$$

and $\theta > 0$ gives $F'(\theta) > 1$. Hence $|y_N - y_*| = |F(y_N) - X| \,/ F'(\theta) \le |F(y_N) - X|$, which is the claim.

The principle is Theorem 16.21: if $|f'| \ge m > 0$ between the approximation and the true point, then the error is at most the residual divided by m; here $m = 1$ works on the whole ray. Two hypotheses are doing real work, and both are analytic: the derivative bound, and the fact that y_N and y_* lie in a common interval on which it holds.

(2) The same θ now satisfies $y_0 \le \theta$, because θ lies between y_N and y_* and both are $\ge y_0$. Since $F'(y) = 1 + 1/y$ is decreasing, $1 < F'(\theta) \le 1 + 1/y_0$, and dividing $F(y_N) - X = F'(\theta)(y_N - y_*)$ by $F'(\theta)$ gives both inequalities at once. So the residual determines the error to within the factor $1 + 1/y_0$, which tends to 1 as $y_0 \to \infty$. This is the two-sided enclosure of Theorem 25.2, with $m = 1$ and $M = 1 + 1/y_0$; that theorem states it on a compact interval and adds that injectivity need not be assumed, being a conclusion.

(3) For $F(y) = y - \log y$ on $(0, 1)$ one has $F'(y) = 1 - 1/y < 0$, and $F'(y) \to 0$ as $y \to 1^-$. So $m = \inf |F'| = 0$ on any interval reaching to 1, and the bound degenerates. That is not an artefact: $y = 1$ is the critical point of $y - \log y$, the two branches of its inverse meet there, and near it the error is of order the square root of the residual — the same phenomenon as at the Lambert branch point (Theorem E.50). ☐

Exercise E.52 (Trap: a formal tail read as an analytic estimate)**.** The following argument is offered for the asymptotic validity of the Wright omega truncations.

> "By Theorem 20.13, $\mathrm{Err}_{\mathrm{rev}}(X + L, \omega_N) = -\mathsf{L}^{\mathrm{Lam}}_{N+1}(L)t^{N+1} + \mathcal{O}^{\mathrm{form}}_t\big(t^{N+2}\big)$. Substituting x for X and $\log x$ for L, the residual of $\omega_N(x)$ is $-\mathsf{L}^{\mathrm{Lam}}_{N+1}(\log x)x^{-(N+1)} + \mathcal{O}_{x\to+\infty}\big(x^{-(N+2)}\big)$. Since $\left|\frac{\mathrm{d}}{\mathrm{d}y}(y + \log y)\right| > 1$ for $y > 0$, Theorem 16.21 converts this into
>
> $$\omega(x) - \omega_N(x) = \mathsf{L}^{\mathrm{Lam}}_{N+1}(\log x)x^{-(N+1)} + \mathcal{O}_{x\to+\infty}\Big(x^{-(N+2)}\Big).$$
>
> Hence the truncations are asymptotically valid."

Locate every unjustified step, say which parts of the conclusion are nevertheless true, and state what has to be proved to repair it.

Solution. There is exactly one unjustified step, and it is the second sentence.

The step. "Substituting x for X and $\log x$ for L" in an identity of $\mathbb{K}[L][[t]]$ is not an operation on functions. The symbol $\mathcal{O}^{\mathrm{form}}_t\big(t^{N+2}\big)$ records that a formal series has outer valuation at least $N + 2$; it asserts no inequality, contains no constant, and names no domain. There is no theorem in the volume, and no theorem at all, that converts it into $\mathcal{O}_{x\to+\infty}\big(x^{-(N+2)}\big)$ without further hypotheses. Two independent obstructions:

- the formal tail is an *infinite* sum of blocks, and its substitution need not converge or even be defined pointwise;
- even if a bound exists, its blocks carry powers of L, so the honest analytic remainder is of the form $\mathcal{O}_{x\to+\infty}\big(L^K x^{-(N+2)}\big)$ for some K, *not* $\mathcal{O}_{x\to+\infty}\big(x^{-(N+2)}\big)$. The two differ by an unbounded factor.

The third and fourth sentences are then fine *given* the second: the derivative bound and Theorem 16.21 do transfer a residual estimate to an error estimate, as in Theorem E.51.

What is true. The conclusion is correct in shape and wrong in detail. Theorem 20.17 establishes

$$\omega(x) - \omega_N(x) = \frac{\mathsf{L}^{\mathrm{Lam}}_{N+1}(\log x)}{x^{N+1}} + \mathcal{O}_{x\to+\infty}\left(\frac{(\log x)^K}{x^{N+2}}\right), \qquad K = (N+2)^2,$$

and hence (20.17). The logarithmic power K cannot be removed. The next block alone contributes the term $\mathsf{L}^{\mathrm{Lam}}_{N+2}(\log x)\, x^{-(N+2)}$. Its exact size is

$$\asymp (\log x)^{N+2} x^{-(N+2)}, \qquad \text{which is } \textit{not} \qquad \mathcal{O}_{x\to+\infty}\left(x^{-(N+2)}\right).$$

The repair. What must be supplied, and what the formal identity cannot supply, is a genuine analytic estimate for the substituted tail: an explicit finite sum of blocks plus a remainder bounded on an explicit ray, with the constant and the exponent K named. That is item (2) of the proof of Theorem 20.17, and in general it is Theorem 25.5, which converts a *finite* truncation's formal residual order into an inequality $|\varphi(h(x)) - x| \le Cx^{-N}(\log x)^M$ on a ray — note that its hypothesis makes B a finite sum of blocks, precisely so that substitution is a finite computation with numbers. The general lesson is the one printed at the head of this appendix: $\mathcal{O}^{\mathrm{form}}_t(\cdot)$ and $\mathcal{O}_{X\to+\infty}(\cdot)$ are different assertions, and no amount of correct formal algebra produces the second. □

Exercise E.53 (Counterexample: what the minimal Poincaré condition fails to do)**.** A source proposes to define "f has the asymptotic expansion $\sum_n p_n(\log x)x^{-n}$" by the minimal Poincaré requirement that, for each N, the remainder after the term $n = N$ be $o_{x\to+\infty}\big(p_N(\log x)x^{-N}\big)$.

1. Show that at a *single* index N the requirement pins p_N only modulo polynomials of strictly smaller degree.
2. Show that, imposed at *every* index, it does nevertheless pin the blocks — so nonuniqueness is not the objection to it.
3. Give the two objections that are real: the requirement is degenerate as soon as one block vanishes, and at any single index it controls the remainder only within its own outer order, not at the next one.
4. Show that the definition actually used (Theorem 8.43) has neither defect and determines the blocks.

Solution. (1) Fix N and take the two candidate blocks

$$p_N(u) = u^5, \qquad q_N(u) = u^5 + u^3.$$

They differ by u^3, and

$$\big(q_N(\log x) - p_N(\log x)\big)x^{-N} = (\log x)^3 x^{-N} = o_{x\to+\infty}\big((\log x)^5 x^{-N}\big).$$

So the remainder after q_N differs from the remainder after p_N by something that is little-o of *either* of the two last retained terms: the requirement read at the index N alone cannot distinguish the two. The same computation works with u^3 replaced by any polynomial of degree < 5. The reason is that the requirement compares the remainder with the last retained *term*, and on a two-generator scale a term is a whole block, whose parts have different growth. On a one-generator scale $\{x^{-n}\}$ the terms are monomials and the classical condition [25] suffers from none of this.

(2) Now impose the requirement at every index, which is what the proposed definition says. Suppose (p_n) and (q_n) both work for the same f, and let n_0 be least with $d := q_{n_0} - p_{n_0} \neq 0$. Write R_p and R_q for the two remainders after the index $n_0 + 1$. Since $p_n = q_n$ for $n < n_0$,

$$R_q - R_p = d(\log x)\, x^{-n_0} + \big(p_{n_0+1} - q_{n_0+1}\big)(\log x)\, x^{-n_0-1}.$$

Multiply by x^{n_0}. The requirement at n_0+1 makes R_p and R_q little-o of a fixed block times x^{-n_0-1}, so $x^{n_0}\,|R_p|$ and $x^{n_0}\,|R_q|$ are $o_{x\to+\infty}\big((\log x)^K/x\big)$ for some K, and so is the last displayed summand. Hence $|d(\log x)|\to 0$. But $d\neq 0$ is a polynomial, so $|d(\log x)|$ is bounded below by a positive constant for large x. Contradiction; $d=0$, and the blocks agree. It is the clause at n_0+1, not the one at n_0, that does the work — which is exactly the observation part (1) makes, read the other way round.

(3) Two defects survive part (2), and they are what disqualify the requirement.

It degenerates at a vanishing block. "$g=o_{x\to+\infty}(0)$" says $|g|\le\varepsilon\cdot 0$ eventually for every $\varepsilon>0$, i.e. that g vanishes identically for large x. So as soon as one block p_N is zero, the requirement at that index demands an exactly vanishing remainder. Take $f(x)=1+x^{-2}$, which has the finite expansion $F=1+t^2$ and no expansion problem whatsoever. Its blocks are $p_0=1$, $p_1=0$, $p_2=1$, and the requirement at $N=1$ reads $x^{-2}=o_{x\to+\infty}(0)$, which is false. The proposed definition therefore does not apply to f at all, although f is a polynomial in x^{-1}.

It gives no next-order control. At the index N, and provided $\deg_{\mathrm{L}} p_N\ge 1$, the requirement permits a remainder as large as $(\log x)^{\deg_{\mathrm{L}} p_N-1}x^{-N}$, which dominates the whole next block $p_{N+1}(\log x)x^{-N-1}$ by a factor of order x up to powers of $\log x$. A reader who takes "expansion through $n=N$" to mean "correct to the order of the next block" is therefore reading in an estimate that the requirement never made — the same confusion, one level down, as Theorem E.52.

(4) Neither defect afflicts Theorem 8.43, which never divides by a block and never compares the remainder with the last retained term.

What does determine them. Theorem 8.43, and its extension to rational blocks in Theorem 24.4, compare the remainder not with the last term but with the *next outer order*: for every N there are M_N, C_N, x_N with

$$\big|f(x)-\big(\mathrm{Trunc}_{<N}F\big)(x)\big|\le C_N\,x^{-N}(\log x)^{M_N}\qquad(x\ge x_N).$$

Suppose F and G both satisfy this and $F\neq G$; let $n_0=\nu_t(F-G)$ and $p=[t^{n_0}](F-G)\neq 0$. Applying the condition at $N=n_0+1$ to both and subtracting — the two truncations $\mathrm{Trunc}_{<n_0+1}F$ and $\mathrm{Trunc}_{<n_0+1}G$ retain the blocks of index $\le n_0$, and those agree below n_0, so their difference is the *single* term $p(\log x)x^{-n_0}$ —

$$|p(\log x)|\;x^{-n_0}\le C\,x^{-n_0-1}(\log x)^M$$

for large x. Dividing by x^{-n_0}, the left side tends to ∞ or to a nonzero constant in absolute value if $p\neq 0$ — indeed $|p(\log x)|\ge c(\log x)^{\deg_{\mathrm{L}} p}$ for large x — while the right side is $Cx^{-1}(\log x)^M\to 0$. Contradiction; so $F=G$. Uniqueness holds — this is Theorem 24.8, which also covers rational blocks and closed subsectors — and the expansion map of Theorem 8.45 is therefore well defined. □

Exercise E.54 (Trap: a guard that was not taken in the analytic estimate)**.** Let $f(X)=X^2$ exactly, so f realises $F=t^{-2}$ with no error at all, and let g satisfy $g-\big(\mathrm{Trunc}_{<N}G\big)=\mathcal{O}_{X\to+\infty}\big(X^{-N}\big)$ for some $G\in\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$. A draft concludes:

> "$fg-\big(\mathrm{Trunc}_{<N}F\big)\big(\mathrm{Trunc}_{<N}G\big)=f\big(g-\mathrm{Trunc}_{<N}G\big)=\mathcal{O}_{X\to+\infty}\big(X^{-N}\big)$, since the first factor is exact and the second remainder is $\mathcal{O}_{X\to+\infty}\big(X^{-N}\big)$."

Find the error, give the correct exponent, and state the general precision rule for an analytic product.

Solution. The error is that the first factor is not $\mathcal{O}_{X\to+\infty}(1)$; it is X^2. Multiplying a remainder of size X^{-N} by X^2 gives

$$f\big(g-\mathrm{Trunc}_{<N}G\big)=\mathcal{O}_{X\to+\infty}\big(X^{-N+2}\big)\,,$$

two orders worse than claimed. To obtain $\mathcal{O}_{X\to+\infty}\big(X^{-N}\big)$ in the product one must know g to precision $N+2$, i.e. $g-\mathrm{Trunc}_{<N+2}G=\mathcal{O}_{X\to+\infty}\big(X^{-N-2}\big)$.

General rule. Let $f \approx F$, $g \approx G$ with $a = \nu_t F$, $b = \nu_t G$. Suppose

$$f - \operatorname{Trunc}_{<P} F = \mathcal{O}_{X\to+\infty}\big(X^{-P}(\log X)^{M}\big)\,, \qquad g - \operatorname{Trunc}_{<Q} G = \mathcal{O}_{X\to+\infty}\Big(X^{-Q}(\log X)^{M'}\Big)\,.$$

Write

$$fg - \operatorname{Trunc}_{<P} F\ \operatorname{Trunc}_{<Q} G = \operatorname{Trunc}_{<P} F\big(g - \operatorname{Trunc}_{<Q} G\big) + \big(f - \operatorname{Trunc}_{<P} F\big)g.$$

The truncation $\operatorname{Trunc}_{<P} F$ has outer order a, and g has outer order b; so as functions on the ray they are of size X^{-a} and X^{-b} up to powers of $\log X$. The two summands are therefore of sizes

$$\mathcal{O}_{X\to+\infty}\big(X^{-a-Q}(\log X)^{*}\big) \qquad \text{and} \qquad \mathcal{O}_{X\to+\infty}\Big(X^{-P-b}(\log X)^{*}\Big)$$

respectively. So the product is known to absolute order N if and only if

$$P \;\geq\; N - b \qquad \text{and} \qquad Q \;\geq\; N - a,$$

which is verbatim the formal truncated-product rule (7.3) of Theorem 7.4, and its error-tolerant version Theorem 8.14. The analytic guard and the formal guard coincide; what does *not* coincide is the naive symmetric rule $P = Q = N$, which is valid only when $a, b \geq 0$. In the example $a = -2$, $b \geq 0$, so $Q \geq N + 2$, which is the two orders that went missing.

Two riders. First, the logarithmic powers accumulate: the exponent in the product's remainder is at least $M + \deg_{\mathrm{L}}$ of the retained blocks, so a statement of the form $\mathcal{O}_{X\to+\infty}\big(X^{-N}\big)$ with no logarithmic factor is almost never available (Theorem E.52). Second, the same arithmetic with two *inexact* factors is where a draft of the composition transfer went wrong: bounding a product of remainders by the smaller of the two exponents is legitimate only when each remainder is multiplied by a factor of nonpositive outer order, and the hypothesis $N \geq a + \tilde{a}$ is exactly what makes that so. ☐

Exercise E.55 (Realizations form an algebra, and the map that forgets)**.** Let $\mathbb{K} \subseteq \mathbb{C}$, $S = [X_0, +\infty)$ with the real logarithm, and $\mathcal{A}(S) = \{f\colon S \to \mathbb{C} : \exists F \in \mathscr{P\!L}_{\mathrm{poly},\mathbb{K}},\ f \approx F\}$ in the sense of Theorem 8.43.

1. Show F is uniquely determined by f.
2. Show $\mathcal{A}(S)$ is a $\mathbb{K}$-algebra and $f \mapsto F$ is a $\mathbb{K}$-algebra homomorphism into $\mathscr{P\!L}_{\mathrm{poly},\mathbb{K}}$. (For surjectivity, quote Theorem 24.13; do not attempt it by hand.)
3. Identify its kernel and show that it is not zero, so that the homomorphism is not injective; give an explicit nonzero element.
4. Deduce that "$f \approx F$" can never be upgraded to "f is the sum of F" without extra hypotheses.

Solution. (1) This is the uniqueness argument of Theorem E.53: if $f \approx F$ and $f \approx G$ with $F \neq G$, let $n_0 = \nu_t(F - G)$ and $p = [t^{n_0}](F - G) \neq 0$. Subtracting the two estimates at cutoff $N = n_0 + 1$ gives $\big|\big(\operatorname{Trunc}_{<n_0+1} F - \operatorname{Trunc}_{<n_0+1} G\big)(X)\big| \leq CX^{-n_0-1}(\log X)^{M}$ for large X, and that difference is the single term $p(\log X)X^{-n_0}$; dividing by X^{-n_0} makes the left side bounded below by a positive constant times $(\log X)^{\deg_{\mathrm{L}} p}$ and the right side tend to 0. Contradiction.

(2) Linearity is immediate from the triangle inequality and linearity of $\operatorname{Trunc}_{<N} \cdot$. For products, Theorem 8.45 gives $fg \approx FG$; its proof is the guard computation of Theorem E.54 run at $P = N - b$, $Q = N - a$, which is legitimate because $f \approx F$ supplies an estimate at *every* cutoff, in particular at those two. The constant function 1 realises 1, so the map is unital. Surjectivity is the Borel–Ritt theorem on this scale, Theorem 24.13: every $F \in \mathscr{P\!L}_{\mathrm{poly},\mathbb{K}}$ is realised by some C^{∞} function, real valued when $\mathbb{K} \subseteq \mathbb{R}$. It is worth noticing what that theorem does *not* say — it produces a smooth realisation, not an analytic one, and it produces no canonical choice, since by (3) below any two realisations differ by a flat function.

(3) The kernel is the ideal $\mathcal{N}$ of *flat* functions of Theorem 24.11, $\{f : f = \mathcal{O}_{X\to+\infty}(X^{-N})$ for every $N\}$, which is exactly the condition $f \approx 0$. It is an ideal, being the kernel of a ring homomorphism, and it is not zero: $f(X) = \mathrm{e}^{-X}$ is nonzero everywhere on S and satisfies $\mathrm{e}^{-X}X^N \to 0$ for every N, so $\mathrm{e}^{-X} \approx 0$. Hence e^{-X} and 0 have the same expansion, and the map is not injective. A second witness, of a different character: $\sin(\mathrm{e}^X)$ is bounded and oscillatory, but $\mathrm{e}^{-X}\sin(\mathrm{e}^X)$ is again flat and nonzero.

(4) Consequently the expansion F determines f only modulo flat functions, and $\mathcal{N}$ is large: by Theorem 24.12 it is prime but not maximal, is not stable under differentiation, and contains no smallest nonzero scale, since $f \in \mathcal{N}$ eventually nonvanishing gives $f^2 \in \mathcal{N}$ with $f^2 = o_{X\to+\infty}(f)$. "f is the sum of F" would assert both that the series converges and that its sum is f; the first is a claim about $\sum_n p_n(\log X)X^{-n}$ as a numerical series and the second would force injectivity, which (3) refutes. What $\approx$ does assert is exactly (8.37): a family of finite inequalities, one per cutoff, each on its own ray. Any statement stronger than that needs a summation method and its own theorem. ☐

Exercise E.56 (Exactness at a point, and what the numbers do and do not show).

1. Show that every truncation is exact at the single point $x = 1$, that is, that $\omega_N(1) = \omega(1) = 1$ for every N; and explain why this is not a convergence statement.
2. Take $x = 20$, so that $L = \log 20 = 2.995732\ldots$. The terms of the series, and the errors after each of them, are, to four significant figures,

n	$\mathsf{L}_n^{\mathrm{Lam}}(L)\,20^{-n}$	$\omega(20) - \omega_n(20)$
1	1.498×10^{-1}	3.507×10^{-3}
2	3.729×10^{-3}	-2.220×10^{-4}
3	-1.880×10^{-4}	-3.395×10^{-5}
4	-3.267×10^{-5}	-1.282×10^{-6}
5	-1.433×10^{-6}	1.503×10^{-7}
6	1.259×10^{-7}	2.443×10^{-8}
7	2.353×10^{-8}	8.975×10^{-10}
8	1.047×10^{-9}	-1.496×10^{-10}

   Check the table against (20.16), and against the sign prediction of Theorem E.44.

3. The terms decrease monotonically through $n = 8$ and beyond. Does that show that the series $\sum_n \mathsf{L}_n^{\mathrm{Lam}}(\log x)\,x^{-n}$ converges at $x = 20$? What does this volume actually prove?

Solution. (1) At $x = 1$ one has $L = \log 1 = 0$, and $\mathsf{L}_n^{\mathrm{Lam}}(0) = 0$ for every n by (21.3). By Theorem 20.12, $\omega_N = X + \mathrm{Trunc}_{<N+1}\,\mathcal{P}$, so

$$\omega_N(1) = 1 + \sum_{n=0}^{N} \mathsf{L}_n^{\mathrm{Lam}}(0)\cdot 1^{-n} = 1 + 0 = 1.$$

And $\omega(1) = 1$ because $1 + \log 1 = 1$. So every truncation is exact at that one point.

This says nothing about convergence, and it is worth being explicit about why. The statement "$\omega_N(1) = \omega(1)$ for all N" is a family of identities between numbers at a single x. An asymptotic statement $\mathcal{O}_{x\to+\infty}(\cdot)$ has no content at a single point — multiplying a function by any factor that equals 1 at $x = 1$ changes nothing about it. And a convergence statement would be about the tail $\sum_{n>N}$, which here is identically zero at $x = 1$ for a trivial reason: every summand vanishes. The same phenomenon in Lambert coordinates is the exactness of the branch expansion at $z = \mathrm{e}$, where $\xi_0 = 1$ and $\ell_0 = 0$; this is Theorem 20.16 and Theorem 22.20, and it is a consequence of $\mathsf{L}_n^{\mathrm{Lam}}(0) = 0$, of nothing else. Indeed the convergence criterion of the Lambert part fails outright at that point.

(2) Theorem 20.17 asserts $\omega(x) - \omega_N(x) = \mathsf{L}_{N+1}^{\mathrm{Lam}}(L)x^{-(N+1)} + \mathcal{O}_{x\to+\infty}(L^K x^{-(N+2)})$. Reading the table, the error after N should be close to the term at $n = N+1$, and it is, in every row: 3.507×10^{-3}

against 3.729×10^{-3}; -2.220×10^{-4} against -1.880×10^{-4}; -3.395×10^{-5} against -3.267×10^{-5}; and so on down to 8.975×10^{-10} against 1.047×10^{-9}. Both magnitude and sign agree throughout, which is (20.17) in action.

The signs are themselves informative. Here $x = 20$ lies above the crossing point $x_* = 8.7465\ldots$ of Theorem E.44 (2), and $\mathsf{L}_2^{\mathrm{Lam}}(L) > 0$ since $L = 2.9957 > 2$; so ω_1 undershoots, as that exercise predicts. The later signs follow the sign of $\mathsf{L}_n^{\mathrm{Lam}}(L)$, which changes as L passes the real roots of the successive Lambert polynomials. There is no alternating pattern in n to be read off, and none should be expected: the sign is the sign of a polynomial value, not of a coefficient.

(3) No — monotone decrease of finitely many computed terms is evidence, not proof, and in an asymptotic series the terms typically decrease for a long time before turning. The series does converge at $x = 20$, but for a reason that has nothing to do with the table. This volume proves four separate things, which must be kept apart:

- the *formal* identity (20.11), an equality in $\mathcal{G}_{\mathrm{poly},\mathbb{K}}$;
- the *asymptotic* statement (20.16), valid as $x \to +\infty$ with an explicit logarithmic loss $K = (N+2)^2$;
- the *frozen* convergence of Theorem 20.19: for each fixed u_0 the series $\sum_{n\geq 1} \mathsf{L}_n^{\mathrm{Lam}}(u)s^n$ has a positive radius in s, uniformly near u_0;
- the *substituted* convergence of Theorem 22.13: with $\xi = x$, $K = x$ and $\log K = L$ that theorem has $M = L$, its seating condition (22.16) reads $2(1+L) < x$, and under it the series $\sum_{n\geq 1} \mathsf{L}_n^{\mathrm{Lam}}(L)x^{-n}$ converges absolutely, with sum $\omega(x) - x + L$.

The fourth item settles the question the table raises, and settles it in the affirmative: at $x = 20$ the seating condition reads $7.991 < 20$, so the series converges absolutely there. The two statements must still not be confused. Convergence at a fixed x is a statement about one point, proved by a Cauchy estimate on a two-variable holomorphic function; the sharp order of (2) is a statement about the limit $x \to +\infty$, with $u = \log x$ and $s = 1/x$ linked. Neither implies the other, and Theorem 20.19 implies neither: it freezes u and its radius $r(u)$ is not claimed to be bounded below as $|u| \to \infty$ (Theorem 20.20). What the table alone shows is the sharp-order statement of (2), and nothing more; the convergence is a theorem imported from Part 22, not a reading of the numbers. $\square$

Remark E.57 (Where each exercise came from). Of the 56 exercises above, 31 descend from a source problem — in most cases from several at once, since the corpus's 83 problems are heavily duplicated, and in most cases with the statement strengthened, the hypotheses printed, or the supplied answer corrected; 9 were built from worked examples in reports 4, 5 and 6, which set no exercises at all; and 16 are new, written because a technique of the body had no exercise anywhere in the corpus. Every trap and every counterexample exercise — 8 and 11 respectively — is in the last two groups.

The source problems that left no descendant were of three kinds: duplicates of a retained problem, arithmetic drill on an operation already exercised elsewhere, and problems whose *supplied* solution is wrong. Of the last kind there are five, all corrected above and all recorded in Part G: the class in which $(L+t)^{-1}$ lives, the universal third inverse block, the sign weight in the lower Lambert branch, the range of the Stirling sum for $\Delta_{0,k}$, and the passage from a formal residual identity to an analytic error bound.

Appendix F

Notation, and a concordance with the six sources

F.1 The normative source

Every mathematical symbol in this volume is defined normatively in the corpus notation catalogue, *FabiusFunction Mathematical Notation Catalogue*, whose entries `polynomial-logarithmic-ambient`, `polynomial-logarithmic-leading-data`, `polynomial-logarithmic-truncation`, `polynomial-logarithmic-differential`, `polynomial-logarithmic-composition`, `polynomial-logarithmic-reversion`, `wright-omega-lambert-polynomials`, and `lambert-branch-expansions` cover exactly the material of this volume. This document introduces no new mathematical notation. Its preamble defines short local aliases — $\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$, $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$, $\mathscr{PL}_{\mathrm{rat},\mathbb{K}}$, $\mathscr{PL}_{\mathrm{itLaur},\mathbb{K}}$, $\mathcal{G}_{\mathrm{poly},\mathbb{K}}$, ν_t, lm, lc and the rest — and every one of them expands to a catalogue command; the aliases exist to keep displayed formulas readable, not to name anything the catalogue does not already name.

F.2 Symbols used throughout

Table F.1: Principal notation.

Symbol	Alias	Meaning
X	–	the large variable; the default regime is $X \to +\infty$ along the positive real ray
t	–	X^{-1}, the outer generator
L	–	$\log X$, the logarithmic generator, formally independent of t
$\mathbb{K}$	`\K`	the coefficient field, of characteristic zero
$\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$	`\PLpow`	$\mathbb{K}[L][[t]]$, the polynomial-log power-series ring
$\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$	`\PLlau`	$\mathbb{K}[L]((t))$, its Laurent extension: finitely many positive powers of X
$\mathscr{PL}_{\mathrm{rat},\mathbb{K}}$	`\PLrat`	$\mathbb{K}(L)((t))$, the rational-log Laurent field
$\mathscr{PL}_{\mathrm{itLaur},\mathbb{K}}$	`\PLit`	$\mathbb{K}((L^{-1}))((t))$, the iterated Laurent-log field
$\mathcal{G}_{\mathrm{poly},\mathbb{K}}$	`\Gnear`	$\{X + A : A \in \mathscr{PL}_{\mathrm{pow},\mathbb{K}}\}$, the near-identity group under composition
$\nu_t F$	`\ordt`	the outer valuation, with $\nu_t 0 = +\infty$

Symbol	Alias	Meaning
$\operatorname{lm}(F)$, $\operatorname{lc}(F)$	`\lm`, `\lc`	leading monomial and scalar leading coefficient; the leading *block* is the whole polynomial $p_{\nu_t F}$, a third and different object
$\deg_{\mathrm{L}} p$	`\degL`	degree in L
$d_F(n)$	`\dprofile`	the log-degree profile, $-\infty$ where the block vanishes
$\operatorname{Trunc}_{<N} F$	`\Trunc`	the *exclusive* truncation $\sum_{n<N} p_n(L)t^n$
$\mathcal{O}_t^{\mathrm{form}}(t^N)$	–	a formal valuation tail: coefficient agreement below outer order N, and nothing analytic
$\mathcal{O}_{X\to\infty}(A(X))$	–	an analytic estimate, with the regime printed
D_X	`\DX`	the distinguished derivation $t\partial_L - t^2\partial_t$
$\Delta_{n,k}$	`\shiftop`	$\prod_{j=0}^{k-1}(\partial_L - n - j)$, empty and equal to the identity at $k=0$
$F^{\langle -1\rangle\circ}$	`\rev`	compositional inverse, $F \circ F^{\langle -1\rangle\circ} = F^{\langle -1\rangle\circ} \circ F = X$
F^{-1}	`\recip`	multiplicative inverse, $FF^{-1} = 1$
$\operatorname{Err}_{\mathrm{rev}}(F,H)$	`\RevErr`	the reversion residual $F \circ H - X$; *both* arguments are part of the object
ω	–	Wright omega, the inverse of $Y \mapsto Y + \log Y$
$\mathsf{L}_n^{\mathrm{Lam}}$	`\LamP`	the canonical zero-based Lambert polynomials
$\mathsf{L}_n^{\mathrm{Lam,sc}}$	`\LamPsc`	the scaled family $n!\mathsf{L}_n^{\mathrm{Lam}}$, for $n \geq 1$
$\left[{n \atop k}\right]$	`\stirlingone`	unsigned Stirling numbers of the first kind (cycle numbers)
$s(n,k)$	`\stirlingsigned`	signed Stirling numbers of the first kind
$\prec, \succ$	–	the dominance order on monomials

F.3 Concordance with the absorbed sources

The six sources used mutually incompatible local abbreviations, and in two cases the *same* abbreviation for different objects. A reader returning to a source through the Git history needs the following table.

Table F.2: Local abbreviations of the absorbed sources and their meaning here. S1–S6 are as identified in Table I.1.

Source	Local	This volume
S1	`\Plog`, `\Rlog`, `\Hlog`	$\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$, $\mathscr{PL}_{\mathrm{rat},\mathbb{K}}$, $\mathscr{PL}_{\mathrm{itLaur},\mathbb{K}}$
S1	`\val`, `\trunc`	ν_t, exclusive $\operatorname{Trunc}_{<N}\cdot$
S1	`\invcomp`	$\cdot^{\langle -1\rangle\circ}$
S2	`\val`	ν_t
S2	`\rev`, `\recip`	$\cdot^{\langle -1\rangle\circ}$, $\cdot^{-1}$ (already the catalogue commands)
S2	`\Dop` $= \mathrm{D}$	D_X
S2	`\normalexp`	normal-ordered shifted exponential; expanded inline here
S3	`\Rring`, `\Pring`, `\Fring`, `\Ggroup`	$\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$, $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$, $\mathscr{PL}_{\mathrm{itLaur},\mathbb{K}}$, $\mathcal{G}_{\mathrm{poly},\mathbb{K}}$
S3	`\ordt`, `\D` $= \mathrm{D}$	ν_t, D_X
S4	`\PL`	$\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$ — *not* $\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$; see below
S4	`\RL`, `\HL`, `\Gnear`	$\mathscr{PL}_{\mathrm{rat},\mathbb{K}}$, $\mathscr{PL}_{\mathrm{itLaur},\mathbb{K}}$, $\mathcal{G}_{\mathrm{poly},\mathbb{K}}$
S4	`\ordx`, `\degell`	ν_t, $\deg_{\mathrm{L}}$
S5	`\PL`	$\mathscr{PL}_{\mathrm{pow},\mathbb{K}}$ — *not* $\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$; see below
S5	`\PLL`, `\QPL`	$\mathscr{PL}_{\mathrm{poly},\mathbb{K}}$, $\mathscr{PL}_{\mathrm{rat},\mathbb{K}}$
S5	`\D` $= \mathcal{D}$	D_X

Source	Local	This volume
S6	`\PLpoly`, `\PLfield`	$\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$, $\mathscr{P}\mathscr{L}_{\mathrm{itLaur},\mathbb{K}}$
S6	`\EulerDerivation`	$\vartheta = XD_X$
S6	`\trunc{F}{N}` $= [F]_{\leq N}$	*inclusive*; equals $\mathrm{Trunc}_{<N+1} F$
S6	`\compinv`	$.^{\langle -1\rangle_\circ}$

Two abbreviations that collide across sources

1. `\PL` denotes the *Laurent* ring $\mathbb{K}[L]((t))$ in S4 and the *power-series* ring $\mathbb{K}[L][[t]]$ in S5. A statement transported from one to the other without checking which ring is meant can silently acquire or lose the finitely many positive powers of X; the unit criterion and the antiderivative obstruction both change. In S4 the power-series subring was written $[\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}]_{\geq 0}$, which is $\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$.

2. `\trunc` is the exclusive $\mathrm{Trunc}_{<N}$ in S1 and the inclusive $[\,\cdot\,]_{\leq N}$ in S6. Every cutoff index transported from S6 into this volume has been shifted by one block. A residual certificate stated with the wrong convention is off by exactly one order, which is precisely the amount that makes a correct algorithm look wrong and an incorrect one look right.

F.4 Conventions that are not merely notational

Three conventions are recorded here because getting them wrong changes theorems, not only glyphs.

1. **Empty products and sums.** $\Delta_{n,0} = 1$; the Lagrange–Bürmann coefficient product is empty at $r = 1$; $C_{0,0} = 1$ and $C_{m,0} = 0$ for $m > 0$; $s(0, 0) = 1$. Every statement below is written so that these conventions make the boundary case correct rather than exceptional.

2. **The three $\mathcal{O}$ roles.** A formal valuation tail, an analytic estimate, and an algorithmic cost bound are three different assertions and are printed with three different tokens. None may be inferred from another. In particular, $\mathcal{O}_t^{\mathrm{form}}(t^N)$ carries no uniformity in L whatsoever, and a bound uniform in L is a separate theorem with separate hypotheses.

3. **$t^N \mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ is not an ideal of $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$.** In the Laurent ring t is a unit, so the principal ideal it generates is the whole ring. Quotient-algebra language is therefore available on $\mathscr{P}\mathscr{L}_{\mathrm{pow},\mathbb{K}}$ and not on $\mathscr{P}\mathscr{L}_{\mathrm{poly},\mathbb{K}}$; on the latter, use $\nu_t(F - G) \geq N$, the formal tail token, or equality of exclusive truncations.

Appendix G

Ledger of editorial repairs

Every correction this consolidation made to its sources is marked at the point of repair in this document's own LaTeX source with an `% ed.:` comment. The table below is generated from those marks, so it cannot drift out of step with the text. Repairs of pure notation — a symbol renamed to its catalogue command — are not listed; those are recorded once, in the concordance of Table F.2. What is listed is every place where a source's mathematical statement, proof, hypothesis, index range, or cost claim was wrong, incomplete, or overstated.

Three kinds of defect recur often enough to name. The first is an *omitted hypothesis*: a statement true under conditions the source did not print, most often a missing empty-product convention, a missing guard-precision requirement, or a missing restriction to $n \geq 1$. The second is *overclaiming*: an analytic conclusion drawn from formal algebra alone, typically a convergence or uniformity assertion that the argument does not support. The third is a *cost or sharpness claim* asserted without the count that would establish it.

Table G.1: Editorial repairs, in document order.

Location	Repair
Theorem 4.8	source draft 5 justifies the lexicographic order only in the case alpha > alpha', leaving the tie-breaking case alpha = alpha' unproved. The display above proves both cases.
Theorem 4.10	the draft said X^{1/2} "arises from inverting a non-integer leading power". That is backwards, and contradicts the reading of the Fabius endpoint law in the remark on its status: a half-integer exponent is produced by inverting an *integer* leading power other than the first (X^2 inverts to X^{1/2}); an inner argument that is already Puiseux has left the class before the inversion begins.
Theorem 4.10	source draft 6 prints the refined scale as "..., X^{-n}L^2, X^{-n}L, X^{-n}, X^{-n}L^{-1}, ...", which silently places negative powers of L inside the polynomial-logarithmic class. They are not there: coefficient blocks of K[L]((t)) are polynomials, so L-exponents are nonnegative, and L^{-1} belongs to the enlargements named above.
Theorem 4.14	the draft called PLlau "the t-adic completion of the Laurent ring in (iii)". That is exactly the error item 7 of the normalizations remark warns against: t is a unit in K[t,t^{-1}][L], so every power of the ideal (t) is the whole ring, the t-adic filtration is constant, and the associated completion collapses to zero. The correct construction completes the polynomial subring K[L][t] and only then inverts t.

Location	Repair
Why powers and logarithms belong in one	the draft attached the exponentiated form to the full display, where it is false — exponentiating \eqref{plt:eq:mot-generic-implicit} exponentiates the higher blocks as well. The equivalence holds for the two-term equation only, and is stated for it.
Theorem 4.21	the draft named the two parts of this estimate A and B, colliding with the B = log(1/y) of the corollary and the status remark that quote this proposition. The internal quantities are renamed A_1 and A_2; B is reserved for the inverse-endpoint variable throughout the subsection.
Why powers and logarithms belong in one	source draft 1 asserts convergence "on the principal real branch for z >= e", supported only by a general citation. That threshold is not established anywhere in the six drafts, and the assertion also conflates a convergence claim with the separate, elementary observation that the series is exact at z = e. We keep the coefficient statement, proved below, and defer the convergence domain to Part VI with its own citation. the draft called the asymptotic display \eqref{plt:eq:mot-W-two-orders} "exact at z = e". A display whose remainder is an analytic O as z -> +infty asserts nothing at a single point, so that formulation commits precisely the conflation the box above forbids. The correct elementary statement is a coefficient identity for the underlying formal series, and it is proved here.
Reader's guide and the structure of this	the draft called "Part V may be read directly after Part IV" an exception to the rule that each part depends only on those before it. It is not an exception at all — V does come after IV — and it also contradicts reading path 1 below, which does draw two statements from Parts II and III. The intended and correct content is the lightness of that dependency.
Reader's guide and the structure of this	source draft 2 tabulates the output class of differentiation as "t K[L]((t)) up to the leading power". That set equals K[L]((t)), because t is a unit there, so the row asserts nothing. The correct statement is the valuation inequality nu_t(D_X F) >= nu_t(F) + 1, proved in Part II, which can be strict only when a constant leading block is annihilated.
Reader's guide and the structure of this	source draft 4 prints the coefficient condition of its defining display as "p_n in K[X]", naming the large variable where the logarithmic one is meant. The intended and correct condition is p_n in K[L].
The formal polynomial–logarithmic rings	the chain \eqref{plt:eq:ring-chain} exhibits three inclusions, not four; the count was wrong as written.
Theorem 5.6	the first inclusion is not a change of coefficient ring (the printed chain \(\K[L]\subset\K[L]\subset\K(L)\) made it look like one); it is the inclusion of a power-series ring in the corresponding Laurent ring.
Theorem 5.6	this appeal was unsupported and mildly circular: the geometric-series corollary was stated only for coefficient rings whose list already contained \(\K((L^{-1}))\), the very object being constructed here. \cref{plt:cor:ring-geometric} is now proved for an arbitrary commutative coefficient ring, so the instance \(E=\K\), generator \(u\), is legitimate.
Theorem 5.11	S4 calls \(t\K[\ell][[t]]\) "the strictly small ideal" of the Laurent ring \(\K[\ell]((t))\); it is neither small in that ring nor an ideal of it.
Theorem 5.14	the statement imposed the standing hypothesis \(\alpha>\alpha'\) and then appended the case \(\alpha=\alpha'\), which that hypothesis excludes; the two cases are now alternatives of one hypothesis. The regime \(X>\EulerE\) is printed here rather than produced inside the proof.

Location	Repair
Theorem 5.14	the sources state this without the hypothesis \(X>\EulerE\); for \(1<X<\EulerE\) one has \(0<\log X<1\) and high logarithmic powers are small, so the comparison is genuinely eventual.
Valuation, leading data, and the degree	"\(\ordt\) is a discrete valuation on \(R\) with valuation ring \(E[[t]]\)" overstates the case for two of the four classes: \(\PLpow\) and \(\PLlau\) are not fields, and on \(\PLpow\) the image of \(\ordt\) is \(\NaturalNumbers\), not \(\IntegerNumbers\). The claim is split accordingly.
Theorem 6.3	the domain hypothesis enters only through the reverse inequality in (i). For an arbitrary commutative ring \(E\) the subadditivity \(\ordt(FG)\ge\ordt F+\ordt G\), together with (ii) and (iii), still holds in \(E[[t]]\) and \(E((t))\), by the first half of the computation in (i); this weaker form is what \cref{plt:cor:ring-geometric} uses, and stating it here is what keeps that corollary free of a circular appeal. S1 and S5 record the strictness of (ii) as happening "exactly when the leading blocks cancel", omitting the hypothesis \(\ordt F=\ordt G\); with unequal valuations no cancellation is possible and equality always holds.
Theorem 6.5	the display "\(p_\nu=aL^d+\text{lower powers of }L\)" is literally a polynomial expansion, but \(R\) here ranges over all four classes, where a block need not be a polynomial. The reading is fixed explicitly.
Valuation, leading data, and the degree	"equivalently" was wrong: additivity of \(v\) is equivalent to multiplicativity of \(\lm\) alone; multiplicativity of \(\lc\) is a further assertion (proved below), not a restatement.
Theorem 6.8	"with value group \(\Gamma\)" is false as stated for two of the four classes: on \(\PLpow\) and \(\PLlau\) a block is a polynomial, so \(\degL\ge0\) and the image of \(v\) never meets \(\IntegerNumbers\times \PositiveIntegers\). The image and the group it generates are separated.
Theorem 6.11	S2 states the prefactor form \(A=cX^{\rho}L^{\beta}(1+U)\) "whenever available"; the example \(L+t\) shows that over \(\K[L]\) it is available only after passing to \(\K(L)\), so no argument may assume it inside \(\PLlau\).
Theorem 6.15	the monotonicity clause \(\mathcal D_\sigma\subseteq\mathcal D_{\sigma'}\) was asserted without proof, and it is not the trivial inequality \(\sigma n\le\sigma'n\): that fails for \(n<0\). The argument below uses that \(\dprofile{F}\) is \(-\infty\) below \(\ordt F\).
Theorem 6.15	S2 states \(\deg c_n\le A+B+\max(\alpha,\beta)n\) for the product. That form is correct only when the factor of smaller slope has nonnegative outer order; the counterexample of \cref{plt:ex:lead-slope-fails} violates it, and \eqref{plt:eq:lead-slope-bound} is the corrected statement, attained there with equality.
Theorem 7.3	S6 uses the inclusive marker throughout; the catalogue makes the exclusive operator normative, so every transferred formula is shifted by one block.
Theorem 7.6	S5 states \eqref{plt:eq:trunc-product-pow} without the hypothesis \(\ordt\ge0\); on \(\PLlau\) it is false, and \eqref{plt:eq:trunc-product} is the correct general form.
Theorem 7.9	\cref{plt:def:trunc-formal-o} defines the token only for an integer cutoff, so (ii) and (iii) below were type-incorrect when \(H\) or \(G\) vanishes and the printed exponent becomes \(+\infty\). The reading of that degenerate case is now fixed explicitly.

Location	Repair
Theorem 7.11	the displayed analytic statement was printed with none of \(f\), \(S\) or \(n_0\) introduced, contrary to the standing requirement that a statement name its regime and domain. They are named here.
Theorem 7.11	S1 offers instead the minimal Poincar\'e requirement that the remainder be little-o of the last retained term. That condition does not determine the blocks: with \(p_n=L^5\) and \(q_n=L^5+L^3\) the two candidate blocks differ by \(L^3=\LittleOAt{L^5}{X\to+\infty}\), so uniqueness fails. The displayed form, and S6's \(\LittleOAt{X^{-N}}{X\to+\infty}\) form, both do determine them.
Theorem 7.12	the label kind "box" is not in the normative list of kinds; these boxed statements are remarks, so each label is renamed to the "rmk" kind.
Theorem 7.15	the label kind "box" is not in the normative list of kinds; these boxed statements are remarks, so each label is renamed to the "rmk" kind.
Theorem 7.19	S2's completeness proof asserts the common lower bound because "the sequence is already Laurent-bounded at some initial precision". That is not a hypothesis of the statement; the bound must be extracted, as above, from the Cauchy condition at cutoff \(N=0\).
Theorem 7.19	the word "exactly" in (iv) asserts that the common lower bound is also necessary; the sources supply only sufficiency, so the converse is proved here.
Theorem 7.20	S3 asserts that \(t\)-adic convergence is "equivalent" to coefficientwise stabilization and that both the power-series and the Laurent ring are complete "in this sense". The equivalence is false on the Laurent ring, as follows; the completeness assertion itself is correct, but needs the proof of \cref{plt:thm:trunc-completeness}\,(iii).
Theorem 7.23	the identification of \(\mu\) was printed as an equality. It is only a lower bound: if every \(\ordt S_i>0\) the right-hand side is \(0\) while \(\mu>0\). Finiteness is all the argument needs, and all it gives.
Theorem 7.24	the corollary was stated only for \(E\in\SetOf{\K[L],\K(L),\K((L^{-1}))}\), yet it is invoked inside \cref{plt:prop:ring-chain} with \(E=\K\) and generator \(L^{-1}\), precisely in order to prove that \(\K((L^{-1}))\) is a field — the fact that puts \(\K((L^{-1}))\) on that list. Stating it for an arbitrary commutative coefficient ring removes the circle at no cost: the proof never needs a domain hypothesis.
Theorem 7.28	the transfer was claimed only for the four ambient classes, but \cref{plt:cor:ring-geometric} needs it for \(E((t))\) with \(E\) an arbitrary commutative ring (the case \(E=\K\), generator \(L^{-1}\), is used in \cref{plt:prop:ring-chain}). The scope of the transfer is widened, and the two places where the domain hypothesis entered are named.
Theorem 7.28	the label kind "box" is not in the normative list of kinds; these boxed statements are remarks, so each label is renamed to the "rmk" kind.
Theorem 7.28	the volume regime is \(X\to+\infty\) on the positive ray; the generic row printed the weaker \(X\to\infty\).
Theorem 8.8	the four classes were displayed on one line, which overfills the text block by about 45pt; set as a two-by-two array instead.

Location	Repair
Theorem 8.13	the bracketed title duplicated that of \cref{plt:prop:trunc-product}. The two statements are not the same result: the ring-chapter proposition is the law on \(\PLlau\) at the canonical cutoffs, this one is the law over an arbitrary logarithmically graded coefficient ring at arbitrary sufficient cutoffs, together with optimality and sharpness. The titles now say so; see \cref{plt:rmk:mul-leading-restated}.
Multiplication: convolution of logarithm	the optimality of $(N-b,N-a)$ was asserted unconditionally. For $N\le a+b$ both sides of \eqref{plt:eq:mul-truncated-product} vanish whatever P,Q are, so no pair is forced and there is nothing to be optimal about; the clause is now stated under $N>a+b$, which is what the argument actually establishes.
Theorem 8.14	the source bounds the third term by $P+Q$ and asserts $P+Q\ge N$. That inequality does not follow from $P\ge N-b$ and $Q\ge N-a$: for $a=b=0$ and $N=P=Q=-5$ one has $P+Q=-10<N$. The repair is to bound $\ordt H$ by b rather than by Q, which is legitimate because every block of H is a block of G.
Theorem 8.14	S6, Theorem "Closure and finite determination", eq. (mult-error-rule). The source states the conclusion as O(t^{min(p+m_0, q+n_0)}) without any hypothesis relating p,q to the valuations n_0,m_0, and its proof silently bounds ord(E*Gtilde) by p+m_0 and ord(Ftilde*H) by q+n_0, which requires ord(Gtilde) >= m_0 and ord(Ftilde) >= n_0. Neither follows from the stated hypotheses. The corrected statement carries the third entry p+q, and the hypotheses p >= n_0, q >= m_0 under which the source's form is recovered. Note that these hypotheses are sufficient but not necessary: the third entry is redundant exactly when p >= a or q >= b.
Theorem 8.18	retitled to distinguish it from \cref{plt:def:lead-leading-data}, which is the same data on the four ambient classes; only the coefficient ring differs. See \cref{plt:rmk:mul-leading-restated}.
Theorem 8.19	the leading data, the rank-two valuation, the log-degree profile, the exclusive truncation law and the summability calculus are all introduced in the ring chapter. Restating them here without saying so would leave two definitions of the same symbols standing in one volume. This remark records that the restatements are verbatim generalizations over the coefficient ring and agree with the originals.
Theorem 8.21	the first equivalence of the display was left unargued. It is not a theorem of this chapter at all: $\succdom$ is defined by the right-hand condition in \cref{plt:def:ring-dominance}, so only the analytic equivalence needs an argument, and that argument restates \cref{plt:lem:ring-dominance-analytic} for integer exponents.
Theorem 8.23	the statement asserts that neither $\PLpow$ nor $\PLlau$ is a field, but the argument as given covers only $\PLlau$. One line closes it.
Theorem 8.24	retitled to distinguish it from \cref{plt:def:lead-profile}, which is the same function on \(\PLlau\), where the blocks are polynomials and the codomain is \(\NaturalNumbers\cup\SetOf{-\infty}\). See \cref{plt:rmk:mul-leading-restated}.

Location	Repair
Theorem 8.26	the criterion was stated as an unconditional “if and only if”. It fails in the degenerate case $P_N=\emptyset$ (equivalently $M=-\infty$): there $[t^N](FG)=0$, so equality holds although the empty sum \eqref{plt:eq:mul-top-coefficient} is zero. The criterion is now confined to M finite and the degenerate case is recorded separately.
Theorem 8.26	S6, Proposition "Degree and support bounds", asserts equality "whenever the maximizing blocks are nonzero" under the hypothesis that all coefficients are nonnegative over an ordered field. That hypothesis is far stronger than necessary and the quoted sufficient condition is stated ambiguously (a zero block has degree -infinity and cannot maximize at all). The criterion below is exact and the positivity hypothesis has been reduced to the leading coefficients of the maximizing blocks only.
Theorem 8.27	condition (3) as first drafted omitted $P_N\ne\emptyset$; over an ordered field its hypotheses are satisfied vacuously when P_N is empty, and the conclusion would then be the false assertion that the sum \eqref{plt:eq:mul-top-coefficient} is nonzero.
Theorem 8.30	the inclusion $\mathfrak A_\alpha\subseteq\mathfrak A_{\alpha”}$ for $\alpha\le\alpha”$ was used without proof. On the Laurent ring it is not immediate: for $n<0$ a larger slope gives a \emph{smaller} bound, so the constant has to be increased, and that is possible only because the support is bounded below. The argument is supplied here.
Theorem 8.33	retitled, and its relation to \cref{plt:def:trunc-summable} made explicit: the condition is verbatim the one of the ring chapter, read over the coefficient ring R instead of $\K[L]$, which \cref{plt:rmk:trunc-summability-classes} already licenses. Only the two clauses of \cref{plt:lem:mul-summable} are used below, so they are the only ones recorded here.
Theorem 8.35	the source calls the two infima “finite by (1)”; part~(1) gives only $>-\infty$, and the value $+\infty$ does occur, namely when every member of the family is zero. That case is disposed of separately.
Theorem 8.44	retitled: “Basic properties of a scale” in the differential part (\cref{plt:lem:dif-scale-basic}) is about the coefficient field of a power–log scale, not about the relation $\approx$ between a function and a series. The two lemmas share nothing but the word “basic”.
Theorem 8.46	the three-term estimate was set on one display line, overfilling the text block by about 53pt; broken across two lines.
Theorem 9.4	S5 and S6 assert the nonunit direction by citing the general power-series unit criterion; the additivity step that makes it valid on the Laurent ring is supplied here explicitly.
Theorem 9.5	the direct-product step was justified by "the three subgroups pairwise intersect trivially", which does not imply an internal direct product for three subgroups (the Klein four-group refutes it); the isomorphism is read off instead from surjectivity plus the uniqueness just proved.
Theorem 9.8	\cref{plt:cor:div-quotient} invokes this theorem "over the coefficient ring E", a generality the statement did not record; the proof uses only commutativity and invertibility of B_0, so the clause is supplied here.
Theorem 9.10	S6 states this correctness theorem with its local inclusive truncation [F]_{<=N}; the statement above is the exclusive normalisation demanded by the catalogue, and the two differ by exactly one block.

Location	Repair
Theorem 9.12	S6 states a "guard precision" principle in words ("compute the quotient beyond the final requested block") but never says by how much; the exact guard is the content of the proposition above.
Theorem 9.13	the negative-guard case was explained by "the denominator has a Laurent principal part", which accounts only for the numerator guard b; the denominator guard 2b-a is negative under the unrelated condition a > 2b.
Theorem 9.15	S1 prints the block Q_3 of this example immediately after inputs given only modulo t^3; the hypothesis that the omitted blocks vanish is a genuine extra assumption and is stated here.
Theorem 9.19	pointwise finiteness alone does not place the sum in the field: one must also check that each block is a Laurent series in L^{-1}, i.e. that its support is bounded below. The same bound gives this at once, and no source records the step.
Theorem 9.29	the input-requirement statement (4) is absent from S2, S3, S5 and S6, each of which asserts doubling without saying which truncations suffice.
Theorem 9.29	the precondition N >= 1 was not printed; with N = 0 the loop body never runs and the routine returns the scalar B_0^{-1} rather than Trunc_{<0}(1/B) = 0.
Theorem 9.30	the induction covers only the exact powers of two, while \cref{plt:alg:div-newton} caps every step at N; the capped step needs its own line before the "in particular" clause is justified.
Theorem 9.30	S2 justifies the doubling by "the valuation is multiplicative"; the valuation is additive on products, which is what the argument uses.
Theorem 9.32	the convention "multiplications by the scalar B_0^{-1} are not counted" was stated loosely enough to exclude also the products B_j C_0 and C_0 E_j, under which reading the two schemes cost exactly the same and the N-1 gap claimed below disappears. The counted set is pinned down here and the other reading is recorded at the end of \cref{plt:prop:div-cost-newton}.
Theorem 9.34	the N-1 gap is fragile: it is created entirely by charging for products against the scalar block C_0. Recording this keeps the comparison from being read as a robust separation, which it is not.
Theorem 9.34	S3 asserts that Newton inversion "is often faster than coefficient recursion for large N even with naive multiplication"; the exact counts above show the opposite at the level of coefficient multiplications, and the honest statement is the corollary. the corollary was headed by the iff "improves ... exactly when block multiplication is subquadratic". Only the two implications below are proved (subquadratic wins asymptotically; schoolbook loses by N-1), and neither the converse nor the intermediate regime M(N) = cN^2 is covered; the claim is downgraded to what the two cost counts give.
Theorem 9.35	S6 gives the cost as O(N^2 d^2) "if the typical coefficient degree is d"; under the affine log-degree profile that actually occurs there is no such constant d, and the honest count is quartic in N.
Units, reciprocals, and division	the catalogue asserts that the two iteration orders differ "in general"; no source proves it, and the witnesses above supply the proof.
Units, reciprocals, and division	"beats the recurrence only when block multiplication is subquadratic" asserts the unproved converse; only the two extreme regimes are established.

Location	Repair
Theorem 10.1	retitled. Three “Summable families” statements stood in the volume with the same bracketed title. This one is the blockwise assembly principle for a sequence in a Laurent ring over any of the coefficient rings used here; \cref{plt:lem:cmp-summable} is the corresponding statement for an arbitrary index set in a Hahn ring with real exponents, and \cref{plt:thm:trunc-summability} is the full calculus on \(\PLlau\). On \(\PLlau\) the present lemma is the sequence case of \cref{plt:thm:trunc-summability}\,(i),(iv),(v); it is stated again because it is applied below in the \(L^{-1}\)-adic order and in a Hahn ambient, neither of which that theorem covers. The hypothesis \(\inf_{r}\ordt F_{r}>-\infty\) is redundant for a sequence with \(\ordt F_{r}\to+\infty\) and is kept only because the proof uses it.
Theorem 10.2	retitled. This is substitution of a small series into a \emph{scalar} power series, \(\K[[z]]\to\PLpow\). The composition part has a different theorem under the same name, \cref{plt:thm:cmp-substitution-hom}, which substitutes a monomial-leading transseries into a transseries; neither implies the other.
Theorem 10.3	retitled: the composition part states a different theorem under the same name (\cref{plt:thm:cmp-finite-determination}, finite determination of \(F\compose G\)). The present one is about \(\Phi(U)\) with \(\Phi\) a scalar power series.
Theorem 10.7	S3 states the analogous expansion with U^4 = O(t^4) as the reason for stopping at r = 3; the correct reason is that r*q >= N kills the term, which is the general statement proved above and matters as soon as q > 1.
Theorem 10.12	retitled to separate it from \cref{plt:lem:mul-asymptotic-basic}, which collects the basic properties of the relation \(\approx\) between a function and a series.
Theorem 10.15	S4 and S6 note only that a "large part" other than an integral multiple of L may require new monomials. The decisive point, which no source states, is that a logarithmic derivative has positive L^{-1}-order, whereas the derivative of a nonaffine polynomial has order at most -1.
Theorem 10.24	no source identifies the image of \partial_L on Laurent-log coefficients, although S5 and S6 both assert closure of the class under antiderivatives and then remark vaguely that "new logarithmic levels can be forced" once inverse powers of L are admitted. The converse of the preceding lemma is what makes that remark precise, so it is proved here and used both for the logarithm boundary and for the antiderivative obstruction.
Theorem 10.28	S6 states this example with a leading factor L^{-2}, which is not an element of K[L]((t)); the phenomenon is the same, and the general statement is that the coefficient of Lambda is -omega of the leading coefficient.
Theorem 11.3	the notation catalogue and four of the six sources assert that D_X is "the unique derivation" with the two generator values, with no hypothesis. The argument above needs a bound on delta on the power-series part; without one, the passage from the dense subring to the completion is not available.
Theorem 11.8	notation reconciliation. Two sources write Delta_n for the single factor partial_L - n and never introduce the second index; the normative two-index operator is used here throughout.

Location	Repair
Theorem 11.10	S1 prints the identity as X^k d^k/dX^k P(log X) = sum_j s(k,j) P^{(j)} with the sum starting at j = 1; that is correct only for k >= 1, since s(0,0) = 1. The range j = 0..k with the empty-product convention is uniform.
Theorem 11.17	the motivation part proves the same dichotomy in the weaker form it needs there (\cref{plt:lem:mot-block-antiderivative}, on \(\K[L]\) only, with no closed form). Rather than restate it, the two lemmas above are identified with it here, so that a reader meets the resonance once as a mechanism and once in the form the antiderivative theorem consumes.
Theorem 11.19	S6 concludes only that "the polynomial-logarithmic ring is closed under termwise antiderivatives, up to an arbitrary constant", and S3 writes the weaker (and redundant) inclusion int P dX subset P + K. The correct statement is surjectivity of D_X on K[L]((t)) with kernel K, which is what the block system actually proves and what is needed downstream.
Theorem 12.3	retitled. This is the summability criterion in a Hahn ring with real exponents, for an arbitrary index set, together with the notion of a map compatible with summable families that the rest of the part uses. It is not \cref{plt:lem:dif-summable} (sequences in a Laurent ring) and not \cref{plt:thm:trunc-summability} (the calculus on \(\PLlau\)); for \(\Gamma=\IntegerNumbers\) it restricts to the latter.
Theorem 12.4	the catalogue warns that a t-adically Cauchy *sequence* in a Laurent ring may fail to converge without a common lower support bound. For *families* satisfying (plt:eq:cmp-summability-criterion) the bound is automatic, as the proof shows; the two statements are not in conflict.
Theorem 12.7	retitled to separate it from \cref{plt:prop:dif-substitution}, which substitutes a small series into a scalar power series. The present theorem substitutes a monomial-leading transseries into a transseries.
Theorem 12.8	the draft deduced "compatible with summable families" from (plt:lem:cmp-summable) directly, but that lemma only supplies the valuation half of the definition; the interchange Phi_G(sum S_i) = sum Phi_G(S_i) needs the argument below. It is used by (plt:lem:tay-determined) and by the associativity proof, so it is supplied here.
Theorem 12.10	retitled: \cref{plt:thm:dif-finite-determination} carries the same name for a different statement, the finite determination of \(\Phi(U)\) with \(\Phi\) a scalar power series.
Theorem 12.15	S2 writes the operator in (its) power-leading formula as ":e^{delta(D-n)}:" with D previously declared to be d/dL, while the identity requires differentiation in the *argument* of the coefficient polynomial. We print partial_z for the dummy variable and evaluate afterwards. an earlier draft of this warningbox described the discrepancy as "a spurious factor rho^k". That is false as soon as n != 0: the chain rule turns (d_z - n)^k into (rho d_z - n)^k, and the two are proportional only when n = 0. Corrected below, with a numerical witness.
Theorem 12.23	the sources state the rule in words; the numerical demonstration of the two standard mistakes is supplied here so that the size of the error is visible.
Theorem 12.33	the exclusive normalization is used throughout; a source using the inclusive cutoff states the same condition as f - sum_{n<=N} = o(X^{-N}), which is (plt:eq:cmp-realization) at N+1.

Location	Repair
Theorem 12.35	the draft omitted this reduction and then asserted that the product of the two remainders is O(X^{-N+eps}); that product is only O(X^{-2N+a+atilde+eps/2}), which is the required bound precisely when N >= a + atilde. The reduction below makes the omission harmless.
The formal Taylor formula	as in (plt:thm:cmp-substitution-hom), the earlier text checked the interchange only for the family of blocks of a single F; the definition of compatibility quantifies over all summable families, and that is what (plt:lem:tay-determined) consumes. Argument supplied.
Theorem 13.5	this multiplicativity is exactly the step that S3 asserts ("both sides are continuous algebra homomorphisms") and that S6 replaces by "the equality then extends to polynomials, products, inverses"; it is not automatic, because T_H is a *normal-ordered* exponential and not exp(H D_X).
Theorem 13.10	S4 and S5 prove the Taylor formula by "inserting a scalar parameter s, forming F(x+sA), and differentiating in s". That argument is circular: the object F(x+sA) is defined only by the composition being constructed, and "formal differentiation in s" of it presupposes the chain rule for that composition. The homomorphism argument above avoids the circle.
Theorem 13.13	the two index orders in the corpus are transposes of each other; the collision is silent because both symbols are two-index arrays of polynomials.
Theorem 13.20	the source that supplies this example specifies F only modulo Y^{-3} and G only modulo t^2, yet quotes h_3, which depends on the coefficient of Y^{-3} in F and on the coefficients of t^2, t^3 in G. The data are restated here as exact finite series, which is what the quoted answer actually assumes.
Theorem 13.22	this proposition restated three results of the elementary-series part under a different normalisation of the outer coefficients: the substitution homomorphism (\cref{plt:prop:dif-substitution}), its finite determination (\cref{plt:thm:dif-finite-determination}), and the Bell-polynomial coefficient formula \eqref{plt:eq:dif-bell} of \cref{plt:prop:dif-bell}, which the displayed formula reproduces verbatim once \(c_{k}=\varphi_{k}/k!\) is substituted. The restatement and its proof are deleted in favour of the pointer below; the label is kept so that \cref{plt:ex:tay-standard-outer} and any citation from elsewhere in the volume continue to resolve. Only the translation clause was genuinely new, and it is retained with its one-line proof, as is the second route to the coefficient formula.
Theorem 14.2	S6 writes $F/X = 1+tA = 1+o(1)$ "formally". A little-o symbol with no regime attached is not a formal statement; the three O-roles must not be conflated. The correct formal assertion is $tA \in t\PLpow$, equivalently $1+tA-1=\FormalBigOAt{t}{t}$, equivalently $tA\precdom1$.
Theorem 14.2	this lemma was stated and proved twice in the volume, here and as \cref{plt:lem:cmp-exp-log} in the composition part, with the same transport-of-identities proof. The composition-part statement is the stronger of the two — it is set in an arbitrary polynomial-logarithmic Hahn ring, and it adds the additivity of \(\exp\), the law \(\log(P^{n})=n\log P\) for all \(n\in\IntegerNumbers\), and the commutation with any homomorphism compatible with summable families, all of which are used below. The restatement and its proof are deleted and replaced by this pointer; the label is kept so that the citations to it elsewhere in the volume continue to resolve.

Location	Repair
Theorem 14.5	the exact formal Taylor formula was stated and proved twice, here and as \cref{plt:thm:tay-formula} in the composition part, where an entire subsection (the Taylor operator, its summability budget, its multiplicativity, and the generator-determination lemma) is built to prove it. The statements are identical, so the restatement and its proof are deleted in favour of a pointer; the label is kept because the reversion part and the algorithms part cite it. The repair recorded here — S3 concludes from agreement on the generators without proving that the normally ordered operator is multiplicative — is carried out at the surviving site, in \cref{plt:lem:tay-homomorphism}.
Theorem 14.6	the Lipschitz bound was printed as \(\ordt(B-C)+1\), dropping the summand \(\ordt(A)\) that the proof below already produces. The sharper form is what the reversion part needs (it is what makes the Newton estimate of \cref{plt:thm:rev-newton-doubling} come out), and it was stated there a second time as a separate lemma. The inequality is corrected here and the restatement in the reversion part is deleted; every use of the weaker form below is unaffected, since \(\ordt(A)\ge0\) on \(\PLpow\).
Theorem 14.7	S4 proves the contraction step with a formal "mean value" identity A o (X+B) - A o (X+C) = (B-C)\int_0^1 A' o (X+C+s(B-C)) ds. The integrand is a composition with a series carrying an auxiliary parameter s, and neither its existence nor the legitimacy of the term-by-term integration is established there. Corollary \ref{plt:cor:grp-gain} replaces it by the exact Taylor difference, which needs nothing beyond Theorem \ref{plt:thm:grp-taylor}.
Theorem 14.8	S5 proves associativity by "introducing finite truncations mod t^N", for which "ordinary substitution is associative". Truncation is not a ring retraction compatible with substitution, and the claim that a truncated composite equals the truncation of the composite is precisely what needs proof. S1 proposes the correct route (check on the generators) but does not carry out the verification for L; the display above is that verification.
Theorem 14.8	the Picard construction of the inverse was carried out twice in the volume: here, and again — with the sharp rate \(\ordt(B_{j+1}-B_{j})\ge(j+1)\alpha+j\), the first-order form of the inverse, and the uniqueness of the two-sided inverse — as \cref{plt:thm:rev-existence}. The two constructions are the same iteration; the reversion part proves strictly more. The duplicate is deleted and item (4) is derived from the reversion theorem, which makes the dependency of the group's inverses on Part~IV explicit instead of hiding it behind two independent proofs. There is no circularity: \cref{plt:thm:rev-existence} uses items (1)–(3) of the present theorem and the gain estimate \cref{plt:cor:grp-gain}, and nothing else from this chapter.
Theorem 14.10	this remark previously asserted that the section is independent of Part~IV and that inverses are established here. That was true only because the same Picard construction was written out twice. With the duplicate removed, the dependency is one-directional and is stated as such.
Theorem 14.13	S6 states the two continuity relations for the contact filtration but justifies the right-composition one by "H' = 1 + O(t)". That is the reason for a gain in the *left* argument; the right-hand estimate is the exact isometry of Proposition \ref{plt:prop:grp-substitution}(4) and needs no derivative hypothesis at all.

Location	Repair
Theorem 14.15	S6 calls the group "pro-unipotent at finite truncation". No algebraic group structure is available here (the graded pieces are the infinite dimensional space K[L]), and none is needed: Propositions \ref{plt:prop:grp-limit} and \ref{plt:prop:grp-commutator} give the exact provable statement, namely that Gnear is the inverse limit of nilpotent quotients.
Theorem 14.20	S6 introduces log(S_F) as a power series in S_F - I and writes the fractional iterate as "F^{[s]} = X S_F^s". Three things are missing there: that S_F - I raises the filtration (true: (S_F-I)H = H o F - H has ordt >= ordt(H)+1 by Corollary \ref{plt:cor:grp-gain}); that S_F^s is again a *substitution* operator, i.e. that F^{[s]} exists in Gnear at all; and the notation, which should be S_F^s(X), not a product. Theorem \ref{plt:thm:grp-exp} and Corollary \ref{plt:cor:grp-iteration} supply all three.
Theorem 15.4	S4's closure table records only the rational case ("c x^{r/s}: Puiseux-log series") and S1 says "arbitrary real beta require a Hahn support", neither separating the three regimes nor proving that each enlargement is necessary. Corollary \ref{plt:cor:grp-power-regimes} does both.
Theorem 15.14	S2 says only that "the rational field K(L_1,...,L_r), or a Laurent-monomial field, repairs this defect". Proposition \ref{plt:prop:grp-tower-derivation} identifies the minimal repair inside the Laurent-monomial family and shows that L_r itself need not be inverted.
Theorem 16.3	report 6 states its reversion algorithms with the inclusive cutoff [F]_{\le N}; every cutoff index taken from that source is shifted by one here.
Theorem 16.4	the closure of \(\Gnear\) under composition was stated and proved a second time in this part, as a separate lemma. It is item (1) of the group theorem, proved there from the substitution proposition, so it is imported here with the other three standing facts. Note that only items (1)–(3) of that theorem are imported: its item (4) cites \cref{plt:thm:rev-existence} below, and importing it here would be circular.
Theorem 16.6	the closure lemma stated here duplicated \cref{plt:thm:grp-group}\,(1), which is proved in the composition-group part with the same one-line argument. It is deleted; closure is now imported as part of (I3) and appears as \eqref{plt:eq:rev-closure}. For the record, the argument is that \(F\compose G=G+A\compose G\) and that \(A\compose G=\sum_{k\ge0}B^k\DX^kA/k!\) has all summands in \(\PLpow\) by \cref{plt:lem:rev-deriv-gain}.
Theorem 16.9	this estimate was proved twice in the volume: here from the two-point Taylor expansion, and in the composition-group part from the exact Taylor formula, where it is \eqref{plt:eq:grp-gain-lipschitz}. The two arguments are the same computation, and the group-part statement was missing the summand \(\ordt(A)\) that its own proof produces; that omission is repaired at \cref{plt:cor:grp-gain} and the duplicate proof is deleted here. The inequality is recorded below as a fifth standing import, in the instantiated form used throughout this part.
Theorem 16.11	all six sources call this a contraction; none says in what sense. The statement is about nu_t, not about coefficient size, and the constant term alpha = nu_t(A) is absent from every source formulation.

Location	Repair
Theorem 16.12	this corollary restated \cref{plt:thm:grp-group} in full. The group statement now stands once, in the composition-group part, and its item (4) cites \cref{plt:thm:rev-existence}; the restatement and its proof are replaced by the sentence below, which records what has just been supplied.
Theorem 16.19	report 1 asserts that at finite order the two residuals "need not have identical first omitted blocks"; for an exact G_N in the group that is false.
Theorem 16.21	report 1 claims the derivative residual "often detects a wrong block one order earlier"; it detects it one level later, and is blind at level 0.
Theorem 17.7	report 4 states the rule as "for a leading slope F = ax + o(x) the correction is divided by a", with no hypothesis on a and no discussion of a nonconstant leading block of D_X F.
Theorem 17.13	the sources give five mutually inconsistent exponents; one contradicts its own proof, and two slide from residual order to correct-block count.
Theorem 17.19	reports 1, 3, 4 and 6 assert Newton is "asymptotically superior" with no cost model stated; the claim is model-dependent and is weakened here.
Theorem 18.10	S1, S2, S3, S4 and S5 all assert only the sufficiency direction ("the r-th term has t-order at least r-1"), and all state it with the hypothesis ord_t(A) >= 0. The correct threshold is ord_t(A) > -1; the two agree on K[L]((t)), whose value group is Z, but they differ on the Puiseux–log enlargement, where the weaker hypothesis is exactly what the reduction of a ramified reversion needs (plt:ex:lag-puiseux). None of the sources observes that the condition is also necessary, which is what makes the misuse in plt:warn:lag-scale-trap detectable.
Theorem 18.12	S3's appendix proves the unshifted formula by a formal residue change of variables, which is valid only when the constant term of the auxiliary series is invertible, and repairs this by a universal-coefficient localisation argument. S6 replaces that argument by the phrase "an integration by parts reduces the resulting residue". The proof above needs neither device.
Theorem 18.16	S3 states this "whenever both sides are interpreted to a fixed finite t-precision" and S6 "for any H for which the family is summable"; both hypotheses are vacuous or unverifiable. The order estimate plt:eq:lag-transport-order shows that no hypothesis beyond membership in the frame is needed, since nu(phi) is finite for phi nonzero.
Theorem 18.25	S4 gives only b_0, b_1, b_2 in this universal form; the b_3 line plt:eq:lag-b3 is derived here and verified against the two independent worked examples plt:ex:lag-mixed and plt:ex:lag-abc.
Theorem 18.28	this proposition restated \cref{plt:cor:rev-finite-certificate}, and its proof re-derived \cref{plt:prop:rev-residual-order} along the way. The statements are identical — the hypothesis \(F=X+A\) with \(A\in\PLpow\) is membership in \(\Gnear\), and the two candidate truncations \(G_N\) are the same object. The restatement and its proof are deleted in favour of the pointer below; every citation of the deleted label in this part now cites the reversion part.
Theorem 19.2	the requirement log d in K is stated in the catalogue and is omitted in S1, S2 and S6, which write log c freely over an unspecified coefficient field. Over K = Q it fails already for d = 2; see plt:warn:lag-scale-trap.

Location	Repair
Theorem 19.8	S6 derives plt:eq:lag-u-expansion by inserting an undetermined ansatz u = rho^{-1}(S - sigma log S + C + (a log S + b)/S + ...) and matching three constants by hand. The derivation above obtains the same coefficients from the near-identity coefficient formula, so the pattern continues to all orders with no further ansatz, and the branch and field hypotheses (log c, log rho in K) become visible.
Theorem 19.12	S4 asserts that after the substitution y = d x^{1/q}(1+V) "the formal implicit-function recursion is triangular" for any leading exponent q, which is true, but it does not record that the intermediate normalisation Phi = F o rev(F_0) can leave the near-identity class, as it does in plt:ex:lag-puiseux. The correct general statement is the converse half of plt:thm:lag-rigidity, where Phi = X(1 + U o rev(F_0)) because F_0 is the exact leading monomial of F, not merely its leading power.
Theorem 20.7	S1's prose ("the same P_n occur, multiplied by (-1)^n") contradicts its own displayed expansion of the inverse of X - log X, where P_1 = L occurs with a plus sign. The correct weight is (-1)^{n+1}; the weight (-1)^n belongs to the negated quantity -rev(X-L), which is what enters W_{-1}.
Theorem 20.10	S5 states uniqueness only inside -L + t K[L][[t]]; the contraction estimate holds on all of K[L][[t]], so uniqueness is unconditional there and membership in -L + t K[L][[t]] is a conclusion, not a hypothesis.
Theorem 20.11	the citation named a label "plt:thm:lagrange-burmann" that is defined nowhere in the volume; the theorem meant is the one labelled plt:thm:lag-burmann in the Lagrange–Buermann part, whose polynomial–logarithmic form is equation plt:eq:lag-burmann-pl.
Theorem 20.14	retitled. This is not the general finite certificate (\cref{plt:cor:rev-finite-certificate}, which it invokes) but its specialisation to the single map \(X+L\), where the criterion becomes a characterisation of the Lambert polynomials themselves.
Theorem 20.17	S5 asserts that the residual-to-error principle "proves the asymptotic validity of every truncation directly from" the FORMAL residual identity, and S1 asserts an analytic O-bound for the branch residual "by construction". Neither is a proof: a formal t-adic identity supplies no inequality. The missing ingredient is item (2), supplied below.
Theorem 21.4	S3 and S6 start the recurrence at P_1 = u and impose P_{n}(0)=0 as a side condition of the theorem rather than proving it; S4 derives the linear recurrence by "equating coefficients" in a NONLINEAR generating ODE, which does not produce it. The proof above is direct from the definition and proves the boundary condition instead of assuming it.
Theorem 21.12	this corollary carried its own argument inside the statement, so it read as an assertion with no proof. The claim and the argument are separated.
Theorem 22.11	the hypotheses of the two theorems invoked here were not printed; both hold once |z| is large, which is the regime of the claim.

Location	Repair
Theorem 22.11	the degree law and leading coefficient quoted here for X + aL^2 were printed for all n >= 0, but they fail at n = 0: the t^0 coefficient is -aL^2, of degree 2, and the formula 2a^{n+1}/n is undefined at n = 0. Restricted to n >= 1 and the n = 0 value recorded; verified by explicit reversion through n = 4 (c_1 = 2a^2L^3, c_2 = a^3L^5 - 5a^3L^4, c_3 = (2/3)a^4L^7 - 7a^4L^6 + 14a^4L^5). independently re-verified through n = 9 in exact rational arithmetic, with the residual of the reversion confirmed to vanish identically through t^9; that computation also produced the Catalan observation recorded below.
Theorem 22.17	S1, S3, S4 and S5 all state the branch expansion "in an admissible sector" or "in a branch-compatible sector" without ever defining one, and none proves that the constructed solution is the branch conventionally labelled W_k. Repaired: the estimate is proved on an explicit region with no sector at all (Theorem lw-uniform-remainder), and the labelling is a separate proposition with a printed distance-to-the-cut hypothesis (Proposition lw-branch-label).
Theorem 22.18	the exhibited S_k was written with non-strict inequalities, hence was not open, while a coordinate sector is required to be open by Definition lw-coordinates. Made strict.
Theorem 22.18	S6 deduces convergence of the Lambert series from (a) convergence at frozen u and (b) $\lvert s u\rvert$ -> 0 along the Lambert path. That inference is invalid: the radius of convergence in s depends on u and shrinks as $\lvert u\rvert$ -> infinity. Repaired by proving the two-variable statement (Lemma lw-two-variable), which is what the argument actually needs.
Theorem 22.19	S1, S3 and S5 each assert, as though it were established here, that the principal-branch series converges for every real z >= e. Downgraded to an attributed literature report: nothing in this volume proves it, and S5's supporting observation (all terms vanish at z = e) is vacuous.
Theorem 22.24	every source displays the lower-branch expansion with the factor (-1)^n and explains it only by analogy ("the same polynomials reappear with alternating block signs"). Repaired: the alternation is derived as the single value a = -1 of the scalar a^{n+1} in Theorem lw-affine-log, and the box states explicitly that no second polynomial family exists.
Theorem 22.25	S1 and S3 assert the Puiseux expansion at the branch point without constructing the local inverse; S1 says only "ordinary series reversion in p produces" the coefficients, and neither source proves that a power-logarithmic expansion is impossible there. Repaired: the local biholomorphism is constructed, the coefficients are rederived (and carried one order further than any source, to p^6 = -221/8505, verified against a high-precision evaluation), and the impossibility is proved from the order of the local monodromy in Corollary lw-branch-point-monodromy.
Theorem 22.33	the four source tables are individually correct but use two incompatible column conventions (S1 and S4 index by x = log z, S3 and S5 by z), which makes them look contradictory when read side by side. All four were recomputed to 60 digits and merged into one table indexed by the intrinsic variable xi_0 = log z; the concordance is recorded below, together with the observation that the S3/S5 columns at z = 10 and z = 100 lie outside the region where any convergence statement is proved here.

Location	Repair
Theorem 23.3	S3, S4, S5 and S6 all give the closed form y = (alpha/beta) W_k((beta/alpha) x^{1/alpha}) with the disclaimer "branches chosen consistently", which hides a genuine ambiguity: for non-integer alpha the symbol x^{1/alpha} is not determined by x. Repaired by Proposition lw-power-exp, which makes the three choices explicit and states the correspondence as a bijection.
Theorem 23.7	the proof establishes the residual bound only "for z large", while the statement attaches it to the same z_r that governs existence. The quantifier is repaired here by enlarging z_r once, which is legitimate because the statement is existential in z_r.
Theorem 23.11	the drafted sentence said rho appears "only in the leading term and inside ell", which the displayed formula contradicts: rho also sits in the overall prefactor mu/rho. What is true is that rho is absent from the ratio a/X = mu/(xi - beta) that drives the series.
Theorem 23.13	the remainder in this display was printed as an analytic estimate BigO(L^4/Y^3) although the three blocks are computed by formal Lagrange–Burmann; per the volume's own rule the three O-roles must not be conflated. Downgraded to the formal tail, with the analytic upgrade stated as a separate sentence carrying its hypothesis.
Theorem 23.13	S6 states that "the coefficient of Y^{-2} is no longer a polynomial in L and M alone". That is true but not sharp: expanding the Y^{-1} block gives A*varpi = alpha^2 L + alpha*gamma M + alpha*gamma + gamma^2 M/L, so a negative power of L is already present one block earlier whenever gamma is nonzero. The corrected statement and its proof are below.
Theorem 23.17	the proof as drafted read the two leading blocks off Equation lw-ex3-X, which the template produces only FORMALLY; using it as an analytic asymptotic statement here would be circular. Replaced by a direct three-line derivation from the defining equation, which needs nothing beyond X -> infinity.
Generalized Lambert equations and reusab	as drafted this corollary wrote the reseated sum as "w", silently asserting that the reseated base point returns the SAME solution as the canonical one, and its proof discharged the point in one clause. That does not follow from the stated hypotheses: Theorem lw-basepoint attaches to each base point SOME solution of w e^w = e^xi, and the reseated one is only known to satisfy |w~- (xi - ell)| <= |M~| + log 2, which may exceed the radius 1 of the uniqueness disc of Proposition lw-branch-label. Repaired: the identification is now a separate clause with the printed hypothesis 4(1 + |ell_k|) <= |xi_k|, under which the disc argument does apply, and the real case is noted to need nothing at all.
Theorem 23.25	the canonical threshold was printed as 5.3569...; the root of xi = 2 + 2 log xi is 5.356693980..., i.e. 5.3567 to four decimals. Corrected, and the threshold of the reseated condition added.
Theorem 23.25	the drafted reading said reseating "is numerically better at xi_0 = 10 and 30", which its own table contradicts at xi_0 = 10, N = 10 (6.4e-10 canonical against 1.7e-9 reseated). Corrected to what the table shows.

Location	Repair
Theorem 24.2	S2 states that "the monomials X^{-n}L^m form an asymptotic scale under the lexicographic order", and S6 opens its analytic chapter by letting (phi_nu) be "a totally ordered asymptotic scale, so that phi_{nu+1} = o(phi_nu) in the chosen enumeration" and then applying this to the polynomial-logarithmic monomials. The pairwise dominance comparisons those drafts display are all correct; what fails is the existence of the enumeration, by the proposition above. The repair is to fix a degree bound first, which is what the block formulation does automatically.
Theorem 24.4	this definition carried the same title as \cref{plt:def:mul-asymptotic}, which it extends rather than repeats: the blocks here are rational in L and may have poles, so the evaluation needs the large-X caveat below. Retitled so the two are distinguishable in the list of results.
Theorem 24.8	S6 writes the blockwise condition as f - sum_{n<=N} p_n(log X) X^{-n} = o(X^{-N}(log X)^{d_N}) and remarks that "one must specify that inner ordering" of the monomials inside the last block. Form (v) of the theorem shows the logarithmic weight is harmless once all N are quantified, and form (vi) shows the inner ordering never has to be specified: the blockwise and monomialwise conditions coincide for every admissible profile.
Theorem 24.17	S5 states products and quotients together with a "proof sketch" and, in its part (ii), adds the hypothesis "and g is eventually nonzero". That hypothesis is redundant: it is a consequence of the leading block being nonzero, as (plt:eq:an-g-lower) shows. S5's part (iii) requires instead that "Q(log x) stays nonzero eventually", which is likewise automatic for a nonzero rational function. The full proof above replaces the sketch, and in particular exhibits the logarithmic degree k of the leading block explicitly, which the sketch suppresses; it is what makes the remainder exponent M'-k, not M'.
Theorem 24.18	the right-hand side was attributed to \cref{plt:prop:tay-scalar-outer}, which is stated for arguments in t\PLpow. Here U has rational blocks, so the ambient is \(\K(L)[[t]]\); that case is covered by the last sentence of \cref{plt:prop:dif-substitution}, which is the citation used now.
Theorem 24.19	the right-hand sides were identified outright with the formal operations of \cref{plt:lem:cmp-exp-log} and \cref{plt:thm:mul-binomial}, both of which are stated over \PLpow; for a genuinely rational \(U\) the formal object is the scalar substitution of \cref{plt:prop:dif-substitution} in \(\K(L)[[t]]\), and the identification with the named operations is asserted only where they are defined. Restricted accordingly.
Theorem 24.21	S6 writes, in the chapter that this part consolidates, "For f(X) = X + O(log X), one has f'(X) = 1 + o(1) under the relevant differentiability hypotheses". No such hypotheses are named, and the implication is false without them, by the display in the box above. S6 uses the conclusion to justify choosing m close to 1 in its residual-transfer theorem. The repair is (plt:prop:an-omega-computable) and (plt:thm:an-inverse-transfer), which take the derivative bound as a hypothesis and verify it directly for the map at hand instead of deducing it from an expansion.
Theorem 24.23	the hypothesis "locally integrable" was missing here although the proof invokes \cref{plt:thm:an-integration}, which requires it; without it the integrals below need not exist. Restored.

Location	Repair
Theorem 24.28	the displayed constant was written (1-\delta)^{-N-1}, which bounds |w|^{-N-1} by ((1-\delta)|z|)^{-N-1} only when N+1\ge0. Cutoffs here run over all of Z (\ordt F may be negative), and for N\le-2 the maximum of |w|^{-N-1} on the circle is attained at the outer end instead. Replaced by the two-sided constant \kappa, which covers both signs.
Theorem 25.2	\cref{plt:prop:rev-analytic-transfer} is printed with the analytic inverse written \rev{f}. In this volume that symbol is reserved for the FORMAL compositional inverse – and the whole content of (plt:thm:an-inverse-transfer) is that the formal and the analytic inverse agree asymptotically, a statement that is vacuous if the two are written with one glyph. This is the same collision the ed. note before (plt:sec:an-omega-constants) records against S2. The recollection below therefore names x_* by the equation it solves.
Theorem 25.5	S5 writes, immediately after its residual-to-error principle: "For f(x) = x + log x, this proves the asymptotic validity of every truncation omega_N directly from [the formal residual identity]." It does not: the principle converts an analytic residual into an analytic error, and the formal identity provides no analytic residual. The missing step is the one carried out in (plt:thm:om-analytic) for this map and in (plt:thm:an-residual-analytic) in general. The same gap is present in S1 ("by construction") and is repaired in the same way.
Theorem 25.5	the coefficient field was left at the ambient \(\K\subseteq\ComplexNumbers\) of this part, but the proof applies \cref{plt:thm:cmp-analytic-transfer}, whose hypothesis (A2) asks for an inner realization with REAL coefficients, and both \(\varphi\) and \(h\) are real valued here. The real coefficient field is now printed, as house rule 5 requires.
Theorem 25.7	coefficient field printed: \cref{plt:thm:an-residual-analytic} is applied below, and it now carries \(\K\subseteq\RealNumbers\).
Theorem 25.8	this sentence read "Item (2) of the list preceding \cref{plt:thm:om-analytic} is the specialisation of (H2) to \varphi(Y)=Y+\log Y", which is wrong in an instructive way: (H2) specialised to that map is the trivial statement \varphi\approx X+L (the remainder vanishes identically), whereas item (2) asks for an analytic remainder for log applied to the TRUNCATION. Item (2) is the conclusion of \cref{plt:thm:an-residual-analytic}, not its hypothesis. Corrected.
Theorem 25.8	S2's transfer theorem lists the hypotheses "F(y) = y + o(y)" and "F'(y) = 1 + o(1)" and then assumes an ANALYTIC residual bound F(G_N(X)) - X = O(X^{-N-1}L^M) outright. Assuming it is legitimate; the surrounding text, however, obtains it from the formal reversion, so the theorem as used is circular. The first hypothesis is never used in S2's proof. (H2) above is the honest replacement, and (plt:thm:an-residual-analytic) discharges it. S2 also writes the analytic inverse as \rev{F}; in this volume that symbol denotes the FORMAL compositional inverse, and the two must not share a glyph in a statement whose whole content is that they agree.
Theorem 25.9	the step "for x>1 the envelope gives omega(x) > x - log x" skipped the verification of that envelope's own hypothesis, which is omega(x)>1 and not x>1. Supplied: omega is strictly increasing with omega(1)=1.

Location	Repair
Theorem 25.10	the proof invoked \cref{plt:prop:an-omega-computable}, which is stated for x>1, while this proposition is stated for x\ge1. The endpoint is dispatched separately rather than left to a proposition that does not cover it.
Theorem 25.12	the relative width was quoted as \mu_3^{-1}. Width divided by the upper end is 1-(1+\mu^{-1})^{-1}=(1+\mu)^{-1}, which is slightly smaller; recomputed, it is 1.0374 per cent against \mu_3^{-1}=1.0483 per cent.
Theorem 25.14	S1 asserts, of its own error table at x = 10, that "the N = 2 truncation is slightly worse than N = 1". Its table gives 1.7466787e-3 at N = 1 and 1.7369609e-3 at N = 2, which is smaller, and recomputation at 40 digits confirms both table entries. The prose contradicts the table it is describing. What is true at x = 10 is that the N = 2 correction almost exactly cancels the N = 1 error and flips its sign, improving it by only 0.6 per cent, and that genuine non-monotonicity occurs later, at N = 6 -> 7 and N = 11 -> 12, as recorded above. S3's table at log z = 2.3026 makes the analogous claim correctly, for the pair N = 2, 3.
Residual-to-error transfer	"with relative width 1/(x-\log x)": the exact relative width is (1+x-\log x)^{-1}; the quoted quantity is an upper bound for it, so the sentence is corrected to say so.
Theorem 26.2	this sentence originally attributed the derivation to all four classes. Unlike its mirror \PLpow at infinity, the power-series class K[Lambda][[x]] is NOT stable under d/dx: d(Lambda)/dx = -1/x already leaves it. This is trap (T6) in its sharpest form, and it is why the near-identity class of plt:def:du-near-identity is described inside K[Lambda]((x)).
Theorem 26.2	S2 sets ell = log(1/x); S1 and S6, following the normal-form literature, set ell = -1/log x. The two are reciprocal, and no single displayed formula distinguishes them. We fix Lambda for the large log and never write a bare ell for it.
Power–log series near a finite point an	this item said "every nonzero f in tau(PLlau), evaluated at the generators". A general element of K[Lambda_2]((Lambda^{-1})) is a formal, typically divergent, series and has no evaluation; and the proof uses only asymptotic equivalence to the leading monomial. Restated for functions.
Power–log series near a finite point an	S6 displays the expansion with a bare remainder O(M^3/S^3) and no argument. The estimate below is proved; the sharp form of S6's remainder is recovered in the last sentence.
Power–log series near a finite point an	"asymptotic expansion on the scale of all monomials Lambda^{-n} Lambda_2^j" would presuppose that this doubly indexed family is a Poincare scale, which it is not (plt:prop:an-no-enumeration). The precise assertion is the relation of plt:def:mul-asymptotic read in the variable X=Lambda, which is exactly what plt:eq:du-lower-branch-error states.
Theorem 26.12	S2 gives the product form of (13.7) but never names the operator; the shift index n-k+1 is the trap, since the naive mirror of plt:eq:dif-repeated would be Delta_{n,k}, which is wrong for k >= 2.
Theorem 26.14	S2's eq:Dulac-repeated-derivative is correct as printed; the repair is that it is stated without proof, without the empty-product convention, and without identifying the operator, so its k >= 2 index shift is invisible.

Location	Repair
Theorem 26.16	S2 (eq:small-Lagrange) states the near-zero Lagrange-Burmann formula as having "exactly the same form" and gives no proof; S1 and S6 do the same for the fixed-point construction. The formula is true (Theorem plt:thm:du-lagrange-zero) but it is not the transport of plt:thm:lag-burmann: the transported frame has a different derivation and a different distinguished element, and no rescaling of nu_x repairs (F3).
Theorem 26.19	the appeal to the enlarged exponent group also needs plt:lem:du-taylor-zero there, which was stated for m_0; its proof uses nu_x(B) >= 2 only through nu_x(B) - 1 > 0, so it generalises as stated.
Theorem 26.22	S1's eq:dulac-inverse-first is correct, but it is written in the reciprocal convention ell = -1/log x, so its exponents of the logarithm are the negatives of ours; transporting it without noting this produces the wrong sign on every logarithmic exponent.
Theorem 26.26	this read "has no finite truncation Trunc_{<1} other than 0", which is doubly wrong: Trunc_{<N} of that series IS finite (and in general nonzero) for every N < 1, and Trunc_{<1} is the whole infinite series, not 0. The correct statement is that truncation stops being finite at the cutoff 1.
Power–log series near a finite point an	this item asserted that f |-> f-hat is a ring homomorphism ONTO D(R). Surjectivity is a Borel–Ritt statement for left-finite real exponent sets; it is not proved here and is not needed for anything below, so the claim is downgraded to "into" and the gap is named.
Theorem 26.35	"P-hat is independent of the parametrisation" would be false as literally read – a change of chart conjugates P-hat. What is invariant is the membership and the leading data, which is what cor:du-chart-independence supplies.
Theorem 26.37	"is the semigroup generated by r and 1" asserts an equality that the grid bookkeeping of plt:thm:du-composition-closure(ii) does not give; it gives a containment, which is all that is used.
Power–log series near a finite point an	the statement here read "it lies in none of sigma(PLlau), sigma(PLrat), sigma(PLit), nor in any Puiseux-log or Dulac class". That is false for rho a negative integer, where item (i) exhibits the element inside sigma(PLrat) and sigma(PLit); the exclusion from those two classes needs rho not an integer, and only the exclusion from the polynomial-block classes holds for every rho outside the natural numbers. Split accordingly.
Power–log series near a finite point an	the second clause read "for every alpha with p_alpha nonzero and gamma*alpha not a natural number". Only the summand of the SMALLEST alpha is protected from interference: for alpha' < alpha the summand of alpha' also contributes at the x-exponent alpha*beta (through its u-tail), with Lambda-exponents in gamma*alpha' + N, and these can meet gamma*alpha, so no cancellation argument is available for a general alpha. Restricted to alpha_0 = nu_x(f-hat), where the leading block stands alone.
Power–log series near a finite point an	the argument here claimed that injectivity of alpha |-> alpha*beta makes the summands non-interfering. It does not: the summand of alpha' has x-support in alpha'*beta + Gamma_u, so for alpha' < alpha it also reaches the x-exponent alpha*beta. What is true, and all that is needed, is that only summands with alpha' <= alpha reach it.

Location	Repair
Theorem 27.4	S4 stores the least allowed exponent n_min as an independent field and then warns that a stale zero at that exponent corrupts order and unit tests. Lemma plt:lem:alg-denotation-coset shows the field is redundant: the order bound is a function of the canonical form, so the failure is eliminated by deriving it rather than by documenting it.
Data structures and finite-order algorit	the previous version of (v) asserted that sharpness fails at N_1=0 because DX annihilates the unknown constant block. That is an artifact of scalar power series and is false over R=K[L]: the unknown block of index N_1 is an arbitrary polynomial, and a=t^{N_1}L already has DX a = t^{N_1+1}(1-N_1 L) \ne 0 for every N_1, including N_1=0. What is true, and is what the sources were reaching for, is that the output ORDER bound can fail to be attained – exactly when the leading block is a nonzero constant sitting at index 0 – and that this is a loss of relative precision, not of absolute precision.
Data structures and finite-order algorit	the criterion was stated without the hypothesis v_1 != 0, at which it fails: there v_1(m_1)=0, yet the returned representative is (0,N_1+1), whose order bound is N_1+1=m_1+1, so the bound is attained.
Data structures and finite-order algorit	the sharpness witnesses below were originally exhibited only at the zero representative (sum) or for one of the two competing bounds (product, quotient), which proves sharpness of the LAW but not of each instance. Since Definition plt:def:alg-sound quantifies sharpness per instance, the witnesses are now chosen so that the block that differs is exactly the declared cutoff, for every instance.
Data structures and finite-order algorit	the witness that perturbs the first argument produces a difference in the block N_1+m_2, which is the declared cutoff only when that bound is the smaller of the two. The previous version used it unconditionally and so proved sharpness of the law, not of each instance as Definition plt:def:alg-sound requires. The witness must be selected by which bound realises the minimum.
Data structures and finite-order algorit	"follows from the two witnesses used in (ii)" was too quick: as in (ii) the witness must be the one realising the minimum. Here, unlike (ii), both witnesses exist for every instance, because the numerator is allowed to be zero and may then be replaced by t^{N_1}.
Theorem 27.11	the proof still carried the discarded claim that sharpness fails at N_1=0, contradicting the corrected statement above. The witness a=t^{N_1} is the wrong one: it is annihilated exactly at N_1=0. Over R=K[L] the undetermined block is an arbitrary polynomial, and a=t^{N_1}L works for every N_1.
Theorem 27.11	the witness was a single pair (F=t^{m_1} with m_1 nonzero), which proves sharpness of the law but not the per-instance claim actually stated. Under the printed hypothesis ordt(DX v_1)=m_1+1 the generic witness works, and the hypothesis is exactly what makes the k=1 term dominate.

Location	Repair
Theorem 27.12	the inequality was stated as rho(DX F) >= rho_1. It runs the other way: the absolute cutoff gains one block, but in the exceptional case of part (v) the output ORDER gains more than one, so the difference – which is rho – can only fall. The extreme case v_1 = c in K^x returns the zero representative, whose relative precision is 0. two further repairs to this clause. (a) The equality criterion for the derivation fails at the zero representative: there \(v_1(m_1)=0\) while \(m_{\mathrm{out}}=N_1+1=m_1+1\), so equality holds although the criterion does not; the clause is restricted to \(v_1\ne0\). (b) "no bound in either direction" was an overclaim for the sum: an upper bound does hold, \(\rho\le\max\{\rho_1,\rho_2\}\) (proved below), so what the witness shows is that no LOWER bound holds – and \(\rho\) can drop to \(0\), not merely to \(1\).
Theorem 27.12	the upper bound for the sum was asserted to be absent; it is not.
Theorem 27.13	"only addition can destroy it" overlooked the annihilating case of the derivation, which the corrected corollary now records.
Theorem 27.15	the source sentence asked for G "through block N-1", which is the datum already assumed and yields the product only through block N-2. Since multiplication by t^{-1} shifts indices down by one, the block of index n of the product is read off the block of index n+1 of G.
Theorem 27.15	S2 states only that inputs known modulo t^N with nonnegative valuation give a sum and product known modulo t^N. That is sound but not sharp: by part (ii) the product is known modulo t^{N+min(m_1,m_2)}, which is strictly better whenever a factor has positive order, and strictly worse – the case S2's formulation hides – as soon as a factor is Laurent.
Data structures and finite-order algorit	"four times the naive quadratic guess in the exponent" is not a statement: the exponent is four rather than two. Also, the profile of the inverse of X+L is d(0)=1, d(n)=n for n>=1, which differs from case (ii) in the block of index 0; the exact count is corrected below.
Data structures and finite-order algorit	"the profile of the compositional inverse of X+L" names the wrong object: that inverse has a block of index -1, namely X itself. The profile in question is the profile of its CORRECTION.
Theorem 27.29	the two recurrences do not cost the same: the logarithm sums j<=n-1 and the exponential sums j<=n, so the exact counts differ by N-1. The single figure binom(N,2) was right only for the exponential.
Theorem 27.30	the claim was "shiftop{n}{k}p = 0 as soon as k > deg p and n = 0", presumably by analogy with iterated partial_L. It is false: only the single factor with index j = -n is partial_L, all the others are bijective, so the operator drops the degree by exactly one and no further. Already shiftop{0}{2}L = (partial_L - 1)(1) = -1, nonzero with k = 2 > 1 = deg L. The corrected kernel statement is what the annihilation clause of the precision calculus needs.
Data structures and finite-order algorit	the count was quoted without the hypothesis N>a under which it is valid; for N<=a the ceiling is nonpositive and counts nothing, which is the right answer only because both sides of the displayed identity then vanish. The case is separated.
Data structures and finite-order algorit	the witness F=t^a fails whenever a<0 and K-1>|a|, because then shiftop{a}{K-1} annihilates the constant 1 – e.g. a=-1, h=0, N=10 needs k=9 while DX^k t^{-1}=0 already for k=2. Taking the block to be L instead of 1 removes the exception, by the kernel description in plt:prop:alg-iterated-derivative.

Location	Repair
Theorem 27.33	S1 records the budget as k <= floor((N-r)/(h+1)). Under the exclusive cutoff adopted here that admits one Taylor term too many; the sharp exclusive bound is k <= floor((N-1-a)/(h+1)). The discrepancy is exactly the inclusive/exclusive one-block conversion of the catalogue and is harmless for correctness but not for cost.
Theorem 27.33	the running power was truncated at the output cutoff N. That is wrong as soon as m_F<0: the term P D contributes to a block of index below N through every block of P of index below N-ordt(D), and ordt(D)>=m_F+k can be negative. Truncating P at N-ordt(D), with the CURRENT D, repairs it and never demands a block of H beyond N_H, since N-ordt(D)<=N-m_F-k<=N_H+1-k.
Data structures and finite-order algorit	the output line used a cutoff N that no line of the algorithm defined, and attributed it to plt:cor:cmp-precision-budget, which furnishes the INTERNAL budget for the auxiliary powers, not the output cutoff. The output cutoff must be a function of the input cutoffs – item (2) of plt:cor:cert-reproducible – and the one the stability estimates deliver is min{varrho N_F, varrho m_F + N_U}.
Data structures and finite-order algorit	ALGORITHM BUG, the generator-engine twin of the one repaired in plt:alg:compose-taylor. The cap on the powers of V was min{D,N-1}, justified by ordt(V^k)>=k alone. That ignores the factor t^{varrho n} multiplying block n: the summand for the outer index n contributes to a block of index below N whenever varrho n + k < N, so for a Laurent outer series (m_F<0) powers with k>=N are still needed. Concretely, F=t^{-1}L^5, varrho=1, N=1 has cap min{5,0}=0 and returns a wrong block of index -1, although V^1 is required. The correct cap is min{D, N-varrho*m_F-1}, which reduces to at most N-1 exactly when m_F>=0.
Theorem 27.36	the two step counts inherited the wrong cap and the wrong index range; the retained outer indices run from m_F to N-1, of which there are N-m_F, not N.
Data structures and finite-order algorit	the loop tested a variable mu that the step list never initialised.
Data structures and finite-order algorit	S4's driver hard-codes the schedule as q := min(N, max(1, 2p+2)). The additive +2 is a fudge for an unstated guard; the guard actually required is the one computed by plt:thm:rev-newton-guard together with the composition law of plt:thm:alg-precision-calculus(vi), and the abort test above makes any residual shortfall visible instead of silently absorbing it.
Theorem 27.42	the cutoff carried by the running power was left as an ellipsis, which is exactly the datum a precision contract may not omit. It is N-r+1: the block of index n of DX^{r-1}P is built from the block of index n-(r-1) of P, so the blocks of P of index < N-r+1 are the ones the accumulator can see, and no earlier truncation is thereby too coarse, since the cutoff demanded of P decreases by one at each step while the product law (plt:thm:alg-precision-calculus(ii)) delivers min{(N-r+2)+h, N+(r-1)h} >= N-r+2 from the previous power and A.
Data structures and finite-order algorit	the product count was quoted as the ceiling itself; the stopping rule of plt:alg:reverse-lagrange runs r=1..R with R = ceil((N+1)/(h+1)) - 1.

Location	Repair
Data structures and finite-order algorit	"only addition destroys it, and only differentiation improves it" is wrong twice: differentiation improves the ABSOLUTE cutoff by one block while leaving the relative precision unchanged, and in the annihilating case it destroys the relative precision outright.
Theorem 28.1	"every primitive has a faithful equation, with exactly one exception" overstated what plt:tab:cert-gauges contains: the table covers the primitives that invert a cheaper one, and lists no equation at all for the sum, the product and the derivation, which invert nothing. For those the natural test is the Leibniz identity, and it is unfaithful in precisely the way plt:rmk:cert-derivative-defect records. The claim is restricted to what is proved.
Theorem 28.12	S2 (eq. recip-residual-test) and S3 (eq. divisionresidual) both state the residual test as "AB_N - 1 = O(t^N)" or "BC_{<N} - A = O(t^N)" and read off correctness through block N-1. That reading is the gauge-zero one and is correct only for a denominator of order 0. With a denominator of order m the same residual certifies through block N-m-1, which is fewer blocks when m>0 and more when the denominator has a pole.
Theorem 28.16	the hypothesis was "whose leading scalar equals b_0^{p/q}", but the leading scalar of D is by plt:def:lead-leading-data the coefficient of its leading MONOMIAL t^0 L^j, which for a block of positive L-degree is not the block; the proof uses the block. The hypothesis is restated in the form the proof needs and the counterexample clause requires.
Theorem 28.20	"four mutually independent identities" is not what is meant and is not true: all four characterise the same family, so they are logically equivalent, not independent. What is claimed, and what matters, is that they share no code path.
Theorem 28.23	T3 had the gauge subtracted where it must be added. By plt:prop:cert-reciprocal, ordt(BQ-A)=ordt(Q-A/B)+ordt(B); a quotient sound at cutoff N has ordt(Q-A/B)>=N, so its residual vanishes below N+ordt(B). The printed N-ordt(B) is the threshold a WRONG answer already passes when ordt(B)>0, so the test as printed was vacuous in exactly the case the remark plt:rmk:cert-division-gauge is about.
Residual certificates and reproducible v	same overclaim as in the opening list: the table has no entry for the sum, the product or the derivation.
Theorem 29.3	the crossover was quoted without the hypothesis D>=3 under which the root exists, and the asymptotic x^{*}(D) ~D\log D was asserted; a one-line proof is supplied, since the volume states nothing else about this root.
Failure modes, diagnostics, and a decisi	three repairs. (a) The clause was stated for every D>=1, but x=D\log x has NO real root for D=1,2 (the function x-D\log x attains its minimum D(1-\log D)>0 there), so x^{*}(D) does not exist and the "if and only if" cannot be read; for those D the blocks decrease at every X>1. (b) Even for D>=3 the equivalence as printed was false near X=1: x-D\log x has TWO roots, and the blocks also decrease below the smaller one – at X=2 and D=10, L^{10}=0.025<2. What x^{*}(D) is, is the threshold beyond which they decrease for good. (c) The printed factor 2.42... is not the truncated decimal expansion of L^{10}/X=2.41584..., only its rounding.

Location	Repair
Failure modes, diagnostics, and a decisi	the proof chose only X >= max{X_0, e^2}, at which the series can still diverge (D=10, X=e^2 has L^{D}=1024>X) and f(X) is then undefined; the choice must also put X in the convergence range, which is possible because L^{D}/X -> 0.
Theorem 29.4	the existence of the threshold was asserted for every D and the second root was overlooked; both are repaired here.
Failure modes, diagnostics, and a decisi	the list ran to nine items against eight failure modes, the branch being the extra one; it is folded into the item it travels with.
Theorem 30.4	the draft claimed that no argument of \cref{plt:sec:ext-depth} needs more than parts (i)–(ii). That is not accurate: the statements of \cref{plt:thm:ext-tower-strict} and \cref{plt:cor:ext-depth-unbounded} call H_r a differential FIELD, and the field property of a Hahn series ring over a non-archimedean monomial group is exactly Neumann's lemma. What is true is that every argument carried out here needs only (i)–(ii); the field property is imported, not reproved.
Theorem 30.6	S1 ("What changes is the proof that every target coefficient receives only finitely many contributions ... generalized supports make it a central axiom") and S3 ("The well-order condition guarantees that each coefficient receives only an admissible family of contributions") both present well-ordering as if it were equivalent to local finiteness of a single product. It is sufficient, and it is necessary for the class, but it is not necessary for an individual product; the witness below shows this.
Theorem 30.6	the naive bound "k,l <= 2/(gamma-2)" is false – for gamma-2 = 1/2 the pair (3,6) is a representation with l = 6 > 4. Only the smaller index is bounded above; the larger is then determined.
Theorem 30.6	the draft said left-finiteness "is literally \cref{plt:thm:ring-support}(d) with Z replaced by Gamma". That is false and contradicts \cref{plt:prop:ext-hahn-trunc-fails}(b) below, which observes that clause (d) read verbatim over a general Gamma is strictly weaker: it asks only for an exhausting enumeration of order type at most omega and does not demand that the enumeration be cofinal. Over Z cofinality is automatic, which is why the two conditions are indistinguishable in Parts I–IV.
Generalized power–log classes, nested l	the draft cited a nonexistent label plt:thm:grp-taylor; the formal Taylor formula of this volume is \cref{plt:thm:tay-formula}.
Theorem 30.12	the draft used an undefined macro \Gfilt here; the volume preamble provides no such macro, so the filtration steps are written out.
Generalized power–log classes, nested l	S1 ("The arithmetic formulas from Part II therefore survive almost unchanged"), S3 ("The differential rule is again ... Near-identity composition remains controlled") and S5 ("Near-identity composition and reversion continue to work when the support conditions ensure coefficientwise finiteness") all extend Parts I-IV to Hahn supports without recording that the finite bookkeeping of Part VI – truncation, sparse representation, precision calculus – requires strictly more than well-basedness. \cref{plt:prop:ext-hahn-trunc-fails} isolates what is lost, and \cref{plt:thm:ext-lf-completion} isolates the class where nothing is.

Location	Repair
Generalized power–log classes, nested l	the draft recorded only "+ o(1)" here, which is too weak to produce the error term displayed below: an additive o(1) inside \log X contributes a relative error o(1/\log Y), larger than (\log\log Y)^2(\log Y)^{-2}. The sharper form is what the two-line computation actually gives, and it is what the displayed remainder needs.
Theorem 30.19	the draft concluded that the display "is exactly the sigma = 1/2 instance" of a theorem stated only for sigma in N. That is a claim about a proof belonging to another part, and it is not made here: the display above is a self-contained computation, and the agreement with the theorem is recorded as a check, not used as an input.
Theorem 30.19	\cref{plt:thm:lag-log-normal-form} restricts to sigma in N although its proof uses only that sigma is a scalar of K; the half-integral instance is needed here and is verified directly above rather than by widening the hypothesis of a theorem belonging to another part. The displayed coefficient 1/4 agrees with \eqref{plt:eq:lag-X-expansion} at c=1, rho=1, sigma=1/2.
Theorem 30.20	the draft pointed at an equation label plt:eq:grp-tower-derivative that the volume does not carry; the statement it needs is the enclosing proposition, which does exist.
Theorem 30.24	the draft proved incomparability only against the identity nesting, while \cref{plt:rmk:ext-completion-order} asserted (r+1)! pairwise distinct iterated fields. The argument already gives the general statement, because it uses nothing about the identity beyond the relative order of the two chosen levels; the last paragraph records this.
Theorem 30.26	the draft cited a label plt:rmk:div-embedding-warning that the volume does not carry; the statement being invoked is the expansion map itself.
Theorem 30.26	the draft wrote "an identity proved after expansion need not descend". That contradicts the injectivity it cites in the same sentence: if \iota(a)=\iota(b) then a=b. What non-surjectivity actually costs is that a computation carried out after expansion may leave the image, and that an exact cancellation in K(L) is, after expansion, visible only in the limit and at no finite truncation. Both claims are restated correctly.
Theorem 30.27	the draft obtained the display by asserting that the proof of \cref{plt:thm:lag-log-normal-form}, which is stated only for sigma in N, works verbatim for every scalar beta. That is a claim about a proof belonging to another part and is not made here; the two-line derivation is given instead, and the agreement with that theorem is recorded as a check.
Theorem 31.3	"a \notin \Gamma" is a statement about the image of \Gamma in \K under the embedding of \cref{plt:def:dif-scale}; the exponent a is read as a scalar of \K, which is what the proof needs and what makes the enlargement of the exponent group in the draft's proof unnecessary.
Theorem 31.3	the draft passed to \Gamma' = \Gamma + (1/2)Z without checking that \Gamma' is admissible, i.e. that the embedding \Gamma \hookrightarrow \K of \cref{plt:def:dif-scale} extends to it injectively. It does, and the verification is one line, given here.

Location	Repair
Theorem 31.3	the draft took \Gamma’ = \Gamma + aZ here. That enlargement is neither needed nor in general admissible – for K = Q, \Gamma = Z and a irrational the group Z + aZ has rank two and admits no injective homomorphism into Q. No enlargement is required: a is a coefficient, not an exponent, and u = at already lies in H_r(K,\Gamma).
Theorem 31.4	the draft pointed at an equation label plt:eq:dif-exp-value that the volume does not carry; the statement meant is the classification of which elements are exponentials.
Theorem 31.5	the draft assumed only "E nonzero and c a nonzero constant". In a ring with zero divisors that does not force cE \ne 0, and the proof needs \partial E = cE \ne 0 in order to apply the hypothesis on \nu at G = E. The hypothesis is stated in the form the proof uses; it is satisfied whenever \mathcal T is a domain.
Relation with the full transseries field	the draft opened this list with a Dahn–G\"oring item citing the key dahn-goering-1987, which is not in the merged bibliography of this volume. Rather than cite a key that does not resolve, the item is dropped; it carried no content used anywhere else.
Theorem 31.7	the draft referred to an equation label plt:eq:cmp-realization that the volume does not carry; the condition invoked is the defining one of the realization relation.
Theorem 31.8	three appeals to a label plt:rmk:om-open, which the volume does not carry, are replaced by the statement they were standing in for; see also \cref{plt:rmk:ext-open-convergence} and \cref{plt:rmk:ext-open-uniformity}.
Theorem 31.8	the draft also cited salvy-shackell-1992, a key that is not in the merged bibliography; the item it carried (functional inverses) is covered by salvy-shackell-1999, which is.
Theorem 31.9	the draft attributed the singularity locus to a label plt:rmk:om-open that the volume does not carry. The locus is recomputed here from the generating equation, and it does come out as the draft stated it.
Theorem A.1	the second group is a restatement inside the volume, not an error, but the draft called its two members "the same statement" and declared them interchangeable. They are not: plt:prop:tay-scalar-outer restates plt:prop:dif-bell and then adds a clause about expanding the outer function about a base point z_0 \ne 0. Nothing below leans on that clause – see the editorial note at plt:lem:bell-chain, where the base point is made part of the datum instead – but the two statements are not interchangeable.
Theorem A.7	the draft wrote "$\BigO(N^{3})$ coefficient multiplications" without saying what a coefficient is here. The count $\BigO(N^{3})$ is correct only for the operation "multiply one indeterminate into one stored entry"; the entries are not scalars, and the scalar count is genuinely superpolynomial. Left unqualified this is exactly the conflation that plt:cor:alg-cost-not-quadratic warns against, so the two levels are now separated.

Location	Repair
Theorem A.14	the draft wrote \varphi=\sum(\varphi_k/k!)z^k – coefficients at the origin – and then set \varphi(F)=\sum(\varphi_k/k!)W^k with W=F-z_0. That is \varphi evaluated at F only when z_0=0; in general the coefficients of the k-th power of W are \varphi^{(k)}(z_0)/k!, not \varphi_k/k!. (Take \varphi(z)=z, z_0=1, F=1+t: the displayed sum returns t, not F.) The justification offered, "apply the translate z\mapsto\varphi(z_0+z)", is also unavailable: re-expanding a formal element of \K[[z]] about a centre z_0\ne0 requires an infinite sum in \K for every coefficient, and \K carries no topology here. Both defects are repaired at once by making the centre part of the datum – the outer function IS its expansion at z_0. Nothing downstream changes: the proof only ever uses c_k=\varphi_k/k! and the relation \varphi'(F)=\sum_{k\ge1}kc_kW^{k-1}.
Theorem B.4	the draft said that for h_0=0 the formula "reduces to plt:eq:tay-C-bell", which is stated only for m>=r>=1; at r=0 it reduces instead to the boundary convention C_{m,0}=\delta_{m,0}. Range added.
Theorem B.5	the draft asserted "then F^\sigma = c^\sigma t^{\sigma n_0}(1+W)^\sigma" as though it were an identity. For non-integer \sigma the symbol F^\sigma has no prior meaning in the volume, and the right-hand side depends on the chosen value of c^\sigma; it is a definition, and is now printed as one.
Theorem B.13	S3's formula sheet displays this identity as [t^n](rev F - X + P) = ((-1)^{n+1}/(n+1)!) D(D-1)...(D-n+1) P^{n+1}. The added "+P" is a slip: it is correct for n >= 1, where P (a level-zero block) contributes nothing, but false at n = 0, where the left side is -P + P = 0 while the right side is -P. Removing the "+P", as in plt:eq:cext-level-zero-reversion, makes the formula correct for every n >= 0 and makes the Lambert case a specialization rather than an exception.
Operational formula sheet	this row printed the division recurrence with no hypothesis on B_0, which repeats the defect of S1's sheet ("B_0 \ne 0"). In K[L]((t)) a nonzero polynomial block is not enough: B_0 must be a unit of K[L], i.e. a nonzero scalar, or the quotient blocks leave K[L]. Hypothesis printed.
Operational formula sheet	this row read "\PLrat is the smallest field of the tower with polynomial-log coefficients", which inverts the content of plt:thm:div-minimal-field: \PLrat has RATIONAL-log coefficients, and the theorem says E((t)) is a field iff E = K(L). Restated as proved.
Exercises with solutions	source 1 asks only for a jump "by at least two", i.e. the case k = 2 of part (2), and does not ask for the converse. The stronger form is proved above because part (3) is what the sum rule of \cref{plt:thm:ring-valuation-laws} actually asserts.
Exercises with solutions	source 2 proves the common lower bound by asserting that "the sequence is already Laurent-bounded at some initial precision". That is not a hypothesis of completeness; the bound has to be extracted from the Cauchy condition at cutoff N = 0, as in part (3).
Exercises with solutions	the sentence printed here was garbled ("ordt(...)-differences are ..."). What is meant is the outer order of the difference of two consecutive partial sums, which is what Cauchyness is about.
Exercises with solutions	source 2 states the prefactor form "whenever available" without an instance of unavailability; L + t is one, and it is the reason the log-degree profile of \cref{plt:def:lead-profile} is defined blockwise rather than through a normal form.

Location	Repair
Exercises with solutions	source 3 answers this item with "a natural home is K((L^{-1}))((t))", i.e. \PLit. That overshoots by a whole class: the coefficients in \eqref{plt:eq:exr-recip-L-plus-t} are Laurent *monomials* in L, so even the localisation K[L,L^{-1}]((t)) suffices, and among the four ambient classes the answer is \PLrat. \PLit is strictly larger (\cref{plt:prop:ring-PLit-strict}), and the next item is what actually needs it.
Exercises with solutions	source 3 poses this product only through t^2 and source 1 poses a different pair only through t^3 without saying that the undisplayed blocks must be assumed zero. The two are merged here and the hypothesis is printed, since \cref{plt:thm:mul-finite-determination} is the point of the exercise.
Exercises with solutions	source 5 prints the block Q_3 of this example immediately after inputs given only modulo t^3. The hypothesis B_3 = 0 is a genuine extra assumption, and \cref{plt:prop:div-relative-precision} shows exactly how much is missing; it is stated here rather than assumed silently.
Exercises with solutions	source 1 prints the Stirling identity with the sum starting at j = 1. That is correct only for k >= 1; the uniform range j = 0..k with the empty-product convention covers k = 0 as well, and is used above.
Exercises with solutions	this sentence cited \cref{plt:prop:alg-shift}, the polynomial-shift formula, which has nothing to do with the truncation convention. The discrepancy is recorded with the sharp budget itself.
Exercises with solutions	source 1 records the budget as k <= floor((N - r)/(h + 1)) without saying which truncation convention its N refers to. Under the exclusive cutoff, which the catalogue makes normative, that form admits one Taylor term too many. Part (3) prints the conversion rather than silently shifting the index, because the same formula is what a routine would apply in reverse to size its inputs, where the extra term is not free.
Exercises with solutions	the exercise as inherited asked for the failure "for every F whose coefficient support is infinite". That quantifier is too strong, and the following paragraph is the counterexample to it.
Exercises with solutions	source 1 states this with the hypothesis q >= 1. The hypothesis is unnecessary: the Picard estimate \eqref{plt:eq:rev-picard-accuracy} holds for every alpha = q >= 0, and the case q = 0 – the Wright omega case – is exactly the one in which the statement is sharp.
Exercises with solutions	the check was printed as "a = aL with a in K", using the letter a for both the coefficient a(L) and the scalar; renamed.
Exercises with solutions	sources 2 and 3 both quote a general third-order inverse block; source 3's version (the b = c = 0 case above) is correct, source 2's is not (\cref{plt:exer:exr-trap-universal-formula}). Source 3 checks the degree law only at N = 1, 2; the block b_3 above is computed here and extends the check to N = 3.
Exercises with solutions	part (3) compares moduli, so it needs an absolute value on the coefficient field; the inherited statement left K unnamed.
Exercises with solutions	the O-term in \eqref{plt:eq:exr-xlogx} is an ANALYTIC estimate, and the computation above is formal. What licenses it is the explicit remainder bound of the Lambert part, not \cref{plt:thm:om-expansion}; the citation is supplied rather than left to the reader, in line with \cref{plt:exer:exr-trap-formal-to-analytic}.

Location	Repair
Exercises with solutions	the two numbers in \cref{plt:warn:lag-scale-trap} were computed at two different cutoffs. Recomputed in exact arithmetic at Y = 20 + log 10: N <= 2 -> correct 10.0000781540, naive 9.6702691781 N <= 3 -> correct 10.0000043732, naive 9.6702070480 The warningbox says "through Y^{-3}" and then quotes 10.000004 (which is the N <= 3 value) alongside 9.67026 (which is the N <= 2 value). Both truncations are quoted at N <= 3 above. The conclusion of the warningbox is unaffected; only the second numeral changes.
Exercises with solutions	"exactly item (i)" contradicted the sentence just above, which already observed that (v) also evaluates correctly at n = 0.
Exercises with solutions	the inherited solution asserted that the sign of omega - Omega_1 itself changes at x = e^2. That is the sign change of the LEADING TERM only. The true difference crosses zero at x = 8.7465..., recomputed here; the gap between the two numbers is the whole moral of the exercise, so printing e^2 as the exact threshold defeats it.
Exercises with solutions	the inherited solution claimed both errors agree "to one digit" with LamP_2(L)x^{-2}. At x = 10 the leading term is 3.484e-3 against a true error of 1.747e-3 – a factor of two, not one digit. Only the signs agree at that point, and the reason is printed below.
Exercises with solutions	the inherited text added "and nothing analogous is claimed for WrightOmega". That is false: the same lemma (plt:lem:lw-two-variable) applied at sigma = +1/x rather than -1/S gives the omega statement, and it is printed in the volume as plt:thm:lw-basepoint / plt:cor:lw-canonical. The claim is corrected here and in plt:exer:exr-omega-numerics, which repeated it.
Exercises with solutions	source 1's prose ("the same P_n occur, multiplied by (-1)^n") contradicts its own displayed expansion of the inverse of X - log X, in which P_1 = L occurs with a plus sign. The weight (-1)^{n+1} belongs to rev(X - L); the weight (-1)^n belongs to its negative, which is what W_{-1} is. Part (3) separates them.
Exercises with solutions	item (vi) of \cref{plt:cor:lambert-coefficients} is indexed from the top, [u^{n-2}], not at the fixed exponent 3. The two coincide at n = 5 and only there; read as "[u^3]" the check would fail at n = 6, where [u^3]LamP_6 = -85/6 while the right-hand side is 75/8 = [u^4]LamP_6. The inherited solution printed the fixed exponent and verified (vi) on LamP_5 only, although the exercise asks for both.
Exercises with solutions	source 5 asserts that the residual-to-error principle "proves the asymptotic validity of every truncation directly from" the formal residual identity, and source 1 asserts an analytic O-bound for the branch residual "by construction". Neither is a proof; the missing ingredient is the substituted-tail estimate, which is why this exercise exists.
Exercises with solutions	the inherited version of this exercise asserted outright that the minimal condition "does not determine the blocks", and proved it by changing one block and checking the condition at that index alone. The check at index N+1 then fails, so the modified family does not satisfy the definition and the claimed nonuniqueness is not there. Part (2) below proves uniqueness; the defects the exercise exists to expose are the two in part (3), and they are genuine.

Location	Repair
Exercises with solutions	this step was written as an inequality on "(F-G)(X)", which is a formal series and not a function. What the two estimates bound is the difference of the two TRUNCATIONS, which is a polynomial expression.
Exercises with solutions	the display here was garbled ("1 - LamP_0(0)(-1) + ..."). By \cref{plt:def:om-truncation}, Omega_N = X + Trunc(P)_{N+1}, so the evaluation is the single sum written below.
Theorem E.57	the inherited solution stopped at the third item and concluded that "this volume does not determine whether that series converges". That is false. \cref{plt:thm:lw-basepoint}, taken at K = x and log K = L, gives M = L and a seating condition 2(1+L) < x, under which the substituted series converges absolutely. At x = 20 the condition reads 7.99 < 20. Recomputed: the partial sum through n = 40 agrees with omega(20) - 20 + log 20 to fifteen digits.
(front matter)	the six sources cited this work inconsistently as a 2010 preprint and as a 2013 journal article; the published version is given, with the preprint retained as a secondary pointer.

Remark G.1 (What is not in this ledger). Three classes of change are deliberately excluded. Deduplication is not a repair: where several sources proved the same result, the strongest statement and the best proof were kept and the rest collapsed, which the provenance section records in aggregate rather than line by line. Restructuring is not a repair. And a source claim that this volume could neither prove nor refute has not been silently dropped: it is stated here as an explicitly labelled open problem or conjecture, with the source's assertion quoted as an assertion.

Appendix H

Formalization register

H.1 What this register claims

The proofs printed in this volume are human proofs. Separately, the hosting corpus now contains Lean declarations that verify some source results exactly and strict portions of others. The honest question is therefore claim-by-claim: which statement is Exact, which compound statement is only Partial, which declaration is merely a Neighbour, and what remains Absent. This register and the inline status paragraphs answer that question. Every Lean name below was checked to resolve in the corpus; a name that no longer resolves is a defect, and the corpus audit `scripts/audit_crosswalk_names.py` enforces it.

Four statuses are used. **Exact** (called **Crosswalked** in older entries) means a Lean declaration states the same mathematical fact, possibly in a different but faithfully translated setting. **Partial** means Lean proves a proper clause, range, or specialization but not the whole labelled source statement. **Neighbouring** means a genuinely different theorem that a reader might mistake for coverage. **Absent** means nothing in the corpus is close.

H.2 The formal algebra: partial foundations

The blanket claim that nothing in Parts I–IV had a Lean counterpart is now obsolete. Exact foundations include the sequence-indexed scale, Poincaré expansion, coefficient recovery, and uniqueness of Theorems 1.1 to 1.3; flatness and the corrected invisible-function proposition Theorems 1.6 and 1.7; Dickson's and Neumann's lemmas, with the latter instantiated on the order dual matching the volume's negligibility convention; the three power–logarithmic ratio limits and every chosen lexicographically decreasing sequence scale; and the polynomial block operators. The abstract differential bridge proves the block law for every integer exponent and nonresonant primitive existence; source-shaped uniqueness is conditional on injective evaluation at L. The exact names and the remaining boundaries are recorded at the statements themselves.

Those pieces do not yet assemble the polynomial–logarithmic calculus. The corpus still has no single construction carrying the concrete $\mathbb{K}[L][[t]]$/Laurent ring, its outer valuation, its distinguished derivation, composition, and reversion at infinity. The following table summarizes the finite and sequence-level statements that now have exact or explicitly delimited partial crosswalks in the body of the volume.

Result	Status	Lean surface and boundary
Theorems 1.1 to 1.3 and Equation (1.3)	Exact	`TransseriesScale.lean`; sequence-indexed, with `NeBot` for uniqueness.

Result	Status	Lean surface and boundary
Theorem 2.5	Exact	`TransseriesWellBased.lean`; finite antichains in $\mathbb{N}_0^k$.
Theorem 2.4	Exact	`TransseriesWellBased.lean`; the public `OrderDual` wrappers encode the printed greatest-element convention, and the printed total order specializes their stronger partial-order hypotheses.
Theorem 2.10	Partial	Both displayed analytic little-o laws are exact; the general nested height/depth prose has no datatype.
Theorems 1.6 and 1.7	Exact	`TransseriesFlat.lean`; abstract flatness, its submodule and absorbing multipliers, the power-scale exponential example, and the sharp same-expansion iff.
Theorem 4.7	Partial	`TransseriesScaleDominance.lean` and `TransseriesPolyLogScale.lean`; the analytic real-exponent trichotomy and ordered sequence scales are Exact, but the unordered full-set packaging and reverse generator implication are not named as the compound claim.
Theorem 4.8	Exact	`TransseriesBlockClasses.lean`; the printed integer-exponent logarithmic-comparability equivalence follows from a real-exponent theorem.
Theorem 4.14	Partial	`TransseriesMonomialUniqueness.lean` and `WrightOmegaTwoOrders.lean` prove its monomial-uniqueness and analytic residual inputs; their assembly from an arbitrary three-term scale is absent.
Theorem 4.1	Exact	`WrightOmega.lean`; all clauses including `Fabius.analyticAt_wrightOmega`, over the reals only, with no complex branch or holomorphy assertion.
Theorem 4.3	Partial	`WrightOmegaTwoOrders.lean` proves both concluding asymptotic equivalences; the explicit second-order block, error, and envelope are absent.
Theorem 4.12	Partial	The integer block law and the three abstract `IsLeast` minimality statements are exact in `TransseriesDifferentialBlock.lean` and `TransseriesDifferentialClosure.lean`; the germ-model growth and algebraic-independence clauses remain absent.
Equations (4.14) and (11.2)	Exact	`Fabius.derivation_zpow_block`; the integer block identity in an ambient field with a nonzero power generator.

Result	Status	Lean surface and boundary
Theorems 4.16 and 11.3	Partial	`TransseriesBlockAntiderivative.lean` and `TransseriesDifferentialBlock.lean`; polynomial operator, integer law, and conditional nonresonant primitive existence/uniqueness are exact, but the concrete Laurent model and remaining ambient links are absent.
Theorem 4.17	Partial	`TransseriesHarmonicIncrement.lean` proves the Stolz leading term; the logarithmic correction, constant, and little-$o(1)$ term remain absent.
Theorem A.4	Exact	`OrdinaryPartialBell.lean`; ordinary powers and the exact factorial normalization bridge to exponential partial Bell polynomials.
Theorems J.12, J.14 and J.16	Exact	All sixteen declarations of `UnitSeriesBellCoefficients.lean`; purely formal coefficient identities.
Theorem 13.24	Exact	`BellLeibnizTower.lean`; the abstract derivation tower and its partial Bell-polynomial recurrence.
Theorem J.23	Partial	`LinLogCoreInversion.lean` proves the real substitution, both real branch rules, uniqueness, threshold, and slope; the asymptotic and general complex-branch clauses remain absent.
Theorem J.43	Partial	`StaircaseInversion.lean` proves the five order/rounding/jump mechanisms (including residue classes); the printed Fourier layer and analytic interpolation construction remain absent.
Theorem J.51	Partial	`RemainderTransport.lean` proves the sharp displacement bound and the displayed first-order law with explicit error; only the closing asymptotic clause remains absent.
Theorem Q.8	Partial	`PowerLogCoreInversion.lean` proves the real $r = 1$ Lambert inversion and slope; general r, complex powers, and branch bookkeeping remain absent.
Theorem Q.20	Exact	`QuadraticCoreCatalan.lean`; Catalan/falling-product and finite coefficient recurrence.
Theorem Q.19	Partial	Its coefficient recurrence is checked; assembled power-series existence, uniqueness, square root, and denominator depth remain absent.

Result	Status	Lean surface and boundary
Theorem S.2	Partial	`DerangementNearestInteger.lean` proves the integer defect identity, strict half-unit error, and rounding; the real-argument bound and the branch-splitting family remain absent.

Together these thirty-five modules expose 304 explicit public commands. The original eighteen-module surface, extended in place where noted, is: `TransseriesScale.lean` 9, `TransseriesScaleDominance.lean` 8, `TransseriesPolyLogScale.lean` 4, `TransseriesWellBased.lean` 7 written declarations, `TransseriesHeight.lean` 3, `TransseriesBlockAntiderivative.lean` 15, `TransseriesDifferentialBlock.lean` 12, `UnitSeriesBellCoefficients.lean` 16, `QuadraticCoreCatalan.lean` 11, `TransseriesFlat.lean` 26, `WrightOmega.lean` 14 (one definition and thirteen theorems), `TransseriesHarmonicIncrement.lean` 2, `OrdinaryPartialBell.lean` 6, `LinLogCoreInversion.lean` 22, `PowerLogCoreInversion.lean` 9, `RemainderTransport.lean` 4, `StaircaseInversion.lean` 7, and `DerangementNearestInteger.lean` 8. Fourteen incoming leaves add `BackwardErrorExistence.lean` 7, `BellLeibnizTower.lean` 5, `CayleyKernel.lean` 10, `CayleyLocalCoordinate.lean` 7, `CayleyTreeFunction.lean` 7, `DivisorTransform.lean` 9, `ExpSeriesRecurrence.lean` 4, `LambertCorrectionEquation.lean` 8, `LambertShiftConcavity.lean` 5, `LeastTermIndex.lean` 7, `TouchardEulerOperator.lean` 9, `TransseriesBlockClasses.lean` 3, `TransseriesDifferentialClosure.lean` 12, and `TransseriesMonomialUniqueness.lean` 2. Three later focused leaves add `StirlingSeriesCoefficients.lean` 15, `WrightOmegaTwoOrders.lean` 8, and `UnitSeriesPowerRecurrence.lean` 3. (The fifteenth incoming leaf, `NewtonInterpolation.lean` with 22 declarations, belongs to a different focused package.) In addition, the named `to_additive` attributes in `TransseriesWellBased.lean` generate two public additive twins, for 306 named inventory/API entries under this convention. The corpus's lexical documentation census intentionally counts explicit commands only; automatically generated structure projections are outside both tallies.

The last of these leaves also closes the coefficient-calculus recurrence for an arbitrary formal power. In `UnitSeriesPowerRecurrence.lean`, `Fabius.coeff_recurrence_of_mul_derivative_eq` extracts a triangular coefficient law from $FG' = \beta F'G$ over any commutative ring, while `Fabius.mul_derivative_fallingSeries_subst_sub_one` and `Fabius.coeff_fallingSeries_subst_sub_one_recurrence` specialize it to the falling-factorial definition of a unit-series power over a commutative $\mathbb{Q}$-algebra. This is purely formal: it adds no convergence or analytic branch assertion.

H.3 The remaining two-generator formal algebra: absent

The finite and abstract fragments above do not yet assemble the volume's two-generator transseries calculus. The corpus has no completed polynomial–logarithmic Laurent-series ring, outer valuation, composition of such transseries, or reversion at infinity. Its abstract derivation block now covers integer powers when the power generator is a unit, but does not construct the ambient Laurent ring. Concretely, a formalization campaign would have to build, in order:

1. the ring $\mathscr{PL}_{\mathrm{pow},\mathbb{K}} = \mathbb{K}[L][[t]]$ with t and L as independent generators, and its Laurent extension (Part 5);
2. the outer valuation and the leading-data trio, which are three different objects and are easy to conflate in a formal setting (Part 6);

3. the concrete derivation D_X characterized by $D_X t = -t^2$, $D_X L = t$, instantiate the existing integer-power bridge in the free Laurent model, prove injectivity of evaluation and the resonant kernel statement there, and construct the repeated-derivative operator $\Delta_{n,k}$ (Part 11);
4. composition by generator substitution, which is where an informal treatment most often goes wrong, since $t \mapsto t + H$ is *not* the substitution meant (Part 12);
5. the exact formal Taylor formula (Theorem 13.6), whose proof here is an algebra-homomorphism argument on the two generators and should formalize without the coefficient bookkeeping;
6. the reversion fixed point and its uniqueness (Theorem 16.11).

Only after those is the Lagrange–Bürmann theorem at infinity (Theorem 18.14) even statable.

H.4 Lagrange–Bürmann: neighbouring, not crosswalked

The corpus *does* formalize Lagrange–Bürmann inversion — for formal power series at the origin, over a commutative $\mathbb{Q}$-algebra. The relevant declarations are `Fabius.Lagrange.solution`, `Fabius.Lagrange.coeff_solution`, and the division-free identity `Fabius.Lagrange.coeff_subst_mul_derivative`.

That is a different theorem from Theorem 18.14, and the difference is not cosmetic:

- the formalized version inverts $g = z\varphi(g)$ in $R[[z]]$, a one-generator power-series ring with the ordinary derivative; the version here inverts $X + A$ in the two-generator polynomial–logarithmic scale with the derivation D_X;
- the formalized version is a statement about coefficients of z^n; the version here is a statement about coefficient *blocks* $p_n(L)$, each a polynomial whose degree is unbounded across n;
- summability is automatic at the origin and is a theorem here (Theorem 18.9).

So the corpus's Lagrange inversion cannot be cited as proving the at-infinity formula, and this volume does not do so. It is, however, the natural model for a future formalization, and its proof strategy — purely algebraic, avoiding formal residues, which Mathlib does not have — is the same strategy the proof here uses.

H.5 The Lambert W series: neighbouring

`Fabius.lambertW` and `Fabius.coeff_lambertW` formalize the *small-argument* expansion $W(z) = \sum_{n\geq1}(-n)^{n-1}z^n/n!$, obtained from the formalized Lagrange inversion above. This volume expands W at *infinity* (Part 22). The two expansions share a name, a function, and nothing else: one is a convergent Taylor series at the origin with radius e^{-1}, the other an asymptotic polynomial–logarithmic expansion at infinity that converges nowhere. Confusing them is the single most likely misreading of this volume's Part V, which is why the entry is here.

H.6 The real branches: crosswalked

The analytic facts about the two real branches that this volume *uses* — as opposed to derives — are kernel-verified:

Table H.2: Analytic branch facts used in Part V and their Lean declarations.

Fact	Declaration
The two real branches exist and are characterized by the defining equation	`Fabius.principalLambertW`, `Fabius.lowerLambertW`, `Fabius.principalLambertW_unique`, `Fabius.lowerLambertW_unique`
Every real solution of $we^w = z$ is one of the two branches	`Fabius.eq_principalLambertW_or_eq_lowerLambertW`
Continuity on the full closed domains, branch point included	`Fabius.principalLambertW_continuousOn_Ici`, `Fabius.lowerLambertW_continuousOn_Ico`
The derivative of the principal branch, and $W_0''(0) = -2$	`Fabius.deriv_principalLambertW`, `Fabius.deriv_deriv_principalLambertW_zero`
Strict concavity of the principal branch	`Fabius.strictConcaveOn_principalLambertW`
The square-root scale at the branch point, and the signed leading laws	`Fabius.lambertWBranchPointScale`, `Fabius.principalLambertW_add_one_isEquivalent_branchPoint`, `Fabius.lowerLambertW_add_one_isEquivalent_branchPoint`

The branch-point entry is the sharpest of these: it confirms, in the kernel, the assertion of Part 22 that the inverse near $z = -e^{-1}$ is *not* a power–logarithmic series but carries a square-root scale, which is exactly the boundary of this volume's class.

H.7 The lower-branch expansion near zero: partially crosswalked

One expansion statement of Part V has a genuine Lean counterpart. `Fabius.tendsto_lowerLambertW_expansion` proves

$$W_{-1}(-\varepsilon) = \log\varepsilon - \log|\log\varepsilon| + o_{\varepsilon\downarrow 0}(1)\,,$$

which in the coordinates of Part 22, with $S = \log(1/\varepsilon)$ and $L = \log S$, is $W_{-1}(-\varepsilon) = -S - L + o_{\varepsilon\downarrow 0}(1)$: precisely the first two blocks of the lower-branch expansion derived there.

A second, sharper declaration pins the next block as well. `Fabius.lowerLambertW_near_zero_bounds` proves, for $x \in (-e^{-1}, 0)$ and $\eta = \log(1/(-x))$,

$$-\eta - \frac{\eta\log\eta}{\eta - 1} \;\le\; W_{-1}(x) \;<\; -\eta - \log\eta.$$

In the coordinates above, $\eta = S$ and $\log\eta = L$, so the bracket says that the deviation of W_{-1} from the two-block truncation $-S - L$ lies in $\big(-L/(S-1),\, 0\big]$. That is a two-sided, non-asymptotic statement: it fixes the sign of the third block and its order L/S, which is exactly what Part 22 derives formally. The strictness of the upper bound is the formal statement that the third block is nonzero and negative.

The supporting declaration is worth naming separately, because it is the piece most likely to be reusable: `Fabius.SubLog.bracket_of_sub_log_eq` brackets the inverse of $y \mapsto y - \log y$ for $y > 1$ with no Lambert content at all, and `Fabius.SubLog.strictMonoOn_sub_log` supplies the monotonicity that makes that inverse well defined. Since this volume's lower-branch treatment *runs* on the inverse of $u \mapsto u - \log u$ rather than on W_{-1} directly, that lemma — not the branch-specific one

— is the natural target for a future crosswalk of the whole lower-branch section. The side condition $\eta > 1$ is itself a lemma (`Fabius.one_lt_log_one_div_neg`) rather than a hypothesis the caller must discharge, so the $(\eta - 1)$ denominator cannot be reached at zero.

Two elementary bounds on the principal branch are also available and are used here only as sanity constraints: `Fabius.principalLambertW_elementary_bounds` gives $x/(1+x) \le W_0(x) \le \log(1+x) \le x$ for $x \ge 0$, resting on the Lambert-free inequality `Fabius.exp_le_one_add_mul_exp_self`, valid for every real argument.

Beyond the third block — the $\mathsf{L}_n^{\mathrm{Lam}}$ coefficients for $n \ge 2$, the remainder order at every order, and any uniformity — everything is human-proved here and unformalized.

H.8 All derivatives of the real branches: crosswalked

The derivative formulas used in Part 22 are kernel-verified to every order. `Fabius.iteratedDeriv_principalLambertW` proves, for $x > -\mathrm{e}^{-1}$ and every $n \in \mathbb{N}_0$,

$$\frac{\mathrm{d}^{n+1}}{\mathrm{d}x^{n+1}} W_0(x) = \frac{\mathrm{e}^{-(n+1)W}\, P_n(W)}{(1+W)^{2n+1}}, \qquad W = W_0(x),$$

and `Fabius.iteratedDeriv_lowerLambertW` proves the same formula for W_{-1} on $(-\mathrm{e}^{-1}, 0)$. The polynomial family is `Fabius.lambertDerivPoly`, defined by $P_0 = 1$ and the recurrence (`Fabius.lambertDerivPoly_succ`)

$$P_{n+1}(w) = (1+w)P_n'(w) - \big((n+1)w + (3n+2)\big)P_n(w),$$

whose first members, recomputed here in exact arithmetic from that recurrence, are

$$P_1 = -(w+2), \quad P_2 = 2w^2+8w+9, \quad P_3 = -(6w^3+36w^2+79w+64), \quad P_4 = 24w^4+192w^3+622w^2+974w+625,$$

the first two being the kernel's own evaluated cases (`Fabius.eval_lambertDerivPoly_one`, . . . `_two`). Two conventions matter. The Lean index is shifted by one against the classical one: the classical P_n of [5] is Lean's P_{n-1}, and the formula above is for the $(n+1)$-st derivative. And $P_1 = -(w+2)$ is exactly what one obtains by differentiating $W' = \mathrm{e}^{-W}/(1+W)$ once, which fixes the sign convention. The engine behind the theorem carries no Lambert content and is the reusable piece: `AutonomousODE.iteratedDeriv_eq_comp` proves that if $W' = \varphi \circ W$ on an open set then every iterated derivative is $G_n \circ W$ with $G_0 = \mathrm{id}$ and $G_{n+1} = G_n'\varphi$. This entry is crosswalked, not neighbouring: it is the same statement in the same normalization, and the Lambert W guide's own crosswalk appendix records the identical index convention.

H.9 Summary

The former conclusion that none of the volume's own results was crosswalked is obsolete. Exact results include sequence-scale coefficient recovery and Poincaré uniqueness, flatness and the corrected invisible-function proposition, Dickson and Neumann support lemmas, the three real power–log limits and chosen sequence scales, the real Wright-omega proposition, both displayed integer block equations, the unit-series Bell coefficient calculus, and the quadratic Catalan identity. The general height prose, the compound unordered power–log scale claim, the ambient Laurent block-antiderivative and differential propositions, and the assembled quadratic core remain Partial at their stated boundaries. The real Lambert branch facts and lower-branch asymptotics have only the precise coverage recorded above.

The assembled two-generator Laurent-series calculus, its composition operation, and its reversion at infinity remain absent, while the small-argument Lagrange and Lambert series remain Neighbouring. This register is deliberately conservative: a strict fragment or a neighbour is never reported as whole-label coverage.

Appendix I

Provenance of the consolidation

I.1 What was merged

This volume is an editorial consolidation, carried out on 2 September 2026, of six independently written article packages that arrived together as bare directories in the direct-arrival commit `730e1763291099cd50ca1e20ed2c62c38d95ab4f` and were filed on 1 September 2026 as standalone quick-gate receipts. No archive or checksum ledger accompanied them. All six treat the same subject – the algebra, division, composition, and asymptotic reversion of polynomial–logarithmic transseries, with Lambert W as the guiding example – from independent starting points, with substantial overlap and no containment relation among them.

Editorial merge means: every result is stated once, in the strongest form that the six sources jointly support and that we can prove; the best available proof is retained and gaps are filled; conflicting conventions are resolved against the corpus notation catalogue; errors are corrected and the corrections are marked in the source with `% ed.:` comments; and every source-specific layer that is not redundant is preserved. Nothing was discarded merely for being repeated: repetition was collapsed, content was not.

I.2 Absorbed sources and their receipts

Each source is recorded twice: as filed at intake, and as actually absorbed. The two differ by exactly three bytes per file because a regrouping on 2 September 2026 moved the packages one directory level deeper and the shared notation input path was repaired from four to five levels (Table I.1). No mathematical content changed between the two states.

The six retained arrival PDFs, which are historical artefacts and are not renders of the absorbed sources, had SHA-256 values `a4fc4af0...69e886` (S1, 119 custom 522-by-738-point pages), `5e9ff596...bfc68e` (S2, 102 custom pages), `3f7c4bc1...58a4af` (S3, 87 Letter pages), `c2d75b35...45a3eb` (S4, 47 A4 pages), `189e95ab...58a2db` (S5, 44 Letter pages), and `b5142bad...467aa` (S6, 100 A4 pages). Git history is the archive for all twelve files.

I.3 What each source contributed

No source was a superset of the others, and each contributed at least one layer that no other supplied.

- **S1** contributed the block-algebra presentation of the scale, the systematic comparison of four reversal algorithms, the residual-certificate and reproducible-verification discipline, the Dulac and power–log normal-form literature, and the largest exercise set.
- **S2** contributed the t-adic completeness treatment, the normal-ordered composition formalism,

Table I.1: The six absorbed sources. "Intake" is the receipt recorded in the group manifest on 1 September 2026; "absorbed" is the state merged into this volume. The retained arrival PDFs were historical publication checkpoints that already predated their own sources and were not rebuilt for this merge.

Source	Lines	Bytes	SHA-256
S1 = `Polynomial-Logarithmic-Transseries-1.tex`			
intake	4,020	187,071	`01a03e0934d05b8b94369da9656fc0da4dde9b7c559b4e07` `38075484e14f4e61`
absorbed	4,020	187,074	`4fd42b7e344c485f8d878279dc3b9a6d5dc326a9c739114e` `443fd9c07a5eb2b8`
S2 = `polynomial_logarithmic_transseries-2.tex`			
intake	5,006	173,396	`aa25baa03099472df765d35c03a2bb265dbe9af5aab500ec` `0a4a04a378ef83e5`
absorbed	5,006	173,399	`d172b81453a805366f056f6114bb6df6fdf7811466413201` `85530940754bdefe`
S3 = `Polynomial_Logarithmic_Transseries-3.tex`			
intake	4,249	150,182	`0962c15683cb4610d45961c227f6e9b102d66ee37073432d` `1943d31a1e6ad348`
absorbed	4,249	150,185	`d2a8bd25d71cc41fc81c1c75fb66ca75496b8425ac496ce1` `05d3220e1ea07e3f`
S4 = `Polynomial-Logarithmic-Transseries-4.tex`			
intake	3,132	120,607	`387ca51f94292a996fb339ec6dfb3477f68575255b2aaaf5` `39fcffb1c32ff564`
absorbed	3,132	120,610	`37026b625cf7b2fd106005b8c89f720b8b724f99a5446ed8` `4ca52531c6e2b18f`
S5 = `Polynomial_Logarithmic_Transseries-5.tex`			
intake	2,443	106,141	`1149ae6884e63dc77b747c786d0455419b6c79cdd3b33dc9` `15bf3875cc8d26cc`
absorbed	2,443	106,144	`187bd510f33ea4296f0247b286d0fcae6200ce0b5bd175c6` `c8706a37652dc17a`
S6 = `polynomial_logarithmic_transseries-6.tex`			
intake	4,388	155,846	`df4e4bc5932a59e4ed265b32d8463edc65f67cbd2735b349` `02acf479647f8491`
absorbed	4,388	155,849	`16978aae6fb1e458e236cac9807ab81bb5314fdb087a2504` `a416867a4d43b4c4`

the power–log series near a finite point with the Dulac conventions, the derivation of the universal composition formulas, and the pitfalls-and-diagnostics decision procedure.

- **S3** contributed the formal summability criterion, the degree law for polynomial perturbations, the Lagrange–Bürmann transport formula, the fully formal proof of Lagrange–Bürmann inversion at infinity, and the reusable inverse templates beyond Lambert.
- **S4** contributed the sharpest statement of the unit criterion and the field property, the finite reversal certificate, and closed-form coefficient extraction.
- **S5** contributed the tightest and most completely proved core – it is the spine of Parts I–IV – together with the Laplace transform of the Lambert polynomials, the resulting exponential-moment cancellations, and the generalized Lambert expansion.
- **S6** contributed the filtered Lipschitz gain underlying the fixed-point construction, the composed-observable form of Lagrange–Bürmann, the systematic change of variable between infinity and a finite point, and the failure-mode catalogue.

I.4 Conventions reconciled

The six sources used mutually incompatible local conventions. All are resolved here in favour of the corpus notation catalogue; the concordance in Part F records the mapping in full. The reconciliations that change a formula rather than only a glyph are:

- **Truncation.** S6 used the inclusive cutoff $[F]_{\le N}$; every other source and the catalogue use the exclusive $\mathrm{Trunc}_{<N} F$. The relation is $[F]_{\le N} = \mathrm{Trunc}_{<N+1} F$, so every S6 statement transported here has had its cutoff index shifted by one.
- **Lambert polynomial indexing.** The sources split between a zero-based family with $\mathsf{L}_0^{\mathrm{Lam}}(u) = -u$ and a one-based family that separates the logarithmic block. The catalogue's zero-based normalization is used throughout; the degree law $\deg \mathsf{L}_n^{\mathrm{Lam}} = n$ is therefore stated only for $n \ge 1$, since $\mathsf{L}_0^{\mathrm{Lam}} = -u$ has degree 1.
- **Coefficient extraction.** Two sources printed $[M]F$ and two printed $[F]_M$ for the same operation. This volume writes the series to the right of the bracket throughout.
- **The three $\mathcal{O}$ roles.** Formal valuation tails, analytic estimates, and algorithmic cost bounds are printed with three different tokens and are never interchanged. Several source statements that used one glyph for all three have been split accordingly.

I.5 Status of the claims

Every theorem, proposition, lemma, and corollary in this volume carries a proof. Where a source asserted a result without proof, the proof is supplied here or the assertion is demoted to an explicitly labelled conjecture or open problem. Where a source claimed analytic validity on the strength of formal algebra alone, the claim has been weakened to what the algebra establishes and the missing analytic hypothesis is named. Corrections to the sources are marked in this document's own source with `% ed.:` comments at the point of repair, and are collected in Part G.

Each section was drafted and then reviewed by an independent adversarial pass instructed to refute rather than confirm, with two exceptions that are recorded here rather than hidden: the Lambert W branch section (Part 22) and the exercises (Part E) did not receive that pass. In its place, every claim in those two sections that admits exact or high-precision computation was recomputed independently of the text: the branch-point Puiseux coefficients through p^6 by exact series reversion; the transverse coefficient identity through $a = 5$; the uniform remainder bound (22.18) at six base points, canonical and off-centre, for $N \leq 8$, with a worst observed error-to-bound ratio below 0.1; the Cauchy bound (22.11) through $a + b = 12$; the two-variable fixed point on 160 complex sample points; the $X + aL^2$ reversion through $n = 9$; and the numerical values quoted in the exercises to every printed digit. What those computations cannot test — the prose of the proofs — has been read but not adversarially reviewed, and a reader should weight those two sections accordingly.

The proofs printed here remain human proofs, but several of their statements now have separate Exact or Partial Lean counterparts. The inline crosswalks and Part H give their precise scope; in particular, a machine-checked component is not evidence for an unlisted clause of its source lemma. This includes the finite formal algebra and the precisely enumerated real Lambert and branch-point results. Statements not identified there as Exact remain human-proved frontier mathematics or are explicitly marked Partial, Neighbouring, conjectural, or open.

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Part X

Inversion: the apparatus for a rapidly growing function

Appendix J

Apparatus for inverting a rapidly growing function

All of the first twelve filed articles carry a private copy of the material of this chapter. The presentation below follows, in the main, the two sections *A universal coefficient calculus for exponential–power inversion* and *Lambert-normalized inverse transseries* of Swing_Factorial_Transseries_Article, which state the reversion in the greatest generality of the first twelve. Four things have been changed. (i) The Lagrange fixed-point lemma is reproved: the swing article's proof appeals to Lagrange–Bürmann inversion in a setting where the required invertibility hypothesis fails, and the gap is closed here by a universality argument (Theorem J.18). (ii) The exponential–power model is separated into a *formal* and an *analytic* reading, and every asymptotic claim is given an explicit remainder hypothesis; the sources routinely pass from a coefficient recurrence to a Poincaré statement without one. (iii) The dominant core is axiomatized (Theorem J.8) so that the factorial-type phase of Part N — which does *not* belong to the exponential–power model — is covered by the same reversion theorem. (iv) The discrete-inverse material of Butcher_Tree_Transseries-2, double_factorial_transseries-2 and the three partition articles is merged into one statement with a residue-class lattice.

J.1 Standing conventions

Throughout the volume R denotes a commutative $\mathbb{Q}$-algebra, and $R[[t]]$, $R[[t,u]]$ formal power series rings over it. For a formal series f we write $[t^n]\,f$ for the coefficient of t^n. The symbols $\mathrm{e}, \mathrm{i}, \mathrm{d}$ are the upright constants and differential; $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$ are the usual number systems, and $\mathbb{N} = \{0, 1, 2, \dots\}$. Asymptotic statements use $\mathcal{O}$ and o in the Landau sense as the stated variable tends to $+\infty$; an implied constant is always allowed to depend on every quantity held fixed in the statement (in particular on the truncation index N) and on nothing else.

The sources disagree on three points of bookkeeping. We fix them once.

Remark J.1 (Conventions adopted, and where the sources differ).

1. *Ordinary versus exponential Bell polynomials.* The partition articles and Butcher_Tree_Transseries-2 take the *exponential* partial Bell polynomials $B_{n,k}$ as primitive and convert; Swing_Factorial_Transseries_Article and double_factorial_transseries-2 take the *ordinary* ones $\widehat{B}_{n,k}$ as primitive. The volume defines both (Theorems J.9 and J.10), records the conversion (Theorem J.12), and uses $\widehat{B}$ in every reversion formula and B in every exponentiation or logarithm of a *unit* series. The reason is not taste: reversion multiplies a correction series whose coefficients carry no factorials, and $\widehat{B}$ is the transform that leaves them alone.

2. *Sign of the logarithmic phase.* We write the dominant phase as
$$L = aX + b \log X,$$
with b a *signed* constant. Butcher_Tree_Transseries-2 writes $Z = X - p \log X$ and the partition articles write $\rho - 2 \log \rho = L$; both are the case $b = -p < 0$. Every formula below is stated in the signed convention, and the dictionary of Table J.1 gives b for each chapter. One dividend of the signed convention is Theorem J.25: the Lambert branch is W_0 when $b > 0$ and W_{-1} when $b < 0$, with no case analysis beyond the sign.

3. *Truncation index.* A truncation "of order N" retains the terms with index $n = 1, \dots, N$ *inclusive*, together with the leading term. Swing_Factorial_Transseries_Article uses the exclusive convention for its forward table ("the truncation labelled N retains $A_0, \dots, A_{N-1}$") and the inclusive one for its inverse table ("the first N correction terms are retained"); the two tables of that article are therefore not on a common scale. The volume is inclusive throughout.

J.2 The exponential–power model

J.2.1 Formal and analytic readings

Definition J.2 (The exponential–power model)**.** Fix constants $C > 0$, $a > 0$, $\alpha \in (0, 1]$, $\delta > 0$, $b \in \mathbb{R}$, and a formal series without constant term,
$$Q(t) = \sum_{j \geq 1} q_j t^j \in t\,\mathbb{R}[[t]].$$
The associated *exponential–power model* is the expression
$$F(x) = C \exp\bigl(ax^{\alpha}\bigr)\, x^b \, \exp Q\bigl(x^{-\delta}\bigr), \tag{J.1}$$
understood through its logarithm
$$\log F(x) = \log C + ax^{\alpha} + b \log x + Q\bigl(x^{-\delta}\bigr). \tag{J.2}$$
We call $(C, a, \alpha, b, \delta, Q)$ the *model data*.

A positive function $F \in C^1([x_0, \infty))$ is said to be *of exponential–power type with data* $(C, a, \alpha, b, \delta, Q)$ when, for every $M \in \mathbb{N}$, there is a constant K_M with
$$\left| \log F(x) - \log C - ax^{\alpha} - b \log x - \sum_{j=1}^{M} q_j x^{-j\delta} \right| \leq K_M x^{-(M+1)\delta} \qquad (x \geq x_0), \tag{J.3}$$
and *differentiably* of that type when in addition
$$(\log F)'(x) = a\alpha x^{\alpha - 1} + \frac{b}{x} + \mathcal{O}\bigl(x^{-1-\delta}\bigr) \qquad (x \to \infty). \tag{J.4}$$

Remark J.3 (Why the two readings are kept apart). The series Q is *not* assumed convergent. In Parts M, N and P it is divergent of Gevrey type, which is what makes Theorem J.54 bite there. It is *not* divergent in Part O: for the partition numbers $Q(t) = \log(1 - t)$ has radius of convergence 1, every algebraic expansion of that chapter converges, and Theorem J.54 correspondingly has no content there — the error floor of Part O is set by the exponentially small Rademacher remainder and not by a least term. This is precisely why the two readings are kept apart: convergence of Q is a property of the subject, not of the frame. Every purely algebraic assertion below — coefficient recurrences, closed coefficient formulae, degree bounds — is a statement about (J.2) as a formal object and needs no analytic hypothesis at all. Every assertion with an error term uses (J.3), and every assertion that inverts uses (J.4) as well: an asymptotic expansion may not be differentiated without a hypothesis, and the inversion of F is controlled by $(\log F)'$. In each chapter (J.4) is verified directly from a digamma, Laplace or Rademacher representation, never inferred from (J.3).

J.2.2 Reduction to a linear phase

Lemma J.4 (Monomial α-reduction). *Let F be of exponential–power type with data $(C, a, \alpha, b, \delta, Q)$ on $[x_0, \infty)$, and put*

$$\xi = x^{\alpha}, \qquad x = \xi^{1/\alpha}, \qquad G(\xi) := F\big(\xi^{1/\alpha}\big). \tag{J.5}$$

Then G is of exponential–power type on $[x_0^{\alpha}, \infty)$ with data

$$\Big(C,\ a,\ 1,\ \frac{b}{\alpha},\ \frac{\delta}{\alpha},\ Q\Big), \tag{J.6}$$

that is,

$$\log G(\xi) = \log C + a\xi + \frac{b}{\alpha}\log\xi + Q\big(\xi^{-\delta/\alpha}\big). \tag{J.7}$$

The coefficients q_j are unchanged. If moreover F is differentiably of its type then G is differentiably of its type. Finally $x \mapsto x^{\alpha}$ is an increasing bijection of $[x_0, \infty)$ onto $[x_0^{\alpha}, \infty)$, so F is strictly increasing if and only if G is, and then

$$F^{-1}(y) = \big(G^{-1}(y)\big)^{1/\alpha}. \tag{J.8}$$

Proof. Substituting $x = \xi^{1/\alpha}$ into (J.2) and using $x^{\alpha} = \xi$, $\log x = \alpha^{-1}\log\xi$ and $x^{-j\delta} = \xi^{-j\delta/\alpha}$ gives (J.7) term by term; the truncated inequality (J.3) transforms into the corresponding inequality for G with the same constants K_M, because $x^{-(M+1)\delta} = \xi^{-(M+1)\delta/\alpha}$. For the derivative, $\mathrm{d}x/\mathrm{d}\xi = \alpha^{-1}\xi^{1/\alpha-1}$, so

$$(\log G)'(\xi) = (\log F)'\big(\xi^{1/\alpha}\big)\cdot\frac{\xi^{1/\alpha-1}}{\alpha} = \Big(a\alpha\xi^{1-1/\alpha} + \frac{b}{\xi^{1/\alpha}} + \mathcal{O}\big(\xi^{-(1+\delta)/\alpha}\big)\Big)\frac{\xi^{1/\alpha-1}}{\alpha} = a + \frac{b/\alpha}{\xi} + \mathcal{O}\big(\xi^{-1-\delta/\alpha}\big),$$

which is (J.4) for G with α replaced by 1. The last two assertions are immediate from strict monotonicity of $x \mapsto x^{\alpha}$ on $(0, \infty)$. □

Remark J.5 (Normalization used from here on). By Theorem J.4 every statement about the model (J.1) reduces to the case $\alpha = 1$, and we assume $\alpha = 1$ from Section J.4 onwards. We assume in addition $\delta = 1$; this is the value of δ/α in every chapter of the volume (Table J.1). Theorem J.31 states the general-δ reversion, on the two-generator monomial grid it requires.

Remark J.6 (The affine shift is taken *before* the reduction). Chapters N and O apply the model not to the combinatorial index n but to a shifted variable $x = n - x_*$ ($x_* = 1/24$ for the partition numbers, $x_* = -1$ for the double factorial). This is not a loss of generality but a genuine choice of normal form, and the order of operations matters. If F has data $(C, a, \alpha, b, \delta, Q)$ in x and one re-expands in $n = x + x_*$, then for $\alpha < 1$

$$a(n - x_*)^{\alpha} = an^{\alpha} + a\sum_{k\ge 1}\binom{\alpha}{k}(-x_*)^k n^{\alpha-k},$$

so the phase acquires an entire new tower of powers $n^{\alpha-k}$: the re-expanded object no longer has model data with a single exponent grid. The partition articles say this explicitly — their infinite algebraic coefficient arrays in the variable $n^{-1/2}$ "arise entirely from re-expanding the shift $N = n - 1/24$". The volume therefore normalizes the shift first, inverts in x, and returns to the index by $n = x + x_*$, an exact identity that costs nothing.

Remark J.7 (Which chapters the model covers, and which it does not). Chapters M, O and P are governed by Theorem J.2: the rooted-tree counts and the swing factorial with $\alpha = 1$, the partition numbers with $\alpha = 1/2$ (Table J.1). Part N is not.

Claim repaired. It is natural, and it is asserted in the merge brief for this volume, that the case $\alpha = 1$ of Theorem J.2 covers the double factorial alongside the swing factorial and the rooted-tree counts. It does not. For the swing factorial $n!/\lfloor n/2 \rfloor!^2$ the growth is $2^n n^{\pm 1/2}$ and the model applies with $a = \log 2$. For the double factorial,

$$\log\big((x+1)!!\big) = \frac{x+1}{2}\big(\log(x+1) - 1\big) + \mathcal{O}(\log x),$$

so the dominant phase is $\frac{1}{2} s \log s$ with $s = x+1$, not ax: no choice of the data $(C, a, \alpha, b, \delta, Q)$ with $\alpha \leq 1$ reproduces it, since $s \log s / s^{\alpha} \to \infty$ for every $\alpha \leq 1$. The three double factorial articles are consistent with this; they never claim membership of the exponential–power family, and they invert the phase $\frac{s}{2}(\log s - 1)$ directly. **Correction.** The chapter states the reversion for an axiomatized *dominant core* (Theorems J.8 and J.33) of which both the linear-plus-logarithmic phase $aX + b \log X$ and the factorial phase $X(\kappa \log X + d)$ are instances, both exactly invertible by Lambert W (Theorems J.23 and J.27). The theorem named in the Part 0 interface, Theorem J.29, is stated for the exponential–power model as specified, and Theorem J.33 is the version Part N cites.

Definition J.8 (Admissible dominant core)**.** Let $X_* \geq 0$. A function $\Phi_0 \in C^2((X_*, \infty))$ is an *admissible core* when

1. $\Phi_0'(X) > 0$ for $X > X_*$ and $\Phi_0(X) \to +\infty$ as $X \to +\infty$;
2. Φ_0' is eventually monotone and $\Phi_0'(X) \to \Lambda_\infty \in (0, +\infty]$;
3. $X\Phi_0'(X) \to +\infty$.

Its *core slope* is Φ_0'. Given a target L in the range of Φ_0, the *core solution* $X = X(L)$ is the unique large solution of $\Phi_0(X) = L$.

The two cores used in this volume are

$$\Phi_0(X) = aX + b \log X \quad \text{(Part M, O, P)}, \qquad \Phi_0(X) = X\,(\kappa \log X + d) \quad \text{(Part N)}, \tag{J.9}$$

with $a, \kappa > 0$. Both are admissible on a suitable half-line, and both are inverted exactly by W in Section J.4.

J.3 The coefficient calculus

J.3.1 Two families of partial Bell polynomials

Definition J.9 (Ordinary partial Bell polynomials)**.** For a sequence $\gamma = (\gamma_1, \gamma_2, \dots)$ in R and $n, k \in \mathbb{N}$ put

$$\widehat{B}_{n,k}(\gamma) := [t^n]\left(\sum_{r\geq 1} \gamma_r t^r\right)^k. \tag{J.10}$$

Explicitly,

$$\widehat{B}_{n,k}(\gamma) = \sum_{\substack{m_1, m_2, \dots \geq 0 \\ \sum_r m_r = k,\ \sum_r r m_r = n}} \frac{k!}{\prod_{r\geq 1} m_r!} \prod_{r \geq 1} \gamma_r^{m_r}. \tag{J.11}$$

In particular $\widehat{B}_{0,0} = 1$, $\widehat{B}_{n,0} = 0$ for $n \geq 1$, $\widehat{B}_{n,k} = 0$ for $k > n$, $\widehat{B}_{n,n} = \gamma_1^n$, and $\widehat{B}_{n,k}$ depends only on $\gamma_1, \dots, \gamma_{n-k+1}$.

Definition J.10 (Exponential partial Bell polynomials)**.** For $x = (x_1, x_2, \dots)$ in R and $n \geq k \geq 0$ put

$$B_{n,k}(x_1, x_2, \dots) := n! \sum_{\substack{m_1, m_2, \dots \geq 0 \\ \sum_j m_j = k,\ \sum_j j m_j = n}} \prod_{j \geq 1} \frac{1}{m_j!}\left(\frac{x_j}{j!}\right)^{m_j}, \tag{J.12}$$

and let $B_n := \sum_{k=0}^{n} B_{n,k}$ be the complete polynomial. These are characterized by the generating identities

$$\frac{1}{k!}\Big(\sum_{j\geq 1} x_j \frac{t^j}{j!}\Big)^k = \sum_{n\geq k} B_{n,k}(x_1, x_2, \dots)\frac{t^n}{n!}, \tag{J.13}$$

$$\exp\Big(\sum_{j\geq 1} x_j \frac{t^j}{j!}\Big) = \sum_{n\geq 0} B_n(x_1, \dots, x_n)\frac{t^n}{n!}. \tag{J.14}$$

Remark J.11 (These are already formal, but not by this definition). The corpus carries exactly one exponential partial Bell polynomial, `Fabius.partialBell` in `PartialBellPolynomials.lean`, built during the combinatorial-coefficient-calculus work rather than for this volume. It is defined by the block recurrence, *not* by the multinomial sum (J.12).

Both generating identities offered above as the characterization are formal: `Fabius.bellWeightSeries_pow` is (J.13) and `Fabius.exp_subst_bellWeightSeries` is (J.14), both in `BellGeneratingFunctions.lean`. Moreover `Fabius.partialBell_eq_of_bellWeightSeries_pow_eq_egf` shows that those identities *determine* the coefficients, so any family satisfying the characterization above is that object. Adjacent formal facts, should a proof in this part want them: `Fabius.partialBell_bihomogeneous` (in `BellHomogeneity.lean`), `Fabius.partialBell_self`, `Fabius.partialBell_eq_zero_of_lt`, `Fabius.partialBell_one` (Stirling numbers of the second kind), `Fabius.partialBell_factorial_pred` (unsigned Stirling numbers of the first kind), `Fabius.partialBell_factorial` (Lah numbers), `Fabius.bell_complete_eq_sum_partialBell`, `Fabius.bell_complete_one` and `Fabius.bell_complete_factorial_pred`.

The gap, stated plainly because it is easy to overclaim here: the equivalence of the multinomial sum (J.12) with the recurrence is *not* formalized, in this notation or in the Equation (A.1) one. Neither sum appears formally anywhere in the corpus. What is formal is that the two generating identities pin down a unique object, and that the corpus's recurrence-defined family satisfies them — which is enough to transfer any consequence of the characterization, and not enough to call (J.12) itself proved.

One naming trap, recorded because the crosswalk name audit cannot catch it: the complete Bell polynomial is `Bell.complete`, in a *top-level* namespace `Bell`, not in `Fabius`. The audit checks only `Fabius.*` names, so a citation written `Fabius.Bell.complete` would pass unexamined while naming nothing.

Lemma J.12 (Conversion)**.** *For all $n \geq k \geq 0$,*

$$\widehat{B}_{n,k}(\gamma_1, \gamma_2, \dots) = \frac{k!}{n!}\, B_{n,k}(1!\,\gamma_1,\, 2!\,\gamma_2, \dots). \tag{J.15}$$

Proof. Set $x_j = j!\,\gamma_j$ in (J.13). The left-hand side becomes $(k!)^{-1}\big(\sum_j \gamma_j t^j\big)^k$, whose coefficient of t^n is $\widehat{B}_{n,k}(\gamma)/k!$; the right-hand side has coefficient $B_{n,k}(1!\gamma_1, \dots)/n!$. □

Remark J.13 (Formal crosswalk). Theorem J.12 is Exact in `UnitSeriesBellCoefficients.lean`. `Fabius.ordPartialBell_eq_factorialRatio_partialBell` is (J.15) over every commutative rational algebra, and `Fabius.factorial_mul_ordPartialBell_eq_factorial_mul_partialBell` is its denominator-cleared companion. Both are polynomial identities; the second form is the robust one to transport to coefficient rings in which a displayed quotient would be inappropriate.

J.3.2 Powers and logarithms of a unit series

Lemma J.14 (Powers and logarithms)**.** *Let $w = \sum_{r\geq 1} \gamma_r t^r \in tR[[t]]$ and $n \geq 1$.*

1. *For every β in R (or in any commutative $\mathbb{Q}$-algebra containing R),*

$$[t^n]\,(1+w)^\beta = \sum_{k=1}^{n} \binom{\beta}{k} \widehat{B}_{n,k}(\gamma), \qquad \binom{\beta}{k} = \frac{\beta(\beta-1)\cdots(\beta-k+1)}{k!}. \tag{J.16}$$

2. For an integer $j \geq 1$,

$$[t^n]\,(1+w)^{-j} = \sum_{k=0}^{n} (-1)^k \binom{j+k-1}{k} \widehat{B}_{n,k}(\gamma). \tag{J.17}$$

3.

$$[t^n]\,\log(1+w) = \sum_{k=1}^{n} \frac{(-1)^{k+1}}{k} \widehat{B}_{n,k}(\gamma). \tag{J.18}$$

Equivalently, in exponential form, for $U(t) = 1 + \sum_{j\geq 1} u_j t^j$,

$$[t^N]\,U(t)^\beta = \frac{1}{N!}\sum_{k=0}^{N} \beta^{\underline{k}}\, B_{N,k}(1!u_1, 2!u_2, \dots), \qquad \beta^{\underline{k}} = \beta(\beta-1)\cdots(\beta-k+1), \tag{J.19}$$

$$[t^N]\,\log U(t) = \frac{1}{N!}\sum_{k=1}^{N} (-1)^{k-1}(k-1)!\, B_{N,k}(1!u_1, 2!u_2, \dots). \tag{J.20}$$

Proof. All five displayed identities are the substitution of w into a formal series in one variable. Since $w \in tR[[t]]$, the family $(w^k)_{k\geq 0}$ is summable in the t-adic topology and $[t^n]\,w^k = 0$ for $k > n$, so every sum displayed is finite and the substitution is legitimate. Now $(1+w)^\beta = \sum_{k\geq 0}\binom{\beta}{k} w^k$ gives (J.16) on applying Theorem J.9; $\binom{-j}{k} = (-1)^k\binom{j+k-1}{k}$ gives (J.17); and $\log(1+w) = \sum_{k\geq 1}(-1)^{k+1}w^k/k$ gives (J.18). For the exponential forms write $U = 1 + w$ with $\gamma_j = u_j$ and substitute (J.15) into (J.16) and (J.18), using $\binom{\beta}{k} = \beta^{\underline{k}}/k!$ and $(-1)^{k+1}/k = (-1)^{k-1}(k-1)!/k!$. □

Remark J.15 (Formal crosswalk). Theorem J.14 is Exact, coefficient by coefficient, in `UnitSeriesBellCoefficients.lean`. The ordinary-Bell power formulas are `Fabius.coeff_fallingSeries_subst_eq_sum_ordPartialBell` and its literal $1 \leq k \leq n$ specialization `Fabius.coeff_fallingSeries_subst_eq_sum_ordPartialBell_of_pos`; the exponential-Bell form is `Fabius.coeff_fallingSeries_subst_eq_sum_partialBell`. Negative integral powers are `Fabius.coeff_negBinomSeries_subst_eq_sum_ordPartialBell` and `Fabius.coeff_negBinomSeries_subst_eq_sum_ordPartialBell_of_pos`. The two logarithmic forms are `Fabius.coeff_logOf_eq_sum_ordPartialBell` and `Fabius.coeff_logOf_eq_sum_partialBell`. Here `fallingSeries.subst` is the formal meaning of $(1+w)^\beta$, and `negBinomSeries.subst` is the canonical formal negative integral power. No analytic convergence or logarithm-branch claim is involved.

Corollary J.16 (Exponential and logarithmic jets). *Let $D(t) = \sum_{j\geq 1} d_j t^j \in tR[[t]]$ and $A(t) = \exp D(t) = \sum_{n\geq 0} A_n t^n$. Then $A_0 = 1$,*

$$A_n = \frac{1}{n!} B_n(1!d_1, 2!d_2, \dots, n!d_n) = \sum_{\substack{k_1,k_2,\dots\geq 0\\ \sum_j jk_j = n}} \prod_{j\geq 1} \frac{d_j^{k_j}}{k_j!}, \tag{J.21}$$

and the coefficients obey the triangular recurrence

$$nA_n = \sum_{j=1}^{n} j\, d_j\, A_{n-j} \qquad (n \geq 1). \tag{J.22}$$

Conversely, if $A = 1 + \sum_{n\geq 1} A_n t^n$ is a unit series then $d_n = [t^n]\,\log A$ is given by (J.20).

Proof. The constant term is $A_0 = 1$ because the substituted formal exponential has constant coefficient one. The first equality is (J.14) with $x_j = j!d_j$; the second is the coefficientwise-finite expansion of $\exp$ as a product over j. For (J.22) differentiate $A = \exp D$ to get $A' = D'A$ and compare coefficients of t^{n-1}. The converse is (J.20). □

Lean status. Exact, as formal power-series identities, in `UnitSeriesBellCoefficients.lean`. The normalization Theorem J.12 is `Fabius.ordPartialBell_eq_factorialRatio_partialBell`, with the denominator-cleared form `Fabius.factorial_mul_ordPartialBell_eq_factorial_mul_partialBell`. The arbitrary-power clauses of Theorem J.14 are `Fabius.coeff_fallingSeries_subst_eq_sum_ordPartialBell`, `Fabius.coeff_fallingSeries_subst_eq_sum_ordPartialBell_of_pos`, and `Fabius.coeff_fallingSeries_subst_eq_sum_partialBell`; the negative-integral-power clause is `Fabius.coeff_negBinomSeries_subst_eq_sum_ordPartialBell`, with manuscript indexing in `Fabius.coeff_negBinomSeries_subst_eq_sum_ordPartialBell_of_pos`; and the two logarithmic normalizations are `Fabius.coeff_logOf_eq_sum_ordPartialBell` and `Fabius.coeff_logOf_eq_sum_partialBell`.

For Theorem J.16, the boundary value $A_0 = 1$ is `Fabius.constantCoeff_exp_subst`, and coefficient reweighting is pinned by `Fabius.egfA_factorialDenormalize_coeff_eq` and `Fabius.bellWeightSeries_factorialDenormalize_coeff_eq`; the complete-Bell formula is `Fabius.coeff_exp_subst_eq_completeBell`; the finite partition forms are `Fabius.coeff_exp_subst_eq_partitionExpSum`, `Fabius.coeff_exp_subst_eq_sum_weightedPartitions`, and the field-valued `Fabius.coeff_exp_subst_eq_sum_div_weightedPartitions`; and (J.22) is `Fabius.coeff_exp_subst_recurrence`. These are algebraic statements over commutative rational algebras (with the displayed division wrapper over a characteristic-zero field); they assert no analytic convergence or logarithm branch.

J.3.3 Lagrange–Bürmann inversion and the fixed-point form

Theorem J.17 (Lagrange–Bürmann). *Let $\varphi \in R[[w]]$ with $\varphi(0)$ invertible in R. There is a unique $W \in tR[[t]]$ with*

$$W = t\,\varphi(W), \tag{J.23}$$

and for every $G \in R[[w]]$ and every $n \geq 1$,

$$[t^n]\,G\big(W(t)\big) = \frac{1}{n}[w^{n-1}]\,\Big(G'(w)\,\varphi(w)^n\Big). \tag{J.24}$$

In particular

$$[t^n]\,W(t) = \frac{1}{n}[w^{n-1}]\,\varphi(w)^n. \tag{J.25}$$

Proof. Existence and uniqueness: (J.23) determines $[t^n]\,W$ from $[t^{<n}]\,W$, because $t\varphi(W)$ modulo t^{n+1} depends only on W modulo t^n.

For (J.24) work in the field of formal Laurent series $R((w))$ and let Res denote the coefficient of w^{-1}. Two facts are needed.

(a) $\mathrm{Res}\, f' = 0$ *for every* $f \in R((w))$, since w^{-1} has no antiderivative among the monomials: $(w^k)' = kw^{k-1}$ is a multiple of w^{-1} only for $k = 0$, when it vanishes.

(b) Change of variables. Put $\psi(w) := w/\varphi(w)$, a series in $wR[[w]]$ with $\psi'(0) = \varphi(0)^{-1}$ invertible; then W is the compositional inverse of ψ, so $\psi(W(t)) = t$. For $f \in R((w))$,

$$\mathrm{Res}_w\big(f(w)\big) = \mathrm{Res}_t\big(f(W(t))\,W'(t)\big). \tag{J.26}$$

Indeed both sides are R-linear and continuous, so it suffices to check $f = w^k$. For $k \neq -1$, $W^kW' = \big(W^{k+1}/(k+1)\big)'$ and (a) gives $0 = \mathrm{Res}_w\, w^k$. For $k = -1$, write $W = t\vartheta(t)$ with ϑ a unit; then $W'/W = t^{-1} + \vartheta'/\vartheta$ and the second summand lies in $R[[t]]$, so $\mathrm{Res}_t(W'/W) = 1 = \mathrm{Res}_w\, w^{-1}$.

Now apply (J.26) to $f(w) := G(w)\psi(w)^{-n-1}\psi'(w)$, noting that $f(W(t))W'(t) = G(W(t))\,t^{-n-1}\,\frac{\mathrm{d}}{\mathrm{d}t}\psi(W(t)) = G(W(t))\,t^{-n-1}$ because $\psi(W(t)) = t$:

$$[t^n]\,G(W(t)) = \mathrm{Res}_t\big(G(W(t))\,t^{-n-1}\big) = \mathrm{Res}_w\big(G(w)\,\psi(w)^{-n-1}\psi'(w)\big).$$

Since $\psi^{-n-1}\psi' = -\frac{1}{n}\big(\psi^{-n}\big)'$, fact (a) and the product rule give

$$\operatorname{Res}_w\big(G\,\psi^{-n-1}\psi'\big) = -\frac{1}{n}\operatorname{Res}_w\big(G\cdot(\psi^{-n})'\big) = \frac{1}{n}\operatorname{Res}_w\big(G'\,\psi^{-n}\big) = \frac{1}{n}\operatorname{Res}_w\big(G'(w)\,\varphi(w)^n w^{-n}\big),$$

which is (J.24). Taking $G(w) = w$ gives (J.25). □

The reversions of this volume all take the form $U = \Psi(t, U)$ in which the rôle of $\varphi(0)$ is played by $\Psi(t, 0)$, an element of $tR[[t]]$, which is *never* invertible. Theorem J.17 therefore does not apply directly, and the following universality lemma is what makes the passage legitimate.

Lemma J.18 (Lagrange's formula without an invertibility hypothesis)**.** *Let S be any commutative $\mathbb{Q}$-algebra and $\varphi \in S[[w]]$* arbitrary *(in particular $\varphi(0)$ need not be invertible). Let $W \in zS[[z]]$ be the unique solution of $W = z\varphi(W)$. Then for every $m \geq 1$,*

$$[z^m]\,W = \frac{1}{m}[w^{m-1}]\,\varphi(w)^m. \tag{J.27}$$

Proof. Existence and uniqueness of W hold as in Theorem J.17 (the recursion is z-adic and uses no invertibility). Fix m. Iterating $W = z\varphi(W)$ shows that $[z^m]\,W$ is a universal polynomial with integer coefficients in $\varphi_0, \varphi_1, \ldots, \varphi_{m-1}$, where $\varphi = \sum_i \varphi_i w^i$; the right-hand side of (J.27) is a universal polynomial with rational coefficients in the same variables. Both sides are therefore elements $P_m,\ \widetilde{P}_m$ of the polynomial ring $\mathcal{P} := \mathbb{Q}[\varphi_0, \ldots, \varphi_{m-1}]$, and they are obtained from φ by the unique $\mathbb{Q}$-algebra homomorphism $\mathcal{P} \to S$ sending $\varphi_i \mapsto \varphi_i$. It thus suffices to prove $P_m = \widetilde{P}_m$ in $\mathcal{P}$. The localization map $\mathcal{P} \to \mathcal{P}[\varphi_0^{-1}]$ is injective ($\mathcal{P}$ is an integral domain), and in $\mathcal{P}[\varphi_0^{-1}]$ the element φ_0 is invertible, so Theorem J.17 applies and gives $P_m = \widetilde{P}_m$ there. Injectivity transports the equality back to $\mathcal{P}$, and the homomorphism transports it to S. □

Lemma J.19 (Lagrange fixed-point formula)**.** *Let $\Psi \in tR[[t,u]]$, that is, a formal series in t and u each of whose monomials has t-degree at least one. Then the equation*

$$U(t) = \Psi\big(t, U(t)\big) \tag{J.28}$$

has a unique solution $U \in tR[[t]]$, and for every $n \geq 1$

$$[t^n]\,U = \sum_{m=1}^{n} \frac{1}{m}[t^n u^{m-1}]\,\Psi(t,u)^m. \tag{J.29}$$

Proof. Uniqueness and existence: $\Psi(t, U)$ modulo t^{n+1} depends only on U modulo t^n because Ψ is divisible by t; so (J.28) determines the coefficients of U recursively, and the resulting series solves the equation by t-adic continuity.

For (J.29) apply Theorem J.18 over the ring $S := R[[t]]$ with $\varphi(w) := \Psi(t, w) \in S[[w]]$ and marker z: the unique $\widetilde{U} \in zS[[z]]$ with $\widetilde{U} = z\varphi(\widetilde{U})$ satisfies

$$[z^m]\,\widetilde{U} = \frac{1}{m}[w^{m-1}]\,\Psi(t,w)^m \in S.$$

Since $\Psi \in tR[[t,u]]$, we have $\Psi^m \in t^m R[[t,u]]$, whence $[t^n]\,[w^{m-1}]\,\Psi(t,w)^m = 0$ for $m > n$. Consequently the family $\big([z^m]\,\widetilde{U}\big)_{m\geq 1}$ is summable in the t-adic topology of S, the specialization $z = 1$ is defined coefficientwise in t, and the resulting series $U := \sum_{m\geq 1}[z^m]\,\widetilde{U}$ satisfies $U = \Psi(t, U)$ by the same summability. Uniqueness identifies it with the solution of (J.28), and extracting $[t^n]$ gives (J.29) with the sum truncated at $m = n$. □

Source claim. Swing_Factorial_Transseries_Article, Lemma *Lagrange fixed-point formula*, proves (J.29) by writing "Ordinary Lagrange–Bürmann inversion in z gives $U = \sum_{m\geq 1} z^m m^{-1}[u^{m-1}]\Psi(t,u)^m$". The same step, with the same silence, appears in Partition_Number_Transseries_and_Inverse ("Introduce $u = z\Phi_\rho(u)$. Lagrange–Bürmann gives …"), in Partition_Numbers_Transseries_and_Inverse, and in double_factorial_transseries-2. **Defect.** The classical Lagrange–Bürmann theorem in the form (J.25) requires $\varphi(0)$ to be invertible in the coefficient ring. Here $\varphi(0) = \Psi(t,0) \in tR[[t]]$, a non-unit, and the residue proof of Theorem J.17 breaks at the change of variables: $\psi = w/\varphi$ is not a well-defined element of $wR[[t]][[w]]$. **Repair.** Theorem J.18: the identity is a polynomial identity in the coefficients of φ over $\mathbb{Q}$, hence holds in the polynomial ring by injectivity of localization at φ_0, hence in every $\mathbb{Q}$-algebra. double_factorial_transseries-2 comes closest to noticing the issue — "Although A is a divergent Gevrey series, it is a valid formal power series with nonzero linear coefficient" — but that remark addresses convergence, not invertibility, and its own G_Λ does have an invertible linear coefficient, so the gap there is only in the inherited lemma.

Lemma J.20 (Fixed-point formula with a quadratic self-interaction)**.** *Let $\Psi = h + f$ with $h \in u^2R[[u]]$ and $f \in tR[[t,u]]$. Then $U = \Psi(t,U)$ has a unique solution $U \in tR[[t]]$, and for every $n \geq 1$*

$$[t^n]\,U = \sum_{m=1}^{2n-1} \frac{1}{m}[t^n u^{m-1}]\,\Psi(t,u)^m. \tag{J.30}$$

Proof. Existence and uniqueness. Write $U = U_{<n} + e_n t^n + \cdots$. Then $h(U) = h(U_{<n}) + h'(U_{<n})\,e_n t^n + \cdots$, and $h'(U_{<n}) \in tR[[t]]$ because $h' \in uR[[u]]$ and $U_{<n} \in tR[[t]]$; so e_n does not occur in $[t^n]\,h(U)$. It does not occur in $[t^n]\,f(t,U)$ either, since f is divisible by t. Hence the equation determines e_n, as $[t^n]\,\Psi(t,U_{<n})$, from $U_{<n}$; the resulting series solves the equation by t-adic continuity.

The closed formula. Introduce a marker z and let $\widetilde{U} \in zS[[z]]$, $S := R[[t]]$, solve $\widetilde{U} = z\Psi(t,\widetilde{U})$. Theorem J.18 gives $[z^m]\,\widetilde{U} = m^{-1}[u^{m-1}]\,\Psi(t,u)^m$. We claim

$$[t^n u^{m-1}]\,\Psi(t,u)^m = 0 \qquad \text{whenever } m > 2n-1. \tag{J.31}$$

Expand $\Psi^m = (h+f)^m$ into 2^m products; a product using h in σ of the m slots and f in the remaining $m-\sigma$ has u-degree at least 2σ and t-degree at least $m-\sigma$. For the u-degree to equal $m-1$ we need $2\sigma \leq m-1$, hence the t-degree is at least $m - \frac{m-1}{2} = \frac{m+1}{2}$; so $[t^n]$ of that product vanishes unless $n \geq (m+1)/2$, which is (J.31). Consequently the family $([z^m]\,\widetilde{U})_{m\geq 1}$ is t-adically summable, the specialization $z=1$ is defined coefficientwise, and, again by summability, the resulting series solves $U = \Psi(t,U)$. Uniqueness identifies it with U, and extracting $[t^n]$ gives (J.30). □

Theorem J.19 is the case $h = 0$ of Theorem J.20, in which the argument above sharpens the range of m from $2n-1$ to n.

J.3.4 Perturbed inversion around an exactly invertible core

The reversions carried out in Chapters 1–4 all reduce to one step: a core F that can be inverted exactly is perturbed by a correction E, and the inverse of $F + \varepsilon E$ is required to all orders in ε. The answer is a single differential formula, stated here once. It is the tool behind Theorem O.62 in the ordinate of the partition chapter and behind Theorem P.39 in the phase coordinate of the swing chapter, and those two are the same identity in different letters.

Theorem J.21 (Perturbed inversion)**.** *Let F and E be analytic in a neighbourhood of $\mu_0 \in \mathbb{C}$, let $F'(\mu_0) \neq 0$, and put $y := F(\mu_0)$. Write $\mu_0(\cdot)$ for the local inverse of F, so that $\mu_0(y) = \mu_0$ and $\mu_0'(y) = 1/F'(\mu_0(y))$, and let*

$$\mathcal{D} := \frac{1}{F'(\mu)}\frac{\mathrm{d}}{\mathrm{d}\mu}, \tag{J.32}$$

which is the derivation $\mathrm{d}/\mathrm{d}y$ *read in the coordinate* $y = F(\mu)$. *Then there are a neighbourhood* $\mathcal{U}$ *of* μ_0 *and an* $\varepsilon_0 > 0$ *such that for* $|\varepsilon| < \varepsilon_0$ *the equation*

$$F(\mu) + \varepsilon E(\mu) = y \tag{J.33}$$

has exactly one root $\mu(y;\varepsilon)$ *in* $\mathcal{U}$*; it is analytic in* ε *there, and*

$$\mu(y;\varepsilon) = \mu_0(y) + \sum_{M\geq 1} \frac{(-\varepsilon)^M}{M!}\,\mathcal{D}^{M-1}\left(\frac{E(\mu_0(y))^M}{F'(\mu_0(y))}\right), \tag{J.34}$$

the series converging for $|\varepsilon| < \varepsilon_0$, *with* M*th coefficient*

$$\frac{(-1)^M}{M!}\,\mathcal{D}^{M-1}\left(\frac{E(\mu_0)^M}{F'(\mu_0)}\right) = \frac{(-1)^M}{M}\operatorname*{Res}_{\mu=\mu_0}\frac{E(\mu)^M}{\big(F(\mu)-y\big)^M}. \tag{J.35}$$

If moreover $E = \sum_j \varepsilon_j E_j$ *with finitely many* E_j *analytic near* μ_0, *then for every multi-index* $\boldsymbol{m} \neq \mathbf{0}$ *with* $M := |\boldsymbol{m}|$ *the coefficient of* $\boldsymbol{\varepsilon}^{\boldsymbol{m}}$ *in* $\mu - \mu_0$ *is*

$$V_{\boldsymbol{m}} = \frac{(-1)^M}{\boldsymbol{m}!}\,\mathcal{D}^{M-1}\left(\frac{E^{\boldsymbol{m}}(\mu_0)}{F'(\mu_0)}\right) = \frac{(-1)^M(M-1)!}{\boldsymbol{m}!}\operatorname*{Res}_{\mu=\mu_0}\frac{E^{\boldsymbol{m}}(\mu)}{\big(F(\mu)-F(\mu_0)\big)^M}, \tag{J.36}$$

where $E^{\boldsymbol{m}} := \prod_j E_j^{m_j}$ *and* $\boldsymbol{m}! := \prod_j m_j!$.

Proof. Let γ be a small positively oriented circle about μ_0, contained with its interior $\mathcal{U}$ in the common domain of analyticity and enclosing no zero of $F(\cdot)-y$ other than μ_0; such a γ exists because $F'(\mu_0) \neq 0$ makes μ_0 a simple zero, hence isolated. Put $m_\gamma := \min_\gamma |F-y| > 0$ and $M_\gamma := \max_\gamma |E|$, and let $\varepsilon_0 := m_\gamma/(M_\gamma+1)$. For $|\varepsilon| < \varepsilon_0$ we have $|\varepsilon E| < |F-y|$ on γ, so Rouché gives exactly one zero $\mu(y;\varepsilon)$ of $F - y + \varepsilon E$ inside γ, and it is simple. The argument principle, applied to that function, expresses it as

$$\mu(y;\varepsilon) = \frac{1}{2\pi \mathrm{i}}\int_\gamma \zeta\,\mathrm{d}_\zeta \log\big(F(\zeta) - y + \varepsilon E(\zeta)\big), \tag{J.37}$$

which is analytic in ε on $|\varepsilon| < \varepsilon_0$ because the integrand is and γ is compact.

Subtract from (J.37) the same expression at $\varepsilon = 0$, whose value is μ_0, and expand

$$\log\left(1 + \varepsilon\frac{E}{F-y}\right) = \sum_{M\geq 1}\frac{(-1)^{M-1}}{M}\,\varepsilon^M\,\frac{E^M}{(F-y)^M},$$

uniformly convergent on γ since $|\varepsilon E/(F-y)| < 1$ there. Integrating by parts once turns $\zeta\,\mathrm{d}_\zeta(\cdot)$ into $-(\cdot)\,\mathrm{d}\zeta$ (the boundary term vanishes on a closed contour), so the coefficient of ε^M is

$$\frac{(-1)^M}{M}\operatorname*{Res}_{\zeta=\mu_0}\frac{E(\zeta)^M}{\big(F(\zeta)-F(\mu_0)\big)^M},$$

which is the right-hand side of (J.35).

It remains to identify that residue with the differential expression. Change coordinate by $w = F(\zeta)$, $\zeta = \mu_0(w)$, $\mathrm{d}\zeta = \mu_0'(w)\,\mathrm{d}w$, legitimate on a neighbourhood of μ_0 because $F'(\mu_0) \neq 0$:

$$\operatorname*{Res}_{\zeta=\mu_0}\frac{E(\zeta)^M}{(F(\zeta)-y)^M}\,\mathrm{d}\zeta = \operatorname*{Res}_{w=y}\frac{E(\mu_0(w))^M\,\mu_0'(w)}{(w-y)^M}\,\mathrm{d}w = \frac{1}{(M-1)!}\frac{\mathrm{d}^{M-1}}{\mathrm{d}y^{M-1}}\left(\frac{E(\mu_0(y))^M}{F'(\mu_0(y))}\right),$$

the last step being the residue of a pole of order M. Multiplying by $(-1)^M/M$ and using $\mathrm{d}/\mathrm{d}y = \mathcal{D}$ gives (J.34) and (J.35).

For (J.36) replace E by $\sum_j \varepsilon_j E_j$ and expand the Mth power by the multinomial theorem: the monomial $\boldsymbol{\varepsilon}^{\boldsymbol{m}}$ occurs with coefficient $M!/\boldsymbol{m}!$, which cancels the $M!$ in (J.34). ☐

The identity (J.34) is needed in Chapter 4 with a correction E that is a *divergent* formal series, where no contour is available. What survives is the coefficientwise identity, and it survives for the reason the Lagrange fixed-point formula does.

Corollary J.22 (Formal form of the reversion)**.** *Fix $M \geq 1$ and let $\mathcal{P} := \mathbb{Q}[e_0, \dots, e_{M-1}, f_2, \dots, f_M][f_1^{-1}]$. Both of the following are elements of $\mathcal{P}$, obtained by writing out the indicated recursions with e_i and f_i standing for $E^{(i)}(\mu_0)$ and $F^{(i)}(\mu_0)$:*

1. *the coefficient μ_M of ε^M produced by solving* (J.33) *by formal Taylor iteration, i.e. by substituting $\mu = \mu_0 + \sum_{k\geq 1} \mu_k \varepsilon^k$, expanding F and E in their Taylor jets at μ_0, and equating coefficients of ε^M;*
2. *the expression $\frac{(-1)^M}{M!}\mathcal{D}^{M-1}\big(E(\mu_0)^M/F'(\mu_0)\big)$ of* (J.34)*, expanded by the Leibniz rule.*

They are equal in $\mathcal{P}$. Consequently (J.34) *and* (J.36) *hold coefficientwise in any commutative $\mathbb{Q}$-algebra carrying jets e_i, f_i with f_1 invertible — in particular when F and E are formal, possibly divergent, series, and when $\mathcal{D}$ is the derivation* (J.32) *written in any coordinate.*

Proof. Each recursion visibly terminates and involves only $e_0, \dots, e_{M-1}, f_2, \dots, f_M$ and divisions by f_1: in (i) the coefficient of ε^M in $F(\mu_0 + \sum_{k\geq 1} \mu_k \varepsilon^k) + \varepsilon E(\mu_0 + \sum_{k\geq 1} \mu_k \varepsilon^k)$ equals $f_1 \mu_M$ plus a polynomial in $f_2, \dots, f_M, e_0, \dots, e_{M-1}$ and $\mu_1, \dots, \mu_{M-1}$, so μ_M is determined by one division by f_1; in (ii) each application of $\mathcal{D} = f_1^{-1}\mathrm{d}/\mathrm{d}\mu$ lowers the available jet order by one and contributes one factor f_1^{-1}, and $M-1$ applications to a function of $e_0, \dots$ and $f_1, \dots$ reach at most e_{M-1} and f_M. So both lie in $\mathcal{P}$.

Choose N with $f_1^N \mu_M$ and $f_1^N \cdot$ (ii) both in $\mathcal{A} := \mathbb{Q}[e_0, \dots, e_{M-1}, f_1, \dots, f_M]$, and regard them as polynomial functions on $\mathbb{C}^{2M}$. Every point of $\mathbb{C}^{2M}$ with $f_1 \neq 0$ is realized by a pair of *polynomials*: take $\mu_0 = 0$, $E(\mu) := \sum_{i=0}^{M-1} e_i \mu^i / i!$ and $F(\mu) := \sum_{i=1}^{M} f_i \mu^i / i!$, which are entire with $F'(0) = f_1 \neq 0$ and with the prescribed jets. For such a pair Theorem J.21 applies, and the Taylor coefficients in ε of the analytic root $\mu(y; \varepsilon)$ satisfy the recursion of (i) — substituting the convergent expansion into (J.33) and equating coefficients is legitimate — so (i) and (ii) agree at that point. Two elements of $\mathcal{A}$ agreeing on the Zariski-dense open set $\{f_1 \neq 0\}$ of $\mathbb{C}^{2M}$ are equal in $\mathbb{C}[e_\bullet, f_\bullet]$, hence equal in $\mathcal{A}$ and, after dividing by f_1^N, in $\mathcal{P}$. The final assertion follows by applying the unique $\mathbb{Q}$-algebra homomorphism $\mathcal{P} \to S$ determined by the given jets, exactly as in Theorem J.18. □

J.4 The Lambert core

J.4.1 Exact inversion of the linear–logarithmic phase

Theorem J.23 (Exact solution of the dominant block)**.** *Let $a > 0$, $b \in \mathbb{R}$, and consider, for $X > 0$,*

$$\Phi_0(X) := aX + b \log X = L. \tag{J.38}$$

1. *If $b = 0$ the unique solution is $X = L/a$.*
2. *If $b \neq 0$, put $W := (a/b)X$. Then* (J.38) *holds if and only if*

$$W\mathrm{e}^W = z(L), \qquad z(L) := \frac{a}{b}\,\mathrm{e}^{L/b}, \tag{J.39}$$

   so that the solutions of (J.38) *are exactly*

$$X = \frac{b}{a}\, W_k\Big(\frac{a}{b}\,\mathrm{e}^{L/b}\Big) \tag{J.40}$$

   as k ranges over the branches of the Lambert function for which $W_k(z(L))$ is defined.
3. Branch rule. *Suppose $b \neq 0$.*

- *If* $b > 0$ *then* $z(L) > 0$*, equation* (J.38) *has exactly one positive solution for every real* L*,* Φ_0 *is strictly increasing on* $(0, \infty)$*, and* $X = (b/a)W_0(z(L))$*.*
- *If* $b < 0$ *then* $z(L) = -(a/|b|)\mathrm{e}^{-L/|b|}$ *lies in* $(-\mathrm{e}^{-1}, 0)$ *precisely when*

$$L > L_c := |b|\Big(1 - \log\frac{|b|}{a}\Big) = \Phi_0\Big(\frac{|b|}{a}\Big), \tag{J.41}$$

and then (J.38) *has exactly two positive solutions, namely* $(b/a)W_0(z(L)) < |b|/a < (b/a)W_{-1}(z(L))$*.* Φ_0 *is strictly increasing on* $(|b|/a, \infty)$ *and the large solution is*

$$X = \frac{b}{a}\,W_{-1}\Big(\frac{a}{b}\,\mathrm{e}^{L/b}\Big) = \frac{|b|}{a}\Big(-W_{-1}\big(z(L)\big)\Big). \tag{J.42}$$

In both cases the large solution satisfies $X \to +\infty$ *and* $X = L/a + \mathcal{O}(\log L)$ *as* $L \to +\infty$*.*

4. Slope identities. *With* $W = (a/b)X$ *and* $b \neq 0$*,*

$$\Phi_0'(X) = a + \frac{b}{X} = a\,\frac{1+W}{W}, \qquad \frac{\mathrm{d}X}{\mathrm{d}L} = \frac{1}{\Phi_0'(X)} = \frac{1}{a}\,\frac{W}{1+W}. \tag{J.43}$$

Proof. (1) is immediate. For (2), multiply (J.38) by $1/b$ and exponentiate:

$$\mathrm{e}^{L/b} = \mathrm{e}^{(a/b)X}\,X = \mathrm{e}^{W}\cdot\frac{b}{a}\,W,$$

because $X = (b/a)W$. Multiplying by a/b gives (J.39); the step is reversible, and no logarithm of a negative number occurs, which is what makes the argument uniform in the sign of b. By definition, the solutions of $W\mathrm{e}^W = z$ are the values $W_k(z)$, whence (J.40).

(3) We have $\Phi_0'(X) = a + b/X$. For $b > 0$ this is positive on $(0, \infty)$ and Φ_0 increases from $-\infty$ to $+\infty$, so there is exactly one solution; correspondingly $W = (a/b)X > 0$ and W_0 is the unique real branch on $(0, \infty)$. For $b < 0$, Φ_0' vanishes only at $X_* = |b|/a$, is negative on $(0, X_*)$ and positive on (X_*, ∞); Φ_0 decreases from $+\infty$ to $\Phi_0(X_*) = |b| - |b|\log(|b|/a) = L_c$ and then increases back to $+\infty$. Hence there are two positive solutions exactly when $L > L_c$, one in each monotonicity interval. On the W side $W = (a/b)X < 0$, and the two real branches of W on $(-\mathrm{e}^{-1}, 0)$ are W_0, with values in $(-1, 0)$, and W_{-1}, with values in $(-\infty, -1)$; the map $W \mapsto (b/a)W$ reverses order, so W_{-1} produces the larger X, namely the one exceeding $|b|/a$. Also $z(L) > -\mathrm{e}^{-1}$ is equivalent to $(a/|b|)\mathrm{e}^{-L/|b|} < \mathrm{e}^{-1}$, that is to (J.41). Finally $aX = L - b\log X$ with $X \to \infty$ gives $aX = L + \mathcal{O}(\log L)$.

(4) $a + b/X = a + b\cdot\dfrac{a}{bW} = a\big(1 + W^{-1}\big)$, and $\mathrm{d}X/\mathrm{d}L$ is the reciprocal of Φ_0' by the inverse function theorem, applicable because $\Phi_0' \neq 0$ on the relevant interval. ☐

Remark J.24 (Formalization of the linear–logarithmic core). Parts (2) and (4), and the $b > 0$ half of the branch rule, are machine-checked over $\mathbb{R}$ in `LinLogCoreInversion.lean`.

The part worth isolating is (2), because the formal statement is stronger than the prose suggests. `Fabius.linLogCore_eq_iff` is an *equivalence*

$$aX + b\log X = L \iff \Big(\frac{a}{b}X\Big)\mathrm{e}^{\frac{a}{b}X} = \frac{a}{b}\mathrm{e}^{L/b} \tag{J.44}$$

holding for *every* nonzero a and b and every $X > 0$, with no branch chosen and no sign condition on b. That is exactly the uniformity the proof above claims when it observes that no logarithm of a negative number occurs, and making it an iff separates the algebra of the substitution from the analysis of which solutions exist — which is where the sign of b genuinely matters, and which is (3).

For (3) with $b > 0$ the module proves the whole first bullet. `Fabius.linLogCoreArg` is $z(L)$, `Fabius.linLogCoreRoot` is $X = (b/a)W_0(z(L))$ — the *principal* branch, which is the correct one here because $z(L) > 0$ — and `Fabius.linLogCoreRoot_pos`, `Fabius.linLogCore_linLogCo` `reRoot` and `Fabius.linLogCoreRoot_unique` give positivity, that it solves (J.38), and that it is the only positive solution. Uniqueness comes from `Fabius.strictMonoOn_linLogCore`, strict monotonicity of Φ_0 on $(0,\infty)$, rather than from any property of the branch. Part (4) is `Fabius.hasDerivAt_linLogCore` for $\Phi_0'(X) = a + b/X$ and `Fabius.linLogCore_slope_eq` for the Lambert form $a(1+W)/W$.

The $b < 0$ half of (3) is formalized as well, on the corpus's lower real branch W_{-1}. `Fabius.linLogCoreThreshold` is L_c, and `Fabius.linLogCore_critical` states what (J.41) asserts in passing but does not display: L_c is the *critical value*, $L_c = \Phi_0(|b|/a)$, an identity in a and b alone. `Fabius.linLogCoreArg_mem_Ioo_iff` is the threshold characterization itself — an equivalence, $z(L) \in (-\mathrm{e}^{-1}, 0) \iff L > L_c$ — so the condition is proved necessary as well as sufficient, which is what the word *precisely* in Theorem J.23(3) claims.

The two roots are `Fabius.linLogCoreRoot` on the principal branch and `Fabius.linLogCoreRootLower` on the lower branch; `Fabius.linLogCoreRoot_pos_of_neg` and `Fabius.linLogCoreRootLower_pos` give positivity, `Fabius.linLogCore_linLogCoreRoot_of_neg` and `Fabius.linLogCore_linLogCoreRootLower` that both solve (J.38), and `Fabius.linLogCoreRoot_lt_critical` together with `Fabius.critical_lt_linLogCoreRootLower` the separation $X_0 < |b|/a < X_{-1}$, whence `Fabius.linLogCoreRoot_ne_linLogCoreRootLower`. The separation is where the order reversal is used, and the formal proof makes the mechanism visible: it is $-1 < W_0(z)$ against $W_{-1}(z) < -1$ multiplied by the *negative* number b/a, so the branch with the larger W yields the smaller X.

What remains unformalized in Theorem J.23 is only the asymptotic clause $X = L/a + \mathcal{O}(\log L)$ as $L \to +\infty$, and the complex reading with a general W_k.

Corollary J.25 (Branch rule)**.** *For $b \neq 0$ the branch of the Lambert function carrying the large solution of $L = aX + b\log X$ is determined by the sign of b alone: W_0 if $b > 0$, W_{-1} if $b < 0$.*

Remark J.26 (Why the branch is not a detail)*.* Three of the four chapters have $b < 0$ and therefore use W_{-1}; the odd phase of the swing factorial has $b = +1/2$ and uses W_0. Swing_Factorial_Transseries_Article puts the point sharply: the principal branch "would produce the small root of the simplified equation and is irrelevant to the large-index inverse". For $b < 0$ the two roots are the two sides of a genuine fold at $X_* = |b|/a$, and choosing wrongly yields an object all of whose asymptotic corrections are then computed correctly about the wrong point.

J.4.2 The factorial core

Proposition J.27 (Exact inversion of a log-linear phase)**.** *Let $\kappa > 0$ and $d \in \mathbb{R}$, and consider*

$$\Phi_0(X) := X\big(\kappa \log X + d\big) = L, \qquad X > 0. \tag{J.45}$$

For $L > 0$ put

$$v := W_0\!\left(\frac{L}{\kappa}\,\mathrm{e}^{d/\kappa}\right), \qquad X := \exp\!\Big(v - \frac{d}{\kappa}\Big). \tag{J.46}$$

Then X is the unique solution of (J.45) *with $\kappa \log X + d > 0$, and*

$$\Phi_0'(X) = \kappa \log X + d + \kappa = \kappa\,(v+1), \qquad \frac{\mathrm{d}X}{\mathrm{d}L} = \frac{1}{\kappa(v+1)}. \tag{J.47}$$

Moreover Φ_0 is an admissible core in the sense of Theorem J.8 on $\big(\mathrm{e}^{-(d+\kappa)/\kappa}, \infty\big)$, with $\Phi_0'(X) \to +\infty$.

Proof. Put $s = \log X$. Then (J.45) reads $\mathrm{e}^s(\kappa s + d) = L$, that is $\big(s + \frac{d}{\kappa}\big)\mathrm{e}^{s+d/\kappa} = (L/\kappa)\mathrm{e}^{d/\kappa}$. The right-hand side is positive, so W_0 is the unique real branch and $s + d/\kappa = v$, which is (J.46). Uniqueness under the stated side condition holds because $\kappa s + d > 0$ forces $s + d/\kappa > 0$, and $\sigma \mapsto \sigma \mathrm{e}^\sigma$ is injective on $(0,\infty)$. Differentiating, $\Phi_0'(X) = \kappa\log X + d + \kappa = \kappa(s + d/\kappa) + \kappa = \kappa(v+1)$. Admissibility: Φ_0' is increasing, tends to $+\infty$, is positive exactly for $\log X > -(d+\kappa)/\kappa$, and $X\Phi_0'(X) \to \infty$. □

Remark J.28 (The double factorial normalization). Part N uses $\kappa = \frac{1}{2}$, $d = -\frac{1}{2}$ and the centred variable $s = x + 1$, so that $\Phi_0(s) = \frac{s}{2}(\log s - 1)$, $(L/\kappa)\mathrm{e}^{d/\kappa} = 2L/\mathrm{e}$, $X = \mathrm{e}^{v+1}$ with $v = W_0(2L/\mathrm{e})$, and $\Phi_0'(X) = \frac{1}{2}(v+1) = \frac{1}{2}\log X$. This reproduces the core of all three double factorial articles; the quantity they call Λ is $v + 1 = \log X$.

J.5 All-orders reversion around the core

J.5.1 The Lambert-centred inverse

Theorem J.29 (All-orders reversion around the Lambert core). *Let R be a commutative $\mathbb{Q}$-algebra, let $a \in R$ be invertible, let $b \in R$, and let $Q = \sum_{j\geq 1} q_j t^j \in tR[[t]]$. Define*

$$\Psi(t,u) := -\frac{1}{a}\left[b\log(1+tu) + Q\!\left(\frac{t}{1+tu}\right)\right]. \tag{J.48}$$

1. *$\Psi \in tR[[t,u]]$.*
2. *There is a unique $U = \sum_{n\geq 1} c_n t^n \in tR[[t]]$ with $U = \Psi(t,U)$, and*

$$c_n = \sum_{m=1}^{n} \frac{1}{m}[t^n u^{m-1}]\,\Psi(t,u)^m \qquad (n \geq 1). \tag{J.49}$$

3. Triangular Bell recurrence. *Put $\gamma_1 := 0$ and $\gamma_r := c_{r-1}$ for $r \geq 2$. Then, for $n \geq 1$,*

$$c_n = -\frac{1}{a}\left[b\sum_{k=1}^{n}\frac{(-1)^{k+1}}{k}\widehat{B}_{n,k}(\gamma) + \sum_{j=1}^{n} q_j \sum_{k=0}^{n-j}(-1)^k\binom{j+k-1}{k}\widehat{B}_{n-j,k}(\gamma)\right], \tag{J.50}$$

   and only $c_1, \ldots, c_{n-1}$ occur on the right-hand side. In particular c_n lies in the $\mathbb{Q}$-subalgebra of R generated by a^{-1}, b and $q_1, \ldots, q_n$; no logarithm appears.

4. Formal exactness. *In $t^{-1}R[[t]]$ put*

$$\mathsf{x} := t^{-1} + U(t) = t^{-1} + \sum_{n\geq 1} c_n t^n. \tag{J.51}$$

   Then $t\mathsf{x} = 1 + tU \in 1 + tR[[t]]$ and $1/\mathsf{x} = t/(1+tU) \in tR[[t]]$, and the following identity holds in $R[[t]]$:

$$a\big(\mathsf{x} - t^{-1}\big) + b\log(t\mathsf{x}) + Q\big(1/\mathsf{x}\big) = 0. \tag{J.52}$$

   Equivalently: writing $t = X^{-1}$ and $x = X + \sum_{n\geq 1} c_n X^{-n}$, the quantity $ax + b\log x + Q(1/x)$ equals $aX + b\log X$ to all orders.

5. Analytic version. *Let F be differentiably of exponential–power type with data $(C, a, 1, b, 1, Q)$ on $[x_0, \infty)$ (Theorem J.2), strictly increasing there, with $F(x) \to \infty$. For y in the range of F set $L = \log(y/C)$ and let $X = X(y)$ be the large core solution of $L = aX + b\log X$ furnished by Theorem J.23. Then $X(y) \to \infty$ and, for every fixed $N \geq 0$,*

$$F^{-1}(y) = X + \sum_{n=1}^{N}\frac{c_n}{X^n} + \mathcal{O}\big(X^{-N-1}\big) \qquad (y \to \infty). \tag{J.53}$$

Proof. (1) $\log(1+tu) = \sum_{k\geq 1}(-1)^{k+1}(tu)^k/k \in tR[[t,u]]$, and $t/(1+tu) = t\sum_{k\geq 0}(-tu)^k \in tR[[t,u]]$, so $(t/(1+tu))^j \in t^j R[[t,u]]$ and the sum over j is summable; hence $Q(t/(1+tu)) \in tR[[t,u]]$.

(2) is Theorem J.19.

(3) Put $w := tU = \sum_{r\geq 1} \gamma_r t^r$ with γ as stated; $\gamma_1 = 0$ because $U \in tR[[t]]$. Then $[t^n]\, \log(1+w)$ is given by (J.18), while

$$[t^n]\, Q\Big(\frac{t}{1+w}\Big) = \sum_{j\geq 1} q_j\, [t^{n-j}]\, (1+w)^{-j} = \sum_{j=1}^{n} q_j \sum_{k=0}^{n-j} (-1)^k \binom{j+k-1}{k} \widehat{B}_{n-j,k}(\gamma)$$

by (J.17). Since $c_n = [t^n]\, \Psi(t, U)$, this is (J.50). For triangularity, $\widehat{B}_{n,k}(\gamma)$ depends only on $\gamma_1, \ldots, \gamma_{n-k+1}$, that is on $c_1, \ldots, c_{n-k}$; in the first sum $k \geq 1$ and in the second $j \geq 1$, so c_n never occurs. (In fact $\widehat{B}_{n,k}(\gamma) = 0$ for $2k > n$, because $\gamma_1 = 0$.)

(4) With $\mathsf{x} = t^{-1}(1+w)$, $w = tU$, one has $a(\mathsf{x} - t^{-1}) = aU$, $\log(t\mathsf{x}) = \log(1+w)$ and $1/\mathsf{x} = t/(1+w)$, so the left-hand side of (J.52) equals $aU + b\log(1+tU) + Q(t/(1+tU)) = a\big(U - \Psi(t,U)\big) = 0$.

(5) Fix $N \geq 0$, set $t := 1/X$, $U_N(t) := \sum_{n=1}^{N} c_n t^n$ and $x_N := X + U_N(t) = X\big(1 + tU_N(t)\big)$. By Theorem J.23(3), $X \to \infty$ and $X \sim L/a$; since $tU_N(t) = \mathcal{O}(t^2)$ we get $x_N = X\big(1 + \mathcal{O}(X^{-2})\big)$, so $x_N \to \infty$ and $x_N \geq x_0$ for all large y.

Residual. Apply (J.3) with $M = N$:

$$\log F(x_N) = \log C + ax_N + b\log x_N + \sum_{j=1}^{N} q_j x_N^{-j} + E_N, \qquad |E_N| \leq K_N x_N^{-N-1}.$$

Since $\log y = \log C + aX + b\log X$ and $\log x_N = \log X + \log(1 + tU_N(t))$,

$$r_N := \log F(x_N) - \log y = g(t) + E_N, \qquad g(t) := aU_N(t) + b\log\big(1 + tU_N(t)\big) + \sum_{j=1}^{N} q_j \left(\frac{t}{1+tU_N(t)}\right)^j.$$

Here g is real-analytic on a neighbourhood of $t = 0$, being a finite composition of analytic maps with $1 + tU_N(t) \to 1$; its Taylor series at $t = 0$ is therefore the corresponding formal series. By (4) the formal series $aU + b\log(1+tU) + Q(t/(1+tU))$ vanishes identically. Replacing U by U_N changes it by an element of $t^{N+1}R[[t]]$, because $U - U_N \in t^{N+1}R[[t]]$ and the maps $u \mapsto \log(1+tu)$, $u \mapsto (t/(1+tu))^j$ are t-adically continuous; and discarding the terms of Q with $j > N$ changes it by an element of $t^{N+1}R[[t]]$ as well. Hence the Taylor series of g vanishes through order t^N, so $g(t) = \mathcal{O}(t^{N+1})$ and

$$|r_N| \leq \big|g(1/X)\big| + K_N x_N^{-N-1} = \mathcal{O}\big(X^{-N-1}\big).$$

Slope. By (J.4) with $\alpha = 1$, $(\log F)'(x) = a + \mathcal{O}(x^{-1})$, so there is $x_1 \geq x_0$ with $(\log F)' \geq a/2$ on $[x_1, \infty)$. For all large y both x_N and $F^{-1}(y)$ lie in $[x_1, \infty)$. Theorem J.47, applied to $\log F$ on that half-line, gives

$$\big|x_N - F^{-1}(y)\big| \leq \frac{|r_N|}{a/2} = \mathcal{O}\big(X^{-N-1}\big),$$

which is (J.53). □

Remark J.30 (No logarithms in the coefficients). Part (3) is the structural reason for centring at the Lambert core: the whole logarithmic tower has been resummed into X, so the c_n are constants of the coefficient ring. The alternative expansion of Section J.6 has coefficients that are polynomials in $\log L$ whose degree grows with the order. This is an algebraic statement and by itself says nothing about numerical accuracy; the numerical comparison is a separate, chapterwise matter (see Theorem J.40).

Proposition J.31 (General exponent grid). *Let $\delta > 0$ and keep the remaining hypotheses of Theorem J.29. Let s be a second formal variable and put*

$$\Psi_\delta(t, s, u) := -\frac{1}{a}\left[b\log(1+tu) + Q\bigg(\frac{s}{(1+tu)^\delta}\bigg)\right], \tag{J.54}$$

where $(1+tu)^{-\delta}$ *is the formal binomial series* (J.16)*. Then* Ψ_δ *lies in the ideal* (t,s) *of* $R[[t,s,u]]$*, the equation* $U=\Psi_\delta(t,s,U)$ *has a unique solution* $U=\sum_{i+j\geq 1} c_{i,j}t^is^j$ *in the ideal* (t,s) *of* $R[[t,s]]$*, and the substitution* $t=X^{-1}$*,* $s=X^{-\delta}$ *reverts* (J.2)*: formally*

$$ax+b\log x+Q\big(x^{-\delta}\big)=aX+b\log X,\qquad x=X+\sum_{i+j\geq 1} c_{i,j}X^{-i-j\delta}. \tag{J.55}$$

For $\delta=1$ *the two markers coincide and* $\Psi_\delta(t,t,u)=\Psi(t,u)$ *of* (J.48).

Proof. The membership $\Psi_\delta\in(t,s)$ and the existence and uniqueness of U follow exactly as in parts (1) and (2) of Theorem J.29, with the t-adic filtration replaced by the (t,s)-adic one: Ψ_δ lies in (t,s), so $\Psi_\delta(t,s,U)$ modulo $(t,s)^{d+1}$ depends only on U modulo $(t,s)^d$. The identity (J.55) is the computation of part (4) with $1/\mathsf{x}=t/(1+tU)$ replaced by $\mathsf{x}^{-\delta}=s(1+tU)^{-\delta}$. Finally the substitution is legitimate because, for each real exponent θ, only finitely many pairs (i,j) with $i,j\geq 0$ satisfy $i+j\delta=\theta$ when $\delta>0$. □

Remark J.32 (When the grid degenerates). If δ is irrational the exponents $i+j\delta$ are pairwise distinct and the monomial scale is a free monoid on two generators. If $\delta=p/q$ in lowest terms, different pairs (i,j) can give the same exponent and the correct statement is that the reversion lives in $R[[X^{-1/q}]]$. Every chapter of this volume has $\delta/\alpha=1$ after Theorem J.4, so the degeneracy is never met here.

J.5.2 Reversion about an arbitrary admissible core

Theorem J.33 (Reversion about an arbitrary core)**.** *Let* R *be a commutative* $\mathbb{Q}$*-algebra and* $\Lambda\in R$ *invertible. Consider*

$$\Lambda E+h(E)+f(t,E)=0,\qquad h\in u^2R[[u]],\quad f\in tR[[t,u]]. \tag{J.56}$$

Then (J.56) *has a unique solution* $E=\sum_{n\geq 1}e_nt^n\in tR[[t]]$*; it satisfies the triangular recurrence*

$$e_n=-\frac{1}{\Lambda}[t^n]\Big[h\big(E_{<n}\big)+f\big(t,E_{<n}\big)\Big],\qquad E_{<n}:=\sum_{j=1}^{n-1}e_jt^j, \tag{J.57}$$

and the closed formula

$$e_n=\sum_{m=1}^{2n-1}\frac{1}{m}[t^nu^{m-1}]\Big(-\Lambda^{-1}\big[h(u)+f(t,u)\big]\Big)^m. \tag{J.58}$$

If moreover $h(u)=\sum_{m\geq 2}h_mu^m$ *and* $f(t,u)=\sum_{k\geq 1}\phi_k(t)\,(1+u)^{-\mu_k}$ *with* $\phi_k\in tR[[t]]$ *and* $\mu_k\in\mathbb{N}$*, then*

$$e_n=-\frac{1}{\Lambda}\left[\sum_{m=2}^{n}h_m\,\widehat{B}_{n,m}(e)+\sum_{k\geq 1}\sum_{i=1}^{n}\big([t^i]\,\phi_k\big)\sum_{m=0}^{n-i}(-1)^m\binom{\mu_k+m-1}{m}\widehat{B}_{n-i,m}(e)\right], \tag{J.59}$$

in which only $e_1,\dots,e_{n-1}$ *occur.*

Proof. Existence, uniqueness and (J.58) are Theorem J.20 applied to $\Psi:=-\Lambda^{-1}[h+f]$. For (J.57), note as in the proof of that lemma that e_n occurs neither in $[t^n]\,h(E)$ nor in $[t^n]\,f(t,E)$, so both may be computed from $E_{<n}$. For (J.59) apply Theorem J.9 to h and (J.17) to each $(1+E)^{-\mu_k}$, then convolve with ϕ_k; the displayed ranges are those outside which the ordinary Bell polynomials vanish. □

Remark J.34 (How the chapters instantiate it). For the linear–logarithmic core the shorter route of Theorem J.29 is available, because measuring the displacement in units of X makes the entire perturbation divisible by t. Part N needs the general form: there Λ is the normalized core slope $\log X$ of Theorem J.27, $h(E)=(1+E)\log(1+E)-E$ and $f(t,E)=\sum_{k\geq 1}\alpha_kt^{2k}(1+E)^{-(2k-1)}$, and (J.59) is exactly the Bell recurrence of double_factorial_transseries-2. The coefficient ring there is $\mathbb{Q}[\Lambda^{-1}]$: the e_n are *not* constants but polynomials in Λ^{-1}, of degree at most $2n-1$ and with no constant term.

J.5.3 The perturbation-operator form

Proposition J.35 (Inverse perturbation operator). *Let Φ_0 be a phase with Φ_0' nonvanishing near the point considered, let X solve $\Phi_0(X) = L$, and consider*

$$L = \Phi_0(x) + \varepsilon\, \varrho(x). \tag{J.60}$$

Put $\mathscr{D}_0 := \Phi_0'(X)^{-1}\, \mathrm{d}/\mathrm{d}X$. Then, as a formal power series in ε,

$$x = X + \sum_{m \geq 1} \frac{(-\varepsilon)^m}{m!}\, \mathscr{D}_0^{m-1}\left[\frac{\varrho(X)^m}{\Phi_0'(X)}\right]. \tag{J.61}$$

Proof. Change coordinate to $\lambda := \Phi_0(x)$, so that $x = \Phi_0^{-1}(\lambda)$ and (J.60) becomes $L = \lambda + \varepsilon\varrho_0(\lambda)$ with $\varrho_0 := \varrho \circ \Phi_0^{-1}$. This is a near-identity reversion in λ with small parameter ε; Lagrange–Bürmann in the form (J.24), with the observable $G = \Phi_0^{-1}$, gives

$$\Phi_0^{-1}(\lambda) = \Phi_0^{-1}(L) + \sum_{m \geq 1} \frac{(-\varepsilon)^m}{m!}\left(\frac{\mathrm{d}}{\mathrm{d}L}\right)^{m-1}\Big[(\Phi_0^{-1})'(L)\, \varrho_0(L)^m\Big].$$

Finally $(\Phi_0^{-1})'(L) = 1/\Phi_0'(X)$, $\varrho_0(L) = \varrho(X)$ and $\mathrm{d}/\mathrm{d}L = \mathscr{D}_0$ by the chain rule. □

Remark J.36 (What the operator form is for). This is the form in which an *exponentially small* forward correction is transported through the inverse map: with ϱ a Stokes discontinuity or a terminant remainder rather than a power series, $-\varepsilon\varrho(X)/\Phi_0'(X)$ is the leading displacement and the higher terms generate the nonlinear sectors. As a statement about a formal series in ε it is exact; as an asymptotic statement about a particular ϱ it needs the hypotheses of Theorem J.51. Expanding ϱ in inverse powers of X reproduces (J.49).

J.6 Flattening the core

Theorem J.29 hides the explicit scale of the answer inside X. Unfolding X produces the polynomial–logarithmic transseries in which the four chapters state their headline formulae. Throughout this section H is an indeterminate; in the analytic application it is specialized to $\log(L/a)$.

Theorem J.37 (Direct polynomial–logarithmic reversion). *Let R be a commutative $\mathbb{Q}$-algebra, $a \in R$ invertible, $b \in R$, $Q = \sum_{j \geq 1} q_j t^j \in tR[[t]]$, and let H be an indeterminate. Set*

$$\Phi(t, u; H) := -b \log\big(1 + t(u - bH)\big) - Q\left(\frac{at}{1 + t(u - bH)}\right) \in t\,R[H][[t, u]]. \tag{J.62}$$

1. *There is a unique $V = \sum_{n \geq 1} P_n(H)\, t^n \in t\,R[H][[t]]$ with $V = \Phi(t, V; H)$, and*

$$P_n(H) = \sum_{m=1}^{n} \frac{1}{m}[t^n u^{m-1}]\, \Phi(t, u; H)^m. \tag{J.63}$$

2. Triangular Bell recurrence. *Put $\gamma_1 := -bH$ and $\gamma_r := P_{r-1}(H)$ for $r \geq 2$. Then*

$$P_n(H) = -b \sum_{k=1}^{n} \frac{(-1)^{k+1}}{k} \widehat{B}_{n,k}(\gamma) - \sum_{j=1}^{n} q_j a^j \sum_{k=0}^{n-j} (-1)^k \binom{j+k-1}{k} \widehat{B}_{n-j,k}(\gamma), \tag{J.64}$$

in which only $P_1, \ldots, P_{n-1}$ occur.

3. Shape. $P_n \in R[H]$ *with*

$$\deg_H P_n \le n, \qquad [H^n]\, P_n = \frac{b^{\,n+1}}{n}, \qquad P_n(0) = a^{\,n+1} c_n, \tag{J.65}$$

where c_n *are the Lambert-centred coefficients of Theorem J.29. In particular* $\deg_H P_n = n$ *exactly when* b *is not a zero divisor, and the coefficients of* P_n *lie in the* $\mathbb{Q}$*-subalgebra generated by* a, a^{-1}, b, $q_1, \ldots, q_n$.

4. Formal exactness and uniqueness. *Put* $t = L^{-1}$ *and* $H = \log(L/a)$, *and*

$$x := \frac{1}{a}\left[L - bH + \sum_{n\ge 1} \frac{P_n(H)}{L^n}\right]. \tag{J.66}$$

Then $ax + b\log x + Q(1/x) = L$ *as an identity of formal series in* L^{-1} *with polynomial coefficients in* H. *Conversely,* (J.66) *is the* only *expansion of the form* $x = a^{-1}\bigl[L - bH + \sum_{n\ge 1} \Pi_n(H) L^{-n}\bigr]$, *with* $\Pi_n \in R[H]$, *that solves the equation coefficientwise.*

5. Analytic version. *Under the hypotheses of Theorem J.29(5), with* $L = \log(y/C)$ *and* $H = \log(L/a)$, *for every fixed* $N \ge 0$,

$$F^{-1}(y) = \frac{1}{a}\left[L - bH + \sum_{n=1}^{N} \frac{P_n(H)}{L^n}\right] + \mathcal{O}\left(\frac{(1+H)^{N+1}}{L^{N+1}}\right) \qquad (y \to \infty). \tag{J.67}$$

Proof. (1) Membership of Φ in $tR[H][[t,u]]$ is checked as in Theorem J.29(1), the extra additive term $-bH$ inside $1 + t(u - bH)$ being harmless because it is multiplied by t. Theorem J.19 over the ring $R[H]$ then gives existence, uniqueness and (J.63).

(2) Put $w := t(V - bH) = \sum_{r\ge 1} \gamma_r t^r$ with γ as stated; apply (J.18) to $\log(1 + w)$ and (J.17) to $(1 + w)^{-j}$, and convolve the latter with the factor $(at)^j$. Only $\gamma_1, \ldots, \gamma_{n-k+1}$, that is H and $P_1, \ldots, P_{n-k}$, occur.

(4) Set $t = L^{-1}$. With x as in (J.66),

$$x = \frac{1}{at}\bigl(1{+}t(V{-}bH)\bigr) = \frac{1+w}{at}, \qquad ax = \frac{1}{t}{+}(V{-}bH), \qquad \log x = H{+}\log(1{+}w), \qquad \frac{1}{x} = \frac{at}{1+w},$$

where $\log x$ uses $\log\bigl(1/(at)\bigr) = \log(L/a) = H$. Hence

$$ax + b\log x + Q(1/x) - L = (V - bH) + b\bigl(H + \log(1+w)\bigr) + Q\left(\frac{at}{1+w}\right) = V - \Phi(t, V; H) = 0.$$

For uniqueness, substituting a general $\Pi = \sum \Pi_n(H) t^n$ in place of V turns the equation into $\Pi = \Phi(t, \Pi; H)$, and Theorem J.19 has a unique solution.

(3) We show by strong induction that $\deg_H P_n \le n$ and, in addition, that $\deg_H \gamma_r \le r - 1$ for $r \ge 2$. The base is $\gamma_1 = -bH$, of degree 1. Assume $\deg_H P_j \le j$ for all $j < n$, so $\deg_H \gamma_r = \deg_H P_{r-1} \le r - 1$ for $2 \le r \le n$. A monomial of $\widehat{B}_{n,k}(\gamma)$ is a product $\gamma_{r_1} \cdots \gamma_{r_k}$ with $r_1 + \cdots + r_k = n$ and $r_\nu \ge 1$, hence of H-degree at most $\sum_\nu r_\nu = n$; so the first sum of (J.64) has H-degree at most n. In the second sum the ordinary Bell polynomial has first index $n - j$ with $j \ge 1$, so its H-degree is at most $n - 1$. Hence $\deg_H P_n \le n$, completing the induction.

A monomial of H-degree exactly n requires $\deg_H \gamma_{r_\nu} = r_\nu$ for every ν, which forces $r_\nu = 1$ for all ν because $\deg_H \gamma_r \le r - 1$ for $r \ge 2$. Thus $k = n$, the monomial is $\gamma_1^n = (-bH)^n$, and $[H^n]\, \widehat{B}_{n,k}(\gamma) = 0$ for $k < n$ while $\widehat{B}_{n,n}(\gamma) = (-bH)^n$. Only the term $k = n$ of the first sum of (J.64) therefore survives, and

$$[H^n]\, P_n = -b \cdot \frac{(-1)^{n+1}}{n} \cdot (-b)^n = \frac{(-1)^{2n+2}\, b^{\,n+1}}{n} = \frac{b^{\,n+1}}{n}.$$

For the constant term put $H = 0$ in (J.62):

$$\Phi(t,u;0) = -b\log(1+tu) - Q\left(\frac{at}{1+tu}\right).$$

If $U(t) = \sum c_n t^n$ solves $U = \Psi(t,U)$ with Ψ of (J.48), put $\widetilde{V}(t) := a\,U(at)$. Then $t\widetilde{V}(t) = (at)U(at)$, so

$$\Phi\big(t,\widetilde{V}(t);0\big) = -b\log\big(1+(at)U(at)\big) - Q\left(\frac{at}{1+(at)U(at)}\right) = a\,\Psi\big(at,U(at)\big) = a\,U(at) = \widetilde{V}(t),$$

so $\widetilde{V}$ solves the $H = 0$ fixed-point equation; by uniqueness $P_n(0) = [t^n]\,\widetilde{V} = a\cdot a^n c_n = a^{n+1}c_n$.

(5) Fix N. Put x_N for the truncation displayed in (J.67). Exactly as in the proof of Theorem J.29(5), the residual $r_N := \log F(x_N) - \log y$ splits as $g(t) + E_N$ with $|E_N| \le K_N x_N^{-N-1}$ and g analytic in $t = 1/L$ *with coefficients polynomial in H*; by (4) the Taylor coefficients of g vanish through order N, and the coefficient of t^{N+1} is a polynomial in H of degree at most $N+1$. Hence $|r_N| = \mathcal{O}\big((1+H)^{N+1}L^{-N-1}\big)$. The slope bound $(\log F)' \ge a/2$ and Theorem J.47 finish the proof, after noting $L \asymp aX$ and $H = \log(L/a)$ so that $x_N \to \infty$. $\square$

Theorem J.38 (The pure Lambert block in closed form). *Let $\left[{n \atop m}\right]$ be the* unsigned *Stirling numbers of the first kind, so that the signed ones are $(-1)^{n-m}\left[{n \atop m}\right]$ and $\log(1+z)^m = m!\sum_{n\ge m}(-1)^{n-m}\left[{n \atop m}\right]z^n/n!$. When $Q = 0$ the polynomials of Theorem J.37 are*

$$\boxed{P_n^{(0)}(H) = b^{n+1}\sum_{m=1}^{n}(-1)^{m+1}\frac{\left[{n \atop m}\right]}{(n-m+1)!}\,H^{\,n-m+1}} \tag{J.68}$$

for $n \ge 1$. Equivalently, clearing the binomial,

$$P_n^{(0)}(H) = \frac{b^{n+1}}{n!}\sum_{m=1}^{n}(-1)^{m+1}(m-1)!\left[{n \atop m}\right]\binom{n}{m-1}H^{\,n-m+1}. \tag{J.69}$$

In particular

$$[H^n]\,P_n^{(0)} = \frac{b^{n+1}}{n}, \qquad P_n^{(0)}(0) = 0, \tag{J.70}$$

and the first four are

$$P_1^{(0)} = b^2H,\quad P_2^{(0)} = \frac{b^3}{2}\big(H^2-2H\big),\quad P_3^{(0)} = \frac{b^4}{6}\big(2H^3-9H^2+6H\big),\quad P_4^{(0)} = \frac{b^5}{12}\big(3H^4-22H^3+36H^2-12H\big). \tag{J.71}$$

Consequently the Lambert core itself unfolds as

$$X = \frac{1}{a}\left[L - bH + \sum_{n\ge1}\frac{P_n^{(0)}(H)}{L^n}\right], \qquad H = \log\frac{L}{a}, \tag{J.72}$$

formally, and with remainder $\mathcal{O}\big((1+H)^{N+1}L^{-N-1}\big)$ after truncation at $n = N$.

Proof. With $Q = 0$, (J.63) reads

$$P_n^{(0)}(H) = \sum_{m=1}^{n}\frac{(-b)^m}{m}[t^n u^{m-1}]\,\log\big(1+t(u-bH)\big)^m.$$

Insert $\log(1+z)^m = m!\sum_{r\ge m}(-1)^{r-m}\left[{r \atop m}\right]z^r/r!$ with $z = t(u-bH)$; the coefficient of t^n is $m!\,(-1)^{n-m}\left[{n \atop m}\right](u-bH)^n/n!$, and $[u^{m-1}]\,(u-bH)^n = \binom{n}{m-1}(-bH)^{n-m+1}$. Hence

$$P_n^{(0)}(H) = \sum_{m=1}^{n}\frac{(-b)^m}{m}\cdot\frac{m!\,(-1)^{n-m}\left[{n \atop m}\right]}{n!}\binom{n}{m-1}(-b)^{n-m+1}H^{n-m+1},$$

and $(-b)^m(-b)^{n-m+1}(-1)^{n-m} = (-1)^{n+1}(-1)^{n-m}b^{n+1} = (-1)^{m+1}b^{n+1}$, which gives (J.69). Since $(m-1)!\binom{n}{m-1}/n! = 1/(n-m+1)!$, the two displayed forms agree and (J.68) follows. Every term carries H^{n-m+1} with $m \le n$, so no constant term occurs and $P_n^{(0)}(0) = 0$; the term $m = 1$ has $\left[{n \atop 1}\right] = (n-1)!$ and gives $[H^n]\,P_n^{(0)} = b^{n+1}(n-1)!/n! = b^{n+1}/n$. Finally (J.72) is Theorem J.37(4) and (5) with $Q = 0$, in which case the unique solution of $ax+b\log x = L$ is the core X of Theorem J.23. $\square$

Remark J.39 (Three source formulae, one identity). The closed form (J.68) appears in the sources in three different dresses, all with the signed Stirling numbers $s(n,m) = (-1)^{n-m}\left[{n \atop m}\right]$. Swing_Factorial_Transseries_Article states (J.69); Partition_Number_Transseries_and_Asymptotic_Inverse states $P_j(L) = 2^{j+1}\sum_{q\le j}s(j,q)L^{j+1-q}/(j+1-q)!$; Partition_Number_Transseries_and_Inverse states $P_m(\ell) = \frac{1}{m!}\sum_j 2^j(j-1)!\,s(m,j)\binom{m}{j-1}(2\ell)^{m-j+1}$. The last two are the specialization $a = 1$, $b = -2$ of (J.68), and all three agree with it as exact rational polynomials for $1 \le n \le 8$, which was checked in exact rational arithmetic (Theorem J.57). The volume uses (J.68): it is the shortest of the four, it avoids the letter s, which Part O needs for Dedekind sums, and it makes the leading coefficient (J.70) a one-line consequence.

Remark J.40 (Both normalizations, and what is and is not claimed). The Lambert-centred expansion (J.53) and the flattened expansion (J.67) are two truncations of the same object: by (J.72), expanding X and then each power X^{-n} in the flattened scale returns (J.67), and conversely collecting in the flattened series all terms generated by $b\log x$ alone rebuilds X. What differs is the coefficient ring: the c_n are constants, the P_n are polynomials of degree n in $H = \log(L/a)$. Several of the sources add that Lambert centring is numerically more accurate at equal truncation order and support this with tables. Part 0 makes no such claim: the comparison depends on the size of H relative to L and on the particular data, it is an empirical statement about specific truncations, and the supporting tables belong to the chapters, where they are reported with their provenance. What Part 0 does assert is the algebraic statement of Theorem J.30, and the remainder bounds (J.53) and (J.67), whose right-hand sides differ by the factor $(1+H)^{N+1}$ — and that factor is the only rigorous difference between the two at fixed order.

J.7 The three inverses of a discrete sequence

Definition J.41 (The three inverse objects)**.** Let $(A_n)_{n\ge n_0}$ be real numbers, strictly increasing for $n \ge n_1 \ge n_0$, with $A_n \to +\infty$.

1. The *integer staircase*, or generalized inverse, is

$$N_*(y) := \min\{n \ge n_1 : A_n \ge y\}, \qquad y \le \sup_n A_n. \tag{J.73}$$

   It is an integer-valued, nondecreasing step function, constant on each interval $(A_{n-1}, A_n]$.

2. Given a continuous, strictly increasing $\mathcal{A} : [n_1,\infty) \to \mathbb{R}$ with $\mathcal{A}(n) = A_n$ for every integer $n \ge n_1$ — an *admissible interpolation* — the *interpolated inverse* is $\nu_{\mathcal{A}} := \mathcal{A}^{-1}$, defined on $[A_{n_1},\infty)$.

3. The *asymptotic inverse* is the formal transseries produced by Theorem J.29 or Theorem J.37 from a model F for which the chapter in question proves an asymptotic identity along the sequence. It is attached to no particular interpolation.

Remark J.42 (The three are genuinely different). Object (1) is canonical but not smooth; object (2) is smooth but not canonical — distinct admissible interpolations agree at the integers and differ in between, hence have different inverses; object (3) is canonical as a formal series but determines a function only after a summation convention is fixed. The three coincide only in the following precise senses: $N_* = \lceil \nu_{\mathcal{A}} \rceil$ for *every* admissible $\mathcal{A}$ (Theorem J.43(1)), and $\nu_{\mathcal{A}}(A_n) = n$ for every admissible $\mathcal{A}$ and every integer $n \geq n_1$. The asymptotic series (3) is independent of the choice in $\mathcal{A}$ at every algebraic order, but the exponentially small layers are not: the partition articles make this explicit for the Rademacher amplitudes, which are periodic in the index and therefore genuinely depend on how the sequence is continued off the lattice.

Theorem J.43 (Staircase recovery and the separation condition)**.** *Notation as in Theorem J.41, with $\mathcal{A}$ admissible and $\nu := \nu_{\mathcal{A}}$.*

1. Exact rounding. *For every y in the range of $\mathcal{A}$, that is for $A_{n_1} \leq y < \sup_n A_n$,*

$$N_*(y) = \lceil \nu(y) \rceil, \tag{J.74}$$

with $\lceil m \rceil = m$ for integers m. The identity holds for every admissible interpolation.

2. Separation condition. *Let $x_* \in \mathbb{R}$ and $\eta \geq 0$ satisfy $|x_* - \nu(y)| \leq \eta$. If*

$$\lceil x_* - \eta \rceil = \lceil x_* + \eta \rceil, \tag{J.75}$$

then $N_*(y)$ equals that common value. Condition (J.75) *holds if and only if the interval $[x_* - \eta, x_* + \eta]$ is contained in a single half-open interval $(m-1, m]$; it can fail for arbitrarily small $\eta > 0$, namely when $\nu(y)$ is within η of an integer from above.*

3. Inverse along the range. *If $y = A_n$ for an integer $n \geq n_1$ and $|x_* - \nu(y)| \leq \eta < \frac{1}{2}$, then $n = \lfloor x_* + \frac{1}{2} \rfloor$. This case is unconditional: no separation hypothesis is needed, because $\nu(A_n) = n$ exactly.*

4. Residue classes. *Suppose $r \geq 1$ and that for each $\rho \in \{0, 1, \ldots, r-1\}$ the subsequence $(A_n)_{n \equiv \rho\ (r)}$ is strictly increasing for $n \geq n_1$ and admits an admissible interpolation $\mathcal{A}_\rho$ on the class, with inverse ν_ρ. Then, for y large enough,*

$$N_*(y) = \min_{0 \leq \rho < r} \left(\rho + r \left\lceil \frac{\nu_\rho(y) - \rho}{r} \right\rceil \right). \tag{J.76}$$

5. No uniform smooth representation. *For every Y in the range and every continuous $g : [Y, \infty) \to \mathbb{R}$,*

$$\sup_{y \geq Y} \bigl| N_*(y) - g(y) \bigr| \ \geq \ \tfrac{1}{2}. \tag{J.77}$$

Equivalently, the $\mathcal{O}(1)$ periodic layer $\lceil x \rceil - x - \frac{1}{2} = \pi^{-1} \sum_{m \geq 1} m^{-1} \sin(2\pi m x)$ (valid for $x \notin \mathbb{Z}$) is not a refinement that can be dropped: it is the leading discrete transmonomial of the generalized inverse.

Proof. (1) For an integer $n \geq n_1$, $A_n \geq y$ is equivalent to $\mathcal{A}(n) \geq y$, hence to $n \geq \nu(y)$ by strict monotonicity of $\mathcal{A}$. The least such integer is $\lceil \nu(y) \rceil$.

(2) The function $\lceil \cdot \rceil$ is constant on each $(m-1, m]$ and nondecreasing; since $\nu(y) \in [x_* - \eta, x_* + \eta]$, (J.75) forces $\lceil \nu(y) \rceil$ to equal the common value, and (1) applies. The characterization of (J.75) and the failure mode are immediate from the same description of $\lceil \cdot \rceil$.

(3) $\nu(A_n) = n$ because $\mathcal{A}(n) = A_n$; hence $|x_* - n| \leq \eta < 1/2$ and rounding to the nearest integer returns n.

(4) Fix ρ. Among the integers $n \equiv \rho \pmod r$ with $n \geq n_1$, the condition $A_n \geq y$ is, by (1) applied to the class, equivalent to $n \geq \nu_\rho(y)$. Writing $n = \rho + rk$ with $k \in \mathbb{Z}$, the least admissible k is

$\lceil(\nu_\rho(y)-\rho)/r\rceil$, so the least admissible n in the class is $\rho+r\lceil(\nu_\rho(y)-\rho)/r\rceil$. Taking the minimum over ρ gives the least admissible n without a congruence restriction, which is $N_*(y)$.

(5) Fix n with $A_n > Y$. On $(A_{n-1}, A_n]$ we have $N_* = n$ and on $(A_n, A_{n+1}]$ we have $N_* = n+1$. For $\epsilon > 0$ small,

$$1 = \big|N_*(A_n+\epsilon)-N_*(A_n)\big| \le \big|N_*(A_n+\epsilon)-g(A_n+\epsilon)\big|+\big|g(A_n+\epsilon)-g(A_n)\big|+\big|g(A_n)-N_*(A_n)\big|.$$

Letting $\epsilon \downarrow 0$ and using continuity of g gives $1 \le 2\sup_{y\ge Y}|N_*-g|$. For the Fourier identity, the function $x \mapsto \lceil x\rceil - x - \frac{1}{2}$ is 1-periodic, equal to $-(x-\lfloor x\rfloor)+\frac{1}{2}$ shifted, i.e. minus the standard sawtooth, whose Fourier series is the displayed one and converges pointwise off $\mathbb{Z}$. □

Remark J.44 (Formalization of the staircase). All five parts are order theory plus a single application of the intermediate value theorem; none of the analytic apparatus of this chapter is used. That is visible in the formalization, `StaircaseInversion.lean`, which is a Mathlib-only leaf and reduces the phrase *admissible interpolation* to the two hypotheses actually consumed: $\mathcal{A}$ strictly monotone, and $\mathcal{A}(\nu) = y$.

Part (1) is `Fabius.staircase_ceil`, stated as an `IsLeast` rather than an equation — $\lceil\nu\rceil$ is the least integer n with $y \le \mathcal{A}(n)$ — which is what (J.74) means and what its uses need; the underlying fact that $\lceil\nu\rceil$ is least above ν is `Fabius.isLeast_ceil`. Part (2) is `Fabius.staircase_separation`. Part (3) is `Fabius.staircase_round`, and the formal statement confirms that it is unconditional: its only hypotheses are $|x-n| \le \eta$ and $\eta < \frac{1}{2}$.

The arithmetic of part (4) is `Fabius.isLeast_residue_class`: for $r > 0$, $\rho + r\lceil(\nu-\rho)/r\rceil$ is the least integer at or above ν in the residue class of ρ modulo r. Stated that way it is independent of the sequence entirely, which is why the same identity serves the parity instances of Theorem J.46 and the floor variant for the threshold query — the volume observes that those have "the same proof", and here they have the same lemma.

Part (5) is `Fabius.exists_half_error_of_jump`, in the local form that carries the content: across a single unit jump of an integer-valued function, every continuous g satisfies $|N-g| \ge \frac{1}{2}$ somewhere on the interval. The proof is the intermediate value theorem forcing g through the half-integer, where every integer is at distance at least $\frac{1}{2}$; the supremum statement (J.77) follows by applying it at a jump.

One thing the formalization adds. Theorem J.43(2) remarks that the separation condition can fail for arbitrarily small η; that is proved, as `Fabius.staircase_separation_fails`, so the hypothesis is demonstrably not removable rather than merely not removed.

Not formalized: the Fourier expansion of the periodic layer, and the construction of admissible interpolations, which is where this chapter's analysis actually lives.

Proposition J.45 (Certified discrete inversion)**.** *In the setting of Theorem J.43(4), suppose that for each ρ an approximation x_ρ with $|x_\rho - \nu_\rho(y)| \le \eta_\rho$ is available. Then $N_*(y)$ is determined by at most $\sum_\rho\big(\lfloor 2\eta_\rho/r\rfloor + 2\big)$ exact evaluations of the sequence: for each ρ list the integers $n \equiv \rho \pmod r$ in $[x_\rho-\eta_\rho,\ x_\rho+\eta_\rho+r]$, evaluate A_n, and take the least n over all classes with $A_n \ge y$.*

Proof. By Theorem J.43(4) the class-ρ candidate is $\rho+r\lceil(\nu_\rho(y)-\rho)/r\rceil$, and $\nu_\rho(y)$ lies in $[x_\rho-\eta_\rho, x_\rho+\eta_\rho]$; the least integer of the class that is $\ge \nu_\rho(y)$ therefore lies in $[x_\rho-\eta_\rho, x_\rho+\eta_\rho+r]$. Since the class subsequence is increasing, the test $A_n \ge y$ identifies it, and the outer minimum over ρ is (J.76). □

Remark J.46 (The parity instances). Part P and Part N are the case $r = 2$. The formula $\rho + r\lceil(\nu_\rho - \rho)/r\rceil$ specializes to $2\lceil\nu_0/2\rceil$ and $2\lceil(\nu_1-1)/2\rceil+1$, which is exactly the pair of parity candidates of double_factorial_transseries-2, and $N_* = \min$ of the two is its "exact generalized inverse" theorem. Swing_Factorial_Transseries_Article states the corresponding *floor* formulae for the threshold query "the greatest index in the class ρ whose sequence value is at most y"; those are $\rho + r\lfloor(\nu_\rho-\rho)/r\rfloor$, obtained from (J.76) by replacing $\lceil\cdot\rceil$ with $\lfloor\cdot\rfloor$ and $\min$ with $\max$, with the same proof. The partition chapter is the case $r = 1$, where (J.76) reduces to (J.74).

Source claim. Several sources write, or invite the reader to write, $N_*(y) = \nu(y) + o(1)$. **Defect.** This is false uniformly in y, by Theorem J.43(5). Butcher_Tree_Transseries-2 already flags it ("An expansion such as $N_*(y) = \nu(y) + o(1)$ is false uniformly in y"), and Partition_Number_Transseries_and_Asymptotic_Inverse likewise separates the three objects; but the qualification is absent from the summary statements of the other articles. **Correction.** The volume uses only the three statements Theorem J.43(1), (3), (4): an exact ceiling identity valid for every admissible interpolation, an unconditional nearest-integer recovery *along the range* $y = A_n$, and a residue-class refinement. Nothing else about N_* is asserted.

J.8 Error control, transport, and optimal truncation

J.8.1 Backward error certifies forward error

Theorem J.47 (Residual-to-root conversion)**.** *Let $I \subseteq \mathbb{R}$ be an interval and $f : I \to \mathbb{R}$ differentiable with*

$$0 < m \le f'(\xi) \le M \qquad (\xi \in I). \tag{J.78}$$

Let $x_\dagger \in I$ satisfy $f(x_\dagger) = \lambda$, let $x_* \in I$, and set the residual $r_* := f(x_*) - \lambda$. *Then*

$$\frac{|r_*|}{M} \;\le\; |x_* - x_\dagger| \;\le\; \frac{|r_*|}{m}. \tag{J.79}$$

Moreover, if f is continuous on I and $[x_* - \varrho,\, x_* + \varrho] \subseteq I$ with $\varrho := |r_*|/m$, then f takes the value λ somewhere in that interval; the residual is therefore a certificate*, not merely a bound conditional on the existence of a nearby root.*

Proof. By the mean value theorem, $r_* = f(x_*) - f(x_\dagger) = f'(\xi)(x_* - x_\dagger)$ for some ξ between the two points, and (J.78) gives (J.79). For the certificate, suppose $r_* > 0$ (the case $r_* < 0$ is symmetric and $r_* = 0$ is trivial). Then $f(x_*) = \lambda + r_* > \lambda$, while

$$f(x_* - \varrho) \le f(x_*) - m\varrho = \lambda + r_* - |r_*| = \lambda,$$

using $f' \ge m$. The intermediate value theorem supplies a point of $[x_* - \varrho, x_*]$ where $f = \lambda$. □

Remark J.48 (Formalization: the bound, and the certificate). Both halves are machine-checked, in two modules, and the split between them follows the split in the statement.

The two-sided bound (J.79) is `MeanValueBracket.lean`: `Fabius.abs_sub_right_inverse_le_div` is the upper half and `Fabius.div_le_abs_sub_right_inverse` the lower, resting on the brackets `Fabius.abs_sub_le_of_le_deriv` and `Fabius.mul_abs_sub_le_abs_sub_of_le_deriv`. The hypothesis $0 < m$ in (J.78) is genuinely needed for the *upper* bound and not merely for the interpretation; Theorem L.6 gives the counterexample.

The certificate clause is `BackwardErrorExistence.lean`. Its hypothesis is deliberately weaker than a derivative bracket: `Fabius.HasSlopeLowerBound` asks only that $m(v - u) \le f(v) - f(u)$ for $u \le v$ in the domain, a condition on *slopes* rather than on a derivative, so the theorem applies where one has monotonicity data and no differentiability at all. `Fabius.hasSlopeLowerBound_of_le_deriv` recovers it from $f' \ge m$, and `Fabius.exists_eq_of_le_deriv` is the differentiable form matching the statement above. The certificate itself is `Fabius.exists_eq_of_residual`; `Fabius.exists_eq_and_abs_sub_le` packages it with the bound, so that existence is *discharged* rather than assumed.

Two things the formalization adds. First, the root is unique: `Fabius.injOn_of_hasSlopeLowerBound` and `Fabius.existsUnique_eq_of_residual` upgrade "f takes the value λ somewhere in that interval" to "at exactly one point of the domain", which the theorem above does not claim and every use of it wants.

Second, and worth stating because the proof above does it without saying so: the *sign* of the residual does two different jobs. In the bound (J.79) it is discarded — only $|r_*|$ appears — and tells the reader at most whether the candidate overshoots. In the certificate it is what *selects* the half-interval on which the intermediate value theorem is applied: the proof above takes $[x_* - \varrho, x_*]$ precisely because $r_* > 0$, the slope bound guaranteeing that f has already fallen to λ by the far end. Same quantity, two roles, and only the second is essential.

Remark J.49 (Formal residual certificate). The "moreover" clause is machine-checked by `Fabius.exists_eq_in_residual_interval` in `MeanValueBracket.lean`. The Lean theorem is slightly stronger and more structural than the displayed specialization: for an arbitrary convex set D, continuity on D, differentiability on its interior, and the sole lower derivative bound $m \le f'$ with $m > 0$, containment of $[x - |f(x) - z|/m, x + |f(x) - z|/m]$ in D produces a point y in that same interval with $f(y) = z$. It assumes neither a previously known root nor the unused upper derivative bound M, and includes the zero-residual and degenerate-interval cases. The two-sided error inequalities remain the separate theorems `Fabius.abs_sub_right_inverse_le_div` and `Fabius.div_le_abs_sub_right_inverse`, which appropriately require a chosen right-inverse value.

Remark J.50 (Use in the logarithmic variable). In every application $f = \log F$ and $\lambda = \log y$: F itself overflows any fixed-precision arithmetic long before the asymptotic regime begins, whereas $\log F$ is computable from a digamma or Laplace representation throughout. The lower bound in (J.79) makes the residual test sharp — a small residual is necessary as well as sufficient.

J.8.2 Transport of a controlled forward remainder

Theorem J.51 (Remainder transport). *Let $I = [x_1, \infty)$, let $f_0 : I \to \mathbb{R}$ be differentiable with $f_0' \ge m > 0$, and let $\varepsilon : I \to \mathbb{R}$ be differentiable with $\sup_I |\varepsilon'| \le \theta < m$. Put $f := f_0 + \varepsilon$. For λ in the common range let $x_0 = x_0(\lambda) \in I$ solve $f_0(x_0) = \lambda$ and $x = x(\lambda) \in I$ solve $f(x) = \lambda$. Then*

1. Rigorous bound.

$$\big|x(\lambda) - x_0(\lambda)\big| \le \frac{\big|\varepsilon\big(x_0(\lambda)\big)\big|}{m - \theta}. \tag{J.80}$$

2. First-order law with explicit error. *If moreover $f_0 \in C^2(I)$ with $\sup_I |f_0''| \le M_2$, then*

$$x(\lambda) - x_0(\lambda) = -\frac{\varepsilon(x_0)}{f_0'(x_0)} + E, \qquad |E| \le \frac{|\varepsilon(x_0)|}{m\,(m-\theta)} \left(\frac{M_2\,|\varepsilon(x_0)|}{m - \theta} + \theta \right). \tag{J.81}$$

In particular, if $\varepsilon(x) = \mathcal{O}(x^{-M-1})$ and $\varepsilon'(x) = o(1)$ as $x \to \infty$, then $x(\lambda) - x_0(\lambda) = -\varepsilon(x_0)/f_0'(x_0)\,(1 + o(1)) = \mathcal{O}(x_0^{-M-1})$.

Proof. Since $f' = f_0' + \varepsilon' \ge m - \theta > 0$, f is strictly increasing and $x(\lambda)$ is unique. The residual of x_0 in f is $f(x_0) - \lambda = f_0(x_0) + \varepsilon(x_0) - \lambda = \varepsilon(x_0)$, so Theorem J.47 applied to f with m replaced by $m - \theta$ gives (J.80).

For (2), write $d := x - x_0$. By the mean value theorem there is ξ between x_0 and x with $0 = f(x) - \lambda = \varepsilon(x_0) + f'(\xi)d$, so $d = -\varepsilon(x_0)/f'(\xi)$ and

$$d + \frac{\varepsilon(x_0)}{f_0'(x_0)} = \varepsilon(x_0) \left(\frac{1}{f_0'(x_0)} - \frac{1}{f'(\xi)} \right) = \varepsilon(x_0)\, \frac{f'(\xi) - f_0'(x_0)}{f_0'(x_0)\, f'(\xi)}.$$

Now $|f'(\xi) - f_0'(x_0)| \le |f_0'(\xi) - f_0'(x_0)| + |\varepsilon'(\xi)| \le M_2|d| + \theta \le M_2|\varepsilon(x_0)|/(m - \theta) + \theta$ by (J.80), while $f_0'(x_0) \ge m$ and $f'(\xi) \ge m - \theta$. Substituting gives (J.81). The final assertion follows since then $\theta \to 0$ and $|\varepsilon(x_0)| \to 0$. ☐

Remark J.52 (Formalization of the transport bound). Part (1), (J.80), is machine-checked in `RemainderTransport.lean`: `Fabius.transport_bound`, with the division-free form `Fabius.transport_bound_mul` beneath it. It is built on the lower mean-value bracket `Fabius.mul_abs_sub_le_abs_sub_of_le_deriv` of `MeanValueBracket.lean`, which is the formal content of Theorem J.47, so the two results compose in Lean the way they do on the page.

Formalizing it exposes an asymmetry between the two hypotheses that the statement above hides behind the common phrase "derivative bound". The hypothesis on f_0 is a one-sided bracket, $f_0' \geq m$, and the lower mean-value bracket applies to it with no sign condition. The hypothesis on ε is two-sided, $|\varepsilon'| \leq \theta$, and it is *not* usable as a bracket: the corresponding bracket is $-\theta \leq \varepsilon' \leq \theta$, whose lower end is negative, and the upper mean-value bracket requires its lower end to be nonnegative — necessarily so, as Theorem L.6 records. What $|\varepsilon'| \leq \theta$ delivers is a Lipschitz bound, a different theorem with a different proof, and that is what the argument actually uses when it writes $|\varepsilon(x)| \leq |\varepsilon(x_0)| + \theta|x - x_0|$. In Lean the two enter as `Fabius.mul_abs_sub_le_abs_sub_of_le_deriv` and `Fabius.lipschitzOn_of_abs_deriv_le` respectively.

The bound is close to sharp, not merely valid: over four hundred random instances with ε a sinusoid of exactly the permitted Lipschitz constant, the ratio of the true displacement to the bound reached 0.9999. So the denominator $m - \theta$ is the right one and cannot be improved to m.

Part (2), (J.81), is formalized as well, as `Fabius.transport_first_order` — and formalizing it showed that it does *not* need the second-order Taylor estimate the proof above reaches for. The mean value point already gives

$$x - x_0 = -\frac{\varepsilon(x_0)}{f'(\xi)}, \tag{J.82}$$

which is *exact*, so the whole of E is the mismatch between $f'(\xi)$ and $f_0'(x_0)$ — a perturbation $\varepsilon'(\xi)$ and a curvature $M_2|\xi - x_0|$. Two consequences. The hypothesis $f_0 \in C^2$ is not used: what the argument consumes is a Lipschitz bound on f_0', which is weaker, and which `Fabius.lipschitzOn_of_abs_deriv_le` supplies from $|f_0''| \leq M_2$. And part (1) is not an input to part (2): the same ξ that gives (J.82) gives (J.80) too, so the two parts are siblings rather than a chain.

The bound is close to sharp. Over four hundred random instances with f_0 of controlled curvature and ε a sinusoid at exactly the permitted Lipschitz constant, $|E|$ reached 0.92 of the stated bound, mean 0.28, with no violations.

Still not formalized: the closing asymptotic clause, that $\varepsilon(x) = \mathcal{O}(x^{-M-1})$ and $\varepsilon'(x) = o(1)$ give $x - x_0 = -\varepsilon(x_0)/f_0'(x_0)\,(1 + o(1))$.

Corollary J.53 (Forward truncation to inverse accuracy). *Let F be differentiably of exponential–power type with data $(C, a, 1, b, 1, Q)$, strictly increasing on $[x_1, \infty)$ with $(\log F)' \geq m > 0$ there. Let x_* solve exactly the truncated equation*

$$\log C + ax_* + b\log x_* + \sum_{j=1}^{M} q_j x_*^{-j} = \log y.$$

Then $|x_* - F^{-1}(y)| \leq K_M x_*^{-M-1}/m$ with K_M the constant of (J.3).

Proof. The residual of x_* in $\log F$ is exactly the forward remainder, of modulus at most $K_M x_*^{-M-1}$; apply Theorem J.47. □

J.8.3 Least-term truncation and the beyond-all-orders floor

Theorem J.54 (Optimal truncation and the exponential floor). *Let $A > 0$, $\beta \in \mathbb{R}$, $K > 0$, $x_0 \geq 1$, and let $\varphi : [x_0, \infty) \to \mathbb{R}$ satisfy the* Gevrey-1 remainder bound

$$\left|\varphi(x) - \sum_{j=1}^{M} q_j x^{-j}\right| \leq K\,\Gamma(M + \beta)\,A^{-M} x^{-M} \qquad (M \geq 1,\ x \geq x_0). \tag{J.83}$$

1. Least-term envelope. *There are* $x_1 \geq x_0$ *and* $K' = K'(K, A, \beta)$ *such that, choosing* $M = M(x) := \lfloor Ax \rfloor$*,*

$$\Big|\varphi(x) - \sum_{j=1}^{M(x)} q_j x^{-j}\Big| \;\leq\; K'\, x^{\beta-1/2}\, \mathrm{e}^{-Ax} \qquad (x \geq x_1). \tag{J.84}$$

2. Transported inverse error. *Let F be as in Theorem J.53, with* $\log F(x) = \log C + ax + b\log x + \varphi(x)$ *and* $(\log F)' \geq m > 0$ *on* $[x_1, \infty)$*. If x_* solves the optimally truncated equation exactly, then*

$$\big|x_* - F^{-1}(y)\big| \;\leq\; \frac{K'}{m}\, x_*^{\beta-1/2}\, \mathrm{e}^{-Ax_*}. \tag{J.85}$$

3. The floor is real. *Suppose in addition that there are* $M_0 \geq 1$ *and* $0 < k \leq K''$ *with*

$$k\,\Gamma(M+\beta)A^{-M} \;\leq\; |q_M| \;\leq\; K''\,\Gamma(M+\beta)A^{-M} \qquad (M \geq M_0). \tag{J.86}$$

Then

$$\log \min_{M \geq M_0} \big|q_M\big| x^{-M} \;=\; -Ax + \mathcal{O}(\log x) \qquad (x \to \infty), \tag{J.87}$$

so that no *truncation of the algebraic series achieves an error of smaller exponential order than* e^{-Ax}*.*

Proof. (1) Stirling's formula gives constants C_β and $M_\beta \geq 1$ with $\Gamma(M+\beta) \leq C_\beta M^{M+\beta-1/2}\mathrm{e}^{-M}$ for all $M \geq M_\beta$. With $M = \lfloor Ax \rfloor$ and x so large that $M \geq M_\beta$,

$$\Gamma(M+\beta)A^{-M}x^{-M} \leq C_\beta M^{\beta-1/2}\mathrm{e}^{-M}\left(\frac{M}{Ax}\right)^M \leq C_\beta M^{\beta-1/2}\mathrm{e}^{-M},$$

because $M \leq Ax$. Since $M > Ax - 1$, we have $\mathrm{e}^{-M} < \mathrm{e}\cdot\mathrm{e}^{-Ax}$, and $M^{\beta-1/2} \leq C'_\beta x^{\beta-1/2}$ for $x \geq x_1$ (with C'_β depending only on A and β, since $Ax - 1 \leq M \leq Ax$). Combining with (J.83) gives (J.84).

(2) The residual of x_* in $\log F$ is exactly the left-hand side of (J.84) evaluated at x_*; apply Theorem J.47.

(3) Enlarging M_0 we may assume $M_0 \geq M_\beta$. Write $u := Ax$ and $g(M) := (M/u)^M\mathrm{e}^{-M}$, so that by Stirling there are $0 < c_\beta \leq C_\beta$ with

$$c_\beta M^{\beta-1/2}g(M) \;\leq\; \Gamma(M+\beta)(Ax)^{-M} \;\leq\; C_\beta M^{\beta-1/2}g(M) \qquad (M \geq M_\beta). \tag{J.88}$$

Upper bound. Take $M = \lfloor Ax \rfloor$ in the right-hand inequality of (J.86) and repeat the estimate of part (1); this gives $\min_{M \geq M_0} |q_M|x^{-M} \leq K'''x^{\beta-1/2}\mathrm{e}^{-Ax}$ once $\lfloor Ax \rfloor \geq M_0$.

Lower bound. Note first that $\log g(M) = M\log(M/u) - M$ has derivative $\log(M/u)$, so g decreases on $(0, u)$, increases on (u, ∞), and

$$g(M) \;\geq\; g(u) = \mathrm{e}^{-u} = \mathrm{e}^{-Ax} \qquad (M > 0). \tag{J.89}$$

Set $h(M) := M^{\beta-1/2}g(M)$; then $(\log h)'(M) = (\beta - \frac{1}{2})/M + \log(M/u)$, which is positive as soon as $M \geq 2u$ and $2u \geq 2|\beta - \frac{1}{2}|/\log 2$, so h is increasing on $[2u, \infty)$ for all large x. Consequently

$$\min_{M \geq M_0} h(M) = \min_{M_0 \leq M \leq 2u} h(M) \;\geq\; \Big(\min_{M_0 \leq M \leq 2u} M^{\beta-1/2}\Big)\mathrm{e}^{-Ax} \;\geq\; c\, x^{-|\beta-1/2|}\, \mathrm{e}^{-Ax}$$

for $x \geq x_1$, using (J.89) and $\min_{M_0 \leq M \leq 2u} M^{\beta-1/2} \geq \min\big(M_0^{\beta-1/2}, (2Ax)^{\beta-1/2}\big)$. Combining with the left-hand inequalities of (J.86) and (J.88), $|q_M|x^{-M} \geq k\, c_\beta\, c\, x^{-|\beta-1/2|}\mathrm{e}^{-Ax}$ for every $M \geq M_0$. Taking logarithms of the two bounds gives (J.87). □

Remark J.55 (The least-term mechanism, formalized without Stirling). The three displayed conclusions of Theorem J.54 all rest on Stirling's formula, and none of them is formalized here. What *is* formalized, in `LeastTermIndex.lean`, is the mechanism underneath them: why $M(x) = \lfloor Ax \rfloor$ is the right truncation index, and why there is no better one further out. That part needs no asymptotics at all.

Write $T(M) = \Gamma(M+\beta)/s^M$ with $s = Ax$, the profile of the coefficient bound in (J.83). Then

$$\frac{T(M+1)}{T(M)} = \frac{M+\beta}{s}, \tag{J.90}$$

which is `Fabius.gevreyTerm_succ` — one application of $\Gamma(z+1) = z\Gamma(z)$. Everything follows from it and the positivity of Γ on $(0,\infty)$: the terms strictly decrease while $M+\beta < s$ (`Fabius.gevreyTerm_succ_lt`) and strictly increase once $M+\beta > s$ (`Fabius.gevreyTerm_lt_succ`), so the profile is unimodal with its minimum exactly where $M+\beta$ crosses Ax. The two monotone stretches are `Fabius.gevreyTerm_antitoneOn` and `Fabius.gevreyTerm_monotoneOn`.

Two things this makes precise. First, the choice $M(x) = \lfloor Ax \rfloor$ is the crossing index up to the shift β, and it is optimal for an *exact* reason, not an asymptotic one — the ratio law is an identity. Second, unimodality is what rules out a second, lower, minimum further out; without it "least term" would name a local rather than a global choice, and the floor (J.87) would not follow from a single truncation. The volume uses both facts implicitly.

Not formalized: the size of the least term — the envelope (J.84), the transported error (J.85), and the floor (J.87). All three need Stirling asymptotics for $\Gamma(M+\beta)$, which is exactly the content this module does without.

Remark J.56 (What may and may not be appended). Theorem J.54 bounds the size of the optimal remainder; it does not produce it. In particular, adding a term $c\,\mathrm{e}^{-Ax}$ with an arbitrarily chosen constant c to the divergent series is *not* a valid transseries completion. The double factorial and swing sources make the point explicitly: on the positive real axis the Borel singularities of these phases lie at $\pm\mathrm{i}\pi m$, so the exponentially small sectors are terminant-weighted combinations of $\mathrm{e}^{\mp\mathrm{i}\pi m x}$, whose optimally truncated magnitude is $\mathrm{e}^{-\pi x}$ but which are not real exponentials. The unambiguous completion on the positive axis is the exact Borel–Laplace representation, when the chapter provides one. Where a chapter does supply a rigorously normalized exponential sector, its transport through the inverse map is Theorem J.35 at first order and Theorem J.51 with an explicit error.

J.9 The parameter dictionary

Table J.1: Model data of the four chapters, in the normal form of Theorem J.2 and after the α-reduction of Theorem J.4. "Core" names the admissible core of Theorem J.8 and the Lambert branch selected by Theorem J.25. The row for Part N is *not* an instance of Theorem J.2; see the repair box in Section J.2.

Chapter	sequence	variable x	α	b	core, branch
M	rooted trees, A000081	n	1	$-3/2$	$ax + b\log x$, W_{-1}
N	double factorial, A006882	$n+1$	—	—	$\frac{x}{2}(\log x - 1)$, W_0
O	partitions, A000041	$n - \frac{1}{24}$	1/2	-1	$ax^{1/2} + b\log x$, W_{-1}
P	swing factorial, A056040	n	1	$\sigma/2$	$ax + b\log x$, W_0 or W_{-1}

The constants a are: $a = \log\alpha_{\text{Otter}}$ for Part M; $a = \pi\sqrt{2/3}$ for Part O, in which $\alpha = 1/2$ and Theorem J.4 substitutes $\xi = (n - 1/24)^{1/2}$, after which $b/\alpha = -2$ and the core equation is $\xi \mapsto a\xi - 2\log\xi$ — the equation the three partition articles write as $\rho - 2\log\rho = L$ after rescaling a to 1; and $a = \log 2$ for Part P, with $\sigma = (-1)^{n+1}$ the parity transmonomial. Each chapter verifies (J.4) directly and states its own q_j.

Remark J.57 (Reproducibility of the Part 0 identities). Every coefficient identity of this chapter was checked in exact rational arithmetic, with no floating-point step, over six independent choices of the data $(a, b, q_1, \ldots, q_N)$ with $a, b, q_j \in \mathbb{Q}$, $N \in \{10, 12\}$, using a truncated-series engine over $\mathbb{Q}[H]$ built on Python's `fractions.Fraction`. The identities checked were: the Lagrange closed formula (J.49) against the defining fixed point; the ordinary-Bell recurrence (J.50); the vanishing of the residual $ax + b \log x + Q(1/x) - L$ under the flattened ansatz (J.66) through order L^{-N}; the shape statements (J.65); and the closed form (J.68) together with its agreement with the three source formulae listed in Theorem J.39. All checks passed. No numerical constant is asserted anywhere in this chapter.

Appendix K

Concordance: the calculus and the inversion apparatus

K.1 Why this chapter exists

The calculus parts of this volume and the inversion apparatus of Part J were written independently, as separate consolidations of separate groups of articles, and the group they came from recorded as an open question whether their machinery coincides. This chapter answers it, pair by pair.

The answer is more interesting than either "yes" or "no". A crude comparison — matching the two apparatuses by the titles of their results — suggests heavy duplication: Lagrange–Bürmann appears on both sides, so do Bell polynomials, so does residual-to-error transfer, so does a closed-form Lambert block. On inspection, most of those pairs are *not* the same statement. They are different theorems about the same subject, and the volume needs both. Exactly one pair is a genuine duplicate, and one pair is a strict strengthening. Table K.1 records the verdicts and the rest of this chapter justifies them.

Table K.1: Apparent overlaps between the calculus parts and Part J, with the verdict for each.

Calculus	Inversion apparatus	Verdict
Theorem A.1 exponential partial and complete Bell polynomials	Theorem J.10 exponential partial Bell polynomials	**same definition**; see Theorem K.1
— (no counterpart)	Theorem J.9 *ordinary* partial Bell polynomials	**new**; the calculus defines only the exponential family
Theorem 16.21 analytic residual transfer	Theorem J.47 residual-to-root conversion	**strengthening**; see Theorem K.3
Theorem 18.14 and Theorem 26.18, near-identity operator form	Theorem J.17, classical coefficient form	**different statements**; Theorem K.5
Theorem 22.7 reversion of $X + aL$ by Lambert polynomials	Theorem J.38 the pure Lambert block in closed form	**different objects**; Theorem K.6
Theorem 6.14 slope transport	Theorem J.51 remainder transport	**different**; one is a leading-order rule, the other a bound with an explicit second-order error

K.2 The one genuine duplicate

Proposition K.1 (The two Bell definitions coincide). *For all $n \geq k \geq 0$ and all indeterminates $x_1, x_2, \ldots,$*

$$B_{n,k}(x_1, x_2, \ldots) = \mathcal{B}^{\exp}_{n,k}(x_1, x_2, \ldots), \tag{K.1}$$

and the two complete polynomials agree as well. That is, Theorem J.10 and Theorem A.1 define the same object in different notation.

Proof. Equation (A.1) reads

$$\mathcal{B}^{\exp}_{n,k} = \sum \frac{n!}{\prod_j m_j!\,(j!)^{m_j}} \prod_j x_j^{m_j},$$

summed over (m_j) with $\sum_j m_j = k$ and $\sum_j jm_j = n$, while (J.12) reads

$$B_{n,k} = n! \sum \prod_j \frac{1}{m_j!} \Big(\frac{x_j}{j!}\Big)^{m_j} = \sum \frac{n!}{\prod_j m_j!\,(j!)^{m_j}} \prod_j x_j^{m_j}$$

over the same index set. The summands are identical term by term. Both sources then define the complete polynomial as $\sum_{k=0}^{n}$ of the partial one, so those agree too. ☐

Lean status: the question does not arise in the corpus, and that is itself the answer. There is one partial Bell polynomial in the corpus, `Fabius.partialBell`, and it is defined by neither of the multinomial sums compared here: its definition is the block-of-the-largest-element recurrence (A.9). What ties it to *both* sides of (K.1) is that the corpus proves the generating characterizations that each of Theorem J.10 and Theorem A.1 offers as its own characterization: `Fabius.bellWeightSeries_pow` is (J.13) and (A.5) at once, and `Fabius.exp_subst_bellWeightSeries` is (J.14) and (A.6) at once. Since `Fabius.partialBell_eq_of_bellWeightSeries_pow_eq_egf` shows those identities determine the coefficients, any object satisfying either characterization is `Fabius.partialBell`, which is the content of the proposition.

The gap to be honest about: what is *not* formalized is the equivalence of the multinomial sum with the recurrence, in either notation. The corpus takes the recurrence as primitive and derives the generating identity from it; neither (J.12) nor (A.1) appears formally. So the formal statement is that the two *characterizations* pin down one object, not that the two displayed sums are termwise equal.

Remark K.2 (What is done about it). Nothing is deleted. Theorem J.10 additionally records the two generating identities (J.13) and (J.14), which the later chapters use constantly and which Theorem A.1 does not state; and the calculus parts use the long normative names throughout, while the inversion chapters use the short B. Collapsing one into the other would cost more in readability than the duplication costs in length. What Theorem K.1 buys is that a reader who meets both knows they are the same object and may use a result about either.

K.3 The one strengthening

Proposition K.3 (The inversion apparatus proves more). *Theorem 16.21 is the implication*

$$|f'| \geq m > 0 \text{ between } x_N \text{ and } x_* \implies |x_N - x_*| \leq \frac{|f(x_N) - y|}{m}, \tag{K.2}$$

and is exactly the upper bound of Theorem J.47. The latter proves two further things: the matching lower *bound $|r_*|/M \leq |x_* - x_\dagger|$ under a two-sided slope bracket $m \leq f' \leq M$, and that the residual is a* certificate *— if $[x_* - \varrho, x_* + \varrho]$ lies in the interval, with $\varrho = |r_*|/m$, then a root exists within that distance, so no separate existence argument is needed. Hence Theorem 16.21 is a corollary of Theorem J.47, and not conversely.*

Proof. For the first sentence, take $M = \infty$ in Theorem J.47, or read its proof, which is the same mean-value argument. The lower bound and the existence clause are proved there and are not stated in Theorem 16.21; that the converse fails is immediate, since (K.2) says nothing when f' is unbounded above and asserts no existence. □

Lean status. Formal, and the comparison the proposition draws is visible in the module structure. The one-sided implication (K.2) is `Fabius.abs_sub_right_inverse_le_div` in `MeanValueBracket.lean`, and the matching lower bound under a two-sided bracket is `Fabius.div_le_abs_sub_right_inverse`; both are stated for an arbitrary function on an arbitrary convex set, with the exact solution supplied as a right inverse.

The third clause — that the residual is a *certificate*, so that no separate existence argument is needed — is `BackwardErrorExistence.lean`, and it is the one place where the two statements genuinely differ rather than merely differing in strength. `Fabius.exists_eq_of_residual` produces the root: if f is continuous on $[a, b]$, its increments grow at least at rate $m > 0$ (`Fabius.HasSlopeLowerBound`), and the ball of radius $|r|/m$ about the candidate lies in $[a, b]$, then $f = y$ somewhere in that ball. `Fabius.exists_eq_and_abs_sub_le` is (K.2) with the existence hypothesis *discharged* rather than assumed, which is precisely the proposition's claim, and `Fabius.existsUnique_eq_of_residual` adds uniqueness, since `Fabius.injOn_of_hasSlopeLowerBound` makes a positive slope bound force injectivity.

Two remarks on the formal shape. The hypothesis is a slope condition, not a derivative bound, so no differentiability is needed anywhere in the existence argument; `Fabius.hasSlopeLowerBound_of_le_deriv` and `Fabius.exists_eq_of_le_deriv` recover the differentiable case from the mean value theorem. And the sign of the residual does more work here than it does in Theorem L.4: there it is diagnostic, telling one whether the candidate overshoots; here it *selects the half-interval* on which the intermediate value theorem is applied, since the slope bound guarantees that f has already crossed y by the time it reaches the far end of that half.

Remark K.4 (Where the difference is used). The lower bound is what makes a residual a two-sided measurement rather than a one-sided guarantee, and Theorem L.4 uses exactly that: for $x + W(x)$ the slope bracket is the constant pair $(1, 2)$, so the residual determines the error to within a factor of two, uniformly in z. The existence clause is used in the proofs of Theorems Q.25 and Q.33, where it removes the obligation to prove separately that the truncated expansion is near an actual root.

K.4 The pairs that only look alike

Remark K.5 (Lagrange–Bürmann, twice, differently). The calculus proves the *operator* form: for a near-identity map $F = X + A$ in a reversion frame,

$$F^{\langle -1\rangle \circ} = X + \sum_{r \geq 1} \frac{(-1)^r}{r!} \partial^{r-1}\big(A^r\big),$$

at infinity in Theorem 18.14 and at a finite point in Theorem 26.18, in both cases with a summability statement that is the real content. Theorem J.17 proves the *coefficient* form: if $W = t\varphi(W)$ with $\varphi(0)$ invertible then $[t^n]\, G(W) = n^{-1}[w^{n-1}]\,(G'\varphi^n)$.

These are the two classical faces of the same theorem, and each is the natural tool for a different job: the operator form reverses a perturbation of the identity, which is what the calculus parts need; the coefficient form extracts a single coefficient of a composite, which is what the reversion recurrences of the application chapters need. Deriving either from the other is a genuine argument, not a rewriting, so neither is redundant and neither proof is removed. What *is* avoided is a third copy: the application chapters cite Theorem J.17 and the $x + W(x)$ chapter cites the calculus, rather than either re-deriving.

Remark K.6 (Affine-logarithmic reversion is not the Lambert block). Theorem 22.7 reverses $X + aL$, a perturbation of the identity by a multiple of the logarithm, and its answer is a Lambert-polynomial series

$X - aL + \sum_{n\geq 1} a^{n+1} \mathsf{L}_n^{\mathrm{Lam}}(L) t^n$. Theorem J.38 evaluates the pure Lambert block of the exponential–power model in closed form. The first is a statement about the near-identity group; the second is a closed-form evaluation of a specific reversion. They meet only in that both involve W, and neither implies the other.

K.5 What this comparison changes

Remark K.7 (A correction to a claim made earlier in this merge). An earlier stage of this consolidation, working from the title-level concordance alone, recorded that the calculus "already contains" Lagrange–Bürmann at infinity and at a finite point, residual-to-error transfer, both Bell families, affine-logarithmic reversion and the pure Lambert block, and that the inversion apparatus was therefore largely redundant. That overstated the overlap, and this chapter is the correction. The accurate statement is the one in Table K.1: of six apparent overlaps, one is a genuine duplicate of a *definition*, one is a strengthening in the inversion apparatus's favour, one item exists only in the inversion apparatus, and the remaining three are distinct theorems that the volume needs separately.

The moral is worth keeping, because the same trap recurs whenever two independently written treatments are compared: shared vocabulary is a weak signal. Two results called "Lagrange–Bürmann" may be different theorems, and a mechanical concordance will report them as the same. Only reading the statements settles it.

Remark K.8 (What the inversion apparatus really adds). Setting the false positives aside, the apparatus of Part J contributes, over and above the calculus: the exponential–power model and its axiomatized dominant core; the monomial α-reduction that brings a fractional phase into the linear case; perturbed inversion around an exactly invertible core, in both an analytic and a formal-jet form; the ordinary partial Bell polynomials and their conversion to the exponential family; the two-sided backward-error certificate; and — with no analogue anywhere in the calculus — the theory of inverting a *sequence*: the three distinct inverse objects, the staircase theorem relating them, and the separation condition under which an asymptotic inverse determines an integer one. The calculus is a theory of functions on a scale; that last group of results is about the passage from a function to a sequence, and it is the reason the two halves of this volume are complementary rather than redundant.

Part XI

One map in depth: reversing $x + W(x)$

Appendix L

Reversing $x + W(x)$

L.1 What this chapter does

The map

$$f(x) = x + W(x), \qquad x \geq 0, \tag{L.1}$$

is the smallest interesting perturbation of the identity by a logarithm. It is strictly increasing from $[0, \infty)$ onto itself, so its inverse $g = f^{-1}$ is globally defined; and because $W(x) = \mathcal{O}(\log x) = o(x)$, reversing it looks like a routine near-identity problem of the kind the reversal part of this volume settles in general.

It is worth a chapter of its own for one reason. *Three* scales occur, and the middle one hides the interesting behaviour:

1. the leading scale z;
2. the slow logarithmic scale generated by $\ell_1 = \log z$ and $\ell_2 = \log \ell_1$;
3. the scale $q = W(z)/z = \mathrm{e}^{-W(z)}$, whose powers form separate transseries sectors.

A computation carried out only in powers of $1/\log z$ determines g to *every* inverse-logarithmic order and still cannot see the first genuine correction, which is of size q. That is the phenomenon Theorem L.9 makes explicit, and it is the same ordering caveat that Theorem Q.41 raises for the Lambert-cored inversions and that the scale part of this volume states abstractly.

Three filed articles treat this map: `lambert_inverse_transseries` (L1), `lambert_inverse_transseries_bundle` (L2) and `reversing_x_plus_lambert_w_transseries` (L3). Between them they state 27 named results, of which 18 are already theorems of the calculus developed in the earlier parts of this volume — near-identity reversion, Lagrange–Bürmann at infinity, triangular reversal, Newton acceleration, residual-to-error transfer, the r-Lambert function, and the pure polynomial–logarithmic perturbation lemma among them. Those are cited, not restated. What remains, and what this chapter contains, is the mathematics specific to $x + W(x)$: the exact reduction, the sharp residual bracket, the explicit coefficient blocks and their integer-polynomial structure, and the ordering of the three scales.

L.2 Exact inversion

Theorem L.1 (Exact inverse)**.** *For every $z \geq 0$,*

$$g(z) = z - \mathrm{W}_1(z) = \mathrm{W}_1(z)\,\mathrm{e}^{\mathrm{W}_1(z)}, \tag{L.2}$$

where W_1 is the r-Lambert function of Theorem R.16 at $\varrho = 1$, that is the inverse of $y \mapsto y\mathrm{e}^y + y$. Equivalently, with $y = \mathrm{W}_1(z)$ one has $g(z) = y\mathrm{e}^y$ and $W(g(z)) = y$.

Proof. Put $y = W(x)$, so that $x = ye^y$. Then $z = x + W(x)$ reads $z = ye^y + y$, which is exactly the defining equation of W_1; hence $y = W_1(z)$ and $x = z - y$. The second form is $x = ye^y$, and $W(g(z)) = W(x) = y$. On $[0, \infty)$ the map $y \mapsto ye^y + y$ is strictly increasing from 0, so the principal real branch is the relevant one throughout. □

Remark L.2 (Formalization status). Theorem L.1 is machine-checked in this repository. In `LambertShiftInverse.lean`, W_1 is `Fabius.shiftedLambertW` (the inverse on $[0, \infty)$ of `Fabius.shiftedMulExp`, $u \mapsto ue^u + u$) and g is `Fabius.lambertShiftInv`; the three identities of (L.2) are `Fabius.lambertShift_lambertShiftInv`, `Fabius.lambertShiftInv_eq_mul_exp` and `Fabius.principalLambertW_lambertShiftInv`, and g is the two-sided inverse of f by `Fabius.lambertShiftInv_lambertShift` together with `Fabius.lambertShift_eq_iff`.

The differential layer is formal as well. g is strictly increasing and continuous on the open half-line (`Fabius.lambertShiftInv_strictMonoOn`, `Fabius.lambertShiftInv_continuousAt`, the latter read off the exact image `Fabius.lambertShiftInv_image_Ioi`), so the inverse-function rule applies: `Fabius.lambertShiftInv_hasDerivAt` differentiates g at every $z > 0$, and `Fabius.deriv_lambertShiftInv_bounds` is the strict bracket $1/2 < g'(z) < 1$, both inequalities strict because $0 < W' < 1$ once $W > 0$ (`Fabius.inv_exp_mul_add_one_lt_one`). From the strict growth of g' (`Fabius.deriv_lambertShiftInv_strictMonoOn`) it follows that g is strictly convex there, `Fabius.strictConvexOn_lambertShiftInv`. The endpoint value $g'(0) = 1/2$, a one-sided derivative, is not formalized, and neither is the concavity of f as such — what the corpus carries is the strict concavity of the branch itself, `Fabius.strictConcaveOn_principalLambertW`, proved from $W'' < 0$.

Within the present volume, this chapter's formal counterparts include this theorem and its differential layer, the numerator recursion of Theorem L.11, and the general bracket of Theorem L.5 — and they are recorded alongside the formal counterparts now identified in other chapters because the corpus keeps a register of what is proved and what is merely written down.

Proposition L.3 (Brackets). *For $x \geq 0$ and $z \geq 0$,*

$$1 \leq f'(x) \leq 2, \qquad \tfrac{1}{2} \leq g'(z) < 1, \qquad z - W(z) \;\leq\; g(z) \;\leq\; z. \tag{L.3}$$

Moreover f is concave and g convex on $(0, \infty)$.

Proof. Differentiating $W(x)e^{W(x)} = x$ gives

$$W'(x) = \frac{W(x)}{x\big(1 + W(x)\big)} = \frac{e^{-W(x)}}{1 + W(x)} \in (0, 1], \tag{L.4}$$

with the limiting value 1 at $x = 0$; hence $1 \leq f' = 1 + W' \leq 2$ and, by the inverse-function rule, $\frac{1}{2} \leq g' < 1$. For the bracket, the defining relation may be written $g(z) = z - W(g(z))$; since $0 \leq g(z) \leq z$ and W increases, $W(g(z)) \leq W(z)$, which gives the lower bound. Finally $W'' < 0$ on $(0, \infty)$ makes f concave, and the inverse of a concave increasing function is convex. □

Lean status. Formal in `LambertShiftInverse.lean` and `LambertShiftConcavity.lean`, with every inequality *strict* in the interior, which is more than the display asserts. The bracket on f' is `Fabius.one_le_deriv_lambertShift` and `Fabius.deriv_lambertShift_le_two`; the bracket on g' is `Fabius.deriv_lambertShiftInv_bounds`, in the strict form $\frac{1}{2} < g'(z) < 1$ for $z > 0$. The envelope $z - W(z) \leq g(z) \leq z$ is `Fabius.sub_principalLambertW_lt_lambertShiftInv` and `Fabius.lambertShiftInv_lt_self`, again strictly for $z > 0$; at $z = 0$ all three quantities are 0, so the weak form stated here is the correct one at the endpoint and the strict form is the correct one off it.

The endpoint derivatives $f'(0) = 2$ and $g'(0) = \frac{1}{2}$ are one-sided and are not formalized. What replaces them, and is in some ways better, is the finite-difference form of the same bracket, which needs no differentiability at 0 and holds on the closed half-line: `Fabius.sub_le_lambertShift_sub` and `Fabius.lambertShift_sub_le_two_mul_sub`, with the absolute-value versions `Fabius.abs_sub_le_abs_lambertShift_sub` and `Fabius.abs_lambertShift_sub_le_two_mul`. These are what the certificate of Theorem L.4 is actually built from.

Convexity of g is `Fabius.strictConvexOn_lambertShiftInv`, obtained from the strict growth of g' rather than from the concavity of f. Concavity of f is `Fabius.strictConcaveOn_lambertShift`, and it holds on the *full* closed domain $[-\mathrm{e}^{-1}, \infty)$ of the principal branch, not merely on $(0, \infty)$: adding the identity to a strictly concave function leaves it strictly concave, the linear part contributing an equality to the concavity inequality, so no positivity enters. The restriction the volume states is `Fabius.strictConcaveOn_lambertShift_Ici_zero`.

Proposition L.4 (A sharp residual certificate). *Let $\widetilde{g}(z) \geq 0$ be any approximation and put*

$$R(z) = \widetilde{g}(z) + W\big(\widetilde{g}(z)\big) - z. \tag{L.5}$$

Then R and $\widetilde{g} - g$ have the same sign, and

$$\tfrac{1}{2}\,|R(z)| \;\leq\; \big|\widetilde{g}(z) - g(z)\big| \;\leq\; |R(z)|. \tag{L.6}$$

Proof. This is Theorem J.47 with f in place of the general map, sharpened by the two-sided bound $1 \leq f' \leq 2$ of Theorem L.3: the mean value theorem applied to $f(\widetilde{g}) - f(g) = R$ gives $R = f'(\xi)(\widetilde{g} - g)$ for some intermediate ξ, and dividing by $f'(\xi) \in [1, 2]$ gives both inequalities and the equality of signs. □

Lean status. Formal, both clauses, in `LambertShiftInverse.lean` with the volume's display assembled in `LambertShiftConcavity.lean`. The sandwich (L.6) is `Fabius.abs_sub_lambertShiftInv_sandwich`, whose two halves are `Fabius.abs_sub_lambertShiftInv_le_abs_residual` (the upper bound, from $f' \geq 1$) and `Fabius.abs_residual_le_two_mul_abs_sub_lambertShiftInv` (the lower, from $f' \leq 2$). The sign agreement is `Fabius.residual_sign_agrees`, packaging `Fabius.residual_pos_iff` and `Fabius.residual_neg_iff`, with the exact case `Fabius.residual_eq_zero_iff`: a vanishing residual pins the candidate at $g(z)$ and nowhere else, so the certificate is sharp at the root as well as away from it.

The proposition is stated for an *arbitrary* approximation $\widetilde{g} \geq 0$, and so is the formal version: the hypotheses are $z \geq 0$ and $\widetilde{g}(z) \geq 0$, nothing else. The underlying two-sided mean-value bracket is `MeanValueBracket.lean`, independent of the Lambert material — `Fabius.mul_abs_sub_right_inverse_le` and `Fabius.abs_sub_le_mul_abs_sub_right_inverse` are the general statements, and one hypothesis there is worth flagging because it is easy to get wrong: a two-sided slope bracket $m \leq f' \leq M$ gives $|f(x) - f(y)| \leq M|x - y|$ only when $0 \leq m$, the conclusion being false otherwise.

Remark L.5 (Why this one is two-sided). Theorem J.47 gives a two-sided bound whenever the slope is bracketed above and below, and here the bracket is the best possible constant pair $(1, 2)$, independent of z. That is unusual: in every Lambert-cored chapter of this volume the slope grows, so the residual buys a bound that improves with z but with a z-dependent constant. Here the certificate is uniform, which is what makes numerical work on this map unusually comfortable. The sandwich is machine-checked as well, as `Fabius.abs_sub_lambertShiftInv_le_abs_residual` and `Fabius.abs_residual_le_two_mul_abs_sub_lambertShiftInv`; and so is the general two-sided statement, for an arbitrary function on a convex set, in `MeanValueBracket.lean` — the brackets are `Fabius.mul_abs_sub_le_abs_sub_of_le_deriv` and `Fabius.abs_sub_le_of_le_deriv`, and the transfer against a right inverse is `Fabius.abs_sub_right_inverse_le_div` together with `Fabius.div_le_abs_sub_right_inverse`.

Remark L.6 (The sign hypothesis is not decoration). Theorem J.47 assumes $0 < m \leq f' \leq M$, and the positivity of m is needed for the *upper* bracket, not merely for the interpretation. Dropping it fails at once: with $f' \equiv -10$ the bracket $m \le f' \le M$ holds for $m = -10$, $M = 0$, while $|f(x) - f(y)| = 10|x - y|$ exceeds $M|x - y| = 0$. What the sign supplies is monotonicity, which is what makes $|f(x) - f(y)|$ the signed increment. The lower bracket needs no sign condition. This is worth stating because a theorem quoted as "f strictly increasing with $m \le f' \le M$" leaves the role of the sign implicit, and the two-sided form is the one this volume uses most.

L.3 The small parameter, and reduction to an ordinary reversion

Set

$$L = W(z), \qquad z = Le^L, \qquad q = e^{-L} = \frac{L}{z}, \tag{L.7}$$

and write $\mathrm{W}_1(z) = L - v$, so that by Theorem L.1

$$g(z) = z - L + v. \tag{L.8}$$

Proposition L.7 (The exact equation for the correction)**.** *The quantity v of* (L.8) *is characterized exactly by*

$$q = B_L(v) := \frac{L}{L - v} - e^{-v}, \tag{L.9}$$

and $B_L(0) = 0$ with $B'_L(0) = 1 + L^{-1} > 0$. Writing $u = L^{-1}$,

$$B(v, u) = \frac{1}{1 - uv} - e^{-v} = \sum_{k \ge 1} b_k(u) v^k, \qquad b_k(u) = u^k - \frac{(-1)^k}{k!}, \tag{L.10}$$

so that $b_1(u) = 1 + u$. Consequently the large-z problem is an ordinary *analytic reversion at the origin, in the variable v with parameter u.*

Proof. Substituting $\mathrm{W}_1(z) = L - v$ into its defining equation $\mathrm{W}_1 e^{\mathrm{W}_1} + \mathrm{W}_1 = z = Le^L$ gives $(L - v)e^{L-v} + L - v = Le^L$; dividing by e^L and using $e^{-L} = q$ yields $(L - v)(e^{-v} + q) = L$, that is $q = L/(L - v) - e^{-v}$, which is (L.9). Then $B_L(0) = 1 - 1 = 0$ and $B'_L(v) = L/(L - v)^2 + e^{-v}$, so $B'_L(0) = L^{-1} + 1$. Expanding $(1 - uv)^{-1} = \sum_k u^k v^k$ and $e^{-v} = \sum_k (-1)^k v^k / k!$ and subtracting gives (L.10); the $k = 0$ terms cancel. □

Lean status. The analytic content is formal in `LambertCorrectionEquation.lean`, a module with no dependence on the rest of the corpus, since it is a statement about one explicit two-variable function. `Fabius.corrB` is $B(v, u) = (1 - uv)^{-1} - e^{-v}$ and `Fabius.corrCoeff` is $b_k(u) = u^k - (-1)^k/k!$. The two facts that make the reversion ordinary are `Fabius.corrB_zero`, that $B(0, u) = 0$, and `Fabius.hasDerivAt_corrB_zero`, that $\partial_v B(0, u) = 1 + u$ — a `HasDerivAt` statement, so the derivative is established and not merely computed from a `deriv` that might be junk; `Fabius.hasDerivAt_corrB` is the derivative at a general v with $uv \neq 1$, and `Fabius.deriv_corrB_zero_pos` the positivity for $u \ge 0$. The coefficient expansion (L.10) is `Fabius.corrB_eq_tsum`, proved for $|uv| < 1$ by subtracting the exponential series from the geometric one, with `Fabius.tsum_corrExpTerm` and `Fabius.summable_corrExpTerm` the supporting lemmas.

One point where the formal statement is cleaner than the display. The volume sums from $k = 1$, which requires knowing separately that the constant terms cancel; the formal sum runs over every $k \ge 0$ and needs no such side condition, because $b_0(u) = u^0 - (-1)^0/0! = 1 - 1 = 0$ identically — `Fabius.corrCoeff_zero`. So the vanishing of the constant term is a consequence of the coefficient formula rather than an extra hypothesis, and $B(0, u) = 0$ is that same $k = 0$ term read off. `Fabius.corrCoeff_one` records $b_1(u) = 1 + u$.

What is *not* formalized: that the v of (L.8) satisfies $q = B_L(v)$. That is the characterization the proposition is about, and it depends on the reduction leading to (L.8) rather than on the identity above; the module proves everything the proposition asserts *about* B, and nothing about where B came from.

Remark L.8 (What has been gained). Theorem L.7 is the whole trick. Before it, one faces a reversion at infinity across three scales; after it, an analytic function vanishing at the origin with invertible derivative is to be reverted, which is Theorem J.17 in its convergent form. The exponentially small parameter q has become the independent variable, and the slow parameter $u = 1/L$ rides along as a coefficient. This is the same manoeuvre as the substitution $u = \mathrm{e}^{-\lambda x}$ in Theorem T.6, and as the passage to the Lambert coordinate in Section Q.3: normalize until what is left is an ordinary reversion.

L.4 The coefficient blocks

Theorem L.9 (All-orders expansion)**.** *As $z \to +\infty$, with $L = W(z)$,*

$$g(z) = z - L + \sum_{n\geq 1} \frac{A_n(L)}{z^n}, \tag{L.11}$$

and the first four coefficients are

$$A_1(L) = \frac{L^2}{1+L}, \tag{L.12}$$

$$A_2(L) = \frac{L^3\big(L^2-2\big)}{2\,(1+L)^3}, \tag{L.13}$$

$$A_3(L) = \frac{L^4(2L-1)\big(L^3-6L-6\big)}{6\,(1+L)^5}, \tag{L.14}$$

$$A_4(L) = \frac{L^5\big(6L^6-8L^5-69L^4-40L^3+96L^2+72L-24\big)}{24\,(1+L)^7}. \tag{L.15}$$

Proof. By Theorem L.7 the correction v solves $B(v,u) = q$ with $B(0,u) = 0$ and $\partial_v B(0,u) = 1+u$ a unit, so Lagrange inversion (Theorem J.17, in the convergent analytic form) determines v as a power series in q whose coefficients are rational in $u = 1/L$; by (L.8) the same is true of $g(z) - z + L$. Converting q^n to $L^n z^{-n}$ and clearing u gives (L.11) with A_n rational in L. The displayed $A_1, \ldots, A_4$ follow from the first four Lagrange coefficients; they were recomputed independently for this volume and checked numerically against the exact inverse (Theorem L.18). ☐

Proposition L.10 (Integer-polynomial structure)**.** *For every $m \geq 1$ there is $P_m \in \mathbb{Z}[v]$ with*

$$A_m(v) = \frac{v^{m+1}P_m(v)}{m!\,(1+v)^{2m-1}}, \tag{L.16}$$

and

$$\deg P_m = 2m-2, \qquad [v^{2m-2}]\,P_m = (m-1)!, \qquad P_m(0) = (-1)^{m+1}m!. \tag{L.17}$$

Consequently

$$A_m(v) \sim \frac{v^m}{m} \qquad (v \to +\infty). \tag{L.18}$$

Proof. The shape (L.16) follows by induction from the Lagrange recursion of Theorem L.9: each step divides once by $b_1(u) = 1+u = (1+v)/v$ evaluated at $v = L$, contributing the powers of $(1+v)$, while the numerators stay integral because the b_k of (L.10) have denominators $k!$ only, absorbed by the $m!$. The three data in (L.17) are read off the same recursion: the top coefficient comes from the pure u^k part of b_k, the constant term from the pure e^{-v} part. For (L.18), substituting $P_m(v) \sim (m-1)!\,v^{2m-2}$ into (L.16) gives

$$A_m(v) \sim \frac{v^{m+1}\,(m-1)!\,v^{2m-2}}{m!\,v^{2m-1}} = \frac{v^m}{m}.$$

The four instances $P_1 = 1$, $P_2 = v^2 - 2$, $P_3 = 2v^4 - v^3 - 12v^2 - 6v + 6$ and $P_4 = 6v^6 - 8v^5 - 69v^4 - 40v^3 + 96v^2 + 72v - 24$ read off (L.12)–(L.15) satisfy (L.17) exactly, as recorded in Theorem L.18. ☐

Proposition L.11 (A recursion for the numerators). *Define $R_{n,k} \in \mathbb{Z}[v]$ by*

$$R_{n,1} = n, \qquad R_{n,k+1} = v(1+v)\,R'_{n,k} + \Big(n(1+v) - (2k-1)v - k(1+v)^2\Big) R_{n,k}. \tag{L.19}$$

Then $R_{n,k} \in n\,\mathbb{Z}[v]$, $\deg R_{n,k} = 2(k-1)$, $[v^{2(k-1)}]\,R_{n,k} = n(-1)^{k-1}(k-1)!$ and $R_{n,k}(0) = n(n-1)\cdots(n-k+1)$; and the numerators of Theorem L.10 are the diagonal

$$P_m = \frac{(-1)^{m+1}}{m+1}\,R_{m+1,m}. \tag{L.20}$$

The three data (L.17) *are exactly the three displayed facts read on that diagonal.*

Proof. Divisibility by n is immediate from (L.19) by induction, since $R_{n,1} = n$ and every step is $\mathbb{Z}[v]$-linear in $R_{n,k}$. For the degree, the derivative term $v(1+v)R'_{n,k}$ has degree at most $2 + \big(2(k-1)-1\big) = 2k-1$, strictly below the degree $2+2(k-1) = 2k$ of the multiplier term, whose leading coefficient is that of $-k(1+v)^2$ against $R_{n,k}$, namely $-k$ times $n(-1)^{k-1}(k-1)!$; and $-k\cdot(-1)^{k-1}(k-1)! = (-1)^k k!$, which is the claim at $k+1$. The value at $v = 0$ satisfies $R_{n,k+1}(0) = (n-k)R_{n,k}(0)$, giving the falling factorial. Substituting $n = m+1$, $k = m$ into the three then gives $\deg P_m = 2m-2$, $[v^{2m-2}]\,P_m = (m-1)!$ and $P_m(0) = (-1)^{m+1}m!$, which is (L.17); the normalization (L.20) is fixed by matching $P_1 = 1$. □

Remark L.12 (Formalization status of the numerator family). Theorem L.11 is machine-checked, in `LambertInverseCoefficients.lean`, a module of pure polynomial algebra with no dependence on the rest of the corpus. With indices shifted so the recursion starts at 0, `Fabius.lambertNumerator` is $R_{n,j+1}$; `Fabius.lambertNumerator_succ_eq_draft` is (L.19), `Fabius.natCast_dvd_lambertNumerator` is $R_{n,k} \in n\mathbb{Z}[v]$, `Fabius.natDegree_lambertNumerator` the degree, `Fabius.leadingCoeff_lambertNumerator` the leading coefficient, and `Fabius.eval_zero_lambertNumerator` the value at 0.

What is *not* formalized, and should not be read as though it were: the closed form for $(\mathrm{d}/\mathrm{d}z)^k v^n$, and hence the identification of this polynomial family with the coefficients A_m of Theorem L.9; and the asymptotic reading $A_m \sim v^m/m$ of (L.18). Both need the operator calculus on functions rather than the recursion, and the formalized statements are about $\mathbb{Z}[v]$ alone.

Theorem L.13 (The error law, and how slowly it is approached). *Let $G_N(z) = z - L + \sum_{n\le N} A_n(L)z^{-n}$. Then, for each fixed $N \ge 0$,*

$$g(z) - G_N(z) \;\sim\; \frac{1}{N+1}\left(\frac{W(z)}{z}\right)^{N+1} \qquad (z \to +\infty). \tag{L.21}$$

The relative approach to this limit is only $\mathcal{O}(1/L) = \mathcal{O}(1/\log z)$.

Proof. By (L.11) the first omitted term is $A_{N+1}(L)z^{-N-1}$, and by (L.18) $A_{N+1}(L) \sim L^{N+1}/(N+1)$; hence the difference is $\sim L^{N+1}/\big((N+1)z^{N+1}\big)$, which is (L.21) since $q = L/z$. The correction to (L.18) is $\mathcal{O}(1/v)$, because P_m has degree exactly $2m-2$ and its next coefficient is finite while $(1+v)^{2m-1}$ contributes $1 + \mathcal{O}(1/v)$; so the relative error in (L.21) is $\mathcal{O}(1/L)$. □

Remark L.14 (The caveat is not academic). Because $L = W(z)$ grows like $\log z$, the $\mathcal{O}(1/L)$ in Theorem L.13 is genuinely slow. Numerically, the ratio of $g - G_N$ to the right-hand side of (L.21) at $N = 1$ is 0.55 at $z = 10^3$, 0.72 at 10^5, 0.82 at 10^8, 0.88 at 10^{12} and 0.93 at 10^{20} — converging to 1, but not fast enough for (L.21) to be a useful predictor at any argument one is likely to meet. The certificate of Theorem L.4, which is uniform and two-sided, is the tool to use instead. A source that quotes the error law without this caveat invites the reader to trust it at $z = 50$, where it is wrong by a factor of three.

L.5 The three scales, and what the logarithmic expansion cannot see

Proposition L.15 (Pure logarithmic asymptotics). *With $\ell_1 = \log z$ and $\ell_2 = \log \ell_1$,*

$$g(z) = z - \ell_1 + \ell_2 - \frac{\ell_2}{\ell_1} + \frac{\ell_2(2-\ell_2)}{2\ell_1^2} - \frac{\ell_2\big(6 - 9\ell_2 + 2\ell_2^2\big)}{6\ell_1^3} + \frac{\ell_2\big(12 - 36\ell_2 + 22\ell_2^2 - 3\ell_2^3\big)}{12\ell_1^4} + \cdots, \quad \text{(L.22)}$$

the displayed series being z minus the classical iterated-logarithm expansion of $W(z)$. Every *term of* (L.22) *is a term of $z - W(z)$: the entire block structure of Theorem L.9 is invisible at this scale.*

Proof. By Theorem L.3, $0 \le z - W(z) - g(z) \le W(z) - W(g(z))$, and by (L.11) the difference is $A_1(L)/z + \mathcal{O}(L^3/z^2) = \mathcal{O}(L/z)$. Since $L/z \to 0$ faster than every power of $1/\ell_1$ — indeed $z^{-1} L\, \ell_1^N \to 0$ for every fixed N, because z grows exponentially in L — the two functions $g(z)$ and $z - W(z)$ have the same asymptotic expansion to all orders on the scale $\{\ell_1^{-k}\ell_2^j\}$. Substituting the classical expansion of W then gives (L.22). □

Remark L.16 (The moral, and where else it applies). This is the cleanest instance in the volume of a general trap. One may compute a correct inverse-logarithmic expansion to any order and still have computed nothing about the correction that actually distinguishes g from its leading approximation, because that correction lies beyond every order of the scale one is working on. The same ordering governs the additive constant $\frac{1}{2}$ in Theorem Q.25, the shift $+1$ in the Barnes hierarchy, and the Bell sector of Theorem S.22; Theorem Q.41 states the rule. What makes $x + W(x)$ the right teaching example is that here both scales are elementary, so the gap between them can be displayed in full rather than described.

Remark L.17 (A normalization clash between the sources). L3 writes the expansion as $g(z) = z - L + \sum_{n\ge1} \mathcal{A}_n(L) q^n$ with $q = L/z$, while L1 writes it as $g(z) = z - L + \sum_{n\ge1} A_n(L) z^{-n}$ and states the structure theorem (L.16) for *its* A_n. The two families differ by $\mathcal{A}_n = A_n/L^n$, so a reader who carries the structure theorem across without the substitution will find it fails at once — $A_1 = L^2/(1+L)$ against $\mathcal{A}_1 = L/(1+L)$. This chapter uses the z^{-n} normalization throughout, which is the one in which (L.16) is true.

Remark L.18 (Independent verification). Recomputed for this volume at 50–60 digits, from the definitions and independently of the sources: the exact inversion (L.2) and the identity $W(g(z)) = \mathrm{W}_1(z)$, at $z = 1, 5, 20, 100, 1000$; the brackets (L.3), including $\frac{1}{2} \le g' < 1$ by numerical differentiation; the residual sandwich (L.6), at three arguments and three trial approximations each, signs included; the four blocks (L.12)–(L.15), by comparing G_N against the root of $x + W(x) = z$ and confirming that the error falls by the predicted factor at $z = 500$ and $z = 5000$; the three structural data (L.17) for $m \le 4$, read off the displayed numerators; and the error law (L.21) together with the rate of Theorem L.14, at $z = 10^3, 10^5, 10^8, 10^{12}, 10^{20}$. The logarithmic expansion (L.22) was checked against W at $z = 10^6$ and 10^{12}.

L.6 What is cited rather than reproved

For the record, and because the three sources each prove these independently: the near-identity reversion theorem, the triangular reversal algorithm, Newton acceleration with precision doubling, the Lagrange–Bürmann formula at infinity, the coefficient-operator form, the residual-to-error transfer, the r-Lambert function itself, and the pure polynomial–logarithmic perturbation lemma are all results of the calculus parts of this volume, and are used here by citation. The general “method for reversing logarithmic transseries” that each of the three articles extracts in its second half is, in each case, the reversal part of this volume specialized to a slow coefficient field; the three extractions agree with each other and with it.

Remark L.19 (What is left open). The family $x + \alpha W(x)$ and the more general $x + h(W(x))$ are treated in the sources only to leading order. The reduction of Theorem L.7 generalizes — one obtains $q = L/(L - v) - \mathrm{e}^{-v}$ with L replaced by the appropriate combination — but the analogue of the integer-polynomial structure Theorem L.10 is not established here, and it is not obvious that the numerators remain integral for general α.

Part XII

Four combinatorial sequences

Appendix M

Butcher–Pólya rooted trees (A000081)

This chapter merges the three independently written treatments *Butcher_Tree_Transseries*, *Butcher_Tree_Transseries-2* and *Butcher_Tree_Transseries-3* of the special-function-inversion draft set. Below they are called S1, S2 and S3. S2 is a book-length treatment and is the most complete on the inverse and on the inherited singularity at $-\sqrt{\rho}$; S3 is the most careful about what is and is not proved beyond the dominant sector; S1 has the shortest route to the universal Cayley coefficients and the only treatment of the compositional inverse of the generating function. Where they disagree the disagreement is stated in the text. Every numerical constant retained here has been recomputed from the exact divisor recurrence at 60–110 working decimal digits; see Section M.11.3.

The subject of this chapter is the sequence

$$0,\ 1,\ 1,\ 2,\ 4,\ 9,\ 20,\ 48,\ 115,\ 286,\ 719,\ 1842,\ 4766,\ 12486,\ 32973,\ 87811,\ \ldots \tag{M.1}$$

(OEIS A000081), where for $n \geq 1$ the term a_n counts isomorphism classes of rooted, unlabelled, non-plane trees on n vertices. These are the *Pólya trees*; in numerical analysis the same shapes index Butcher series, elementary differentials and Runge–Kutta order conditions, which is the origin of the name *Butcher tree numbers.* We set $a_0 = 0$ throughout.

The sequence grows like $C\alpha^n n^{-3/2}$ with $\alpha = \rho^{-1}$ and ρ the Otter singularity. Thus, in the language of Theorem J.2, its asymptotic interpolation is an exponential–power model with

$$\boxed{\alpha_{\text{model}} = 1, \qquad a_{\text{model}} = \lambda := \log(1/\rho), \qquad b_{\text{model}} = -\tfrac{3}{2}, \qquad \delta = 1.} \tag{M.2}$$

This chapter is therefore the *linear-phase* instance of the Part 0 apparatus: no α-reduction (Theorem J.4) is needed, the Lambert core (Theorem J.23) applies verbatim, and the entire reversion is a substitution into Theorem J.29. Everything specific to rooted trees is the computation of the model's data C, λ and Q; that computation is the substance of Sections M.1 to M.5, and the inversion in Section M.7 is short precisely because Part 0 has already been proved.

M.1 The functional equation and the exact recurrence

M.1.1 Pólya's multiset equation

Let $\mathcal{T}$ be the class of unlabelled non-plane rooted trees and let $\mathcal{Z}$ be one atom (one vertex). Deleting the root of a tree leaves an unordered multiset of rooted trees, and grafting a multiset of rooted trees onto a fresh root inverts this. Hence

$$\mathcal{T} = \mathcal{Z} \times \operatorname{MSET}(\mathcal{T}). \tag{M.3}$$

Proposition M.1 (Pólya's equation and the Euler product). *Let $T(z) = \sum_{n\geq 1} a_n z^n \in \mathbb{Z}[[z]]$. Then, as identities of formal power series,*

$$T(z) = z \prod_{r\geq 1} (1 - z^r)^{-a_r}, \tag{M.4}$$

$$T(z) = z \exp\left(\sum_{k\geq 1} \frac{T(z^k)}{k} \right). \tag{M.5}$$

Proof. A multiset of rooted trees that contains m_j copies of the jth shape has size $\sum_j m_j |j|$ and contributes $z^{\sum_j m_j |j|}$. For one fixed shape of size r, unbounded multiplicity contributes $\sum_{m\geq 0} z^{rm} = (1 - z^r)^{-1}$; there are a_r shapes of size r, and distinct shapes are chosen independently, so the forest hanging from the root has generating function $\prod_{r\geq 1}(1 - z^r)^{-a_r}$. Every factor is a unit of $\mathbb{Z}[[z]]$ and the product is z-adically convergent because the rth factor is $1 + \mathcal{O}(z^r)$. Multiplying by the root gives Equation (M.4). Taking the formal logarithm and using $-\log(1 - w) = \sum_{k\geq 1} w^k / k$,

$$\log \frac{T(z)}{z} = \sum_{r\geq 1} a_r \sum_{k\geq 1} \frac{z^{rk}}{k} = \sum_{k\geq 1} \frac{1}{k} \sum_{r\geq 1} a_r (z^k)^r = \sum_{k\geq 1} \frac{T(z^k)}{k},$$

the rearrangement being legitimate because for each fixed power of z only finitely many pairs (r, k) contribute. Exponentiating gives Equation (M.5). □

Remark M.2. Equation (M.5) is purely formal; no analytic hypothesis is used, and none is available at this stage. All three siblings state it correctly, but only S2 says explicitly that the identity holds in $\mathbb{Z}[[z]]$ before any radius of convergence has been produced. We keep that discipline: Section M.2 is the first place where analysis enters.

M.1.2 The divisor recurrence

Definition M.3 (Divisor transform). For $m \geq 1$ put

$$b_m := \sum_{d|m} d\, a_d. \tag{M.6}$$

Lean status. Formal as `Fabius.divisorTransform` in `DivisorTransform.lean`, for a sequence valued in an arbitrary commutative ring, since nothing in (M.6) is about rooted trees. The splitting that the next lemma needs is `Fabius.divisorTransform_eq_add_properDivisors`, that $b_m = m\, a_m + \sum_{d|m,\, d<m} d\, a_d$ — the divisors of m are its proper divisors together with m itself, and the term m contributes is exactly $m\, a_m$.

Theorem M.4 (Otter–Pólya recurrence). *With $a_1 = 1$ one has, for every $n \geq 2$,*

$$(n - 1)\, a_n = \sum_{j=1}^{n-1} b_j\, a_{n-j}. \tag{M.7}$$

Proof. Write $G(z) = T(z)/z = \sum_{n\geq 0} a_{n+1} z^n$, a unit of $\mathbb{Q}[[z]]$ with $G(0) = 1$. From Equation (M.4) and the computation in the proof of Theorem M.1,

$$\log G(z) = \sum_{r\geq 1} a_r \sum_{k\geq 1} \frac{z^{rk}}{k} = \sum_{m\geq 1} \frac{z^m}{m} \sum_{d|m} d\, a_d = \sum_{m\geq 1} \frac{b_m}{m} z^m,$$

where the inner rearrangement groups the pairs (r,k) with $rk=m$ and uses $1/k=d/m$ for $d=r$. Formal logarithmic differentiation gives $G'(z)=G(z)\sum_{m\ge1} b_m z^{m-1}$. Extracting $[z^{n-1}]$,

$$n\,a_{n+1}=\sum_{j=1}^{n} b_j\,a_{n-j+1},$$

and replacing n by $n-1$ yields Equation (M.7). □

Remark M.5 (Why the recurrence is triangular, and its cost). Although b_n contains the summand $n\,a_n$, the right-hand side of Equation (M.7) uses only b_j with $j<n$, so a_n is determined by $a_1,\ldots,a_{n-1}$. Implemented as a sieve — after computing a_n, add $n\,a_n$ to $b_n, b_{2n}, b_{3n},\ldots$ — the divisor transform costs $\mathcal{O}(N\log N)$ additions and the convolutions cost $\mathcal{O}(N^2)$ integer operations for $a_1,\ldots,a_N$. The integrality assertion $(n-1)\mid\sum_j b_j a_{n-j}$ is a free exact self-test on every step, and was used as such in the computations of Section M.11.3.

M.1.3 The self-interaction and its coefficients

Separating the $k=1$ term of Equation (M.5) is the structural move on which everything else rests.

Definition M.6 (Disturbance and Cayley argument)**.** Put

$$R(z):=\sum_{k\ge2}\frac{T(z^k)}{k},\qquad \zeta(z):=z\,\mathrm{e}^{R(z)}. \tag{M.8}$$

With this notation Equation (M.5) reads

$$T(z)=\zeta(z)\,\mathrm{e}^{T(z)}. \tag{M.9}$$

Lemma M.7 (Coefficients of the disturbance). *$R(z)=\sum_{r\ge2}\varrho_r z^r$ with*

$$\varrho_r=\frac{1}{r}\sum_{\substack{d\mid r\\ d<r}} d\,a_d=\frac{b_r-r\,a_r}{r}\in\mathbb{Q}_{\ge0}. \tag{M.10}$$

In particular ϱ_r depends only on a_d with $d\le r/2$, and $\varrho_2=\frac12$, $\varrho_3=\frac13$, $\varrho_4=\frac34$, $\varrho_5=\frac15$, $\varrho_6=\frac32$, $\varrho_7=\frac17$, $\varrho_8=\frac{19}{8}$.

Proof. $R(z)=\sum_{k\ge2}\sum_{d\ge1}\frac{a_d}{k}z^{kd}$; collecting the terms with $kd=r$ gives $\varrho_r=\sum_{d\mid r,\,d<r} a_d/(r/d)=\frac1r\sum_{d\mid r,\,d<r} d\,a_d$, which is $(b_r-ra_r)/r$ by Equation (M.6). Every divisor $d<r$ of r satisfies $d\le r/2$. The listed values follow from $a_1=a_2=1$, $a_3=2$, $a_4=4$: e.g. $\varrho_4=(1\cdot1+2\cdot1)/4=\frac34$ and $\varrho_6=(1+2\cdot1+3\cdot2)/6=\frac32$. □

Lean status: the arithmetic clauses are formal; the numerical values are not. In `DivisorTransform.lean`. The coefficient itself is `Fabius.disturbanceCoeff`, defined by the first expression of (M.10), and the equality with the second is `Fabius.disturbanceCoeff_eq_divisorTransform`, an immediate consequence of `Fabius.divisorTransform_sub`. Nonnegativity, the $\mathbb{Q}_{\ge0}$ of the display, is `Fabius.disturbanceCoeff_nonneg` under the hypothesis that the sequence is nonnegative.

The clause that ϱ_r depends only on a_d with $d\le r/2$ is formal in the form that makes it usable: `Fabius.disturbanceCoeff_congr` says that two sequences agreeing at every d with $2d\le r$ give the same ϱ_r. The arithmetic underneath is `Fabius.two_mul_le_of_mem_properDivisors`, that a proper divisor is at most half of its multiple.

The seven numerical values are treated unevenly, and deliberately. Four of them are not four facts but one: a prime has $\{1\}$ for its proper divisors, so `Fabius.disturbanceCoeff_prime` gives

$\varrho_p = a_1/p$ for *every* prime p, which with $a_1 = 1$ is $\varrho_2 = \frac{1}{2}$, $\varrho_3 = \frac{1}{3}$, $\varrho_5 = \frac{1}{5}$ and $\varrho_7 = \frac{1}{7}$ at once. The remaining three, $\varrho_4 = \frac{3}{4}$, $\varrho_6 = \frac{3}{2}$ and $\varrho_8 = \frac{19}{8}$, depend on the tree numbers a_2, a_3, a_4 themselves; the module carries no particular sequence, so they are not formalized, and neither is Theorem M.4, which is what would supply them.

Theorem M.7 is due to S2; S1 and S3 carry R only through the double sum $\sum_{k\geq 2}\sum_n a_n\rho^{kn}(\cdots)$. The single-sum form is used throughout this chapter because it turns every quantity below into one geometrically convergent series in the exact integers a_r (Theorem M.20).

M.2 The Cayley–Lambert normal form and the Otter singularity

M.2.1 Factorisation through the Cayley tree function

Definition M.8 (Cayley tree function). Let $\mathcal{C}(w)$ be the unique formal solution of $\mathcal{C} = w\mathrm{e}^{\mathcal{C}}$ with $\mathcal{C}(0) = 0$. It is analytic on $|w| < \mathrm{e}^{-1}$ and

$$\mathcal{C}(w) = -W_0(-w), \tag{M.11}$$

where W_0 is the principal Lambert branch.

Lean status. Formal in `CayleyTreeFunction.lean`, with (M.11) taken as the *definition*: `Fabius.cayleyTree` $w = -W_0(-w)$. That makes the identification with the principal branch a rewriting step in both directions — `Fabius.principalLambertW_eq_neg_cayleyTree` reads it as a statement about W_0 — and leaves the rest of the definition as consequences of the Lambert API: the functional equation $\mathcal{C} = w\mathrm{e}^{\mathcal{C}}$ is `Fabius.cayleyTree_eq_mul_exp`, the initial condition `Fabius.cayleyTree_zero`, positivity `Fabius.cayleyTree_nonneg`, strict growth `Fabius.cayleyTree_strictMonoOn` and the bound `Fabius.cayleyTree_le_one`. The boundary value $\mathcal{C}(\mathrm{e}^{-1}) = 1$ is `Fabius.cayleyTree_exp_neg_one`: the tree function reaches 1 exactly where W_0 reaches its branch point, which is why that is also where $\mathcal{C}$ ceases to be analytic.

Two limitations of the formal statement, both deliberate. The domain is real: the corpus carries the real principal branch on $[-\mathrm{e}^{-1}, \infty)$, so $\mathcal{C}(w)$ is available for $w \leq \mathrm{e}^{-1}$, and the analyticity on the complex disc $|w| < \mathrm{e}^{-1}$ asserted here is not formalized. And uniqueness carries the branch condition explicitly: `Fabius.cayleyTree_unique` asks $c \leq 1$, which is $-c \geq -1$, the principal side of W_0. That hypothesis is not decoration — for $0 < w < \mathrm{e}^{-1}$ the equation $c = w\mathrm{e}^{c}$ has a second, larger root coming from the lower branch, so the statement without it is false, not merely unproved.

Comparing Equation (M.9) with Theorem M.8,

$$\boxed{T(z) = \mathcal{C}\big(\zeta(z)\big) = -W_0\big(-\zeta(z)\big).} \tag{M.12}$$

This holds as formal series, and as an identity of analytic functions on the common domain of the branches selected at the origin. The point of the factorisation is a clean separation of labour: *all* of the singular mechanism is the universal square-root branch point of $\mathcal{C}$ at $w = \mathrm{e}^{-1}$, and *all* of the tree-specific information is packed into the map ζ, which — as we now show — is analytic on a strictly larger disc than T itself.

Lemma M.9 (Elementary bounds on the radius). *The radius of convergence ρ of T satisfies $\frac{1}{4} \leq \rho \leq \mathrm{e}^{-1}$.*

Proof. Since $a_n \geq 1$ for $n \geq 1$ we have $\rho \leq 1$, so for $0 < x < \rho$ every x^k with $k \geq 2$ lies in $(0, \rho)$ as well. Upper bound: for such x the series $T(x)$ and $R(x)$ converge, all terms are nonnegative, and $\zeta(x) = x\mathrm{e}^{R(x)} \geq x$; so Equation (M.9) gives $T(x) \geq x\mathrm{e}^{T(x)}$, i.e. $x \leq T(x)\mathrm{e}^{-T(x)} \leq \sup_{u\geq 0} u\mathrm{e}^{-u} = \mathrm{e}^{-1}$. Letting $x \uparrow \rho$ gives $\rho \leq \mathrm{e}^{-1}$. Lower bound: forgetting the identification of isomorphic children embeds the shapes counted by a_n into the rooted *plane* trees on n vertices, of which there are $\frac{1}{n}\binom{2n-2}{n-1} \leq 4^n$; hence $\limsup a_n^{1/n} \leq 4$ and $\rho \geq \frac{1}{4}$. ☐

Proposition M.10 (Analyticity of the disturbance)**.** *R, and hence ζ, is analytic on $|z| < \sqrt{\rho}$, and the series $\sum_{k\geq 2} T(z^k)/k$ converges normally on compact subsets together with all its derivatives. In particular R and ζ are analytic at $z = \rho$, since $\rho < \sqrt{\rho}$ by Theorem M.9.*

Proof. Fix $r < \sqrt{\rho}$, so $r^2 < \rho$. Since T has radius ρ and $T(w) = w + \mathcal{O}(w^2)$, there are $M > 0$ and $r_0 \in (0, 1)$ with $|T(w)| \leq M|w|$ for $|w| \leq r_0$. For $|z| \leq r$, only finitely many k have $r^k > r_0$, and for each such k we have $r^k \leq r^2 < \rho$, so $T(z^k)$ is defined and bounded there. For the remaining k, $|T(z^k)/k| \leq Mr^k/k$, which is summable. Differentiating j times multiplies the kth tail term by a factor $\mathcal{O}(k^j r^{-j})$, still summable. Weierstrass' theorem gives analyticity and termwise differentiation. □

Proposition M.11 (Critical characterisation of ρ)**.** *With $\rho \in [\frac{1}{4}, \mathrm{e}^{-1}]$ the radius of convergence of T,*

$$T(\rho) = 1, \qquad \zeta(\rho) = \mathrm{e}^{-1}, \qquad \log\rho + 1 + R(\rho) = 0. \tag{M.13}$$

Moreover ζ is strictly increasing on $(0, \sqrt{\rho})$ and

$$\theta_1 := \mathrm{e}\rho\,\zeta'(\rho) = 1 + \rho R'(\rho) > 0. \tag{M.14}$$

Proof. All coefficients of T, hence of R and of $\zeta = z\mathrm{e}^R$, are nonnegative, and $\zeta(z) = z + \frac{1}{2}z^3 + \frac{1}{3}z^4 + \cdots$ has positive linear coefficient; therefore $\zeta'(x) > 0$ on $(0, \sqrt{\rho})$, and ζ is strictly increasing there.

Step 1: $T(x) < 1$ on $[0, \rho)$, and $T = \mathcal{C}\circ\zeta$ there. On $[0, \rho)$ the function T is continuous, increasing, with $T(0) = 0$, and Equation (M.9) gives $\zeta(x) = T(x)\mathrm{e}^{-T(x)} \leq \mathrm{e}^{-1}$. Let $\rho_1 := \sup\{x \in [0, \rho) : T(x) < 1\}$. On $[0, \rho_1)$ we have $\zeta(x) < \mathrm{e}^{-1}$, so $\mathcal{C}$ is analytic at $\zeta(x)$ and $\mathcal{C}\circ\zeta$ is an analytic solution of Equation (M.9) agreeing with T at 0; by uniqueness of that germ and analytic continuation, $T = \mathcal{C} \circ \zeta$ on $[0, \rho_1)$. If $\rho_1 < \rho$ then $T(\rho_1) = 1$ and $\zeta(\rho_1) = \mathrm{e}^{-1}$; but ζ is strictly increasing, so $\zeta(x) > \mathrm{e}^{-1}$ for $x \in (\rho_1, \rho)$, contradicting $\zeta(x) \leq \mathrm{e}^{-1}$. Hence $\rho_1 = \rho$.

Step 2: $\zeta(\rho) = \mathrm{e}^{-1}$. By Theorem M.10 the function ζ is analytic at ρ, and by Step 1 and monotonicity $\zeta(\rho) = \lim_{x\uparrow\rho}\zeta(x) \leq \mathrm{e}^{-1}$. If the inequality were strict, $\mathcal{C}$ would be analytic at $\zeta(\rho)$ and $\mathcal{C} \circ \zeta$ would be analytic in a full neighbourhood of ρ and would agree with T on $[0, \rho)$; then T would continue analytically through ρ, contradicting Pringsheim's theorem, which makes the positive point $z = \rho$ a singularity of a series with nonnegative coefficients. Hence $\zeta(\rho) = \mathrm{e}^{-1}$ and, by Theorem M.8, $T(\rho) = \mathcal{C}(\mathrm{e}^{-1}) = 1$.

Step 3. Taking logarithms in $\zeta(\rho) = \rho\mathrm{e}^{R(\rho)} = \mathrm{e}^{-1}$ gives the scalar equation; equivalently $R(\rho) = \lambda - 1$ with $\lambda = -\log\rho$, and $R(\rho) > 0$ independently re-proves $\rho < \mathrm{e}^{-1}$. Finally $\zeta' = (1/z + R')\zeta$, so $\mathrm{e}\rho\zeta'(\rho) = \mathrm{e}\zeta(\rho)\big(1 + \rho R'(\rho)\big) = 1 + \rho R'(\rho)$, positive because R has nonnegative coefficients. □

Proposition M.12 (Uniqueness of the dominant singularity)**.** *$z = \rho$ is the only singularity of T on the circle $|z| = \rho$.*

Proof. Take $z = \rho\mathrm{e}^{\mathrm{i}\vartheta}$ with $\vartheta \not\equiv 0$. The series $\zeta(z) = z + \frac{1}{2}z^3 + \frac{1}{3}z^4 + \cdots$ has nonnegative coefficients, so $|\zeta(z)| \leq \zeta(\rho) = \mathrm{e}^{-1}$, with equality only if the three displayed nonzero terms z, $\frac{1}{2}z^3$, $\frac{1}{3}z^4$ share one argument, i.e. $\mathrm{e}^{2\mathrm{i}\vartheta} = \mathrm{e}^{3\mathrm{i}\vartheta} = 1$, hence $\mathrm{e}^{\mathrm{i}\vartheta} = 1$ and $\vartheta \equiv 0$. So $|\zeta(z)| < \mathrm{e}^{-1}$ strictly, $\mathcal{C}$ is analytic at $\zeta(z)$, and ζ is analytic at z because $\rho < \sqrt{\rho}$ (Theorem M.10). Hence $T = \mathcal{C} \circ \zeta$ is analytic at z. □

Remark M.13. This is S3's argument, and it is the only one of the three that is self-contained: S1 and S2 appeal to "positivity and aperiodicity" without exhibiting the aperiodicity witness. The three coefficients z, z^3, z^4 of ζ furnish that witness explicitly, because $\gcd(3 - 1, 4 - 1) = 1$.

M.2.2 The constants

Convention M.14 (Growth constants)**.** Throughout the chapter

$$\alpha := \rho^{-1}, \qquad \lambda := \log\alpha = -\log\rho, \qquad p := \tfrac{3}{2}. \tag{M.15}$$

Notation collision between the siblings. S2 and S3 write $\alpha = \rho^{-1}$ and $\lambda = \log\alpha$; S1 writes exactly the opposite, $\lambda = \rho^{-1}$ and $\alpha = -\log\rho$. The symbols therefore cross-denote between the drafts, and S1's Ω_-/α, $\mathrm{e}^{\alpha n}$ and d_m formulas must be read with $\alpha = \lambda_{\text{here}}$. This volume adopts the two-to-one majority reading Equation (M.15); every formula imported from S1 has been translated.

Solving $\log\rho + 1 + R(\rho) = 0$ from Equation (M.10) with 700 exact tree numbers at 110 working digits (see Section M.11.3) gives

$$\rho = 0.3383218568992076951961126257170170531837746075329677955723037 76\ldots, \quad \text{(M.16)}$$

$$\alpha = 2.955765285651994974714817524123194588375492304663596595350472478\ldots, \quad \text{(M.17)}$$

$$\lambda = 1.083757597244624660482711980896174279049662124202672234914427065\ldots, \quad \text{(M.18)}$$

$$R(\rho) = 0.083757597244624660482711980896174279049662124202672234914427065\ldots, \quad \text{(M.19)}$$

and, as an independent consistency check on the whole pipeline,

$$\lambda = 1 + R(\rho), \tag{M.20}$$

which is nothing but Equation (M.13) rearranged; the two computed values agree to all 70 digits carried. Two further constants are recorded here because they are quoted repeatedly:

$$\theta_1 = 1.2160045618240489221572126192281874 44\ldots, \qquad \zeta'(\rho) = 1.3222411426969302644512848 89\ldots. \tag{M.21}$$

M.3 The universal Cayley branch germ

The next result is universal: it depends on the Cayley function alone, not on rooted trees, and applies verbatim to *any* class satisfying $T = \zeta\mathrm{e}^T$ with analytic ζ.

Definition M.15 (Cayley kernel). Put

$$\Phi(v) := \frac{2\big(1-(1-v)\mathrm{e}^v\big)}{v^2} = \sum_{m\ge 0}\frac{2(m+1)}{(m+2)!}\,v^m = \sum_{m\ge 0}\frac{2}{(m+2)\,m!}\,v^m = 1+\frac{2}{3}v+\frac{1}{4}v^2+\frac{1}{15}v^3+\cdots. \tag{M.22}$$

Lean status. Formal in `CayleyKernel.lean`, all three displays of (M.22). That the two coefficient formulas agree — $2(m+1)/(m+2)! = 2/((m+2)m!)$, since $(m+2)! = (m+2)(m+1)m!$ — is `Fabius.cayleyKernelCoeff_eq_factorial`; the numerical values $1,\ 2/3,\ 1/4,\ 1/15$ are `Fabius.cayleyKernelCoeff_zero` through `Fabius.cayleyKernelCoeff_three`. The closed form is `Fabius.tsum_cayleyKernelTerm`, and its denominator-free version `Fabius.tsum_cayleyKernelTerm_mul_sq`, $v^2\Phi(v) = 2(1-(1-v)\mathrm{e}^v)$, holds with no hypothesis on v at all. The proof is a subtraction of two shifted exponential series, `Fabius.tsum_exp_shift_one` and `Fabius.tsum_exp_shift_two`, after the termwise identity `Fabius.cayleyKernelCoeff_mul_pow_eq_sub`, which is where the two coefficient formulas do their work. The kernel is entire: `Fabius.summable_cayleyKernelTerm` holds for every real v.

Which display is the definition matters, and the formal treatment differs from the one the notation above suggests. The closed form $2(1-(1-v)\mathrm{e}^v)/v^2$ is undefined at $v = 0$; the series is not, and $\Phi(0) = 1$. So the series is primitive in the corpus, the closed form is a theorem for $v \neq 0$, and the removable singularity is recorded separately as `Fabius.tsum_cayleyKernelTerm_zero`. Taking the closed form as the definition would have been actively harmful: Lean's convention $x/0 = 0$ would have made $\Phi(0) = 0$ rather than 1, and every statement about the kernel at the origin — which is where it is used — would have been silently wrong, with nothing in the type checker to object.

The two displayed coefficient forms agree because $2(m+1)/(m+2)! = 2/((m+2)\,m!)$; S1 and S3 use the first, S2 the second.

Theorem M.16 (Universal Cayley coefficients). *Set $w = (1-v)/\mathrm{e}$ and $\mathcal{C}(w) = 1 - v$ on the branch reached from $0 < w < \mathrm{e}^{-1}$. Then $v = \frac{1}{2}v^2\Phi(v)$, the inverse germ is $v = \sum_{m\geq 1} \omega_m (2v)^{m/2}$, and for every $m \geq 1$*

$$\boxed{\omega_m = \frac{1}{m}\,[v^{m-1}]\,\Phi(v)^{-m/2} = \frac{1}{m}\sum_{k=0}^{m-1}\binom{-m/2}{k}\widehat{B}_{m-1,k}\big(\tfrac{2}{3},\tfrac{1}{4},\tfrac{1}{15},\ldots\big) \in \mathbb{Q}.} \tag{M.23}$$

Consequently, on that branch,

$$\mathcal{C}\left(\frac{1-v}{\mathrm{e}}\right) = 1 + \sum_{r\geq 1} c_r\, v^{r/2}, \qquad c_r = -2^{r/2}\,\omega_r. \tag{M.24}$$

Proof. From $\mathcal{C} = w\mathrm{e}^{\mathcal{C}}$ we get $\mathrm{e}w = \mathcal{C}\mathrm{e}^{1-\mathcal{C}}$, so with $\mathcal{C} = 1 - v$,

$$1 - v = \mathrm{e}w = (1-v)\mathrm{e}^{v}, \qquad \text{hence} \qquad v = 1-(1-v)\mathrm{e}^{v} = \sum_{m\geq 2}\frac{m-1}{m!}v^m = \frac{v^2}{2}\Phi(v),$$

the middle identity by expanding e^v and collecting. Put $\varsigma = \sqrt{2v}$ with the determination $\varsigma/v \to 1$; then $\varsigma = v\,\Phi(v)^{1/2}$ is an ordinary power series in v with nonzero linear coefficient, so Theorem J.17 applies with $\phi(v) = \Phi(v)^{1/2}$ and yields

$$[\varsigma^m]\,v(\varsigma) = \frac{1}{m}[v^{m-1}]\,\phi(v)^{-m} = \frac{1}{m}[v^{m-1}]\,\Phi(v)^{-m/2} = \omega_m.$$

The Bell form is Theorem J.14 applied to the unit series Φ, whose coefficients are $2(m+1)/(m+2)!$ for $m \geq 1$. Rationality follows because $\binom{-m/2}{k} \in \mathbb{Q}$ and $\widehat{B}_{m-1,k}$ is a polynomial with rational coefficients in rational arguments. Finally $\mathcal{C} = 1 - v = 1 - \sum_m \omega_m\varsigma^m = 1 - \sum_m \omega_m 2^{m/2}v^{m/2}$, which is Equation (M.24). ☐

Lean status: partial. In `CayleyLocalCoordinate.lean`. What is formal is the setup, and it is formal in both of its parts. The identity $v = \frac{1}{2}v^2\Phi(v)$ is `Fabius.cayleyUpsilon_eq_half_mul_kernel`, where `Fabius.cayleyUpsilon` $v = 1-(1-v)\mathrm{e}^v$; it is the kernel's defining relation `Fabius.tsum_cayleyKernelTerm_mul_sq` rearranged, and stating it against the series rather than the closed form makes it hold at $v = 0$ too, where the closed form does not exist. `Fabius.cayleyUpsilon_eq_half_mul_sq_kernel` is the closed-form version for $v \neq 0$.

The substitution itself is not free and is proved separately: `Fabius.cayleyTree_eq_one_sub` says that $\mathcal{C}\big((1-v(v))/\mathrm{e}\big) = 1 - v$ for $v \geq 0$, so that v is genuinely the local coordinate of the branch rather than a function that happens to have the right series. Two things make it work. The argument never leaves the domain, by `Fabius.one_sub_upsilon_div_exp_one_le`, which rests on `Fabius.one_sub_mul_exp_le_one`: $(1-v)\mathrm{e}^v \leq 1$ for *every* real v, in one step from $1 - v \leq \mathrm{e}^{-v}$ scaled by $\mathrm{e}^v > 0$, with no case split on the sign of $1 - v$. And the hypothesis $v \geq 0$ is exactly the branch condition $1 - v \leq 1$ of `Fabius.cayleyTree_unique` — which is the formal content of the phrase "on the branch reached from $0 < w < \mathrm{e}^{-1}$". Positivity of the local coordinate, `Fabius.cayleyUpsilon_nonneg`, is the same inequality again.

What is *not* formalized: everything after the setup. The reversion $v = \sum_{m\geq 1}\omega_m(2v)^{m/2}$, the master formula (M.23) for ω_m through the ordinary partial Bell polynomials, and the Puiseux expansion (M.24) all need Lagrange reversion at a square-root branch point, which the corpus does not carry. The ordinary Bell polynomials the master formula is written in are themselves unformalized; see the note at Theorem A.4.

Remark M.17 (Two normalisations, one object). S2 works with ω_m (expanding in $\varsigma = \sqrt{2v}$) while S1 and S3 work with c_r (expanding in $v^{1/2}$). The dictionary is $c_r = -2^{r/2}\omega_r$, verified here for $1 \leq r \leq 18$ by exact rational arithmetic. The ω_m are rational, the c_r alternate between rationals and rational multiples of $\sqrt{2}$; this is the only reason to prefer ω for tables.

Table M.1: Exact universal Cayley coefficients ω_m of Equation (M.23), recomputed by rational arithmetic from Equation (M.22).

m	ω_m	m	ω_m
1	1	10	$-5776369/1515591000$
2	$-1/3$	11	$169709463197/69528040243200$
3	$11/72$	12	$-1118511313/709296588000$
4	$-43/540$	13	$667874164916771/650782456676352000$
5	$769/17280$	14	$-500525573/744761417400$
6	$-221/8505$	15	$103663334225097487/234281684403486720000$
7	$680863/43545600$	16	$-466901817532379/1595278956070800000$
8	$-1963/204120$	17	$212352941850863050 43/109242202556140093440000$
9	$226287557/37623398400$	18	$-106040742894306601/818378104464320400000$

The corresponding c_r are

$$\begin{aligned} c_1 &= -\sqrt{2}, & c_2 &= \tfrac{2}{3}, & c_3 &= -\tfrac{11\sqrt{2}}{36}, & c_4 &= \tfrac{43}{135}, \\ c_5 &= -\tfrac{769\sqrt{2}}{4320}, & c_6 &= \tfrac{1768}{8505}, & c_7 &= -\tfrac{680863\sqrt{2}}{5443200}, & c_8 &= \tfrac{3926}{25515}. \end{aligned} \tag{M.25}$$

Remark M.18 (On the observed sign pattern). All three siblings display ω_m with strictly alternating signs and none proves it. The data in Table M.1 are consistent with $\operatorname{sgn}\omega_m = (-1)^{m+1}$ for $1 \le m \le 18$, and this is what one expects from the fact that the inverse function $v(\varsigma)$ of $\varsigma = v\Phi(v)^{1/2}$ has its nearest singularity on the negative ς-axis; but no proof is offered here and the statement is recorded as an observation, not as a result.

M.4 The Puiseux expansion at the Otter singularity

M.4.1 The local disturbance series

Definition M.19 (Local coordinate and disturbance)**.** Write

$$s := 1 - \frac{z}{\rho}, \qquad \theta(s) := 1 - \mathrm{e}\,\zeta\big(\rho(1-s)\big) = \sum_{m\ge 1} \theta_m s^m. \tag{M.26}$$

The constant term vanishes by Equation (M.13). By Theorem M.10, θ is analytic on the image of $|z| < \sqrt{\rho}$ under $z \mapsto s$, namely the disc $|s-1| < \rho^{-1/2}$; in particular its Taylor series at $s = 0$ converges for $|s| < \rho^{-1/2} - 1 = 0.71923\ldots$.

Taylor's formula gives the derivative form

$$\theta_m = \frac{(-1)^{m+1}\mathrm{e}\rho^m}{m!}\,\zeta^{(m)}(\rho), \qquad m \ge 1, \tag{M.27}$$

consistent with Equation (M.14): $\theta_1 = \mathrm{e}\rho\zeta'(\rho)$. For computation the following coefficient form is preferable, since it needs nothing but the integers a_r and the single scalar ρ.

Lemma M.20 (Local coefficients of R from the tree numbers)**.** *Let* $\widehat{R}_m := [s^m]\, R\big(\rho(1-s)\big)$*. Then*

$$\boxed{\widehat{R}_m = (-1)^m \sum_{r\ge 2} \varrho_r \binom{r}{m} \rho^r = (-1)^m\, \frac{\rho^m R^{(m)}(\rho)}{m!},} \tag{M.28}$$

with ϱ_r as in Equation (M.10)*. The series converges geometrically: for fixed m the general term is $\mathcal{O}\big(r^m \rho^{r/2}\big)$.*

Proof. By Theorem M.7, $R(\rho(1-s)) = \sum_{r\geq 2} \varrho_r \rho^r (1-s)^r$, and $[s^m]\,(1-s)^r = (-1)^m\binom{r}{m}$. The middle expression is a Taylor coefficient of R at ρ in the variable ρs, giving the right-hand form. For the convergence rate, every divisor $d < r$ of r satisfies $d \leq r/2$, so $\varrho_r \leq a_{\lfloor r/2 \rfloor}\, d(r)$ with $d(r)$ the number of divisors of r; since $\limsup_n a_n^{1/n} = \rho^{-1}$, for every $\epsilon > 0$ there is K_ϵ with $\varrho_r \rho^r \leq K_\epsilon(\sqrt{\rho}+\epsilon)^r$. With $\binom{r}{m} \leq r^m$ the claim follows. □

Remark M.21 (A dedup). S1 (its ℓ_m) and S3 (its $[s^m]\,R(\rho(1-s))$) both give the *double* sum

$$(-1)^m \sum_{k\geq 2} \frac{1}{k} \sum_{n\geq 1} a_n \binom{kn}{m} \rho^{kn}.$$

Equation (M.28) is the same number after the inner divisor collection of Theorem M.7, and is one sum rather than two. The two forms were checked against each other numerically to 70 digits for $0 \leq m \leq 30$.

Lemma M.22 (The disturbance coefficients). *With $\widehat{R}_m$ as in Theorem M.20, put $\widehat{R}_+(s) := \sum_{m\geq 1} \widehat{R}_m s^m$ and $E(s) := \exp \widehat{R}_+(s) = \sum_{m\geq 0} E_m s^m$. Then $E_0 = 1$,*

$$E_m = \sum_{k=0}^{m} \frac{1}{k!}\, \widehat{B}_{m,k}\big(\widehat{R}_1, \widehat{R}_2, \dots\big) = \frac{1}{m} \sum_{k=1}^{m} k\, \widehat{R}_k\, E_{m-k}, \tag{M.29}$$

and

$$\boxed{\theta_m = E_{m-1} - E_m, \qquad m \geq 1.} \tag{M.30}$$

Proof. By Equation (M.13), $\mathrm{e}\zeta(\rho) = 1$, so

$$\mathrm{e}\,\zeta\big(\rho(1-s)\big) = \frac{\rho(1-s)\mathrm{e}^{R(\rho(1-s))}}{\rho\,\mathrm{e}^{R(\rho)}} = (1-s)\,\mathrm{e}^{\widehat{R}_+(s)} = (1-s)E(s),$$

because $\widehat{R}_0 = R(\rho)$. Hence $\theta(s) = 1-(1-s)E(s)$ and $\theta_m = -(E_m - E_{m-1})$ for $m \geq 1$, while the constant term is $1 - E_0 = 0$. The two expressions for E_m are the ordinary Bell expansion of an exponential (Theorem J.9) and the standard triangular exponential recurrence obtained from $E' = \widehat{R}'_+ E$. □

Remark M.23 (The jet route). S2 computes θ_m instead from the *scaled logarithmic jets*

$$\ell_j := \rho^j \left.\frac{\mathrm{d}^j}{\mathrm{d}z^j} \log\zeta(z)\right|_{z=\rho} = (-1)^{j-1}(j-1)! + \sum_{r\geq j} \varrho_r\, r^{\underline{j}}\, \rho^r, \tag{M.31}$$

via $\theta_m = \frac{(-1)^{m+1}}{m!} B_m(\ell_1,\dots,\ell_m)$, where B_m is the complete exponential Bell polynomial and $r^{\underline{j}} = r(r-1)\cdots(r-j+1)$. This is equivalent to Theorem M.22 through $\widehat{R}_m = (-1)^m \rho^m R^{(m)}(\rho)/m!$ and $\ell_j = (-1)^j j!\,\widehat{R}_j$ for $j \geq 1$ up to the $\log z$ term; both routes were implemented and agree. Theorem M.22 is preferred because it avoids the factorial rescaling and its intermediate quantities stay $\mathcal{O}(1)$. S2's normalisation $s_j = 2\theta_j$ (it expands $2(1-\mathrm{e}\zeta)$ rather than $1-\mathrm{e}\zeta$) accounts for every factor of 2 that differs between the siblings' formulas for t_r.

The first disturbance coefficients, recomputed, are

$$\begin{aligned}
\theta_1 &= 1.2160045618240489221572126l9\dots, & \theta_2 &= -0.4584157298563917191808477l7\dots,\\
\theta_3 &= 0.4481047743895350436657341 22\dots, & \theta_4 &= -0.3995616799562096774062319 80\dots,\\
\theta_5 &= 0.3852513419466130007195463 96\dots, & \theta_6 &= -0.3898065619592033419491646 40\dots.
\end{aligned} \tag{M.32}$$

M.4.2 All local coefficients

Theorem M.24 (Every Puiseux coefficient at ρ). *Factor $\theta(s) = \theta_1 s\big(1+U(s)\big)$ with $U(s) = \sum_{j\ge1} u_j s^j$, $u_j = \theta_{j+1}/\theta_1$. Then in a slit neighbourhood of $z = \rho$ there is a convergent Puiseux expansion*

$$T(z) = 1 + \sum_{n\ge1} t_n\, s^{n/2}, \qquad s = 1 - \frac{z}{\rho}, \tag{M.33}$$

and for every $n \ge 1$

$$\boxed{t_n = \sum_{\substack{1\le r\le n\\ r\equiv n\ (2)}} c_r\, \theta_1^{r/2}\, [s^{(n-r)/2}]\, \big(1+U(s)\big)^{r/2} = \sum_{\substack{1\le r\le n\\ r\equiv n\ (2)}} c_r\, \theta_1^{r/2} \sum_{k=0}^{(n-r)/2} \binom{r/2}{k} \widehat{B}_{\frac{n-r}{2},k}(u_1, u_2, \ldots).} \tag{M.34}$$

Thus t_n is a finite expression in $\omega_1, \ldots, \omega_n$, $\theta_1^{\pm1/2}$ and $\theta_2, \ldots, \theta_{\lceil (n+1)/2\rceil}$.

Proof. Near $z = \rho$ we have $\mathrm{e}\zeta = 1 - \theta(s)$ with $\theta(0) = 0$, so by Equation (M.12) and Equation (M.24), $T = 1 + \sum_{r\ge1} c_r\theta(s)^{r/2}$, the branch being the one with $T \to 1^-$ as $s \to 0^+$. Substituting $\theta(s)^{r/2} = \theta_1^{r/2} s^{r/2}(1+U(s))^{r/2}$, a term of index r can contribute to $s^{n/2}$ only when $n - r$ is a nonnegative even integer. Extracting $[s^{(n-r)/2}]$ and applying Theorem J.14 to the unit series $1 + U$ gives Equation (M.34). Convergence: θ is analytic and has a simple zero at $s = 0$ with $\theta_1 > 0$ (Equation (M.14)), so U is analytic with $U(0) = 0$; choosing a slit neighbourhood on which $|U| < 1$ and a consistent square root, the convergent series Equation (M.24) may be composed with the analytic map $s \mapsto \theta(s)$. □

Corollary M.25. *$t_1 = -\sqrt{2\theta_1}$, $t_2 = \frac{2}{3}\theta_1$, and $t_3 = -\frac{11\sqrt{2}}{36}\theta_1^{3/2} - \frac{\sqrt{2}}{2}\theta_2\theta_1^{-1/2}$.*

Proof. Take $n = 1, 2, 3$ in Equation (M.34) with $c_1 = -\sqrt{2}$, $c_2 = \frac{2}{3}$, $c_3 = -\frac{11\sqrt{2}}{36}$ and $\widehat{B}_{0,0} = 1$, $\widehat{B}_{1,1}(u) = u_1$, $\binom{1/2}{1} = \frac{1}{2}$. □

Remark M.26 (Sibling cross-check). In S2's normalisation $s_1 = 2\theta_1$, $s_2 = 2\theta_2$, the same coefficient reads $t_3 = -(11s_1^2 + 36s_2)/(72\sqrt{s_1})$; substituting $s_j = 2\theta_j$ reproduces Theorem M.25 exactly. Likewise $t_1 = -\sqrt{s_1}$ and $t_2 = s_1/3$. Both were checked numerically.

Table M.2: Puiseux coefficients in $T(z) = 1 + \sum_{n\ge1} t_n(1 - z/\rho)^{n/2}$, recomputed from Equation (M.34) at 90 working digits.

n	t_n	n	t_n
1	−1.559490020374640885542	10	0.205984831277907418736
2	0.810669707882699281438	11	−0.052724708498190562315
3	−0.285487021612845605304	12	0.017026568754953669446
4	0.165372365712083893477	13	−0.152370624366325396286
5	−0.342459970402154201042	14	0.173702883299850462894
6	0.317407225946528563568	15	−0.014473703739527044857
7	−0.107778800291631009182	16	−0.021899517611215562233
8	0.061384957055835104689	17	−0.144547193570909705477
9	−0.195212383597556463613	18	0.176077108885017779062

Remark M.27 (Analytic versus singular local terms). The even part $\sum_j t_{2j}s^j$ of Equation (M.33) is analytic at $z = \rho$. A fixed analytic monomial $(1 - z/\rho)^j$ is a polynomial in z and contributes nothing to $[z^n]$ once $n > j$; hence only the odd coefficients t_{2j+1} enter the algebraic $\rho^{-n}n^{-\bullet}$ asymptotics of Section M.5. This is not a discarding of information: the analytic germ continues through ρ and reappears in the exponentially smaller sectors of Section M.9. The sign pattern of Table M.2 is not monotone and should not be over-read; these are local analytic data, not the final coefficient constants.

In Otter's coordinate $\rho - z = \rho s$ the leading behaviour reads $T(z) = 1 - \mathsf{b}\,(\rho - z)^{1/2} + \mathcal{O}(\rho - z)$ with

$$\mathsf{b} = -\frac{t_1}{\sqrt{\rho}} = 2.681128147267112238577328 78\ldots. \tag{M.35}$$

Theorem M.28 (Local expansion with an explicit remainder). *For every $N \geq 1$ there are a dented ("Δ-domain") neighbourhood Δ of ρ and a constant K_N such that*

$$\left|T(z) - 1 - \sum_{n=1}^{N} t_n \left(1 - \frac{z}{\rho}\right)^{n/2}\right| \leq K_N \left|1 - \frac{z}{\rho}\right|^{(N+1)/2}, \qquad z \in \Delta. \tag{M.36}$$

Proof. By Theorem M.10 the map $s \mapsto \theta(s)$ is analytic on a disc $|s| < \varepsilon_0$; by Equation (M.14) it has a simple zero at $s = 0$ with $\theta_1 > 0$. Shrinking ε_0, $|U(s)| < \frac{1}{2}$ there. The series Equation (M.24) converges for $|v|$ small; a Cauchy estimate on the analytic function $v \mapsto \mathcal{C}((1 - v)/\mathrm{e})$ restricted to a slit disc gives $|c_r| \leq K\varepsilon_1^{-r/2}$. Composing convergent expansions and using Taylor's theorem with remainder on $(1 + U)^{r/2}$ gives Equation (M.36) on the slit disc $\{|s| < \varepsilon,\ |\arg s| < \pi - \eta\}$, which is a Δ-domain at ρ in the z-variable. $\square$

M.5 Transfer to the all-orders coefficient asymptotics

M.5.1 Exact transfer of a singular monomial

For $\beta \notin \mathbb{N}_0$ and $n \geq 0$,

$$[z^n] \left(1 - \frac{z}{\rho}\right)^{\beta} = \rho^{-n}(-1)^n \binom{\beta}{n} = \rho^{-n} \frac{\Gamma(n - \beta)}{\Gamma(-\beta)\Gamma(n + 1)}. \tag{M.37}$$

For $\beta \in \mathbb{N}_0$ the left-hand side vanishes for $n > \beta$ — the analytic-germ statement of Theorem M.27. It remains to expand the gamma ratio to all orders.

Lemma M.29 (Gamma-ratio coefficients). *Let $B_m(x)$ be the Bernoulli polynomials, and for $\beta \in \mathbb{C}$ set*

$$\varkappa_m(\beta) := \frac{(-1)^{m+1}}{m(m+1)}\big(B_{m+1}(-\beta) - B_{m+1}(1)\big), \qquad m \geq 1, \tag{M.38}$$

and define $g_r(\beta)$ by $\exp\big(\sum_{m\geq 1} \varkappa_m(\beta) v^m\big) = \sum_{r\geq 0} g_r(\beta) v^r$, equivalently

$$g_0(\beta) = 1, \qquad g_r(\beta) = \frac{1}{r}\sum_{m=1}^{r} m\,\varkappa_m(\beta)\,g_{r-m}(\beta) = \sum_{k=0}^{r} \frac{1}{k!}\widehat{B}_{r,k}\big(\varkappa_1(\beta), \varkappa_2(\beta), \ldots\big). \tag{M.39}$$

Then, for fixed β and $n \to \infty$ in any sector avoiding the negative real axis,

$$\frac{\Gamma(n - \beta)}{\Gamma(n + 1)} \sim n^{-\beta-1} \sum_{r\geq 0} \frac{g_r(\beta)}{n^r}. \tag{M.40}$$

Each g_r is a polynomial of degree $2r$ in β with rational coefficients; the first five are

$$g_0 = 1, \qquad g_1(\beta) = \frac{\beta(\beta+1)}{2}, \quad g_2(\beta) = \frac{\beta(\beta+1)(\beta+2)(3\beta+1)}{24},$$
$$g_3(\beta) = \frac{\beta^2(\beta+1)^2(\beta+2)(\beta+3)}{48}, \tag{M.41}$$

$$g_4(\beta) = \frac{\beta(\beta+1)(\beta+2)(\beta+3)(\beta+4)\big(15\beta^3 + 30\beta^2 + 5\beta - 2\big)}{5760}. \tag{M.42}$$

In particular $\big(g_r(\tfrac{1}{2})\big)_{r\geq 0} = 1, \frac{3}{8}, \frac{25}{128}, \frac{105}{1024}, \frac{1659}{32768}, \frac{6237}{262144}, \ldots$

Proof. The Euler–Maclaurin/Bernoulli form of Stirling's expansion gives, for fixed a, b,

$$\log \frac{\Gamma(n+a)}{\Gamma(n+b)\, n^{a-b}} \sim \sum_{m \geq 1} \frac{(-1)^{m+1}}{m(m+1)} \frac{B_{m+1}(a) - B_{m+1}(b)}{n^m}$$

as $n \to \infty$ in the stated sectors. Taking $a = -\beta$, $b = 1$ yields $\log\bigl(\Gamma(n-\beta)/\Gamma(n+1)\bigr) \sim -(\beta+1)\log n + \sum_m \varkappa_m(\beta) n^{-m}$, and exponentiating an asymptotic series term by term is legitimate, producing exactly the coefficients of $\exp\bigl(\sum \varkappa_m v^m\bigr)$. The recurrence follows from differentiating that generating identity; the Bell form is Theorem J.9. Degree: $\varkappa_m$ has degree $m+1$ in β and vanishes at $\beta = 0$, so a monomial $\prod \varkappa_{m_i}$ with $\sum m_i = r$ has degree $\leq r + \#\{i\} \leq 2r$; the displayed $g_1, \ldots, g_4$ were recomputed here from Equation (M.38) by exact rational arithmetic. □

Lean status: partial, the coefficient recurrence only. The recurrence in (M.39) is formal in `ExpSeriesRecurrence.lean`, and in the generality it deserves: it has nothing to do with gamma ratios. For an arbitrary power series K without constant term over a commutative $\mathbb{Q}$-algebra, `Fabius.expCoeff` is the coefficient sequence of $\exp K$, `Fabius.expCoeff_zero` is $g_0 = 1$, and `Fabius.natCast_mul_expCoeff` is $n\, g_n = \sum_{m \leq n} m\, \varkappa_m\, g_{n-m}$ with $\varkappa_m$ the coefficients of K. `Fabius.succ_mul_expCoeff_succ` is the same fact in the shape the Cauchy product produces, before reindexing.

Two remarks on the formal statement. It is the coefficient form of $G' = K'G$, so `derivative_subst` and `derivative_exp` give it at once and all the work is the reindexing — the convolution runs the other way round from the display above. And the sum is taken over $0 \leq m \leq n$ rather than $1 \leq m \leq n$; the $m = 0$ term carries the factor m and vanishes, which costs nothing and makes the identity true at $n = 0$ as well, where the volume's indexing would compare an empty sum with $0 \cdot g_0$.

What is *not* formalized: everything else in the lemma. The definition of $\varkappa_m(\beta)$ through Bernoulli polynomials, the closed forms (M.41) and (M.42) and the degree-$2r$ statement, the ordinary Bell form of (M.39) — which would need the ordinary partial Bell polynomials, themselves unformalized, see Theorem A.4 — and the asymptotic (M.40) itself, which needs Stirling's expansion to all orders in a sector. The formalized content is the recurrence that turns $\varkappa$ into g, and nothing about where $\varkappa$ comes from.

Remark M.30 (Part 0 candidate). Theorem M.29 is not specific to rooted trees and is used verbatim wherever a chapter transfers an algebraic singularity to coefficient asymptotics. It is listed in the report as a candidate for promotion into Part 0. All three siblings state it; S2 and S3 give the same recurrence, S1 the same object under the name $\gamma_m(\beta)$ with $\eta_r(\beta)$ for $\varkappa_r(\beta)$.

M.5.2 The master coefficient formula

Theorem M.31 (All-orders dominant expansion of A000081). *For every fixed $M \geq 0$,*

$$\boxed{a_n = \frac{\rho^{-n}}{\sqrt{\pi}\, n^{3/2}} \left(\sum_{m=0}^{M} \frac{\tau_m}{n^m} + \mathcal{O}(n^{-M-1}) \right), \qquad n \to \infty,} \tag{M.43}$$

where

$$\boxed{\tau_m = \sum_{j=0}^{m} \Delta_j\, t_{2j+1}\, g_{m-j}\bigl(j + \tfrac{1}{2}\bigr), \qquad \Delta_j := \frac{(-1)^{j+1}(2j+1)!!}{2^{j+1}} = \frac{\sqrt{\pi}}{\Gamma(-j-\frac{1}{2})}.} \tag{M.44}$$

Every ingredient is a finite coefficient extraction: t_{2j+1} by Equation (M.34)*, g_r by Equation* (M.39)*, θ_m by Equation* (M.30)*, ω_m by Equation* (M.23)*, ϱ_r by Equation* (M.10)*.*

Proof. Fix M and apply Theorem M.28 with $N = 2M+2$: on a Δ-domain at ρ,

$$T(z) = A(z) + \sum_{j=0}^{M} t_{2j+1}\left(1-\frac{z}{\rho}\right)^{j+1/2} + \mathcal{O}\left(\left|1-\frac{z}{\rho}\right|^{M+3/2}\right),$$

with A the polynomial collecting the even (analytic) terms. By Theorem M.12 the only singularity on $|z| = \rho$ is ρ itself, so the Cauchy contour may be deformed into the standard Flajolet–Odlyzko Δ-contour. The polynomial A contributes nothing to $[z^n]$ for $n > \deg A$. Each retained monomial contributes exactly $\rho^{-n} t_{2j+1}\Gamma(n-j-\frac{1}{2})/\big(\Gamma(-j-\frac{1}{2})\Gamma(n+1)\big)$ by Equation (M.37), whose asymptotic expansion is Equation (M.40); the $\mathcal{O}$-term transfers to $\mathcal{O}(\rho^{-n}n^{-M-5/2})$ by the standard Hankel-contour estimate. Collecting the coefficient of $\rho^{-n}n^{-m-3/2}$ from the pairs $j+r=m$ and multiplying through by $\sqrt{\pi}$ gives Equation (M.44), the identity $\Gamma(-j-\frac{1}{2}) = (-4)^{j+1}(j+1)!\sqrt{\pi}/(2j+2)! = (-1)^{j+1}2^{j+1}\sqrt{\pi}/(2j+1)!!$ supplying Δ_j. ☐

Corollary M.32 (Otter's constant). *With $C := \tau_0/\sqrt{\pi}$,*

$$\tau_0 = -\frac{t_1}{2} = \sqrt{\frac{\theta_1}{2}}, \qquad C = \sqrt{\frac{\theta_1}{2\pi}} = \sqrt{\frac{\mathrm{e}\rho\,\zeta'(\rho)}{2\pi}} = \sqrt{\frac{1+\rho R'(\rho)}{2\pi}}, \qquad a_n \sim C\,\alpha^n n^{-3/2}. \quad \text{(M.45)}$$

Numerically

$$\tau_0 = 0.77974501018732044277110369 79\ldots, \qquad C = 0.4399240125710253040409033914345447648\ldots. \quad \text{(M.46)}$$

Proof. Set $m = 0$ in Equation (M.44): $\Delta_0 = -\frac{1}{2}$, $g_0 = 1$, so $\tau_0 = -t_1/2 = \sqrt{2\theta_1}/2 = \sqrt{\theta_1/2}$ by Theorem M.25. Divide by $\sqrt{\pi}$ and use Equation (M.14). ☐

The identity $C^2 = \theta_1/(2\pi)$ is a genuinely independent check: the left side comes through the whole Puiseux/transfer chain, the right side directly from one derivative of ζ. The two agree to all 55 digits carried.

Expanding Equation (M.44) for small m gives the transfer identities

$$\begin{aligned}
\tau_0 &= -\tfrac{1}{2}t_1,\\
\tau_1 &= -\tfrac{3}{16}\big(t_1 - 4t_3\big) = -\tfrac{3}{16}t_1 + \tfrac{3}{4}t_3,\\
\tau_2 &= -\tfrac{5}{256}\big(5t_1 - 72t_3 + 96t_5\big) = -\tfrac{25}{256}t_1 + \tfrac{45}{32}t_3 - \tfrac{15}{8}t_5,\\
\tau_3 &= -\tfrac{105}{2048}\big(t_1 - 44t_3 + 160t_5 - 128t_7\big) = -\tfrac{105}{2048}t_1 + \tfrac{1155}{512}t_3 - \tfrac{525}{64}t_5 + \tfrac{105}{16}t_7,\\
\tau_4 &= -\tfrac{21}{65536}\big(79t_1 - 10800t_3 + 81600t_5 - 161280t_7 + 92160t_9\big).
\end{aligned} \quad \text{(M.47)}$$

S2 states the right-hand forms, S3 the factored left-hand forms; they are identical after clearing denominators, and both were re-derived here symbolically from Equation (M.44). For any m beyond 4 the compact convolution is far preferable to the expanded form.

Higher values continue smoothly; for reference $\tau_{15} = 8.078305401468884289 \times 10^{13}$, $\tau_{18} = 2.073828085209893749 \times 10^{18}$.

Writing

$$F(v) := 1 + \sum_{m\geq 1} f_m v^m, \qquad f_m = \frac{\tau_m}{\tau_0}, \quad \text{(M.48)}$$

the inversion-ready normal form is

$$\boxed{a_n \sim C\,\mathrm{e}^{\lambda n}\, n^{-p}\, F(1/n), \qquad p = \tfrac{3}{2}.} \quad \text{(M.49)}$$

Table M.3: Coefficients of the dominant expansion $a_n \sim \rho^{-n}(\pi n^3)^{-1/2}\sum_{m\geq 0}\tau_m n^{-m}$ and their normalisations $f_m = \tau_m/\tau_0$, recomputed here.

m	τ_m	$f_m = \tau_m/\tau_0$
0	0.7797450101873204427711	1
1	0.07828911261061096206128	0.1004034800964014347637
2	0.3929402676631860184743	0.5039343151023035331022
3	1.537879315978838091102	1.972284908382278469206
4	8.200844090435596159220	10.51734090413157399274
5	57.29291473494343787740	73.47647498402047106417
6	503.0445050262735818248	645.1397552456612928153
7	5359.600933884326033403	6873.530274463408619941
8	67342.06920114653048341	86364.21948371140797700
9	975425.4970695924732213	1250954.458605978293018
10	15996932.49293173312961	20515594.56480361813667
11	292822531.3353392231146	–
12	5914523441.293932519117	–

Remark M.33 (Divergence and least-term truncation). Equation (M.43) asserts a Poincaré expansion at every fixed order; it does *not* assert convergence, and Table M.3 shows the coefficients growing rapidly. The optimal truncation and the size of the resulting floor are governed by Theorem J.54; the relevant nearest competing exponential scale is identified in Section M.9 as $\rho^{-n/2}$, so the least-term floor at index n is $\mathcal{O}(\rho^{n/2})$ relative to the leading term — consistent with the error tables of Section M.11.2, where the improvement stops at $n = 20$ around $M = 6$. S3 states this correctly as a numerical strategy rather than a theorem; S1 and S2 call the late-term growth "the perturbative shadow of farther singularities", which is the right heuristic but is not proved anywhere in the three drafts and is not proved here.

M.6 The multiplicative reciprocal

If "inverse" is read as the reciprocal sequence, the answer is immediate from Equation (M.49) and requires no inversion theory at all.

Proposition M.34 (All-orders reciprocal). *Write* $1/F(v) = 1 + \sum_{m\geq 1}\bar{f}_m v^m$. *Then*

$$\bar{f}_m = \sum_{k=1}^{m}(-1)^k\,\widehat{B}_{m,k}(f_1, f_2, \ldots) = -\sum_{k=1}^{m} f_k\,\bar{f}_{m-k} \qquad (\bar{f}_0 = 1), \tag{M.50}$$

and for every fixed $M \geq 0$

$$\frac{1}{a_n} = \frac{\sqrt{\pi}\,\rho^n n^{3/2}}{\tau_0}\left(\sum_{m=0}^{M}\frac{\bar{f}_m}{n^m} + \mathcal{O}(n^{-M-1})\right). \tag{M.51}$$

Proof. The first equality in Equation (M.50) is Theorem J.14 with exponent -1; the second is the Cauchy product $F \cdot F^{-1} = 1$. For Equation (M.51), note that $a_n > 0$ and, by Equation (M.43), $a_n = C\alpha^n n^{-p}(1+\varepsilon_n)$ with $\varepsilon_n \to 0$; hence $1/a_n$ is defined and $1/(1+\varepsilon_n)$ has the asymptotic expansion obtained by formally inverting the (asymptotic) series for $1+\varepsilon_n$ — a legitimate operation because the leading term is $1 \neq 0$ and each truncation error is $\mathcal{O}(n^{-M-1})$. □

The first reciprocal coefficients, recomputed, are

$$\begin{aligned} \bar{f}_1 &= -0.1004034800964014347637, & \bar{f}_2 &= -0.4938534562868350540384,\\ \bar{f}_3 &= -1.872103543737176348103, & \bar{f}_4 &= -9.882481221441551170916,\\ \bar{f}_5 &= -69.51082491336461883211, & \bar{f}_6 &= -616.9168645978433816315. \end{aligned} \tag{M.52}$$

Remark M.35 (Reciprocal of the full transseries). Writing $a_n = \mathcal{M}_n(1+\mathcal{E}_n)$ with $\mathcal{M}_n$ the complete dominant sector and $\mathcal{E}_n$ the sum of relative exponentially small sectors of Section M.9, one has $1/a_n = \mathcal{M}_n^{-1}\sum_{q\geq 0}(-\mathcal{E}_n)^q$; the products generate exactly the additive semigroup of the singular actions, with coefficients given by finite Cauchy products. This is a formal statement about the transseries, and inherits whatever status Theorem M.55 has on the chosen continuation domain.

Warning M.36. The reciprocal is *not* a functional inverse. Its exponential scale is ρ^n, while the functional inverse of Section M.7 grows only logarithmically in its argument. All three siblings make this point; it is repeated here because the ambiguity of the word "inverse" is exactly what Theorem J.41 exists to remove.

M.7 The asymptotic inverse: the Part 0 dictionary

M.7.1 The parameter dictionary

Everything needed to invert Equation (M.49) has been proved in Part 0. What this chapter must supply is only the dictionary and the verification of Part 0's hypotheses.

Proposition M.37 (A000081 as an instance of the exponential–power model)**.** *Let*

$$\mathcal{A}(x) := C\,\mathrm{e}^{\lambda x}x^{-p}F(x^{-1}), \qquad p = \tfrac{3}{2}, \tag{M.53}$$

with C from Theorem M.32 and F from Equation (M.48)*. Then $\mathcal{A}$ is an instance of Theorem J.2 under*

Part 0	C	a	α	b	δ	$Q(t)$
here	$C = \sqrt{\theta_1/(2\pi)}$	$\lambda = \log(1/\rho)$	1	$-\frac{3}{2}$	1	$H(t) = \log F(t)$

(M.54)

with $H \in t\mathbb{R}[[t]]$. In particular $a = \lambda > 0$ and $\alpha = 1$, so $\mathcal{A}$ is already in the linear-phase normal form and Theorem J.4 is the identity substitution.

Proof. Equation (M.49) is Theorem J.2 with the stated substitutions; $\lambda > 0$ because $\rho < 1$ (Theorem M.11), $C > 0$ because $\theta_1 > 0$, and $H(0) = \log F(0) = \log 1 = 0$ so $H \in t\mathbb{R}[[t]]$. ☐

Definition M.38 (Additive phase coefficients)**.** Write $H(v) = \log F(v) = \sum_{m\geq 1}\varphi_m v^m$. By Theorem J.14,

$$\varphi_m = \sum_{k=1}^{m}\frac{(-1)^{k-1}}{k}\widehat{B}_{m,k}(f_1,f_2,\ldots) = f_m - \frac{1}{m}\sum_{k=1}^{m-1}k\,\varphi_k\,f_{m-k}. \tag{M.55}$$

Taking logarithms in Equation (M.43) gives the form used for inversion,

$$\log a_n = \lambda n - \tfrac{3}{2}\log n + \log C + \sum_{m=1}^{M}\frac{\varphi_m}{n^m} + \mathcal{O}(n^{-M-1}), \tag{M.56}$$

with, recomputed here,

$$\begin{aligned} \varphi_1 &= 0.1004034800964014347637, & \varphi_2 &= 0.4988938856945692935703,\\ \varphi_3 &= 1.922025533841821821591, & \varphi_4 &= 10.19739642337376887053,\\ \varphi_5 &= 71.47146706207073604412, & \varphi_6 &= 630.8599376073372478899. \end{aligned} \tag{M.57}$$

M.7.2 The exact Lambert carrier

Corollary M.39 (Carrier; specialisation of Theorem J.23). *Let* $y > y_* := C\,(\mathrm{e}\lambda/p)^p = 1.2108232025837572603\ldots$. *Then the equation* $y = C\mathrm{e}^{\lambda x}x^{-p}$ *has exactly one solution* $x_0 = x_0(y) > p/\lambda$, *namely*

$$\boxed{x_0(y) = -\frac{p}{\lambda}\,W_{-1}\left(-\frac{\lambda}{p}\left(\frac{C}{y}\right)^{1/p}\right) = -\frac{3}{2\lambda}\,W_{-1}\left(-\frac{2\lambda}{3}\left(\frac{C}{y}\right)^{2/3}\right).} \tag{M.58}$$

Proof. Raise $y = C\mathrm{e}^{\lambda x}x^{-p}$ to the power $-1/p$ and multiply by $-\lambda/p$:

$$-\frac{\lambda}{p}\left(\frac{C}{y}\right)^{1/p} = -\frac{\lambda x}{p}\,\mathrm{e}^{-\lambda x/p} = \xi\mathrm{e}^{\xi}, \qquad \xi := -\frac{\lambda x}{p}.$$

Write $\epsilon := -\frac{\lambda}{p}(C/y)^{1/p}$. The map $x \mapsto C\mathrm{e}^{\lambda x}x^{-p}$ is strictly increasing on $(p/\lambda, \infty)$ (its logarithmic derivative is $\lambda - p/x > 0$ there) with infimum y_* attained as $x \downarrow p/\lambda$, so $y > y_*$ gives a unique root $x_0 > p/\lambda$, i.e. $\xi < -1$; and $y > y_*$ is exactly $\epsilon \in (-\mathrm{e}^{-1}, 0)$. On that range $\xi \mapsto \xi\mathrm{e}^{\xi}$ has the two real inverse branches $W_0 \in (-1, 0)$ and $W_{-1} < -1$, and the required root is the second. This is Theorem J.23 with $L = \log(y/C)$, $a = \lambda$, $b = -p$ and branch index $k = -1$; the branch rule there selects $k = -1$ under exactly the condition $x > -b/a$. □

Remark M.40. The principal branch W_0 returns the other, small root of the same equation, which is the spurious solution: all three siblings say so, and the threshold y_* makes the dichotomy quantitative. For A000081 the threshold lies below $a_3 = 2$, so Equation (M.58) may be used at every sequence value from $n = 3$ on.

M.7.3 All algebraic corrections

Theorem M.41 (Corrections; specialisation of Theorem J.29). *Let*

$$\Psi(t, u) := \frac{p}{\lambda}\log(1 + tu) - \frac{1}{\lambda}H\left(\frac{t}{1 + tu}\right) \in t\mathbb{R}[[t, u]]. \tag{M.59}$$

Then there is a unique $D(t) = \sum_{m\ge1} d_m t^m \in t\mathbb{R}[[t]]$ *with* $D(t) = \Psi\big(t, D(t)\big)$, *its coefficients are given by Theorem J.19,*

$$\boxed{d_m = \sum_{k=1}^{m}\frac{1}{k}\,[t^m u^{k-1}]\,\Psi(t, u)^k,} \tag{M.60}$$

equivalently by the triangular extraction

$$d_m = [t^m]\left\{\frac{p}{\lambda}\log\big(1 + tD_{<m}(t)\big) - \frac{1}{\lambda}H\left(\frac{t}{1 + tD_{<m}(t)}\right)\right\}, \qquad D_{<m} = \sum_{j<m} d_j t^j, \tag{M.61}$$

or by the completely explicit ordinary-Bell form

$$d_m = \frac{p}{\lambda}\sum_{r=1}^{m}\frac{(-1)^{r-1}}{r}\widehat{B}_{m-r,r}(d_1, d_2, \ldots) - \frac{1}{\lambda}\sum_{k=1}^{m}\varphi_k\sum_{r=0}^{m-k}(-1)^r\binom{k+r-1}{r}\widehat{B}_{m-k-r,r}(d_1, d_2, \ldots), \tag{M.62}$$

in which the index constraints force only $d_1, \ldots, d_{m-1}$ *to occur on the right. With* x_0 *from Equation* (M.58) *the complete algebraic inverse is*

$$\boxed{\nu(y) \sim x_0(y) + \sum_{m\ge1}\frac{d_m}{x_0(y)^m}.} \tag{M.63}$$

Proof. Theorem J.29 states exactly this for a model $C\mathrm{e}^{ax}x^b\mathrm{e}^{Q(x^{-\delta})}$ with $\Psi(t,u) = -\frac{1}{a}\big[b\log(1+tu) + Q\big(t/(1+tu)\big)\big]$; substituting the dictionary Equation (M.54) ($a = \lambda$, $b = -p$, $\delta = 1$, $Q = H$) gives Equation (M.59). Concretely: putting $x = x_0 + D$ into $\log(y/C) = \lambda x - p\log x + H(x^{-1})$ and subtracting the carrier equation $\log(y/C) = \lambda x_0 - p\log x_0$ gives, with $t = x_0^{-1}$,

$$\lambda D - p\log(1+tD) + H\left(\frac{t}{1+tD}\right) = 0,$$

which is $D = \Psi(t,D)$. Since $\Psi \in t\mathbb{R}[[t,u]]$, the formal implicit-function theorem gives a unique solution in $t\mathbb{R}[[t]]$; Theorem J.19 converts it into Equation (M.60), and Equation (M.61) is the same statement read as a recursion. Equation (M.62) follows by expanding $\log(1+tD) = \sum_r \frac{(-1)^{r-1}}{r}t^rD^r$ and $H\big(t/(1+tD)\big) = \sum_k \varphi_k t^k \sum_r (-1)^r\binom{k+r-1}{r}t^rD^r$ and applying Theorem J.9 to the powers D^r. □

A formal-to-analytic slide in S1. S1's "General coefficient formula for the inverse" is stated as $x(y) \sim X(y) + \sum_m d_m X(y)^{-m}$, an asymptotic assertion, but its proof ends with "substitution … proves formal inversion order by order" and supplies no remainder. As stated, Theorem M.41 is a theorem about formal series only; the asymptotic reading requires Theorem M.42 below, whose argument is S3's — the only one of the three drafts that separates the two statements. The volume keeps them separate.

Theorem M.42 (Remainder; specialisation of Theorem J.51)**.** *Fix $M \geq 1$ and set $\nu_M(y) := x_0(y) + \sum_{m=1}^{M-1} d_m x_0(y)^{-m}$. Let $f(x) = \log C + \lambda x - p\log x + H(x^{-1})$ be the (asymptotic) logarithm of the forward interpolant. Then*

$$f\big(\nu_M(y)\big) - \log y = \mathcal{O}\big(x_0(y)^{-M}\big), \qquad \nu(y) - \nu_M(y) = \mathcal{O}\big(x_0(y)^{-M}\big) = \mathcal{O}\big((\log y)^{-M}\big). \tag{M.64}$$

Proof. By Theorem M.41 the coefficients of $t^1,\dots,t^{M-1}$ in $D - \Psi(t,D)$ vanish, so truncating leaves a residual $\mathcal{O}(t^M)$; this is the first claim. Since $f'(x) = \lambda - p/x + \mathcal{O}(x^{-2})$ is bounded away from 0 for large x, Theorem J.47 converts the residual into a forward error of the same order divided by f'; this is Theorem J.51 with $a = \lambda$. Finally $x_0(y) = \lambda^{-1}\log y\,(1 + o(1))$ by Equation (M.58). □

M.7.4 Coefficients

Expanding Equation (M.61) through t^5 — and verifying the result symbolically — gives the closed forms

$$d_1 = -\frac{\varphi_1}{\lambda}, \tag{M.65}$$

$$d_2 = -\frac{p\varphi_1 + \lambda\varphi_2}{\lambda^2}, \tag{M.66}$$

$$d_3 = -\frac{\lambda\varphi_1^2 + p^2\varphi_1 + p\lambda\varphi_2 + \lambda^2\varphi_3}{\lambda^3}, \tag{M.67}$$

$$d_4 = -\frac{1}{2\lambda^4}\Big(5p\lambda\varphi_1^2 + 6\lambda^2\varphi_1\varphi_2 + 2p^3\varphi_1 + 2p^2\lambda\varphi_2 + 2p\lambda^2\varphi_3 + 2\lambda^3\varphi_4\Big), \tag{M.68}$$

$$\begin{aligned} d_5 = -\frac{1}{2\lambda^5}\Big(&4\lambda^2\varphi_1^3 + 9p^2\lambda\varphi_1^2 + 14p\lambda^2\varphi_1\varphi_2 + 8\lambda^3\varphi_1\varphi_3 + 4\lambda^3\varphi_2^2 \\ &+ 2p^4\varphi_1 + 2p^3\lambda\varphi_2 + 2p^2\lambda^2\varphi_3 + 2p\lambda^3\varphi_4 + 2\lambda^4\varphi_5\Big). \end{aligned} \tag{M.69}$$

Beyond $m = 5$ the expanded expressions grow fast, and Equations (M.60) and (M.61) are both shorter and less error-prone.

Table M.4: Additive phase coefficients φ_m and Lambert-carrier corrections d_m, recomputed here. The last column is S2's normalisation $\mathrm{D}_m = \lambda^{m+1} d_m$ (see Theorem M.43).

m	φ_m	d_m	$\mathrm{D}_m = \lambda^{m+1} d_m$
1	0.1004034800964014348	−0.09264385352561312984	−0.1088130343442745145
2	0.4988938856945692936	−0.5885630399313472628	−0.7491856512881932495
3	1.922025533841821822	−2.596680167407531021	−3.582177326832277789
4	10.19739642337376887	−13.14905339996706569	−19.65872121138776004
5	71.47146706207073604	−85.49870151462556652	−138.5327478518560861
6	630.8599376073372479	−711.2629148329465609	−1248.979326371692854
7	6753.629059387619229	−7306.788270284967257	−13905.40884474814835
8	85171.04383262353790	−89561.76088263724860	−184719.1911265080905
9	1236987.135642947528	−1274829.503083708903	—
10	20325777.10708850337	−20641099.00032138544	—

Numerical error in S2's audit appendix. S2's appendix table of η_j and D_j reports $D_3 = -3.582177326832123\ldots$, disagreeing with its own main text ($-3.582177326832277789103009844$) from the fourteenth significant digit. Recomputation confirms the main-text value, $\mathrm{D}_3 = -3.58217732683227778910301\ldots$; the appendix entry is wrong. Every other entry of that table was checked and is correct to the digits shown.

Remark M.43 (Three normalisations of the same inverse). The siblings work in three different variables, which is why their coefficient lists look different:

- S1 and S3 expand in the index variable x itself and obtain the d_m of Table M.4; their carriers agree with Equation (M.58).
- S2 rescales to $X = \lambda x$, so that its carrier is $X_0 = -pW_{-1}\big(-\frac{1}{p}\mathrm{e}^{-Z/p}\big)$ with $Z = \log\big(y/(C\lambda^p)\big)$, its corrections are $\mathrm{D}_m = \lambda^{m+1} d_m$, and its phase coefficients are $\eta_m^{\mathrm{S2}} = \lambda^m \varphi_m$. Both dictionaries were verified numerically for $1 \le m \le 8$, and $X_0 = \lambda x_0$ holds identically because the two carrier equations differ by that substitution.
- S3 also uses x but writes the polynomial–logarithmic form in terms of $\mathsf{q} = p/\lambda$ and $\eta_m^{\mathrm{S3}} = \varphi_m/\lambda$.

The constant appearing in S2's Z is $C\lambda^p = 0.4963361416595997474727369464\ldots$.

M.7.5 Polynomial–logarithmic form

Eliminating the Lambert function gives the expansion in the canonical logarithmic scale. That is Theorem J.37; only the dictionary is recorded here.

Corollary M.44 (Flattened inverse). *Put*

$$\mathcal{X} := \frac{1}{\lambda}\log\frac{y}{C}, \qquad L := \log\mathcal{X}, \qquad \mathsf{q} := \frac{p}{\lambda}, \qquad \eta_k := \frac{\varphi_k}{\lambda}. \tag{M.70}$$

Then the inverse problem is $x - \mathsf{q}\log x + \sum_{k\ge1}\eta_k x^{-k} = \mathcal{X}$, and Theorem J.37 gives

$$\boxed{\nu(y) \sim \mathcal{X} + \mathsf{q}L + \sum_{m\ge1}\frac{P_m(L)}{\mathcal{X}^m},} \tag{M.71}$$

where each P_m is a polynomial of degree m over $\mathbb{Q}[\mathsf{q}, \eta_1, \ldots, \eta_m]$ with

$$[L^m]\,P_m(L) = \frac{(-1)^{m+1}}{m}\mathsf{q}^{m+1}, \qquad P_m(0) = d_m. \tag{M.72}$$

The identity $P_m(0) = d_m$ is not an accident: setting $L = 0$ in the flattening recursion reduces it to the fixed-point equation solved by the Lambert carrier, and formal uniqueness in Theorem J.29 identifies the two. The first two polynomials are

$$P_1(L) = \mathsf{q}^2 L - \eta_1 = 1.9156590172195452616\, L - 0.0926438535256131298, \tag{M.73}$$

$$\begin{aligned} P_2(L) &= -\tfrac{1}{2}\mathsf{q}^3 L^2 + \big(\mathsf{q}^3 + \mathsf{q}\eta_1\big) L - \mathsf{q}\eta_1 - \eta_2 \\ &= -1.3257062894576032164\, L^2 + 2.7896427837264004647\, L - 0.5885630399313472628. \end{aligned} \tag{M.74}$$

Remark M.45 (The two flattenings are not the same polynomials). S2 flattens in $Z = \log\big(y/(C\lambda^p)\big)$ and $\log Z$; S3 flattens in $\mathcal{X} = \lambda^{-1}\log(y/C)$ and $\log \mathcal{X}$. The two recursions are formally the *same* recursion with different parameters — S3's is S2's under $(p, \eta_k^{\mathrm{S2}}) \mapsto (\mathsf{q}, \eta_k^{\mathrm{S3}})$ — but they expand the same function in *different* variables, so their coefficient polynomials genuinely differ. Neither draft is in error; the caution is only that the two lists must not be compared term by term. We adopt S3's normalisation because it keeps the index variable x throughout and so matches Equation (M.63).

Retaining only the first two scales,

$$\nu(y) = \frac{\log y}{\lambda} + \frac{p}{\lambda}\log\log y - \frac{1}{\lambda}\log\big(C\lambda^p\big) + o(1), \tag{M.75}$$

that is

$$\nu(y) = 0.9227155616186015248 \log y + 1.3840733424279022872 \log\log y + 0.6463639827002088073 + o(1). \tag{M.76}$$

Equation (M.63) is preferable numerically because W_{-1} resums the entire logarithmic tower exactly before any truncation is made.

M.8 The discrete inverse

The objects distinguished in Theorem J.41 are, for this sequence: the smooth inverse ν of Section M.7; the integer staircase

$$N(y) := \min\{n \geq 1 : a_n \geq y\}; \tag{M.77}$$

and the round-to-nearest generalized inverse. Applying Theorem J.43 requires its separation hypothesis, which for A000081 is supplied by the next two lemmas.

Lemma M.46 (Strict monotonicity). *$a_1 = a_2 = 1$ and $a_{n+1} > a_n$ for every $n \geq 2$.*

Proof. Monotonicity: the map sending a rooted tree $\mathfrak{t}$ on n vertices to the rooted tree whose root has the single child subtree $\mathfrak{t}$ is an injection from the n-vertex shapes into the $(n+1)$-vertex shapes, so $a_{n+1} \geq a_n$. Strictness for $n \geq 2$: when $n+1 \geq 3$, the star $K_{1,n}$ rooted at its centre has $n \geq 2$ children and is therefore not in the image of that injection, whence $a_{n+1} \geq a_n + 1$. The values $a_1 = a_2 = 1$ are immediate. □

Lemma M.47 (Separation). *$a_{n+1}/a_n \to \alpha = 2.9557\ldots > 1$; more precisely*

$$\log\frac{a_{n+1}}{a_n} = \lambda - \frac{3}{2n} + \frac{\frac{3}{4} - \varphi_1}{n^2} + \frac{-\frac{1}{2} + \varphi_1 - 2\varphi_2}{n^3} + \mathcal{O}(n^{-4}). \tag{M.78}$$

Proof. Subtract Equation (M.56) at n from the same relation at $n+1$, and expand $\log(n+1) = \log n + n^{-1} - \frac{1}{2}n^{-2} + \frac{1}{3}n^{-3} - \cdots$ and $(n+1)^{-m} = n^{-m}(1 - mn^{-1} + \cdots)$. Collecting powers of n^{-1} gives Equation (M.78); the limit statement is its leading term. □

Corollary M.48 (Staircase recovery). *Let ν_M be as in Theorem M.42. For every $M \geq 2$ and all sufficiently large n, $\left|\nu_M(a_n) - n\right| < \frac{1}{2}$, so n is recovered from a_n by rounding. For a general real target $y > 1$, $N(y) = \nu(y) + \mathcal{O}(1)$, and certifying $N(y)$ exactly requires the two-sided test $a_{m-1} < y \leq a_m$, which Equation* (M.63) *reduces to a bounded neighbourhood of $m \approx \nu(y)$.*

Proof. The first claim is Theorem M.42 with $M \geq 2$, since $\mathcal{O}\big((\log a_n)^{-2}\big) = \mathcal{O}(n^{-2})$. For the second, apply Theorem J.43, whose separation hypothesis holds by Theorems M.46 and M.47: the logarithmic derivative of the interpolation tends to $\lambda > 0$, so multiplying y by a bounded factor moves $\nu(y)$ by a bounded amount, while consecutive sequence values differ by the asymptotically fixed factor α. The final sentence is the definition of N. □

Warning M.49 (The $\mathcal{O}(1)$ rounding layer is not a small correction). An assertion of the form $N(y) = \nu(y) + o(1)$ *uniformly in real y* is false, and no descending power–logarithmic series can be true uniformly: arbitrarily small changes of y across a value a_n change $N(y)$ by one. Once a monotone interpolation $\mathcal{A}_{\mathrm{ex}}$ with $\mathcal{A}_{\mathrm{ex}}(n) = a_n$ has been fixed, one has exactly $N(y) = \lceil \nu_{\mathrm{ex}}(y) \rceil$ with ν_{ex} its ordinary inverse, and for non-integral χ

$$\lceil \chi \rceil = \chi + \frac{1}{2} + \frac{1}{\pi} \sum_{m \geq 1} \frac{\sin(2\pi m \chi)}{m}, \tag{M.79}$$

an $\mathcal{O}(1)$ periodic layer that never tends to zero. The correct statements are: about ν; or at the special values $y = a_n$, where $\nu(a_n) = n + \mathcal{O}(n^{-M})$ for every fixed M; or about a rounded exact interpolation. S2 states this sharply and S3 restates it; S1's phrase "can never be resolved more accurately than one unit" is the same point stated loosely, and is corrected here. The layer Equation (M.79) must not be confused with the exponentially small oscillation of Section M.9: the latter is genuinely $o(1)$ and has an entirely different origin.

Remark M.50 (A definitional disagreement). S1 and S2 define the staircase as $\min\{n \geq 2 : a_n \geq y\}$, S3 as $\min\{n \geq 1 : a_n \geq y\}$. The two differ only at $y \leq 1$, where the tie $a_1 = a_2 = 1$ makes the choice a pure convention. This volume uses Equation (M.77) and confines all statements to $y > 1$, where the definitions agree.

M.9 Exponentially small sectors

M.9.1 Why one exponential scale is not the whole story

Equation (M.43) is complete in inverse powers of n at the scale ρ^{-n}. It cannot see a term σ^{-n} with $|\sigma| > \rho$, because $(\rho/|\sigma|)^n$ decays faster than every power of n^{-1}. Such terms come from singularities of the analytic continuation of T, and the functional equation exposes exactly two mechanisms for producing them:

1. **inherited preimages**: if T is singular at τ and $\sigma^k = \tau$ for some $k \geq 2$, the summand $T(z^k)/k$ of R can make R, hence T, singular at σ;
2. **new Cayley critical points**: on a continued sheet where R is analytic, any σ with $\zeta(\sigma) = \mathrm{e}^{-1}$ and $\zeta'(\sigma) \neq 0$ is a square-root branch point of T by Theorem M.16.

Iterating the first mechanism from the dominant point produces the *root-lift skeleton*

$$\mathscr{R} := \big\{\rho^{1/m}\mathrm{e}^{2\pi \mathrm{i} j/m} \ : \ m \geq 1,\ 0 \leq j < m\big\}, \tag{M.80}$$

whose radii increase to 1 and whose arguments become dense. Relative to the dominant sector the level-m points carry the exponential weight

$$\sigma_{m,j}^{-n} = \rho^{-n/m}\mathrm{e}^{-2\pi \mathrm{i} jn/m}, \qquad \text{i.e.} \qquad \exp\left[-\lambda\left(1 - \frac{1}{m}\right)n\right] \text{ relative to } \rho^{-n}. \tag{M.81}$$

Warning M.51 (What the skeleton is and is not). Equation (M.80) is a list of *candidate* inherited germs generated by the recursion. It is not a theorem that every listed point is singular on a given sheet, that every singularity is listed, or that a listed point is reachable along a prescribed path. Continuation can also meet solutions of $\zeta(z) = \mathrm{e}^{-1}$ that are not in $\mathscr{R}$, and a critical point can coalesce with an inherited germ (which, by Equation (M.89) below, halves an exponent and can create quarter powers).

Remark M.52 (A disagreement about the unit circle). S1 and S2 assert that $|z| = 1$ is a natural boundary for T and that the continued singularities accumulate at every point of it, citing Drmota–Gittenberger. S3 makes no such assertion and states only the candidate skeleton with the caveat of Theorem M.51. This volume records the natural-boundary claim as *reported* and does not use it: every statement in this section is conditioned on the explicit finite-radius hypothesis $\mathsf{H}(\mathcal{R})$ of Theorem M.54, and nothing below depends on what happens as $\mathcal{R} \uparrow 1$.

M.9.2 Transfer from a general singular germ

Theorem M.53 (Universal sector coefficients). *Let $\sigma \neq 0$ be an accessible singularity on a specified sheet, put $\mathsf{u} = 1 - z/\sigma$, and suppose that in a Δ-domain at σ*

$$T(z) = A_\sigma(\mathsf{u}) + \sum_{\beta \in E_\sigma} c_{\sigma,\beta}\, \mathsf{u}^\beta + \mathcal{O}\big(|\mathsf{u}|^{\beta_{\max}}\big), \tag{M.82}$$

with A_σ analytic, $E_\sigma \subset \mathbb{Q}_{>0} \setminus \mathbb{N}_0$ finite and $\beta_{\max} > \max E_\sigma$. Then the contribution of Equation (M.82) *to $[z^n]\,T$ is*

$$\boxed{\mathfrak{S}_\sigma(n) \sim \sigma^{-n} \sum_{\beta \in E_\sigma} \frac{c_{\sigma,\beta}}{\Gamma(-\beta)}\, n^{-\beta-1} \sum_{r \geq 0} \frac{g_r(\beta)}{n^r},} \tag{M.83}$$

so the coefficient of $\sigma^{-n} n^{-\beta-1-r}$ is exactly $c_{\sigma,\beta}\, g_r(\beta)/\Gamma(-\beta)$.

Proof. Apply Equation (M.37) with ρ replaced by σ to each displayed monomial and expand the gamma ratio by Theorem M.29. The analytic part A_σ has no Hankel jump and contributes no large-n tail; the $\mathcal{O}$-term transfers by the usual Hankel estimate. ☐

Equation (M.83) is the exact analogue of Equation (M.44) and allows arbitrary ramification exponents, not only half-integers. Relative to the dominant sector one introduces the *action*

$$\Omega_\sigma := \operatorname{Log} \frac{\sigma}{\rho}, \qquad \text{so that} \qquad \sigma^{-n} = \rho^{-n} \mathrm{e}^{-n\Omega_\sigma}. \tag{M.84}$$

Definition M.54 (Continuation hypothesis $\mathsf{H}(\mathcal{R})$). Fix $\rho < \mathcal{R} < 1$. We say $\mathsf{H}(\mathcal{R})$ holds for a chosen continuation of T if that continuation is analytic on $|z| < \mathcal{R}$ after removing finitely many pairwise disjoint cuts issuing from a finite set $\Sigma_{\mathcal{R}}$; every $\sigma \in \Sigma_{\mathcal{R}}$ carries a germ of the form Equation (M.82) in a Δ-domain; and the deformed coefficient contour may be taken on a circle of radius at least $\mathcal{R}$, up to harmless indentations, on which T has at most polynomial growth.

Theorem M.55 (Finite-level coefficient transseries). *Assume $\mathsf{H}(\mathcal{R})$ and let $\mathfrak{S}_\sigma$ be the multiplier-weighted sector of Equation* (M.83) *attached to the local Hankel cycle actually traversed. Then, for any finite truncation of the exponent and inverse-power sums,*

$$a_n = \sum_{\sigma \in \Sigma_{\mathcal{R}}} \mathfrak{S}_\sigma(n) + \mathcal{O}\big(\mathcal{R}^{-n} n^K\big) \tag{M.85}$$

for a finite K determined by the boundary growth.

Proof. Deform the Cauchy contour from a small circle about 0 into the union of small Hankel contours around the cuts issuing from $\Sigma_{\mathcal{R}}$ plus an outer contour of radius $\mathcal{R}$. On each Hankel contour apply Theorem M.53; the outer contour is bounded by $\mathcal{R}^{-n}$ times the allowed polynomial factor. Only finitely many singularities and finitely many local terms are retained, so the deformation and the term-by-term summation are legitimate. Passing to a different sheet changes the homology coefficients of the local cycles; those coefficients are the Stokes/intersection multipliers absorbed into $\mathfrak{S}_\sigma$. □

S1's global transseries is not a theorem. S1 displays $a_n \sim \mathcal{A}_\rho(n) + \sum_{\sigma\in\mathfrak{S}\setminus\{\rho\}} S_\sigma \mathcal{A}_\sigma(n)$ as a numbered displayed result over "the singular set encountered by the chosen contour continuation", with no hypothesis attached, while conceding in its introduction that globally the formula "is best understood as a formal resurgent/Darboux transseries". The volume replaces it by Theorem M.55, which is S3's formulation and the only one of the three carrying an explicit hypothesis; the unrestricted sum over an infinite $\mathfrak{S}$ is not asserted anywhere here. The correct reading of a "global" transseries in the presence of accumulating singularities is the compatible family of the finite-radius expansions Equation (M.85) as $\mathcal{R}$ grows, which is exactly what S2's "projective meaning" remark says.

M.9.3 The local recursion produces every germ

Lemma M.56 (Pull-back coordinates)**.** *Let $z = \sigma(1-\mathsf{u})$ and $\tau = \sigma^k$. Then*

$$1 - \frac{z^k}{\tau} = 1 - (1-\mathsf{u})^k = k\,\mathsf{u}\left(1 + \sum_{r\geq 1} p_{k,r}\mathsf{u}^r\right), \qquad p_{k,r} = \frac{(-1)^r}{k}\binom{k}{r+1}, \tag{M.86}$$

and consequently, for a parent monomial,

$$\left(1 - \frac{z^k}{\tau}\right)^\beta = (k\mathsf{u})^\beta \sum_{N\geq 0} \mathsf{u}^N \sum_{j=0}^{N} \binom{\beta}{j} \widehat{B}_{N,j}(p_{k,1}, p_{k,2}, \ldots). \tag{M.87}$$

Proof. $1-(1-\mathsf{u})^k = -\sum_{r\geq 1}\binom{k}{r}(-\mathsf{u})^r = k\mathsf{u} - \sum_{r\geq 2}\binom{k}{r}(-1)^r\mathsf{u}^r$; dividing by $k\mathsf{u}$ and re-indexing $r \mapsto r+1$ gives $p_{k,r}$. Equation (M.87) is Theorem J.14 applied to the unit series $1 + \sum_r p_{k,r}\mathsf{u}^r$. □

Theorem M.57 (Recursive completeness of the local calculus)**.** *Fix a sheet and a point $\sigma \neq 0$, and choose a local uniformiser $z = \sigma(1 - v^{\mathsf{L}})$ in which every parent germ needed in $R(z)$ is an ordinary power series in v. Then the local germ of T at σ is computable to any finite order by the following finite operations, and only these:*

1. *pull back each singular summand $T(z^k)/k$ by Equation* (M.87)*;*
2. *add the regular Taylor parts to obtain R, and exponentiate by Equation* (M.29) *to obtain $\zeta = z\mathrm{e}^R$;*
3. *if $\zeta_0 := \zeta(\sigma) \neq \mathrm{e}^{-1}$ on the reached branch, so that the selected Cayley branch $\mathcal{C}_\flat$ is analytic at ζ_0, compose regularly:*

$$T = \sum_{r\geq 0} \frac{\mathcal{C}_\flat^{(r)}(\zeta_0)}{r!}(\zeta - \zeta_0)^r, \qquad \mathcal{C}_\flat'(w) = \frac{\mathcal{C}_\flat(w)}{w\bigl(1 - \mathcal{C}_\flat(w)\bigr)}, \tag{M.88}$$

   the higher derivatives following by Faà di Bruno;

4. *if $\zeta_0 = \mathrm{e}^{-1}$ and the reached branch has $T(\sigma) = 1$, compose critically with the universal germ of Theorem M.16:*

$$T = 1 + \sum_{r\geq 1} c_r\,\theta_\sigma^{r/2}, \qquad \theta_\sigma := 1 - \mathrm{e}\zeta. \tag{M.89}$$

If θ_σ vanishes to order ς in v, then $\theta_\sigma^{1/2}$ begins at $v^{\varsigma/2}$ and the ramification index doubles when ς is odd; this is the only new ramification mechanism.

Proof. Induction on the depth of σ in the pull-back graph. At depth 0, Theorem M.24 gives the dominant germ. At the induction step, $z \mapsto z^k$ converts each already-known germ into a Puiseux series in a common uniformiser by Equation (M.87); the ring $\mathbb{C}[[v]]$ is closed under the sum defining R, under exponentiation and under multiplication by $z = \sigma(1 - v^{\mathsf{L}})$; and the outer composition is either Equation (M.88) or Equation (M.89), both of which determine every coefficient at finite order by Theorems J.9 and J.10. The derivative formula in Equation (M.88) follows by differentiating $\mathcal{C} = w\mathrm{e}^{\mathcal{C}}$. □

Remark M.58 (A dedup). S1 introduces a "generalized Bell algebra" $\mathfrak{B}_{\gamma,m}$ on a semigroup-graded power-series ring for this step. Theorem M.57 avoids it: after a common uniformiser v has been chosen, everything is an *ordinary* power series in v and the standard $\widehat{B}$, B of Theorems J.9 and J.10 suffice. That is S3's organisation and is adopted here; nothing is lost, because the choice of L is exactly what the semigroup grading was encoding.

M.9.4 The first inherited sector, at $-\sqrt{\rho}$

The point

$$\sigma_- := -\sqrt{\rho} = -0.5816544136333942538101\ldots \tag{M.90}$$

is reached by continuing the germ at 0 along the negative real axis, which never meets the dominant positive singularity. At $z = \sigma_-$ the summand $T(z^2)/2$ of R reaches $T(\rho)/2$ and inherits the dominant square-root branch, while every $T(z^k)$ with $k \geq 3$ is evaluated at $|\sigma_-|^k \leq \rho^{3/2} < \rho$, strictly inside the original disc. Take the uniformiser

$$z = \sigma_-(1 - v^2), \qquad v = \Big(1 - \tfrac{z}{\sigma_-}\Big)^{1/2}, \qquad \text{so} \qquad 1 - \frac{z^2}{\rho} = v^2(2 - v^2). \tag{M.91}$$

Proposition M.59 (Complete local data at σ_-). *Write $R\big(\sigma_-(1 - v^2)\big) = \sum_{m\geq 0} r_m^- v^m$ and $\zeta\big(\sigma_-(1 - v^2)\big) = \sum_{m\geq 0} \zeta_m^- v^m$. Then, with $t_0 := 1$,*

$$r_m^- = \frac{1}{2} \sum_{\substack{0\leq r\leq m\\ r\equiv m\,(2)}} t_r (-1)^{\frac{m-r}{2}} 2^{\frac{r}{2} - \frac{m-r}{2}} \binom{r/2}{\frac{m-r}{2}} + \mathbf{1}_{2|m} (-1)^{m/2} \sum_{k\geq 3} \frac{1}{k} \sum_{n\geq 1} a_n \sigma_-^{kn} \binom{kn}{m/2}, \tag{M.92}$$

both parts converging geometrically. Defining $\mathcal{E}_m^-$ by $\sum_{m\geq 0} \mathcal{E}_m^- v^m = \exp\big(\sum_{m\geq 0} r_m^- v^m\big)$, so that $\mathcal{E}_m^- = \mathrm{e}^{r_0^-} \sum_{k\geq 0} \frac{1}{k!} \widehat{B}_{m,k}(r_1^-, r_2^-, \ldots)$, one has

$$\zeta_m^- = \sigma_-\big(\mathcal{E}_m^- - \mathcal{E}_{m-2}^-\big), \qquad \mathcal{E}_{<0}^- := 0. \tag{M.93}$$

Setting $y_- := \mathcal{C}_0(\zeta_0^-) = -W_0(-\zeta_0^-)$, the outer composition is regular because $y_- \neq 1$, and writing $T\big(\sigma_-(1 - v^2)\big) = y_- + \sum_{m\geq 1} t_m^- v^m$ one has the closed form

$$\boxed{r_1^- = \frac{t_1}{\sqrt{2}}, \qquad t_1^- = \frac{y_-}{1 - y_-} \cdot \frac{t_1}{\sqrt{2}}.} \tag{M.94}$$

Proof. The $k = 2$ contribution to R is $\frac{1}{2}T\big(\rho(1 - v^2(2 - v^2))\big)$; inserting Equation (M.33) with $s = v^2(2 - v^2)$, so $s^{r/2} = v^r 2^{r/2}(1 - v^2/2)^{r/2}$, and expanding $(1 - v^2/2)^{r/2}$ binomially gives the first sum. The $k \geq 3$ contributions are $\sum_{k\geq 3} \frac{1}{k} \sum_n a_n \sigma_-^{kn}(1 - v^2)^{kn}$, whose v^{2j} coefficient is the second sum; convergence is geometric already at $k = 3$ because $|\sigma_-|^3 < \rho$. Equation (M.93) is $\zeta = \sigma_-(1 - v^2)\mathrm{e}^R$. For Equation (M.94): only $r = 1$ contributes to v^1 in the first sum, giving $r_1^- = \frac{1}{2}t_1\sqrt{2} = t_1/\sqrt{2}$; hence $\zeta_1^- = \zeta_0^- r_1^-$ and, by Equation (M.88) with $\mathcal{C}_0'(w) = \mathcal{C}_0(w)/\big(w(1 - \mathcal{C}_0(w))\big)$, $t_1^- = \mathcal{C}_0'(\zeta_0^-)\zeta_1^- = \frac{y_-}{\zeta_0^-(1-y_-)}\zeta_0^- r_1^-$, which is the claim. □

Recomputed at 80 working digits (see the repair below),

$$\begin{aligned} r_0^- &= 0.4686366279248455480872, & \zeta_0^- &= -0.9293757352012050888501, \\ y_- &= -0.5410329414204534926205, & t_1^- &= 0.3871501110302429034772, \end{aligned} \tag{M.95}$$

and the first local coefficients are

$$\begin{aligned} t_2^- &= -0.0325649185156476341473\,1, & t_3^- &= 0.0172004216031220914830\,1, \\ t_4^- &= 0.1040770395183844385254\,6, & t_5^- &= 0.1619473607727023611895\,2, \\ t_6^- &= -0.0629259981879930868969\,4, & t_7^- &= -0.2440310392314213513551\,0. \end{aligned} \tag{M.96}$$

S2's minus-sector constants are printed past their accuracy. S2 prints r_0^-, z_0^-, y_- and b_1^- to 40 significant digits, e.g. $r_0^- = 0.4686366279248455480872026033545827113234$. Two independent recomputations here — one through the germ recursion at 90 digits, one by direct summation of $\frac{1}{2} + \sum_{k\geq 3} T(\sigma_-^k)/k$ at 80 digits with 1400 exact tree numbers and no truncation of the inner sums — agree with each other and give $r_0^- = 0.4686366279248455480872026219318193 14\ldots$, which departs from S2's value at the *twenty-sixth* significant digit. The same 25-digit agreement, and no more, is found for ζ_0^-, y_-, t_1^- and every derived quantity, consistent with a single truncation of the $k \geq 3$ tail in S2's computation. All minus-sector constants are therefore quoted here to at most 22 digits.

Only the odd t_{2j+1}^- are singular in z at σ_-, so Theorem M.53 with $\beta = j + \frac{1}{2}$ gives

$$\boxed{a_n^{(-)} \sim \mathfrak{s}_-\,\sigma_-^{-n} n^{-3/2} \sum_{k\geq 0} \frac{B_k^-}{n^k}, \qquad B_k^- = \sum_{j=0}^{k} \frac{t_{2j+1}^-}{\Gamma(-j-\frac{1}{2})} g_{k-j}\big(j+\tfrac{1}{2}\big)\,,} \tag{M.97}$$

where $\mathfrak{s}_-$ is the multiplier of the local cycle; for the direct negative-axis continuation it is 1, and it is displayed only to keep the contour dependence visible in larger deformations. Since $\sigma_-^{-n} = (-1)^n \rho^{-n/2}$, the sector is alternating, with

$$\begin{aligned} B_0^- &= -0.1092130299563101757054\,9, & B_1^- &= -0.0336766622077828533671\,9, \\ B_2^- &= -0.1790000901173046316389\,03, & B_3^- &= -1.642342004953279983\,00, \\ B_4^- &= -18.9266458632920839673, & B_5^- &= -277.723089846678995649. \end{aligned} \tag{M.98}$$

Relative to the leading dominant term its amplitude is

$$\frac{B_0^-}{C} = -0.2482543049151649529916\ldots, \qquad \frac{a_n^{(-)}}{a_n^{(0)}} \sim -0.2482543049\ldots\,\big(-\sqrt{\rho}\big)^n, \tag{M.99}$$

which is smaller than every power of $1/n$: this is the first explicitly computed genuinely beyond-all-orders correction to A000081. Its action is

$$\Omega_- = \operatorname{Log}\frac{\sigma_-}{\rho} = \frac{\lambda}{2} + \mathrm{i}\pi \pmod{2\pi\mathrm{i}}, \qquad \frac{\Omega_-}{\lambda} = \frac{1}{2} + \mathrm{i}\frac{\pi}{\lambda}, \qquad \frac{\pi}{\lambda} = 2.898796429733978737 7\ldots. \tag{M.100}$$

Remark M.60. The point $+\sqrt{\rho}$ lies behind the dominant branch point on real continuation and belongs to sheet-dependent continuations; complex critical points and overlapping root lifts may occur farther out. Their local data are produced by Theorem M.57 and their multipliers depend on the chosen contour. Equation (M.97) is one explicit sector, not an assertion that it exhausts modulus $\sqrt{\rho}$.

M.9.5 Transport of the sectors through the inverse

Write the completed forward interpolation as

$$\mathfrak{a}(x) = \mathcal{A}(x)\big(1 + \mathcal{E}(x)\big), \qquad \mathcal{E}(x) = \sum_{\Omega \neq 0} \mathrm{e}^{-\Omega x} x^{\mu_\Omega} \sum_{r \geq 0} e_{\Omega,r} x^{-r}, \tag{M.101}$$

with $\mathcal{A}$ from Equation (M.53) and the support of $\mathcal{E}$ contained in the additive semigroup generated by the actions Equation (M.84). For non-integer x a factor σ^{-x} requires a choice of $\operatorname{Log}\sigma$; different choices agree at integers and differ between them, so the exponentially small part of a *functional* inverse is not canonical until a continuation and pairing convention has been fixed. Fix one, pairing conjugate branches so that $\mathfrak{a}$ is real on the positive axis.

Proposition M.61 (Sector transport). *Let ν_0 be the complete algebraic inverse of $\mathcal{A}$ (Equation* (M.63)*) and put $\nu_{\text{full}} = \nu_0 + \Delta$. Then Δ is the unique solution, in the transseries field ordered by the semigroup of actions, of*

$$\Delta = \Psi_{\nu_0}(\Delta), \qquad \Psi_{\nu_0}(w) := -\frac{\log\big(1 + \mathcal{E}(\nu_0 + w)\big)}{\mathcal{Q}_{\nu_0}(w)}, \qquad \mathcal{Q}_x(w) := \frac{f(x+w) - f(x)}{w}, \tag{M.102}$$

with $f = \log \mathcal{A}$ and $\mathcal{Q}_x(0) = f'(x)$; its coefficients are given by Theorem J.19, so that in particular

$$\Delta = -\frac{\mathcal{E}(\nu_0)}{f'(\nu_0)} + \mathcal{O}(\mathcal{E}^2) \sim -\frac{\mathcal{E}(\nu_0)}{\lambda}, \tag{M.103}$$

and a single relative forward sector $\mathcal{E}_\Omega \sim e_{\Omega,0} \mathrm{e}^{-\Omega x} x^{\mu_\Omega}$ maps to

$$\Delta_\Omega(y) \sim -\frac{e_{\Omega,0}}{\lambda}\, C^{\Omega/\lambda}\, y^{-\Omega/\lambda}\, \nu_0^{\mu_\Omega - p\Omega/\lambda}. \tag{M.104}$$

Proof. $f(\nu_0 + \Delta) - f(\nu_0) + \log(1 + \mathcal{E}(\nu_0 + \Delta)) = 0$ is the exact equation, and $f(\nu_0 + w) - f(\nu_0) = w\, \mathcal{Q}_{\nu_0}(w)$ by definition; dividing gives Equation (M.102). Since $f'(\nu_0)$ has leading term $\lambda \neq 0$ it is invertible in the coefficient field, and every occurrence of Δ on the right is either nonlinear or multiplied by a positive transmonomial, so the support is well ordered and each requested transmonomial receives finitely many contributions; Theorem J.19 then gives the coefficients. Equation (M.103) is the first term. For Equation (M.104), use $y \sim C\mathrm{e}^{\lambda\nu_0}\nu_0^{-p}$ to write $\mathrm{e}^{-\Omega\nu_0} = (y/C)^{-\Omega/\lambda}\nu_0^{-p\Omega/\lambda}$. □

For the first inherited sector, $\Omega_- = \frac{\lambda}{2} + \mathrm{i}\pi$ and $e_{-,0} = B_0^-/C$. Pairing the two lateral determinations $\pm\mathrm{i}\pi$ replaces $\mathrm{e}^{-\mathrm{i}\pi x}$ by $\cos(\pi x)$, and $\mathrm{e}^{-\lambda\nu_0/2} = C^{1/2}y^{-1/2}\nu_0^{-3/4}$, so

$$\boxed{\Delta_-(y) \sim 0.15193347365953098826\ldots\; y^{-1/2}\, \nu_0(y)^{-3/4} \cos\big(\pi\nu_0(y)\big),} \tag{M.105}$$

the amplitude being $\sqrt{C}\cdot\big(-B_0^-/(\lambda C)\big)$ with $-B_0^-/(\lambda C) = 0.22906811038403197857\ldots$. The phase is

$$\begin{aligned}
\pi\nu_0(y) &= \frac{\pi}{\lambda}\log y + \frac{3\pi}{2\lambda}\log\log y + \mathcal{O}(1),\\
\frac{\pi}{\lambda} &= 2.89879642973397873771\ldots,\\
\frac{3\pi}{2\lambda} &= 4.34819464460096810657\ldots,
\end{aligned} \tag{M.106}$$

so the first nonperturbative correction to the inverse has envelope $y^{-1/2}(\log y)^{-3/4}$ and a logarithmic oscillation with a slower $\log\log y$ drift.

Warning M.62 (Two different oscillations). Equation (M.105) tends to zero and comes from a complex singular action. The sawtooth of Equation (M.79) is $\mathcal{O}(1)$, never tends to zero, and comes from integer rounding. They have different origins, different scales, and must not be conflated.

M.10 Other inverse notions: the compositional inverse of T

Since $T(z) = z + \mathcal{O}(z^2)$ with integer coefficients and unit linear coefficient, T has a compositional inverse $Z := T^{\langle -1\rangle} \in z\mathbb{Z}[[z]]$, unrelated to the growth inversion of Section M.7.

Proposition M.63 (Coefficients of Z). *For $n \geq 1$,*

$$[w^n]\, Z(w) = \frac{1}{n}[z^{n-1}] \left(\frac{z}{T(z)}\right)^n = \frac{1}{n}[z^{n-1}] \exp\left(-n \sum_{k\geq 1} \frac{T(z^k)}{k}\right), \tag{M.107}$$

a finite ordinary-Bell expression in $a_1, \dots, a_{n-1}$. The first sixteen values, recomputed by exact rational arithmetic, are

$$1,\ -1,\ 0,\ 1,\ -1,\ 1,\ -4,\ 11,\ -18,\ 18,\ 0,\ -60,\ 189,\ -360,\ 453,\ -373. \tag{M.108}$$

Proof. The first equality is Theorem J.17 with $\phi(z) = T(z)/z$, a unit series. The second is Equation (M.5) rearranged, and Theorem J.9 converts the exponential into a finite Bell sum. Integrality is automatic for the compositional inverse of a series in $z + z^2\mathbb{Z}[[z]]$. □

Remark M.64. S1 displays $Z(w) = w - w^2 + w^4 - w^5 + w^6 - 4w^7 + 11w^8 - \cdots$, which agrees with Equation (M.108), and identifies the coefficients with OEIS A050395. The expansion is confirmed here; the OEIS identification is reported from S1 and has not been re-checked in this volume.

Near the dominant value $w = 1$ the branch of Z through $Z(1) = \rho$ is *analytic* in $1 - w$, because it undoes the square root: from $T = \zeta e^T$ one has $\zeta = Te^{-T}$, hence

$$Z(w) = \zeta^{\langle -1\rangle}\big(we^{-w}\big) \tag{M.109}$$

on that branch, ζ being injective on $(0, \sqrt{\rho})$ by Theorem M.11. Since $we^{-w} - e^{-1}$ has a double zero at $w = 1$, the local expansion of Z at $w = 1$ has no linear term. Its coefficients follow from one more Lagrange extraction: writing $\zeta(\rho + x) - e^{-1} = x\,\psi(x)$ with $\psi(0) = \zeta'(\rho) \neq 0$,

$$[\mathsf{v}^m]\,\big(\zeta^{\langle -1\rangle}(e^{-1} + \mathsf{v}) - \rho\big) = \frac{1}{m}[x^{m-1}]\,\psi(x)^{-m}, \tag{M.110}$$

followed by composition with $(1-\varsigma)e^{-(1-\varsigma)} - e^{-1} = -e^{-1}\sum_{m\geq 2}\frac{m-1}{m!}\varsigma^m = -e^{-1}\frac{\varsigma^2}{2}\Phi(\varsigma)$, the same kernel Equation (M.22) that governed the forward branch.

M.11 Algorithms, validation and audit

M.11.1 The pipeline

Algorithm M.65 (All dominant and inverse coefficients through order M).

1. Generate $a_1, \dots, a_N$ and $b_1, \dots, b_N$ exactly by Equation (M.7) with a divisor sieve, checking the divisibility assertion at each step; form ϱ_r by Equation (M.10).
2. Solve $\log x + 1 + \sum_{r=2}^{N} \varrho_r x^r = 0$ for ρ by safeguarded Newton or interval bisection. Choose N so that $\rho^{N/2}$ is below the target precision; the omitted tail is geometrically small because R is analytic on $|z| < \sqrt{\rho}$ (Theorem M.10).
3. Compute $\widehat{R}_0, \dots, \widehat{R}_{2M+1}$ by Equation (M.28); exponentiate by Equation (M.29) and form $\theta_1, \dots, \theta_{2M+2}$ by Equation (M.30), then $u_j = \theta_{j+1}/\theta_1$.
4. Compute $\omega_1, \dots, \omega_{2M+1}$ exactly by Equation (M.23) (rationals), set $c_r = -2^{r/2}\omega_r$, and form $t_1, t_3, \dots, t_{2M+1}$ by Equation (M.34).

5. Compute $g_r(j+\frac{1}{2})$ exactly for $j+r \le M$ by Equation (M.39), then $\tau_0, \ldots, \tau_M$ by Equation (M.44), then $C = \tau_0/\sqrt{\pi}$ and $f_m = \tau_m/\tau_0$.
6. Reciprocal: $\bar{f}_m$ by Equation (M.50).
7. Inverse: φ_m by Equation (M.55), then d_m by Equation (M.61), or $P_m(L)$ by the flattening recursion of Theorem J.37.
8. Sectors: apply Theorem M.57 outward in modulus, transfer each germ by Equation (M.83), and invert by Theorem M.61.

At every stage the coefficient of order m depends only on data of order at most m; no nonlinear global solve occurs after ρ is known. The useful power-series primitives are the three dense recurrences

$$[v^n]\,F^\gamma = \frac{1}{n}\sum_{k=1}^{n}\big((\gamma+1)k-n\big)F_k[v^{n-k}]\,F^\gamma, \quad E_n = \frac{1}{n}\sum_{k=1}^{n} kA_kE_{n-k}, \quad H_n = F_n - \frac{1}{n}\sum_{k=1}^{n-1} kH_kF_{n-k}, \tag{M.111}$$

for F^γ with $F_0 = 1$, for $E = \mathrm{e}^A$ with $A_0 = 0$, and for $H = \log F$ with $F_0 = 1$. They are equivalent to the Bell forms used in the statements and are what an implementation should actually run.

Warning M.66 (Precision discipline). The coefficients grow rapidly and Equation (M.44) combines alternating contributions from several Puiseux coefficients, so obtaining d reliable digits of τ_M can require substantially more than d working digits. The tables in this chapter were produced with 90–110 decimal working digits, with universal objects (ω_m, $g_r(\beta)$, a_n) held in exact rational or integer arithmetic. The repair recorded at Section M.9.4 is exactly what happens when this discipline lapses.

M.11.2 Numerical validation

Let $A_M(n)$ denote Equation (M.43) truncated after $\tau_M n^{-M}$. Table M.5 reports the signed relative error $\big(A_M(n) - a_n\big)/a_n$ against exact integers from Equation (M.7); Table M.6 reports the signed index error $\nu_M(a_n) - n$ for the Lambert-normalised inverse $\nu_M = x_0 + \sum_{m\le M} d_m x_0^{-m}$. Both tables were regenerated here and agree entry by entry with the corresponding tables of S1 and S2 at the displayed precision.

Table M.5: Signed relative error $(A_M(n) - a_n)/a_n$ of the forward expansion.

n	$M=0$	$M=2$	$M=4$	$M=6$	$M=8$	$M=10$
10	$-1.52\cdot10^{-2}$	$-3.49\cdot10^{-4}$	$2.63\cdot10^{-3}$	$3.99\cdot10^{-3}$	$5.52\cdot10^{-3}$	$8.77\cdot10^{-3}$
20	$-6.58\cdot10^{-3}$	$-3.42\cdot10^{-4}$	$-3.18\cdot10^{-5}$	$1.04\cdot10^{-6}$	$9.73\cdot10^{-6}$	$1.41\cdot10^{-5}$
50	$-2.22\cdot10^{-3}$	$-1.77\cdot10^{-5}$	$-2.88\cdot10^{-7}$	$-1.20\cdot10^{-8}$	$-9.84\cdot10^{-10}$	$-1.36\cdot10^{-10}$
100	$-1.06\cdot10^{-3}$	$-2.08\cdot10^{-6}$	$-8.06\cdot10^{-9}$	$-7.88\cdot10^{-11}$	$-1.50\cdot10^{-12}$	$-4.73\cdot10^{-14}$
200	$-5.15\cdot10^{-4}$	$-2.53\cdot10^{-7}$	$-2.40\cdot10^{-10}$	$-5.73\cdot10^{-13}$	$-2.66\cdot10^{-15}$	$-2.04\cdot10^{-17}$
500	$-2.03\cdot10^{-4}$	$-1.59\cdot10^{-8}$	$-2.39\cdot10^{-12}$	$-9.02\cdot10^{-16}$	$-6.62\cdot10^{-19}$	$-8.01\cdot10^{-22}$

Table M.6: Signed index error $\nu_M(a_n) - n$ of the Lambert-normalised inverse; $M = 0$ is the bare carrier x_0.

n	$M=0$	$M=1$	$M=2$	$M=4$	$M=6$	$M=8$
10	$1.64\cdot10^{-2}$	$7.15\cdot10^{-3}$	$1.29\cdot10^{-3}$	$-2.61\cdot10^{-3}$	$-4.16\cdot10^{-3}$	$-5.76\cdot10^{-3}$
20	$6.55\cdot10^{-3}$	$1.91\cdot10^{-3}$	$4.44\cdot10^{-4}$	$3.76\cdot10^{-5}$	$-2.02\cdot10^{-7}$	$-9.39\cdot10^{-6}$
50	$2.11\cdot10^{-3}$	$2.59\cdot10^{-4}$	$2.32\cdot10^{-5}$	$3.32\cdot10^{-7}$	$1.26\cdot10^{-8}$	$9.98\cdot10^{-10}$
100	$9.88\cdot10^{-4}$	$6.16\cdot10^{-5}$	$2.74\cdot10^{-6}$	$9.34\cdot10^{-9}$	$8.35\cdot10^{-11}$	$1.53\cdot10^{-12}$
200	$4.78\cdot10^{-4}$	$1.50\cdot10^{-5}$	$3.33\cdot10^{-7}$	$2.79\cdot10^{-10}$	$6.09\cdot10^{-13}$	$2.71\cdot10^{-15}$
500	$1.88\cdot10^{-4}$	$2.38\cdot10^{-6}$	$2.10\cdot10^{-8}$	$2.78\cdot10^{-12}$	$9.59\cdot10^{-16}$	$6.75\cdot10^{-19}$

Two features are worth naming. First, at $n = 10$ and $n = 20$ the error stops improving and then worsens: the expansion is asymptotic, and the least-term truncation of Theorem J.54 has been passed.

Second, the inverse errors track the forward relative errors divided by $f' \approx \lambda$, exactly as Theorem M.42 predicts; this is why Tables M.5 and M.6 are so nearly proportional.

Remark M.67 (On comparisons with the literature). S1 states that its first five coefficients "agree with Genitrini's independently computed values" and S3 that its Puiseux table "agrees with the independent table in [Genitrini 2016]". Those comparisons are reported from the drafts and have *not* been re-performed here; what has been done is a full independent recomputation of every constant from the exact recurrence, described in Section M.11.3.

M.11.3 Audit

Every numerical value printed in this chapter was produced by an independent implementation of Theorem M.65 written for this volume: exact integer generation of $a_1, \ldots, a_N$ with N between 700 and 1400; exact rational ϱ_r, ω_m and $g_r(\beta)$ (the last from an exact Bernoulli-number table); and 60–110 decimal working digits for everything else. The following independent identities were checked and hold to all digits carried.

1. $(n-1) \mid \sum_{j<n} b_j a_{n-j}$ at every step, for $n \leq 1400$; $a_{10} = 719$, $a_{20} = 12\,826\,228$, $a_{50} = 425\,976\,989\,835\,141\,038\,353$.
2. $\lambda = 1 + R(\rho)$, i.e. Equation (M.20).
3. $\theta_0 = 0$, the vanishing of the constant term in Equation (M.26) predicted by $\mathrm{e}\zeta(\rho) = 1$.
4. $C^2 = \theta_1/(2\pi)$, i.e. Equation (M.45) — the two sides travel through disjoint parts of the pipeline.
5. $c_r = -2^{r/2}\omega_r$ for $1 \leq r \leq 18$ in exact arithmetic, and $t_1 = -\sqrt{2\theta_1}$, $t_2 = \frac{2}{3}\theta_1$.
6. Equation (M.47) re-derived symbolically from Equation (M.44), and the $g_r(\beta)$ of Equations (M.41) and (M.42) re-derived from Equation (M.38).
7. $\mathrm{D}_m = \lambda^{m+1} d_m$ and $\eta_m^{\mathrm{S2}} = \lambda^m \varphi_m$ for $1 \leq m \leq 8$, reconciling S1/S3 with S2.
8. $P_1(0) = d_1$, $P_2(0) = d_2$ (a case of Equation (M.72)).
9. $r_1^- = t_1/\sqrt{2}$ and the closed form Equation (M.94), each against the germ recursion; and r_0^- by a second, wholly independent summation (see the repair in Section M.9.4).
10. The single-sum Equation (M.28) against the siblings' double sum, for $0 \leq m \leq 30$.

M.12 Summary

The Pólya equation becomes transparent after the factorisation $T = \mathcal{C} \circ \zeta$ with $\mathcal{C}(w) = -W_0(-w)$: the universal layer is one Lagrange coefficient $[v^{r-1}]\,\Phi(v)^{-r/2}$, the problem-specific layer is ordinary Bell composition of the analytic disturbance ζ, and coefficient transfer is Bernoulli polynomials in the gamma ratio. The chain is

$$a_n \;\to\; \varrho_r \;\to\; \rho \;\to\; \widehat{R}_m \;\to\; \theta_m \;\to\; t_n \;\to\; \tau_m \;\to\; f_m \;\to\; \varphi_m \;\to\; d_m \text{ or } P_m(L), \tag{M.112}$$

every arrow a finite Bell- or Bernoulli-polynomial formula. Inversion adds nothing new: A000081 is the $\alpha = 1$ case of Theorem J.2 with $a = \lambda$, $b = -\frac{3}{2}$, $Q = H$, so Theorem J.23 supplies the carrier exactly, Theorem J.29 every algebraic correction, Theorem J.37 the logarithmic form, Theorem J.51 the error, and Theorem J.43 the passage to the integer inverse. Beyond all orders, the recursion supplies a singularity grammar whose first explicit instance is the alternating sector of relative size $0.2483\ldots\,(-\sqrt{\rho})^n$ at $\sigma_- = -\sqrt{\rho}$, transported by Theorem M.61 into an inverse correction of envelope $y^{-1/2}(\log y)^{-3/4}$ with a $\log y$-periodic phase. What remains open is not coefficient algebra but analysis: the classification

Table M.7: Principal symbols of this chapter.

Symbol	Meaning
a_n, $T(z)$	A000081 and its ordinary generating function.
b_m, ϱ_r	Divisor transform $\sum_{d\mid m} da_d$; coefficients of R, $\varrho_r = (b_r - ra_r)/r$.
R, ζ	Disturbance $\sum_{k\geq 2} T(z^k)/k$ and Cayley argument $z\mathrm{e}^R$; $T = -W_0(-\zeta)$.
ρ, α, λ, p	Otter singularity; $\alpha = \rho^{-1}$, $\lambda = \log\alpha$, $p = \frac{3}{2}$.
Φ, ω_m, c_r	Universal Cayley kernel and its two coefficient normalisations, $c_r = -2^{r/2}\omega_r$.
s, θ, θ_m, u_j	Local coordinate $1 - z/\rho$; disturbance $1 - \mathrm{e}\zeta$; its coefficients; $u_j = \theta_{j+1}/\theta_1$.
$\widehat{R}_m$, ℓ_j	$[s^m]\, R(\rho(1-s))$; scaled logarithmic jets $\rho^j(\log\zeta)^{(j)}(\rho)$.
t_n	Puiseux coefficients, $T = 1 + \sum t_n s^{n/2}$.
$\varkappa_m(\beta)$, $g_r(\beta)$	Bernoulli increments and gamma-ratio transfer coefficients.
τ_m, C, f_m, $\bar{f}_m$	Forward coefficients; $C = \tau_0/\sqrt{\pi}$; $f_m = \tau_m/\tau_0$; reciprocal coefficients.
F, H, φ_m	$F = 1 + \sum f_m v^m$; $H = \log F = \sum \varphi_m v^m$; this H is the Part 0 datum Q.
x_0, d_m, ν	Lambert carrier; its corrections; the smooth asymptotic inverse.
$\mathcal{X}$, L, q, η_k, P_m	Flattening variables and coefficient polynomials.
$N(y)$	Integer staircase $\min\{n \geq 1 : a_n \geq y\}$.
σ_-, t_m^-, B_k^-, Ω_σ	The inherited singularity $-\sqrt{\rho}$, its local and sector coefficients, and the actions.

of accessible sheets, of critical preimages, and of their Stokes connections — and, prior to that, a proof of $\mathsf{H}(\mathcal{R})$ for $\mathcal{R}$ close to 1.

Appendix N

The double factorial

This chapter merges three independent treatments of $n!!$: *Double_Factorial_Transseries* (henceforth S1), *Double_Factorial_Transseries-3* (S2), and *double_factorial_transseries-2* (S3). All three reach the same forward and inverse coefficients; they differ in organisation, in what is proved, and — decisively — in notation, where the letter C carries three incompatible meanings. The chapter adopts S3's *exact* centred normal form as its spine, S2's Borel kernel and sharp remainder, S2's graded closed coefficient formula, and S1/S2's parity-mean centring of the periodic interpolation. Every displayed rational number below was regenerated with exact rational arithmetic (`sympy`) from the defining recurrences, and every numerical table was recomputed at 220-digit precision (`mpmath`); see Section N.14.

N.1 Orientation: which inverse?

For $n \in \mathbb{N}$ the double factorial is

$$0!! = 1!! = 1, \qquad n!! = n\,(n-2)!! \quad (n \geq 2), \tag{N.1}$$

so that $n!!$ is the product of the positive integers not exceeding n of the same parity as n. The sequence $1, 1, 2, 3, 8, 15, 48, 105, 384, 945, \ldots$ is OEIS A006882. Its elementary closed forms already contain the whole difficulty:

$$(2m)!! = 2^m\, m!, \qquad (2m+1)!! = \frac{2^{m+1}\Gamma\left(m + \frac{3}{2}\right)}{\sqrt{\pi}}. \tag{N.2}$$

The even subsequence is a gamma function of the halved argument; the odd subsequence is the *same* gamma function of the halved argument times a different constant. The two constants do not agree, and no amount of reindexing makes them agree.

Warning N.1. $n!!$ is not the restriction to $\mathbb{N}$ of one canonical slowly varying function. Its even and odd subsequences are two gamma-type branches whose quotient is a nonzero constant $\sqrt{2/\pi} \neq 1$. Consequently *there is no canonical real inverse of* $n!!$ *until an interpolation is fixed*, and different admissible interpolations give inverses that differ at an order *larger* than the entire Bernoulli correction grid (Theorem N.8). Every statement in this chapter that begins "the inverse" names its interpolation first.

Following Theorem J.41 we therefore distinguish, for a target $y \to +\infty$:

(i) *the two parity-resolved functional inverses*, obtained by inverting the canonical gamma continuation of one parity subsequence (Section N.6);

(ii) *the functional inverse of a single all-integer interpolation*, for instance the periodic one recorded by OEIS (Section N.8);

(iii) *the integer staircase inverses*, the least n with $n!! \geq y$ and the greatest n with $n!! \leq y$ (Section N.9).

Objects (i) and (iii) are intrinsic to the sequence; object (ii) is not. This is the chapter's spine, and Theorem N.8 makes it quantitative.

Two asymptotic scales must also be kept apart. Taking logarithms converts the growth of $n!!$ into a phase $\frac{1}{2}u(\log u - 1)$ whose inversion is *exactly* a Lambert expression (Theorem J.27); expanding that Lambert function produces an *outer* layer in powers of $1/\log\log y$ with polynomial dependence on $\log\log\log y$ (Theorem J.37). The Bernoulli corrections generate a far finer *inner* grid in odd inverse powers of the Lambert core. The fixed shift -1 lies strictly between the two levels. Writing all of this as one informal series in $1/\log y$ destroys the ordering; see Theorem N.42.

N.2 Exact continuations and the parity split

N.2.1 The two parity branches

Definition N.2 (Parity branches). For the parity label $\delta \in \{0, 1\}$ put

$$\kappa_\delta = \left(\frac{2}{\pi}\right)^{\delta/2}, \qquad \mathcal{D}_\delta(x) = \kappa_\delta\, 2^{x/2}\, \Gamma\Big(\frac{x}{2} + 1\Big) \qquad (x > -2), \tag{N.3}$$

and write

$$A_\delta = \log\big(\kappa_\delta \sqrt{\pi}\big) = \begin{cases} \frac{1}{2}\log \pi, & \delta = 0, \\ \frac{1}{2}\log 2, & \delta = 1, \end{cases} \qquad \mathrm{e}^{A_0} = \sqrt{\pi}, \quad \mathrm{e}^{A_1} = \sqrt{2}. \tag{N.4}$$

Here $\delta = 0$ labels the even subsequence and $\delta = 1$ the odd one.

Proposition N.3 (Exact parity interpolation and monotonicity). *For every integer $n \geq 0$ with $n \equiv \delta \pmod 2$ one has $\mathcal{D}_\delta(n) = n!!$. Moreover each $\mathcal{D}_\delta$ is positive and strictly increasing on $[0, \infty)$, so it has a single-valued inverse $\mathcal{D}_\delta^{-1}$ on its range; and*

$$\mathcal{D}_1(x) = \sqrt{\frac{2}{\pi}}\, \mathcal{D}_0(x), \qquad \mathcal{D}_1^{-1}(y) = \mathcal{D}_0^{-1}\left(\sqrt{\frac{\pi}{2}}\, y\right). \tag{N.5}$$

Proof. If $n = 2m$ then $\mathcal{D}_0(2m) = 2^m\Gamma(m+1) = 2^m m! = (2m)!!$ by Equation (N.2). If $n = 2m+1$ then

$$\mathcal{D}_1(2m+1) = \sqrt{\frac{2}{\pi}}\, 2^{m+1/2}\Gamma\big(m + \tfrac{3}{2}\big) = \frac{2^{m+1}}{\sqrt{\pi}}\Gamma\big(m + \tfrac{3}{2}\big) = (2m+1)!!.$$

Positivity is clear because $\Gamma > 0$ on $(0, \infty)$. For monotonicity,

$$\frac{\mathcal{D}_\delta'(x)}{\mathcal{D}_\delta(x)} = \frac{1}{2}\left(\log 2 + \psi\Big(\frac{x}{2} + 1\Big)\right), \qquad \psi = \Gamma'/\Gamma, \tag{N.6}$$

and ψ is strictly increasing on $(0, \infty)$ because $\psi'(z) = \sum_{m \geq 0}(z+m)^{-2} > 0$. At $x = 0$ the bracket equals $\log 2 - \gamma = 0.11593\ldots > 0$, hence it is positive for all $x \geq 0$. Equation (N.5) is immediate from Equation (N.3), since $\kappa_1/\kappa_0 = \sqrt{2/\pi}$. □

Remark N.4. The constant κ_δ disappears from Equation (N.6). Both branches therefore have the same local conditioning, the same forward correction series, and — as Theorem N.29 shows — literally the same inverse coefficient polynomials. Parity survives only as the additive constant A_δ subtracted from $\log y$.

N.2.2 The periodic OEIS interpolation, and why it is not canonical

Definition N.5 (Periodic interpolation)**.** Put

$$\mathscr{D}(x) = 2^{x/2}\,\Gamma\Big(\frac{x}{2}+1\Big)\left(\frac{2}{\pi}\right)^{(1-\cos\pi x)/4}. \tag{N.7}$$

Because $\frac{1}{2}\sin^2(\pi x/2) = \frac{1}{4}(1-\cos\pi x)$, this is the interpolation recorded in the OEIS entry for A006882, written there with the half-argument sine.

Proposition N.6 (Interpolation and eventual monotonicity)**.** *$\mathscr{D}(n) = n!!$ for every integer $n \geq 0$, and $\mathscr{D}$ is strictly increasing on $[\,1.04,\infty)$. Its inverse on that ray, denoted $\mathscr{D}^{-1}$, is what "the inverse of $\mathscr{D}$" means below; it is not a globally single-valued inverse near the origin.*

Proof. The periodic factor equals 1 at even integers and $\sqrt{2/\pi}$ at odd integers, so $\mathscr{D}$ agrees with $\mathcal{D}_0$ at even and with $\mathcal{D}_1$ at odd integers; Theorem N.3 finishes the first claim. Next,

$$\frac{\mathscr{D}'(x)}{\mathscr{D}(x)} = \frac{1}{2}\left(\log 2 + \psi\Big(\frac{x}{2}+1\Big)\right) + \frac{\pi}{4}\log\left(\frac{2}{\pi}\right)\sin\pi x. \tag{N.8}$$

For $z > \frac{1}{2}$ one has $\psi(z) > \log\big(z-\frac{1}{2}\big)$, hence $\psi(x/2+1) > \log((x+1)/2)$ and the first summand exceeds $\frac{1}{2}\log(x+1)$. The second summand has modulus at most πa, where

$$a := \frac{1}{4}\log\frac{\pi}{2} = 0.112895676322\ldots > 0. \tag{N.9}$$

Thus $\mathscr{D}'/\mathscr{D} > \frac{1}{2}\log(x+1) - \pi a$, which is positive as soon as $x+1 > \mathrm{e}^{2\pi a} = 2.03265\ldots$, i.e. for $x \geq 1.04$. ☐

Remark N.7. The bound in the proof is not sharp: numerically the last zero of Equation (N.8) is at $x = 0.6970947\ldots$, so $\mathscr{D}$ is in fact strictly increasing on $[0.70,\infty)$. Only the proved threshold is used below.

The next theorem is the precise form of Theorem N.1. None of the three sources states it; all three assert the phenomenon informally.

Theorem N.8 (The inverse transseries is not determined by the sequence)**.** *For $\lambda \in \mathbb{R}$ set $\mathscr{D}_\lambda(x) = \mathscr{D}(x)\,\mathrm{e}^{\lambda\sin\pi x}$. Then:*

(a) *$\mathscr{D}_\lambda(n) = n!!$ for every integer $n \geq 0$, and $\mathscr{D}_\lambda$ is real-analytic on $\mathbb{R}$;*

(b) *$\mathscr{D}_\lambda$ is strictly increasing on $[\,\mathrm{e}^{2\pi(a+|\lambda|)}-1,\infty)$, so it has a large real inverse;*

(c) *with T and Λ the Lambert data of Equation* (N.32) *built from $R_* = \log y - \frac{1}{4}\log(2\pi)$, the leading displacement of that inverse beyond the core is*

$$\mathscr{D}_\lambda^{-1}(y) - (T-1) = \frac{2\sqrt{a^2+\lambda^2}}{\Lambda}\cos\left(\pi T - \arctan\frac{\lambda}{a}\right) + \mathcal{O}\big(\Lambda^{-2}\big)\,. \tag{N.10}$$

In particular the amplitude $2\sqrt{a^2+\lambda^2}/\Lambda$ of the leading oscillatory sector is an unbounded function of the interpolation and is larger, by the factor T, than the entire Bernoulli grid $\mathcal{O}(T^{-1}\Lambda^{-1})$ *of Theorem N.29.*

Proof. (a) $\sin\pi n = 0$ for integer n, and $\mathrm{e}^{\lambda\sin\pi x}$ is entire.

(b) Differentiating logarithmically adds $\lambda\pi\cos\pi x$ to Equation (N.8); repeating the estimate in the proof of Theorem N.6 gives $\mathscr{D}_\lambda'/\mathscr{D}_\lambda > \frac{1}{2}\log(x+1) - \pi(a+|\lambda|)$.

(c) By Theorem N.44 below, with $u = x + 1$,

$$\log \mathscr{D}_\lambda(u-1) = \frac{u}{2}(\log u - 1) + \frac{1}{4}\log(2\pi) - a\cos\pi u - \lambda\sin\pi u + \Omega(u),$$

because $\cos\pi x = -\cos\pi u$ and $\sin\pi x = -\sin\pi u$. Subtracting the core identity $\frac{1}{2}T(\log T - 1) = R_*$, putting $u = T + \Delta$ and multiplying by 2 exactly as in Theorem N.45 gives

$$\Lambda\Delta + T\,\varphi(\Delta/T) + 2\Omega(T+\Delta) - 2a\cos(\theta + \pi\Delta) - 2\lambda\sin(\theta+\pi\Delta) = 0, \qquad \theta = \pi T.$$

All terms except the last two are $\mathcal{O}(T^{-1})$ once $\Delta = \mathcal{O}(\Lambda^{-1})$, so at leading order $\Lambda\Delta_0 = 2a\cos(\theta + \pi\Delta_0) + 2\lambda\sin(\theta+\pi\Delta_0) = 2\sqrt{a^2+\lambda^2}\cos(\theta + \pi\Delta_0 - \arctan(\lambda/a))$. Since $\Delta_0 = \mathcal{O}(\Lambda^{-1})$, replacing $\theta + \pi\Delta_0$ by θ costs $\mathcal{O}(\Lambda^{-2})$, which is Equation (N.10). □

Remark N.9. The family $e^{\lambda\sin\pi x}$ is only the simplest witness. Any real-analytic g vanishing on $\mathbb{N}$ and with g' bounded produces the same conclusion, with $2a\cos\theta$ replaced by the value of $-2g$ transported to the core. What *is* intrinsic is the pair of parity-branch transseries and the integer staircase; the periodic sector is a property of the chosen interpolation and nothing else.

N.2.3 The centred coordinate and the exact normal form

Definition N.10 (Centred coordinate)**.** Throughout, $u := x + 1$ and

$$\Phi(u) := \frac{u}{2}(\log u - 1), \qquad \Omega(u) := \log\Gamma\left(\frac{u+1}{2}\right) - \frac{u}{2}\log\frac{u}{2} + \frac{u}{2} - \frac{1}{2}\log(2\pi). \tag{N.11}$$

Theorem N.11 (Exact centred normal form)**.** *For every $x > -1$ and every $\delta \in \{0,1\}$, with $u = x+1$,*

$$\boxed{\mathcal{D}_\delta(x) = e^{A_\delta}\left(\frac{u}{e}\right)^{u/2} e^{\Omega(u)}, \qquad \textit{equivalently} \qquad \log\mathcal{D}_\delta(x) = \Phi(u) + A_\delta + \Omega(u).} \tag{N.12}$$

This is an identity, not an asymptotic statement, and Ω does not depend on δ.

Proof. Insert $x = u - 1$ into Equation (N.3):

$$\log\mathcal{D}_\delta(x) = \log\kappa_\delta + \frac{u-1}{2}\log 2 + \log\Gamma\left(\frac{u+1}{2}\right).$$

Subtract $A_\delta + \Phi(u)$ and use $A_\delta = \log\kappa_\delta + \frac{1}{2}\log\pi$:

$$\begin{aligned}\log\mathcal{D}_\delta(x) - A_\delta - \Phi(u) &= \frac{u-1}{2}\log 2 + \log\Gamma\left(\frac{u+1}{2}\right) - \frac{1}{2}\log\pi - \frac{u}{2}\log u + \frac{u}{2}\\ &= \log\Gamma\left(\frac{u+1}{2}\right) - \frac{u}{2}(\log u - \log 2) + \frac{u}{2} - \frac{1}{2}\log 2 - \frac{1}{2}\log\pi,\end{aligned}$$

which is exactly $\Omega(u)$ as defined in Equation (N.11). □

Remark N.12 (Why $u = x+1$)**.** The shift is not cosmetic. It turns the gamma argument into $\frac{1}{2}(u+1) = \frac{u}{2} + \frac{1}{2}$, and Bernoulli polynomials evaluated at $\frac{1}{2}$ annihilate every odd index; consequently Ω has only *odd* inverse powers of u (Theorem N.16) and the explicit $\frac{1}{2}\log x$ of the uncentred Stirling form (Theorem N.23) is absorbed into the dominant phase. This sparse support is precisely why the inverse lives on the grid $T^{-1}, T^{-3}, T^{-5}, \ldots$.

At an integer n the two constants A_0, A_1 combine into one parity transmonomial: writing $\chi_n = (-1)^n$,

$$A(n) = \frac{1}{4}\log(2\pi) + \frac{\chi_n}{4}\log\frac{\pi}{2}, \qquad e^{A(n)} = (2\pi)^{1/4}\left(\frac{\pi}{2}\right)^{\chi_n/4}, \tag{N.13}$$

so that for every integer $n \geq 0$

$$n!! = \left(\frac{n+1}{e}\right)^{(n+1)/2} \exp\bigl(A(n) + \Omega(n+1)\bigr). \tag{N.14}$$

Equation (N.14) is exact. The factor $(-1)^n$ is a genuine transmonomial of the sequence's leading behaviour, not an error term.

Remark N.13 (Notation across the three sources). The letter C means three different things in the siblings, and two Greek letters collide:

this chapter	S1	S2	S3
$\kappa_\delta = (2/\pi)^{\delta/2}$	κ_δ	κ_r	C_ε
$e^{A_\delta} \in \{\sqrt{\pi}, \sqrt{2}\}$	C_δ	A_r	—
$A_\delta \in \{\frac{1}{2}\log\pi, \frac{1}{2}\log 2\}$	$\log C_\delta$	C_r	A_ε
c_k (centred log. coefficient)	c_k	β_k	c_k
$a_k = 2c_k$	$2c_k$	α_k	a_k
$a = \frac{1}{4}\log(\pi/2)$	$\rho/2$	a	—
$P_r(L_2)$ (Lambert polynomial)	P_r	u_{r+1}	α_{r+1}
$\tau_m(L_2)$ (reciprocal poly.)	h_m	τ_m	β_m

Part 0 adds one further collision: Theorem J.8 writes the factorial core as $\Phi_0(X) = X(\kappa\log X + d)$, and that κ is *not* the branch prefactor κ_δ of S1 and S2. This chapter therefore never abbreviates the core parameters, always writing $\Phi_0(u) = u\bigl(\frac{1}{2}\log u - \frac{1}{2}\bigr)$ in full; see Theorem J.28. In particular S2's α_k, β_k are Bernoulli data while S3's α_j, β_n are outer Lambert polynomials. Readers comparing the sources should translate before comparing formulae; the numbers themselves agree everywhere (Section N.14).

N.3 Parameter dictionary for the Part 0 apparatus

The chapter uses the Part 0 toolkit rather than re-deriving it. Its dominant core is the *factorial* one of Theorem J.8, not the exponential–power model Theorem J.2; see Theorem N.14. The specialisation is the following.

Part 0 datum	even branch $\delta = 0$	odd branch $\delta = 1$
interpolation	$2^{x/2}\Gamma(\frac{x}{2}+1)$	$\sqrt{2/\pi}\, 2^{x/2}\Gamma(\frac{x}{2}+1)$
affine shift x_* of Theorem J.8	$x_* = -1$, taken first	$x_* = -1$, taken first
centred coordinate $u = x - x_*$	$u = x + 1$	$u = x + 1$
admissible core Φ_0 of Theorem J.8	$\Phi_0(u) = u\bigl(\frac{1}{2}\log u - \frac{1}{2}\bigr)$	same
additive constant	$A_0 = \frac{1}{2}\log\pi$	$A_1 = \frac{1}{2}\log 2$
perturbation $Q(s) = \sum_{k\geq 1} a_k s^{2k-1}$	$a_k = 2c_k$, Equation (N.19)	same
target after constant removal	$R_0 = \log y - \frac{1}{2}\log\pi$	$R_1 = \log y - \frac{1}{2}\log 2$
core (Theorem J.27)	$T_\delta = 2R_\delta/W_0(2R_\delta/e)$	same formula
core slope parameter	$\Lambda_\delta = \log T_\delta$	same formula
inner grid	$q = T_\delta^{-1}$, $z = q^2$	same

Concretely: Theorems J.9 and J.10 supply the Bell polynomials used throughout; Theorem J.14 supplies the coefficients of $(1+w)^{-j}$ and $\log(1+w)$; Theorem J.27 inverts the dominant phase exactly by W_0; Theorem J.33, whose formal engine is Theorem J.19, generates the correction blocks and is specialised once in Theorem N.28; Theorem J.37 unfolds W_0 into the outer layer of Section N.7; Theorems J.41 and J.43 govern Section N.9; Theorems J.47 and J.51 convert forward residuals into inverse errors; and Theorem J.54 fixes the least-term scale of Section N.10.

Warning N.14 (The double factorial is not of exponential–power type). It is tempting — and wrong — to file $\mathcal{D}_\delta$ under Theorem J.2, $F(x) = C\exp(ax^\alpha)x^b\exp Q(x^{-\delta})$ with $\alpha \in (0,1]$. By Theorem N.11, $\log\mathcal{D}_\delta(x) = \frac{u}{2}(\log u - 1) + A_\delta + \Omega(u)$ with $u = x+1$, so the dominant phase is $\frac{1}{2}u\log u$. For every $\alpha \le 1$,

$$\frac{u\log u}{u^\alpha} \longrightarrow \infty \qquad (u \to +\infty),$$

so no admissible choice of $(C, a, \alpha, b, \delta, Q)$ reproduces it: the double factorial lies outside the exponential–power family, in the *factorial* class of Theorem J.8, with the parameters that Theorem J.28 records for this chapter. This is the structural difference from Parts M, O and P, whose sequences have $\log a_n = an^\alpha + b\log n + \cdots$ with $\alpha \le 1$. What survives from the exponential–power theory is the *shape* of the reversion, not the model: Theorem J.33 covers both cores, and the resulting fixed-point map is Equation (N.37). Note also that the shift $x_* = -1$ must be applied *before* any further reduction: it is what removes the $\frac{1}{2}\log x$ of Equation (N.30) and produces the sparse odd-power perturbation (Theorem N.12). None of S1, S2, S3 claims membership of the exponential–power family; all three invert $\frac{1}{2}u(\log u - 1)$ directly.

N.4 Borel structure of the centred correction

The exact Borel kernel is due to S2 and S3, independently and in the same form; S1 has only the formal Bernoulli series. The sharp first-omitted-term bound of Theorem N.17 is S2's; S3 proves the weaker bound with $\eta(2N+2)$ replaced by 1, and that weaker form is not used here.

Theorem N.15 (Borel–Laplace representation)**.** *For $\Re u > 0$,*

$$\Omega(u) = \int_0^\infty e^{-us}\,\widehat{\Omega}(s)\,ds, \qquad \widehat{\Omega}(s) = \frac{1}{2s^2}\left(\frac{s}{\sinh s} - 1\right), \tag{N.15}$$

the singularity of $\widehat{\Omega}$ at $s = 0$ being removable. Moreover

$$\widehat{\Omega}(s) = \sum_{m\ge1} \frac{(-1)^m}{s^2 + \pi^2 m^2}, \tag{N.16}$$

so the Borel singularities of Ω are exactly the simple poles $s = \pm i\pi m$, $m \ge 1$, with residues

$$\operatorname{Res}_{s=i\pi m}\widehat{\Omega}(s) = \frac{(-1)^{m+1}i}{2\pi m}, \qquad \operatorname{Res}_{s=-i\pi m}\widehat{\Omega}(s) = \frac{(-1)^m i}{2\pi m}. \tag{N.17}$$

Proof. Put $t = u/2$ and $\widetilde{\Omega}(t) = \log\Gamma(t+\frac{1}{2}) - t\log t + t - \frac{1}{2}\log(2\pi)$, so that $\Omega(u) = \widetilde{\Omega}(u/2)$ by Equation (N.11). Binet's first integral and Frullani's formula give, for $t > 0$,

$$\psi\big(t+\tfrac{1}{2}\big) - \log t = \int_0^\infty e^{-tr}\left(\frac{1}{r} - \frac{1}{2\sinh(r/2)}\right)dr,$$

the integrand being $\mathcal{O}(r)$ at the origin and exponentially damped at infinity. On the other hand

$$\frac{d}{dt}\int_0^\infty e^{-tr}\,\frac{r/(2\sinh(r/2)) - 1}{r^2}\,dr = \int_0^\infty e^{-tr}\left(\frac{1}{r} - \frac{1}{2\sinh(r/2)}\right)dr,$$

differentiation under the integral being justified by the same bounds. Since $\widetilde{\Omega}'(t) = \psi(t+\frac{1}{2}) - \log t$ and both $\widetilde{\Omega}(t)$ and the displayed integral tend to 0 as $t \to +\infty$, the two functions coincide. Substituting $r = 2s$ and $t = u/2$ turns the integral into Equation (N.15); analytic continuation in u extends it to $\Re u > 0$. The expansion $s/\sinh s = 1 - s^2/6 + \mathcal{O}(s^4)$ removes the origin.

For Equation (N.16), the elementary Mittag–Leffler expansion $\dfrac{s}{\sinh s} = 1 + 2s^2\sum_{m\ge1}\dfrac{(-1)^m}{s^2+\pi^2m^2}$ gives the claim after division by $2s^2$. The residues follow from $1/(s^2+\pi^2m^2) = \big[(s - i\pi m)(s + i\pi m)\big]^{-1}$. ∎

Theorem N.16 (The centred logarithmic coefficients). *As $u \to \infty$ in any closed subsector of $|\arg u| < \pi/2$,*

$$\Omega(u) \sim \sum_{k\ge 1} \frac{c_k}{u^{2k-1}}, \tag{N.18}$$

where the following three expressions coincide:

$$\boxed{c_k = \frac{2^{2k-1}B_{2k}\left(\frac{1}{2}\right)}{2k(2k-1)} = \frac{\left(1-2^{2k-1}\right)B_{2k}}{2k(2k-1)} = (-1)^k \frac{(2k-2)!\,\eta(2k)}{\pi^{2k}}} \tag{N.19}$$

with $B_m(z)$ the Bernoulli polynomials, $B_m = B_m(0)$, and $\eta(s) = (1-2^{1-s})\zeta(s)$ the Dirichlet eta function. There are no even inverse powers.

Proof. Expand each summand of Equation (N.16) by

$$\frac{1}{s^2+A^2} = \sum_{j=0}^{N-1} \frac{(-1)^j s^{2j}}{A^{2j+2}} + \frac{(-1)^N s^{2N}}{A^{2N}(s^2+A^2)}, \qquad A > 0, \tag{N.20}$$

with $A = \pi m$, sum over m, and integrate term by term against e^{-us}, using $\int_0^\infty \mathrm{e}^{-us}s^{2j}\mathrm{d}s = (2j)!\,u^{-2j-1}$. The coefficient of $u^{-(2j+1)}$ is

$$(2j)! \sum_{m\ge 1} \frac{(-1)^m(-1)^j}{(\pi m)^{2j+2}} = (-1)^{j+1}\frac{(2j)!\,\eta(2j+2)}{\pi^{2j+2}},$$

which is the third expression in Equation (N.19) with $k = j+1$. Euler's evaluation $\zeta(2k) = (-1)^{k+1}(2\pi)^{2k}B_{2k}/\big(2(2k)!\big)$ together with $\eta(2k) = (1-2^{1-2k})\zeta(2k)$ and $B_{2k}(\frac{1}{2}) = (2^{1-2k}-1)B_{2k}$ converts it into the first two. Only odd powers $u^{-(2k-1)}$ occur because $\widehat{\Omega}$ is even. □

Theorem N.17 (Sharp alternating remainder). *For real $u > 0$ and $N \ge 0$ put $R_N(u) = \Omega(u) - \sum_{k=1}^N c_k u^{-(2k-1)}$ (empty sum for $N = 0$). Then $R_N(u)$ has the sign of the first omitted term and is smaller than it in modulus:*

$$0 < (-1)^{N+1}R_N(u) < \frac{|c_{N+1}|}{u^{2N+1}} = \frac{(2N)!\,\eta(2N+2)}{\pi^{2N+2}u^{2N+1}}. \tag{N.21}$$

Proof. Insert Equation (N.20) with $A = \pi m$ into Equation (N.16) and Laplace-transform. The residual kernel is $(-1)^{N+1}s^{2N}S_N(s)$ with

$$S_N(s) = \sum_{m\ge 1} \frac{(-1)^{m-1}}{(\pi m)^{2N}\big((\pi m)^2+s^2\big)}.$$

Writing $\big((\pi m)^2+s^2\big)^{-1} = \int_0^\infty \mathrm{e}^{-s^2v}\mathrm{e}^{-\pi^2m^2v}\mathrm{d}v$ gives

$$S_N(s) = \int_0^\infty \mathrm{e}^{-s^2v} \sum_{m\ge 1} \frac{(-1)^{m-1}\mathrm{e}^{-\pi^2m^2v}}{(\pi m)^{2N}}\,\mathrm{d}v.$$

For each $v \ge 0$ the inner series is alternating with strictly decreasing positive terms, hence strictly positive. Therefore $S_N(s) > 0$ and S_N is strictly decreasing in s^2, so $0 < S_N(s) \le S_N(0) = \eta(2N+2)/\pi^{2N+2}$. Multiplying by s^{2N}, integrating against e^{-us} and using $\int_0^\infty \mathrm{e}^{-us}s^{2N}\mathrm{d}s = (2N)!u^{-(2N+1)}$ gives Equation (N.21). □

Corollary N.18 (Certified forward logarithm). *For $x \to +\infty$, $u = x + 1$ and either parity,*

$$\log \mathcal{D}_\delta(x) = \Phi(u) + A_\delta + \sum_{k=1}^{N} \frac{c_k}{u^{2k-1}} + \theta_N \frac{c_{N+1}}{u^{2N+1}} \qquad \text{for some } \theta_N = \theta_N(u) \in (0,1). \tag{N.22}$$

Thus truncating after any number of terms overshoots or undershoots by strictly less than the first omitted term, with the sign of that term. For the integer sequence replace A_δ by $A(n)$ of Equation (N.13).

Proof. Combine Theorems N.11 and N.17. ☐

Table N.1: The centred logarithmic coefficients c_k of Equation (N.19) and the doubled coefficients $a_k = 2c_k$ used in the inverse problem. Regenerated in exact rational arithmetic from B_{2k}.

k	c_k	$a_k = 2c_k$
1	$-1/12$	$-1/6$
2	$7/360$	$7/180$
3	$-31/1260$	$-31/630$
4	$127/1680$	$127/840$
5	$-511/1188$	$-511/594$
6	$1414477/360360$	$1414477/180180$
7	$-8191/156$	$-8191/78$
8	$118518239/122400$	$118518239/61200$
9	$-5749691557/244188$	$-5749691557/122094$

Remark N.19 (Late coefficients). From the third form in Equation (N.19) and $1 < \eta(2k) < 1 + 2^{-2k} \cdot \frac{4}{3}$,

$$c_k = (-1)^k \frac{(2k-2)!}{\pi^{2k}} \big(1 + \mathcal{O}(4^{-k})\big), \tag{N.23}$$

so Equation (N.18) is Gevrey-1 in the variable u^{-2}. The ratio of consecutive term moduli is $\sim 2k(2k-1)/(\pi u)^2$; the least term occurs near $2k = \pi u$ and has size $\asymp u^{-1/2}\mathrm{e}^{-\pi u}$, matching the distance π from the origin to the nearest Borel poles $\pm \mathrm{i}\pi$. This is the instance of Theorem J.54 relevant here; see Section N.10.

N.5 The forward transseries

N.5.1 Multiplicative, reciprocal, and arbitrary-power coefficients

Exponentiation of Equation (N.18) is an instance of the exponential formula, so Theorem J.10 does all the work at once for every power.

Definition N.20. Embed the c_k sparsely on the odd indices,

$$\gamma_r = \begin{cases} c_{(r+1)/2}, & r \text{ odd}, \\ 0, & r \text{ even}, \end{cases} \qquad r \geq 1, \tag{N.24}$$

and for a scalar α define $s_m^{(\alpha)}$ by

$$\exp\left(\alpha \sum_{r\geq 1} \gamma_r\, u^{-r}\right) = \sum_{m\geq 0} \frac{s_m^{(\alpha)}}{u^m}, \qquad s_m := s_m^{(1)}. \tag{N.25}$$

Proposition N.21 (All multiplicative coefficients at once). *For every α and every $m \geq 0$,*

$$s_m^{(\alpha)} = \frac{1}{m!}\, B_m\big(\alpha\, 1!\, \gamma_1, \alpha\, 2!\, \gamma_2, \ldots, \alpha\, m!\, \gamma_m\big) \tag{N.26}$$

$$= \sum_{\substack{k_1,k_2,\ldots\geq 0\\ \sum_{r\geq 1} rk_r = m}} \prod_{r\geq 1} \frac{(\alpha\gamma_r)^{k_r}}{k_r!} = \sum_{\substack{k_1,k_2,\ldots\geq 0\\ \sum_{j\geq 1}(2j-1)k_j = m}} \prod_{j\geq 1} \frac{(\alpha c_j)^{k_j}}{k_j!}, \tag{N.27}$$

and equivalently

$$s_0^{(\alpha)} = 1, \qquad m\, s_m^{(\alpha)} = \alpha \sum_{1\leq j\leq (m+1)/2} (2j-1)\, c_j\, s_{m-2j+1}^{(\alpha)} \qquad (m \geq 1). \tag{N.28}$$

Consequently, for $x \to +\infty$, $u = x+1$ and every fixed $M \geq 0$,

$$\mathcal{D}_\delta(x)^\alpha = \mathrm{e}^{\alpha A_\delta} \left(\frac{u}{\mathrm{e}}\right)^{\alpha u/2} \left(\sum_{m=0}^{M} \frac{s_m^{(\alpha)}}{u^m} + \mathcal{O}\Big(u^{-(M+1)}\Big)\right). \tag{N.29}$$

The cases $\alpha = 1$ and $\alpha = -1$ give $\mathcal{D}_\delta$ itself and its reciprocal.

Proof. Put $t = u^{-1}$ and $S_\alpha(t) = \exp(\alpha \sum_r \gamma_r t^r)$. Equation (N.26) is the complete-Bell generating identity of Theorem J.10; multinomial expansion of the exponential gives Equation (N.27), and the second form there uses $\gamma_{2j-1} = c_j$, $\gamma_{2j} = 0$. Logarithmic differentiation $tS'_\alpha = \alpha\big(\sum_r r\gamma_r t^r\big) S_\alpha$ and comparison of coefficients give Equation (N.28). For Equation (N.29), take N with $2N-1 \geq M$; by Theorems N.11 and N.17, $\alpha\Omega(u) = \alpha \sum_{k\leq N} c_k u^{1-2k} + \mathcal{O}(u^{-(2N+1)})$, and exp of a finite sum of $\mathcal{O}(u^{-1})$ terms plus a $\mathcal{O}(u^{-(2N+1)})$ error equals $\sum_{m\leq M} s_m^{(\alpha)} u^{-m} + \mathcal{O}(u^{-(M+1)})$, the exponential being Lipschitz on bounded sets. □

Corollary N.22 (Reciprocal sign symmetry). $s_m^{(-\alpha)} = (-1)^m s_m^{(\alpha)}$ *for all $m \geq 0$; in particular the reciprocal coefficients are $s_m^{(-1)} = (-1)^m s_m$.*

Proof. Only odd r carry a nonzero γ_r, so $\sum_r \gamma_r(-t)^r = -\sum_r \gamma_r t^r$ and hence $S_\alpha(-t) = S_{-\alpha}(t)$. Reading off $[t^m]$ gives the claim. □

Table N.2: Centred multiplicative coefficients $s_m = s_m^{(1)}$ of Equation (N.29) and the uncentred coefficients g_m of Theorem N.23. Regenerated from Equation (N.28) in exact rational arithmetic.

m	s_m (centred)	g_m (uncentred)
0	1	1
1	$-1/12$	$1/6$
2	$1/288$	$1/72$
3	$1003/51840$	$-139/6480$
4	$-4027/2488320$	$-571/155520$
5	$-5128423/209018880$	$163879/6531840$
6	$168359651/75246796800$	$5246819/1175731200$
7	$68168266699/902961561600$	$-534703531/7054387200$
8	$-587283555451/86684309913600$	$-4483131259/338610585600$

N.5.2 The uncentred normal form and the centre-shift identity

Proposition N.23 (Uncentred expansion). *As $x \to +\infty$,*

$$\log \mathcal{D}_\delta(x) \sim \frac{x}{2}(\log x-1)+\frac{1}{2}\log x+A_\delta+\sum_{k\geq 1}\frac{b_k}{x^{2k-1}}, \qquad b_k=\frac{2^{2k-1}B_{2k}}{2k(2k-1)}=(-1)^{k-1}\frac{(2k-2)!\,\zeta(2k)}{\pi^{2k}}, \tag{N.30}$$

with $b_1=\frac{1}{6}, b_2=-\frac{1}{45}, b_3=\frac{8}{315}, b_4=-\frac{8}{105}, b_5=\frac{128}{297}$. *Exponentiating,* $\mathcal{D}_\delta(x)\sim \mathrm{e}^{A_\delta}\sqrt{x}\,(x/\mathrm{e})^{x/2}\sum_m g_m x^{-m}$, *where g_m is obtained from Equation* (N.28) *on replacing c_j by b_j; this is the familiar form, with amplitudes $\sqrt{\pi}$ (even) and $\sqrt{2}$ (odd). The exact Borel kernel of the uncentred correction is* $\widehat{\mathcal{U}}(s)=\frac{\coth s}{2s}-\frac{1}{2s^2}=\sum_{m\geq 1}\bigl(s^2+\pi^2m^2\bigr)^{-1}$, *with the* same *pole lattice $\pm \mathrm{i}\pi m$ but all residues of one sign pattern.*

Proof. Apply the standard Stirling expansion to $\log\Gamma(z+1)$ with $z=x/2$ and add $(x/2)\log 2+\log\kappa_\delta$; the elementary terms give the first three summands of Equation (N.30), and the corrections scale as $z^{-(2k-1)}=2^{2k-1}x^{-(2k-1)}$. Euler's evaluation of $\zeta(2k)$ gives the second form of b_k. For the kernel, the uncentred correction is $\mathcal{U}(x)=\Omega_\Gamma(x/2)$ with Ω_Γ Binet's function; substituting $t=2s$ in $\Omega_\Gamma(z)=\int_0^\infty \mathrm{e}^{-zt}\bigl(\frac{\coth(t/2)}{2t}-\frac{1}{t^2}\bigr)\mathrm{d}t$ gives the stated kernel, and $\frac{\coth s}{s}=\frac{1}{s^2}+2\sum_{m\geq 1}(s^2+\pi^2m^2)^{-1}$ gives the partial fractions. □

Warning N.24. The two statements $n!!\sim \mathrm{e}^{A(n)}\sqrt{n}\,(n/\mathrm{e})^{n/2}$ and $n!!\sim \mathrm{e}^{A(n)}\bigl((n+1)/\mathrm{e}\bigr)^{(n+1)/2}$ are both correct and are not in conflict: they use different normal forms. The first leaves a factor $\sqrt{n}$ and an *uncentred* correction series; the second absorbs that factor into the phase. Only the centred form leads to the sparse inverse grid, and it is the one used from Section N.6 onwards.

Proposition N.25 (Centre-shift identity). *For every $r\geq 1$,*

$$\frac{(-1)^{r+1}}{2r(r+1)}+(-1)^{r+1}\sum_{k=1}^{\lfloor (r+1)/2\rfloor}\binom{r-1}{2k-2}c_k=\begin{cases} b_{(r+1)/2}, & r \text{ odd},\\ 0, & r \text{ even}.\end{cases} \tag{N.31}$$

Proof. Substitute $u=x+1=x(1+w)$, $w=x^{-1}$, into $\Phi(u)+\sum_k c_k u^{1-2k}$ and compare with Equation (N.30). First, with $\log u=\log x+\log(1+w)$,

$$\Phi(u)=\frac{x}{2}(\log x-1)+\frac{1}{2}(\log x-1)+\frac{x+1}{2}\log(1+w).$$

Expanding $\log(1+w)=\sum_{j\geq 1}(-1)^{j-1}w^j/j$ and splitting the prefactor $\frac{x+1}{2}=\frac{x}{2}+\frac{1}{2}$ gives

$$\frac{x+1}{2}\log(1+w)=\frac{1}{2}+\frac{1}{2}\sum_{r\geq 1}(-1)^r x^{-r}\left(\frac{1}{r+1}-\frac{1}{r}\right)=\frac{1}{2}+\sum_{r\geq 1}\frac{(-1)^{r+1}}{2r(r+1)}x^{-r},$$

so that $\Phi(u)=\frac{x}{2}(\log x-1)+\frac{1}{2}\log x+\sum_{r\geq 1}(-1)^{r+1}\bigl(2r(r+1)\bigr)^{-1}x^{-r}$; the two halves $\pm\frac{1}{2}$ cancel. Second,

$$\sum_{k\geq 1}\frac{c_k}{u^{2k-1}}=\sum_{k\geq 1}c_k\,x^{-(2k-1)}(1+w)^{-(2k-1)}=\sum_{r\geq 1}x^{-r}\sum_{k\leq (r+1)/2}c_k\binom{-(2k-1)}{r-2k+1},$$

and $\binom{-(2k-1)}{r-2k+1}=(-1)^{r-2k+1}\binom{r-1}{r-2k+1}=(-1)^{r+1}\binom{r-1}{2k-2}$. Adding the two coefficient lists and matching against Equation (N.30) — where the coefficient of x^{-r} is $b_{(r+1)/2}$ for odd r and 0 for even r — gives Equation (N.31). □

Remark N.26. Equation (N.31) was verified symbolically for $1\leq r\leq 9$: the even cases return 0 and the odd cases return $\frac{1}{6}, -\frac{1}{45}, \frac{8}{315}, -\frac{8}{105}, \frac{128}{297}$. It is the cheapest available certificate that the centring is consistent, because it fails immediately under any sign or index slip in Equation (N.19). S3 states the identity without proof; the proof above is supplied here.

N.6 The branchwise functional inverse

All three siblings reach the same objects in this section. The presentation follows S3 for the exact normal form and S2 for the graded closed formula and the sign law; S1's "scaling from the half-shifted inverse Gamma problem" is subsumed by Theorem N.61. The proofs of Theorems N.35 and N.37 are new: the corresponding source arguments are sketches (see the chapter report).

N.6.1 The Lambert core

Lemma N.27 (Exact inversion of the dominant phase)**.** *Let $R > 0$. The equation $\Phi(T) = \frac{1}{2}T(\log T - 1) = R$ has a unique solution $T > \mathrm{e}$, namely*

$$w = W_0\left(\frac{2R}{\mathrm{e}}\right), \qquad T = \frac{2R}{w} = \mathrm{e}^{w+1}, \qquad \Lambda := w + 1 = \log T, \tag{N.32}$$

and then

$$\Phi'(T) = \frac{\Lambda}{2}, \qquad \frac{\mathrm{d}T}{\mathrm{d}R} = \frac{2}{\Lambda}, \qquad T(\Lambda - 1) = 2R. \tag{N.33}$$

Proof. This is Theorem J.27 for the parameters of Theorem J.28; for completeness, setting $w = \log T - 1$ turns $\Phi(T) = R$ into $w\mathrm{e}^{w+1} = 2R$, i.e. $w\mathrm{e}^{w} = 2R/\mathrm{e} > 0$, whose unique real solution is $w = W_0(2R/\mathrm{e}) > 0$, the principal branch being the only real branch on $(0, \infty)$. Φ is strictly increasing on $(1, \infty)$ with range $(-\frac{1}{2}, \infty)$, which gives uniqueness of T. The identities in Equation (N.33) follow from $\Phi'(T) = \frac{1}{2}\log T$, implicit differentiation of $\Phi(T) = R$, and $T = 2R/w = 2R/(\Lambda - 1)$. $\square$

For the parity branch δ one takes

$$R_\delta = \log y - A_\delta, \qquad w_\delta,\ T_\delta,\ \Lambda_\delta \text{ as in Equation (N.32).} \tag{N.34}$$

The subscript δ is suppressed inside coefficient calculations, where it never appears.

N.6.2 The normalised displacement equation

Lemma N.28 (The factorial-core reversion kernel)**.** *Write $u = T(1 + E)$, $q = T^{-1}$, $z = q^2$, and*

$$\varphi(v) = (1 + v)\log(1 + v) - v = \sum_{m\geq 2} \frac{(-1)^m}{m(m-1)} v^m, \qquad a_k = 2c_k. \tag{N.35}$$

Then the equation $\Phi(u) + \Omega(u) = R$, with Ω replaced by its asymptotic series, is equivalent to the formal normal form

$$\boxed{\Lambda E + \varphi(E) + \sum_{k\geq 1} a_k\, z^k (1 + E)^{-(2k-1)} = 0.} \tag{N.36}$$

Equivalently, in absolute displacement $U = TE$ with $t = q$,

$$U = \Psi(t, U), \qquad \Psi(t, v) = -\frac{1}{\Lambda}\left[\frac{\varphi(tv)}{t} + \sum_{k\geq 1} a_k\, t^{2k-1}(1 + tv)^{-(2k-1)}\right] \in t\,\mathbb{Q}(\Lambda)[[t, v]], \tag{N.37}$$

which is the specialisation of Theorem J.33 to the factorial core, so that Theorem J.19 applies verbatim. There is a unique formal solution, and it has the form

$$E = \sum_{n\geq 1} e_n(\Lambda)\, z^n, \qquad \textit{hence} \qquad u - T = \sum_{n\geq 1} \frac{e_n(\Lambda)}{T^{2n-1}}: \tag{N.38}$$

only odd *inverse powers of the Lambert core occur.*

Proof. Because $\Phi(T) = R$,

$$\Phi\big(T(1+E)\big) - \Phi(T) = \frac{T(1+E)}{2}\big(\log T + \log(1+E) - 1\big) - \frac{T}{2}(\log T - 1) = \frac{T}{2}\big[\Lambda E + \varphi(E)\big],$$

using $\Lambda = \log T$ and $(1+E)\log(1+E) = \varphi(E) + E$. Also $\Omega(u) = \sum_k c_k T^{-(2k-1)}(1+E)^{-(2k-1)}$ formally. Multiplying the whole equation by $2/T = 2q$ and using $a_k = 2c_k$ and $q \cdot q^{2k-1} = z^k$ gives Equation (N.36). Dividing by T and solving for $U = TE$ gives Equation (N.37); the bracket lies in $t\,\mathbb{Q}[[t,v]]$ because $\varphi(tv) = t^2v^2/2 + \mathcal{O}(t^3)$, so $\varphi(tv)/t = \mathcal{O}(t)$, and every Bernoulli term carries t^{2k-1} with $k \geq 1$. Existence and uniqueness of the fixed point are Theorem J.19. Finally, if E contains only even powers of q then so does every term of Equation (N.36); an induction on the q-valuation therefore forces $E \in z\,\mathbb{Q}(\Lambda)[[z]]$. □

N.6.3 The all-orders inverse and its remainder

Theorem N.29 (Complete branchwise inverse transseries)**.** *Fix $\delta \in \{0,1\}$ and let $R_\delta, T_\delta, \Lambda_\delta$ be as in Equation* (N.34)*. Write $d_{2n-1} := e_n$ for the coefficients of Equation* (N.38)*. Then for every fixed $N \geq 0$, as $y \to +\infty$,*

$$\boxed{\mathcal{D}_\delta^{-1}(y) = T_\delta - 1 + \sum_{n=1}^{N} \frac{d_{2n-1}(\Lambda_\delta)}{T_\delta^{2n-1}} + \mathcal{O}\left(\frac{1}{T_\delta^{2N+1}\Lambda_\delta}\right),} \tag{N.39}$$

with an implied constant that may be chosen uniformly in $\delta \in \{0,1\}$. All parity dependence sits in $R_\delta = \log y - A_\delta$; the coefficient functions d_{2n-1} are universal.

Proof. Let $u_N = T + \sum_{n=1}^{N} e_n(\Lambda)T^{1-2n}$ and $E_N = (u_N - T)/T$. By construction, substituting E_N into Equation (N.36) annihilates the coefficients of $z^1, \ldots, z^N$, so the formal residual is $\mathcal{O}(z^{N+1})$ with coefficients rational in Λ and bounded as $\Lambda \to \infty$. Undoing the normalisation (multiplying by $T/2$) therefore bounds the exact forward residual

$$\rho_N := \Phi(u_N) + A_\delta + \Omega(u_N) - \log y$$

by $\mathcal{O}\big(T \cdot z^{N+1}\big) = \mathcal{O}\big(T^{-(2N+1)}\big)$, where replacing Ω by its N-term truncation costs at most $|c_{N+1}|u_N^{-(2N+1)}$ by Theorem N.17, of the same order. This is the hypothesis of Theorem J.47. For the slope, on the interval between u_N and the exact root,

$$\frac{\mathrm{d}}{\mathrm{d}u}\big[\Phi(u) + \Omega(u)\big] = \frac{1}{2}\log u + \Omega'(u) = \frac{\Lambda}{2}\big(1 + o(1)\big),$$

because $u = T\big(1 + \mathcal{O}(T^{-2})\big)$ and, differentiating Equation (N.15) under the integral sign, $\Omega'(u) = -\int_0^\infty s\mathrm{e}^{-us}\widehat{\Omega}(s)\mathrm{d}s = \mathcal{O}(u^{-2})$. The mean-value theorem now divides ρ_N by a quantity asymptotic to $\Lambda/2$, which is the instance of Theorem J.51 for the factorial core and gives Equation (N.39). The parity constant A_δ disappears on differentiation, so the estimate is uniform in δ. □

N.6.4 Two generators for the coefficients

Equation (N.36) is exactly Equation (J.56), the reversion problem of Theorem J.33, with the instance data

$$h(v) = \varphi(v), \quad h_m = \frac{(-1)^m}{m(m-1)}, \qquad f(z,v) = \sum_{k\geq 1} \phi_k(z)(1+v)^{-\mu_k}, \quad \phi_k(z) = a_k z^k, \quad \mu_k = 2k-1. \tag{N.40}$$

Its triangular recurrence Equation (J.57) and its Bell form Equation (J.59) therefore apply verbatim; written out, the former is

$$e_n(\Lambda) = -\frac{1}{\Lambda}\,[z^n]\left\{\varphi(E_{n-1}) + \sum_{k=1}^{n} a_k\, z^k \big(1+E_{n-1}\big)^{-(2k-1)}\right\}, \qquad E_{n-1} = \sum_{j<n} e_j z^j. \tag{N.41}$$

Only the coefficient *structure* is subject-specific.

Theorem N.30 (Structure of the inverse blocks)**.** *For every $n \ge 1$,*

$$e_n(\Lambda) \in \Lambda^{-1}\mathbb{Q}[\Lambda^{-1}], \qquad \deg_{\Lambda^{-1}} e_n \le 2n-1, \qquad e_n(\Lambda) = -\frac{a_n}{\Lambda} + \mathcal{O}(\Lambda^{-2}). \tag{N.42}$$

Proof. Specialising Equation (J.59) to Equation (N.40) gives

$$e_n = -\frac{1}{\Lambda}\left[\sum_{m=2}^{n} \frac{(-1)^m}{m(m-1)}\,\widehat{B}_{n,m}(e) + \sum_{k=1}^{n} a_k \sum_{m=0}^{n-k} (-1)^m \binom{2k+m-2}{m} \widehat{B}_{n-k,m}(e)\right], \tag{N.43}$$

in which every occurring e_j has $j<n$. A monomial of $\widehat{B}_{r,m}$ is a product $e_{j_1}\cdots e_{j_m}$ with $j_1+\cdots+j_m = r$; by induction its Λ^{-1}-degree lies between m and $\sum_\nu (2j_\nu - 1) = 2r-m$, and the outer factor Λ^{-1} raises this by one. In the first sum $r = n$ and $m \ge 2$, giving degree at most $2n-1$; in the second $r = n-k$, giving at most $2(n-k)-m+1 \le 2n-1$. Rationality is preserved throughout. Finally the only contribution of Λ^{-1}-degree exactly 1 is the term $k=n$, $m=0$, namely $-a_n/\Lambda$, because every Bell monomial contains at least one $e_j = \mathcal{O}(\Lambda^{-1})$. □

Theorem N.31 (Graded closed formula)**.** *Let $A(z) = \sum_{k\ge1} a_k z^k$ and put*

$$A_{n,p} := [z^n]\,A(z)^p = \widehat{B}_{n,p}(a), \qquad K_{n;j,p} := [v^{j-1}]\,\varphi(v)^{\,j-p}(1+v)^{-(2n-p)}. \tag{N.44}$$

Expand $d_{2n-1}(\Lambda) = \sum_{j\ge1} c_{n,j}\Lambda^{-j}$. Then

$$\boxed{c_{n,j} = \frac{(-1)^j}{j} \sum_{p=\lceil (j+1)/2\rceil}^{\min(j,n)} \binom{j}{p} A_{n,p}\, K_{n;j,p}} \qquad (1 \le j \le 2n-1), \tag{N.45}$$

and $c_{n,j} = 0$ for $j > 2n-1$. The right-hand side is a finite expression in Bernoulli numbers, binomial coefficients and ordinary partial Bell polynomials; it computes any single scalar coefficient without generating the lower blocks.

Proof. Put $H(z,v) = \varphi(v) + \sum_{k\ge1} a_k z^k (1+v)^{-(2k-1)}$, so that Equation (N.36) reads $E = -\Lambda^{-1}H(z,E)$. The closed formula Equation (J.58) of Theorem J.33 gives

$$e_n = \sum_{j\ge1} \frac{1}{j}\,[z^n v^{j-1}]\left(-\frac{H(z,v)}{\Lambda}\right)^j = \sum_{j\ge1} \frac{(-1)^j}{j\Lambda^j}\,[z^n v^{j-1}]\,H(z,v)^j,$$

which already exhibits the grading by powers of Λ^{-1}. It remains to evaluate $[z^n v^{j-1}]\,H^j$. Expand H^j binomially, taking p of the j factors from the Bernoulli sector:

$$H^j = \sum_{p=0}^{j} \binom{j}{p}\,\varphi(v)^{\,j-p}\left(\sum_{k\ge1} a_k z^k (1+v)^{-(2k-1)}\right)^p.$$

Extracting z^n from the p-th power, the surviving terms have $k_1+\cdots+k_p = n$; their coefficients sum to $A_{n,p}$ and they share the factor $(1+v)^{-\sum_i (2k_i-1)} = (1+v)^{-(2n-p)}$. Hence $[z^n]\,H^j =$

$\sum_p \binom{j}{p} A_{n,p}\varphi(v)^{j-p}(1+v)^{-(2n-p)}$, and extracting v^{j-1} gives Equation (N.45). (Equation (J.58) caps the outer sum at $j \le 2n-1$; the support argument below recovers that bound independently.)

For the summation range: $A_{n,p} = 0$ unless $1 \le p \le n$, while $\varphi(v)^{j-p}$ has v-valuation $2(j-p)$, so a nonzero v^{j-1}-coefficient forces $2(j-p) \le j-1$, i.e. $p \ge \lceil (j+1)/2 \rceil$. Combining, $j \le 2p-1 \le 2n-1$; this re-proves the degree bound of Equation (N.42) and shows the expansion terminates. □

Remark N.32. Equation (N.45) was checked against Equation (N.41) for all $1 \le n \le 5$ and all $1 \le j \le 2n-1$ in exact rational arithmetic: all 25 scalars agree. For explicit evaluation, $K_{n;j,p}$ unfolds by Theorems J.9 and J.14 into

$$K_{n;j,p} = \sum_{h=2(j-p)}^{j-1} \widehat{B}_{h,\,j-p}(\nu)\,(-1)^{j-1-h}\binom{2n-p+j-2-h}{j-1-h}, \qquad \nu_1 = 0, \quad \nu_m = \frac{(-1)^m}{m(m-1)}\,(m \ge 2),$$

with the conventions $\widehat{B}_{0,0} = 1$ and $\widehat{B}_{h,0} = 0$ for $h > 0$.

Proposition N.33 (A second, ungraded closed formula). *Let $S = A^{\langle -1 \rangle}$ be the formal compositional inverse of $A(z) = \sum_k a_k z^k$, which exists because $a_1 = -\frac{1}{6} \neq 0$, with*

$$[w^m]\,S(w) = \frac{1}{m}\,[t^{m-1}]\left(\frac{t}{A(t)}\right)^m. \tag{N.46}$$

Put

$$G_\Lambda(E) = (1+E)^2\,S\biggl(-\frac{\Lambda E + \varphi(E)}{1+E}\biggr) \in E\,\mathbb{Q}(\Lambda)[[E]], \qquad G'_\Lambda(0) = -\frac{\Lambda}{a_1} = 6\Lambda \neq 0. \tag{N.47}$$

Then

$$e_n(\Lambda) = \frac{1}{n}\,[E^{n-1}]\left(\frac{E}{G_\Lambda(E)}\right)^n. \tag{N.48}$$

Proof. Since $\sum_k a_k z^k (1+E)^{-(2k-1)} = (1+E)\,A\bigl(z/(1+E)^2\bigr)$, Equation (N.36) reads $A\bigl(z/(1+E)^2\bigr) = -\bigl(\Lambda E + \varphi(E)\bigr)/(1+E)$. Applying S and multiplying by $(1+E)^2$ gives $z = G_\Lambda(E)$; differentiating at $E = 0$, where $\varphi(0) = \varphi'(0) = 0$, gives $G'_\Lambda(0) = S'(0)\cdot(-\Lambda) = -\Lambda/a_1$. Equations (N.46) and (N.48) are then two applications of Theorem J.17. Although A is a divergent Gevrey series, every finite jet uses only finitely many a_k, so this is a formal identity requiring no convergence hypothesis. □

Remark N.34. Equation (N.48) was verified against Equation (N.41) for $n \le 4$. It is less useful than Equation (N.45) in practice: it does not separate the powers of Λ^{-1}, and it requires reverting A first. Its interest is conceptual — it exhibits the inverse blocks as a *double* Lagrange–Bürmann extraction, the inner one reverting the pure Bernoulli series.

N.6.5 Boundary coefficients and the global sign law

Corollary N.35 (The two edges of the coefficient triangle). *For every $n \ge 1$,*

$$c_{n,1} = -a_n = -2c_n, \qquad c_{n,2n-1} = \frac{1}{3^n}\binom{1/2}{n} = -\frac{1}{2^{n-1}(2n-1)}\binom{2n-1}{n}a_1^n. \tag{N.49}$$

In particular the two closed forms recorded separately by S1 and S2 for the deepest coefficient are the same number.

Proof. For $j = 1$ the range in Equation (N.45) forces $p = 1$; then $A_{n,1} = a_n$, $K_{n;1,1} = [v^0]\,(1+v)^{-(2n-1)} = 1$, and the prefactor is -1.

For $j = 2n-1$, the range forces $p = n$; then $A_{n,n} = a_1^n$ and

$$K_{n;2n-1,n} = [v^{2n-2}]\,\varphi(v)^{\,n-1}(1+v)^{-n} = [v^{2n-2}]\,\left(\frac{v^2}{2}\right)^{n-1} = 2^{1-n},$$

because $\varphi(v)^{n-1}$ has valuation $2n-2$. With $\frac{(-1)^{2n-1}}{2n-1}\binom{2n-1}{n}$ as prefactor this is the second expression in Equation (N.49).

Equality of the two expressions is elementary. Using $\binom{1/2}{n} = (-1)^{n-1}\binom{2n}{n}\bigl(4^n(2n-1)\bigr)^{-1}$,

$$\frac{1}{3^n}\binom{1/2}{n} = (-1)^{n-1}\frac{\binom{2n}{n}}{12^n(2n-1)},$$

while $a_1 = -\frac{1}{6}$ and $\binom{2n-1}{n} = \frac{1}{2}\binom{2n}{n}$ give

$$-\frac{\binom{2n-1}{n}a_1^n}{2^{n-1}(2n-1)} = (-1)^{n+1}\frac{\binom{2n}{n}}{2\cdot 2^{n-1}6^n(2n-1)} = (-1)^{n+1}\frac{\binom{2n}{n}}{12^n(2n-1)}.$$

□

Remark N.36 (The deepest edge as a square root). Let $t_n = c_{n,2n-1}$. Only the quadratic part of φ and the single coefficient a_1 can reach $\Lambda^{-(2n-1)}$: in Equation (N.43) the degree bound $2r-m+1$ attains $2n-1$ only for $r = n$, $m = 2$ in the first sum, since in the second sum $2(n-k)-m+1 = 2n-1$ forces $k = 1$, $m = 0$, where $\widehat{B}_{n-1,0} = 0$ for $n \geq 2$. Hence $t_1 = -a_1 = \frac{1}{6}$ and $t_n = -\frac{1}{2}\sum_{j=1}^{n-1} t_j t_{n-j}$ for $n \geq 2$, i.e. $T(z) = \sum_n t_n z^n$ satisfies $T + \frac{1}{2}T^2 = \frac{1}{6}z$, whence $T(z) = -1 + \sqrt{1+z/3}$ and $t_n = 3^{-n}\binom{1/2}{n}$. Equivalently, the maximal-pole part of Equation (N.36) is $\Lambda E + \frac{1}{2}E^2 - \frac{1}{6}z = 0$ with root $E = -\Lambda + \sqrt{\Lambda^2 + z/3}$. S1 asserts this without proof; the degree bookkeeping above supplies it.

Theorem N.37 (Coefficientwise sign law)**.** *For every* $n \geq 1$,

$$(-1)^{n+1}d_{2n-1}(\Lambda) \in \Lambda^{-1}\mathbb{Q}_{>0}[\Lambda^{-1}], \qquad \deg_{\Lambda^{-1}} d_{2n-1} = 2n-1 \textit{ exactly}. \tag{N.50}$$

That is, $c_{n,j} \neq 0$ *with sign* $(-1)^{n+1}$ *for every* $1 \leq j \leq 2n-1$, *and* $c_{n,j} = 0$ *beyond.*

Proof. Write $a_k = (-1)^k a_k^+$ with $a_k^+ = 2(2k-2)!\,\eta(2k)\pi^{-2k} > 0$ by Equation (N.19). Substituting $z = -\zeta$ and $E = -P$ into Equation (N.36) and using $\varphi(-P) = \sum_{m\geq 2} P^m/\bigl(m(m-1)\bigr)$ turns it into

$$\Lambda P = \sum_{m\geq 2}\frac{P^m}{m(m-1)} + \sum_{k\geq 1} a_k^+\,\zeta^k(1-P)^{-(2k-1)}. \tag{N.51}$$

Every coefficient on the right is *strictly positive*: $1/(m(m-1)) > 0$, $a_k^+ > 0$, and $(1-P)^{-(2k-1)} = \sum_{m\geq 0}\binom{2k+m-2}{m}P^m$ has positive coefficients. Writing $P = \sum_n p_n(\Lambda)\zeta^n$, the triangular recurrence obtained from Equation (N.51) expresses p_n as Λ^{-1} times a positive-coefficient polynomial in $p_1, \ldots, p_{n-1}$; by induction each $p_n \in \Lambda^{-1}\mathbb{Q}_{\geq 0}[\Lambda^{-1}]$, with no cancellation possible anywhere. Since $E(z) = -P(-z)$, we get $d_{2n-1} = e_n = (-1)^{n+1}p_n$, which gives the sign statement once we show that p_n has a *strictly* positive coefficient at every Λ^{-j} with $1 \leq j \leq 2n-1$.

We prove that by induction, exhibiting one contributing configuration for each j; because all contributions are nonnegative, one suffices. For $n = 1$, $p_1 = a_1^+/\Lambda$, and $2n-1 = 1$. Let $n \geq 2$ and assume the claim for all smaller indices.

- $j = 1$: take the forcing term $k = n$, $m = 0$ in Equation (N.51), contributing $a_n^+\Lambda^{-1}$.
- $2 \leq j \leq 2n-2$: take the forcing term $k = 1$ together with the single factor $m = 1$ of $(1-P)^{-1}$, so that the contribution is $\Lambda^{-1}a_1^+\binom{1}{1}$ times a coefficient of p_{n-1}. By induction p_{n-1} has strictly positive coefficients at $\Lambda^{-j'}$ for every $1 \leq j' \leq 2n-3$, and $j-1$ runs over exactly that range.

- $j = 2n - 1$: take the nonlinear term $m = 2$, i.e. a product $p_{n_1} p_{n_2}$ with $n_1 + n_2 = n$, $n_i \geq 1$, at their top degrees $2n_i - 1$; the resulting Λ^{-1}-degree is $1 + (2n_1 - 1) + (2n_2 - 1) = 2n - 1$.

Hence every $c_{n,j}$, $1 \leq j \leq 2n - 1$, is nonzero of sign $(-1)^{n+1}$; Theorem N.31 already showed $c_{n,j} = 0$ for $j > 2n - 1$. □

Remark N.38. The sign law was confirmed symbolically for $n \leq 7$: after multiplication by $(-1)^{n+1}$ all $1, 3, 5, 7, 9, 11, 13$ coefficients of the respective blocks are positive rationals, and the degree is exactly $2n - 1$ in each case. S2 states the theorem but its proof gestures at "a positive rooted-tree contribution" without exhibiting one; the attainability induction above replaces that step. The law is also the cheapest runtime invariant for a symbolic implementation (Theorem N.64).

N.6.6 The explicit blocks

Repeated use of Equation (N.41) gives, in exact rational arithmetic,

$$d_1(\Lambda) = \frac{1}{6\Lambda}, \tag{N.52}$$

$$d_3(\Lambda) = -\frac{14\Lambda^2 + 10\Lambda + 5}{360\Lambda^3}, \tag{N.53}$$

$$d_5(\Lambda) = \frac{2232\Lambda^4 + 1176\Lambda^3 + 714\Lambda^2 + 350\Lambda + 105}{45360\Lambda^5}, \tag{N.54}$$

$$\begin{aligned} d_7(\Lambda) = -\frac{1}{5443200\Lambda^7}\big(&822960\Lambda^6 + 292536\Lambda^5 + 136956\Lambda^4 \\ &+ 68040\Lambda^3 + 32970\Lambda^2 + 12250\Lambda + 2625\big), \end{aligned} \tag{N.55}$$

$$\begin{aligned} d_9(\Lambda) = \frac{1}{359251200\Lambda^9}\big(&309052800\Lambda^8 + 77920128\Lambda^7 + 26024592\Lambda^6 \\ &+ 10019592\Lambda^5 + 4516028\Lambda^4 + 2023560\Lambda^3 \\ &+ 812350\Lambda^2 + 242550\Lambda + 40425\big), \end{aligned} \tag{N.56}$$

$$\begin{aligned} d_{11}(\Lambda) = -\frac{1}{5884534656000\Lambda^{11}}\big(&46195687238400\Lambda^{10} + 8854370366400\Lambda^9 \\ &+ 2171643318000\Lambda^8 + 605515022112\Lambda^7 + 212869408752\Lambda^6 \\ &+ 84136852800\Lambda^5 + 35589153600\Lambda^4 + 14234019800\Lambda^3 \\ &+ 4868113250\Lambda^2 + 1213962750\Lambda + 165540375\big). \end{aligned} \tag{N.57}$$

The rapidly growing numerators are an argument for regenerating blocks from Equation (N.41) rather than storing a table; d_{13}, which S3 also displays, is omitted here for that reason. The alternating block signs and the coefficientwise positivity after multiplication by $(-1)^{n+1}$ illustrate Theorem N.37; the leading and trailing coefficients illustrate Theorem N.35. In separated form,

$$\begin{aligned} d_1 &= \frac{1}{6\Lambda}, \qquad d_3 = -\frac{7}{180\Lambda} - \frac{1}{36\Lambda^2} - \frac{1}{72\Lambda^3}, \\ d_5 &= \frac{31}{630\Lambda} + \frac{7}{270\Lambda^2} + \frac{17}{1080\Lambda^3} + \frac{5}{648\Lambda^4} + \frac{1}{432\Lambda^5}, \end{aligned} \tag{N.58}$$

$$d_7 = -\frac{127}{840\Lambda} - \frac{4063}{75600\Lambda^2} - \frac{11413}{453600\Lambda^3} - \frac{1}{80\Lambda^4} - \frac{157}{25920\Lambda^5} - \frac{35}{15552\Lambda^6} - \frac{5}{10368\Lambda^7}. \tag{N.59}$$

The common-denominator forms are less prone to transcription error; the separated form makes $c_{n,1} = -a_n$ visible. Through three blocks,

$$\begin{aligned} \mathcal{D}_\delta^{-1}(y) = T - 1 + \frac{1}{6T\Lambda} &- \frac{14\Lambda^2 + 10\Lambda + 5}{360\,T^3\Lambda^3} \\ &+ \frac{2232\Lambda^4 + 1176\Lambda^3 + 714\Lambda^2 + 350\Lambda + 105}{45360\,T^5\Lambda^5} + \mathcal{O}\left(\frac{1}{T^7\Lambda}\right). \end{aligned} \tag{N.60}$$

Remark N.39 (Lambert-only form, and a correction to the usual first guess). Because $T = 2R_\delta/w$ and $\Lambda = w + 1$, the expansion can be written without T. The first three terms are

$$\mathcal{D}_\delta^{-1}(y) \sim \frac{2R_\delta}{w} - 1 + \frac{w}{12R_\delta(w+1)} - \frac{w^3\big(14(w+1)^2 + 10(w+1) + 5\big)}{2880\,R_\delta^3(w+1)^3} + \frac{w^5\big(2232(w+1)^4 + 1176(w+1)^3 + 714(w+1)^2 + 350(w+1) + 105\big)}{1451520\,R_\delta^5(w+1)^5} + \cdots, \tag{N.61}$$

with $w = W_0(2R_\delta/\mathrm{e})$. A frequently used rough inversion is $x \approx 2R_\delta/W_0(2R_\delta/\mathrm{e}) - 1$; its leading correction is *not* of order T^{-1} alone but

$$\frac{1}{6\Lambda T} = \frac{w}{12R_\delta(w+1)} = \frac{1}{12R_\delta}\left(1 - \frac{1}{w+1}\right) \sim \frac{1}{12\log y}.$$

The factor Λ^{-1} is forced by the slope $\Phi'(T) = \Lambda/2$ of Equation (N.33).

N.7 The outer iterated-logarithmic layer

Keeping W_0 unexpanded is both more accurate and algebraically cleaner. To place the result inside a conventional polynomial–logarithmic transseries one applies Theorem J.37. Set

$$Z = \frac{2R_\delta}{\mathrm{e}}, \qquad L_1 = \log Z, \qquad L_2 = \log L_1. \tag{N.62}$$

Proposition N.40 (Outer data). *With $\left[{n \atop k}\right]$ the unsigned Stirling numbers of the first kind, Theorem J.37 gives, for $Z \to +\infty$,*

$$W_0(Z) \sim L_1 - L_2 + \sum_{r\ge 1} \frac{P_r(L_2)}{L_1^r}, \qquad P_r(t) = \sum_{m=1}^{r} (-1)^{r-m} \left[{r \atop r-m+1}\right] \frac{t^m}{m!}. \tag{N.63}$$

Write $W_0(Z)/L_1 = 1 + \sum_{j\ge 1} \upsilon_j(L_2) L_1^{-j}$ with $\upsilon_1 = -L_2$ and $\upsilon_j = P_{j-1}$ for $j \ge 2$, and define the reciprocal polynomials

$$\tau_0 = 1, \qquad \tau_m = -\sum_{j=1}^{m} \upsilon_j\, \tau_{m-j} \quad (m \ge 1). \tag{N.64}$$

Then

$$T \sim \frac{2R_\delta}{L_1} \sum_{m\ge 0} \frac{\tau_m(L_2)}{L_1^m}, \qquad \Lambda = 1 + W_0(Z). \tag{N.65}$$

The first data, recomputed from Equations (N.63) and (N.64), are

$$P_1 = t, \qquad P_2 = \tfrac{1}{2}t^2 - t, \qquad P_3 = \tfrac{1}{3}t^3 - \tfrac{3}{2}t^2 + t, \tag{N.66}$$

$$\begin{aligned} &\tau_0 = 1, && \tau_1 = L_2, && \tau_2 = L_2^2 - L_2, \\ &\tau_3 = L_2^3 - \tfrac{5}{2}L_2^2 + L_2, && \tau_4 = L_2^4 - \tfrac{13}{3}L_2^3 + \tfrac{9}{2}L_2^2 - L_2, \\ &\tau_5 = L_2^5 - \tfrac{77}{12}L_2^4 + \tfrac{71}{6}L_2^3 - 7L_2^2 + L_2. \end{aligned} \tag{N.67}$$

Theorem N.41 (Every coefficient of the fully expanded transseries). *Put $t = L_1^{-1}$,*

$$\mathcal{W}(t; L_2) = 1 + \sum_{j\ge 1} \upsilon_j(L_2) t^j, \qquad \mathcal{L}(t; L_2) = \mathcal{W}(t; L_2) + t, \tag{N.68}$$

so that $W_0/L_1 = \mathcal{W}$, $\Lambda/L_1 = \mathcal{L}$ *and* $T = (2R_\delta/L_1)\mathcal{W}^{-1}$. *With* $e_n(\Lambda) = \sum_{p=1}^{2n-1} c_{n,p}\Lambda^{-p}$,

$$\mathcal{D}_\delta^{-1}(y) \sim \frac{2R_\delta}{L_1}\sum_{r\geq 0}\frac{\tau_r(L_2)}{L_1^r} - 1 + \sum_{n\geq 1}\left(\frac{L_1}{2R_\delta}\right)^{2n-1}\sum_{r\geq 1}\frac{D_{n,r}(L_2)}{L_1^r}, \tag{N.69}$$

where every $D_{n,r} \in \mathbb{Q}[L_2]$ *is given by the finite extraction*

$$\boxed{D_{n,r}(L_2) = [t^r]\left\{\mathcal{W}(t;L_2)^{2n-1}\sum_{p=1}^{2n-1} c_{n,p}\, t^p\, \mathcal{L}(t;L_2)^{-p}\right\}.} \tag{N.70}$$

Together with Equations (N.45), (N.63) *and* (N.64) *this is a general formula for every coefficient at every level. The leading term of the* n*-th inverse-power sector is* $-a_n L_1^{2n-2}(2R_\delta)^{-(2n-1)}$.

Proof. The core term is Equation (N.65), and Equation (N.64) is reciprocal series division. For the n-th block,

$$e_n(\Lambda)\, T^{-(2n-1)} = \left(\frac{L_1}{2R_\delta}\right)^{2n-1}\mathcal{W}^{2n-1}\sum_{p=1}^{2n-1} c_{n,p}L_1^{-p}\mathcal{L}^{-p},$$

and setting $t = L_1^{-1}$ and extracting $[t^r]$ gives Equation (N.70); at fixed r only finitely many v_j and $c_{n,p}$ occur, so the extraction is finite. The first nonzero power is $r = 1$, from $p = 1$ and the constant terms $\mathcal{W}(0) = \mathcal{L}(0) = 1$, with $c_{n,1} = -a_n$ by Theorem N.35. □

Proposition N.42 (Ordering of the scales)**.** *For every fixed* M, $T/L_1^M \to \infty$ *as* $y \to \infty$. *Consequently*

$$\frac{T}{L_1^M} \gg 1 \gg \frac{1}{T\Lambda} \gg \frac{1}{T^3\Lambda} \gg \frac{1}{T^5\Lambda} \gg \cdots, \tag{N.71}$$

so the constant shift -1 *is beyond every fixed order of the outer* L_1^{-1}*-expansion, yet dominates the entire inner grid. In particular the first Bernoulli displacement satisfies* $1/(6T\Lambda) \asymp 1/(12R_\delta) \asymp 1/(12\log y)$.

Proof. $T = \mathrm{e}^\Lambda$ with $\Lambda = 1 + W_0(Z) \sim L_1$, while $L_1^M = \mathrm{e}^{ML_2}$ with $L_2 = \log L_1 = o(L_1)$; hence $\log(T/L_1^M) = \Lambda - ML_2 \to \infty$. The remaining comparisons are immediate from $T \to \infty$ and Equation (N.33), which give $T\Lambda = 2R_\delta\Lambda/(\Lambda - 1) \sim 2R_\delta$. □

Remark N.43. This ordering is destroyed if all terms are written as one informal series in $1/\log y$: the shift -1, the outer polynomials $\tau_m(L_2)/L_1^m$ and the inner blocks $d_{2n-1}(\Lambda)T^{1-2n}$ then become indistinguishable. S1, S2 and S3 all make this point; the quantitative version is Equation (N.71).

N.8 The periodic interpolation and its oscillatory inverse

S1 and S2 centre the periodic problem at the parity-mean constant $\frac{1}{4}\log(2\pi)$; S3 centres it at the even-branch constant $\frac{1}{2}\log\pi$ and consequently reports a leading displacement $2a(1+\cos\theta)/\Lambda$ rather than $2a\cos\theta/\Lambda$. The two are the same function expanded about different cores (they differ by $2a/\Lambda + \mathcal{O}(\Lambda^{-2})$, which is exactly $h'\cdot a$ for the core map h). This chapter adopts the mean centring, which makes the carrier a mean-zero oscillation with the cleanest harmonic content.

Lemma N.44 (Centred exact form of $\mathscr{D}$)**.** *With* $u = x + 1$, $a = \frac{1}{4}\log(\pi/2)$ *as in Equation* (N.9) *and*

$$A_* := \frac{1}{4}\log(2\pi) = 0.4594692666\ldots, \qquad \mathrm{e}^{A_*} = (2\pi)^{1/4} = 1.5832334871\ldots, \tag{N.72}$$

one has, exactly and for all $x > -1$,

$$\log \mathscr{D}(x) = \Phi(u) + A_* - a\cos(\pi u) + \Omega(u). \tag{N.73}$$

At an integer $x = n$, $\cos \pi u = (-1)^{n+1}$ and $A_* - a\cos \pi u = A(n)$ of Equation (N.13)*: the periodic factor freezes to the correct parity constant.*

Proof. By Equation (N.7) and Theorem N.11 with $\delta = 0$, $\log \mathscr{D}(x) = \Phi(u) + A_0 + \Omega(u) + P(x)$ where $P(x) = \frac{1}{4}(1 - \cos \pi x)\log(2/\pi) = -a(1 - \cos \pi x)$. Since $\cos \pi x = \cos \pi(u-1) = -\cos \pi u$, $P = -a(1 + \cos \pi u) = -a - a\cos \pi u$, and $A_0 - a = \frac{1}{2}\log \pi - \frac{1}{4}\log(\pi/2) = \frac{1}{4}\log(2\pi) = A_*$. □

Lemma N.45 (Exact displacement equation). *Let $R_* = \log y - A_*$ and let T, Λ be the core data of Equation* (N.32) *built from R_*; put $q = T^{-1}$, $\theta = \pi T$. Then $x = \mathscr{D}^{-1}(y)$ satisfies $x = T - 1 + \Delta$ where Δ is the unique small root of*

$$\Lambda\Delta + \frac{\varphi(q\Delta)}{q} + 2\,\Omega(T + \Delta) - 2a\cos(\theta + \pi\Delta) = 0. \tag{N.74}$$

Replacing Ω by its asymptotic series turns Equation (N.74) *into the formal equation $\mathcal{F}(q, \Delta) = 0$ with*

$$\mathcal{F}(q, v) = \Lambda v + \sum_{m\geq 2} \frac{(-1)^m}{m(m-1)} v^m q^{m-1} - 2a\cos(\theta + \pi v) + 2\sum_{k\geq 1} c_k q^{2k-1}(1 + qv)^{-(2k-1)}. \tag{N.75}$$

Since the periodic term is $\mathcal{O}(1)$ while every other nonlinear term carries a positive power of q, one has $\Delta = \mathcal{O}(\Lambda^{-1})$, not *$\mathcal{O}(q\Lambda^{-1})$.*

Proof. Evaluate Equation (N.73) at $u = T + \Delta$, set it equal to $\log y = R_* + A_*$, subtract $\Phi(T) = R_*$, and multiply by 2. By the computation in Theorem N.28 with $E = q\Delta$, $2\big[\Phi(T + \Delta) - \Phi(T)\big] = T\big[\Lambda q\Delta + \varphi(q\Delta)\big] = \Lambda\Delta + \varphi(q\Delta)/q$, since $T = q^{-1}$. This gives Equation (N.74). The last sum in Equation (N.75) is $2\Omega(T(1 + qv))$ expanded. Finally $\varphi(q\Delta)/q = q\Delta^2/2 + \mathcal{O}(q^2)$, so at $q = 0$ the equation reduces to $\Lambda\Delta_0 = 2a\cos(\theta + \pi\Delta_0)$, forcing $\Delta_0 \asymp \Lambda^{-1}$. □

N.8.1 The periodic carrier

Theorem N.46 (Existence, uniqueness, and the closed carrier expansion). *Fix $\theta \in \mathbb{R}$ and put $\mu = 2a/\Lambda$.*

(a) *If $\Lambda > 2\pi a = 0.70934\ldots$, the equation*

$$\Delta_0 = \mu\cos(\theta + \pi\Delta_0) \tag{N.76}$$

has exactly one real solution, and $|\Delta_0| \leq \mu$.

(b) *If moreover $\mu \leq \frac{1}{5}$, equivalently $\Lambda \geq 10a = 1.12896\ldots$, that solution is the sum of the convergent Lagrange–Bürmann series*

$$\boxed{\Delta_0 = \sum_{m\geq 1} \frac{\mu^m}{m!} \frac{\mathrm{d}^{m-1}}{\mathrm{d}v^{m-1}} \cos^m(\theta + \pi v)\bigg|_{v=0} = \sum_{m\geq 1} \frac{a^m}{m!\,\Lambda^m} \sum_{j=0}^{m} \binom{m}{j} \big(\mathrm{i}\pi(m - 2j)\big)^{m-1} \mathrm{e}^{\mathrm{i}(m-2j)\theta}.} \tag{N.77}$$

The second expression is real. At inverse-logarithmic order m only the Fourier modes $m, m - 2, \ldots, -m$ occur.

(c) *The first three terms are*

$$\Delta_0 = \mu\cos\theta - \frac{\pi}{2}\mu^2\sin 2\theta + \frac{\pi^2}{2}\mu^3\cos\theta\,(2 - 3\cos^2\theta) + \mathcal{O}(\mu^4). \tag{N.78}$$

Proof. (a) The map $g(v) = \mu\cos(\theta + \pi v)$ satisfies $|g'| \le \pi\mu = 2\pi a/\Lambda < 1$ and $|g| \le \mu$, so it is a contraction of $\mathbb{R}$ into $[-\mu, \mu]$; Banach's theorem gives a unique fixed point there, and any real fixed point lies in $[-\mu, \mu]$.

(b) Equation (N.76) is $v = \mu\phi(v)$ with $\phi(v) = \cos(\theta + \pi v)$ entire. By the classical convergence criterion for Lagrange's series, if there is $r > 0$ with $|\mu| \max_{|v|=r} |\phi(v)| < r$, then the series $\sum_m \frac{\mu^m}{m!} \left(\frac{\mathrm{d}}{\mathrm{d}v}\right)^{m-1} \phi(v)^m|_{v=0}$ converges and its sum is the unique root of $v = \mu\phi(v)$ in $|v| < r$. Here $|\cos(\theta + \pi v)| \le \cosh(\pi|\Im v|) \le \cosh(\pi r)$ on $|v| = r$; with $r = 0.38$ and $\mu \le \frac{1}{5}$, $\mu\cosh(0.38\pi) \le 0.3603 < 0.38$. (The optimal choice $\max_{r>0} r/\cosh(\pi r) = 0.21095\ldots$ at $r = 0.382$ shows $\mu < 0.2109$ already suffices.) For the Fourier form, insert $\cos^m v = 2^{-m}\sum_j \binom{m}{j} \mathrm{e}^{\mathrm{i}(m-2j)v}$ into the derivative:

$$\frac{\mathrm{d}^{m-1}}{\mathrm{d}v^{m-1}} \cos^m(\theta + \pi v)\bigg|_{v=0} = 2^{-m} \sum_{j=0}^{m} \binom{m}{j} \big(\mathrm{i}\pi(m - 2j)\big)^{m-1} \mathrm{e}^{\mathrm{i}(m-2j)\theta},$$

and $\mu^m 2^{-m} = a^m\Lambda^{-m}$. Pairing j with $m - j$ replaces $m - 2j$ by its negative, conjugating each summand, so the sum is real. Only $|m - 2j| \le m$ occurs, with the same parity as m.

(c) For $m = 1$ the derivative is $\cos\theta$. For $m = 2$, $\frac{\mathrm{d}}{\mathrm{d}v}\cos^2 = -2\pi\cos\sin = -\pi\sin 2(\cdot)$, giving $-\frac{\pi}{2}\mu^2 \sin 2\theta$. For $m = 3$, $\frac{\mathrm{d}^2}{\mathrm{d}v^2}\cos^3(\theta + \pi v)|_0 = 3\pi^2\cos\theta(2 - 3\cos^2\theta)$, and division by 3! gives the third term. □

S1 states the carrier expansion as "convergent for sufficiently large Λ" with no threshold and no criterion, and S2 proves only the contraction estimate, from which convergence of the Lagrange series does not follow. Theorem N.46 separates the two claims and supplies an explicit sufficient condition, $\mu \le \frac{1}{5}$. Numerically at $\Lambda = 3$, $\theta = 2$, the partial sums $m \le 1, \ldots, 5$ of Equation (N.77) approximate the exact fixed point -0.0256993556 with errors $5.6 \cdot 10^{-3}, 1.1 \cdot 10^{-3}, 1.8 \cdot 10^{-4}, 1.9 \cdot 10^{-5}, 1.9 \cdot 10^{-6}$, consistent with a convergence ratio $\approx \mu/0.211$.

N.8.2 Algebraic corrections around the carrier

Theorem N.47 (Triangular periodic-grid recurrence). *Let*

$$M := \partial_v\mathcal{F}(0, \Delta_0) = \Lambda + 2\pi a\sin(\theta + \pi\Delta_0), \tag{N.79}$$

which satisfies $M \ge \Lambda - 2\pi a > 0$ for $\Lambda > 2\pi a$. Seek $\Delta = \Delta_0 + \sum_{n\ge1}\Delta_n q^n$ with each Δ_n a formal series in Λ^{-1} with trigonometric-polynomial coefficients, and put $\Delta_{<n} = \Delta_0 + \sum_{j<n}\Delta_j q^j$. Then the solution of $\mathcal{F} = 0$ is unique and generated by

$$\boxed{\Delta_n = -\frac{1}{M}\,[q^n]\;\mathcal{F}\big(q, \Delta_{<n}(q)\big), \qquad n \ge 1.} \tag{N.80}$$

The first two are

$$\Delta_1 = \frac{\frac{1}{6} - \frac{1}{2}\Delta_0^2}{M}, \qquad \Delta_2 = -\frac{1}{M}\left[a\pi^2\cos(\theta + \pi\Delta_0)\,\Delta_1^2 + \Delta_0\Delta_1 + \frac{\Delta_0 - \Delta_0^3}{6}\right]. \tag{N.81}$$

Proof. In $[q^n]\,\mathcal{F}(q, \Delta)$ the unknown Δ_n appears only linearly, through the v-derivative of the q-free part $\Lambda v - 2a\cos(\theta + \pi v)$ evaluated at $v = \Delta_0$, which is M; every other contribution involves only Δ_j with $j < n$. This is Theorem J.19 applied in the variable q over the coefficient ring generated by Λ^{-1} and $\mathrm{e}^{\pm\mathrm{i}\theta}$, and M is invertible there because $M = \Lambda(1 + \mathcal{O}(\Lambda^{-1}))$.

For $n = 1$: the $m = 2$ term of Equation (N.75) gives $\frac{1}{2}\Delta_0^2$, the cosine gives $2a\pi\sin(\theta + \pi\Delta_0)\Delta_1$, and the $k = 1$ Bernoulli term gives $2c_1 = -\frac{1}{6}$; collecting gives $M\Delta_1 + \frac{1}{2}\Delta_0^2 - \frac{1}{6} = 0$. For $n = 2$: the $m = 2$ term gives $\Delta_0\Delta_1$, the $m = 3$ term gives $-\frac{1}{6}\Delta_0^3$, the second-order Taylor term of the cosine gives $a\pi^2\cos(\theta + \pi\Delta_0)\Delta_1^2$, and expanding $2c_1 q(1 + q\Delta)^{-1}$ gives $-2c_1\Delta_0 = \frac{1}{6}\Delta_0$; collecting gives Equation (N.81). □

Corollary N.48 (Asymptotic meaning). *Fix $N, K \geq 0$. Truncate Δ after q^N and expand each Δ_n in Λ^{-1} through Λ^{-K}, obtaining $\widetilde{\Delta}$. Then, as $y \to +\infty$,*

$$\mathscr{D}^{-1}(y) = T - 1 + \widetilde{\Delta} + \mathcal{O}\left(\frac{1}{T^{N+1}\Lambda}\right) + \mathcal{O}\left(\frac{1}{\Lambda^{K+2}}\right), \tag{N.82}$$

uniformly in the phase $\theta = \pi T$.

Proof. By construction Equation (N.80) cancels the residual of $\mathcal{F}$ through q^N, and Theorem N.46 cancels it through Λ^{-K} at each fixed q-order; the replacement of Ω by its truncation is controlled by Theorem N.17. The exact v-derivative of the left side of Equation (N.74) equals $M + \mathcal{O}(q)$ and is bounded below by $\Lambda - 2\pi a - \mathcal{O}(q)$, a positive multiple of Λ for large T; all trigonometric coefficients and their derivatives are bounded, so Theorem J.47 converts the residual into a root error after division by a quantity $\asymp \Lambda$, uniformly in θ. ☐

Proposition N.49 (The parity branches resum the periodic sector). *At an integer $u = n + 1$ one has $A_* - a\cos\pi u = A_\delta$ with $\delta \equiv n \pmod 2$. The branchwise normalisation therefore absorbs the whole $\mathcal{O}(1)$ periodic constant into the target R_δ before the core is formed, so the $\mathcal{O}(\Lambda^{-1})$ carrier disappears and the first correction is the much smaller $1/(6T\Lambda)$. In this precise sense the two parity transseries are subsequence resummations of the periodic one: they build the parity phase into the core instead of expanding it perturbatively.*

Proof. Immediate from Theorem N.44 and Theorems N.29 and N.46. ☐

Proposition N.50 (Phase-coordinate reversion, formal). *Let $h(R) = 2R/W_0(2R/\mathrm{e})$ be the inverse of Φ (Theorem N.27), and set $\mathcal{A}(R) = -a\cos\big(\pi h(R)\big) + \Omega\big(h(R)\big)$. Then the equation $\Phi(u) + \mathcal{A}(\Phi(u)) = R_*$ — which is Equation* (N.73) *rewritten in the phase coordinate $\sigma = \Phi(u)$ — has the formal reversion*

$$\sigma = R_* + \sum_{m\geq 1} \frac{(-1)^m}{m!}\left(\frac{\mathrm{d}}{\mathrm{d}R_*}\right)^{m-1} \mathcal{A}(R_*)^m, \qquad \mathscr{D}^{-1}(y) = h(\sigma) - 1. \tag{N.83}$$

Proof. Write $\sigma = R_* + \varsigma$; then $\varsigma = -\mathcal{A}(R_* + \varsigma)$, and Lagrange's formula for $\varsigma = \tau\phi(\varsigma)$ with $\phi(\varsigma) = -\mathcal{A}(R_* + \varsigma)$, $\tau = 1$, together with the translation invariance of $\mathrm{d}/\mathrm{d}\varsigma$, gives Equation (N.83). ☐

S3 states Equation (N.83) as a boxed asymptotic expansion, writing "setting $\tau = 1$ as an asymptotic expansion gives". That is a formal-to-analytic slide: $\mathcal{A}$ is not small in any norm that makes the reversion series converge, and m-fold differentiation with respect to R_* produces factors $h'(R_*) = 2/\Lambda$ whose accumulated effect is exactly what needs to be estimated. The statement is demoted here to a *formal* proposition. An asymptotic claim would require, at each order m, a bound on $\big(\mathrm{d}/\mathrm{d}R_*\big)^{m-1}\mathcal{A}^m$ uniform in the oscillatory phase — available for the truncated grid of Theorem N.48, which is therefore the statement to use. Comparing the two routes at leading order: the $m = 1$ term of Equation (N.83) contributes $h'(R_*) \cdot (-\mathcal{A}) = 2a\cos(\pi T)/\Lambda + \mathcal{O}(T^{-1}\Lambda^{-1})$, in agreement with Equation (N.78).

Warning N.51. The carrier of Theorem N.46 records the choice Equation (N.7), not a property of $n!!$. By Theorem N.8 another interpolation, agreeing with $n!!$ at every nonnegative integer, changes the amplitude of this sector from $2a$ to $2\sqrt{a^2 + \lambda^2}$ and shifts its phase, while leaving every integer value and both parity transseries untouched.

N.9 The discrete inverses

Objects (i) and (iii) of Section N.1 are intrinsic to the sequence. They are the residue-class staircase of Theorems J.41 and J.43 at modulus $r = 2$, and nothing new has to be proved: Theorem J.46 already records this chapter as that instance. Only the dictionary and the arithmetic caveat are subject-specific.

Proposition N.52 (Specialisation of the staircase). *Take $r = 2$, residue classes $\rho \in \{0, 1\}$, $A_n = n!!$, and $\nu_\rho(y) = \mathcal{D}_\rho^{-1}(y)$, which is legitimate because $\mathcal{D}_\rho$ interpolates the class-ρ subsequence and is strictly increasing (Theorem N.3). Then Equation* (J.76) *reads*

$$N(y) := \min\{n \in \mathbb{N} : n!! \geq y\} = \min\left\{2\left\lceil\frac{\nu_0(y)}{2}\right\rceil,\ 2\left\lceil\frac{\nu_1(y) - 1}{2}\right\rceil + 1\right\} \qquad (y > 1), \tag{N.84}$$

and its floor counterpart, obtained from Theorem J.43 by replacing $\lceil\cdot\rceil$ with $\lfloor\cdot\rfloor$ and $\min$ with $\max$, is

$$J(y) := \max\{n \in \mathbb{N} : n!! \leq y\} = \max_{\rho\in\{0,1\}}\left\{\rho + 2\left\lfloor\frac{\nu_\rho(y) - \rho}{2}\right\rfloor\right\}. \tag{N.85}$$

Replacing ν_ρ by the truncation of Equation (N.39) *with a certified error bound η_ρ (for instance from Equation* (N.100)*) determines $N(y)$ and $J(y)$ exactly under the separation condition of Theorem J.43, and Theorem J.45 bounds the number of exact evaluations of $n!!$ needed by $\sum_\rho(\lfloor\eta_\rho\rfloor + 2)$. Those evaluations should be made as $\log\mathcal{D}_\rho(n) = \log\kappa_\rho + \frac{n}{2}\log 2 + \log\Gamma(\frac{n}{2} + 1)$, or by Equation* (N.1) *in exact integers.*

Proof. Everything is Theorem J.43 and Theorem J.45 with $r = 2$; the hypotheses hold by Theorem N.3 and, for the asymptotic input, by Theorem N.29. ☐

Warning N.53 (Never form y). Equation (N.84), evaluated with four inverse blocks, was tested at $y = n!!$ for every $6 \leq n \leq 199$ and returned the correct $N(y) = n$ in all 194 cases — but only when the comparison $A_n \geq y$ is made on exact integers or on $\log y$. Converting $n!!$ to a fixed-precision float first destroys the comparison long before the asymptotics fail: at 80 working digits this already produces spurious off-by-one answers from $n \approx 120$ onwards, because 120!! has more than 100 digits. The failure is arithmetic, not asymptotic, and it is easy to misdiagnose as a defect of the expansion.

N.10 Beyond all algebraic orders

Proposition N.54 (Optimal truncation, forward and inverse). *Let $u > 0$. The terms of Equation* (N.18) *have moduli $|c_k|u^{1-2k}$ with consecutive ratio $\sim 2k(2k-1)/(\pi u)^2$; the least term occurs at $2k \approx \pi u$ and*

$$\min_k\left|\frac{c_k}{u^{2k-1}}\right| = \mathcal{O}\left(u^{-1/2}\mathrm{e}^{-\pi u}\right). \tag{N.86}$$

Consequently, by Theorem J.54 and the slope $\Phi'(T) = \Lambda/2$, the optimally truncated branchwise inverse has error

$$\mathcal{O}\left(\frac{\mathrm{e}^{-\pi T}}{\sqrt{T}\,\Lambda}\right). \tag{N.87}$$

Proof. Equation (N.23) gives $|c_k| = (2k-2)!\pi^{-2k}(1 + \mathcal{O}(4^{-k}))$, whence the ratio. At $2k = \pi u$, Stirling's formula for $(2k-2)!$ gives Equation (N.86). For the inverse, Theorem N.17 bounds the forward logarithmic residual at optimal truncation by Equation (N.86) with $u = T(1 + \mathcal{O}(T^{-2}))$; dividing by the slope $\frac{1}{2}\Lambda + \mathcal{O}(T^{-2})$ via Theorems J.47 and J.51 gives Equation (N.87). ☐

Proposition N.55 (Transport of a forward exponential sector). *Suppose an exponentially improved representation of the centred forward phase retains the algebraic terms $c_1, \ldots, c_K$ and has an additional residual $\mathcal{R}$, and let u_{alg} solve the correspondingly truncated algebraic inverse problem. Then*

$$u - u_{\mathrm{alg}} = -\frac{2\,\mathcal{R}(u_{\mathrm{alg}})}{\log u_{\mathrm{alg}} - 2\sum_{k=1}^{K}(2k-1)c_k\,u_{\mathrm{alg}}^{-2k}} + \mathcal{O}\left(\frac{\sup|F''|\,\mathcal{R}^2}{\Lambda^3} + \frac{|\mathcal{R}\,\mathcal{R}'|}{\Lambda^2}\right), \tag{N.88}$$

where F is the doubled algebraic phase and the supremum is over the interval between u_{alg} and the exact root; the estimate assumes the displayed denominator is $\asymp \Lambda$ and $|\mathcal{R}'| = o(\Lambda)$. At leading order this is $-2\mathcal{R}(T)/\Lambda$.

Proof. Let F be the doubled algebraic phase minus $2R$, so $F(u_{\text{alg}}) = 0$ and the exact phase equals $F + 2\mathcal{R}$. Taylor expansion of $0 = F(u_{\text{alg}} + \Delta) + 2\mathcal{R}(u_{\text{alg}} + \Delta)$ about u_{alg}, with $F'(u_{\text{alg}})$ the stated denominator, gives the first Newton correction; replacing $F' + 2\mathcal{R}'$ by F' and iterating once produces the two displayed error terms. □

Warning N.56 (What the exponential sector is, and is not). The Borel poles of Equation (N.16) lie at $s = \pm i\pi m$, so the exponentials associated with them are $e^{\mp i\pi m u}$ together with direction-dependent terminant multipliers, not real exponentials. On the positive axis the optimally truncated remainder has modulus $\asymp u^{-1/2}e^{-\pi u}$, but this is the size of a *terminant-weighted combination*: appending a bare term $\text{const} \cdot e^{-\pi u}$ to the divergent Bernoulli series is not a valid transseries completion, and conflating the two gives incorrect Stokes statements. The unambiguous positive-real completion is the exact integral Equation (N.15) itself, which by Theorem N.17 even provides two-sided bounds at every order; Equation (N.88) then carries any rigorously normalised forward completion through the inverse map. All three sources make this point; S3 states it most sharply.

N.11 Complex inverse sheets

Proposition N.57 (Local sheets). *Fix a branch of* $\log y$ *and integers* ℓ *(logarithmic winding) and* k *(Lambert branch). Put*

$$R_{\delta,\ell} = \log y + 2\pi i\ell - A_\delta, \quad w_{k,\ell} = W_k\left(\frac{2R_{\delta,\ell}}{e}\right), \quad T_{k,\ell} = \frac{2R_{\delta,\ell}}{w_{k,\ell}}, \quad \Lambda_{k,\ell} = w_{k,\ell} + 1. \tag{N.89}$$

In any sector in which $T_{k,\ell} \to \infty$*, the Stirling expansion for* $\log\Gamma$ *is uniform, and the corresponding solution branch of* $\mathcal{D}_\delta(x) = y$ *is simple, the* same *universal polynomials give*

$$x_{\delta;k,\ell}(y) \sim T_{k,\ell} - 1 + \sum_{n\geq 1} \frac{d_{2n-1}(\Lambda_{k,\ell})}{T_{k,\ell}^{2n-1}}. \tag{N.90}$$

Proof. Every step of Theorems N.28 and N.30 is an identity of formal series over $\mathbb{C}(\Lambda)$ and uses only $\Phi(T) = R$, $\Lambda = \log T$; Equation (N.89) secures those. The analytic statement is the sectorial version of Theorem N.29, with the mean-value step replaced by the standard implicit-function estimate in the sector. □

Remark N.58 (Branch caveats). The pair (k, ℓ) is an asymptotic label, not a global classification: distinct labels can be joined by analytic continuation around the poles of Γ or the logarithmic cut. Branch collisions occur exactly where $\mathcal{D}_\delta' = 0$, i.e. where $\log 2 + \psi(x/2 + 1) = 0$; by Theorem N.3 this never happens for real $x \geq 0$, but it does for complex and for negative real x, and near such points Equation (N.90) must be replaced by a Puiseux-type local analysis.

N.12 Multifactorials

Only S3 treats multifactorials. Its normal form is reproduced here with a proof, and Theorem N.61 is new: it subsumes the "half-shifted inverse Gamma" scaling that S1 and S2 record separately, by identifying the Gamma case as the member $p = 1$.

Proposition N.59 (Residue-class branches). *Let* $p \geq 1$ *and* $r \in \{1, \dots, p\}$*. For integers* $x \equiv r \pmod p$*,* $x \geq r$*, set* $x!_{(p)} = x(x-p)(x-2p)\cdots r$*. Then, with*

$$\mathcal{D}_{p,r}(x) = C_{p,r}\, p^{x/p}\,\Gamma\left(\frac{x}{p} + 1\right), \qquad C_{p,r} = \frac{p^{1-r/p}}{\Gamma(r/p)}, \tag{N.91}$$

one has $\mathcal{D}_{p,r}(x) = x!_{(p)}$ *for every such* x*. For* $p = 2$*,* $C_{2,2} = 1 = \kappa_0$ *and* $C_{2,1} = \sqrt{2/\pi} = \kappa_1$*; for* $p = 1$*,* $C_{1,1} = 1$ *and* $\mathcal{D}_{1,1}(x) = \Gamma(x+1)$*.*

Proof. Write $x = r+mp$. Then $x!_{(p)} = \prod_{i=0}^{m}(r+ip) = p^{m+1}\prod_{i=0}^{m}(\frac{r}{p}+i) = p^{m+1}\Gamma(\frac{r}{p}+m+1)/\Gamma(\frac{r}{p})$, while $\mathcal{D}_{p,r}(x) = C_{p,r}p^{r/p+m}\Gamma(\frac{r}{p}+m+1)$. Equating gives $C_{p,r} = p^{1-r/p}/\Gamma(r/p)$. $\square$

Theorem N.60 (Centred multifactorial normal form). *Put $u = x+\frac{p}{2}$ and $A_{p,r} = \log C_{p,r}+\frac{1}{2}\log(2\pi/p)$. Then, as $x \to +\infty$,*

$$\log \mathcal{D}_{p,r}(x) = \frac{u}{p}(\log u-1)+A_{p,r}+\Omega_p(u), \qquad \Omega_p(u) \sim \sum_{k\geq 1}\frac{c_k^{(p)}}{u^{2k-1}}, \quad c_k^{(p)} = \frac{p^{2k-1}B_{2k}(\frac{1}{2})}{2k(2k-1)}, \tag{N.92}$$

and Ω_p has the exact Borel–Laplace representation

$$\Omega_p(u) = \int_0^\infty \mathrm{e}^{-us}\,\frac{\dfrac{ps}{2\sinh(ps/2)}-1}{ps^2}\,\mathrm{d}s, \tag{N.93}$$

whose singularities are the simple poles $s = \pm 2\pi\mathrm{i}m/p$. Hence the optimally truncated forward remainder has scale $\mathrm{e}^{-2\pi u/p}$. For $p = 2$ this is Theorems N.11 and N.15; for $p = 1$ it is the half-shifted Stirling expansion of $\log\Gamma(x+1) = \log\Gamma(u+\frac{1}{2})$.

Proof. Put $t = u/p$, so $x = pt - \frac{p}{2}$ and $\log \mathcal{D}_{p,r} = \log C_{p,r} + (t-\frac{1}{2})\log p + \log\Gamma(t+\frac{1}{2})$. The half-shifted Stirling expansion $\log\Gamma(t+\frac{1}{2}) \sim t\log t - t + \frac{1}{2}\log(2\pi) + \sum_k B_{2k}(\frac{1}{2})\big(2k(2k-1)\big)^{-1}t^{1-2k}$ — the case $a = \frac{1}{2}$ of $\log\Gamma(t+a)$, in which every odd-index term vanishes because $B_{2j+1}(\frac{1}{2}) = 0$ — gives

$$\log \mathcal{D}_{p,r} \sim \log C_{p,r} - \tfrac{1}{2}\log p + \tfrac{1}{2}\log(2\pi) + \frac{u}{p}(\log u - 1) + \sum_k \frac{p^{2k-1}B_{2k}(\frac{1}{2})}{2k(2k-1)u^{2k-1}},$$

since $(t-\frac{1}{2})\log p + t\log t = (u/p)\log u - \frac{1}{2}\log p$ and $t^{1-2k} = p^{2k-1}u^{1-2k}$. The constant is $A_{p,r}$. For Equation (N.93), $\Omega_p(u) = \widetilde{\Omega}(u/p)$ with $\widetilde{\Omega}$ as in the proof of Theorem N.15, where $\widetilde{\Omega}(t) = \int_0^\infty \mathrm{e}^{-tr}\big(r/(2\sinh(r/2))-1\big)r^{-2}\mathrm{d}r$; substituting $r = ps$ gives Equation (N.93), and $\sinh(ps/2) = 0$ at $s = 2\pi\mathrm{i}m/p$. $\square$

Theorem N.61 (Universal multifactorial inverse and its scaling). *Let $R = \log y - A_{p,r}$, $v = W_0(pR/\mathrm{e})$, $X = pR/v$, $\Lambda = v+1 = \log X$, so that $\frac{X}{p}(\log X - 1) = R$. With $u = X(1+E)$, $q = X^{-1}$, $z = q^2$, the normalised equation is*

$$\Lambda E + \varphi(E) + \sum_{k\geq 1} a_k^{(p)} z^k(1+E)^{-(2k-1)} = 0, \qquad a_k^{(p)} := p\,c_k^{(p)} = p^{2k}\frac{B_{2k}(\frac{1}{2})}{2k(2k-1)}. \tag{N.94}$$

Every statement of Theorems N.30, N.31, N.33, N.35 and N.37 holds verbatim with a_k replaced by $a_k^{(p)}$, and for every fixed N

$$\mathcal{D}_{p,r}^{-1}(y) = X - \frac{p}{2} + \sum_{n=1}^{N}\frac{d_{2n-1}^{(p)}(\Lambda)}{X^{2n-1}} + \mathcal{O}\left(\frac{1}{X^{2N+1}\Lambda}\right). \tag{N.95}$$

Moreover the coefficient polynomials of all multifactorials are one family:

$$\boxed{d_{2n-1}^{(p)}(\Lambda) = p^{2n}\,d_{2n-1}^{(1)}(\Lambda) \qquad \textit{for every } p \geq 1,\ n \geq 1,} \tag{N.96}$$

where $d^{(1)}$ are the blocks of the half-shifted inverse-Gamma problem. In particular $d_{2n-1} = d_{2n-1}^{(2)} = 4^n\,d_{2n-1}^{(1)}$.

Proof. The derivation of Equation (N.94) repeats Theorem N.28 with $\frac{2}{T}$ replaced by $\frac{p}{X}$: the phase difference is $\frac{X}{p}[\Lambda E + \varphi(E)]$, and $\frac{p}{X}\Omega_p(u) = \sum_k p\, c_k^{(p)} q^{2k}(1+E)^{-(2k-1)}$. Since Equation (N.94) differs from Equation (N.36) only in the values of the forcing coefficients, all structural statements transfer. Now $a_k^{(p)} = p^{2k} a_k^{(1)}$ by Equation (N.92), so substituting $q \mapsto pq$ (equivalently $z \mapsto p^2 z$) carries the $p = 1$ equation into the general one with Λ untouched. Uniqueness of the formal solution gives $E^{(p)}(q) = E^{(1)}(pq)$, i.e. $e_n^{(p)} = p^{2n} e_n^{(1)}$, which is Equation (N.96). The remainder in Equation (N.95) follows from Theorems J.47 and J.51 exactly as in Theorem N.29, the slope now being $\frac{1}{p}\log X = \Lambda/p$. $\square$

Remark N.62. Equation (N.96) explains and replaces the "scaling from the half-shifted inverse Gamma problem" of S1 ($p_n = 4^n d_{2n-1}$) and the identity $d^{!!}_{2n-1} = 4^n d^{\Gamma}_{2n-1}$ of S2: both are the case $(p,1) = (2,1)$. It was verified symbolically for $n \le 4$ and $p \in \{2,3\}$. As a coefficient-polynomial identity it says nothing about the numerical Lambert data, which differ between the two problems; the identity is in Λ, not in y. It also supplies a strong regression test: an implementation storing only $d^{(1)}$ reproduces every multifactorial.

p	object	centre u	phase	$a_1^{(p)}$	Borel poles
1	$\Gamma(x+1)$	$x+\frac{1}{2}$	$u(\log u - 1)$	$-\frac{1}{24}$	$\pm 2\pi \mathrm{i} m$
2	$n!!$	$x+1$	$\frac{u}{2}(\log u - 1)$	$-\frac{1}{6}$	$\pm \pi \mathrm{i} m$
3	$n!!!$	$x+\frac{3}{2}$	$\frac{u}{3}(\log u - 1)$	$-\frac{3}{8}$	$\pm \frac{2\pi}{3} \mathrm{i} m$
p	$x!_{(p)}$	$x+\frac{p}{2}$	$\frac{u}{p}(\log u - 1)$	$-\frac{p^2}{24}$	$\pm \frac{2\pi}{p} \mathrm{i} m$

N.13 Algorithms and certification

Algorithm N.63 (Forward coefficient table)**.** Given M and a power $\alpha \in \mathbb{Q}$: compute c_k from Equation (N.19) for $1 \le k \le \lceil M/2 \rceil$; set $\gamma_{2k-1} = c_k$, $\gamma_{2k} = 0$; run Equation (N.28). This costs $\mathcal{O}(M^2)$ rational operations and never forms a symbolic exponential.

Algorithm N.64 (Inverse blocks)**.** On the variable $z = q^2$, with Λ symbolic:

```
input: N
E := 0
for n := 1,...,N:
    H := (1+E)*log(1+E) - E
    for k := 1,...,n:
        H := H + a[k]*z^k*(1+E)^(-(2*k-1))
    H := truncate(H, z^(n+1))
    e[n] := - coefficient(H, z^n) / Lambda
    E := E + e[n]*z^n
    assert truncate(residual(E), z^(n+1)) = 0      # (2.residual)
    assert sign_invariant(e[n], n)                 # Theorem sign law
return e[1],...,e[N]
```

Every product, logarithm and negative power may be truncated above z^n at step n; no full symbolic logarithm is ever needed. Representing each e_n as a dense polynomial in Λ^{-1} of degree exactly $2n-1$ (Theorem N.37) gives an exact allocation size.

The two assertions are worth more than a comparison against a stored table. The first is the *formal residual certificate*: with $E_N = \sum_{n \le N} e_n z^n$,

$$\Lambda E_N + \varphi(E_N) + \sum_{k=1}^{N+1} a_k z^k (1+E_N)^{-(2k-1)} = \mathcal{O}\big(z^{N+1}\big)\,, \tag{N.97}$$

and the coefficient of z^{N+1}, divided by $-\Lambda$, *is* the next block. This single identity catches sign, indexing and centring mistakes at once. The second is Theorem N.37: after multiplication by $(-1)^{n+1}$ every one of the $2n-1$ coefficients must be a positive rational. Independent checks available in addition:

(1) expanding the exact gamma ratio of Theorem N.11 reproduces c_k;

(2) expanding the Borel kernel Equation (N.16) reproduces c_k from η-values, and Theorem N.23 reproduces b_k from ζ-values;

(3) the centre-shift identity Equation (N.31) links the two;

(4) Equation (N.45) and Equation (N.48) must agree with Equation (N.41) coefficient by coefficient;

(5) Equation (N.96) compares every block with the independently generated $p=1$ family.

Every displayed block $d_1, \dots, d_{13}$ of this chapter passes all five.

Algorithm N.65 (Stable evaluation from a logarithmic target)**.** Never form y; accept $\ell = \log y$. For branch δ:

$$R = \ell - A_\delta, \quad w = W_0(2R/\mathrm{e}), \quad T = 2R/w, \quad \Lambda = w+1, \quad x \leftarrow T - 1 + \sum_{n=1}^{N} \frac{d_{2n-1}(\Lambda)}{T^{2n-1}}. \tag{N.98}$$

Refine by Newton on the logarithmic residual $F_\delta(x) = \log \kappa_\delta + \frac{x}{2}\log 2 + \log\Gamma(\frac{x}{2}+1) - \ell$,

$$x_{\text{new}} = x - \frac{F_\delta(x)}{\frac{1}{2}\big(\log 2 + \psi(\frac{x}{2}+1)\big)}, \tag{N.99}$$

whose denominator is positive by Equation (N.6); convergence is quadratic once the asymptotic value is in the basin, which a few blocks always achieve in the ranges tested. For $\mathscr{D}$, add $\frac{1}{4}(1-\cos\pi x)\log(2/\pi)$ to F_0 and its derivative to the denominator, and start from Equation (N.82).

Proposition N.66 (A posteriori certificate)**.** *Let $\widetilde{x}$ be any approximation and $\rho = F_\delta(\widetilde{x})$. If I is an interval containing $\widetilde{x}$ and $\mathcal{D}_\delta^{-1}(y)$, then*

$$\big|\widetilde{x} - \mathcal{D}_\delta^{-1}(y)\big| \le \frac{2|\rho|}{\inf_{\xi\in I}\big(\log 2 + \psi(\xi/2+1)\big)}, \tag{N.100}$$

and since ψ is increasing the infimum is attained at the left endpoint. At large arguments the denominator is $\sim \Lambda$, matching Equation (N.39).

Proof. This is Theorem J.47 with the explicit slope Equation (N.6); the mean-value theorem applied to F_δ on I gives Equation (N.100). □

N.14 Numerical validation

All tables in this section were recomputed at 220-digit precision with `mpmath`: the exact target $\ell = \log \mathcal{D}_\delta(n)$ is formed from $\log\Gamma$, the Lambert data from Equation (N.98), and the blocks from Equation (N.41) in exact rational arithmetic. No rounded value of y is ever formed. Entries are *signed* (approximation minus exact), which the source tables partly suppress.

Remark N.67. The near-geometric improvement at these moderate arguments is not a claim of convergence: at fixed x the Bernoulli-driven expansion diverges, and by Theorem N.54 the attainable accuracy is bounded below by $\asymp \mathrm{e}^{-\pi T}/(\sqrt{T}\Lambda)$. In high-precision work, truncate at the least term or finish with Equation (N.99). Note also that the parity dependence is confined to the core: once the correct A_δ is used, the two subsequences show nearly identical error patterns, as Theorem N.4 predicts.

Table N.3: Forward: centred logarithmic expansion, $\Phi(u)+A_\delta+\sum_{k\le N} c_k u^{1-2k}-\log\mathcal{D}_\delta(n)$, $u=n+1$. Signs alternate, confirming Theorem N.17.

δ	n	$N=0$	$N=1$	$N=2$	$N=3$	$N=4$
0	20	$3.96616\cdot 10^{-3}$	$-2.09362\cdot 10^{-6}$	$5.98269\cdot 10^{-9}$	$-4.14413\cdot 10^{-11}$	$5.30661\cdot 10^{-13}$
1	21	$3.78606\cdot 10^{-3}$	$-1.82137\cdot 10^{-6}$	$4.74399\cdot 10^{-9}$	$-2.99567\cdot 10^{-11}$	$3.49749\cdot 10^{-13}$
0	100	$8.25064\cdot 10^{-4}$	$-1.88702\cdot 10^{-8}$	$2.34020\cdot 10^{-12}$	$-7.04697\cdot 10^{-16}$	$3.92938\cdot 10^{-19}$
1	101	$8.16975\cdot 10^{-4}$	$-1.83207\cdot 10^{-8}$	$2.22773\cdot 10^{-12}$	$-6.57742\cdot 10^{-16}$	$3.59602\cdot 10^{-19}$

Table N.4: Forward: relative error $|F_M/\mathcal{D}_\delta-1|$ of the centred multiplicative expansion Equation (N.29) truncated after $s_M u^{-M}$, with $\delta\equiv x \pmod 2$.

x	δ	$M=0$	$M=1$	$M=3$	$M=5$	$M=9$
10	0	$7.590\cdot 10^{-3}$	$4.330\cdot 10^{-5}$	$2.599\cdot 10^{-7}$	$4.974\cdot 10^{-9}$	$1.388\cdot 10^{-11}$
11	1	$6.957\cdot 10^{-3}$	$3.538\cdot 10^{-5}$	$1.751\cdot 10^{-7}$	$2.782\cdot 10^{-9}$	$5.454\cdot 10^{-12}$
20	0	$3.974\cdot 10^{-3}$	$9.988\cdot 10^{-6}$	$1.432\cdot 10^{-8}$	$6.756\cdot 10^{-11}$	$1.314\cdot 10^{-14}$
21	1	$3.793\cdot 10^{-3}$	$9.014\cdot 10^{-6}$	$1.166\cdot 10^{-8}$	$4.972\cdot 10^{-11}$	$7.956\cdot 10^{-15}$
50	0	$1.635\cdot 10^{-3}$	$1.483\cdot 10^{-6}$	$3.106\cdot 10^{-10}$	$2.113\cdot 10^{-13}$	$9.594\cdot 10^{-19}$
51	1	$1.604\cdot 10^{-3}$	$1.424\cdot 10^{-6}$	$2.861\cdot 10^{-10}$	$1.866\cdot 10^{-13}$	$7.800\cdot 10^{-19}$

Table N.5: Branchwise inverse: x_N-x with $y=\mathcal{D}_\delta(x)$ exact and x_N the truncation of Equation (N.39) after N blocks. Signs alternate, as Theorem N.37 predicts for the dominant term.

δ	x	$N=0$	$N=1$	$N=2$	$N=3$	$N=5$
0	10	$-6.3074\cdot 10^{-3}$	$1.6444\cdot 10^{-5}$	$-1.6097\cdot 10^{-7}$	$3.6890\cdot 10^{-9}$	$1.1386\cdot 10^{-11}$
1	11	$-5.5808\cdot 10^{-3}$	$1.2104\cdot 10^{-5}$	$-9.9815\cdot 10^{-8}$	$1.9338\cdot 10^{-9}$	$4.2632\cdot 10^{-12}$
0	20	$-2.6055\cdot 10^{-3}$	$1.7520\cdot 10^{-6}$	$-4.7751\cdot 10^{-9}$	$3.1029\cdot 10^{-11}$	$7.6506\cdot 10^{-15}$
1	21	$-2.4497\cdot 10^{-3}$	$1.4956\cdot 10^{-6}$	$-3.7170\cdot 10^{-9}$	$2.2041\cdot 10^{-11}$	$4.5233\cdot 10^{-15}$
0	50	$-8.3109\cdot 10^{-4}$	$8.9794\cdot 10^{-8}$	$-4.1941\cdot 10^{-11}$	$4.7156\cdot 10^{-14}$	$3.4429\cdot 10^{-19}$
1	51	$-8.1110\cdot 10^{-4}$	$8.4219\cdot 10^{-8}$	$-3.7845\cdot 10^{-11}$	$4.0942\cdot 10^{-14}$	$2.7669\cdot 10^{-19}$
0	100	$-3.5755\cdot 10^{-4}$	$9.5805\cdot 10^{-9}$	$-1.1469\cdot 10^{-12}$	$3.3159\cdot 10^{-16}$	$1.5896\cdot 10^{-22}$
1	101	$-3.5329\cdot 10^{-4}$	$9.2785\cdot 10^{-9}$	$-1.0892\cdot 10^{-12}$	$3.0878\cdot 10^{-16}$	$1.4232\cdot 10^{-22}$

Table N.6: Inverse of the periodic interpolation: $\left(T-1+\widetilde{\Delta}\right)-x$ with $y=\mathscr{D}(x)$ exact, successively adding the carrier Δ_0 (computed as the exact fixed point of Equation (N.76)), Δ_1/T and Δ_2/T^2 from Equation (N.81). The core error matches the predicted carrier amplitude $2a/\Lambda$: at $x=20$, $2a/\Lambda=0.074080$ against an observed 0.071518; at $x=100$, 0.048919 against 0.048564. No source validates this sector numerically.

x	core $T-1$	$+\Delta_0$	$+\Delta_1/T$	$+\Delta_2/T^2$
20	$7.1518\cdot 10^{-2}$	$-2.5603\cdot 10^{-3}$	$-1.2693\cdot 10^{-5}$	$1.6637\cdot 10^{-6}$
21	$-7.5539\cdot 10^{-2}$	$-2.4124\cdot 10^{-3}$	$1.4359\cdot 10^{-5}$	$1.4221\cdot 10^{-6}$
20.5	$-2.5254\cdot 10^{-3}$	$-3.2850\cdot 10^{-3}$	$2.0627\cdot 10^{-6}$	$2.2808\cdot 10^{-6}$
100	$4.8564\cdot 10^{-2}$	$-3.5495\cdot 10^{-4}$	$-2.3071\cdot 10^{-7}$	$9.4103\cdot 10^{-9}$
100.25	$3.4218\cdot 10^{-2}$	$-3.9845\cdot 10^{-4}$	$-1.8912\cdot 10^{-7}$	$1.0779\cdot 10^{-8}$

Table N.7: Multifactorial inverse Equation (N.95), using only the $p = 1$ blocks rescaled by Equation (N.96). Entries are $x_N - x$ with $y = \mathcal{D}_{p,r}(x)$ exact; the $p = 2$ row reproduces Table N.5.

p	r	x	$N=0$	$N=1$	$N=2$	$N=3$
2	2	20	$-2.60549 \cdot 10^{-3}$	$1.75197 \cdot 10^{-6}$	$-4.77514 \cdot 10^{-9}$	$3.10287 \cdot 10^{-11}$
3	1	22	$-5.05001 \cdot 10^{-3}$	$6.03949 \cdot 10^{-6}$	$-2.95265 \cdot 10^{-8}$	$3.44131 \cdot 10^{-10}$
3	2	20	$-5.67882 \cdot 10^{-3}$	$8.16471 \cdot 10^{-6}$	$-4.75923 \cdot 10^{-8}$	$6.60178 \cdot 10^{-10}$
4	3	23	$-8.27262 \cdot 10^{-3}$	$1.54424 \cdot 10^{-5}$	$-1.18248 \cdot 10^{-7}$	$2.15627 \cdot 10^{-9}$
5	2	27	$-1.04166 \cdot 10^{-2}$	$2.15596 \cdot 10^{-5}$	$-1.85311 \cdot 10^{-7}$	$3.79755 \cdot 10^{-9}$

N.15 Logical status

Three kinds of statement have been used, and the chapter keeps them apart.

Exact identities. Equations (N.3), (N.7) and (N.91) and the interpolation statements Theorems N.3, N.6 and N.59; the centred normal forms Equations (N.12), (N.14) and (N.73); the Lambert identities Equation (N.33); the Borel–Laplace representations Equations (N.15) and (N.93) and the Mittag–Leffler decomposition Equation (N.16). These hold wherever the functions are defined.

Formal coefficient identities. The Bell formulae Equations (N.26) and (N.43), the recurrences Equations (N.28), (N.41) and (N.80), the closed formulae Equations (N.45), (N.48) and (N.70), the boundary and sign statements Theorems N.35 and N.37, the centre-shift identity Equation (N.31), the scaling Equation (N.96), and Theorem N.50. These are identities of formal series over $\mathbb{Q}(\Lambda)$, or over the ring generated by Λ^{-1} and $e^{\pm i\theta}$ in the periodic case. No convergence of the Bernoulli series is asserted anywhere.

Poincaré asymptotics with remainders. Theorems N.16 to N.18 for the forward direction — where the remainder is in fact two-sided and sharp; Theorems N.29 and N.48 and Equation (N.95) for the inverse, each obtained from a forward residual by Theorems J.47 and J.51 together with the explicit slope Equation (N.6); and Theorems N.54 and N.55 for the beyond-all-orders layer. Theorem N.46(b) is a genuine convergence statement, with an explicit threshold.

Remark N.68 (Practical summary). For symbolic work, centre at $u = x + 1$, keep W_0 exact, generate c_k from Bernoulli numbers, exponentiate with complete Bell polynomials, and invert with the triangular $z = T^{-2}$ recurrence, checking Equation (N.97) and the sign law at every step. Expand W_0 into iterated logarithms only as a final presentation layer. When the parity of n is known, use the branch inverse; when it is not, state the interpolation — by Theorem N.8 its choice changes the leading sector of the answer.

Appendix O

The partition numbers (A000041)

This chapter merges three independent treatments of the partition numbers filed in the `special-function-inversion` collection: *Partition_Number_Transseries_and_Asymptotic_Inverse* (below: **S1**), *Partition_Number_Transseries_and_Inverse* (**S2**), and *Partition_Numbers_Transseries_and_Inverse* (**S3**). All three take Rademacher's convergent series as analytic input and build a coefficient calculus on top of it; they differ in notation, in how the non-integral arithmetic phase is defined, and in which tail estimate is proved. Everything below that is not attributed is common to all three, restated once. Numerical constants, coefficient tables and error tables have been recomputed here (exact rationals through `fractions`/`sympy`; 140-digit `mpmath` for the tables); the recomputation is described in Section O.12.

O.1 Orientation: what is specific to $p(n)$

The partition function $p(n)$ is defined by Euler's product

$$\sum_{n\geq 0} p(n)q^n = \prod_{m\geq 1} \frac{1}{1-q^m}, \qquad |q| < 1, \tag{O.1}$$

and is OEIS A000041. Hardy and Ramanujan proved

$$p(n) \sim \frac{1}{4\sqrt{3}\,n} \exp\left(\pi\sqrt{\frac{2n}{3}}\right), \tag{O.2}$$

and Rademacher replaced the asymptotic formula by an exactly convergent series. That single fact makes this chapter structurally different from Parts M, N and P: the exponentially small sectors are *given*, not reconstructed from the late behaviour of a divergent perturbative series. Three features are genuinely specific to $p(n)$ and are the subject of this chapter.

1. **An arithmetic sector index.** The sectors are labelled by a positive integer k (a root of unity $\mathrm{e}^{2\pi\mathrm{i}h/k}$) and a sign σ, and their amplitudes are the Rademacher sums $A_k(n)$ built from Dedekind sums. These amplitudes are *periodic* in n with period k; they are not constants, and after inversion they become bounded oscillations whose phase is quadratic in $\log y$.

2. **Non-well-ordered actions.** Relative to the dominant sector, the exponential actions are $1-1/k$ (increasing to 1) and $1+1/k$ (decreasing to 1). The second family has no least element. The support of the completed object is therefore not a locally finite Hahn grid, and the two signs may not be separated before summing over k.

3. **Convergence where one expects divergence.** Every fixed sector's algebraic series in $n^{-1/2}$ has radius of convergence exactly $\sqrt{24}$ in the variable $n^{-1/2}$, and the reversion series of the dominant sector converges as well. Nothing in this chapter is a divergent asymptotic series; consequently the least-term machinery of Theorem J.54 degenerates here, as recorded in Theorem O.37.

Everything else — the Lambert core, the reversion around it, the flattening into logarithms, the passage from a smooth inverse to an integer staircase, and the transport of remainders — is Part J material and is *cited*, never re-proved.

O.1.1 The exponential–power reduction and the Part 0 dictionary

Fix throughout

$$N = n - \frac{1}{24}, \qquad \lambda = \pi\sqrt{\frac{2}{3}} = \frac{\pi\sqrt{6}}{3}, \qquad \mu = \lambda\sqrt{N} = \frac{\pi}{6}\sqrt{24n-1}, \qquad \kappa = \frac{\pi^2}{6\sqrt{3}}. \tag{O.3}$$

The dominant Rademacher sector, derived in Theorem O.14, is

$$P_{1,+}(n) = \frac{e^{\lambda\sqrt{N}}}{4\sqrt{3}\,N}\left(1 - \frac{1}{\lambda\sqrt{N}}\right). \tag{O.4}$$

Read as a function of N, Equation (O.4) is exactly an instance of the exponential–power model Theorem J.2, $F(x) = C\exp(ax^{\alpha})x^{b}\exp Q(x^{-\delta})$, with

$$x = N, \quad C = \frac{1}{4\sqrt{3}}, \quad a = \lambda, \quad \alpha = \frac{1}{2}, \quad b = -1, \quad \delta = \frac{1}{2}, \quad Q(t) = \log\left(1 - \frac{t}{\lambda}\right) = -\sum_{r\geq 1}\frac{t^r}{r\lambda^r}. \tag{O.5}$$

So the partition problem is the case $\alpha = 1/2$: a *square-root* phase, not a linear one. It is reduced to the linear case in two steps: the monomial substitution $\xi = x^{\alpha} = \sqrt{N}$ of Theorem J.4, followed by the elementary linear rescaling $\mu = \lambda\xi$ that normalises the exponential rate to 1.

Lemma O.1 (The α-reduction for $p(n)$)**.** (i) *Theorem J.4 applied to Equation* (O.5) *with* $\xi = \sqrt{N}$ *gives an* $\alpha = 1$ *model with data* $\left(\frac{1}{4\sqrt{3}},\ \lambda,\ 1,\ -2,\ 1,\ Q\right)$, $Q(t) = \log(1 - t/\lambda)$; *note* $b/\alpha = -2$ *and* $\delta/\alpha = 1$, *so this is the normalisation* $\delta = 1$ *assumed from Part J onwards.* (ii) *The further substitution* $\mu = \lambda\xi$ *rescales any* $\alpha = 1$ *model with data* $(C, a, 1, b, 1, Q)$ *to one with data* $\left(Ca^{-b},\ 1,\ 1,\ b,\ 1,\ Q(a\,\cdot)\right)$. *Here the two steps give*

$$P_{1,+} = \kappa\,\frac{e^{\mu}}{\mu^2}\left(1 - \frac{1}{\mu}\right) = C' e^{a'\mu}\mu^{b'}\exp Q'(\mu^{-\delta'}), \tag{O.6}$$

with the dictionary

$$C' = \kappa = \frac{\pi^2}{6\sqrt{3}}, \qquad a' = 1, \qquad \alpha' = 1, \qquad b' = -2, \qquad \delta' = 1, \qquad Q'(t) = \log(1-t).$$

The inverse map back to the index is exact and quadratic:

$$n = \frac{1}{24} + \frac{3\mu^2}{2\pi^2}. \tag{O.7}$$

Proof. (i) is Theorem J.4 at $\alpha = \delta = 1/2$, $b = -1$, which leaves C, a and the coefficients of Q unchanged and replaces (b, δ) by $(b/\alpha, \delta/\alpha)$. For (ii), substituting $\xi = \mu/a$ in $Ce^{a\xi}\xi^{b}\exp Q(\xi^{-1})$ gives $Ca^{-b}e^{\mu}\mu^{b}\exp Q(a/\mu)$, and $t \mapsto Q(at)$ again lies in $t\,\mathbb{R}[[t]]$. Here $a = \lambda$, $b = -2$, so $Ca^{-b} = \lambda^2/(4\sqrt{3}) = \pi^2/(6\sqrt{3}) = \kappa$ (using $\lambda^2 = 2\pi^2/3$) and $Q'(t) = Q(\lambda t) = \log(1-t)$. Composing, $\mu = \lambda\sqrt{N}$ and Equation (O.4) become Equation (O.6). Inverting $\mu = \lambda\sqrt{n - 1/24}$ gives Equation (O.7), again by $\lambda^2 = 2\pi^2/3$. ☐

Remark O.2 (The model hypotheses hold exactly, and Q converges). Theorem J.2 asks for the analytic estimates Equation (J.3) and Equation (J.4) in addition to the formal shape. For the partition envelope both hold *exactly and trivially*: $F(\mu) = \kappa e^{\mu}\mu^{-2}(1 - 1/\mu)$ is an elementary closed form, so $\log F(\mu) - \log\kappa - \mu + 2\log\mu = \log(1 - 1/\mu)$ is literally the sum of the series Q', and $(\log F)' = \Phi'$ is computed in closed form in Theorem O.49; no asymptotic estimate is being differentiated. In particular $Q'(t) = \log(1 - t)$ is *convergent*, with radius 1. Theorem J.3 states that Q is divergent of Gevrey type in all four chapters of this volume; that is not so here, and the discrepancy should be read as a further instance of Theorem O.37 — the partition chapter is the one place in the volume where the whole apparatus applies to a convergent object.

Remark O.3 (Why this is the point of the whole construction). After Theorem O.1 nothing in the inversion of the dominant sector is partition-specific. The three sources each rediscover, in their own notation, the Lambert core of Theorem J.23, the centred reversion of Theorem J.29 and the logarithmic flattening of Theorem J.37. We record the dictionary once (Table O.1) and cite. What is *not* generic — and occupies the rest of the chapter — is the arithmetic: the sector index, the Dedekind-sum amplitudes, the accumulation of actions at 1, and the interaction between the periodic amplitudes and the $\mathcal{O}(1)$ shift produced by the reversion.

Table O.1: Parameter dictionary from Part J to the partition chapter. All Part 0 statements are used at these values.

Part 0 object	Partition value	Reference
model variable x	$\mu = \pi\sqrt{2(n - 1/24)/3}$	Equation (O.3)
C	$\kappa = \pi^2/(6\sqrt{3})$	Theorem O.1
$a,\ \alpha$	$1,\ 1$ (after reduction and rescaling; $\lambda,\ 1/2$ before)	Theorem O.1
b	-2	Theorem O.1
$Q(t),\ \delta$	$\log(1 - t),\ 1$	Theorem O.1
Lambert core X	$-2W_{-1}\left(-\frac{1}{2}\sqrt{\kappa/y}\right)$	Theorem O.46
$\Psi(t, u)$ of Theorem J.29	$3\log(1 + tu) - \log(1 - t + tu)$	Theorem O.50
reversion coefficients c_m	$1, \frac{5}{2}, \frac{13}{3}, \frac{53}{12}, -\frac{14}{5}, \ldots$	Table O.2
index map $x \mapsto n$	$n = \frac{1}{24} + 3\mu^2/(2\pi^2)$	Equation (O.7)
staircase separation	unit spacing of $\mathbb{Z}$; see Theorem O.83	Theorem J.43

Remark O.4 (Notation across the three sources). The sources disagree on symbols in a way that has to be stated once, because two of the clashes are dangerous.

1. **The letter x.** S1 writes $N = n - 1/24$ for the shifted index and x for the phase $\pi\sqrt{2N/3}$; S2 writes $x = n - 1/24$ for the shifted index and μ for the phase; S3 writes N and μ. This volume uses N for the shifted index and μ for the phase, i.e. the S3 convention, which is the intersection of the other two.

2. **The shift itself.** Every displayed *exact* identity in this chapter is in the shifted variable $N = n - 1/24$. Expansions in $n^{-1/2}$ (Section O.5) re-expand the shift and are the only place where unshifted n appears inside a coefficient. Confusing the two is the single most common error in this subject: e.g. replacing N by n in Equation (O.4) costs a relative $\mathcal{O}(n^{-1/2})$ and destroys every statement about exponentially small sectors.

3. **The normalising constant** is $\mathcal{C}$ in S1, κ in S2, D in S3; all equal $\pi^2/(6\sqrt{3})$. We write κ, reserving D for nothing and Φ' for the slope.

4. **The logarithmic variable.** S1 and S2 expand in $R = \log(y/\kappa)$; S3 expands in $H = \log(4y/\kappa) = R + 2\log 2$. The two expansions have the *same* coefficient polynomials (see Theorem O.48) but are different series and must not be mixed.

O.2 Dedekind sums and the arithmetic amplitudes

O.2.1 Definitions

Definition O.5 (Sawtooth and Dedekind sum)**.** For $u \in \mathbb{R}$ put

$$((u)) = \begin{cases} u - \lfloor u \rfloor - \frac{1}{2}, & u \notin \mathbb{Z}, \\ 0, & u \in \mathbb{Z}, \end{cases} \tag{O.8}$$

and for $k \geq 1$, $h \in \mathbb{Z}$,

$$s(h,k) = \sum_{r=1}^{k-1} \left(\left(\frac{r}{k}\right)\right) \left(\left(\frac{hr}{k}\right)\right), \tag{O.9}$$

with the empty-sum convention $s(h,1) = 0$.

Throughout, $U_k = \{h : 0 \leq h < k,\ \gcd(h,k) = 1\}$, so that $U_1 = \{0\}$.

Definition O.6 (Rademacher amplitude)**.** For $k \geq 1$ and $n \in \mathbb{Z}$,

$$A_k(n) = \sum_{h \in U_k} \exp\left(\pi \mathrm{i}\, s(h,k) - \frac{2\pi \mathrm{i} hn}{k}\right). \tag{O.10}$$

Lemma O.7 (Elementary properties)**.** *For every $k \geq 1$ and $n \in \mathbb{Z}$:*

1. $A_k(n) \in \mathbb{R}$*, and* $A_k(n) = \sum_{h \in U_k} \cos\bigl(\pi s(h,k) - 2\pi hn/k\bigr)$*;*
2. $A_k(n+k) = A_k(n)$*;*
3. $|A_k(n)| \leq \varphi(k) \leq k$*;*
4. $A_1(n) = 1$ *and* $A_2(n) = (-1)^n$*.*

Proof. (i) For $k \geq 2$ the involution $h \mapsto k - h$ is a bijection of U_k (note $k - h \neq h$ unless $k = 2h$, impossible for $\gcd(h,k) = 1$, $k \geq 3$; for $k = 2$ the set is $\{1\}$ and $s(1,2) = 0$ makes the single term real). The reciprocity $s(-h,k) = -s(h,k)$, immediate from $((-u)) = -((u))$, shows that the terms at h and $k - h$ are complex conjugates. Hence the sum is real and equals the sum of the real parts, which is the stated cosine sum. (ii) Replacing n by $n + k$ changes each exponent by $-2\pi \mathrm{i} h$, an integer multiple of $2\pi\mathrm{i}$. (iii) Triangle inequality on $\varphi(k)$ unimodular terms. (iv) $U_1 = \{0\}$ and $s(0,1) = 0$ give $A_1 \equiv 1$; $U_2 = \{1\}$, $s(1,2) = 0$ give $A_2(n) = \mathrm{e}^{-\pi \mathrm{i} n} = (-1)^n$. □

Only the trivial bound (iii) is used anywhere in this chapter. Weil-type bounds for these Kloosterman-like sums would sharpen the constants in Theorems O.23 and O.78 but change no structure; see Theorem O.26.

Proposition O.8 (The first amplitudes)**.** *With* $s(1,k) = \dfrac{(k-1)(k-2)}{12k}$,

$$\begin{aligned} A_1(n) &= 1, & A_2(n) &= (-1)^n, \\ A_3(n) &= 2\cos\left(\frac{\pi}{18} - \frac{2\pi n}{3}\right), & A_4(n) &= 2\cos\left(\frac{\pi}{8} - \frac{\pi n}{2}\right), \\ A_5(n) &= 2\cos\left(\frac{\pi}{5} - \frac{2\pi n}{5}\right) + 2\cos\frac{4\pi n}{5}, & A_6(n) &= 2\cos\left(\frac{5\pi}{18} - \frac{\pi n}{3}\right). \end{aligned} \tag{O.11}$$

Proof. $s(1,k) = \sum_{r=1}^{k-1}((r/k))^2 = \sum_{r=1}^{k-1}(r/k - 1/2)^2 = \frac{(k-1)(2k-1)}{6k} - \frac{k-1}{2} + \frac{k-1}{4} = \frac{(k-1)(k-2)}{12k}$. Hence $s(1,3) = \frac{1}{18}$, $s(1,4) = \frac{1}{8}$, $s(1,5) = \frac{1}{5}$, $s(1,6) = \frac{5}{18}$, and $s(k-1,k) = -s(1,k)$. For $k = 3,4,6$ the set U_k is $\{1, k-1\}$, and the two terms are conjugate, giving twice the real part of the $h = 1$ term after reducing $2\pi(k-1)n/k \equiv -2\pi n/k \pmod{2\pi}$. For $k=5$, $U_5=\{1,2,3,4\}$ and a direct evaluation of Equation (O.9) gives $s(2,5)=s(3,5)=0$: with $((1/5)),((2/5)),((3/5)),((4/5)) = (-\frac{3}{10},-\frac{1}{10},\frac{1}{10},\frac{3}{10})$, the four products in $s(2,5)$ are $\frac{3}{100},-\frac{3}{100},-\frac{3}{100},\frac{3}{100}$. Pairing $h=1$ with $h=4$ and $h=2$ with $h=3$ gives the stated two cosines. □

O.2.2 The real-analytic interpolation of the amplitudes

The inverse problem needs the amplitudes at non-integral arguments: the smooth inverse of a smooth interpolation evaluates A_k at a real point. The finite Fourier shape of Equation (O.10) supplies a canonical choice.

Definition O.9 (Entire amplitude interpolation)**.** For $k \ge 1$ and $\nu \in \mathbb{C}$ set

$$\widetilde{A}_k(\nu) = \sum_{h\in U_k} \cos\left(\pi s(h,k) - \frac{2\pi h\nu}{k}\right). \tag{O.12}$$

Lemma O.10. *$\widetilde{A}_k$ is entire, k-periodic, real on $\mathbb{R}$, satisfies $\widetilde{A}_k(n) = A_k(n)$ for $n \in \mathbb{Z}$, and $|\widetilde{A}_k^{(r)}(\nu)| \le \varphi(k)(2\pi/k)^r \cosh(2\pi|\Im\nu|) \le k\,(2\pi)^r e^{2\pi|\Im\nu|}$ for every $r \ge 0$. Moreover $\widetilde{A}_1 \equiv 1$ and $\widetilde{A}_2(\nu) = \cos\pi\nu$, while*

$$\widetilde{A}_3(\nu) = 2\cos(\pi\nu)\,\cos\left(\frac{\pi}{18} + \frac{\pi\nu}{3}\right). \tag{O.13}$$

Proof. Each summand is entire and k-periodic in ν; realness on $\mathbb{R}$ is manifest; agreement at integers is Theorem O.7(i). The derivative bound is termwise, using $|h/k| \le 1$ and $|\cos^{(r)}(z)| \le \cosh(\Im z)$. For $k=1$, $U_1=\{0\}$ makes the single term $\cos 0 = 1$ identically. For $k=3$, sum-to-product on $A = \frac{\pi}{18} - \frac{2\pi\nu}{3}$ and $B = -\frac{\pi}{18} - \frac{4\pi\nu}{3}$ gives $\frac{A+B}{2} = -\pi\nu$ and $\frac{A-B}{2} = \frac{\pi}{18} + \frac{\pi\nu}{3}$, whence Equation (O.13). □

S2, Definition "Real Rademacher amplitude". S2 defines the interpolation as a sum over $1 \le h \le k$ with $\gcd(h,k)=1$. For $k \ge 2$ this is the same index set as U_k, but for $k=1$ it produces the single term $\cos(2\pi\nu)$ instead of the constant 1. S2 nevertheless uses $\mathcal{A}_1 \equiv 1$ throughout (its dominant sector carries no amplitude, and its relative-correction formula separates the $k=1$ terms with amplitude 1), so the definition contradicts its own use: with S2's index set the interpolating function $\mathcal{P}$ would not be an interpolation of p — its dominant term would oscillate off-lattice with period 1 at full relative strength, and the "eventual monotonicity" proposition would be false. **Fix.** Take h over $U_k = \{0 \le h < k : \gcd(h,k)=1\}$ with $U_1=\{0\}$, i.e. Theorem O.9; equivalently, decree $\widetilde{A}_1 \equiv 1$ separately, as S3 does. All of S2's later formulas are then correct as printed.

Remark O.11 (Non-uniqueness, honestly stated)*.* Theorem O.9 is *a* choice, not *the* choice. Any strictly increasing smooth $\mathcal{P}$ with $\mathcal{P}(n) = p(n)$ has an inverse, and the algebraic (Lambert–logarithmic) part of that inverse is the same for all of them, because it is determined by the dominant sector, which is interpolation-independent once N is treated as a real variable. The exponentially small sectors are *not* interpolation-independent. Theorem O.9 is singled out only because it is entire, finite-Fourier, real on $\mathbb{R}$, and literally the analytic continuation of the amplitudes already present in Rademacher's theorem. Section O.10 gives the alternative that avoids interpolation altogether.

O.3 Rademacher's formula in exact transseries normal form

O.3.1 The analytic input

Theorem O.12 (Hardy–Ramanujan–Rademacher; quoted)**.** *For every integer $n \ge 1$,*

$$p(n) = \frac{1}{\pi\sqrt{2}} \sum_{k=1}^{\infty} A_k(n)\sqrt{k}\,\frac{d}{dn}\left[\frac{\sinh\left(\frac{\lambda}{k}\sqrt{n-\frac{1}{24}}\right)}{\sqrt{n-\frac{1}{24}}}\right], \tag{O.14}$$

the series converging when summed in increasing k.

This is the only unproved input of the chapter. It is Rademacher's theorem (1937, 1938, 1943); a textbook proof is in Apostol, and a machine-checked proof exists in Isabelle/HOL (Eberl–Paulson, 2026); see Theorem O.100. All three sources take it as given and so do we. Note the factor $\sqrt{k}$: some normalisations absorb it into the definition of the arithmetic sum, and mixing conventions is a standard source of a spurious factor k. Theorem O.87 records the cross-checks that detect such a slip.

O.3.2 Exact two-exponential sectors

Lemma O.13 (Elementary derivative). *For $a > 0$ and $v > 0$,*

$$\frac{\mathrm{d}}{\mathrm{d}v}\Big(v^{-1/2}\sinh(a\sqrt{v})\Big) = \frac{1}{4v^{3/2}}\Big[(a\sqrt{v}-1)\mathrm{e}^{a\sqrt{v}} + (a\sqrt{v}+1)\mathrm{e}^{-a\sqrt{v}}\Big]. \tag{O.15}$$

Proof. With $w = a\sqrt{v}$ one has $\mathrm{d}w/\mathrm{d}v = w/(2v)$ and $v^{-1/2}\sinh w = a\,w^{-1}\sinh w$, so

$$\frac{\mathrm{d}}{\mathrm{d}v}\big(v^{-1/2}\sinh(a\sqrt{v})\big) = a\,\frac{\mathrm{d}}{\mathrm{d}w}\left(\frac{\sinh w}{w}\right)\frac{w}{2v} = \frac{a}{2v}\left(\cosh w - \frac{\sinh w}{w}\right) = \frac{1}{2v^{3/2}}\big(w\cosh w - \sinh w\big),$$

using $a = w/\sqrt{v}$ in the last step. Finally $2(w\cosh w - \sinh w) = (w-1)\mathrm{e}^{w} + (w+1)\mathrm{e}^{-w}$. □

Theorem O.14 (Exact sector decomposition). *Let N, λ, μ, κ be as in Equation* (O.3) *and set, for $k \geq 1$ and $\sigma \in \{+1, -1\}$,*

$$P_{k,\sigma}(n) = \frac{A_k(n)}{4\sqrt{3k}\;N}\,\mathrm{e}^{\sigma\mu/k}\left(1 - \frac{\sigma k}{\mu}\right) = \frac{\kappa}{\mu^2}\frac{A_k(n)}{\sqrt{k}}\,\mathrm{e}^{\sigma\mu/k}\left(1 - \frac{\sigma k}{\mu}\right). \tag{O.16}$$

Then, for every integer $n \geq 1$,

$$p(n) = \sum_{k\geq 1}\big(P_{k,+}(n) + P_{k,-}(n)\big), \tag{O.17}$$

the two signs being paired before the sum over k is formed. Equivalently, with the entire paired kernel

$$g(z) = \cosh z - \frac{\sinh z}{z} = \sum_{m\geq 1}\frac{2m}{(2m+1)!}z^{2m}, \qquad g(0) = 0, \tag{O.18}$$

one has the manifestly convergent form

$$p(n) = \frac{2\kappa}{\mu^2}\sum_{k\geq 1}\frac{A_k(n)}{\sqrt{k}}\,g\Big(\frac{\mu}{k}\Big). \tag{O.19}$$

Proof. Apply Theorem O.13 with $v = N$ and $a = \lambda/k$, so that $a\sqrt{v} = \mu/k$, inside Equation (O.14). The kth summand becomes

$$\frac{A_k(n)\sqrt{k}}{\pi\sqrt{2}}\cdot\frac{1}{4N^{3/2}}\Big[\Big(\frac{\mu}{k}-1\Big)\mathrm{e}^{\mu/k}+\Big(\frac{\mu}{k}+1\Big)\mathrm{e}^{-\mu/k}\Big] = \frac{A_k(n)\sqrt{k}}{\pi\sqrt{2}}\cdot\frac{\mu}{4kN^{3/2}}\Big[\mathrm{e}^{\mu/k}\Big(1-\frac{k}{\mu}\Big)+\mathrm{e}^{-\mu/k}\Big(1+\frac{k}{\mu}\Big)\Big].$$

Now $\mu/(\pi\sqrt{2}\sqrt{N}) = \lambda/(\pi\sqrt{2}) = 1/\sqrt{3}$, so the prefactor is $A_k(n)/(4\sqrt{3k}\,N)$, which is Equation (O.16) summed over σ. Since $\mu^2 = \lambda^2 N = (2\pi^2/3)N$, we get $1/(4\sqrt{3k}\,N) = \kappa/(\sqrt{k}\,\mu^2)$, the second form of Equation (O.16). Adding the two signs and using $\mathrm{e}^{z}(1-1/z)+\mathrm{e}^{-z}(1+1/z) = 2g(z)$ with $z = \mu/k$ gives Equation (O.19). All manipulations are termwise algebra, so the sum is unchanged. □

Remark O.15 (The three sources' normal forms are the same). S1 writes $p(n) = \frac{\mathcal{C}}{x^2}\sum_k \frac{A_k}{\sqrt{k}}\{\mathrm{e}^{x/k}(1 - k/x) + \mathrm{e}^{-x/k}(1+k/x)\}$ (its x is our μ); S2 writes the sectors $P_{k,\sigma} = \frac{A_k}{4\sqrt{3k}\,x}\mathrm{e}^{\sigma\mu/k}(1-\sigma k/\mu)$ (its x is our N); S3 writes $p(n) = D\mu^{-3}\sum_k A_k\sqrt{k}[(\frac{\mu}{k}-1)\mathrm{e}^{\mu/k} + (\frac{\mu}{k}+1)\mathrm{e}^{-\mu/k}]$. Factoring μ/k out of S3's bracket and using $\sqrt{k}\cdot(1/k) = 1/\sqrt{k}$ turns S3 into S1, and $\kappa/\mu^2 = 1/(4\sqrt{3k}\,N)\cdot\sqrt{k}$ turns S1 into S2. All three are Equation (O.16). We adopt the second form of Equation (O.16) because it displays the Part 0 model $\kappa\mathrm{e}^{\mu}\mu^{-2}(1-1/\mu)$ at $k=1$, $\sigma = +$ verbatim.

O.3.3 The dominant envelope and the exact relative factorisation

Definition O.16 (Dominant envelope, actions)**.** Put

$$F(\mu) = P_{1,+} = \kappa \frac{e^{\mu}}{\mu^2}\left(1 - \frac{1}{\mu}\right) = \kappa\, e^{\mu}\, \frac{\mu - 1}{\mu^3}, \tag{O.20}$$

and for $k \geq 1$, $\sigma = \pm 1$ (excluding $(1,+)$) the *action*

$$\alpha_{k,\sigma} = 1 - \frac{\sigma}{k} \qquad \text{so that} \qquad \alpha_{1,-} = 2, \quad \alpha_{k,+} = 1 - \frac{1}{k}, \quad \alpha_{k,-} = 1 + \frac{1}{k}. \tag{O.21}$$

Proposition O.17 (Exact relative factorisation)**.** *For $\mu > 1$,*

$$p(n) = F(\mu)\big(1 + \mathcal{R}(\mu; n)\big), \tag{O.22}$$

where, with the amplitude ratios

$$r_{k,\sigma}(\mu; n) = \frac{A_k(n)}{\sqrt{k}}\, \frac{1 - \sigma k/\mu}{1 - 1/\mu}, \tag{O.23}$$

one has the finite-level resolved form, valid for each $K \geq 1$,

$$\mathcal{R} = r_{1,-}\, e^{-2\mu} + \sum_{k=2}^{K} \sum_{\sigma=\pm 1} r_{k,\sigma}(\mu; n)\, e^{-\alpha_{k,\sigma}\mu} + \mathcal{R}_{>K}(\mu; n), \tag{O.24}$$

and the exact paired form

$$\mathcal{R}(\mu; n) = \frac{1 + \mu^{-1}}{1 - \mu^{-1}} e^{-2\mu} + \frac{2e^{-\mu}}{1 - \mu^{-1}} \sum_{k \geq 2} \frac{A_k(n)}{\sqrt{k}}\, g\Big(\frac{\mu}{k}\Big), \qquad \mathcal{R}_{>K} = \frac{2e^{-\mu}}{1 - \mu^{-1}} \sum_{k > K} \frac{A_k(n)}{\sqrt{k}} g\Big(\frac{\mu}{k}\Big). \tag{O.25}$$

Proof. Divide Equation (O.16) by Equation (O.20): $P_{k,\sigma}/F = \frac{A_k}{\sqrt{k}} \frac{1-\sigma k/\mu}{1-1/\mu} e^{(\sigma/k-1)\mu}$, which is Equation (O.23) times $e^{-\alpha_{k,\sigma}\mu}$; the case $k = 1, \sigma = -1$ has $A_1 = 1$ and gives the first term of Equation (O.25). For $k \geq 2$ the paired sum $P_{k,+} + P_{k,-} = \frac{2\kappa}{\mu^2} \frac{A_k}{\sqrt{k}} g(\mu/k)$ from Equation (O.19), divided by $F = \kappa e^{\mu} \mu^{-2}(1 - 1/\mu)$, gives the second term. □

Remark O.18 (What the Hardy–Ramanujan formula omits). Equation (O.2) is $F(\mu)$ after replacing N by n and discarding $1 - 1/\mu$. Each of those two simplifications costs a relative $\mathcal{O}(n^{-1/2})$; together they account for the entire algebraic half-power series of Section O.5. Equation (O.22) loses nothing: it records every Ford-circle arc, in both signs, with its exact algebraic amplitude.

O.4 Ordering of the sectors and tail estimates

O.4.1 The action-one accumulation

The relative actions of Equation (O.21) are

$$\tfrac{1}{2},\ \tfrac{2}{3},\ \tfrac{3}{4},\ \ldots \uparrow 1 \qquad \text{and} \qquad 2,\ \tfrac{3}{2},\ \tfrac{4}{3},\ \ldots \downarrow 1.$$

The first family is increasing with supremum 1; the second is decreasing with infimum 1 and *has no least element.* Two consequences must be stated before any transseries language is used.

Definition O.19 (Rademacher transseries)**.** A *Rademacher transseries* for p is the datum of

1. for each $K \geq 1$, the finite family of resolved sectors $\{(k, \sigma) : k \leq K\}$ with the coefficients of Section O.5, treated as independent formal exponential variables (so that coefficient extraction takes place in a finitely generated multivariate power-series ring), together with

2. the exact paired remainder $\mathcal{R}_{>K}$ of Equation (O.25), retained as one analytic block.

The limit $K \to \infty$ is taken analytically, through the convergence in Theorem O.12, and never by claiming that the whole support is a locally finite Hahn grid.

Remark O.20 (Why the pairing is mandatory). For $k \gg \mu$ each of $P_{k,+}$ and $P_{k,-}$ separately contains a term proportional to $\sqrt{k}/\mu$, and $\sum_k \sqrt{k}/\mu$ diverges. Only in the sum do those pieces cancel: $2g(\mu/k) = \frac{2}{3}(\mu/k)^2 + O((\mu/k)^4)$, which makes the kth paired term $\mathcal{O}(k^{-3/2})$. Thus Equation (O.17) prescribes an order of summation; it is not an absolutely rearrangeable double series. All three sources say this; S2 and S3 say it most sharply, and Theorem O.19 follows S2.

Remark O.21 (Truncating "through $\mathrm{e}^{-\mu}$" is not a prescription). Because the positive actions accumulate at 1, an instruction such as "retain all terms down to relative order $\mathrm{e}^{-\mu}$" selects infinitely many sectors and is meaningless. A legitimate truncation names either a Rademacher level K, or an action cutoff $\eta < 1$ (which then names finitely many positive sectors, namely $k \leq (1-\eta)^{-1}$), or a target accuracy together with Theorem O.23.

O.4.2 The tail bound

Lemma O.22 (Kernel bound)**.** *For $z \geq 0$,*

$$0 \leq g(z) \leq \frac{z^2}{3}\cosh z \leq \frac{z^2}{3}\mathrm{e}^z, \qquad \textit{hence} \qquad g(z) \leq \frac{\cosh 1}{3} z^2 \quad \textit{for } 0 \leq z \leq 1. \tag{O.26}$$

The factor $\cosh z$ *may not be dropped:* $g(1) = \mathrm{e}^{-1} = 0.3679\ldots > \frac{1}{3}$.

Proof. By Equation (O.18), $g(z) = \sum_{m\geq 1} \frac{2m}{(2m+1)!} z^{2m}$ has non-negative coefficients, whence $g \geq 0$. For the upper bound compare with $\frac{z^2}{3}\cosh z = \sum_{m\geq 1} \frac{z^{2m}}{3(2m-2)!}$ termwise: the claim is $\frac{2m}{(2m+1)!} \leq \frac{1}{3(2m-2)!}$, i.e. $6m(2m-2)! \leq (2m+1)!$, i.e. $6m \leq (2m+1)2m(2m-1)$, which holds for all $m \geq 1$ with equality at $m = 1$. Monotonicity of cosh gives the last inequality on $[0,1]$, and $g(1) = \cosh 1 - \sinh 1 = \mathrm{e}^{-1}$ shows the factor cannot be dropped. $\square$

Theorem O.23 (Explicit finite-K tail bound)**.** *Let $\mu > 1$ and $K \geq 1$. Then*

$$\left|\mathcal{R}_{>K}(\mu; n)\right| \leq \frac{4\mu^2}{3\,(1-\mu^{-1})\sqrt{K}}\,\exp\!\left[-\left(1 - \frac{1}{K+1}\right)\mu\right]. \tag{O.27}$$

The same bound holds with $A_k(n)$ replaced by $\widetilde{A}_k(\nu)$ for any real ν, since $|\widetilde{A}_k| \leq \varphi(k) \leq k$ as well.

Proof. From Equation (O.25), $|A_k| \leq k$ and Theorem O.22,

$$|\mathcal{R}_{>K}| \leq \frac{2\mathrm{e}^{-\mu}}{1-\mu^{-1}} \sum_{k>K} \sqrt{k}\, g\!\left(\frac{\mu}{k}\right) \leq \frac{2\mathrm{e}^{-\mu}}{1-\mu^{-1}} \cdot \frac{\mu^2}{3} \sum_{k>K} k^{-3/2}\mathrm{e}^{\mu/k} \leq \frac{2\mu^2\mathrm{e}^{-\mu+\mu/(K+1)}}{3(1-\mu^{-1})} \sum_{k>K} k^{-3/2},$$

and $\sum_{k>K} k^{-3/2} \leq \int_K^\infty t^{-3/2}\mathrm{d}t = 2K^{-1/2}$. $\square$

Theorem O.24 (Sharper fixed-K tail)**.** *For each fixed $K \geq 1$, as $\mu \to \infty$,*

$$\sum_{k>K} \left(P_{k,+} + P_{k,-}\right) = \mathcal{O}_K\!\left(\mu^{-1/2}\mathrm{e}^{\mu/(K+1)}\right), \qquad \mathcal{R}_{>K} = \mathcal{O}_K\!\left(\mu^{3/2}\mathrm{e}^{-(1-1/(K+1))\mu}\right). \tag{O.28}$$

Proof. Split at $k = \mu$. For $K < k \le \mu$ use $|A_k| \le k$ and $2g(\mu/k) \le 2\cosh(\mu/k) \le 2e^{\mu/(K+1)}$ in Equation (O.19), so those terms contribute at most $\frac{2\kappa}{\mu^2} \cdot 2e^{\mu/(K+1)} \sum_{k\le\mu} \sqrt{k} = \mathcal{O}(\mu^{-1/2} e^{\mu/(K+1)})$ since $\sum_{k\le\mu} \sqrt{k} = \mathcal{O}(\mu^{3/2})$. For $k > \mu$ put $z = \mu/k \in (0, 1)$; then $g(z) \le \frac{1}{3} z^2 \cosh 1$ by Theorem O.22, and the kth term is $\mathcal{O}(\mu^{-2}\sqrt{k} \cdot \mu^2 k^{-2}) = \mathcal{O}(k^{-3/2})$, whose tail is $\mathcal{O}(\mu^{-1/2})$. Dividing by $F \asymp \kappa\mu^{-2} e^{\mu}$ gives the relative statement. □

Remark O.25 (Which bound to use). Theorem O.23 (from S2) is explicit and uniform in K; Theorem O.24 (from S1 and S3) is sharper in μ by a factor $\mu^{-1/2}$ but only for fixed K. Both are needed: the first to certify a computation, the second to state clean asymptotic orders. S1 and S3 state only the second, S2 only the first.

Remark O.26 (Sharper amplitude bounds). All constants above come from $|A_k| \le k$. Lehmer's explicit remainder estimates and the effective error bounds of Banerjee–Paule–Radu–Schneider (Theorem O.100) are much sharper, and Weil-type bounds for the Kloosterman-like sums A_k give $|A_k(n)| \ll_\varepsilon k^{1/2+\varepsilon}$ for many k. Substituting any such bound in the proofs of Theorems O.23 and O.24 improves the powers of μ but changes no statement of this chapter, all of which are of the form "a fixed Rademacher level is exponentially accurate". For a *specific* residue class some A_k vanish, so the first non-zero omitted sector can be much smaller than the generic estimate.

O.4.3 An exact condensation at the accumulation level

Right at the accumulation action 1, an alternative to sector-by-sector bookkeeping is to sum the kernel expansion first. This is S3's contribution and has no counterpart in S1 or S2.

Definition O.27 (Rademacher–Kloosterman zeta)**.** For $\Re s > 2$ and $n \in \mathbb{Z}$ put $\mathcal{Z}_n(s) = \displaystyle\sum_{k\ge1} \frac{A_k(n)}{k^s}$; the series converges absolutely there by Theorem O.7(iii). The same definition with $\widetilde{A}_k(\nu)$ defines $\mathcal{Z}_\nu(s)$ for real ν.

Theorem O.28 (Exact zeta condensation)**.** *For every integer $n \ge 1$,*

$$p(n) = \kappa \sum_{r\ge1} \frac{4r}{(2r+1)!}\, \mu^{2r-2}\, \mathcal{Z}_n\!\left(2r + \frac{1}{2}\right), \tag{O.29}$$

and the double series obtained by expanding $\mathcal{Z}_n$ converges absolutely.

Proof. Insert $2g(z) = \sum_{r\ge1} \frac{4r}{(2r+1)!} z^{2r+1} \cdot z^{-1}$ — more precisely $2g(z) = \sum_{r\ge1} \frac{4r}{(2r+1)!} z^{2r}$ — into Equation (O.19) with $z = \mu/k$:

$$p(n) = \frac{\kappa}{\mu^2} \sum_{k\ge1} \frac{A_k(n)}{\sqrt{k}} \sum_{r\ge1} \frac{4r}{(2r+1)!} \frac{\mu^{2r}}{k^{2r}} = \kappa \sum_{k,r} \frac{4r}{(2r+1)!} \mu^{2r-2} \frac{A_k(n)}{k^{2r+1/2}}.$$

Absolute convergence: $\sum_k |A_k| k^{-2r-1/2} \le \sum_k k^{-2r+1/2} < \infty$ for $r \ge 1$, and $\sum_{r\ge1} \frac{4r}{(2r+1)!} \mu^{2r-2} \zeta(2r - \frac{1}{2})$ converges for every fixed μ because $\zeta(2r - \frac{1}{2}) \to 1$ and $\sum_r \frac{4r\mu^{2r-2}}{(2r+1)!} < \infty$. Tonelli's theorem justifies the interchange, giving Equation (O.29). □

Remark O.29. Equation (O.29) orders nothing by asymptotic size — it is an exact analytic rearrangement, not a transseries. Its use is precisely where the transseries language fails: it packages the whole action-one accumulation into Dirichlet values $\mathcal{Z}_n(2r + \frac{1}{2})$, which can be used as a single block in the inverse formulas of Section O.9.

O.5 All-order coefficients in the unshifted variable

The shifted phase μ makes each sector elementary; the price is that the conventional scale is $n^{-1/2}$. The entire infinite algebraic array of this subject comes from re-expanding the single shift $N = n - 1/24$. This section gives all of it, in one formula valid for every sector.

O.5.1 The sector generating function

Theorem O.30 (Sectorwise convergent half-power expansion)**.** *Fix $k \geq 1$ and $\sigma \in \{+1, -1\}$, and set $t = n^{-1/2}$. Then*

$$P_{k,\sigma}(n) = \frac{A_k(n)}{4\sqrt{3k}\,n} \exp\left(\frac{\sigma\lambda\sqrt{n}}{k}\right) \sum_{m \geq 0} c_m^{(k,\sigma)}\, n^{-m/2}, \tag{O.30}$$

where the $c_m^{(k,\sigma)}$ are the Taylor coefficients at $t = 0$ of

$$\Omega_{k,\sigma}(t) = \frac{1}{1 - t^2/24} \exp\left[\frac{\sigma\lambda}{kt}\left(\sqrt{1 - \frac{t^2}{24}} - 1\right)\right]\left(1 - \frac{\sigma kt}{\lambda\sqrt{1 - t^2/24}}\right). \tag{O.31}$$

The function $\Omega_{k,\sigma}$ is holomorphic on $|t| < \sqrt{24}$, so the series in Equation (O.30) *converges for every real $n > 1/24$; in particular $c_0^{(k,\sigma)} = 1$.*

Proof. With $t = n^{-1/2}$ we have $N = n(1 - t^2/24)$ and $\mu = \lambda\sqrt{N} = (\lambda/t)\sqrt{1 - t^2/24}$. Substituting in Equation (O.16),

$$P_{k,\sigma} = \frac{A_k(n)}{4\sqrt{3k}\,n} \cdot \frac{1}{1 - t^2/24} \cdot \mathrm{e}^{\sigma\mu/k}\left(1 - \frac{\sigma kt}{\lambda\sqrt{1 - t^2/24}}\right),$$

and $\mathrm{e}^{\sigma\mu/k} = \mathrm{e}^{\sigma\lambda/(kt)} \exp[\frac{\sigma\lambda}{kt}(\sqrt{1 - t^2/24} - 1)]$ with $\mathrm{e}^{\sigma\lambda/(kt)} = \mathrm{e}^{\sigma\lambda\sqrt{n}/k}$. This is Equation (O.30) with Equation (O.31). Holomorphy: the only candidate singularities of Equation (O.31) in $|t| < \sqrt{24}$ are $t = 0$ and the zeros of $\sqrt{1 - t^2/24}$, and the latter lie on $|t| = \sqrt{24}$. At $t = 0$ the exponent is $\frac{\sigma\lambda}{kt}(\sqrt{1 - t^2/24} - 1) = -\frac{\sigma\lambda t}{48k} + \mathcal{O}(t^3)$, so the singularity is removable, and the last factor is holomorphic there as well. Setting $t = 0$ gives $\Omega_{k,\sigma}(0) = 1$. □

S1, Theorem "Universal sector coefficient generator". S1 states only that the series "converges for all sufficiently small $|t|$ and supplies an asymptotic expansion to arbitrary order". This understates the truth and, read literally, invites the error the next remark forbids: it suggests the half-power series is a merely asymptotic object. S2 and S3 both identify the radius as exactly $\sqrt{24}$, i.e. convergence for every real $n > 1/24$; Theorem O.30 adopts their statement with the branch-point argument made explicit. **Consequence.** Nothing in this chapter is a divergent series in $n^{-1/2}$, and no least-term truncation is available; see Theorem O.37.

O.5.2 A finite closed formula for every sector coefficient

Write

$$\beta = \beta_{k,\sigma} = \frac{\sigma\pi}{6k}, \qquad \text{so that} \qquad \frac{\sigma\lambda}{k} = 2\sqrt{6}\,\beta, \qquad \frac{\sigma k}{\lambda} = \frac{1}{12\beta} \cdot \sqrt{6}\,, \tag{O.32}$$

the two identities following from $\lambda = \pi\sqrt{6}/3$ and $k = \sigma\pi/(6\beta)$. Thus each sector is the *same* function of the single parameter β; the sector label enters nowhere else.

Theorem O.31 (Finite coefficient formula, every sector). *For every $k \geq 1$, $\sigma = \pm 1$ and $m \geq 0$,*

$$c_m^{(k,\sigma)} = \frac{1}{(-4\sqrt{6})^m} \sum_{j=0}^{\lfloor (m+1)/2 \rfloor} \binom{m+1}{j} \frac{m+1-j}{(m+1-2j)!} \beta_{k,\sigma}^{m-2j}, \tag{O.33}$$

terms with a negative factorial index being absent. In particular $c_m^{(k,\sigma)} \in \mathbb{Q}[\beta, \beta^{-1}] \cdot \sqrt{6}^{\,m \bmod 2}$, and the dependence on the sector is only through $\beta_{k,\sigma} = \sigma\pi/(6k)$.

Proof. Put $u = -t/(4\sqrt{6})$, so that $t^2/24 = 4u^2$ and, by Equation (O.32), $\frac{\sigma\lambda}{kt} = \frac{2\sqrt{6}\beta}{-4\sqrt{6}\,u} = -\frac{\beta}{2u}$ and $\frac{\sigma kt}{\lambda} = -\frac{4\sqrt{6}\,u\sqrt{6}}{12\beta} = -\frac{2u}{\beta}$. Writing $s = \sqrt{1-4u^2}$, Equation (O.31) becomes

$$G(u) := \Omega_{k,\sigma}(-4\sqrt{6}\,u) = \frac{\mathrm{e}^{\beta T}}{s^2}\left(1 + \frac{2u}{\beta s}\right), \qquad T = T(u) = \frac{1-s}{2u}, \tag{O.34}$$

where we used $-\frac{\beta}{2u}(s-1) = \beta\frac{1-s}{2u} = \beta T$ and $s^2 = 1-4u^2$. The Catalan variable T satisfies

$$T = u(1+T^2), \qquad u = \frac{T}{1+T^2}, \qquad s = \frac{1-T^2}{1+T^2}, \qquad \mathrm{d}u = \frac{1-T^2}{(1+T^2)^2}\,\mathrm{d}T. \tag{O.35}$$

(The first identity follows from $2uT = 1-s$ and $s^2 = 1-4u^2$; the rest are algebra.) Substituting Equation (O.35) into Equation (O.34) gives the exact differential identity

$$G(u)\,\mathrm{d}u = \mathrm{e}^{\beta T}\left(\frac{1}{1-T^2} + \frac{2T}{\beta(1-T^2)^2}\right)\mathrm{d}T = \frac{1}{\beta}\,\mathrm{d}\!\left(\frac{\mathrm{e}^{\beta T}}{1-T^2}\right), \tag{O.36}$$

because $\frac{\mathrm{d}}{\mathrm{d}T}\frac{\mathrm{e}^{\beta T}}{1-T^2} = \mathrm{e}^{\beta T}\big(\frac{\beta}{1-T^2} + \frac{2T}{(1-T^2)^2}\big)$. Now let $D_m = [u^m]\,G(u) = (-4\sqrt{6})^m c_m^{(k,\sigma)}$. Then

$$D_m = \operatorname*{Res}_{u=0} \frac{G(u)\mathrm{d}u}{u^{m+1}} = \frac{1}{\beta} \operatorname*{Res}_{T=0} \frac{(1+T^2)^{m+1}}{T^{m+1}}\,\mathrm{d}\!\left(\frac{\mathrm{e}^{\beta T}}{1-T^2}\right),$$

using $u^{-m-1} = T^{-m-1}(1+T^2)^{m+1}$. The residue of an exact differential vanishes, so integrating by parts,

$$D_m = -\frac{1}{\beta}\operatorname*{Res}_{T=0} \frac{\mathrm{e}^{\beta T}}{1-T^2}\,\mathrm{d}\big(T^{-m-1}(1+T^2)^{m+1}\big) = \frac{m+1}{\beta}\operatorname*{Res}_{T=0} \mathrm{e}^{\beta T}(1+T^2)^m\,\frac{\mathrm{d}T}{T^{m+2}},$$

where we used $\mathrm{d}(T^{-m-1}(1+T^2)^{m+1}) = (1+T^2)^m T^{-m-2}\big(-(m+1)(1+T^2) + 2(m+1)T^2\big)\mathrm{d}T = -(m+1)(1+T^2)^m(1-T^2)T^{-m-2}\mathrm{d}T$, and the factor $1-T^2$ cancels. Hence

$$D_m = \frac{m+1}{\beta}[T^{m+1}]\,\mathrm{e}^{\beta T}(1+T^2)^m = \frac{m+1}{\beta}\sum_{j\geq 0}\binom{m}{j}\frac{\beta^{m+1-2j}}{(m+1-2j)!},$$

the sum running over $0 \leq j \leq \lfloor (m+1)/2 \rfloor$. Finally $(m+1)\binom{m}{j} = (m+1-j)\binom{m+1}{j}$ and division by $(-4\sqrt{6})^m$ gives Equation (O.33). □

Remark O.32 (Why a finite sum exists). A square-root shift would normally give an infinite convolution at each order. The Catalan coordinate T absorbs the square root, and the derivative already present in Rademacher's formula turns the integrand into the exact differential Equation (O.36). After the change-of-variable Jacobian cancels the denominator $1-T^2$, only the polynomial $(1+T^2)^m$ survives.

Remark O.33 (Provenance and the scope of the generalisation). For $(k,\sigma)=(1,+1)$, Equation (O.33) at $\beta=\pi/6$ is O'Sullivan's coefficient formula, also used by Banerjee, Paule, Radu and Schneider in their error analysis. S1 proves only that case, by the same Catalan substitution. S2 and S3 both observe that the proof never uses $k=1$ or $\sigma=+1$ — the sector enters only through β — and state the general formula; S3 gives a second, generating-function proof through the identity $\sum_{j\ge 0}\binom{m+2j}{j}q^j = C(q)^m/\sqrt{1-4q}$ for the Catalan series C. The residue proof above is S2's, written out. We record the generalisation as the principal coefficient statement of the chapter, since it covers every exponentially small sector at no extra cost.

The first coefficients, as functions of β, are

$$c_0=1,\qquad c_1=-\frac{\sqrt{6}\,(\beta^2+2)}{24\beta},\qquad c_2=\frac{\beta^2+12}{192},$$
$$c_3=-\frac{\sqrt{6}\,(\beta^4+36\beta^2+72)}{13824\,\beta},\quad c_4=\frac{\beta^4+80\beta^2+720}{221184},\quad c_5=-\frac{\sqrt{6}\,(\beta^6+150\beta^4+3600\beta^2+7200)}{26542080\,\beta},\tag{O.37}$$

$$c_6=\frac{\beta^6+252\beta^4+12600\beta^2+100800}{637009920},\qquad c_7=-\frac{\sqrt{6}\big(\beta^8+392\beta^6+35280\beta^4+705600\beta^2+1411200\big)}{107017666560\,\beta}.\tag{O.38}$$

These were recomputed here by expanding Equation (O.31) symbolically to order t^9 and comparing with Equation (O.33) coefficient by coefficient in exact arithmetic; the two agree identically. The parity pattern visible in Equation (O.37) — c_m is $\sqrt{6}$ times an odd Laurent polynomial in β for odd m, and a polynomial in β^2 for even m — is immediate from Equation (O.33), since $m-2j\equiv m \pmod 2$ and the prefactor contributes $\sqrt{6}^{\,m}$.

O.5.3 The Bell-polynomial route

Equation (O.33) is the fastest generator. For symbolic systems, or when the sector coefficients are wanted as a byproduct of a general composition engine, the following route uses only the primitives of Theorems J.9 and J.10.

Proposition O.34 (Logarithmic coefficients of a sector)**.** *Write* $\log\Omega_{k,\sigma}(t)=\sum_{j\ge1}\ell_j t^j$ *(suppressing the sector labels) and put* $a=1/24$*. Then, for* $j\ge1$*,*

$$\ell_j=\mathbf{1}_{2|j}\,\frac{a^{j/2}}{j/2}+\mathbf{1}_{2\nmid j}\,\frac{\sigma\lambda}{k}\binom{1/2}{(j+1)/2}(-a)^{(j+1)/2}-\sum_{\substack{1\le q\le j\\ 2|(j-q)}}\frac{1}{q}\left(\frac{\sigma k}{\lambda}\right)^{q}\frac{(q/2)_{(j-q)/2}}{\big((j-q)/2\big)!}\,a^{(j-q)/2},\tag{O.39}$$

where $(x)_r=x(x+1)\cdots(x+r-1)$*. Consequently the sector coefficients are the complete exponential Bell polynomials*

$$c_m=\frac{1}{m!}\,B_m\big(1!\,\ell_1,2!\,\ell_2,\ldots,m!\,\ell_m\big),\tag{O.40}$$

equivalently the triangular recurrence

$$c_0=1,\qquad c_m=\frac{1}{m}\sum_{j=1}^{m} j\,\ell_j\,c_{m-j}\quad (m\ge1).\tag{O.41}$$

Proof. Taking logarithms in Equation (O.31),

$$\log\Omega=-\log(1-at^2)+\frac{\sigma\lambda}{kt}\big(\sqrt{1-at^2}-1\big)+\log\left(1-\frac{\sigma k}{\lambda}t(1-at^2)^{-1/2}\right).$$

The first term contributes a^r/r at degree $2r$. For the second, $\sqrt{1-at^2}-1=\sum_{r\geq 1}\binom{1/2}{r}(-a)^r t^{2r}$, so dividing by t contributes $\frac{\sigma\lambda}{k}\binom{1/2}{r}(-a)^r$ at degree $2r-1$. For the third, expand $\log(1-w) = -\sum_{q\geq 1} w^q/q$ with $w=\frac{\sigma k}{\lambda}t(1-at^2)^{-1/2}$ and use $(1-at^2)^{-q/2}=\sum_{r\geq 0}\frac{(q/2)_r}{r!}a^r t^{2r}$; the term of degree $j=q+2r$ is the displayed sum. Equation (O.40) is the exponential formula of Theorem J.10, and Equation (O.41) follows by comparing coefficients in $\Omega' = (\log\Omega)'\,\Omega$. ☐

For orientation, at general (k,σ) and $a=1/24$,

$$\ell_1 = -\frac{\sigma\lambda a}{2k} - \frac{\sigma k}{\lambda}, \qquad \ell_2 = a - \frac{k^2}{2\lambda^2},$$
$$\ell_3 = -\frac{\sigma\lambda a^2}{8k} - \frac{\sigma k a}{2\lambda} - \frac{\sigma k^3}{3\lambda^3}, \qquad \ell_4 = \frac{a^2}{2} - \frac{k^2 a}{2\lambda^2} - \frac{k^4}{4\lambda^4}. \tag{O.42}$$

O.5.4 The dominant Poincaré expansion, and what it does not say

Definition O.35. $\omega_m = c_m^{(1,+1)}$, i.e. Equation (O.33) at $\beta=\pi/6$.

Corollary O.36. *For each fixed $M\geq 1$,*

$$p(n) = \frac{\mathrm{e}^{\lambda\sqrt{n}}}{4\sqrt{3}\,n}\left(\sum_{m=0}^{M-1}\omega_m n^{-m/2} + \mathcal{O}_M(n^{-M/2})\right), \tag{O.43}$$

with

$$\omega_0 = 1, \quad \omega_1 = -\frac{\sqrt{6}\,(\pi^2+72)}{144\pi}, \quad \omega_2 = \frac{\pi^2+432}{6912}, \quad \omega_3 = -\frac{\sqrt{6}\,(\pi^4+1296\pi^2+93312)}{2985984\,\pi}, \tag{O.44}$$

$$\omega_4 = \frac{\pi^4+2880\pi^2+933120}{286654464}, \tag{O.45}$$

numerically $\omega_1 = -0.4432879768735823 91\ldots, \omega_2 = 0.063927894155250197\ldots, \omega_3 = -0.027730933907464250\ldots,$ $\omega_4 = 0.0033547074632899 19\ldots.$

Proof. By Theorem O.30 with $(k,\sigma)=(1,+1)$, the displayed sum plus a convergent tail is exactly $P_{1,+}(n)$ divided by $\mathrm{e}^{\lambda\sqrt{n}}/(4\sqrt{3}\,n)$; the tail after M terms is $\mathcal{O}_M(n^{-M/2})$ because $\Omega_{1,+}$ is holomorphic at 0. The difference $p(n)-P_{1,+}(n)$ is $\mathcal{O}(\mu^{3/2}\mathrm{e}^{-\mu/2})$ relative to $P_{1,+}$ by Theorem O.24 with $K=1$, hence beyond every algebraic order. The closed forms follow from Equation (O.33) at $\beta=\pi/6$; they were recomputed here in exact arithmetic and agree with the tables of S1 and S2 (which print the same numbers in two different but equal groupings). ☐

Convergent sector versus asymptotic total. This is the trap of the subject and all three sources are, at some point, loose about it. The three statements below are *different*:

1. $\sum_{m\geq 0}\omega_m n^{-m/2}$ *converges* for every $n>1/24$ (Theorem O.30);
2. its sum is $P_{1,+}(n)\cdot 4\sqrt{3}\,n\,\mathrm{e}^{-\lambda\sqrt{n}}$, *not* $p(n)\cdot 4\sqrt{3}\,n\,\mathrm{e}^{-\lambda\sqrt{n}}$;
3. Equation (O.43) is an *asymptotic* statement about $p(n)$, whose error at every fixed order is a sum of two utterly different things: a convergent algebraic tail of size $\mathcal{O}(n^{-M/2})$ and an exponentially small Rademacher remainder of size $\mathcal{O}(\mu^{3/2}\mathrm{e}^{-\mu/2})$ relative to the dominant sector.

S1's "Dominant Poincaré expansion" subsection writes $p(n)\sim\cdots$ without separating (ii) from (iii) and, in its Theorem on the sector generating function, weakens (i) to "sufficiently small $|t|$". S3's remark after its dominant Poincaré display states the correct trichotomy in one sentence. S2 states it as a standalone remark and is the presentation adopted here. **The practical consequence:**

summing the ω_m series to convergence does *not* compute $p(n)$ to arbitrary accuracy; it computes $P_{1,+}(n)$, and the residual error stalls at the $\mathrm{e}^{-\mu/2}$ floor.

Remark O.37 (Least-term truncation degenerates here). Theorem J.54 determines the optimal truncation point of a *factorially divergent* algebraic series and the size of the resulting beyond-all-orders floor. Its hypothesis fails for $p(n)$: by Theorem O.30 the algebraic series of each sector has a finite positive radius of convergence, so there is no least term and no truncation-induced floor. The floor that does exist — the $\mathrm{e}^{-\mu/2}$ Rademacher gap — is not produced by resummation ambiguity but is supplied explicitly by the next sector. The partition function is thus the degenerate, best possible case of Theorem J.54: the exponentially small sectors are known exactly and independently of the perturbative data, and the optimal algebraic truncation is simply "as many terms as one wishes". This is the structural reason the numerical accuracy in Table O.3 improves so violently with K but not at all with extra algebraic orders past the $\mathrm{e}^{-\mu/2}$ level.

O.6 The multiplicative reciprocal $1/p(n)$

The phrase "inverse of the partition numbers" is ambiguous. The reciprocal is the easy reading and is disposed of first; it needs no reversion at all.

Proposition O.38 (Exact reciprocal transseries). *Let $\mu > 1$ be large enough that $|\mathcal{R}(\mu; n)| < 1$, with $\mathcal{R}$ as in Equation* (O.25). *Then*

$$\frac{1}{p(n)} = \frac{\mu^2}{\kappa}\,\frac{\mathrm{e}^{-\mu}}{1-\mu^{-1}}\sum_{r\ge 0}(-1)^r \mathcal{R}(\mu; n)^r, \tag{O.46}$$

the series converging. As a formal identity in Theorem O.19 it holds without the inequality. If $\mathcal{R}$ is written at finite level as a linear form $\sum_{j\in J_K} r_j(\mu; n)\, q_j$ in independent exponential variables $q_j = \mathrm{e}^{-\alpha_j \mu}$, then for every multi-index $\boldsymbol{\alpha} = (\alpha_j)_{j\in J_K} \neq \mathbf{0}$,

$$[\boldsymbol{q}^{\boldsymbol{\alpha}}]\,\big(1+\mathcal{R}\big)^{-1} = (-1)^{|\boldsymbol{\alpha}|}\,\frac{|\boldsymbol{\alpha}|!}{\boldsymbol{\alpha}!}\prod_{j\in J_K} r_j(\mu; n)^{\alpha_j}. \tag{O.47}$$

Proof. Invert Equation (O.22) and expand $(1+\mathcal{R})^{-1}$ geometrically; $1/F = \mu^2 \mathrm{e}^{-\mu}/(\kappa(1-\mu^{-1}))$. For Equation (O.47), only the term $(-1)^{|\boldsymbol{\alpha}|}\mathcal{R}^{|\boldsymbol{\alpha}|}$ of the geometric series can produce $\boldsymbol{q}^{\boldsymbol{\alpha}}$, and the multinomial theorem supplies the coefficient. ☐

Corollary O.39 (Reciprocal Poincaré coefficients). *Define η_m by $\big(\sum_{m\ge 0}\omega_m t^m\big)^{-1} = \sum_{m\ge 0}\eta_m t^m$, i.e.*

$$\eta_0 = 1, \qquad \eta_m = -\sum_{j=1}^{m}\omega_j \eta_{m-j} \quad (m\ge 1), \qquad \textit{equivalently} \qquad \eta_m = \frac{1}{m!}B_m\big(-1!\ell_1, \dots, -m!\ell_m\big) \tag{O.48}$$

with $\ell_j = \ell_j^{(1,+1)}$ of Equation (O.39)*. Then for each fixed M,*

$$\frac{1}{p(n)} = 4\sqrt{3}\,n\,\mathrm{e}^{-\lambda\sqrt{n}}\left(\sum_{m=0}^{M-1}\eta_m n^{-m/2} + \mathcal{O}_M(n^{-M/2})\right), \tag{O.49}$$

and $\sum_{m\ge 0}\eta_m t^m$ converges on $|t| < \sqrt{24}$ minus the zero set of $\Omega_{1,+}$, to $1/\Omega_{1,+}(t)$.

Proof. The recurrence is Cauchy inversion of a power series with unit constant term; the Bell form is the exponential formula applied to $\Omega_{1,+}^{-1} = \exp(-\sum \ell_j t^j)$. Convergence is inherited from Theorem O.30 and $\Omega_{1,+}(0) = 1$. The asymptotic statement follows from Theorem O.36 exactly as there; the same caveats of the repair box after Theorem O.36 apply verbatim, with "$P_{1,+}$" replaced by "$1/P_{1,+}$". ☐

Remark O.40. The reciprocal therefore requires no new analytic input and no series reversion. It is recorded because two of the three sources devote a section to it, and because it is the natural place to notice that the multinomial formula Equation (O.47) is the $|\boldsymbol{\alpha}|$-linear shadow of the far harder reversion formulas of Section O.9.

O.7 What "the inverse" of a discrete sequence means here

Proposition O.41 (Strict monotonicity)**.** $p(n) < p(n+1)$ *for every* $n \geq 1$.

Proof. Adjoining a part 1 injects the partitions of n into those of $n+1$, and the one-part partition $(n+1)$ is not in the image. □

By Theorem J.41 there are three distinct objects, and this chapter needs all three.

1. **The integer staircase** (the generalised inverse)
$$p^{\leftarrow}(y) = \min\{n \geq 1 : p(n) \geq y\}, \qquad y \geq 1. \tag{O.50}$$
Well defined by Theorem O.41. It is integer valued with unit jumps, so no smooth expansion can represent it to $o(1)$.
2. **The functional inverse of a chosen interpolation.** Any strictly increasing smooth $\mathcal{P}$ with $\mathcal{P}(n) = p(n)$ has an inverse $\mathcal{N}$, and $p^{\leftarrow}(y) = \lceil \mathcal{N}(y) \rceil$ (Theorem O.44). Theorem O.9 supplies the canonical choice used here.
3. **The inverse along the range.** If one knows in advance that $y = p(n)$ for an integer n, the target is that integer. A smooth formula that is accurate to $o(1/2)$ recovers it by rounding (Theorem O.83), and the accuracy available is exponentially small, not merely $o(1)$.

Definition O.42 (The Rademacher interpolation)**.** For real $\nu > 1/24$ put $\mu(\nu) = \frac{\pi}{6}\sqrt{24\nu - 1}$ and
$$\mathcal{P}(\nu) = \frac{2\kappa}{\mu(\nu)^2} \sum_{k \geq 1} \frac{\widetilde{A}_k(\nu)}{\sqrt{k}}\, g\!\left(\frac{\mu(\nu)}{k}\right), \tag{O.51}$$
with g as in Equation (O.18) and $\widetilde{A}_k$ as in Theorem O.9.

Proposition O.43 (Interpolation, regularity, eventual monotonicity)**.** *The series Equation* (O.51) *and all its derivatives converge locally uniformly on* $(1/24, \infty)$*;* $\mathcal{P}$ *is real analytic there and* $\mathcal{P}(n) = p(n)$ *for every integer* $n \geq 1$*. Moreover there is* ν_0 *with* $\mathcal{P}'(\nu) > 0$ *for* $\nu \geq \nu_0$*, so* $\mathcal{P}$ *has a real inverse* $\mathcal{N}$ *on* $[\mathcal{P}(\nu_0), \infty)$.

Proof. Fix a compact $I \subset (1/24, \infty)$ and a complex neighbourhood of it on which μ is holomorphic and bounded. By Theorem O.10 every ν-derivative of $\widetilde{A}_k$ is $\mathcal{O}_r(k)$ uniformly in k there, while $g(\mu/k) = \mathcal{O}(k^{-2})$ uniformly with all parameter derivatives (from Equation (O.18) and μ bounded). Hence the kth summand of Equation (O.51) and each of its derivatives is $\mathcal{O}(k \cdot k^{-1/2} \cdot k^{-2}) = \mathcal{O}(k^{-3/2})$, and the Weierstrass M-test gives locally uniform convergence and legitimises termwise differentiation. Equality at integers is Theorem O.10 combined with Theorem O.14.

Monotonicity: write $\mathcal{P}(\nu) = F(\mu(\nu))(1 + \mathcal{R}(\mu(\nu); \nu))$ as in Theorem O.17 with $\widetilde{A}_k$ in place of A_k. We have $\mathrm{d}\mu/\mathrm{d}\nu = \lambda^2/(2\mu) > 0$ and $\mathrm{d}\log F/\mathrm{d}\mu = 1 - 3/\mu + 1/(\mu - 1) \to 1$. Differentiating $\mathcal{R}$ term by term and applying Theorem O.23 with $K = 1$ together with Theorem O.10 shows $\mathcal{R} + \partial_\mu \mathcal{R} = \mathcal{O}(\mu^3 \mathrm{e}^{-\mu/2})$; note that differentiating $\widetilde{A}_k(\nu(\mu))$ costs a factor $\mathcal{O}(k\mu)$, absorbed by replacing $k^{-3/2}$ with $k^{-1/2}$ in the tail sum and hence by one more power of μ. Thus $\mathcal{P}'(\nu) = F'(\mu)\mu'(\nu)\big(1 + \mathcal{O}(\mu^3 \mathrm{e}^{-\mu/2})\big)$, which is positive for large ν. □

Theorem O.44 (Staircase from the smooth inverse)**.** *For all sufficiently large real y,*

$$p^{\leftarrow}(y) = \lceil \mathcal{N}(y) \rceil, \tag{O.52}$$

and $\mathcal{N}(p(n)) = n$ exactly for every integer $n \geq \nu_0$.

Proof. This is the specialisation of Theorem J.43 to the unit-spaced lattice $\mathbb{Z}$, whose separation hypothesis is Theorem O.41. Concretely: let $m = p^{\leftarrow}(y)$, so $p(m-1) < y \leq p(m)$. Since $\mathcal{P}$ is increasing on $[\nu_0, \infty)$ and $\mathcal{P}(j) = p(j)$, we get $m - 1 < \mathcal{N}(y) \leq m$, whence $\lceil \mathcal{N}(y) \rceil = m$. □

Remark O.45 (The unavoidable sawtooth). For general real y the difference $p^{\leftarrow}(y) - \mathcal{N}(y)$ is the ceiling defect, of size $\mathcal{O}(1)$, with Fourier expansion

$$\lceil v \rceil - v = \frac{1}{2} + \frac{1}{\pi} \sum_{r \geq 1} \frac{\sin(2\pi r v)}{r} \qquad (v \notin \mathbb{Z}), \tag{O.53}$$

the series converging to the midpoint $1/2$ at integers where the true defect is 0. Since the first arithmetic correction to $\mathcal{N}$ is of size $y^{-1/2}$ (Theorem O.73), *exponential precision in $\mathcal{N}$ buys nothing for $p^{\leftarrow}$ at generic y*: the order-one sawtooth is an intrinsic final sector of the discrete problem, not an error to be removed. It matters only along the range $y = p(n)$, where the defect vanishes identically. This warning is S3's and is the sharpest of the three.

O.8 The Lambert core and the algebraic inverse

Throughout this section and the next, y denotes a large ordinate and

$$R = \log \frac{y}{\kappa}, \qquad \ell = \log R. \tag{O.54}$$

O.8.1 The carrier

Proposition O.46 (Lambert carrier; specialisation of Theorem J.23)**.** *The large solution $X = X(y)$ of $\kappa \mathrm{e}^X X^{-2} = y$, equivalently of*

$$X - 2\log X = R, \tag{O.55}$$

is

$$X(y) = -2\, W_{-1}\!\left(-\frac{1}{2}\sqrt{\frac{\kappa}{y}}\right) = -2\, W_{-1}\!\left(-\frac{\pi}{\sqrt{24\sqrt{3}\, y}}\right). \tag{O.56}$$

Proof. This is Theorem J.23 at the dictionary values $a = 1$, $b = -2$, $C = \kappa$ of Table O.1: $X = (b/a) W_k((a/b)\mathrm{e}^{R/b}) = -2W_k(-\frac{1}{2}\mathrm{e}^{-R/2})$ and $\mathrm{e}^{-R/2} = \sqrt{\kappa/y}$. The branch rule of Theorem J.23 selects $k = -1$ because $X \to +\infty$ forces $-X/2 \leq -1$. The second form uses $\frac{1}{2}\sqrt{\kappa} = \frac{1}{2}\pi/\sqrt{6\sqrt{3}} = \pi/\sqrt{24\sqrt{3}}$. □

O.8.2 Flattening into logarithms

Theorem O.47 (Carrier coefficient polynomials)**.** *As $y \to \infty$,*

$$X \sim R + 2\ell + \sum_{j \geq 1} \frac{\Pi_j(\ell)}{R^j}, \tag{O.57}$$

in the sense of Theorem J.37, where

$$\Pi_j(\ell) = \frac{2^{j+1}}{j+1} [v^j] \left(\ell + \log(1+v)\right)^{j+1} = 2^{j+1} \sum_{q=1}^{j} \frac{s(j,q)}{(j+1-q)!}\, \ell^{j+1-q}, \tag{O.58}$$

with $s(j,q)$ the signed Stirling numbers of the first kind, $v(v-1)\cdots(v-j+1) = \sum_q s(j,q)v^q$. For each fixed J,

$$X = R + 2\ell + \sum_{j=1}^{J} \frac{\Pi_j(\ell)}{R^j} + \mathcal{O}\left(\frac{\ell^{\,J+1}}{R^{J+1}}\right). \tag{O.59}$$

Moreover $\deg \Pi_j = j$ *with leading coefficient* $(-1)^{j-1}2^{j+1}/j$, *and* $\Pi_j(0) = 0$.

Proof. Write $X = R(1+v)$ in Equation (O.55): $R + Rv - 2\log R - 2\log(1+v) = R$, i.e.

$$v = z\big(\ell + \log(1+v)\big), \qquad z = \frac{2}{R}, \tag{O.60}$$

a Lagrange fixed point $v = z\phi(v)$ with $\phi(v) = \ell + \log(1+v)$ and $\phi(0) = \ell \neq 0$. Theorem J.17 gives $v = \sum_{m\geq 1} \frac{z^m}{m}[v^{m-1}]\,\phi(v)^m$; multiplying by R and putting $m = j+1$ (so $z^m R = 2^{j+1}R^{-j}$) yields the first equality of Equation (O.58). For the second, expand $(\ell+\log(1+v))^{j+1} = \sum_q \binom{j+1}{q}\ell^{\,j+1-q}\log(1+v)^q$ and use $\log(1+v)^q = q!\sum_{m\geq q} s(m,q)v^m/m!$; extracting $[v^j]$ keeps only $1 \leq q \leq j+1$, and the term $q = j+1$ contributes $s(j,j+1) = 0$. The surviving coefficient of $\ell^{\,j+1-q}$ is $\frac{2^{j+1}}{j+1}\binom{j+1}{q}q!\,\frac{s(j,q)}{j!} = 2^{j+1}\frac{s(j,q)}{(j+1-q)!}$, as claimed. Since $q \geq 1$, every monomial carries a positive power of ℓ, so $\Pi_j(0) = 0$; the top power is $q = 1$, with $s(j,1) = (-1)^{j-1}(j-1)!$, giving leading coefficient $2^{j+1}(-1)^{j-1}(j-1)!/j! = (-1)^{j-1}2^{j+1}/j$. The remainder statement Equation (O.59) is the general estimate of Theorem J.37 applied on the scale $\ell/R \to 0$; concretely, Equation (O.60) has an analytic solution $v = v(z,\ell)$ near $z = 0$ for ℓ in any fixed compact set, and $\ell/R \to 0$ lets Taylor's theorem with remainder be applied at order J. □

The first polynomials, recomputed here from the Lagrange form and independently from the Stirling form (they agree identically), are

$$\begin{aligned} &\Pi_1 = 4\ell, && \Pi_2 = -4\ell(\ell-2),\\ &\Pi_3 = \tfrac{8}{3}\ell\,(2\ell^2 - 9\ell + 6), && \Pi_4 = -\tfrac{8}{3}\ell\,(3\ell^3 - 22\ell^2 + 36\ell - 12),\\ &\Pi_5 = \tfrac{16}{15}\ell\,(12\ell^4 - 125\ell^3 + 350\ell^2 - 300\ell + 60), && \Pi_6 = -\tfrac{16}{15}\ell\,(20\ell^5 - 274\ell^4 + 1125\ell^3 - 1700\ell^2 + 900\ell - 120), \end{aligned} \tag{O.61}$$

$$\Pi_7 = \tfrac{32}{105}\ell\left(120\ell^6 - 2058\ell^5 + 11368\ell^4 - 25725\ell^3 + 24500\ell^2 - 8820\ell + 840\right), \tag{O.62}$$

so that, explicitly,

$$X = R + 2\ell + \frac{4\ell}{R} + \frac{-4\ell^2 + 8\ell}{R^2} + \frac{\frac{16}{3}\ell^3 - 24\ell^2 + 16\ell}{R^3} + \frac{-8\ell^4 + \frac{176}{3}\ell^3 - 96\ell^2 + 32\ell}{R^4} + \cdots. \tag{O.63}$$

Remark O.48 (Three printings of one polynomial family). The sources print Equation (O.58) in three different shapes, and it is worth recording once that they are the same family, because a reader comparing them will otherwise suspect an error.

- S1: $P_j(L) = 2^{j+1}\sum_{q=1}^{j}\frac{s(j,q)}{(j+1-q)!}L^{j+1-q}$ — identical to Equation (O.58).
- S2: $P_m(\ell) = \frac{1}{m!}\sum_{q=1}^{m}2^q(q-1)!\,s(m,q)\binom{m}{q-1}(2\ell)^{m-q+1}$. The coefficient of $\ell^{\,m-q+1}$ is $\frac{2^q(q-1)!}{m!}\binom{m}{q-1}2^{m-q+1} = \frac{2^{m+1}s(m,q)}{(m-q+1)!}$ after cancelling $\binom{m}{q-1} = \frac{m!}{(q-1)!(m-q+1)!}$. Identical.
- S3: $P_r(\ell) = \frac{1}{r!}\sum_{q=1}^{r}(q-1)!\,s(r,q)\binom{r}{q-1}\ell^{\,r-q+1}$, appearing in the expansion as $2^{r+1}P_r(\ell)$. The same cancellation gives $2^{r+1}P_r = \Pi_r$. Identical.

But the variables differ, and this is a genuine disagreement. S1 and S2 expand in $R = \log(y/\kappa)$ with $\ell = \log R$; S3 expands in $H = \log(4y/\kappa) = R + 2\log 2$ with $\ell_{\mathrm{S3}} = \log(H/2)$, which is the natural variable for the substitution $X = 2w$, $w - \log w = H/2$. The two series

$$X = R + 2\log R + \sum_j \frac{\Pi_j(\log R)}{R^j} \qquad \text{and} \qquad X = H + 2\log\tfrac{H}{2} + \sum_j \frac{\Pi_j(\log\frac{H}{2})}{H^j}$$

are *different* asymptotic expansions of the same function, agreeing order by order only after re-expanding $H = R + 2\log 2$. Both are correct; mixing them is not. This volume uses the S1/S2 variables R, ℓ throughout, because R makes the phase equation exactly Equation (O.55) with no additive constant. We verified all of S3's displayed coefficients (its u- and N_0-expansions in H, through H^{-4} and H^{-3} respectively) by symbolic re-expansion; they are correct as printed in S3's own variables.

O.8.3 Restoring the dominant amplitude: the centred reversion

The carrier X solves $y = \kappa e^X X^{-2}$; the *dominant core* $\mu_0 = \mu_0(y)$ solves the full dominant sector,

$$y = F(\mu_0) = \kappa\, e^{\mu_0} \mu_0^{-2} \left(1 - \frac{1}{\mu_0}\right), \qquad \text{i.e.} \qquad \Phi(\mu_0) = R, \tag{O.64}$$

where the *dominant phase* is

$$\Phi(\mu) = \mu - 2\log\mu + \log\left(1 - \frac{1}{\mu}\right) = \mu - 3\log\mu + \log(\mu - 1), \qquad \mu > 1. \tag{O.65}$$

Lemma O.49 (Derivatives of the dominant phase)**.** *For $\mu > 1$,*

$$\Phi'(\mu) = 1 - \frac{3}{\mu} + \frac{1}{\mu - 1} = 1 - \frac{2}{\mu} + \frac{1}{\mu(\mu-1)} = \frac{\mu^2 - 3\mu + 3}{\mu(\mu - 1)} > 0, \tag{O.66}$$

and for $m \geq 1$

$$\Phi^{(m)}(\mu) = \mathbf{1}_{m=1} + (-1)^{m-1}(m-1)! \left(\frac{1}{(\mu-1)^m} - \frac{3}{\mu^m}\right). \tag{O.67}$$

In particular $\Phi'(\mu) = 1 + \mathcal{O}(\mu^{-1})$, and $\mu^2 - 3\mu + 3 > 0$ for all real μ since its discriminant is -3. Writing $\varrho_\pm = \frac{1}{2}(3 \pm i\sqrt{3})$ for the roots of $\mu^2 - 3\mu + 3$,

$$F'(\mu) = \kappa\, e^{\mu} \frac{\mu^2 - 3\mu + 3}{\mu^4}, \qquad \frac{d^m}{d\mu^m} \log F'(\mu) = \mathbf{1}_{m=1} + (-1)^{m-1}(m-1)! \left(\frac{1}{(\mu - \varrho_+)^m} + \frac{1}{(\mu - \varrho_-)^m} - \frac{4}{\mu^m}\right). \tag{O.68}$$

Proof. Differentiate Equation (O.65) in the form $\mu - 3\log\mu + \log(\mu - 1)$ and clear denominators for the three displayed shapes of Φ'; Equation (O.67) is $m - 1$ further differentiations of $-3\mu^{-1} + (\mu - 1)^{-1}$. For Equation (O.68), $F = \kappa e^{\mu}(\mu - 1)\mu^{-3}$ gives $F' = F\big(1 + \frac{1}{\mu-1} - \frac{3}{\mu}\big) = F\,\Phi'(\mu)$, and $F\Phi' = \kappa e^{\mu} \frac{\mu-1}{\mu^3} \cdot \frac{\mu^2-3\mu+3}{\mu(\mu-1)}$. Then $\log F' = \log\kappa + \mu + \log(\mu - \varrho_+) + \log(\mu - \varrho_-) - 4\log\mu$, and differentiate. □

Theorem O.50 (The dominant-core correction; specialisation of Theorem J.29)**.** *Put $z = X^{-1}$. There is a unique $U \in z\,\mathbb{Q}[[z]]$ with*

$$U = 3\log(1 + zU) - \log(1 - z + zU), \tag{O.69}$$

and U is analytic in a neighbourhood of $z = 0$. The dominant core is

$$\mu_0(y) = X(y) + U\big(X(y)^{-1}\big), \qquad U(z) = \sum_{m\geq 1} c_m z^m. \tag{O.70}$$

Moreover Equation (O.69) *is exactly the fixed-point equation $U = \Psi(z, U)$ of Theorem J.29 at the dictionary values of Table O.1, so*

$$c_m = \sum_{q=1}^{m} \frac{1}{q}\, [z^m v^{q-1}] \Big(3\log(1 + zv) - \log(1 - z + zv)\Big)^q \tag{O.71}$$

by Theorem J.19, and equivalently

$$c_m = [z^m]\Big\{3\log\big(1 + zU_{<m}\big) - \log\big(1 - z + zU_{<m}\big)\Big\}, \qquad U_{<m}(z) = \sum_{j=1}^{m-1} c_j z^j, \tag{O.72}$$

a triangular recurrence over $\mathbb{Q}$*.*

Proof. Put $\mu = X + U$ and divide Equation (O.64) by the carrier identity $y = \kappa e^X X^{-2}$:

$$e^U (1 + zU)^{-2} \left(1 - \frac{z}{1 + zU}\right) = 1, \qquad z = \frac{1}{X},$$

since $\mu = X(1 + zU)$ and $1/\mu = z/(1 + zU)$. As $1 - \frac{z}{1+zU} = \frac{1-z+zU}{1+zU}$, taking logarithms gives $U - 3\log(1 + zU) + \log(1 - z + zU) = 0$, i.e. Equation (O.69). Write $G(z, U)$ for the left side: $G(0, 0) = 0$ and $\partial_U G(0, 0) = 1$, so the analytic implicit function theorem gives a unique analytic solution near 0, necessarily with rational Taylor coefficients since G has rational ones.

For the identification with Theorem J.29: that theorem's fixed-point map at parameters a, b, Q is $\Psi(t, u) = -\frac{1}{a}\big[b\log(1 + tu) + Q\big(\frac{t}{1+tu}\big)\big]$. At $a = 1$, $b = -2$, $Q(s) = \log(1 - s)$ this is

$$\Psi(t, u) = 2\log(1+tu) - \log\left(1 - \frac{t}{1 + tu}\right) = 2\log(1+tu) - \log(1-t+tu) + \log(1+tu) = 3\log(1+tu) - \log(1-t+tu),$$

which is the right side of Equation (O.69). Equation (O.71) is then Theorem J.19 verbatim. Equation (O.72) holds because in Equation (O.69) every occurrence of U on the right is multiplied by an explicit z, so $[z^m]$ of the right side involves only $c_1, \dots, c_{m-1}$. □

Table O.2: The centred-reversion coefficients c_m of Equation (O.70), recomputed here in exact rational arithmetic from Equation (O.72) (truncated power-series arithmetic over $\mathbb{Q}$, $M = 20$). All eighteen agree with the tables printed by the three sources, which call them b_m (S1), a_m (S2) and d_r (S3).

$c_1 = 1$	$c_2 = \frac{5}{2}$	$c_3 = \frac{13}{3}$	$c_4 = \frac{53}{12}$
$c_5 = -\frac{14}{5}$	$c_6 = -\frac{763}{30}$	$c_7 = -\frac{2179}{35}$	$c_8 = -\frac{42289}{630}$
$c_9 = \frac{132973}{1260}$	$c_{10} = \frac{3449711}{5040}$	$c_{11} = \frac{29813143}{18480}$	$c_{12} = \frac{496569589}{356400}$
$c_{13} = -\frac{4334124833}{926640}$	$c_{14} = -\frac{509723063851}{21621600}$	$c_{15} = -\frac{298798016297}{5896800}$	$c_{16} = -\frac{933244850273}{33633600}$
$c_{17} = \frac{4993013022640963}{23156733600}$	$c_{18} = \frac{208843210189406957}{231567336000}$		

Thus

$$\mu_0 = X + \frac{1}{X} + \frac{5}{2X^2} + \frac{13}{3X^3} + \frac{53}{12X^4} - \frac{14}{5X^5} - \frac{763}{30X^6} - \frac{2179}{35X^7} + \cdots. \tag{O.73}$$

Remark O.51 (Implementation check). The cheapest correctness test for a table of c_m is the residual identity: with $U_M = \sum_{m \le M} c_m z^m$,

$$U_M - 3\log(1 + zU_M) + \log(1 - z + zU_M) = \mathcal{O}(z^{M+1}). \tag{O.74}$$

We verified Equation (O.74) at $M = 12$ symbolically: the residual is $\mathcal{O}(z^{13})$ exactly. This test is far more reliable than comparing printed digits, and it is the one we used to certify Table O.2.

O.8.4 The algebraic inverse index

Definition O.52. $N_0(y) = \dfrac{1}{24} + \dfrac{3\,\mu_0(y)^2}{2\pi^2}$, the *algebraic* (phase-independent) inverse index; and $N_X(y) = \dfrac{1}{24} + \dfrac{3X(y)^2}{2\pi^2}$, the carrier index.

Corollary O.53 (All algebraic coefficients of the inverse index). *With c_m as in Table O.2, put*

$$\gamma_j = 3c_{j+1} + \frac{3}{2}\sum_{r=1}^{j-1} c_r c_{j-r} \qquad (j \geq 1). \tag{O.75}$$

Then

$$N_0(y) = \frac{3X^2}{2\pi^2} + \left(\frac{1}{24} + \frac{3}{\pi^2}\right) + \frac{1}{\pi^2}\sum_{j\geq 1}\frac{\gamma_j}{X^j}, \tag{O.76}$$

with

$$\gamma_1 = \tfrac{15}{2}, \quad \gamma_2 = \tfrac{29}{2}, \quad \gamma_3 = \tfrac{83}{4}, \quad \gamma_4 = \tfrac{559}{40}, \quad \gamma_5 = -\tfrac{611}{20}, \quad \gamma_6 = -\tfrac{112459}{840}, \tag{O.77}$$

$$\gamma_7 = -\tfrac{101329}{420}, \quad \gamma_8 = -\tfrac{228677}{3360}, \quad \gamma_9 = \tfrac{1709167}{1680}, \quad \gamma_{10} = \tfrac{325098619}{92400}. \tag{O.78}$$

Proof. Insert $\mu_0 = X + U$ into Theorem O.52. The cross term $\frac{3}{2\pi^2}\cdot 2XU = \frac{3}{\pi^2}\sum_{m\geq 1} c_m X^{1-m}$ contributes $3c_1/\pi^2 = 3/\pi^2$ to the constant and $3c_{j+1}/(\pi^2 X^j)$ at order X^{-j}; the square $\frac{3}{2\pi^2}U^2$ contributes $\frac{3}{2\pi^2}\left(\sum_{r=1}^{j-1} c_r c_{j-r}\right)X^{-j}$ and nothing at order X^0 or above. The listed values were recomputed from Equation (O.75); $\gamma_1, \ldots, \gamma_8$ agree with the tables of all three sources, and γ_9, γ_{10} are new here. □

Corollary O.54 (The carrier bias). *$N_0(y) - N_X(y) = \dfrac{3}{\pi^2} + \mathcal{O}(X^{-1})$, and*

$$\frac{3}{\pi^2} = 0.30396355092701331433\ldots \tag{O.79}$$

Proof. Subtract $N_X = \frac{1}{24} + 3X^2/(2\pi^2)$ from Equation (O.76): the $3X^2/(2\pi^2)$ and $\frac{1}{24}$ cancel, leaving $3/\pi^2 + \pi^{-2}\sum_{j\geq 1}\gamma_j X^{-j}$. The constant was evaluated here to 20 digits. □

Remark O.55 (Phase locking: do not expand the arithmetic phase about the carrier). This is the single most consequential structural point of the inverse problem, and only S2 states it explicitly.

The amplitudes $\widetilde{A}_k$ are *periodic on the index scale*, with period k. The core correction is small in the phase variable, $\mu_0 - X = \mathcal{O}(X^{-1})$; but $\mathrm{d}n/\mathrm{d}\mu = 3\mu/\pi^2 \asymp X$, so the induced change in the index is $\mathcal{O}(1)$ — exactly $3/\pi^2$ in the limit, by Theorem O.54. A shift of 0.304 in the index is *not* a small perturbation of $\cos(\pi\nu)$. Consequently:

- Taylor-expanding $\widetilde{A}_k(N(\mu_0))$ around N_X does not begin with a small correction, and produces the wrong phase already at the leading exponentially small order.
- Every arithmetic amplitude must be evaluated at the *full algebraic core* $N_0(y)$, not at $N_X(y)$ and not at any truncation of N_0 coarser than $o(1)$. By Equation (O.76), truncating the γ-series after M terms leaves an index error $\mathcal{O}(X^{-M})$, so $M \geq 1$ already suffices for $o(1)$; $M = 0$ (i.e. dropping the constant $3/\pi^2$) does not.
- Only the *subsequent* exponentially small displacement may be Taylor-expanded; that is what Section O.9 does.

The numerical signature of the effect is the first column of Table O.4, which tends to $-3/\pi^2$ and not to 0.

Remark O.56 (Elementary-logarithm form). Substituting Equation (O.57) into Equation (O.76) gives the pure logarithmic block

$$N_0(y) = \frac{3R^2}{2\pi^2} + \frac{6R\ell}{\pi^2} + \frac{6\ell^2 + 12\ell + 3}{\pi^2} + \frac{1}{24} + \frac{3(8\ell^2 + 16\ell + 5)}{2\pi^2 R} + \mathcal{O}\left(\frac{\ell^3}{R^2}\right), \tag{O.80}$$

with $R = \log(y/\kappa)$ and $\ell = \log R$. We recomputed this block symbolically; it agrees with S2's display (which stops at the constant) and, after the change of variable $H = R + 2\log 2$, with S3's. For *numerical* work Equation (O.80) is markedly inferior to evaluating Equation (O.56) and Equation (O.73) directly, and it is recorded only to exhibit the transseries shape: N_0 is a polynomial of degree 2 in R with coefficients polynomial in ℓ, plus a descending series in R^{-1} with $\deg_\ell$ growing by one per order.

O.9 The full arithmetic inverse transseries

Everything so far is phase-independent: X, μ_0, N_0 are functions of y alone. The remaining correction is exponentially small and arithmetically oscillatory, and the oscillation must be frozen before "analytic inverse" has a meaning. The three sources do this in two different ways, both correct.

O.9.1 Two ways to freeze the phase

Definition O.57 (Residue-frozen finite level; S1)**.** For $K \geq 1$ let $\Lambda_K = \operatorname{lcm}(1, 2, \dots, K)$ and fix a residue $r \in \mathbb{Z}/\Lambda_K\mathbb{Z}$. Since A_k has period $k \mid \Lambda_K$ for $k \leq K$, the numbers $A_k(r)$ are well defined, and

$$P_{K,r}(\mu) = \frac{\kappa}{\mu^2} \sum_{k=1}^{K} \frac{A_k(r)}{\sqrt{k}} \left[e^{\mu/k}\Big(1 - \frac{k}{\mu}\Big) + e^{-\mu/k}\Big(1 + \frac{k}{\mu}\Big) \right] \tag{O.81}$$

is a genuine analytic function of μ with constant coefficients. For large μ it is increasing (the $k = 1$ positive term dominates), and we write $\mu_{K,r}(y)$ for its large inverse branch.

Definition O.58 (Interpolated level; S2, S3)**.** $\mathcal{P}_K(\mu)$ is Equation (O.81) with $A_k(r)$ replaced by $\widetilde{A}_k(n(\mu))$, $n(\mu) = \frac{1}{24} + 3\mu^2/(2\pi^2)$; its large inverse branch is written $\mu_K(y)$.

Proposition O.59 (The two conventions agree below action one)**.** *Fix K, let y be large, and let $r \equiv \lfloor N_0(y) \rceil \pmod{\Lambda_K}$, where $\lfloor\cdot\rceil$ is nearest-integer rounding. Assume $|N_0(y) - r| \leq \frac{1}{2}$ modulo Λ_K has been resolved (which holds along the range $y = p(n)$ for large n by Theorem O.83). Then every inverse transmonomial of total action strictly less than 1 is the same for $\mu_{K,r}$ and μ_K; the two first differ at total action ≥ 1.*

Proof. Both inverses are constructed as $\mu_0 + v$ with v exponentially small, and in both the coefficient of the transmonomial $e^{-\alpha_j \mu_0}$ at first order is $-r_j(\mu_0)/\Phi'(\mu_0)$ (Theorem O.69 below), where r_j differs between the two conventions only in the amplitude, $A_k(r)$ versus $\widetilde{A}_k(n(\mu_0)) = \widetilde{A}_k(N_0(y))$. Now $\widetilde{A}_k$ is entire and k-periodic with $\widetilde{A}_k(r) = A_k(r)$, so $|\widetilde{A}_k(N_0) - A_k(r)| \leq \sup|\widetilde{A}'_k| \cdot |N_0 - r|$. Along the range, $|N_0(p(n)) - n| = \mathcal{O}(\mu^{5/2} e^{-\mu/2})$ by Theorem O.79, so the amplitude discrepancy is $\mathcal{O}(k\mu^{5/2} e^{-\mu/2})$ and its contribution to the sector carries the extra factor $e^{-\mu/2}$, i.e. additional action $\frac{1}{2}$. Since the smallest primary action is $\frac{1}{2}$, the first affected total action is $\frac{1}{2} + \frac{1}{2} = 1$. Higher-order terms of the reversion have total action ≥ 1 already. ☐

Remark O.60 (Which convention to prefer). The residue-frozen convention is intrinsic: it invents no interpolation, and at each finite level it reduces the problem to an ordinary analytic reversion with *constant* coefficients. Its cost is that the inverse is defined only after the residue class is known, which Theorem O.83 supplies. The interpolated convention defines a single function $\mathcal{N}$ for all real y and connects directly to the staircase (Theorem O.44); its cost is the interpolation dependence of Theorem O.11 and the appearance of the amplitude's own μ-derivatives in every higher coefficient. This chapter develops the reversion in a form that covers both: all formulas below are stated for a family of *blocks* $E_j(\mu)$, and the two conventions are two choices of block.

O.9.2 Blocks, multi-indices, and the reversion equation

Definition O.61 (Sector blocks)**.** Fix $K \geq 1$ and put

$$J_K = \{(1, -1)\} \cup \{(k, \sigma) : 2 \leq k \leq K,\ \sigma = \pm 1\}. \tag{O.82}$$

For $j = (k, \sigma) \in J_K$ set $\alpha_j = 1 - \sigma/k$ as in Equation (O.21), and define the *block*

$$E_j(\mu) = \frac{\kappa}{\mu^2} \frac{a_k(\mu)}{\sqrt{k}} \left(1 - \frac{\sigma k}{\mu}\right) e^{\sigma\mu/k} = F(\mu)\, r_j(\mu)\, e^{-\alpha_j \mu}, \qquad r_j(\mu) = \frac{a_k(\mu)}{\sqrt{k}} \frac{1 - \sigma k/\mu}{1 - 1/\mu}, \tag{O.83}$$

where $a_k(\mu)$ is either the constant $A_k(r)$ (residue-frozen convention, Theorem O.57) or $\widetilde{A}_k(n(\mu))$ (interpolated convention, Theorem O.58); $a_1 \equiv 1$ in both. A block may also be the whole paired tail $\mathcal{R}_{>K} \cdot F$ of Equation (O.25), or the zeta-condensation block of Theorem O.28, treated as one indivisible symbol.

Introduce independent formal variables ε_j, $j \in J_K$, and write $\boldsymbol{\varepsilon}^{\boldsymbol{m}} = \prod_j \varepsilon_j^{m_j}$, $|\boldsymbol{m}| = \sum_j m_j$, $\boldsymbol{m}! = \prod_j m_j!$ for multi-indices $\boldsymbol{m} \in \mathbb{N}^{J_K}$. The forward object at level K is

$$\mathcal{P}_K(\mu; \boldsymbol{\varepsilon}) = F(\mu) + \sum_{j \in J_K} \varepsilon_j E_j(\mu), \tag{O.84}$$

and the inverse problem is to solve $\mathcal{P}_K(\mu; \boldsymbol{\varepsilon}) = y$ for μ near $\mu_0 = F^{-1}(y)$, as a formal series in $\boldsymbol{\varepsilon}$ which is finally specialised at $\varepsilon_j = 1$. Two equivalent engines are available; both are finite at every prescribed multi-index.

O.9.3 Engine I: additive Lagrange–Bürmann

Theorem O.62 (Additive reversion). *Let F and E be analytic near μ_0 with $F'(\mu_0) \neq 0$ and $F(\mu_0) = y$. Write $\mathcal{D} = \frac{1}{F'(\mu)} \frac{\mathrm{d}}{\mathrm{d}\mu}$ evaluated along $\mu = \mu_0(y)$, i.e. $\mathcal{D} = \mathrm{d}/\mathrm{d}y$ in the coordinate y. Then the solution of $F(\mu) + \varepsilon E(\mu) = y$ near μ_0 is*

$$\mu(y; \varepsilon) = \mu_0(y) + \sum_{M \geq 1} \frac{(-\varepsilon)^M}{M!} \mathcal{D}^{M-1} \left(\frac{E(\mu_0)^M}{F'(\mu_0)} \right). \tag{O.85}$$

Consequently, for the multivariate problem Equation (O.84)*, the coefficient of $\boldsymbol{\varepsilon}^{\boldsymbol{m}}$ with $M = |\boldsymbol{m}| \geq 1$ in $\mu - \mu_0$ is*

$$V_{\boldsymbol{m}}(\mu_0) = \frac{(-1)^M}{\boldsymbol{m}!} \mathcal{D}^{M-1} \left(\frac{E^{\boldsymbol{m}}(\mu_0)}{F'(\mu_0)} \right) = \frac{(-1)^M (M-1)!}{\boldsymbol{m}!} \operatorname*{Res}_{\mu=\mu_0} \frac{E^{\boldsymbol{m}}(\mu)}{(F(\mu) - F(\mu_0))^M}, \tag{O.86}$$

where $E^{\boldsymbol{m}} = \prod_j E_j^{m_j}$.

Proof. Both displays are Theorem J.21 in the same letters: Equation (O.85) is (J.34) and Equation (O.86) is (J.36).

What that theorem supplies is convergence for $|\varepsilon| < \varepsilon_0$, with ε_0 read off a small circle about μ_0; here the specialisation $\varepsilon_j = 1$ is wanted, and for the partition blocks it is legitimate. With $E = \sum_j E_j + \mathcal{R}_{>K} F$ we have $E/F = \mathcal{O}(\mu^C \mathrm{e}^{-\mu/2})$ together with the same bound for finitely many derivatives, by Theorem O.23 and Theorem O.10; so on a fixed relative neighbourhood of a large μ_0 the Rouché comparison made in the proof of Theorem J.21 holds for all $|\varepsilon| \leq 1 + \eta$ with some $\eta > 0$, and the Taylor series in ε converges at $\varepsilon = 1$. □

Remark O.63 ($\mathcal{D}$ is iterated, not a plain derivative). In Equation (O.85) the operator $\mathcal{D}^{M-1}$ is the $(M-1)$st power of $\frac{1}{F'} \frac{\mathrm{d}}{\mathrm{d}\mu}$. Replacing it by $\frac{1}{(F')^{M-1}} \frac{\mathrm{d}^{M-1}}{\mathrm{d}\mu^{M-1}}$ after "pulling out" the powers of F' is wrong from $M = 3$ onwards, because F' is not constant. S3 flags this explicitly; it is the most common implementation bug in this formula.

Corollary O.64 (Bell realisation). *Let $H_{\boldsymbol{m}} = E^{\boldsymbol{m}}/F'$ and $\Lambda_s^{(\boldsymbol{m})} = \frac{\mathrm{d}^s}{\mathrm{d}\mu^s} \log H_{\boldsymbol{m}}$, and let $\mu_0^{(s)} = \mathrm{d}^s \mu_0 / \mathrm{d}y^s$ be generated by*

$$\mu_0^{(1)} = \frac{1}{F'(\mu_0)}, \qquad \mu_0^{(s)} = -\frac{1}{F'(\mu_0)} \sum_{q=2}^{s} F^{(q)}(\mu_0)\, B_{s,q}\big(\mu_0^{(1)}, \ldots, \mu_0^{(s-q+1)}\big), \quad s \geq 2. \tag{O.87}$$

Then, with $M = |\boldsymbol{m}| \geq 1$,

$$V_{\boldsymbol{m}} = \frac{(-1)^M H_{\boldsymbol{m}}(\mu_0)}{\boldsymbol{m}!} \sum_{s=0}^{M-1} B_{M-1,s}\big(\mu_0^{(1)}, \mu_0^{(2)}, \dots\big)\, B_s\big(\Lambda_1^{(\boldsymbol{m})}, \dots, \Lambda_s^{(\boldsymbol{m})}\big), \tag{O.88}$$

with the conventions $B_{0,0} = B_0 = 1$.

Proof. Faà di Bruno (Theorem J.10) gives $\frac{\mathrm{d}^{M-1}}{\mathrm{d}y^{M-1}} H_{\boldsymbol{m}}(\mu_0(y)) = \sum_s H_{\boldsymbol{m}}^{(s)}(\mu_0) B_{M-1,s}(\mu_0^{(1)}, \dots)$, and writing $H_{\boldsymbol{m}} = \exp(\log H_{\boldsymbol{m}})$ gives $H_{\boldsymbol{m}}^{(s)} = H_{\boldsymbol{m}}\, B_s(\Lambda_1^{(\boldsymbol{m})}, \dots)$. Substitute in Equation (O.86). Equation (O.87) is obtained by differentiating $F(\mu_0(y)) = y$ s times and solving the Faà di Bruno identity for the term containing $\mu_0^{(s)}$. □

The point of Equation (O.88) is that every ingredient is elementary and closed-form for the partition blocks. Indeed, splitting each aggregate block into its primitive Fourier constituents

$$E_{k,h,\sigma,\epsilon}(\mu) = \frac{\kappa}{2\sqrt{k}} \frac{\mu - \sigma k}{\mu^3} \exp\left[\frac{\sigma\mu}{k} + \mathrm{i}\epsilon\Big(\pi s(h,k) - \frac{2\pi h}{k} n(\mu)\Big)\right], \qquad h \in U_k,\ \epsilon = \pm 1, \tag{O.89}$$

and using $n'(\mu) = 3\mu/\pi^2$, $n''(\mu) = 3/\pi^2$, $n^{(s)} = 0$ for $s \geq 3$, one gets the closed logarithmic jets

$$\frac{\mathrm{d}^s}{\mathrm{d}\mu^s} \log E_{k,h,\sigma,\epsilon} = (-1)^{s-1}(s-1)! \left(\frac{1}{(\mu - \sigma k)^s} - \frac{3}{\mu^s}\right) + \mathbf{1}_{s=1}\frac{\sigma}{k} - \mathrm{i}\epsilon \frac{2\pi h}{k}\, n^{(s)}(\mu), \tag{O.90}$$

while $\frac{\mathrm{d}^s}{\mathrm{d}\mu^s} \log F'$ is Equation (O.68). Hence $\Lambda_s^{(\boldsymbol{m})} = \sum_j m_j$ (Equation (O.90)) − (Equation (O.68)) with no hidden differentiation anywhere. This is S3's presentation and is the most explicit of the three.

O.9.4 Engine II: the triangular Newton–Hensel recurrence

The reversion may equally be organised in the phase coordinate, which is cheaper when many coefficients are wanted.

Theorem O.65 (Triangular recurrence)**.** *Put $\mu = \mu_0 + v$ and*

$$Q(v; \boldsymbol{\varepsilon}) = \log\Big(1 + \sum_{j \in J_K} \varepsilon_j B_j(v)\Big), \qquad B_j(v) = r_j(\mu_0 + v)\, \mathrm{e}^{-\alpha_j v}, \tag{O.91}$$

the bookkeeping variable ε_j standing for the actual exponential $\mathrm{e}^{-\alpha_j \mu_0}$, which is substituted only at the very end; the residual v-dependence of $\mathrm{e}^{-\alpha_j(\mu_0 + v)}$ is the factor $\mathrm{e}^{-\alpha_j v}$ inside B_j. Keeping the ε_j independent until the last step is what makes resonant exponent sums ($\alpha_j + \alpha_{j'} = \alpha_{j''}$, which does occur here) come out right. Then the reversion equation is

$$\Phi'(\mu_0)\, v + \sum_{s \geq 2} \frac{\Phi^{(s)}(\mu_0)}{s!} v^s + Q(v; \boldsymbol{\varepsilon}) = 0, \tag{O.92}$$

and its unique formal solution $v = \sum_{\boldsymbol{m} \neq \boldsymbol{0}} V_{\boldsymbol{m}} \boldsymbol{\varepsilon}^{\boldsymbol{m}}$ obeys

$$V_{\boldsymbol{m}} = -\frac{1}{\Phi'(\mu_0)} [\boldsymbol{\varepsilon}^{\boldsymbol{m}}] \left\{ \sum_{s \geq 2} \frac{\Phi^{(s)}(\mu_0)}{s!}\, v_{<\boldsymbol{m}}^s + Q(v_{<\boldsymbol{m}}; \boldsymbol{\varepsilon}) \right\}, \tag{O.93}$$

where $v_{<\boldsymbol{m}}$ collects the already-computed coefficients. Each right-hand side is a finite expression in μ_0, $(\mu_0 - 1)^{-1}$, the actions α_j and the amplitudes a_k.

Proof. Taking logarithms in $\mathcal{P}_K(\mu;\boldsymbol{\varepsilon}) = y$ and subtracting $\Phi(\mu_0) = R$ gives $\Phi(\mu_0+v) - \Phi(\mu_0) + Q(v;\boldsymbol{\varepsilon}) = 0$; Taylor expansion of Φ at μ_0 is Equation (O.92), legitimate as a formal identity because v has no constant term in $\boldsymbol{\varepsilon}$. For triangularity: at a fixed $\boldsymbol{m}$, the unknown $V_{\boldsymbol{m}}$ occurs linearly only in the term $\Phi'(\mu_0)v$; each v^s with $s \geq 2$ needs a decomposition of $\boldsymbol{m}$ into at least two non-zero parts and so uses only strictly smaller multi-indices; and every monomial of Q carries at least one explicit ε_j, so an occurrence of $V_{\boldsymbol{m}}$ inside Q would be multiplied into a strictly larger multi-index. Solving the coefficient equation gives Equation (O.93). Finiteness holds because a fixed multi-index has finitely many additive decompositions. □

Corollary O.66 (Multinomial expansion of Q). *For $\boldsymbol{m} \neq \boldsymbol{0}$,*

$$[\boldsymbol{\varepsilon}^{\boldsymbol{m}}]\, Q(v;\boldsymbol{\varepsilon})\Big|_{v \text{ fixed}} = (-1)^{|\boldsymbol{m}|+1} \frac{(|\boldsymbol{m}|-1)!}{\boldsymbol{m}!} \prod_j B_j(v)^{m_j}. \tag{O.94}$$

Proof. Only $S^{|\boldsymbol{m}|}$ in $\log(1+S) = \sum_{d\geq 1}(-1)^{d+1} S^d/d$ contributes, with multinomial coefficient $|\boldsymbol{m}|!/\boldsymbol{m}!$. □

Corollary O.67 (Closed multi-index Lagrange form in the phase). *Let $G(v) = \big(\Phi(\mu_0+v) - \Phi(\mu_0)\big)/v$ for $v \neq 0$ and $G(0) = \Phi'(\mu_0)$. Then $v = -Q(v;\boldsymbol{\varepsilon})/G(v)$, and by Theorem J.17,*

$$V_{\boldsymbol{m}} = \sum_{q=1}^{|\boldsymbol{m}|} \frac{(-1)^q}{q}\, [v^{q-1}]\, \frac{1}{G(v)^q} \sum_{\substack{\boldsymbol{m}_1+\cdots+\boldsymbol{m}_q=\boldsymbol{m} \\ \boldsymbol{m}_i \neq \boldsymbol{0}}} \prod_{i=1}^{q} [\boldsymbol{\varepsilon}^{\boldsymbol{m}_i}]\, Q(v;\boldsymbol{\varepsilon}), \tag{O.95}$$

both sums being finite.

Proof. G is analytic near 0 with $G(0) = \Phi'(\mu_0) \neq 0$, and Equation (O.92) reads $vG(v) + Q(v;\boldsymbol{\varepsilon}) = 0$. Apply Theorem J.17 to $v = z \cdot \big(-Q(v;\boldsymbol{\varepsilon})/G(v)\big)$ and set $z = 1$; since every monomial of Q has positive $\boldsymbol{\varepsilon}$-degree, only $q \leq |\boldsymbol{m}|$ contributes. Expanding Q^q and using Equation (O.94) gives Equation (O.95). □

Remark O.68 (Three engines, one answer). Theorem O.62 works in the ordinate y and is S3's; Theorem O.67 works in the phase μ and is S2's; Theorem O.65 is S1's. They compute the same coefficients: all three solve the same analytic equation with the same normalisation $V_{\boldsymbol{0}} = 0$, and the solution of a triangular system is unique. In practice Equation (O.93) is preferable when many multi-indices are wanted (it reuses everything), Equation (O.86) when a single high-order multi-index is wanted (it needs no lower ones), and Equation (O.88) when the answer is wanted symbolically in μ_0.

O.9.5 First-order sectors and the leading parity chirp

Theorem O.69 (The first correction from each sector). *Let $\boldsymbol{e}_j$ be the unit multi-index at $j = (k,\sigma) \in J_K$. Then*

$$V_{\boldsymbol{e}_j}(\mu_0) = -\frac{r_j(\mu_0)}{\Phi'(\mu_0)} = -\frac{a_k(\mu_0)}{\sqrt{k}} \frac{\mu_0(\mu_0-\sigma k)}{\mu_0^2 - 3\mu_0 + 3}, \tag{O.96}$$

so that the corresponding displacement of the inverse phase is $V_{\boldsymbol{e}_j}(\mu_0)\, \mathrm{e}^{-\alpha_j \mu_0}$, and the corresponding displacement of the inverse index is

$$\delta N^{(1)}_{k,\sigma}(y) = -\frac{3\, a_k(\mu_0)}{\pi^2 \sqrt{k}} \frac{\mu_0^2(\mu_0-\sigma k)}{\mu_0^2 - 3\mu_0 + 3}\, \mathrm{e}^{-(1-\sigma/k)\mu_0}. \tag{O.97}$$

Proof. Take $\boldsymbol{m} = \boldsymbol{e}_j$ in Equation (O.93): the $s \geq 2$ terms and all higher Q-monomials vanish, leaving $V_{\boldsymbol{e}_j} = -[\varepsilon_j]\, Q(0;\boldsymbol{\varepsilon})/\Phi'(\mu_0) = -B_j(0)/\Phi'(\mu_0) = -r_j(\mu_0)/\Phi'(\mu_0)$. Insert $r_j = \frac{a_k}{\sqrt{k}} \frac{1-\sigma k/\mu}{1-1/\mu} =$

$\frac{a_k}{\sqrt{k}}\frac{\mu-\sigma k}{\mu-1}$ and Equation (O.66): the factors $\mu_0 - 1$ cancel, giving Equation (O.96). Finally $n(\mu) = \frac{1}{24} + \frac{3\mu^2}{2\pi^2}$ is exactly quadratic, so

$$n(\mu_0 + v) = N_0(y) + \frac{3\mu_0}{\pi^2}v + \frac{3}{2\pi^2}v^2, \tag{O.98}$$

and the linear term at first order gives Equation (O.97). □

Corollary O.70 (Index coefficient of every transmonomial). *For every $\boldsymbol{m} \neq \boldsymbol{0}$, the coefficient of $\boldsymbol{\varepsilon}^{\boldsymbol{m}}$ in the inverse index is*

$$\mathfrak{N}_{\boldsymbol{m}}(\mu_0) = \frac{3\mu_0}{\pi^2}V_{\boldsymbol{m}} + \frac{3}{2\pi^2}\sum_{\substack{\boldsymbol{a}+\boldsymbol{b}=\boldsymbol{m}\\ \boldsymbol{a},\boldsymbol{b}\neq\boldsymbol{0}}} V_{\boldsymbol{a}}V_{\boldsymbol{b}}. \tag{O.99}$$

Proof. Expand Equation (O.98) with $v = \sum V_{\boldsymbol{m}}\boldsymbol{\varepsilon}^{\boldsymbol{m}}$ and compare coefficients. No higher powers occur because n is quadratic. □

The first two positive sectors, with $a_2 = \widetilde{A}_2(N_0) = \cos(\pi N_0)$ and $a_3 = \widetilde{A}_3(N_0) = 2\cos(\pi N_0)\cos(\frac{\pi}{18} + \frac{\pi N_0}{3})$ (interpolated convention) or $a_2 = (-1)^r$, $a_3 = 2\cos(\frac{\pi}{18} - \frac{2\pi r}{3})$ (residue-frozen), are

$$\delta N^{(1)}_{2,+} = -\frac{3\,a_2}{\pi^2\sqrt{2}}\,\frac{\mu_0^2(\mu_0-2)}{\mu_0^2 - 3\mu_0 + 3}\,\mathrm{e}^{-\mu_0/2}, \tag{O.100}$$

$$\delta N^{(1)}_{3,+} = -\frac{3\,a_3}{\pi^2\sqrt{3}}\,\frac{\mu_0^2(\mu_0-3)}{\mu_0^2 - 3\mu_0 + 3}\,\mathrm{e}^{-2\mu_0/3}. \tag{O.101}$$

Thus *parity of the index appears at relative scale* $\mathrm{e}^{-\mu_0/2}$*, the residue modulo* 3 *at* $\mathrm{e}^{-2\mu_0/3}$*, and the residue modulo k first appears in the growing kth sector at* $\mathrm{e}^{-(1-1/k)\mu_0}$.

O.9.6 The fractional-power staircase in the ordinate

The core equation converts exponentials in μ_0 into fractional powers of y exactly:

$$\mathrm{e}^{-\alpha\mu_0} = \left(\frac{\kappa(\mu_0-1)}{y\,\mu_0^3}\right)^{\alpha} \qquad \text{for every } \alpha, \tag{O.102}$$

directly from $y = \kappa\mathrm{e}^{\mu_0}(\mu_0-1)\mu_0^{-3}$.

Proposition O.71 (Every first-order sector in closed y-form). *With $\alpha = \alpha_{k,\sigma}$ and $z = 1/\mu_0$,*

$$\delta N^{(1)}_{k,\sigma} = -\frac{3\kappa^{\alpha}}{\pi^2\sqrt{k}}\,a_k(\mu_0)\;y^{-\alpha}\;\mu_0^{1-2\alpha}\sum_{s\geq 0} g_s^{(k,\sigma)}\,\mu_0^{-s}, \tag{O.103}$$

where

$$\sum_{s\geq 0} g_s^{(k,\sigma)} z^s = \frac{(1-\sigma kz)(1-z)^{\alpha}}{1-3z+3z^2}, \qquad g_s^{(k,\sigma)} = \sum_{i=0}^{s}(-1)^i\binom{\alpha}{i}\big(\chi_{s-i} - \sigma k\,\chi_{s-i-1}\big), \tag{O.104}$$

with $\chi_{-1} = 0$ and $\chi_s = \dfrac{\varrho_+^{s+1} - \varrho_-^{s+1}}{\varrho_+ - \varrho_-}$ the coefficients of $(1-3z+3z^2)^{-1}$ ($\varrho_\pm = \frac{1}{2}(3 \pm \mathrm{i}\sqrt{3})$; $\chi_0 = 1, \chi_1 = 3, \chi_2 = 6, \chi_3 = 9, \chi_4 = 9, \chi_5 = 0, \ldots$).

Proof. Insert Equation (O.102) into Equation (O.97) and factor $\mu_0^{1-2\alpha}$ out of $\dfrac{\mu_0^2(\mu_0-\sigma k)(\mu_0-1)^{\alpha}}{(\mu_0^2-3\mu_0+3)\mu_0^{3\alpha}}$; with $z = 1/\mu_0$ the remaining factor is $\dfrac{(1-\sigma kz)(1-z)^{\alpha}}{1-3z+3z^2}$. The coefficient formula is the Cauchy product of $(1-z)^{\alpha} = \sum_i(-1)^i\binom{\alpha}{i}z^i$ with $(1-\sigma kz)/(1-3z+3z^2)$, and χ_s is the standard solution of $\chi_s = 3\chi_{s-1} - 3\chi_{s-2}$, $\chi_0 = 1$, $\chi_1 = 3$, whose closed form is as stated since $\varrho_\pm$ are the roots of $\varrho^2 = 3\varrho - 3$. □

Corollary O.72 (The inverse staircase of scales). *Since $\mu_0 \sim \log y$, the primary positive sectors contribute, in order,*

$$y^{-1/2}, \qquad y^{-2/3}(\log y)^{-1/3}, \qquad y^{-3/4}(\log y)^{-1/2}, \qquad \ldots, \qquad y^{-(1-1/k)}(\log y)^{-1+2/k}, \tag{O.105}$$

accumulating at the scale y^{-1}, where they meet the square of the first sector and the whole negative family.

Proof. By Equation (O.103) the $(k,+)$ sector has size $y^{-\alpha}\mu_0^{1-2\alpha}(1+\mathcal{O}(\mu_0^{-1}))$ times a bounded amplitude, with $\alpha = 1-1/k$, so $1-2\alpha = -1+2/k$; and $\mu_0 \sim \log y$ because $\Phi(\mu_0) = \log(y/\kappa)$ with $\Phi(\mu) = \mu(1+o(1))$. Substituting $k = 2,3,4$ gives the first three displayed scales. Since $\alpha_{k,+} \uparrow 1$, the scales accumulate at y^{-1}; the square of the $(2,+)$ sector has action $\frac{1}{2}+\frac{1}{2} = 1$ and every negative action is ≥ 1. □

Theorem O.73 (The leading parity chirp). *In the interpolated convention,*

$$\mathcal{N}(y) = N_0(y) - \frac{3\sqrt{\kappa}}{\pi^2\sqrt{2}}\,\frac{\cos\big(\pi N_0(y)\big)}{\sqrt{y}}\left(1+\mathcal{O}\Big(\frac{1}{\log y}\Big)\right) + \mathcal{O}\Big(y^{-2/3}(\log y)^{-1/3}\Big), \tag{O.106}$$

with the constant

$$\frac{3\sqrt{\kappa}}{\pi^2\sqrt{2}} = \frac{3^{1/4}}{2\pi} = 0.2094596846361762626265841\ldots \tag{O.107}$$

Since $N_0(y) = \frac{3}{2\pi^2}(\log y)^2(1+o(1))$, the cosine is a quadratic logarithmic chirp, *not a log-periodic oscillation.*

Proof. Take $k=2$, $\sigma=+1$, $\alpha=\frac{1}{2}$ in Equation (O.103): the power $\mu_0^{1-2\alpha} = \mu_0^0$ is absent, $g_0^{(2,+)} = 1$, and $a_2 = \cos(\pi N_0)$ by Theorem O.10. The constant is $3\kappa^{1/2}/(\pi^2\sqrt{2})$, and $\sqrt{\kappa} = \pi/\sqrt{6\sqrt{3}}$ gives $\frac{3\pi}{\pi^2\sqrt{2}\sqrt{6\sqrt{3}}} = \frac{3}{\pi\sqrt{12\sqrt{3}}} = \frac{3}{2\pi\cdot 3^{3/4}} = \frac{3^{1/4}}{2\pi}$, which we evaluated to 25 digits. The relative $\mathcal{O}(1/\log y)$ is the $s \geq 1$ tail of Equation (O.103), and the next omitted contribution is the $k=3$ sector at scale $y^{-2/3}(\log y)^{-1/3}$ by Theorem O.72; all products of two blocks have action ≥ 1. □

Remark O.74 (Constant check). S2 prints this constant as 0.2094596846361762626; the closed form $3^{1/4}/(2\pi)$ is S3's. The two agree, and both agree with our independent 25-digit evaluation Equation (O.107). S1 does not state the constant, giving instead the equivalent phase-variable form Equation (O.100); the three are algebraically identical, as the cancellation of $\mu_0 - 1$ in the proof of Theorem O.69 shows.

O.10 The profinite arithmetic phase

The residue-frozen convention of Theorem O.57 organises itself into a projective system, which is the cleanest way to say what data an "arithmetically complete" inverse of p actually depends on.

Theorem O.75 (Compatibility and the profinite parameter). *Let $\Lambda_K = \operatorname{lcm}(1,\ldots,K)$, so $\Lambda_K \mid \Lambda_{K+1}$. Suppose $r_K \in \mathbb{Z}/\Lambda_K\mathbb{Z}$ satisfy $r_{K+1} \equiv r_K \pmod{\Lambda_K}$, and let μ_{K,r_K} be the corresponding inverse branches of Theorem O.57. Then for every K, the transseries of $\mu_{K+1,r_{K+1}}$ obtained by deleting all monomials containing the variables of level $K+1$ coincides with the transseries of μ_{K,r_K}. Consequently the complete arithmetic inverse is a projective family indexed by the profinite integers $\widehat{r} \in \widehat{\mathbb{Z}} = \varprojlim \mathbb{Z}/m\mathbb{Z}$, and an actual partition index n supplies its own phase through the diagonal embedding $n \mapsto \widehat{n}$.*

Proof. Deleting the level-$(K+1)$ variables from the defining equation Equation (O.92) at level $K+1$ produces the defining equation at level K *provided* the retained amplitudes agree, i.e. provided $A_k(r_{K+1}) = A_k(r_K)$ for $k \leq K$. That is exactly the compatibility hypothesis, since A_k has period $k \mid \Lambda_K$ and $r_{K+1} \equiv r_K \pmod{\Lambda_K}$. Both truncated equations are triangular with the same normalisation

$V_0 = 0$, and a triangular system has a unique solution, so the solutions agree coefficient by coefficient. The projective limit statement is the definition of $\widehat{\mathbb{Z}}$: compatible systems (r_K) are exactly the elements of $\varprojlim \mathbb{Z}/\Lambda_K\mathbb{Z} = \widehat{\mathbb{Z}}$, because $\{\Lambda_K\}$ is cofinal in the divisibility order on $\mathbb{N}$. □

Remark O.76 (Why a profinite parameter and not a real one). At every algebraic order all indices share one expansion. Parity enters at relative order $\mathrm{e}^{-\mu/2}$; the residue mod 3 at $\mathrm{e}^{-2\mu/3}$; the residue mod k at $\mathrm{e}^{-(1-1/k)\mu}$. No finite real-valued smooth parameter records all of these congruence classes simultaneously, and no single real analytic function of n interpolates all A_k canonically (Theorem O.11). A profinite integer does so canonically and finitely at every exponential resolution: each finite action cutoff $\eta < 1$ sees only A_k for $k \le (1-\eta)^{-1}$, i.e. only $\widehat{r}$ modulo $\Lambda_{\lfloor (1-\eta)^{-1}\rfloor}$. This is S1's observation and is absent from S2 and S3.

Remark O.77 (A structure suggested, not proved). S1 proposes an "arithmetic transseries sheaf" with base $\widehat{\mathbb{Z}}$ and fibres the finite-level analytic transseries algebras, and calls Theorem O.75 its elementary case. We record this as a suggestion. Nothing beyond Theorem O.75 — namely, projective compatibility of the coefficient systems — is established here or in the sources: no topology on the fibres, no gluing statement, and no continuity of $\widehat{r} \mapsto \mu_{K,\widehat{r}}$ in any topology on $\widehat{\mathbb{Z}}$ has been proved. It is stated as Theorem O.96 in Section O.15, not as a theorem.

O.11 Validity, truncation control, and integer recovery

O.11.1 Remainder transport

Theorem O.78 (Inverse error after K Rademacher levels)**.** *Fix $K \ge 1$. Let $y = p(n)$, let $\mu = \lambda\sqrt{n - 1/24}$, and let μ_K denote either inverse branch of Theorem O.57 (with r $\equiv n$) or of Theorem O.58. Then*

$$\mu_K(y) - \mu = \mathcal{O}_K\Big(\mu^{3/2}\,\mathrm{e}^{-(1-\frac{1}{K+1})\mu}\Big), \qquad n_K(y) - n = \mathcal{O}_K\Big(\mu^{5/2}\,\mathrm{e}^{-(1-\frac{1}{K+1})\mu}\Big), \tag{O.108}$$

where $n_K = \frac{1}{24} + 3\mu_K^2/(2\pi^2)$. Explicitly, with the constant of Theorem O.23, for $\mu \ge \mu_*$ (any threshold at which $\Phi' \ge \frac{1}{2}$ and the omitted tail is $\le \frac{1}{2}$ in absolute value),

$$|n_K(y) - n| \le \frac{3\mu}{\pi^2}\cdot\frac{2}{1}\cdot\frac{8\mu^2}{3(1-\mu^{-1})\sqrt{K}}\exp\Big[-\Big(1 - \frac{1}{K+1}\Big)\mu\Big]\,\big(1 + o(1)\big). \tag{O.109}$$

Proof. This is Theorem J.51 with the forward remainder supplied by Theorems O.23 and O.24. In detail: let $\mathcal{F}_K = \log \mathcal{P}_K$ and $\mathcal{F} = \log p$ regarded as functions of the phase. By Theorem O.17, $\mathcal{F}(\mu) - \mathcal{F}_K(\mu) = \log\big(1 + \mathcal{R}_{>K}/(1 + \mathcal{R}_{\le K})\big)$, which by Theorem O.24 and $|\log(1+u)| \le 2|u|$ for $|u| \le \frac{1}{2}$ is $\mathcal{O}_K(\mu^{3/2}\mathrm{e}^{-(1-1/(K+1))\mu})$. Since $\mathcal{F}_K'(\mu) = \Phi'(\mu)(1 + o(1)) \to 1$, the mean value theorem converts this logarithmic residual into the phase error, which is the first half of Equation (O.108). Finally $\mathrm{d}n/\mathrm{d}\mu = 3\mu/\pi^2$ gives the index error. The explicit form Equation (O.109) replaces Theorem O.24 by the explicit Theorem O.23 (losing $\mu^{-1/2}$, gaining a constant) and $\mathcal{F}_K' \ge \frac{1}{2}$. □

Corollary O.79 (Beyond-all-orders accuracy of the core inverse)**.** *For $y = p(n)$,*

$$N_0(y) - n = \mathcal{O}\Big(\mu^{5/2}\mathrm{e}^{-\mu/2}\Big) = \mathcal{O}\Big(n^{5/4}\exp\Big(-\tfrac{1}{2}\pi\sqrt{2n/3}\Big)\Big), \tag{O.110}$$

so $N_0(p(n)) - n \to 0$ faster than every power of n^{-1}.

Proof. Theorem O.78 at $K = 1$; note $N_0 = n_1$ because the $K = 1$ level of Theorem O.57 contains the sectors $(1, \pm)$, and the $(1, -)$ sector has action 2, hence contributes below the $K = 1$ error bound. □

Proposition O.80 (A posteriori certificate; specialisation of Theorem J.47)**.** *Let* $\mathcal{F}_K(\mu) = \log \mathcal{P}_K(\mu)$ *and let* $\rho(\widetilde{\mu}) = \mathcal{F}_K(\widetilde{\mu}) - \log y$ *be the residual of a candidate* $\widetilde{\mu}$*. If* $\mathcal{F}_K' \geq m > 0$ *on an interval* I *containing both* $\widetilde{\mu}$ *and the exact root* $\mu_K(y)$*, then*

$$|\widetilde{\mu} - \mu_K(y)| \leq \frac{|\rho(\widetilde{\mu})|}{m}, \qquad |\widetilde{n} - n_K(y)| \leq \frac{3}{2\pi^2}\big(|\widetilde{\mu}| + |\mu_K(y)|\big)\frac{|\rho(\widetilde{\mu})|}{m}. \tag{O.111}$$

Proof. Mean value theorem applied to $\mathcal{F}_K$, then the factorisation $\widetilde{\mu}^2 - \mu_K^2 = (\widetilde{\mu} - \mu_K)(\widetilde{\mu} + \mu_K)$. Since $\mathcal{F}_K' \to 1$, one may take $m = \frac{1}{2}$ beyond an effectively checkable threshold. □

Remark O.81 (Why a residual certificate rather than term counting). Theorem O.80 separates the *local inversion* error (how well $\widetilde{\mu}$ solves the level-K equation) from the *Rademacher tail* (how far the level-K equation is from the truth). Only the second needs Theorem O.23. This modularity is what makes it possible to certify a truncated symbolic transseries and a numerically refined Newton root by the same inequality.

O.11.2 Algebraic truncation

Proposition O.82. *With* $U_M(z) = \sum_{m \leq M} c_m z^m$ *as in Theorem O.50,*

$$\mu_0 - \big(X + U_M(X^{-1})\big) = \mathcal{O}(X^{-M-1}), \qquad N_0 - \left[\frac{1}{24} + \frac{3}{2\pi^2}\big(X + U_M(X^{-1})\big)^2\right] = \mathcal{O}(X^{-M}), \tag{O.112}$$

the loss of one order coming from $\mathrm{d}n/\mathrm{d}\mu \asymp X$*. In particular* $M \geq 1$ is necessary *for the index to be correct to* $o(1)$ *and hence for the arithmetic phases of Section O.9 to be evaluated at the right point (Theorem O.55).*

Proof. U is analytic at 0 by Theorem O.50, so its Taylor remainder after M terms is $\mathcal{O}(z^{M+1})$; substitute $z = X^{-1}$ and use $N_0 - \widetilde{N} = \frac{3}{2\pi^2}(\mu_0 - \widetilde{\mu})(\mu_0 + \widetilde{\mu})$ with $\mu_0 + \widetilde{\mu} = \mathcal{O}(X)$. □

O.11.3 Integer recovery

Theorem O.83 (Eventual exact recovery by rounding)**.** *There is* n_0 *such that for every integer* $n \geq n_0$*,*

$$n = \big\lfloor N_0(p(n)) \big\rceil, \tag{O.113}$$

$\lfloor\cdot\rceil$ *denoting nearest-integer rounding. Consequently the arithmetic phase* $\widehat{n}$ *of Theorem O.75 is recoverable from* $y = p(n)$ before *any exponentially small sector is evaluated.*

Proof. By Equation (O.110), $|N_0(p(n)) - n| \to 0$; it is therefore $< \frac{1}{2}$ for all large n, at which point rounding is exact. The last sentence is the observation that $A_k(n)$ depends only on $n \bmod k$. □

Remark O.84 (The threshold is very small in practice). Theorem O.83 is asymptotic and claims no optimal n_0. We computed $N_0(p(n)) - n$ for all $1 \leq n \leq 59$ in 140-digit arithmetic; the largest absolute value is $0.1723\ldots$ at $n = 1$, and the sequence decreases in magnitude thereafter (through 2.8×10^{-2} at $n = 10$, 4.5×10^{-4} at $n = 50$, 1.5×10^{-5} at $n = 100$, 4.3×10^{-17} at $n = 1000$). So $n_0 = 1$ is consistent with the data, though we prove only the asymptotic statement; producing a certified n_0 requires effective constants in Theorem O.78, i.e. effective Rademacher remainder bounds, which is listed as Theorem O.95.

Remark O.85 (Circularity, avoided). The arithmetic transseries needs the residue r of n, which appears to require knowing n. The order of operations resolves this: *(i)* compute X from Equation (O.56); *(ii)* compute μ_0 and N_0, which are phase-independent; *(iii)* round, obtaining n by Theorem O.83 for y on the range; *(iv)* now every $A_k(n)$ is known, and the exponentially small sectors refine an already identified integer. The arithmetic layer is thus a *refinement* of a known answer, not a device for guessing the phase. For y off the range, step (iii) yields $\lceil \mathcal{N}(y) \rceil$ by Theorem O.44 instead, and the sectors describe $\mathcal{N}$ itself.

O.12 Numerical verification

Every number in this chapter was recomputed for this volume. Partition values were generated exactly by Euler's pentagonal recurrence

$$p(n) = \sum_{m\geq 1}(-1)^{m-1}\left[p\Big(n-\tfrac{m(3m-1)}{2}\Big)+p\Big(n-\tfrac{m(3m+1)}{2}\Big)\right], \qquad p(0)=1,\ p(r)=0\ (r<0), \tag{O.114}$$

in exact integer arithmetic (checked against $p(10) = 42$, $p(50) = 204226$, $p(100) = 190569292$); Dedekind sums exactly as rationals from Equation (O.9); the transcendental factors in 140-digit floating point. Coefficient identities were verified in exact rational or exact symbolic arithmetic, never by floating-point spot checks. The tables check normalisation, signs, phases and the hierarchy of actions; they are used in no proof.

Table O.3: Relative error $\big(S_K(n)-p(n)\big)/p(n)$ of the paired truncation $S_K=\sum_{k\leq K}(P_{k,+}+P_{k,-})$, recomputed at 140 digits. Accuracy need not improve monotonically in K at small n: the amplitudes $A_k(n)$ change sign and can suppress or enhance an individual sector.

n	$p(n)$	$K=1$	$K=2$	$K=3$	$K=5$
10	42	$-8.8621\cdot 10^{-3}$	$1.6606\cdot 10^{-3}$	$3.7391\cdot 10^{-4}$	$3.4722\cdot 10^{-4}$
25	1 958	$1.0717\cdot 10^{-3}$	$3.0045\cdot 10^{-6}$	$-6.1015\cdot 10^{-5}$	$-3.4866\cdot 10^{-6}$
50	204 226	$-7.3074\cdot 10^{-5}$	$3.9028\cdot 10^{-6}$	$2.1050\cdot 10^{-7}$	$-1.1350\cdot 10^{-7}$
100	190 569 292	$-1.8220\cdot 10^{-6}$	$8.6838\cdot 10^{-9}$	$-4.9491\cdot 10^{-9}$	$3.1454\cdot 10^{-10}$
250	$2.3079\cdot 10^{14}$	$-1.0760\cdot 10^{-9}$	$7.2267\cdot 10^{-13}$	$4.3021\cdot 10^{-14}$	$2.1256\cdot 10^{-15}$
500	$2.3002\cdot 10^{21}$	$-2.4389\cdot 10^{-13}$	$1.7537\cdot 10^{-17}$	$-2.0017\cdot 10^{-19}$	$-3.5933\cdot 10^{-22}$
1000	$2.4061\cdot 10^{31}$	$-1.6997\cdot 10^{-18}$	$1.2574\cdot 10^{-24}$	$-3.4744\cdot 10^{-27}$	$3.9453\cdot 10^{-30}$

Table O.4: Inverse errors at exact ordinates $y = p(n)$; entries are (approximation) $- n$. N_X is the bare Lambert carrier index, N_0 the dominant core, N_K the Newton inverse of the level-K interpolated sum (Theorem O.58). The residue-frozen inverses (Theorem O.57) reproduce the last three columns to all five digits shown for $n \geq 100$; at $n = 50$ they differ in the fifth digit of the $K = 3, 5$ entries ($-1.2992\cdot 10^{-6}$, $7.0057\cdot 10^{-7}$), at $n = 25$ in the fourth ($2.7926\cdot 10^{-4}$, $1.5958\cdot 10^{-5}$), at $n = 10$ in the third ($-5.3338\cdot 10^{-3}$, $-1.2006\cdot 10^{-3}$, $-1.1150\cdot 10^{-3}$). That is exactly the pattern Theorem O.59 predicts.

n	N_X-n	N_0-n	N_2-n	N_3-n	N_5-n
10	$-3.9910\cdot 10^{-1}$	$2.8447\cdot 10^{-2}$	$-5.3290\cdot 10^{-3}$	$-1.1863\cdot 10^{-3}$	$-1.1015\cdot 10^{-3}$
25	$-3.7877\cdot 10^{-1}$	$-4.9052\cdot 10^{-3}$	$-1.3751\cdot 10^{-5}$	$2.7902\cdot 10^{-4}$	$1.5941\cdot 10^{-5}$
50	$-3.5044\cdot 10^{-1}$	$4.5102\cdot 10^{-4}$	$-2.4089\cdot 10^{-5}$	$-1.2993\cdot 10^{-6}$	$7.0059\cdot 10^{-7}$
100	$-3.3600\cdot 10^{-1}$	$1.5378\cdot 10^{-5}$	$-7.3293\cdot 10^{-8}$	$4.1772\cdot 10^{-8}$	$-2.6548\cdot 10^{-9}$
250	$-3.2364\cdot 10^{-1}$	$1.3942\cdot 10^{-8}$	$-9.3644\cdot 10^{-12}$	$-5.5747\cdot 10^{-13}$	$-2.7544\cdot 10^{-14}$
500	$-3.1768\cdot 10^{-1}$	$4.4042\cdot 10^{-12}$	$-3.1668\cdot 10^{-16}$	$3.6146\cdot 10^{-18}$	$6.4888\cdot 10^{-21}$
1000	$-3.1356\cdot 10^{-1}$	$4.2960\cdot 10^{-17}$	$-3.1780\cdot 10^{-23}$	$8.7815\cdot 10^{-26}$	$-9.9717\cdot 10^{-29}$

Three features of Table O.4 confirm the theory point by point. *(i)* The first column tends to $-3/\pi^2 = -0.30396\ldots$, not to 0: this is Theorem O.54 and the numerical signature of Theorem O.55. (S1 and S2 tabulate $N_X = \frac{1}{24}+3X^2/(2\pi^2)$ and report $\to -0.304$; S3 tabulates $3X^2/(2\pi^2)$ without the $\frac{1}{24}$ and reports $\to -0.346$. The columns differ by exactly $1/24$ and each is correct for its own definition; we keep the $\frac{1}{24}$ so that N_X and N_0 are directly comparable.) *(ii)* Restoring the amplitude factor $1-1/\mu$, i.e. passing from X to μ_0, removes that entire $\mathcal{O}(1)$ bias and leaves an exponentially small error. *(iii)* The residual after the core alternates with the parity of n: at $n = 49, 50, 51$ it is $-5.2643\cdot 10^{-4}$, $+4.5102\cdot 10^{-4}$, $-4.0920\cdot 10^{-4}$, exactly as Equation (O.100) predicts through $\widetilde{A}_2 = \cos\pi N_0$. Adding the single first-order $k = 2$ sector reduces these to $-1.0890 \cdot 10^{-5}$, $-2.4087 \cdot 10^{-5}$, $+2.9029 \cdot 10^{-5}$; adding the first-order sectors through $k = 10$ reduces them to $-2.24 \cdot 10^{-7}$, $-9.95 \cdot 10^{-8}$, $-5.81 \cdot 10^{-8}$. This checks both the sign and the phase of the leading chirp Equation (O.106).

Remark O.86 (What was verified in exact arithmetic). (1) Equation (O.33) against the Taylor expansion of Equation (O.31), symbolically in β through order t^9: identical; the closed forms Equations (O.37) and (O.38) were produced from Equation (O.33) and match S2 (through c_5) and S3 (through c_8). (2) $\omega_0, \dots, \omega_4$ against both the S1 and the S2 grouping of the same numbers: identical (they look different only because S1 splits each into partial fractions in π). (3) $c_1, \dots, c_{18}$ of Table O.2 from Equation (O.72) in exact rationals, plus the residual test Equation (O.74) at $M = 12$; all eighteen agree with the sources. (4) $\gamma_1, \dots, \gamma_8$ against all three sources; γ_9, γ_{10} are new here. (5) $\Pi_1, \dots, \Pi_8$ from the Lagrange form against the Stirling form and against each of the three printed shapes, confirming Theorem O.48; S1's printed P_6, P_7 are correct. (6) S3's u-expansion in H through H^{-4} and its N_0-expansion through H^{-3}, and S2's N_0-block in R through the constant term: correct as printed. (7) Equation (O.11) against direct evaluation of Equation (O.10) from exact Dedekind sums. (8) The chirp constant Equation (O.107) to 25 digits in both forms. No discrepancy with any source's arithmetic was found: every repair recorded in this chapter is structural, not numerical.

Remark O.87 (Normalisation cross-checks). Because Equation (O.14) carries a factor $\sqrt{k}$ that some normalisations absorb into A_k, a missing or spurious power of k is the easiest error to make and the hardest to see. Three identifications of one quantity detect it: the kth derivative factor of Equation (O.14) equals $\frac{\mu}{4kN^{3/2}}[\mathrm{e}^{\mu/k}(1 - k/\mu) + \mathrm{e}^{-\mu/k}(1 + k/\mu)]$, and after multiplication by $\sqrt{k}/(\pi\sqrt{2})$ it may be written either as $\frac{1}{4\sqrt{3k}\,N}[\cdots]$ or as $\frac{\kappa}{\sqrt{k}\,\mu^2}[\cdots]$; at $k = 1$ the positive term is $\frac{\mathrm{e}^{\mu}}{4\sqrt{3}\,N}(1 - \frac{1}{\mu})$, which becomes Equation (O.2) after replacing N by n and dropping $1 - 1/\mu$. Any of these disagreeing with the others by a power of k localises the slip.

O.13 Algorithms

All formulas of this chapter are constructive and need only formal power-series arithmetic over $\mathbb{Q}(\pi)$ together with exact rational Dedekind sums.

Algorithm O.88 (Forward sector coefficients). Given k, σ, M: set $\beta = \sigma\pi/(6k)$ and evaluate the finite sum Equation (O.33) for $0 \le m \le M$ ($\mathcal{O}(m)$ field operations each, no series manipulation; keep β symbolic if sector-uniform output is wanted). If the whole array is wanted at once, evaluate $\ell_1, \dots, \ell_M$ from Equation (O.39) and run the triangular recurrence Equation (O.41) instead. Assemble Equation (O.30), pairing the two signs before summing over large k (Theorem O.20).

Algorithm O.89 (Phase-independent inverse). Given a large y and an order M: (1) compute X from Equation (O.56) with a standard W_{-1} routine — do *not* substitute the logarithmic expansion Equation (O.57), which is numerically far inferior; (2) form $\mu_0 = X + \sum_{m \le M} c_m X^{-m}$ from the universal rationals of Table O.2, optionally refined by one Newton step on $\Phi(\mu) = R$; (3) return $N_0 = \frac{1}{24} + 3\mu_0^2/(2\pi^2)$; (4) certify by evaluating $\rho = \mu_0 - 2\log\mu_0 + \log(1 - \mu_0^{-1}) - \log(y/\kappa)$ and applying Theorem O.80; (5) round (Theorem O.83) if y is known to lie on the range, else take $\lceil\cdot\rceil$ for the staircase (Theorem O.44).

Algorithm O.90 (Arithmetic exponential correction). Given y, a level K and a finite set of multi-indices: (1) compute μ_0, N_0 by Theorem O.89 and use N_0, never N_X, to evaluate amplitudes (Theorem O.55); (2) compute exact rational $s(h, k)$ for $k \le K$ and hence $A_k(r)$ (frozen, with r from Theorem O.83) or $\widetilde{A}_k(N_0)$ (interpolated); (3) build the actions and amplitudes of Equation (O.83); (4) traverse the multi-indices in a graded monomial order using Equation (O.93), or, for one isolated $\boldsymbol{m}$, use Equation (O.86) or the Bell form Equation (O.88) with the closed jets Equation (O.90); (5) substitute $\varepsilon_j = \mathrm{e}^{-\alpha_j\mu_0}$ and convert to the index by Equation (O.99); (6) certify by Theorem O.80 on $\mathcal{P}_K$, adding Theorem O.23 if the comparison is with the exact sequence.

Remark O.91 (Modular structure of the computation). The layers are independent and should be cached separately: the carrier polynomials Π_j depend only on the exponent data $(a,b)=(1,-2)$; the rationals c_m only on the amplitude $1-1/\mu$; the inverse-derivative polynomials $\mu_0^{(s)}$ only on Φ; and the arithmetic enters solely through the numbers a_k in Equation (O.83). Universal tables are computed once and residue-dependent evaluation is then cheap.

O.14 A template for Rademacher-type sequences

Nothing above used the partition function except through four properties. S3 has the most careful version of this observation and S1 the most usable one; what follows merges them into a hypothesis list and a theorem, so that the template is a statement one can check rather than a slogan.

Definition O.92 (Rademacher-type representation)**.** A sequence $(a(n))_{n\geq n_1}$ of positive reals is of *Rademacher type* if:

(T1) there is a real analytic, strictly increasing *phase* $\phi:(\nu_1,\infty)\to(0,\infty)$ with $\phi\to\infty$ and real analytic inverse, giving the large variable $\mu=\phi(n)$;

(T2) the *envelope* $F(\mu)=C\mathrm{e}^{a\mu^\alpha}\mu^b\exp Q(\mu^{-\delta})$ is an exponential–power model Theorem J.2 with $a>0$, $\alpha\in(0,1]$;

(T3) there are actions $\alpha_j>0$ ($j\in J$) with $\inf_j\alpha_j>0$, amplitudes $a_j(n)$ taking values in a fixed finite-dimensional space at each truncation level (e.g. periodic in n), and g_j analytic at ∞, with

$$a(n)=F(\mu)\Big(1+\sum_{j\in J'}a_j(n)g_j(\mu)\mathrm{e}^{-\alpha_j\mu}+\mathcal{T}_{J'}(\mu;n)\Big)\quad\text{for every finite } J'\subset J;\qquad\text{(O.115)}$$

(T4) $\{j:\alpha_j\leq\eta\}$ is finite for every $\eta<\sup_j\alpha_j$, and either $\mathcal{T}_{J'}=\mathcal{O}(\mu^{C'}\mathrm{e}^{-\eta'\mu})$ for some $\eta'>\max_{j\in J'}\alpha_j$, or — if the actions accumulate — the grouped remainder is retained as one analytic block.

Theorem O.93 (Template)**.** *Let* $(a(n))$ *be of Rademacher type. Then:*

1. (Reduction.) $\mu=\xi^{1/\alpha}$ *of Theorem J.4 brings* (T2) *to the linear-phase model with computable* $(C',1,b',Q')$.
2. (Carrier.) $C'\mathrm{e}^{\xi}\xi^{b'}=y$ *is solved exactly by Theorem J.23; equivalently, for* $y=C\mathrm{e}^{a\mu^\alpha}\mu^{-\beta}$,
$$\mu=\left(-\frac{\beta}{a\alpha}W_{-1}\left(-\frac{a\alpha}{\beta}\Big(\frac{C}{y}\Big)^{\alpha/\beta}\right)\right)^{1/\alpha}.\qquad\text{(O.116)}$$
3. (Amplitude.) *The envelope inverse is* $\xi_0=X+U(X^{-1})$ *with* U *the unique solution of the fixed-point equation of Theorem J.29 at* $(1,b',Q')$*; coefficients by Theorem J.19 or by isolating the linear occurrence of* U.
4. (Phase locking.) *If some* a_j *is periodic on the index scale and* $\mathrm{d}n/\mathrm{d}\mu\to\infty$*, the amplitudes must be evaluated at* ξ_0*, not at* X*: the carrier bias* $n(\xi_0)-n(X)$ *is* $\mathcal{O}(1)$ *in general.*
5. (Sectors.) *With the phase frozen, every inverse transmonomial is given by Equation* (O.86) *(in* y*) or Equation* (O.95)*/Equation* (O.93) *(in* μ*), with first-order term*
$$\mu=\mu_0-\frac{a_jg_j(\mu_0)}{(\log F)'(\mu_0)}\mathrm{e}^{-\alpha_j\mu_0}+\mathcal{O}\big(\mathrm{e}^{-2\alpha_{\min}\mu_0}\big).\qquad\text{(O.117)}$$
6. (Transport.) *A relative forward remainder* ϵ *transports to a phase error* $\epsilon/(\log F)'(\mu_0)$ *and an index error* $\epsilon\, n'(\mu_0)/(\log F)'(\mu_0)$*, by Theorem J.51.*

7. (Staircase.) *If a is integer valued with gaps bounded below, the generalised inverse is the ceiling of the smooth one (Theorem J.43), with the order-one sawtooth of Theorem O.45 as an irreducible final sector.*

Proof. (i)–(iii) are Theorems J.4, J.23 and J.29 at the stated parameters. For the display in (ii): raising $y = C\mathrm{e}^{a\mu^\alpha}\mu^{-\beta}$ to the power α/β and rearranging gives $\left(\frac{a\alpha}{\beta}\mu^\alpha\right)\mathrm{e}^{-(a\alpha/\beta)\mu^\alpha} = \frac{a\alpha}{\beta}(C/y)^{\alpha/\beta}$, which is the Lambert equation with the stated argument, the branch -1 forced by $\mu \to \infty$. (iv) is the computation of Theorem O.55 with $3\mu/\pi^2$ replaced by $n'(\mu_0)$: what matters is only that $n' \to \infty$ while $\xi_0 - X \to 0$, so their product need not vanish. (v) is Theorems O.62, O.65 and O.67, whose proofs used no property of F or of the blocks beyond analyticity and $F' \neq 0$; the first-order display is Equation (O.96) in that generality. (vi) is the mean-value argument of Theorem O.78, and (vii) is Theorem O.44. □

Remark O.94 (The partition case, and the scope of the template). Taking $\phi(n) = \pi\sqrt{2(n-1/24)/3}$, (C, a, α, b, Q) as in Equation (O.5), $J = \{(k,\sigma)\}$, $\alpha_{k,\sigma} = 1 - \sigma/k$ and $a_k = A_k$, hypotheses (T1)–(T3) hold by Theorems O.14 and O.17, and (T4) holds in its second form: the actions *do* accumulate at 1, and the grouped block is the paired tail Equation (O.25) controlled by Theorem O.23. With $(\alpha,\beta) = (1,2)$ in μ, Equation (O.116) reduces to Equation (O.56). The template applies verbatim to coefficient sequences of weakly holomorphic modular forms and eta-quotients with Rademacher-type expansions; the partition function is unusually clean because the Bessel index $3/2$ makes the kernel two exponentials with amplitudes linear in μ^{-1}, so step (iii) is the explicit rational recurrence Equation (O.72) rather than a general implicit solve. *No theorem about any other specific sequence is claimed here*: verifying (T1)–(T4) for a given family is exactly the work the template does not do.

O.15 Structural consequences and open questions

O.15.1 Convergent blocks inside an exponentially complete object

The exact sector sum Equation (O.19) converges; each sector's algebraic series in $n^{-1/2}$ converges on $|n^{-1/2}| < \sqrt{24}$ (Theorem O.30); and the reversion series U converges near 0 (Theorem O.50). Divergence is therefore *not* what creates the non-perturbative sectors here: they are supplied by modularity and the circle method, independently of the perturbative data. The useful trichotomy is that a *Poincaré expansion* records one dominant sector and forgets all exponentially small information; a *transseries completion* records the other sectors, whether or not the series in each diverges; and an *exact transseries representation*, as here, may itself converge. Compare Parts M, N and P, where the algebraic series do diverge and Theorem J.54 is the operative tool.

O.15.2 Arithmetic amplitudes are not Stokes constants

The $A_k(n)$ weight distinct exponential sectors and govern exponentially small corrections to the inverse, which is formally the role of Stokes constants. They are not Stokes constants: they are arithmetic, periodic in n rather than fixed, and no natural analytic continuation in the index produces them canonically (Theorem O.11). Theorem O.75 says what replaces the Stokes data: a profinite integer, of which every finite exponential resolution sees only a finite quotient.

O.15.3 Open questions

Conjecture O.95 (Effective rounding threshold)**.** *There is an explicit small n_0 — the evidence of Theorem O.84 suggests $n_0 = 1$ — with $n = \lfloor N_0(p(n)) \rceil$ for all $n \geq n_0$, provable by inserting effective Rademacher remainder bounds (Lehmer; Banerjee–Paule–Radu–Schneider) in place of $|A_k| \leq k$ in Theorem O.78.*

Conjecture O.96 (Arithmetic transseries sheaf)**.** *The projective family of Theorem O.75 extends to a sheaf of transseries algebras over $\widehat{\mathbb{Z}}$, with $\widehat{r} \mapsto$ (level-K inverse transseries) continuous for the profinite topology and the finite-level algebras gluing over the standard clopen basis. Only the projective compatibility is proved here (Theorem O.77).*

Conjecture O.97 (Uniform treatment at the accumulation action)**.** *The zeta condensation Equation* (O.29) *can serve as a single block in Theorem O.62, yielding a description of the inverse uniform across the accumulation action* 1*, with error bounds depending on the analytic behaviour of $\mathcal{Z}_n(s)$ rather than on a sector cutoff. This needs continuation and vertical growth information for the Kloosterman-type Dirichlet series $\mathcal{Z}_n$, which is not established here. S3 raises it as an optimisation problem rather than a gap, correctly: the exact Rademacher representation is always available as a fallback.*

Remark O.98 (Classifying interpolations). Different strictly increasing smooth interpolations of p agree at every algebraic order of the inverse and on the integer range, but differ in the exponentially small off-lattice sectors: candidates include Theorem O.9, piecewise-logarithmic interpolation through $(n, \log p(n))$, Newton or cardinal series with prescribed growth, and continuations built from Poincaré series. Which part of the inverse transseries is interpolation-independent? Theorem O.59 answers this for one pair of choices — they agree strictly below action 1 — and a general statement is open.

Remark O.99 (Formalisation targets). Rademacher's theorem is now machine-checked (Eberl and Paulson, Archive of Formal Proofs, 2026). The natural next steps, in dependency order, are: (a) the normalisation Theorem O.14; (b) the generating function Equation (O.31) and the recurrence Equation (O.41); (c) the Catalan/residue proof of Theorem O.31; (d) the implicit construction of U and Equation (O.72); (e) the triangular multivariate inversion Equation (O.93); (f) residual-to-root transfer, Theorem O.80, under an explicit derivative lower bound. Steps (b), (d), (e) are Bell-polynomial infrastructure shared with Part J.

O.16 Summary

1. In the shifted phase $\mu = \pi\sqrt{2(n-1/24)/3}$, Rademacher's series is the exact sector sum Equation (O.17), each sector having a two-term amplitude linear in μ^{-1}. At $(k, \sigma) = (1, +1)$ it is the exponential–power model Theorem J.2 with $\alpha = 1/2$, reduced by Theorem J.4 to $\kappa e^{\mu}\mu^{-2}(1-\mu^{-1})$ (Theorem O.1 and Table O.1).
2. Re-expansion in $n^{-1/2}$ is governed by one analytic function Equation (O.31); its coefficients have the finite closed form Equation (O.33) in *every* sector, which enters only through $\beta = \sigma\pi/(6k)$. Each such series converges while the total is only asymptotic — confusing the two is the characteristic error of the subject.
3. The actions $1 \mp 1/k$ accumulate at 1 from both sides, so the completed object is not a Hahn series and the two signs may not be separated (Theorems O.19 and O.20); Theorem O.23 bounds the tail explicitly and Theorem O.28 packages the accumulation exactly.
4. The inverse carrier is $-2W_{-1}(-\frac{1}{2}\sqrt{\kappa/y})$ with Stirling flattening polynomials Equation (O.58); the amplitude is restored by Theorem J.29 at $\Psi(t,u) = 3\log(1+tu) - \log(1-t+tu)$, whose coefficients are Table O.2.
5. Since $\mathrm{d}n/\mathrm{d}\mu \asymp \mu$, that reversion produces an $\mathcal{O}(1)$ index shift $3/\pi^2$ which must be resummed *before* the periodic amplitudes are evaluated (Theorem O.55).
6. With the phase frozen — by a residue class mod $\operatorname{lcm}(1,\ldots,K)$ or by the entire interpolation Theorem O.9, which agree strictly below action 1 (Theorem O.59) — every inverse transmonomial is given by Equation (O.86), Equation (O.88), Equation (O.95) or Equation (O.93). The leading correction is the parity chirp Equation (O.106) with constant $3^{1/4}/(2\pi)$, and the primary sectors form the staircase Equation (O.105).

7. Along the range the core alone recovers n by rounding (Theorem O.83); for general y the staircase is the ceiling of the smooth inverse (Theorem O.44), whose order-one sawtooth dominates every arithmetic sector (Theorem O.45).

Remark O.100 (Works referred to). Analytic input: Rademacher, Proc. Nat. Acad. Sci. USA **23** (1937), 78–84; Proc. London Math. Soc. (2) **43** (1938), 241–254; Ann. of Math. **44** (1943), 416–422; textbook treatment in Apostol, *Modular Functions and Dirichlet Series in Number Theory*, 2nd ed., Springer GTM 41, 1990; machine-checked in Eberl and Paulson, *Rademacher's series for the partition function*, Archive of Formal Proofs, 2026. Leading asymptotic: Hardy and Ramanujan, Proc. London Math. Soc. (2) **17** (1918), 75–115. Remainder estimates: Lehmer, J. London Math. Soc. **12** (1937), 171–176 and Trans. Amer. Math. Soc. **43** (1938), 271–295, **46** (1939), 362–373; Banerjee, Paule, Radu and Schneider, arXiv:2209.07887. The dominant-sector case of Equation (O.33) is O'Sullivan's, arXiv:2205.13468 and arXiv:2203.02868. Lambert W: Corless, Gonnet, Hare, Jeffrey and Knuth, Adv. Comput. Math. **5** (1996), 329–359; Jeffrey, Corless, Hare and Knuth, arXiv:math/9512230. The sequence is OEIS A000041.

Appendix P

The swing factorial (A056040)

This chapter merges three independent treatments of the swinging factorial: *All-Orders Asymptotic Transseries for the Swing Factorial and Its Two Inverse Branches* (cited below as S1), *The Full Asymptotic Transseries of the Swing Factorial and Its Inverse* (S2), and *Parity-Resolved Transseries for the Swing Factorial and Its Inverse* (S3). The general apparatus of S2 — its universal coefficient calculus for exponential–power inversion and its Lambert-normalized reversion theorem — was promoted to Part J; here it is only *cited* and *specialized.* S1 works in the half index m with the carrier constant $\log 4$; S2 and S3 work in the original index with $\log 2$ and differ only in the sign convention for the parity marker. The chapter adopts the S2/Part J normalization throughout and gives the translation dictionary in Theorem P.12. S1's genuinely distinct contribution — the centred (translation-symmetric) normal form — is kept, but recast so that it too is a single σ-uniform statement.

Every rational, polynomial and algebraic coefficient displayed in this chapter was recomputed from the stated recurrences in exact arithmetic (sympy, rational Bernoulli input, truncated power series over $\mathbb{Q}[\lambda, \lambda^{-1}, \sigma]/(\sigma^2 - 1)$); every decimal in the validation tables was recomputed at 50 significant digits. Where a recomputation reproduced a source table, that is said; where it did not, the discrepancy is recorded in a repairbox.

P.1 The sequence, the parity obstruction, and what “inverse” can mean

P.1.1 Definition and exact parity identities

Definition P.1 (Swing factorial)**.** For $n \in \mathbb{N}_0$ put

$$\mathrm{sw}(n) = \frac{n!}{\lfloor n/2 \rfloor!^{\,2}}. \tag{P.1}$$

This is OEIS A056040, the *swinging factorial.*

Splitting on the parity of n gives at once

$$\mathrm{sw}(2m) = \binom{2m}{m}, \qquad \mathrm{sw}(2m+1) = (2m+1)\binom{2m}{m}, \tag{P.2}$$

whence the one-step recurrence

$$\mathrm{sw}(0) = 1, \qquad \mathrm{sw}(n) = \begin{cases} n\ \mathrm{sw}(n-1), & n \text{ odd},\\ \dfrac{4}{n}\ \mathrm{sw}(n-1), & n \text{ even}, \end{cases} \tag{P.3}$$

and the closed product form

$$\mathrm{sw}(n) = 2^{n-(n \bmod 2)} \prod_{k=1}^{n} k^{(-1)^{k+1}}. \tag{P.4}$$

The first values are

$$1,\ 1,\ 2,\ 6,\ 6,\ 30,\ 20,\ 140,\ 70,\ 630,\ 252,\ 2772,\ 924,\ 12012,\ 3432,\ 51480,\ \ldots$$

Proposition P.2 (Generating functions).

$$\sum_{n\geq 0} \mathrm{sw}(n)\frac{z^n}{n!} = (1+z)\, I_0(2z), \tag{P.5}$$

$$\sum_{n\geq 0} \mathrm{sw}(n) z^n = \frac{1}{\sqrt{1-4z^2}} + \frac{z}{(1-4z^2)^{3/2}}, \tag{P.6}$$

where I_0 is the modified Bessel function of order zero.

Proof. By Equation (P.2) the even exponential part is $\sum_{m\geq 0} z^{2m}/(m!)^2 = I_0(2z)$ and the odd exponential part is $z\, I_0(2z)$, which is Equation (P.5). For the ordinary series put $F(v) = \sum_{m\geq 0} \binom{2m}{m} v^m = (1-4v)^{-1/2}$; then the even part is $F(z^2)$ and the odd part is $z\big(2vF'(v)+F(v)\big)\big|_{v=z^2} = z(1-4z^2)^{-3/2}$. □

Remark P.3 (The generating function already shows the obstruction). The two singularities $z = \pm\frac{1}{2}$ of Equation (P.6) have equal modulus. Co-dominant singularities of equal modulus at $\pm\rho$ produce a period-two oscillation in the coefficient asymptotics, and no single non-oscillatory singular expansion can represent the whole sequence. Even/odd projection separates the two contributions exactly. This is the generating-function shadow of everything that follows.

P.1.2 The sequence is not eventually monotone

Proposition P.4 (Exact adjacent ratios). *For $m \geq 0$,*

$$\frac{\mathrm{sw}(2m+1)}{\mathrm{sw}(2m)} = 2m+1, \qquad \frac{\mathrm{sw}(2m+2)}{\mathrm{sw}(2m+1)} = \frac{2}{m+1}. \tag{P.7}$$

Consequently sw *rises strictly from every even index to the next odd index, and falls strictly from every odd index to the next even index as soon as $m \geq 2$. In particular* sw *is not eventually monotone and has no eventually defined single-valued inverse.*

Proof. Both ratios are Equation (P.3). The second is < 1 exactly when $m+1 > 2$. □

Warning P.5 (There is no "the" inverse swing factorial). Any formula presented as *the* inverse of sw without naming a parity sheet or a generalized-inverse convention is incomplete: of the three inverse objects of Theorem J.41, none exists globally for sw and all three exist branchwise (Section P.10).

P.1.3 The two analytic branches

All three sources resolve the obstruction the same way: interpolate the two bisections separately by gamma quotients. We use the parity marker $\sigma \in \{-1, +1\}$ of S2, with

$$\sigma = -1 \text{ (even phase)}, \qquad \sigma = +1 \text{ (odd phase)}, \qquad \sigma_n := (-1)^{n+1}, \qquad \lambda := \log 2. \tag{P.8}$$

The coefficient ring of the whole chapter is the *parity algebra*

$$\mathscr{R} := \mathbb{Q}[\lambda, \lambda^{-1}, \sigma]/(\sigma^2 - 1), \tag{P.9}$$

in which every formula below is a single element carrying both sheets. This is not a notational economy only: σ_n is a discrete transmonomial of modulus one, and it must remain visible at every order of every expansion.

Definition P.6 (Parity-resolved gamma branches). For $x > 0$ and $\sigma \in \{-1, +1\}$ set

$$\mathcal{S}_\sigma(x) := \frac{\Gamma(x+1)}{\Gamma\big(x/2 + (3-\sigma)/4\big)^2}. \tag{P.10}$$

Explicitly $\mathcal{S}_-(x) = \Gamma(x+1)/\Gamma(x/2+1)^2$ and $\mathcal{S}_+(x) = \Gamma(x+1)/\Gamma\big((x+1)/2\big)^2$, and

$$\mathcal{S}_-(2m) = \mathrm{sw}(2m), \qquad \mathcal{S}_+(2m+1) = \mathrm{sw}(2m+1) \qquad (m \in \mathbb{N}_0). \tag{P.11}$$

Proposition P.7 (Duplication normal form and duality). *For $x > 0$,*

$$\mathcal{S}_\sigma(x) = \frac{2^x}{\sqrt{\pi}} \left(\frac{\Gamma(x/2+1)}{\Gamma(x/2+1/2)} \right)^{\sigma}, \tag{P.12}$$

and consequently

$$\mathcal{S}_+(x)\,\mathcal{S}_-(x) = \frac{4^x}{\pi}, \qquad \frac{\mathcal{S}_\sigma(x+2)}{\mathcal{S}_\sigma(x)} = \frac{16(x+1)(x+2)}{(2x+3-\sigma)^2} = \begin{cases} \dfrac{4(x+1)}{x+2}, & \sigma = -1, \\ \dfrac{4(x+2)}{x+1}, & \sigma = +1. \end{cases} \tag{P.13}$$

Proof. Legendre duplication in the form $\Gamma(x+1) = 2^x \pi^{-1/2} \Gamma\big((x+1)/2\big)\Gamma(x/2+1)$ turns Equation (P.10) into Equation (P.12); multiplying the two values of σ makes the gamma ratio cancel and leaves $4^x/\pi$. For the two-step ratio, $\Gamma(x+3)/\Gamma(x+1) = (x+1)(x+2)$ while the denominator contributes $\big(x/2 + (3-\sigma)/4\big)^{-2} = 16(2x+3-\sigma)^{-2}$. □

Proposition P.8 (Strict monotonicity and existence of the real inverses). *For $x > 0$,*

$$\frac{\mathrm{d}}{\mathrm{d}x} \log \mathcal{S}_\sigma(x) = \lambda + \frac{\sigma}{2}\Big[\psi\big(\tfrac{x}{2}+1\big) - \psi\big(\tfrac{x}{2}+\tfrac{1}{2}\big)\Big], \tag{P.14}$$

where $\psi = \Gamma'/\Gamma$. The bracket decreases strictly from 2λ at $x = 0$ to 0 at $x = \infty$. Hence $\mathcal{S}_+$ has strictly positive derivative on $[0,\infty)$, $\mathcal{S}_-$ has derivative 0 at $x = 0$ and strictly positive derivative on $(0,\infty)$, and both are strictly increasing bijections of $[0,\infty)$ onto $[1,\infty)$. Write

$$\mathcal{I}_\sigma := \mathcal{S}_\sigma^{-1} : [1,\infty) \to [0,\infty). \tag{P.15}$$

Proof. Differentiating Equation (P.12) logarithmically gives Equation (P.14). Put $g(x) = \psi(x/2+1) - \psi(x/2+1/2)$. Then $g'(x) = \frac{1}{2}\big[\psi'(x/2+1) - \psi'(x/2+1/2)\big] < 0$ because the trigamma function $\psi'(t) = \sum_{k\ge 0}(t+k)^{-2}$ is strictly decreasing on $(0,\infty)$; and $g(x) \to 0$ as $x \to \infty$ by $\psi(t+\frac{1}{2}) - \psi(t) \to 0$. Finally $g(0) = \psi(1) - \psi(1/2) = 2\log 2 = 2\lambda$. Thus $0 < g(x) < 2\lambda$ for $x > 0$. For $\sigma = +1$, Equation (P.14) equals $\lambda + g/2 > 0$ on $[0,\infty)$; for $\sigma = -1$ it equals $\lambda - g/2$, which vanishes only at $x = 0$ and is positive afterwards. Both branches take the value 1 at $x = 0$ and tend to $+\infty$, so each is a continuous increasing bijection onto $[1,\infty)$. □

P.1.4 Why one cannot simply interpolate through both parities

Proposition P.9 (Every smooth interpolation of the whole sequence oscillates). *With $R(x) = \Gamma(x/2+1)/\Gamma\big((x+1)/2\big)$, put*

$$\widehat{\mathcal{S}}(x) = \frac{2^x}{\sqrt{\pi}}\, R(x)^{-\cos \pi x}. \tag{P.16}$$

Then $\widehat{\mathcal{S}}(n) = \mathrm{sw}(n)$ for every $n \in \mathbb{N}_0$, yet $\widehat{\mathcal{S}}$ is not eventually monotone: its logarithmic derivative

$$\frac{\mathrm{d}}{\mathrm{d}x}\log\widehat{\mathcal{S}}(x) = \lambda + \pi\sin(\pi x)\log R(x) - \cos(\pi x)\,\frac{R'(x)}{R(x)} \tag{P.17}$$

has an oscillating term of amplitude $\sim \frac{\pi}{2}\log x$ against remaining terms that are $O(1)$, so it changes sign infinitely often. The same holds for the convex interpolation $\frac{1+\cos\pi x}{2}\mathcal{S}_- + \frac{1-\cos\pi x}{2}\mathcal{S}_+$. Consequently the phase-resolved pair $(\mathcal{S}_-, \mathcal{S}_+)$*, and not any single function, is the asymptotically stable continuation of* sw*; the corresponding inverse object is the pair $(\mathcal{I}_-, \mathcal{I}_+)$.*

Proof. At $x = n$ one has $-\cos\pi n = (-1)^{n+1} = \sigma_n$, so $\widehat{\mathcal{S}}(n) = (2^n/\sqrt{\pi})R(n)^{\sigma_n} = \mathcal{S}_{\sigma_n}(n) = \mathrm{sw}(n)$ by Equation (P.12) and Equation (P.11). Differentiating $\log\widehat{\mathcal{S}} = x\lambda - \frac{1}{2}\log\pi - \cos(\pi x)\log R(x)$ gives Equation (P.17); by Theorem P.15 below, $\log R(x) = \frac{1}{2}\log(x/2) + O(x^{-1})$ and $R'/R = O(x^{-1})$, so the oscillating term has amplitude $\frac{\pi}{2}\log x + O(1) \to \infty$ and dominates. The convex interpolation is treated the same way after writing it as $\mathcal{S}_-(x)$ times a factor whose logarithmic derivative oscillates with amplitude $\asymp \log x$. □

P.2 Exact normal form, Binet reduction, and Borel structure

P.2.1 The single scalar correction

Definition P.10 (The gamma-ratio correction)**.** For $x > 0$ set

$$D(x) := \log\frac{\Gamma(x/2+1)}{\Gamma(x/2+1/2)} - \frac{1}{2}\log\frac{x}{2}. \tag{P.18}$$

Theorem P.11 (Exact σ-uniform normal form)**.** *For $x > 0$ and $\sigma \in \{-1, +1\}$,*

$$\boxed{\mathcal{S}_\sigma(x) = C_\sigma\,\mathrm{e}^{\lambda x}\,x^{\sigma/2}\,\mathrm{e}^{\sigma D(x)}, \qquad C_\sigma = \frac{2^{-\sigma/2}}{\sqrt{\pi}}.} \tag{P.19}$$

At integer arguments this is the exact parity formula

$$\log\mathrm{sw}(n) = n\lambda - \tfrac{1}{2}\log\pi + \sigma_n\Big(\tfrac{1}{2}\log\tfrac{n}{2} + D(n)\Big), \qquad n \ge 1. \tag{P.20}$$

Proof. By Equation (P.18), $\Gamma(x/2+1)/\Gamma(x/2+1/2) = (x/2)^{1/2}\mathrm{e}^{D(x)}$. Substituting into Equation (P.12) gives $\mathcal{S}_\sigma = (2^x/\sqrt{\pi})(x/2)^{\sigma/2}\mathrm{e}^{\sigma D}$, and $(x/2)^{\sigma/2} = 2^{-\sigma/2}x^{\sigma/2}$. For Equation (P.20), take logarithms of Equation (P.19) at $x = n$, $\sigma = \sigma_n$, and use $\frac{1}{2}\log n - \frac{\lambda}{2} = \frac{1}{2}\log\frac{n}{2}$. □

Equation (P.19) is exactly Theorem J.2 with

$$\boxed{\alpha = 1, \qquad a = \lambda = \log 2, \qquad b = \frac{\sigma}{2}, \qquad C = C_\sigma = \frac{2^{-\sigma/2}}{\sqrt{\pi}}, \qquad q_j = \sigma d_j,} \tag{P.21}$$

where d_j are the coefficients of the asymptotic series of D (Theorem P.18). Because $\alpha = 1$ no monomial substitution is needed and Theorem J.4 is not invoked. We record the dictionary once and use it in every specialization below.

Remark P.12 (Dictionary between the three sources)*.* Let $\varepsilon_{\mathsf{S3}} \in \{+1, -1\}$ be the marker of S3 and $\epsilon_{\mathsf{S1}} \in \{0, 1\}$ that of S1. Then

$$\sigma = -\varepsilon_{\mathsf{S3}}, \qquad \sigma = 2\epsilon_{\mathsf{S1}} - 1, \qquad \mathsf{S3}: \ Q = -D, \quad q_r^{\mathsf{S3}} = -d_{2r-1}, \qquad \mathsf{S1}: \ \alpha_{\mathsf{S1}} = \log 4 = 2\lambda.$$

S1 parametrizes each sheet by the half index m with $S^{\mathsf{S1}}_\epsilon(m) = \mathrm{sw}(2m+\epsilon)$; as functions this is the reparametrization $x = 2m+\epsilon$ of Theorem P.6, so

$$S^{\mathsf{S1}}_\epsilon(m) = \mathcal{S}_{2\epsilon-1}(2m+\epsilon), \qquad \mathcal{I}_\sigma(y) = 2M^{\mathsf{S1}}_\epsilon(y) + \epsilon.$$

The reparametrization is invisible in the forward coefficients up to the substitution $x \mapsto 2m+\epsilon$, but it is *not* invisible in the inverse coefficients: the two normalizations invert different leading phases ($\lambda x + \frac{\sigma}{2}\log x$ against $2\lambda m + \frac{\sigma}{2}\log m$) and therefore have different Lambert cores. Only for the even sheet do the two cores agree exactly, through $X_-(y) = 2w^{\mathsf{S1}}_0(y)$; see Theorem P.37.

P.2.2 The Binet kernel collapses to a hyperbolic tangent

Theorem P.13 (Exact Borel–Laplace representation)**.** *For $\Re x > 0$,*

$$\boxed{D(x) = \int_0^\infty \mathrm{e}^{-xt}\,\widehat{D}(t)\,\mathrm{d}t, \qquad \widehat{D}(t) = \frac{\tanh(t/2)}{2t},} \tag{P.22}$$

the singularity of $\widehat{D}$ at $t=0$ being removable with $\widehat{D}(0) = \frac14$.

Proof. Put $z = x/2$ and $h(z) = \log\Gamma(z+1) - \log\Gamma(z+\frac12) - \frac12\log z$, so $h(x/2) = D(x)$. From the standard integral for a digamma difference,

$$\psi(z+1) - \psi(z+\tfrac12) = \int_0^\infty \frac{\mathrm{e}^{-(z+1/2)s} - \mathrm{e}^{-(z+1)s}}{1-\mathrm{e}^{-s}}\,\mathrm{d}s = 2\int_0^\infty \frac{\mathrm{e}^{-2zv}}{\mathrm{e}^v+1}\,\mathrm{d}v,$$

the second equality by $s = 2v$ and $(\mathrm{e}^{-v} - \mathrm{e}^{-2v})/(1-\mathrm{e}^{-2v}) = 1/(\mathrm{e}^v+1)$. Since $1/(2z) = \int_0^\infty \mathrm{e}^{-2zv}\mathrm{d}v$,

$$h'(z) = \psi(z+1) - \psi(z+\tfrac12) - \frac{1}{2z} = \int_0^\infty \mathrm{e}^{-2zv}\Big(\frac{2}{\mathrm{e}^v+1} - 1\Big)\mathrm{d}v = -\int_0^\infty \mathrm{e}^{-2zv}\tanh\frac{v}{2}\,\mathrm{d}v.$$

Both $h(z)$ and the right-hand side of Equation (P.22) tend to 0 as $\Re z \to +\infty$, so integrating from $+\infty$ down to z,

$$h(z) = \int_z^\infty\int_0^\infty \mathrm{e}^{-2\zeta v}\tanh\tfrac{v}{2}\,\mathrm{d}v\,\mathrm{d}\zeta = \int_0^\infty \mathrm{e}^{-2zv}\frac{\tanh(v/2)}{2v}\,\mathrm{d}v,$$

which is Equation (P.22) after $x = 2z$. Fubini applies because the double integral is absolutely convergent for $\Re z > 0$. ☐

S3 reaches Equation (P.22) from Binet's first formula, through the elegant identity $b(t) - 2b(2t) = -\frac12\tanh(t/2)$ for $b(t) = (\mathrm{e}^t-1)^{-1} - t^{-1} + \frac12$; S2 reaches it from the digamma integral, as above. S1 keeps Binet's *second* formula (the $\arctan$ kernel) instead and never simplifies the difference of the two Binet integrals. The $\tanh$ collapse is the decisive simplification: it is what makes the Borel plane elementary, and it is adopted here.

Corollary P.14 (Sign, monotonicity, and elementary bounds)**.** *For $x > 0$,*

$$0 < D(x) < \frac{1}{4x}, \qquad 0 < -D'(x) < \frac{1}{4x^2}, \qquad (-1)^k D^{(k)}(x) > 0 \quad (k \ge 0). \tag{P.23}$$

In particular D is completely monotone and strictly decreasing.

Proof. $\widehat{D} > 0$ on $(0,\infty)$, so $(-1)^k D^{(k)}(x) = \int_0^\infty \mathrm{e}^{-xt}t^k\widehat{D}(t)\mathrm{d}t > 0$. From $0 < \tanh(t/2) < t/2$ we get $0 < \widehat{D}(t) < \frac14$, and integrating $\mathrm{e}^{-xt}t^k$ gives the two displayed bounds. ☐

Corollary P.15 (Asymptotics of the gamma ratio)**.** *With $R(x) = \Gamma(x/2+1)/\Gamma((x+1)/2)$ as in Equation* (P.16)*,*

$$\log R(x) = \tfrac12\log(x/2) + D(x) = \tfrac12\log(x/2) + O(x^{-1}), \qquad \frac{R'(x)}{R(x)} = \frac{1}{2x} + D'(x) = O(x^{-1}).$$

Proof. Immediate from Equation (P.18) and Equation (P.23). ☐

P.2.3 The Borel singularity lattice

Proposition P.16 (Mittag–Leffler expansion and poles). *The Borel transform of Equation* (P.22) *satisfies*

$$\widehat{D}(t) = 2\sum_{k\geq 0} \frac{1}{t^2+\pi^2(2k+1)^2}, \tag{P.24}$$

is even and meromorphic on $\mathbb{C}$, *and has exactly the simple poles*

$$t = \pm \mathrm{i}\pi(2k+1), \qquad k\geq 0, \qquad \operatorname{Res}_{t=t_k}\widehat{D}(t) = \frac{1}{t_k}. \tag{P.25}$$

The positive real ray is free of singularities; the nearest singularities lie at distance π*, so the formal series of* D *is Gevrey-*1 *with singulant modulus* π*.*

Proof. $\tanh(w) = \sum_{k\geq 0} 2w/\big(w^2+\pi^2(k+\frac12)^2\big)$ is the classical Mittag–Leffler expansion; putting $w=t/2$ and dividing by $2t$ gives Equation (P.24), both sides being even, meromorphic, with the same poles and residues and the same value $\frac14$ at $t=0$ (using $\sum_{k\geq 0}(2k+1)^{-2}=\pi^2/8$). At $t=t_k$ the function $\tanh(t/2)$ has residue 2, so $\widehat{D}$ has residue $2/(2t_k)=1/t_k$. □

Remark P.17 (What the exponentially small scale is, and is not). The quantity $\mathrm{e}^{-\pi x}$ that governs optimal truncation (Section P.9) is the envelope set by the distance π of the nearest *off-axis* Borel poles; it is not an ambiguity of the positive-axis sum, since the ray $\arg t = 0$ meets no singularity and Equation (P.22) is exact. Genuine Stokes jumps appear only when the Laplace direction is rotated onto the imaginary rays. S1, retaining an unsimplified Binet integral, makes the weaker but correct statement that no Stokes constant need be assigned on the real axis; the Borel picture explains why.

P.3 The logarithmic coefficients and a sharp enveloping remainder

P.3.1 All coefficients

We use the Bernoulli convention $z/(\mathrm{e}^z-1)=\sum_{n\geq 0} B_n z^n/n!$, so $B_1=-\frac12$, and the Bernoulli polynomials $t\mathrm{e}^{wt}/(\mathrm{e}^t-1)=\sum_n B_n(w)t^n/n!$.

Theorem P.18 (All logarithmic coefficients). *As* $x\to\infty$ *in any closed subsector of* $|\arg x|<\pi/2$*,*

$$D(x)\sim\sum_{j\geq 1}\frac{d_j}{x^j}, \qquad d_{2r}=0, \qquad d_{2r+1}=\frac{(2^{2r+2}-1)B_{2r+2}}{(2r+1)(2r+2)}. \tag{P.26}$$

Equivalently

$$d_{2r+1} = (-1)^r\,\frac{2\,(2r)!}{\pi^{2r+2}}\big(1-2^{-2r-2}\big)\zeta(2r+2) = (-1)^r\,\frac{2\,(2r)!}{\pi^{2r+2}}\sum_{k\geq 0}\frac{1}{(2k+1)^{2r+2}}, \tag{P.27}$$

and the formal Borel transform of the series is exactly the Taylor expansion of $\widehat{D}$*:*

$$\widehat{D}(t)=\sum_{r\geq 0} d_{2r+1}\frac{t^{2r}}{(2r)!}=\frac{\tanh(t/2)}{2t}. \tag{P.28}$$

Proof. Insert into Equation (P.24) the finite geometric identity

$$\frac{1}{t^2+a^2}=\sum_{r=0}^{N-1}\frac{(-1)^r t^{2r}}{a^{2r+2}}+\frac{(-1)^N t^{2N}}{a^{2N}(t^2+a^2)}, \qquad a=a_k=\pi(2k+1),$$

and integrate the finite sum against e^{-xt} using $\int_0^\infty \mathrm{e}^{-xt}t^{2r}\mathrm{d}t = (2r)!x^{-2r-1}$. This gives the second expression in Equation (P.27), hence the first by $\zeta(2r+2)(1-2^{-2r-2}) = \sum_{k\geq 0}(2k+1)^{-(2r+2)}$, and hence Equation (P.26) by Euler's evaluation $B_{2m} = (-1)^{m-1}2(2m)!\zeta(2m)/(2\pi)^{2m}$. The unexpanded last term is the remainder controlled in Theorem P.20, which in particular proves the asymptotic statement on $\Re x > 0$; the identity Equation (P.28) is the Taylor expansion of $\widehat{D}$ at the origin. $\square$

Remark P.19 (Sector of validity: a disagreement between the sources). S2 and S3 state Equation (P.26) on closed subsectors of $|\arg x| < \pi/2$, which is what the Laplace representation Equation (P.22) gives. S1 states the corresponding expansion on $|\arg m| \leq \pi - \delta$. Both are correct: the wider sector follows not from the Laplace integral but from the Tricomi–Erdélyi expansion of a gamma ratio,

$$\log\frac{\Gamma(z+a)}{\Gamma(z+b)} \sim (a-b)\log z + \sum_{n\geq 1}\frac{(-1)^{n+1}}{n(n+1)z^n}\big(B_{n+1}(a) - B_{n+1}(b)\big), \qquad |\arg z| \leq \pi - \delta, \tag{P.29}$$

applied with $a = 1, b = \frac{1}{2}, z = x/2$. The volume states the Laplace sector where an *exact* representation and a signed remainder are wanted, and the wider sector where only a Poincaré statement is wanted. The two routes give the same coefficients: with $B_m(\frac{1}{2}) = (2^{1-m}-1)B_m$ and $B_m(1) = B_m$ for $m \neq 1$, Equation (P.29) yields $d_n = 2^n\,\frac{(-1)^{n+1}}{n(n+1)}\big(2-2^{-n}\big)B_{n+1}$, which is Equation (P.26).

Table P.1: The logarithmic coefficients d_{2r+1} of Theorem P.18, recomputed from Equation (P.26) in exact rational arithmetic and cross-checked against the Taylor expansion of $\tanh(t/2)/(2t)$.

j	1	3	5	7	9	11	13
d_j	$\frac{1}{4}$	$-\frac{1}{24}$	$\frac{1}{20}$	$-\frac{17}{112}$	$\frac{31}{36}$	$-\frac{691}{88}$	$\frac{5461}{52}$

Thus

$$D(x) \sim \frac{1}{4x} - \frac{1}{24x^3} + \frac{1}{20x^5} - \frac{17}{112x^7} + \frac{31}{36x^9} - \frac{691}{88x^{11}} + \frac{5461}{52x^{13}} - \cdots \tag{P.30}$$

P.3.2 An enveloping remainder, not merely a Poincaré one

Theorem P.20 (Exact remainder and strict next-term envelope). *For $x > 0$ and $N \geq 0$ write $D(x) = \sum_{r=0}^{N-1} d_{2r+1}x^{-(2r+1)} + R_N(x)$ (empty sum for $N = 0$). Then, with $a_k = \pi(2k+1)$,*

$$R_N(x) = 2(-1)^N\sum_{k\geq 0} a_k^{-2N}\int_0^\infty \frac{\mathrm{e}^{-xt}t^{2N}}{t^2+a_k^2}\,\mathrm{d}t, \tag{P.31}$$

and consequently

$$\boxed{0 < (-1)^N R_N(x) < \frac{|d_{2N+1}|}{x^{2N+1}}} \qquad (x > 0,\ N \geq 0). \tag{P.32}$$

Every truncation therefore brackets $D(x)$ together with its successor.

Proof. Substitute the finite geometric identity of the previous proof into Equation (P.24) and then into Equation (P.22); Tonelli's theorem justifies the interchange since all terms are positive. This gives Equation (P.31), in which every integral is positive. Replacing $(t^2+a_k^2)^{-1}$ by the strict upper bound a_k^{-2} and using $\int_0^\infty \mathrm{e}^{-xt}t^{2N}\mathrm{d}t = (2N)!x^{-2N-1}$ gives

$$0 < (-1)^N R_N(x) < 2(2N)!\,x^{-(2N+1)}\sum_{k\geq 0} a_k^{-(2N+2)} = \frac{|d_{2N+1}|}{x^{2N+1}},$$

by the last expression in Equation (P.27). $\square$

Remark P.21. Theorem P.20 is strictly stronger than the Poincaré statement, and it is what makes every later analytic claim in this chapter effective: all remainder constants below are explicit rational multiples of the first omitted $|d_{2N+1}|$. S2 and S3 both prove it; S1 does not have it, because it never simplifies the Binet kernel. It was verified numerically at $x = 5$ and $x = 10$ for $0 \le N \le 4$: the bracketing Equation (P.32) holds with the two sides agreeing to within a factor 1.33 at $x = 5$ and 1.09 at $x = 10$.

P.3.3 Large order and least term

Proposition P.22 (Least term and its envelope). *As $r \to \infty$, $d_{2r+1} \sim (-1)^r 2(2r)!\pi^{-2r-2}$, so the ratio of consecutive nonzero term magnitudes in Equation* (P.30) *is $\sim (2r+2)(2r+1)/(\pi^2 x^2)$. The least term therefore occurs at $r \approx \pi x/2$ and has magnitude*

$$\frac{2\sqrt{2}}{\pi\sqrt{x}}\,\mathrm{e}^{-\pi x}\big(1 + O(x^{-1})\big). \tag{P.33}$$

By Theorem P.20, truncation at the least term attains an error of exactly this scale, with no hidden constant.

Proof. The asymptotic form of d_{2r+1} is Equation (P.27) with $\zeta(2r+2) \to 1$. The ratio statement is immediate. Setting $2r = \pi x$ and applying Stirling's formula to $(2r)!$ gives $2(2r)!\pi^{-2r-2}x^{-2r-1} \sim 2\sqrt{2\pi^2 x}\,\mathrm{e}^{-\pi x}/(\pi^2 x) = 2\sqrt{2}\,\mathrm{e}^{-\pi x}/(\pi\sqrt{x})$. □

Remark P.23 (Numerical confirmation). Recomputed: at $x = 10$ the least nonzero term of Equation (P.30) occurs at $r = 15$ (against $\pi x/2 = 15.708$) with magnitude 6.5448×10^{-15}, while Equation (P.33) predicts 6.4659×10^{-15}; at $x = 20$ the least term occurs at $r = 31$ (against 31.416) with magnitude 1.03859×10^{-28} against a predicted 1.03837×10^{-28}.

P.4 The forward transseries

P.4.1 Exponentiation

Theorem P.24 (All forward coefficients). *Define $A_0 = 1$ and, for $n \ge 1$,*

$$A_n = [t^n]\,\exp\Big(\sum_{j\ge 1} d_j t^j\Big) = \frac{1}{n!}\,B_n\big(1!\,d_1, 2!\,d_2, \ldots, n!\,d_n\big) = \sum_{\substack{k_1,k_2,\ldots\ge 0\\ \sum_j jk_j = n}} \prod_{j\ge 1} \frac{d_j^{k_j}}{k_j!}, \tag{P.34}$$

with B_n the complete exponential Bell polynomial of Theorem J.10. Equivalently and most cheaply,

$$A_0 = 1, \qquad nA_n = \sum_{j=1}^{n} j d_j A_{n-j} = \sum_{\substack{1\le j\le n\\ j \text{ odd}}} j d_j A_{n-j}. \tag{P.35}$$

Then, for every fixed $N \ge 1$ and uniformly on closed subsectors of $\Re x > 0$,

$$\boxed{\mathcal{S}_\sigma(x) = C_\sigma\, 2^x x^{\sigma/2}\Big(\sum_{n=0}^{N-1} \frac{\sigma^n A_n}{x^n} + O_N(x^{-N})\Big).} \tag{P.36}$$

Proof. The three expressions in Equation (P.34) are the definition of B_n, its partition form, and the expansion of the exponential as a product; Equation (P.35) follows from $A'(t) = A(t)\sum_j j d_j t^{j-1}$ for

$A(t) = \exp \sum_j d_j t^j$, together with $d_{2r} = 0$. The parity of the σ-dependence is exact: since $\sum_j d_j t^j$ is an odd formal series,

$$\exp\Big(-\sum_j d_j t^j\Big) = \exp\Big(\sum_j d_j (-t)^j\Big) = \sum_{n\geq 0} (-1)^n A_n t^n,$$

so $\exp(\sigma D)$ has coefficients $\sigma^n A_n$ in either sheet. For the remainder, Theorem P.25 below gives an explicit bound. □

Proposition P.25 (Effective forward remainder)**.** *Fix $N \geq 1$ and let $x \geq x_N := \max\{1, 2N\}$. Then*

$$\Big|\mathrm{e}^{\sigma D(x)} - \sum_{n=0}^{N-1} \frac{\sigma^n A_n}{x^n}\Big| \leq \frac{K_N}{x^N}, \qquad K_N := \mathrm{e}^{1/4}\Big(|d_{2\lceil N/2\rceil+1}| + \sum_{n\geq N} \frac{|A_n|}{x_N^{n-N}}\Big) < \infty, \tag{P.37}$$

where the tail sum converges because $A_n = O(C^n n!^{1/2})$ for some C.

Proof. Write $D_{<M}(x) = \sum_{r<M} d_{2r+1} x^{-(2r+1)}$ with $M = \lceil N/2\rceil$, so that by Theorem P.20 $|D(x) - D_{<M}(x)| \leq |d_{2M+1}| x^{-(2M+1)} \leq |d_{2M+1}| x^{-N}$ for $2M+1 \geq N$. Since $|D(x)| \leq \frac{1}{4}$ and $|D_{<M}(x)| \leq \frac{1}{4}$ for $x \geq 1$ by Equation (P.32) with $N = 0$, the mean value theorem gives $|\mathrm{e}^{\sigma D} - \mathrm{e}^{\sigma D_{<M}}| \leq \mathrm{e}^{1/4}|D - D_{<M}|$. Finally $\mathrm{e}^{\sigma D_{<M}(x)}$ is an absolutely convergent power series in $1/x$ for $x \geq x_N$ whose coefficients agree with $\sigma^n A_n$ for $n < N$ up to terms already accounted for, and whose tail is bounded by $\sum_{n\geq N} |A_n| x^{-n}$. Collecting the two bounds gives Equation (P.37). The growth bound on A_n follows from $d_{2r+1} = O((2r)!\pi^{-2r})$ and the Bell-polynomial partition form. □

Source claim. S1 obtains its multiplicative remainder by writing "$e^{R_N(m)} = 1+O(m^{-N})$, so formal exponentiation also gives the claimed analytic remainder". **Defect.** That sentence exponentiates a *formal* truncation error and silently assumes that the tail of the exponentiated series is of the same order; no bound on the coefficients A_n enters, although the series $\sum A_n x^{-n}$ diverges. **Repair.** Theorem P.25 supplies the missing ingredients (a bound on $D - D_{<M}$ from the enveloping theorem, a uniform bound on the exponential's Lipschitz constant, and an explicit tail bound), so the constant in Equation (P.36) is effective.

P.4.2 Coefficient table and the two sectors

Table P.2: The universal forward coefficients A_n of Equation (P.34), recomputed by Equation (P.35) in exact rational arithmetic. The table reproduces the corresponding tables of all three sources (S3 tabulates $C_m = (-1)^n A_n$; S1 tabulates $A_n/2^n$ in its half index).

n	A_n	n	A_n
0	1	8	−334477/8388608
1	1/4	9	28717403/33554432
2	1/32	10	59697183/268435456
3	−5/128	11	−8400372435/1073741824
4	−21/2048	12	−34429291905/17179869184
5	399/8192	13	7199255611995/68719476736
6	869/65536	14	14631594576045/549755813888
7	−39325/262144	15	−4251206967062925/2199023255552

Explicitly, with $C_- = \sqrt{2/\pi}$ and $C_+ = 1/\sqrt{2\pi}$,

$$\mathcal{S}_-(x) \sim \sqrt{\frac{2}{\pi}}\,\frac{2^x}{\sqrt{x}}\Big(1 - \frac{1}{4x} + \frac{1}{32x^2} + \frac{5}{128x^3} - \frac{21}{2048x^4} - \frac{399}{8192x^5} + \frac{869}{65536x^6} + \cdots\Big), \tag{P.38}$$

$$\mathcal{S}_+(x) \sim \frac{2^x\sqrt{x}}{\sqrt{2\pi}}\Big(1+\frac{1}{4x}+\frac{1}{32x^2}-\frac{5}{128x^3}-\frac{21}{2048x^4}+\frac{399}{8192x^5}+\frac{869}{65536x^6}+\cdots\Big), \tag{P.39}$$

and for the integer sequence itself,

$$\boxed{\mathrm{sw}(n) \sim \frac{2^n}{\sqrt{\pi}}\,2^{-\sigma_n/2}\,n^{\sigma_n/2}\sum_{k\ge 0}\frac{\sigma_n^k A_k}{n^k}, \qquad \sigma_n=(-1)^{n+1}.} \tag{P.40}$$

Remark P.26 (Period-two, not oscillatory error). Equation (P.40) is a two-phase (period-two) transseries: the discrete transmonomial σ_n has modulus one, so it is never absorbed into a remainder, and it changes the algebraic scale from $2^n n^{-1/2}$ to $2^n n^{+1/2}$. Equivalently, with the idempotents $\chi_\sigma(n)=(1+\sigma\sigma_n)/2$,

$$\mathrm{sw}(n) \sim \sum_{\sigma=\pm1}\chi_\sigma(n)\,C_\sigma 2^n n^{\sigma/2}\sum_{k\ge0}\sigma^k A_k n^{-k}.$$

Corollary P.27 (Central binomial coefficient and the reciprocal)**.** *Putting $x=2m$ in Equation* (P.38),

$$\binom{2m}{m} \sim \frac{4^m}{\sqrt{\pi m}}\Big(1-\frac{1}{8m}+\frac{1}{128m^2}+\frac{5}{1024m^3}-\frac{21}{32768m^4}-\frac{399}{262144m^5}+\cdots\Big), \tag{P.41}$$

that is, the half-index coefficients of S1 are $A_n/2^n$ on the even sheet. Moreover

$$\frac{1}{\mathcal{S}_\sigma(x)} \sim \frac{1}{C_\sigma}\,2^{-x}x^{-\sigma/2}\sum_{n\ge0}\frac{(-\sigma)^n A_n}{x^n}. \tag{P.42}$$

Proof. For Equation (P.41) substitute $x=2m$ and $C_-2^{2m}(2m)^{-1/2}=4^m/\sqrt{\pi m}$. For Equation (P.42) note that the reciprocal of Equation (P.19) carries $\mathrm{e}^{-\sigma D}$, whose coefficients are $(-\sigma)^n A_n$ by the oddness argument in the proof of Theorem P.24. ☐

P.5 The centred normal form

This section is S1's distinctive contribution: a translation of the index that makes the correction series *sparse in a second way* and equalizes the two leading constants. S1 states it as two parallel statements in its half index; it is recast here as a single σ-uniform identity in the original index, and the coefficients are recomputed in that normalization.

Theorem P.28 (Centred exact normal form)**.** *Put*

$$u := \frac{x}{2}+\frac{1}{4}. \tag{P.43}$$

Then, exactly and for both parities,

$$\boxed{\mathcal{S}_\sigma(x) = \frac{4^u}{\sqrt{2\pi}}\left(\frac{\Gamma(u+3/4)}{\Gamma(u+1/4)}\right)^{\sigma},} \tag{P.44}$$

and, as $u\to\infty$ in $|\arg u|\le\pi-\delta$,

$$\mathcal{S}_\sigma(x) \sim \frac{4^u}{\sqrt{2\pi}}\,u^{\sigma/2}\exp\Big(\sigma\sum_{q\ge1}\frac{\nu_{2q}}{u^{2q}}\Big), \qquad \nu_{2q}=\frac{B_{2q+1}(1/4)}{q(2q+1)}. \tag{P.45}$$

In particular every odd inverse power is absent *from the centred correction, and the leading constant $1/\sqrt{2\pi}$ is the same on both sheets.*

Proof. From $u = x/2 + 1/4$ we get $x/2 + 1 = u + 3/4$ and $x/2 + 1/2 = u + 1/4$, while $2^x = 2^{2u-1/2} = 4^u/\sqrt{2}$; substituting in Equation (P.12) gives Equation (P.44). For Equation (P.45) apply Equation (P.29) with $a = \frac{3}{4}$, $b = \frac{1}{4}$, $z = u$: the logarithmic term is $(a - b)\log u = \frac{1}{2}\log u$, and the n-th coefficient is $\frac{(-1)^{n+1}}{n(n+1)}\big(B_{n+1}(\frac{3}{4}) - B_{n+1}(\frac{1}{4})\big)$. Since $a + b = 1$, the reflection formula $B_m(1 - w) = (-1)^m B_m(w)$ gives $B_{n+1}(\frac{1}{4}) = (-1)^{n+1}B_{n+1}(\frac{3}{4})$, so the difference vanishes for odd n and equals $2B_{n+1}(\frac{3}{4}) = -2B_{n+1}(\frac{1}{4})$ for even $n = 2q$. This yields $\nu_{2q} = B_{2q+1}(1/4)/(q(2q + 1))$. □

Table P.3: Centred logarithmic coefficients ν_{2q} of Equation (P.45), recomputed from Bernoulli polynomials at $1/4$. They agree with S1's $\widehat{\lambda}^{(0)}_{2q} = -\nu_{2q}$ after the marker translation of Theorem P.12.

$2q$	2	4	6	8	10
ν_{2q}	$\frac{1}{64}$	$-\frac{5}{2048}$	$\frac{61}{49152}$	$-\frac{1385}{1048576}$	$\frac{50521}{20971520}$

Remark P.29 (Two different sparsities). The uncentred correction $\sigma D(x)$ contains only *odd* powers of $1/x$; the centred correction contains only *even* powers of $1/u$. There is no contradiction: the two statements concern different remainder functions, because the $\frac{\sigma}{2}\log$ term is attached to x in one and to u in the other. Both are instances of one rule, recorded in Theorem P.30 below, and the swing shifts are precisely the optimally centred ones.

Lemma P.30 (Optimal centring of a gamma ratio)**.** *Let $a, b \in \mathbb{R}$ with $\beta := a - b \neq 0$, and set $s = (a + b - 1)/2$ and $z = w + s$. Then $\Gamma(w + a)/\Gamma(w + b) = \Gamma(z + \frac{1+\beta}{2})/\Gamma(z + \frac{1-\beta}{2})$, the two shifts sum to* 1, *and*

$$\log\frac{\Gamma(z + \frac{1+\beta}{2})}{\Gamma(z + \frac{1-\beta}{2})} \sim \beta\log z - \sum_{q\geq 1}\frac{B_{2q+1}\big(\frac{1-\beta}{2}\big)}{q(2q + 1)}\, z^{-2q}, \qquad |\arg z| \leq \pi - \delta, \tag{P.46}$$

with no odd inverse powers.

Proof. The shift identity is immediate. Apply Equation (P.29) and the argument of Theorem P.28: complementary shifts make the Bernoulli difference vanish in odd degree by $B_m(1-w) = (-1)^m B_m(w)$. □

For the swing factorial $a = 1$, $b = \frac{1}{2}$, $w = x/2$, so $s = \frac{1}{4}$ and $z = u$, and $\beta = \frac{1}{2}$: Equation (P.45) is exactly Equation (P.46) raised to the power σ. The first centred expansions are

$$\mathcal{S}_-(x) \sim \frac{4^u}{\sqrt{2\pi u}}\Big(1 - \frac{1}{64u^2} + \frac{21}{8192u^4} - \frac{671}{524288u^6} + \cdots\Big), \tag{P.47}$$

$$\mathcal{S}_+(x) \sim \frac{4^u\sqrt{u}}{\sqrt{2\pi}}\Big(1 + \frac{1}{64u^2} - \frac{19}{8192u^4} + \frac{631}{524288u^6} + \cdots\Big), \tag{P.48}$$

whose coefficients were recomputed from $\exp\big(\sigma\sum_q \nu_{2q}z^q\big)$ and agree with S1.

P.6 The two Lambert cores, and why the branch matters

P.6.1 Specialization of the core theorem

Write

$$L_\sigma(y) := \log\frac{y}{C_\sigma} = \log y + \tfrac{1}{2}\log\pi + \tfrac{\sigma}{2}\lambda. \tag{P.49}$$

Dropping the correction in Equation (P.19) leaves the *core equation*

$$L_\sigma(y) = \lambda X + \frac{\sigma}{2}\log X, \qquad \text{equivalently} \qquad C_\sigma\, 2^X X^{\sigma/2} = y. \tag{P.50}$$

Theorem P.31 (Explicit Lambert cores). *Theorem J.23 with the dictionary Equation* (P.21) *gives*

$$X_{\sigma,k}(y) = \frac{\sigma}{2\lambda}\, W_k\big(2\sigma\lambda\, \mathrm{e}^{2\sigma L_\sigma(y)}\big), \tag{P.51}$$

and the unique large positive real solutions are

$$\boxed{X_+(y) = \frac{W_0\big(4\pi\lambda y^2\big)}{2\lambda}, \qquad X_-(y) = -\frac{W_{-1}\big(-4\lambda/(\pi y^2)\big)}{2\lambda}.} \tag{P.52}$$

X_+ is defined for all $y > 0$; X_- is defined exactly for

$$y \geq y_\star := 2\sqrt{\frac{\mathrm{e}\lambda}{\pi}} = 1.548870223880\ldots, \tag{P.53}$$

and satisfies $X_-(y) \geq 1/(2\lambda) = 0.72134752\ldots$ there.

Proof. With $a = \lambda$, $b = \sigma/2$, Theorem J.23 reads $X = (b/a)W_k\big((a/b)\mathrm{e}^{L/b}\big)$, and $1/\sigma = \sigma$ gives Equation (P.51). For $\sigma = +1$, $C_+ = 1/\sqrt{2\pi}$ so $\mathrm{e}^{2L_+} = 2\pi y^2$ and the argument is $4\pi\lambda y^2 > 0$, on which W_0 is the only real branch. For $\sigma = -1$, $C_- = \sqrt{2/\pi}$ so $\mathrm{e}^{-2L_-} = 2/(\pi y^2)$ and the argument is $-4\lambda/(\pi y^2) < 0$; it lies in $[-1/\mathrm{e}, 0)$ exactly when $y^2 \geq 4\lambda\mathrm{e}/\pi$, i.e. $y \geq y_\star$. On that interval both W_0 and W_{-1} are real; $W_{-1} \leq -1$ gives $X_- \geq 1/(2\lambda)$, and $W_{-1}(z) \to -\infty$ as $z \to 0^-$ gives $X_-(y) \to \infty$. Theorem P.32 identifies the two roots and shows that W_0 gives the wrong one. □

P.6.2 Why W_0 is the wrong root on the even sheet

Proposition P.32 (The even carrier is not injective). *Let $g_-(X) = C_- 2^X X^{-1/2}$ be the even carrier of Equation* (P.50). *Then g_- is strictly decreasing on $(0, 1/(2\lambda))$, strictly increasing on $(1/(2\lambda), \infty)$, and*

$$\min_{X>0} g_-(X) = g_-\Big(\frac{1}{2\lambda}\Big) = 2\sqrt{\frac{\mathrm{e}\lambda}{\pi}} = y_\star. \tag{P.54}$$

Hence for $y > y_\star$ the equation $g_-(X) = y$ has exactly two positive roots $X^{\mathrm{small}}(y) < 1/(2\lambda) < X^{\mathrm{large}}(y)$; they coalesce at $y = y_\star$. In terms of the Lambert branches,

$$X^{\mathrm{large}}(y) = -\frac{W_{-1}\big(-4\lambda/(\pi y^2)\big)}{2\lambda} = X_-(y) \xrightarrow[y\to\infty]{} \infty, \qquad X^{\mathrm{small}}(y) = -\frac{W_0\big(-4\lambda/(\pi y^2)\big)}{2\lambda} = \frac{2}{\pi y^2}\big(1+O(y^{-2})\big). \tag{P.55}$$

The principal branch therefore returns a root that tends to zero*; it is not a large-index inverse at any order, and no amount of asymptotic correction can repair the choice. On the odd sheet the carrier $g_+(X) = C_+ 2^X X^{1/2}$ is a strictly increasing bijection of $(0,\infty)$ onto $(0,\infty)$, its Lambert argument is positive, and W_0 is the only real branch.*

Proof. $(\log g_-)'(X) = \lambda - 1/(2X)$ vanishes only at $X = 1/(2\lambda)$ and changes sign there from $-$ to $+$; the minimum value is $C_-\mathrm{e}^{\lambda/(2\lambda)}(2\lambda)^{1/2} = \sqrt{2/\pi}\cdot\sqrt{\mathrm{e}}\cdot\sqrt{2\lambda} = 2\sqrt{\mathrm{e}\lambda/\pi}$, which is Equation (P.54), and equals $y_\star$ of Equation (P.53). Strict monotonicity on each side gives exactly two roots for $y > y_\star$. Both are given by Equation (P.51) with $\sigma = -1$ and a real branch of W; since $W_{-1} \leq -1 \leq W_0 \leq 0$ on $[-1/\mathrm{e}, 0)$, the branch W_{-1} gives the root $\geq 1/(2\lambda)$ and W_0 the root $\leq 1/(2\lambda)$. Finally $W_0(z) = z + O(z^2)$ as $z \to 0$, so $X^{\mathrm{small}} = -\big(-4\lambda/(\pi y^2) + O(y^{-4})\big)/(2\lambda) = 2/(\pi y^2) + O(y^{-4})$. □

Remark P.33 (Numerical confirmation and the branch point). Recomputed at 50 digits: for $y = 10^2, 10^3, 10^5$ the principal-branch root is $6.366759642 \times 10^{-5}$, $6.366203342 \times 10^{-7}$, $6.366197724 \times 10^{-11}$, against $2/(\pi y^2) = 6.366197724 \times 10^{-5}$, $6.366197724 \times 10^{-7}$, $6.366197724 \times 10^{-11}$. The threshold $y_\star$ where the two roots coalesce is exactly the value at which the Lambert argument reaches the branch point $-1/\mathrm{e}$; the agreement of these two descriptions is a useful consistency check on the whole normalization. Note also $\mathcal{S}_-\big(1/(2\lambda)\big) = 1.15213\ldots < y_\star$, a fact used in Theorem P.35.

Remark P.34 (The branch rule is forced by $\operatorname{sign} b$). All three sources agree on the assignment: W_{-1} on the even sheet, W_0 on the odd sheet. In the model Theorem J.2 the rule is $b < 0 \Rightarrow W_{-1}$, $b > 0 \Rightarrow W_0$ for the large real root, and $b = \sigma/2$; so the branch distinction is forced by the sign of the power of x in the forward asymptotics — that is, by the fact that the even bisection carries $n^{-1/2}$ and the odd bisection $n^{+1/2}$. It is a consequence of the parity split, not an independent convention.

P.6.3 Two exact inequalities

The sources deduce the *sign* of the first inverse correction from $D > 0$. The following sharpening is exact at all y, not only to first order, and costs nothing.

Proposition P.35 (The core brackets the true inverse)**.** *For every* $y > 1$,

$$\mathcal{I}_+(y) < X_+(y), \tag{P.56}$$

and for every $y > y_\star$,

$$\mathcal{I}_-(y) > X_-(y). \tag{P.57}$$

Proof. By Equation (P.19) and Theorem P.14, $\mathcal{S}_+(x) = g_+(x)\mathrm{e}^{D(x)} > g_+(x)$ for all $x > 0$, where g_+ is the odd carrier. Both $\mathcal{S}_+$ and g_+ are strictly increasing. Put $x_1 = \mathcal{I}_+(y)$; then $g_+(x_1) < \mathcal{S}_+(x_1) = y = g_+(X_+(y))$, and g_+ increasing gives $x_1 < X_+(y)$.

For $\sigma = -1$, $\mathcal{S}_-(x) = g_-(x)\mathrm{e}^{-D(x)} < g_-(x)$. Let $x_0 = \mathcal{I}_-(y)$ with $y > y_\star$. Since $\mathcal{S}_-$ is increasing and $\mathcal{S}_-(1/(2\lambda)) = 1.15213\ldots < y_\star < y$, we have $x_0 > 1/(2\lambda)$; also $X_-(y) \geq 1/(2\lambda)$ by Theorem P.31. On $[1/(2\lambda), \infty)$ the carrier g_- is strictly increasing (Theorem P.32), and $g_-(x_0) > \mathcal{S}_-(x_0) = y = g_-(X_-(y))$, whence $x_0 > X_-(y)$. $\square$

Remark P.36. This explains without computation the signs $c_1^{(+1)} = -1/(4\lambda) < 0$ and $c_1^{(-1)} = +1/(4\lambda) > 0$ found in Section P.7, and supplies a rigorous one-sided bracket at every y, used as a check in Theorem P.63.

P.6.4 A scaled form, and the half-index cores

Putting $w_\sigma = 2\lambda X_\sigma$ turns Equation (P.50) into the single scalar equation

$$w + \sigma \log w = \Lambda_\sigma, \qquad \Lambda_\sigma := \log(\pi y^2) + \sigma \log(4\lambda), \tag{P.58}$$

so that $w_+ = W_0(\mathrm{e}^{\Lambda_+})$ and $w_- = -W_{-1}(-\mathrm{e}^{-\Lambda_-})$. This is S3's normalization, with its $\varepsilon = -\sigma$; it is convenient in Section P.8.

Remark P.37 (Half-index cores). S1's carriers, in the half index and with $\alpha = \log 4$, are $w_0^{\mathrm{S1}} = -\frac{1}{2\alpha}W_{-1}(-2\alpha/(\pi y^2))$ and $w_1^{\mathrm{S1}} = \frac{1}{2\alpha}W_0(\pi\alpha y^2/2)$. On the even sheet $n = 2m$ exactly, and one checks directly that $X_-(y) = 2w_0^{\mathrm{S1}}(y)$; the two normalizations then agree, and their inverse coefficients are related by $c_n^{(-1)} = 2^{n+1}d_n^{\mathrm{S1},(0)}$, which we verified symbolically for $1 \leq n \leq 8$. On the odd sheet $n = 2m+1$, so the two carriers solve *different* equations ($C_+2^X X^{1/2} = y$ against $K_1 4^w w^{1/2} = y$) and their coefficient lists are *not* termwise comparable, although both expansions converge asymptotically to the same function $\mathcal{I}_+$. Failing to notice this is the easiest way to “discover” a contradiction between S1 and S2/S3 where there is none.

P.7 All-orders reversion around the core

P.7.1 Specialization of the reversion theorem

Theorem P.38 (Swing specialization of Theorem J.29). *Let $X = X_\sigma(y)$ solve Equation* (P.50)*, let $t = X^{-1}$, and put*

$$\Psi_\sigma(t,u) = -\frac{1}{\lambda}\left[\frac{\sigma}{2}\log(1+tu) + \sigma\sum_{j\ge 1} d_j\Big(\frac{t}{1+tu}\Big)^j\right]. \tag{P.59}$$

Then the unique $U \in t\,\mathscr{R}[[t]]$ with $U = \Psi_\sigma(t,U)$ has coefficients

$$c_n^{(\sigma)} = [t^n]\,U = \sum_{m=1}^{n}\frac{1}{m}\,[t^n u^{m-1}]\,\Psi_\sigma(t,u)^m \tag{P.60}$$

by Theorem J.19, and equivalently, with $\gamma_1 = 0$ and $\gamma_r = c_{r-1}^{(\sigma)}$ for $r \ge 2$,

$$c_n^{(\sigma)} = -\frac{1}{\lambda}\left[\frac{\sigma}{2}\sum_{k=1}^{n}\frac{(-1)^{k+1}}{k}\widehat{B}_{n,k}(\gamma) + \sigma\sum_{\substack{1\le j\le n\\ j\text{ odd}}} d_j\sum_{k=0}^{n-j}(-1)^k\binom{j+k-1}{k}\widehat{B}_{n-j,k}(\gamma)\right], \tag{P.61}$$

with $\widehat{B}$ the ordinary partial Bell polynomials of Theorem J.9 and the expansions of Theorem J.14. Only $c_1^{(\sigma)},\dots,c_{n-1}^{(\sigma)}$ occur on the right. The coefficients lie in the parity algebra $\mathscr{R}$ of Equation (P.9)*; no logarithm of L_σ occurs.*

Proof. This is Theorem J.29 with the dictionary Equation (P.21): $a = \lambda$, $b = \sigma/2$, $q_j = \sigma d_j$, which turns the general Ψ into Equation (P.59). The restriction of the inner sum to odd j is $d_{2r} = 0$. ☐

P.7.2 Recomputed coefficients

The coefficients below were recomputed from Equation (P.59) by truncated power-series arithmetic over $\mathbb{Q}[\lambda,\lambda^{-1}]$, separately for $\sigma = +1$ and $\sigma = -1$, and then recombined into $\mathscr{R}$ by $c = \frac{1}{2}(c^+ + c^-) + \sigma\cdot\frac{1}{2}(c^+ - c^-)$. They agree with S2 for $1 \le n \le 10$ and with S3 for $1 \le n \le 8$ after the marker translation $\sigma = -\varepsilon_{\mathrm{S3}}$.

$$c_1^{(\sigma)} = -\frac{\sigma}{4\lambda},\qquad c_2^{(\sigma)} = \frac{1}{8\lambda^2}, \tag{P.62}$$

$$c_3^{(\sigma)} = \frac{2\sigma\lambda^2 - 3\lambda - 3\sigma}{48\lambda^3},\qquad c_4^{(\sigma)} = \frac{-4\lambda^2 + 15\sigma\lambda + 6}{192\lambda^4}, \tag{P.63}$$

$$c_5^{(\sigma)} = -\frac{96\sigma\lambda^4 - 80\lambda^3 + 40\sigma\lambda^2 + 135\lambda + 30\sigma}{1920\lambda^5}, \tag{P.64}$$

$$c_6^{(\sigma)} = \frac{96\lambda^4 - 180\sigma\lambda^3 + 215\lambda^2 + 210\sigma\lambda + 30}{3840\lambda^6}, \tag{P.65}$$

$$\begin{aligned} c_7^{(\sigma)} = \frac{1}{53760\lambda^7}\big(&8160\sigma\lambda^6 - 4312\lambda^5 + 1428\sigma\lambda^4 + 1050\lambda^3 \\ &- 4025\sigma\lambda^2 - 2100\lambda - 210\sigma\big), \end{aligned} \tag{P.66}$$

$$\begin{aligned} c_8^{(\sigma)} = \frac{1}{645120\lambda^8}\big(&-48960\lambda^6 + 56056\sigma\lambda^5 - 40908\lambda^4 + 15015\sigma\lambda^3 \\ &+ 50610\lambda^2 + 17010\sigma\lambda + 1260\big), \end{aligned} \tag{P.67}$$

$$c_9^{(\sigma)} = -\frac{1}{1290240\lambda^9}\big(1111040\sigma\lambda^8 - 413184\lambda^7 + 79616\sigma\lambda^6 + 43512\lambda^5$$

$$- 83748\sigma\lambda^4 + 83475\lambda^3 + 92610\sigma\lambda^2 + 22050\lambda + 1260\sigma\big), \tag{P.68}$$

$$c_{10}^{(\sigma)} = \frac{1}{5160960\lambda^{10}}\big(2222080\lambda^8 - 1756032\sigma\lambda^7 + 779152\lambda^6 - 203280\sigma\lambda^5 - 162582\lambda^4 + 487305\sigma\lambda^3 + 309960\lambda^2 + 55440\sigma\lambda + 2520\big). \tag{P.69}$$

Setting $\sigma = \pm 1$ and $\lambda = \log 2$,

$$\mathcal{I}_+(y) \sim X_+ - \frac{1}{4\lambda X_+} + \frac{1}{8\lambda^2 X_+^2} + \frac{2\lambda^2 - 3\lambda - 3}{48\lambda^3 X_+^3} + \frac{-4\lambda^2 + 15\lambda + 6}{192\lambda^4 X_+^4} + \cdots, \tag{P.70}$$

$$\mathcal{I}_-(y) \sim X_- + \frac{1}{4\lambda X_-} + \frac{1}{8\lambda^2 X_-^2} + \frac{-2\lambda^2 - 3\lambda + 3}{48\lambda^3 X_-^3} + \frac{-4\lambda^2 - 15\lambda + 6}{192\lambda^4 X_-^4} + \cdots. \tag{P.71}$$

The signs of the first corrections are those forced by Theorem P.35.

P.7.3 A differential form of the same coefficients

All three sources also give an operator form of the reversion, which is the convenient one for the Stokes analysis of Section P.9. We state it as a chapter-local lemma; it is a candidate for promotion, being independent of the swing factorial.

Lemma P.39 (Perturbed inverse in phase coordinates)**.** *Let F_0 be invertible near the point considered with $F_0' \neq 0$, let g be a formal correction, and let X satisfy $F_0(X) = z$. Introduce*

$$\mathscr{D} := \frac{1}{F_0'(X)}\frac{\mathrm{d}}{\mathrm{d}X}. \tag{P.72}$$

Then the solution x of $F_0(x) + \eta\, g(x) = z$ with $x|_{\eta=0} = X$ is, as a formal power series in the marker η,

$$x = X + \sum_{k\geq 1} \frac{(-\eta)^k}{k!}\,\mathscr{D}^{k-1}\left(\frac{g(X)^k}{F_0'(X)}\right). \tag{P.73}$$

Proof. This is Theorem J.21 under the dictionary $F \mapsto F_0$, $E \mapsto g$, $\varepsilon \mapsto \eta$, $\mu_0 \mapsto X$, $y \mapsto z$ and $\mathcal{D} \mapsto \mathscr{D}$, which carries (J.32) to Equation (P.72) and (J.34) to Equation (P.73). Since g is here a formal and in general divergent series, no contour is available and the statement invoked is the coefficientwise one, Theorem J.22, whose only hypothesis is that $F_0'(X)$ be invertible. □

Proposition P.40 (Differential formula for $c_n^{(\sigma)}$)**.** *Let $\mathcal{Q}_\sigma(t) = \sigma \sum_{j\geq 1} d_j t^j$ and*

$$\mathscr{L}_\sigma := -\frac{t^2}{\lambda + \sigma t/2}\frac{\mathrm{d}}{\mathrm{d}t} \qquad (t = X^{-1}), \tag{P.74}$$

which is Equation (P.72) *for $F_{0,\sigma}(X) = \lambda X + \frac{\sigma}{2}\log X$ expressed in the variable t. Then, for every $n \geq 1$,*

$$\boxed{c_n^{(\sigma)} = [t^n] \sum_{k=1}^{\lfloor (n+1)/2 \rfloor} \frac{(-1)^k}{k!}\,\mathscr{L}_\sigma^{k-1}\left(\frac{\mathcal{Q}_\sigma(t)^k}{\lambda + \sigma t/2}\right),} \tag{P.75}$$

a finite *sum requiring only $d_1, \ldots, d_n$.*

Proof. Apply Theorem P.39 with $F_0 = F_{0,\sigma}$, $g = \sigma D$ and $\eta = 1$; the substitution $t = X^{-1}$ turns $F_{0,\sigma}'(X) = \lambda + \sigma/(2X)$ into $\lambda + \sigma t/2$ and $\mathrm{d}/\mathrm{d}X$ into $-t^2\mathrm{d}/\mathrm{d}t$. Setting the marker η to 1 is legitimate coefficientwise because $\mathcal{Q}_\sigma(t)^k = O(t^k)$ while each application of $\mathscr{L}_\sigma$ raises the order in t by one, so the k-th summand begins at t^{2k-1} and can contribute to $[t^n]$ only when $2k - 1 \leq n$. □

Source claim. S2 writes its operator expansion as an *equality*, $\mathcal{I}_\sigma(y) = X + \sum_{m\ge 1}\frac{(-\sigma)^m}{m!}\mathscr{D}^{m-1}\big[D(X)^m/F_0'(X)\big]$, after setting its coupling parameter to 1. **Defect.** With the marker removed there is no small parameter, the series over m does not converge, and the left-hand side is a number while the right-hand side is a divergent series; the statement as printed is false. **Repair.** The correct statement is the asymptotic one: the m-th summand is $O(X^{-(2m-1)})$, so for each fixed M the partial sum through $m = M$ differs from $\mathcal{I}_\sigma(y)$ by $O(X^{-(2M+1)})$, and regrouping by powers of X^{-1} reproduces Equation (P.75). S1 and S3 state the same formula correctly, as a coefficient identity.

P.7.4 The analytic statement, with a proof

Source claim. All three sources assert the Poincaré statement "$\mathcal{I}_\sigma(y) = X_\sigma(y) + \sum_{n<M} c_n^{(\sigma)} X_\sigma(y)^{-n} + O(X_\sigma(y)^{-M})$". **Defect.** S3 gives no proof at all ("Likewise, for each fixed $M, \dots$"); S1 gives a two-line sketch that appeals to the mean value theorem but never bounds the residual of the truncated ansatz, and in particular never uses any control on the forward remainder at the *perturbed* point; S2 proves the ingredients (a residual-to-index bound and derivative lower bounds) but does not assemble them. This is the one place in the three sources where a formal computation is upgraded to an analytic conclusion without an argument. **Repair.** Theorem P.41 below supplies the proof, and makes the constant effective by using the enveloping remainder Theorem P.20.

Theorem P.41 (Inverse transseries, analytically)**.** *Fix $M \ge 1$ and $\sigma \in \{-1, +1\}$. Then, as $y \to +\infty$,*

$$\boxed{\mathcal{I}_\sigma(y) = X_\sigma(y) + \sum_{n=1}^{M-1} \frac{c_n^{(\sigma)}}{X_\sigma(y)^n} + O\big(X_\sigma(y)^{-M}\big)\,,} \tag{P.76}$$

with an effective implied constant depending only on M and σ. Since $X_\sigma(y) = \big(\log y\big)/\lambda\,(1 + o(1))$, the expansion is one in inverse powers of $\log y$ once the dominant power–logarithmic phase has been inverted exactly.

Proof. Write $X = X_\sigma(y)$ and $x_M := X + \sum_{n=1}^{M-1} c_n^{(\sigma)} X^{-n}$; for y large, $x_M = X(1 + O(X^{-2}))$, so $x_M > 1$ and $x_M \asymp X$. Consider the phase residual

$$\rho := \log \mathcal{S}_\sigma(x_M) - \log y = \lambda(x_M - X) + \frac{\sigma}{2}\log\frac{x_M}{X} + \sigma D(x_M),$$

where we used Equation (P.19) and $\log y = \log C_\sigma + \lambda X + \frac{\sigma}{2}\log X$. Put $t = X^{-1}$ and $U_{M-1}(t) = \sum_{n<M} c_n^{(\sigma)} t^n$, so that $x_M = X(1 + tU_{M-1})$ and

$$\rho = \lambda\, X \Big[\underbrace{tU_{M-1} + \frac{\sigma t}{2\lambda}\log(1 + tU_{M-1}) + \frac{\sigma t}{\lambda} D_{<M}(x_M)}_{=: t\,\mathcal{E}(t)}\Big] + \sigma\big(D(x_M) - D_{<M}(x_M)\big),$$

where $D_{<M}$ denotes the truncation of the series Equation (P.26) after the terms of order $x^{-(M-1)}$. Two estimates finish the proof.

(i) The algebraic residual. Substituting $x_M = X(1 + tU_{M-1})$ and $D_{<M}(x_M) = \mathcal{Q}_\sigma\big(t/(1 + tU_{M-1})\big)/\sigma$ truncated, the bracket $\mathcal{E}(t)$ is exactly $U_{M-1}(t) - \Psi_\sigma\big(t, U_{M-1}(t)\big)$ modulo terms of order t^M coming from the truncation of $\mathcal{Q}_\sigma$. By Theorem P.38 the coefficients of $t^1, \dots, t^{M-1}$ in $U - \Psi_\sigma(t, U)$ vanish when U is replaced by U_{M-1} — this is the triangularity of the recurrence — so $\mathcal{E}(t) = O(t^M)$ with an explicit rational-in-λ constant obtained from $c_1^{(\sigma)}, \dots, c_{M-1}^{(\sigma)}$ and $d_1, \dots, d_M$.

Hence $\lambda X \cdot t\,\mathcal{E}(t) = O(X^{-M+1} \cdot X^{-1}) \cdot X = O(X^{-M+1})$; more carefully, $X\,t\,\mathcal{E}(t) = \mathcal{E}(t) = O(X^{-M})$.

(ii) The transcendental residual. By Theorem P.20 with $N = \lceil M/2 \rceil$, $|D(x_M) - D_{<M}(x_M)| \le |d_{2N+1}|\, x_M^{-(2N+1)} \le c_M\, X^{-M}$ for an explicit c_M, since $2N+1 \ge M$ and $x_M \asymp X$.

Therefore $|\rho| \le C_M X^{-M}$ with C_M explicit. Finally, $\big(\log \mathcal{S}_\sigma\big)'(x) = \lambda + \frac{\sigma}{2x} + \sigma D'(x) \ge \lambda - \frac{1}{2x} - \frac{1}{4x^2}$ by Theorem P.14, which exceeds $\lambda/2$ once $x \ge 2/\lambda$. Theorem J.47 (residual-to-root conversion) then gives

$$|x_M - \mathcal{I}_\sigma(y)| \le \frac{|\rho|}{\lambda/2} \le \frac{2C_M}{\lambda}\, X^{-M},$$

which is Equation (P.76). The final sentence follows from $\lambda X + \frac{\sigma}{2}\log X = L_\sigma(y)$ and $L_\sigma(y) = \log y + O(1)$. □

Remark P.42 (An exact algebraic certificate). Step (i) of the proof is also the recommended *implementation* check. Given a candidate truncation U_{M-1}, substitute it into $U - \Psi_\sigma(t, U)$ and expand: the coefficients of $t^1, \ldots, t^{M-1}$ must vanish identically in $\mathbb{Q}[\lambda, \lambda^{-1}, \sigma]/(\sigma^2 - 1)$. This is an exact algebraic certificate for every table of $c_n^{(\sigma)}$, and it is how the tables in Equation (P.62)–Equation (P.69) were verified here, independently of the recurrence that produced them.

P.7.5 The centred reversion

Applying Theorem J.29 to the centred normal form Equation (P.45) — that is, with $a = 2\lambda$, $b = \sigma/2$, $C = 1/\sqrt{2\pi}$ and $q_j = \sigma\nu_j$ ($\nu_j = 0$ for odd j) — gives the centred cores

$$\mathcal{U}_+(y) = \frac{W_0\big(8\pi\lambda y^2\big)}{4\lambda}, \qquad \mathcal{U}_-(y) = -\frac{W_{-1}\big(-2\lambda/(\pi y^2)\big)}{4\lambda}, \tag{P.77}$$

and the centred inverse expansion

$$\boxed{\mathcal{I}_\sigma(y) \sim 2\mathcal{U}_\sigma(y) - \frac{1}{2} + 2\sum_{n\ge 2} \frac{\widehat{c}_n^{(\sigma)}}{\mathcal{U}_\sigma(y)^n}, \qquad \widehat{c}_1^{(\sigma)} = 0.} \tag{P.78}$$

The vanishing of $\widehat{c}_1^{(\sigma)}$ is immediate from $\nu_1 = 0$. Recomputed in the present normalization,

$$\widehat{c}_2^{(\sigma)} = -\frac{\sigma}{128\lambda}, \qquad \widehat{c}_3^{(\sigma)} = \frac{1}{512\lambda^2}, \qquad \widehat{c}_4^{(\sigma)} = \frac{\sigma(5\lambda^2 - 2)}{4096\lambda^3}, \tag{P.79}$$

$$\widehat{c}_5^{(\sigma)} = -\frac{7\lambda^2 - 2}{16384\lambda^4}, \qquad \widehat{c}_6^{(\sigma)} = -\frac{\sigma\big(244\lambda^4 - 57\lambda^2 + 12\big)}{393216\lambda^5}, \qquad \widehat{c}_7^{(\sigma)} = \frac{334\lambda^4 - 75\lambda^2 + 12}{1572864\lambda^6}, \tag{P.80}$$

and

$$\widehat{c}_8^{(\sigma)} = \frac{\sigma\big(4155\lambda^6 - 460\lambda^4 + 96\lambda^2 - 12\big)}{6291456\lambda^7}. \tag{P.81}$$

These agree with S1's centred table after the substitution $\alpha = 2\lambda$ and the marker translation of Theorem P.12; observe that $\widehat{c}_n$ is σ-even for odd n and σ-odd for even n, a reflection symmetry of Equation (P.44) that the uncentred coefficients do not have.

Source claim. S1 concludes that "the centred one is often the preferable computational normal form". **Defect.** Its own table already contradicts the unqualified form of this claim: at $m = 200$ the centred core beats the uncentred core by 10^3 (2.8×10^{-7} against 4.5×10^{-4}), but at $R = 4$ the *uncentred* expansion is slightly better (7.3×10^{-16} against 1.1×10^{-15}). **Repair.** The correct statement, confirmed by the recomputation in Table P.5 and Table P.6, is: centring removes the order-X^{-1} term entirely, so it improves the *bare core* by a factor $\asymp X$ and is the right choice when no corrections are to be carried; once several corrections are carried, the two normalizations are comparable, with neither dominating uniformly.

P.8 Flattening the core into a polynomial–logarithmic transseries

P.8.1 The specialization

Set

$$H_\sigma(y) := \log \frac{L_\sigma(y)}{\lambda}, \tag{P.82}$$

and recall from Theorem J.37 that the inverse of the model Theorem J.2 has the unique power–logarithmic form $x \sim a^{-1}\big[L - bH + \sum_{n\geq 1} P_n(H)L^{-n}\big]$ with $\deg P_n \leq n$ and leading term $b^{n+1}H^n/n$. With the dictionary Equation (P.21),

$$\boxed{\mathcal{I}_\sigma(y) \sim \frac{1}{\lambda}\left[L_\sigma - \frac{\sigma}{2}H_\sigma + \sum_{n\geq 1} \frac{P_n^{(\sigma)}(H_\sigma)}{L_\sigma^n}\right],} \tag{P.83}$$

where the $P_n^{(\sigma)}$ are generated by Theorem J.37 from $b = \sigma/2$, $a = \lambda$, $q_j = \sigma d_j$; the fixed-order meaning of Equation (P.83) is the analytic clause of Theorem J.37, namely an error $O\big((1 + H_\sigma)^{N+1} L_\sigma^{-(N+1)}\big)$ after N correction terms, whose hypotheses hold here by Theorem P.41. The recomputed polynomials are

$$P_1^{(\sigma)}(H) = \frac{H - \sigma\lambda}{4}, \tag{P.84}$$

$$P_2^{(\sigma)}(H) = \frac{\sigma H^2 - 2(\lambda + \sigma)H + 2\lambda}{16}, \tag{P.85}$$

$$P_3^{(\sigma)}(H) = \frac{1}{96}\Big[2H^3 - (6\sigma\lambda + 9)H^2 + (18\sigma\lambda + 6)H + 4\sigma\lambda^3 - 6\lambda^2 - 6\sigma\lambda\Big], \tag{P.86}$$

$$\begin{aligned} P_4^{(\sigma)}(H) = \frac{1}{384}\Big[&3\sigma H^4 - (12\lambda + 22\sigma)H^3 + (66\lambda + 36\sigma)H^2 \\ &+ (24\lambda^3 - 36\sigma\lambda^2 - 72\lambda - 12\sigma)H - 8\lambda^3 + 30\sigma\lambda^2 + 12\lambda\Big], \end{aligned} \tag{P.87}$$

$$\begin{aligned} P_5^{(\sigma)}(H) = \frac{1}{3840}\Big[&12H^5 - (60\sigma\lambda + 125)H^4 + (500\sigma\lambda + 350)H^3 \\ &+ (240\sigma\lambda^3 - 360\lambda^2 - 1050\sigma\lambda - 300)H^2 \\ &+ (-280\sigma\lambda^3 + 780\lambda^2 + 600\sigma\lambda + 60)H \\ &- 192\sigma\lambda^5 + 160\lambda^4 - 80\sigma\lambda^3 - 270\lambda^2 - 60\sigma\lambda\Big]. \end{aligned} \tag{P.88}$$

These reproduce S2 exactly for $1 \leq n \leq 5$; they were recomputed for $\sigma = \pm 1$ separately and recombined in $\mathscr{R}$, and independently re-derived through $n = 6$. They also satisfy the cross-normalization identity $P_n^{(\sigma)}(0) = \lambda^{n+1}c_n^{(\sigma)}$ of Theorem J.37: for instance $P_1^{(\sigma)}(0) = -\sigma\lambda/4 = \lambda^2 c_1^{(\sigma)}$ and $P_2^{(\sigma)}(0) = \lambda/8 = \lambda^3 c_2^{(\sigma)}$, which ties Equation (P.84)–Equation (P.88) to Equation (P.62)–Equation (P.69) and catches any transcription error in either list.

P.8.2 The pure Lambert block in closed form

The pure block — the case $Q \equiv 0$, in which the whole correction series is suppressed and only the Lambert core survives — is Theorem J.38, proved there in the shortest of the three forms the sources use. Specializing its data to the swing branches, $a = \lambda$ and $b = \sigma/2$, gives

$$P_n^{(0)}(H) = \left(\frac{\sigma}{2}\right)^{n+1} \sum_{m=1}^{n} (-1)^{m+1} \begin{bmatrix} n \\ m \end{bmatrix} \frac{H^{n-m+1}}{(n-m+1)!}, \tag{P.89}$$

with $\left[{n \atop m}\right]$ the unsigned Stirling numbers of the first kind. Reducing with $\sigma^2 = 1$, the two parities differ only by the overall sign σ^{n+1}: the even and odd pure blocks coincide for odd n and are opposite for even n.

Writing $b = \sigma/2$, the first few are

$$P_1^{(0)} = b^2 H, \quad P_2^{(0)} = \tfrac{b^3}{2}(H^2 - 2H), \quad P_3^{(0)} = \tfrac{b^4}{6}(2H^3 - 9H^2 + 6H), \quad P_4^{(0)} = \tfrac{b^5}{12}(3H^4 - 22H^3 + 36H^2 - 12H),$$

and Equation (P.89) was recomputed here against the recurrence of Theorem J.37 for $1 \le n \le 8$, in exact arithmetic with b symbolic. In particular the leading coefficient is b^{n+1}/n, which is Theorem J.37's degree statement. Theorem J.38 is the precise bridge between Lambert asymptotics and the first-kind Stirling calculus, and it is stated there rather than here because all four chapters use it.

P.8.3 The core series and its convergence

The flattening can also be organized as "expand the core, then compose", which is S3's route and which is constructive at every order.

Proposition P.43 (All coefficients of the scaled core). *Let w solve $w + \sigma \log w = \Lambda$ as in Equation* (P.58), *and put $\ell = \log \Lambda$. Define*

$$A_k(\ell) := \frac{1}{k+1} [z^k] \big(\ell + \log(1+z)\big)^{k+1}, \qquad k \ge 0. \tag{P.90}$$

Then

$$w \;=\; \Lambda + \sum_{k \ge 0} \frac{(-\sigma)^{k+1} A_k(\ell)}{\Lambda^k}, \tag{P.91}$$

and, with $s(k, j)$ the signed Stirling numbers of the first kind,

$$A_0(\ell) = \ell, \qquad A_k(\ell) = \frac{1}{(k+1)\,k!} \sum_{j=1}^{k} \binom{k+1}{j} j!\, s(k,j)\, \ell^{k+1-j} \quad (k \ge 1). \tag{P.92}$$

The first polynomials are

$$A_0 = A_1 = \ell, \quad A_2 = \tfrac{\ell(2-\ell)}{2}, \quad A_3 = \tfrac{\ell(2\ell^2 - 9\ell + 6)}{6}, \quad A_4 = -\tfrac{\ell(3\ell^3 - 22\ell^2 + 36\ell - 12)}{12},$$

$$A_5 = \tfrac{\ell(12\ell^4 - 125\ell^3 + 350\ell^2 - 300\ell + 60)}{60}, \quad A_6 = -\tfrac{\ell(20\ell^5 - 274\ell^4 + 1125\ell^3 - 1700\ell^2 + 900\ell - 120)}{120}.$$

Proof. Write $w = \Lambda(1+v)$ and $z = \Lambda^{-1}$. The equation becomes $\Lambda v = -\sigma\big(\ell + \log(1+v)\big)$, i.e. $v = -\sigma z\big(\ell + \log(1+v)\big)$. Lagrange inversion gives $v = \sum_{k \ge 0} \frac{(-\sigma z)^{k+1}}{k+1} [v^k] \big(\ell + \log(1+v)\big)^{k+1}$; multiplying by $\Lambda = z^{-1}$ yields Equation (P.91) with A_k as in Equation (P.90). Expanding the $(k+1)$-st power and using $\big(\log(1+z)\big)^j = j! \sum_{k \ge j} s(k,j) z^k / k!$ gives Equation (P.92). □

The sign in Equation (P.91) matches S3 after $\varepsilon = -\sigma$; explicitly $w_- = \Lambda_- + \ell + \ell/\Lambda_- + \cdots$ on the even sheet (W_{-1}) and $w_+ = \Lambda_+ - \ell + \ell/\Lambda_+ - \cdots$ on the odd sheet (W_0).

Proposition P.44 (An explicit convergence criterion). *Fix $\rho \in (0,1)$ and put $M_\rho(\ell) = |\ell| + \log \frac{1}{1-\rho}$. Then*

$$|A_k(\ell)| \le \frac{M_\rho(\ell)^{k+1}}{(k+1)\rho^k}, \tag{P.93}$$

so the series in Equation (P.91) *converges absolutely whenever $M_\rho(\ell) < \rho|\Lambda|$ for some $\rho \in (0,1)$, and its sum is then the solution branch $w \sim \Lambda$. In particular, for real $\Lambda \ge 10$ the choice $\rho = \frac{1}{2}$ already gives convergence.*

Proof. On $|z| = \rho$ one has $|\log(1+z)| \le \sum_{k\ge1} \rho^k/k = \log \frac{1}{1-\rho}$, so $|\ell + \log(1+z)| \le M_\rho(\ell)$ there; Cauchy's estimate applied to Equation (P.90) gives Equation (P.93). Summing the geometric-type majorant gives absolute convergence for $M_\rho(\ell) < \rho|\Lambda|$. The sum solves $w + \sigma \log w = \Lambda$ because Lagrange's theorem identifies the series with the unique analytic solution $v(z)$ of $v = -\sigma z(\ell + \log(1+v))$ vanishing at $z = 0$; the branch with $w \sim \Lambda$ is the one selected. For $\Lambda = 10$, $\ell = \log 10 = 2.3026$ and $M_{1/2} = 2.3026 + 0.6931 = 2.9957 < 5$. □

Source claim. S3 asserts that "the pure Lambert logarithmic expansion has a nonzero convergence domain for sufficiently large argument, as in the classical analysis of Lambert W", citing Corless et al. but giving no criterion. **Defect.** The cited classical result concerns the doubly-indexed series in $(\log z)^{-1}$ and $\log\log z$; the single series Equation (P.91), whose coefficients are polynomials in the slowly varying ℓ, needs its own statement, since convergence depends on the ratio ℓ/Λ and fails for ℓ comparable to Λ. **Repair.** Theorem P.44 supplies an explicit, proved criterion, which also delimits the regime where Equation (P.91) may be summed rather than truncated.

P.8.4 Composition, and the equivalence of the two normalizations

Theorem P.45 (Constructive equivalence of the two inverse normalizations). *With $v_j = (-\sigma)^j A_{j-1}(\ell)$ and*

$$R_r^{(m)}(\ell) = [z^r]\Big(1 + \sum_{j\ge1} v_j z^j\Big)^{-m} = \sum_{k=0}^{r} (-1)^k \binom{m+k-1}{k} \widehat{B}_{r,k}(v), \qquad R_0^{(m)} = 1, \tag{P.94}$$

the Lambert-normalized expansion Equation (P.76) *and the flattened expansion Equation* (P.83) *are term-by-term equivalent: in the scaled variables of Equation* (P.58),

$$\mathcal{I}_\sigma(y) \sim \frac{\Lambda_\sigma}{2\lambda} - \frac{\sigma\,\ell_\sigma}{2\lambda} + \sum_{N\ge1} \frac{\Pi_N^{(\sigma)}(\ell_\sigma)}{\Lambda_\sigma^N}, \qquad \Pi_N^{(\sigma)}(\ell) = \frac{(-\sigma)^{N+1} A_N(\ell)}{2\lambda} + \sum_{m=1}^{N} c_m^{(\sigma)} (2\lambda)^m R_{N-m}^{(m)}(\ell), \tag{P.95}$$

with $\deg \Pi_N^{(\sigma)} = N$ and leading coefficient $(-1)^{N-1}(-\sigma)^{N+1}/(2\lambda N)$.

Proof. Insert $X_\sigma = w_\sigma/(2\lambda)$ into Equation (P.76), write $w = \Lambda(1+V)$ with $V = \sum_j v_j z^j$ from Equation (P.91), and collect the coefficient of Λ^{-N}: the core contributes the first term of Equation (P.95) and the m-th correction contributes $c_m^{(\sigma)}(2\lambda)^m [z^{N-m}]\,(1+V)^{-m}$. The Bell form of $R_r^{(m)}$ is Theorem J.14. Degrees: A_N has degree N with leading coefficient $(-1)^{N-1}/N$ by Equation (P.92), while every correction with $m \ge 1$ contributes degree at most $N - m$. □

Source claim. S1 asserts the equivalence of its two inverse normalizations in prose: "expanding the Lambert carrier in the scale $X^{-1}\mathbb{R}[\ell]$ and then substituting that expansion … reproduces the recurrence coefficient by coefficient." **Defect.** No proof and no formula; the assertion is exactly the sort of "successive matching" that the coefficient calculus is supposed to eliminate. **Repair.** Theorem P.45, which follows S3, makes the equivalence a finite explicit composition with a proof, and identifies the degrees and leading coefficients.

Remark P.46 (Three flattening normalizations, and which one the volume uses). The three sources flatten in three different variables: S2 in $L_\sigma = \log(y/C_\sigma)$ with $H_\sigma = \log(L_\sigma/\lambda)$; S3 in $\Lambda_\sigma = \log(\pi y^2) + \sigma\log(4\lambda)$ with $\ell_\sigma = \log\Lambda_\sigma$; S1 in $X_\epsilon = \log(y\alpha^{\beta_\epsilon}/K_\epsilon)$ in the half index. Since $\Lambda_\sigma = 2L_\sigma + \sigma(\log(4\lambda) - \lambda) \neq 2L_\sigma$, these are genuinely different variables, the three coefficient lists are not comparable termwise, and none is in error — all were recomputed and are correct in their own normalization. The volume adopts S2's variables in Equation (P.83) (they are those of Theorem J.37) and records S3's in Equation (P.95), where the composition proof is cleanest.

Corollary P.47 (Leading elementary forms)**.** *As $y \to \infty$,*

$$\mathcal{I}_\sigma(y) = \frac{1}{\lambda}\Big(L_\sigma(y) - \frac{\sigma}{2}\log\frac{L_\sigma(y)}{\lambda} + O\Big(\frac{\log L_\sigma(y)}{L_\sigma(y)}\Big)\Big). \tag{P.96}$$

The opposite signs of the $\frac{1}{2}\log L$ term on the two sheets are the inverse-side signature of the forward powers $n^{-1/2}$ and $n^{+1/2}$.

P.9 Asymptotic meaning, optimal truncation, and Stokes structure

P.9.1 What each expansion means at fixed order

For each fixed M the forward statement is Equation (P.36) with the effective constant of Theorem P.25, and the inverse statement is Equation (P.76). Both are Poincaré statements about divergent series: the coefficients d_j grow factorially (Theorem P.22), hence so do the A_n and, through the reversion, the $c_n^{(\sigma)}$. The divergence is not a defect but a datum: the Borel transform and the exact sum of the logarithmic series are known in closed form (Theorem P.13, Theorem P.16). By contrast the pure core series Equation (P.91) *converges* in the regime delimited by Theorem P.44; the full inverse is asymptotic only because of the gamma-ratio correction.

P.9.2 Optimal truncation and its image in the inverse variable

Proposition P.48 (Least-term truncation, forward and inverse)**.** *Theorem J.54 applied to Equation* (P.30) *with the singulant modulus π of Theorem P.16 gives the least-term index $r \approx \pi x/2$ and the beyond-all-orders floor*

$$\big|R_N(x)\big|_{N\approx\pi x/2} = \frac{2\sqrt{2}}{\pi\sqrt{x}}\,\mathrm{e}^{-\pi x}\big(1 + O(x^{-1})\big), \tag{P.97}$$

which by Theorem P.20 is an equality up to the stated factor and not merely an upper bound. Transporting it through the inverse by Theorem J.51 and $\big(\log\mathcal{S}_\sigma\big)' = \lambda + O(x^{-1})$ gives the inverse floor

$$O\Big(X_\sigma^{-1/2}\mathrm{e}^{-\pi X_\sigma}\Big) = O\Big(y^{-\pi/\lambda}\,(\log y)^{-1/2+\sigma\pi/(2\lambda)}\Big), \qquad \frac{\pi}{\lambda} = 4.5323601418\ldots \tag{P.98}$$

Proof. Equation (P.97) is Theorem P.22. For the image, use the core equation in the form $\mathrm{e}^{\lambda X} = (y/C_\sigma)X^{-\sigma/2}$, so that $\mathrm{e}^{-\pi X} = (y/C_\sigma)^{-\pi/\lambda}X^{\sigma\pi/(2\lambda)}$; multiply by $X^{-1/2}$ and use $X \asymp \log y$. □

Remark P.49. Exponential smallness in the index becomes *algebraic* smallness in the level: an $\mathrm{e}^{-\pi x}$ forward sector maps to a $y^{-\pi/\lambda}$ inverse sector. This is why hyperasymptotic corrections to the inverse live on a strictly finer grid than the algebraic powers of $1/\log y$ produced by Equation (P.83) — they are invisible to that scale at every order.

P.9.3 Lateral sums and the aggregate Stokes multipliers

For a direction θ let $\mathcal{S}_{\theta\pm} D(x) = \int_0^{\mathrm{e}^{\mathrm{i}(\theta\pm 0)}\infty} \mathrm{e}^{-xt}\widehat{D}(t)\,\mathrm{d}t$ whenever the rotated Laplace integral converges, and let $\operatorname{Disc}_\theta$ denote the lower-minus-upper lateral difference.

Proposition P.50 (Bridge equations)**.** *With the poles and residues of Equation* (P.25)*,*

$$\operatorname{Disc}_{\pi/2} D(x) = 2\sum_{k\geq 0} \frac{\mathrm{e}^{-\mathrm{i}\pi(2k+1)x}}{2k+1} = 2\operatorname{artanh}\bigl(\mathrm{e}^{-\mathrm{i}\pi x}\bigr) \qquad (\Im x < 0), \tag{P.99}$$

$$\operatorname{Disc}_{-\pi/2} D(x) = -2\sum_{k\geq 0} \frac{\mathrm{e}^{\mathrm{i}\pi(2k+1)x}}{2k+1} = -2\operatorname{artanh}\bigl(\mathrm{e}^{\mathrm{i}\pi x}\bigr) \qquad (\Im x > 0), \tag{P.100}$$

the closed forms giving the analytic continuations beyond those half-planes. Consequently $\operatorname{Disc}_\theta \log \mathcal{S}_\sigma = \sigma \operatorname{Disc}_\theta D$*, and the multiplicative Stokes factor of the forward branch across the ray* $\theta = \pm\pi/2$ *is* $\exp\bigl(\sigma \operatorname{Disc}_{\pm\pi/2} D(x)\bigr)$*, that is*

$$\mathfrak{M}_+^{(\sigma)}(x) = \left(\frac{1+\mathrm{e}^{-\mathrm{i}\pi x}}{1-\mathrm{e}^{-\mathrm{i}\pi x}}\right)^{\sigma}, \qquad \mathfrak{M}_-^{(\sigma)}(x) = \left(\frac{1-\mathrm{e}^{\mathrm{i}\pi x}}{1+\mathrm{e}^{\mathrm{i}\pi x}}\right)^{\sigma}. \tag{P.101}$$

On the positive real axis no lateral choice arises, because $\arg t = 0$ *is a regular direction.*

Proof. Rotating the Laplace contour past the ray through $t_k = \mathrm{i}\pi(2k+1)$ picks up $2\pi\mathrm{i}$ times the residue of $\mathrm{e}^{-xt}\widehat{D}(t)$ there, namely $2\pi\mathrm{i}\,\mathrm{e}^{-t_k x}/t_k = 2\mathrm{e}^{-\mathrm{i}\pi(2k+1)x}/(2k+1)$; summing over k and using $\operatorname{artanh} z = \sum_{k\geq 0} z^{2k+1}/(2k+1)$ gives Equation (P.99). The series converges for $|\mathrm{e}^{-\mathrm{i}\pi x}| = \mathrm{e}^{\pi\Im x} < 1$, that is $\Im x < 0$. The lower ray is the conjugate computation. The last two displays follow from $\log \mathcal{S}_\sigma = \log C_\sigma + \lambda x + \frac{\sigma}{2}\log x + \sigma D$, in which only D is discontinuous, together with $2\operatorname{artanh} z = \log\frac{1+z}{1-z}$. □

Remark P.51 (Convention dependence). S2 defines the discontinuity as lower minus upper lateral sum; S3 instead adopts the residue convention in which crossing a singular ray contributes $+2\pi\mathrm{i}$ times the enclosed residues, and works with $Q = -D$. The two sets of formulae differ by an overall sign, hence by a reciprocal in the multipliers, and both are internally consistent. A Stokes generator is only defined relative to a stated convention, so the volume states the convention explicitly in Theorem P.50 and does not declare either source's signs canonical.

P.9.4 Transport of an exponential sector through inversion

Proposition P.52 (Sector transport)**.** *Let* F *be one lateral branch of* $\log \mathcal{S}_\sigma$ *and let the other be* $F + \Delta$*. As a formal expansion in the sector* Δ*,*

$$x_{\mathrm{new}} = x + \sum_{k\geq 1} \frac{(-1)^k}{k!}\left(\frac{1}{F'(x)}\frac{\mathrm{d}}{\mathrm{d}x}\right)^{k-1}\left(\frac{\Delta(x)^k}{F'(x)}\right), \tag{P.102}$$

with $\Delta = \sigma \operatorname{Disc}_\theta D$ *and* $F'(x) = \lambda + \frac{\sigma}{2x} + \sigma D'(x)$*. Quantitatively, and with no formal series at all: if* $|\Delta| \leq \delta$ *on the interval joining* x *and* x_{new} *and* $F' \geq \mu > 0$ *there, then*

$$|x_{\mathrm{new}} - x| \leq \frac{\delta}{\mu}, \qquad x_{\mathrm{new}} - x = -\frac{\Delta(x)}{F'(x)} + O(\delta^2), \tag{P.103}$$

the implied constant depending only on $\sup |(F^{-1})''|$ *on that interval.*

Proof. Equation (P.102) is Theorem P.39 with $F_0 = F$, $g = \Delta$ and $\eta = 1$, read as an identity of formal series in the sector marker. The first inequality in Equation (P.103) is Theorem J.47; the second is Taylor's theorem for F^{-1} at the point $F(x)$. □

Source claim. S1 states its transport rule as a *theorem* (“Transseries-sector transport through inversion”) whose hypothesis is that “E is smaller than the algebraic scale retained in Φ_{alg}”, whose conclusion is prefaced by the word “formally”, and whose displayed first terms carry an unjustified error symbol $O(E^3, E^2E', E(E')^2)$. **Defect.** The hypothesis is never used; the conclusion is a formal-series identity, not an analytic one; and the error term is asserted rather than proved. A statement that holds only formally must not stand in a `theorem` environment. **Repair.** Theorem P.52 separates the two claims: the all-orders identity is formal (and is exactly Theorem P.39), while the quantitative first-order statement is proved by the mean value theorem under stated hypotheses. S2 and S3 already state their versions correctly, as formal expansions.

Source claim. S1’s section on complex inverse sheets asserts that for every Lambert branch with $|w_{\epsilon,k}| \to \infty$ in a suitable sector, “the same coefficients $d_r^{(\epsilon)}$ give $M_{\epsilon,k}(y) \sim w_{\epsilon,k}(y) + \sum_r d_r^{(\epsilon)} w_{\epsilon,k}(y)^{-r}$”. **Defect.** For $k \neq 0, -1$ nothing has been said, let alone proved, about $M_{\epsilon,k}$ being an inverse of anything: the existence of an analytic branch of $\mathcal{S}_\sigma^{-1}$ along the corresponding path is a hypothesis, not a consequence, and it fails when the path meets a pole of the gamma quotient or a Stokes direction. **Repair.** The statement is demoted to Theorem P.53, with the two missing hypotheses displayed. S2 states the same thing correctly, as a *formal* complex branch whose analytic validity is a separate requirement.

Remark P.53 (Complex branches, formally). For $k \in \mathbb{Z}$ let $X_{\sigma,k}$ be as in Equation (P.51) after fixing branches of the powers and logarithms in L_σ. Suppose that along a path with $|X_{\sigma,k}(y)| \to \infty$ *(i)* the forward expansion Equation (P.36) is valid, which requires the path to stay in a sector free of the poles of $\Gamma(x/2+(3-\sigma)/4)^{-2}$ and away from the Stokes directions of Theorem P.50, and *(ii)* an analytic branch $\mathcal{I}_{\sigma,k}$ of the inverse exists along it. Then that branch satisfies $\mathcal{I}_{\sigma,k}(y) \sim X_{\sigma,k}(y) + \sum_{n\geq 1} c_n^{(\sigma)} X_{\sigma,k}(y)^{-n}$ with the *same* coefficients, by the proof of Theorem P.41. The coefficient sequence does not depend on k: all branch dependence sits in W_k and in the chosen logarithm. Without hypotheses (i) and (ii) the display is a formal statement only.

P.10 Discrete inversion and index recovery

P.10.1 The three inverse objects, parity by parity

Theorem J.41 attaches three objects to a discrete sequence: the functional inverse of a chosen interpolation, the integer staircase, and the round-to-nearest generalized inverse.

Warning P.54 (None of the three exists for the unsplit sequence). By Theorem P.4 the set $\{n : \mathrm{sw}(n) \leq y\}$ is not an initial segment of $\mathbb{N}_0$ — for instance $\mathrm{sw}(5) = 30 > 20 = \mathrm{sw}(6)$ — so the staircase of sw is not monotone and does not invert it; the round-to-nearest inverse is ambiguous whenever two indices of opposite parity are nearly equidistant in value; and by Theorem P.9 no eventually monotone interpolation exists at all. All three objects exist, are unique, and obey Theorem J.43 *on each parity class separately.*

Proposition P.55 (Exact branchwise staircases). *For $y \geq 1$ put*

$$N_-^{\leq}(y) = \max\{n \textit{ even} : \mathrm{sw}(n) \leq y\}, \qquad N_+^{\leq}(y) = \max\{n \textit{ odd} : \mathrm{sw}(n) \leq y\}, \tag{P.104}$$

both sets being non-empty because $\mathrm{sw}(0) = \mathrm{sw}(1) = 1$. Then

$$\boxed{N_-^{\leq}(y) = 2\left\lfloor \frac{\mathcal{I}_-(y)}{2} \right\rfloor, \qquad N_+^{\leq}(y) = 2\left\lfloor \frac{\mathcal{I}_+(y)-1}{2} \right\rfloor + 1,} \tag{P.105}$$

and the least index of each parity that reaches the level y is

$$N_-^{\geq}(y) = 2\left\lceil \frac{\mathcal{I}_-(y)}{2} \right\rceil, \qquad N_+^{\geq}(y) = 2\left\lceil \frac{\mathcal{I}_+(y)-1}{2} \right\rceil + 1. \tag{P.106}$$

These identities are exact, *not asymptotic. The quantity Equation* (P.106) *is the staircase* N_* *of Theorem J.41, and the displayed formula for it is Theorem J.43's exact-rounding identity* $N_* = \lceil \nu \rceil$*, applied to the bisection* $(\mathrm{sw}(2m+\epsilon))_{m\ge 0}$ *with its admissible interpolation* $m \mapsto \mathcal{S}_\sigma(2m+\epsilon)$*.*

Proof. By Equation (P.11) and Theorem P.8, $\mathcal{S}_-$ is a strictly increasing bijection agreeing with sw at the even integers, so for even n we have $\mathrm{sw}(n) \le y \iff \mathcal{S}_-(n) \le y \iff n \le \mathcal{I}_-(y)$. The largest such even n is $2\lfloor \mathcal{I}_-(y)/2 \rfloor$ and the least even $n \ge \mathcal{I}_-(y)$ is $2\lceil \mathcal{I}_-(y)/2 \rceil$. The odd case is identical after the shift $n \mapsto n-1$. The boundary case in which $\mathcal{I}_\sigma(y)$ is an integer of the relevant parity is covered, because then y is exactly a sequence value and floor and ceiling return it. ☐

The round-to-nearest inverses on the two parity classes are

$$n_-^{\mathrm{near}}(y) = 2\left\lfloor \frac{\mathcal{I}_-(y)}{2} + \frac{1}{2} \right\rfloor, \qquad n_+^{\mathrm{near}}(y) = 2\left\lfloor \frac{\mathcal{I}_+(y)-1}{2} + \frac{1}{2} \right\rfloor + 1. \tag{P.107}$$

Remark P.56 (The rounding must be certified, not trusted). Equation (P.105)–Equation (P.107) are exact statements about the real numbers $\mathcal{I}_\sigma(y)$, whereas a truncated transseries supplies only an approximation to them. When y lies exponentially close to a sequence value the floor can flip. The remedy is cheap: use the transseries to produce a candidate integer, then decide the inequality $\mathrm{sw}(n) \le y$ exactly, either in integer arithmetic or by comparing $\log \mathcal{S}_\sigma(n)$ with $\log y$ at the candidate and its same-parity neighbour, working through $\log \Gamma$ to stay in range. Theorem J.43 is the general separation condition and Theorem J.47 converts the logarithmic residual into a certified bracket for $\mathcal{I}_\sigma(y)$.

P.10.2 Which parity wins, and by how much

Theorem P.57 (The sheet gap). *Put* $Y = \log(y\sqrt{\pi})$ *and* $h = \log(Y/\lambda)$*, so that* $L_\sigma = Y + \sigma\lambda/2$*. Then, as* $y \to \infty$*,*

$$\boxed{\mathcal{I}_-(y) - \mathcal{I}_+(y) = \frac{h}{\lambda} - 1 + \frac{1}{2Y} + O\!\left(\frac{h^2}{Y^2}\right) = \log_2\!\left(\frac{\log(y\sqrt{\pi})}{\log 2}\right) - 1 + o(1).} \tag{P.108}$$

In particular the gap tends to $+\infty$*: every sufficiently large level is reached first on the* odd *sheet, about* $\log_2 \log y$ *indices before the even sheet catches up.*

Proof. Subtract the two instances of Equation (P.83) truncated after $n = 1$. The leading terms contribute $(L_- - L_+)/\lambda = -1$. Next, $H_\sigma = \log\big((Y + \sigma\lambda/2)/\lambda\big) = h + \sigma\lambda/(2Y) + O(Y^{-2})$, and the two contributions $-(\sigma/2\lambda)H_\sigma$ combine to $(H_- + H_+)/(2\lambda) = h/\lambda + O(Y^{-2})$, the σ-odd parts cancelling. Finally, with $P_1^{(\sigma)} = (H - \sigma\lambda)/4$ from Equation (P.84),

$$\frac{1}{\lambda}\left[\frac{P_1^{(-1)}(H_-)}{L_-} - \frac{P_1^{(+1)}(H_+)}{L_+}\right] = \frac{1}{4\lambda}\left[\frac{H_- + \lambda}{L_-} - \frac{H_+ - \lambda}{L_+}\right] = \frac{1}{4\lambda}\cdot\frac{2\lambda}{Y} + O\!\left(\frac{h}{Y^2}\right) = \frac{1}{2Y} + O\!\left(\frac{h}{Y^2}\right).$$

Collecting the three displays gives Equation (P.108); the second form follows from $h/\lambda = \log_2(Y/\lambda)$ and $1/(2Y) \to 0$. The error term $O(h^2/Y^2)$ absorbs the $n = 2$ terms of Equation (P.83), whose polynomials have degree 2 in H_σ. ☐

Remark P.58 (Numerical confirmation, and agreement with S1). Recomputed with eight Lambert-normalized corrections on each sheet: for $\log y = 10, 30, 100, 400$ the gap $\mathcal{I}_-(y) - \mathcal{I}_+(y)$ equals $2.97806, 4.47863, 6.18565, 8.17591$, against the prediction $h/\lambda - 1 + 1/(2Y)$ equal to $2.97829, 4.47928, 6.18583, 8.17593$. S1 states the same result in its half index as $N_0 - N_1 = \frac{2}{\alpha}\log\log y + \frac{2}{\alpha}\log\frac{2}{\alpha} - 1 + o(1)$ with $\alpha = 2\lambda$; since $\frac{2}{\alpha} = \frac{1}{\lambda}$ and $\log\frac{2}{\alpha} = -\log\lambda$, this reads $\frac{1}{\lambda}(\log\log y - \log\lambda) - 1 + o(1)$, which is Equation (P.108). Directly from the sequence, the least index reaching $y = 10, 10^3, 10^6, 10^{12}$ is $5, 11, 21, 39$ on the odd sheet against $6, 14, 24, 44$ on the even sheet: gaps $1, 3, 3, 5$, consistent with the very slow $\log_2 \log y$ growth.

Corollary P.59 (Counting function). *Let* $N(y) := \#\{n \in \mathbb{N}_0 : \mathrm{sw}(n) \le y\}$. *Then*

$$N(y) = \frac{\mathcal{I}_-(y) + \mathcal{I}_+(y)}{2} + O(1) = \log_2\big(y\sqrt{\pi}\big) + O(1). \tag{P.109}$$

Proof. By Theorem P.55 the even indices counted are $0, 2, \ldots, 2\lfloor \mathcal{I}_-(y)/2 \rfloor$, that is $\mathcal{I}_-(y)/2 + O(1)$ of them, and similarly there are $\mathcal{I}_+(y)/2 + O(1)$ odd ones; this is the first equality. For the second, average the two instances of Equation (P.83): as in the proof of Theorem P.57 the σ-odd term $-(\sigma/2\lambda)H_\sigma$ cancels in the average, leaving $\frac{1}{2\lambda}(L_- + L_+) + O(\log\log y/\log y) = Y/\lambda + O(1)$, and $Y/\lambda = \log_2(y\sqrt{\pi})$. □

P.11 Algorithms, certification, and validation

P.11.1 Generating the coefficients

Algorithm P.60 (Exact coefficient generation). Every coefficient family of this chapter is produced by a finite exact recurrence over $\mathbb{Q}[\lambda, \lambda^{-1}, \sigma]/(\sigma^2 - 1)$:

1. d_{2r+1} from Equation (P.26) (rational Bernoulli input; $d_{2r} = 0$);
2. A_n from the triangular recurrence Equation (P.35);
3. $c_n^{(\sigma)}$ from Equation (P.61), or by solving $U = \Psi_\sigma(t, U)$ coefficientwise, or from the differential formula Equation (P.75);
4. $P_n^{(\sigma)}(H)$ from Theorem J.37 with the dictionary Equation (P.21), or from Equation (P.95) after Theorem P.43;
5. the exact certificate of Theorem P.42 after every step.

Truncating every intermediate series at order n is exact for the n-th coefficient, because all three recurrences are triangular; at order N the cost is $\widetilde{O}(N^2)$ coefficient operations. No numerical fitting and no order-specific ansatz occurs anywhere.

Remark P.61 (How this chapter's tables were produced). The three sources supply three implementations of the above. For this chapter every coefficient was regenerated by a fourth, independent one: truncated power-series arithmetic over exact rationals with λ symbolic, run separately at $\sigma = \pm 1$ and recombined in $\mathscr{R}$. Every displayed coefficient agreed with the source that printed it; the corrections recorded in this chapter concern statements, never numbers.

P.11.2 Certified numerical inversion

Proposition P.62 (Derivative bounds for certification). *For* $x \ge 1$,

$$\big(\log \mathcal{S}_+\big)'(x) > \lambda + \frac{1}{2x} - \frac{1}{4x^2} > \lambda, \qquad \big(\log \mathcal{S}_-\big)'(x) > \lambda - \frac{1}{2x}, \tag{P.110}$$

the exact value being Equation (P.14). *Hence if* $r_* = \log \mathcal{S}_\sigma(x_*) - \log y$ *and* $\big(\log \mathcal{S}_\sigma\big)' \ge \mu > 0$ *on the interval joining* x_* *and* $\mathcal{I}_\sigma(y)$, *then* $|x_* - \mathcal{I}_\sigma(y)| \le |r_*|/\mu$ *by Theorem J.47.*

Proof. $\big(\log \mathcal{S}_\sigma\big)'(x) = \lambda + \frac{\sigma}{2x} + \sigma D'(x)$ by Equation (P.19); apply $0 < -D'(x) < 1/(4x^2)$ from Theorem P.14, and $1/(2x) - 1/(4x^2) > 0$ for $x > 1/2$. □

Algorithm P.63 (Branchwise inversion of a level). Given a large y and a parity σ:

1. Evaluate the Lambert core: $X_+ = W_0(4\pi\lambda y^2)/(2\lambda)$ or $X_- = -W_{-1}(-4\lambda/(\pi y^2))/(2\lambda)$. *The branch choice is not optional* (Theorem P.32).

2. Add corrections $\sum_n c_n^{(\sigma)} X^{-n}$ by Horner nesting in $1/X$, stopping at or before the least term rather than summing indefinitely; alternatively use the centred form Equation (P.78), whose bare core is already accurate to $O(\mathcal{U}^{-2})$.

3. Certify: compute $r_* = \log \mathcal{S}_\sigma(x_*) - \log y$ through $\log \Gamma$ and bound the error by Theorem P.62. The inequalities Equation (P.56) and Equation (P.57) give an independent one-sided check that costs nothing.

4. For higher accuracy run Newton's method on the logarithmic equation,

$$x_{j+1} = x_j - \frac{\log \mathcal{S}_\sigma(x_j) - \log y}{\lambda + \frac{\sigma}{2}\big[\psi(x_j/2+1) - \psi(x_j/2+1/2)\big]},$$

   which is stable even when $\mathcal{S}_\sigma(x)$ is far outside floating-point range. Theorem P.8 guarantees a positive denominator, and the transseries seed lies in the basin for large y.

5. For a discrete answer, round by Equation (P.105) and verify the final inequality exactly (Theorem P.56).

P.11.3 Validation

The tables below were recomputed at 50 significant digits against exact gamma-quotient reference values, with y generated as $\mathcal{S}_\sigma(x_0)$ so that the exact inverse is x_0. They reproduce the corresponding tables of S2 and S3 to every printed digit, and, after the reparametrization of Theorem P.12, those of S1 as well.

Table P.4: Relative error of the forward expansion Equation (P.36) after the term $\sigma^M A_M x^{-M}$. Recomputed.

x	σ	$M=0$	$M=2$	$M=4$	$M=6$	$M=8$
10	-1	2.5273×10^{-2}	3.8527×10^{-5}	4.7159×10^{-7}	1.4186×10^{-8}	7.8542×10^{-10}
10	$+1$	2.4650×10^{-2}	3.8626×10^{-5}	4.7375×10^{-7}	1.4233×10^{-8}	7.8709×10^{-10}
20	-1	1.2573×10^{-2}	4.8642×10^{-6}	1.5087×10^{-8}	1.1546×10^{-10}	1.6336×10^{-12}
20	$+1$	1.2417×10^{-2}	4.8704×10^{-6}	1.5121×10^{-8}	1.1565×10^{-10}	1.6353×10^{-12}
50	-1	5.0122×10^{-3}	3.1226×10^{-7}	1.5560×10^{-10}	1.9152×10^{-13}	4.3651×10^{-16}
50	$+1$	4.9872×10^{-3}	3.1242×10^{-7}	1.5573×10^{-10}	1.9164×10^{-13}	4.3668×10^{-16}

Table P.5: Absolute error $\left|\mathcal{I}_\sigma^{[K]}(\mathcal{S}_\sigma(x_0)) - x_0\right|$ of the Lambert-normalized inverse Equation (P.76) with K corrections retained; $K=0$ is the bare core. Recomputed.

x_0	σ	$K=0$	$K=2$	$K=4$	$K=6$	$K=8$
10	-1	3.8813×10^{-2}	1.6744×10^{-5}	2.9057×10^{-7}	2.4840×10^{-8}	1.7249×10^{-9}
10	$+1$	3.3589×10^{-2}	2.2677×10^{-4}	3.9282×10^{-6}	5.5615×10^{-8}	2.0051×10^{-9}
20	-1	1.8701×10^{-2}	1.2039×10^{-6}	6.9374×10^{-9}	1.8267×10^{-10}	3.2855×10^{-12}
20	$+1$	1.7399×10^{-2}	3.0220×10^{-5}	1.3176×10^{-7}	4.7142×10^{-10}	4.3075×10^{-12}
50	-1	7.3186×10^{-3}	4.2970×10^{-8}	5.6559×10^{-11}	2.8529×10^{-13}	8.3851×10^{-16}
50	$+1$	7.1104×10^{-3}	2.0095×10^{-6}	1.4043×10^{-9}	8.1104×10^{-13}	1.1816×10^{-15}
100	-1	3.6329×10^{-3}	3.9487×10^{-9}	1.6091×10^{-12}	2.1882×10^{-15}	1.6191×10^{-18}
100	$+1$	3.5808×10^{-3}	2.5440×10^{-7}	4.4455×10^{-11}	6.4401×10^{-15}	2.3379×10^{-18}

Warning P.64 (These tables are not convergence). The decreasing entries at fixed x_0 do not indicate convergence as $K \to \infty$. Both the forward and the inverse series are divergent, with the beyond-all-orders floor Equation (P.98); the tables stop far short of the least term (at $x = 20$ the least term of

Table P.6: The same errors for the centred inverse Equation (P.78). The bare core is better by a factor $\asymp X$, because $\widehat{c}_1^{(\sigma)} = 0$; once corrections are retained the two normalizations are comparable. Recomputed.

x_0	σ	$K=0$	$K=2$	$K=4$	$K=6$	$K=8$
10	-1	8.7330×10^{-4}	5.5308×10^{-5}	1.1593×10^{-7}	2.7056×10^{-9}	4.4842×10^{-11}
10	$+1$	7.6102×10^{-4}	5.6715×10^{-5}	2.3489×10^{-7}	7.9121×10^{-9}	4.0347×10^{-10}
20	-1	2.2206×10^{-4}	7.4914×10^{-6}	5.2119×10^{-9}	3.7661×10^{-11}	3.3070×10^{-13}
20	$+1$	2.0696×10^{-4}	7.5956×10^{-6}	7.4297×10^{-9}	6.3564×10^{-11}	8.0880×10^{-13}
50	-1	3.5860×10^{-5}	5.0352×10^{-7}	6.5130×10^{-11}	8.3102×10^{-14}	1.4333×10^{-16}
50	$+1$	3.4850×10^{-5}	5.0641×10^{-7}	7.5153×10^{-11}	1.0250×10^{-13}	2.0275×10^{-16}
100	-1	8.9913×10^{-6}	6.3982×10^{-8}	2.1684×10^{-12}	7.1217×10^{-16}	3.2358×10^{-19}
100	$+1$	8.8632×10^{-6}	6.4167×10^{-8}	2.3300×10^{-12}	7.9120×10^{-16}	3.8477×10^{-19}

Equation (P.30) sits at $r = 31$, of size 10^{-28}). The non-monotone entries in the odd column at $x_0 = 10$ are ordinary behaviour of an asymptotic series at small argument, not an error. Beyond the least term one should use the exact Laplace representation Equation (P.22), or a Newton step against $\log\Gamma$, or an exponentially improved representation transported by Theorem P.52.

P.12 Summary of the coefficient hierarchy

The whole calculation is a chain of constructive coefficient transformations, each with a closed extraction formula and a triangular recurrence:

$$\boxed{B_{2r+2} \xrightarrow{\text{gamma ratio}} d_{2r+1} \xrightarrow{\exp,\,B} A_n \xrightarrow{\text{Lagrange, }\widehat{B}} c_n^{(\sigma)} \xrightarrow{\text{unfold }W} P_n^{(\sigma)}(H),} \tag{P.111}$$

while the Borel kernel $\tanh(t/2)/(2t)$ supplies, in one stroke, the exact sum of the first stage, the sign and size of every remainder (Theorem P.20), the singulant π, the least-term scale, and the complete Stokes data. Two facts are specific to the swing factorial and cannot be relegated to the general theory of Part J:

1. *Parity is leading-order data.* The forward object is a period-two transseries Equation (P.40) whose discrete transmonomial σ_n switches the algebraic scale between $2^n n^{-1/2}$ and $2^n n^{+1/2}$; the inverse object is a *pair* of monotone branches, and Theorem P.9 shows that this is forced, not chosen.

2. *The two sheets need different Lambert branches.* W_{-1} on the even sheet, W_0 on the odd sheet, the assignment being forced by $\operatorname{sign} b = \sigma$ in Theorem J.23. The principal branch on the even sheet does not merely lose accuracy: by Theorem P.32 it returns the root $2/(\pi y^2)\big(1 + O(y^{-2})\big) \to 0$, so that choice is fatal before any correction is computed.

Part XIII

Four special functions

Appendix Q

Gamma and Barnes G

Q.1 What this chapter does

The four subjects of Chapters M–P are *sequences*; the next two are *special functions*, and the change matters in exactly one place. A sequence has three inverse objects (Theorem J.41) and the staircase theorem is needed to pass between them; a function analytic and eventually strictly monotone on a real ray has one, and the inverse asked for is the honest compositional inverse of a restriction. Everything else — the exponential–power bookkeeping, the exact inversion of a dominant block, the all-orders reversion around it, the backward-error certificate — is the apparatus of Chapter 0 and is cited, not repeated.

This chapter consolidates three filed articles, listed in Table Q.1. They were written independently and they agree: every coefficient, table and decimal any of them prints was recomputed here in exact arithmetic or to 60 digits, and all of it reproduced. What the three do *not* share is a normalization — one absorbs the Barnes constant $\zeta'(-1)$ into the target and two do not, and the three use three different letters for the slope — so Section Q.8 gives the dictionary and the rest of the chapter fixes one convention.

Tag	Directory	Distinctive content
S1	`inverse_gamma_barnesG_transseries`	Hahn supports; Newton valuation doubling; the $\sqrt{T^2+1/6}$ subsector
S2	`inverse_gamma_barnes_transseries`	the pole branches of Γ^{-1}; universal C_0, C_1, C_2 for general p; $\zeta'(-1)$ left unabsorbed
S3	`Asymptotic_Inversion_Gamma_Barnes_G`	the kernel coefficients $\kappa_n(p, v)$; the slope denominator bound; resonance; b_3, b_4, b_5

Two features distinguish this chapter from the first four.

The dominant block is not the one Chapter 0 inverts. Theorem J.23 solves $aX + b\log X = L$ exactly; here the unknown *multiplies* the logarithm, and the block to be inverted is $ax^p(\log x - \beta)$. That block is still solved exactly by W, but by a different substitution, and the resulting normalized slope is $v+1$ rather than $a + b/X$. Section Q.3 proves the core lemma in the generality the two subjects and the next chapter need, namely for $ax^p(\log x - \beta)^r$.

The forward phase is a Stirling series, not an exponential–power model. As in Part N, the phase $\frac{1}{2}z^2\log z$ grows faster than z^α for every $\alpha \le 1$ and Theorem J.2 does not apply. What does apply is the axiomatized dominant core; we say precisely which axioms hold in Theorem Q.4 and do not force the model.

The chapter also proves two things none of the three sources states. The first is that the slope-denominator bound of Theorem Q.18 is *attained*, with the deepest pole given in closed form (Theo-

Table Q.1: Parameter dictionary for this chapter.

	Γ	G
inverted map	$x \mapsto \Gamma(x)$, increasing branch	$z \mapsto G(z+1)$, eventual branch
centred variable	$t = x - \frac{1}{2}$	$z = x - 1$
core exponent p	1	2
core scale a	1	$\frac{1}{2}$
core shift β	1	$\frac{3}{2}$
absorbed constant C	$\frac{1}{2}\log(2\pi)$	$\zeta'(-1)$
perturbation support	$2\mathbb{N}$	$\mathbb{N}$
first perturbation	$a_1 t^{-1} = -1/(24t)$	αz, $\alpha = \frac{1}{2}\log(2\pi)$
core	$T = e^{\Lambda}$, $\Lambda = w + 1$	$U = e^{\lambda+1}$, $\lambda = (v+1)/2$
inverse support	$T^{-1}, T^{-3}, T^{-5}, \ldots$	$U^0, U^{-1}, U^{-2}, \ldots$

rem Q.22); the second is that the two square roots the sources find — S1's and S3's $\sqrt{U^2 + 1/6}$ for Barnes, and the previously unnoticed $\sqrt{\Lambda^2 T^2 + 1/12}$ for Gamma — are the same phenomenon, the exactly solvable quadratic truncation of the normalized equation (Theorem Q.19).

Q.2 Forward normal forms

Bernoulli numbers and polynomials follow the conventions of Chapter 0: $B_n = B_n(0)$ and $z e^{xz}/(e^z - 1) = \sum_{n \ge 0} B_n(x) z^n / n!$, so that $B_1 = -\frac{1}{2}$, $B_2 = \frac{1}{6}$, $B_4 = -\frac{1}{30}$, and

$$B_n\big(\tfrac{1}{2}\big) = \big(2^{1-n} - 1\big) B_n. \tag{Q.1}$$

Q.2.1 The half-shifted Gamma phase

Lemma Q.1 (Symmetry-adapted Gamma normal form). *Put $t = x - \frac{1}{2}$. In every closed subsector of $|\arg t| < \pi$, as $t \to \infty$,*

$$\log \Gamma\big(t + \tfrac{1}{2}\big) \sim t(\log t - 1) + \tfrac{1}{2}\log(2\pi) + \sum_{k \ge 1} \frac{a_k}{t^{2k-1}}, \tag{Q.2}$$

where

$$a_k = \frac{B_{2k}(1/2)}{2k(2k-1)} = \frac{\big(2^{1-2k} - 1\big) B_{2k}}{2k(2k-1)}. \tag{Q.3}$$

Only odd inverse powers of t occur. The first values are

$$a_1 = -\frac{1}{24}, \quad a_2 = \frac{7}{2880}, \quad a_3 = -\frac{31}{40320}, \quad a_4 = \frac{127}{215040}, \quad a_5 = -\frac{511}{608256}. \tag{Q.4}$$

Proof. The shifted Stirling expansion is, for fixed h and in the stated sectors,

$$\log \Gamma(t + h) \sim \big(t + h - \tfrac{1}{2}\big)\log t - t + \tfrac{1}{2}\log(2\pi) + \sum_{k \ge 2} \frac{(-1)^k B_k(h)}{k(k-1)\, t^{k-1}}. \tag{Q.5}$$

Set $h = \frac{1}{2}$. The coefficient of $\log t$ collapses to t exactly, which is the entire point of the shift. By (Q.1) and $B_{2m+1} = 0$ for $m \ge 1$, we get $B_{2m+1}(\frac{1}{2}) = 0$ for all $m \ge 0$, so only even k survives; writing $k = 2n$ and noting $(-1)^{2n} = 1$ gives (Q.2)–(Q.3). The five values follow from $B_2 = \frac{1}{6}$, $B_4 = -\frac{1}{30}$, $B_6 = \frac{1}{42}$, $B_8 = -\frac{1}{30}$, $B_{10} = \frac{5}{66}$. $\square$

Remark Q.2 (The shift is not cosmetic). Expanding in x^{-1} instead of t^{-1} leaves both parities in the forward phase, hence a perturbation support $\mathbb{N}$ rather than $2\mathbb{N}$, hence — by Theorem Q.14 — an inverse displacement with every integer power of T^{-1} instead of only the odd ones. The shift is the cheapest available instance of the normalization rule of Theorem Q.11: translate first, and only then declare what is small.

Q.2.2 The Barnes phase in the centred variable

Lemma Q.3 (Simplified Barnes normal form)**.** *In every closed subsector of* $|\arg z| < \pi$*, as* $z \to \infty$*,*

$$\log G(z+1) \sim \frac{1}{2}z^2 \log z - \frac{3}{4}z^2 + \alpha z - \frac{1}{12}\log z + \zeta'(-1) + \sum_{k\geq 1} \frac{c_k}{z^{2k}}, \tag{Q.6}$$

where

$$\alpha = \tfrac{1}{2}\log(2\pi), \qquad c_k = \frac{B_{2k+2}}{4k(k+1)}, \tag{Q.7}$$

so that $c_1 = -\frac{1}{240}$, $c_2 = \frac{1}{1008}$, $c_3 = -\frac{1}{1440}$, $c_4 = \frac{1}{1056}$.

Proof. Start from the classical form

$$\log G(z+1) \sim \frac{1}{4}z^2 + z\log\Gamma(z+1) - \left(\frac{1}{2}z(z+1) + \frac{1}{12}\right)\log z - \log A + \sum_{k\geq 1} \frac{B_{2k+2}}{2k(2k+1)(2k+2)z^{2k}},$$

A being the Glaisher–Kinkelin constant, and substitute

$$\log\Gamma(z+1) \sim \left(z + \tfrac{1}{2}\right)\log z - z + \tfrac{1}{2}\log(2\pi) + \sum_{j\geq 1} \frac{B_{2j}}{2j(2j-1)z^{2j-1}}.$$

Collect by type. The $z^2 \log z$ terms give $z^2\log z - \frac{1}{2}z^2\log z = \frac{1}{2}z^2 \log z$; the $z\log z$ terms cancel, $\frac{1}{2}z\log z - \frac{1}{2}z\log z = 0$; the z^2 terms give $-z^2 + \frac{1}{4}z^2 = -\frac{3}{4}z^2$; the linear term is αz; and the lone $\log z$ term is $-\frac{1}{12}\log z$.

Two constants remain. Multiplying the Gamma tail by z moves its $j = 1$ term to $B_2/2 = \frac{1}{12}$, a constant; together with $-\log A$ and $\log A = \frac{1}{12} - \zeta'(-1)$ this is $\frac{1}{12} - \frac{1}{12} + \zeta'(-1) = \zeta'(-1)$.

At z^{-2k} two contributions meet: the Gamma tail with $j = k+1$, multiplied by z, contributes $B_{2k+2}/\big((2k+2)(2k+1)\big)$, and the explicit Barnes tail contributes $B_{2k+2}/\big(2k(2k+1)(2k+2)\big)$. Their sum is

$$\frac{B_{2k+2}}{(2k+1)(2k+2)}\left(1 + \frac{1}{2k}\right) = \frac{B_{2k+2}}{(2k+2)(2k)} = \frac{B_{2k+2}}{4k(k+1)},$$

which is (Q.7). □

Remark Q.4 (Not an exponential–power model). Neither phase is of the form of Theorem J.2. For Γ the phase is $t\log t$ and for G it is $\frac{1}{2}z^2\log z$; in both cases phase$/x^\alpha \to \infty$ for every $\alpha \leq 1$, by the argument already given for the double factorial in Theorem N.14. What both phases do satisfy is the axiomatized dominant core: each is C^∞ and eventually strictly increasing on a real ray, its derivative is eventually positive and regularly varying, and the residual after removing the core admits a Poincaré expansion with a differentiable remainder in sectors. Those are the only properties used below, and they are what Theorems J.47 and J.51 require.

Remark Q.5 (Which inverse). On $(0,\infty)$ the digamma function ψ is strictly increasing, because $\psi'(x) = \sum_{n\geq 0}(n+x)^{-2} > 0$, and it runs from $-\infty$ to $+\infty$; so it has exactly one zero and $\Gamma' = \Gamma\psi$ changes sign exactly once. Every sufficiently large $y > 0$ therefore has exactly two positive preimages. We write Γ_+^{-1} for the branch tending to $+\infty$, which is the subject of Section Q.5, and Γ_0^{-1} for the branch tending to 0^+, treated in Section Q.6. These are genuinely different objects: the first has a divergent asymptotic transseries, the second a convergent power series in $1/y$.

For Barnes, $(\log G(z+1))' = \frac{1}{2} + z + \frac{1}{2}\log(2\pi) - z\psi(z+1) + \ldots$; rather than track that expression we use the cleaner statement of Theorem Q.6, which gives an explicit ray on which $z \mapsto G(z+1)$ is strictly increasing.

Proposition Q.6 (An explicit ray of monotonicity). *For $z > 0$,*

$$\frac{\mathrm{d}}{\mathrm{d}z}\log G(z+1) = \frac{1}{2}\log(2\pi) - \frac{1}{2} - z + z\,\psi(z+1) = \frac{1}{2}\log(2\pi) + \frac{1}{2} - z + z\,\psi(z). \tag{Q.8}$$

Consequently $z \mapsto G(z+1)$ is strictly increasing on $[3,\infty)$, and

$$\frac{\mathrm{d}}{\mathrm{d}z}\log G(z+1) \;\geq\; \tfrac{1}{2}\,z\log z \qquad (z \geq 7). \tag{Q.9}$$

It therefore has a well-defined inverse $\mathcal{B} : [G(4),\infty) \to [3,\infty)$, and $G_+^{-1}(y) = 1 + \mathcal{B}(y)$.

Proof. Write the Weierstrass form

$$\log G(1+z) = \frac{z}{2}\log(2\pi) - \frac{z}{2} - \frac{z^2(1+\gamma)}{2} + \sum_{k\geq 1}\left[k\log\Big(1+\frac{z}{k}\Big) - z + \frac{z^2}{2k}\right],$$

which converges locally uniformly, so it may be differentiated termwise:

$$\frac{\mathrm{d}}{\mathrm{d}z}\log G(1+z) = \frac{1}{2}\log(2\pi) - \frac{1}{2} - z(1+\gamma) + \sum_{k\geq 1}\left[\frac{k}{k+z} - 1 + \frac{z}{k}\right] = \frac{1}{2}\log(2\pi) - \frac{1}{2} - z(1+\gamma) + z\sum_{k\geq 1}\left[\frac{1}{k} - \frac{1}{k+z}\right].$$

The last sum is $\psi(z+1)+\gamma$, which gives the first form in (Q.8); the second follows from $\psi(z+1) = \psi(z) + 1/z$.

Now $\psi(z+1) = \psi(z) + 1/z > \log z - 1/z + 1/z = \log z$, using the standard bound $\psi(z) > \log z - 1/z$. Hence, with $c := \frac{1}{2}\log(2\pi) - \frac{1}{2} = 0.41893\ldots$,

$$\frac{\mathrm{d}}{\mathrm{d}z}\log G(z+1) \;>\; z(\log z - 1) + c \qquad (z > 0). \tag{Q.10}$$

The right-hand side has derivative $\log z$, so it increases on $[1,\infty)$; at $z = 3$ its value is $3(\log 3 - 1) + c = 0.7147\ldots > 0$. This proves strict monotonicity on $[3,\infty)$.

For (Q.9) it suffices, by (Q.10), that $z(\log z - 1) + c \geq \frac{1}{2}z\log z$, i.e. that $\frac{1}{2}\log z \geq 1 - c/z$. The left side increases and the right side decreases, so it is enough to check $z = 7$: $\frac{1}{2}\log 7 = 0.9729\ldots$ and $1 - c/7 = 0.9401\ldots$. ☐

Remark Q.7 (How sharp the threshold is). The bound (Q.10) is lossy. Solving (Q.8) numerically, the derivative has its only positive zero at $z = 1.55766\ldots$, so $G(\,\cdot+1)$ is in fact strictly increasing on $[1.56,\infty)$, and $\frac{1}{2}z\log z$ is dominated from $z = 5.5$ onward. The thresholds 3 and 7 are what the elementary argument gives; they are stated because a proof is wanted, not because they are best possible.

All three sources assert the existence of the eventual Barnes inverse with the words "$(\log G)'(x) \sim x\log x$, hence strictly increasing for all sufficiently large x". That is true but ineffective, and none of the three supplies a threshold; yet every numerical table in all three evaluates the inverse at arguments as small as $z = 5$, inside the range their own justification does not cover. Theorem Q.6 supplies a threshold, and $3 < 5$, so the tables are legitimate.

Q.3 The power–logarithmic core

Theorem J.23 inverts $aX + b\log X$. The block needed here is different and is inverted by a different substitution.

Lemma Q.8 (Exact inversion of a power–logarithmic block)**.** *Let $a \neq 0$, $p \neq 0$, $r \neq 0$ and $\beta \in \mathbb{C}$, and put*

$$\Phi_0(x) = a\, x^p\, (\log x - \beta)^r. \tag{Q.11}$$

Fix R and compatible determinations of the logarithm, of the r-th root and of a Lambert branch W_k, and set

$$v = W_k\left(\frac{p}{r}\left(\frac{R}{a\,\mathrm{e}^{p\beta}}\right)^{1/r}\right), \qquad u = \frac{r}{p}\,v, \qquad X = \mathrm{e}^{\beta+u}. \tag{Q.12}$$

Then $\Phi_0(X) = R$, and

$$\log X - \beta = u, \qquad \Phi_0'(X) = a\, X^{p-1} u^{r-1}\,(pu + r). \tag{Q.13}$$

In the case $r = 1$ these read

$$v = W_k\left(\frac{pR}{a}\mathrm{e}^{-p\beta}\right), \qquad X = \mathrm{e}^{\beta+v/p} = \left(\frac{pR}{av}\right)^{1/p}, \qquad \Phi_0'(X) = a\, X^{p-1}(v+1). \tag{Q.14}$$

Proof. By construction $\log X - \beta = u$, so $\Phi_0(X) = a\mathrm{e}^{p\beta}\mathrm{e}^{pu}u^r$. Hence $\Phi_0(X) = R$ is equivalent to $u^r\mathrm{e}^{pu} = R/(a\mathrm{e}^{p\beta})$, and taking the r-th root, to $u\,\mathrm{e}^{pu/r} = \big(R/(a\mathrm{e}^{p\beta})\big)^{1/r}$. Multiplying by p/r turns this into $(pu/r)\mathrm{e}^{pu/r} = (p/r)\big(R/(a\mathrm{e}^{p\beta})\big)^{1/r}$, i.e. $v\mathrm{e}^v = (p/r)(\cdots)^{1/r}$ with $v = pu/r$, which is (Q.12).

For the slope, differentiate (Q.11):

$$\Phi_0'(x) = ax^{p-1}(\log x - \beta)^{r-1}\big(p(\log x - \beta) + r\big),$$

and evaluate at $x = X$, where $\log X - \beta = u$. For $r = 1$, $u = v/p$ and $pu + r = v + 1$. □

Remark Q.9 (Formalization of the core at $r = 1$). The case $r = 1$ — the one used for Γ in Section Q.5 and for the K-function, and the case in which no determination of an r-th root has to be made — is machine-checked over $\mathbb{R}$, in `PowerLogCoreInversion.lean`. `Fabius.powerLogCore` is $\Phi_0(x) = ax^p(\log x - \beta)$ with the real power; `Fabius.powerLogCoreArg` is the Lambert argument $pR\mathrm{e}^{-p\beta}/a$ and `Fabius.powerLogCoreRoot` the root $X = \mathrm{e}^{\beta+v/p}$; `Fabius.powerLogCore_powerLogCoreRoot` is $\Phi_0(X) = R$; `Fabius.log_powerLogCoreRoot_sub` is $\log X - \beta = v/p$; and `Fabius.hasDerivAt_powerLogCore_root` is the slope $aX^{p-1}(v+1)$, the third formula of (Q.14). The Lambert branch is the corpus's real principal branch `Fabius.principalLambertW` of `PrincipalLambertW.lean`, so the hypothesis is the one that branch needs, $-\mathrm{e}^{-1} \leq pR\mathrm{e}^{-p\beta}/a$.

Two features of the formal statement are worth naming, because they say what the lemma actually depends on. First, the inversion is proved in the form `Fabius.powerLogCore_of_lambert`, which assumes only that v solves $v\mathrm{e}^v = pR\mathrm{e}^{-p\beta}/a$ and concludes $\Phi_0(\mathrm{e}^{\beta+v/p}) = R$; the principal branch enters only in the corollary. So the lemma is a statement about the Lambert *equation*, not about any branch of its solution, and it transfers to the lower real branch, or to W_k, unchanged. Second, nothing analytic is used beyond that equation: the substitution $x = \mathrm{e}^{\beta+u}$ makes $\log x - \beta = u$ and $x^p = \mathrm{e}^{p\beta}\mathrm{e}^{pu}$ by construction (`Fabius.powerLogCore_exp`), and the rest is field arithmetic.

Not formalized: general r, which needs a determination of the r-th root and is where the branch bookkeeping of (Q.12) lives; and the complex-analytic reading with a general W_k.

Remark Q.10 (The universal divisor). The factor $v + 1$ that divides every inverse coefficient below is not an algebraic accident: by (Q.14) it is the normalized slope $\Phi_0'(X)/(aX^{p-1})$ of the exact core. Its vanishing at $v = -1$ is the Lambert branch point, where the core map has a critical point and no inverse of this shape can exist. Every statement in this chapter that requires "$v + 1$ invertible" is requiring distance from that turning point, and every error bound degrades as $v + 1 \to 0$.

Remark Q.11 (What must be absorbed). A term of the *same* algebraic power as the core is not small and must be absorbed before anything is called a perturbation: $ax^p \log x + cx^p = ax^p(\log x - \beta)$ with

$\beta = -c/a$. Likewise an additive constant C in the forward phase is moved to the target, $R := Y - C$. Failing to absorb makes the normalized perturbation have valuation zero and destroys the hypothesis of Theorem Q.13; failing to absorb the constant merely shifts the core, but by an amount that is not small compared with the corrections one is trying to compute. A phase $x^p\big(a_r(\log x)^r + a_{r-1}(\log x)^{r-1} + \cdots\big)$ with $r \geq 2$ needs a genuine outer inversion in the logarithmic scale first; only its leading logarithmic power is covered by Theorem Q.8.

Q.4 Normalized displacement and triangular inversion

Q.4.1 The universal kernel

Take $r = 1$ from here on, and consider a phase

$$\Phi(x) = a\, x^p(\log x - \beta) + C + \Psi(x), \tag{Q.15}$$

to be solved for $\Phi(x) = Y$. Put $R = Y - C$, construct v and X by (Q.14), write $q = X^{-1}$, and substitute

$$x = X(1+\epsilon). \tag{Q.16}$$

Dividing $\Phi(x) - Y$ by aX^p turns the equation into

$$\mathcal{K}_{p,v}(\epsilon) + \mathcal{R}(q, v, \epsilon) = 0, \tag{Q.17}$$

where

$$\mathcal{K}_{p,v}(\epsilon) = (1+\epsilon)^p\left(\frac{v}{p} + \log(1+\epsilon)\right) - \frac{v}{p}, \qquad \mathcal{R}(q, v, \epsilon) = \frac{\Psi\big(X(1+\epsilon)\big)}{aX^p}. \tag{Q.18}$$

Lemma Q.12 (Kernel coefficients). *Write $\mathcal{K}_{p,v}(\epsilon) = \sum_{n\geq 1} \kappa_n(p,v)\epsilon^n$. Then*

$$\kappa_n(p,v) = \frac{v}{p}\binom{p}{n} + \frac{\partial}{\partial p}\binom{p}{n}, \tag{Q.19}$$

a polynomial in p and v; in particular

$$\kappa_1(p,v) = v + 1. \tag{Q.20}$$

Where $p, p-1, \ldots, p-n+1$ are all nonzero one may write $\partial_p\binom{p}{n} = \binom{p}{n}\sum_{j=0}^{n-1}(p-j)^{-1}$, but (Q.19) *is the form valid at every p. For $p = 1$,*

$$\mathcal{K}_{1,v}(\epsilon) = v\epsilon + (1+\epsilon)\log(1+\epsilon) = (v+1)\epsilon + \sum_{m\geq 2}\frac{(-1)^m}{m(m-1)}\epsilon^m, \tag{Q.21}$$

and for $p = 2$,

$$\mathcal{K}_{2,v}(\epsilon) = (v+1)\epsilon + \frac{v+3}{2}\epsilon^2 + \frac{1}{3}\epsilon^3 - \frac{1}{12}\epsilon^4 + \frac{1}{30}\epsilon^5 - \cdots. \tag{Q.22}$$

Proof. Since $(1+\epsilon)^p\log(1+\epsilon) = \partial_p(1+\epsilon)^p$, we have $[\epsilon^n]\big((1+\epsilon)^p\log(1+\epsilon)\big) = \partial_p\binom{p}{n}$, while $[\epsilon^n]\big((v/p)(1+\epsilon)^p\big) = (v/p)\binom{p}{n}$; the constant $-v/p$ affects only $n = 0$. This is (Q.19). For $n = 1$, $\binom{p}{1} = p$ and $\partial_p\binom{p}{1} = 1$, so $\kappa_1 = v + 1$. Both displayed specializations follow by direct expansion; note that the coefficient of ϵ^2 in (Q.22) is $(v/2)\binom{2}{2}\cdot 2 + \partial_p\binom{p}{2}|_{p=2} = \frac{v}{2} + \frac{3}{2} = \frac{v+3}{2}$. □

Q.4.2 Triangular inversion over a well-ordered support

Theorem Q.13 (Triangular inversion on a Hahn grid)**.** *Let $\mathscr{C}$ be a field of characteristic zero containing p, v and $(v+1)^{-1}$, let $S \subset \mathbb{R}_{\geq 0}$ be a* well-ordered *additive submonoid, and let $\mathscr{C}[[q^S]]$ be the corresponding Hahn ring. Assume*

$$\mathcal{R}(q, v, E) = \sum_{\rho \in S,\ \rho > 0} \sum_{m \geq 0} r_{\rho,m}(v)\, q^{\rho} E^{m}, \tag{Q.23}$$

with every monomial of positive q-exponent. Then (Q.17) *has a unique solution ϵ of positive valuation in $\mathscr{C}[[q^S]]$, say $\epsilon = \sum_{\gamma \in S,\, \gamma > 0} e_\gamma(v) q^\gamma$, and writing $\epsilon_{<\gamma}$ for its truncation below γ,*

$$e_\gamma(v) = -\frac{1}{v+1}\,[q^\gamma]\left(\sum_{n \geq 2} \kappa_n(p, v)\,\epsilon_{<\gamma}^{n} + \mathcal{R}(q, v, \epsilon_{<\gamma})\right). \tag{Q.24}$$

Proof. Well-ordering is exactly what makes the coefficient extraction in (Q.24) a finite sum. By Neumann's lemma, if $A, B \subset \mathbb{R}$ are well ordered then $A + B$ is well ordered and each of its elements has only finitely many representations $a + b$ with $a \in A$, $b \in B$; by induction the same holds for any finite number of summands. Since $\mathrm{val}(\epsilon) > 0$, we get $\mathrm{val}(\epsilon^n) \geq n\,\mathrm{val}(\epsilon) \to \infty$, so at a prescribed exponent γ only finitely many n contribute, and for each such n only finitely many decompositions. The same applies to $\mathcal{R}$ because every one of its monomials has $\rho > 0$.

Now fix γ and suppose e_δ is known for all $0 < \delta < \gamma$. In the left-hand side of (Q.17), the coefficient of q^γ receives $\kappa_1 e_\gamma = (v+1)e_\gamma$ from the linear term, and from nothing else: a monomial of ϵ^n with $n \geq 2$ that involves e_γ has q-exponent at least $\gamma + \mathrm{val}(\epsilon) > \gamma$, and a monomial $q^\rho E^m$ of $\mathcal{R}$ involving e_γ has q-exponent at least $\rho + \gamma > \gamma$. Solving the resulting linear equation gives (Q.24); transfinite induction along the well-ordered support constructs ϵ, and the same computation run on the difference of two solutions shows the first exponent at which they could differ satisfies $(v+1)(e_\gamma - \widetilde{e}_\gamma) = 0$, so there is none. □

S1 states the same theorem under the hypothesis that the support semigroup is *locally finite*, meaning that every bounded interval meets it in a finite set. That is strictly stronger than well-ordering — $\{1 - 1/n\} \cup \{1\}$ is well ordered and not locally finite — and it is more than the proof uses. What the argument needs is precisely the finite-decomposition property, which Neumann's lemma supplies for every well-ordered set. The hypothesis is weakened accordingly here. For the two subjects of this chapter the distinction is invisible, since $S = 2\mathbb{N}$ and $S = \mathbb{N}$; it matters for the fractional and mixed grids of Section Q.12.

Corollary Q.14 (Support semigroup)**.** *If the q-support of $\mathcal{R}(q, v, 0)$ lies in $A \subset \mathbb{R}_{>0}$ and no monomial of $\mathcal{R}$ has a q-exponent outside the monoid generated by A, then* $\operatorname{supp}\epsilon$ *lies in the additive semigroup generated by A.*

Proof. Equation (Q.24) produces e_γ from sums of products of earlier coefficients and of base exponents from A; the set of exponents so reachable is closed under addition and contains A. Induction along the support. □

Example Q.15 (The two supports)**.** For Γ we have $p = a = \beta = 1$, and the perturbation $a_k t^{-(2k-1)}$ normalized by $aX^p = T$ becomes $a_k q^{2k}$ up to a factor $(1+\epsilon)^{-(2k-1)}$; so $A = \{2, 4, 6, \dots\}$, the generated semigroup is $2\mathbb{N}$, and the displacement $t - T = T\epsilon$ contains only the odd powers $T^{-1}, T^{-3}, \dots$. For G we have $p = 2$, $a = \frac{1}{2}$, $\beta = \frac{3}{2}$, and the linear forward term αz normalized by $\frac{1}{2}U^2$ becomes $2\alpha q$; so $1 \in A$, the semigroup is all of $\mathbb{N}$, and every integer power occurs.

Q.4.3 The other engine: formal Newton iteration

The triangular recurrence (Q.24) produces one coefficient per step. When many are wanted, a second engine is better, and it needs no new hypothesis — only that the slope be a unit, which is what Theorem Q.13 already assumes.

Theorem Q.16 (Valuation doubling)**.** *Let $\mathcal{E}(\epsilon) := \mathcal{K}_{p,v}(\epsilon) + \mathcal{R}(q, v, \epsilon)$ be the left-hand side of* (Q.17)*, and assume the hypotheses of Theorem Q.13. Then $\mathcal{E}'$ is a unit of valuation* 0*, the formal Newton step*

$$\epsilon^{[m+1]} = \epsilon^{[m]} - \frac{\mathcal{E}\left(\epsilon^{[m]}\right)}{\mathcal{E}'\left(\epsilon^{[m]}\right)} \tag{Q.25}$$

is defined, and

$$\operatorname{val}\mathcal{E}\left(\epsilon^{[m]}\right) \geq \varrho \quad \Longrightarrow \quad \operatorname{val}\mathcal{E}\left(\epsilon^{[m+1]}\right) \geq 2\varrho. \tag{Q.26}$$

Consequently the number of correct blocks doubles at each step, and N of them cost $\mathcal{O}(\log N)$ steps of truncated-series arithmetic instead of N triangular steps.

Proof. By Theorem Q.12, $\mathcal{K}'_{p,v}(0) = \kappa_1 = v + 1$, and every monomial of $\mathcal{R}$ has positive q-valuation; so $\mathcal{E}'(\epsilon) = (v+1) + $ (positive valuation), which is a unit of valuation 0 because $v + 1$ is invertible in $\mathscr{C}$ and the Hahn ring is complete. Hence (Q.25) is defined and, writing $\Delta = -\mathcal{E}(\epsilon)/\mathcal{E}'(\epsilon)$,

$$\operatorname{val}\Delta = \operatorname{val}\mathcal{E}(\epsilon) \geq \varrho.$$

Taylor expansion of $\mathcal{E}$ about ϵ — legitimate because $\mathcal{E}$ is a formal series in ϵ with coefficients in the complete ring — gives

$$\mathcal{E}(\epsilon + \Delta) = \mathcal{E}(\epsilon) + \mathcal{E}'(\epsilon)\Delta + \sum_{j\geq 2} \frac{\mathcal{E}^{(j)}(\epsilon)}{j!}\Delta^j.$$

The first two terms cancel by the definition of Δ, and every remaining term has valuation at least $2\operatorname{val}\Delta \geq 2\varrho$, since the $\mathcal{E}^{(j)}$ have valuation at least 0. This is (Q.26), and iterating from $\operatorname{val}\mathcal{E}(0) > 0$ gives the count. □

Remark Q.17 (Which engine to use). The triangular recurrence exposes exact coefficient formulas and is what the proofs in this chapter and the next use; Newton is preferable in a computer algebra system at high order, because one inversion of a unit replaces many scalar recurrence steps. They also check each other: run to the same valuation and the truncated residuals must agree, which catches an error in either. The same doubling holds verbatim for the accelerated equation (R.30) and the refined equation (R.42) of Part R, whose linearizations are S and $4L_*$.

Q.4.4 How deep the slope divisor goes

Every coefficient is produced by one division by $v + 1$, but earlier coefficients enter nonlinearly and carry their own divisions. The accumulation is bounded, and — this is new — the bound is attained.

Proposition Q.18 (Slope-denominator bound)**.** *Suppose $S = \mathbb{N}$, that the coefficients $r_{\rho,m}$ of* (Q.23) *lie in $A[v]$ for a commutative ring $A \supseteq \mathbb{Q}$, and that none of them involves $(v+1)^{-1}$. If $\epsilon = \sum_{n\geq 1} e_n q^n$ is the solution of Theorem Q.13, then*

$$(v+1)^{2n-1} e_n \in A[v] \qquad (n \geq 1). \tag{Q.27}$$

Proof. Induction on n. For $n = 1$, (Q.24) gives $e_1 = -r_{1,0}/(v+1)$, and $(v+1)e_1 \in A[v]$.

Let $n \geq 2$ and assume the claim below n. In (Q.24) the bracket is a sum of two kinds of terms. A term of $\kappa_m(p, v)\epsilon^m$ with $m \geq 2$ contributing to q^n is κ_m times a product $e_{j_1} \cdots e_{j_m}$ with $j_1 + \cdots + j_m = n$

and every $j_i \geq 1$; since $\kappa_m \in \mathbb{Q}[p, v]$ carries no divisor, the inductive hypothesis bounds its denominator exponent by

$$\sum_{i=1}^{m}(2j_i - 1) = 2n - m \;\leq\; 2n - 2.$$

A term $r_{\rho,m}q^{\rho}E^{m}$ contributing to q^n has $j_1 + \cdots + j_m = n - \rho$ with $\rho \geq 1$, so its denominator exponent is at most $2(n-\rho) - m \leq 2n - 2$ as well (and at most 0 when $m = 0$). Hence the bracket has denominator exponent at most $2n - 2$, and the final division by $v + 1$ in (Q.24) gives at most $2n - 1$. □

The next two statements are the chapter's first new contribution. They say that the bound of Theorem Q.18 is exactly attained, and identify the coefficient of the deepest pole in closed form, in both subjects at once.

Lemma Q.19 (The quadratic core). *Let $\mathbb{K}$ be a field of characteristic zero, let $A, c, \sigma \in \mathbb{K}$, and assume $A \neq 0$ and $c \neq 0$. Let $\epsilon = \sum_{n\geq 1} \epsilon_n q^n$ be the unique solution of positive valuation of the* quadratic model

$$A\epsilon + c\,\epsilon^2 + \sigma q = 0. \tag{Q.28}$$

Then

$$\epsilon = \frac{A}{2c}\left(\sqrt{1 - \frac{4c\sigma q}{A^2}} - 1\right), \qquad \epsilon_n = \frac{(-4c\sigma)^n}{2c}\binom{1/2}{n}A^{-(2n-1)}. \tag{Q.29}$$

Here the square root denotes the unique formal square root with constant term 1*. In the universal case, with A an indeterminate over a characteristic-zero field containing nonzero c and σ, the A-valuation of ϵ_n is exactly $-(2n-1)$. Under an arbitrary field specialization,* (Q.29) *is the claim; no intrinsic* exact denominator in *A is inferred from the specialization.*

Proof. Because A is invertible, comparison of coefficients determines a positive-valuation solution uniquely: the coefficient of q gives $A\epsilon_1 + \sigma = 0$, and for $n \geq 2$ the coefficient of q^n gives

$$A\epsilon_n + c\sum_{j=1}^{n-1}\epsilon_j\epsilon_{n-j} = 0,$$

whose right-hand side involves only earlier coefficients. In $\mathbb{K}[[q]]$ the formal binomial series with constant term 1 is the square root of $1 - 4c\sigma q/A^2$. Substitution in the quadratic identity, choosing the root with zero constant term, gives

$$\epsilon = \frac{A}{2c}\left(\sqrt{1 - \frac{4c\sigma q}{A^2}} - 1\right),$$

which is the first formula and supplies existence. Expanding the formal identity $\sqrt{1+w} = \sum_{n\geq 0}\binom{1/2}{n}w^n$ at $w = -4c\sigma q/A^2$ and multiplying by $A/(2c)$ gives the second. In the universal indeterminate case the remaining factor is nonzero and independent of A, while $\binom{1/2}{n} \neq 0$ in characteristic zero, so the A-valuation is exactly $-(2n-1)$. □

Proposition Q.20 (The coefficients of the quadratic core are Catalan numbers). *For every $n \geq 1$,*

$$\binom{1/2}{n} = (-1)^{n-1}\,\mathrm{Cat}_{n-1}\,2^{-(2n-1)}, \tag{Q.30}$$

where $\mathrm{Cat}_k = \frac{1}{k+1}\binom{2k}{k}$ is the kth Catalan number, and consequently the convolution that (Q.28) *imposes on its coefficients,*

$$\sum_{j=1}^{n-1}\binom{1/2}{j}\binom{1/2}{n-j} = -2\binom{1/2}{n} \qquad (n \geq 2), \tag{Q.31}$$

is precisely the Catalan recursion $\mathrm{Cat}_{m+1} = \sum_{i+j=m}\mathrm{Cat}_i\mathrm{Cat}_j$.

Proof. Both sides of (Q.30) satisfy the same first-order recursion and agree at $n = 1$: writing g_n for the right-hand side, $\binom{1/2}{n+1} = \binom{1/2}{n}(1/2 - n)/(n+1)$, while $(n+2)\mathrm{Cat}_{n+1} = 2(2n+1)\mathrm{Cat}_n$ — the latter obtained by cancelling $n+1$ from the central binomial recursion $(n+1)\binom{2n+2}{n+1} = 2(2n+1)\binom{2n}{n}$ — and these two ratios agree after the sign and the power of 2 are accounted for. Given (Q.30), substituting into the left side of (Q.31), with $j = i + 1$ and $n = m + 2$, gives $(-1)^m 2^{-(2m+2)} \sum_{i+j=m} \mathrm{Cat}_i \mathrm{Cat}_j$, and the right side is $-2(-1)^{m+1}\mathrm{Cat}_{m+1}2^{-(2m+3)} = (-1)^m \mathrm{Cat}_{m+1}2^{-(2m+2)}$; the two agree exactly when $\mathrm{Cat}_{m+1} = \sum_{i+j=m} \mathrm{Cat}_i \mathrm{Cat}_j$, which is the Catalan recursion. □

Remark Q.21 (What the reduction buys, and the formalization). The proof of Theorem Q.19 above is formal: its square root is a power series, so no convergence claim is involved. Theorem Q.20 replaces even that formal infinite-series packaging by a finite coefficient recursion. The only property of the coefficient family the quadratic model actually uses is the convolution (Q.31), and under (Q.30) that convolution is a finite combinatorial recursion. Thus the Catalan identities and the coefficient recurrence rest on no analysis; assembling them into the unique formal square-root solution and identifying the coefficients as the deepest poles of the two inversions are separate steps.

Lean status. Theorem Q.20 is Exact in the Mathlib-only leaf `QuadraticCoreCatalan.lean`. Its complete three-definition, eight-theorem surface is `Fabius.catalan_two_step`, `Fabius.quadHalf`, `Fabius.quadHalf_zero`, `Fabius.quadHalf_antidiagonal`, `Fabius.halfBinom`, `Fabius.halfBinom_step`, `Fabius.quadHalf_rat`, `Fabius.quadCoef`, `Fabius.quadCoef_rat`, `Fabius.quadCoef_zero`, and `Fabius.quadCoef_rec`. The first seven declarations prove the Catalan ratio, closed falling-product bridge, and antidiagonal convolution; the last four package the displayed shifted coefficient family and verify (Q.28) coefficient by coefficient under $A \neq 0$ and $c \neq 0$. The index is shifted so that the convolution runs over `Finset.antidiagonal`; the empty boundary case at $n = 1$ is carried by `Fabius.quadCoef_zero` rather than folded into the recurrence.

The larger Theorem Q.19 is only Partial. The Lean leaf does not construct the assembled positive-valuation power series, prove its existence or uniqueness, identify it with the formal square root, or state the exact denominator-exponent/nonvanishing conclusion. Nor does it identify this family with the deepest poles in the Gamma or Barnes inversions: Theorem Q.22 remains unformalized. The field hypotheses above are essential for the displayed divisions; a general commutative rational algebra with merely $c \neq 0$ would not suffice in the presence of nonunits or zero divisors.

Theorem Q.22 (The deepest pole is attained, with an explicit coefficient)**.**

1. Gamma. *For* $n \geq 1$ *let* $d_{2n-1}(\Lambda)$ *be the coefficients of Theorem Q.25. Then*

$$[\Lambda^{-(2n-1)}]\ d_{2n-1}(\Lambda) = \binom{1/2}{n} 12^{-n} \neq 0, \tag{Q.32}$$

so the bound of Theorem Q.18 is attained at every n*, and the sum of the deepest-pole terms of the displacement is the convergent series*

$$\sum_{n\geq 1} \binom{1/2}{n} 12^{-n} (\Lambda T)^{1-2n} = \sqrt{\Lambda^2 T^2 + \tfrac{1}{12}} - \Lambda T \qquad \left(\Lambda^2 T^2 > \tfrac{1}{12}\right). \tag{Q.33}$$

2. Barnes. *For* $n \geq 1$ *let* $e_n(\lambda)$ *be the coefficients of Theorem Q.33. Then*

$$[\lambda^{-(2n-1)}]\ e_n(\lambda) = \binom{1/2}{n} (-2\alpha)^n \neq 0, \tag{Q.34}$$

and the deepest-pole part of the displacement $z - U$ *sums to*

$$\sum_{n\geq 1} \binom{1/2}{n} (-2\alpha)^n \lambda^{1-2n} U^{1-n} = \sqrt{\lambda^2 U^2 - 2\alpha U} - \lambda U \qquad (\lambda^2 U > 2\alpha). \tag{Q.35}$$

In both cases the deepest-pole family is the coefficient family of the quadratic model (Q.28)*, with*

$$(A, c, \sigma) = (\Lambda,\ \tfrac{1}{2},\ a_1)\ \ \textit{for}\ \Gamma, \qquad (A, c, \sigma) = (2\lambda,\ 1,\ 2\alpha)\ \ \textit{for}\ G. \tag{Q.36}$$

Proof. Both parts have the same shape. Write π_n for the coefficient of the deepest possible pole, $\pi_n := [A^{-(2n-1)}]\, e_n$, which is well defined by Theorem Q.18; the claim is that (π_n) obeys the recursion of the quadratic model. Everything turns on which terms of the normalized equation can reach pole order $2n-2$ after the coefficient e_n has been isolated — for the recurrence reads $Ae_n = -(\text{bracket})$, and $A \cdot \pi_n A^{-(2n-1)} = \pi_n A^{-(2n-2)}$.

Gamma. Here $A = \Lambda = v + 1$ and, by (Q.21), the normalized equation on the grid q^2 is

$$\Lambda\epsilon + \tfrac{1}{2}\epsilon^2 - \tfrac{1}{6}\epsilon^3 + \cdots + \sum_{k\ge 1} a_k q^{2k}(1+\epsilon)^{-(2k-1)} = 0. \tag{Q.37}$$

Index by n, so that $\epsilon = \sum_n e_n q^{2n}$ and e_n has pole order at most $2n-1$. The term $\frac{1}{2}\epsilon^2$ contributes at q^{2n} the sum $\frac{1}{2}\sum_{j=1}^{n-1} e_j e_{n-j}$, of pole order at most $(2j-1) + (2(n-j)-1) = 2n-2$ with top coefficient $\frac{1}{2}\sum_{j=1}^{n-1}\pi_j\pi_{n-j}$. A term ϵ^m with $m \ge 3$ has pole order at most $2n - m \le 2n-3$. In the Bernoulli sum, the summand $k = n$ contributes the constant a_n, of pole order 0, which is $2n-2$ only when $n = 1$; and a summand with $1 \le k \le n-1$ contributes a_k times a product of $m \ge 1$ coefficients with indices summing to $n - k$, of pole order at most $2(n-k) - m \le 2n-3$. Hence for $n \ge 2$

$$\pi_n = -\tfrac{1}{2}\sum_{j=1}^{n-1}\pi_j\pi_{n-j}, \qquad \text{while} \qquad \pi_1 = -a_1 = \tfrac{1}{24}.$$

Let $g(w) = \sum_{n\ge 1}\pi_n w^n$. The recursion says $g = -a_1 w - \frac{1}{2}g^2$, i.e. $g^2 + 2g + 2a_1 w = 0$, whence $g = -1 + \sqrt{1 - 2a_1 w} = -1 + \sqrt{1 + w/12}$ and $\pi_n = \binom{1/2}{n}12^{-n}$, which is (Q.32); it is nonzero because $\binom{1/2}{n} \ne 0$ for all n. This is exactly Theorem Q.19 with $(A, c, \sigma) = (\Lambda, \frac{1}{2}, a_1)$: indeed $(-4 \cdot \frac{1}{2} \cdot a_1)^n / (2 \cdot \frac{1}{2}) = (1/12)^n$.

Since $t - T = T\epsilon$, the deepest-pole terms of the displacement are

$$\pi_n \Lambda^{-(2n-1)} T^{1-2n} = \binom{1/2}{n} 12^{-n} (\Lambda T)^{1-2n},$$

and summing the binomial series at $w = 1/(12\Lambda^2 T^2)$ gives

$$\Lambda T\left(\sqrt{1 + \frac{1}{12\Lambda^2 T^2}} - 1\right) = \sqrt{\Lambda^2 T^2 + \tfrac{1}{12}} - \Lambda T,$$

convergent for $12\Lambda^2 T^2 > 1$. This is (Q.33).

Barnes. Here $A = 2\lambda$ and, by (Q.22) with $v = 2\lambda - 1$ so that $(v+3)/2 = \lambda + 1$, the normalized equation on the grid q is

$$2\lambda\epsilon + (\lambda+1)\epsilon^2 + \tfrac{1}{3}\epsilon^3 - \cdots + 2\alpha q(1+\epsilon) + 2b\, q^2\big(\lambda + 1 + \log(1+\epsilon)\big) + 2\sum_{k\ge 1} c_k q^{2k+2}(1+\epsilon)^{-2k} = 0, \tag{Q.38}$$

with $b = -\frac{1}{12}$. Pole orders are now taken in λ, and e_n has order at most $2n-1$; we must collect everything of order $2n-2$.

The quadratic term $(\lambda+1)\epsilon^2$ splits. Its part $1 \cdot \epsilon^2$ contributes $\sum_{j=1}^{n-1}\pi_j\pi_{n-j}$ at order $2n-2$. Its part $\lambda\epsilon^2$ multiplies a quantity of order at most $2n-2$ by λ, giving order at most $2n-3$; it therefore does *not* reach $2n-2$. Terms ϵ^m with $m \ge 3$ reach at most $2n-3$.

In the perturbation, $2\alpha q \cdot 1$ contributes only at $n = 1$, where its order 0 equals $2n - 2 = 0$; $2\alpha q\epsilon$ contributes $2\alpha e_{n-1}$, of order at most $2n-3$. The term $2b\lambda q^2$ has no pole at all, and $2bq^2$ and

$2bq^2 \log(1+\epsilon)$ have order at most $2(n-2)-1$, while multiplying the logarithm by λ lowers the order further; none reaches $2n-2$. The same holds for every c_k term. Hence for $n \geq 2$

$$2\pi_n = -\sum_{j=1}^{n-1} \pi_j \pi_{n-j}, \qquad \text{while} \qquad 2\pi_1 = -2\alpha, \;\; \pi_1 = -\alpha.$$

With $g(w) = \sum \pi_n w^n$ this reads $2g = -2\alpha w - g^2$, so $g = -1 + \sqrt{1-2\alpha w}$ and $\pi_n = \binom{1/2}{n}(-2\alpha)^n$, which is (Q.34) and agrees with Theorem Q.19 at $(A, c, \sigma) = (2\lambda, 1, 2\alpha)$. Since $z - U = U\epsilon$, the deepest-pole terms are $\pi_n \lambda^{-(2n-1)} U^{1-n}$, and

$$\sum_{n\geq 1} \binom{1/2}{n} (-2\alpha)^n \lambda^{1-2n} U^{1-n} = \lambda U \sum_{n \geq 1} \binom{1/2}{n} \left(\frac{-2\alpha}{\lambda^2 U}\right)^n = \lambda U \left(\sqrt{1 - \frac{2\alpha}{\lambda^2 U}} - 1\right),$$

which is (Q.35). □

Remark Q.23 (Two square roots, one mechanism). S1 and S3 both isolate a square root in the Barnes problem, namely the coefficientwise limit $\sqrt{U^2 + 1/6}$ of Theorem Q.35 below, and both present it as a curiosity of Barnes G caused by the resonant $-\frac{1}{12}\log z$ term. Theorem Q.22 shows it is not special to Barnes and not special to resonance. Every problem in this class has a distinguished exactly solvable truncation — keep the slope, the quadratic kernel coefficient, and the lowest forcing monomial, discard the rest — and that truncation is a quadratic, so it always resums to a square root. Which *extraction* of the coefficient family it computes depends on the subject: for Barnes, discarding by λ-degree gives the resonant limit (Q.72); discarding by pole depth gives (Q.35); for Gamma there is no resonance and only the second extraction exists. The three square roots

$$\sqrt{U^2 + \tfrac{1}{6}}, \qquad \sqrt{\lambda^2 U^2 - 2\alpha U}, \qquad \sqrt{\Lambda^2 T^2 + \tfrac{1}{12}}$$

are the same lemma applied three times.

Remark Q.24 (What the deepest pole is good for). Two things. First, it is a sharp regression test: a symbolic implementation that produces d_{2n-1} or e_n can check the single rational number $[A^{-(2n-1)}]$ against (Q.32) or (Q.34) at negligible cost, and that number is exactly the one most sensitive to an error committed early in the recursion, since it is the accumulation of all previous divisions. Second, it locates the breakdown: the expansions of this chapter are asymptotic in q but their coefficients degrade like $(v+1)^{-(2n-1)}$, so the useful range in n at a given distance from the branch point $v = -1$ is governed by $\Lambda T \gg 1$ resp. $\lambda\sqrt{U} \gg 1$ — and by Theorem Q.22 those are precisely the conditions under which the displayed square roots converge.

Q.5 The increasing inverse of Gamma

Q.5.1 Core variables

Let $y \to +\infty$ and put

$$R = \log y - \tfrac{1}{2}\log(2\pi), \qquad w = W_0(R/\mathrm{e}), \qquad T = \frac{R}{w} = \mathrm{e}^{w+1}, \qquad \Lambda = \log T = w + 1. \quad \text{(Q.39)}$$

These are Theorem Q.8 at $p = a = \beta = r = 1$: the core equation is

$$T(\log T - 1) = R, \quad \text{(Q.40)}$$

its slope is Λ, and the three identities in (Q.39) are equivalent to $w\mathrm{e}^w = R/\mathrm{e}$. Write $q = T^{-1}$ and $t = T(1+\epsilon)$. Subtracting (Q.40) from Theorem Q.1 and dividing by T gives the exact normalized equation

$$\Lambda\epsilon + \big[(1+\epsilon)\log(1+\epsilon) - \epsilon\big] + \sum_{k\geq 1} a_k q^{2k}(1+\epsilon)^{-(2k-1)} = 0. \quad \text{(Q.41)}$$

Q.5.2 The all-orders theorem

Theorem Q.25 (All-orders expansion of Γ_+^{-1}). *Let Γ_+^{-1} be the branch of Theorem Q.5 tending to $+\infty$, and let T, Λ be as in* (Q.39). *There are uniquely determined $d_{2n-1} \in \Lambda^{-1}\mathbb{Q}[\Lambda^{-1}]$ such that, for every fixed $N \geq 0$,*

$$\Gamma_+^{-1}(y) = \frac{1}{2} + T + \sum_{n=1}^{N} \frac{d_{2n-1}(\Lambda)}{T^{2n-1}} + \mathcal{O}_N\left(\frac{1}{\Lambda\, T^{2N+1}}\right), \qquad y \to +\infty. \tag{Q.42}$$

Writing $\epsilon = \sum_{n\geq 1} d_{2n-1}(\Lambda) q^{2n}$ and $E_{n-1} = \sum_{j<n} d_{2j-1} q^{2j}$, they are computed by

$$d_{2n-1}(\Lambda) = -\frac{1}{\Lambda}[q^{2n}] \left[(1+E_{n-1})\log(1+E_{n-1}) - E_{n-1} + \sum_{k=1}^{n} a_k q^{2k} (1+E_{n-1})^{-(2k-1)} \right], \tag{Q.43}$$

and they satisfy

$$\deg_{\Lambda^{-1}} d_{2n-1} \leq 2n-1, \qquad d_{2n-1}(\Lambda) = -\frac{a_n}{\Lambda} + \mathcal{O}(\Lambda^{-2}). \tag{Q.44}$$

Only the odd powers $T^{-1}, T^{-3}, T^{-5}, \ldots$ occur.

Proof. Formal part. Equation (Q.41) is (Q.17) with $p = a = \beta = 1$, support $S = 2\mathbb{N}$ by Theorem Q.15, and linear coefficient $\kappa_1 = v + 1 = \Lambda$; so Theorem Q.13 gives a unique formal solution and (Q.43). The parity claim is Theorem Q.14 together with $t - T = T\epsilon$: an exponent q^{2n} in ϵ is an absolute power T^{1-2n}.

For (Q.44), induct. The bracket in (Q.43) has rational Taylor coefficients, so if every d_{2j-1} with $j < n$ lies in $\Lambda^{-1}\mathbb{Q}[\Lambda^{-1}]$ with degree at most $2j-1$, the bound of Theorem Q.18 applies verbatim and the bracket has degree at most $2n-2$; dividing by Λ gives degree at most $2n-1$ and no constant term. At order q^{2n} the only contribution not involving an earlier coefficient is a_n, whence $d_{2n-1} = -a_n/\Lambda + \mathcal{O}(\Lambda^{-2})$.

Analytic part. Fix N and set $M = N+1$. We must know that (Q.2) truncated after $a_M t^{-(2M-1)}$ has remainder $\mathcal{O}(t^{-(2M+1)})$ on the positive real axis, with a differentiable remainder. This follows from the Legendre duplication formula

$$\Gamma(t)\Gamma\big(t+\tfrac{1}{2}\big) = 2^{1-2t}\sqrt{\pi}\,\Gamma(2t), \quad \text{i.e.} \quad \log\Gamma\big(t+\tfrac{1}{2}\big) = \log\Gamma(2t) - \log\Gamma(t) + (1-2t)\log 2 + \tfrac{1}{2}\log\pi,$$

applied to two ordinary Stirling expansions on the positive real axis, each of which has the classical remainder bound with a remainder that may be differentiated once (Olver). Subtracting term by term reproduces (Q.2) with a remainder of the stated order.

Now let $x_N = \frac{1}{2} + T + \sum_{n\leq N} d_{2n-1} T^{1-2n}$ and $f(x) = \log\Gamma(x)$, and let $\lambda = \log y$. By construction the formal substitution cancels the normalized equation through q^{2N}, so the residual satisfies $f(x_N) - \lambda = \mathcal{O}(T^{-(2N+1)})$; the forward truncation error just bounded is of the same order or smaller. On an interval around x_N of length $\mathcal{O}(T^{-2N})$ we have $f'(x) = \psi(x) \geq \frac{1}{2}\Lambda > 0$ for all large y, since $\psi(x) = \log x + \mathcal{O}(1/x)$ and $\log x_N = \Lambda + o(1)$. Theorem J.47 applied with $m = \frac{1}{2}\Lambda$ converts the residual into $|x_N - \Gamma_+^{-1}(y)| \leq 2|f(x_N) - \lambda|/\Lambda = \mathcal{O}(\Lambda^{-1}T^{-(2N+1)})$, which is (Q.42). The second half of Theorem J.47 also guarantees that a root exists within that distance, so no separate existence argument is needed. ∎

Q.5.3 Explicit blocks

Proposition Q.26 (The first five blocks). *With $\Lambda = w + 1$,*

$$d_1 = \frac{1}{24\Lambda}, \tag{Q.45}$$

$$d_3 = -\frac{14\Lambda^2 + 10\Lambda + 5}{5760\,\Lambda^3}, \tag{Q.46}$$

$$d_5 = \frac{2232\Lambda^4 + 1176\Lambda^3 + 714\Lambda^2 + 350\Lambda + 105}{2903040\,\Lambda^5}, \tag{Q.47}$$

$$d_7 = -\frac{822960\Lambda^6 + 292536\Lambda^5 + 136956\Lambda^4 + 68040\Lambda^3 + 32970\Lambda^2 + 12250\Lambda + 2625}{1393459200\,\Lambda^7}, \tag{Q.48}$$

$$d_9 = \frac{\begin{array}{l}309052800\Lambda^8 + 77920128\Lambda^7 + 26024592\Lambda^6 + 10019592\Lambda^5\\ \quad + 4516028\Lambda^4 + 2023560\Lambda^3 + 812350\Lambda^2 + 242550\Lambda + 40425\end{array}}{367873228800\,\Lambda^9}. \tag{Q.49}$$

Equivalently, separated into powers of Λ^{-1},

$$d_3 = -\frac{7}{2880\Lambda} - \frac{1}{576\Lambda^2} - \frac{1}{1152\Lambda^3}, \tag{Q.50}$$

$$d_5 = \frac{31}{40320\Lambda} + \frac{7}{17280\Lambda^2} + \frac{17}{69120\Lambda^3} + \frac{5}{41472\Lambda^4} + \frac{1}{27648\Lambda^5}. \tag{Q.51}$$

Proof. Run (Q.43). The first two steps are short enough to display. Since $(1+\epsilon)\log(1+\epsilon) - \epsilon = \frac{1}{2}\epsilon^2 - \frac{1}{6}\epsilon^3 + \cdots$, the coefficient of q^2 is $\Lambda d_1 + a_1$, so $d_1 = -a_1/\Lambda = 1/(24\Lambda)$. At q^4,

$$\Lambda d_3 + \tfrac{1}{2}d_1^2 - a_1 d_1 + a_2 = 0,$$

and substituting $a_1 = -\frac{1}{24}$, $a_2 = \frac{7}{2880}$, $d_1 = \frac{1}{24\Lambda}$ gives (Q.46). The remaining three were computed by truncated-series arithmetic over $\mathbb{Q}(\Lambda)$ and are recorded in Theorem Q.27. □

Remark Q.27 (Independent verification). Every rational number displayed in this chapter was recomputed for this volume, in exact arithmetic, by an implementation written from the normalized equations (Q.41) and (Q.59) and not from any source's coefficient list: the forward data a_k and c_k from Bernoulli numbers, then $d_1, \ldots, d_9$ over $\mathbb{Q}(\Lambda)$ and $b_0, \ldots, b_5$ over $\mathbb{Q}(\alpha, \lambda)$ by (Q.43) and (Q.62). All agreed with the three sources, in every digit of every numerator, including the 24-term numerator of b_5. The structural predictions were checked as well: the parity (Q.63), the resonant limits (Q.71) for $m \le 3$, the deepest-pole formulas (Q.32) and (Q.34) for $n \le 6$, and the two outer families of Theorem Q.39. Separately, the nine high-precision roots that S2 prints were recomputed to 60 digits and agree in all 25 digits displayed.

Remark Q.28 (Prior art). The block d_1 is classical, and the expansion through d_3 — in the separated form (Q.50) — was proved by Gerhold, Hubalek and Tomovski. What the recurrence adds is that the expansion continues to all orders with an explicit triangular rule, a proof of the parity, and the coefficient ring. The leading Lambert approximation $\Gamma_+^{-1}(y) \approx \frac{1}{2} + T$ itself goes back further; see Borwein–Corless and Jeffrey–Watt for its history and for the branch geometry of the complex inverse.

Q.6 The branches of Γ^{-1} at the poles

The Lambert transseries describes one branch. Every pole of Γ carries another, and those are convergent — a useful contrast, since it isolates exactly what makes the large branch divergent.

Theorem Q.29 (The pole branches are convergent). *Fix* $n \in \{0, 1, 2, \ldots\}$ *and put*

$$\Phi_n(\varepsilon) = \frac{1}{\Gamma(1+\varepsilon)} \prod_{k=1}^{n} \left(1 - \frac{\varepsilon}{k}\right), \qquad \Phi_n(0) = 1. \tag{Q.52}$$

Then $1/\Gamma(-n+\varepsilon) = (-1)^n n!\,\varepsilon\,\Phi_n(\varepsilon)$, *and for all sufficiently large* $|y|$ *the branch of* Γ^{-1} *tending to* $-n$ *is given by the* convergent *series*

$$\Gamma^{-1}_{[-n]}(y) = -n + \sum_{m\ge 1} \frac{\tau^m}{m} [\varepsilon^{m-1}]\, \Phi_n(\varepsilon)^{-m}, \qquad \tau = \frac{(-1)^n}{n!\,y}. \tag{Q.53}$$

With $\psi_n = \psi(n+1) = H_n - \gamma$ and $H_n^{(r)} = \sum_{k\le n} k^{-r}$, the first four terms are

$$\Gamma^{-1}_{[-n]}(y) = -n+\tau+\psi_n\tau^2+\frac{3\psi_n^2 + H_n^{(2)} + \zeta(2)}{2}\,\tau^3+\frac{8\psi_n^3 + 6\psi_n\big(H_n^{(2)} + \zeta(2)\big) + H_n^{(3)} - \zeta(3)}{3}\,\tau^4+\mathcal{O}(\tau^5). \tag{Q.54}$$

Proof. From $\Gamma(-n+\varepsilon) = \Gamma(1+\varepsilon)/\big(\varepsilon(\varepsilon-1)\cdots(\varepsilon-n)\big)$ we get $1/\Gamma(-n+\varepsilon) = \varepsilon\prod_{k=1}^n(\varepsilon-k)/\Gamma(1+\varepsilon) = (-1)^n n!\,\varepsilon\Phi_n(\varepsilon)$. Thus $\tau = \varepsilon\Phi_n(\varepsilon)$, an analytic function of ε near 0 with derivative 1 there; ordinary Lagrange inversion (Theorem J.17, in its convergent analytic form) gives (Q.53), convergent on a disc because Φ_n is analytic and nonvanishing near 0.

For the coefficients, expand

$$\log\Phi_n(\varepsilon) = \sum_{k=1}^n \log\Big(1-\frac{\varepsilon}{k}\Big) - \log\Gamma(1+\varepsilon) = -\psi_n\varepsilon - \frac{H_n^{(2)} + \zeta(2)}{2}\varepsilon^2 + \frac{\zeta(3) - H_n^{(3)}}{3}\varepsilon^3 - \cdots,$$

using $\log\Gamma(1+\varepsilon) = -\gamma\varepsilon + \sum_{j\ge2}(-1)^j\zeta(j)\varepsilon^j/j$ and $H_n - \gamma = \psi_n$. Then $\Phi_n^{-m} = \exp(-m\log\Phi_n)$ and $[\varepsilon^{m-1}]$ of it gives, for $m = 1,2,3,4$, the values 1, $2\psi_n$, $\frac{9}{2}\psi_n^2 + \frac{3}{2}(H_n^{(2)} + \zeta(2))$ and $\frac{32}{3}\psi_n^3 + 8\psi_n(H_n^{(2)} + \zeta(2)) - \frac{4}{3}(\zeta(3) - H_n^{(3)})$; dividing by m gives (Q.54). □

Corollary Q.30 (The branch tending to zero)**.** *For $n = 0$, $\tau = r := 1/y$ and, with γ Euler's constant,*

$$\begin{aligned}\Gamma_0^{-1}(y) &= r - \gamma r^2 + \frac{18\gamma^2 + \pi^2}{12}r^3 - \frac{8\gamma^3 + \pi^2\gamma + \zeta(3)}{3}r^4 \\ &\quad + \frac{7500\gamma^4 + 1500\pi^2\gamma^2 + 2400\gamma\zeta(3) + 29\pi^4}{1440}r^5 \\ &\quad - \left(\frac{54\gamma^5}{5} + 3\gamma^3\pi^2 + 6\gamma^2\zeta(3) + \frac{17\gamma\pi^4}{120} + \frac{\pi^2\zeta(3)}{6} + \frac{\zeta(5)}{5}\right)r^6 + \mathcal{O}(r^7),\end{aligned} \tag{Q.55}$$

a convergent expansion. Equivalently, in closed form,

$$\Gamma_0^{-1}(y) = \sum_{m\ge1}\frac{B_{m-1}\big(m\ell_1,\dots,m\ell_{m-1}\big)}{m!\,y^m}, \qquad \ell_1 = -\gamma, \quad \ell_j = (-1)^j(j-1)!\,\zeta(j)\ (j\ge2), \tag{Q.56}$$

where B_{m-1} is the complete exponential Bell polynomial of Theorem J.10.

Proof. For $n = 0$ the product in (Q.52) is empty, so $\tau = r = \varepsilon/\Gamma(1+\varepsilon)$ and $\psi_0 = -\gamma$, $H_0^{(r)} = 0$. Substituting into (Q.54) and using $\zeta(2) = \pi^2/6$ gives the terms through r^4; the r^5 and r^6 terms were computed the same way and use in addition $\zeta(4) = \pi^4/90$. For (Q.56), Lagrange inversion of $r = \varepsilon/\Gamma(1+\varepsilon)$ gives $[r^m] = \frac{1}{m!}\big(\mathrm{d}/\mathrm{d}\varepsilon\big)^{m-1}\Gamma(1+\varepsilon)^m\big|_0$, and $\Gamma(1+\varepsilon)^m = \exp\big(m\sum_j \ell_j\varepsilon^j/j!\big)$, whose $(m-1)$st derivative at 0 is the complete exponential Bell polynomial $B_{m-1}(m\ell_1,\dots,m\ell_{m-1})$. □

Remark Q.31 (Why one branch converges and the other does not). The contrast is instructive and is the cleanest illustration in the volume of the distinction Chapter 0 insists on. Theorem Q.29 inverts an *analytic* map with nonvanishing derivative at a point, so its Lagrange series converges. Theorem Q.25 inverts an *asymptotic* series at infinity: the identity it establishes is an equality of formal objects plus a transported remainder bound at each fixed order, and the full series $\sum_n d_{2n-1}T^{1-2n}$ diverges for every y, because the a_n grow factorially. No amount of care with the algebra can change that; what can be done is to truncate optimally, which is Theorem J.54 and Section Q.11.

Q.7 The eventual positive inverse of Barnes G

Q.7.1 Core variables and the normalized equation

Let $y \to +\infty$ and put

$$R = \log y - \zeta'(-1), \qquad v = W_0\left(\frac{4R}{\mathrm{e}^3}\right), \qquad U = 2\sqrt{\frac{R}{v}} = \mathrm{e}^{(v+3)/2}, \qquad \lambda = \log U - 1 = \frac{v+1}{2}. \tag{Q.57}$$

This is Theorem Q.8 at $p = 2$, $a = \frac{1}{2}$, $\beta = \frac{3}{2}$, $r = 1$; the core equation is

$$\frac{1}{2}U^2\left(\log U - \frac{3}{2}\right) = R, \tag{Q.58}$$

and by (Q.14) its slope is $\frac{1}{2}U(v+1) = U\lambda$. Write $q = U^{-1}$ and $z = U(1+\epsilon)$. Subtracting (Q.58) from Theorem Q.3 and dividing by $\frac{1}{2}U^2$ gives

$$\mathcal{K}_\lambda(\epsilon) + 2\alpha q(1+\epsilon) + 2b\,q^2\big(\lambda+1+\log(1+\epsilon)\big) + 2\sum_{k\geq 1} c_k q^{2k+2}(1+\epsilon)^{-2k} = 0, \qquad b = -\tfrac{1}{12}, \tag{Q.59}$$

where, by (Q.22) with $v = 2\lambda - 1$,

$$\mathcal{K}_\lambda(\epsilon) = (1+\epsilon)^2\left(\lambda - \tfrac{1}{2} + \log(1+\epsilon)\right) - \left(\lambda - \tfrac{1}{2}\right) = 2\lambda\epsilon + (\lambda+1)\epsilon^2 + \tfrac{1}{3}\epsilon^3 - \tfrac{1}{12}\epsilon^4 + \tfrac{1}{30}\epsilon^5 - \cdots. \tag{Q.60}$$

Remark Q.32 (Absorbing $\zeta'(-1)$). Taking $R = \log y - \zeta'(-1)$ rather than $R = \log y$ removes the constant from the perturbation, in accordance with Theorem Q.11. It is not mandatory — S2 does not do it — but it makes the coefficients shorter, and it is what makes the Glaisher constant disappear from them. Section Q.8 records the exact price of the other choice.

Q.7.2 The all-orders theorem

Theorem Q.33 (All-orders expansion of the Barnes inverse). *Let $\mathcal{B}$ be the inverse of Theorem Q.6, and let U, λ be as in* (Q.57). *There are uniquely determined $e_n \in \mathbb{Q}[\alpha, \lambda, \lambda^{-1}]$, $n \geq 1$, such that with $b_n := e_{n+1}$ one has, for every fixed $N \geq 0$,*

$$\mathcal{B}(y) = U + \sum_{n=0}^{N} \frac{b_n(\lambda)}{U^n} + \mathcal{O}_N\left(U^{-N-1}\right), \qquad y \to +\infty, \tag{Q.61}$$

and $G_+^{-1}(y) = 1 + \mathcal{B}(y)$. With $E_{n-1} = \sum_{j<n} e_j q^j$ they are computed by

$$\begin{aligned} e_n(\lambda) = \; -\frac{1}{2\lambda}[q^n]\Big[&\mathcal{K}_\lambda(E_{n-1}) - 2\lambda E_{n-1} \\ &+ 2\alpha q(1+E_{n-1}) + 2bq^2\big(\lambda+1+\log(1+E_{n-1})\big) \\ &+ 2\sum_{2k+2\leq n} c_k q^{2k+2}(1+E_{n-1})^{-2k}\Big], \end{aligned} \tag{Q.62}$$

they obey the parity rule

$$e_n(-\alpha) = (-1)^n e_n(\alpha), \tag{Q.63}$$

and each is a bounded function of λ as $\lambda \to +\infty$.

Proof. Formal part. Equation (Q.59) has support $S = \mathbb{N}$ by Theorem Q.15 and linear coefficient $\kappa_1 = v + 1 = 2\lambda$; Theorem Q.13 applies and yields (Q.62) and uniqueness. The coefficient ring follows by induction, since the Taylor coefficients of $\log(1+\epsilon)$ and of the binomial powers are rational, α enters only through $2\alpha q$, and the only division is by 2λ.

For (Q.63), observe that the substitution $(q, \alpha, \epsilon) \mapsto (-q, -\alpha, \epsilon)$ leaves (Q.59) invariant: $\mathcal{K}_\lambda(\epsilon)$ has no q; $2\alpha q(1+\epsilon)$ is invariant because both signs flip; and q^2, q^{2k+2} are even. Uniqueness therefore forces $\epsilon(-q, -\alpha) = \epsilon(q, \alpha)$, i.e. $(-1)^n e_n(-\alpha) = e_n(\alpha)$, which is (Q.63).

Boundedness in λ is Theorem Q.35 below, which exhibits the coefficientwise limit.

Analytic part. Let $x_N = U + \sum_{n \le N} b_n U^{-n}$ and $f(z) = \log G(z+1)$, $\lambda_* = \log y$. Retain the forward expansion (Q.6) far enough that its remainder is $\mathcal{O}(U^{-N-1})$ after division by the slope; this is legitimate because (Q.6) is a Poincaré expansion with differentiable remainder on the positive axis, with explicit bounds available from Nemes. By construction the formal substitution cancels (Q.59) through q^{N+1}, so the forward residual $f(x_N) - \lambda_*$ is $\mathcal{O}(\lambda U^{-N})$ — one factor $\frac{1}{2}U^2$ is restored on undoing the normalization, and the first omitted normalized term is $2\lambda e_{N+2} q^{N+2}$. By Theorem Q.6, $f' \ge \frac{1}{2} z \log z$ on $[7, \infty)$, and along the relevant interval $f' = U\lambda(1 + o(1))$. Theorem J.47 with $m = \frac{1}{2}U\lambda$ turns the residual into $|x_N - \mathcal{B}(y)| = \mathcal{O}(U^{-N-1})$, which is (Q.61), and again certifies that a root lies that close. □

Q.7.3 Explicit blocks

Proposition Q.34 (The first six blocks)**.** *With* $\alpha = \frac{1}{2}\log(2\pi)$ *and* $\lambda = (v+1)/2$*,*

$$b_0 = -\frac{\alpha}{\lambda}, \tag{Q.64}$$

$$b_1 = \frac{6\alpha^2\lambda - 6\alpha^2 + \lambda^3 + \lambda^2}{12\lambda^3}, \tag{Q.65}$$

$$b_2 = \frac{\alpha\big(8\alpha^2\lambda - 6\alpha^2 + \lambda^2\big)}{12\lambda^5}, \tag{Q.66}$$

$$b_3 = -\frac{P_3(\lambda, \alpha)}{1440\lambda^7}, \qquad b_4 = -\frac{\alpha P_4(\lambda, \alpha)}{1440\lambda^9}, \qquad b_5 = \frac{P_5(\lambda, \alpha)}{362880\lambda^{11}}, \tag{Q.67}$$

where

$$\begin{aligned} P_3 &= 180\alpha^4\lambda^3 + 120\alpha^4\lambda^2 - 1500\alpha^4\lambda + 900\alpha^4 + 60\alpha^2\lambda^5 + 120\alpha^2\lambda^4 + 60\alpha^2\lambda^3 - 180\alpha^2\lambda^2 \\ &\quad + 5\lambda^7 - \lambda^6 + 5\lambda^5 + 5\lambda^4, \end{aligned} \tag{Q.68}$$

$$\begin{aligned} P_4 &= 576\alpha^4\lambda^3 + 480\alpha^4\lambda^2 - 2520\alpha^4\lambda + 1260\alpha^4 + 80\alpha^2\lambda^5 + 280\alpha^2\lambda^4 + 200\alpha^2\lambda^3 - 300\alpha^2\lambda^2 \\ &\quad - 22\lambda^7 - 11\lambda^6 + 10\lambda^5 + 15\lambda^4, \end{aligned} \tag{Q.69}$$

$$\begin{aligned} P_5 &= 22680\alpha^6\lambda^5 + 40824\alpha^6\lambda^4 - 349272\alpha^6\lambda^3 - 352800\alpha^6\lambda^2 + 1111320\alpha^6\lambda - 476280\alpha^6 \\ &\quad + 11340\alpha^4\lambda^7 + 34020\alpha^4\lambda^6 + 3780\alpha^4\lambda^5 - 138600\alpha^4\lambda^4 - 132300\alpha^4\lambda^3 + 132300\alpha^4\lambda^2 \\ &\quad + 1890\alpha^2\lambda^9 + 8568\alpha^2\lambda^8 + 15120\alpha^2\lambda^7 + 11718\alpha^2\lambda^6 - 3150\alpha^2\lambda^5 - 9450\alpha^2\lambda^4 \\ &\quad + 105\lambda^{11} - 668\lambda^{10} - 378\lambda^9 - 21\lambda^8 + 175\lambda^7 + 105\lambda^6. \end{aligned} \tag{Q.70}$$

Proof. Run (Q.62); see Theorem Q.27. The first step is $2\lambda e_1 + 2\alpha = 0$, giving $b_0 = e_1 = -\alpha/\lambda$. The parity (Q.63) is visible: b_0, b_2, b_4 are odd in α and b_1, b_3, b_5 even, since $b_n = e_{n+1}$. The denominators are λ^{2n-1} for e_n, in agreement with Theorem Q.18, and the numerators' extreme coefficients $-\alpha$, $-6\alpha^2/12$, $-6\alpha^3/12$, $-900\alpha^4/1440$, $-1260\alpha^5/1440$, $-476280\alpha^6/362880$ are $-\alpha, -\frac{\alpha^2}{2}, -\frac{\alpha^3}{2}, -\frac{5}{8}\alpha^4, -\frac{7}{8}\alpha^5, -\frac{21}{16}\alpha^6$, which is (Q.34). □

Q.7.4 Resonance: the surviving algebraic subseries

The forward term $-\frac{1}{12}\log z$ is small relative to the core, but its *logarithm* is of the same size as the slope. It therefore survives the large-slope limit, and it is the only perturbation that does.

Proposition Q.35 (Coefficientwise large-slope limit). *For each fixed n, the limit of $e_n(\lambda)$ as $\lambda \to +\infty$ exists and*

$$\lim_{\lambda\to\infty} e_{2m}(\lambda) = \binom{1/2}{m} 6^{-m}, \qquad \lim_{\lambda\to\infty} e_{2m+1}(\lambda) = 0. \tag{Q.71}$$

Equivalently $\lim b_{2m-1} = \binom{1/2}{m}6^{-m}$ *and* $\lim b_{2m} = 0$*; the first values are* $b_1 \to \frac{1}{12}$, $b_3 \to -\frac{1}{288}$, $b_5 \to \frac{1}{3456}$. *The limiting coefficients are those of*

$$E_0(q) = \sqrt{1 + \tfrac{q^2}{6}} - 1, \qquad \text{i.e.} \qquad U\big(1 + E_0(U^{-1})\big) = \sqrt{U^2 + \tfrac{1}{6}}. \tag{Q.72}$$

Proof. Divide (Q.59) by 2λ and let $\lambda \to \infty$ with q formal. By (Q.60), $\mathcal{K}_\lambda(E)/(2\lambda) \to \big((1+E)^2 - 1\big)/2$, because the λ-free part of $\mathcal{K}_\lambda$ is bounded coefficientwise. The term $2\alpha q(1+E)/(2\lambda) \to 0$; each c_k term $\to 0$; and

$$\frac{2bq^2\big(\lambda + 1 + \log(1+E)\big)}{2\lambda} = bq^2\left(1 + \frac{1 + \log(1+E)}{\lambda}\right) \longrightarrow bq^2.$$

So the limiting equation is $\big((1+E_0)^2 - 1\big)/2 + bq^2 = 0$, i.e. $(1+E_0)^2 = 1 - 2bq^2 = 1 + q^2/6$, whose branch with $E_0(0) = 0$ is (Q.72). That the coefficientwise limit may be taken inside the recursion is the triangular structure: at each n the coefficient equation has limiting linear divisor $1 \neq 0$ and involves only finitely many lower coefficients, so induction commutes with the limit. Expanding $\sqrt{1 + q^2/6}$ gives (Q.71), and $z = U(1+\epsilon)$ turns E_0 into $\sqrt{U^2 + 1/6}$. □

Remark Q.36 (Two limits, not one). Theorem Q.35 is a statement about each coefficient separately. It does *not* say that $\mathcal{B}(y) - \sqrt{U^2 + 1/6}$ is small: at any actual y the parameter λ is finite and the neglected terms, headed by $b_0 = -\alpha/\lambda$, are numerically the largest corrections in the whole expansion. S1 writes that the subsector "resums" to $\sqrt{T^2 + 1/6}$ and then, in the following remark, correctly warns that it does not replace the expansion; the two statements sit awkwardly together. The accurate form is the one above: a limit of coefficients, useful as an exact structural check on a symbolic computation and as the identification of one coherent subseries, not an approximation to the inverse.

Q.8 The three normalizations reconciled

The three sources use three coordinate systems. They agree, and the dictionary is worth recording because the disagreement is entirely in the choice of what to absorb.

Proposition Q.37 (Dictionary). *Let $C_n(D)$ denote the Barnes coefficients obtained from* (Q.59) *when $\zeta'(-1)$ is* not *absorbed, so that $R = \log y$, $Z = \mathrm{e}^{(v+3)/2}$, $D = v + 1$, and the constant $\kappa = \zeta'(-1) = \frac{1}{12} - \log A$ appears as the extra perturbation $2\kappa q$. Then the two families are related by*

$$b_n(\lambda, \alpha) = C_n(D)\Big|_{\log A \,\mapsto\, 1/12,\ \log(2\pi) \,\mapsto\, 2\alpha,\ D \,\mapsto\, 2\lambda}, \tag{Q.73}$$

that is, the absorbed family is obtained from the unabsorbed one by setting $\kappa = 0$ and halving the slope variable. In particular

$$C_1(D) = \frac{1}{12} + \frac{2\log A}{D} + \frac{\log^2(2\pi)}{2D^2} - \frac{\log^2(2\pi)}{D^3}, \qquad b_1(\lambda) = \frac{1}{12} + \frac{1}{12\lambda} + \frac{\alpha^2}{2\lambda^2} - \frac{\alpha^2}{2\lambda^3}, \tag{Q.74}$$

and these agree under (Q.73).

Proof. Both families solve the same normalized equation, the second with the extra term $2\kappa q$ and with λ replaced by $D/2$. Since κ enters only through that term and only linearly, and since $\kappa = \frac{1}{12} - \log A$ vanishes exactly when $\log A = \frac{1}{12}$, the substitution $\log A \mapsto \frac{1}{12}$ turns the unabsorbed equation into the absorbed one; uniqueness of the solution does the rest. The two displayed forms of the first coefficient were checked directly, as recorded in Theorem Q.27. □

Remark Q.38 (A cancellation worth seeing). In C_1 the constant $\log A$ appears and $\frac{1}{6D}$ does not; in b_1 the reverse. This is not a discrepancy. The forward term $-\frac{1}{12}\log z$ contributes $\frac{1}{12}+\frac{1}{12\lambda}$ to both, and the unabsorbed constant contributes $-2\kappa/D = -\frac{1}{6D}+\frac{2\log A}{D}$, whose first half exactly cancels the $\frac{1}{6D}=\frac{1}{12\lambda}$. A reader comparing the two articles term by term without tracking the normalization will conclude that one of them has a sign error. Neither does.

Q.9 Outer expansions in iterated logarithms

Retaining W exactly is both shorter and numerically better, but an answer purely in elementary logarithms is sometimes wanted. It is obtained by reverting the core equation directly, without first invoking a generic W-series.

Proposition Q.39 (The two outer families)**.**

1. Gamma. *Put $L=\log R$, $\ell=\log L$, $D_\Gamma=\ell+1$, and write $T=(R/L)F$. Then F is the unique solution with $F(\infty)=1$ of $F\bigl(1-D_\Gamma/L+L^{-1}\log F\bigr)=1$, and $F=1+\sum_{n\ge1} f_nL^{-n}$ with*

$$\begin{aligned}
&f_1=D,\quad f_2=D(D-1),\quad f_3=\tfrac12 D(D-2)(2D-1),\\
&f_4=\tfrac16 D(6D^3-26D^2+27D-6),\\
&f_5=\tfrac1{12} D(12D^4-77D^3+142D^2-84D+12),\\
&f_6=\tfrac1{30} D(30D^5-261D^4+725D^3-775D^2+300D-30),
\end{aligned}\tag{Q.75}$$

abbreviating $D=D_\Gamma$.

2. Barnes. *Put $L=\log(4R)$, $\ell=\log L$, $D_G=\ell+3$, and write $U=2\sqrt{R/L}\,F$. Then F solves $F^2\bigl(1-D_G/L+2L^{-1}\log F\bigr)=1$ and $F=1+\sum_{n\ge1} g_nL^{-n}$ with*

$$\begin{aligned}
&g_1=\tfrac12 D,\quad g_2=\tfrac18 D(3D-4),\quad g_3=\tfrac1{16} D(5D^2-16D+8),\\
&g_4=\tfrac1{384} D(105D^3-568D^2+720D-192),\\
&g_5=\tfrac1{768} D(189D^4-1488D^3+3296D^2-2304D+384),\\
&g_6=\tfrac1{15360} D(3465D^5-36516D^4+120400D^3-150400D^2+67200D-7680),
\end{aligned}\tag{Q.76}$$

abbreviating $D=D_G$.

Proof. For (i), $T(\log T-1)=R$ and $\log T=L-\ell+\log F$ give $(R/L)F\,(L-\ell-1+\log F)=R$, i.e. the displayed equation. Substituting $F=1+\sum f_nL^{-n}$ and equating coefficients of L^{-n} determines f_n uniquely from $f_1,\dots,f_{n-1}$, because the coefficient of f_n is 1. For (ii), write $L=\log(4R)$, so that $\log(4R/L)=L-\ell$ and $\log U=\frac12(L-\ell)+\log F$. Substituting $U=2\sqrt{R/L}\,F$ into $\frac12U^2(\log U-\frac32)=R$ gives

$$\frac{2R}{L}F^2\left(\frac{L-\ell}{2}-\frac32+\log F\right)=R,\qquad\text{i.e.}\qquad F^2\left(1-\frac{\ell+3}{L}+\frac{2\log F}{L}\right)=1,$$

which is the displayed equation with $D_G=\ell+3$. Both families were recomputed independently (Theorem Q.27). ☐

Remark Q.40 (The outer variable is part of the answer). The polynomials g_n in (Q.76) are attached to the choice $L=\log(4R)$, $D_G=\log L+3$. Choosing instead $\widetilde L=\log(4Re^{-3})=L-3$ and $U=2\sqrt{R/\widetilde L}\,F$ leads to the *same* polynomials evaluated at $D=\log\widetilde L$ — the shift $+3$ migrates from the argument into the variable. S1 uses the first convention and S3 the second, and both are correct; but the two lists of coefficients are only equal after the substitution is made, and comparing them term by term without it produces a spurious disagreement at every order. The same caution applies to the Gamma family, where the corresponding shift is $+1$.

Remark Q.41 (The hierarchy, and why it must not be merged). For every fixed N, $T/L^N \to \infty$; so the constant $\frac{1}{2}$ in Theorem Q.25 is smaller than every term of the outer expansion of T, and the first Bernoulli block d_1/T is smaller again. The correct reading is lexicographic,

$$T \gg \frac{T}{L} \gg \frac{T}{L^2} \gg \cdots \gg 1 \gg \frac{1}{T\Lambda} \gg \frac{1}{T^3\Lambda} \gg \cdots, \tag{Q.77}$$

and for Barnes, with the extra intermediate level supplied by b_0,

$$U \gg \frac{U}{L} \gg \cdots \gg 1 \gg \frac{1}{\lambda} \gg \frac{1}{U}. \tag{Q.78}$$

An expansion that interleaves the outer L^{-1} series with the additive constants and the q-grid is not merely inelegant, it is wrong: no finite truncation of the outer series can reproduce even the first block d_1/T. This is the same phenomenon as the flattening caveat of Theorem J.37, and the reason the volume keeps W unexpanded by default.

Q.10 Numerical verification

The tables below test the entire chain and involve no numerical inversion. An exact abscissa is chosen, the special function is evaluated at 80 digits, the core variables are reconstructed from that value alone, and the truncations are compared with the abscissa. Errors are signed, approximation minus exact.

Table Q.2: Γ_+^{-1}: signed error of $\frac{1}{2} + T$ and of the truncations including $d_1, \ldots, d_9$.

x	$\frac{1}{2}+T$	$+d_1$	$+d_3$	$+d_5$	$+d_7$	$+d_9$
5	-6.1414×10^{-3}	2.8694×10^{-5}	-4.1862×10^{-7}	1.3452×10^{-8}	-8.2694×10^{-10}	8.4558×10^{-11}
10	-1.9470×10^{-3}	1.7430×10^{-6}	-5.7560×10^{-9}	4.4485×10^{-11}	-6.5578×10^{-13}	1.5870×10^{-14}
20	-7.1924×10^{-4}	1.4126×10^{-7}	-1.1183×10^{-10}	2.1113×10^{-13}	-7.5612×10^{-16}	4.4191×10^{-18}
50	-2.1572×10^{-4}	6.1957×10^{-9}	-7.6816×10^{-13}	2.2927×10^{-16}	-1.2910×10^{-19}	1.1809×10^{-22}
100	-9.1031×10^{-5}	6.2869×10^{-10}	-1.9388×10^{-14}	1.4440×10^{-18}	-2.0231×10^{-22}	4.5949×10^{-26}

Table Q.3: $\mathcal{B}$: signed error of U and of the truncations including $b_0, \ldots, b_5$, at $y = G(z+1)$.

z	U	$+b_0$	$+b_1$	$+b_2$	$+b_3$	$+b_4$	$+b_5$
5	1.1187	-1.3867×10^{-2}	-7.8343×10^{-3}	-1.5935×10^{-3}	-1.8345×10^{-4}	5.8515×10^{-6}	8.9807×10^{-6}
10	6.5273×10^{-1}	-2.0081×10^{-2}	-8.4026×10^{-4}	1.5249×10^{-5}	3.2299×10^{-6}	4.1936×10^{-8}	-1.3999×10^{-8}
20	4.4669×10^{-1}	-8.7205×10^{-3}	-6.6821×10^{-5}	2.3736×10^{-6}	3.4133×10^{-8}	-1.5564×10^{-9}	-1.9947×10^{-11}
100	2.5342×10^{-1}	-1.2959×10^{-3}	-3.9481×10^{-7}	8.1326×10^{-9}	-5.3660×10^{-12}	-8.9993×10^{-14}	6.0482×10^{-16}

Three things are visible. The Gamma expansion gains roughly two orders per block, as (Q.42) predicts, and does so already at $x = 5$. The Barnes core error is much larger, because the linear forward term αz produces the slowly decaying displacement $b_0 = -\alpha/\lambda$; once that single block is included the improvement is again rapid. And at $z = 5$ the last Barnes column is *worse* than the one before it (8.98×10^{-6} against 5.85×10^{-6}): the expansion is asymptotic, not convergent, and at a fixed moderate argument it has reached its useful limit. That is the behaviour Theorem J.54 describes and Section Q.11 quantifies; it is not a defect of the coefficients, each of which was verified exactly.

Remark Q.42 (What the tables do and do not establish). They check the algebra, the signs, the shifts and the branch choices, end to end from the special function to the inverse; a transcription error in any numerator would show immediately. They are not a substitute for the analytic remainder arguments in the proofs of Theorems Q.25 and Q.33, and conversely those theorems, which control only the order of the remainder, would not catch a wrong rational number. The two kinds of evidence are complementary, and both were obtained.

Q.11 Beyond all orders

The expansions above are complete relative to a Poincaré forward input. They do not contain the exponentially small sectors that the late Bernoulli terms encode.

Proposition Q.43 (Transport of an exponentially small sector)**.** *Suppose the forward phase splits as $\Phi = \Phi_{\mathrm{alg}} + E$, where x_{alg} solves $\Phi_{\mathrm{alg}}(x) = R$ to the algebraic order considered and E is smaller than every term retained. If $\Phi'_{\mathrm{alg}}(x_{\mathrm{alg}}) \neq 0$, then the induced displacement of the root is*

$$\Delta x = -\frac{E(x_{\mathrm{alg}})}{\Phi'_{\mathrm{alg}}(x_{\mathrm{alg}})} + \mathcal{O}\left(\frac{|E|\,|E'|}{|\Phi'_{\mathrm{alg}}|^2} + \frac{|E|^2\,|\Phi''_{\mathrm{alg}}|}{|\Phi'_{\mathrm{alg}}|^3}\right). \tag{Q.79}$$

Proof. This is Theorem J.51 with $f_0 = \Phi_{\mathrm{alg}}$ and $\varepsilon = E$: part (ii) of that theorem gives exactly (Q.79), the two error terms being its θ and M_2 contributions. □

Proposition Q.44 (Least term and the inverse floor)**.** *The half-shifted Stirling coefficients satisfy*

$$a_n = (-1)^n\,\frac{2\,(2n-2)!}{(2\pi)^{2n}}\big(1 + \mathcal{O}(4^{-n})\big), \tag{Q.80}$$

so the Gamma inverse blocks $d_{2n-1} \sim -a_n/\Lambda$ decrease until $n \approx \pi T$ and grow thereafter. Consequently the least term of (Q.42) *has size $\exp\big(-2\pi T + \mathcal{O}(\log T)\big)$, and by Theorem J.47 the attainable accuracy of the inverse is*

$$\frac{\exp\big(-2\pi T + \mathcal{O}(\log T)\big)}{\Lambda}. \tag{Q.81}$$

The Barnes analogue, from $c_k = B_{2k+2}/(4k(k+1))$, has least term near $k \approx \pi U$ and floor $\exp\big(-2\pi U + \mathcal{O}(\log U)\big)/(U\lambda)$.

Proof. From $B_{2n} = (-1)^{n+1}2(2n)!\,\zeta(2n)/(2\pi)^{2n}$ and $\zeta(2n) = 1 + \mathcal{O}(4^{-n})$, together with $a_n = (2^{1-2n}-1)B_{2n}/\big(2n(2n-1)\big)$ and $2^{1-2n}-1 = -1+\mathcal{O}(4^{-n})$, we get (Q.80) using $(2n)!/(2n(2n-1)) = (2n-2)!$. The ratio of consecutive terms $|a_{n+1}|T^{-2n-1}/\big(|a_n|T^{1-2n}\big)$ is $2n(2n-1)/\big((2\pi)^2T^2\big)(1 + o(1))$, which equals 1 at $n \sim \pi T$; substituting that index into (Q.80) and using Stirling gives the stated size, and dividing by the slope Λ transports it to the inverse. The Barnes computation is identical with B_{2k+2} in place of B_{2n}. □

Remark Q.45 (What is not claimed). Theorem Q.44 locates the floor; it does not compute the Stokes multipliers, and this chapter does not. A genuine hyperasymptotic inverse must start from an exponentially improved *forward* expansion on the same branch and in the same sector — Nemes supplies these for both functions — and then apply (Q.79). Appending terms of the shape $\mathrm{e}^{-2\pi T}$ to the real formula by pattern matching is not a branch-complete construction, and none of the three sources does more than indicate the first sector. This is recorded as open in Section Q.13.

Q.12 The general family

Nothing above used more about Γ and G than the shape of their phases. The following is the class the chapter actually treats, and it is what Part R will reuse.

Theorem Q.46 (Inversion of a power–logarithmic phase)**.** *Let*

$$\Phi(x) = a\,x^p(\log x - \beta) + C + \sum_{\rho\in A} x^{p-\rho}P_\rho(\log x) + \mathcal{E}(x), \tag{Q.82}$$

where $a \neq 0$, $p > 0$, $A \subset \mathbb{R}_{>0}$ generates a well-ordered monoid S, each P_ρ is a polynomial, and $\mathcal{E}$ is smaller than every retained term. Put $R = Y - C$ and let v, X, q be given by (Q.14)*. If $v+1$ is invertible then the equation $\Phi(x) = Y$ has a unique formal solution*

$$x = X\Big(1 + \sum_{\mu \in S,\ \mu > 0} e_\mu(v)\, X^{-\mu}\Big), \tag{Q.83}$$

with e_μ given by (Q.24)*; the coefficients lie in $K[v, (v+1)^{-1}]$ where K is any field containing the coefficients of the P_ρ and β; and their denominators obey Theorem Q.18 when $S = \mathbb{N}$. If in addition Φ is real, eventually strictly increasing, with a differentiable Poincaré remainder at each truncation order, then each truncation of* (Q.83) *approximates the actual inverse with the error obtained by dividing the forward residual by $aX^{p-1}(v+1)$.*

Proof. Substituting $x = X(1+\epsilon)$ into (Q.82) and dividing by aX^p gives (Q.17) with

$$\mathcal{R}(q, v, \epsilon) = \frac{1}{a} \sum_{\rho \in A} q^\rho (1+\epsilon)^{p-\rho} P_\rho\Big(\beta + \frac{v}{p} + \log(1+\epsilon)\Big) + \frac{\mathcal{E}\big(X(1+\epsilon)\big)}{aX^p},$$

every monomial of which has q-exponent in S and positive. Powers of v in the coefficients are harmless: $q^\rho v^M \to 0$ for fixed M, and the Hahn ring is graded by the q-exponent alone. Theorem Q.13 gives existence, uniqueness and the recurrence; Theorem Q.14 the support; the coefficient ring follows because the only division performed is by $v+1$ and every other operation is a rational combination of the data. The analytic clause is Theorem J.47 applied with the slope $\Phi'(X) = aX^{p-1}(v+1)(1+o(1))$ from (Q.14). □

Example Q.47 (The two subjects, and a third)**.** Γ is $(a, p, \beta) = (1, 1, 1)$ with $A = \{2, 4, 6, \dots\}$; G is $(\frac{1}{2}, 2, \frac{3}{2})$ with $A = \{1, 2, 4, 6, \dots\}$, the exponent 1 coming from αz. The hyperfactorial of Part R is the next member, and the higher multiple gamma functions continue the family with $p = 3, 4, \dots$, their perturbation supports read off from the drops in algebraic degree in the known forward expansions.

Remark Q.48 (Where the closed form stops). Two boundaries deserve naming. A leading block $ax^p(\log x - \beta)^r$ with $r \neq 1$ is still exactly invertible, by Theorem Q.8, with normalized slope $(pu+r)/u$ in place of $v+1$. A leading block $ax^p(\log x)^r(\log\log x)^s$ with $s \neq 0$ is not: there is no one-step elementary reduction, and the honest procedure is to invert the first two factors exactly, treat the third as a slow multiplier, and solve the residual slow equation by iteration before the q-grid theorem is applied. The method's real hypothesis is not "Lambert" but "the dominant transmonomial can be inverted exactly and its normalized slope is a unit".

Q.13 What is left open

1. *Closed forms for the coefficient families.* The recurrences compute d_{2n-1} and b_n to any order and Theorem Q.22 gives the extreme coefficient of each in closed form, but no closed description of the full numerators is known. For Gamma the pattern $d_{2n-1} = P_n(\Lambda)/\big(\gamma_n \Lambda^{2n-1}\big)$ with P_n of degree $2n-2$ invites a Bell-polynomial formula decorated by Bernoulli values at $\frac{1}{2}$; for Barnes the grading by powers of α is a $\mathbb{Z}/2$-grading whose elementary proof is (Q.63) but whose finer structure — which powers α^j actually occur in b_n, and with what λ-degrees — is not determined here.

2. *Hyperasymptotics.* Theorem Q.45.

3. *Uniformity near the branch point.* Everything degrades as $v+1 \to 0$, and at $v = -1$ the core map has a critical point where the correct local scale is a square root rather than a power series. A uniform approximation matching a branch-point expansion to the transseries developed here

is not attempted. It is irrelevant for the positive real branches, which stay far from $v = -1$, and unavoidable for a global treatment of the complex sheets.

4. *Complex branches.* Replacing $\log y$ by $\operatorname{Log} y + 2\pi \mathrm{i} m$ and W_0 by W_k produces, in every sector where the forward expansion holds and the branch choices are continued consistently, a formal inverse sheet with the *same* coefficient functions. That is a local asymptotic chart, not a classification: distinct pairs (m, k) can describe continuations that meet, and the critical point of Γ creates branch geometry the leading Lambert model does not see. No global statement is made.

5. *Formalization.* The triangular recurrence is well suited to a proof assistant, since every requested coefficient depends on finitely many earlier ones; the natural decomposition separates the algebra of well-ordered supports, the formal implicit-function theorem (Theorem Q.13), the exact rational certificates, and the analytic import of the Stirling remainder bounds. Those assembled layers are not formalized. The isolated quadratic coefficient core is described in Theorem Q.21: its Catalan identity is Exact, while its exact order-by-order recurrence supplies only Partial coverage of the full quadratic-core lemma. The triangular transseries, Hahn- support, rational-certificate, and analytic-remainder apparatus listed in this item remains unformalized.

Appendix R

The hyperfactorial K-function

R.1 What this chapter does

The K-function interpolates the hyperfactorial, $K(n+1) = \prod_{j=1}^{n} j^j$, and $\log K(x) \sim \frac{1}{2}x^2 \log x - \frac{1}{4}x^2$. Its dominant block is therefore of exactly the type inverted in Theorem Q.8, with the same exponent $p = 2$ as Barnes G and a different shift; and the whole apparatus of Section Q.4 applies unchanged. What makes the subject worth a chapter of its own is a single feature: after the correct centring the entire perturbation of power gap two collapses to one term,

$$-\tfrac{1}{24} \log q, \tag{R.1}$$

and *that* term is resonant — its logarithm is of the same size as the core slope, so it does not shrink relative to the core the way an ordinary correction does. The three source articles deal with it in three different ways, and this chapter's main contribution is to prove that the three are one construction seen from three sides.

The three filed articles are listed in Table R.1. As in Part Q, they agree: every coefficient and constant any of them prints was recomputed for this volume in exact arithmetic and reproduced, including the eleven-digit numerators of the refined block b_5. Their difference is one of strategy, not of arithmetic.

Tag	Directory	Strategy for (R.1)
T1	`inverse_k_function_transseries`	leave it in the perturbation, *and* absorb it into an algebraic shift $T = r^2 - \frac{1}{12}$
T2	`K_Function_Inverse_Transseries`	absorb it into the core, using the r-Lambert function; also the arbitrary-centre expansion
T3	`K_function_inverse_transseries_article`	leave it in the perturbation; slow-depth and ordered-block formulas, sectorial version

Table R.1: The K-function in the notation of Theorem Q.46.

inverted map	$x \mapsto K(x)$ on $[2, \infty)$
centred variable	$r = x - \frac{1}{2}$
core $ax^p(\log x - \beta)$	$a = \frac{1}{2}$, $p = 2$, $\beta = \frac{1}{2}$
absorbed constant	$C = \log A - \frac{1}{8}$, A the Glaisher–Kinkelin constant
resonant term	$-\frac{1}{24} \log r$
perturbation support	$2\mathbb{N}$
core	$\rho = \mathrm{e}^L$, $L = \frac{1}{2}(w+1)$, $w = W_0(4Y/\mathrm{e})$
inverse support	$\rho^{-1}, \rho^{-3}, \rho^{-5}, \ldots$

Compare Table Q.1. Barnes G has $p=2$, $\beta=\frac{3}{2}$, a linear term αz generating the full integer support, and a resonant term $-\frac{1}{12}\log z$. The K-function has $p=2$, $\beta=\frac{1}{2}$, *no* linear term — so the support stays even, as for Γ — and a resonant term $-\frac{1}{24}\log r$. It is, in a precise sense made in Section R.7, the cleanest member of the family: the only obstruction it has is the resonance.

R.2 The function and its increasing branch

Lemma R.1 (Exact logarithmic derivatives). *Write $\kappa(x)=\log K(x)$. For $x>0$,*

$$\kappa'(x)=\log\Gamma(x)+x-\tfrac{1}{2}-\tfrac{1}{2}\log(2\pi),\qquad \kappa''(x)=\psi(x)+1. \tag{R.2}$$

Proof. Use $K(z)=\exp\big(\zeta'(-1,z)-\zeta'(-1)\big)$ with the Hurwitz zeta function. Differentiating $\partial_z\zeta(s,z)=-s\zeta(s+1,z)$ with respect to s gives $\partial_z\zeta'(s,z)=-\zeta(s+1,z)-s\zeta'(s+1,z)$, and at $s=-1$, $\partial_z\zeta'(-1,z)=-\zeta(0,z)+\zeta'(0,z)$. Substituting $\zeta(0,z)=\frac{1}{2}-z$ and $\zeta'(0,z)=\log\Gamma(z)-\frac{1}{2}\log(2\pi)$ gives the first identity, and differentiating it again gives the second, since $(\log\Gamma)'=\psi$ and the derivative of $x-\frac{1}{2}$ is 1. □

Proposition R.2 (The increasing branch). *K is strictly increasing on $[2,\infty)$, with $K(2)=1$ and $K(x)\to+\infty$; write $K_+^{-1}:[1,\infty)\to[2,\infty)$ for the inverse. On $(0,\infty)$, K has exactly two stationary points,*

$$x_{\max}=0.29095698669918\ldots,\qquad x_{\min}=1.53768886373648\ldots, \tag{R.3}$$

with $K(x_{\max})=1.29867230693177\ldots$ and $K(x_{\min})=0.87978684305939\ldots$; so below $K(x_{\max})$ there are further real branches, none of which survives as $y\to\infty$.

Proof. By Theorem R.1, $\kappa''=\psi+1$, and ψ is strictly increasing with $\psi(x)\to-\infty$ as $x\to0^+$ and $\psi(x)\to+\infty$; so κ'' has exactly one zero and κ' is strictly convex with a unique minimum. Since $\psi(2)=1-\gamma>0$ we get $\kappa''>0$ on $[2,\infty)$, and $\kappa'(2)=\log\Gamma(2)+2-\frac{1}{2}-\frac{1}{2}\log(2\pi)=\frac{3}{2}-\frac{1}{2}\log(2\pi)=0.5811\ldots>0$; hence $\kappa'>0$ on $[2,\infty)$ and K increases there. $K(2)=1$ is (R.4) at $n=1$, and $K\to\infty$ follows from Theorem R.4. Because κ' has a unique minimum and tends to $+\infty$ at both ends of $(0,\infty)$ while $\kappa'(1)=1-\frac{1}{2}-\frac{1}{2}\log(2\pi)<0$, it has exactly two zeros; these are (R.3), located numerically. □

Remark R.3 (The notation is not cosmetic). K is not monotone on $(0,\infty)$, so "the inverse of K" is ambiguous below $K(x_{\max})$. Everything in this chapter concerns K_+^{-1}, the branch tending to $+\infty$. The functional equation $K(z+1)=z^zK(z)$ with $K(1)=1$ gives

$$K(n+1)=\prod_{j=1}^{n}j^j, \tag{R.4}$$

so K_+^{-1} restricted to the values $\prod_{j\le n}j^j$ inverts the hyperfactorial exactly.

R.3 Forward expansion, and why the centre is forced

The following puts every centring on the same footing; the distinguished one then identifies itself.

Theorem R.4 (Expansion about an arbitrary centre). *Fix $\alpha\in\mathbb{C}$ and let $x=r+\alpha$. As $r\to\infty$ in a closed subsector of $|\arg r|<\pi$,*

$$\log K(r+\alpha)\sim\frac{r^2}{2}\log r-\frac{r^2}{4}+\Big(\alpha-\frac{1}{2}\Big)r\log r+\frac{B_2(\alpha)}{2}\log r+\log A+\frac{\alpha^2-\alpha}{2}+\sum_{n\ge1}\frac{(-1)^{n+1}B_{n+2}(\alpha)}{n(n+1)(n+2)\,r^n}. \tag{R.5}$$

Truncation after r^{-N} leaves a remainder $\mathcal{O}(r^{-N-1})$.

Proof. Start from the expansion at $\alpha = 0$, which follows by integrating (R.2): substituting Stirling's series into $\kappa'(x) = \log\Gamma(x) + x - \frac{1}{2} - \frac{1}{2}\log(2\pi)$ gives

$$\kappa'(x) \sim \Big(x - \frac{1}{2}\Big)\log x - \frac{1}{2} + \sum_{m\geq 1} \frac{B_{2m}}{2m(2m-1)x^{2m-1}}, \tag{R.6}$$

and termwise integration gives

$$\log K(x) \sim \Big(\frac{x^2}{2} - \frac{x}{2} + \frac{1}{12}\Big)\log x - \frac{x^2}{4} + \log A - \sum_{m\geq 1} \frac{B_{2m+2}}{2m(2m+1)(2m+2)x^{2m}}, \tag{R.7}$$

the integration constant being $\log A$ by the classical hyperfactorial asymptotic.

Now substitute $x = r+\alpha$ and expand $\log(r+\alpha) = \log r + \sum_{j\geq 1}(-1)^{j+1}\alpha^j/(jr^j)$ and $(r+\alpha)^{-2m} = r^{-2m}\sum_{j\geq 0}(-1)^j\binom{2m+j-1}{j}\alpha^j r^{-j}$. At each fixed inverse power only finitely many terms contribute. The polynomial and logarithmic part collects into the first line, using $B_2(\alpha) = \alpha^2 - \alpha + \frac{1}{6}$ for the coefficient of $\log r$: the three sources of $\log r$ are $\frac{1}{2}r^2 \cdot 0$, the term $(\frac{\alpha^2}{2} - \frac{\alpha}{2} + \frac{1}{12})\log r$ from the quadratic prefactor, which is $\frac{1}{2}B_2(\alpha)\log r$. For $n \geq 1$ the coefficient of r^{-n} is a binomial convolution of Bernoulli numbers against powers of α, and the identity $B_{n+2}(\alpha) = \sum_j \binom{n+2}{j}B_j\alpha^{n+2-j}$ collapses it to $(-1)^{n+1}B_{n+2}(\alpha)/\big(n(n+1)(n+2)\big)$. Translation by a fixed α preserves closed proper subsectors, so the Poincaré remainder survives. □

Corollary R.5 (The midpoint is the unique good centre)**.** *The coefficient of* $r\log r$ *in* (R.5) *vanishes if and only if* $\alpha = \frac{1}{2}$*; and at that same value* $B_{n+2}(\frac{1}{2}) = 0$ *for every odd* n*, so the tail becomes even. With* $r = x - \frac{1}{2}$ *and* $C = \log A - \frac{1}{8}$*,*

$$\kappa\Big(r + \frac{1}{2}\Big) \sim \Phi(r) - \frac{1}{24}\log r + C + \sum_{m\geq 1}\frac{c_m}{r^{2m}}, \qquad \Phi(r) = \frac{r^2}{2}\log r - \frac{r^2}{4}, \tag{R.8}$$

where

$$c_m = -\frac{B_{2m+2}(1/2)}{2m(2m+1)(2m+2)} = -\frac{(2^{-2m-1}-1)B_{2m+2}}{2m(2m+1)(2m+2)}. \tag{R.9}$$

The first values are

$$c_1 = -\frac{7}{5760}, \quad c_2 = \frac{31}{161280}, \quad c_3 = -\frac{127}{1290240}, \quad c_4 = \frac{511}{4866048}, \quad c_5 = -\frac{1414477}{7380172800}, \quad c_6 = \frac{8191}{15335424}. \tag{R.10}$$

Proof. The coefficient of $r\log r$ is $\alpha - \frac{1}{2}$. At $\alpha = \frac{1}{2}$ the constant is $\log A + (\frac{1}{4} - \frac{1}{2})/2 = \log A - \frac{1}{8} = C$ and the coefficient of $\log r$ is $\frac{1}{2}B_2(\frac{1}{2}) = \frac{1}{2}(-\frac{1}{12}) = -\frac{1}{24}$. Odd-index Bernoulli polynomials vanish at $\frac{1}{2}$ by $B_n(1-\alpha) = (-1)^n B_n(\alpha)$, so only even $n = 2m$ survives, and $B_{2m+2}(\frac{1}{2}) = (2^{-2m-1}-1)B_{2m+2}$ gives (R.9). □

Remark R.6 (Two coincidences that are one). That the shift removing $r\log r$ is also the shift killing the odd tail looks like luck. It is not: both are the reflection symmetry $B_n(1-\alpha) = (-1)^n B_n(\alpha)$ acting at its fixed point $\alpha = \frac{1}{2}$. The same remark applies to Γ in Theorem Q.2, where the half-shift does the same two jobs at once. Theorem R.4 makes this visible by displaying every centre at the same time, instead of verifying the good one after the fact; this is the main reason it is stated in that generality.

R.4 The Lambert anchor and the canonical inverse

Proposition R.7 (Core variables)**.** *Let* $Y = \log y$ *and*

$$w = W_0\Big(\frac{4Y}{\mathrm{e}}\Big), \qquad \rho = \mathrm{e}^{(w+1)/2} = 2\sqrt{\frac{Y}{w}}, \qquad L = \log\rho = \frac{w+1}{2}. \tag{R.11}$$

Then $\Phi(\rho) = Y$ *exactly, and* $\Phi'(\rho) = \rho \log \rho = \rho L$.

Proof. This is Theorem Q.8 with $a = \frac{1}{2}, p = 2, \beta = \frac{1}{2}, r = 1$: there $v = W_k\big((pY/a)\mathrm{e}^{-p\beta}\big) = W_0(4Y/\mathrm{e})$ and $X = \mathrm{e}^{\beta+v/p} = \mathrm{e}^{(1+w)/2}$, and the slope is $aX^{p-1}(v+1) = \frac{1}{2}\rho(w+1) = \rho L$. □

Write $q = \rho^{-2}$, $r = \rho(1+v)$, and

$$\mathcal{D}_L(v) := \frac{1}{2}(1+v)^2\big(L+\log(1+v)\big) - \frac{1}{4}(1+v)^2 - \frac{L}{2} + \frac{1}{4} = Lv + \frac{L+1}{2}v^2 + \sum_{k\ge 3} \frac{(-1)^{k+1}}{k(k-1)(k-2)} v^k, \tag{R.12}$$

which is $\mathcal{K}_{2,v}$ of (Q.22) rescaled; its linear coefficient is L because the normalization here is by ρ^2 rather than $2\rho^2$. Subtracting $\Phi(\rho) = Y$ from (R.8) and dividing by ρ^2 gives the exact normalized equation

$$\mathcal{D}_L(v) + q\left[E - \frac{1}{24}\log(1+v) + \sum_{m\ge 1} c_m q^m (1+v)^{-2m}\right] = 0, \qquad E := C - \frac{L}{24}. \tag{R.13}$$

Theorem R.8 (Canonical inverse transseries)**.** *Equation* (R.13) *has a unique solution* $v = \sum_{n\ge 1} V_n(L) q^n$ *with* $V_n \in \mathbb{Q}[C, L, L^{-1}]$, *given by*

$$V_n(L) = -\frac{1}{L}[q^n]\left\{\mathcal{D}_L(v_{<n}) + q\left[E - \frac{1}{24}\log(1+v_{<n}) + \sum_{m\ge 1} c_m q^m (1+v_{<n})^{-2m}\right]\right\}, \tag{R.14}$$

$v_{<n} = \sum_{j<n} V_j q^j$*; only* $c_1, \ldots, c_{n-1}$ *can occur in* V_n*. For every fixed* $N \ge 1$*, as* $y \to +\infty$*,*

$$K_+^{-1}(y) = \frac{1}{2} + \rho + \sum_{n=1}^{N-1} V_n(L)\rho^{1-2n} + \mathcal{O}\big(\rho^{1-2N}\big). \tag{R.15}$$

The first coefficients are

$$V_1 = -\frac{E}{L} = \frac{1}{24} - \frac{C}{L}, \tag{R.16}$$

$$V_2 = -\frac{1}{L}\left(\frac{L+1}{2}V_1^2 - \frac{V_1}{24} + c_1\right) = \frac{7L^2 - 240EL - 2880(L+1)E^2}{5760L^3}, \tag{R.17}$$

$$V_3 = -\frac{1}{L}\left((L+1)V_1V_2 + \frac{V_1^3}{6} - \frac{1}{24}\Big(V_2 - \frac{V_1^2}{2}\Big) - 2c_1V_1 + c_2\right), \tag{R.18}$$

$$V_4 = -\frac{1}{L}\Big[(L+1)\Big(V_1V_3 + \frac{V_2^2}{2}\Big) + \frac{V_1^2V_2}{2} - \frac{V_1^4}{24} - \frac{1}{24}\Big(V_3 - V_1V_2 + \frac{V_1^3}{3}\Big) + c_1\big(-2V_2 + 3V_1^2\big) - 4c_2V_1 + c_3\Big]. \tag{R.19}$$

Proof. The formal part is Theorem Q.13 with $S = 2\mathbb{N}$ and linear coefficient L: by (R.12) the coefficient of v in $\mathcal{D}_L$ is L, which is invertible in $\mathbb{Q}[C, L, L^{-1}]$, and every monomial of the bracket in (R.13) carries at least one factor q. Hence a unique solution and (R.14). The displayed $V_1, \ldots, V_4$ follow by running it; they were recomputed independently (Theorem R.9).

For the analytic statement, (R.8) is a Poincaré expansion with differentiable remainder on the positive ray, obtained by integrating the sectorial Stirling remainder in (R.6); retain it far enough that its own error is below the first omitted inverse term. The formal substitution kills the normalized equation through q^{N-1}, so the forward residual is $\mathcal{O}(\rho^2 q^N) = \mathcal{O}(\rho^{2-2N})$. By Theorem R.1 and (R.6), $\kappa'(r + \frac{1}{2}) = r\log r + \mathcal{O}(r^{-1}) \asymp \rho L$ on the relevant interval, and this is bounded below by $\frac{1}{2}\rho L > 0$ for large y. Theorem J.47 converts the residual into an error $\mathcal{O}(\rho^{2-2N}/(\rho L)) = \mathcal{O}(\rho^{1-2N}/L)$, which is (R.15), and certifies that a root lies within it. □

Remark R.9 (Independent verification). As in Theorem Q.27, every rational number in this chapter was recomputed for this volume from the defining equations, in exact arithmetic, by an implementation independent of the sources: the tail c_m from Bernoulli polynomials at $\frac{1}{2}$ through $m = 6$; $V_1, \ldots, V_4$ from (R.14), including the closed form (R.17) for V_2; the accelerated forward blocks $d_1, \ldots, d_6$ both from the closed formula (R.27) and, independently, by direct expansion of the identity in the proof of Theorem R.12; the accelerated inverse blocks $U_2, \ldots, U_5$; and the refined blocks $b_1, \ldots, b_5$ together with the limits (R.44). All agreed with all three sources. The numerical tables of Section R.6 were computed afresh at 60 digits from $K(x) = \exp\big(\zeta'(-1, x) - \zeta'(-1)\big)$, and reproduce $K(5) = 27648 = 1^1 2^2 3^3 4^4$ exactly.

R.5 The resonance and its three resolutions

The term $-\frac{1}{24}\log r$ enters (R.13) at q^1 carrying the factor $E = C - L/24$, which *grows* with the slope. It is therefore resonant in the sense of Theorem Q.35: it survives the large-slope limit, while every c_m and the constant C do not. The consequence is visible in $V_1 = \frac{1}{24} - C/L$, whose limit is not zero.

R.5.1 First resolution: leave it, and take the limit

Proposition R.10 (Coefficientwise limit)**.** *For each fixed* n*,*

$$\lim_{L\to\infty} V_n(L) = \binom{1/2}{n} 12^{-n}, \tag{R.20}$$

so $V_1 \to \frac{1}{24}$, $V_2 \to -\frac{1}{1152}$, $V_3 \to \frac{1}{27648}$. *The limiting coefficients are those of*

$$v_0(q) = \sqrt{1 + \tfrac{q}{12}} - 1, \qquad \textit{equivalently} \qquad \rho\big(1 + v_0(\rho^{-2})\big) = \sqrt{\rho^2 + \tfrac{1}{12}}. \tag{R.21}$$

Proof. Divide (R.13) by L and let $L \to \infty$ with q formal. By (R.12), $\mathcal{D}_L(v) = L \cdot \frac{1}{2}\big((1+v)^2 - 1\big) + \big(\text{terms free of } L\big)$, so $\mathcal{D}_L(v)/L \to \frac{1}{2}\big((1+v)^2 - 1\big)$. In the bracket, $qE/L = q(C/L - \frac{1}{24}) \to -q/24$, while $q\log(1+v)/(24L) \to 0$ and every c_m term $\to 0$. The limiting equation is therefore

$$\tfrac{1}{2}\big((1+v_0)^2 - 1\big) - \tfrac{q}{24} = 0, \qquad \text{i.e.} \qquad (1+v_0)^2 = 1 + \tfrac{q}{12},$$

whose branch with $v_0(0) = 0$ is (R.21). The limit may be taken inside the recursion because at each order the coefficient equation has limiting divisor 1 and involves only finitely many earlier coefficients. Expanding the square root gives (R.20), and $r = \rho(1+v)$ turns v_0 into $\sqrt{\rho^2 + 1/12}$. □

R.5.2 Second resolution: absorb it into an algebraic shift

Lemma R.11 (Absorbing a logarithmic correction)**.** *Let* $a \neq 0$, $p \neq 0$, *and suppose*

$$F(r) = ar^p(\log r - b) + \beta\log r + \gamma + \mathcal{O}(r^{-p}). \tag{R.22}$$

Put $T = r^p + \alpha$ *and* $\Psi(T) = \frac{a}{p}T(\log T - pb)$. *Then*

$$\Psi(r^p + \alpha) = ar^p(\log r - b) + a\alpha\log r + a\alpha\Big(\frac{1}{p} - b\Big) + \mathcal{O}(r^{-p}), \tag{R.23}$$

so the unique shift absorbing the term $\beta\log r$ *is*

$$\alpha = \frac{\beta}{a}. \tag{R.24}$$

Proof. Insert $T = r^p + \alpha$ and use $\log T = p\log r + \alpha r^{-p} + \mathcal{O}(r^{-2p})$:

$$\Psi(T) = \frac{a}{p}(r^p+\alpha)\big(p\log r+\alpha r^{-p}-pb\big)+\mathcal{O}(r^{-p}) = ar^p\log r+\frac{a\alpha}{p}-abr^p+a\alpha\log r-ab\alpha+\mathcal{O}(r^{-p}),$$

which is (R.23). Matching the coefficient of $\log r$ gives $a\alpha = \beta$. $\square$

For the K-function $a = \frac{1}{2}$, $p = 2$, $b = \frac{1}{2}$ and $\beta = -\frac{1}{24}$, so $\alpha = -\frac{1}{12}$ and, remarkably, the constant correction $a\alpha(1/p - b)$ vanishes because $p^{-1} = b$. The new coordinate is

$$T = r^2 - \frac{1}{12} = x^2 - x + \frac{1}{6}, \qquad x = \frac{1}{2} + \sqrt{T + \frac{1}{12}}. \tag{R.25}$$

Theorem R.12 (Quadratic normal form). *With $\Psi(T) = \frac{1}{4}T(\log T - 1)$ and $T = x^2 - x + \frac{1}{6}$, as $T \to \infty$,*

$$\log K(x) \sim \Psi(T) + C + \sum_{n\geq 1}\frac{d_n}{T^n}, \tag{R.26}$$

where, writing $a_0 = \frac{1}{12}$,

$$d_n = \frac{(-1)^n a_0^{n+1}}{4(n+1)} + \sum_{m=1}^{n}(-1)^{n-m}\binom{n-1}{m-1}a_0^{n-m}c_m. \tag{R.27}$$

The first values are

$$d_1 = -\frac{1}{480}, \quad d_2 = \frac{31}{90720}, \quad d_3 = -\frac{103}{725760}, \quad d_4 = \frac{179}{1330560}, \quad d_5 = -\frac{6480899}{28021593600}, \quad d_6 = \frac{3485299}{5604318720}. \tag{R.28}$$

Proof. Since $r^2 = T + a_0$ and $\frac{1}{2}r^2 - \frac{1}{24} = \frac{T}{2}$, the elementary part of (R.8) satisfies the *exact* identity

$$\left(\frac{r^2}{2} - \frac{1}{24}\right)\log r - \frac{r^2}{4} = \frac{T}{4}\log(T+a_0) - \frac{T}{4} - \frac{a_0}{4} = \Psi(T) + \frac{T}{4}\log\left(1+\frac{a_0}{T}\right) - \frac{a_0}{4},$$

using $\log r = \frac{1}{2}\log(T+a_0)$. The first term of $\frac{T}{4}\log(1+a_0/T)$ is $a_0/4$, which cancels the constant $-a_0/4$ exactly; this is why no constant is generated and C is unchanged. Expanding the remaining logarithm and $(T+a_0)^{-m} = T^{-m}(1+a_0/T)^{-m}$ and collecting the coefficient of T^{-n} gives (R.27). $\square$

The gain is one full sector. In (R.8) the largest unresolved term was $-\frac{1}{24}\log r$, of size $\log r$; in (R.26) it is d_1/T. Inverting exactly as before, with

$$\eta = Y - C, \qquad \omega = W_0\left(\frac{4\eta}{\mathrm{e}}\right), \qquad \tau = \frac{4\eta}{\omega} = \mathrm{e}^{\omega+1}, \qquad S = \log\tau = \omega + 1, \tag{R.29}$$

so that $\Psi(\tau) = \eta$ exactly, and writing $T = \tau(1+u)$, $\nu = \tau^{-1}$, the normalized equation becomes

$$\mathcal{E}_S(u) + 4\sum_{m\geq 1}d_m\nu^{m+1}(1+u)^{-m} = 0, \qquad \mathcal{E}_S(u) = Su + \sum_{k\geq 2}\frac{(-1)^k}{k(k-1)}u^k. \tag{R.30}$$

Theorem R.13 (Accelerated inverse). *Equation* (R.30) *has a unique solution $u = \sum_{n\geq 2}U_n(S)\nu^n$ in $\nu^2\mathbb{Q}[S,S^{-1}][[\nu]]$, with*

$$U_n(S) = -\frac{1}{S}[\nu^n]\left\{\mathcal{E}_S(u_{<n}) + 4\sum_{m\geq 1}d_m\nu^{m+1}(1+u_{<n})^{-m}\right\}, \tag{R.31}$$

and

$$U_2 = \frac{1}{120S}, \quad U_3 = -\frac{31}{22680S}, \quad U_4 = \frac{1030S^2 - 126S - 63}{1814400S^3}, \quad U_5 = -\frac{16110S^2 - 1023S - 341}{29937600S^3}. \tag{R.32}$$

Consequently, as $y \to +\infty$,

$$K_+^{-1}(y) \sim \frac{1}{2} + \sqrt{\frac{1}{12} + \tau\Big(1 + \sum_{n\geq 2} U_n(S)\tau^{-n}\Big)}, \tag{R.33}$$

and truncating after $U_N\tau^{-N}$ leaves an error $\mathcal{O}(\tau^{-N-1/2})$ in x. Expanding the root,

$$K_+^{-1}(y) = \frac{1}{2} + R + \frac{1}{240S\tau R} - \frac{31}{45360S\tau^2 R} + \mathcal{O}(\tau^{-7/2}), \qquad R = \sqrt{\tau + \tfrac{1}{12}}. \tag{R.34}$$

Proof. $\mathcal{E}_S$ has linear coefficient S, a unit, and the forcing begins at ν^2 because $d_m\nu^{m+1}$ starts at $m = 1$; Theorem Q.13 with $S = \mathbb{N}$ gives uniqueness and (R.31), and $u = \mathcal{O}(\nu^2)$. The values (R.32) were computed and independently verified (Theorem R.9). The analytic step is Theorem J.47 exactly as in Theorem R.8, with slope $\Psi'(\tau) = \frac{1}{4}(\log\tau) = \frac{S}{4}$ in the T variable; converting to x through $x = \frac{1}{2} + \sqrt{T + 1/12}$ costs the factor $\frac{1}{2}T^{-1/2}$, which is the source of the half-integer exponent. □

R.5.3 The first two resolutions are the same

Theorem R.14 (Acceleration is resummation). *The coefficientwise limit of Theorem R.10 and the algebraic substitution of Theorem R.11 are the same construction:*

$$\frac{1}{2} + \rho + \sum_{n\geq 1}\Big(\lim_{L\to\infty} V_n(L)\Big)\rho^{1-2n} = \frac{1}{2} + \sqrt{\rho^2 + \frac{1}{12}} = \frac{1}{2} + \sqrt{T + \frac{1}{12}}\Big|_{T=\rho^2-1/12}, \tag{R.35}$$

and the map $T \mapsto \frac{1}{2} + \sqrt{T + \frac{1}{12}}$ appearing on the right is precisely the inverse of the accelerated coordinate (R.25)*. In other words, the leading term of the accelerated expansion* (R.33) is *the sum of the resonant subsector of the canonical expansion* (R.15)*, and the residual series $\sum U_n\tau^{-n}$ is what remains of the canonical coefficients after that subsector is removed.*

Proof. By Theorem R.10 the limiting subsector is $\sum_{n\geq 1}\binom{1/2}{n}12^{-n}\rho^{1-2n} = \rho\big(\sqrt{1 + 1/(12\rho^2)} - 1\big) = \sqrt{\rho^2 + 1/12} - \rho$, so adding $\frac{1}{2} + \rho$ gives the middle expression in (R.35). The right-hand equality is (R.25) read backwards. For the last sentence: Theorem R.11 removes from the forward expansion exactly the term $\beta\log r$ that is resonant, and by Theorem Q.35 a perturbation is resonant precisely when it survives division by the slope in the large-slope limit; the remaining forward terms d_nT^{-n} have no $\log T$ and are therefore non-resonant, so the accelerated coefficients U_n all tend to 0 — indeed $U_n = \mathcal{O}(S^{-1})$ by (R.32) and the recurrence. □

Remark R.15 (The same statement for Barnes G). Neither Barnes source performs this absorption, and it costs nothing. In Theorem Q.3 the data are $a = \frac{1}{2}$, $p = 2$, $b = \frac{3}{2}$, $\beta = -\frac{1}{12}$, so Theorem R.11 gives $\alpha = -\frac{1}{6}$, the coordinate $T = z^2 - \frac{1}{6}$, and the normal form

$$\log G(z+1) = \frac{1}{4}T(\log T - 3) + \alpha_*\sqrt{T + \tfrac{1}{6}} + \Big(\zeta'(-1) - \frac{1}{12}\Big) + \mathcal{O}(T^{-1}), \qquad \alpha_* = \tfrac{1}{2}\log(2\pi), \tag{R.36}$$

the constant $-\frac{1}{12}$ coming from $a\alpha(1/p - b)$ in (R.23). The square root $\sqrt{T + \frac{1}{6}}$ is exactly the resummed subsector $\sqrt{U^2 + 1/6}$ that Theorem Q.35 identifies as a coefficientwise limit — so Theorem Q.36, which warns that the limit is not an approximation, is answered here: it becomes one as soon as it is moved into the coordinate. The price for Barnes, and the reason its sources may have balked, is the half-integer power $\sqrt{T + 1/6}$ forced by the linear term $\alpha_* z$, which the K-function does not have. That is the precise sense in which the K-function is the cleaner subject.

R.5.4 Third resolution: absorb it into the core

The shift of Theorem R.11 absorbs $\beta\log r$ only modulo $\mathcal{O}(r^{-p})$. It can instead be absorbed *exactly*, at the cost of replacing W by a one-parameter relative.

Definition R.16 (The r-Lambert function). For fixed ϱ, let $W^{(\varrho)}$ denote a branch of the inverse of $s\mapsto se^s+\varrho s$, so that

$$W^{(\varrho)}(z)\,\mathrm{e}^{W^{(\varrho)}(z)}+\varrho\,W^{(\varrho)}(z)=z. \tag{R.37}$$

For $\varrho=0$ this is W.

Theorem R.17 (Exact inversion of an affine logarithmic power core). *Let*

$$\Psi_*(x)=ax^p\log x+b_*x^p+c_*\log x+d_*,\qquad ap\neq 0. \tag{R.38}$$

Fix compatible branches and set

$$\varrho=\frac{c_*}{a}\exp\!\left(\frac{pb_*}{a}\right),\qquad z=\frac{p}{a}\exp\!\left(\frac{pb_*}{a}\right)\left(Y-d_*+\frac{c_*b_*}{a}\right). \tag{R.39}$$

If $s=W^{(\varrho)}(z)$, then

$$x=\exp\!\left(\frac{s}{p}-\frac{b_*}{a}\right) \tag{R.40}$$

solves $\Psi_*(x)=Y$, and every local branch arises from compatible choices of the logarithm and of a branch of $W^{(\varrho)}$.

Proof. Put $s=p\log x+pb_*/a$, so that $\log x=s/p-b_*/a$ and $x^p=\exp(s-pb_*/a)$. The two power terms collapse:

$$ax^p\log x+b_*x^p=x^p\big(a\log x+b_*\big)=\exp\!\left(s-\frac{pb_*}{a}\right)\cdot\frac{as}{p}=\frac{a}{p}\exp\!\left(-\frac{pb_*}{a}\right)s\,\mathrm{e}^s,$$

where the middle step uses $a\log x+b_*=a(s/p-b_*/a)+b_*=as/p$. The logarithmic term is $c_*(s/p-b_*/a)$. Hence $\Psi_*(x)=Y$ reads

$$\frac{a}{p}\mathrm{e}^{-pb_*/a}s\mathrm{e}^s+\frac{c_*}{p}s=Y-d_*+\frac{c_*b_*}{a},$$

and multiplying by $(p/a)\mathrm{e}^{pb_*/a}$ gives $se^s+\varrho s=z$ with the data (R.39). □

Corollary R.18 (The refined K anchor). *Apply Theorem R.17 to $\Psi_*(r)=\frac12 r^2\log r-\frac14 r^2-\frac{1}{24}\log r+C$, that is to $a=\frac12$, $p=2$, $b_*=-\frac14$, $c_*=-\frac{1}{24}$, $d_*=C$. Then*

$$\varrho=-\frac{1}{12\mathrm{e}},\qquad z=\frac{48(Y-C)+1}{12\mathrm{e}}, \tag{R.41}$$

and the resulting $\rho_*=\exp\big(\frac12 W^{(\varrho)}(z)+\frac12\big)$ satisfies $\Psi_*(\rho_*)=Y$ exactly, with slope $\Psi_*'(\rho_*)=\rho_*\log\rho_*-\frac{1}{24\rho_*}$. The residual perturbation is only the Bernoulli tail $\sum_{m\ge1}c_m\rho_*^{-2m}$.

Proof. Substitute the data into (R.39): $pb_*/a=-1$, so $\varrho=(-\frac{1}{24})\cdot 2\cdot \mathrm{e}^{-1}=-1/(12\mathrm{e})$ and $z=4\mathrm{e}^{-1}\big(Y-C+(-\frac{1}{24})(-\frac14)\cdot 2\big)=4\mathrm{e}^{-1}\big(Y-C+\frac{1}{48}\big)$, which is (R.41). The formula for ρ_* is (R.40) with $p=2$ and $b_*/a=-\frac12$; the slope is immediate. □

Theorem R.19 (Refined blocks). *With* $t = \rho_*^{-2}$, $L_* = \log \rho_*$, $r = \rho_* H$ *and* $H = 1+\sum_{n\geq 1} b_n(L_*)t^{n+1}$, *the normalized equation is*

$$H^2\big(2L_* - 1 + 2\log H\big) - (2L_* - 1) - \frac{t}{6}\log H + 4\sum_{m\geq 1} c_m t^{m+1} H^{-2m} = 0, \tag{R.42}$$

its linearization at $H = 1$, $t = 0$ *is* $4L_*$, *and*

$$b_n(L) = -\frac{1}{4L}[t^{n+1}]\,\big(\textit{left side of (R.42) at } H = H^*_{n-1}\big), \qquad n \geq 1. \tag{R.43}$$

Each b_n *is rational in* L *with* $b_n = \mathcal{O}(L^{-1})$, *uses only* $c_1, \ldots, c_n$, *and satisfies*

$$\lim_{L\to\infty} L\, b_n(L) = -c_n. \tag{R.44}$$

For every fixed N,

$$K_+^{-1}(y) = \frac{1}{2} + \rho_* + \sum_{n=1}^{N} \frac{b_n(L_*)}{\rho_*^{2n+1}} + \mathcal{O}\left(\frac{\rho_*^{-2N-3}}{L_*}\right). \tag{R.45}$$

The fourth and fifth blocks are

$$b_4 = -\frac{12877200L^3 - 703428L^2 + 57552L + 539}{122624409600\, L^4}, \tag{R.46}$$

$$b_5 = \frac{1539856241280L^4 - 45140228640L^3 + 1666058823L^2 - 38145107L - 25162137}{8034351316992000\, L^5}. \tag{R.47}$$

Proof. The equation is (R.13) with the resonant term and the constant moved into the core, so only the tail remains; the linearization is $4L_*$ because the normalization is by $\frac{1}{4}\rho_*^2$. Theorem Q.13 gives uniqueness and (R.43). Since the forcing begins at t^2, so does $H - 1$, which is why the ansatz uses t^{n+1}. For (R.44): at order t^{n+1} the only contribution independent of earlier blocks is $4c_n t^{n+1}$, so $4Lb_n = -4c_n + \mathcal{O}(L^0 \cdot L^{-1})$, giving $Lb_n \to -c_n$; equivalently this is the linear response $-S(\rho_*)/\Psi_*'(\rho_*)$. The two displayed blocks were computed and independently verified (Theorem R.9). The analytic clause is again Theorem J.47, with the slope $\Psi_*'(\rho_*) \sim \rho_* L_*$. ☐

Remark R.20 (Which resolution to use). All three are correct and they answer different questions. The canonical expansion is the one to state, because its coefficients live in $\mathbb{Q}[C, L, L^{-1}]$ and it needs only W_0. The accelerated coordinate is the one to compute with: Table R.3 shows its leading term alone beating four canonical blocks at $x = 5$. The r-Lambert anchor is the one that explains: it shows that the obstruction is not deep, merely misplaced, since a single one-parameter special function removes it exactly and leaves nothing but the Bernoulli tail — which is the same tail that Γ and G leave, and which is beyond repair because it is divergent.

R.6 Numerical verification

The protocol is that of Section Q.10: choose an exact abscissa x, evaluate $K(x) = \exp\big(\zeta'(-1, x) - \zeta'(-1)\big)$ to 60 digits, reconstruct the core variables from that value alone, and compare the truncations with x. Errors are signed.

The comparison is the point. At $x = 5$ the accelerated *leading* term (1.5×10^{-5}) is already as accurate as the canonical expansion carried through its first block (1.8×10^{-5}), and the pattern persists down the table: at every argument, accelerated column j tracks canonical column $j + 1$. This is Theorem R.14 in numbers. The accelerated leading term is not a better approximation of the same kind — it is the canonical expansion with an entire infinite subsector already summed, so it starts one block ahead and stays there. The accelerated form is also the cheaper of the two to evaluate, since $U_2, \ldots, U_5$ are shorter expressions than $V_1, \ldots, V_4$ and involve no C.

Table R.2: Canonical expansion (R.15): signed error of $\frac{1}{2}+\rho$ and of the truncations including $V_1,\ldots,V_4$.

x	$\frac{1}{2}+\rho$	$+V_1$	$+V_2$	$+V_3$	$+V_4$
5	9.0012×10^{-3}	1.8437×10^{-5}	1.2665×10^{-7}	-5.9396×10^{-10}	6.4854×10^{-11}
10	1.3996×10^{-3}	-1.9257×10^{-7}	8.4998×10^{-10}	-5.1976×10^{-12}	6.4879×10^{-14}
20	-2.7600×10^{-7}	-5.5164×10^{-8}	2.0887×10^{-11}	-2.9637×10^{-14}	8.4762×10^{-17}
50	-2.0103×10^{-4}	-2.9310×10^{-9}	1.9068×10^{-13}	-3.7123×10^{-17}	1.5892×10^{-20}
100	-1.4839×10^{-4}	-2.6924×10^{-10}	5.4398×10^{-15}	-2.5162×10^{-19}	2.6077×10^{-23}

Table R.3: Accelerated expansion (R.33): signed error of $\frac{1}{2}+\sqrt{\frac{1}{12}+\tau}$ and of the truncations including $U_2,\ldots,U_5$.

x	$\frac{1}{2}+\sqrt{\frac{1}{12}+\tau}$	$+U_2$	$+U_3$	$+U_4$	$+U_5$
5	-1.5162×10^{-5}	1.2198×10^{-7}	-2.3300×10^{-9}	1.0850×10^{-10}	-9.0171×10^{-12}
10	-1.0786×10^{-6}	1.9569×10^{-9}	-8.6886×10^{-12}	9.1990×10^{-14}	-1.7528×10^{-15}
20	-9.4572×10^{-8}	4.0777×10^{-11}	-4.3490×10^{-14}	1.0944×10^{-16}	-4.9719×10^{-19}
50	-4.4020×10^{-9}	2.9465×10^{-13}	-4.9102×10^{-17}	1.9149×10^{-20}	-1.3502×10^{-23}
100	-4.5974×10^{-10}	7.6166×10^{-15}	-3.1503×10^{-19}	3.0371×10^{-23}	-5.2971×10^{-27}

Remark R.21 (The anomalous row). The entry -2.76×10^{-7} at $x=20$ in Table R.2 is far smaller than its neighbours at $x=10$ and $x=50$, and the column changes sign there. This is not an accident of the tabulation and not a numerical artefact: by (R.16) the first correction vanishes exactly when

$$V_1(L)=\frac{1}{24}-\frac{C}{L}=0, \qquad \text{i.e.} \qquad L=24C=2.97010\ldots, \tag{R.48}$$

which corresponds to $\rho=\mathrm{e}^{24C}=19.4944\ldots$, that is to $x=\frac{1}{2}+\rho=19.994\ldots$. The tabulated argument $x=20$ is within 0.006 of the zero of V_1, so at that one point the leading correction is almost exactly absent and the uncorrected core is anomalously good. The effect is entirely local; it says nothing about the quality of the expansion and everything about the numerical value of the Glaisher constant.

R.7 Comparison with Γ and G

Table R.4: The three power–logarithmic subjects of Parts Q and R in the notation of Theorem Q.46.

	$\Gamma(t+\frac{1}{2})$	$K(r+\frac{1}{2})$	$G(z+1)$
a	1	$\frac{1}{2}$	$\frac{1}{2}$
p	1	2	2
β	1	$\frac{1}{2}$	$\frac{3}{2}$
linear term	—	—	$\frac{1}{2}\log(2\pi)\,z$
resonant term	—	$-\frac{1}{24}\log r$	$-\frac{1}{12}\log z$
support	$2\mathbb{N}$	$2\mathbb{N}$	$\mathbb{N}$
tail	$a_k=\dfrac{B_{2k}(1/2)}{2k(2k-1)}$	$c_m=\dfrac{-B_{2m+2}(1/2)}{2m(2m+1)(2m+2)}$	$c_k=\dfrac{B_{2k+2}}{4k(k+1)}$
square root	$\sqrt{\Lambda^2T^2+\frac{1}{12}}$	$\sqrt{\rho^2+\frac{1}{12}}$	$\sqrt{U^2+\frac{1}{6}}$
its source	deepest pole	resonance	resonance

Three remarks are worth making from the merged view; none is in any of the six sources.

Remark R.22 (Where the $\frac{1}{12}$ comes from, twice). The constant $\frac{1}{12}$ appears in the last row of Table R.4 twice, for two apparently unrelated reasons: for Γ it is the deepest pole in the slope (Theorem Q.22, where it arises as $-2a_1$), and for K it is the resonant limit (Theorem R.10, where it arises as $-24^{-1}\cdot(-2)$).

The coincidence is not one. Both are $-2 \cdot B_2(\frac{1}{2})/2 = \frac{1}{12}$: for Γ, $a_1 = B_2(\frac{1}{2})/2 = -\frac{1}{24}$ is the first tail coefficient; for K, $\frac{1}{2}B_2(\frac{1}{2}) = -\frac{1}{24}$ is the coefficient of the resonant logarithm produced by Theorem R.5. The same Bernoulli value at the same reflection centre is doing both jobs, in one case as a tail coefficient and in the other as a logarithmic coefficient.

Remark R.23 (The square root is always the quadratic model). All three square roots in Table R.4, and the fourth one $\sqrt{\lambda^2 U^2 - 2\alpha U}$ of (Q.35), are instances of Theorem Q.19: truncate the normalized equation to its linear and quadratic terms in the displacement together with the lowest forcing monomial, and solve. What differs between the cases is only which *extraction* of the coefficient family the truncation computes — the deepest pole in the slope, or the coefficientwise large-slope limit — and that in turn is decided by whether the forcing monomial carries a factor of the slope, i.e. by whether it is resonant. Resonance and pole depth are thus two readings of one quadratic.

Remark R.24 (What Γ and G are missing). The K-function sources supply two devices the Γ/G sources do not: the absorption lemma Theorem R.11 and the r-Lambert anchor Theorem R.17. Both apply to G. The absorption is carried out in Theorem R.15. The r-Lambert anchor does *not* apply to G as it stands, because Theorem R.17 requires the core to be $ax^p \log x + b_* x^p + c_* \log x + d_*$ and the Barnes linear term $\alpha_* z$ is of degree $p-1$, outside that shape; absorbing it would need a two-parameter relative of W for which no closed theory is invoked here. Neither device applies to Γ, which has $p = 1$ and no resonant term at all — for Γ the logarithmic coefficient $\frac{1}{2}B_2(h)$ at $h = \frac{1}{2}$ enters the *tail*, not the phase. So the three subjects are not three copies of one calculation: they are three different positions of a single obstruction, and the K-function is the one where it sits in the phase and can be removed.

R.8 What is left open

1. *A two-parameter anchor.* Theorem R.24 identifies the gap: a core of the shape $ax^p \log x + b_* x^p + e_* x^{p-1} + c_* \log x + d_*$, which would cover Barnes G exactly as $W^{(\varrho)}$ covers K. Whether its inverse is expressible through a named function, or whether the right statement is simply that the implicit equation defines one, is not settled here.

2. *Closed forms.* As in Section Q.13, the recurrences compute V_n, U_n and b_n to any order and their extreme coefficients are known — (R.20) and (R.44) — but no closed description of the full numerators is available. The refined blocks are the most promising, since (R.44) says their leading behaviour is exactly the forward tail with a sign, so the question is the structure of the correction to that.

3. *Hyperasymptotics.* Unchanged from Theorem Q.45. The K-function tail c_m inherits the factorial growth of B_{2m+2}, so the least term of (R.8) is near $m \approx \pi r$ and the inverse floor is $\exp(-2\pi\rho + \mathcal{O}(\log \rho))/(\rho L)$ by the argument of Theorem Q.44. The Stokes multipliers are not computed.

4. *The other real branches.* Theorem R.2 locates the two stationary points and hence the additional real branches below $K(x_{\max})$. Those branches are governed by the local behaviour at a critical point — a square-root, not a Lambert, scale — and are not treated.

5. *Formalization.* Most of both chapters remains unformalized. The Catalan identity of Theorem Q.20 is Exact in Lean, and the recurrence check for Theorem Q.19 is Partial at the boundary stated in Theorem Q.21; neither promotes the chapter-level inverse results. The r-Lambert function would need its own development.

Appendix S

The subfactorial

S.1 What this chapter does

The derangement numbers $D_n = !n = n! \sum_{j \le n} (-1)^j / j!$ are the first subject in this volume whose inverse is *not* governed by its own dominant block. The block is $\Gamma(x+1)/\mathrm{e}$, whose inverse is the subject of Section Q.5; everything that is specifically about derangements lives in a correction so small that it is beyond every fixed exponential scale relative to the inverse variable. The chapter is therefore organized around that separation, and its principal economy is that the carrier is *cited*, not recomputed.

All three sources rederive the inverse of the gamma envelope from scratch, each proving its own Lambert-centred reversion theorem and each printing the blocks

$$\frac{1}{24s}, \quad -\frac{14s^2+10s+5}{5760s^3}, \quad \frac{2232s^4+1176s^3+714s^2+350s+105}{2903040s^5}, \quad \ldots$$

These are *identical*, coefficient for coefficient, to d_1, d_3, d_5, d_7 of Theorem Q.26 — three more copies of a computation the volume already contains twice over. They are not repeated here. Theorem S.10 states the dictionary and cites Theorem Q.25; the general reversion theorem each source proves in order to get them is Theorem J.21, and is cited too.

The three filed articles are `inverse_subfactorial_transseries` (S1), `inverse_subfactorial_transseries-2` (S2) and `inverse_subfactorial_transseries-3` (S3). Their distinctive contributions are: S1, the exact core–tail decomposition, the median interpolation, and the threshold analysis of Section S.9; S2, the signed-remainder Bell expansion with derivatives and the a posteriori error bound; S3, the branch-indexed continuation, the single-frequency sector factorization of Theorem S.14, and the multi-frequency grid. They agree on everything they share; every constant and coefficient was recomputed for this volume and reproduced.

S.2 Which inverse?

D_n is defined on $\mathbb{N}_0$, so Theorem J.41 applies: there is an integer staircase, there are interpolated inverses — one for each admissible interpolation — and there is an asymptotic inverse attached to no interpolation. The subfactorial adds a fourth issue that the earlier chapters did not face: its natural analytic continuation is *branch-dependent*, and different branches disagree away from the integers while agreeing on them. Section S.3 settles this before any inverse is taken.

S.3 The exact core–tail decomposition

Theorem S.1 (Branch splitting). *For $\Re z > -1$ put*

$$F(z) = \frac{\Gamma(z+1)}{\mathrm{e}}, \qquad C(z) = \frac{1}{\mathrm{e}}\int_0^1 t^z \mathrm{e}^t \,\mathrm{d}t = \frac{1}{\mathrm{e}}\sum_{j\geq 0}\frac{1}{j!\,(z+j+1)}, \tag{S.1}$$

and for $\kappa \in \mathbb{Z}$ set $\omega_\kappa = (2\kappa+1)\pi\mathrm{i}$. Then the continuation of $!z$ obtained by running the incomplete-gamma path from 0 to -1 with argument $(2\kappa+1)\pi$ is

$$\mathcal{S}_\kappa(z) = F(z) + \mathrm{e}^{\omega_\kappa z} C(z), \tag{S.2}$$

and $\mathcal{S}_\kappa(n) = D_n$ for every $n \in \mathbb{N}_0$ and every κ.

Proof. Inclusion–exclusion gives $D_n = n!\sum_{j\leq n}(-1)^j/j!$, and the upper incomplete gamma identity $\Gamma(n+1,w) = n!\,\mathrm{e}^{-w}\sum_{j\leq n} w^j/j!$ at $w=-1$ gives $D_n = \Gamma(n+1,-1)/\mathrm{e}$. Continue: with $s = z+1$ and the path $t = \varrho \mathrm{e}^{\mathrm{i}(2\kappa+1)\pi}$, $0 \leq \varrho \leq 1$, on which $\mathrm{e}^{\mathrm{i}(2\kappa+1)\pi} = -1$,

$$\gamma(s,-1) = \mathrm{e}^{\mathrm{i}(2\kappa+1)\pi s}\int_0^1 \varrho^{s-1}\mathrm{e}^{\varrho}\,\mathrm{d}\varrho,$$

so that $\Gamma(s,-1) = \Gamma(s) - \gamma(s,-1)$ yields

$$\frac{\Gamma(z+1,-1)}{\mathrm{e}} = F(z) - \frac{\mathrm{e}^{\mathrm{i}(2\kappa+1)\pi(z+1)}}{\mathrm{e}}\int_0^1 \varrho^z \mathrm{e}^{\varrho}\,\mathrm{d}\varrho = F(z) + \mathrm{e}^{\omega_\kappa z}C(z),$$

the sign coming from $\mathrm{e}^{\mathrm{i}(2\kappa+1)\pi} = -1$. Expanding e^t under the integral and integrating termwise gives the partial-fraction form in (S.1). At $z = n$ the phase $\mathrm{e}^{\omega_\kappa n}$ equals $(-1)^n$ for every κ, so all branches agree there and equal $n!/\mathrm{e} + (-1)^n C(n)$, which is D_n. □

Corollary S.2 (The nearest-integer property). *For every $n \geq 0$, $D_n = n!/\mathrm{e} + (-1)^n C(n)$, and*

$$0 < C(x) < \frac{1}{x+1} \qquad (x > -1). \tag{S.3}$$

Hence for $n \geq 1$ the correction is less than $\frac{1}{2}$ in absolute value and D_n is the nearest integer to $n!/\mathrm{e}$.

Proof. The identity is Theorem S.1 at $z = n$. For the bound, $C(x) = \int_0^1 \mathrm{e}^{t-1}t^x \mathrm{d}t$ and $0 < \mathrm{e}^{t-1} < 1$ on $[0,1)$, so $0 < C(x) < \int_0^1 t^x \mathrm{d}t = 1/(x+1)$. For $n \geq 1$ this is at most $\frac{1}{2}$, with equality impossible by strictness. □

Remark S.3 (The nearest-integer property needs no continuation). Theorem S.2 is stated here as a corollary of Theorem S.1, that is, of continuing the incomplete gamma function along a ray through $\mathrm{e}^{\mathrm{i}(2\kappa+1)\pi}$. That route is convenient once the branch family is in hand, but it is not necessary, and it is worth recording the cheaper argument because it is the one that formalizes.

Both sides of $D_n = n!/\mathrm{e}+(-1)^n C(n)$ satisfy the same first-order recursion. On the right, integrating $\mathrm{d}(t^{n+1}\mathrm{e}^{t-1})$ over $[0,1]$ gives

$$C(n+1) + (n+1)C(n) = 1, \tag{S.4}$$

which is one application of the fundamental theorem of calculus; on the left, $D_{n+1} = (n+1)D_n + (-1)^{n+1}$ is the classical recursion. The two agree at $n = 0$, where $D_0 = 1$ and $C(0) = 1 - 1/\mathrm{e}$. So the identity is an induction, and no complex analysis — no continuation, no branch family, no incomplete gamma function — enters at all. What the branch-splitting theorem adds is the whole family $\mathcal{S}_\kappa$ and the interpolations built from it; for the nearest-integer property alone it is more than is needed.

This cheaper route is machine-checked, in `DerangementNearestInteger.lean`. `Fabius.subfactorialDefect` is $C(n)$ as the interval integral $\int_0^1 \mathrm{e}^{t-1} t^n \mathrm{d}t$; `Fabius.subfactorialDefect_succ` is (S.4); `Fabius.numDerangements_sub_eq` is the identity $D_n - n!\mathrm{e}^{-1} = (-1)^n C(n)$; `Fabius.subfactorialDefect_pos` and `Fabius.subfactorialDefect_lt` are the two halves of (S.3) at integer argument; `Fabius.abs_numDerangements_sub_lt_half` is the bound $|D_n - n!/\mathrm{e}| < \frac{1}{2}$ for $n \geq 1$; and `Fabius.round_factorial_mul_exp_neg_one` is the nearest-integer property itself, $\mathrm{round}(n!/\mathrm{e}) = D_n$. Mathlib knows the derangement numbers and that $D_n/n! \to \mathrm{e}^{-1}$, but carries no quantitative form of that limit and no nearest-integer statement, so this is new to the library as well as to the corpus.

Two things in Theorem S.2 are *not* formalized. The bound (S.3) is proved only at integer argument, because the formalization uses the monoid power t^n rather than t^x; the real-x statement needs the real power and is otherwise the same three lines. And Theorem S.1 itself, which produces $\mathcal{S}_\kappa$ and hence Theorem S.4, is not formalized — only the numerical consequence drawn from it here.

Definition S.4 (Two real interpolations)**.** The branches $\kappa = 0$ and $\kappa = -1$ are complex conjugate on the real axis; their mean is the *median* (or symmetrized) interpolation

$$\mathcal{S}_{\mathrm{med}}(x) = F(x) + C(x)\cos(\pi x), \tag{S.5}$$

which is real, agrees with D_n at every $n \in \mathbb{N}_0$, and is entire. The *parity envelopes* are

$$G_\sigma(x) = F(x) + \sigma C(x), \qquad \sigma \in \{+1, -1\}, \tag{S.6}$$

so that $G_+(n) = D_n$ for even n and $G_-(n) = D_n$ for odd n.

Remark S.5 (The median is canonical, not unique)*.* Any entire function vanishing on $\mathbb{N}_0$ may be added to $\mathcal{S}_{\mathrm{med}}$ without disturbing the interpolation property, so (S.5) is not determined by interpolation alone. What distinguishes it is that it is the median of the two *nearest* lateral determinations of the same incomplete-gamma continuation, so it inherits their asymptotics rather than importing new ones. This is the same situation as Part N, where no canonical real inverse of the double factorial exists until an interpolation is fixed; the resolution here is cleaner only because the two candidate branches are conjugates.

Proposition S.6 (Eventual monotonicity)**.** *There is X such that $\mathcal{S}_{\mathrm{med}}$, G_+, G_- and F are all strictly increasing on $[X, \infty)$.*

Proof. $F'(x) = F(x)\psi(x+1)$, and $\psi(x+1) \sim \log x \to \infty$, so $F' > 0$ and $F' \to \infty$ eventually. From (S.3) and $C'(x) = \int_0^1 \mathrm{e}^{t-1} t^x \log t \,\mathrm{d}t$ we get $C(x) = \mathcal{O}(x^{-1})$ and $|C'(x)| = \mathcal{O}(x^{-2})$ — for the latter, $|\log t|\, t^x$ integrates to $(x+1)^{-2}$ on $[0,1]$. Hence

$$\left|\frac{\mathrm{d}}{\mathrm{d}x}\big(C(x)\cos \pi x\big)\right| \leq |C'(x)| + \pi |C(x)| = \mathcal{O}(x^{-1}),$$

while $F'(x) \to \infty$; the same bound covers $\sigma C'$. So the envelope derivative dominates. □

S.4 The Bell expansion of the tail

C is a Stieltjes transform, and its moments are Bell numbers. This gives the tail expansion together with an exact remainder — one of the few places in this volume where a divergent expansion comes with a two-sided sign statement.

Theorem S.7 (Bell expansion with signed remainder)**.** *Let $\mathfrak{B}_m$ denote the Bell numbers, $\mathfrak{B}_1 = 1$, $\mathfrak{B}_2 = 2$, $\mathfrak{B}_3 = 5$, $\mathfrak{B}_4 = 15$, $\mathfrak{B}_5 = 52$, $\mathfrak{B}_6 = 203$. For every real $x > 0$ and integer $M \geq 1$,*

$$C(x) = \sum_{m=1}^{M} (-1)^{m-1} \frac{\mathfrak{B}_m}{x^m} + R_M(x), \qquad R_M(x) = \frac{(-1)^M}{\mathrm{e}\, x^M} \sum_{j \geq 0} \frac{(j+1)^M}{j!\,(x+j+1)}. \tag{S.7}$$

In particular $(-1)^M R_M(x) > 0$ *and*

$$|R_M(x)| \le \frac{\mathfrak{B}_{M+1}}{x^{M+1}}, \tag{S.8}$$

so the remainder has the sign of the first omitted term and is no larger than it. Consequently $C(x) \sim \sum_{m\ge1}(-1)^{m-1}\mathfrak{B}_m x^{-m}$*, an expansion that diverges for every* x*.*

Proof. For $b > 0$ the finite geometric identity

$$\frac{1}{x+b} = \sum_{m=1}^{M}(-1)^{m-1}\frac{b^{m-1}}{x^m} + (-1)^M \frac{b^M}{x^M(x+b)}$$

is exact. Put $b = j+1$, divide by $\mathrm{e}\, j!$, and sum over $j \ge 0$; the partial-fraction form of C in (S.1) gives the left side. The m-th coefficient is

$$\frac{1}{\mathrm{e}}\sum_{j\ge0}\frac{(j+1)^{m-1}}{j!} = \mathfrak{B}_m, \tag{S.9}$$

which is Dobiński's formula combined with the Bell recurrence $\mathfrak{B}_m = \sum_k \binom{m-1}{k}\mathfrak{B}_k$: expanding $(j+1)^{m-1}$ binomially and applying Dobiński's $\mathfrak{B}_k = \mathrm{e}^{-1}\sum_j j^k/j!$ to each power gives exactly that sum. The last term of the geometric identity gives R_M, whose summand is positive, so $(-1)^M R_M > 0$. Finally $x + j + 1 \ge x$ gives

$$|R_M(x)| \le \frac{1}{\mathrm{e}x^{M+1}}\sum_{j\ge0}\frac{(j+1)^M}{j!} = \frac{\mathfrak{B}_{M+1}}{x^{M+1}},$$

again by (S.9). Divergence follows from $\mathfrak{B}_{2m} \ge (2m-1)!!$, so $\mathfrak{B}_m^{1/m} \to \infty$. □

Corollary S.8 (Derivatives)**.** *For every* $r \ge 0$*,*

$$C^{(r)}(x) = \frac{1}{\mathrm{e}}\int_0^1 t^x(\log t)^r \mathrm{e}^t\,\mathrm{d}t, \qquad (-1)^r C^{(r)}(x) > 0, \tag{S.10}$$

and termwise differentiation of the expansion is legitimate at every fixed order:

$$C^{(r)}(x) \sim \sum_{m\ge1}(-1)^{m+r-1}(m)_r\,\mathfrak{B}_m\,x^{-m-r}, \qquad (m)_r = m(m+1)\cdots(m+r-1). \tag{S.11}$$

In particular $C^{(r)}(x) = \mathcal{O}(x^{-r-1})$*.*

Proof. Differentiation under the integral sign is justified by dominated convergence on $[0,1]$, and $(\log t)^r$ has sign $(-1)^r$ there. For (S.11), differentiate the exact identity (S.7) r times: the finite sum differentiates term by term, and $R_M^{(r)}$ obeys the same kind of bound because differentiating the integral representation of R_M only inserts a bounded power of $\log t$. □

Remark S.9 (Numerical check)*.* The moment identity (S.9) was verified symbolically for $m \le 6$, giving $\mathfrak{B}_1, \ldots, \mathfrak{B}_6 = 1, 2, 5, 15, 52, 203$ as the coefficients in (S.7). The two representations of C in (S.1) were checked against each other to 30 digits at $x = 5, 10, 20$, and at each of those points the truncation after $\mathfrak{B}_5$ was confirmed to have an error of the predicted sign and within the bound (S.8) — for instance $|R_5(20)| = 2.616 \times 10^{-6}$ against the bound $\mathfrak{B}_6/20^6 = 3.172 \times 10^{-6}$. The general reversion coefficients of Theorem S.10 were recomputed and, specialized to the centred Stirling tail, reproduce d_1, d_3 of Theorem Q.26 exactly.

S.5 The carrier

Proposition S.10 (The carrier is Chapter 5's inverse). *Let $u(Y)$ denote the eventual inverse of $F(x) = \Gamma(x+1)/\mathrm{e}$, so that $\Gamma(u+1) = \mathrm{e}Y$. Then*

$$u(Y) = \Gamma_+^{-1}(\mathrm{e}Y) - 1, \tag{S.12}$$

and therefore, by Theorem Q.25, with

$$R = 1 + \log Y - \tfrac{1}{2}\log(2\pi), \qquad w = W_0(R/\mathrm{e}), \qquad Q = \frac{R}{w} = \mathrm{e}^{w+1}, \qquad s = w+1, \tag{S.13}$$

one has, for every fixed N,

$$u(Y) = -\frac{1}{2} + Q + \sum_{n=1}^{N} \frac{d_{2n-1}(s)}{Q^{2n-1}} + \mathcal{O}\left(\frac{1}{s\,Q^{2N+1}}\right), \tag{S.14}$$

with d_1, d_3, d_5, d_7 exactly as in Theorem Q.26. Moreover $u(Y) \asymp \log Y/\log\log Y$.

Proof. $F(x) = Y$ means $\Gamma(x+1) = \mathrm{e}Y$, i.e. $x + 1 = \Gamma_+^{-1}(\mathrm{e}Y)$, which is (S.12); the branch is the increasing one of Theorem Q.5, which exists eventually by Theorem S.6. Substituting $y = \mathrm{e}Y$ into (Q.39) gives $R = \log(\mathrm{e}Y) - \frac{1}{2}\log(2\pi) = 1 + \log Y - \frac{1}{2}\log(2\pi)$, and (Q.42) then reads $\Gamma_+^{-1}(\mathrm{e}Y) = \frac{1}{2} + Q + \sum d_{2n-1}Q^{1-2n} + \cdots$; subtracting 1 gives (S.14). The order statement follows from $Q = R/w$ with $w = W_0(R/\mathrm{e}) \sim \log R$. □

Remark S.11 (The shift, again). The $-\frac{1}{2}$ in (S.14) is the half-shift of Theorem Q.2 transported through (S.12): $\frac{1}{2} - 1 = -\frac{1}{2}$. It is what makes the correction series proceed in steps of Q^{-2}, and all three sources remark on it independently.

At $x = u(Y)$ the derivatives of the carrier are needed; write

$$\lambda = \lambda(u) = \psi(u+1), \qquad P_k(u) = \frac{F^{(k)}(u)}{F(u)}, \tag{S.15}$$

so that $F^{(k)}(u) = Y\,P_k(u)$ and $F'(u) = Y\lambda$. Differentiating $F' = F\lambda$ repeatedly gives

$$P_0 = 1, \qquad P_{k+1} = P_k' + \lambda P_k, \tag{S.16}$$

so $P_k = B_k(\lambda, \lambda', \ldots, \lambda^{(k-1)})$ is the complete exponential Bell polynomial of Theorem J.10; explicitly $P_1 = \lambda$, $P_2 = \lambda^2 + \lambda'$, $P_3 = \lambda^3 + 3\lambda\lambda' + \lambda''$. Since $\lambda = \psi(u+1) \sim \log u$ and $\lambda^{(j)} = \mathcal{O}(u^{-j})$ for $j \geq 1$, we have $P_k = \mathcal{O}\big((\log u)^k\big)$.

S.6 Inverting a perturbation of the carrier

Every inverse in this chapter solves $F(x) + E(x) = Y$ for a correction E that is small compared with F'. That is exactly Theorem J.21, and no separate reversion theorem is needed.

Proposition S.12 (The additive reversion in carrier coordinates). *Let E be analytic near $u = u(Y)$ with $E = o(F')$ there. The solution of $F(x) + E(x) = Y$ near u is*

$$x = u + \sum_{n\geq 1} \frac{(-1)^n}{n!}\mathscr{D}_u^{n-1}\left(\frac{E(u)^n}{F'(u)}\right), \qquad \mathscr{D}_u = \frac{1}{F'(u)}\frac{\mathrm{d}}{\mathrm{d}u} = \frac{1}{Y\lambda}\frac{\mathrm{d}}{\mathrm{d}u}, \tag{S.17}$$

convergent for $|E|$ small and, in the formal-jet sense, valid whenever $F'(u)$ is invertible.

Proof. This is Theorem J.21 with $F \mapsto F$, $E \mapsto E$, $\varepsilon \mapsto 1$, $\mu_0 \mapsto u$, $y \mapsto Y$ and $\mathcal{D} \mapsto \mathscr{D}_u$; the operator identification uses $F'(u) = Y\lambda$ from (S.15). The formal version is Theorem J.22. □

Remark S.13 (Why the amplitude series converges here). Theorem J.21 gives convergence for $|\varepsilon|$ below some ε_0, and one wants $\varepsilon = 1$. For the subfactorial that is legitimate, and for a stronger reason than in Part O: in the coordinate $v = F(x)$ the map $v \mapsto Y - \varepsilon\widetilde{E}(v)$, with $\widetilde{E} = E \circ F^{-1}$, has

$$\widetilde{E}'(Y) = \frac{E'(u)}{F'(u)} = \mathcal{O}\left(\frac{1}{Y\, u \log u}\right), \tag{S.18}$$

by Theorem S.8 and $F'(u) = Y\lambda$. So for each sufficiently large fixed Y the map is a contraction on a small disc for *all* $|\varepsilon| \le 1$, and the Lagrange series converges at $\varepsilon = 1$. What remains merely asymptotic is the subsequent expansion of the coefficients in powers of u^{-1} and λ^{-1}; the sector expansion in Y^{-1} converges while the Bell and Stirling expansions inside each coefficient diverge. This is a genuinely different situation from every earlier chapter and is worth stating plainly.

S.7 Sector factorization

The correction is not merely small: it carries an exponential factor $\mathrm{e}^{\omega z}$, and that factor organizes the whole inverse.

Theorem S.14 (Single-frequency sector factorization)**.** *Let $E(x) = \mathrm{e}^{\omega x}C(x)$ with ω fixed and C as in* (S.1)*. Then the solution of $F(x) + E(x) = Y$ has the formal expansion*

$$x \sim u(Y) + \sum_{n\ge 1} \frac{\mathrm{e}^{n\omega u(Y)}}{Y^n}\, \Phi_n^{(\omega)}\big(u(Y)\big), \tag{S.19}$$

where, writing

$$\mathcal{L}_{n,r}^{(\omega)} H = \frac{H' + (n\omega - r\lambda)H}{\lambda}, \tag{S.20}$$

the coefficients are the finite operator products

$$\Phi_n^{(\omega)} = \frac{(-1)^n}{n!}\left(\mathcal{L}_{n,n-1}^{(\omega)} \circ \cdots \circ \mathcal{L}_{n,1}^{(\omega)}\right)\left(\frac{C^n}{\lambda}\right), \tag{S.21}$$

the empty product being the identity. Explicitly

$$\Phi_1^{(\omega)} = -\frac{C}{\lambda}, \tag{S.22}$$

$$\Phi_2^{(\omega)} = \frac{C\, D_\omega C}{\lambda^2} - \frac{C^2 P_2}{2\lambda^3}, \qquad D_\omega := \frac{\mathrm{d}}{\mathrm{d}u} + \omega, \tag{S.23}$$

$$\Phi_3^{(\omega)} = -\frac{C(D_\omega C)^2}{\lambda^3} - \frac{C^2 D_\omega^2 C}{2\lambda^3} + \frac{3C^2(D_\omega C)P_2}{2\lambda^4} + \frac{C^3 P_3}{6\lambda^4} - \frac{C^3 P_2^2}{2\lambda^5}. \tag{S.24}$$

Moreover

$$\Phi_n^{(\omega)}(u) = \mathcal{O}\left(\frac{u^{-n}}{\log u}\right), \tag{S.25}$$

so the grading by powers of Y^{-1} is asymptotically strict even when ω is purely imaginary.

Proof. Start from (S.17). Its n-th summand begins with

$$\frac{E(u)^n}{F'(u)} = \frac{\mathrm{e}^{n\omega u} C(u)^n}{Y\lambda(u)},$$

and the key observation is that $\mathscr{D}_u$ maps objects of this shape to objects of the same shape. Let H be any function of u and $r \geq 1$. Along the differentiation Y is *not* constant — it is $F(u)$ — so $\mathrm{d}(Y^{-r})/\mathrm{d}u = -rY^{-r-1}F'(u) = -rY^{-r}\lambda$, and therefore

$$\mathscr{D}_u\left(\frac{\mathrm{e}^{n\omega u}H(u)}{Y^r}\right) = \frac{1}{Y\lambda}\left(\frac{\mathrm{e}^{n\omega u}\big(H' + n\omega H\big)}{Y^r} - \frac{r\lambda\,\mathrm{e}^{n\omega u}H}{Y^r}\right) = \frac{\mathrm{e}^{n\omega u}}{Y^{r+1}}\cdot\frac{H' + (n\omega - r\lambda)H}{\lambda} = \frac{\mathrm{e}^{n\omega u}}{Y^{r+1}}\,\mathcal{L}^{(\omega)}_{n,r}H. \tag{S.26}$$

Applying this $n-1$ times to $H = C^n/\lambda$ starting at $r = 1$ gives (S.21), and the prefactor $(-1)^n/n!$ is the one in (S.17). The displayed Φ_1, Φ_2, Φ_3 follow by expanding the products and using $P_2 = \lambda^2 + \lambda'$, $P_3 = \lambda^3 + 3\lambda\lambda' + \lambda''$.

For (S.25): by Theorem S.8, $C^{(j)} = \mathcal{O}(u^{-j-1})$, so $C^n = \mathcal{O}(u^{-n})$; $\lambda \sim \log u$ and $P_k = \mathcal{O}((\log u)^k)$; and each application of $\mathcal{L}^{(\omega)}_{n,r}$ divides by λ while either differentiating (which lowers the power of u but not below u^{-n} overall, since the leading behaviour is set by C^n) or multiplying by the bounded quantity $n\omega - r\lambda$, whose size $\mathcal{O}(\log u)$ cancels the division. Induction gives (S.25). □

Corollary S.15 (Multi-frequency grids). *If $E(x) = \sum_{\alpha=1}^{M} \mathrm{e}^{\omega_\alpha x}C_\alpha(x)$, then the inverse support lies on the grid $\big\{Y^{-|\boldsymbol{r}|}\mathrm{e}^{(\boldsymbol{r}\cdot\boldsymbol{\omega})u} : \boldsymbol{r} \in \mathbb{N}^M \setminus \{\boldsymbol{0}\}\big\}$ and*

$$x \sim u(Y) + \sum_{\boldsymbol{r}\neq\boldsymbol{0}} Y^{-|\boldsymbol{r}|}\mathrm{e}^{(\boldsymbol{r}\cdot\boldsymbol{\omega})u(Y)}\Phi_{\boldsymbol{r}}(u(Y)), \tag{S.27}$$

ordered first by total degree $|\boldsymbol{r}|$. Purely imaginary frequencies may give equal magnitudes or resonant phases, but the factor $Y^{-|\boldsymbol{r}|}$ keeps the total-degree grading well founded.

Proof. Expand $E(u)^n$ in (S.17) by the multinomial theorem and apply (S.26) to each resulting monomial; the frequency $\boldsymbol{r}\cdot\boldsymbol{\omega}$ and the power $Y^{-|\boldsymbol{r}|}$ are preserved by $\mathscr{D}_u$, as the computation there shows. □

S.8 The inverses

Theorem S.16 (Branch inverse). *Fix $\kappa \in \mathbb{Z}$ and let $z_\kappa(Y)$ be the solution of $\mathcal{S}_\kappa(z) = Y$ with $z_\kappa(Y) - u(Y) \to 0$. Then, for every fixed N,*

$$z_\kappa(Y) = u(Y) + \sum_{n=1}^{N} \frac{\mathrm{e}^{n\omega_\kappa u(Y)}}{Y^n}\Phi_n^{(\omega_\kappa)}(u(Y)) + \mathcal{O}\left(\frac{Y^{-N-1}u^{-N-1}}{\log u}\right), \tag{S.28}$$

with Φ_n from (S.21) *and $\omega_\kappa = (2\kappa+1)\pi\mathrm{i}$. In particular the leading displacement is*

$$z_\kappa(Y) - u(Y) = -\frac{\mathrm{e}^{\omega_\kappa u}C(u)}{Y\lambda(u)}\big(1 + o(1)\big), \tag{S.29}$$

which is $\mathcal{O}\big(1/(Y\,u\log u)\big)$ in modulus.

Proof. Existence and uniqueness near u: by Theorem S.1, $\mathcal{S}_\kappa(u) - Y = \mathrm{e}^{\omega_\kappa u}C(u) = \mathcal{O}(u^{-1})$ while $\mathcal{S}_\kappa'(u) = Y\lambda + \mathcal{O}(u^{-1})$, so Theorem J.47 places a root within $\mathcal{O}\big(1/(Y\lambda u)\big)$ of u and Theorem S.13 makes it unique in a disc of that size. The expansion is Theorem S.14 with $\omega = \omega_\kappa$, and the remainder is (S.25) applied to the first omitted sector together with the geometric domination of the convergent tail from (S.18). □

Theorem S.17 (The real median inverse). *Let $\rho(Y) = \mathcal{S}_{\mathrm{med}}^{-1}(Y)$ be the eventual real inverse of* (S.5), *and put $H(u) = C(u)\cos(\pi u)$. Then*

$$\rho(Y) \sim u(Y) + \sum_{n\geq1} \frac{(-1)^n}{n!}\mathscr{D}_u^{n-1}\left(\frac{H(u)^n}{Y\lambda}\right), \tag{S.30}$$

whose n-th term contains the harmonics $\mathrm{e}^{\mathrm{i}m\pi u}$ *with* $m = -n, -n+2, \dots, n$. *Explicitly,*

$$\rho(Y) = u - \frac{H}{Y\lambda} + \frac{1}{Y^2}\left[\frac{HH'}{\lambda^2} - \frac{H^2(\lambda^2+\lambda')}{2\lambda^3}\right] + \mathcal{O}\left(\frac{Y^{-3}u^{-3}}{\log u}\right), \tag{S.31}$$

and the leading displacement is

$$\rho(Y) - u(Y) \sim -\frac{\cos\bigl(\pi u(Y)\bigr)}{Y\,u(Y)\log u(Y)}. \tag{S.32}$$

At every derangement ordinate, $\rho(D_n) = n$ *exactly.*

Proof. Apply Theorem S.12 with $E = H$; (S.30) is (S.17). Writing $\cos\pi u = \frac{1}{2}(\mathrm{e}^{\mathrm{i}\pi u} + \mathrm{e}^{-\mathrm{i}\pi u})$ and expanding H^n shows that the n-th term is a combination of $\mathrm{e}^{\mathrm{i}m\pi u}$ with $m \equiv n \pmod 2$ and $|m| \le n$, which is Theorem S.15 for the two frequencies $\pm\pi\mathrm{i}$. For (S.32), $\Phi_1 = -H/\lambda$ with $C(u) \sim 1/u$ from Theorem S.7 and $\lambda \sim \log u$. The interpolation statement is $\mathcal{S}_{\mathrm{med}}(n) = D_n$ from Theorem S.4. □

Corollary S.18 (Parity envelopes). *For* $\sigma = \pm 1$ *the eventual inverse of* $G_\sigma = F + \sigma C$ *is*

$$x_\sigma(Y) \sim u(Y) + \sum_{n\ge 1} \frac{\sigma^n}{Y^n}\Phi_n^{(0)}(u(Y)), \tag{S.33}$$

which is Theorem S.14 at $\omega = 0$ *— no oscillation, and the coefficients are the same operator products with* $D_0 = \mathrm{d}/\mathrm{d}u$. *In particular* $x_\sigma(Y) = u - \sigma C/(Y\lambda) + \mathcal{O}\bigl(Y^{-2}u^{-2}/\log u\bigr)$.

Proof. Immediate from Theorem S.14 with $\omega = 0$ and the amplitude σ, which enters as σ^n through E^n. □

S.9 The staircase and the thresholds

The discrete inverse is governed by Theorem J.43, and the only subject-specific input is where the jumps sit.

Definition S.19 (Thresholds). For $n \ge 2$ let α_n be defined by $F(\alpha_n) = D_n$, that is $\alpha_n = u(D_n)$. Since F is increasing, the generalized inverse of the sequence (D_n) is

$$N_*(y) = \min\{n \ge 2 : u(y) \le \alpha_n\}. \tag{S.34}$$

Theorem S.20 (Threshold displacement). *Write* $\varsigma_n = (-1)^n$, $J_n = C(n)$ *and* $\Pi_n = \psi(n+1)$. *Then*

$$\alpha_n - n = \frac{\varsigma_n J_n}{F'(n)} - \frac{F''(n)J_n^2}{2F'(n)^3} + \frac{\varsigma_n\bigl(3F''(n)^2 - F'(n)F'''(n)\bigr)J_n^3}{6F'(n)^5} + \mathcal{O}\left(\frac{J_n^4}{F(n)^4\Pi_n}\right), \tag{S.35}$$

and in particular

$$\alpha_n = n + (-1)^n\frac{\mathrm{e}\,C(n)}{n!\,\psi(n+1)} + \mathcal{O}\left(\frac{1}{(n!)^2 n^2\log n}\right) = n + (-1)^n\frac{\mathrm{e}}{n!\,(H_n-\gamma)}\left(\frac{1}{n} - \frac{2}{n^2} + \frac{5}{n^3} - \cdots\right). \tag{S.36}$$

Consequently $|\alpha_n - n|$ *is factorially small, and for every* $n \ge 2$ *the separation condition* (J.75) *holds with room to spare.*

Proof. By Theorem S.2, $D_n = F(n) + \varsigma_n J_n$, so α_n is the value of the local inverse of F at $F(n) + \varsigma_n J_n$. Taylor expanding that inverse, with

$$(F^{-1})' = \frac{1}{F'}, \quad (F^{-1})'' = -\frac{F''}{(F')^3}, \quad (F^{-1})''' = \frac{3(F'')^2 - F'F'''}{(F')^5},$$

and substituting the increment $\varsigma_n J_n$ gives (S.35). Now $F(n) = n!/\mathrm{e}$ and $F'(n) = F(n)\Pi_n$, so the first term is $\varsigma_n J_n \mathrm{e}/(n!\,\Pi_n)$, which with $\Pi_n = \psi(n+1) = H_n - \gamma$ and the Bell expansion of $J_n = C(n)$ from Theorem S.7 gives (S.36); the second term is $\mathcal{O}(J_n^2/(F(n)^2\Pi_n)) = \mathcal{O}\big((n!)^{-2}n^{-2}(\log n)^{-1}\big)$ because $F'' = F(\Pi^2 + \Pi')$ and $J_n = \mathcal{O}(n^{-1})$. Since $|\alpha_n - n| \le \mathrm{e}/(n!\,(H_n - \gamma)\,n)$ is far below $\frac{1}{2}$ for $n \ge 2$, Theorem J.43(ii) applies with any η up to $\frac{1}{3}$, say. □

Remark S.21 (What the staircase costs). Theorem S.20 says the discrete inverse is easy: the thresholds sit factorially close to the integers, so rounding $u(y)$ correctly needs only $|u(y) - \nu(y)| < \frac{1}{2}$, which (S.14) delivers already at $N = 0$. In this respect the subfactorial is far better behaved than the sequences of Parts M and O, where the interpolated inverse and the staircase separate only slowly. The reason is structural: the correction that distinguishes D_n from $n!/\mathrm{e}$ is the same correction that displaces the threshold, and it is factorially small in both roles.

S.10 The four scales

Proposition S.22 (Hierarchy). *As $Y \to +\infty$, with $u = u(Y)$, the inverse of any of the interpolations of Theorem S.4 decomposes into four strictly ordered layers:*

1. *the* Lambert–logarithmic *layer, the carrier Q of* (S.13), *of size u;*
2. *the* Stirling *layer, the blocks $d_{2n-1}(s)Q^{1-2n}$ of* (S.14), *of sizes $u^{-1}, u^{-3}, \ldots$ divided by powers of s;*
3. *the* hyperasymptotic gamma *layer, below the least term of the Stirling series, of size $u^{-1/2}\mathrm{e}^{-2\pi u}$ by Theorem Q.44;*
4. *the* Bell *layer, the genuinely subfactorial correction, of size*

$$\frac{1}{Y\,u\log u} = \exp\big(-u\log u + \mathcal{O}(u)\big). \tag{S.37}$$

Layer (iv) *is smaller than e^{-cu} for every fixed $c > 0$.*

Proof. Only the last statement needs an argument. By Theorem S.10, $Y = F(u) = \Gamma(u+1)/\mathrm{e}$, so $\log Y = u\log u - u + \mathcal{O}(\log u)$ by Stirling; hence $1/(Yu\log u) = \exp\big(-u\log u + u + \mathcal{O}(\log u)\big)$, which is (S.37). Since $u\log u - u \ge cu$ eventually for every fixed c, the claim follows. The ordering of (i)–(iii) is Theorem Q.41 transported by (S.12), and (iii) precedes (iv) because $\mathrm{e}^{-2\pi u}$ is larger than $\mathrm{e}^{-u\log u + \mathcal{O}(u)}$. □

Remark S.23 (The consequence for practice). Theorem S.22 says something uncomfortable and worth saying: for any Y at which one could actually compute, the subfactorial correction to the inverse is far below the accuracy that the Stirling series can deliver at all. Recovering it numerically requires either exact arithmetic on $\Gamma(u+1)/\mathrm{e}$ or a hyperasymptotic treatment of layer (iii). What the Bell sector *is* good for is exact statements at integers — Theorem S.2 and Theorem S.20 — where the correction is not competing with a divergent series but is the whole content.

S.11 What is left open

1. *Closed forms for Φ_n.* Equation (S.21) is a finite operator product, hence computable, but its expansion in the differential ring generated by C, λ and their derivatives has no closed description beyond $n = 3$. Because P_k is a complete Bell polynomial (S.16) and C has Bell-number coefficients, a double-Bell formula is plausible; it is not derived here.

2. *The hyperasymptotic gap.* Layer (iii) of Theorem S.22 separates the algebraic inverse from the Bell sector, and nothing in this chapter bridges it. A complete numerical theory of the real inverse below $\mathrm{e}^{-2\pi u}$ needs the exponentially improved inverse gamma of Theorem Q.45, which is itself open.

3. *Uniqueness of the real interpolation.* Theorem S.5: the median is canonical among the lateral determinations but not singled out by interpolation alone. A characterization — for instance by a growth or complete-monotonicity condition in the style of the Bohr–Mollerup theorem — is not attempted, and would be the right way to make "the" real inverse subfactorial well defined.

4. *Complex branches.* Theorem S.16 is stated for fixed κ along the positive real Y-axis. How the sheets z_κ are connected, and what happens when κ grows with Y, is not addressed.

Appendix T

A real-argument Fibonacci function

T.1 What this chapter does, and what it cannot use

This is the one subject in the volume with no Lambert core. Its dominant block is a pure exponential, φ^x, so the equation to be inverted is $e^{\lambda x} \approx R$ and the leading inverse is an honest logarithm — no W, no slow coefficient field, no $(v+1)^{-1}$. What replaces the apparatus of Section Q.3 is a different and, in one respect, much better structure: the inverse equation *factors exactly* into conjugate factors, the reversion is an ordinary analytic reversion at an ordinary point, and the resulting expansion *converges.*

The chapter is therefore organized around what carries over and what does not.

Applies unchanged	Does not apply
Theorem J.21 and its formal form: the correction is an analytic perturbation of an exactly invertible core.	Theorem J.2: the phase is λx, not ax^α with a subdominant $b \log x$; the model's $\log x$ slot is empty and its Q slot carries oscillation instead of a power series.
Theorem J.17 and Theorem J.18, in their multivariate Lagrange–Good form (T.8).	Theorem J.23 and Theorem Q.8: nothing needs inverting beyond an exponential.
Theorem J.47, for the residual certificate.	Theorem J.37: there is no W to unfold, and the expansion is already elementary.
Theorem J.41 and Theorem J.43: F_n is a sequence.	Theorem J.54: the expansion converges, so there is no least term and no beyond-all-orders floor.

The three filed articles are `Fibonacci_Inverse_LogPeriodic_Transseries` (V1), `fibonacci_inverse_transseries_article` (V2) and `fibonacci_inverse_transseries_article-2` (V3). V1 supplies the conjugate factorization and the Lagrange–Good closed formula; V2 the structural laws for the real blocks, the residual certificate and the convergence radius; V3 the general block-reversion formula for products of exponentials. They agree, and every constant, coefficient and checkpoint was recomputed for this volume (Theorem T.19).

T.2 The interpolation and its branches

Definition T.1 (The even Fibonacci modulus). Let

$$\varphi = \frac{1+\sqrt{5}}{2}, \qquad \lambda = 2\log\varphi, \qquad \omega = \pi, \qquad \rho = \frac{\omega}{\lambda}, \tag{T.1}$$

and define, for $x \in \mathbb{R}$,

$$\mathcal{F}(x) = \sqrt{\tfrac{2}{5}}\,\sqrt{\cosh(\lambda x) - \cos(\omega x)} \;\; \geq 0. \tag{T.2}$$

Numerically $\lambda = 0.9624236501192068949955\ldots$ and $\rho = 3.2642513026364969\ldots$.

Proposition T.2 (Interpolation, evenness, and the modulus form). *$\mathcal{F}$ is even, real, nonnegative, vanishes only at $x = 0$, and satisfies $\mathcal{F}(n) = F_n$ for every integer $n \geq 0$, where $F_0 = 0$, $F_1 = 1$, $F_{n+2} = F_{n+1} + F_n$. Equivalently*

$$\mathcal{F}(x)^2 = \tfrac{1}{5}\big(\mathrm{e}^{\lambda x} + \mathrm{e}^{-\lambda x} - 2\cos(\omega x)\big) \qquad \text{and} \qquad \mathcal{F}(x) = \frac{1}{\sqrt{5}}\big|\varphi^x - \mathrm{e}^{\mathrm{i}\pi x}\varphi^{-x}\big|. \tag{T.3}$$

Near the origin $\mathcal{F}(x) = \sqrt{(\lambda^2+\omega^2)/5}\;|x| + \mathcal{O}(|x|^3)$, so the interpolation has a cusp there.

Proof. Evenness and nonnegativity are clear from (T.2), and $\cosh(\lambda x) \geq 1 \geq \cos(\omega x)$ with equality only at $x = 0$. Expanding $\cosh$ gives the first identity in (T.3), and since $\mathrm{e}^{\lambda x} = \varphi^{2x}$ that expression is $\frac{1}{5}|\varphi^x - \mathrm{e}^{\mathrm{i}\pi x}\varphi^{-x}|^2$, which is the second. For the interpolation, Binet's formula $F_n = (\varphi^n - (-\varphi^{-1})^n)/\sqrt{5}$ gives

$$F_n^2 = \tfrac{1}{5}\big(\varphi^{2n} + \varphi^{-2n} - 2(-1)^n\big) = \tfrac{1}{5}\big(\mathrm{e}^{\lambda n} + \mathrm{e}^{-\lambda n} - 2\cos(\pi n)\big) = \mathcal{F}(n)^2,$$

and $F_n \geq 0$ for $n \geq 0$, so taking nonnegative roots gives $\mathcal{F}(n) = F_n$. The behaviour at the origin follows from $\cosh(\lambda x) - \cos(\omega x) = \frac{1}{2}(\lambda^2+\omega^2)x^2 + \mathcal{O}(x^4)$. □

Remark T.3 (Which extension this is not). $\mathcal{F}$ is even, so $\mathcal{F}(-n) = F_n$, whereas the usual signed convention has $F_{-n} = (-1)^{n+1}F_n$. The choice matters when the two outer inverse branches are named, and it is what makes the negative branch trivially $-X_+$.

Proposition T.4 (The outer branches). *Let*

$$x_* = \frac{1}{\lambda}\operatorname{arsinh}\Big(\frac{\omega}{\lambda}\Big) = 1.972992734136994\ldots. \tag{T.4}$$

Then $\mathcal{F}'(x) > 0$ for every $x > x_*$; in particular $\mathcal{F}$ maps $[2,\infty)$ strictly increasingly onto $[1,\infty)$. Write $X_+(y)$ for the inverse of that restriction and $X_-(y) = -X_+(y)$. Every sufficiently large y has exactly the two real preimages $X_\pm(y)$.

Proof. Differentiating (T.3),

$$\mathcal{F}'(x) = \frac{\lambda\sinh(\lambda x) + \omega\sin(\omega x)}{5\,\mathcal{F}(x)}, \tag{T.5}$$

so the sign is that of the numerator. Since $\sin(\omega x) \geq -1$, the numerator is at least $\lambda\sinh(\lambda x) - \omega$, which is positive exactly for $x > x_*$. As $x_* < 2$ and $\mathcal{F}(2) = F_2 = 1$ with $\mathcal{F} \to \infty$, the stated bijection follows. Finally $[0,2]$ is compact, so $\mathcal{F}$ is bounded there and every y above that bound has no preimage in $[0,2]$; with evenness this leaves exactly $X_\pm(y)$. □

Remark T.5 (Inner branches). $\mathcal{F}$ is not monotone on $(0,2)$: it has two critical points there, at $x = 1.132379068\ldots$ and $x = 1.719250886\ldots$, with values $1.013859885\ldots$ and $0.911172684\ldots$. So values slightly above 1 have extra preimages. These are governed by a simple critical point, where the local inverse is a Puiseux series in $\sqrt{y - y_{\text{crit}}}$ rather than anything in this chapter's scale; V2 treats them and we do not. They are irrelevant to the large-y problem.

T.3 Exact factorization

Everything in this chapter rests on one identity.

Theorem T.6 (Conjugate factorization). *Fix the outer branch, let $y \geq 1$, and put*

$$R = 5y^2, \qquad u = \mathrm{e}^{-\lambda x} \in (0,1), \qquad \alpha = 1 - \mathrm{i}\rho, \qquad \bar{\alpha} = 1 + \mathrm{i}\rho. \tag{T.6}$$

Then $\mathcal{F}(x) = y$ is exactly *equivalent to*

$$R\,u = (1 - u^{\alpha})(1 - u^{\bar{\alpha}}), \tag{T.7}$$

where $u^{\alpha} := \mathrm{e}^{\alpha\log u}$ with the real logarithm of $u \in (0,1)$.

Proof. By (T.3), $\mathcal{F}(x) = y$ is $R = \mathrm{e}^{\lambda x} + \mathrm{e}^{-\lambda x} - 2\cos(\omega x)$. With $u = \mathrm{e}^{-\lambda x}$ we have $\log u = -\lambda x$, so $\mathrm{e}^{\mathrm{i}\omega x} = \mathrm{e}^{-\mathrm{i}\rho \log u} = u^{-\mathrm{i}\rho}$ and likewise $\mathrm{e}^{-\mathrm{i}\omega x} = u^{\mathrm{i}\rho}$; hence $2\cos(\omega x) = u^{\mathrm{i}\rho} + u^{-\mathrm{i}\rho}$ and

$$R = u^{-1} + u - u^{\mathrm{i}\rho} - u^{-\mathrm{i}\rho}.$$

Multiplying by u gives $Ru = 1 + u^2 - u^{1+\mathrm{i}\rho} - u^{1-\mathrm{i}\rho}$, and expanding the right side of (T.7),

$$(1 - u^{1-\mathrm{i}\rho})(1 - u^{1+\mathrm{i}\rho}) = 1 - u^{1-\mathrm{i}\rho} - u^{1+\mathrm{i}\rho} + u^2,$$

which is the same. □

Remark T.7 (Why this is the decisive step). Reverting (T.2) directly is possible but hides the structure: one is inverting a square root of a difference of two oscillating exponentials. After squaring and substituting $u = \mathrm{e}^{-\lambda x}$, the problem becomes the reversion of an *ordinary analytic germ at the origin* in the two variables u^α and $u^{\bar\alpha}$. That is why the answer converges and why a closed coefficient formula exists at all. It is also the precise sense in which this subject is easier than the other seven, not harder: the exponents are complex, but nothing is divergent.

T.4 The closed coefficient formula

Theorem T.8 (Lagrange–Good form of the outer inverse)**.** *For all sufficiently large y, with $R = 5y^2$,*

$$X_+(y) = \frac{\log R}{\lambda} + \sum_{\substack{j,k\geq 0\\ j+k\geq 1}} \frac{(-1)^{j+k+1}}{\lambda\, s_{jk}} \binom{s_{jk}}{j}\binom{s_{jk}}{k} R^{-s_{jk}}, \qquad s_{jk} = j + k - \mathrm{i}\rho(j-k), \tag{T.8}$$

the series converging absolutely. Truncating at $j + k \leq N$ leaves an error $\mathcal{O}(R^{-N-1}) = \mathcal{O}(y^{-2N-2})$. The sum is real, because the terms (j,k) and (k,j) are complex conjugates.

Proof. Write $u = \mathrm{e}^{-\lambda x}$, so $x = -\lambda^{-1}\log u$ and, by Theorem T.6, u is the small root of

$$u = \frac{(1-u^\alpha)(1-u^{\bar\alpha})}{R} =: \frac{1}{R}\,\Psi(u). \tag{T.9}$$

Set $v = R^{-1}$. Then $u = v\Psi(u)$ with Ψ analytic at $u = 0$ and $\Psi(0) = 1$; so $u = u(v)$ is the unique solution with $u(0) = 0$, analytic near $v = 0$, and Lagrange inversion (Theorem J.17 in its convergent analytic form) applies to the observable $G(u) = -\lambda^{-1}\log u$. Since G has a logarithmic singularity at 0 we use the standard variant: for $n \geq 1$,

$$[v^n]\,\big(G(u(v)) - G_{\mathrm{sing}}\big) = \frac{1}{n}[u^{n-1}]\,\big(G'(u)\,\Psi(u)^n\big) = -\frac{1}{\lambda n}[u^n]\,\Psi(u)^n,$$

using $G'(u)u = -1/\lambda$. Summing and adding the singular part $-\lambda^{-1}\log v = \lambda^{-1}\log R$ gives

$$X_+(y) = \frac{\log R}{\lambda} - \frac{1}{\lambda}\sum_{n\geq 1}\frac{1}{n}[u^n]\,\Psi(u)^n\,R^{-n}. \tag{T.10}$$

It remains to extract $[u^n]\,\Psi^n$. Here the exponents are not integers, and the clean way to say what "$[u^n]$" means is to treat u^α and $u^{\bar\alpha}$ as independent variables — which is the content of Good's multivariate form. Expanding

$$\Psi(u)^n = (1-u^\alpha)^n(1-u^{\bar\alpha})^n = \sum_{j,k\geq 0}(-1)^{j+k}\binom{n}{j}\binom{n}{k}u^{\alpha j + \bar\alpha k},$$

the monomial $u^{\alpha j+\bar{\alpha}k} = u^{j+k-\mathrm{i}\rho(j-k)} = u^{s_{jk}}$ contributes to the coefficient of u^n exactly when $n = s_{jk}$. Making that substitution in (T.10) — i.e. summing over the pairs (j,k) and setting $n = s_{jk}$ throughout — gives

$$X_+(y) = \frac{\log R}{\lambda} - \frac{1}{\lambda}\sum_{j+k\geq 1} \frac{(-1)^{j+k}}{s_{jk}}\binom{s_{jk}}{j}\binom{s_{jk}}{k} R^{-s_{jk}},$$

which is (T.8).

Convergence: $|R^{-s_{jk}}| = R^{-(j+k)}$ because $\Re s_{jk} = j+k$, and $\left|\binom{s_{jk}}{j}\binom{s_{jk}}{k}/s_{jk}\right|$ grows at most geometrically in $j+k$ — each binomial is a product of j resp. k factors of modulus at most $j+k+\rho|j-k|+1 \leq (1+\rho)(j+k)+1$, divided by $j!$ resp. $k!$, so the modulus is at most $\big((1+\rho)(j+k)+1\big)^{j+k}/(j!\,k!)$, and by $j!\,k! \geq ((j+k)/2)!^2/\binom{j+k}{j}$ and Stirling this is $\mathcal{O}(c^{j+k})$ for a constant c depending only on ρ. Hence the series converges absolutely for $R > c$, and the tail beyond $j+k = N$ is $\mathcal{O}(R^{-N-1})$.

Reality: $\overline{s_{jk}} = s_{kj}$ and $\overline{R^{-s_{jk}}} = R^{-s_{kj}}$ since R is real and positive, while $\binom{\bar{s}}{j} = \overline{\binom{s}{j}}$; so the (j,k) and (k,j) terms are conjugate and the total is real. □

Remark T.9 (On the coefficient extraction). The step "$u^{s_{jk}}$ contributes to $[u^n]$ when $n = s_{jk}$" deserves the caution V1 gives it. The function Ψ is not a power series in u with integer exponents; it is a series in the two monomials $u^\alpha, u^{\bar{\alpha}}$, whose real exponents are $j+k$ but whose imaginary parts are not zero. The correct formal setting is a Hahn series in the group generated by α and $\bar{\alpha}$, ordered by real part, and Lagrange inversion in that setting is Good's theorem applied to the two variables $u^\alpha, u^{\bar{\alpha}}$ with the diagonal specialization. The reordering used to reach (T.8) is legitimate because the series converges absolutely, which the proof establishes independently.

T.5 The real form: log-periodic sectors

Theorem T.10 (Real sectors)**.** *Put*

$$\xi = \frac{\log R}{\lambda} = \log_\varphi\big(\sqrt{5}\,y\big), \qquad \theta = \rho\log R = \pi\xi, \qquad q = R^{-1} = \frac{1}{5y^2}. \tag{T.11}$$

Then

$$X_+(y) = \xi + \sum_{n\geq 1} D_n(\theta)\,q^n, \tag{T.12}$$

where each D_n is a real trigonometric polynomial in θ whose harmonics all have the same parity as n and whose degree is at most n. The first three are

$$D_1(\theta) = \frac{2}{\lambda}\cos\theta, \tag{T.13}$$

$$D_2(\theta) = -\frac{1}{\lambda}\big(2+\cos 2\theta + 2\rho\sin 2\theta\big), \tag{T.14}$$

$$D_3(\theta) = \frac{1}{\lambda}\Big[(6-\rho^2)\cos\theta + 5\rho\sin\theta + \Big(\tfrac{2}{3} - 3\rho^2\Big)\cos 3\theta + 3\rho\sin 3\theta\Big]. \tag{T.15}$$

In particular the leading correction is

$$X_+(y) = \log_\varphi\big(\sqrt{5}\,y\big) + \frac{\cos\big(\pi\log_\varphi(\sqrt{5}\,y)\big)}{5\,(\log\varphi)\,y^2} + \mathcal{O}(y^{-4}). \tag{T.16}$$

Proof. Group the terms of (T.8) by $n = j+k$. Since $R^{-s_{jk}} = R^{-n}\mathrm{e}^{\mathrm{i}\rho(j-k)\log R} = q^n\mathrm{e}^{\mathrm{i}(j-k)\theta}$, the n-th group is q^n times a finite combination of $\mathrm{e}^{\mathrm{i}h\theta}$ with $h = j-k$ running over $-n, -n+2, \ldots, n$; so the harmonics have the parity of n and degree at most n, and by the conjugation of Theorem T.8 the combination is real. This is (T.12).

For $n = 1$ the pairs are $(1, 0)$ and $(0, 1)$ with $s = 1 \mp i\rho$; since $\binom{s}{1} = s$ and $\binom{s}{0} = 1$, each term is $(-1)^2\lambda^{-1}s^{-1} \cdot s \cdot R^{-s} = \lambda^{-1}R^{-s}$, and the two add to $\lambda^{-1}q\big(e^{i\theta} + e^{-i\theta}\big) = 2\lambda^{-1}q\cos\theta$, which is (T.13). Substituting $q = 1/(5y^2)$, $\lambda = 2\log\varphi$ and $\theta = \pi\xi$ gives (T.16). The coefficients (T.14) and (T.15) follow from the pairs with $j + k = 2$ and $j + k = 3$ by the same computation; they were also obtained independently and checked against (T.8) numerically (Theorem T.19). □

Remark T.11 (No limiting normalized coefficient). Every earlier chapter had a distinguished large-parameter limit of its coefficient family — the resonant limit of Theorem Q.35 and Theorem R.10, or the deepest pole of Theorem Q.22 — and in each case that limit summed to a square root. Here there is no such limit. The coefficients $D_n(\theta)$ do not converge as $y \to \infty$: $\theta = \pi\xi$ increases without bound and D_n is a nonconstant periodic function of it, so $D_n(\theta)$ oscillates forever between fixed bounds. This is not a defect of the expansion — (T.12) converges, which none of the others do — but it does mean that the resummation technique of Theorem Q.19 has no analogue here, and that statements like "the n-th coefficient tends to …" must be replaced by statements about the mean or the extremes of D_n over a period.

T.6 Convergence, and a residual certificate

Proposition T.12 (Residual certificate). *Let $\widehat{x} \geq 2$ and $y \geq 1$. Then*

$$\big|X_+(y) - \widehat{x}\big| \leq \frac{\big|\mathcal{F}(\widehat{x}) - y\big|}{\min\limits_{[\widehat{x}\wedge X_+,\ \widehat{x}\vee X_+]} \mathcal{F}'}, \qquad \mathcal{F}'(x) \ \geq\ \frac{\lambda\sinh(\lambda x) - \omega}{5\,\mathcal{F}(x)} > 0 \quad (x > x_*). \tag{T.17}$$

In particular, for $x \geq 2$ one may use $\mathcal{F}'(x) \geq \lambda\big(\mathcal{F}(x) - \frac{1}{\mathcal{F}(x)}\big)/2$ for $\mathcal{F}(x)$ large, so the inverse is well conditioned: a residual of size ε certifies an error $\mathcal{O}(\varepsilon/(\lambda y))$.

Proof. The first display is Theorem J.47 applied to $f = \mathcal{F}$ on the interval joining $\widehat{x}$ to the root, together with the lower bound on $\mathcal{F}'$ from (T.5) and $\sin \geq -1$, valid past x_* by Theorem T.4. For the conditioning, $5\mathcal{F}^2 = e^{\lambda x} + e^{-\lambda x} - 2\cos\omega x$ gives $e^{\lambda x} = 5\mathcal{F}^2/2 + \mathcal{O}(1)$, hence $\lambda\sinh(\lambda x) = \lambda e^{\lambda x}/2 + \mathcal{O}(1) = \mathcal{O}(\lambda\mathcal{F}^2)$ and $\mathcal{F}' \asymp \lambda\mathcal{F}$; dividing gives the stated error scale. □

Remark T.13 (What is different here). Theorem T.8 is a *convergent* expansion of an analytic function, not a Poincaré asymptotic statement. There is no least term, no beyond-all-orders floor, and Theorem J.54 has nothing to say: one simply takes more terms. Every other chapter in this volume inverts a divergent Stirling- or Bernoulli-type series and must stop somewhere. The reason for the difference is visible in Theorem T.6: the forward function is, after squaring, an *entire* combination of exponentials, with no asymptotic series anywhere in it.

T.7 Is the log-periodicity intrinsic?

The synthesis chapter asks this of every subject with an unexpected feature, and here it has a clean answer in two halves.

Proposition T.14 (The oscillation is forced; its shape is not). *Let $\mathcal{G}$ be any function that interpolates the Fibonacci numbers, $\mathcal{G}(n) = F_n$ for $n \geq 0$, and is eventually strictly increasing with an inverse $X_{\mathcal{G}}$. Then*

$$X_{\mathcal{G}}(F_n) = n = \log_\varphi\big(\sqrt{5}\,F_n\big) + \frac{(-1)^n}{\sqrt{5}\,\varphi^n\log\varphi\,F_n}\big(1 + \mathcal{O}(\varphi^{-2n})\big), \tag{T.18}$$

so along the sequence the displacement from the pure logarithm is necessarily *of size F_n^{-2} and* necessarily *alternates in sign. By contrast, the values of $X_{\mathcal{G}}$ between the F_n, and hence the harmonic content of the correction as a function of* $\log y$, *depend on $\mathcal{G}$ and are not determined by the sequence.*

Proof. Binet gives $\sqrt{5}F_n = \varphi^n - (-1)^n\varphi^{-n}$, so

$$\log_\varphi(\sqrt{5}F_n) = n + \log_\varphi\bigl(1 - (-1)^n\varphi^{-2n}\bigr) = n - \frac{(-1)^n\varphi^{-2n}}{\log\varphi} + \mathcal{O}(\varphi^{-4n}).$$

Rearranging and using $\varphi^{-2n} = \bigl(\sqrt{5}F_n\varphi^n\bigr)^{-1}\bigl(1+\mathcal{O}(\varphi^{-2n})\bigr)$ gives (T.18). Since $\mathcal{G}(n) = F_n$ forces $X_{\mathcal{G}}(F_n) = n$ for every interpolation, the displayed relation is a statement about the *sequence*, independent of $\mathcal{G}$. For the second half, let χ be any nonzero entire function vanishing on $\mathbb{N}_0$ and small enough that $\mathcal{G}+\chi$ is still eventually increasing; it interpolates the same numbers and its inverse differs from $X_{\mathcal{G}}$ between the integers. $\square$

Remark T.15 (Reading the answer). So the log-periodicity is intrinsic in the only sense that a statement about a *sequence* can be: the second Binet term $(-1)^n\varphi^{-n}$ is φ^{-2n} relative to the first, which is R^{-1}, and it alternates. Every interpolation must reproduce that. What $\mathcal{F}$ contributes is the specific choice $\cos(\pi x)$ — the minimal-growth real interpolation of $(-1)^n$ — and *that* is what makes the correction a pure cosine at first order rather than some other function with the same values at integers. An article that says "the inverse Fibonacci function is log-periodic" is therefore making a statement half about Fibonacci and half about its own choice of continuation; the volume records both halves.

T.8 The general mechanism: inverting a product of exponentials

The factorization of Theorem T.6 is an instance of a general shape, and the closed formula of Theorem T.8 generalizes with it.

Theorem T.16 (Inverse of a finite exponential product)**.** *Let $\lambda > 0$, let $\alpha_1,\ldots,\alpha_m$ have positive real part, and let $c_1,\ldots,c_m$ and $\beta_1,\ldots,\beta_m$ be complex. Suppose*

$$R = \mathrm{e}^{\lambda x}\prod_{r=1}^{m}\bigl(1 - c_r\mathrm{e}^{-\lambda\alpha_r x}\bigr)^{\beta_r}, \tag{T.19}$$

with the principal branch of each factor near 1*. Then for R large the solution with $x\to\infty$ is*

$$x = \frac{\log R}{\lambda} - \frac{1}{\lambda}\sum_{\boldsymbol{\jmath}\in\mathbb{N}^m\setminus\{\mathbf{0}\}}\frac{1}{s_{\boldsymbol{\jmath}}}\prod_{r=1}^{m}(-c_r)^{j_r}\binom{s_{\boldsymbol{\jmath}}\beta_r}{j_r}R^{-s_{\boldsymbol{\jmath}}}, \qquad s_{\boldsymbol{\jmath}} = \sum_{r=1}^{m} j_r\alpha_r, \tag{T.20}$$

the series converging absolutely for R large.

Proof. Put $u = \mathrm{e}^{-\lambda x}$, so that (T.19) reads $Ru = \Psi(u) := \prod_r(1 - c_r u^{\alpha_r})^{\beta_r}$, i.e. $u = v\Psi(u)$ with $v = R^{-1}$ and $\Psi(0) = 1$. As in Theorem T.8, Lagrange inversion for the observable $-\lambda^{-1}\log u$ gives

$$x = \frac{\log R}{\lambda} - \frac{1}{\lambda}\sum_{n\ge 1}\frac{1}{n}[u^n]\,\Psi(u)^n\,R^{-n},$$

and the generalized binomial theorem gives

$$\Psi(u)^n = \prod_r(1 - c_r u^{\alpha_r})^{n\beta_r} = \sum_{\boldsymbol{\jmath}\in\mathbb{N}^m}\prod_r(-c_r)^{j_r}\binom{n\beta_r}{j_r}\,u^{\sum_r j_r\alpha_r}.$$

The monomial $u^{s_{\boldsymbol{\jmath}}}$ contributes at $n = s_{\boldsymbol{\jmath}}$, which gives (T.20). Absolute convergence follows as before: $|R^{-s_{\boldsymbol{\jmath}}}| = R^{-\Re s_{\boldsymbol{\jmath}}}$ with $\Re s_{\boldsymbol{\jmath}} \ge \delta|\boldsymbol{\jmath}|$ for $\delta = \min_r\Re\alpha_r > 0$, while the binomial products grow at most geometrically in $|\boldsymbol{\jmath}|$. $\square$

Corollary T.17 (The Fibonacci case)**.** *Taking $m = 2$, $\alpha_1 = \alpha = 1 - \mathrm{i}\rho$, $\alpha_2 = \bar{\alpha}$, $c_1 = c_2 = \beta_1 = \beta_2 = 1$ in Theorem T.16 returns* (T.8)*, with $s_{(j,k)} = j\alpha + k\bar{\alpha} = j + k - \mathrm{i}\rho(j-k) = s_{jk}$ and $\prod_r(-c_r)^{j_r} = (-1)^{j+k}$.*

Proof. Substitute those values into (T.20). The exponent is $s_{(j,k)} = j\alpha + k\bar{\alpha} = j + k - \mathrm{i}\rho(j-k) = s_{jk}$; each $\binom{s\beta_r}{j_r}$ becomes $\binom{s_{jk}}{j}$ resp. $\binom{s_{jk}}{k}$ because $\beta_r = 1$; and $\prod_r (-c_r)^{j_r} = (-1)^{j+k}$, so the overall sign is $-(-1)^{j+k} = (-1)^{j+k+1}$, which is the sign in (T.8). □

Remark T.18 (Where the shape comes from, and where it stops). A forward function of the form (T.19) arises whenever a dominant exponential is multiplied by finitely many exponentially small corrections — which is exactly what a Binet-type formula with finitely many characteristic roots gives. Conjugate pairs $\alpha, \bar{\alpha}$ produce the log-periodicity, since R^{-s} then carries $\mathrm{e}^{\mathrm{i}\Im(s)\log R}$. Two boundaries are worth naming. If the correction is an *infinite* product or a general analytic germ, the closed formula is replaced by a Lagrange–Good expansion whose coefficients are no longer products of two binomials; V1 and V3 both develop that case. And if the carrier is mixed, $\mathrm{e}^{\lambda x}x^{\mu}$, then W reappears and one is back in the world of Section Q.3 — which is the precise point at which this chapter's subject rejoins the rest of the volume.

T.9 Numerical verification

Remark T.19 (What was checked). All of the following were recomputed for this volume at 50 digits, from the definitions and independently of the sources. The constants $\lambda = 0.9624236501192068949955\ldots$, $\rho = 3.2642513026364969\ldots$ and $x_* = 1.972992734136994\ldots$ of (T.1) and (T.4). The interpolation $\mathcal{F}(n) = F_n$ for $0 \le n \le 10$, to 40 digits. The integer checkpoints $X_+(2) = 3$, $X_+(5) = 5$ and $X_+(55) = 10$, each to 40 digits — these test the branch convention as well as the formula. The exact factorization (T.7), at $y = 3, 10, 100$, to 35 digits. The closed formula (T.8): its imaginary part vanishes to 54 digits or better, and its truncations converge at the predicted rate. And the agreement between (T.8) truncated at $j + k \le 3$ and the trigonometric sectors (T.13)–(T.15), which matched to the full working precision at $y = 100$ and $y = 1000$.

Table T.1: Absolute error of (T.12) truncated after D_N, against the root of $\mathcal{F}(x) = y$ computed to 50 digits.

y	$N = 0$ (i.e. ξ)	$N = 1$	$N = 2$	$N = 3$
10	5.4520×10^{-4}	1.1350×10^{-5}	1.5430×10^{-7}	2.1470×10^{-9}
100	3.0100×10^{-5}	3.5620×10^{-9}	2.9720×10^{-13}	5.7170×10^{-18}
1000	4.1410×10^{-7}	1.7020×10^{-13}	2.5800×10^{-19}	4.2100×10^{-25}

Each additional block gains a factor $R^{-1} = 1/(5y^2)$, as Theorem T.8 predicts: from $y = 100$ to $y = 1000$ the $N = 3$ error drops by about six orders of magnitude per block, and R^{N+1} times the error stays bounded across the table. Unlike every other error table in this volume, the columns would continue to improve indefinitely.

T.10 Comparison, and what is left open

Remark T.20 (Where this subject sits). Against the seven others: it is the only one whose inverse expansion converges; the only one with no Lambert core and no slow coefficient field; the only one whose coefficient family has no large-parameter limit (Theorem T.11); and the only one where the interesting structure — oscillation — comes from a *pair of complex conjugate exponents* rather than from a logarithm. What it shares with the rest is the organizing principle: invert a dominant block exactly, and revert around it. That is Theorem J.21, and it is the only piece of Chapter 0 this chapter genuinely needs.

1. *Closed forms for D_n.* Theorem T.10 gives the first three and a rule for the harmonics, but no closed formula for the coefficients of $\cos h\theta$ and $\sin h\theta$ inside D_n. They are finite explicit sums by (T.8); a generating function in θ is not derived.

2. *The exact radius.* The proof of Theorem T.8 gives convergence for R larger than some constant depending on ρ, obtained from a crude bound on the binomials. The sharp radius — equivalently, the nearest singularity of $v \mapsto u(v)$, which is where $\Psi'(u)v = 1$ — is not computed here; V1 states one and we did not verify it, so no value is quoted.

3. *Inner branches.* Theorem T.5: the preimages coming from the two critical points in $(1, 2)$ are Puiseux, not log-periodic, and are not treated.

4. *Complex branches.* Replacing $\log R$ by $\log R + 2\pi \mathrm{i} m$ gives formal sheets, but $\mathcal{F}$ is built from a modulus and is not analytic, so the continuation question is genuinely different from that of Section Q.13(iv) and is not addressed.

5. *Other interpolations.* Theorem T.14 shows the answer depends on the continuation. Which interpolation is canonical — by minimal growth, by complete monotonicity of some transform, or by agreement with the signed convention $F_{-n} = (-1)^{n+1} F_n$ rather than with evenness — is open, and is the same question left open for the subfactorial in Section S.11(iii) and for the double factorial in Part N.

Part XIV

Two classical sequences by their generating function

Appendix U

The Bell numbers: a Lambert saddle

U.1 What this chapter does

The Bell numbers $\mathfrak{B}_n$, counting set partitions, have exponential generating function $\exp(\mathrm{e}^z - 1)$ and a saddle point at

$$re^r = n, \qquad r = W_0(n). \tag{U.1}$$

That the saddle is a Lambert point is what places this subject in the present volume: r is the same object the Lambert parts study, and the natural transmonomial turns out to be $\mathrm{e}^{-r} = r/n$, which is the reciprocal of the Lambert scale.

The leading Moser–Wyman–de Bruijn approximation is classical. What the two filed sources supply, and what is kept here, is the data that turns it into a usable all-orders formula: a named asymptotic scale, a finite rule computing every coefficient, the algebraic structure of those coefficients, and a remainder theorem.

Two articles: `Bell_Number_Asymptotic_Transseries` (B1) and `Bell_Number_Transseries_Article` (B2). They agree; every coefficient either prints was recomputed for this volume against exact Bell numbers. B1 carries the finite partition rule and the explicit c_1, c_2, c_3; B2 the Stirling interface and the singulant ordering. Both flag their nonprincipal-saddle sections as formal, and that labelling is preserved.

U.2 The saddle expansion to all orders

Definition U.1 (Touchard cumulants). For $j \geq 1$ put

$$T_j(r) = \mathrm{e}^{-r}(r\partial_r)^j \mathrm{e}^r = \sum_{k=1}^{j} \left\{ {j \atop k} \right\} r^k, \qquad q_j(r) = \frac{T_j(r)}{r\,(1+r)^{j/2}}. \tag{U.2}$$

Lean status. Formal in `TouchardEulerOperator.lean`. The polynomial side of (U.2) is `Fabius.touchardPolynomial`, which has been in the corpus since the combinatorial-calculus campaign; what this module adds is the *analytic* characterization, which is the one the saddle-point argument actually uses, and the bridge between the two. The Euler operator $r\partial_r$ is `Fabius.eulerOp`, `Fabius.iterate_eulerOp_exp` identifies its j-th iterate on e^r as $\mathrm{e}^r T_j(r)$ — stated as an equality of functions, since that is what the induction on j requires — and `Fabius.exp_neg_mul_iterate_eulerOp_exp` is both halves of (U.2) at once, $\mathrm{e}^{-r}(r\partial_r)^j \mathrm{e}^r = \sum_k \left\{ {j \atop k} \right\} r^k$.

The whole content is one polynomial identity, `Fabius.touchardPolynomial_succ_eq_X_mul_add_derivative`: $T_{n+1} = X(T_n + T_n')$, which is the differential form of the Stirling recurrence $\left\{ {n+1 \atop k+1} \right\} = (k+1)\left\{ {n \atop k+1} \right\} + \left\{ {n \atop k} \right\}$. Given it, the analytic statement is an induction whose step is the product rule. The supporting coefficient identity is `Fabius.coeff_touchardPolynomial`, valid at every index rather than only below the degree.

Two points of care. The formal sum runs over $0 \le k \le j$, not $1 \le k \le j$, and in that form it is correct for *every* j including $j = 0$, where the volume's sum from $k = 1$ would be empty while $\mathrm{e}^{-r}\mathrm{e}^{r} = 1$; the volume's indexing is recovered for $j \ge 1$ by `Fabius.eval_touchardPolynomial_succ_eq_sum_Icc`, which drops the vanishing $k = 0$ term. And nothing in the analytic statement needs $r > 0$: it holds at every real r, the only hypothesis anywhere being the nonvanishing of the denominator in the normalized form. That normalized form is `Fabius.touchardQ`, with `Fabius.touchardQ_mul` the defining relation cleared of its denominator and `Fabius.touchardQ_one` the central value $q_j(1) = B_j/2^{j/2}$, which exhibits the normalization as the one that divides the Bell number by the Gaussian factor.

Lemma U.2 (Central and complementary arcs)**.** *Write $\Phi_r(\theta)$ for the exponent of the Cauchy integrand on the circle $|z| = r$, normalized so that $\Phi_r(0) = 0$. There are absolute constants $\delta, c_1, c_2 > 0$ such that, for all large n and $r = W_0(n)$,*

$$\Re\Phi_r(\theta) \le -c_1 n(1+r)\theta^2, \qquad |\theta| \le \delta/r, \tag{U.3}$$

$$\Re\Phi_r(\theta) \le -c_2\, n/r^2, \qquad \delta/r \le |\theta| \le \pi. \tag{U.4}$$

Consequently the complementary arc contributes

$$\int_{\delta/r \le |\theta| \le \pi} \mathrm{e}^{\Phi_r(\theta)}\mathrm{d}\theta = \mathcal{O}\big(\mathrm{e}^{-c_2 n/r^2}\big), \tag{U.5}$$

which is smaller than $(r/n)^M$ for every fixed M.

Proof. On $|z| = r$ the exponent is $\mathrm{e}^{r\cos\theta}\cos(r\sin\theta) - \mathrm{e}^{r} - \mathrm{i}n\theta$ up to the terms that cancel at $\theta = 0$; its real part is $\mathrm{e}^{r\cos\theta}\cos(r\sin\theta) - \mathrm{e}^{r}$. For $|\theta| \le \delta/r$ with δ small, expanding to second order gives $\Re\Phi_r(\theta) = -\frac{1}{2}\mathrm{e}^{r} r(1+r)\theta^2\big(1 + \mathcal{O}(\delta)\big)$, and $\mathrm{e}^{r} r = n$, which is (U.3). For $\delta/r \le |\theta| \le \pi$ the factor $\cos(r\sin\theta)$ is bounded away from 1 except near $\theta = 0$, while $\mathrm{e}^{r\cos\theta} \le \mathrm{e}^{r}$; a short computation — splitting at $|\theta| = \pi/2$ and using $\cos\theta \le 1 - 2\theta^2/\pi^2$ on $[0, \pi/2]$ — gives $\Re\Phi_r \le -c_2\mathrm{e}^{r}/r = -c_2 n/r^2$, which is (U.4). Integrating (U.4) over an arc of length at most 2π gives (U.5), and $\mathrm{e}^{-c_2 n/r^2}$ beats every power of $r/n = \mathrm{e}^{-r}$ because $n/r^2 = \mathrm{e}^{r}/r$ grows faster than any multiple of r. ☐

Theorem U.3 (All-orders saddle expansion)**.** *Let $r = W_0(n)$. For every fixed $M \ge 1$, as $n \to \infty$,*

$$\mathfrak{B}_n = \frac{n!\,\exp(\mathrm{e}^{r} - 1)}{r^n\sqrt{2\pi n(1+r)}}\left(\sum_{m=0}^{M-1}\frac{c_m(r)}{n^m} + \mathcal{O}_M\big((r/n)^M\big)\right), \tag{U.6}$$

where $c_0 = 1$ and, for $m \ge 1$,

$$c_m(r) = \sum_{\substack{k_1,\dots,k_{2m}\ge 0\\ \sum_s s\,k_s = 2m}} (-1)^{m+K}(2m+2K-1)!!\prod_{s=1}^{2m}\frac{q_{s+2}(r)^{k_s}}{\big((s+2)!\big)^{k_s}k_s!}, \qquad K = \sum_s k_s. \tag{U.7}$$

Equivalently, c_m is the Gaussian average of a complete exponential Bell polynomial in the q_j.

Proof. The Cauchy integral $\mathfrak{B}_n/n! = (2\pi\mathrm{i})^{-1}\oint\exp(\mathrm{e}^{z} - 1)z^{-n-1}\mathrm{d}z$ is put on the circle $|z| = r$ with r the saddle (U.1); writing $z = r\mathrm{e}^{\mathrm{i}\theta/\sqrt{n(1+r)}}$ normalizes the quadratic term, and the higher Taylor coefficients of the exponent are exactly the q_j of (U.2). Expanding $\exp(\text{cubic and higher})$ and integrating the resulting Gaussian moments term by term gives (U.7), the double factorial being the Gaussian moment $\mathbb{E}Z^{2m+2K} = (2m+2K-1)!!$.

Two steps make this an asymptotic statement rather than a formal one, and both are Theorem U.2. First, (U.5) discards the complementary arc at a cost smaller than every retained term, so the integral may be replaced by the central one. Second, (U.3) bounds the central integrand by a Gaussian, which justifies integrating the expansion term by term and bounds the tail after M terms by the first omitted Gaussian moment; that is the $\mathcal{O}_M\big((r/n)^M\big)$ of (U.6). What (U.7) adds to the classical mechanism is that every coefficient is a *finite* explicit sum, which is what Theorem U.4 then exploits. □

Proposition U.4 (Structure of the coefficients). *For every $m \geq 1$,*

$$c_m(r) = \frac{P_m(r)}{(1+r)^{3m}}, \qquad P_m \in \mathbb{Q}[r], \qquad \deg P_m \leq 4m, \qquad c_m(r) = \mathcal{O}(r^m) \;\; (r \to \infty). \tag{U.8}$$

The first three are

$$c_1(r) = -\frac{2r^4 + 9r^3 + 16r^2 + 6r + 2}{24\,(1+r)^3}, \tag{U.9}$$

$$c_2(r) = \frac{4r^8 - 12r^7 - 143r^6 - 384r^5 - 588r^4 - 636r^3 + 100r^2 + 24r + 4}{1152\,(1+r)^6}, \tag{U.10}$$

$$c_3(r) = \frac{\begin{aligned}&1112r^{12} + 12708r^{11} + 56082r^{10} + 118683r^9 + 97176r^8 - 74394r^7\\ &- 259046r^6 - 456444r^5 - 635424r^4 + 115308r^3 + 39432r^2 + 10008r + 1112\end{aligned}}{414720\,(1+r)^9}. \tag{U.11}$$

Proof. By (U.2), $T_j = \sum_{k=1}^{j} {j \brace k} r^k$ is a polynomial of degree j with $T_j(0) = 0$; write $T_j = r\,\widetilde{T}_j$ with $\widetilde{T}_j \in \mathbb{Z}[r]$ of degree $j-1$. Fix a term of (U.7), indexed by $(k_1, \ldots, k_{2m})$ with $\sum_s s k_s = 2m$ and $K = \sum_s k_s$. Then

$$\prod_s q_{s+2}^{k_s} = \frac{\prod_s T_{s+2}^{k_s}}{r^K\,(1+r)^{\sum_s k_s(s+2)/2}} = \frac{\prod_s \widetilde{T}_{s+2}^{k_s}}{(1+r)^{m+K}},$$

because $\prod_s T_{s+2}^{k_s} = r^K \prod_s \widetilde{T}_{s+2}^{k_s}$ — the factor r^K cancels exactly — and $\sum_s k_s(s+2)/2 = \big(\sum_s s k_s + 2K\big)/2 = m + K$. The surviving numerator has degree $\sum_s k_s(s+1) = 2m + K$.

Now $s \geq 1$ forces $K \leq \sum_s s k_s = 2m$, so $(1+r)^{m+K}$ divides $(1+r)^{3m}$; and on putting the term over the common denominator $(1+r)^{3m}$ its numerator acquires the factor $(1+r)^{2m-K}$, reaching degree exactly $(2m+K) + (2m-K) = 4m$, independently of the index. Summing over the finitely many indices gives (U.8) with $\deg P_m \leq 4m$. Since the denominator has degree $3m$, this also gives $c_m = \mathcal{O}(r^{4m-3m}) = \mathcal{O}(r^m)$. The three displayed coefficients follow from (U.7) at $m = 1, 2, 3$; they were recomputed and checked against exact Bell numbers (Theorem V.11). □

Remark U.5 (Why the Poincaré form hides the right scale). Because $c_m(r) = \mathcal{O}(r^m)$, the m-th term of (U.6) is of size $(r/n)^m$, not n^{-m}. The genuine transmonomial is therefore

$$\frac{r}{n} = \mathrm{e}^{-r}, \tag{U.12}$$

and rewriting the expansion in powers of e^{-r} — which requires absorbing the Stirling series for $n!$ — turns every coefficient into a *bounded* rational function of r. Then

$$\log \mathfrak{B}_n \sim n\Big(r - 1 + \frac{1}{r}\Big) - 1 - \frac{1}{2}\log(1+r) + \sum_{m \geq 1} \mathcal{D}_m(r)\,\mathrm{e}^{-mr}, \tag{U.13}$$

with each $\mathcal{D}_m$ bounded and given by finite coefficient operations. This is the same distinction the volume draws repeatedly: an expansion may be correct and still be written on the wrong scale, in which case its coefficients grow and its ordering is misleading.

Remark U.6 (Three ranked sectors). Substituting the iterated-logarithm expansion of $r = W_0(n)$, with $L = \log n$ and $\lambda = \log L$, splits $\log \mathfrak{B}_n$ into

$$nL\,\mathbb{Q}[\lambda][[L^{-1}]], \qquad \log L + \mathbb{Q}[\lambda][[L^{-1}]], \qquad \sum_{m\geq 1} n^{-m}L^m\mathbb{Q}[\lambda][[L^{-1}]],$$

the third family lying beyond every power of L^{-1} relative to the first two. Nonprincipal complex saddles lie beyond all three. Both sources label their treatment of those saddles *formal*, because identifying the Stokes multipliers of the original Cauchy contour is a separate global steepest-descent problem; that labelling is kept here, and the sectors are not claimed as asymptotics.

Appendix V

The Fubini numbers: an exact pole lattice

V.1 What this chapter does

The Fubini (ordered Bell) numbers $\mathfrak{F}_n = \sum_j j!\left\{ {n \atop j} \right\}$ count weak orderings. Their exponential generating function is

$$\sum_{n \geq 0} \mathfrak{F}_n \frac{z^n}{n!} = \frac{1}{2 - \mathrm{e}^z}, \tag{V.1}$$

which is meromorphic with *simple* poles at every solution of $\mathrm{e}^z = 2$, that is at

$$\zeta_k = \rho + 2\pi \mathrm{i} k, \qquad \rho = \log 2, \qquad k \in \mathbb{Z}. \tag{V.2}$$

This is the one subject in the volume whose transseries is exact and convergent rather than asymptotic — as is, for a different reason, the Fibonacci inverse of Part T. The contrast with the Bell numbers, whose saddle expansion diverges, is the point of putting the two chapters side by side, and Section V.5 draws it.

> Three articles: `Fubini_Number_Full_Transseries`, `Fubini_Number_Transseries` and `Fubini_Number_Transseries_Article`. They agree. Their common content is the pole-lattice formula and the factorial normalization; the first adds the meromorphic-generating-function principle and the higher-order-pole analysis, which is kept because it is what explains the shape of the answer.

V.2 The exact formula

Theorem V.1 (Exact pole-lattice transseries). *For every integer $n \geq 1$,*

$$\mathfrak{F}_n = \frac{n!}{2} \sum_{k \in \mathbb{Z}} \zeta_k^{-n-1} = \frac{n!}{2\rho^{n+1}} \left[1 + 2 \sum_{k \geq 1} \left(\frac{\rho}{|\zeta_k|} \right)^{n+1} \cos\bigl((n+1) \arg \zeta_k\bigr) \right], \tag{V.3}$$

both series converging absolutely. This is an identity, not an asymptotic statement.

Proof. Every zero of $2 - \mathrm{e}^z$ is simple, since $(2 - \mathrm{e}^z)' = -\mathrm{e}^z = -2 \neq 0$ there, and the zeros are exactly the ζ_k of (V.2). The residue of $(2 - \mathrm{e}^z)^{-1}$ at ζ_k is $1/(-\mathrm{e}^{\zeta_k}) = -\frac{1}{2}$. The Mittag–Leffler expansion of the meromorphic function $(2 - \mathrm{e}^z)^{-1}$, which converges because the poles have spacing 2π in the imaginary direction and the function is bounded on the intervening horizontal strips, therefore gives

$$\frac{1}{2 - \mathrm{e}^z} = -\frac{1}{2} \sum_{k \in \mathbb{Z}} \frac{1}{z - \zeta_k} + \text{entire},$$

and the entire part vanishes by the decay of both sides as $\Re z \to -\infty$. Extracting $[z^n/n!]$ from $-\frac{1}{2}(z-\zeta_k)^{-1} = \frac{1}{2}\sum_{m\geq 0} z^m \zeta_k^{-m-1}$ gives $\mathfrak{F}_n/n! = \frac{1}{2}\sum_k \zeta_k^{-n-1}$, which is the first equality; absolute convergence holds for $n \geq 1$ because $|\zeta_k| \sim 2\pi|k|$ and $n+1 \geq 2$. Pairing k with $-k$, and using $\zeta_{-k} = \overline{\zeta_k}$, turns the sum into the real cosine form. □

Remark V.2 (What a simple pole does and does not contribute). A simple pole contributes *one* exact sector with a *constant* amplitude. It does not generate an independent $1/n$ fluctuation series. In the elementary normalization the inverse powers of n appear solely because $n!$ has been replaced by Stirling's expansion; keep $n!$ exact and there are none. This is worth stating because the opposite is easy to assume: a singularity-analysis habit expects each singularity to carry its own algebraic series. Higher-order poles are the case where that expectation is correct, and there the amplitudes are polynomial in n, governed by generalized Bernoulli numbers.

Corollary V.3 (Elementary-scale form). *Let $\Lambda_k = \operatorname{Log}(\zeta_k/\rho)$ with the principal logarithm, and let c_j be defined by the Stirling series of the scaled gamma factor,*

$$\exp\left(\sum_{r\geq 1}\frac{B_{r+1}}{r(r+1)}t^r\right) = \sum_{j\geq 0} c_j t^j, \qquad c_0 = 1,\ c_1 = \tfrac{1}{12},\ c_2 = \tfrac{1}{288},\ c_3 = -\tfrac{139}{51840}. \tag{V.4}$$

Then

$$\mathfrak{F}_n \sim \frac{\sqrt{2\pi n}}{2\rho}\left(\frac{n}{e\rho}\right)^n \sum_{k\in\mathbb{Z}} e^{-n\Lambda_k} \sum_{j\geq 0}\frac{C_{j,k}}{n^j}, \qquad C_{j,k} = c_j\,\frac{\rho}{\zeta_k}, \tag{V.5}$$

where the k-sum converges for every $n \geq 1$ and the j-sum is the Poincaré expansion of the gamma factor.

Proof. Substitute $n! = \sqrt{2\pi n}\,(n/e)^n \sum_j c_j n^{-j}$, which is (V.4) exponentiated, into (V.3), and write $\zeta_k^{-n-1} = \rho^{-n-1}e^{-n\Lambda_k}\cdot \rho/\zeta_k$. The amplitude $C_{j,k}$ is then the product of the j-th Stirling coefficient and the constant ρ/ζ_k of Theorem V.2; no k-dependence enters the j-sum, which is why the same series multiplies every sector. □

Lean status: partial, the Stirling coefficients only. The definition (V.4) and its four listed values are formal in `StirlingSeriesCoefficients.lean`. `Fabius.stirlingKernelCoeff` is $\varkappa_r = B_{r+1}/(r(r+1))$, `Fabius.stirlingKernel` the series it assembles, `Fabius.constantCoeff_stirlingKernel` the fact that makes exp of it a substitution, and `Fabius.stirlingCoeff` the coefficients c_j. Their values are `Fabius.stirlingCoeff_zero`, `Fabius.stirlingCoeff_one`, `Fabius.stirlingCoeff_two` and `Fabius.stirlingCoeff_three`, giving $1, \frac{1}{12}, \frac{1}{288}, -\frac{139}{51840}$ – constants that this volume quotes rather than derives, so it is worth having them machine-checked from the definition rather than from a table.

The derivation is the general recurrence of `ExpSeriesRecurrence.lean` read at this kernel (`Fabius.stirlingCoeff_recurrence`); the only arithmetic inputs are $B_2 = \frac{1}{6}$, $B_3 = 0$ and $B_4 = -\frac{1}{30}$, the last of which is derived formally from $\sum_{k<5}\binom{5}{k}B_k = 0$ rather than assumed (`Fabius.bernoulli_four`). The vanishing of B_3 is why c_2 receives no contribution from $\varkappa_2$ and equals $\varkappa_1^2/2$.

Not formalized: the asymptotic (V.5) itself, the convergence of the k-sum, and the identification of $\sum_j c_j t^j$ with the Stirling series of the gamma factor. What is formal is the coefficient family, defined exactly as the corollary defines it.

V.3 Certified truncation

The lattice sum converges, so truncating it is a numerical choice rather than an asymptotic one, and the cost can be bounded exactly. Write

$$T_n = \sum_{k\in\mathbb{Z}} u_k^{n+1}, \qquad u_k = \frac{\rho}{\zeta_k}, \qquad T_{n,K} = \sum_{|k|\le K} u_k^{n+1}, \tag{V.6}$$

so that $\mathfrak{F}_n = \frac{1}{2} n!\, \rho^{-n-1} T_n$ by Theorem V.1.

Theorem V.4 (Pole-tail bounds). *For $n \ge 1$ and $K \ge 0$,*

$$|T_n - T_{n,K}| \;\le\; 2\sum_{k>K} |u_k|^{n+1} \;\le\; 2\Big(\frac{\rho}{2\pi}\Big)^{n+1} \zeta(n+1, K+1), \tag{V.7}$$

$\zeta(s,a)$ being the Hurwitz zeta function, and for $K \ge 1$ also

$$|T_n - T_{n,K}| \;\le\; \frac{2}{n}\Big(\frac{\rho}{2\pi}\Big)^{n+1} K^{-n}. \tag{V.8}$$

For the leading approximation $K = 0$ one has the uniform estimate

$$|T_n - 1| \;\le\; \frac{\rho^2}{12} = 0.04003775115985011\ldots \qquad (n \ge 1). \tag{V.9}$$

Proof. $|\zeta_k|^2 = \rho^2 + 4\pi^2 k^2 \ge 4\pi^2 k^2$, so $|u_k| \le \rho/(2\pi|k|)$, and pairing k with $-k$ gives the first inequality in (V.7); summing $|u_k|^{n+1} \le (\rho/2\pi)^{n+1} k^{-n-1}$ over $k > K$ gives the second, by the definition of the Hurwitz zeta function. For (V.8), compare that sum with $\int_K^\infty t^{-n-1}\mathrm{d}t = K^{-n}/n$. For (V.9), take $K = 0$ in (V.7): $|T_n - 1| \le 2(\rho/2\pi)^{n+1}\zeta(n+1)$, which decreases in n, so its value at $n = 1$ bounds it; that value is $2(\rho/2\pi)^2\zeta(2) = 2\cdot\rho^2/(4\pi^2)\cdot\pi^2/6 = \rho^2/12$. ☐

Proposition V.5 (A sufficient linear pole budget). *Fix $\delta > 0$ and choose $K = K_n$ with*

$$K_n + 1 \;\ge\; \frac{(1+\delta)\,n}{2\pi\mathrm{e}}. \tag{V.10}$$

Then the bound (V.8)*, transported to $\mathfrak{F}_n$, is $\mathcal{O}_\delta\big(n^{-1/2}(1+\delta)^{-n}\big)$ relative to $\mathfrak{F}_n$. In particular a number of poles* linear *in n suffices to determine $\mathfrak{F}_n$ to within less than $\frac{1}{2}$, so the integer is recovered by rounding.*

Proof. By (V.8) the relative error is at most $\frac{2}{n}(\rho/2\pi K_n)^n \rho/(2\pi)$ divided by $T_n \ge 1 - \rho^2/12$, which by (V.10) is at most a constant times $n^{-1}\big(\mathrm{e}/(1+\delta)\big)^n \rho/(2\pi)\cdot(\rho/2\pi)^n\cdot(2\pi)^n$; collecting the powers and using $\rho\mathrm{e} < 2\pi$ leaves $\mathcal{O}_\delta((1+\delta)^{-n})$, and the elementary Stirling upper bound $n! \le C\sqrt{n}\,(n/\mathrm{e})^n$ contributes the remaining $n^{-1/2}$. ☐

V.4 Weighted Fubini polynomials

Everything above is the case $x = 1$ of a one-parameter family, and the parameter costs nothing.

Theorem V.6 (Full weighted pole formula). *For $x > 0$ let $\mathfrak{F}_n(x) = \sum_j j!\left\{ {n \atop j} \right\} x^j$, whose exponential generating function is $\big(1 - x(\mathrm{e}^z - 1)\big)^{-1}$. Put*

$$\rho_x = \log\Big(1 + \frac{1}{x}\Big), \qquad \zeta_k(x) = \rho_x + 2\pi\mathrm{i}k. \tag{V.11}$$

Then, for every $n \ge 1$,

$$\mathfrak{F}_n(x) = \frac{n!}{1+x} \sum_{k\in\mathbb{Z}} \zeta_k(x)^{-n-1}, \tag{V.12}$$

an exact convergent identity, which at $x = 1$ is Theorem V.1.

Proof. The poles of $\big(1 - x(\mathrm{e}^z - 1)\big)^{-1}$ are the solutions of $\mathrm{e}^z = 1 + 1/x$, that is the $\zeta_k(x)$ of (V.11), and each is simple because the derivative of the denominator is $-x\mathrm{e}^z = -(1+x) \neq 0$ there; the residue is therefore $-1/(1+x)$, constant in k. The Mittag–Leffler argument of Theorem V.1 applies verbatim, with $\frac{1}{2}$ replaced by $1/(1+x)$, and gives (V.12). At $x = 1$, $\rho_1 = \log 2 = \rho$ and $1/(1+x) = \frac{1}{2}$. □

Remark V.7 (What the parameter shows). The residue is constant in k for every x, which is Theorem V.2 again: a simple pole contributes one constant-amplitude sector whatever the weight. What the weight moves is the *location* of the lattice, through $\rho_x = \log(1 + 1/x)$; as $x \to \infty$ the lattice spacing stays 2π but $\rho_x \to 0$, so the sectors become comparable in size and the leading-pole approximation degrades. The uniformity of (V.9) in n is therefore *not* uniform in x, and a statement about $\mathfrak{F}_n(x)$ on a compact x-interval needs the bound of Theorem V.4 with ρ replaced by $\min_x \rho_x$.

V.5 Two sequences, two kinds of expansion

Remark V.8 (The contrast). The Bell and Fubini numbers are both counted by set partitions and both have an entire or meromorphic exponential generating function, yet their expansions are of opposite type. $\exp(\mathrm{e}^z - 1)$ is entire, so there is no singularity to expand about and the answer comes from a *saddle*; saddle expansions are asymptotic and the Bell series diverges. $(2 - \mathrm{e}^z)^{-1}$ is meromorphic, so the answer comes from its *poles*; the pole sum is exact and converges. The dividing line is not the combinatorics but the location of the singularities of the generating function — which is the same principle that makes Part T the one convergent inversion in this volume.

Proposition V.9 (One coefficient family serves both). *Let $\gamma_m = \lim_{r\to\infty} c_m(r)/r^m$ be the leading large-saddle coefficients of the Bell expansion* (U.6)*, and let c_j be the Stirling coefficients* (V.4) *that multiply every Fubini sector in* (V.5)*. Then, for $m = 1, 2, 3$,*

$$\gamma_m = (-1)^m c_m, \tag{V.13}$$

that is $\gamma_1 = -\frac{1}{12}$, $\gamma_2 = \frac{1}{288}$, $\gamma_3 = \frac{139}{51840}$ against $c_1 = \frac{1}{12}$, $c_2 = \frac{1}{288}$, $c_3 = -\frac{139}{51840}$.

Proof. Take $r \to \infty$ in (U.9)–(U.11): the numerator's leading term over $(1+r)^{3m}$ gives $\gamma_1 = -2/24 = -\frac{1}{12}$, $\gamma_2 = 4/1152 = \frac{1}{288}$ and $\gamma_3 = 1112/414720 = \frac{139}{51840}$. Comparison with the expansion of (V.4) gives (V.13). □

Lean status: one side of the comparison is formal. The Stirling coefficients c_1, c_2, c_3 of the right-hand side are machine-checked — $\frac{1}{12}$, $\frac{1}{288}$, $-\frac{139}{51840}$, by `Fabius.stirlingCoeff_one`, `Fabius.stirlingCoeff_two` and `Fabius.stirlingCoeff_three` in `StirlingSeriesCoefficients.lean`, derived from the definition rather than quoted. So the numbers this proposition compares against are no longer taken on trust, and the displayed values agree with the formal ones under the sign convention $\gamma_m = (-1)^m c_m$.

The other side is not formalized: $\gamma_m = \lim_{r\to\infty} c_m(r)/r^m$ requires the Bell saddle expansion of Theorem U.3 and the coefficient structure of Theorem U.4, neither of which the corpus carries. So the proposition's content — that the two families coincide up to sign — remains unformalized; what is settled is that one of the two families is what the volume says it is.

Remark V.10 (Why that is not a coincidence, and what is not proved). Both families come from the same place: the gamma factor. In the Bell expansion the factor $n!$ sits in front of (U.6) and its Stirling series is what Theorem U.5 absorbs when passing to the e^{-r} scale; in the Fubini expansion the same $n!$ is what Theorem V.3 replaces. The sign alternation is the inversion of the scale — r/n against $1/n$. Neither source states (V.13), because neither treats both sequences. What is established here is the identity for $m \leq 3$, verified in exact arithmetic; the general statement $\gamma_m = (-1)^m c_m$ is *not* proved, and is recorded as a conjecture in Theorem V.12.

Remark V.11 (Independent verification). Recomputed for this volume, independently of the sources. Fubini: the sequence $1, 1, 3, 13, 75, 541, 4683, 47293, 545835, \ldots$ from $\sum_j j!\left\{ {n \atop j} \right\}$; the exact formula (V.3) against those integers at $n = 2, 3, 5, 8, 12, 20$, agreeing to relative error 5×10^{-15} at $n = 2$ and below 10^{-40} from $n = 8$ on, the residual being the truncation of the lattice sum and not an error in the identity — at $n = 1$ the tail of the k-sum is only $\mathcal{O}(K^{-1})$ and a truncation at $K = 4000$ leaves 6×10^{-6}, exactly the observed discrepancy; the cosine form against the same integers; and the coefficients $c_0 = 1$, $c_1 = 1/12$, $c_2 = 1/288$, $c_3 = -139/51840$ from (V.4). Bell: the weighted formula (V.12) against $\sum_j j!\left\{ {n \atop j} \right\} x^j$ at $x = \frac{1}{2}, 1, 2, 3$ and $n = 3, 6, 10$, exact to 10^{-17} or better in every one of the twelve cases; the constant $\rho^2/12 = 0.04003775115985011\ldots$ of (V.9); and the expansion (U.6) against exact Bell numbers at $n = 20, 50, 200, 1000$, the relative error falling from 1.5×10^{-2} to 7.6×10^{-7} at $n = 20$ and from 5.6×10^{-4} to 1.5×10^{-12} at $n = 1000$ as c_1, c_2, c_3 are included; the structural claims (U.8) for $m \leq 3$, degree and denominator exactly as stated; and (V.13).

Remark V.12 (What is left open).

1. *The identity $\gamma_m = (-1)^m c_m$ for all m.* Verified for $m \leq 3$ (Theorem V.9). A proof should come from tracking the gamma factor through (U.7) rather than from computing both sides.
2. *Nonprincipal Bell saddles.* Formal, as both sources say; the Stokes multipliers of the Cauchy contour are not determined.
3. *Higher-order Fubini poles.* The meromorphic principle of Theorem V.2 predicts polynomial-in-n amplitudes governed by generalized Bernoulli numbers; the general statement is quoted from the source and not reproved here.

Part XV

Synthesis

Appendix W

Synthesis: what the eight subjects share, and where they part

This chapter belongs to the consolidation rather than to any source. It compares the eight inversions carried out in Parts M to T, states precisely how much of each is an instance of Part J, and records what the twenty-four articles left open. Nothing here is asserted beyond what the earlier chapters prove.

W.1 One inversion, four dictionaries

The first four subjects of this volume were written about independently, and their *forward* analyses have almost nothing in common: a singularity analysis of a Pólya functional equation, a pair of gamma-quotient branches, a convergent Rademacher series over root-of-unity sectors, and a second, quite different pair of gamma quotients. Their *inverse* analyses are, to a degree that the individual articles could not see, the same computation.

The reason is Theorem J.8. In each subject one first isolates a *dominant core* Φ_0 — an increasing function whose inverse is available in closed form — and writes the exact equation for the index x in terms of the observed value y as

$$\Phi_0(x) + (\text{a correction that is small relative to } \Phi_0') = L(y).$$

The core is then inverted exactly by the Lambert W function, and the correction is removed to all orders by one triangular recurrence, Theorem J.33 (in the exponential–power case, Theorem J.29). The recurrence does not know which sequence it came from: it sees only the core data and the coefficients q_j of the correction.

Table J.1 records the resulting dictionary. Three points in it deserve emphasis, because each was invisible from inside a single subject.

The partition numbers are the same problem after one substitution. Of the four, the partition numbers look least like the others: the Hardy–Ramanujan phase is $\pi\sqrt{2N/3}$, a *square root* of the variable, where the other three carry a phase linear in the index or in $x \log x$. The three partition articles accordingly build their inversion by hand around the equation they write as $\rho - 2\log\rho = L$. By Theorem J.4 that equation *is* the linear-phase core $a\xi + b\log\xi$ in the variable $\xi = \sqrt{N}$, with $b/\alpha = -2$; the substitution costs nothing and removes the apparent exception. What the substitution does *not* do is commute with the arithmetic shift, and Theorem J.6 shows why the order matters: re-expanding $N = n - \frac{1}{24}$ in powers of n after the reduction manufactures an entire second tower of exponents $n^{\alpha-k}$, which is exactly the infinite algebraic array that one source reports and attributes, correctly, to the shift alone.

The two "factorials" are not the same kind of object. It is tempting — and this volume's own brief made the mistake — to expect the double factorial and the swing factorial to sit side by side. They do not. The swing factorial $n!/\lfloor n/2\rfloor!^2$ has its factorials cancel to leave growth $2^n n^{\pm 1/2}$, so it is an exponential–power model with $a = \log 2$ (Part P). The double factorial does not: its logarithm is $\frac{s}{2}(\log s - 1) + \mathcal{O}(\log s)$ with $s = n + 1$, a phase of factorial type, and since $s \log s/s^\alpha \to \infty$ for every $\alpha \leq 1$, no choice of model data reproduces it (Part N). What survives the difference is exactly the axiomatized core: both $aX + b\log X$ and $X(\kappa \log X + d)$ are admissible in the sense of Theorem J.8, both are inverted exactly by W (Theorems J.23 and J.27), and the reversion above them is one theorem, Theorem J.33. The right statement is therefore not "four instances of one model" but "four instances of one *core calculus*, of which three share a model".

The Lambert branch is determined by a sign. Each of the first twelve articles verifies by hand, for its own case, which branch of W produces the large root; several remark that the appearance of W_{-1} is essential and surprising. With the signed convention of Theorem J.1 it is neither: by Theorem J.25 the branch is W_0 when $b > 0$ and W_{-1} when $b < 0$, with the fold at $X_* = |b|/a$ and the threshold $L_c = |b|(1 - \log(|b|/a))$. Reading Table J.1 down the b column now predicts the branch in every chapter, including the two branches of Part P, which differ only in the sign of σ.

W.2 Where the common frame stops

The shared apparatus covers the passage from an exact forward description to an all-orders inverse. It covers neither end of that passage, and it is worth being explicit about which parts of the four chapters are irreducibly their own.

(a) *Getting the forward description at all.* The four forward analyses share no method. Part M needs a singularity analysis of $T(z) = z \exp \sum_{k\geq 1} T(z^k)/k$, the location and nature of the Otter singularity, and a transfer theorem; Parts N and P need Binet's representation and Stirling's series for gamma quotients; Part O needs the circle method, or rather its exact endpoint, Rademacher's convergent series. Nothing in Part J produces any of these, and nothing in one of them helps with another.

(b) *The arithmetic layer.* The sectors of Part O are indexed by a denominator k and carry Kloosterman-type amplitudes built from Dedekind sums. This structure has no analogue in the other three subjects, whose exponentially small corrections come from a competing singularity (Part M) or from Borel geometry (Parts N and P). The frame transports each sector through the inversion once the sector is given; it does not explain where the sectors come from.

(c) *Parity.* Two of the four sequences are not the restriction of any single smooth interpolation, and in both cases this is forced, not accidental: the even and odd subsequences have different exponents ($n^{\pm 1/2}$ for the swing factorial) or different gamma representations (the double factorial). Theorem J.41 and Theorem J.43(4) handle the consequence uniformly — the integer inverse is a minimum over residue classes — but the *fact* of the split is a property of the sequence.

(d) *Which interpolation.* For a discrete sequence the asymptotic inverse is well defined only after an interpolation is chosen, and the choice is not innocent. Theorem J.42 separates the three inverse objects; Part N shows what changes when the OEIS periodic interpolation is used instead of the branchwise gamma continuation, namely a log-periodic oscillation in the inverse that is absent from either branch. One partition source records the sharper version of the same phenomenon: the algebraic part of the inverse is interpolation-independent, the nonperturbative part is not.

W.3 The three inverse objects, subject by subject

Theorem J.41 distinguishes the staircase inverse N_*, the inverse of a chosen interpolation, and the asymptotic inverse. The four chapters do not stand in the same relation to them, and the differences are not cosmetic.

Chapter	interpolation available	parity split	exact staircase recovery
M rooted trees	from the singular expansion	no	along the range
N double factorial	two branches, or OEIS	yes	per residue class
O partitions	from Rademacher	no	along the range
P swing factorial	two branches	yes	per residue class

Two cautions apply to the whole table. First, by Theorem J.43(5) the staircase N_* is *not* uniformly approximable by any continuous function: $\sup_{y\geq Y}|N_*(y) - g(y)| \geq \frac{1}{2}$ for every continuous g and every Y. Statements of the form “$N_*(y) = \nu(y) + o(1)$” are therefore false as uniform statements, however natural they look, and the volume asserts only the exact ceiling identity, recovery along the range $y = A_n$, and the residue-class refinement. Second, the error floor of the asymptotic inverse is not a defect of the reversion but is inherited: by Theorem J.54 the forward series has a least-term envelope, and Theorem J.47 converts it into a floor for the inverse. No amount of further reversion crosses it.

W.4 What this consolidation establishes

Several statements in this volume are not in any of its first twelve sources. They are listed here because a reader who knows the sources should be able to see quickly what has changed; the corrections themselves are boxed where they occur and collected in Part D.

(1) *A correct proof of the Lagrange fixed-point formula.* Five of the first twelve articles derive $[t^n]\,U = \sum_{m\leq n} \frac{1}{m}[t^n u^{m-1}]\,\Psi^m$ by introducing a marker z and applying Lagrange–Bürmann in z. That application is not available: Lagrange–Bürmann needs $\varphi(0)$ invertible, and here $\varphi(0) = \Psi(t, 0) \in tR[[t]]$ is a non-unit. The identity is nonetheless true, and Theorem J.18 proves it by universality — it is a polynomial identity over $\mathbb{Q}$ in the coefficients of φ, valid in a localization into which the relevant domain injects. Every reversion in this volume rests on this lemma, so the repair is not cosmetic.

(2) *The branch rule* Theorem J.25, with the fold point and threshold, replacing four case analyses.

(3) *The normalization constant of the flattened blocks.* The sources state $P_n(0) = D_n$ in a case where $a = 1$; the correct general statement is $P_n(0) = a^{n+1}c_n$ (Theorem J.37(3)), and since $a \neq 1$ in every chapter of this volume the factor is needed throughout.

(4) *A quantitative obstruction for the staircase*, Theorem J.43(5), in place of an $o(1)$ claim that is false uniformly.

(5) *Backward error as a certificate.* Theorem J.47 upgrades the sources’ conditional bound — which presupposes that a root exists — to an existence statement, by a sign change and the intermediate value theorem.

(6) *One closed form where the sources had three.* Theorem J.38 is proved equal to all three published forms of the pure Lambert block, and the equality was verified in exact rational arithmetic for $1 \leq n \leq 8$.

(7) *A non-recursive formula for the factorial-core corrections.* Theorem J.20 extends the fixed-point formula to $\Psi = h + f$ with $h \in u^2R[[u]]$, which is what Part N needs; the source reaches the same coefficients only through an ad hoc compositional inverse.

W.5 The second four

Parts Q to T add four subjects that are not sequences of the first kind, and they change the picture in three ways.

A second core, and a second family. Theorem J.23 inverts $aX + b\log X$, in which the logarithm is *added*. Gamma, Barnes G and the K-function need $ax^p(\log x - \beta)$, in which the unknown *multiplies* the logarithm; that block is inverted exactly by W too, but by a different substitution, and its normalized slope is $v+1$ rather than $a + b/X$. Theorem Q.8 supplies it, in the generality $ax^p(\log x - \beta)^r$, and Theorem Q.46 is the corresponding inversion theorem. Those three subjects form a family of their own, indexed by (a, p, β) and by where a resonant logarithm sits; the comparison is Table R.4.

A subject that borrows its core from another chapter. The subfactorial has no core of its own: $D_n = n!/\mathrm{e} + (-1)^n C(n)$, and the inverse of the first term is the inverse of Γ, which is Theorem Q.25. Everything specifically about derangements lives in a correction of size $\exp(-u\log u + \mathcal{O}(u))$, below every fixed exponential scale e^{-cu} — Theorem S.22. This is the first subject in the volume whose interesting content is *not* in its dominant block, and it is the reason Part S is the shortest chapter with the largest citation load.

A subject the frame does not cover. The real-argument Fibonacci function has a pure exponential core, no slow coefficient field, and no Lambert function anywhere. Its inverse expansion *converges*. Section T.1 tabulates exactly which parts of Chapter 0 survive: essentially only Theorem J.21 and Theorem J.47. It is included because the purpose is the same — invert a rapidly growing function at infinity to all orders — and because the contrast is informative: it isolates which features of the other seven come from the *logarithm* in their phase rather than from inversion as such.

Table W.1: The eight subjects. "Core" is the block inverted exactly; "slope" is the quantity every correction is divided by.

Chapter	Core	Slope	Correction scale
M rooted trees	$aX + b\log X$	$a + b/X$	X^{-1}, $\alpha = 1$
N double factorial	$X(\kappa\log X + d)$	$\kappa\log X + \kappa + d$	X^{-1}
O partitions	$aX + b\log X$	$a + b/X$	X^{-1}, $\alpha = \frac{1}{2}$
P swing factorial	$aX + b\log X$	$a + b/X$	X^{-1}
Q Γ	$t(\log t - 1)$	$\Lambda = w + 1$	T^{-2} (even)
Q Barnes G	$\frac{1}{2}z^2(\log z - \frac{3}{2})$	$U\lambda$	U^{-1} (all)
R K	$\frac{1}{2}r^2(\log r - \frac{1}{2})$	ρL	ρ^{-2} (even)
S subfactorial	$\Gamma(x+1)/\mathrm{e}$	$Y\lambda$	$\mathrm{e}^{\omega u}/Y$
T Fibonacci	$\mathrm{e}^{\lambda x}$	λ	R^{-1}, oscillatory

W.6 Three things only the merge shows

Every subject with a logarithmic phase hides a square root. Four apparently unrelated square roots appear in the second half of the volume:

$$\sqrt{\Lambda^2T^2 + \tfrac{1}{12}}, \qquad \sqrt{\lambda^2U^2 - 2\alpha U}, \qquad \sqrt{U^2 + \tfrac{1}{6}}, \qquad \sqrt{\rho^2 + \tfrac{1}{12}}.$$

The first two are the deepest pole in the slope, for Γ and for G (Theorem Q.22); the third and fourth are coefficientwise large-slope limits, for G and for K (Theorems Q.35 and R.10). All four are the same lemma: truncate the normalized equation to its linear and quadratic terms in the displacement together with the lowest forcing monomial, and the truncation is a quadratic, hence solvable in closed form (Theorem Q.19). Which *extraction* of the coefficient family the quadratic computes — deepest pole or large-slope limit — is decided by whether the forcing monomial carries a factor of the slope, that is by whether it is resonant. Resonance and pole depth are two readings of one quadratic, and no source saw either connection.

Acceleration is resummation. The K-function sources offer an “accelerated coordinate” $T = x^2 - x + \frac{1}{6}$ that gains a full asymptotic sector, and present it as a change of variable. Theorem R.14 proves it is the resummation of the resonant subsector of the unaccelerated expansion — the same square root, arrived at from the other side. The absorption lemma behind it (Theorem R.11) applies verbatim to Barnes G, which no Barnes source attempted; the resulting normal form is (R.36).

One number, two jobs. The constant $\frac{1}{12}$ occurs as the Γ deepest pole and again as the K resonant limit, for reasons that look unrelated. Both are $-2 \cdot B_2(\frac{1}{2})/2$: for Γ, $B_2(\frac{1}{2})/2 = -\frac{1}{24}$ is the first Stirling tail coefficient; for K, the same value is the coefficient of the resonant logarithm. The identical Bernoulli value at the identical reflection centre is doing both jobs (Theorem R.22). The same reflection symmetry is what makes the half-shift simultaneously remove $r \log r$ and kill the odd tail, in Theorem Q.2 and Theorem R.6.

W.7 What the second wave removed

The first twelve articles carried twelve copies of the Chapter 0 apparatus. The second twelve carry, in addition, copies of *each other*:

- The single block $d_5 = (2232\Lambda^4 + 1176\Lambda^3 + 714\Lambda^2 + 350\Lambda + 105)/(2903040\Lambda^5)$ of Theorem Q.26 is printed, in one letter or another, in *seven* of the twelve second-wave sources: the three whose subject it is, all three subfactorial articles — which need the inverse of $\Gamma(x+1)/\mathrm{e}$ as their carrier and rederive it from scratch, each proving its own Lambert-centred reversion theorem to get there — and one K-function article. Four of those seven copies are removed here; Theorem S.10 cites Theorem Q.25 instead.
- Eight of the twelve state a general triangular or formal-inversion theorem over a graded coefficient ring. They are all Theorem Q.13, and one of them states it under a hypothesis strictly stronger than its own proof uses.
- Seven of the twelve state a named residual-to-error or a posteriori transfer result. They are all Theorem J.47, which the first wave had already needed.
- Seven of the twelve prove a Lagrange–Bürmann reversion adapted to their setting. In every case the content is Theorem J.21 or, where the perturbation is formal, Theorem J.22.

That the second wave's sources duplicate the first wave's is the strongest single piece of evidence that the consolidation was worth doing: the twelve new articles were written without knowledge of the twelve old ones, and converged on the same apparatus independently.

W.8 Open questions

(i) *A Stokes-transport theorem.* The mechanism by which an exponentially small forward sector becomes an exponentially small inverse sector is uniform — it is the perturbation-operator form Theorem J.35 — but a theorem needs each subject's Borel geometry as a hypothesis. The four chapters supply four separate instances; the general statement is not proved here.

(ii) *The counting function.* One source states that the number of indices with $A_n \leq y$ equals the average of the branch inverses plus $\mathcal{O}(1)$. The statement is plausible for every subject with a parity split, but it needs a per-subject monotonicity hypothesis and is not proved in this volume.

(iii) *Rademacher-type sequences in general.* Part O gives a template for inverting a sequence presented as a convergent sum over arithmetic sectors. Whether the template applies to the other classical Rademacher-type expansions is untested here.

(iv) *Multifactorials.* One double-factorial source sketches the m-fold generalization $n!^{(m)}$. The core is again of factorial type, so Theorem J.33 applies verbatim; what is not done is the residue-class analysis of the staircase for general m, which Theorem J.43(4) reduces to a finite computation but does not carry out.

(v) *Relation to the formalized Lambert layer of this repository.* The surrounding development contains a Lean-verified treatment of both real branches of W, a lower-branch bracket with an explicit remainder rate, and the saddle expansions of the Fabius function. The Lambert core these four chapters are built on is therefore already formalized to a degree none of the sources assumes. Which of the reversion machinery above is within reach of that layer — Theorem J.23 and Theorem J.25 look closest — is an open and, in this repository, a natural question.

Appendix A

Provenance

This volume consolidates *forty-two* independently written articles, filed in this repository between 1 and 3 September 2026. Thirty of them had already been consolidated once, into the two volumes that became the calculus parts and the inversion parts; those two carry their own provenance records, which are reproduced unchanged inside this volume. The remaining twelve are merged here for the first time and are listed below.

Source directory	Source	PDF	Absorbed into
Orientation, from `transseries-tutorials/`			
`Transseries_Tutorial-1`	5 159 lines	143 pp, Letter	Part 1, Part 2
`Transseries_Tutorial-2`	7 749 lines	164 pp, Letter	Part 1, Part 2
`Transseries_Tutorial-3`	4 410 lines	121 pp, Letter	Part 1
`Transseries_Tutorial-4`	8 781 lines	217 pp, $522 \times 738\,\mathrm{pt}$	Part 1, Part 2
Reversing $x + W(x)$, from `lambert-inverse-transseries/`			
`lambert_inverse_transseries`	1 907 lines	25 pp, A4	Part L
`lambert_inverse_transseries_bundle`	1 384 lines	21 pp, Letter	Part L
`reversing_x_plus_lambert_w_transseries`	1 918 lines	29 pp, A4	Part L
Bell and Fubini numbers, from `sequence-transseries/`			
`Bell_Number_Asymptotic_Transseries`	2 211 lines	32 pp, A4	Part U
`Bell_Number_Transseries_Article`	1 714 lines	23 pp, A4	Part U
`Fubini_Number_Full_Transseries`	2 465 lines	33 pp, A4	Part V
`Fubini_Number_Transseries`	1 881 lines	25 pp, A4	Part V
`Fubini_Number_Transseries_Article`	1 472 lines	25 pp, Letter	Part V

The other thirty are documented in place. The six sources of the calculus parts are listed, with a notation concordance and a repair ledger, in the final part of the calculus itself; the twenty-four sources of the inversion parts are listed in Part C below, which is the provenance appendix of that volume reproduced unchanged. All forty-two remain in git history.

A.1 Counts

Group	Articles	Source lines	Absorbed as
`transseries-tutorials/`	4	26 099	Orientation
`polynomial-logarithmic-transseries/`	6	36 033	the calculus parts
`special-function-inversion/`	24	16 771	the inversion parts
`lambert-inverse-transseries/`	3	5 209	$x + W(x)$
`sequence-transseries/`	5	9 743	Bell and Fubini
total	42	—	this volume

The line counts in the middle column are those of the *immediate* sources: for the two groups that had already been consolidated, the figure is the size of the consolidated volume, not of the six or twenty-four articles behind it.

Appendix B

Repairs made in this merge

Corrections made while merging the twelve newly consolidated sources. Repairs made in the two earlier consolidations are listed in their own ledgers, which this volume reproduces: the calculus's in its final part, the inversion volume's in Part D.

An error law quoted without its rate. All three $x + W(x)$ sources state $g(z) - G_N(z) \sim (N+1)^{-1}\big(W(z)/z\big)^{N+1}$ and stop there. The law is correct — Theorem L.13 proves it — but it is approached at relative rate only $\mathcal{O}(1/\log z)$, so it is a poor predictor at any accessible argument: the ratio of the true error to the stated law is 0.55 at $z = 10^3$, 0.72 at 10^5, 0.82 at 10^8, 0.88 at 10^{12} and 0.93 at 10^{20}, and at $z = 50$ the law is wrong by a factor of three. Theorem L.14 states the rate and recommends the uniform two-sided certificate instead.

A structure theorem stated in the wrong normalization. Two of the same three sources normalize the correction coefficients differently — one in powers of $q = W(z)/z$, the other in powers of z^{-1} — and the integer-polynomial structure theorem is true only of the second family. Carried across unchanged it fails at the first coefficient. Theorem L.17 records the dictionary $\mathcal{A}_n = A_n/L^n$, and the chapter uses throughout the normalization in which the theorem holds.

Five theorems stated without proof, demoted. The tutorials state real closedness of the transseries field, closure under antidifferentiation, a formal Taylor principle, a formal contraction principle and generalized Borel reconstruction as named theorems, three of them explicitly labelled "stated without proof" or "schematic". They are recorded in Section 2.4 as attributed quotations rather than as theorems of this volume, together with the observation — checked — that no result proved here depends on any of them.

A hand-wavy structure argument, replaced. The Bell sources assert that the saddle coefficients have the form $c_m = P_m/(1+r)^{3m}$ with $\deg P_m \leq 4m$ without exhibiting the cancellation that makes it true. Theorem U.4 proves it by tracking the factor r^K that cancels between the Touchard numerators and the denominators of the q_j, after which the degree count is exact and index independent.

An overstatement made by this merge, and corrected in it. Working from a title-level concordance, an intermediate stage of this consolidation recorded that the calculus already contained most of the inversion apparatus. Reading the statements showed otherwise, and Part K is the corrected comparison: of six apparent overlaps only one is a genuine duplicate, one is a strengthening in the inversion apparatus's favour, one has no counterpart at all, and three are distinct theorems. The episode is recorded in Theorem K.7 rather than quietly fixed, because the failure mode — treating shared vocabulary as evidence of shared content — is one a reader of a merged volume should be warned about.

Appendix C

Provenance of the inversion sources

This volume consolidates twenty-four independently written articles, three on each of its eight subjects. All twenty-four were filed in this repository on 3 September 2026 as a single quick-intake batch, at `drafts/series-and-transseries/special-function-inversion/`; the directories below are relative to that path. Nothing in the twenty-four was edited in place: this volume is a new document, and the sources remain available in git history.

Source directory	Source	PDF	Absorbed into
Chapter M: Butcher–Pólya rooted trees			
`Butcher_Tree_Transseries`	1 674 lines	21 pp, A4	Chapter M
`Butcher_Tree_Transseries-2`	2 853 lines	62 pp, A4	Chapter M
`Butcher_Tree_Transseries-3`	2 382 lines	35 pp, A4	Chapter M
Chapter N: the double factorial			
`Double_Factorial_Transseries`	1 552 lines	18 pp, A4	Chapter N
`Double_Factorial_Transseries-3`	2 287 lines	28 pp, A4	Chapter N
`double_factorial_transseries-2`	2 138 lines	44 pp, A4	Chapter N
Chapter O: the partition numbers			
`Partition_Number_Transseries_and_Asymptotic_Inverse`	1 901 lines	32 pp, A4	Chapter O
`Partition_Number_Transseries_and_Inverse`	2 020 lines	27 pp, A4	Chapter O
`Partition_Numbers_Transseries_and_Inverse`	1 905 lines	33 pp, Letter	Chapter O
Chapter P: the swing factorial			
`Swing_Factorial_Transseries`	1 885 lines	28 pp, A4	Chapter P
`Swing_Factorial_Transseries_Article`	1 824 lines	25 pp, A4	Chapters J and P
`Swing_Factorial_Transseries_and_Inverse`	1 389 lines	22 pp, A4	Chapter P
Chapter Q: Gamma and Barnes G			
`Asymptotic_Inversion_Gamma_Barnes_G`	2 376 lines	29 pp, A4	Chapters Q and R
`inverse_gamma_barnesG_transseries`	1 655 lines	28 pp, A4	Chapter Q
`inverse_gamma_barnes_transseries`	1 827 lines	25 pp, A4	Chapter Q
Chapter R: the hyperfactorial K-function			
`inverse_k_function_transseries`	2 259 lines	29 pp, A4	Chapter R
`K_Function_Inverse_Transseries`	2 644 lines	33 pp, A4	Chapter R
`K_function_inverse_transseries_article`	2 581 lines	30 pp, A4	Chapter R
Chapter S: the subfactorial			
`inverse_subfactorial_transseries`	2 631 lines	38 pp, A4	Chapter S
`inverse_subfactorial_transseries-2`	2 188 lines	27 pp, A4	Chapter S
`inverse_subfactorial_transseries-3`	1 878 lines	35 pp, Letter	Chapter S
Chapter T: a real-argument Fibonacci function			
`Fibonacci_Inverse_LogPeriodic_Transseries`	2 554 lines	33 pp, A4	Chapter T
`fibonacci_inverse_transseries_article`	2 857 lines	36 pp, A4	Chapter T
`fibonacci_inverse_transseries_article-2`	1 884 lines	24 pp, A4	Chapter T

The shared apparatus of Chapter J is drawn from all twenty-four. Its organizing normal form, and the Lagrange fixed-point route to the reversion coefficients, follow the presentation in `Swing_Factorial_`

`Transseries_Article` most closely; the α-reduction that brings the partition numbers into the same frame is this volume's, since no source needed it.

Appendix D

Repairs to the inversion sources

Every correction this volume makes to one of its twenty-four sources is boxed in the text where it occurs and listed here. A repair is recorded when a source states something false, states an asymptotic conclusion that its argument does not support, or asserts a named result without proof. Differences of convention are not repairs and are recorded in the text as remarks instead.

D.1 Chapter J: the common apparatus

A proof that does not apply (five sources). Swing_Factorial_Transseries_Article, Partition_Number_Transseries_and_Inverse (twice), Partition_Numbers_Transseries_and_Inverse, Partition_Number_Transseries_and_Asymptotic_Inverse and double_factorial_transseries-2 all prove the Lagrange fixed-point formula $[t^n]\, U = \sum_{m \le n} \frac{1}{m}[t^n u^{m-1}]\, \Psi^m$ by introducing an auxiliary marker z, solving $U = z\Psi(t, U)$, and applying Lagrange–Bürmann in z. Lagrange–Bürmann requires the relevant $\varphi(0)$ to be invertible; here $\varphi(0) = \Psi(t, 0) \in tR[[t]]$ is a non-unit, and the residue proof genuinely breaks, since $\psi = w/\varphi(w)$ is then not an element of $w\, R[[t]][[w]]$. The identity is true; Theorem J.18 proves it by universality instead. Every reversion in this volume depends on the lemma, so the repair is structural rather than cosmetic.

An asymptotic claim that is false uniformly. Several sources state, or leave the reader to infer, that the integer inverse satisfies $N_*(y) = \nu(y) + o(1)$. No continuous function approximates a staircase uniformly: Theorem J.43(5) shows $\sup_{y \ge Y} |N_*(y) - g(y)| \ge \frac{1}{2}$ for every continuous g and every Y. The volume asserts only the exact ceiling identity, nearest-integer recovery *along the range*, and the residue-class refinement.

A normalization constant. The sources record $P_n(0) = D_n$ for the flattened blocks, in a case where $a = 1$. The general statement is $P_n(0) = a^{n+1} c_n$ (Theorem J.37(3)); since $a \ne 1$ in every chapter of this volume the factor is needed throughout.

Two truncation conventions in one article. Swing_Factorial_Transseries_Article uses the exclusive convention in its forward error table and the inclusive one in its inverse table, so those two tables are not on a common scale. The volume is inclusive throughout (Theorem J.1); tables inherited from that article's forward section are re-indexed.

A claim of this volume's own, corrected. The brief under which Part J was written asserted that the case $\alpha = 1$ of Theorem J.2 covers the double factorial. It does not: $\log\big((x + 1)!!\big) = \frac{s}{2}(\log s - 1) + \mathcal{O}(\log s)$ with $s = x + 1$, and $s \log s / s^\alpha \to \infty$ for every $\alpha \le 1$. The sources never claimed membership. Theorem J.8 axiomatizes the dominant core instead, and Theorem J.27 inverts the factorial phase exactly.

Corrupt source text, not reproduced. Swing_Factorial_Transseries_Article equation `eq:inverse-Stokes-leading` and Butcher_Tree_Transseries-2 lines 2482–2483 contain corrupt control sequences. The affected statements are re-derived here rather than copied.

D.2 Chapter N: the double factorial

Convergence asserted without a threshold. The first source states that the periodic carrier "has the convergent local Lagrange–Bürmann expansion" for sufficiently large Λ, with no threshold and no proof of convergence as opposed to formal solvability; the second proves only a contraction estimate, from which convergence of the Lagrange series does not follow. Theorem N.46 separates the two: existence and uniqueness of the real root by contraction for $\Lambda > 2\pi a$, and convergence of the Lagrange series under the explicit sufficient condition $\mu = 2a/\Lambda \leq 1/5$.

Two assertions in place of proofs. The deepest-pole coefficient ("only the quadratic kernel and the first Bernoulli coefficient can contribute") is exactly the step needing the degree bookkeeping, and is supplied here by induction on the Λ^{-1}-degree. The coefficientwise sign law is argued in one source with "use a positive rooted-tree contribution …; such a tree exists", exhibiting no such object; Theorem N.37 keeps the correct positivity reduction and replaces the strictness step with an explicit attainability induction, verified symbolically for $n \leq 7$.

A formal-to-analytic slide. The third source presents an all-orders phase reversion as a theorem with an asymptotic conclusion, introduced by "setting $\tau = 1$ as an asymptotic expansion". The identity is formal; making it asymptotic needs estimates, uniform in the oscillatory phase, that are neither stated nor proved. It is demoted here to the formal Theorem N.50, with a proved remainder supplied separately.

A silent three-way inconsistency. One source centres the periodic inverse at the even-branch constant, the other two at the parity mean; the leading displacements consequently differ by $2a/\Lambda$. Both are correct for their own centring, but no source says so, and a reader comparing them would conclude that one is wrong. The volume adopts the mean centring and states the relation.

A notation collision across three sources. The letter C denotes the multiplicative amplitude, the logarithmic constant, and the branch prefactor in the three siblings respectively, and α, β carry two incompatible meanings. A crosswalk table fixes the volume's usage; the core parameters are never abbreviated, to avoid a fourth collision with κ in Theorem J.8.

A qualitative warning made quantitative. All three sources warn that a different smooth interpolation "can change this sector" without exhibiting one. Theorem N.8 exhibits the family $\mathcal{D}(x)\mathrm{e}^{\lambda \sin \pi x}$, which interpolates $n!!$ at every non-negative integer and changes the leading oscillatory amplitude from $2a/\Lambda$ to $2\sqrt{a^2 + \lambda^2}/\Lambda$ — a change larger than the entire Bernoulli grid. This is what makes the chapter's organizing claim a theorem rather than a caution.

D.3 Chapter M: the rooted-tree numbers

Digits printed beyond the accuracy of the computation. One source prints the constants of the inherited $-\sqrt{\rho}$ sector to forty decimal digits, but only about twenty-five of them are correct. Two independent recomputations here — including a from-scratch direct summation at eighty digits — give

$$r_0^- = 0.4686366279248455480872026219318193l4\ldots,$$

and the volume prints only digits it has verified. This is the one place in the twenty-four articles where a printed numeral is wrong; it is a precision failure rather than a mistake in a formula.

An audit appendix that contradicts its own main text. The same source's audit appendix reports $D_3 = -3.582177326832123\ldots$, disagreeing with the value its main text derives. The main text is correct; the appendix entry is not, and the volume uses the recomputed value.

A symbol used for two different constants. One source writes α for the growth constant where the other two write λ, and the volume's Chapter J uses α for the phase exponent of Theorem J.2. The chapter fixes λ for the growth constant throughout and never uses α for it.

An asymptotic statement with a formal proof. A source states its inverse-coefficient theorem asymptotically but proves it only formally, and separately displays a global transseries with no hypothesis under which it could hold. The first is proved here in its asymptotic form; the second is replaced by a conditional theorem with a finite radius and an explicit hypothesis.

A circular derivation of the singularity bounds. The elementary bounds $\frac{1}{4} \leq \rho \leq \mathrm{e}^{-1}$ are used in the sources in a way that presupposes what they are used to prove. They are established here independently, from $x \leq T\mathrm{e}^{-T} \leq \mathrm{e}^{-1}$ and the Catalan bound, and the critical characterization is then obtained non-circularly.

D.4 Chapter P: the swing factorial

A divergent series written as an equality. One source sets its bookkeeping coupling to 1 and prints the resulting operator expansion as an equality. There is then no small parameter, the sum diverges, and a number is equated to a divergent series; as printed the statement is false. The m-th summand is $\mathcal{O}(X^{-(2m-1)})$, so the correct statement is asymptotic, and regrouping gives the finite differential form used here. The other two sources state the same expansion correctly.

A theorem whose hypothesis is never used. A sector-transport result is labelled a theorem, its hypothesis is not used in the argument, and its conclusion is prefaced "formally". It is split here into the formal all-orders identity, which is what the proof gives, and a quantitative first-order statement proved by the mean value theorem under stated hypotheses.

An overclaim about complex branches. One source asserts that every Lambert branch with $|w| \to \infty$ yields an inverse branch with the same coefficients. For $k \neq 0, -1$ the existence of an analytic branch of the inverse along the path is a hypothesis, not a consequence, and it fails at gamma poles and Stokes directions. The claim is demoted to a remark with the two hypotheses displayed.

The analytic inverse statement, unproved in all three sources. One source writes "likewise, for each fixed M, …" with no proof; the second sketches an argument that never bounds the residual of the truncated ansatz nor uses forward control at the perturbed point; the third proves the ingredients but never assembles them. This is the one formal-to-analytic slide common to all three. Theorem P.41 proves the statement in full, with an effective constant.

A recommendation contradicted by its own table. A remark asserts that the centred normalization is "often the preferable computational normal form". The same article's table shows the centred core better by a factor $\approx 10^3$ at one end and the uncentred form slightly better at the other. The corrected statement is that centring kills the X^{-1} term and so improves the *bare core* by a factor $\asymp X$, while with several corrections retained the two are comparable and neither dominates uniformly.

Two convergence claims without criteria. An equivalence of the Lambert and logarithmic normalizations is asserted in prose with no formula and no proof; it is proved here as a finite explicit composition. A convergence claim appeals to "the classical analysis of Lambert W", but the classical result concerns a doubly indexed series, whereas the series at hand has coefficients polynomial in ℓ and converges only when ℓ/L is small; an explicit criterion is proved instead.

D.5 Chapter O: the partition numbers

A definition that contradicts its own use. The source abbreviated S2 in Part O defines the real Rademacher amplitude by a sum over $1 \leq h \leq k$ with $\gcd(h, k) = 1$. For $k \geq 2$ that is the intended index set, but at $k = 1$ it returns $\cos(2\pi\nu)$ rather than the constant 1, while the same article uses $A_1 \equiv 1$ throughout. With the printed definition the interpolant would not interpolate p: its dominant term would oscillate off-lattice at full relative strength, and the article's own eventual-monotonicity statement would be false. The volume takes $h \in U_k$ with $U_1 = \{0\}$; every later formula of that source is then correct as printed.

An understated radius. S1 states only that its universal sector-coefficient series "converges for all sufficiently small $|t|$". The radius is exactly $\sqrt{24}$ in $t = n^{-1/2}$, the nearest singularities being the branch points of $\sqrt{1 - t^2/24}$; convergence therefore holds for every real $n > 1/24$.

A convergent series and an asymptotic total, conflated. All three partition sources are loose about the distinction. The algebraic series $\sum_m \omega_m n^{-m/2}$ converges, but its sum is the leading Rademacher block, *not* $p(n)$; the Poincaré statement about $p(n)$ carries an error that is the sum of a convergent algebraic tail and an exponentially small Rademacher remainder. Summing the algebraic series to convergence does not compute $p(n)$: the residual stalls at the $\mathrm{e}^{-\mu/2}$ floor.

A false intermediate bound. S1's kernel estimate asserts $g(z) \leq z^2/3$ on $[0, 1]$, which fails at $z = 1$ where $g(1) = \mathrm{e}^{-1} = 0.3679\ldots > 1/3$. The chapter proves $g(z) \leq (z^2/3)\cosh z$ instead, which is what the tail bound actually needs.

An overclaim in this volume's own Chapter J. Theorem J.3 originally said that the correction series Q is divergent of Gevrey type "in all four chapters". It is not: for the partition numbers $Q(t) = \log(1 - t)$ has radius of convergence 1. The remark now distinguishes the three divergent chapters from Part O, where Theorem J.54 has no content and the error floor is set by the Rademacher remainder instead.

The second wave: Chapters Q–T

An existence claim that no source makes effective. All six power–logarithmic sources justify the eventual inverse of Barnes G and of the K-function with the words "$(\log G)'(x) \sim x \log x$, hence strictly increasing for all sufficiently large x". That is true and ineffective: none of the six supplies a threshold, and yet every numerical table in all six evaluates the inverse at arguments as small as $z = 5$, inside the range their own justification does not cover. Theorem Q.6 proves that $z \mapsto G(z + 1)$

is strictly increasing on $[3, \infty)$ with slope at least $\frac{1}{2} z \log z$ on $[7, \infty)$, and Theorem R.2 does the same for K on $[2, \infty)$ with the two stationary points located. The tabulated arguments are then legitimate.

A hypothesis stronger than the proof needs. S1 of Part Q states its triangular inversion theorem under the hypothesis that the support semigroup be *locally finite*, meaning that every bounded interval meets it in a finite set. That is strictly stronger than well-ordering — $\{1 - 1/n\} \cup \{1\}$ is well ordered and not locally finite — and more than the argument uses, which is only the finite-decomposition property. Neumann's lemma supplies that for every well-ordered set, so Theorem Q.13 is stated with the weaker hypothesis. For the two subjects of that chapter the distinction is invisible, since the supports are $2\mathbb{N}$ and $\mathbb{N}$; it matters for the fractional and mixed grids of Section Q.12.

A limit presented as a resummation. S1 of Part Q writes that the leading constant subsector of the Barnes coefficients "resums" to $\sqrt{T^2 + 1/6}$, and its next remark correctly warns that this does not replace the expansion; the two statements sit awkwardly together. The accurate form is Theorem Q.35: a limit of coefficients, each taken separately as the slope grows, which is a structural identity and an exact check on a symbolic computation but not an approximation to the inverse — at any actual argument the neglected terms, headed by $b_0 = -\alpha/\lambda$, are the largest corrections present. Theorem Q.36 states this, and Theorem R.15 then shows what does turn the limit into an approximation: moving it into the coordinate.

A carrier rederived three times. All three subfactorial sources need the inverse of $\Gamma(x+1)/\mathrm{e}$ as the carrier for everything else, and all three derive it from scratch, each proving its own Lambert-centred reversion theorem and each printing the blocks $1/(24s)$, $-(14s^2 + 10s + 5)/(5760s^3)$, $(2232s^4 + \cdots + 105)/(2903040s^5)$. These are identical, coefficient for coefficient, to d_1, d_3, d_5 of Theorem Q.26. The block d_5 is in fact printed in seven of the twelve second-wave sources. Theorem S.10 states the dictionary $u(Y) = \Gamma_+^{-1}(\mathrm{e}Y) - 1$ and cites Theorem Q.25; the general reversion theorem the three sources prove to get there is Theorem J.21. This is not an error in any source — each is correct in isolation — but it is the largest single duplication the consolidation removes.

An unverified radius, quoted by neither. V1 of Part T states an exact radius of convergence for the Lagrange–Good series. The proof of Theorem T.8 establishes convergence for R beyond a constant depending only on ρ, by a crude bound on the binomial factors, and the sharp radius was not verified for this volume. No value is therefore quoted, and the question is recorded as open in Section T.10(ii). This is listed here because a reader comparing the volume with its source will find a number in one and not the other, and the reason is deliberate.